MODULE-1: BLOOD - FUNCTIONS AND PROPERTIES

Learning objectives

- This module deals with
  - composition and functions of blood
  - properties of blood

BLOOD

- Blood is a fluid connective tissue that flows all over the body in the vessels of the cardiovascular system.

FUNCTIONS OF BLOOD

- **Transportation**
  - Transport oxygen from lungs to tissues and carbon dioxide from tissues to lungs
  - Transport nutrients from GI tract to liver and other cells
  - Transport hormones from endocrine glands to target cells
  - Transport waste products from cells to excretory organs
- **Regulation**
  - Maintains homeostasis
  - Regulates pH / acid base balance
  - Regulates body temperature and water balance
  - Regulates osmotic pressure
- **Protection**
  - Provides immunity

COMPOSITION OF BLOOD

- Blood is composed of fluid in which the cells and cell fragments are suspended. The fluid portion is known as the *plasma*, and the suspended cellular elements are the
  - Erythrocytes (RBCs)
  - Leukocytes (WBCs) and
  - Thrombocytes (platelets)
- Plasma constitutes about 55 to 70% of the blood volume, while the cellular components account roughly about 30-45% of the total blood. It performs numerous functions that are essential for life and health.

SPECIFIC GRAVITY OF BLOOD

- Specific gravity is the ratio of weight of a given volume of a fluid to the weight of the same volume of distilled water measured at 25°C.
- Plasma protein concentration is largely responsible for the specific gravity of the plasma. The cellular elements called the corpuscles have higher specific gravity than the plasma.

Species	Specific gravity
Goat	1.042 (1.036 - 1.051)
Pig	1.045 (1.035 - 1.055)
Dog	1.048 (1.045 - 1.052)
Cat	1.050 (1.045 - 1.057)
Sheep	1.051 (1.041 - 1.061)
Cattle	1.052 (1.046 - 1.061)
Horse	1.053 (1.046 - 1.059)

- The copper sulphate drop falling method is commonly used for the assessment of specific gravity of blood and plasma. Various factors influence the specific gravity of the blood.

Specific gravity is increased in the following conditions

- Polycythemia: High altitude, polycythemia vera, newborns etc.
- Severe dehydration: Excessive fluid loss such as in vomiting, diarrhoea.
- Hemoconcentration: Loss of plasma as in burns.

Specific gravity is decreased in the following conditions

- Pregnancy: Increase in plasma volume causes hemodilution.
- Anaemia.
- Renal diseases: Loss of albumin, oedema etc.
- Hemodilution: Hypersecretion or prolonged treatment with glucocorticoids.
- Starvation and malnutrition: Decrease in plasma protein.
- Intravenous fluid transfusions.

VISCOSITY OF BLOOD

- Viscosity is influenced by concentration of RBCs and plasma proteins. Among plasma proteins, the viscosity is mainly contributed by the *gamma globulins* which provide the resistance to blood flow and helps the normal pumping activity of the heart.
- It is normally about five times greater than water. Viscosity of blood is determined using **viscosimeter**.
- An *increase* in the viscosity of blood is seen in conditions like polycythemia, congestive heart failure, jaundice, vomiting, diarrhoea etc, whereas a *decrease* in the viscosity is commonly noticed in anemia, oedema etc.

PLASMA

- It is the fluid portion of the blood. It is yellow to colourless depending on the quantity, species of the animal and animal's diet.
- The plasma colour is golden yellow in dog, colourless or slightly yellow in sheep and goat, while it is highly yellow coloured in horse and cow which is chiefly due to bilirubin, and to some extent by the carotene, xanthine and other pigments.

Icterus index

- It is a measure of the *yellow colour of the plasma*. It is measured by comparing the colour of the plasma with a standard solution of potassium dichromate.
- Plasma is obtained by adding anticoagulant to whole blood to prevent clotting and centrifuging it for settling down of the cells.

Composition of Plasma

- Plasma is made up of water to the extent of 91 - 92%; Solids : 8 - 9%; the solid portion of plasma includes proteins: 6 - 8 g %: Organic and inorganic compounds: 2 - 3 %

Organic	Carbohydrates : Glucose, Lactate, Pyruvate
Inorganic	Macro level **:** Fe, Cl, SO_4, PO_4, Micro level : Mn, Co, Cu, Zn, Iodide.
Plasma proteins	Albumin, Globulin, and Fibrinogen
Non Protein Nitrogenous (NPN) compounds	Urea, Uric acid, Creatine, Creatinine, Salts of Ammonia
Lipids	Neutral fat, Cholesterol, Phospholipids, Lecithin
Others	Enzymes, Hormones, and Vitamins

SERUM

- In the absence of the anticoagulants, the blood comes out of the blood vessels and gets coagulated to form a blood clot. On shrinking it discharges a clear watery liquid called the *serum***.**
- Serum is also obtained by collecting the blood without the addition of anticoagulant and allowing it to clot. Later, upon shrinkage of the blood clot the serum is extruded.
- It differs from the plasma in lacking fibrinogen, prothrombin and other coagulation factors which are involved in blood coagulation.

REACTION OF THE BLOOD

- The normal pH of the blood is 7.4.
- The venous blood is slightly towards acidic side because of the increased content of CO_2, whereas the arterial blood is slightly more towards alkaline side.
- The plasma is more alkaline than the corpuscles.
- The blood pH range:
    - Dog : 7.32- 7.68
    - Cattle : 7.35 - 7.50
    - Horse : 7.35 - 7.43
    - Fowl : 7.56.
- Excessive production of metabolic acid products causes abnormal reduction in the alkaline reserve leading to the condition referred to as ***acidosis*****,** whereas abnormal increase in the alkaline reserve is called as ***alkalosis***.

MODULE-2: PLASMA PROTEINS AND LIPIDS

Learning objectives

- This module deals with
  - types of plasma proteins
  - functions and normal concentration of plasma proteins

PLASMA PROTEINS

- Plasma proteins contribute about 7% of the total organic molecules of the plasma. The major proteins present in plasma are:
  - Albumin (55%),
  - Globulins (38 %) and
  - Fibrinogen (7 %).
- Liver functions as a chief site of synthesis of plasma proteins, albumin, fibrinogen, prothrombin and some of the globulins. While the gamma globulins are synthesised extra-hepatically in the lymph nodes and in the Mononuclear Phagocytic System (MPS) of spleen and bone marrow.
- The ratio of albumin to globulin is 1:0.7. There are species variations in the ratio of plasma proteins *i.e.* albumin: globulin ratio. In humans, sheep, goats and dogs, the albumin predominates over globulin.
- In horses, pigs, cow and cats albumin globulin ratio is equal. In new born animals (except rodents and primates plasma gamma globulin is absent or found in minute quantity.

Separation of plasma proteins

- *Salting out method:* Using different concentrations of ammonium sulphate solution.
- *Electrophoresis:* Paper, starch gel, polyacrylamide gel electrophoresis are used for the separation of plasma protein fractions.

PLASMA PROTEIN FRACTIONS

Prealbumin

- Transports thyroxine and Vitamin A.

Albumin

- Two fifth is intravascular and the rest occur extravascularly. It provides colloidal osmotic pressure because of its high concentration and low molecular weight and non-diffusible property through blood vessels, thus prevents excessive passage of fluid from the blood into the interstitial tissue and serves to control the fluid balance between blood and tissues.
- It contributes to the amino acid pool with tissue protein and helps in transport of some anions and cations.

Globulins

- Includes fractions α1 and α2, β1 and β2 and γ globulins which include IgA, IgD,IgE, IgG and IgM.
    - *α - globulins* act as carrier proteins of bilirubin, lipids, steroids and thyroxine.
    - *α_1 globulins* are protease inhibitors and functions to inhibit WBC elastase and other WBC proteases and activated factor XI. The α_1 lipoproteins transport lipids, fat soluble vitamins and hormones.
    - *α_2 globulins* are the macroglobulins and inhibit plasmin, thrombin and kallikrein.
    - *β - globulins* are*:*
        - *Ceruloplasmin:* Transports copper.
        - *Ferroxidase:* Transferrin transports iron.
        - *Haptaglobulin*: Transports Hb to liver after haemolysis.
        - *Apolipoproteins:* Transports triglycerides, phospholipids and cholesterol.
        - *Hemopexin:* Aids the transport of heme from lysed RBCs.
        - *Transcortin:* Transports cortisol.
        - *Transcobalamins I & II:* Transports cobalt.
    - *γ - globulins*
        - They provide viscosity to the blood, thus maintain normal blood pressure. Synthesised by -lymphocytes and plasma cells, functions to provide immunity and also helps transport of vitamin D.
        - The γ globulins are the immunoglobulins (Ig) and are classified into the following types:
        - *IgG* is responsible for most of the *humoral immunity* of the organism. It can cross placenta.
        - *IgM* is the second major immunoglobulin of the serum and it is typically the first immunoglobulin increase in concentration in serum during *primary immune response*. These are naturally occurring antibodies against erythrocytes in certain incompatible blood types.
        - *IgA* is a glycoprotein found in external secretion such as saliva, tears, colostrum etc. It forms the primary immunoglobulin in the colostrum and is responsible for natural passive immunity in the neo-natal calf, foal, lamb, kid and piglets. Plays an important role in local defence by protecting various body surfaces e.g., the intestinal, respiratory and uro-genital tracts, mammary gland and the eyes from bacterial and viral invasions. IgA does not cross the placenta. It is most abundantly found in normal animals.
        - *IgD* is involved in B-cell differentiation to form 'clones'.
        - *IgE* is involved in hypersensitivity and allergic responses. It causes the release of histamine from basophils and mast cells.

Fibrinogen

- Fraction from the liver, functions as a precursor to form a mesh work of fibrin threads and play a key role in blood coagulation.
- It influences the suspension stability of the erythrocytes. Increased concentration of fibrinogen and globulins alter the colloidal state of the blood, hastens agglutination of RBCs and settling.

FUNCTIONS OF PLASMA PROTEINS

1. Function as source of amino acids for the synthesis of tissue proteins. The amino acids of plasma protein and tissue proteins are in a state of dynamic equilibrium.
2. Provide colloidal osmotic pressure and helps to regulate fluid balance. Around 80% of the colloidal osmotic pressure is contributed by albumin.
3. They act as blood buffer and regulate acid-base balance, thus maintains *normal* pH of *7.4*.
4. Fibrinogen and various clotting factors are essential for coagulation of blood.
5. Influence the suspension stability of RBCs in the blood.
6. Contributes to the *viscosity* of plasma (by *gamma globulins*), thereby providing for peripheral resistance, which is essential for efficient cardiac function.
7. As carrier proteins they are invloved in transport of copper, iron, heme, bilirubin, thyroxine, cortisol, sex hormones, vitamin A, vitamin D, fatty acids, triglycerides, phospholipids and cholesterol.
8. Immunoglobulins provide specific antibody against specific antigen.

PLASMA PROTEIN CONCENTRATION

Species	Total protein	Albumin	Globulin	Fibrinogen
Horse	6.0 - 8.0	2.8 - 3.8	2.8 - 3.8	0.2-0.4
Cow	7.0 - 8.5	3.0 - 3.8	3.6 - 4.4	0.2-0.5
Sheep	6.0 - 8.0	3.5 - 4.5	2.5 - 3.5	0.2-0.4
Goat	6.5 - 7.5	3.7 - 4.5	2.4 - 3.2	0.2-0.5
Pig	6.5 - 8.5	3.5 - 4.0	3.5 - 4. 0	0.2-0.4
Dog	6.0 - 7.8	3.4 - 4.4	2.7 - 3.2	0.1-0.4
Cat	6.0 - 7.4	3.0 - 3.8	2.5 – 3.5	0.1 – 0.4
Fowl	4.0 - 5.2	1.2 - 3.8	2.3 - 3.3	--

PLASMA LIPIDS

- **Lipids** are heterogenous molecules soluble in organic solvents but not in water. They are esters of fatty acids, formed by the reaction of fatty acids with glycerol.
- **Blood lipids** (or blood fats) are lipids in the blood and they are present either free or bound to other molecules. Blood lipids are mainly fatty acids, triglycerides, lipoproteins and cholesterol.
- Since lipids are insoluble in water they are mostly transported in a protein covering, and the density of the lipids and type of protein determines the fate of the lipid and its influence on metabolism.

- The concentration of blood lipids depends on dietary intake, absorption from the intestine and excretion and uptake and secretion from cells.

FATTY ACIDS

- Fatty acids are present in different forms {as in chylomicrons, Very low density lipoproteins (VLDL), Low density lipoproteins (LDL)} in blood. In addition, the fatty acids released from adipocytes exist in the blood as free fatty acids.
- Short- and medium chain fatty acids are absorbed directly into the blood via intestine capillaries and travel through the portal vein. Long-chain fatty acids are too large to be directly released into the intestine capillaries. Instead, they are coated with cholesterol and protein (protein coat of lipoproteins) into a compound called as *chylomicrons*. The chylomicrons enter into lymphatic capillary and then into the bloodstream (having bypassed the liver).
- The concentration of blood fatty acids increase temporarily after a meal which increases the uptake of fatty acids in different cells of the body like liver cells, adipocytes and muscle cells. This uptake is stimulated by insulin from the pancreas.
- Some of the fatty acids taken up by the liver is converted into VLDL and again secreted into the blood.
- When the concentration of fatty acids in the blood decreases, this triggers adipocytes to release stored fatty acids into the blood as free fatty acids, in order to supply the energy for the muscle cells and other cells.

LIPOPROTEINS

- Lipoproteins are complex aggregates of lipids and proteins that increases the solubility of lipids and enable their transport throughout the body.
- Lipoproteins are synthesised mainly in the liver and intestines.
- The most abundant lipid constituents of lipoproteins are triacylglycerols, free cholesterol, cholesterol esters and phospholipids (phosphatidylcholine and sphingomyelin); fat-soluble vitamins and anti-oxidants are also transported in this way.
- The lipoproteins contain different protein components called apoproteins (or apolipoproteins). Apoproteins are required to solubilise the non-polar lipids in the circulation. These proteins determine the overall structures and metabolism, and their uptake in liver and peripheral tissues.
- Lipoproteins are classified as chylomicrons (CM), very-low-density lipoproteins (VLDL), low-density lipoproteins (LDL) and high-density lipoproteins (HDL), based on the relative densities on ultracentrifugation.
- Based on the relative mobilities on electrophoresis, lipoproteins can also be classified into α, pre-β and β lipoproteins that correspond to HDL, VLDL and LDL, respectively.

PHYSICAL PROPERTIES AND LIPID COMPOSITIONS OF LIPOPROTEIN CLASSES

	CM	VLDL	LDL	HDL
Density (g/ml)	0.94	0.94-1.006	1.006-1.063	1.063-1.210
Total lipids (wt%)	99	91	80	44
Triacylglycerol	85	55	10	6

Cholesterol esters	3	18	50	40
Cholesterol	2	7	11	7
Phospholipids	8	20	29	46

- Lipoproteins are spherical in shape with the core containing non-polar lipids, triacylglycerols and cholesterol esters, and a surface layer consisting of phospholipids and non-esterified cholesterol, which also serve to present a hydrophobic face to the aqueous phase.
- The principal role of the chylomicrons and VLDL is to transport triacylglycerols as a source of fatty acids from the intestines or liver to the peripheral tissues.
- *Chylomicrons:* They are largest of the lipoproteins; they are formed in the intestinal mucosa and transport triacylglycerols from the intestinal mucosa to the liver or to other tissues; they carry mostly fats in the form of triglycerides and cholesterol. In the liver, chylomicron particles release triglycerides and some cholesterol.
- *VLDL:* They are synthesised in liver. The liver converts the excess fatty acids (dietary or synthesised) into very low density lipoproteins (VLDL) and secretes them into plasma; in the plasma they are converted to intermediate density lipoproteins(IDL), and then to low-density lipoprotein (LDL) particles. VLDL transports triacylglycerols from liver to peripheral tissues.
- *LDL:* It is the main transporter of cholesterol to the peripheral tissues. When they are present in excess they are strongly associated with the formation of atheromatous (plaque formation within arterial wall) disease within the arteries. For this reason, LDL is referred to as "bad cholesterol".
- *HDL:* They are synthesised in liver. The HDL removes excess cholesterol from peripheral tissues and delivers them to the liver for excretion in bile in the form of bile acids, a process that has been termed 'reverse cholesterol transport', helps to lower blood cholesterol level and hence HDL is commonly called "good cholesterol".
- After being transported to the liver by HDL, cholesterol is delivered to the intestines via bile production. However, 92-97% is reabsorbed in the intestines and recycled via enterohepatic circulation

CHOLESTEROL

- It is lipid, waxy steroid found in the cell membranes and transported in the blood plasma of all animals.
- It is an essential component of mammalian cell membranes where it is required to establish proper membrane permeability and fluidity.
- In addition, cholesterol is an important precursor molecule for the biosynthesis of bile acids, and several fat soluble vitamins, Vitamin D_3 and the steroid hormones, including the adrenal gland hormones such as cortisol and aldosterone as well as the sex hormones like progesterone, estrogens, and testosterone and their derivatives.
- Within the cell membrane, cholesterol also functions in intracellular transport, cell signalling and nerve conduction.
- Since cholesterol is essential for life, it is primarily synthesized *de novo,* within the body, with smaller contributions from the diet.
- Excessive levels of cholesterol in blood circulation however are strongly associated with progression of atherosclerosis.

- The fate of cholesterol in the blood is highly determined by its constitution of lipoproteins - some types (LDL) favour transport towards body tissues and others (HDL) towards the liver for excretion into the intestines.
- The average amount of *blood cholesterol* varies with age, typically rising gradually as age advances.
- *Hyperlipidemia* refers to the elevated or abnormal levels of cholesterol and triacylglycerols in the blood. One of the most clinically relevant lipid substances is cholesterol, especially on atherosclerosis and cardiovascular disease.
- *Lipemia* is a term used to refer marked hyperlipidemic condition resulting in plasma presenting a milky white appearance.
- Postprandial hyperlipidemia is most common in animals consuming fatty diet; hence to evaluate lipid status, blood samples have to be taken from fasting animals.
- Hyperlipidemia is noticed in dogs and man with hypothyroidism – main lipid increased is cholesterol. Pancreatitis, uncontrolled diabetes, cholestasis, hyperadrenocorticism are some other conditions causing hyperlipidemia.
- *Hypercholesterolemia* is the presence of high levels of cholesterol in the blood. It is not a disease but a metabolic derangement that can be secondary to many diseases and can contribute to many forms of disease, most notably cardiovascular disease.

Sources

- Animal fats are complex mixtures of triglycerides with phospholipids and cholesterol; hence, all foods containing animal fat contain cholesterol to varying extents.
- Cholesterol is not present in plant based food sources. However, plant products such as flax seeds and peanuts contain cholesterol-like compounds called *phytosterols*, which are suggested to help lower serum cholesterol levels.
- About 20–25% of total daily cholesterol production occurs in the liver; other sites of high synthesis rates include the intestines, adrenal glands and reproductive organs.
- Synthesis within the body starts with one molecule of acetyl CoA and one molecule of acetoacetyl-CoA. HMG-CoA reductase is an important enzyme in the cholesterol synthesis. Total cholesterol is the sum of HDL cholesterol, LDL cholesterol and 20% of triglycerides.

Blood cholesterol levels in human beings

Type	Level	Remarks
Total cholesterol	< 200 mg/dl	Desirable level. Low risk for coronary disease
	240 & above	Hypercholesterolemia, high risk for coronary heart disease
HDL cholesterol	<40 mg/dl	Low HDL levels is the risk factor for cardiac disease
	60 & above	High HDL, protective to heart
LDL cholesterol	< 100 mg/dl	Desirable level
	> 160 mg/dl	High, risk of heart disease
Triglycerides	< 150 mg/dl	Normal
	>200 mg/dl	High

Total blood cholesterol levels (mg/100 ml) in animals

Species	Blood cholesterol levels (mg/100 ml)
Horse	75-150
Cow	60-190
Sheep	52-90
Goat	80-130
Pig	80-134
Dog	135-270

MODULE-3: ERYTHROCYTE (RBC)

Learning objectives

- This module deals with
  - Functions and composition of erythrocytes
  - Shape, structure and size of erythrocytes

FUNCTIONS OF ERYTHROCYTES

- Transport hemoglobin which in turn carries oxygen from the lungs to the tissues.
- RBCs contain large quantity of carbonic anhydrase which catalyzes the reversible reaction between carbon dioxide and water. Thereby makes it possible for the water of the blood to transport enormous quantities of carbon dioxide from the tissues to the lungs in the form of bicarbonate.
- Hemoglobin in the cells is an excellent acid base buffer. Therefore, the RBCs are responsible for most of the acid base buffering power of whole blood

SHAPE OF RBC

- The mammalian RBCs are usually non-nucleated and non-motile cells, biconcave circular disc with central pale spot. Its shape differs in various species of animals:
  - *Dog, Cow, Sheep:* Markedly biconcave.
  - *Horse and cat:* Shallow concaving,
  - *Goat:* Very shallow or flat surfaced erythrocytes.
  - *Camel and Deer:* Elliptical and sickle shaped.
  - *Cold blooded animals (amphibians and birds):* Elliptical and nucleated.

Significance of biconcavity of RBCs

- It increase the surface area thus facilitating the exchange of oxygen and carbon dioxide carried by the RBCs.

STRUCTURE OF RBC

- The cell membrane of RBC is made up of lipids (*lecithin, cephalin and cholesterol*) and glycoprotein encloses a spongy inner structure called the *stroma*.
- The RBC membrane protein, *Spectrin* forms the inner lining of the membrane, whereas the outer layer is formed by glycoproteins, have the blood group antigens. The cell membrane is highly permeable to lipid soluble substances, glucose, urea and water.
- Hemoglobin is deposited in the inter-spaces of the spongy stroma. The surface of mature erythrocyte is smooth, while the immature RBCs have relatively rough surface.

SIZE OF RBC

- Average diameter of RBC ranges from 4.1 to 7.5 µm .

Species	Size (µm)
Goat	4.1
Sheep	5.0
Cattle	5.6
Horse	5.6
Pig	6.2
Cat	6.5
Dog	7.3
Man	7.5

- Surface area, varies from 57-67m^2 / kg body weight in mammals. It is lowest in goat (lesser diameter) and highest in man (greater diameters).

COMPOSITION OF ERYTHROCYTES

- Erythrocyte contains 62 - 72% water and 35% solids. Of the solids, 95% is contributed by Hb and the remaining 5% by cell and stromal protein, lipids, phospholipids, cholesterol, cholesterol esters, neutral fat and vitamins.

RBC METABOLISM

Energy is required for RBCs to

- To maintain the shape and flexibility of the cell membrane.
- To preserve high K^+, low Na^+ and low Ca^{++} ions within the RBCs against the concentration gradient of these ions of plasma.
- To maintain iron in ferrous (Fe^{++}) state (to reduce ferric to ferrous state, NADH and NADPH are required).
- To generate reduced glutathione (anti-oxidant); this helps to maintain the ferrous state.

- To generate 2,3 Diphospho glycerate (DPG) for O_2 dissociation.
- *Mitochondria are absent in mature erythrocytes*. These cells derive their energy from glucose metabolism via anaerobic Embden- Meyerhoff (EM) pathway (90%) and oxidative pentose cycle (10%) which produce NADH and NADPH. Kreb's cycle is very much reduced in activity.

CONCENTRATION OF RBCs

- The concentration of RBC depends on various factors such as interspecies, intraspecies and diurnal variation, age, sex, environment, exercise, nutritional status, climate and altitude.

Concentration of RBC in domestic animals (millions/ mm^3 of blood)

Species	Concentration of RBC
Fowl	3.0 (2.8 – 3.2)
Pig	6.5 (5.8- 8.0)
Dog	6.8 (5.5-8.5)
Sheep	12.0 (8.0-16.0)
Cattle	7.0 (5.0-10.0)
Goat	13.0 (8.0-18.0)
Horse	6.5 (6.5-12.5)
Cat	7.5 (5.0-10.0)
Man	5.4 (5.0-6.0)
Women	4.8 (4.0-5.0)

MODULE-4: ABNORMALITIES OF RBC CONCENTRATION

Learning objectives

- This module deals with
  - Abnormalities of RBC concentration
  - Erythrocyte indices

ABNORMALITIES OF RBC CONCENTRATION

1. Polcythemia
2. Oligocythemia
3. Anaemia

POLYCYTHEMIA

- It is otherwise known as *erythrocytosis*. It is a condition of increased number of RBCs in the circulation. It is of two types.
  - Physiological (secondary) polycythemia

- An increase in RBCs occurs as a compensatory measure (in high altitude of 14000 to 17000 feet to compensate low PO_2). Whenever tissues become hypoxic because of too little oxygen in the atmosphere, for example at high altitude or because of failure of delivery of oxygen in the tissues as in cardiac failure, then the blood forming organs automatically produce large quantities of extra RBCs i.e 30% above the normal.
- Increased Hb requirement during heavy muscular exercise to meet increased oxygen demand. In sports animals (racehorse, hunting dogs) RBC elevation is a normal feature.
- Increased environmental stress / temperature, the spleenic contraction, and increased RBC synthesis by the bone marrow cause increased number of RBCs into the circulation.
- Hemoconcentration due to water loss that occurs in vomiting, diarrhoea, prolonged high fever and burns also causes polycythemia.

o Pathological polycythemia
- Due to decreased O_2 supply to the tissue, chronic carbon monoxide poisoning, myeloid (bone marrow) cancer, pulmonary emphysema, repeated hemorrhage.
- Polycythemia vera is the condition due to bone marrow cancer (myeloid leukemia). It occurs as a result of genetic aberration in the hemocytoblastic cell line that produces the blood cells.

OLIGOCYTHEMIA

- Reduction in the number of erythrocytes in the circulation is called as oligocythemia
- *Physiological oligocythemia* occurs due to hemodilution; RBC number per unit volume is reduced. **Example:** *pregnancy*.
- *Pathological oligocythemia* is also known as *anaemia*.

ANEMIA

- Abnormal reduction in the number of the erythrocytes or the hemoglobin content in the blood or both.

Causes

- Excessive whole blood loss occurs in hemorrhage or by blood sucking parasites (Hookworms, ticks), increased destruction of RBCs by the reticuloendothelial cells.
- Impaired RBC production and Hb synthesis, due to deficiency of Fe, Cu, Vitamin B12 and folic acid.
- Hemolytic:
  - Disease caused by blood parasites, (babesiosis) or drugs like sulphanamides, antimalarial drugs and high doses of aspirin (analgesic)

Anemia due to defective blood formation:

- Aplastic anemia
  - It occurs due to lack of functional bone marrow caused by excessive x-ray treatment or bone marrow cancer, certain industrial chemicals, drugs etc.

- Anemic anemia
  - *Megaloblastic anemia:*
    - It is due to deficiency of iron, folic acid, Vitamin B_{12} (extrinsic factor) and intrinsic factor of the gastric mucosa.
  - *Microcytic and hypochromic anemia:*
    - It results due to deficiency of iron results in small sized, decreased number of RBCs and low Hb content.
  - *Macrocytic and hyperchromic anemia:*
    - Lack of extrinsic factor, the Vitamin B_{12} causes decreased number of RBCs, large sized RBCs and high Hb content because the erythroblasts cannot proliferate rapidly enough to form normal number of RBCs, the cells that are formed are mostly oversized, bizarre in shape and have a fragile membrane.
  - *Pernicious anemia:*
    - It occurs due to the deficiency of the intrinsic factor of the gastric mucosa that interferes with the Vitamin B_{12} absorption.

Anemia due to excessive blood loss or increased RBC destruction:

- Hemorrhagic anemia:
  - Excessive blood loss due to accident, peptic ulcers etc.
- Hemolytic anemia:
  - Following acute destruction of RBCs (haemolysis) the number of RBCs is below normal, but the RBC size and Hb content are normal, known as normocytic and normochromic anemia.

Causes

- Blood parasites: Eg. babesiosis, theileriosis and trypanosomiasis;
- Chemicals: Copper, lead, nitrate and nitrite.

Anemia due to abnormal structure of RBC :

- In some hereditary diseases the defects are with the RBC membrane - e.g., *sickle cell anemia*, defects in the globin chain structure (*thalassemia*) or its synthesis or the deficiency of the enzymes of the RBCs energy system, the pyruvate kinase and glucose 6 phosphate dehydrogenase (G.6-PD).
  - Sickle cell anemia:
    - In this type of anemia the cells contain an abnormal type of hemoglobin called as *Hb "S"*. It is caused by abnormal composition of β chains of the hemoglobin.
    - When this type of hemoglobin is exposed to low concentration of oxygen it precipitates into long crystals inside the erythrocytes. These crystals elongate the cell and it gives the appearance of sickle shape.
    - The precipitated hemoglobin also damages the cell membrane so that the cells become highly fragile leading to anemia.
  - Thalassemia:
    - It is otherwise known as *Cooley's anemia or Mediterranean anemia*. It occurs due to defect in the synthesis of α or β peptide chains to form

hemoglobin or due to deficiency of enzymes of the RBC energy system, pyruvate kinase and G-6-PD thereby depressing the hemoglobin synthesis.

ERYTHROCYTIC INDICES

- These indices help in the diagnosis various types of anemia (microcytic vs. macrocytic or normocytic).
- Mean Corpuscular Volume (MCV):
  - It expresses the average cell size of the erythrocyte.

- Mean Corpuscular Hemoglobin (MCH):
  - It gives the average weight of Hb present in the erythrocytes.

$$MCH\ (pg) = \frac{Hb\ (g\%) \times 10}{No.\ of\ RBCs\ /\ mm^3 \times 106}$$

- Mean Corpuscular Hemoglobin Concentration (MCHC):
  - It is the average percentage of the mean corpuscular volume that the Hb occupies.

$$MCHC\ (g\%) = \frac{Hb\ (g\%) \times 100}{PCV\ /\ dl}$$

NORMAL RANGE OF ERYTHROCYTE INDICES IN DOMESTIC ANIMALS

Species	MCV (fl)	MCHC (%)	MCH (pg)
Dog	60-77 (70)	32-36 (34)	20-24
Cat	39-55 (45)	30-36 (33)	13-17
Cow	40-60 (52)	30-36 (33)	19
Sheep	23-48 (33)	31-38 (33)	10-14
Goat	15-30 (23)	35-42 (38)	8
Horse	34-58 (46)	31-37 (35)	18
Pig	50-68 (63)	30-34 (32)	16-20

MODULE-5: HAEMOLYSIS

Learning objectives

- This module deals with lifespan and fate of erythrocytes.

LIFE SPAN OF ERYTHROCYTES

- Life span of erythrocytes (days)

Cattle	Sheep	Goat	Horse	Dog	Cat	Pig	Poultry
125-150	140-150	125-150	140-150	100-120	70-80	51-79	20-30

SITE OF DESTRUCTION OF ERYTHROCYTES

- In most of the domestic animals bone marrow functions as a chief site of destruction of erythrocytes, whereas in man it is the spleen.
- In the birds liver acts as an organ of destruction of erythrocytes.

FATE OF ERYTHROCYTES

- The erythrocytes have a remarkable capacity to change their shape when they pass through the capillaries but they become less deformable when they reach the end of their life span.
- Two types of destruction of erythrocytes takes place,
    1. Intravascular hemolysis
    2. Extravascular hemolysis

INTRAVASCULAR HEMOLYSIS

- About 10% of aged RBCs undergo intravascular hemolysis within the capillaries due to loss of compressibility of RBCs caused by increased membrane permeability and osmotic change.
- When this occurs the hemoglobin is released, which combine with haptoglobulin which is removed by the cells of the mononuclear phagocytic system (MPS).

EXTRAVASCULAR HEMOLYSIS

- About 90% of the aged RBCs are directly destroyed by the mononuclear phagocytic system (MPS).

- The Hb and proteins are catabolised by the MPS cells. The MPS (also known as reticulo-endothelial system) includes the histiocyte or macrophages, stellate or Kupffer cells of the sinusoids of the liver, spleen, mononuclear cells of bone marrow and lymph nodes.
- The globin of the Hb is degraded to amino acids and is reutilized. Iron removed from the heme is stored in the MPS cells in the form of ferritin or hemosiderin and utilised for the synthesis of hemoglobin or enters the plasma and combine with apotransferrin to form transferrin. The transferrin enters the bone marrow to produce more erythrocytes.
- The heme is converted to bile pigments, biliverdin (a green pigment) and then reduced to bilirubin (a yellow pigment). The free bilirubin enters the plasma, binds with albumin and transported to liver. In the liver bilirubin is conjugated with glucuronic acid, secreted in bile to enter intestine. Large intestinal bacteria reduce the bilirubin to urobilinogen, most of that are excreted in feces in the oxidised form of urobilin or stercobilin which impart colour to feces.
- Part of the urobilinogen is reabsorbed into the enterohepatic circulation and reexcreted in bile. Some of the urobilinogen in the plasma enters the kidneys to be excreted in urine as urobilin.
- Globin protein portion of hemoglobin is broken down to amino acid and used in the formation of new hemoglobin or other proteins.

Hemolysis caused by external agents like

- *Blood parasites:* Babesiosis, theileriosis, trypanosomiasis and sarcocystosis.
- *Chemicals:* Copper, lead, nitrate and nitrite poisoning.

MODULE-6: ERYTHROPOIESIS

Learning objectives

- This module deals with
  - Erythropoiesis
  - Regulation of erythropoiesis

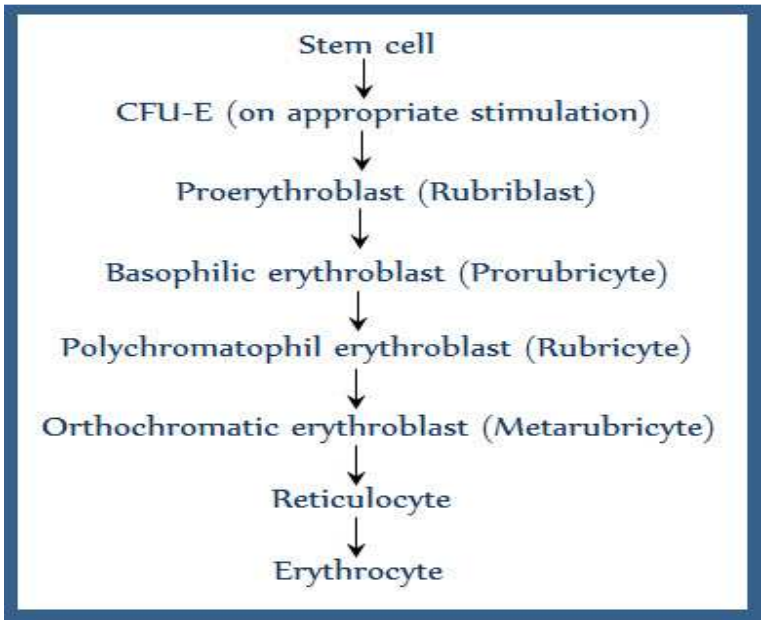
HEMATOPOIESIS

- It is the processes of formation of erythrocytes, leukocytes and platelets in the body. Formation of erythrocytes and leukocytes respectively are known as erythropoiesis and leukopoiesis.
- During embryonic state the blood islands of pander of the yolk sac functions as a site of hematopoiesis. The mesenchymal cells of the liver, spleen, bone marrow and lymph glands are the hemopoietic organs in early fetal life.
- The bone marrow *is* concerned with the production of erythrocytes, granulocytes and platelets during postnatal life, whereas the lymphocytes production occurs in the lymphoid tissues of lymph glands, Payer's patches of intestine, spleen and thymus.
- The lymphoid tissues of the bone marrow and also the spleen are the sites of production for monocytes. In ruminants, hemolymph nodes (hemal) functions as spleen. It takes part in the erythropoiesis during the foetal period, while granulopoiesis, is more prevalent in postnatal life.

- The mesenchymal cells of the yolk sac produce primitive stem cells, which give rise to the pleuripotent stem cells (colony forming units - CFU). These stem cells give rise to five different blast cells, *viz.*
    - Proerythroblast to form RBC
    - Myeloblast to form neutrophils, eosinophils and basophils
    - Monoblast to form monocyte
    - Lymphoblast to form lymphocyte
    - Megakaryoblast to form platelets. Depending on the microenvironment, *i.e.*, the location of the stem cells and the growth factors, the stem cells differentiate into progenitor cells of different blood cells (Committed Stem Cells-CSC). A CSC that produces erythrocytes is called colony-forming unit-erythrocyte (CFU-E). Similarly, CFU that produce granulocytes and monocytes are designated as CFU-GM.
- The stem cells continue to divide throughout the life of the animal and a part of the cells remains as pleuripotent stem cells and retained in the bone marrow to maintain supply of stem cells.
- The pleuripotent stem cells differentiate to form the CSC. Several hemopoietic growth factors and differentiation factors stimulate the growth and differentiation of these stem cells into a particular progenitor cells.
- Cytokinins are the growth factors that regulate the formation of blood cells. Two cytokinins that stimulate red cell and WBC formation are the colony stimulating factors and interleukins. Erythropoietin increases erythrocyte precursor formation.

ERYTHROPOIESIS

- From stem cell, the formation of reticulocyte takes about 72 hours and conversion of reticulocyte to erythrocyte requires 48 hours. Thus RBC formation requires 5 days time.
- Under appropriate stimulation, CFU-E progenitor cells produce proerythroblast. Hb synthesis begins in polychromatophil erythroblast and maximum synthesis occurs in orthochromatic erythroblast.
- The metarubricyte ejects the nucleus to become the reticulocyte that contains some mitochondria, ribosomes and endoplasmic reticulum. In 1-2 days, they develop into erythrocytes and enter circulation.



REGULATION OF ERYTHROPOIESIS

- The level of oxygen in the tissue is the principle regulatory factor of erythropoietic activity of the bone marrow. The kidney cells, during hypoxia, releases erythrogenin (erythropoietin releasing factor) from the glomeruli, which in turn acts on erythropoietinogen, an $\square_2$ globulin of plasma and converts it into free erythropoietin (hemopoietin). Kidney produces 90% of erythropoeitin and liver produces about 10%.
- Erythropoietin as a hormone stimulates hemopoietic stem cells of bone marrow to produce the committed stem cells-proerythroblast, thus initiates erythropoiesis. It stimulates,
    - The proliferation of rubriblast by mitosis in the developing rubricytes.
    - Accelerates maturation of the rubricytic cells.
    - Induces the release of reticulocytes into the circulation.

THE ROLE OF VITAMINS AND MINERALS IN ERYTHROPOIESIS

1. Vitamin B_{12} and folic acid are essential for the maturation of erythrocytes. Vitamin B_{12} is required for DNA synthesis and folic acid for RNA synthesis. Macrocytic anemia is a very common in Vitamin B_{12} and folic acid deficiencies.
2. Thiamine (B1), Pantothenic acid, Nicotinic acid, Vitamin E and pyridoxine (B_6), riboflavin, biotin and ascorbic acid are essential for erythropoiesis. Deficiency of Vitamin B_6 causes *microcytic hypochromic anaemia* in pigs. Pantothenic acid deficiency results in deficiency of ALA synthatase in birds and animals. Normocytic anemia in swine and primates is due to Vitamin E deficiency.
3. Minerals such as iron, copper and cobalt are essential for erythropoiesis. Iron acts as an integral part of Hb which is absolutely essential for Hb synthesis. Copper acts as a co-

factor in ALA dehydrase in Hb synthesis. It is part of the enzyme ferroxidase which is necessary for oxidation of ferrous iron to ferric form and is necessary for the incorporation of iron into Hb. *Copper deficiency is common in pigs*, which may interfere with Fe absorption and utilization. In ruminants cobalt plays a key role for the synthesis of Vitamin B_{12} by the rumen bacteria which in turn is required for the normal production of erythrocytes.

RETICULOCYTE

- A low percentage (1 to 3%) of erythrocytes in circulation has a network of bluish threads within the cell and is called *reticulocytes.*
- These cells are immature RBCs, which have entered into the circulation at times of need from blood forming tissues.
- In some diseases or due to excessive loss of blood or destruction of RBCs, the reticulocytic number increases in circulation. These cells have less or no O_2 carrying capacity.

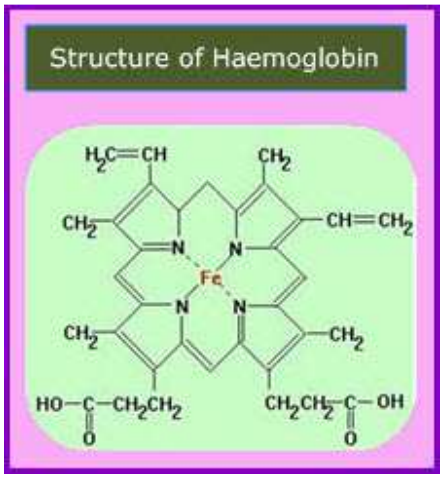
MODULE-7: HAEMOGLOBLIN

Learning objectives

- This module deals with
  - Hemoglobin and its properties
  - Types of hemoglobin
  - Regulation and concentration of hemoglobin

HEMOGLOBIN

- Hemoglobin is the iron-containing oxygen-transport metalloprotein present in the red blood cells of all vertebrates that carries oxygen from the lungs to the tissues and carbon dioxide from the tissues back to the lungs.
- The oxygen carrying capacity of hemoglobin is 60% more than that of plasma.
- It also functions as a buffer in the regulation of acid base balance.
- A hemoglobin molecule is a complex substance consisting of conjugate protein composed of a pigment heme and a protein, globin.
- The protein portion of each of these chains is called "globin". The a and b globin chains are very similar in structure. In this case, **α** and **ß** refer to the two types of globin.
- The globin is a conjugated protein and heme contains iron in ferrous state.
- Hemoglobin contains four polypeptide chains namely two alpha and two beta chains. Each of the four chains unites with a heme group resulting in a hemoglobin molecule.



HEMOGLOBIN BIOSYNTHESIS

- Hemoglobin (Hb) is synthesized in a complex series of steps.
- The heme part is synthesized in a series of steps in the mitochondria and the cytosol of immature red blood cells, while the globin protein parts are synthesized by ribosomes in the cytosol.
- Production of Hb continues in the cell throughout its early development from the proerythroblast to the reticulocyte in the bone marrow. At this point, the nucleus is lost in mammalian red blood cells, but not in birds and many other species. Even after the loss of the nucleus in mammals, residual ribosomal RNA allows further synthesis of Hb until the reticulocyte loses its RNA soon after entering the vasculature.

Heme synthesis

- Heme is synthesized in a complex series of steps involving enzymes in the mitochondrion and in the cytosol of the cell. The first step in heme synthesis takes place in the mitochondrion, with the condensation of succinyl CoA and glycine to form delta-amino levulinate. This molecule is transported to the cytosol where a series of reactions produce a ring structure called protoporphyrin IX.
- Many enzymes concerned with heme synthesis are intra-mitochondrial, limited to erythroid precursors including reticulocytes.
- The ALA synthetase is the rate-limiting enzyme of the Hb synthesis, present within the mitochondria.

Globin synthesis

- After heme is synthesized within the mitochondria, 4 heme molecules combine with 4 globin polypeptides to form one molecule of haemoglobin.
- The globin molecule of haemoglobin differs among the species, whereas there is no difference in the heme portion.

FACTORS INFLUENCING HEMOGLOBIN SYNTHESIS

- The amount of Hb in the blood is influenced by age, sex, muscular activity, season, excitement etc. Erythropoietin stimulates RNA and DNA synthesis, the cell division, heme synthesis and hemoglobin production.
- At an oxygen pressure (PO_2) of 100 mm of Hg in the lung, the Hb forms loose and reversible combination with oxygen, the oxyhemoglobin, but at low oxygen pressure of 40 mm of Hg at tissue level, it readily releases oxygen to the tissues for complex metabolic process.

CONCENTRATION OF HEMOGLOBIN IN VARIOUS SPECIES

Species	Hb (g/dl)
Dog	12-18 (15)
Cat	8-15 (12)
Cow	8-15 (11)
Sheep	8-16 (12)
Goat	8-14 (11)
Horse	11-19 (15)
Pig	10-16 (13)

TYPES OF HEMOGLOBIN

- Based on physiological functions, the hemoglobins are typed as adult hemoglobin and fetal hemoglobins. Electrophoretically, the Hbs are classified as HbA, HbB, HbC and HbF. Human beings show three types of Hb, HbA (98%), HbA2 (2%) in the adult and HbF in fetus and new born. HbA has 2 μ -chains and 2 ß -chains; HbA2 is represented by 2 μ and 2 delta - chains. HbF has 2 μ -chains and 2 γ -chains.
- In adult sheep HbA (2 μ, 2 ß) is electrophoretically fast and has higher O2 affinity than HbB with 4γ chains. Sheep having HbA or HbB under anemia or hypoxic condition develop another type of Hb, the HbC, which partially or completely replaces the HbA. Such a change is also observed in goat. HbC is the naturally occurring Hb in sheep during growth period. HbF has higher affinity for O2. In many animal species fetal hemoglobin (HbF) is replaced by the adult types within 4 to 8 weeks after birth. In adult cat the HbA and HbB are found in the same erythrocyte.
- Some of the Hb variants, HbS, HbC, HbE are associated with specific hematologic disorders. HbS is responsible for sickle-cell anaemia in Negro race. HbC and HbE cause failure of synthesis of alpha or beta chains thus results in alpha or beta thalassemia.

DERIVATIVES OF HAEMOGLOBIN

Oxyhemoglobin

- Hemoglobin has an important physiological relationship with oxygen. Oxygen forms loose and reversible combination with hemoglobin called oxyhemoglobin when the erythrocytes passes through the pulmonary capillaries. Since there are four ferrous atoms in the hemoglobin molecule, four molecules of oxygen are transported by a molecule of hemoglobin.

$$Hb + 4O_2 \longleftrightarrow Hb4O_2$$

- As blood is transported through the systemic capillaries hemoglobin loses its oxygen to the tissues and becomes hemoglobin again. Hemoglobin shows progressive increase in the affinity for O_2 after the first two molecules of O_2 are taken up by the heme.
- The oxygen carrying capacity of the hemoglobin is dependent on the pigment it contains which in turn depends on the iron content for oxygen combining capacity.
- The amount of iron present in the blood is minute about 0.334% of the hemoglobin molecule or 0.04 to 0.05% of the blood itself. Each gram of Hb combines with a maximum of 1.34 ml of oxygen.
- In the lung (PO_2 100 mm Hg) the oxygen binds with Hb which shows 97% saturation. One hundred ml of blood containing approximately 15 grams of Hb, can carry approximately 19.4 ml of oxygen.
- In the tissue capillaries, (PO_2 40 mm Hg) Hb is 72% saturated and contains 14.4 ml of oxygen per 100 ml of blood, which indicates oxygen release from the Hb into the tissues. Thus under normal resting conditions about 5 ml of oxygen is transported by each 100 ml of blood during each cycle to the tissues. During heavy exercise this is increased to about 15 times normal. The oxygen hemoglobin dissociation curve is "S" shaped or sigmoid shaped.

Myoglobin (*Muscle hemoglobin*)

- It is a true hemoglobin and functions to store oxygen in the muscle. It contains only one heme and a polypeptide chain. It contains only one iron atom and can therefore store only one molecule of O_2.
- Its molecular weight is approximately 17,000, which is four times less than Hb. Hence it can pass through glomerulus.
- The oxygen dissociation curve with myoglobin is hyperbolic (very steep). The appearance of myoglobin in the urine is referred to as myoglobinuria or azoturia, which is a very characteristic symptom of monday morning sickness in horse.

Carboxyhemoglobin (*HbCO*)

- Hb has 200 times more affinity for carbon monoxide than oxygen.

Hb+ CO □ HbCO

- Carbon monoxide firmly attaches with the Fe^{++} molecules of heme, thus interferes with the transport of O_2 as oxy Hb. 0.1% of CO in inspired air will convert 20% of Hb into HbCO within 30 to 60 minutes.
- Oxygen under higher partial pressure is the only means to reverse the reaction.

$$HbCO + \uparrow PO_2 \rightarrow Hb\,O_2 + CO$$

Methemoglobin (*ferrihemoglobin*)

- It is formed by the oxidation of ferrous iron to ferric iron. During the circulation of blood about 1% of methemoglobin is formed by the oxidation of ferrous iron to ferric iron. Ferrihemoglobin cannot combine with oxygen, hence is useless as a respiratory pigment in the blood. It produces dark colored blood. Glutathione (GSH) in the erythrocytes prevents the excessive oxidation of ferrous iron into ferric iron.
- Chemicals like nitrates, sulphanamides, aminophenol and acetanilide cause increased concentration of methemoglobin in the blood. Horse blood at normal conditions shows significant amounts of methemoglobin. The normal blood of dog and cat has about 1% of methemoglobin.

ABSORPTION SPECTRA

- When white light is passed through a solution of hemoglobin or one of its derivatives, certain wavelengths are absorbed. The resulting spectrum is termed as absorption spectra; the region of absorption is known as absorption bands.
- They can be seen by examining the solution with a spectroscope.
- When white light is examined spectroscopically, a series of colours known as spectrum (VIBGYOR) is obtained.
- When sun light is examined, certain black vertical lines called as Fraunhofer's lines are found at definite places in the spectrum; these lines are designated as A, B, C, D, E, etc. In lamp light, these lines are not seen.
- When hemoglobin solution or its derivatives are examined in certain concentrations spectroscopically, absorption bands of definite size, appearance, and position are noticed.
- Hence, spectroscopic examination helps to identify these pigments in solution. E.g. Dilute oxy-Hb solution shows two absorption bands between line D and E; adding a reducing agent (produces reduced-Hb) gives one band at line D.
- Carboxy-Hb shows two bands but adding a reducing agent does not produce a single band.
- Met-Hb shows a band between line C and D.

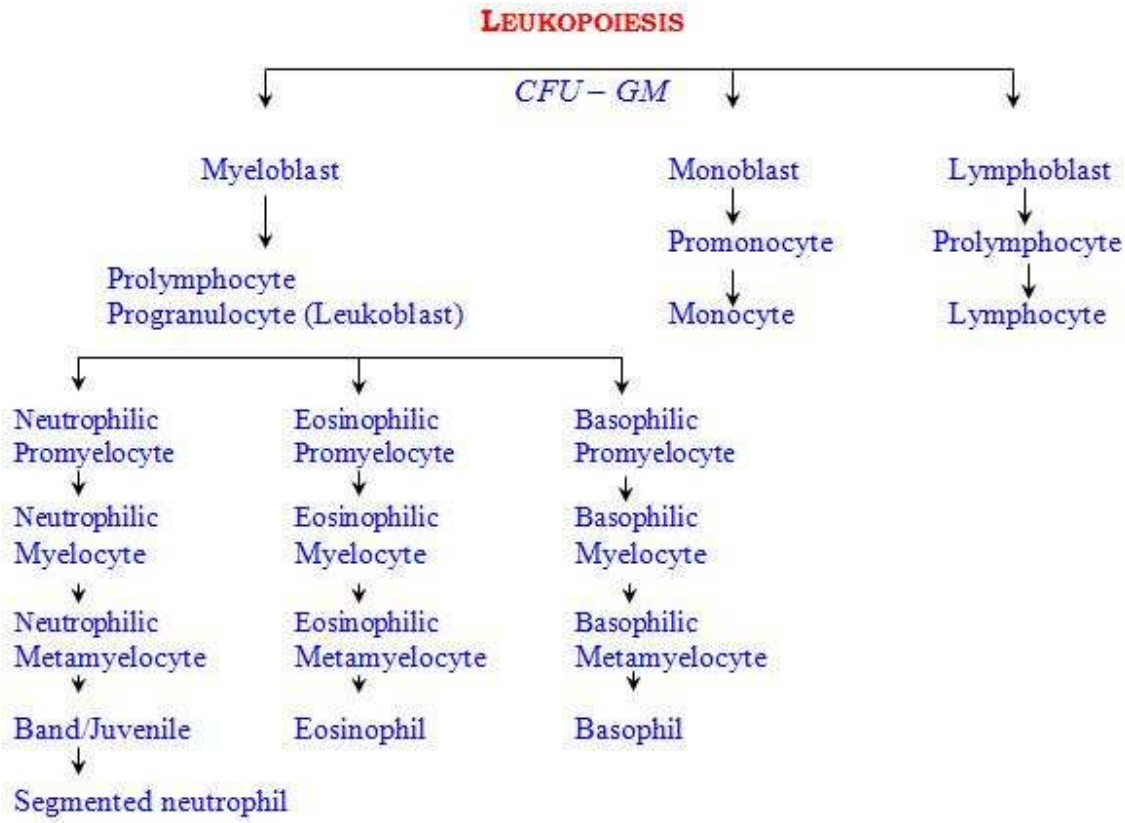
MODULE-8: LEUKOCYTES (WBC)

Learning objectives

- This module deals with
  - production of leukocytes
  - types of leukocytes
  - fate and concentration of leukocytes

LEUKOPOIESIS

- The process of formation of leukocytes is known as *Leukopoiesis*. Leukocytes are produced from the pleuripotent stem cells.



LEUKOCYTES

- Leukocytes are large sized, nucleated cells, lesser in number and do not have hemoglobin content. These cells show polymorphic forms, which differ morphologically and functionally.
- Majority of the WBCs is larger than RBCs; while RBCs are present in millions per ml of blood, the WBCs are present in thousands per ml of blood.
- The ratio of WBC to RBC varies from 1:100 in chicks to 1:1300 in goats, 1:600 in dog and cat, 1:800 in cattle 1:1000 in horse 1:1200 in sheep and 1:700 in man.

Types of leukocytes

- Based on the staining nature of the cytoplasmic granules when subjected to Leishman's stain, leukocytes are classified as granulocytes, which are further classified into three groups the neutrophils, eosinophils and basophils.
- Some of the leukocytes have cytoplasmic granules but they are non-visible even after staining with Leishman's stain, hence they are known as agranulocytes. The lymphocytes and monocytes form two groups of this type.

Site of production of WBCs

- The granulocytes and the megakaryocytes are produced from the myelocytes of the bone marrow hence often referred as myeloid series of blood cells.
- The lymphocytes and the plasma cells are formed from the lymphoid tissues such as lymph nodes, spleen and lymphoid tissues of the bone marrow, gut etc.
- The monocytes are produced by the mononuclear phagocytic system (MPS) cells of spleen and bone marrow (reticuloendothelial system cells).

NEUTROPHILS

- Neutrophils are produced in the bone marrow from the extravascular neutrophilic myelocytes. They are numerously found in the blood of most animals.
- The nucleus of mature cells is divided into lobes, usually five lobes and are stained blue or purple by Leishman's stain. They have abundant and fine granular cytoplasm which take up the neutral stain.
- These granules store lysosomes, which contfain hydrolytic enzymes, proteolytic enzymes and lipases to digest the invading organisms.
- The oxidative enzymes of the lysosome produce hydrogen peroxide, which attack the bacterial cell wall to cause bactericidal effect.
- Neutrophils are highly motile, responds to chemotaxis and actively phagocytic, thus serve as a first line of defence against invading organisms, (bacteria, viruses and cellular remnants).
- At the onset of infection neutrophils produce pyrogens which act on thermo-regulator centre of the brain results in fever. This rise in body temperature slows the reproduction process of bacteria and viruses
- Immature forms of neutrophils are characterized by unsegmented or less number of nuclear lobes and are referred to as *juvenile or band* cells. Pseudo neutrophils and *heterophils* are comparable to neutrophils, which are present in rabbit and poultry respectively.
- Heterophils contain large rod or spindle shaped granules, which are acid in reaction and stain red or pink with eosin.

Abnormalities

Neutrophilia

- It indicates more number of neutrophils in the circulation.
- Physiological neutrophilia occur in conditions like exercise, emotion, pregnancy, lactation and parturition.
- Abnormal or pathological neutrophilia may be due to acute inflammation following injury, surgery, burns, arthritis and acute infection by pyrogenic bacteria**.**
- Shift to left is a term used to describe an increase in the number of immature neutrophils (Band cells) in the circulation which is characteristic of bacterial infections. The shift to left is clinically helpful in diagnosis of traumatic reticulopericarditis (TRP).

Neutropenia

- It indicates reduction in neutrophils in the circulation, which is very common in viral infections and chronic infections like TB, brucellosis and protozoal and fungal infections.
- Injection of antiinflammatory drugs (cortisol) and antibacterial drugs (chloramphenicol and sulphanamides) may result in neutrophilia.

EOSINOPHILS

Eosinophils

- They are large cells which contain large cytoplasmic granules that stain red or purple eosin stain. They have bilobed nucleus, connected by a thin filament.
- The cytoplasmic granules contain enzymes, which are rich in oxidases, and peroxidases. Eosinophils are produced from the bone marrow and are highly motile, but less phagocytic.
- Eosinophils function to detoxify the proteins of the parasites, phagocytise antigen-antibody complexes, the inflammatory products of the mast cells and basophils.
- These cells have antiheparin, and anti- histaminic substances, thus act as an anti-inflammatory and anti- allergic agent. Eosinophils release profibrinolysin, which is then activated to fibrinolysin, and causes the dissolution of old blood clot.

bnormalities

- *Eosinophilia:* It is increased number of eosinophils in the circulation. This condition is common in G.I. parasitic infections, allergic disorders like bronchial asthma, allergic rhinitis, in skin diseases like eczema and dermatitis, drug reactions following penicillin and sulphanamides administration.
- *Eosinopenia:* It is decreased number of eosinophils in circulation. This condition occurs following stress , administration of ACTH or cortisol etc.

BASOPHILS

Basophils

- These cells are found in the blood in lower concentration. They have irregular shaped nucleus and the granules are stained blue by the basic dye of Leishman's stain. They originate in the bone marrow and have slight or no phagocytosis.
- Extravascularly, basophils enlarge and become tissue mast cells. Both basophils and mast cells release heparin (anticoagulant), and vasodilator substances histamine, serotonin and bradykinins. Thesevasodilators causes increased blood flow and reddish colour (hyperemia) at the site of infection followed by increased permeability of the plasma and proteins from the blood vessels, a process calledinflammation. Basophils and mast cells have receptors for immunoglobin E.

Abnormalities

- *Basophilia:* It is increase in basophils, and it is seen in allergic conditions and hematological malignancies.

LYMPHOCYTES

- These cells are characterised by well defined centrally placed nucleus, surrounded by cytoplasm. According to the size and the ratio of nucleus to cytoplasm, these cells are further divided into 3 types.
    - *Small lymphocytes:* These are small sized cells with a large notched nucleus.
    - *Medium lymphocytes:* These are medium sized, which have notched nucleus surrounded by thin rim of cytoplasm.
    - *Large lymphocytes:* These have comparatively small sized nucleus with broad rim of cytoplasm.
- Lymphocytes are actively motile cells with amoeboid movement and are mainly concerned with the development of immunity against specific disease organisms. They have no phagocytic properties.

Abnormalities

- *Lymphocytosis:* This condition indicates *increased* number of lymphocytes in the circulation, which is common in viral infection (eg) infectious hepatitis.
- *Lymphopenia:* This condition indicates *decreased* number of lymphocytes in circulation and is seen in cases of TB, acute stress and glucocorticoid injection.

MONOCYTES

- These are the largest of WBCs, characterised by kidney or bean shaped nucleus.
- Monocytes are actively motile, pinocytic and also highly phagocytic, which may migrate into the tissue spaces and become macrophages. These cells are much more powerful phagocytes than neutrophils by their ability to engulf much large sized particles, necrotic tissues and more number of microorganisms.
- Monocytes leave the blood and are attracted to the tissues by chemotaxis and lymphokines and they become macrophages, example reticular cells of lymph nodes, spleen, bone marrow and Kupffer cells of the liver sinuses.

Abnormalities

Monocytosis:

- This condition indicates increased number of monocytes in the circulation and is seen in hematological malignancies, endocarditis, typhoid, TB, brucellosis and prolonged blood parasitic infections. Eg: trypanosomiasis.

NORMAL VALUES OF LEUCOCYTES IN DOMESTIC ANIMALS

Animal	Total leukocytes (x10³/ cu.mm)	Differential Leukocyte Count (%)					
		Neutrophils		Lymphocytes	Monocytes	Eosinophils	Basophils
		Band	Mature				
Dog	6.0-17.0 (11.5)	0-3(0.8)	60-77 (70)	12-30 (20)	3-10 (5)	2-10 (4)	Rare
Cat	5.5-19.5 (12.5)	0-3 (0.5)	35-75 (59)	20-55 (32)	1-4 (3)	2-12 (5)	Rare
Cow	4-12 (8)	0-2 (0.5)	15-45 (28)	45-75 (58)	2-7 (4)	2-20 (9)	0-2 (0.5)
Sheep	4-12 (8)	Rare	10-50 (30)	40-75 (62)	0-6 (2.5)	0-10 (5)	0-3 (0.5)
Goat	4-13 (9)	Rare	30-48 (36)	50-70 (56)	0-4 (2.5)	1-8 (5)	0-1 (0.5)
Horse	5.5-12.5 (9)	0-2 (0.5)	30-65 (49)	25-70 (44)	1-7 (4)	0-11 (4)	0-3 (0.5)
Pig	11-22 (16)	0-4 (1)	28-47 (37)	39-82 (53)	2-10 (5)	1-11 (3)	0-2 (0.5)
Chicken	20-30	-	25-30	55-60	10	3-8	1-4

FATE OF WBCs

1. It is very difficult to assess the life span of leukocytes because the WBCs move between blood and tissue.
2. The life span of granulocytes is normally 9 days but once they are released into circulation from bone marrow their life span is 4 to 8 hours.
3. During infection, the life span in shortened. “B” lymphocytes live for 3-4 days only but “T” lymphocytes may live for 1-3 years in tissues and return to circulation many times.
4. The monocytes also have short transit time in the blood (24 hours) before migrating to the tissues where they become tissue macrophages and can live for months or even years.

ABNORMALITIES OF LEUKOCYTES

- Leukocytosis:
  - It is an increase in the number of leukocytes which is an indication of presence of some infection.

- Leukocytosis may be physiological (related to time of day, meal, exercise, epinephrine, stress) or pathological. Bacterial infection shows leukocytosis with neutrophilia, whereas viral infections result in neutropenia.

- Leukemia:
  - Pathologically, cancer of leukocyte producing tissues results in abnormally high white cell count which is known as *leukemia*.

PLASMA CELLS

- Plasma cells are white blood cells which produce large volumes of antibodies.
- They are transported by the blood plasma and the lymphatic system and originate from the bone marrow.
- They are otherwise known as plasma B cells , plasmocytes , and effector B cells .
- They are present in lymph nodes ,spleen and diffuse lymphoid tissue of alimentary and respiratory tract.

STRUCTURE OF PLASMA CELLS

- Plasma cells are large lymphocytes round in shape containing a granular cytoplasm which stains with basic dyes.
- The nucleus is eccentric in position and typically represents clumps of chromatin in a radiating manner, resembling a "cart -wheel" or clock face in appearance.
- They have a considerable nucleus-to-cytoplasm ratio.
- The cytoplasm contains a conspicuous Golgi apparatus and abundant endoplasmic reticulum .
- Immunoglobulins are localised in the endoplasmic reticulum where it sometimes forms aggregates called as Russel bodies.

FORMATION OF PLASMA CELLS

- When specific B-lymphocytes are stimulated by specific antigen, they enlarge and become lymphoblast, some of which further differentiate to immature plasmablasts which finally forms, the plasma cells.

FUNCTIONS OF PLASMA CELLS

- These cells are rarely seen in the circulation, which are formed by the lymphoid tissues (spleen).
- They play a very important function in body defence mechanism.
- They are involved in the production of humoral antibodies.
- Plasma cells are concerned with the synthesis, storage and release of immunoglobulins at a very rapid rate.
- A plasma cell can only synthesize an antibody of a single specimen, either IgM or IgG or IgA except in primary immune response when a plasma cell producing IgM initially, may later switch over to the synthesis of IgG antibody.
- Mature plasma cells are end cells and survive only a few weeks and die after a few cell division.

MODULE-9: IMMUNITY

Learning objectives

- This module deals with
  - immunity and its types
  - production of antibody and its mechanism of action

TYPES OF IMMUNITY

- It is the ability of the body to resist the tissue damage by destroying the disease organisms or neutralising their toxins.

Types of immunity

- Innate immunity
- Acquired immunity

INNATE IMMUNITY

- This type of immunity is *present from birth*. This type of immunity is non specific and act on many organisms. It prevents first entry of microorganisms into the tissues or if the organisms has gained entry, eliminates them prior to the occurrence of the disease.
- This type of immunity is not effective on subsequent exposure to the same organisms. Non specific elimination of microorganisms is done by two methods:
  - Phagocytosis: It is ingestion and killing of microorganisms by specialised cells such as neutrophils, monocytes, macrophages etc.
  - Opsonisation: It is the process of coating the microorganisms with some of the proteins found in the plasma to make them more phagocytosable.

ACQUIRED IMMUNITY

- The development of immunity against specific organisms, the bacteria, virus and foreign tissues or toxins is referred to as *acquired immunity*.
- This type of immunity is also known as *adaptive or specific immunity*. It is caused by a special immune system that forms antibodies or activated lymphocytes that attack and destroy the microorganisms.

TYPES OF ACQUIRED IMMUNITY

  - Humoral immunity or B cell immunity
  - Cell mediated immunity or T cell immunity
- In mammals, shortly before or after birth the stem cells or the committed lymphoid progenitor cells migrate either to thymus, liver and bone marrow (in mammals) or to *bursa of Fabricius* (in birds) where they are processed and differentiated.
- The `T' lymphocytes are processed in the thymus and `B' lymphocytes are processed in bursa, liver and bone marrow. These lymphocytes then get into the circulation and localised in spleen and lymph nodes as clones of `T', and `B' lymphocytes.

- Thymus plays a key role in the development of cellular immunity shortly before birth or few months after birth. The T-lymphocytes provide cellular immunity, and the B-lymphocytes provide humoral immunity, by producing specific antibodies. There are different types of B and T cells, each of which responds specifically to one particular antigen.
- All the different B and T cells capable of forming a specific antibody or activated T cells against a specific antigen which are called “clone of lymphocytes”.
- The invading organisms provide proteins, large polysaccharides and large lipoprotein complexes, which function as an antigen. The antigen stimulates specific clone of lymphocytes.
- Macrophages in lymphoid tissues stimulated by antigen releases interleukin I, and it promotes growth of specific lymphocytes. The helper T cells release interleukin II, which stimulate B cells to produce specific antibodies. Those B-lymphocytes specific for an antigen enlarge and form lymphoblasts, which differentiates to plasmablast and finally to plasma cells. The plasma cells produce antibodies very rapidly.
- When these sensitised plasma cells are activated by an antigen, specific antibodies, the gamma globulins- IgG (75%), IgM, IgA, IgD and IgE (during allergy) are produced by them as primary response. However these antibodies have weak potency and short life.
- Some of the lymphoblasts remain dormant and function as memory T and B cells, which on subsequent stimulation by the same antigen (booster) produce more potent secondary response. On exposure to a specific antigen, the T cells of the specific clone proliferate and release large numbers of activated T cells, which neutralises the antigen. The memory T cells are retained in the lymphoid tissues.
- Humoral immunity is developed as specific antibody (gamma globulin) by the “B”-lymphocytes, rarely persists more than few months or utmost few years. On the other hand, the specific clones of “T” lymphocytes on stimulation by the very same antigens are converted into three major groups of “T” lymphocytes.
- Cytotoxic or killer “T” cells: These are the direct attack cells killing the micro-organisms. They have the lysosomal enzymes, the peroxidases, phosphatases and oxygenases and produce H_2O_2, which is lethal to the invading organisms.
- Helper `T' cells: These cells are very sensitive even to very small quantities of the specific antigen, release interleukin-2 (lymphokines), and activate the other two types of “T” cells, the cytotoxic and suppressor “T” cells and also the “B” lymphocytes. Helper “T” cells are not capable of secreting antibodies by themselves. It promotes the production of lymphokines.
- Lymphokines act as a macrophage migration inhibition factor, stop the chemotactically attracted macrophages and neutrophils in the infected area to effect more efficient phagocytosis, and thus limit the spread of the infection.
- Suppressor or regulatory “T” cells: Regulate the activities of both the helper and killer “T” cells. The activated “T” lymphocytes provide the cell-mediated immunity, which is far more persistent for many years than humoral immunity.

ACTIVE IMMUNITY

- Active immunity involves both the cell-mediated and humoral immunity as well as the innate immune system .
- Active immunity is induced in the host itself by the antigen, and lasts for a longerperiod of time.

TYPES OF ACTIVE IMMUNITY

Naturally acquired active immunity

- Naturally acquired active immunity occurs when an individual is exposed to live pathogens, and develops antibodies against specific pathogens which leads to immunological memory.
- In some cases, the immunity may be life-long as in the case of small pox, measles, chicken pox, yellow fever etc. while in some others the immunity may be lost after only a few years (e.g., diphtheria, tetanus) or even for lesser period (e.g., influenza, pneumonia).

Artificially acquired active immunity

- Artificially Acquired Active Immunity occurs when an antigen is intentionally introduced into a body to be immunized, the latter develops immunity .e.g., vaccine, toxoids.
- This immunity is artificial because the antigens are intentionally or purposely introduced, and it is active because the recipients immune system synthesizes antibodies in response.
- Vaccines provide usually long-term immunity.
- Killed and attenuated strains of bacteria and viruses are widely used forms of immunization against many diseases

PASSIVE IMMUNITY

- Passive immunity is the transfer of active immunity, in the form of readymade antibodies, from one individual to another.
- Passive immunity provides immediate protection to the individual, but the duration of passive immunization is relatively short, a few days to several weeks when compared to years for active immunity. This is due to the natural degradation of injected antibody from the circulation without internal replacement.
- Passive immunity can occur naturally, when maternal antibodies are transferred to the fetus through the placenta, and can also be induced artificially, when high levels of human (or horse) antibodies specific for a pathogen or toxin are transferred to non- immune individuals.

TYPES OF PASSIVE IMMUNITY

Naturally acquired passive immunity

- When antibodies produced in the body of an individual (called donor) are naturally transferred into the body of other individual (called recipient), the latter develops immunity, called naturally acquired passive immunity, in its immune system.
- This immunity is natural because the transfer of antibodies from donor to recipient occurs under natural conditions, and it is passive because the recipient does not synthesize antibodies but picks them up from the donor.
- This type of immunity is significant mainly in the survival of the neonates. The neonate passively acquires antibodies from its mother. The antibodies may pass from the immune mother to the fetus across the placental barrier.

- Certain antibodies are also transferred from mother to young ones through colostrum and milk during lactation. These antibodies, called **maternal antibodies,** remain with the young ones for certain period and after the specified time the immune state disappears. The maternal antibodies generally provide resistance against diseases.

Artificially acquired passive immunity

- Artificially acquired passive immunity is a short-term immunization induced by the transfer of antibodies to another individual i.e, produced in one individual or animal and then administered to the other individual, which can be administered in several forms; as human or animal blood plasma, as pooled human immunoglobulin for intravenous (IVIG) or intramuscular (IG) use, and in the form of monoclonal antibodies (MAb).
- Currently passive immunization is mainly used for prophylaxis following exposure to such diseases as rubella and infectious hepatitis. This is usually accomplished by injecting the patient with gammaglobulin which has been extracted from the blood of immune persons.
- Artificially acquired passive immunity is immediate but short-lived (only for two to three weeks .
- Other examples of this type of immunization are injections of hyperimmune serum and antiserum.

ANTIBODY

- These are the gamma globulins or immunoglobulins made up of a combination of light and heavy polypeptide chains. Each heavy chain is paralleled by a light chain at one of its end, which form the variable portion.
- This variable region with both light and heavy chains provides very specific binding site to the specific antigen. Specific Antibodies differ from each other based on the different amino acid sequence in the light and heavy chains of the variable portion.
- On the other hand the free ends of the two heavy chains form the constant portion of the antibody which establishes the properties like diffusibility, adherence to tissue and attachment to complement complex.

MECHANISMS OF ACTION OF ANTIBODIES

- Agglutination in which antigens in the bacteria are bound by antibodies to form a clump.
- Precipitation in which the soluble antigen is rendered insoluble and precipitated.
- Neutralization, to cover the binding site of the toxin thus makes the virus non-virulent.
- Lysis, by the lysosome following the attachment with the antibody with the cell membrane of the invaders.
- Lymphocytes are motile, but not phagocytic in function. During viral infection, these cells come in contact with specific virus and release specific interferons, which inhibits the multiplication of that virus. Interferon also circulates throughout the body and protects other body cells from that virus.

NATURAL KILLER CELLS

- Natural killer cells (also known as NK cells, K cells, and killer cells) are a type of lymphocyte (a white blood cell) and a component of innate immune system.
- NK cells play a major role in the host-rejection of both tumours and virally infected cells.

- These cells are cytotoxic and contain small granules in their cytoplasm with special proteins such as perforin and proteases known as granzymes.
- When NK cells comes in close contact with target cells, perforin forms pores in the cell membrane of the target cell through which the granzymes and associated molecules enter, inducing apoptosis or death of the cell.
- NK cells are activated in response to interferons or macrophage-derived cytokines and control viral infections.

MODULE-10: BLOOD COAGULATION

Learning objectives

- This module deals with
  - mechanism of blood coagulation or blood clotting
  - mechanism of prevention of blood clotting in blood vessels

MEGAKARYOCYTE

- These are the myeloid cells of the bone marrow, rarely seen in the circulation as giant cells.
- Its fragments provide the platelets or thrombocytes.

PLATELETS OR THROMBOCYTES

- They are disk-shaped structures, derived from the megakaryocytes of the bone marrow.
- Mammalian platelets are non-nucleated, whereas they are nucleated in birds and reptiles.
- In mammals, platelets originate from the megakaryocytes of the bone marrow.
- In birds they arise from large mononucleated cells of bone marrow.
- Thrombocytes are about 3 µm in diameter.

STRUCTURE OF PLATELETS

- The thick glycocalyx of the platelet cell membrane contains clotting factors like fibrinogen, factor V, VII and IX. The phospholipid of the cell membrane is the precursors for prostaglandins and platelet factors.
- Platelet cytoplasm have two types of granules; the *alpha and dense granules*. They also have contractile fibrils containing actin and myosin, lysosomes, mitochondria and glycogen.

a - Granules	Functions
Albumin	Maintains colloidal osmotic pressure
β - Thromboglobulin	Inhibitor of Prostacycline formation
Factor V/Proaccelerin	Involved in blood coagulation

Fibronectin	Helps platelet adhesion with extracellular membrane
Platelet derived growth factor	Necessary for the growth of endothelial cells & Fibroblast
TGF – β	Involved in tissue repair
von Willebrand factor (vWF)	Helps platelet adhesion
Factor XIII	Fibrin stabilizing factor

Dense Granules	**Functions**
ADP	Involved in platelet activation
ATP	Provides energy
Calcium	Provides principle stimulus to blood coagulation
Serotonin	Vasoconstriction
Actin & Myosin	Contractile proteins, involved in clot retraction
Mitochondria	Powerhouse to synthesis ATP
Lysosomes	To digest the damaged tissues
Glycogen	Primary source of energy

- Platelets produce thromboxane A_2, which produces platelet aggregation and contraction of platelet (Clot retraction).
- Platelets survive for 8-11 days in circulating blood.

FUNCTIONS OF PLATELETS

- Their main function is in blood coagulation and clot retraction.
- They provide platelet phospholipid for coagulation and carry several clotting factors on their surface.
- They prevent hemorrhage when blood vessels are injured by transforming into platelet plugs, which is an important mechanism for closure of minute ruptures in blood vessels.

NORMAL PLATELET COUNT IN DOMESTIC ANIMALS

- **Mean platelet count /ml of blood**

Dog	Cat	Cow	Sheep	Goat	Horse	Pig
3,00,000	4,50,000	5,00,000	4,00,000	4,50,000	2,25,000	5,20,000

HEMOSTASIS

- It means prevention of blood loss from damaged vessels.

VASCULAR SPASM

- Injury or trauma to the blood vessels stimulates reflex constriction of the blood vessels through the sympathetic division or the local myogenic spasm by the action of ***serotonin (5-hydroxy tryptamine)*** to close even the large blood vessels to prevent excessive blood loss.
- When the injury is severe with extensive tissue damage (Crushing, laceration) this spasm is strong and bleeding is less.
- Smooth cut causes weak spasm and bleeding is severe.

FORMATION OF PLATELET PLUG

- Platelets have contractile proteins, the actin and myosin, factor XIII, the fibrin stabilising factor, enzyme systems for the synthesis of cAMP, ATP, ADP and prostaglandins (PGG_2, PGH_2, $PG1_2$ and PGF_2). Platelets are activated by their contact with collagen, which is present in the subendothelial membrane or by substances like ADP, serotonin and thromboxane A_2 released from damaged cells.
- Immediately after the vascular endothelial damage, the subendothelial collagen attracts the platelets to the site of injury. The platelets attach to the injured surface. They then undergo a series of complex physical and biochemical changes like swelling of the platelets, projection of radiating processes, *(pseudopods)* from the platelets and their adherence with the endothelial wall of the blood vessels. Factors vWF and fibronectin from subendothelium and platelets help in platelet adhesion.
- This reaction in turn stimulates release of Ca^{++} which stimulates the enzyme systems of the platelets and causes the release of ADP and thromboxane-A_2 which activate other platelets resulting in adherence of more number of platelets in the damaged endothelial wall forming the platelet plug.
- This plugs the injury on the blood vessel and prevents blood loss. Thromboxane-A_2 is the most potent platelet-aggregating agent that lowers platelet cAMP and also causes vasoconstriction. PGG_2 and PGH_2 also cause platelet aggregation, whereas PGI_2 produced by the normal endothelium is a vasodilator which acts as a powerful inhibitor of platelet adhesion / aggregation.
- Platelet aggregation is prevented by prostacyclins produced by endothelial cells of the artery and lungs. In normal blood flow, prostacyclin level is more than thromboxane and aggregation is prevented. When arterial walls are damaged, prostacyclin level is reduced and thromboxane level becomes high leading to platelet aggregation.

BLOOD COAGULATION OR BLOOD CLOTTING

- Many substances present in tissue and blood affect coagulation. Substances that promote coagulation are called as procoagulants, and those inhibiting coagulation are anticoagulants.
- Normally in the blood, anticoagulant activity predominates and blood does not coagulate. But, when a vessel is ruptured, the procoagulants activity in the damaged area becomes more and the blood clots.
- Clot is the meshwork of fibrin threads running in all directions to entrap the blood cells, platelets and plasma. This is achieved by the activation of series of clotting factors of the blood by two mechanisms:
    - The intrinsic and
    - Extrinsic systems

THREE STAGES OF CLOT FORMATION

- Formation of prothrombin activator in response to rupture of blood vessel or damage to the blood itself.
- Conversion of prothrombin into thrombin by the catalytic activity of prothrombin activator.
- Conversion of fibrinogen into fibrin thread by the enzymatic activity of thrombin.

INTRINSIC AND EXTRINSIC MECHANISM

Intrinsic or endogenous mechanism

- When the blood come in contact with foreign surface (other than intact vascular endothelium) like disrupted endothelium, negatively charged surface or glass, it causes a sequence of enzymatic reactions to initiate the coagulation mechanism.
- Blood drawn in a glass tube clots by this mechanism.

Extrinsic or exogenous mechanism

- These sequential enzymatic reactions occur when the blood from the damaged blood vessel contacts extra-vascular tissue factor, the tissue thromboplastin, released from the surrounding tissues.
- Tissue thromboplastin is not present normally in the blood but released when the surrounding tissue is damaged.

BLOOD CLOTTING CASCADE

- Most of the clotting factors are inactive proteolytic enzymes and when activated produce enzymatic activities leading to cascading reaction of clotting process.
- The extrinsic mechanism is *fast acting*. The cascade reactions of clotting process involving so many factors helps even a low amount of stimulus (damage) to produce large amounts of final end product (clot).

- In the intrinsic mechanism, injury to the blood alters the factor XII and the platelets of the blood. When the factor XII comes in contact with collagen or a wettable surface, it becomes a proteolytic enzyme called activated XII (XIIa).
- The blood trauma also damages the platelets, release platelet phospholipid, which is essential for subsequent reactions. In the extrinsic mechanism, the damaged tissue releases a complex of factor called *tissue thromboplastin* contains phospholipids and a glycoprotein which acts as proteolytic enzyme.

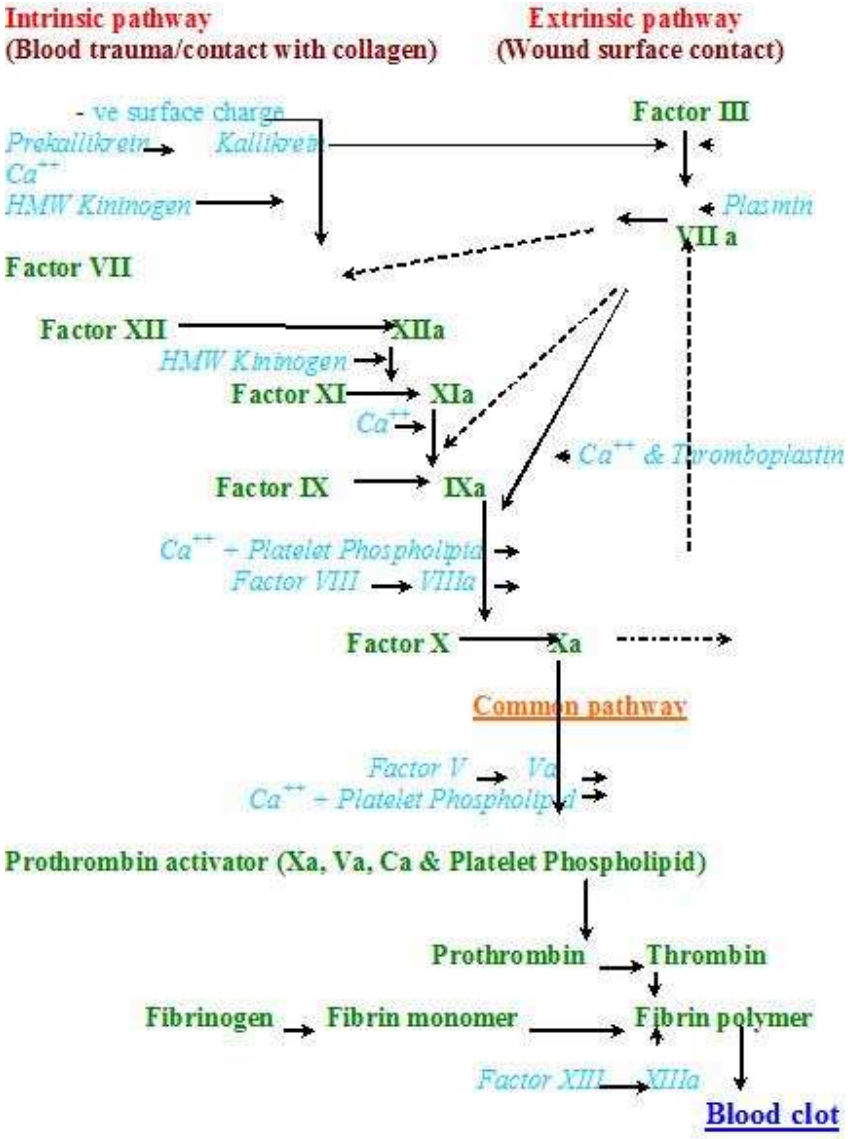
CONVERSION OF PROTHROMBIN TO THROMBIN

- Both intrinsic and extrinsic mechanisms results in the formation of a complex called prothrombin activator complex, which includes activated Factor X, (acts as protease on prothrombin); Factor V (accelerates protease activity) and phospholipid (further activates the whole process).
- The prothrombin activator acts on prothrombin and converts it to thrombin. The thrombin converts fibrinogen to fibrin monomers. It also activates more and more of Factor V, which further accelerates prothrombin activation.

Clotting factors	Proteins except factor IV
Factor I	Fibrinogen
Factor II	Prothrombin
Factor III	Tissue thromboplastin
Factor IV	Ca^{++} ion
Factor V	Labile factor/Proacclerin
Factor VII	Serum prothrombin conversion accelerator/Proconvertin
Factor VIII	Antihemophlic factor C
Factor IX	Christmas factor/Antihemophlic factor B
Factor X	Stuart factor
Factor XI	Plasma thromboplastin antecedent
Factor XII	Hageman factor
Factor XIII	Fibrin stabilizing factor
VWF	Von Willebrand factor
Prekallikrein	Fletcher factor
HMWK	High molecular weight kininogen
Fibronectin	

- Prothrombin is a plasma protein, an $alpha_2$ globulin (molecular weight 68,700) produced from the liver. Vitamin K is required for the formation of prothrombin and clotting factors VII, IX and X. Thrombin is formed from prothrombin by the action of prothrombin activator and Ca^{++}.

BLOOD CLOT FORMATION



CONVERSION OF FIBRINOGEN TO FIBRIN

- Fibrinogen is a high molecular weight plasma protein produced from liver. The proteolytic enzyme thrombin acts on fibrinogen and remove 4 low molecular weight peptides from each molecule of fibrinogen resulting in formation of fibrin monomer which polymerises with other fibrin monomers and forms fibrin threads.
- The fibrin-stabilising factor, present in platelets and to small extent in plasma, causes covalent bonds between the fibrin monomers and cross linkage between adjacent fibrin threads thus adding strength to the fibrin meshwork, the clot.

- Clot, is composed of fibrin threads running in all directions and entrapping blood cells, platelets and plasma. The fibrin threads adhere to damaged surface of blood vessel and close the opening in the blood vessel thereby prevent blood loss.
- *Clot retraction:* Within a few minutes after a clot is formed, it begins to contract and expresses most of the fluid from the clot within 30 to 60 minutes. The fluid that is expressed out of the clot is called serum.
- Platelets are necessary for clot retraction. This is caused by the contraction of actin and myosin of the platelets by using ATP of platelets.

FATE OF CLOT

Fate of clot

- Once a blood clot is formed, it can follow either of the following two courses.
    - The clot may be invaded by fibroblasts (the formation and invasion is stimulated by platelet derived growth factor) within few hours of its formation. The fibroblasts promote connective tissue formation within the clot and within 7-10 days the clot is organised into fibrous tissue.
    - When the hemorrhage occur in large quantities and when the blood clot forms in the tissues, some special substances formed within the clot will dissolve the clot (lysis of clot).

Lysis of blood clot

- The plasma contain a glycoprotein called plasminogen or profibrinolysin. During the formation of clot, this is trapped within the clot.
- This can be activated to plasmin or fibrinolysin by tissue plasminogen activator (tPA) released from the damaged tissues and vascular endothelium. The plasmin acts as proteolytic enzyme and digests fibrin threads leading to dissolution of clot.

PREVENTION OF BLOOD CLOTTING IN BLOOD VESSELS

- The important factors that prevent the process of clotting within the vascular system are:
    - Smooth endothelium, which prevents contact activation of intrinsic system
    - Glycocalyx layer adsorbed to the inner surface of the endothelium that repels clotting factors and platelets
    - Protein bound with endothelium which binds thrombin and removes its effect; this protein also activates a plasma protein called protein C which inactivates the activated factor V and VIII. Damage to endothelium causes loss of smoothness and the glycocalyx layer and thus initiates coagulation.
- Anticoagulants in blood also prevent clotting within the vessels. Important anticoagulants are
    - Fibrin threads formed during clotting
    - A globulin known as antithrombin III, heparin and alpha$_2$ macroglobulin .
- The thrombin formed during clotting becomes adsorbed to fibrin and thus prevents further spreading of the blood clotting. Thrombin not adsorbed to fibrin combines with antithrombin and gets inactivated. Heparin increases the effectiveness of antithrombin in removing thrombin. The heparin and antithrombin also inactivates the activated factors XII, XI, and X. The alpha$_2$ macroglobulin acts similar to antithrombin.

- Clotting time, or coagulation time is the time required for the blood to clot after it is drawn out from the blood vessels. When blood is exposed to air, it clots within a short time. Clotting time is less than 5 min. in most of the domestic animals except cow and horse. Average clotting time (in minutes):
    - Horse : 11.5,
    - Sheep : 2.5,
    - Cow : 6.5,
    - Pig : 3.5,
    - Dog : 2.5,
    - Human : 5.0.
- The bleeding time is the time required for bleeding to stop after puncture of the skin. The blood is blotted away every 30 seconds or the wound is immersed in physiological salt solutions: (ranges from 3- 6 min)
- The prothrombin time is a measure of the clotting time of plasma to which an excess of thromboplastin has been added so that this coagulation factor will not be a limiting one. Since oxalated plasma is used, $CaCl_2$ is also added to the plasma.
- Longer the clotting time, the smaller the prothrombin concentration. Prothrombin time in dogs: 9-12 sec. in sheep 13-25 sec.

CAUSES OF EXCESSIVE BLEEDING

- Vitamin K deficiency: This vitamin is required for hepatic synthesis of prothrombin factors V, VII, IX and X and deficiency leads to reduced synthesis of these coagulation factors resulting in excessive bleeding.
- Hepatitis or cirrhosis of liver: Results in less synthesis of coagulation factors derived from liver.
- Thrombocytopenia: Reduction in platelet count, which may occur in irradiation of bone marrow, drug sensitivity and some infections.
- Hereditary hemorrhagic disorders
    - *Hemophilia A* due to deficiency of factor VIII. It is congenital, seen in dogs and also reported in cat and horse.
    - *Hemophilia B:* caused by reduction in factor IX. Seen in dogs and it is inherited.
    - *Von Willebrand's disease*: Inherited diseases resembles hemophilia A caused by reduction in platelet function and factor VIII. Deficiency of factor IX is reported in cattle.

MODULE-11: BLOOD GROUPS

Learning objectives

- This module deals with
    - blood group systems and blood typing
    - blood group and its significance in animals

BLOOD GROUPS

- Blood is divided into different groups based on the type of antigen present in the cells. The blood group antigen are glycolipids or glycoproteins having a molecular weight of 20,000 to 30,000 present on the surface of the erythrocytes.

- The antigens are highly specific and the specificity of the antigens appears to depend upon carbohydrate portion of the molecule. Based on the antigenic type present on the surface of the RBCs cell membrane blood can be grouped into different groups or types.
- Landsteiner (1900) was the first to identify four blood groups, A, B, AB and O in human beings.
- About 70 blood groups have been identified in man, but the ABO, Rh, MNS, P and Lewis groups are best known. Among these five, ABO and Rh systems are widely used for blood grouping in human beings.

O-A-B BLOOD GROUPS

- There are two related antigens namely type A and type B are present on the surfaces of the RBCs among most of the population. These antigens are inherited and an individual may have either one or both nor do neither of them are present in their cells.
- The genes on two adjacent chromosomes, one gene on each chromosome determine the O-A- B blood groups. There may be any one type of the three types O, A, B and six possible combinations of genes are known. Type O is always functionless as there are no antigens present.
- The antigens (type A and type B) present on the surface of the RBCs are known as Agglutinogens and the plasma antibodies that cause agglutination are known as Agglutinins.
- Based on the presence or absence of agglutinogens in the RBCs blood is grouped.
    - Group A - when only type A antigen is present
    - Group B - when only type B antigen is present
    - Group AB - when type A and B antigens are present
    - Group O – when neither A nor B are present.

Blood Group	Antigens on RBCs	Antibodies in Sera	Genotypes
A	A	Anti-B	*AA or AO*
B	B	Anti-A	*BB or BO*
AB	A and B	Neither	*AB*
O	Neither	Anti-A and anti-B	*OO*

- During blood transfusion, if the blood between recipient and donor are mismatched, there will be reaction of antigen (agglutinogen) with the antibody (agglutinin) causing clumping or agglutination of the erythrocytes. These agglutinated RBCs are carried by the leukocytes to the RE system where they are lysed releasing Hb. Sometimes, mismatching of recipient and donor's blood leads to immediate hemolysis of the RBC's in the circulating blood, caused by the complement system.
- AB group of blood is designated as universal recipient
- O group of blood is commonly referred as universal donor

ANTIGEN ANTIBODY REACTION

- Generally, antibodies present in the donor's plasma, which would be active against the recipient's red cells, do not produce such a reaction because of the rapid dilution in the recipient's circulation.

- Serious problem of antigen-antibody reaction result from antibodies present in the recipients plasma reacting with the donor's RBCs. This agglutination of the donor's erythrocytes may produce systemic thrombosis in the blood vessels.

Blood groups of recipient (Antibody)	Blood groups of the donor(Antigen)			
	A	B	AB	O
A (Anti B)	--	+	+	--
B (Anti A)	+	--	+	--
AB	--	--	--	--
O (Anti A, B)	+	+	+	--

(+ Agglutination or clumping of RBCs and - No agglutination)

BLOOD TYPING

- Determination of the blood groups of the recipient and donor prior to transfusion is done by matching. RBCs are diluted with saline and one portion is mixed with anti A antibody and the other with anti B antibody. The mixture is observed under the microscope.
- If the RBCs have become clumped antigen antibody reaction is said to have taken place which indicates mismatching.

RBCs	Sera	
	Anti A	Anti B
A	+	-
B	-	+
AB	+	+
O	-	-

Rh SYSTEM

- Rh antigens are transmembrane proteins with loops exposed at the surface of red blood cells. They are named after the rhesus monkey in which they were first discovered.
- They are used for the transport of carbon dioxide and/or ammonia across the plasma membrane. In Rh system, spontaneous agglutinins do not occur. There are six Rh antigens (Rh factors) C, D, E, c, d and e. Of these six types D is widely prevalent and is considered to be more antigenic than others. The presence of D antigen indicates Rh^{+}.
- In this system there is no immediate reaction when blood transfusion is effected between Rh^{+} and Rh^{-} individuals. When Rh^{-} individual receives Rh^{+} blood, the development of antibody or agglutinin occurs only after 2 to 4 weeks.
- Hence, the transfusion reaction is usually delayed and mild. If the same person has subsequent blood transfusion with the same antigen, there will be enhanced antigen - antibody reaction.

ERYTHROBLASTOSIS FETALIS

- It is a disease of the fetus and newborn infants characterized by progressive agglutination and phagocytosis of RBC's. The mother is Rh^-, father Rh^+, the baby inherits Rh^+ from the father. Usually the Rh^- mother develops anti-Rh agglutinins only when the Rh $^+$ child develops by inheriting the Rh^+ factor from its father.
- The child's Rh^+ antigen enters the maternal system and causes development of Rh^+ antibodies. Placental diffusion of this antibody causes hemolytic conditions in the subsequent Rh^+ new born infants. This disease condition is characterised by varying degrees of anaemia and jaundice in the newborn infants depending upon antibody reaction by the mother.
- This condition can be prevented by passively immunising the mother against Rh^+ factor.

BLOOD GROUPS IN ANIMALS

- In animals, the antigens representing the blood group are not strongly antigenic and occurrence of the natural antibody in their blood is rare. However, naturally occurring antibodies to some red cell antigens can be found in normal animals that lack the respective antigens.
- In domestic animals, the initial transfusion of whole blood will not result in serious problem. However, subsequent transfusions with the similar isoantigen can cause enhanced antigen-antibody response, which may produce the clinical symptoms like muscular trembling, salivation, dyspnea, and hemoglobinuria.
- In dogs, the blood groups are known as DEA system. They include 1.1, 1.2, and 3-8. DEAs 1.1 and 1.2 are present in 60% of the canine populations and these dogs are considered as A positive and the others in which DEA 1.1 and 1.2 blood groups are not present as A negative. The A negative dogs do have antibodies against A positive blood.
- In horses eight blood *groups* A, C, D, K, P, O, T and U has been identified. Hemolytic icterus occurs in newborn foals due to their erythrocytic destruction by the isoantibodies from the colostrum. This condition is very common in blood containing antibody for A or C.
- In cattle more than sixty erythrocyte antigenic factors, which are divided into 11 groups. They are designated as A, B, C, F- v, J, L, M, N, S, Z and R'-S. Anti- J is the naturally occurring antibody.
- In sheep, there are seven antigenic groups A, B, C, D, M, R–O and X-Z of which anti- R is the naturally occurring antibody of the R - O antigen.
- In the goats, five blood group antigens have been identified, A, B, C, M and J.
- In pigs, thirteen blood group systems, A, B, C, E, F, G, H, I, J, K, L, M and N have been identified . `A' antigen possesses naturally occurring antibody.
- In cats, three blood groups have been identified A, B and AB. Type A is most common and they have antibodies against A isoantigens. Type AB is very rare in occurrence.

SIGNIFICANCE OF BLOOD GROUPS IN ANIMALS

- To identify monozygotic twins from dizygotic twins.
- Some blood group systems correlate with economic traits, eg: milk fat, milk yield. Hence, useful for selection of animals for breeding.
- To solve parentage problem if there is any dispute.

Precautions for blood transfusion in animals

- The blood with anti-isoantibody A in pig, J in cattle, R in sheep and A or C in horse should be avoided for blood transfusion.

MODULE-12: BLOOD VOLUME, PCV & ESR

Learning objectives

- This module deals with
  - volume of blood in various species
  - packed cell volume
  - erythrocytes sedimentation rate

BLOOD VOLUME

- It makes up about 6 to 8% of the body weight. Blood volume measurement plays an important role when blood transfusion is attempted.
- It is also important to interpret PCV, Hb, RBC, and hematological parameters. These values are altered when the blood volume changes e.g. hemoconcentration and hemodilution.
- Blood volume is influenced by body type, body size, age, sex, breed, nutrition, pregnancy, and lactation, physical and metabolic activities. Males show higher blood volume than females.
- Blood volume increases with pregnancy, muscular activity, stress, and high temperature, whereas starvation, hemorrhage, burns, dehydration, anemia and cold decrease the blood volume.

AVERAGE BLOOD VOLUME IN VARIOUS SPECIES

Blood Volume (ml)	Cattle	Horse	Sheep	Goat	Pig	Dog	Cat	Poultry
Blood volume (ml)	40,000	40,000	3,200	2,400	8,000	1,600	240	160
% of body wt.	8	10	8	6	-	7	-	6.5

MEASUREMENT OF BLOOD VOLUME

Principle

- The volume of any fluid compartment of the body can be measured by placing a substance in the compartment and allowing it to disperse through the fluid. The extent to which the substance gets diluted in the fluid is measured.
- The substance used for determining blood volume must be capable of dispersing throughout the blood easily and must remain in the circulatory system for a longer period of time till the measurements are made.
- Blood volume may be measured indirectly by two methods
    - Plasma volume method
    - Erythrocyte volume method

Plasma volume method

- Plasma volume can be measured by dye dilution or by radioisotope. Blood sample from the experimental animal is collected and the plasma of the pre injected anticoagulant-added blood serves as a blank.
- Known quantity of Evan's blue dye/ (T-1824), or radioisotope ^{131}I, when injected, combines with the plasma proteins and disperses throughout the circulatory system in about 10 minutes.
- Blood samples are collected sequentially at 5 minutes interval for the next 15 to 30 minutes. These blood samples are centrifuged to get RBC free plasma. The concentrations of the dye or the radioactivity of isotope in the plasma samples can be measured *spectrophotometrically* at 620 nm or by the *scintillation counter* respectively.

Erythrocyte volume method

- This can be determined by radio active ^{32}P, ^{59}Fe and ^{51}Cr. Blood sample from the experimental animal is collected to get plasma free erythrocytes.
- Small quantity of ^{51}Cr is mixed with these RBCs which are then incubated at 36 °C for 30 minutes to activate the binding process of ^{51}Cr with the RBCs. Wash the erythrocytes with saline to remove the free ^{51}Cr. These radioactive erythrocytes are then injected into the circulation of the experimental animal. After proper mixing of these RBCs in the circulation, blood samples are collected after 10 minutes.
- RBCs from the blood samples are separated and the radioactivity of pre and post injected RBC samples are determined by scintillation counter.

$$\text{Erythrocyte volume (ml)} = \frac{\text{Quantity of radioactivity injected}}{\text{Radioactivity / ml of RBCs}}$$

$$\text{Blood volume (ml)} = \frac{\text{Plasma volume} \times 100}{100 - \text{PCV} \times 0.96} \quad \text{(or)} \quad \frac{\text{Erythrocyte volume} \times 100}{\text{PCV}}$$

- The trapped plasma value may interfere with the PCV. Hence the correction factor for trapped plasma is introduced as 0.96 for blood volume determination.
- Starvation, haemorrhage, hot environment and water deprivation reduce the blood volume.

PACKED CELL VOLUME

Packed cell volume (PCV) or Hematocrit (%)

- It is percent volume of packed blood cells present in whole blood after centrifugation at 3000 rpm for 30 minutes. Hematocrit is measured by two methods.

Wintrobe hematocrit (macro method)

- Wintrobe tubes have uniform 3 mm bore, calibrated by 10 cm scales with millimetre divisions. The wintrobe pipette with a long but narrow delivery tube is used to fill about 1 ml of blood into the wintrobe tube without air bubbles.
- Horse and dog blood require a relative centrifugal force of 2,260 G (generally 3000 rpm) for 30 minutes, while it is for 60 minutes in the case of cattle and pig blood. The blood of sheep and goat require higher relative centrifugal force than 2,260 G.

True PCV = Venous PCV **x** 0.96 (correction factor for trapped plasma)

- The RBCs settle to the bottom of the hematocrit tube leaving the plasma above the packed erythrocytes. A thin white or yellow layer called the buffy coat occurs in between these two and it indicates the WBC or the leukocytes.
- The amount of trapped plasma in the erythrocyte column varies with PCV, size of erythrocytes and the degree of rouleaux formation. Canine blood is characterised by large blood cells and greater tendency for rouleaux formation. Similarly equine blood is characterised by extreme rouleaux formation, hence less plasma is trapped.

Microhematocrit method

- PCV may be estimated by blood filled capillary tubes (32 mm x 0.8 mm) centrifuged at 14,000 G for 2 minutes. In general, the PCV is approximately three times the Hb concentration. PCV values ranges between 38 to 45% in most of the domestic animals.

Factors influencing PCV

- High temperature, muscular exercise, stress, pregnancy, hemorrhage, dehydration and burns, anemia, polycythemia and leukemia.

NORMAL PCV VALUES IN DOMESTIC ANIMALS

-

Species	PCV (%)
Dog	37-55 (45)

Cat	24-45 (37)
Cow	24-46 (35)
Sheep	24-50 (38)
Goat	19-38 (28)
Horse	32-52 (42)
Pig	32-50 (42)

ERYTHROCYTE SEDIMENTATION RATE (ESR)

- It is a measure of the settlement of the erythrocytes in millimetre for a given period usually 30 minutes or one hour when citrated blood filled standard hematocrit tube is placed in an absolutely vertical position.
- The length of storage of the blood sample and temperature of storage may influence ESR. The ESR test is generally done as part of complete hematological investigations.
- Suspension stability or sedimentation of RBCs in the blood is influenced by RBCs and plasma characteristics. Rouleaux formation or the agglutination of RBCs influences the ESR.
- Increasing concentration of fibrinogen or the globulin hastens agglutinations of RBCs. ESR is positively influenced by rouleaux formation and plasma contents of fibrinogen, α_2 globulins and gamma globulins, while it is negatively influenced by reticulocytes and plasma content of the albumin.
- Fibrinogen and a globulin are commonly elevated in variety of inflammatory diseases e.g., pleurisy, pericarditis, peritonitis, whereas hemoconcentration and glomerulonephritis may adversely affects the ESR by altering the influence of albumin fraction.
- The speed of settling of RBCs is inversely proportional to the number of erythrocytes in the given sample. Among the domestic animals, the erythrocytes of ruminants show little or low natural tendency to form rouleaux, whereas the equine blood shows high tendency to form rouleaux formation.
- Increased ESR is very common in increased cholesterol: lecithin ratio, pregnancy, acute general infections (septicaemia), and rheumatic fever, TB, arthritis, toxaemia, malignant tumours, hypothyroidism and glomerulonephritis.
- Decreased ESR is seen in afibrinogenemia, polycythemia, allergy etc.

ESR value (mm)

ESR (mm)	Cattle	Sheep	Goat	Dog	Cat	Horse	Chicken
30 minutes	--	--	--	1-5	--	15-38	0-1
60 minutes	--	--	--	6-10	7–27	--	1 – 3

- The ESR merely helps in evaluating the health status of animals, but it is not helpful in diagnosing any specific diseases

Factors influencing ESR

- Size and shape of erythrocytes, specific gravity of erythrocytes, specific gravity of plasma, viscosity of plasma, fibrinogen and globulin content, temperature.

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MODULE-13: HEART - STRUCTURE, CONDUCTION SYSTEM AND TRANSMISSION OF IMPULSE

Learning objectives

- This module deals with
  - structure of the heart
  - working mechanisms of the heart

ANATOMY OF THE HEART

- Heart is situated in the middle mediastinal space. It has two anterior chambers, the atria and two posterior chambers, the ventricles. The heart is separated into left and right sides by a septum. The atria and ventricles communicate with each other through the atrio-ventricular orifices, which are guarded by the AV valves, the bicuspid or mitral valve and tricuspid valve on the left and right sides, respectively.
- Chorda tendinae are the elastic structures attached to the valves with the internal muscular projections of the heart known as the papillae and prevent the evertion of the valves during ventricular contraction.
- AV valves prevent the back flow of blood into the atria during ventricular contraction (systole). Of the four chambers, the left ventricular wall is much thicker to provide effective force during its contraction to pump its contents throughout the body.
- The venous system provides blood to right atrium of the heart and the blood is pumped from the right ventricle to the lung; hence, it is referred to as venous or pulmonary side.
- The left atrium of the heart receives the oxygenated blood from the lungs and the left ventricle pumps it to the peripheral systemic organ which is known as systemic or arterial side. Blood vessels between aorta and vena cava supplying blood to all organs except to the lungs are called systemic circulation.
- The blood vessels of lungs constitute pulmonary circulation. Pulmonary circulation and heart are collectively termed as central circulation. In the orifices of the aorta and the pulmonary artery the Semilunar valves are present which prevent the back flow of blood from aorta and pulmonary artery into the ventricles during ventricular relaxation (diastole).
- The major function of the cardio vascular system is to transport oxygen, nutrients, water electrolytes and hormones to the tissues and remove the metabolic wastes such as CO_2, urea, creatinine etc from tissues to the lung and excretory organ like kidney

CARDIAC MUSCLE

- Heart is composed of three major types of cardiac muscle namely:
  - Atrial,
  - Ventricular and
  - Special excitatory and conductive muscle fibres.
- The atrial and ventricular muscle fibres contract similarly to skeletal muscle fibres except the duration of contraction is longer. The excitatory and conductive muscle fibres contract only feebly but exhibit rhythmicity and varying rates of conduction providing an excitatory system that controls rhythmical beating of the heart.
- Cardiac muscle fibres are arranged in a lattice network, fibres divide, recombine and spread again. The cardiac muscle fibres contain intercalated discs that contain many gap junctions through which action potential can spread from one to other cardiac muscle cell very easily.
- When one of the cardiac muscle cells is excited, and the action potential spreads to all of them, hence the cardiac muscle obey all or none law.
- The heart is composed of two separate functional syncytia, the atrial and the ventricular walls. They are separated from each other by fibrous tissue surrounding the valves between atria and ventricles, but are connected to each other by specialised conduction system fibres. This type of two functional syncytia allows the atria to contract a short time before ventricular contraction which is important for effectiveness for pumping of the heart.

CONDUCTION SYSTEM OF THE HEART

- The specialized excitatory and conduction fibres show feeble contractions only because of very few contractile fibres, but they conduct impulses very rapidly through out the heart.
- In mammals, S.A. node (Sino auricular node) is the specialized structure of the cardiac muscle located at the junction of the right atrium and cranial vena cava. It has an inherent property of generating its own action potential at periodical interval because of low resting membrane potential (- 55 mV).
- The SA node normally controls the rate of the heart; hence it is called as pace maker. The pacemaker dominates the normal rate and rhythm of the heart. In sub mammalian species, frog, this function is taken up by sinus venosus. This rhythm is called as ectopic rhythm.
- Most of the cardiac fibres including the conduction system have the ability of self-excitation and can produce automatic rhythmical contractions. In some pathological conditions of the mammalian heart, the excitatory impulses originate outside the S. A. node referred as ectopic foci in which the heart rate will be less than normal.
- S.A. node spreads its impulse through atrial muscular wall and interatrial bundles to A.V. node (atrio -ventricular node) which lies in the septal walls of the right atrium cranio dorsal to tricuspid valve.
- It conducts the action potentials to common bundle of His or AV bundle which then runs into the ventricular septum where it divides into a right and left bundle branches that run underneath the septal endocardium. At the ventricular apex, these branches finally terminate as purkinje fibres form a net work of conductive system in the ventricular muscle.
- The AV node and the AV bundle is the only route for the conduction of impulse from atria to ventricles. In the conduction system, the AV node shows a delay in the propagation of the action potential, the nodal delay for a period of 50 - 150 ms.

BASIC MECHANISMS OF MEMBRANE POTENTIAL

Three major factors cause the membrane potential.

- Differential permeability of the membrane to diffusion of ions
  - During resting state the cell membrane is 50 to100 times more permeable to K^+ ions out side the cell membrane through the *leak channels* than Na^+ ions due to their concentration gradient. This contributes accumulation of positive charges immediately outside the membrane.
  - The resting membrane is almost completely impermeable /very slightly permeable to Na^+ ions.
- The Na^+ - K^+ electrogenic pump
  - It generates positive membrane potential out side the cell by actively pumping 3 Na^+ ions out of the cell for every 2 K^+ ions pumped into the cell against their concentration gradient .
- Trapping of negatively charged anions inside the cell
  - Many intracellular anions are large molecules proteins, organic SO_4 and PO_4 are trapped within the cell and are attracted to the inner surface of the cell membrane generate negative electrical potential inside the cell.
  - These three factors are the primary cause of membrane potential. The magnitude of this potential produced by diffusion of ions can be predicted by the *Nernst equation*.

EVENTS OF ACTION POTENTIAL

- The action potential is caused by a sequence of changes or events occurring in the membrane permeability to Na^+ and K^+ ions. When an excitable cell is stimulated the cell membrane permits the inflow of Na^+ ions into the cell to generate the action potential.
- Action potential is the rapid changes in the membrane potential from its normal negativity to positive potential inside the cell membrane which last for few milliseconds and then returns back to its original resting negative potential level.
- *Polarised membrane* is the resting cell membrane with a normal negative resting membrane potential.
- *Depolarization stage*:
  - It is the *first event* of action potential and is characterized by rapid increase in the permeability to Na^+ ions (5000 folds) to interior of the cell generating more positive electrical potential inside of the cell.
  - This is followed by a gradual inactivation of Na^+ channels (closure) that occur within another few milliseconds and the membrane becomes impermeable to Na^+ ions. It is associated with gradual opening of***voltage gated K*$^+$ *channels*** to allow K ions outflow to the exterior of the cell membrane. The potential inside the cell is re-established to its normal resting level (- 90mV). This stage is called as the***repolarisation*** *stage*
  - Higher concentration of K^+ ions in the exterior of the cell towards the end of the action potential continues for a short period creates more negativity inside referred to as ***hyper-polarised*** *state*. At this state, re-excitation of the cell will not occur.
- The *final event* is characterized by ***electrogenic pump mechanism***, which aids in the transport of three Na^+ ions to the exterior for every two K^+ ions to interior of the cell and create the normal resting potential (- 90 mV) on the inside of the cell membrane.

CHANNEL SYSTEMS OF CELL MEMBRANE

- Na – K leakage channels
- Voltage gated Na channels and K channels

Na – K leakage channels

- These channels are 100 times more permeable to K^+ ions than Na^+ ions and are exceedingly important in determining the level of normal resting membrane potential.

Voltage - gated channels of the cell membrane

- The *voltage-gated Na^+ channel* causes both depolarisation and repolarisation of the nerve/muscle membrane during the action potential.
- The *voltage-gated K^+ channel* also plays an important role in establishing the repolarisation of the membrane. These two voltage-gated channels are present in the cell membrane.
    - Voltage - gated Na^+ channel
        - The voltage gated Na^+ channels have two gates, the ***external gate or activation gate which*** opens to outside of the cell. The other gate is at the interior end and opens to inside of the cell referred as***internal gate or inactivation gate.***
        - The activation gate is closed during resting stage (- 90 mV), while the inactivation gate is opened. This prevents free passage of Na^+ ions from outside to interior of the cell.
        - During depolarisation stage, the resting membrane potential drops from – 90 mV or to - 70 to - 50 mV causes opening of the activation gate due to conformational changes results in increased Na^+ ion permeability as much as 500 to 5000 folds into the cell through the channel system.
        - A gradual increase to positivity following the opening of the activation gate also closes the inactivation gate comparatively at a slow speed.
        - The time lapse between the activation and inactivation of the channels causes the passage of Na^+ ions to the interior of the cell for few milliseconds.
        - During repolarisation stage, the inactivation gate is completely closed and prevents Na^+ ion entry from outside to the interior of the cell membrane. The inactivation gate will not reopen until the disturbed membrane potential returns nearly to the original resting membrane potential level of -70 to- 90 mV.
    - Voltage - gated potassium channel
        - This channel has only one gate at the interior of the membrane. It may either close or open to the interior of the cell.
        - During resting stage this gate is *inactivated* and prevents the passage of K^+ ion to the exterior of the cell through this channel system. When the electrical potential drops towards zero, Na^+ channels get inactivated causes the activation of K^+ channels to allow increased K^+ ions diffusion to outside of the cells.
    -

ELECTROCHEMICAL CHANGES DURING ACTION POTENTIAL

Spike potential *(over shoot)*

- It is the steep change in the negativity of the cell from - 90 mV to more positivity of + 40 mV in the membrane potential due to rapid increase in the permeability of the Na^+ ions to the interior of the cell.

Positive after potential

- During repolarisation, the membrane potential drops little more negative than the normal resting value of – 90 mV due to excess of K^+ ion diffusion to out side the cell. This is caused by the prolonged opening of K^+ channels for several milliseconds even after repolarisation.

Resting membrane potential in cardiac muscle

- The resting membrane potential in cardiac muscle is – 90 mV, but has long action potential (250 msec).
  - S.A. node = - 50 to - 55 mV
  - Cardiac muscle = - 85 to - 95 mV
  - Conduction system = - 90 to -100 mV
  - Ventricle muscle = -100 to -105 mV

ACTION POTENTIAL IN CARDIAC MUSCLE

- Cardiac muscle has an inherent property of generating its own action potentials rhythmically, independent of nerve stimulation. This occurs in the peacemaker cells of the S.A. node, which depolarises faster than any other parts of the heart.
- The onset of action potential in cardiac muscle is slower but prolonged than skeletal muscle which lasts for 150 msec. in atria and 300 m.sec. in ventricle. In cardiac muscle, repolarisation does not occur immediately after depolarisation The positivity generated as the spike potential by the depolarisation remains as a plateau near the peak. This plateau lasts for a few-hundred m.sec prolongs the contraction of the cardiac muscle.
- *The reason for the prolonged action potential* in cardiac muscle cells is due to the presence of two separate channel systems
  - Voltage activated Na channel (fast channel)
  - Voltage activated Na - Ca channel (slow channel)
- Activation of the fast Na channels causes the spike potential of the action potential, whereas the slow channel prolongs the passage of Ca^{++} and Na^+ ions into the interior of the cell, thus establishes the plateau in the action potential.
- The inflow of Ca^{++} ions in to the cardiac muscle cells decreases K^+ permeability by about 5 fold. This delays the K^+ ion permeability to outside which in turn delays the re-polarization process of the action potential in cardiac muscle. This prolonged action potential (250 – 300 m.sec) provides longer contraction period in the cardiac muscle cells than the skeletal muscles. Hence, cardiac muscles do not functionally tetanized.

ACTION POTENTIAL IN PACEMAKER CELL

- In the cells of SA node, the resting potential is -55 mv only due to naturally leaky nature of these cells to Na^+ ions cause spontaneous depolarisation and the resting potential gradually rise up to - 40 mv. At this voltage, the fast Na^+ and Na^+- Ca^{++} channels are opened leading to development of self-excitation and heart beat. This is followed by opening of the K^+ channels causes diffusion of greater number K^+ ions to out side the cell causing the hyperpolarization (-55 to -60 mv).
- After this event, the K^+ channels close and Na^+ ion leaks to the interior of the cells to repeat the cycle.
- In cardiac muscles, the action potential is prolonged and it is referred as *plateau of action potential.*

VELOCITY OF CONDUCTION

Atrial muscles	0.05 - 0.1 m/sec
A.V. node	0.02 m/sec
Bundle of His and Purkinje fibres	1.5-4.0 m/sec
Cardiac muscles	0.3-0.5 m/sec

- The right atrium begins to depolarize about 0.01 sec before the left atrium. Conduction velocity is fastest in Purkinje fibres but slowest in A.V. nodal fibres.
- The A.V. node delays allow the atria to discharge their blood into the ventricles before ventricular systole. Of the two syncytia of the heart, the atrial syncytia cause the atrial contractions a short time ahead of the ventricular contraction.

NORMAL HEART RATE IN ANIMALS

-

Species	Cow	Horse	Sheep	Goat	Pig	Dog	Cat
Heart rate/min	48-84	28-40	70-80	70-80	70-120	70-120	120-140

PROPERTIES OF MYOCARDIUM

- Rhythmicity: The cardiac muscles have an inherent ability to generate their own impulses spontaneously at regular intervals known as rhythmicity without the influence of nerves and hormones,. This property can be demonstrated by coronary perfusion of the isolated mammalian heart using *Ringer's solution.*
- Contractility: Contraction of cardiac muscle differs from other contractile tissues in the following aspects:
- All or none principle: If at all the cardiac muscle responds to a stimulation, it contracts with maximum strength (i.e. all contractions of the cardiac muscle are maximum) when stimulated with either minimum or maximum strength of stimulus under the prevailing physical and chemical environment. This is due to the syncytial nature of the cardiac muscle.

- Staircase phenomenon or Treppe: When an effective stimulus is applied repeatedly with very short interval (1-2 seconds), the cardiac muscle responds with increasing degree of strength for the first 3 to 5 stimuli, after that the response becomes constant. When the interval between two stimuli of same strength is very short, the physico chemical changes known as beneficial effects, occurring during the first response persist and facilitate greater response for the second stimuli.
- Refractory period: This period of unresponsiveness of cardiac muscle when it is responding to a stimulus, it may not show another response for a subsequent stimulation for a shorter period. Following first stimulation, there is a total absence of response for a very brief period, called as *absolute refractory period*. This is followed by a short period of partial responsiveness referred as relative refractory period, during which the cardiac muscle can be stimulated by a stronger stimuli.
    - The cardiac muscle is in absolute refractory period (atria 0.15 sec., ventricle 0.25 sec.) during systole and in the relative refractory period (atria 0.03 sec, ventricles 0.05 sec) during early diastole.
    - The refractory period is *longer in cardiac muscle* than skeletal muscles, hence cardiac muscle can not be tetanized by a series of rapid stimuli.

MODULE-14: CARDIAC CYCLE AND HEART SOUNDS

Learning objectives

- This module deals with
    - normal and abnormal heart sounds in animals
    - pressure and volume changes in the atria and ventricles of the heart

CARDIAC CYCLE

- It is defined as sequence of events such as pressure and volume changes, cardiac sounds and changes in the electrical potential occurring during the period between two successive ventricular contractions.
- It is characterised by *one complete contraction (systole) and relaxation (diastole)* and indicates a complete heart beat.
- These periods are not equal in duration, when *heart rate is 75/minute, one cardiac cycle lasts for about 0.8 sec. in which* ventricular systole is 0.3 sec. and diastole is 0.5 sec.
- In atria, the systole lasts for 0.1 sec. and diastole 0.7 sec. During ventricular systole, the atria are not contracting and during atrial systole, ventricles are in diastole.

PRESSURE CHANGES IN THE ATRIA

- The intraatrial pressure records three major positive waves.
- 'a' wave: Results from atrial systole. Atrial systole causes an increase in the atrial pressure resulting in the opening of the AV valves and blood flows into the ventricles.
- 'c' wave: It results due to bulging of the AV valves into the atria during ventricular systole. As ventricular systole is initiated, the AV valves close and this increases the ventricular pressure causing the AV valves to bulge into the atria. This causes a second increase in

atrial pressure. As ventricular ejection takes place there will be a drop in the atrial pressure.

- 'v' wave: It results due to continuous inflow of blood into the atria during ventricular ejection. When blood flows into the atria from the vena cava of pulmonary vein, there is dilatation of the atrial walls and intraatrial pressure increases producing 'v' wave during atrial diastole.

PRESSURE AND VOLUME CHANGES IN THE VENTRICLES

- Isovolumetric contraction
  - Ventricular contraction starts at the peak of R wave in the ECG. This periods marks the onset of ventricular contraction. The ventricular pressure exceeds than that of the atrial pressure causing closure of AV valves. The volume of the ventricles do not change and the pressure rises rapidly. The ventricles remain as a closed chamber whereas AV and SL valves are closed.
- Maximum Ejection Phase
  - Pressure inside the ventricles exceeds the aortic pressure causing opening of the SL valves and ejection of ventricular blood rapidly into the pulmonary artery. It lasts until the peak of the arterial pressure curve.
- Reduced Ejection
  - During the onset of this phase pressure reaches a maximum. The pressure in the left ventricle falls below that of the aortic pressure and there is a decline in the outflow of blood from the ventricles. There is a progressive decrease in ventricular volume.
- Protodiastole
  - This marks the onset of ventricular relaxation and there is a rapid drop in ventricular pressure. The pressure falls below that of the aorta and pulmonary artery, while the atrial pressure gradually increases.
- Isovolumetric Relaxation
  - The AV valves and SL valves closes and these is no blood flow from the atria to the ventricles. Ventricles remain as closed chamber and there is myocardial relaxation with no increase in fibre length. When the intraventricular pressures fall below that of atrial pressure, AV valves open, at the end of this period. During this period ventricles remain partially filled with blood and they eject only 50-65% of the diastolic volume of blood.
- Rapid filling
  - It begins with the opening of AV valves as the atrial pressure rises causing blood to flow rapidly from the atria to the ventricles.
- Reduced filling (diastasis)
  - There is slow filling of blood from the atria to the ventricles. At the end of this period, there is onset of atrial contraction.
- Atrial systole
  - It is marked by the beginning of artial contraction (a) wave in which the two chambers are in communication. The atrial systole pressure wave peaks and begins to decline by the onset of ventricular isovolumetric contraction, thereby completing one cardiac cycle. During atrial systole 70-80% of the ventricles get filled but during rapid heart rate ventricular filling is reduced due to shorter time for diastolic filling. During exercise, atrial systole forces an increased 20-30% of the atrial blood into the ventricles thereby increases the ventricular pressure and volume.

- Pressure changes in the right ventricles are qualitatively similar but quantitatively lower than that of left ventricle. The right ventricle functions as volume pump and the left ventricle as pressure pump.

CARDIAC SOUNDS

- These are the distinct sounds produced during systole (S_1) and diastole (S_2) of the cardiac cycle, which can be recorded by the phonocardiogram.
- The first heart sound "lub", is systolic sound indicates the firm closure of AV valves, occur during ventricular systole. It is associated with the vibration of the valves (valvular factor) and also by the vibrations of the chordae tendinae and contracting ventricular muscle fibres (muscle factors). This sound is louder, low pitched but longer duration.
- The second heart sound "dub" is the diastolic sound developed during the closure of the aortic and pulmonary valves at protodiastole phase of the cardiac cycle.
- In addition to the above two normal sounds, occasional sounds like third heart sound (S_3) is produced early in diastole at the end of rapid ventricular filling due to sudden tension of chordae tendinae. The S_4 sound also referred as atrial sound, is due to the contraction of the atrial muscles which causes tension of the A.V. valves. These two sounds are very common in horses but pathological in dogs and cats. Usually in other animals the S_3 sound indicates congestive heart failure whereas S_4 indicates some degree of A.V. bundle block.

CARDIAC MURMURS

- Prolonged sounds of the heart occurring during normally silent intervals of cardiac cycle are referred as murmurs or adventitious heart sounds, which is due to valvular insufficiency or stenosis (narrowing) of the orifices. There are two types of cardiac murmurs.
- Systolic murmurs:
  - It occurs due to improper or defective closure of the mitral or tricuspid valves or due to stenosis of pulmonary or aortic valves.
- Diastolic murmurs:
  - It is caused by insufficiency of pulmonary or aortic valves or by stenosis of AV valves.
  - Endocarditis and erysepalus infection (in hogs) cause valvular insufficiency, which is very similar to that of rheumatic fever in man.

MODULE-15: CARDIAC OUTPUT AND ITS REGULATION

Learning objectives

- This module deals with
  - cardiac outputs - its determination and regulation
  - regulation of heart rate
  - metabolism and energetics of working myocardial cells

STROKE VOLUME

- The definite volume of blood pumped into the aorta / pulmonary trunk by each ventricular contraction (left / right) is referred to as stroke or pulse volume or the systolic discharge.

- If the stroke volume is expressed based on body weight it is known as stroke index.

$$\text{Stroke index} = \frac{\text{Stroke volume (ml)}}{\text{Body weight (kg)}}$$

- A portion of the blood is left in the ventricles even after the ventricular systole, which is known as end systolic volume. Similarly, the volume of blood present in the ventricles at the end of the diastole is referred to as end diastolic volume.

CARDIAC OUTPUT OR MINUTE VOLUME

- It is the volume of blood ejected by either the left or the right ventricles into the greater vessels, the aorta or pulmonary artery in a minute.

$$\text{Cardiac output (L / min)} = \text{Stroke volume x pulse rate}$$

- In most animals, the cardiac output is 10% of body weight in kgs. Cardiac out put of right ventricle will be normally equal to that of left ventricle.
- If this minute volume is expressed on the basis of body surface it is referred to as cardiac index (L / m^2 / min).

$$\text{Cardiac index} = \frac{\text{Minute volume}}{\text{Surface area of the body}}$$

- Venous return is the quantity of blood flowing from the great veins, vena cava in to the right atrium each minute.

DETERMINATION OF CARDIAC OUTPUT

Cardiac output can be measured by

- Direct Fick method
- Indicator dilution method
- Flowmeter method

Direct Fick method

Fick's principle

- The amount of substance taken up by the circulation per unit time equals the arterial level of the substance minus the venous level times the blood flow (Cardiac output). According to Fick's principle, cardiac output can be determined by measuring oxygen consumption/ minute and arterio-venous oxygen difference.

- Samples of blood are collected from artery and mixed venous blood sample drawn from right ventricle or pulmonary artery using catheter. Oxygen consumption/ minute is determined by spirometry.
- Amount of O_2 present in both arterial and venous samples are determined and the difference in the O_2 content, the atrio-venous O_2 difference is expressed as per cent volume.

$$\text{Cardiac output (L)} = \frac{\text{O2 consumption (ml) / minute}}{\text{Mean arterio-venous o2 difference} / \text{100 ml blood}}$$

Indicator dilution method

- This method is based on the principle that if a known quantity of a substance is mixed with an unknown volume of fluid the volume can be determined from the concentration of the substance in the fluid. The volume equals the amount of substance (in milligrams) divided by concentration of substance (in milligrams/litre).
- A known amount of indicator (Cardio green dye) is injected into a vein; arterial blood is sampled continually or at short intervals, frequently. The average concentration of dye in arterial blood is determined by the dye dilution curve, which is drawn using the sampling time interval and the log of dye concentration for each time sample. The duration of the curve for one complete circulation of the dye is determined from the curve.

$$\text{Cardiac output} = \frac{\text{Amount of dye injected x 60}}{\text{(Mean concentration of the dye / L) x Duration of curve}}$$

Flowmeter method

Magnetic flowmeter method

- The electromagnetic flowmeter used measures the electromotive force generated when the blood flowing within a vessel of known diameter passes through a magnetic field at right angles to the magnetic lines of force. Cardiac output is determined by placing the flowmeter probe around the root of aorta.

FACTORS INFLUENCING CARDIAC OUTPUT

- Muscular exercise (4-6 times increase), pregnancy, hypoxia, active digestion, stress and increased body temperature

Circulation time

- It is the time required for one complete circulation.

$$\text{Circulation time} = \frac{\text{Total blood volume (ml)}}{\text{Minute volume (ml)}}$$

- This can be measured either by injecting dyes like ferrocyanide or fluorescine or injecting radioactive chemicals.

Cardiac reserve

- It is the maximum percentage that the cardiac output can increase above normal. Cardiac reserve can be measured by exercise test.
- It is an adaptive mechanism of the heart by which cardiac output can be increased to maintain a balance between right atrial input and left ventricular output.
- This is achieved by
    - Increase in stroke volume
    - Increase in heart rate
    - Hypertrophy of the left ventricle
- During exercise the cardiac output can rise up to 400% above normal. If cardiac reserve is inadequate congestive heart failure or cardiac decompensation results which causes stasis of blood in the pulmonary system and right atrium.

REGULATION OF CARDIAC OUTPUT

- Heart functions (cardiac output, heart rate) can be regulated by
    - Intrinsic mechanisms, operating within the myocardium itself and
    - Extrinsic mechanisms (by factors outside the heart).

INTRINSIC REGULATION OF HEART

Frank's Starling Mechanism/Heterometric Regulation

- Regulation of the cardiac function is independent of the nervous or humoral control. This was proved by the Starling's experiment using Starling's heart lung preparation i.e., isolating the heart from the body so as to remove humoral and neural influences. The heart is perfused through the aorta with an oxygenated solution containing nutrients essential to maintain normal cardiac function. In this preparation, contractility of the heart is studied by allowing the isolated heart to do work under various degrees of pressure and load made in the pulmonary artery or aorta. Starling's law of heart was derived from these studies and was initially known as Heterometric autoregulation and it states that the force or contraction is a function of the initial length of the muscle fibres i.e upto a physiological limit, greater the initial length, greater is the force of contraction.
- An increase in the venous return to the heart increases the input of blood into the ventricles which stretches the ventricular wall and induces a stronger contraction during systole. According to this law the heart makes adjustments to load upon the ventricles during each beat without any neural or humoral influences.
- Preload is the amount of stretch on the heart prior to contracttion. The load is directly proportional to the volume of blood in the ventricles or end diastolic volume. Greater ventricular filling during diastole, stretches the muscle fibres, causes increased fibre

length and a greater amount of Ca^{++} release during systole resulting in increased force of contraction and stroke volume.

- After Load is the pressure that must be exceeded by the ventricles before blood can be ejected through the semilunar valves is called as after load. The resistance is determined by the diameter of the arterioles and pre-capillary sphincters which is controlled by the sympathetic system which in turn, controls the tone of the arteriole muscles as well as the diameter of the blood vessels. Any factor that increases after load, will increase end systolic volume and decrease stroke volume.

Homeometric autoregulation

- This type of regulation occurs when aortic blood pressure increases due to increase in cardiac output or increase in peripheral resistance to blood flow. When venous return is normal but the arterial pressure is increased by peripheral resistance, causes a reduction in the ventricular discharge for few beats and accumulation of ventricular blood volume. This causes subendocardial ischemia resulting in reduced contractile strength. Subsequently, the subendocardial ischemia is relieved by vascular autoregulation, the contractile strength increases without changing the muscle fibre length to pump normal blood volume against increased arterial resistance known as *homeometric autoregulation* and the ventricular volume returns to normal. The significance of homeometric regulation is to allow the ventricles to eject the same stroke volume against varying degrees of arterial resistance without an increase in end diastolic pressure or increase in myocardial fibre length during diastole.
- Cardiac output will change to match changing metabolic demands of the body. The outputs of both ventricles must be identical, and also equal to the venous return of blood from the body. During exercise of the muscles, the increased blood flow through the blood vessels is due to increased metabolism. This causes increased venous return and preload to right ventricular and pulmonary circulation. Consequently, more blood is delivered to the left ventricle and cardiac output increases. There will also be increased contractility and heart rate from the sympathetic activity associated with exercise to meet tissue requirements.

EXTRINSIC REGULATION OF HEART

- Heart regulates its cardiac output based on the functional blood requirement of different parts of the body by adjusting its rate. Heart rate is directly proportional to metabolic activity, but inversely proportional to the size of the animal. Large animals have a slower heart rate than smaller ones.
- Athletic animals (Hare) and animals trained for exercise (Greyhound) has slower heart rate than rabbit or untrained animals of the same species.
- Heart rate is also slow in male and adult animals than in female and young ones.
- Extrinsic regulation of heart may be achieved by two means:
    - Nervous regulation and
    - Chemical regulation

NERVOUS REGULATION OF HEART RATE

- Sympathetic and parasympathetic divisions of ANS play a key role in the regulation of heart rate.
- The nerves regulate to a larger extent the rate rather than the force of contraction either directly or indirectly.

- The two atria are especially well supplied with large number of both sympathetic and parasympathetic nerves, but the ventricles are supplied mainly by sympathetic nerves and fewer parasympathetic fibres.

DIRECT NERVOUS CONTROL

- Cardiac activity is regulated by cardio accelerator (augmentor) nerves and cardio inhibitory nerves.

Cardio accelerator nerve fibres

- Sympathetic nerves to the heart originate as post ganglionic fibres from the inferior, middle and superior cervical and first 4-5 thoracic ganglia, run along with vagus to form the cardiac nerves and plexus which pass to the heart along with great veins and arteries.
- Sympathetic division of the ANS provides two sets of accelerator fibres, the vago-sympathetic fibres**,** originating from the cervical ganglion and run along the vagal fibres and the cardio sympathetic fibres from the 2, 3, 4, 5th thoracic nerves which function in emergency. Neurotransmitter of the sympathetic system is nor-epinephrine, which mainly act through b_1-receptors.
- Sympathetic fibres supply the atria, S.A. node, A.V. node and the ventricles stimulation of which causes increase heart rate and force of contraction referred to as positive chronotropic and positive inotrophic effects, respectively.

Cardio inhibitory nerve fibres

- Vagus, the Xth cranial nerve is the parasympathetic nerve originates from the dorsal nucleus of the medulla, form cardiac plexus with sympathetic nerves of the cervical ganglia. Vagus descends along with sympathetic fibres and terminates on the S.A. node in mammals and ventricles in amphibians.
- In mammals, vagus supplies fibres to the S.A. node, atrial muscles, A.V. node and A.V. bundle. The vagus nerve on stimulation produces inhibitory effect on heart resulting in negative chronotropic and negative inotropic effects. The inhibitory effect is more by right vagus than the left because SA node receives more fibres from the right vagus.
- Stimulation of vagus results in complete inhibition of heart beat (vagal inhibition) for a short duration. Following this the ventricles, start weak contractions called the vagal escape due to the shifting of the pacemaker from S.A. node to ventricular site, the ectopic foci**.**
- Principally, parasympathetic stimulation causes a marked decrease in heart rate and a slight decrease in contractility. Both the sympathetic and parasympathetic fibres of ANS constantly discharge impulses to the heart to keep the heart in tonic state.
- Under normal resting state, the parasympathetic nerves to heart are in tone. Vagal tone is particularly more in athletic animals. Hence, the heart rate is slower in athletic animals.
- Cutting the vagi accelerates heart rate. The sympathetic nerves are more functional only during emergency to meet excessive O_2 and nutrients demand of the metabolically active tissues. Stimulation of sympathetic fibres increases the heart rate, tachycardia and parasympathetic decreases heart rate, bradycardia.

INDIRECT NERVOUS CONTROL OF HEART

- Indirect nervous control of cardiac activity is principally mediated by reflex mechanisms, which are essential for the maintenance of blood pressure (BP). The reflexes are of two groups.
- Cardio accelerator reflex (*Bain bridge reflex*)
    - When the blood volume in the right side of the heart has increased with accumulation of blood, the stretch receptors of the right atrial wall get excited and the impulses are carried through vagus to the vasomotor centre of the medulla.
    - It causes inhibition of the vagal tone to the heart, but stimulates the vasoconstrictor centre. This result in increased force of contraction of the heart and constriction of the blood vessels, the BP returns to normal.
- Cardio inhibitory reflex
    - The two baroreceptors, the carotid sinus and the aortic body are very sensitive to increased BP and play a key role in the regulation of BP and heart rate.
    - Carotid sinus, pressure sensitive stretch receptors is located at the bifurcation of the internal and external carotid artery sense the BP changes and transmits the sensory impulses to cardio inhibitory area of the vasomotor centre of the medulla through afferent nerve, the sinus nerve, a branch of the glossopharyngeal nerve.
    - Aortic body pressure receptors is the thickened portion of the aortic arch get stimulated by increase in BP and the impulses are transmitted through aortic or cardio depressor nerve, a branch of vagus to the cardio inhibitory area of the medulla.
    - On stimulation of these two receptors by increased BP, inhibits vascular sympathetic nerves resulting in vasodilatation. The efferent impulses from cardio inhibitor centres pass through vagus causes reduction in heart rate and also vasodilatation. The sinus and aortic nerves are called as *buffer nerves*.
    - The signals regarding the drop in BP are transmitted to cardio accelerator area of the vasomotor centres of the medulla causes inhibition of the activities of cardio inhibitor centres, whereas the cardio accelerator centre and the sympathetic division are stimulated resulting in vasoconstriction and increased heart rate.

CHEMICAL REGULATION OF THE HEART

Cations

- Na^+ and K^+ ions favour diastole, whereas Ca^{++} ion favours systole. Sudden infusion of the Ca^{++} ions into the heart causes calcium rigor or systolic arrest.

Catecholamines

- Epinephrine and norepinephrine are the neurotransmitters of sympathetic division, acts on β_1 receptors and exerts positive chronotropic (atria and ventricles) and inotropic effects on heart. β_1 receptors are found in S.A. node, A. V. node and the ventricles.
- Stimulation of α_1 receptor (SA node, myocardium) causes positive inotropic effect in the atrial and ventricular muscles. Catecholamine increases the stroke volume and cardiac output during emergency / stress.

Acetyl choline

- The neurotransmitter of parasympathetic division exhibits negative chronotropic (decreased rate) and inotropic (decreased force of atrial contraction) effects, thus decreases cardiac output.

Metabolic acid end products

- CO_2 and other metabolic acid products, H^+ ions, lactic acid lower the p^H below normal, causes increased rate and force of contraction of the heart.

METABOLISM AND ENERGETICS OF WORKING MYOCARDIAL CELLS

- Cardiac muscle requires a continuous supply of oxygen and therefore myocardial cells under normal conditions, utilize aerobic metabolic system that provides a constant supply of high energy phosphate for mechanical and chemical working of the heart.
- The major fuel required for cardiac metabolism is free fatty acids followed by glucose and lactate with a minor contribution from amino acids, ketones and pyruvates.
- Myocardial metabolism has three phases:
    - Energy Liberation or Production: It is the stage during which energy is released from the carbon-hydrogen bonds of substrates including fatty acids, glucose, lactate and other compounds used by the heart.
    - Energy conservation or storage: It is the transfer of the energy into synthesis of high energy phosphate compounds such as ATP, creatine phosphate, AMP, ADP.
    - Energy Utilization: It is the release of this stored chemical energy into shortening or development of tension of cardiac contraction.
- Cardiac muscles cells contains larger number of mitochondria that are rich in enzymes of citric acid cycle and oxidative phosphorylation for muscle contraction. ATP is utilized by the myocardial cells for muscle contraction which is broken down to ADP and inorganic phosphate (Pi).
- The increased amount of ADP and inorganic phosphate (Pi) formed stimulates oxidative phosphorylation, the citric acid cycle and glycolysis thereby increasing the high energy phosphate bond production. Creatine phosphate also provides an immediate source of high energy phosphate bonds.

MODULE-16: CORONARY CIRCULATION

Learning objectives

- This module deals with
    - coronary circulation and its regulation
    - coronary reflexes
    - metabolic control
    - coronary vasodilators and
    - vasodilator chemicals

CORONARY CIRCULATION

- It starts as coronary orifices in the aorta and supplies the blood to the heart muscles through right and left coronary arteries. These orifices are in free communication with aorta both during systole and diastole.
- The main coronary artery branches into right and left coronary arteries. The left coronary artery is more important in dog, cat and ruminants, whereas in man, swine, horse and some primates the right coronary artery is more prominent.
- The left coronary artery gives of the left circumflex branch which runs along the auriculo-ventricular groove and distributes about 80% of the total blood flow to the heart.
- It supplies mainly the atria and left ventricle, A.V. node, inter ventricular septum. The right coronary artery nourishes the right ventricular muscles.
- The venous system of the heart comprises the great cardiac veins, which empties through the coronary sinus drains the blood supplied by the left coronary artery into the right atrium of the heart, whereas the anterior cardiac veins collect the blood supplied by the right coronary artery and discharge their blood into the right atrium. Small amount of blood is drained through the thebesian veins directly into the right ventricle and other chambers of the heart.
- About 4% of the cardiac output flows through the coronary arteries to meet the very high oxygen demand by the cardiac muscles under normal condition (10-15 ml of O_2 is taken by cardiac muscle from each 100 ml of blood as against 5 ml/100 ml in skeletal muscles). This blood supply may be increased to 40% during heavy muscular exercise.
- In the left coronary artery the blood flow is decreased during systole because of the high coronary resistance developed by the contraction of ventricular muscles. However, this gets reversed during ventricular diastole which shows increased coronary blood flow. On the other hand the right coronary artery shows increased blood flow during systole and lower during diastole.
- If a coronary artery is gradually blocked by stenosis, there may be a progressive development of new blood vessels from the pre-existing capillaries, known as the *collateral vessels*.

REGULATION OF CORONARY CIRCULATION

- Heart rate
  - Higher the heart rates, lesser the duration of diastole, and lowers the coronary flow.
- Aortic blood pressure
  - Increase in aortic pressure causes increased coronary flow.
- Peripheral resistance
  - More the peripheral resistance greater will be the aortic pressure and coronary flow increases.
- Autonomic control
  - Both the sympathetic (ß - adrenergic receptors (vasodilators), α - *receptors* (vasoconstrictors) and parasympathetic vasodilators are the primary autonomic control on the coronary arteries. Stimulation of $ß_1$ and $ß_2$ adrenergic receptors results in vasodilatation in major parts of the large coronary arteries, whereas the vasoconstrictor tone by α adrenergic receptors regulates the resting in coronary blood flow. The overall effect of stimulation of sympathetic nerves is coronary vasodilatation. Vasodilatation by vagus is less important.

- Norepinephrine and neuropeptide Y are the neurotransmitters of sympathetic nerves of which neuropeptide Y is strong vasoconstrictor particularly in coronary vasculature.
- Acetyl choline, the neurotransmitters of the parasympathetic system acts on the cholinergic receptors of the endothelium, causes the release of endothelium derived relaxing factor (EDRF). The EDRF acts on the smooth muscles of the blood vessel causes vasodilatation.

CORONARY REFLEXES

- Neurally mediated coronary vasomotor reflex effects can be elicited from the carotid sinus and aortic body.
- Increased venous inflow reflexly stimulates vagus and causes reflex vasoconstriction in abdominal and thoracic organs to divert increased coronary flow by vasodilatation.

METABOLIC CONTROL

- The most important controlling factor of coronary circulation is oxygen demand. Lack of oxygen in tissue release *adenosine*, which is a potent vasodilator and other metabolic products like CO_2, H^+ and lactic acid also produce vasodilatation.

VASODILATOR CHEMICALS

- Nitro-glycerine,
- Halothane,
- Nikethamide,
- Caffeine,
- Camphor and
- Nitrates.

HORMONES

Vasodilatation

- Glucagon,
- Insulin,
- Thyroxine,
- Serotonin,
- Histamine,
- PGE_2,
- Prostacyclin,
- VIP and
- Substance P

Vascoconstrictor

- Vasopressin,
- PGF_2□,

- Thromboxane A_2,
- Leukotriene D_4 and
- Angiotensin II are the vasoconstrictor hormones**.**
- *Endothelin* is the most potent of the all mammalian vasoconstrictor substances.

MODULE-17: ELECTROCARDIOGRAPHY (ECG)

Learning objectives

- This module deals with
  - recording of electrical activity of the heart
  - significance of ECG

ELECTROCARDIOGRAPHY

- Electrocardiography (ECG) is the recording of the electrical changes occurring during the process of depolarization and repolarization of the heart. The instrument, the electrocardiograph is used to take the recording known as elctrocardiogram.
- Einthoven (1908) first recorded ECG using bipolar leads. The depolarisation and repolarisation of cardiac muscle generate electrical current. It flows in a circuitous route from depolarized to polarized area, which can be measured by the electrodes (leads) which are placed on the surface of the body surrounding the heart.
- ECG is recorded as deflection or waves using lead systems. A lead is connection of two points /parts of the body through electrodes and wires with the electrocardiograph. If the electrical impulse travels towards the positive pole, a positive (upward) deflection occurs. If the impulse moves towards negative lead the deflection is negative (downward).

COMMON LEAD SYSTEMS

Bipolar limb leads (by Einthoven)

  - Lead I: Right arm, (- ve) & Left arm (+ ve)
  - Lead II: Right arm (- ve) & Left leg (+ ve)
  - Lead III: Left arm (- ve) & Left leg (+ ve)
  - The right hind leg is connected to the earth.
- The triangle formed by the three lead systems with the heart at the centre is referred to as *Einthoven's triangle*.

Augmented unipolar limb leads

- Basically, an augmented unipolar limb lead compares the electrical activity at the reference limb (right arm, or left arm or left leg) to the sum of the electrical activity at the other two limbs. Unipolar leads permit more precise location of direction of electrical potential within the myocardium.
  - Lead aVR: Right arm (+ ve) & Left arm and left leg (- ve)
  - Lead aVL: Left arm (+ ve) & Right arm and left leg (- ve)
  - Lead aVF: Left leg (+ve) & Right and left arm (- ve)

- Among the augmented unipolar limb leads only aVR shows inverted deflections when compared to others.

Unipolar chest leads (Exploring/ precordial/ chest leads)

- These leads record the potential from the anterior surface of the heart, from right side to left side. A central chest lead is placed at different locations on the chest and these leads are designated as V leads. The common positions of V leads are
    - V1 – 4th intercostal space at the right sternal border
    - V2 – 4th intercostal space at the left sternal border
    - V3 – equidistant between V2 and V4
    - V4 – at mid clavicular line in the 5th intercostal space
    - V5 – anterior axillary line at the same level as V4.
    - V6 – midaxillary line in the 5th intercostal space
    - V7 – posterior axillary line in the 5th intercostal space
    - V8 – posterior scapular line
- These electrodes are positive poles, which are compared with the average voltage across the three standard leads (Lead I to III). These leads record the potentials from the anterior surface of the heart and provide additional information about right and left heart enlargement.

ECG WAVES

- Deflection of normal ECG are referred as P, QRS and T waves

P wave

- It is a positive wave that indicates the sum of all the electrical potentials produced during the depolarisation of both atria and the spreading of the electrical activity from S.A node throughout the atrial musculature. It slightly precedes atrial systole and initiates the atrial contraction. There is a very brief interval between P wave and atrial systole, which is due to time lag between electrical and mechanical events.

QRS wave

- *QRS complex* shows the spreading depolarization wave through the ventricular muscles, thus initiates ventricular systole (depolarization). QRS wave precedes isovolumetric contraction.

T wave

- The positive (upward) deflection precedes ventricular diastole and indicates the initiation of ventricular repolarization (relaxation). ECG has no separate deflection or wave for atrial relaxation, which is due to the fusion of this deflection with QRS complex.

ECG INTERVALS

- P-Q or P-R interval is the duration of time between the beginning of the P wave and the beginning of the QRS wave. It indicates the time of excitation wave travel from SA node to purkinje system which includes short A.V nodal delay after the atrial contraction to permit complete ventricular filling. P-R interval indicates atrio ventricular conduction time. Normally P-R interval is 0.1 sec.
- Q-T interval is the duration from the beginning of Q wave to the end of T wave which indicates the duration from initiation of ventricular depolarization to completion of repolarization or ventricular contraction.
- S-T interval is the duration between the beginning of S wave and the beginning of T wave during which the ventricles remain depolarized.

AVERAGE HEART RATE PER MINUTE AND ECG INTERVALS

Species	Average heart rate	Intervals in seconds		
		PR	QRS	QT
Cattle	70 (48 – 84)	0.18	0.09	0.39
Horse	35 (28 - 40)	0.30	0.11	0.52
Swine	100 (70-120)	0.13	0.06	0.32
Sheep/goat	100 (72-120)	0.13	0.04	0.28
Dog	100 (70-120)	0.10	-	0.06
Cat	130(120-140)	0.07	0.04	0.17

- The time between two successive P waves is P-P interval, corresponds to time between atrial contractions which can be used to calculate the number of atrial contractions per minute.
- Similarly, R-R interval can be used to calculate the ventricular rate.

SIGNIFICANCE OF ECG

- It is a non-invasive method to evaluate cardiac function.
- To diagnose ventricular hypertrophy.
- To evaluate conduction system blocks, myocardial infarction and drug effects etc.

MODULE-18: ECHOCARDIOGRAPHY

Learning objectives

- This module deals with
  - principles of echocardiography
  - types of echocardiography

ECHOCARDIOGRAPHY

- Echocardiography is a non invasive tool for imaging the heart of the surrounding intrathoracic structures to diagnose various cardiac diseases to assess cardiac function.

Uses

- To evaluate size of cardiac chambers
- To study the thickness and movement of the wall
- To study the structure and movement of the valves
- To detect pericardial and pleural fluid
- To find out congenital cardiac anomalies
- To identify mass lesions within and adjacent to the heart
- To diagnose valvular and myocardial pathology

PRINCIPLE OF ECHOCARDIOGRAPHY

- Echocardiography employs high frequency sound waves greater than 20,000 Hz. These ultrasound waves when passed through the tissues are reflected back from the tissues based on the acoustic impedance of the tissue. The amount of reflection of ultrasound waves depends on the difference in the acoustic impendance between two adjacent tissues. i.e. greater the difference, greater in the reflection. Bone tissue and air/tissue interfaces are highly reflective. Bone has a very high acoustic impedance whereas air has a very low acoustic impedence, as well as the soft tissue.
- Echocardiography is usually performed in the inter spaces within the cardiac windows. Ultrasound waves are transmitted through the tissue at a known speed depending upon the type of tissue through which it travels. Ultrasound obeys the laws of geometric optics with regard to reflection, transmission and refraction.
- When an ultrasound wave meets an interface of differing acoustic impedance, the wave is reflected, refracted and absorbed. The intensity of the ultrasound beam decreases as it travels away from the transducer because of beam divergence, absorption, scatter, and reflection of wave energy at tissue interfaces. The largest ultrasound reflection occurs when the ultrasound beam is perpendicular to the imaged structure, creating a strong reflection or echo.
- These reflected ultrasound waves are then received by the transducer and processed by the ultrasound machine to create an image. The transducer acts as a receiver over 99% of the time. The echocardiographic images obtained are displayed on the monitor and can be recorded on videotape, thermal paper, radiographic film or computer disc.
- The frequency of the ultrasound waves emitted by the transducer markedly influence the quality of the image obtained and the depth of tissue that can be imaged successfully. Higher frequency ultrasound waves have a shorter wavelength and yield better resolution of small structures close to the skin surface.
- However, more energy is absorbed and scattered with high frequency ultrasound and thus, high frequency transducers have less penetrating ability. Conversely, a lower frequency transducer will have a greater depth of penetration but poor resolution. The transducer selected for echocardiography should be the highest frequency available that will penetrate to the depths needed to image the heart in its entirety. Frequencies generally used for veterinary echocardiography range from 2.25-3.5 MHz for adult horses and cattle and 3.5-10.0 MHz for small animals, small ruminants, foals, calves and exotics.

TYPES OF ECHOCARDIOGRAPHY

- There are three types of echocardiography, used clinically:
  - M-mode,
  - Two-dimensional (2-D, B-mode or real time), and
  - Doppler echocardiography.

M-mode Echocardiography

- It is a one-dimensional ("ice-pick") view of the cardiac structures moving over time. The echoes from various tissue interfaces along the axis of the beam are moving during the cardiac cycle and are swept across time, providing the dimension in time. The lines on the recordings correspond to the position of the imaged structures in relation to the transducer and other cardiac structures at any instance in time. More accurate placement of the M-mode cursor within the heart is performed by using the two-dimensional (2-D) real-time image as a guide. The M-mode echocardiogram uses a high sampling rate and can yield cleaner images of cardiac borders, allowing the echocardiographer to obtain more accurate measurements of cardiac dimensions and more critically evaluate cardiac motion.
- Standard M-mode views are obtained from the right parasternal position. The standard M-mode views utilized in the veterinary medicine include the left ventricle (at the level of the chordae tendineae), the mitral valve and the aortic root (aorta/left artrial appendage) view.

Two-Dimensional Echocardiography

- Two-dimensional echocardiography allows a plane of tissue (both depth and width) to be imaged in real time. Thus, the anatomic relationships between various structures are easier to appreciate than with M-mode echocardiographic images. An infinite number of imaging planes through the heart are possible, however, standard views are used to evaluate the intra and extracardiac structures. The standard views are obtained from either the right parasternal window in all species and from the left parasternal window in adult large animals or in other species when imaging the heart from the left side is desirable.

Doppler echocardiography

- Doppler imaging allows evaluation of blood flow patterns, direction, and velocity; thus, it permits documentation and quantification of valvular insufficiency or stenosis and cardiac shunts. Estimation of blood flow and cardiac output can also be made. Doppler echocardiography is based on detection of frequency changes (the Doppler shift) occurring as ultrasound waves reflect off individual blood cells moving either away from or toward the transducer. Calculation of blood flow velocity is possible when the flow is parallel to the angle of the ultrasound beam.
- Two types of Doppler echocardiography are used clinically:
  - Pulsed wave and
  - Continous wave.
  - Pulsed wave

        - Pulsed wave (PW) Doppler uses short bursts of ultrasound transmitted to a point (designated the "sample volume") distant from the transducer.
        - The advantage of this type of Doppler is that blood flow velocity, direction and spectral characteristics from a specified point in the heart or blood vessel can be calculated.
        - The main disadvantage is that the limited.
    - Continuous wave
        - Continuous wave (CW) Doppler uses dual crystals so that ultrasound waves can be simultaneously and continuously sent and received.
        - There is no maximum measurable velocity (Nyquist limit) with CW so high velocity flows can be measured.
        - The disadvantage with CW Doppler is that sampling of blood flow velocity and direction occurs all along the ultrasound beam, not in a specified area.
- Color flow Doppler echocardiography is a form of PW Doppler ultrasonography which combines the M-mode and 2-D modalities with blood flow imaging. With color flow Doppler, multiple sample volumes are analyzed along multiple scan lines. The mean frequency shift obtained from these many sample volumes is color-coded for direction and velocity. Several types of mapping are usually available. Most systems code blood flow toward the transducer as red and flow away as blue. Differences in relative velocity of flow can be accentuated, and the presence of multiple velocities and directions of flow (turbulence) can be indicated by different maps which utilize variations in brightness and color.

MODULE-19: PULSE, REGULATION OF HEART AND CIRCULATION – HEMODYNAMICS

Learning objectives

- This module deals with
    - pulse
    - velocity of pulse wave
    - venous pulse
    - physiology of circulation (Hemodynamics)
    - distribution of blood in the systemic circulation
    - compliance (capacitance)
    - blood velocity Resource

PULSE

- It is a wave of expansion and elongation followed by recoiling of the arterial walls and due to the forceful entry of the blood from the aorta during each heart beat. It originates from aorta spreads throughout arterial system and disappears at the arterioles.
- Systolic rush of blood into aorta causes steep increase in the distension of the arterial walls represents anacrotic limb of the pulse wave, whereas the elastic recoiling of the distended arteries causes a drop in the declining slope in the pulse wave, referred as catacrotic limb.
- During the end of systole and the beginning of isometric relaxation phase of cardiac cycle, there will be drop in the intra-ventricular pressure causes the back flow of blood in to the aorta. This results in closure of the aortic valve and the distension of the root of the aorta. This creates a drop in the intra-arterial pressure characterized by dicrotic notch or incisura of the pulse.

VELOCITY OF PULSE WAVE

- It is greater than blood velocity
- Rigid vessels have more velocity than elastic arteries
- Velocity is less in large central arteries than in small peripheral arteries
    - Aorta : 3-5 m/sec
    - Large artery : 7-9 m/sec
    - Small artery : 15-40 m/sec

VENOUS PULSE

- It is the pulsation seen in the large veins near the heart.
- It is of two types
    - If the wave travels towards heart it is referred as *positive venous pulse*. Downward movement of AV valves, lengthening of the atrium followed by a drop in the intra atrial pressure, gradual atrial filling (v wave of the atria), all result in positive venous pulse.
    - Bulging of AV valves during ventricular systole causes *negative venous pulse* (away from heart) in the jugular vein.

PHYSIOLOGY OF CIRCULATION (HEMODYNAMICS)

- Circulation is a continuous circuit through which the same amount of blood flow through each subdivision of circulatory system is maintained by the heart.
- In a resting animal about 25% of the blood volume is in the *central circulation* (pulmonary and coronary circulation) and the rest 75% in the *systemic circulation*.

DISTRIBUTION OF BLOOD IN THE SYSTEMIC CIRCULATION

- Arteries and arterioles : 15%
- Capillaries : 5%
- Venules and veins : 80%

- Aorta is the biggest blood vessel, originates from the left ventricle of the heart and continues as large arteries. Both the aorta and the large arteries have more elastic tissues with less smooth muscle; hence known as elastic vessels. Arteries are the high-pressure passage way and deliver blood to capillaries. These vessels function in the maintenance of blood pressure during diastole by their elastic recoiling effect.
- The large arteries form the main arterial branches thar give rise to the terminal branches or smaller arteries, which have smooth muscles in their walls. They are continued as the small arterioles or the true arterioles, which are continued as meta-arterioles. Small arteries and arterioles have thick walls with less elastic tissue and more of smooth muscle, called as muscular vessels. These arterioles are gates of the systemic circulation, which may constrict or dilate to regulate blood flow to capillaries.
- Capillaries are tiny tubular network composed of only a thin layer of endothelium that favours selective permeability of water, oxygen, nutrients, CO_2 and other metabolic waste between blood and tissues and they are known as exchange vessels. In capillaries the blood cells flow in single layer with RBCs in the centre and WBCs in the periphery. Capillaries form the highest cross-sectional area in the circulatory system.
- The true capillaries are interconnected which forms the place of nutrient exchange. At the point the capillaries branch off from metarterioles, the capillaries are enriched by smooth

muscle called precapillary sphincter. The metarterioles and the precapillary sphincters are not innervated and are controlled by local conditions in the tissues.

- Blood flows from the arteriole directly into metarteriole then leads to capillaries and capillary like channels which connect directly with veins called as A - V shunts.
- When the tissues are at rest and large amount of blood is not required, the sphincters may be completely closed. When large amount of nutritive blood flow is required for active tissues, the metarterioles and the sphincters dilate to shunt the blood nearly the entire length into the capillary bed. The precapillary sphincter undergoes periodical contraction at intervals of 15 second to 3 minutes, called as vasomotion.
- Vasomotor mechanism causes either vasoconstriction (decreased diameter) or vasodilatation (increased diameter) of the arteries and arterioles. It involves an active contraction of the smooth muscles of arterioles and precapillary sphincters, whereas vasodilatation is a passive event involving only relaxation of smooth muscle.
- Arteriovenous anastomoses or shunt occurs between arteries and veins. It is not involved either in blood flow through true capillaries or in the exchange of materials between blood and extracellular fluid. These shunts are primarily by passing the resistance pathways of the precapillary area in tissues during a period of low requirement for nutritive blood supply. In dogs arteriovenous anastomose is primarily functional in thermoregulation allows increased blood flow to the tongue for evaporatory heat loss through panting. This structure is under separate neural and hormonal control.
- The terminal vascular beds consist of small arterioles, metarterioles, capillaries and venules forms microcirculation. Capillaries continue as venules, a thin walled network which unit to form terminal veins. Venules and veins are thin walled structures contain elastic tissues and smooth muscles. Veins function as a blood reservoir called as capacitance vessels. When dilated, they can store large amount of blood and by constriction shift blood volume and affect cardiac venous return.
- The veins continue as vena cava, which drains its content into the right atrium. Vena cava has the highest diameter.
- Veins in the extremities have valves. These valves prevent back flow of blood, thus permits unidirectional blood flow towards heart. There are no valves in the major abdominal and thoracic veins.
- Hemal nodes are the nodal structure found in cattle and sheep on the course of small blood vessels. They are similar in structure and function like spleen. Erythropoiesis occurs during fetal period and granulopoiesis occur in postnatal life.

COMPLIANCE (CAPACITANCE)

- It is the total storage capacity in a given portion of the circulatory system for every mm Hg of pressure rise.
- Veins have about 24 times more capacitance than arteries since about 80% of the total blood volume is located within the venous system.

TYPES OF ACTIVE IMMUNITY

- The blood flows at a steady rate through a long smooth vessel with each layer of blood remaining in the same distance from the wall i.e it flows in streamlines. The central portion of the blood stays in the centre of the vessel. This type of flow is called as laminar flow or streamline flow. If the blood flows in all direction in the vessel and continually mixing within the vessel it is called as turbulent flow. In turbulent flow, the blood flows cross wise in and along the vessel forming whirls in the blood called as eddy currents.

Turbulent flow occurs when the rate of blood becomes too great or when it passes an obstruction in a vessel or when it passes over a rough surface.

- Blood flow through the blood vessel is determined entirely by two factors
  - The pressure difference between the two ends of the vessel, which is the force that pushes the blood through the vessel
  - The impediment to blood flow the vessel known as the vascular resistance
- Blood flow through the vessel is calculated by Ohm's Law

$$\mathbf{Q = \Delta P/R}$$

  - Where,
  - Q = Blood flow
  - ΔP = Pressure difference between the two ends of the vessel (P_1-P_2)
  - P_1 = Pressure at the beginning of the vessel
  - P_2 = Pressure at the end of the vessel
  - R = Resistance
- According to this formula, the blood flow is directly proportional to the pressure difference but inversely proportional to the resistance

BLOOD VELOCITY

- It indicates the direction and the rate of blood flow in the blood vessels. It is directly proportional to the minute volume but inversely proportional to the cross sectional area of the blood vessel and circulation time. Velocity is high in the arteries and veins but slow in capillaries.
- The arterial portion of the circulatory system shows progressive increase in the cross sectional area due to extensive branching of the arterial tree, thus causes progressive fall in the velocity towards capillary bed. On the contrary, the venous system shows decreasing cross sectional area towards heart that results in progressive increase in blood velocity.
- In *dogs*, the cross-sectional area of aorta is 0.8 cm^2, large arteries is 3.0 cm^2 and in capillaries it is 600 cm^2.

Parts	Velocity of blood flow (cm/sec)
Aorta	13
Large arteries	6
Arterioles	0.3
Capillaries	0.05
Venules	0.1
Veins	1.0
Vena cava	9

- The distance travelled by a column of blood in a given time through a specific blood vessel is known as *linear blood velocity*.

$$\text{Mean velocity (mm/sec)} = \frac{\text{Blood volume / min}}{\text{Cross section}}$$

The peak velocity occurs during the period of maximal ventricular ejection followed by reduced velocity during the period of reduced ventricular ejection.

MODULE-20: BLOOD PRESSURE, VENOUS PRESSURE AND ARTERIA PRESSURE

Learning objectives

- This module deals with
  - blood pressure
  - mean circulatory filling pressure
  - determination of arterial, capillary and venous blood pressure
  - factors influencing production and maintenance of BP
  - normal BP in animals
  - regulation of arterial pressure
  - role of veins in BP regulation
  - hormonal regulation

BLOOD PRESSURE

- It is the pressure exerted by the circulating blood against any unit area of the blood vessel wall. Stephen Hales (1730) demonstrated the existence of pressure in the blood vessels. Average pressure in the aorta is 98 mm Hg and the mean pressure in veins is 3 mm Hg and this pressure difference (98-3 = 95) moves the through the systemic vessels. This pressure difference between the aorta and veins is the perfusion pressure.
- BP is highest in aorta (98 mm Hg), moderate in capillaries and lowest in the vena cava (3 mm Hg). This pressure gradient favours the blood flow through the blood vessels.
- During ventricular systole the arteries show the highest pressure, referred to as systolic pressure. It is due to increased blood volume and distension of the arterial walls. During ventricular diastole, blood continues to flow out of the aorta into small arteries, the volume of blood in the large arteries decreases, the arteries become less distended and blood exert less pressure in the arteries. The minimum pressure reached in the blood vessels before each new ventricular ejection is referred to as diastolic pressure.

Pulse pressure = the difference between systolic and diastolic pressure

- The pulse pressure increases as the blood flows from aorta to distal arteries and then becomes less and less when the blood moves towards periphery. It disappears in the arterioles and capillaries.

- *Systolic pressure:* Indicates the total kinetic energy imparted to the blood by the heart.
- *Diastolic pressure:* Reflects the state of peripheral vessels and load on vascular wall.
- *Pulse pressure:* Ventricular output and measure the variations of kinetic energy of heart.
- *Mean pressure* is the average pressure in a blood vessel over a given period of time. This represents potential energy that draws blood through the systemic circulation.

- This formula is useful to find out mean pressure in major arteries distal to aorta but not in aorta because the pattern of arterial pressure pulsation change as the pulse moves away from the heart. The pressure pulses are more triangular in large arteries not in aorta.
- Arterial BP is always expressed in mm Hg whereas capillary and venous pressure can be expressed as mm H_2O.

MEAN CIRCULATORY FILLING PRESSURE

- This eventual pressure in the static circulatory system when there is no pressure difference between the aorta and the vena cava and the pressure throughout the circulatory system which is called mean circulatory filling pressure, and it is is 7 mm Hg.
- This pressure is caused by the static blood distending the blood vessels; the vessels being elastic, they recoil and this recoiling accounts for the pressure in the static circulation.

DETERMINATION OF ARTERIAL BLOOD PRESSURE

- It helps in diagnosing the defects of heart and circulatory system.
- It can be measured by two means,
    - The direct and
    - The indirect (clinical) methods

Direct method

- Experimentally the carotid artery can be connected to any of the following devices such as, the mercury manometer, membrane manometer or the optical manometer to record BP. Mercury manometer is a `U' glass tube containing mercury in one limb and 10% sodium citrate in the opposite limb to balance the mercury.
- The limb with sodium citrate is connected to carotid artery through a tube with a cannula at its end. The float over the mercury column will record the BP over the kymograph.

Indirect method

- Clinically, BP is measured by sphygmomanometer (mercury manometer). It has an inflatable cuff connected to the manometer through a rubber tube. By another tube, the cuff is connected to an inflating bulb. A deflating valve is attached to the inflating bulb to deflate the air from the cuff.
- The cuff is placed in the arm (brachial artery) in human, thigh region in dog (femoral artery), tail region in cattle (middle coccygeal artery) which may be inflated by the inflating bulb. When the pressure in the cuff exceeds the pressure of the artery, the

blood vessel collapses and no blood flow through it. If the stethoscope is placed below the level of the cuff there won't be hearing of any sound of blood flow. The pulse also completely disappears.

- When the pressure in the cuff is released gradually by the deflating valve the pressure of the cuff slowly drops. When the cuff pressure is less than the pressure of the blood vessel, the blood begins to flow through the partially opened artery producing the sounds that are called Korotkoff sounds. The pulse and sound reappear.
- The pressure indicated by the manometer when the first sound is heard represents the systolic pressure. As the cuff is further deflated, more blood returns to the artery the sound gradually reduces (muffles) and disappears, the normal pulse is re-established. The pressure at muffling or at disappearance of sound indicates diastolic pressure.
- When the BP is measured using stethoscope, the method is called as auscultatory method. In palpatory method pulse is used to find systolic pressure only.
- Capillary pressure can be measured by introducing a micropipette into the capillary. In human, the capillary pressure at arteriolar and venous end is 45 and 22 mm H_2O, respectively.
- Venous pressure is very low and can be estimated by inserting a needle into the vein. Saline is allowed to flow into the vein from a long vertical graduated tube.
- The height of the vertical column is adjusted so that the flow ceases, the height of the top of the saline column above the heart is the venous pressure in mm H_2O. It is very low in great veins and negative at atria. BP determination helps in diagnosing disorders of heart and circulatory system.

Ultrasound Method

- In this method the Korotkoff sounds are amplified by using piezoelectric microphones mounted within or below the cuff. The electric signal obtained is amplified to increase the audibility.

Microphone Method

- In this method ultrasound is used to detect arterial wall movement as pressure is decreased in the blood pressure cuff.

DETERMINATION OF CAPILLARY PRESSURE

- Capillary pressure can be measured by introducing a micropipette into the capillary.
- In human, the capillary pressure at arteriolar and venous end is 45 and 22 mm H_2O, respectively.

DETERMINATION OF VENOUS PRESSURE

- Venous pressure is very low and can be estimated by inserting a needle into the vein.
- Saline is allowed to flow into the vein from a long vertical graduated tube.

- The height of the vertical column is adjusted so that the flow ceases, the height of the top of the saline column above the heart is the venous pressure in mm H_2O.
- It is very low in great veins and negative at atria.
- BP determination helps in diagnosing disorders of heart and circulatory system.

FACTORS INFLUENCING PRODUCTION AND MAINTENANCE OF BLOOD PRESSURE

- Heart beat
- Peripheral resistance
- Elasticity of arteries
- Volume flow
- Diameter of the blood vessels

HEART BEAT

- During systole, the arterial pressure increases.
- The rate at which blood enters the arterial system exceeds the rate at which it drains through arterioles and capillaries, increasing the pressure.
- During diastole, the BP decreases since blood passes out of arteries into capillaries.
- When other three factors remain constant, an increase in rate of the heart raises BP and decrease in heart rate will cause a fall in BP.

PERIPHERAL RESISTANCE

- It is caused by internal friction produced by the viscosity of blood and mostly present in arterioles and capillaries (peripheral circulation).
- BP is directly proportional to peripheral resistance. When most capillaries are opened peripheral resistance decrease causes reduction in BP.
- Resistance varies inversely with the fourth power of the radius of the vessels and directly proportional to the viscosity, and length of the vessel. Total peripheral resistance (TPR) is the resistance to flow of blood in the whole of systemic vessels.

$$\text{TPR} = \frac{\text{Aortic pressure (BP)}}{\text{Cardiac output}}$$

or

BP = TPR x Cardiac output

ELASTICITY OF ARTERIES

- The elastic recoil of large arteries is responsible for the continuous flow of blood in the arterioles and capillaries.
- During systole when more blood enters aorta and large arteries, they expand to store the blood i.e. potential energy is stored and during diastole, the stored blood is released due to elastic recoil of the walls of arteries causing blood to flow into arterioles during diastole.
- Arterial stiffening increases systolic pressure (due to reduced expansion of artery) and decreases diastolic pressure (since less blood is stored by reduced expansion).

VOLUME FLOW

- An increase and decrease in the blood volume through the circulatory system causes either increase or decrease in BP.
- When hemorrhage occurs in a large artery, the BP falls immediately due to drop in the blood volume.
- On moderate hemorrhage and when small arteries are cut, the fall in BP will be negligible because of the compensatory vasomotor mechanism and spleenic constriction.

NORMAL BP IN ANIMALS

BP	Cow	Horse	Sheep	Pig	Dog	Cat	Giraffe	Human
Systolic	140	130	140	140	120	140	260	120 mm.Hg
Diastolic	95	95	90	80	70	90	160	70 mm.Hg

REGULATION OF ARTERIAL PRESSURE

- Two mechanisms regulate the arterial pressure.
- Rapidly acting short term regulatory mechanism operates through nerves or hormones.
- Long term control of arterial pressure.
- When the BP undergoes change, after few hours to few days the nerves loose their power to control pressure because receptors adapt.
- Long-term mechanisms come into play, which involve renal fluid volume-pressure control mechanisms.

SHORT TERM REGULATION

- Nervous regulation
  - Baroreceptor reflex
  - Atrial volume receptor reflex
  - Bain bridge reflex
  - Psychogenic responses

        - Defense alarm reaction
        - Vaso vagal syncope
    - Chemoreceptor reflex
- Hormonal regulation
    - Renin- angiotensin system
    - Vasopressin

BARORECEPTOR REFLEX

- Arterial blood pressure is monitored by pressure sensitive nerve endings known as *baroreceptors; Carotid sinus and aortic body* located in the carotid bodies and aortic arch are sensitive to stretch of the arterial wall. The frequency of impulses generated from baroreceptors is proportional the BP.
- They send afferent impulses through sinus and aortic nerves (Buffer nerves) to vasomotor areas of the CNS which monitor beat to beat changes in BP.
- The CNS reflexly alters cardiac output via sympathetic and parasympathetic divisions of ANS altering the vascular resistance to keep blood pressure at a set point by either vasoconstriction or vasodilatation.
- The vasomotor area of the brain responds to an increase in BP by decreasing sympathetic activity and enhancing parasympathetic activity. Heart rate and force of contraction are decreased and stroke volume is decreased.
- Sympathetic inhibition causes vasodilatation of arterioles results in decreased peripheral resistance. All of these effects restore arterial BP to normal. A decrease in BP produces opposite effect to those considered above.

ATRIAL VOLUME RECEPTOR REFLEX

- This reflex is initiated by *stretch/ volume receptors* located in the walls of the left and right atria which are sensitive to change in atrial blood volume (increase or decrease).
- When blood volume is decreased due to hemorrhage and dehydration the CNS receives fewer impulses from the volume receptors and CNS reflexly increases the sympathetic activity to heart and blood vessels, but reflexly decreases parasympathetic activity.
- The volume receptor reflexly stimulates the thirst center of the hypothalamus to increase blood volume by drinking of water. It also activates renin-angiotensin-aldosterone system to conserve sodium along with water. ADH secretion is also enhanced to facilitate water reabsorption from the renal tubule.

BAIN BRIDGE REFLEX

- This reflex operates when the venous return in increased

PSYCHOGENIC RESPONSES

- Two important psychogenic responses are the

- Defense- alarm reaction
- Vaso-vagal syncope

- Defense alarm reaction
  - It is known as fight or flight and emotional response to a threatening situation such as physical injury or trauma.
  - It involves increased sympathetic and decreased parasympathetic activity which includes increased heart rate and stroke volume, vasoconstriction in kidneys, splanchnic organs and skin and vasodilatation in coronary vessel and skeletal muscles and increased BP.
  - There is enhanced secretion of ADH, angiotensin II and ACTH, retention of water and sodium ions to improve blood volume.
- Vaso vagal syncope
  - It occurs in response to certain emotional situation in which some animals faint due to decrease in BP.
  - It involves a decrease in sympathetic activity and increase in parasympathetic activity causing vasodilatation and decrease in TPR.
  - Heart rate and cardiac output decreases with a drop in BP resulting in inadequate blood flow to brain and fainting.

CHEMORECEPTOR REFLEX

- Carotid bodies and aortic bodies contain chemoreceptors.
- The excess CO_2 or H^+ from the body fluids or lack of oxygen or blood supply to these receptors stimulate the chemical sensory impulses through sinus and aortic nerves to vasomotor centre and BP is reflexly elevated to eliminate excess quantity of CO.

ROLE OF VEINS IN BP REGULATION

- Sympathetic stimulation of veins decreases the capacity of veins, venous return to heart increases and cardiac output is increased. Hence BP elevated.

HORMONAL REGULATION

- The sympathetic system influences the cardiovascular system through the release of neurotransmitter norepinephrine (NE) and also activates adrenal medulla to release epinephrine(EPN).
  The EPN and NE activate membrane receptors called adrenergic receptors on the cardiac muscle cells and on the endothelial or smooth muscle cells of blood vessels.
- The adrenergic receptors include α and β receptors which are subdivided into α1 & α2 andβ1&β2.
  The EPN and NE act through α receptors and causes arteriolar vasoconstriction.
- Resistance to blood flow is increased and total peripheral resistance (TPR) increases.
- These hormones also produce venoconstriction (α receptor activation), blood is displaced to central circulation and ventricular preload increases.

- The hormones also activate β receptors in heart and increases rate and force of cardiac contraction resulting in increased stroke volume, TPR, preload and resulting in elevation of BP.
  EPN has an affinity for both α and β receptors causes increased cardiac output, decreased peripheral resistance in skeletal muscle (vasodilatation) and elevated BP, whereas the NE has high affinity for α receptors but low affinity for β receptors causes vasoconstriction and rise in BP.

RENIN-ANGIOTENSIN SYSTEM

- Decrease in renal arterial pressure, ECF or blood volume, stimulation of sympathetic nerves or disturbance in ECF sodium level causes release of renin from the juxta glomerular apparatus of the kidneys.
- The renin, a proteolytic enzyme, splits angiotensinogen, a plasma α_2 globulin (synthesised from the liver) to form angiotensin-I by the enzymatic action in the kidney.
- Angiotensin–I is converted to angiotensin-II by the enzyme system present in the capillary endothelium of lungs. The angiotensin-II is a very powerful vasoconstrictor and destroyed by angiotensinases in peripheral capillaries.
- Angiotensin-II stimulates aldosterone release from adrenal cortex.
- Aldosterone acts on the thirst centre of the hypothalamus, and renal tubule causes increased water intake and retention of sodium and water from the renal filtrate.
- Aldosterone also causes vasoconstriction.
- All the three actions of aldosterone elevate the BP to normal.

VASOPRESSIN

- Its physiological role is related to long term regulation of BP brought about by water reabsorption from the kidneys.
- In hemorrhage, large amounts of ADH are released to bring about vasoconstriction.
- When the BP undergoes change, after few hours to few days the nerves loose their power to control pressure because receptors adapt.
- Long-term mechanisms come into play, which involve renal fluid volume-pressure control mechanisms.

LONG TERM REGULATION

- Long term control is brought about by renal regulation of blood volume.
- Blood volume has a direct effect on cardiac output and therefore affects blood pressure.

Direct renal mechanism

- An increase in BP or blood volume causes increased filtration rate in the kidney.
- As filtration rate exceeds the reabsorption rate in the kidney tubules, more urine is produced resulting in increased fliud loss and decreased blood volume.

Indirect renal mechanism

- If arterial BP declines, Renin angiotensin system comes into play. Angiotensin II also stimulates ADH release from the posterior pituitary gland.
- ADH increases water reabsorption from the kidney tubules.
- Increased sodium and water reabsorption from the kidneys leads to increased blood volume and BP.

MODULE-21: CONTROL OF BLOOD VESSELS AND VASOMOTOR REFLEXES

Learning objectives

- This module deals with
  - vasomotor mechanism
  - control of arterioles
  - control of capillaries
  - microcirculation
  - control of veins
  - vasomotor reflexes

VASOMOTOR MECHANISM

- It deals with the maintenance of the diameter of the blood vessels (arteries and arterioles) to maintain BP, thus regulates the blood flow to various organs or tissues according to their demand.

Functions

- Regulates blood flow to different organs according to their demand. *i.e.* shifting of blood from one organ to another on demand.
- Regulates peripheral resistance, thus the BP.

CONTROL OF ARTERIOLES

- It involves
  - Neural regulation
  - Hormonal regulation

NEURAL REGULATION BY VASOMOTOR CENTERS

- It is located bilaterally in the medullary reticular formation and lower pons and includes the
  - Vasoconstrictor and
  - Vasodilator centers.

- This centre receives sensory signals through vagus and glossopharyngeal nerves. The output signals from these centers regulate their vasoconstrictor or vasodilator activities on blood vessels through sympathetic and parasympathetic divisions of ANS.
- Vasoconstrictor centre is a bilaterally located centre, extends from the middle pons to upper spinal cord. It projects its vasoconstrictor fibers through sympathetic system. Under normal conditions, the vasoconstrictor area transmits continuous signals through the sympathetic vasoconstrictor nerve fibers which maintain a partial state of contraction of the blood vessels called as sympathetic vasoconstrictor tone.
- Vasodilator centre is located close to the vasoconstrictor area medially in the floor of the ventricles of the medulla oblongata that inhibits the vasoconstrictor area producing vasodilatation.
- Vasoconstrictor and vasodilator centers show reciprocal inhibition. These centers are controlled by higher neurons from hypothalamus and cerebral cortex and also from sensory impulses from the peripheral organs and chemical composition of blood.

VASOCONSTRICTOR NERVES

- The sympathetic nerves (thoraco-lumber outflow) innervate the small arteries, large arterioles and veins.
- They provide peripheral resistance and function to maintain the tonus of the arterioles, thus regulate BP.
- On the other hand, the capillaries and precapillary sphincters are free from sympathetic innervations.
- The venules have fewer adrenergic fibres than large veins which themselves are less richly innervated than the arterioles.
- Sympathetic innervation to veins causes either constriction or dilatation of the veins and regulates BP by altering the blood volume of the circulatory system.
- Norepinephrine is the sympathetic neurotransmitter.
- Sympathetic stimulation also causes the release of epinephrine from adrenal medulla.
- Both epinephrine and norepinephrine acts through α and β receptors on the blood vessels.
- NEP excites mainly α receptors and causes vasoconstriction.
- By its action on α1-adrenergic receptors, causes vasoconstriction in arterioles of all organs of the body.
- Epinephrine acts both on α and β adrenergic receptors.
- Its action on α adrenergic receptors in cutaneous and renal arterioles causes vasoconstriction.
- Epinephrine causes vasodilatation in the cardiac and skeletal muscles via its effect on β receptors.
- Neuropeptide-Y potentiates the vasoconstrictor effect of adrenergic receptors in primates.
- All vasoconstrictor fibers are sympathetic fibers and these fibers are in tone.
- Vasoconstrictor or vasodilator effect can be produced by altering vasoconstrictor tone without involving vasodilator fibers.
- Stimulation of sympathetic fibers to veins causes vasoconstriction.

VASODILATOR FIBRES

- These are of three types, parasympathetic, sympathetic cholinergic and antidromic fibers. These fibers do not exert tonic activity on blood vessels.
- Parasympathetic vasodilator fibers
  - These are the cranio-sacral outflow as chorda tympani (branch of facial nerve), glossopharyngeal, vagus and pelvic nerves.
  - The neurotransmitter of these fibers is acetyl choline, which act through the cholinergic muscarinic M_3 type receptors, located on the endothelial cells and the smooth muscle cells of most arterioles.
  - M_3 receptors are innervated by parasympathetic fibers in the coronary circulation and in the external genitalia and by sympathetic cholinergic fibers in skeletal muscles.
  - Activation of M_3 receptors on the endothelial cells causes vasodilatation and it also releases nitrous oxide from endothelial cells.
  - This nitrous oxide also produces vasodilatation by acting through smooth muscle cells of arterioles.
  - Stimulation of these fibers results in vasodilatation in coronary vessels, tongue, salivary gland, external genitalia, bladder and rectum.
- Sympathetic vasodilators
  - Two types of fibers are adrenergic and cholinergic vasodilator fibers.
    - Sympathetic adrenergic fibers
      - Stimulation of β_1 receptor causes relaxation of renal arterioles.
      - Sympathetic stimulation of β_2 adrenergic receptors causes vasodilatation.
      - Epinephrine acts on β_2 adrenergic receptors to effect vasodilatation of coronary vessels, skeletal muscle, pulmonary and splanchnic arterioles and systemic veins.
      - β_2 receptors are found on the coronary and skeletal muscle arterioles. These receptors are not innervated by sympathetic nerves but they respond to circulating epinephrine released from the adrenal medulla in response to sympathetic stimulation.
      - It causes anticipatory increase in blood flow to heart and skeletal muscles during fight or flight reaction.
    - Cholinergic sympathetic vasodilator fibers
      - They are limited to the arterioles of active skeletal muscles, causes an anticipatory increase in blood flow even before exercise to overcome fatigue (in dog and cat).
      - Sympathetic cholinergic fibers to cholinergic muscarinic M_3 receptors cause cutaneous vasodilatation.
- Antidromic fibers
  - These fibers originate from dorsal roots of spinal cord and show bidirectional conduction of impulses.
  - The antidromic fibers divide at their peripheral end, one branch supplying the receptors of the skin or muscle and the other to the nearby arterioles.

- The receptor are sensitive to trauma, heat, cold and frostbite.
- When the receptor are stimulated, the impulses reache the blood vessel concerned by travelling in opposite direction (antidromic) and result in vasodilatation. Since the reflex operates on the sensory nerve and its branch to the blood vessel without involving CNS, it is referred as the axon reflex.

HUMORAL REGULATION OF ARTERIOLES

- Effect of epinephrine, norepinephrine and renin - angiotensin on arterioles has already been dealt.
- ADH, angiotensin II, dopamine, prostaglandins and high cAMP act on epsilon receptors will result in vasoconstrictions.
- Atrial natriuretic factor (ANF) release is stimulated by stretching of atria due to increased blood volume. In the kidney tubules it causes increased GFR, decreased Na^+ reabsorption (natriuresis), diuresis by inhibition of renin, aldosterone and ADH activities. It also exhibits vasodilator effect through cGMP mechanism.
- Neuropeptide Y has direct vasoconstrictor properties, regulates the release of atrial natriuretic factor and angiotensin-II.
- The vasoconstrictor actions of serotonin and K^+ and norepinephrine are potentiated by neuropeptide Y, whereas it inhibits renin release. It is involved in moment by moment regulation of blood pressure and blood flow.
- Vasoactive inhibitory polypeptide (VIP) causes vasodilatation and renin release.
- Serotonin by its effect on S_1 and S_2 receptors causes vasodilatation and vasoconstriction, respectively. Serotonin and K^+ ions stimulate the release of atrial natriuretic factor; tend to reduce BP by vasodilatation.
- Prostaglandins are local hormones, synthesized by vascular endothelium. PGE_2 and prostacyclin (PGI_2) are vasodilators, whereas thromboxane A_2 and $PGF_{2\alpha}$ are vasoconstrictors.
- Acetylcholine, insulin, glucagon and prostaglandins are coronary vasodilators, whereas vasopressin (ADH) and oxytocin are coronary vasoconstrictors.
- Thyroxine causes increased blood flow through coronary vessels.
- Bradykinin and histamine cause very powerful vasodilatation and play a key role in the regulation of blood flow in the skin and gastrointestinal glands.
- Histamine acts on H_1 and H_2 histamine receptors cause the release of endothelium derived relaxing factor (EDRF) and low cAMP, results in vasodilatation.
- High CO_2, H^+ and lactic acid, the metabolic products cause local vasodilatation.
- Adenosine, ATP, K^+ ions are the vasodilator substances.
- An increase in Ca^{++} ions causes vasoconstriction, whereas increased concentrations of Na^+,K^+ and Mg^+ result in vasodilatation.
- GABA and glycine are the neurotransmitters of δ receptors cause vasodilatation**.**
- Adenosine released during tissue anoxia stimulates adenosine A_2 receptors activates cAMP mechanism results in profound vasodilatation. ADP and ATP causes release of nitrous oxide from endothelial cells, act on P_2 *receptors* cause vasodilatation**.**

CONTROL OF CAPILLARIES

Local control of blood flow

- Capillaries regulate the local blood supply to the tissues according to the need of the tissues.

MICROCIRCULATION

- Metabolic control of blood flow to an organ is the most important local control mechanism.
- Increased blood flow in tissues in response to increased metabolic rate is called as active hyperemia.
- When the metabolic rate of a tissue increases, its O_2 consumption also increases leading to increased rate of production of metabolic products like CO_2, adenosine and lactic acid. Simultaneously, O_2 concentration decreases in the interstitial fluid. This causes the dilatation of arterioles, relaxation of precapillary sphincters and opening of more capillaries to facilitate increased blood flow to the tissues to deliver more O_2 and removal of accumulated metabolic products.
- Autoregulation of blood flow is also a metabolic control phenomenon. If metabolic rate of an organ does not change but arterial-venous pressure is increased above normal, additional blood flow to that organ is achieved due to increased pressure. The additional blood flow accelerates removal of metabolic products and increases oxygen delivery to the tissues. Hence, the concentration of metabolic vasodilator products decreases. These changes cause the arterioles to constrict and blood flow returns to normal.

CONTROL OF VEINS

- On sympathetic stimulation adrenaline causes venous constriction but has no influence on peripheral resistance to increase venous return to the heart to maintain normal cardiac output and BP.
- In dog and cat, the sympathetic vasoconstrictor fibres (spinal nerves from lumbar 2-4) control the vasomotion in the veins of the hind legs.
- The splanchnic nerve supplies both vasoconstrictor and vasodilator fibres to the portal veins.

Factors causing venous blood flow

- Low pressure in the veins
- Unidirectional blood flow by valves
- Higher intra- abdominal pressure
- Tonicity of the skeletal muscles

Venous pump/muscle pump

- The veins contain valves to direct the blood flow only towards the heart.
- Movement of the legs or tension on the leg muscles causes the propulsion of the blood towards the heart and the venous pressure is lowered. This pumping mechanism is referred as venous pump.

VASOMOTOR REFLEXES

- There are two types of vasomotor reflexes.
- Pressor reflex

o When the BP falls below normal, the impulse frequency passing through the buffer nerves decreases causes the stimulation of cardio accelerator and vasoconstrictor centers of the medulla resulting in vasoconstriction to maintain BP to normal level.

- Depressor reflex
  - oA rise in BP above normal causes increased sensory impulse frequency through buffer nerves to the medullary vasomotor centers causes inhibition of vasoconstrictor and sympathetic centre, but stimulates vagal centre. It results in vasodilatation and decreased heart rate to establish normal BP.

Other reflexes

- Hypothalamic reflex
  - oThermal stimuli - causes vasoconstriction during cold and vasodilatation by heat.
- Cerebral cortex reflex
  - oStress, emotion, pain and exercise acts through dilator centre, vasodilatation in skeletal muscles.
- Chemoreceptor reflex
  - oHypoxia, high CO_2 and H^+ ion causes vasodilatation.
- Stretch reflex
  - o Mesenteric stretch causes distension of intestine which in turn causes vasodilatation.

MODULE-22: ADAPTATION OF CIRCULATION DURING EXERCISE AND SHOCK

Learning objectives

- This module deals with
  - o mechanisms of adaptation of circulation during exercises
  - o shock - definition, classification and stages

ADJUSTMENT OF CIRCULATION DURING EXERCISE

- The response of circulatory system to exercise is an integration of nervous system, endocrine, musculoskeletal and cardiovascular system. During exercise the demands on the circulatory system greatly varies. In the most of the human beings there is an increase in oxygen consumption to five fold within in a few minutes of intense exercise. This increase in oxygen consumption is due to increased demand of oxygen by the working muscle.
- During exercise the mechanoreceptors present in the muscles detect changes in the tension of the muscle and send afferent sensory information to the brain, thereby activating the cardiovascular control centre in the medulla.
- The cardiovascular control centre reduces the activity of the parasympathetic nervous system and increases the activity of sympathetic nervous system, changing the efferent signals going to the heart and the arteriolar smooth muscle.
- This change in parasympathetic and sympathetic activity has greater effect on cardiac output. Cardiac output can be increased 4-8 times the value at rest in humans where as in a trained thoroughbred horse a ten fold increase occurs during maximum exercise.
- At the onset, parasympathetic activity decreases causing an increase in heart rate. At the same time, increase in muscular activity and breathing improves the function of respiratory and skeletal muscle pumps causing an increase in venous return to the heart, resulting in increased stroke volume. Therefore, during initial stages of exercise, the cardiac output increase is due to increase in heart rate and stroke volume.
- Next, sympathetic stimulation of the heart increases, causing an increase in both heart rate and contractility. During the later stage of exercise, increase in heart rate, increases cardiac output rather than increase in stroke volume.
- In addition to changes in cardiac output, blood flow pattern also changes during exercise. During rest, skeletal muscle receive only about 20% of cardiac output, whereas during exercise they receive 88% of total cardiac output *i.e.*, blood flow to skeletal muscles is increased from 1.2 l/min at rest to 2.2 l/min during exercise. These changes in blood flow are as a result of vasodilation of the arterioles of skeletal muscle and heart. The increase in activity causes a generalised vasoconstriction of the arterioles of other organs.
- In skeletal muscle, local release of paracrine factors causes vascular smooth muscle to release by opposing vasoconstrictive effects of α-adrenergic receptor stimulation. These factors together cause an intense local vasodilatation thereby increasing the blood flow to the muscles.
- Vasodilatation of the arterioles of skeletal muscles and vasoconstriction of arterioles of other organs causes a decrease in the total peripheral resistance.
- As a result blood pressure increases slightly during exercise. This stimulates the baroreceptor reflex and brings back the BP to the normal by decreasing cardiac output or total peripheral resistance. The afferent signal from the muscle mechanoreceptors also changes the set point of the baroreceptor reflex allowing BP to increase slightly during exercise.

SHOCK

- Shock is an acute circulatory state in which the cardiac output is insufficient to maintain adequate blood flow to the tissues.
- Any condition causes a sudden decrease in circulating blood volume or increase in volume of the vascular system reduces the mean circulatory pressure results in shock.

CAUSES OF SHOCK

- Shock may be caused either by decreased cardiac output or without decreased cardiac output
- Shock caused by decreased cardiac output
  - Cardiac abnormalities include myocardial infarction, severe heart valve dysfunction, and cardiac arrhythmias etc decrease cardiac ability to pump blood, called as *cardiogenic shock*.
  - Factors decreasing venous return due to reduced blood volume, decreased vascular tone and obstruction to blood flow.
- Shock caused by without diminished cardiac output
  - Cardiac output may be normal or may even be greater than normal. This can result from
    - Excessive metabolism of the body so that the normal cardiac output is not adequate.
    - Abnormal tissue perfusion.
  - These conditions are present in septic shock (blood poisoning) due to massive bacterial infection.
- Tissue deterioration is the end stage of circulatory shock

CLASSIFICATION OF SHOCK

- Shock can be classified into
  - *Hemorrhagic shock* due to blood loss (hypovolumic).
  - *Cardiogenic shock* due to myocardial damage.
  - *Septic or endotoxic shock* due to massive bacterial infection. The infection is carried by blood from tissue to tissue causing extensive damage. Septic shock may result in peritonitis, generalized infection by streptococci or staphylococci, generalized gangrenous infections etc.
  - *Anaphylactic shock* is due to immunological reaction. Antigen-antibody reaction provokes the release of harmful chemical substances like histamine, slow reacting substance (SRS), eosinphil chemotactic factor which causes cardiac failure and vascular fluid loss. Histamine and SRS increase capillary permeability and bronchospasm.

STAGES OF SHOCK

- Compensated or recovering shock: Blood volume and pressure are reduced. Compensatory mechanisms try to maintain adequate blood flow to vital organs and the patient can recover without transfusion.
- Progressive or degenerative shock: Blood pressure progressively falls despite compensatory mechanisms like reflex tachycardia, vasoconstriction of skin, muscle and splanchnic areas and fluid shift from interstitial compartment to plasma. When B.P falls to 60 mm Hg or less, functions of vital organs heart, lungs, CNS etc. are impaired. Transfusion and other treatments can arrest this stage to restore normality.
- Irreversible shock: Cardiac output and B.P. are very low; vital functions are depressed. Effects of transfusion are transient, B.P. falls and the patient dies of myocardial failure and cerebral depression.

Positive feedback in progressive shock

- Progressive shock if untreated, leads to irreversible shock because of a vicious cycle in which, with each degree of increase in shock, the cardiac output and B.P. further decreases and death occurs. Five feedback mechanisms account for this vicious cycle response.
- As shock is advancing, cerebral blood flow decreases progressively causing depression of the vasomotor and respiratory centers.
    - Vasomotor failure: Vascular tone is reduced because of reduced blood supply to vasomotor centre. The sympathetic discharge is reduced; this lead to decrease in means circulatory pressure; venous return leads to reduced cardiac output.
    - Increased capillary permeability: After many hours of capillary hypoxia and lack of nutrients leads to increased capillary permeability and fluid loss to ECF. Reduced blood volume further reduces cardiac output.
    - Blockage of minute blood vessels: Addition of acids and deterioration products of tissues causes agglutination of local blood due to sluggish blood flow in capillaries; lead to clotting and further impeding blood flow.
    - Cardiac depression: When arterial pressure falls, coronary blood flow is reduced leads to myocardial hypoxia and reduction in the cardiac output.
    - At an arterial pressure of 50 mm Hg or less the positive feedback exceeds the negative feedback gains and the circulatory system collapses.

Negative feedback resistance to shock

- Regulatory mechanisms of the circulatory system tend to maintain cardiac output and B.P. at a normal level operate as a negative feedback control. These mechanisms have very short response time in restoring the cardiac output and BP.
- This include the baroreceprtor reflex, chemoreceptor reflex, central nervous ischemic responses. The renin-angiotensin-aldosterone and renal fluid balance mechanisms respond gradually over a period of several hours.

- Within minutes to few hours, the renin-angiotensin mechanism shows vasoconstriction and shifting of fluids into blood to maintain the normal blood volume and BP.

MODULE-23: REGIONAL AND FETAL CIRCULATION

Learning objectives

- This module deals with
  - cerebral, cutaneous and skeletal muscle circulation
  - splanchnic circulation
  - blood flow to rumen, spleen and liver
  - fetal circulation
  - changes in circulation after birth

CEREBRAL CIRCULATION

- Cerebral circulation constituents about 14% of the cardiac output. The arteries that supply the brain are the anterior, middle and posterior arteries of both sides through a common arterial pathway, the circle of Willis.
- In most of the species this circle is supplied by the internal carotid arteries and vertebral arteries, whereas in cat, dog, sheep and goat the circle is supplied by a vascular net, the rete mirable. This carotid rete aids in thermoregulatory mechanism of the brain.
- The circle of the Willis freely communicates with the contralateral side of the brain from both the carotid and vertebral arteries. There are no arteriovenous anastomoses in the brain.
- The walls of the capillaries are non porous and are separated by the nerve cells by neuroglia which prevents diffusion of high molecular weight substances thereby forming a blood brain barrier.
- The blood is drained through the internal jugular vein or through the vertebral venous plexus and external jugular vein.
- The cerebral blood vessels receive both sympathetic and parasympathetic nervous system.

Regulation of cerebral circulation

- Cerebral circulation is controlled primarily by auto regulatory mechanisms mediated by local pH changes.
- An increase is arterial PCO_2 or a decrease in PO_2 causes vasodilatation. Potassium ions produce vasodilatation whereas an increase in bicarbonate concentration causes vasoconstriction.
- The cerebral blood flow is decreased in chronic alkalosis and increased in chronic acidosis. Neural regulation of blood flow plays only a secondary role.

CUTANEOUS CIRCULATION

- The skin is supplied by a dense network of cutaneous arterioles under the dermis layer. These arterioles inturn divide into metarterioles which inturn give rise to capillary loops. These capillary loops provide a greater surface area for heat exchange mechanism.
- The venules form an extensive sub papillary venous plexus. Arteriovenous anastomoses communicate between the smaller arteries, arterioles and venous channels and are located in the distal parts of the extremities, nose, lips, ears etc. These vessels are wide and have low resistance connections that serve as shunts and allow blood to bypass superficial capillary tube.
- The cutaneous vessels are innervated by sympathetic adrenergic vasoconstrictor fibres. Alpha and beta adrenergic receptors are found in the cutaneous arterioles and only alpha receptors in the arteriovenous anastomoses. The venous plexus have separate neural connections and undergo a marked vasoconstriction which minimizes the amount of blood to the skin.
- The major function of the cutaneous vessels is the regulation of body temperature. It aids in the heat conservation and heat dissipation mechanisms. Heat conservation is brought about by vasoconstriction thereby reducing the blood flow to the skin.
- Heat dissipation occurs during increased environmental temperature wherein increased blood flow to the skin occurs in order to increase the heat loss from the body.
- It is brought about by vasodilatation in which increased blood temperature stimulates the hypothalamus which inturn inhibits vasoconstriction thereby increasing the blood flow and causes evaporative heat loss.
- Heat exchange mechanisms also occur via radiation, conduction and convection.

SKELETAL MUSCLE CIRCULATION

- Under normal resting condition, blood flow to the skeletal muscle is 15% and this varies according to the activity of the muscle. Blood flow to the skeletal muscle is controlled by the alpha adrenergic sympathetic system tone on the blood vessels.
- The neurotransmitter substance involved is the norephinephrine which stimulates alpha receptors in the vascular smooth muscle. These fibres also produce vasoconstriction and decreases the blood flow. This vasoconstriction is reflexely influenced by arterial baroreceptors, chemoreceptors and cardiac baroreceptors.
- On the other hand, epinephrine, has a slight to moderate vasodilator effect by stimulating the beta receptors. In certain species like cats and lower animals, sympathetic vasodilator fibres are present that have acetylcholine as the neurotransmitter.
- During rest and at exercise, blood flow to the skeletal muscle in autoregulated and is not affected by the level of metabolites. But local metabolites play role in vasodilatation.
- Vasodilator factors activated during muscle contraction are anoxia, increased Co_2 tension, lactic acid, hydrogen ions, bradykinin, histamine, acetyl choline, ATP, adenylic acid, potassium ions etc.

SPLANCHNIC CIRCULATION

- The splanchnic circulation consists of three parts:
  - oThe mesenteric bed, supplying the gastrointestinal tract;
  - oThe splenic bed and
  - oThe hepatic bed.
- A unique feature of this circulatory system is that the combined outflow from the mesenteric and splenic constitutes the major portion of the inflow of the hepatic through the portal vein.
- All of the splanchnic flow reaches the liver, 70 per cent arrives there via the portal vein from the stomach, intestine, spleen, and pancreas and the remaining 30 per cent comes via the hepatic artery. Together these fractions constitute the splanchnic flow, the hepatic blood flow, and the total through the hepatic veins.
- Three mechanisms operate to regulate blood flow at the local level.
  - oEnhanced metabolism of parenchymal cells of the villi which lowers tissue PO_2 and increases vasodilator metabolites, relaxing arteriolar smooth muscle and precapillary sphincters to increase mucosal blood flow,
  - oIncreased muscle activity metabolically produces active hyperemia of the muscularis mucosae and
  - oIntrinsic myogenic response to stretch autoregulates blood flow, when blood pressure fluctuates.

BLOOD FLOW TO RUMEN, SPLEEN AND LIVER

Rumen blood flow

- Feeding and the products of rumen digestion, carbon dioxide and volatile fatty acids, increases rumen blood flow.

Splenic Circulation

- In certain mammals (dog, cat, horse, guinea pig) the spleen serves as a reservoir of blood. Normally blood is stored in the venous sinuses and is released into the circulation when the need arises (as during exercise, anoxia and hemorrhage). This splenic emptying mechanism is under sympathetic regulation.
- Stimulation of splenic nerve fibers results in decrease in arterial inflow and increase in venous outflow. Norepinephrine mediates arteriolar inflow constriction and epinephrine-mediates emptying.

Hepatic Circulation

- The hepatic blood supply from the hepatic artery and portal vein is governed by hepatic arterial vascular resistance, the arterial tone in the vascular beds of the gastrointestinal tract, pancreas, and the spleen and the intrahepatic portal venous vascular resistance.

- The total hepatic blood flow in humans, cats and dogs contributes about 70 to 75% of the total hepatic blood flow.
- Sympathetic nerve stimulation increases hepatic arterial and portal venous tone. Since the liver, is a major blood reservoir, sympathetic stimulation rapidly mobilizes large amounts of blood for redistribution to vital organs under stress conditions.
- Epinephrine decreases hepatic artery flow and total hepatic blood volume. The hepatic vasculature contains both alpha and beta-adrenergic receptors.
- The hepatic arterial vascular bed exhibits relatively weak local control of its blood flow through both myogenic and metabolic types of auto regulation.
- The portal system lacks such intrinsic local control mechanisms. A partial reciprocity between the hepatic artery and the portal vein helps maintain total liver blood flow constant.
- Vasopressin has three effects on the liver circulation,
    - oHepatic arterial constriction,
    - oMesenteric vasoconstriction, and
    - oDilator action on intrahepatic portal resistance vessels. Angiotensin II causes marked vasoconstriction of both hepatic arterial and portal beds.

FETAL CIRCULATION

- The fetal lungs are nonfunctional before birth and therefore the heart pumps large quantities of blood through the placenta.
- Special anatomical arrangements operate in the fetal circulatory system which is different from the adult.
- Blood returns from the placenta through the umbilical vein passes through the ductus venosus, bypassing the liver.
- Most of the blood that enters the right atrium from the inferior vena cava passes across the posterior region of the right atrium and then through the foramen ovale directly into the left atrium.
- Therefore, well oxygenated blood from the placenta enters mainly the left ventricle of the heart rather than the right ventricle and is pumped by the left ventricle into the vessels of the head and forelimbs.
- The blood entering the right atrium from the superior vena cava passed downward through the tricuspid valve into the right ventricle. This blood is deoxygenated and is pumped into the pulmonary artery and through the ductus arteriosus into the descending aorta and through the two umbilical arteries into the placenta where deoxygenated blood becomes oxygenated.

CHANGES IN CIRCULATION AFTER BIRTH

- Primary changes occurs in the pulmonary and systemic vascular resistance at birth.
    - oLoss of tremendous blood flow through the placenta and increases the systemic vascular resistance. This increases the pressure in the aorta, left atrium and left ventricle.

- Decrease in pulmonary vascular resistance due to expansion of lungs. This increases blood flow to lungs with a reduction in the pressures of pulmonary artery, right ventricles and right atrium.

- Closure of foramen ovale.
- Closure of ductus arteriosus.
- Closure of ductus venosus

MODULE-24: STRUCTURE OF KIDNEY

Learning objectives

- This module deals with structure and functions of kidney.

FUNCTIONS OF KIDNEY

- The important function of the kidneys is to rid the body of waste materials that are either ingested or produced by metabolism.
- To control the volume and composition of the body fluids. A balance between the intake and output is maintained in large part by the kidneys. This regulatory function of kidneys maintains the stable environment of cells necessary for them to perform various activities.
- Kidneys perform most of their important function by filtering the plasma and removing the substances from the filtrate at variable rates, depending on the needs of the body. Ultimately, they clear unwanted substances from the filtrate by excreting them in the urine while returning substances that are needed back to the blood.

Multiple Functions of Kidney

- Excretion of metabolic waste products & foreign chemicals, drugs & hormone metabolites -urea, creatinine, uric acid, end products of hemoglobin breakdown and metabolites of various hormones.
- Regulation of water and electrolyte balance.
- Regulation of arterial pressure.
- Regulation of acid-base balance.
- Regulation of RBC production.
- Regulation of 1,25 Dihydroxy Vit D_3 production.
- Gluconeogenesis: Kidneys synthesize glucose from amino acids and other precursors during prolonged fasting.
- Formation of urine.
- Concentration of urine and reabsorption of essential electrolytes.

ANATOMY OF KIDNEY

- The two kidneys are located on the posterior wall of the abdomen, outside the peritoneal cavity.
- The functional unit of the kidney is the nephron. Nephron numbers vary considerably among species and within species and their numbers are relatively constant.
- The kidney consists of two regions such as outer cortex and an inner region, the medulla.

Nephron Components

- The glomerular capsule (Bowman's capsule) is the dilated blind end of the nephron and consists of the invaginated capillary tuft called as the glomerulus.
- In the mammals the blood from the renal artery is delivered to the afferent arteriole which divides into numerous glomerular capillaries.
- The capillaries coalesce to form the efferent arteriole which conducts blood away from the glomerulus and is returned to the systemic circulation through the renal vein.
- The glomerular capsule is lined by a layer of epithelial cells.
- The area between the glomerular tuft and the Bowman's capsule is known as the Bowman's space and it is the site of collection of the glomerular filtrate which is directly funneled into the proximal tubule.
- The nephron is continued from the glomerular capsule by proximal tubule which is composed of proximal convoluted portion and the proximal straight portion.
- Convoluted portion is within the cortex and the straight portion extends about half way into the outer medulla.
- The loop of Henle consist of descending thin limb which is continuous from the proximal straight tubule, the ascending thin limb that terminates at the junction of the inner and outer medulla (cortical nephrons lack a thin ascending limb) and the ascending thick limb that returns to the glomerulus of origin in the cortex and passes between the afferent and efferent arterioles.
- The distal nephron begins at this point and consist of distal tubule, the connecting tubule, cortical collecting tubule and collecting duct (outer medullary and inner medullary).
- The distal tubule, connecting tubule and cortical collecting tubule are collectively referred to as distal convoluted tubule.

BLOOD SUPPLY

- Blood flow to the kidneys is normally 22% of the cardiac output.
- The renal artery enters the kidney and branches to form the interlobar arteries, arcuate arteries, interlobular arteries and afferent arterioles which lead to the glomerular capillaries, where large amount of fluid and solutes (except plasma proteins) are filtered to begin urine formation.
- The distal ends of the capillaries of each glomerular coalesce to form the efferent arteriole which leads to a second capillary network, the peritubular capillaries surrounding the renal tubules
- Renal circulation is unique, in that it has two capillary beds:
    - oGlomerular capillaries and

- Peri-tubular capillaries, separated by efferent arterioles which help to regulate the hydrostatic pressure.

- The glomerulus has a high pressure of 60 mm Hg and peritubular capillaries have a low pressure of 13 mm Hg which helps in rapid fluid filtration.
- The peritubular capillaries empty into vessels of the venous system which run parallel to the arteriolar vessels and progressively form the interlobular vein, arcuate vein, interlobar vein and renal vein.

Vasa recta

- The peritubular capillaries branches to form the vasa recta into the medulla and lie side by side with the loops of henle.
- Like the loops of henle, the vasa recta return toward the cortex and empty into the cortical veins.
- They are associated with long looped nephrons.
- They play an essential role in the formation of concentrated urine.

JUXTAGLOMERULAR (JG) APPARATUS

- When the thick segment of the ascending limb of the loop of Henle returns to its glomerulus of origin in the cortex, it passes in the angle between afferent and efferent arterioles and continues as the distal tubule. The side of the tubule that faces the glomerulus comes in contact with the arterioles, the contact epithelial cells are more dense than the other epithelial cells and are collectively called as macula densa. Macula densa marks beginning of the distal tubule.
- The smooth cells of the afferent and efferent arterioles that make contact with the macula densa are specialized smooth muscle cells and are called as Juxta glomerular cells or Granular cells. Juxta glomerular cells have secretory granules that contain renin, a proteolytic enzyme.
- The space between the macula densa and the afferent and efferent arterioles and the space between the glomerular capillaries is known as mesangial region/Extra glomerular mesangial cells or Lacis cells.
- JG cells are involved in feed back mechanism that assist regulation of renal blood flow and glomerular filtrate rate.

INNERVATION

- Innervation to the kidney is provided by fibres from the sympathetic division of the autonomic nervous system which assists in the regulation of renal blood flow (RBF), glomerular filtration rate (GFR), salt and water reabsorption by the nephron.
- RBF and GFR is brought about by vasoconstriction initiated by reflexes through the vasomotor centre in the medulla and pons.
- Increased sympathetic tone elicits renin secretion from granular cells and enhances sodium reabsorption from nephron segments.

TYPES OF NEPHRONS

- Mammalian kidney has two principal types of nephrons and are classified based on
  - Location of their glomeruli
  - Depth of penetration of the loops of Henle into the medulla.
- Those nephrons with glomeruli in the outer and middle cortices are called ***cortical nephrons.*** They are associated with the loop of Henle that extend to the junction of the cortex and medulla or into the outer zone of the medulla, *e.g.*, Marine aquatic mammals.
- Those nephrons with glomeruli in the cortex close to the medulla are known as ***juxtaxmedullary nephrons***. They are associated with loops of Henle that extend more deep into the medulla, e.g.,mammals in arid region desert animals such as Kangaroo and rat.

MODULE-25: GFR AND RBF

Learning objectives

This module deals with

- how blood is filtered in the kidney and what are the factors that regulate filtration?
- autoregulation of renal blood flow and filtration rate.

GLOMERULAR FILTRATION RATE (GFR)

- It is the quantity of GF formed each minute in all the nephrons of both the kidneys/kg body weight.
- In humans it is about 125 ml/min. Total quantity of GFR formed /day = 180 L. Over 99% of the filtrate is reabsorbed in the tubules, the remainder passing into the urine.

Filtration fraction

- It is the percentage of the renal plasma flow that becomes Glomerular Filtrate.
- The normal plasma flow through both the kidneys is 650 ml/min but the normal GFR in both the kidneys is 125 ml/min, hence the average filtration fraction is 19%

FACTORS AFFECTING GLOMERULAR FILTRATION RATE (GFR)

- Three factors that determine the filtration pressure are
  - Glomerular pressure
  - Plasma colloidal osmotic pressure (COP)
  - Bowman's capsular pressure

- Greater the glomerular pressure, greater is the filtration.
- Greater the plasma COP and Bowman's capsular pressure, lesser is the filtration.
- Effect of renal blood flow
    - oGFR is affected by the rate of blood flow through the nephrons. Since a very large portion of plasma is filtered through the glomerular membrane, the COP in the glomerulus is high which opposes further filtration. Therefore, a portion of plasma fluid is not filtered until new plasma flows into the glomerulus. Greater the plasma flow, greater the filtration rate.
- Effect of afferent arteriolar constriction
    - oAfferent arteriolar constriction decreases the rate of blood flow into the glomerulus and decreases GFR, causing decreased filtration rate whereas dilatation increases glomerular pressure as well as GFR.
- Effect of efferent arteriolar constriction
    - oConstriction of the efferent arteriole increases the resistance and outflow from the glomeruli, increases the glomerular pressure and also GFR initially. But when blood stagnates in the glomerulus for a prolonged period, increase in plasma COP occurs which causes a fall in GFR. Net effect is slight increase in GFR.
- Effect of sympathetic stimulation
    - oMild to moderate stimulation of sympathetic nerves causes afferent arteriolar constriction and decreases GFR. Strong sympathetic stimulation causes great reduction in the glomerular blood flow and glomerular pressure resulting in fall of GFR to zero level.
- Effect of arterial pressure
    - oWhen arterial pressure increases from 100 to 200 mm Hg afferent arteriolar constriction occurs automatically by autoregulation, thus prevents a major rise in glomerular pressure (GP) and GFR increases to only 15-20%.

AUTOREGULATION OF RENAL BLOOD FLOW (RBF) AND GFR

- RBF and GFR remains almost relatively constant when the systemic arterial pressure changes from 75 mm Hg to 160 mm Hg. This ability of the RBF and GFR to resist severe changes in the arterial pressure is called as 'autoregulation of RBF and GFR'. It is an intrinsic mechanism which is independent of the nerve supply.
- Two theories have been proposed to explain autoregulation
    - oMyogenic Theory
    - oJG Theory
- Myogenic theory
    - oAccording to this theory, the increase in BP would expand an artery and it would respond by contracting. In this way, RBF would be decreased and glomerular HP reduced. The reduced glomerular HP reduces GFR.
    - oA reduction in BP causes less tension and blood vessel would dilate to increase RBF and glomerular HP with subsequent increase in GFR. That is, when arterial pressure rises, arterioles are stretched, once stretched, arterioles contract forcefully. This decreases RBF to normal level. A decrease in arterial pressure dilates the blood vessels which increases RBF and GFR.

- JG theory
  - oJuxta glomerular apparatus (JGA) contains renin (a proteolytic enzyme). Renin is released when:-
    - Reduced GFR
    - Reduced glomerular pressure
    - Increased sympathetic stimulation of kidneys
  - oThe last two occur during low BP and always cause reduced GFR. Reduced GFR causes reduced sodium concentration in the tubular fluid as it flows past the macula densa and this low sodium causes release of renin from JGA. Once renin is released from JG cells it diffuses into the blood of afferent arteriole and circulates through out the body.
  - oIn the blood it splits a renin substrate, an alpha 2 globulin to angiotensin I, a decapeptide. Angiotensin I is rapidly converted to angiotensin II by converting enzyme which is present in high concentration in the lungs. Angiotensin II is a powerful vasoconstrictor and causes vasoconstriction through out the body thereby increasing the BP. Some amount of Angiotensin II is formed in the glomerulus, *i.e.* in the JG cells.
  - oAngiotensin II causes marked constriction of efferent arteriole which increase glomerular pressure and also GFR but decreases RBF. Increase in pressure increases GFR but decrease in blood flow decreases GFR and in effect there is a less change in GFR.
  - oDecreased blood flow to peritubular capillaries decreases peritubular pressure, which causes increased tubular reabsorption, so excretion is reduced. When efferent arterioles are constricted, GFR is normal, excretion of waste products such as urea, creatinine is normal. At the same time there is an increase in tubular reabsorption of salt and water. Therefore, effect of angiotensin is to conserve water and salt while allowing normal excretion of waste products.
  - oAngiotensin II constricts the efferent arterioles to a greater extent than that of afferent arteriole. Reabsorption of water and salt by renin-angiotensin system helps to control arterial BP.

Tubulo glomerular feedback

- The mechanism of tubulo glomerular feedback (TGF) is also associated with the JG theory of autoregulation. TGF refers to alteration in GFR with changes in the tubular flow rate.
- It is mediated by the macula densa cells of the JG apparatus. These cells sense changes in the sodium and chloride to their region. If GFR is increased because of increased glomerular HP there will be increase in macula densa flow and sodium and chloride delivery intiates a response that returns GFR and macula densa flow towards the normal by afferent arteriole constriction (which lowers glomerular HP).
- When renal blood flow falls too low it decreases the GFR thereby, decreases in sodium and chloride delivery to the distal tubule which initiate a signal from macula densa and dilates the afferent arterioles and it increases glomerular blood flow and glomerular pressure. This causes the GFR to increase to normal.

MODULE-26: FORMATION OF URINE

Learning objectives

This module deals with

- Formation of urine
- Absorptive capabilities of different tubular segments
- Reabsorption and secretion in different regions of tubules
- Functions of the urinary bladder
- Uremia or Azotemia

URINE FORMATION

- Three processes are involved in the urine formation in the nephrons :
  - Glomerular filtration
  - Tubular reabsorption
  - Tubular secretion

GLOMERULAR FILTRATION

- Glomerular filtrate is the fluid that is filtered through the glomerular membrane into the Bowman's capsule.

Glomerular capillary membrane

- Glomerular capillaries are relatively impermeable to proteins so that the filtered fluid, glomerular fluid (GF) is essentially protein free and devoid of cellular elements.
- The glomerular capillary membrane has three layers:
  - Endothelium of the capillary
  - Basement membrane
  - A layer of epithelial cells (podocytes) surrounding the outer surface of the capillary basement membrane.
- The endothelial cells lining the glomerulus are perforated by thousands of small holes called fenestrae.
- Surrounding the endothelium is the basement membrane composed of a meshwork of collagen and proteoglycan fibrillae which can filter large amount of water and small solutes.
- The final layer contains epithelial cells called podocytes lining the outer surface of the glomerulus. These cells are not continuous but consists of many finger like projections which form slit pores through which glomerular filtrate filters.
- Therefore, GF must pass through three different layers before it enters the Bowman's capsule.

- The permeability of the glomerular membrane is 100 to 1000 times as great as that of the usual capillary and is because of the pores of the endothelium which are approximately 100 nm diameter and also slit pores approximately 25 nm wide.
- Despite the tremendous permeability of glomerular membrane, it has a high degree of selectivity for the sizes of the molecules that it allows to pass.

Molecular weight	Permeability	Example
5000	1.0	Inulin
30,000	0.5	Very small protein
69,000	0.005	Albumin

- Therefore, plasma proteins are completely impermeable.

Reasons for high degree of selectivity

- Size of the molecule: Pores in the membrane allow molecules with a diameter upto 8 nm.
- Pores are lined with a strong negative charges and electrostatic repulsion of the protein molecules (proteins are electronegative) prevent their filtration.

Dynamics of Glomerular filtrate

- The energy for the filtration is provided by the heart in the form of hydrostatic pressure of the blood inside the glomerular capillaries and the colloidal osmotic pressure of the fluid within the Bowman's space through the capillary membrane into the Bowman's capsule. On the other hand, colloid osmotic pressure in the blood and the hydrostatic pressure in the Bowman's space oppose filtration.
- The colloidal osmotic pressure in the Bowman's capsule is negligible due to very low protein content. The colloidal osmotic pressure in the glomerular capillaries increases, since $1/5^{th}$ of the fluid portion of the plasma in the capillaries filters into the capsule increases the protein concentration about 20% as blood passes from arterial to venous end of the glomerular capillaries.
- The colloidal osmotic pressure of the blood entering the capillaries is 28 mm Hg which rises to 36 mm Hg by the time the blood reaches the venous side and so the average colloidal osmotic pressure is 32 mm Hg.

Filtration pressure

- It is the net pressure forcing the fluid through the glomerular membrane equals to the glomerular pressure minus sum of glomerular colloidal osmotic pressure and capsular pressure.
- For example, If Glomerular Hydrostatic Pressure = 60 mm Hg, Colloidal Osmotic Pressure in glomerulus = 32 mm Hg and capsular pressure (Bowmans capsule Hydrostatic Pressure) = 18 mm Hg
- Then, the Filtration pressure = 60-(18+32) = 10 mm Hg.

Tubular transport

- Transport of fluid from the Bowman's capsule to the renal pelvis is accomplished by a difference in the hydrostatic pressure.
- Tubular reabsorption involves transport of water and solutes from the tubular fluid to the peritubular capillaries.

Tubular secretion

- It is the transport of solute from the peritubular capillaries to the tubular fluid.

TUBULAR REABSORPTION

- Reabsorption plays a major role than the secretion in the formation of urine. More than 90% of the water in the GF is reabsorbed as it passes through the tubules. Some substances such as sodium, glucose and amino acids are almost completely reabsorbed so that their concentration decreases almost to zero before the fluid becomes urine so that they are conserved by the body and not excreted and lost by the urine.
- Basic mechanisms of absorption is by two process
    - Active transport
    - Passive transport

Active transport through the tubular wall

- Tubular epithelial cells have a brush border at the luminal surface. It is composed of microvilli which increases surface area of the lumen. The base of the cell rests on the basement membrane. Basal channels in the basement membrane increases the surface area. Epithelial cells are attached to each other near the brush border forming tight junction or zona occludens.
- Sodium, glucose, amino acid, calcium, potassium, phosphate and urate ions are actively transported. They are transported through carrier proteins.

- Uniport
    - Transport by a carrier for a single compound (e.g., sodium) and is unidirectional.
- Symport or co-transport
    - Transport of two compounds on the same carrier in the same direction (e.g., sodium plus glucose, or sodium plus amino acid).
- Antiport or counter transport
    - Movement of a compound in one direction driven by the movement of a second compound in opposite direction (e.g., Na^{2+} and H^{+} antiport).
- Sodium reabsorption
    - About 65% of Na^{2+} is reabsorbed in the proximal tubule. The energy required for this mechanism is derived from the Na^{2+}-K^{+} ATPase (sodium pump) located in the basolateral membrane of the proximal tubule epithelial cells.
- Electrochemical Gradient

- Sodium ions are actively transported from the interior of the tubular epithelial cells to the peritubular space across the basal membrane. Therefore, intra-cellular Na^{2+} is lowered, creating a chemical gradient for Na^{2+} (lumen concentration higher) between the tubular lumen and tubular epithelial cell. It also causes a low negative intracellular electrical potential (-70 mv) which in turn causes the Na^{2+} to diffuse from the tubular lumen into the cell through the brush border. This mechanism is uniport or unidirectional Na^{2+} transport. Chloride ion readily diffuses from the tubular lumen to the peritubular space through tight junction between tubular epithelial cells because of transepithelial electrical potential difference (lumen negative) created by Na^{2+} transport.

- Antiport or counter transport of sodium ion
  - Diffusion of Na^{2+} because of electrochemical gradient is coupled through a carrier protein with H^+ diffusing in the opposite direction from the cell interior to the tubular lumen. HCO_3^- in the cell can diffuse through the basolateral membrane to the peritubular space or move into the tubular lumen in counter transport to Cl^- diffusion into the cell.
- Chloride driven sodium ion transport
  - As more of HCO_3^- is being reabsorbed in to the peritubular space Cl^- gradient favours diffusion of Cl^- through the leaky tight function from the tubular lumen into the peritubular space and is accompanied by diffusion of Na^{2+} in the same direction to maintain electrical neutrality.
- Glucose and amino acid reabsorption
  - These are reabsorbed by symport or co-transport. They are coupled with specific carriers that require Na^{2+} binding and diffuse into the cell. Inside the cell Na^{2+} and glucose or amino acid separates from the carrier. The Na^{2+} is actively transported by Na^{2+}- K^+ ATPase to the peritubular space. Glucose and amino acids are then transported by facilitated diffusion.

Passive transport of water and other solutes

- After the diffusion of solute (Na^{2+}, Cl^- , HCO_3^-, glucose, amino acid) into the peritubular space, an osmotic gradient is established, where by a greater osmotic pressure is present in the peritubular space. Therefore, water diffuses from the peritubular lumen and tubular cells into the peritubular space.
- As 65% of Na^{2+} is reabsorbed, similarly 65% of water is reabsorbed from the proximal tubule (an additional amount for other osmotically active substances such as glucose, amino-acid).
- As water is reabsorbed, urea and other non-actively reabsorbed solutes get concentrated in the tubular lumen.
- A chemical concentration gradient is established for these substances and they are reabsorbed down their concentrated gradient. The extent of their reabsorption depends on the permeability of the tubular epithelium for the solute.
- Urea permeability for the proximal tubule is much less than that of water and more than 50% of the amount of urea in the GF continues beyond the proximal tubule. There is no

permeability of tubular membrane for reabsorption of creatinine, inulin, mannitol and sucrose and therefore once these are filtered, their total quantity appears in the urine.

- Reabsorption of proteins and peptides
  - Proteins with a molecular weight of less than 69,000 are reabsorbed in the proximal tubule and not lost in the urine. They are reabsorbed by endocytosis and subsequently degraded by cellular lysosomes to amino acids. The amino acids move from the cell to the peritubular space by facilitated diffusion. Small peptides are hydrolyzed at the luminal brush border of proximal tubule and the resultant amino acids is taken into the cell by co-transport mechanism.

TUBULAR SECRETION

- Several substances are transported from the peritubular capillaries into the interstitial fluid and then to the tubular lumen via the tubular epithelial cells.
  - Antiport of H^+ along with Na^{2+} reabsorption in the proximal tubule and distal tubule. H^+ is secreted throughout the nephron.
  - K^+ transport is unique, in that it is reabsorbed in some parts of the tubule and secreted in others. It is reabsorbed in the convoluted portion of proximal tubule and secreted in the straight portion of proximal tubule. It is secreted and reabsorbed in the distal nephron.
  - Ammonium ions are synthesised in the epithelial cells and diffuse into the tubular fluid.

ABSORPTIVE CAPABILITIES OF DIFFERENT TUBULAR SEGMENTS

- Proximal tubule:
- 65 % of the reabsorption and secretion take place. Only 35% of the GF leaves the proximal tubule.
- Thin segment of Loop of Henle:
  - Permeability of the epithetical cells of descending limb of Loop of Henle is great and occurs by simple diffusion. It is highly permeable to water and moderately permeable to urea and Na^{2+}.
- Ascending limb of Loop of Henle:
- It is less permeable to water and more permeable to urea.
- Thick segment of Loop of Henle and distal tubule
- They have rudimentary brush border and cells adapted for Na^{2+} transport against concentration gradient. They are impermeable to water and urea. Here, active absorption of Na^{2+} and active secretion of K^+ is controlled by aldosterone.
- Collecting tubule
  - Final concentration of urine takes place in the collecting tubule. It has two functional units, cortical and medullary portion. Cortical portion is impermeable to urea and medullary portion is moderately permeable to urea. Permeability of collecting tubule to water is determined by the concentration of Antidiuretic hormone (ADH) in the blood. Large amount of ADH causes

collecting tubule to be highly permeable to water. In the absence of ADH, very little water is reabsorbed. They can secrete H^+ into the tubule.

REABSORPTION AND SECRETION IN DIFFERENT REGIONS OF TUBULES

- Water transport is by osmotic diffusion.
- Proximal tubule : 65% reabsorption.
- Loop of Henle : 15%.
- Distal tubule : 10%.
- Collecting duct : 9.3%.
- Urine : 0.7%.
- Glucose, proteins, amino acids, vitamins and acetoacetate ions are completely reabsorbed by active process in proximal tubule.
- 50% of urea is reabsorbed.
- Creatinine is not reabsorbed but some quantities are secreted in proximal tubule.
- About 86% of urate ions are reabsorbed.
- Sulphate, phosphate and nitrates are transported similarly like urates.
- K^+ ion is secreted in distal tubule and collecting tubule.
- H^+ ions are actively secreted in proximal tubule, distal tubule and collecting tubule.

Transport maximum

- Substances such as glucose that are actively absorbed by a carrier transport, there is a maximum rate at which they can be reabsorbed. This is known as transport maximum (T_m).
- When T_m is exceeded in the nephron, it appears in the urine, e.g., in Diabetes mellitus, the movement of glucose from the plasma to the cells is impaired because of lack of insulin. Glucose concentration increases causing plasma and tubular loads to increase. When increased tubular load exceeds the availability of the carriers molecules for glucose reabsorption, excess glucose flows through the tubules into to the urine. As glucose is retained within the tubules it contributes to an increase in osmotic pressure and therefore water also remains in the tubular fluid. The point at which the glucose first begins to appear in the urine, 175 mg/dl is known as the renal threshold for glucose. The T_m for the kidney is reached when all the nephrons are reabsorbing to their maximum ability.
- T_m for glucose = 320 mg/min

FUNCTIONS OF THE URINARY BLADDER

- It provides an expandable reservoir for urine, which is continuously flowing from pelvis of the kidney through ureters.

Micturition

- It is the process in which the urinary bladder empties when it becomes filled with urine.

Passage of Urine from the kidney to the bladder

- Urine is secreted continuously though at a varying rate and it passes through the collecting ducts into the pelvis of the kidney and carried to the urinary bladder by the ureters. The ureters contain muscles capable of contraction, which helps in propulsion of urine along the tube into bladder. Each ureter is innervated by both sympathetic and parasympathetic nerves. As urine collects in the pelvis, pressure increases which initiates peristaltic contraction beginning from pelvis and spreading down along the ureters to force urine towards bladder.
- Parasympathetic stimulation increases frequency and sympathetic stimulation decreases frequency of peristalsis of ureters.

Filling and emptying of bladder (Micturition Reflex)

- At the junction of ureter with bladder, an ureterovesicular valve is present which prevents reflux of urine from bladder. The urinary bladder possess the power of accommodation to increase in its contents without increase in internal pressures up to about 150 mm H_2O. So, as urine enters the bladder, its walls becomes distended. Out flow of urine into the urethra is prevented by sphincters at the neck of the bladder (internal and external sphincters).
- When the pressure in the bladder reaches 150 mm water, contraction of bladder wall begins, relaxation of sphincter occurs and urination or micturition occurs. Contraction of abdominal muscles and contraction of diaphragm assist the emptying of the bladder. The urinary reflex may be assisted by voluntary effect; may also be opposed voluntarily. This is accompanied by inhibition of spinal centres of micturition and by contraction of external sphincter which surrounds the external part of the urethra. The desire to urinate arises from stimulation of receptors in the wall of the bladder by stretch and contraction of musculature.
- Nerves from sympathetic and parasympathetic divisions of ANS supply the bladder. The preganglionic sympathetic fibres leave from spinal cord in 2nd to 4th lumbar nerves to the posterior mesentric ganglion and postganglionic fibres reach the bladder through hypogastric nerves.
- In all mammals, parasympathetic nerves cause contraction of whole bladder and they are main motor nerves. They are inhibitory to internal sphincter.
- Complete emptying of the bladder depends upon maintenance of bladder muscle contraction and sphincter relaxation. This is achieved by two reflex systems.
- Receptors are present in the bladder walls that are stimulated during contraction of urinary bladder. Another reflex arises from receptors in the wall of urethra. Urine flow through urethra helps to maintain bladder contraction and relaxation of external sphincter.
- The tone of the bladder decreases as the bladder becomes emptied of its contents and this is accompanied by contraction of sphincter.
- The cessation of micturition involves voluntary control and voluntary regulation.

UREMIA OR AZOTEMIA

- Urea is the chief nitrogenous end product of protein metabolism and is excreted by the kidneys in the urine of mammals. It is also found in the blood and lymph. Uremia is a toxic condition that occurs due to retention of urea is the blood. It results due to
  - Renal failure.
  - Increased production of urea in the liver due to high protein diet, drugs, increased breakdown of protein etc.
  - Decreased elimination of urea due to reduced blood flow to kidney, obstruction of urinary tract etc.
  - Dehydration
  - Chronic infection of the kidney.
- The symptoms of uremia are lethargy, depression, nausea, vomiting, deep breathing, dizziness, coma and convulsions. In these cases, there is usually a smell of urine in all the animal's secretions.

TERMINOLOGIES

- Diuresis: Increased urine formation.
- Polyuria: Increased excretion of urine. It may be due to the deficiency of ADH.
- Oliguria: Reduced excretion of urine
- Anuria: Complete cessation of urine formation.
- Dysuria: Difficult or painful micturition.
- Stranguria: Slow dropwise painful discharge of urine caused by spasm of urethra and bladder.

MODULE-27: RENAL FUNCTION TESTS

Learning objectives

- This module deals with the tests required to detect functional disorders of the kidney.

KIDNEY FUNCTION TESTS

- Renal clearance is the measurement of the kidney's ability to remove substances from the plasma.
- Clearance measurements are used to determine Renal Blood Flow (RBF), Renal Plasma Flow (RPF), Glomerular Filtration Rate (GFR), Filtration Fraction (FF) and how different substances are handled by the kidney tubules (reabsorbed or secreted) and to compare the kidney function values for diagnostic purposes.

PLASMA CLEARANCE

- It is used to express the ability of the kidneys to clean or clear the plasma of various substances. e.g., If the plasma passing through the kidneys contains 1 mg of a substance in each ml and 1 mg of this substance is also excreted into the urine each minute, then 1 ml/min of the plasma is cleared of the substance.

- Plasma clearance is an excellent measure of kidney function and the clearance rate of different substances are determined by analyzing the concentration of substance simultaneously in plasma and urine and measuring the rate of urine formation.

CLEARANCE AS AN ESTIMATE OF GFR

- A substance to measure GFR must be freely filtered at the glomerulus and should not be reabsorbed or secreted by the tubular epithelium after it enters the nephron. Inulin, a fructose polysaccharide is the substance that is most commonly used.

$$GFR = \frac{U_x \times V}{P_x}$$

- Mannitol, is another polysaccharide used to estimate GFR.

CREATININE CLEARANCE

- In clinical conditions it can be used to measure GFR and kidney functions. Creatinine is freely filtered and not reabsorbed by the tubules.
- In some species (not in dogs) about 10% is secreted by the tubules. It can't be used in birds because creatinine is either secreted or reabsorbed from the tubules to a greater extent.

MEASUREMENT OF RPF

- The substance used must be freely filtered at the glomerulus and must not be reabsorbed from the tubular lumen and must be secreted by the tubular epithelium so that all of the substances in the blood perfusing the tubules is removed before the blood leaves the kidney. Therefore, if all the substance in the plasma that perfuses the kidneys is excreted in the urine the rate of its excretion is the same as its plasma load.
- Para Amino Hippuric acid (PAH) is

PLASMA LOAD AND TUBULAR LOAD

- Plasma load of a substance is the total amount of substance in the plasma that passes through the kidneys each minute.

- Eg: If the concentration of glucose in the plasma is 100 mg/100 ml and 600 ml of plasma passes through both the kidneys each minute, then the plasma load of glucose is 600 mg/min.
- A fraction of plasma load that is filtered as GF is referred to as tubular load.
- For example, if 125 ml of GF is formed each minute with a glucose concentration of 100 mg% then the tubular load of glucose is 125 mg (100 x 1.25) glucose/min.
- used to measure RPF.

MODULE-28: COUNTER CURRENT MECHANISM

Learning objectives

This module deals with

- renal mechanism of urine formation
- hyperosmolality of the medullary fluid
- permeability of the tubules
- counter current mechanism
- excretion of dilute urine
- excretion of a concentrated urine

RENAL MECHANISM OF URINE FORMATION

Renal mechanism of concentrated and dilute urine formation

- Concentrated urine is formed by passive water reabsorption from the tubules while many solutes in the tubular fluid are absorbed by active process.
- Dilute urine is formed by absorption of solutes alone and excretion of water in the urine.

Dilution or concentration of urine depends on

- Hypertonicity/osmolality of the interstitial fluid in the renal medulla.
- Dilution of the tubular fluid by the thick ascending limb and distal tubules by solute reabsorption which allows formation of dilute urine.
- Variation in the water permeability of collecting ducts due to ADH, which determines the final concentration of urine.
- For concentration of urine, there is generation of hypertonic medullary fluid and increase water reabsorption in the distal tubule.
- For generating hypertonic medullary fluid, *two factors* are essential:
    - Reabsorption of osmotically active substance by the tubules into the medulla.
    - Removal of water from the interstitium by the vasa recta.
- These two factors are produced by counter current mechanism.

HYPEROSMOLALITY OF THE MEDULLARY FLUID

- Normal osmolality of the GF as it enters the proximal tubule is 300 milli osm/L. But osmolality in the medullary interstitial fluid is higher reaching a maximum in the inner most regions of the medulla, it increases from 300 milli osm/L to 1200 milli osm/L. The cause of this increased osmolality is the active transport of both sodium and chloride out of the Loop of Henle's ascending limb and slight reabsorption of sodium actively from the collecting tubule into the interstitial fluid.

Also the cause for accumulation of sodium chloride in medulla is:

- Sluggish blood flow through vasa recta which helps to prevent removal of sodium chloride from the interstitial fluid.
- Presence of counter current mechanism in the Loop of Henle and vasa recta.

PERMEABILITY OF THE TUBULES

- Descending limb of Loop of Henle: High permeability for water and no permeability for sodium, chloride and urea
- Ascending thin limb of Loop of Henle: No permeability for water, highly permeable to sodium, chloride and moderately permeable to urea.
- Ascending thick limb of Loop of Henle: Permeable to sodium, chloride, low permeability to water and urea.
- Distal tubule: Permeable to sodium, chloride and low permeability for water and urea.
- Cortical collecting tubule, outer medullary collecting duct and inner medullary collecting duct: Sodium reabsorption is stimulated by aldosterone and water and urea reabsorption by ADH.

COUNTER CURRENT MECHANISM

- A counter current system of tubules or vessels exists where the inflow of fluid runs parallel to, counter to, and in close proximity to the outflow for some distance.
- These characteristics are common to the anatomical arrangements of the Loops of Henle and vasa recta. In the kidney, two counter current systems operate.
    - oCounter current multiplier - Loops of Henle
    - oCounter current exchanger – Vasa recta

Counter current multiplier

- The ascending thick limb of the Loop of Henle is permeable to solutes and so the solutes diffuse into the medullary interstitial fluid with retention of water in the tubule, thereby diluting the tubular fluid. This creates a small osmotic gradient between the tubular and peritubular fluids (interstitial fluid). This osmotic gradient is being multiplied vertically

by counter current flow in the descending thin limb (permeable for water and not for solutes).

- Water diffuses from the lumen of the descending thin limb into the interstitial fluid. Therefore, the tubular fluid of the descending thin limb increases in osmotic concentration as it descends to the inner most region of the medulla. When thin tubular fluid enters the ascending thin limb (permeable for solute and not for water), sodium chloride diffuses readily outward into the inner medullary interstitial fluid and urea diffuses inward into the tubular fluid.
- Continued active secretion by the ascending thick limb, concentration of tubular fluid in the descending thin limb and diffusion from the lumen of the ascending thin limb into the medullary interstitial fluid establishes a vertical osmotic gradient. Therefore, each time sodium chloride makes the circuit around the Loop of Henle, this multiples the concentration of sodium chloride in the medulla and hence, Loop of Henle is called as counter current multiplier.

Countercurrent exchanger - Vasa recta

- It is a counter current system in which transport between the outflow and inflow is entirely passive. Vasa recta is permeable to water and solutes through out their length.
- In the descending limb of the vasa recta, water is drawn by osmosis from the plasma of vasa recta to the hyperosmotic peritubular fluid (created by counter current multiplier) and the solutes diffuse from the peritubular fluid into the vasa recta.
- In the ascending limb, solutes diffuse back into the peritubular fluid and water is drawn by osmosis back into the vasa recta. Net result is that the solutes responsible for medullary gradient are mostly retained in the medulla and the vasa recta carry only slightly more solutes than are brought to them.
- Blood flow in the vasa recta is sluggish because an increased rate of medullary blood flow results in decreased time required for diffusion of solute from the ascending limb back to peritubular fluid resulting in gradual loss or wash out of medullary gradient. All the excess salt removed from the interstitial fluid has to be replaced to maintain an osmotic gradient.

Recirculation of urea

- It is a mechanism for concentration of urea in the medulla. Urea is concentrated in collecting tubule and diffuses through the walls of collecting tubule into the medullary interstitial fluid. From there, it is reabsorbed in the Loop of Henle and flows with tubular fluid in the ascending limb through distal tubule into collecting tubule and again out of the collecting tubule into the medullary interstitial fluid. Urea circulates several times before it flows into the urine and it causes urea to accumulate in high concentration in medullary interstitium. This counter current multiplier system helps in concentration of urine and also ensures constant excretion of urea when urine output is low.
-

EXCRETION OF DILUTE URINE

- When there is excess of water in the body and a reduction in plasma osmolarity, kidneys can excrete a dilute urine with a concentration as low as 50 mOsm/L. This can be achieved by reabsorbing only the solutes and not water in the distal parts of the nephron.
- The total amount of solute excreted remains constant but the urine formed is very dilute and urine osmolarity decreases from 600 to about 100 mOsm/L.
- The glomerular filtrate initially formed has the osmolarity similar to that of plasma (300 mOsm/L). Inorder to excrete excess water it is necessary to dilute the filtrate as it passes along the tubule which is achieved by reabsorption of solutes to a greater extent than water.
- As the fluid flows through the proximal tubule solutes and water are reabsorbed in equal amount and there is a little change in osmolarity but when it reaches down the descending limb of loop of Henle water is reabsorbed by osmosis and the tubular fluid reaches the osmolarity of renal medulla, *i.e.* it becomes hyperosmotic.
- In the ascending limb of loop of Henle sodium, potassium and chloride are actively reabsorbed. Therefore, the fluid becomes dilute as it flows to distal tubule with osmolarity decreasing to 100 mOsm/L (osmolarity is one third of that of plasma).
- When the fluid passes to late distal tubule and collecting tubule there is little reabsorption of sodium chloride. In the absence of ADH, collecting tubule is impermeable to water and additional reabsorption of solutes causes tubular fluid to become even more dilute, decreasing its osmolarity to as low as 50 mOsm/L.
- Therefore, failure of reabsorption of water and continued reabsorption of solutes leads to a large volume of dilute urine.

EXCRETION OF A CONCENTRATED URINE

- For formation of a concentrated urine two basic requirements are essential
    - High level of ADH which increases the permeability of distal tubule and collecting tubule to water.
    - High osmolarity of the renal medullary fluid which provides osmotic gradient for reabsorption of water in the presence of ADH.
- When the tubular fluid leaves the loop of Henle and flows into the distal convoluted tubule in the renal medulla, fluid is dilute with a osmolarity of 100 mOsm/L.
- As fluid flows into the cortical collecting tubule, the amount of water reabsorption is dependent on plasma concentration of ADH. In the presence of high concentration of ADH the cortical collecting tubule becomes highly permeable to water and large amounts of water are reabsorbed.
- As the fluid flows into the medullary collecting tubule there is further water reabsorption and when the fluid reaches the end of the collecting ducts the osmolarity reaches 1200 mOsm/L which is similar to renal medullary osmolarity.

- Therefore, by reabsorbing as much as water as possible, kidneys form a highly concentrated urine excreting normal amount of solutes in urine and increasing the ECF volume.

HORMONAL REGULATION OF RENAL FUNCTION

ADH and water conservation

- ADH is synthesised in the cell bodies of hypothalamic nuclei (supraoptic nuclei) and transported to nerve fibre endings in the posterior lobe of the pituitary where it is stored in the secretory granules.
- Its release in the blood is controlled by osmoreceptors in the hypothalamus that are close to the supraoptic nuclei. It regulates water conservation. Increase in plasma osmolality stimulates osmoreceptors in the hypothalamus causing release of ADH which decreases the ECF volume. Fear and pain also causes release of ADH.

Aldosterone

- Adrenal cortex regulates K^+ and Na^+ concentration. It acts on tubules causing Na^+ reabsorption and K^+ excretion. Aldosterone increases Na^+ reabsorption from distal tubules by increasing Na^+ transport protein; salt free diet causes increased aldosterone secretion resulting in increased Na reabsorption.

Renin - Angiotensin system

- Renin is activated by reduced circulating blood volume as in hemorrhage. Decreased sodium concentration in the distal convoluted tubule and sympathetic stimulation also causes release of renin.

Parathyriod hormone

- Ca^{2+} and PO_4 excretion in urine is regulated at the proximal tubule by the action of parathyroid and thyrocalcitonin from thyroid. Parathyroid hormone (PTH) causes decrease in PO_4 reabsorption and increase in PO_4 excretion in urine. PTH increases mobilisation of Ca^{2+} from bone and absorption from intestine resulting in increased plasma Ca^{2+} level and decreased Ca^{++} excretion.

Atrial natriuretic peptide

- Myocardial cells of the atria release the ANP when the atria are stretched during high volume of blood. It has the following functions.
  - Increases the GFR by causing vasodilatation of afferent arterioles and vasoconstriction of efferent arterioles.
  - Inhibits angiotensin II stimulated absorption of Na^+ and water in proximal tubules.
  - Reduces water reabsorption in collecting tubules.

- Inhibits aldosterone release.
- Decreases the response of the collecting tubules and collecting ducts to ADH.

Local hormones

- Erythropoietin produced from kidney regulates erythropoiesis.
- Renin is produced from kidney.
- Prostaglandin from kidney acts as blood pressure lowering agents. PGE is natriuretic.

MODULE-29: ACID BASE BALANCE

Learning objectives

This module deals with

- various buffer systems of the body
- how kidney plays a major role in maintaining pH of the blood?

INTRODUCTION

- The normal blood pH is 7.4. Maintenance of normal blood and extracellular pH within the narrow limits is essential for homeostasis. The pH usually refers to the hydrogen ion (H^+) concentration and has a widespread effect on the function of the body systems. Any disturbance in the H^+ ion concentration leads to imbalance of pH.
- When pH rises above the optimal value it is referred to as alkalosis and if pH drops below optimum level it is referred as acidosis.

Regulation of hydrogen ion concentration

- Three primary buffer systems are involved in regulation of H^+ ion concentration in the body fluids to prevent alkalosis or acidosis.
  - Chemical acid base buffer systems
  - Respiratory regulation of acid base balance
  - Renal control of

CHEMICAL ACID BASE BUFFER SYSTEMS

- When there is a change in the H^+ ion concentration, the body fluids react immediately to minimise the change. Chemical buffers act by converting either strong acids or bases into weaker acids or bases. The various chemical buffers are:
- Bicarbonate
  - This is the most important buffer system in the body. Bicarbonate combines with H^+ ions to form carbonic acid in the tubular fluid, which then dissociates to CO_2

and H_2O. The CO_2 formed is removed by the lungs and the HCO_3^- formed in the cells is reabsorbed from the filtrate to the blood.

- Phosphate buffer
  - It plays a major role in buffering renal tubular fluid and ICF. The two main elements of the phosphate buffer are HPO_4^{2-} (base) and $H_2PO_4^-$ (weak acid). Hydrogen ions from strong acids are captured by converting a weak base to a weak acid and strong base captured by conversion of a weak acid to a weak base.
- Protein buffer
  - It is an intracellular buffer present in high concentrations in the blood. Hemoglobin molecule forms the second most important blood buffer and is present in the form of proteinate ions (Hb^-). These basic ions, with their weak acids (HHb) form a buffer pair. When an acid is added to the blood,

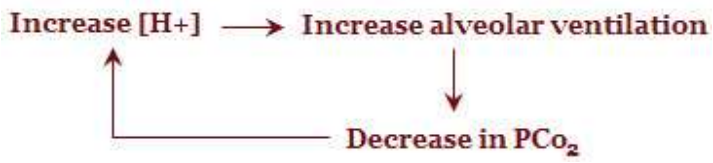
$$H^+ + Hb^- \longrightarrow HHb$$

  - This reaction shifts to the right and the ratio of base to acid is decreased. Hemoglobin is a powerful buffer and it binds with protein, CO_2 and H^+ ions. In the tissues CO_2 passes into the RBC where it combines with water to form carbonic acid which is catalyzed by the enzyme carbonic anhydrase. Carbonic acid then dissociates into HCO_3^- and H^+ ions. The H^+ ions binds to hemoglobin to form HHb and the HCO_3^- ions passes back to the plasma in exchange for chloride ions. In the lungs this process is reversed where in H^+ ions bind to hemoglobin and recombine with bicarbonate to form CO_2 which passes into the alveoli.
- Ammonia buffer system
  - Ammonia is formed by the hydrolysis of glutamine in the presence of enzyme glutaminase in the tubular cells which freely diffuses into luminal fluid and continues with H^+ ions to form NH_4^+ ions. This NH_4^+ ions combines with chloride ions and is excreted as ammonium chloride in the urine.
  - acid base balance

RESPIRATORY REGULATION OF ACID BASE BALANCE

- The respiratory system acts as the second line of defense against acid base disturbances. An increase in PCO_2 of ECF, decreases the pH, while a decrease in PCO_2, increases the pH. Therefore, by adjusting the PCO_2, the lungs effectively regulate the H^+ ion concentration of the ECF.
- An increase in ventilation removes CO_2 from ECF thereby reducing the H^+ ion concentration. Similarly, a decrease in ventilation, increases CO_2 thus increasing H^+ ion concentration in ECF.
- Arterial PCO_2 is inversely proportional to alveolar ventilation, i.e., if alveolar ventilation falls, PCO_2 rises. Therefore, relatively small changes in ventilation has a profound effect on H^+ ion concentration and pH.
- Respiratory system acts as a typical negative feedback controller of H^+ ion concentration.

- An increase in the H^+ concentration above the normal, stimulates the respiratory system and alveolar ventilation increases. This decreases the PCO_2 in ECF and reduces H^+ ions concentration back to normal.



RENAL CONTROL OF ACID BASE BALANCE

- The kidneys regulate acid-base balance by excreting either an acidic or basic urine.
- Excretion of either an acidic or a basic urine removes acids or basic from the ECF. Large numbers of HCO_3^- ions are filtered in the urine and if they are excreted into the urine, it removes base from the blood. Similarly, large numbers of H^+ ions are secreted into the urine, and if they are excreted, it results in loss of acid from the blood. If more HCO_3^- ions are filtered than the H^+ secreted, there will be net loss of base from ECF.
- The kidneys regulate ECF H^+ ions through three basic mechanisms:
  - oTubular Secretion of H^+ ions
  - oReabsorption of filtered bicarbonate ions
  - oCombination of excess H^+ ions with phosphate and amino buffers

Tubular secretion of H^+ ions

- H^+ ions are secreted in the proximal tubule, thick ascending loop of Henle and distal tubule by sodium hydrogen counter transport. This occurs by means of active transport of sodium ions into the cell and H^+ ions from the tubular cell into the tubular lumen against the concentration gradient provided by sodium-potassium ATP pump. For each H^+ ion secreted, one HCO_3^- ion is reabsorbed. When CO_2 diffuses into the tubular cells, formed by metabolism, CO_2 combines with water, forms H_2CO_3 which dissociates into HCO_3^- and H^+ ions. H^+ ions are secreted from the cell into the lumen by sodium-hydrogen counter transport.
- The sodium moves into the cell by concentration gradient established by sodium-potassium ATP ase pump in the basolateral membrane. H^+ ions are also secreted in the distal tubule and collecting ducts and transported through H^+ pump by H^+ ATP ase mechanism.

Reabsorption of filtered bicarbonates

- The filtered bicarbonate ions are not easily reabsorbed across the tubular membrane. The bicarbonates combines with H^+ ions to form H_2CO_3 which then dissociates to form Co_2 and H_2O. This CO_2 moves across the tubular membrane and diffuses immediately into the tubular cell.

- Inside the cell, it combines with H_2O to from H_2CO_3 in the presence of carbonic anhydrase and H_2Co_3 dissociate to HCO_3^- and H^+ ion. Therefore for energy H^+ ion formed in the tubular epithelial cells, a HCO_3^- ion is formed and reabsorbed into the blood.

Combination of excess H^+ with phosphate buffer

- When excess of H^+ ions are secreted, only a small fraction of is excreted in the ionic form (H^+) in the urine and the remaining H^+ ions combines with buffers such as phosphate and ammonia buffer in the tubular fluid as urine can be acidified to a pH of about 4.5.
- The phosphate buffer system is composed of HPO_4^{2-} and $H_2PO_4^-$. Both are concentrated in the tubular fluid because of poor reabsorption. Excess H^+ ions combines with HPO_4^{2-} to form H_2PO_4 which in turn are excreted as sodium salt (NaH_2PO_4).

Combination of excess H^+ with ammonia buffer system

- This buffer system is composed of ammonia (NH_3) and ammonium ion (NH_4^+). Ammonium ion is synthesized from glutamine which is actively transported into the tubular epithelial cells. Inside the cell, glutamine is metabolised to form NH_4^+ and two HCO_3^- ions. The NH_4^+ is secreted into the tubular lumen by countercurrent mechanism in exchange for sodium which is reabsorbed. The HCO_3^- moves across the basolateral membrane along with reabsorbed Na^+ into the blood. Therefore for each molecule of glutamine metabolised in the proximal tubule, two NH_4^+ ions are secreted into the urine and two HCO_3^- ions are reabsorbed into the blood.
- In the collecting tubule, formation of NH_4^+ ions occurs by a combination of NH_3 (ammonia) with H^+ ions and then excreted as NH_4^+. The collecting ducts are permeable to NH_3 and form NH_4^+. Hydrogen ions react with NH_3 and form NH_4^+. For each NH_4^+ excreted, one HCO_3^- reabsorbed in to the blood.

ELECTROLYTES

- Sodium / Potassium: Sodium reabsorption and H^+ ion excretion are interlinked. Sodium reabsorption is controlled by the hormone, aldosterone and ion exchange proteins that exchange sodium for H^+ ions or K+ ions. Therefore, the changes in aldosterone secretion may influence acid-base balance.
- Chloride: The bicarbonate and chloride ions are the two abundant negative ions in the plasma. In order to maintain electroneutrality any change in chloride must be accompanied by opposite change in bicarbonate concentration. Therefore, chloride concentration may also influence acid base balance.

ACID BASE BALANCE DISTURBANCES

- The pH of the ECF is determined by the rate of conjugate base to their weak acids. The total amount of buffer base in whole blood including HCO_3, Hb and other bases of lesser importance is called buffer base (B.B). These bases are known as metabolic components determining blood pH. Acid base disturbance involves either the gain or loss of strong acid or the gain or loss of base (Cl^- or HCO_3^-) by the ECF.

Metabolic acidosis

- The gain of strong acid or loss of base from the ECF is known as metabolic acidosis. Acidemia will be present in metabolic acidosis. It occurs in:
  - Ketosis.
  - Diabetes mellitus in which ß hydroxy butyric acid, acetone, aecto acetic acid are produced.
  - Renal acidosis in which there is failure of HCO_3^- reabsorption and loss in the urine.
  - Diarrhoea where pancreatic juice containing HCO_3 is not reabsorbed and is lost.
- In all these cases, HCO_3^- falls either as a result of a reaction with acid or due to direct loss from ECF and pH falls. This results in fall of all blood buffer bases. Usually there is no change in plasma PCO_2 . However, a fall in pH results in increased alveolar ventilation and therefore a fall in PCO_2. Decreased PCO_2 will bring the ratio of conjugate base to weak acid back to normal. However, the total bases will be less than normal and this requires renal correction - the excretion of H^+ and restoration of plasma $HCO3^-$.
- The acidemia stimulates secretion of H^+ ion by the renal tubule. This ensures reabsorption of all HCO_3 ions from tubular fluid and the excess H^+ ions will begin to acidify the urine. For each H^+ ion secreted, one HCO_3 will be reabsorbed into the plasma. This holds good for short-term stress and in severe conditions, therapeutic action is required.

Metabolic alkalosis

- This process involves the gain of base (OH or HCO_3 ions) or loss of strong acid by ECF.
- Metabolic alkalosis is present in
  - Persistent vomiting, in which gastric acid is lost from the body.
  - K^+ deficiency in which renal tubules secretes large amount of H^+ ions into urine.
  - Injection of HCO_3 solutions.
- In all these cases, there is an increase in $HCO3^-$ in ECF, resulting in increased base content. The response of the body is opposite to the one observed in metabolic acidosis. Alkalemia results in rise in pH which will depress pulmonary ventilation and PCO_2 will rise. This respiratory compensation thus will bring pH back to normal. Renal correction consists of decreased secretion of pH ions and so increased excretion of HCO_3^- ions.

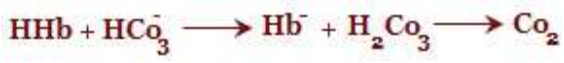
Respiratory acidosis

- If excretion of CO_2 by the lungs falls below the rate of CO_2 production in the body, respiratory acidosis develops.

- There will be an increase in blood PCO_2 (hypercapnia) and the primary defect will be in the inability of lungs to expire CO_2 at a normal rate. This may be due to
  - Depression of respiratory centres in CNS.
  - Abnormality of chest wall or respiratory muscles which prevents enlargement of thorax.
  - Obstruction to gas movement in lungs.
- A rise in PCO_2 causes increase in H_2CO_3 and buffer reaction prevents the fall of pH caused by rise in H_2CO_3.
- Renal compensation then follows. Low pH stimulates secretion of H^+ into urine with a rise in plasma $HCO3^-$.

Respiratory alkalosis

- When alveolar hyperventilation occurs, the expiration of CO_2 may exceed the rate of its production within the body and respiratory alkalosis develops.
- There will be low plasma PCO_2 (hypocapnia) and alkalemia.
- Hyperventilation is caused by abnormal stimulus to respiratory centres either directly as in NH_3 toxicity or through hypoxemia acting through peripheral chemoreceptors.
- Buffer reaction follows:



- Thus HCO_3 falls and Hb rises. The renal compensation begins, alkalemia depresses H^+ ion secretion by renal tubules and excretion of filtered HCO_3 rises. This results in further fall of plasma HCO_3 and the ratio of HCO_3 to H_2CO_3 moves back to normal.

MODULE-30: FLUID AND ELECTROLYTE BALANCE

Learning objectives

This module deals with

- maintainence of fluid and electrolyte balance by the kidneys during dehydration.

INTRODUCTION

- Regulation of fluid by the body is necessary to maintain homeostasis. If the water or electrolyte equilibrium is affected, many body functions fail to proceed at normal rates.

Water

- Water is a major constituent of all living things. Most of the ions and molecules that make up living matter have chemical and physical relationships with water.

- The total body water content varies among different species, age, sex, nutritional factors etc. Water content is highest in the new born animal and declines as age advances.
- An adult contains 60% of water by weight depending upon age and the amount of body fat. Body fat is inversely proportional to the body water content. For example, a very lean animal will contain 70% of water whereas very fat animal will have only 40% of total body water.

Body fluid Compartments

- Body fluid is present in three different compartments namely, intracellular fluid and extracellular fluid, which in turn is divided into interstitial fluid and plasma.
- In a lean animal, 50% of water is present intracellular, 15% in the interstitial spaces and 5% in the blood plasma. Apart from this, water is also present in the transcellular fluids such as in CSF, aqueous humor of the eye, synovial fluids, urine, bile etc.
- Water molecules can rapidly penetrate most of the cell membrane. If an osmotic or hydrostatic pressure gradient exists between body fluid compartments, a shift of water will occur. Addition of an isotonic NaCl solution to the ECF causes equal distribution of water extracellularly and intracellularly.
- If a hypertonic NaCl solution is added, water would begin to shift into the plasma, while, an addition of hypotonic NaCl solution shifts the water into the cell.

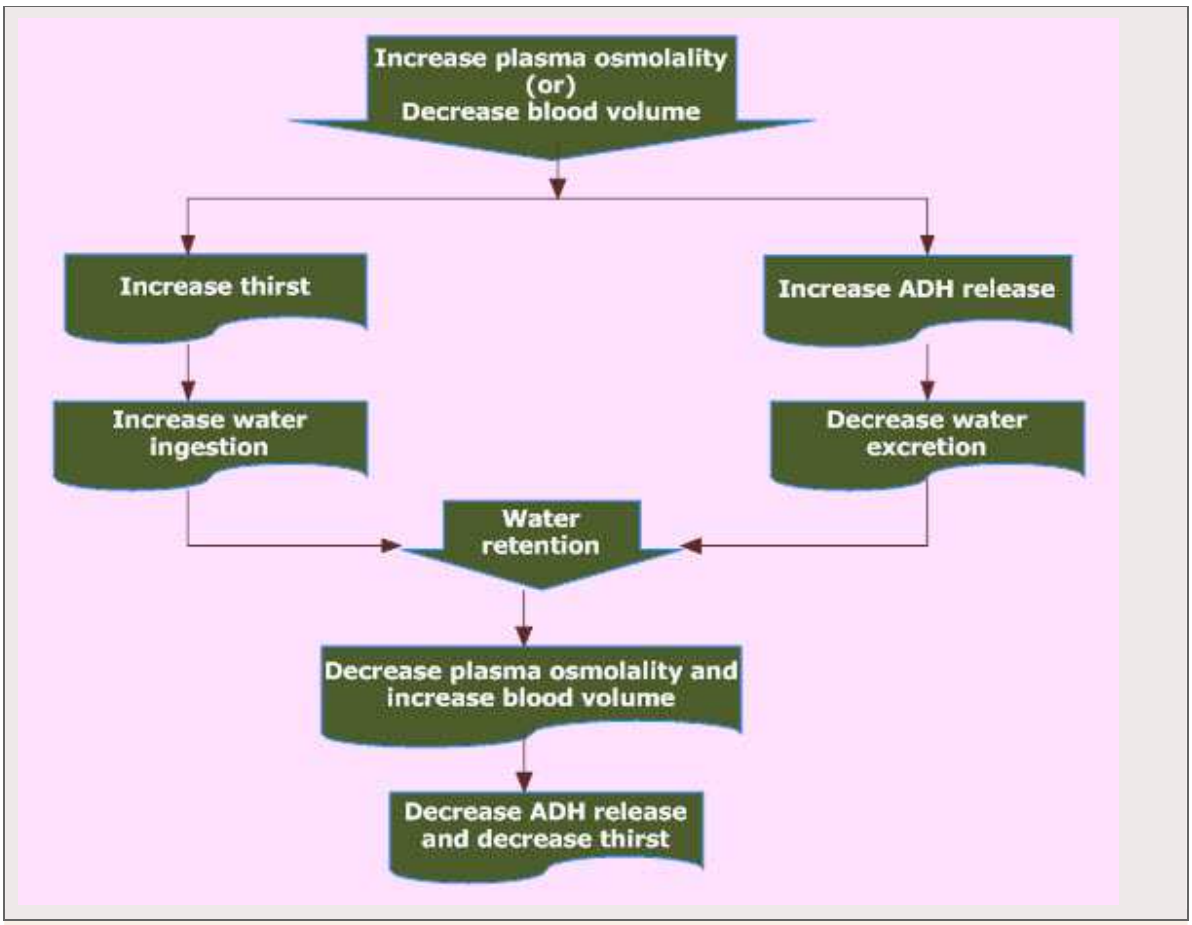
Water balance

- The total amount of water in the body almost remains relatively constant. The body gains water either by ingestion or as end product of cellular metabolism. Similarily loss of water occurs in urine, from the skin, expired gases, faeces etc.

THIRST

- Loss of water from the body is continuous and when there is deficiency of water, specific controls of water intake act to correct the deficit. Water deprivation causes the sensation of thirst and drive to drink water. Several mechanisms aid in the control of the amount of water to be ingested so that it equals the deficit of body water.
- A decrease in body water causes thirst which is characterized by dryness of the throat and mouth due to decreased salivary secretion. The centre for thirst and drinking behaviour is located in the hypothalamus of the brain.
- Water deficiency causes a rise in osmotic and sodium concentration in the ECF and a decrease in the circulating fluid volume together with a fall in BP.
- A fall in the BP is detected by the baroreceptors and the renin relating cells of the kidney which stimulates the thirst centre in the hypothalamus.
- As a result, thirst and drinking of water occurs. A fall in arterial BP alone also causes the release of renin from the juxta glomerular cells of the afferent arterioles of the kidney glomeruli. Renin acts on angiotensinogen to form angiotensin I, which in turn in is converted to angiotensin II by the converting enzyme. In most of the animals angiotensin II causes an increase in drinking water.

- During water deprivation, excretion of water by the kidney is controlled primarily by the Antidiuretic Hormone (ADH) secreted from the posterior pituitary gland. ADH acts on the nephrons of the kidney and causes increased reabsorption of water thereby decreasing the excretion of water with urine.
- The oxidation of food stuffs also yields some amount of water. The oxidation of each gram of carbohydrate, fat and protein yields about 0.6 ml, 1.1 ml and 0.4 ml of water. This metabolic water constitutes 5 to 10% of the total water intake in animals.



DEHYDRATION

- Dehydration in the animals results due to reduced or absence of intake of water and continuous loss of body water.
- Dehydration involves both loss of water and electrolytes. The process of dehydration may be slow or rapid depending on the relative rates of water and electrolyte loss. In a simple dehydration due to lack of water under moderate environmental conditions, causes the animals to seek and drink water. There will be a decrease in urine volume output. These changes can be seen when the dehydration is about 1-2% of the body weight.
- Severe dehydration occurs when the water loss is 10% of the body weight. The immediate source of water lost from the body is the ECF. If the rate of water loss is vey rapid, there

is drastic reduction in ECF volume. In slow dehydration, there will be a shift of water from the cells into the ECF.

- During water deprivation loss of water from the body causes an increase in ECF osmolality and a decrease in ECF volume. This stimulates the drinking behaviour and a reduced urine output. Electrolytes are also excreted from the body in proportion to water loss inorder to prevent a further rise in osmoconcentration.
- In early dehydration, sodium chloride is excreted in the urine. There is a shift of intracellular water to extracellular fluid. Cellular potassium is also excreted in the urine. Therefore, after a long period of dehydration depletion of both water and primary electrolytes occurs. In human, the limit to which the water can be lost is about 15-25% of the body weight before death occurs.

Adaptation to water lack

- Some of the species that are found in the desert regions such as camel, donkey, kangaroo rats etc., have acquired adaptation to resist water lack. During high summer temperature, these animals have to expend increased amount of water to control their body temperature in the desert regions as there is little amount of rainfall and natural vegetation which contains less water.
- Under these conditions, the animals have to obtain water either through vegetation or through metabolic water.
- The water obtained through these mechanisms are small as the actual amount of water formed from the oxidation of fat, carbohydrates and protein are 0.12 g/kcal, 0.14 kg/Kcal and 0.10 g/kcal.
- The camel has a remarkable ability to conserve water and can withstand a water loss of 25%. During the day when the heat stress is greatest, the camel can rise its body temperature thereby storing heat and saving water.
- During the cooler night this stored heat is dissipated and the body temperature is brought back to normal.
- The camel's summer fur also aid in reducing the solar heat gain. Camel conserves water by excreting dry faeces. The water lost through respiration also is very low in camels because of their ability to reduce the relative humidity of expired air to below 100%. This is due to the hygroscopic nature of the nasal secretions. The camel also has the ability to rehydrate rapidly as it can ingest $1/4^{th}$ of its body weight as water in a few minutes. The oval biconcave RBCs of the camel are highly resistant to osmotic hemolysis.
- The donkeys can also withstand dehydration upto 30% and can rehydrate rapidly by drinking 20.5 litres of water in 2.5 minutes.
- Sheep also has remarkable ability to withstand heat stress and water lack. It can resist 30% dehydration and minimize solar heat absorption by increasing the wool surface temperature upto $87^{\circ}C$. It dissipates heat by means of panting.

ELECTROLYTES

- An electrolyte in any chemical that dissociates to ions when dissolved in a solution. Ions can be positively charged (cations) or negatively charged (anions). The major electrolytes found in the body are sodium, potassium, calcium, magnesium, chloride, phosphate, sulphate and bicarbonate. The primary electrolytes of the ECF are sodium, chloride and bicarbonate and in the ICF are potassium and phosphates.

Composition of major electrolytes in the body fluids (mOsm/Kg H_2O)

Ions	Plasma	Interstitial Fluid	ICF
Sodium	146	142	14
Potassium	4.2	4.0	140
Chloride	105	108	4
Bicarbonate	27	28.3	10

Sodium

- It is the major cation of ECF. About 45% of sodium stored is found in ECF, 45% in the bone and the remaining intracelullarly. It plays an important role in the excitability of muscles and neurons and in regulating the fluid balance in the body.
- A constant sodium equilibrium in maintained in the ECF by two mechanisms-long term and short term control. The short term control is achieved by the ADH- thirst control system whereas long term control is by both ingestion and urinary excretion of sodium.

Short term Control (ADH-Thirst Control System)

- If ECF sodium rises, it stimulates the release of ECF and thirst. Water gained by this mechanism, dilutes the ECF thereby restoring normal sodium level. This increases the blood volume and a slight increase in BP, which in turn increases the GFR and excretion of excess sodium and water there by restoring the ECF volume to its normal level.

Longterm Control (Salt hunger/ Sodium ingestion)

- A deficiency of sodium in the ECF causes increased excretion of water due to inhibition of the release of ADH. A decline in osmolality decreases the GFR with a subsequent increase in reabsorption of sodium and water. Many sodium deficient animals have a strong behavioural drive to salt to replace the deficiency of sodium. This is called as salt hunger and is commonly seen in ruminants.
- An increase in angiotensin II stimulates the salt hunger. A decrease in the plasma sodium and BP stimulates the release of aldosterone from the adrenal cortex via the Renin-Angiotensin system. Aldosterone enhances the reabsorption of sodium from the renal tubule. Aldosterone also increases the secretion of potassium ions from the renal tubules into the lumen.

Urinary excretion of Sodium

- Sodium is filtered at the glomerulus and most of it is reabsorbed in the renal tubules. If the plasma sodium or the GFR increases, the amount of sodium filtered into the tubule will also increase but the reabsorption of sodium will not increase.
- An increase in the arterial BP results in the release of Atrial Natriuretic factor (ANF) from the atria of the heart. This hormone inhibits renin and aldosterone release which results in increased excretion of sodium and inhibition of reabsorption of sodium. Along with sodium, water will also be lost thereby bringing the BP back to normal level.

Potassium

- It is the major cation of intra cellular fluid and about 89% of the total body content of potassium is present in the ICF. Potassium is important for the functioning of excitable cells and in the regulation of fluid levels with in the cells.
- Potassium output in usually equal to the potassium input. Almost nearly all the potassium filtered by the kidney glomeruli is reabsorbed by proximal convoluted tubule and is secreted in the distal tubule and the collecting ducts. Potassium reabsorption by the tubular cells occurs by active transport of potassium into the cells in exchange for sodium through the Na/K ATPase system.
- Potassium concentration is regulated by the aldosterone. An increase in ECF potassium results in the aldosterone release, which increases the potassium excretion in the urine and returning of the plasma potassium to its normal level. Potassium is secreted in exchange for the sodium ion in the tubular cells. Hydrogen ion secretion competes with K^+ for Na^+ with which it exchanges. Therefore, an increased H^+ secretion depresses K^+ secretion. Likewise an increased reabsorption of Na^+ facilitates increased K^+ secretion. An excess of aldosterone results in hypokalemia whereas deficiency of aldosterone causes hyperkalemia.

Chloride and bicarbonate

- The sodium ions are balanced electrically with the chloride and bicarbonate ions. The chloride ions are regulated secondarily to sodium and bicarbonate ions.
- The excretion or reabsorption of sodium ions is accompanied by chloride ions. Similarily, chloride ions are excreted along with the bicarbonate ions to maintain electroneutrality in the ECF.
- The bicarbonate ion is unique in that it is formed or removed rapidly by the body.
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- Bicarbonates are formed and removed by carbon dioxide.

MODULE-31: FORMATION AND EXCRETION OF URINE IN BIRDS

Learning objectives

This module deals with

- structure of kidney in birds
- formation of urine in birds

EXCRETION IN BIRDS

- Urine formation in birds is almost similar to the mammals but still then there are some notable differences:
  - Presence of two major types of nephrons which are functionally different.
  - Presence of renal portal system.
  - Formation of uric acid instead of urea as the end product of nitrogen metabolism.
  - Post renal modification of the urine in the ureter.
- In the birds, the ureters transport the urine to the cloaca, which is the common collection site for digestive, reproductive and urinary organs.
- There is no urinary bladder in birds.

NEPHRON TYPES

- Avian kidneys are characterized by having two major types of nephrons:
  - Reptilian type and
  - Mammalian type
- The *reptilian type* nephrons are located in the cortex and it lacks the loop of Henle. It has no capacity to concentrate the urine, *i.e.*, there is no tubular transport system and whatever solute and water is present in the filtrate, directly passes to the cloaca.
- *Mammalian type* of nephrons have well defined loop of Henle. It has the capacity to concentrate the urine. In this tubular transport system is present.

RENAL PORTAL SYSTEM (RPS)

- An unique feature of avian kidney is its Renal Portal System (RPS) which provides an extra branch of blood flow to the renal tubules along with peritubular capillaries.
- Venous blood from portal vein gives one branch to the kidney and this branch provides the microcapillaries which perfuse the tubules along with peritubular capillaries.
- Both the branches of capillaries are interconnected and drained by the venous system to the posterior venacava. When it leaves these capillary network and enters the renal vein there is a valve which regulates the transition of the blood in these capillary network.

URIC ACID FORMATION AND EXCRETION

- The metabolic end product of protein and amino acids in reptiles and birds is the uric acid (instead of urea in mammals).
- Uric acid is formed in liver and also in kidneys from ammonia.
- Uric acid is freely filterable at the glomerulus, and it is also secreted by the tubules.
- Tubule secretion accounts for 90% of total uric acid eliminated.

- The renal portal system may provide a greater quantity of blood to the tubules for the secretion of uric acid by the tubules. Since greater quantity of uric acid is available in the tubules, which exceeds the solubility, the uric acid is precipitated. It passes through the tubules in the precipitated form and appears in the urine as a white coagulum. Since the uric acid is not in solution, it does not contribute to osmotic pressure, and thus avoids obligatory water loss.

MODIFICATION OF URETERAL URINE

- The post renal modification of urine is possible in the birds when the urine reaches the cloaca water is drawn back to colon and cecum due to antiperistaltic movement of colon.
- So when urine is exposed to colon and cecum for water absorption sodium is also reabsorbed.

CONCENTRATION OF AVIAN URINE

- Renal response to ADH in birds is similar to that of the mammals.
- In addition to the action of ADH on the tubular cells it also controls the functioning of reptilian and mammalian type of nephrons.

URINE CHARACTERISTICS AND FLOW

- Birds urine when mixed with faeces is cream coloured and contains thick mucous.
- The precipitated uric acid is mixed with mucus and mucus facilitates the carrying of uric acid in the urine.

MODULE-32: LYMPH AND CSF

Learning objectives

This module deals with

- formation of lymph and cerebrospinal fluid
- composition of lymph and cerebrospinal fluid

LYMPH

- Lymph is a clear colorless fluid somewhat similar to plasma. It is derived as a left out fluid at venous end of the capillary bed which is filtered out of the arterial end.
- It contains few or no RBCs, numerous leukocytes, inorganic salts, glucose, NPN substances and some proteins. Lymph derived from the intestine during absorption contains lipids which gives milky white appearance known as chyle.

- Lymphatic vessels originate as blind lymphatic capillaries in between the tissues to collect the tissue fluids, which are not absorbed by the venous system. Lymph capillaries are highly permeable and have less resistance to the entry of macromolecule proteins.
- The lymphatic capillaries form a complex network throughout the tissues and combine to form small lymphatic vessels. This in turn unites to form large lymphatic vessels.
- All the lymphatic vessels of the left side of head, arm and chest and the posterior portion of the body drain their lymph into the thoracic duct, whereas the right lymphatic duct receives the lymph from the right side of the head, neck, foreleg and thorax.
- These two lymphatic vessels empty their contents into the venous system at the juncture of the right subclavian vein and internal jugular vein.
- Lymphatic vessels resemble veins, but their walls are much thinner than veins. They have valves throughout their course to direct the lymph flow towards heart and prevent back flow of lymph.
- Lymph glands or nodes are discrete nodular structures scattered along the course of lymphatic vessels. These nodes function as filters to foreign agents (bacteria).
- Lymphatic glands have lymphocytes and plasma cells, produce antibodies and macrophages against foreign invaders, thus act as a first line of defense organ against infective organisms. Lymph nodes are absent in brain, bone marrow and also in birds.

FORMATION OF LYMPH

- At venous capillary end, only about 90% of the fluid portion of plasma that comes out from the capillary bed of the blood vessels is reabsorbed due to pressure gradient. The remaining 10% constitute the lymph which enters the lymph channels and return to the circulatory system.
- The movement of the lymph inside the vessel is caused by the pressure difference between two ends of the lymphatic system, contractions of the muscles during movement of the body, contractions of the smooth muscles of the lymphatic vessels and presence of valves which prevent back flow.

Forces causing filtration (at the atrial end)

- Forces tending to move fluid outward
  - oCapillary pressure : 30.0 mm Hg
  - oNegative pressure of interstitial fluid : 3.0 mm Hg
  - oI.F. colloidal osmotic pressure : 8.0 mm Hg
  - oTotal outward force : 41.0 mm Hg
- Forces tending to move fluid inward
  - oPlasma colloidal osmotic pressure : 28.0 mm Hg
  - oNet filtration pressure = Out pressure – In ward pressure
  - o41.0 - 28.0 = 13.0 mm Hg.

Force causing reabsorption (At the venous end)

- Forces tending to move fluid outward

- oCapillary pressure : 10.0 mm Hg
- oNegative pressure of interstitial fluid : 3.0 mm Hg
- oI.F colloidal osmotic pressure : 8.0 mm Hg
- oTotal outward force :21.0 mm Hg

- Forces tending to move fluid inward
  - oPlasma colloidal osmotic pressure : 28.0 mm Hg
  - oNet absorption pressure = Inward pressure - Outward pressure
  - o28.0 - 21.0 = 7.0 mm Hg.

Factors determining fluid movement across the capillary wall

- Capillary pressure or capillary hydrostatic pressure tends to move fluid outward.
- Interstitial fluid pressure tends to move fluid inward when it is positive and outward when it is negative.
- Plasma colloid osmotic pressure (Oncotic pressure) tends to cause osmosis of fluid inward.
- Interstitial fluid colloid osmotic pressure tend to cause osmosis of fluid outward.
-

FUNCTIONS OF LYMPH

- The lymph helps in the reabsorption of the fluid from the tissue spaces and adds it to the circulatory system, thus maintains a constant fluid equilibrium.
- It filters the harmful infectious agents (bacteria) from the body.
- It absorbs and transports the fatty substances from the intestine in the form of chyle.
- It helps in the absorption of macromolecular proteins from the interstitial spaces, which are then added to the blood.

CEREBROSPINAL FLUID

- It is formed by the choroid plexus of the lateral, third and fourth ventricles of the brain.

- CSF is periodically absorbed by the arachnoid villi of the subarachnoid space. This structure is quite large in man and horse, but generally small or microscopic in other domestic animals.
- The piamater covering the surface of the brain and spinal cord also contributes small quantities of CSF. This fluid circulates throughout the sub arachnoid space between the piamater and arachnoid membrane, ventricles of the brain and central canal of spinal cord.
- CSF contains a very small quantity of protein. The concentration of sodium and chloride are higher and potassium, urea and glucose are lower in CSF than plasma. The pH is same as blood. It does not contain cellular elements except very few lymphocytes.
- CSF secretion is an active process and is not affected by the blood pressure or CSF pressure. The choroid plexus provides some selectivity of permeability for certain substances of the blood and the CSF is not identical with the blood plasma. It contains

lower concentration of proteins, K^+ and glucose than plasma, but has higher concentration of Na^+ ions. Thus the choroid plexus functions as blood-cerebrospinal fluid barrier which protects the CNS from the influence of a variety of substances.

- CSF is absorbed by the arachnoid villi of the arachnoid membrane, which project through the venous sinuses of the duramater.
- The CSF serves partly as a nutritive medium for the brain and spinal cord as well as cushioning these structures against shock. It also aids in the transport of some peptide hormones and other substances of the brain into the circulation.

EDEMA

- It is the abnormal accumulation of the tissue fluids due to excessive filtration by increased capillary pressure or poor reabsorption at the venous end of the capillaries.
- Edema occurs in
    - oIncreased capillary pressure
    - oIncreased capillary permeability
    - oDecreased concentration of plasma proteins and
    - oObstruction of lymph vessels.
- The low plasma colloidal osmotic pressure due to hypoproteinemia, obstruction of the lymphatic vessels by tumours or by filarial worms, vasodilator substances of allergic reaction etc causes increased capillary permeability leading to edema.
- Chronic congestive heart failure due to improper drainage of fluids into the right heart results in fluid accumulation in the peritoneum and is referred to as *ascites*.

MODULE-33: JOINTS AND BONE METABOLISM

Learning objectives

This module deals with

- the types of joints and composition of synovial fluid
- hormonal regulation of bone metabolism

JOINTS

- Joint is a junction between two bones. It is also called as articulation. Joints are classified into three types
- Fibrous Joints:
    - oThese are immovable and the bones are firmly united by fibrous tissue. **Eg.:** Joints found between the skull bones.
- Cartilaginous Joints:
    - oThese joints are slightly movable. **Eg.:** Intervertebral disks between the bodies of adjacent vertebrae in the spine.
- Synovial Joints:

- These are freely movable joints, *e.g.*, shoulders joint, stifle joint etc. The movement of a joint is normally controlled and united by the action of muscles and ligaments and tendons. All synovial joints have some common characteristics. They have a capsule with an outer fibrous layer consisting of collagen fibres running from the periosteum of one bone to the other.
- Ligaments are located in and around the capsule. The inner lining layer of the joint surface is the synovial membrane which produces synovial fluid that lubricates the joint surfaces. Synovial fluid is a transparent viscous liquid and it provides lubrication to the joint provides lubrication to the joint.
- Some bones contain special areas of dense fibro cartilage between the articulating cartilage surfaces and are known as menisci. **Eg.:** Bones that bear great weight. The function of the menisci is to cushion compressive forces acting on the ends of the bones.

SYNOVIAL FLUID

- Synovial fluid is a thick, viscous liquid found in joint cavities, tendon sheath and bursae.
- It contains small amounts of albumin, globulin, mucin and hyaluronic acid but no fibrinogen.
- Synovial fluid functions as a lubricant to reduce friction in the joints and help to nourish the articular cartilages.

BONE METABOLISM AND HOMEOSTASIS

- Bone consists of both organic and organic components. The major inorganic compound is calcium phosphate which accounts for 2/3rd of the bone weight. This calcium phosphate interacts with calcium hydroxide to from hydroxyapatite.
- Other inorganic compounds include calcium carbonate, sodium, magnesium and fluoride.
- The remaining organic portion of the bone is made up of cells such as osteoblasts, osteocytes, osteoclasts etc and osteoid such as collagen fibers, proteoglycans, glycoproteins etc. The osteoid is secreted by osteoblasts.
- Bone mineral metabolism is important for calcium homeostasis. During remodelling of the bone, minerals are released into the ECF through the action of osteoclasts and these minerals are removed from the blood by the osteoblasts during bone formation. Under normal conditions the rate of bone resorption is equal to the rate of bone formation.
- There are three major hormones that regulate calcium homeostasis,
  - Parathyroid hormone(PTH),
  - Calcitonin and
  - Vitamin D metabolite (1,25, Dihydroxy cholecalciferol).

EFFECT OF PTH IN THE BONE

- PTH is released from the parathyroid gland in response to low blood calcium levels. The action of PTH on the bone is calcium resorption from the skeletal reserves into the ECF. Release of calcium from the bone occurs in 2 phases. They are:
  - Rapid phase and
  - Slow phase.

- Rapid phase
  - This phase is termed as osteocytic osteolysis. It results from activation of the already existing osteocytes and osteoclasts present in the bone. During this phase there is an increased flow of calcium from the bone fluid to the ECF by osteocyte-osteoblast pump.
  - PTH causes removal of bone salts from two areas in the bone, one from the bone matrix in the vicinity of the osteocytes lying in the bone itself and two, in the vicinity of the osteoblasts along the bone surface. When this osteocyte - osteoblast pump is activated by PTH, bone fluid calcium concentration falls and calcium and phosphorous salts are absorbed from the bone.
  - When the pump is inactivated, bone fluid calcium concentration rises and calcium and phosphorous salts are redeposited in the matrix.
  - PTH has receptors on the cell membrane of both osteoblast and osteocytes When stimulated by PTH, calcium permeability of the osteocytic membrane increases thus allowing calcium to diffuse into the membrane cells from the bone fluid to the ECF.
- Slow phase
  - This phase results from proliferation of the osteoclasts followed by increased osteoclastic reabsorption of the bone itself. PTH do not have receptors on the osteoclasts. Activation of osteoclastic system occurs in two stages,
    - Immediate activation of the osteoblasts that are already formed.
    - Formation of new osteoblasts.
  - PTH stimulates the conversion of osteoprogenitor cells to osteoclasts. PTH binds to the receptors present on the surfaces of osteoblasts causing the cells to contract and thereby exposes the minerals in the matrix to osteoclasts. The minerals and organic components released from the bone are phagocytized by osteoclasts and transported across the cell to be released into the ECF. When the animal is in the positive calcium balance, PTH secretion subsides and osteoblasts finish the reversal and formation phases of bone remodelling.

CALCITONIN

- It is the *second major hormone* that regulates calcium. It is the secreted by the 'C' cells within the thyroid gland in response to high blood calcium levels.
- Calcitonin has receptors on the osteoclasts and directly act on these cells to inhibit bone resorption by blocking osteoclastic osteocytic activity.

1, 25 DIHYDROXY CHOLECALCIFEROL (1,25 DHCC)

- It is the third major hormone involved in calcium metabolism and is a vitamin D metabolite. The action of 1,25 DHCC in bone formation is primarily indirect. It acts on the intestine to increase the efficiency of absorption of calcium and phosphorous absorption via calcium binding protein .
- Osteoblasts have receptors for 1,25 DHCC. It causes osteoclastic resorption and calcium mobilization from the bone.
- Not only bone helps in the maintenance of blood calcium level it also acts as a buffer of the blood pH. It can release cations or anions in response to acidosis or alkalosis.

EFFECT OF ESTROGEN

- Estrogen receptors are present on osteoblasts and osteoclasts. In the osteoclast, estrogen increases cell proliferation and enhances expression of genes that encode for growth factors, enzymes, matrix proteins and cytokines that alter bone remodeling.
- Estrogen inhibit production of cytokines associated with bone resorption such as interlukins (IL–I, IL-6 and IL-11). Estrogen also stimulates synthesis of transforming growth factor-beta (TGF- ß), Insulin like growth factor (IGF)–I and IGF-binding proteins. Proteins involved in bone remodeling, such as bone morphogenic protein and osteoprotegenin are also regulated by estrogen. In estrogen deficiency these is an upward regulation in the maturation of osteoclasts and osteoblasts, with an over all increase in bone remodeling.
- The major action of estrogen on bone is inhibition of bone resorption. Estrogen induces apoptosis and reduces the lifespan of the osteoclast. The synergy of several cytokines enhance osteoclast recruitment, differentiation and activity. The blockage of the productive of cytokines such as IL and tumour necrosis factor μ (TNF μ) by osteoblasts prevents bone resorption in an estrogen deficient state.

Practicals

EXERCISE-1: COLLECTION OF BLOOD

Materials required

- Syringe
- Hypodermic needle of suitable size
- Scissors
- Blood collection vials or test tube with suitable anticoagulants
- Centrifuge

In cattle, horse, sheep and goats, 5 ml blood is adequate for routine blood studies, but in cat, dogs and birds, it is advisable to collect 2 ml blood.

SITE OF COLLECTION

Species	Site	Needle size (gauge)
Horse	Jugular vein	18-20
Cow	Jugular vein	16-18
Sheep & Goat	Jugular vein	18-20
Pig	Anterior vena cava	20
Dog	Cephalic, saphenous or Jugular vein	20-22
Cat	Cephalic or Jugular vein	22-24
Poultry	Wing vein, cutaneous vein	23-24

- Before collection of blood, the site of collection should be clipped of hairs, cleaned and swabbed with antiseptics.

ANTICOAGULANTS

- Anticoagulants are the agents, which prevent coagulation or the clotting of the blood both outside the blood vessels (*in vitro*) and also inside the blood vessels (*in vivo*).

IN VITRO ANTICOAGULANTS

Anticoagulant	Concentration	Mode of action
Sodium citrate	10-20 mg per 10 ml or 3.8% solution @ 1 ml to 9 ml blood	Combines with Calcium to form insoluble salt

EDTA (Na or K salts)	1 to 2 mg per ml or 10% EDTA solution @ one drop/ 5 ml	Forms insoluble Calcium salt (chelating agent)
Heparin	1%solution @ 0.1 ml per 5 ml blood 1-2 mg/ 10 ml blood (not to be used for D.C. of leukocytes)	Antithrombim, antithromboplastin, inhibits Factor IX
Heller-Paul mixture Double oxalate mixture (3:2) 0.1 ml/ml	Ammonium oxalate 1.2 g and potassium oxalate 0.8 g in 100 ml distilled water; 20 mg dried powder/10 ml	Binds calcium to form insoluble Calcium oxalate
Sodium oxalate	2 mg per ml; mainly used for prothrombin time estimation	Same as double oxalate mixture
Sodium/ Potassium fluoride	10 mg fluoride and 1mg thymol per 10 ml; used for blood glucose estimation	Forms weakly dissociated calcium compound

- Sodium fluoride prevents disintegration of the platelets. It is the ideal anti-coagulant for the estimation of blood glucose level, because it prevents the oxidation reaction of glucose.
- EDTA acts as a chelating agent that takes up Ca^{++} ions, useful for blood morphological studies.

NATURAL ANTICOAGULANTS

- *Heparin:* Naturally occurring anticoagulant produced by the mast cell and the basophils, it inhibits the actions of prothrombin, thrombin and thromboplastin. It also inactivates the coagulation factors IX, X, XI and XII, used clinically (*in vivo*) to prevent thrombosis and embolism.
- Concentration: 20-units/ ml of blood. (Or) 0.1 ml of 1% sol/5 ml of blood
- *Hirudin:* Produced by leech, inactivates thrombin thus delays coagulation.
- *Snake venom:* Interferes with the action of thromboplastin or destroys blood fibrinogen.
- *Dicoumarol:* It is vitamin K antagonist, produced in spoiled sweet clover hay and prevents the utilisation of vitamin K by the liver, thus depresses the prothrombin production leading to hypoprothrombinemia.

- Non-wettable surface, silicon coated tubes, plastic containers and rapid chilling of the blood samples to 0 ^{o}C delays the formation of fibrin stabilising factor activator and interferes with thromboplastin formation.

SEPARATION OF PLASMA

- Take anticoagulant added blood sample in a small clean and dry test tube.
- Centrifuge it at 3000 rpm for 30 minutes.
- Take the centrifuged tube and observe the cells packed at the bottom.
- Remove the supernatant plasma carefully and preserve it for further analysis.

SEPARATION OF SERUM

- Collect blood in a wider test tube without adding anticoagulant and place the tube in a slanting position.
- Allow it to clot for 30 minutes at room temperature.
- After clotting, allow it to stand for overnight at 4°C.
- The blood clot will retract and serum will ooze out from the clot to the top.
- Remove the supernatant serum carefully.
- For quicker separation of serum, the clotted blood may be centrifuged for 30 minutes at 3000 rpm.

PREPARATION OF DEFIBRINATED BLOOD

Defibrinated blood

- It is nothing but the whole blood from which fibrin has been separated. The term defibrination refers to a method to prevent blood from clotting after it is withdrawn from the donor without the addition of any chemical additives to the blood.
- It is well known that it is necessary to take action either during or immediately after the collection of blood to prevent clotting of the blood. One such action is the addition of anticoagulant drugs or chemicals to the blood. While the addition of these additives is effective to prevent the blood from clotting, there are instances where blood containing these additives may not be used; i.e., in the preparation of blood agar plates used in microbiology. In such instances it is necessary to defibrinate the blood.

Procedure

- For defibrination, the beaters can be used to whip the blood and thereby remove the fibrin necessary for clotting. Sometimes a glass or other rigid material containers can be used to collect the blood. A sufficient amount of glass beads, marbles, metallic balls, or other formed elements can be placed in the rigid container.
- First draw the blood into the container and shake the container is continuously in a manner to agitate the glass beads or substitutes through the blood for a specified time sufficient to whip out the fibrin.
- In order to obtain the necessary whipping action of the beads within the blood, the container can only be about half full of blood. The reason for this is that the beads will move sluggishly through a container full of blood, but will move rapidly through a

container half full of blood. This movement of the beads is most important since the speed of defibrination is essential if clotting is to be prevented.

EXERCISE-2: ENUMERATION OF ERYTHROCYTES

Microscope

The microscope consists of the following parts

- Base: Supports the microscope.
- Handle: Curved and help to tilt the microscope at the hinged joint.
- Body tube: Attached to handle and can be raised or lowered by *rack and opinion arrangement*.
- Coarse and fine adjustment screws: To bring the object to focus. The body can be raised or lowered quickly by the coarse screw and then finer adjustments for sharp focusing is done with the fine adjuster. One pair of coarse and fine adjusters is located on either side of the handle. If one side is rotated, the other side screw also rotates. Generally, left hand is used for both coarse and fine adjustment screws and the right hand is used for manipulating the mechanical stage.
- Fixed stage: It is the platform with an aperture at its centre and the material to be viewed is kept over the stage on the aperture.
- Mechanical stage: Fixed on the stage and is calibrated. Materials can be held by clips provided with the mechanical stage, two screws move the stage forward and backward or transversely.
- Nosepiece: Attached to the lower end of the body tube, it is revolving and carries the objective lenses.
- Objective lenses: Three or four objectives attached with the nosepiece with magnifying power indicated on the lens –
    - 10x: Low power objective,
    - 40 or 45x: High power objective and
    - 100x: Oil immersion lens.
    - The position of the objective can be brought straight below the body tube to view the object by rotating the nosepiece and indicated by 'click' sound. When an object is viewed with one objective lens, and when viewing is changed to other objective lens, it requires only fine adjustment with the second objective lens, because of parafocal arrangement of lenses.
- Eyepiece: Fitted to the top of the body tube. Eyepiece of varying power is available, however, 10x is commonly used. The final magnification can be obtained by multiplication of objective power with ocular magnification, **e.g.** objective-100x and ocular-10x provides a final magnification of 1000x. Power of resolution is the ability to view finer details of objects and it is due to objective lenses (eyepiece only enlarges the images produced by objective lens).
- Illumination and mirror: A mirror is fitted below the condenser and it reflects light into the condenser. The mirror consists of one side plane mirror and the other side concave mirror. Plane mirror is used for distant light (natural light) and concave mirror is used when light source is nearer).
- Condenser: Located below the fixed stage and consists of two lenses mounted in a short cylinder. An iris diaphragm controls the amount of light entering the microscope. Light is converged into a cone and passes through the aperture and the condenser can be lowered or raised to adjust the intensity of light falling on the object.

- Oil immersion lens requires a drop of special oil to be placed between the objective and the slide. The condenser should be so adjusted as to bring it closer to the slide. Immediately after use of oil immersion lens, the oil has to be removed from the objective by cleaning the lens with special lens paper or xylol or spirit.

MATERIALS REQUIRED

- Hemocytometer with cover slip
- RBC pipette

- Diluting fluids – Hayem's fluid

HEMOCYTOMETER

- It is a rectangular thick glass slide with two raised crossbars, having an upper and lower counting chambers on which a cover slip can be placed. Since the surface of the counting chamber is 0.1mm below the cover slip, the depth of fluid in the chamber when it is filled is 1 x 1 x 0.1 = 0.1 mm^3.
- Counting chamber of the hemocytometer is a ruled area consisting of 9 primary squares; each square has dimension of 1x1mm, i.e. 1mm^2.
- The RBC count can be carried out in the central primary square; this is divided into 25 secondary squares each of that is further subdivided into 16 tertiary squares. The borders of the secondary squares of red cell counting area are marked by triple lines to facilitate RBC counting. Generally the red cells in 5 secondary squares (5 x 16=80 tertiary squares), which are at the four corners (Left top, right top, left bottom and right bottom) and the centre secondary square are counted.
- The four corner primary squares are used for WBC count and each corner primary square is subdivided into 16 secondary squares.

DILUTING PIPETTES

- The diluting pipette consists of a calibrated stem portion and a bulbar mixing chamber that contains a glass bead – red in colour in RBC pipette and white bead in WBC pipette, which facilitates uniform dispersion of the blood cells in the diluting fluid.
- The lower portion of the stem is divided by marks into two equal parts – 0.5 and 1, totaling one unit. On the upper portion of the bulb, the RBC pipette has 101 marking and WBC pipette has 11 marking. These calibrations represent comparable volumes of non-specific amount.
- The upper end of the pipette is attached by a rubber tube fitted with a glass or plastic mouthpiece.

DILUTING FLUID

- Hayem's Fluid (for red cells):

S. No.	Composition	Quantity
1	Mercuric chloride	0.25 g

2	Sodium sulphate	2.5 g
3	Sodium chloride	0.5 g
4	Distilled water	100 ml

PROCEDURE

- By gentle suction, blood is drawn into the lower stem portion of a dry diluting pipette upto the level of 0.5 mark. If blood is drawn beyond this mark, the excess may be removed by touching the tip of the pipette with tissue paper/wet squeezed cotton or tapping the tip on fingernails. Wipe the tip of the pipette with clean cotton to remove blood sticking outside of the pipette.
- Draw the diluting fluid into the pipette upto 101 mark. To avoid error in dilution, flow of fluid should be stopped exactly at the upper marking of 101. When blood is drawn upto 0.5 and diluted to 101 marking in the RBC pipette, the final dilution rate is 1:200.
- The pipette should be gently rotated for 1 to 2 minutes to bring about a uniform dilution of the blood cells in the diluting fluid. Discard few drops (up to half the volume of the bulb) of the contents of the pipette to remove cell free fluid from the lower stem.
- Thoroughly clean the hemocytometer and the cover glass before charging the diluted blood into the counting chamber.

Charging the chamber

- Focus the RBC counting chamber under low power (10 X) and then to high power (45 X).
- Place a cover slip over the counting chamber.
- Touch the tip of the pipette to the edge of the cover slip to allow the diluted blood to flow under the cover slip by capillary action.
- While filling the chamber, care should be taken to avoid spilling of fluid on the cover slip, fluid should not overflow into the side channels.
- Charging of the diluted blood into the counting chamber has to carried out both in the upper and lower counting chambers without air bubbles.

CALCULATION

- Total number of RBCs in 5 secondary squares = A
- Number of red cells in 1mm (area) = 5 x A
- Depth of the chamber = 0.1 mm
- Dilution factor = 200
- Number of erythrocytes = 5 A x 10 x 200
- Erythrocyte Count = Total No. of RBCs in 5 squares x depth x dilution x area = A x 10, 000 / ml or mm^3 of blood

PRECAUTIONS

- Start counting from the upper left square and include all the cells touching the central line in the bottom and left borders and ignore the cells touching the top and right borders of the respective squares.

- Move across the top row of squares from left to right and then drop down to second row and proceed from right to left.
- Repeat this pattern until all the squares are counted as required for WBC and RBC count.

SOURCE OF ERROR

- Improper mixing of blood.
- Uneven distribution of cells in the counting chamber.
- Cells touching the lines of all the four sides included in the count.
- Dust and stain particles counted as cells.
- Delay in counting especially for leukocytes (15% reduction in 24 hours).
- Clumping of cells.
- Overcharging the chamber.

NORMAL VALUES

Species	Erythrocyte count (10^6/ml)
Cattle	6.0- 8.0
Sheep	10.0-13.0
Goat	13.0-14.0
Dog	6.0- 8.0
Horse	5.5- 9.5
Pig	6.0- 8.0
Chicken	2.5- 3.2

EXERCISE-3: ENUMERATION OF LEUKOCYTES

Materials required

- Hemocytometer with cover slip
- WBC pipette
- Diluting fluids – Thomas fluid

ENUMERATION OF LEUKOCYTES - DILUTING FLUID

- **Thomas Fluid** (Turk's Fluid)

S. No.	Composition	Quantity
1	Glacial acetic acid	2.0ml
2	Gentian violet	0.35g (1 ml of 1% solution)
3	Distilled water	100ml

ENUMERATION OF LEUKOCYTES - PROCEDURE

- By gentle suction, blood is drawn into the lower stem portion of a dry diluting pipette upto the level of 0.5 mark. If blood is drawn beyond this mark, the excess may be removed by touching the tip of the pipette with tissue paper/wet squeezed cotton or tapping the tip on fingernails. Wipe the tip of the pipette with clean cotton to remove blood sticking outside of the pipette.
- Draw the diluting fluid into the pipette upto 11 mark. To avoid error in dilution, flow of fluid should be stopped exactly at the upper marking of 11. When blood is drawn upto 0.5 and diluted to 11 marking in the WBC pipette, the final dilution rate is 1:20.
- The pipette should be gently rotated for 1 to 2 minutes to bring about a uniform dilution of the blood cells in the diluting fluid. Discard few drops (up to half the volume of the bulb) of the contents of the pipette to remove cell free fluid from the lower stem.
- Thoroughly clean the hemocytometer and the cover glass before charging the diluted blood into the counting chamber.
- Charging the chamber
    - Focus the WBC counting chamber under low power (10 X) and then to high power (45 X). Place a cover slip over the counting chamber.
    - Touch the tip of the pipette to the edge of the cover slip to allow the diluted blood to flow under the cover slip by capillary action.
    - While filling the chamber, care should be taken to avoid spilling of fluid on the cover slip; fluid should not overflow into the side channels.
    - Charging of the diluted blood into the counting chamber has to carried out both in the upper and lower counting chambers without air bubbles.

ENUMERATION OF LEUKOCYTES - CALCULATION

- Total number of WBC in 4 primary squares = A
- Average number of WBC in one mm^2 = A/4
- Depth of the chamber = 0.1mm
- Dilution factor = 20

$$\frac{\text{Number of WBC}}{\text{mm}^3 \text{ or ml of blood}} = \frac{A}{4} \times 10 \times 20$$

Total number of WBC = A x 50 x (10^3 / ml)

ENUMERATION OF LEUKOCYTES - NORMAL VALUES

Species	Total leukocyte count (10^3/ ml)
Cattle	7-10
Sheep	7-10
Goat	8-12
Dog	9-13
Horse	8-11
Pig	15-22
Chicken	20-30

EXERCISE-4: DIFFERENTIAL LEUKOCYTE COUNT

Definition

It is the percentage distribution of different types of leukocytes that are determined after staining the leukocytes

Materials required

- Blood smeared glass slide
- Cell counter

PROCEDURE

Preparation of blood smear

- Take a clean grease free slide (washing the slides with detergents followed by washing in tap water and rinsing in distilled water then cleaning with muslin cloth).
- Place a very small drop of thoroughly mixed blood and deposit it near one end of the slide.
- Spread the blood as even and thin film by means of another slide. The spreading edge should be perfectly smooth. The spreader slide is brought near the blood drop; as it makes contact with the drop of blood, the blood will spread along the edge of the spreader slide. Keep the spreader slide at an angle of 30° to 45° and move it forward smoothly on the smear slide. The angle at which the spreader slide is held determines the thickness of the blood film; greater the angle thicker the film.
- Dry the film in air by vigorously shacking.
- A good blood smear should have a smooth and even appearance, free from holes and blood cells distributed in a single layer.

Staining

- Place the smear in staining rack.
- Cover the blood smear with Leishman's stain and allow 2 minutes.
- Add double the quantity of buffered distilled water at pH 6.6 - 6.8. (3.0 g disodium hydrogen phosphate and 5.47g potassium dihydrogen phosphate in 1 litre water).
- Gently blow air for uniform mixing, keep for 10 to 12 minutes.
- Drain the fluid from the slide, wash with buffered distilled water and air dry.
- Add one drop of immersion oil on the stained blood smear and focus under oil immersion objective (100 x) of the microscope.
- Make a tabular column or use D.C. cell counter and count 200 cells and express the number of each type of leukocyte in percentage.

- The film is examined systematically by moving three fields along the edge, two fields up, two fields along, two fields down starting at thin end of the smear (Battlement method).

Staining characteristics of a correctly stained normal film

Leukocytes	Nucleus	Granules
Neutrophils	Five lobed in matured cells, less than 5 in Banded/Juvenile cells	Fine purple
Eosinophils	Bilobed, connected by thin filament	Red-orange
Basophils	Irregular	Blue
Monocytes	Kidney/Bean shaped	
Lymphocytes	Rounded and large	

Precautions

- When blood is spread on a slide, there is a tendency for the neutrophils to settle on the edge of the film lymphocytes to remain in the body of the smear, while the monocytes and eosinophils tend to become evenly distributed throughout the smear.
- Thick blood film should not be used for cell counting, as it will cause variation in numbers
- Buffered water should be used for diluting the stain on the blood smear and for final washing of the stain

Calculation

- Absolute count:

$$\text{The absolute number of each leukocyte (per mm}^3\text{)} = \frac{\text{Number of each leukocyte \%}}{100} \times \text{Total leukocyte count}$$

NORMAL VALUES (%)

Species	Neutrophils	Lymphocytes	Monocytes	Eosinophils	Basophils
Cattle	25-30	60-65	5	2-5	<1
Sheep	25-30	60-65	5	2-5	<1
Goat	35-40	50-55	5	2-5	<1
Dog	65-70	20-25	5-6	2-5	<1
Horse	50-60	30-40	5-6	2-5	<1
Pig	30-35	55-60	5-6	2-5	<1
Chicken	25-30	55-60	10	3-8	1-4

EXERCISE-5: PLATELET COUNT

Materials required

- Hemocytometer with cover slip
- RBC pipette
- Petri dish
- Diluting fluid: Rees Ecker's fluid

Composition

- Sodium citrate: 3.8 g
- 40% formalin: 0.2 ml
- Brilliant crystal blue: 0.05 g
- Distilled water: 100 ml

PROCEDURE

- Collect a fresh sample of blood in EDTA in a clean dust free vial. (Counting should be carried out within 2 hours of collection).
- Draw the diluting fluid upto 1 mark in the RBC pipette and blow out the content just to moisten the pipette, which prevents sticking of blood to the glass wall.
- Draw blood upto 0.5 mark and dilute to 101 mark.
- Shake the pipette for 5 minutes.
- Expel the first few drops to remove the cell free fluid content from the pipette, charge both sides of the hemocytometer counting chamber.
- Keep the hemocytometer in a petri dish having moistened filter paper (to prevent drying).
- Allow the cells to settle down for 15 to 20 minutes.
- Focus under high power objective of the microscope and count the platelets in the entire 1 mm^2 of RBC counting area.

Calculation

$$\text{Platelet count} = \text{Number counted} \times \text{Area} \times \text{Depth} \times \text{Dilution} = A \times 1 \times 0.1 \times 200 = \frac{A \times 2000}{\text{ml of blood}}$$

MATERIALS REQUIRED

- Syringe and needle
- Constant temperature Water bath
- Test tube and capillary tube
- Stopwatch

PROCEDURE

Coagulation time

- Coagulation time is the time required for the blood to clot after it is drawn from the blood vessels.
  - Lee-White method:
    - Draw 5 ml of blood from a vein in a dry glass syringe.
    - At the same time start a stopwatch.
    - Transfer 1 ml blood from the syringe into each of 3 clean test tubes (10mm).
    - Keep the tubes in a constant temperature water bath maintained at 37°C.
    - Tilt the tubes in rotation every minute until one tube can be tilted to an angle greater than 90° without spilling the blood, i.e. no blood flows when the tube is inverted.
    - The remaining tubes are examined at half-minute intervals. Record the time of clot formation in each tube. Coagulation time is taken as the average time of the clotting times of all the four tubes.
    - Normal coagulation time is 5 to 11 minutes.
  - Capillary method:
    - Prick the finger or the tip of the ear of an animal with a lancet.
    - Apply the tip of a capillary tube to the drop of blood at an angle and allow filling ¾ of its length by capillary suction. Start a stopwatch immediately when the blood has entered the capillary tube.
    - Tilt the tubes at intervals of 30 seconds till blood flow ceases.
    - Short sections of the tube are then broken off at intervals of 30 seconds until fibrin thread forms between the main tube and broken fragment.
    - Note the time when the broken capillary tube hangs from the main capillary tube. It is the coagulation time.

Bleeding time

- Bleeding time is the time required to arrest bleeding after the blood has come out of skin puncture.
  - Duke's method:
    - Make a deep puncture at the fingertip or at tip of the ear of animal.
    - Start a stopwatch as soon as the blood comes out.
    - Blot the blood every 30 seconds with a filter paper without squeezing the finger. Note the size of the blood spot on the filter paper, which goes on reduced and eventually disappears.
    - Stop the watch when there is no trace of blood in the filter paper i.e. bleeding has stopped.
    - Express the duration of time in minutes as bleeding time.

NORMAL VALUES

Species	Coagulation time in minutes
Sheep, cow and horse	4-15
Dog	3-13
Cat	8

EXERCISE-6: HEMOGLOBIN ESTIMATION

Principle

- Hemoglobin can be estimated by
  - Cyanmethemoglobin method and
  - Oxyhemoglobin method colorimetrically or
  - By converting the hemoglobin to acid/ alkaline hematin by the addition of acid/ alkali and comparing the color of hematin with a standard.

MATERIALS REQUIRED

- Sahli's hemoglobino meter

- It consists of a graduated tube and a fixed, color standard.
- Hemoglobinometer tube has two divisions:
  - One giving the concentration of hemoglobin in g/dl and
  - The other giving the percent hemoglobin i.e. 100% = 14.8g hemoglobin.
- 0.1 N HCl
- 20 microlitre hemoglobin pipette

PROCEDURE

- Place 0.1 N HCl up to the 20 marking in an empty hemometer tube.

- Draw blood up to exactly 20 mark in the hemoglobin pipette, wipe the tip of the pipette with cotton to remove blood sticking to sides.

- Directly blow the blood into the HCl in the hemometer tube, rinse the pipette with the same HCl for thorough washing.

- Hemoglobin in the blood is converted into a brown colored acid hematin.
- Stir well and allow it to stand for 10 minutes.
- Add distilled water drop by drop, each time stirring well.

- Continue this procedure until the color in the hemometer tube matches the standard.
- Note the height of the column of the diluted acid hematin in the hemometer tube and it is the hemoglobin concentration in g/dl. Approximately hemoglobin in g% is 1/3 of PCV.

Cyanmethemoglobin method:

- Mix 0.02 ml blood with 5 ml Drabkin's solution.
- Allow it to stand for 5 minutes.
- The color developed is read in 540 nm in a colorimeter or using a hemoglobinometer, which gives hemoglobin concentration in g/dl directly. (While drawing Drabkin's solution, utmost care should be taken to avoid its contact with skin or mouth because it contains cyanide).

ERYTHROCYTE INDICES

Mean Corpuscular Volume(MCV):

- It indicates the average volume of a single erythrocyte.

$$\text{MCV (fl)} = \frac{\text{PCV x 10}}{\text{RBC number in million per ml}}$$

Mean Corpuscular Haemoglobin (MCH):

- This expresses the average weight of hemoglobin in a single RBC.

$$\text{MCH (pg)} = \frac{\text{Hemoglobin in g\% x 10}}{\text{RBC number in million per ml}}$$

Mean Corpuscular Haemoglobin Concentration (MCHC):

- This measures the ratio of weight of hemoglobin to volume of erythrocytes i.e. the percentage of hemoglobin in 100ml of packed RBCs.

$$\text{MCHC (g/100 ml) or \%} = \frac{\text{Hemoglobin in (g\%) x 100}}{\text{PCV}}$$

NORMAL VALUES

Species	Hb (g/dl)	MCV (fl)	MCH (pg)	MCHC (%)
Cattle	8-15	42-60	11-17	30-36
Sheep	8-16	23-48	9-12	31-38

Goat	8-14	15-30	10-12.6	35-42
Dog	12-18	60-77	19-25	32-36
Horse	8-14	34-58	12-18	31-37

EXERCISE-7: PACKED CELL VOLUME (HAEMATOCRIT)

Definition

- Packed cell volume (PCV) is the percentage of blood cells in 100 ml blood.

Materials required

Wintrobe hematocrit tube with Wintrobe stand	Wintrobe pipette (or Pasteur pipette) with long narrow delivery tube

PROCEDURE

- The Wintrobe hematocrit tube has 3mm bore and 10cms long, which are subdivided into millimeters. The calibration on one side starts with 'zero' from the bottom to 10 cm at the top; this is used for PCV estimation. On the other side, calibration starts from '0' at top to 10 at bottom and it is used for ESR estimation.
- Mix the anticoagulant added blood thoroughly.
- Draw the blood carefully in the Wintrobe pipette.
- Deliver the blood by inserting the tip of the delivery tube into the Wintrobe tube so that it touches the bottom.
- Discharge the blood carefully and slowly by gradually raising the tip of the pipette and fill the Wintrobe tube upto the mark "0" without any air bubbles.
- Centrifuge the tube at 3000 rpm for 30 minutes.
- Take the reading and centrifuge again for 5 minutes or till you gets two identical consecutive values.
- The red cells are packed at the bottom and the plasma is at the top, the leukocytes are located in between cells and plasma (buffy coat).
- The buffy coat layer is not included while taking PCV reading.
- The result is expressed as percentage of packed cells.
-

Microhematocrit

- PCV can also be determined by microhematocrit method in which capillary tubes are used.
    - Draw blood upto 2/3 level into the capillary tube and seal one end.
    - Centrifuge at 12, 000 rpm for 5 minutes (keep the closed end outwards).
    - Read the capillary tube using the microhematocrit reader provided.

NORMAL PCV IN VARIOUS SPECIES

Species	PCV (%)
Cattle	24-46
Sheep	24-50
Goat	19-38
Dog	37-55
Horse	24-44

ERYTHROCYTE SEDIMENTATION RATE (ESR)

Definition

- ESR is the millimeters of clear plasma obtained over a vertical column of blood after a standard time, usually one-hour.

Materials required

- Wintrobe hematocrit tube or Westergren's pipette with stand (Westergren's pipette is 300 x 3m long tube, graduated from "0" to "200"mm at one mm intervals)
- Wintrobe pipette
- Wintrobe stand

PROCEDURE

- Fill the Wintrobe tube as detailed in PCV experiment or the Westergren's pipette upto mark "0".
- Keep the tube vertically in the Wintrobe stand; even a slight slant enhances sedimentation.
- The Westergren's pipette is kept tightly in its stand without any leakage of blood.
- Measure the fall of erythrocyte column at 20 minute intervals for 60 minutes.
- In wintrobe tube, the calibration starting from "0" at top is used for ESR determination.
- ESR is expressed in mm/hour.

PRECAUTIONS

- Excess oxalate concentration (> 2mg/ml blood) may slow ESR.
- Hematocrit tube should be free from dust, since dust will influence the rate of sedimentation.
- Keep the tube absolutely vertical. For ruminant's blood, the tubes can be held at 45°angle to produce maximum sedimentation of RBCs.

- Any air bubbles in the hematocrit tube, while filling, will influence the rate of sedimentation and hence should be avoided.
- Conduct the test immediately after collection at room temperature since higher temperature enhances settling of RBCs.

NORMAL ESR VALUES

Species	ESR in mm
Cattle	0-3 / 8 hrs
Goat	2-3 / 24 hrs
Dog	6-10 / hr
Horse	15-38/ 20 min
Pig	0-6 / 30 min & 1-14 /hr
Cat	7-27 / Hr

EXERCISE-8: ERYTHROCYTE FRAGILITY

Introduction

- When the red cells are suspended in a buffered hypotonic saline, they swell up and finally rupture releasing their intracellular contents –known as *hemolysis*.
- The intensity of hemolysis is proportional to the degree of hypotonicity of the solution.

Materials required

- Test tube and rack
- Graduated pipette: 5 ml
- 10% NaCl buffered stock solution

Contents	Quantity
NaCl	180 g
Na_2HPO_4	27.31 g
NaH_2PO_4	3.74 g
Made it to 2 litres	

- Dilute 5 ml of the stock solution with DW to make 50 ml (1% solution).

PROCEDURE

- Place two racks of 16 serially numbered test tubes in a test tube rack.
- Prepare different concentrations of NaCl solution in these tubes by mixing 1% NaCl buffer with distilled water as follows.

Tube No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
NaCl	4.25	4.0	3.75	3.5	3.25	3.0	2.75	2.5	2.25	2.0	1.75	1.5	1.25	1.0	0.5	0.0
DW	0.75	1.0	1.25	1.5	1.75	2.0	2.25	2.5	2.75	3.0	3.25	3.5	3.75	4.0	4.5	5.0
Final Conc. of NaCl (%)	0.85	0.8	0.75	0.7	0.65	0.6	0.55	0.5	0.45	0.4	0.35	0.3	0.25	0.2	0.1	0.0

- Add 20 m l of blood to each tube using the haemoglobin pipette.
- Mix gently, let it stand at room temperature for 30 minutes.
- Centrifuge all the tubes for 10 minutes.
- Read the extent of haemolysis in each tube by visual observation.
- If there is no haemolysis, the red cells will be seen packed at the bottom of the tube and the saline is clear.
- If some haemolysis has occurred, the saline will be tinged red.
- The colour of the saline will be increasing deeper with decreasing concentration of NaCl (decreasing tonicity).
- In complete hemolysis, saline is uniformly colored and there are no red cells at the bottom.
- Beginning of hemolysis is the point of minimum resistance of red cells at that percent hypotonicity.
- Complete hemolysis indicates point of maximum resistance of red cells at that percent hypotonicity.
- Normally, hemolysis begins at 0.45% NaCl and is complete at 0.3%.

Calculation

- To determine the percentage hemolysis in each tube, 3.5ml of the supernatant fluid from the centrifuged tube are transferred to empty tubes.
- Read the optical density of each sample in a photoelectric colorimeter at 540nm using DW as blank.
- Content of the 16th tube indicates 100% hemolysis.
- Convert the observed O. D. to percent hemolysis as

$$\text{Percent hemolysis} = \frac{\text{O.D. of unknown}}{\text{O.D. of tube 16}} \times 100$$

DETERMINATION OF VISCOSITY IN BLOOD

Introduction

- The internal resistance offered by the fluid to the flow or is the thickness of the fluid.
- The viscosity of blood is due to cells and that of plasma is due to plasma proteins.

PRINCIPLE

- The viscosity of a liquid is measured by noting the rate of flow of the liquid through a capillary tube.

MATERIALS REQUIRED

- Viscosimeter
- Uncoagulated blood sample
- Distilled water
- Stop watch

PROCEDURE

- Attach a rubber tube with a pinch cock to the viscosimeter and suck distilled water up to the mark above the bulb .Close the pinch cock.
- Fix the viscosimeter vertically. Wipe the tip. Release the pinch cock and simultaneously start the stop watch. Stop the watch when the meniscus reaches the mark below the bulb. Note the time taken for the water to flow out.
- Dry the viscosimeter and repeat the same with blood. Note the time for the blood.

Viscosity Ratio = Time ratio x Density Ratio

- Taking Viscosity of water as unit.
    - The relative viscosities of water, plasma and whole blood are 1, 1.8 and 4.7 respectively.

$$\text{Viscosity of blood} = \frac{\text{Blood time}}{\text{Water time}} \times \text{specific gravity (1.06)}$$

EXERCISE-9: DETERMINATION OF BLOOD GROUPS

Introduction

- The human red cells contain antigen (agglutinogen) and the serum contains antibodies (agglutinin).
- On the basis of the antigen content, the human blood is divided into A, B, AB, and O blood groups.
- Agglutination occurs when the blood cells of one group is mixed with the blood of another group due to the reaction of antigen with its specific antibodies.

Materials required

- Glass slides or porcelain slab
- Applicator sticks
- A, B, and O and Rh antisera
- Microscope or magnifying glasses

PROCEDURE

- Keep in the glass slide or porcelain slab a drop of antisera A, B, O and Rh (each drop should be kept separately).
- Make a finger prick.
- Add a small drop of blood near each of the antisera.
- Mix the blood with antisera using the applicator stick (for each antisera, (a separate stick should be used).
- Observe for hemoagglutination (clumping) reaction and based on the clumping, blood groups are identified as follows.
- If clumping is not observed within two minutes, it is considered negative.

EXERCISE-10: RECORDING OF ECG

Introduction

- The ECG is the record of electrical events occurring in a functional heart.
- The ECG records the electrical activity of the heart from different angles and gives a complete picture.
- The depolarisation and repolarisation of the cardiac muscle spreads upto the surface of the body and by keeping electrodes, these electrical activities can be measured.

Materials required

- ECG machine complete with electrodes, lead selector, electrolyte jelly. Electrocardiographic paper is a graph sheet in which horizontal and vertical lines are marked at 1mm intervals and every fifth line is thicker. Time is measured along the horizontal line (1mm=0.04 seconds and 5mm =0.2 seconds). Voltage is measures along the vertical line. Generally, 1mm height represents 1mV. Recording speed can be either 25 or 50mm/second (normally, 25mm/sec. is selected)

- A pair of electrodes kept at a specific point on the surface of the skin, along with its wire connection to the electrocardiographic machine is called as *lead*. One electrode is always connected to right leg, which acts as indifferent electrode and connects the subject to earth.

COMMON LEAD SYSTEMS IN ANIMALS

- Bipolar Standard Leads
    - Lead I is connection between Right Arm and Left Arm
    - Lead II – Right Arm and Left Leg
    - Lead III – Left Leg and Left Arm
- Augmented Unipolar Limb Leads
    - These leads are designated as lead aVR, aVL and aVF. Amplitude of recorded waves is greatly increased in these leads.
        1. Lead aVR compares potential difference between Right Arm and Left Arm+Left Leg
        2. Lead aVL compares Left Arm with Right Arm + Left Leg
        3. Lead aVF compares Left Leg with Right Arm+ Left Arm

- Precordial (chest) leads
  - These leads record the potential from the anterior surface of the heart, from right side to left side. A central chest lead is placed at different locations on the chest and these leads are designated as V leads.
  - The common positions of V leads are
    - V1 – 4th intercostal space at the right sternal border
    - V2 – 4th intercostal space at the left sternal border
    - V3 – equidistant between V1 and V2
    - V4 – 5th - intercostal space in the left clavicular line
  - Leads V5 to V9 are taken in the same horizontal plane as V4
    - V5 – anterior axillary line
    - V6 – midaxillary line
    - V7 – posterior axillary line
    - V8 – posterior scapular line
    - V9 – left border of spine.

PROCEDURE

- Make the subject to lie on a flat table, comfortably and fully relaxed. Remove all metal objects from the subject.
- Connect all the electrodes as per the marking given in the electrode securely by strapping it.
- Before keeping the electrode on the point, apply electrolyte jelly (to reduce skin resistance).
- The electrodes are usually placed just above the wrist and ankles. In dogs, the animal is placed on a non-conductive surface in right lateral recumbent position.
- Keep all the four legs apart in such a way that the legs should not touch each other.
- Apply the jelly on the electrodes and on the skin over point of application of the electrodes and attach the electrode clips to the skin.
- Alligator clips or flat contact electrodes with bands are used to attach the leads. Connect the machine to mains, select appropriate speed of the chart paper and centre the writing point.



- Adjust the sensitivity calibration control in so that 1mV can produce 1mm deflection.
- Put the lead selector knob of the ECG machine to standard leads and obtains leads I, II and III.
- Put the lead selector to unipolar leads and obtain graph.
- Now using the V leads and putting the lead selector to 'V' position take lead V1 to V6.

- Mark the leads on the record.
  Carefully study the different ECG waves, time intervals, voltage of the waves, polarity of the waves and also calculate heart rate.
- As per Einthovan's law, the algebraic sum of all three leads (I+II+III) is equal to 'o' i.e. sum of electrical potential of lead I+III = lead II.

EXERCISE-11: MEASUREMENT OF BLOOD PRESSURE

- **Materials required**
- Sphygmomanometer consists of an inflatable bag, a cuff, manometer (mercury or aneroid type) and an inflating bulb equipped with a deflating valve. (for dogs, modified inflatable bag can be used)

PROCEDURE

Auscultatory method

- In man, the cuff is strapped around the arm on a level with the heart (to exclude hydrostatic effect).
- In case of dogs, put the dog on a table without exciting, anaesthetise if necessary and apply the cuff over the femoral artery.
- Place a stethoscope over the brachial artery at the cubital fossa in man or over the femoral artery in dogs.
- By pressing the inflating bulb, inflate the cuff to create pressure (approximately 180 mmHg) necessary to obliterate the pulse or arrest the blood flow below the level of the cuff.
- When the artery is occluded, no sound is heard in the stethoscope. The cuff is slowly deflated by releasing exhaust - (deflating) valve.
- The blood flow and pulse reappears below the cuff, which can be determined by auscultation.
- Pressure at the appearance of the first "tapping" sound (known as korotkoff sounds) indicates systolic pressure.
- Continue the deflation; the sound become louder, then dull, muffled and finally disappears. Note the pressure at "muffling" or disappearance of sound, and it is the diastolic pressure.

Palpatory method

- Tie the cuff around the arm at the level of heart.
- Inflate the cuff pressure till the pulse disappears in the radial artery at wrist.
- Release the cuff pressure slowly.
- When the pulse reappears, note the pressure. It is the systolic pressure.
- Diastolic pressure cannot be determined.

EXERCISE-12: STUDY OF EFFECT OF HEAT AND COLD ON HEART

Recording of cardiogram of frog heart

Materials required

- Starling recording drum,
- Myograph stand,
- Frog board
- Dissection board,
- Straw lever holder and
- Straw lever.

PITHING OF FROG

- Pith a frog and place it on its ventral side up on a dissection board. Make a median incision through the skin over the sternum. Raise the cartilaginous xiphisternum and immediately behind it make a transverse cut into the abdominal wall. Take care not to cut the anterior abdominal vein.
- Insert the blade of large scissors closely under the pectoral girdle. By holding the blade closely in contact with the bone to prevent injury to the underlying structures continue the V-shaped cut anteriorly through either sides of the pectoral girdle to the end of xiphisternum. Pull the forelimbs apart, lift off the contents of thoracic cavity and fix them. Observe the heart is beating inside the pericardium. Slit open the pericardium and expose the heart.
- Note the position of bulbus arteriosus, which is anterior to ventricle, filled by each ventricular contraction. Beneath this is the auricle. Between auricle and ventricle is the auriculo-ventricular groove.
- Moisten the fingers while handling the heart during the course of experiment. Gently lift up the ventricle and observe the sinus venosus marked off from right auricle by a white crescentric line. The sinus venosus may seem to be formed by the confluence of the inferior vena cava and superior vena cava.

PROCEDURE

- Mount the frog on the frog board. Fit the Starling heart lever on the vertical rod of the stand above the heart.
- Bend a small pin into a hook and tie one end of a piece of strong thread to it. Fix the hook at the apex of the heart.
- Gently lift the heart up. Now note the contraction and relaxation of the heart.
- During contraction the lever is pulled down whereas during diastole the lever is pulled back to its normal position. Record the activity of the heart by moving the drum at slow speed (1.5mm/ sec).

EFFECT OF HEAT AND COLD ON HEART

- Slit open the pericardium and expose the heart.

- Note the position of bulbus arteriosus, which is anterior to ventricle, filled by each ventricular contraction.
- Beneath this is the auricle. Between auricle and ventricle is the auriculo-ventricular groove.
- Bend a small pin into a hook and tie one end of a piece of strong thread to it. Fix the hook at the apex of the heart.
- Moisten the fingers while handling the heart during the course of experiment.
- Gently lift up the ventricle and observe the sinus venosus marked off from right auricle by a white crescentric line.
- The sinus venosus may seem to be formed by the confluence of the inferior vena cava and superior vena cava.
- Mount the frog on the frog board.
- Tie the free end of the thread to the straw lever.
- Set the drum in slow speed (12.5 mm/sec) and record few contractions.
- Pour Ringer's solution at 30°C drop by drop over the heart and record its effect.
- Repeat the same with Ringer's at a temperature of 38-40°C and at 5°C. Observe the effects of heat and cold on cardiac activity.

EFFECT OF HEAT AND COLD ON HEART

Effect of vagal stimulation and vagal escape (on heart)

Materials required

As in previous experiment

- Induction coil,
- Short circuiting key,
- Vagal electrode

PROCEDURE

- Expose the heart as per the previous experiment.
- Place the frog on the dissection board with its dorsal side up and remove the skin over the scapular region.
- Hold the cartilage of the scapula with forceps and carefully remove all attachments and disarticulate the forelimb without damaging the underlying cranial nerves.

The following structures are exposed:

- The thin ptero-hyoid muscle running from the base of the skull to the hyoid bone.
- The glossopharyngeal nerve run forward crossing the ptero-hyoid muscle.
- Along the lower border of the ptero hyoid muscle running from above downward are the laryngeal nerve, the carotid artery and the vagus.
    - Mount the frog on the frog board.
    - Insert the vagal electrode in such a way to place the ptero-hyoid muscle and the vagus and other structures in the special groove of the vagal electrode.
    - Fix the vagal electrode to the board.

- Arrange the electrical connection to give 100 stimuli per second by Neef's hammer and adjust the distance of secondary coil carefully to obtain threshold stimuli.
- During stimulation initially there will be a slight cardiac acceleration followed by inhibition of the heartbeat.
- Observe the ventricular muscle escapes from the inhibiting effect of vagus if the stimulation is continued.

EXERCISE-13: PHYSIOLOGICAL CONSTITUENTS OF URINE

Objective

- To determine physiological constituents of urine and urine analysis.

Principle

- Urinalysis provides information about all the tissue of the body and also assess the functional of the kidneys.

EXERCISE-13: PHYSIOLOGICAL CONSTITUENTS OF URINE

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Principle

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COLLECTION OF URINE

- Urine is collected is a sterile container by means of catheterization.

PHYSICAL EXAMINATION OF URINE

- Urine is examined of volume, color, transparency and specific gravity.

URINE VOLUME

- The amount of urine voided per day depends on species, diet, fluid intake, climate, exercise etc. Urine is inversely related to specific gravity i.e., an increased volume and low specific gravity.

Interpretation

- Increased volume is known as polyuria. It occurs in increased water consumption, diuretics, diabetes, chronic renal failure, hyperadrenocorticism etc.
- Decreased volume is kown as oliguria. It occurs in decreased water consumption, high environmental temperature, dehydration, acute renal failure, reduced BP and obstruction of urinary passage.

Volume of urine voided in various species

Horse	3-18 ml/kg b. wt/day
Cow	17-45 ml/kg b. wt/day
Sheep and Goat	10-40 ml/kg b. wt/day
Pig	5-30 ml/kg b. wt/day
Dog	20-100 ml/kg b. wt/day
Man	8.6-28.6 ml/kg b. wt/day

COLOR

- Record the color of urine of observing it in a test tube. Normal color of urine in pale yellow.

Abnormal colors

- *Red:* Blood in urine seen in haematuria.
- *Dark yellow to brown:* Due to bilirubin.
- *Reddish brown* in haemoglobinuria. Some drugs also discolor the urine.

TRANSPARENCY

- Record the transparency of urine by observing it in a test tube.

Interpretation

- Clear Urine: In all animals freshly voided urine is clear except in horse it is thick and cloudy due to calcium carbonate crystals and mucus.
- Cloudy urine: It may occur due to the presence of blood, leukocytes bacteria, epithelial cells, mucus, crystals etc

SPECIFIC GRAVITY - URINE

Determination

- Specific gravity of urine is determined using urinometer.

Interpretation

- Normal specific gravity of urine in various species:

Horse	1.040
Cow	1.032
Sheep and goat	1.030
Pig	1.012
Dog	1.025
Man	1.020

- The normal kidney can dilute urine to a specific gravity of 1.001 and concentrate urine to specific gravity of 1.060. Various factors that affect specific gravity are diet, fluid intake, climate etc.

REACTION OF URINE

- The p^H of urine is determined using p^H paper. Dip the paper strip into urine several times and match the color of the strip with the chart 1 minute after wetting with urine.

Interpretation

- The p^H varies in different species depending upon the diet and metabolism.

Horse	Alkaline
Cattle	
Sheep	
Dog, Cat, Man	Acidic

EXERCISE-14: PATHOLOGICAL CONSTITUENTS OF URINE

Objective

- To identify any pathological conditions in the body. Commonly analysed pathological constituents of urine are protein, reducing sugars, ketone bodies, blood, bile salts and bile pigments.

HELLERS NITRIC ACID TEST

Objective

- To test protein in the urine sample.

Materials required

- Urine sample.
- Concentrated Nitric acid.

Procedure

- Take 2 ml of urine in a test tube.
- Add 2 ml of Concentrated Nitric acid along the sides of the test tube.
- Appearance of a white ring at the junction of two liquids indicates the presence of protein.

View animation

BIURET TEST

Objective

- To test protein in the urine sample.

Materials Required

- Urine sample.
- 5% sodium hydroxide.
- 1% Copper sulphate.

Procedure

- Take 2 ml of urine in a test tube.
- Add 1 ml of 5% sodium hydroxide and 1-2 drops of 1% copper sulphate.
- Appearance of a violet color is positive for the presence of peptide bonds in proteins.

View animation

Interpretation

- Renal proteinuria occurs is nephritis, chemical poisoining, polycystic kidneys etc.
- Accidental proteinuria occurs in cystitis, urethritis, urolithiasis etc
- Albuminuria occurs in excess consumption of proteins, emotional stress, excess muscle exertion etc.

BENEDICT'S TEST

Objective

- To detect glucose (reducing sugars) in urine.

Principle

- Benedict's reaction is based on the reduction of alkaline copper sulphate by glucose present in the urine.

Materials required

- Urine sample
- Benedict's qualitative reagent

Procedure

- Take 2.5 ml of Benedict's qualitative reagent in a test tube.
- Add four drops of urine.
- Boil for two minutes and cool.

View animation

Read it as follows

- No change in color: Negative
- Green color (+): Positive (0 to 0.5% sugar)
- Yellow color (++): Positive (0.5 to 1% sugar)
- Orange color (+++): (1-2% sugar)
- Red color (++++): (Above 2% sugar)

Interpretation

- Glycosuria occurs in Diabetes mellitus, hyperpituitarism, hyperthyroidism, hyperadrenocortism, acute or chronic pancreatitis etc.
- False positive reaction may also occur during administration of drugs such as antibiotics (streptomycin), ascorbic acid, salicylates etc.

ROTHERA'S TEST

Objective

- To detect ketone bodies (acetone/aceto acetic acid) in the urine.

Principle

- Acetoacetic acid reacts with Sodium nitroprusside to give a violet color.

Materials required

- Urine sample
- Ammonium sulphate
- 5% Sodium nitroprusside
- Ammonium hydroxide or liquor ammonia

Procedure

- Saturate 5 ml of urine with crystalline ammonium sulphate.
- Add a few drops of 5% sodium nitroprusside.
- Add 2 ml of concentrated ammonium hydroxide or liquor ammonia along the walls of the test tube.
- Appearance of violet color is positive for ketone bodies.

View animation

Interpretation

- Increased concentrations of acetone, aceto acetic acid and beta hydroxyl butyric acid in the blood and urine is known as ketosis.
- Ketosis occurs in diabetic mellitus, high fat diet, starvation, milk fever, impaired liver function, certain endocrine disorders etc.

BENZIDINE TEST

Objective

- To detect blood in the urine.

Principle

- Hemoglobin in the blood vessels reacts with hydrogen peroxide to liberate oxygen which oxidizes toluidine (benzidine reagent) to give a blue color.

Materials required

- Urine sample
- Benzidine reagent
- 3% Hydrogen peroxide

Procedure

- Take 2 ml of urine in a test tube.
- Add 1-2 drops of benzidine reagent and 2 ml of 3% H_2O_2.

- Blue color indicates blood present in urine.

View animation

Interpretation

- Differentiation between haematuria and haemoglobinuria has to be done.
- Centrifuge the urine and examine the sediment under microscope.
- If RBCs are intact it is haematuria, otherwise haemoglobinuria. Haematuria occurs in acute nephritis, neoplasm of kidney, canine hepatitis, chemical poisoning, parasitic infections such as Dirofilariasis in dogs.
- Haemogloburia occurs in leptospirosis, babesisois, uncompatible blood transfusion etc.

HAY'S TEST

Objective

- To detect bile salts in urine.

Principle

- It is based on the property of reducing the surface tension of liquids by bile acids.

Materials Required

- Urine sample
- Sulphur powder

Procedure

- Take 5 ml of urine in a test tube.
- Sprinkle sulphur powder and observe the tube without shaking it.
- It sulphur sinks to the bottom of the tube it is positive for presence of bile salts.

FOUCHET'S TEST

Objective

- To detect bile pigments in urine.

Materials required

- Urine sample
- 10% barium chloride solution
- Filter paper

- Fouchet's reagent

Procedure

- Take 5 ml of urine in the test tube.
- Add equal volume of 10% barium chloride solution.
- Filter the contents.
- Spread the filter paper and dry the precipitate.
- Add 2 drops of Fouchet's reagent.
- Appearance of a green or bluish green color is positive for bile pigments.

View animation

Interpretation

- Bilirubinuria occurs in liver disorders, obstruction of bile duct etc.

EXERCISE-15: DETERMINATION OF GLOMERULAR FILTRATION RATE (GFR)

Inulin clearance - Procedure

- Inulin is administered intra-venously to maintain a constant plasma level.
- Initially, 30 ml of 10% inulin solution is diluted with 250 ml isotonic saline is administered @ 20 ml/min.
- Then 70 ml of 10% inulin solution diluted with 500 ml of isotonic saline is administered @ 4 ml/min.
- The patient is then catheterized.
- The bladder is evacuated every 15 min for one hour and urine and blood samples are collected.
- Inulin clearance is calculated over each 15 min period and their mean is obtained.
- GFR is calculated by the formula

$$GFR = \frac{U_x \times V}{P_x}$$

INULIN CLEARANCE - INTERPRETATION

- In healthy adults, inulin clearance is 130 ml/min for a body surface area of 1.73 sq.m.
- A decrease indicates impairment of glomerular function.

CREATININE CLEARANCE - PROCEDURE

1. Creatinine clearance is generally measured over a 24 hr period.
2. 24 hr urine is collected.
3. Creatinine is estimated in the urine and in the blood during the test period.

Calculation

$$GFR = \frac{U_x \times V_2}{P_x}$$

CREATININE CLEARANCE - INTERPRETATION

- Creatinine clearance in normal adult males = 100-120 ml/min.
- Females = 95 – 105 ml/min.
- It decreases if the glomerular function is impaired.

EXERCISE-16: ESTIMATION OF TITRABLE ACIDITY IN URINE

Objective

- To measure titrable acidity of urine.

Principle

- Hydrogen ions present in the urine is titrated against sodium hydroxide to find out acidity.
- Potassium oxalate added to the urine precipitated calcium, as otherwise interfaces with the end point.

ESTIMATION OF TITRABLE ACIDITY IN URINE

Materials required

- Urine sample
- 0.1 N Sodium hydroxide solution
- Potassium oxalate reagent powder
- Phenolphthalein indicator 1%
- Burette
- Conical flasks
- Pipettes

ESTIMATION OF TITRABLE ACIDITY IN URINE

Procedure

- Take 10 ml of urine in a conical flask.
- Add 2 g of potassium oxalate and mix well.
- Add 2 drops of phenolphthalein indicator.
- Take 0.1 N Sodium hydroxide solution in the burette.
- Titrate the acid present in the urine with 0.1 N Sodium hydroxide to a faint pink color.

- The amount of sodium hydroxide required to reach the neutral print is referred to as titrable acidity.

ESTIMATION OF TITRABLE ACIDITY IN URINE

Calculation

- Let the titre value be x ml.

$$\text{Titrable acidity} = \frac{(X \times 10)\text{ ml of } 0.1N \text{ of NaoH}}{100 \text{ ml of urine}}$$

ESTIMATION OF TITRABLE ACIDITY IN URINE - INTERPRETATION

Interpretation

- Normal titrable acidity is between 200-500 ml of 0.1 N NaOH in 24 hrs.
- Acidity is increased in Metabolic acidosis & decreased in Metabolic alkalosis.

ESTIMATION OF URINE AMMONIA

Objective

- To estimate ammonia in the urine.

Principle

- Neutral ammonium salts and aminoacids in the urine react with excess of neutral formaldehyde to give hexamethylene tetramine (hexamine) and liberate equivalent quantity of acids.

$$4NH_4Cl + 6HCHO \text{ ---------> } N_4(CH_2)_6 + 6H_2O + 4HCl$$

- The liberated acid is titrated against 0.1N NaOH using phenolphthalein as indicator. The titre value is proportional to the content of ammonia is urine.

ESTIMATION OF URINE AMMONIA

Procedure

- Take 10 ml of urine in a conical flask.
- Add 5 ml of Neutral formaldehyde.
- Titrate the contents against 0.1 N NaOH till a pale permanent pink color is obtained.

ESTIMATION OF URINE AMMONIA

Calculation

- Let the titre value be 'y' ml.
- 1 ml of 0.1 NaOH = 1.7 mg of Ammonia.
- Therefore, ammonia content in 100 ml of urine = (y x 10 x 1.7) mg.

ESTIMATION OF URINE AMMONIA

Interpretation

- Normal daily excretion of ammonia = 0.7 g on an average diet.
- Ammonia excretion in increased in metabolic acidosis.

EXERCISE-17: ESTIMATION OF INORGANIC PHOSPHORUS

Objective

- To determine inorganic phosphorous is urine

ESTIMATION OF INORGANIC PHOSPHORUS

Materials required

- Urine sample
- Ammonia solution
- Concentration Nitric acid (HNO_3)
- 2.5% ammonium molybdate
- Acetic acid

ESTIMATION OF INORGANIC PHOSPHORUS

Procedure

- Take 10 ml of urine in a conical flask and add a few drops of ammonia. Boil the content and cool.
- A flaky precipitate of calcium and phosphate is formed. It is filtered and the filtrate is discarded. Dissolve the precipitate in 5 ml of hot acetic acid. Add a few drops of Concentrated HNO_3 and 2 ml of 2.5% ammonium molybdate solution warm the solution gently.
- A canary yellow precipitate of ammonium phosphomolybdate is formed.

ESTIMATION OF CREATININE

Objective

- To determine creatinine in the urine.

Principle

- Creatinine reacts with alkaline picrate to form orange red color of creatinine picrate.

ESTIMATION OF CREATININE

Materials required

- Urine sample
- 5% NaOH
- 1% Picric acid

ESTIMATION OF CREATININE

Procedure

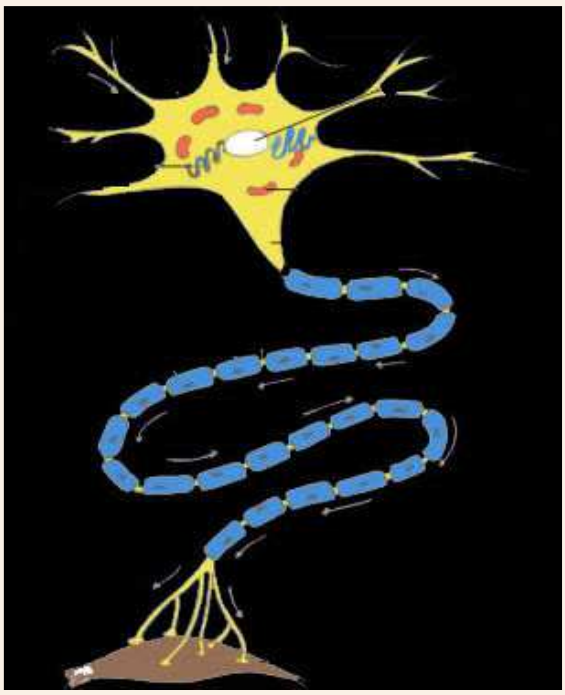
- Take 3 ml of urine in a test tube.
- Add four drops of 5% NaOH and two drops of picric acid solution.
- An orange red color of creatinine picrate is formed.

ESTIMATION OF CREATININE

Interpretation

- The amount of creatinine excreted daily in the urine is constant.
- Creatinine occurs is small amounts in the urine of growing and pregnant animals.
- Determination of creatinine is urine is of little importance.
- An increased blood creatinine value indicates renal impairment.

MODULE-1: ORGANISATION OF CNS
NEURONS, NERVE FIBRES AND ITS CLASSIFICATION



LEARNING OBJECTIVES

This module deals with,

- the components of the nervous system,
- the structure of a neuron,
- Glial cells, its function and their basic types and
- the basic structure of nerve fibres.

NERVOUS SYSTEM - AN INTRODUCTION

- Pioneering discoveries of LUGI GALVANI'S explanation of animal electricity which he showed as a muscle twitch in a dead frog on application of electrical current led away to the modern study of neurophysiology. From his work, a clue was spread about the functions of nervous system, as a unit of control of activities of the body. Based on this revolutionary idea, nerves were thought to be similar to water carrying pipes. Much of such discussion on neuronal communications and cellular cross talk find its way for discoveries and updates theories in modern neurophysiology.
- Nervous system is a structural and functional mechanism involved in the co-ordination of an animal's response to internal and external environment. Nervous system collect information from outside and inside the body, as well as integrate the information. It comprises of collection of neurons to specialize to covey information with the great speed and accuracy. Nerve cells are designated to respond to stimuli and transmit information to various parts of the body. It provides effective responses to stimuli. Nervous system is composed of millions of discrete structural, functional and genetic unit called *neurons*.

Nervous system can be subdivided into different parts for the convenience of study but they function together.

- *Central nervous system (CNS):* brain & spinal cord
- *Peripheral nervous system (PNS):* cranial nerves and spinal nerves.
- *Autonomic nervous system (ANS):* sympathetic and parasympathetic divisions.

- Brain is the major area of central nervous system, is a large structure with the cluster of neurons viz SUPERGANGLION integrates and dictates other systems according to the external and internal stimuli.
- PNS is the communication network that connects CNS with the other parts of the body. Their neural tissue lie outside the CNS they have afferent division (towards) brings sensory information from receptor (centripetal) and efferent division (from/ exit) carries motor commands to effectors. Effectors can be skeletal muscles, smooth muscles or cardiac muscles. Efferent division includes somatic nervous system (skeletal muscle) and autonomic nervous system (smooth muscles, cardiac muscles, glands, brown fat and immune organs).
- Both central and peripheral nervous systems are made up of NEURONS and supporting tissues known as GLIAL CELLS.

EMBRYOLOGICAL DEVELOPMENT AND DIVISION

- Embryonic neural tube which is a hollow structure from epithelial origin, develop into brain and spinal cord.
- Brain is derived from anterior portion and spinal cord is divided from posterior portion of neural tube.
- Anterior portion develop into three swellings viz.
    - Rombencephalon (Hind brain)
    - Mesencephalon (Mid brain)
    - Prosencephalon (Fore brain)

- Cavity that develop in the centre is known as *ventricles* which are filled with cerebrosipinal fluid (CSF) and ependymal cells, a type of glial cell allow the circulation of CSF into ventricular and spinal cord due to their ciliary action.
- Hind brain controls reflex responses and regulate involuntary behaviour (breathing, equilibrium, maintenance of body position).
- Mid brain is predominantly involved in co-ordinating visual, auditory or sensory information from mechanoceptors (touch/pressure) from parts of the body. It acts as a routing centre.
- Fore brain is involved in processing olfactory information, integerates with other sensory information, regulates body vegetative functions. It is designated to perform complex tasks like memory processing etc.
- Size of the brain varies with body size individual structures type of activity pertaining to the environment of the animal.
- Basal nuclei is the cluster of grey matter situated deep into the brain.

Hind brain: (Pons, Medulla and cerebellum)

Location: Between spinal cord mid brain. Functions together to support vital body functions such as respiration cardiac function and movement.

- *Medulla*
    - Located at the top of the spinal cord
    - Has centres that control reflex activities such as respiration, heart rate, vasodilation and blood pressure.
    - Houses neural pathways that connects cerebral cortex and spinal cord seat of some of the cranial nerves (9th, 10th and 12th).

- *Pons*
  - Structures present above the medulla functions as bridging and communicating informations between medulla, cerebellum and fore brain structures.
  - Has neurons that control rate of respiration.
  - Concerned control of alertness.
- *Cerebellum*
  - Encased in cerebellar hemisphere and is located in the back of the brain.
  - Responsible for motor co ordination by integerating sensory inputs from receptors of muscle, eyes and ears with motor orders of the forebrain.
  - Maintain Equilibrium and posture.

Mid brain

- Relay centre
- Roof of midbrain is known as *tectum* which has pair of brain centres namely optic lobes in non mammalian species.
- Superior Colliculi is a similiar structure as optic lobes in mammals functions as reflex optical response (orientation towards visual stimuli, focussing etc).
- Tectum has a pair of inferior colliculi which are neurons concerned with hearing.
- *Tegmentum* is the posterior part of midbrain that possess fine control of muscles.
- Brain stem is the grouping of midbrain with the pons and medulla.

Fore brain

- Involved in processing and integrating sensory information and co ordinates behaviour.
- Structures of the fore brain consists of cerebrum , thalamus, epithalamus and hypothalamus.
- Cerebrum
  - Has outer layer known as cerebral cortex which is divided into two cerebral hemispheres.
  - Hemispheres excercise control over opposite side of the body.
  - Hemispheres functions independently of each other.
  - Two hemispheres are connected by a mass of white matter known as *corpus collosum* through which they communicate.
- Hypothalamus
  - Present in the base of the forebrain, below the thalamus.
  - Maintains homeostasis by controlling internal organs and interacts with autonomic nervous system.
  - Regulates endocrine system and links endocrine and neural systems.
  - Controls vegetative functions of the body.
  - If forms the part of limbic system which influences emotion and behaviour.
  - *Limbic system* is a network of connected structures that lie between the cortex and reset of the brain.
  - Structures associated with limbic systems are hypothalamus, amygdala, hippocampus and olfactory bulbs.
  - *Structures of limbic system*
    - *Amygdala*
      - Concerns with emotional responses of fear and aggressiveness.
      - It is also involved in maintenance of memories of the emotion.
    - *Hippocampus*
      - Structures convert short term memories to long term memories.

        - *Olfactory bulb*
            - Organ of importance for sensing the smell. Behaviours are smell driven in animals. Sensory neurons from the olfactory epithelium are connected to the olfactory bulb.
            - Olfactory information reaches directly here without passing through thalamus.
            - Information are transmitted to cortex for processing.
            - It is connected to amygdala and hippocampus. Emotional behaviours are mediated through odours in animals and human beings.
- Thalamus
    - Act as relay centre, largest sensory ganglion of the brain
    - Has group of grey matter located deep into the forebrain.
    - Receive all sensory information for upward transmission except olfaction.
    - Reticular formation is a set of neurons extend from thalamus to brain stem including pons, medulla and midbrain, that filters incoming sensory information.
- Epithalamus
    - Located above the thalamus harbours pineal complex, that establishes circadian rhythm and secretes melatonin.

NEURONS

- These are basic building blocks for communicating information using both chemical and electrical signals. Neurons are designed with specialized properties to receive process and transmit the information. Neurons like any cell have a cell membrane boundary, cellular organelles and extensions which perform conduction of information and reception.
- Cell membrane is electrically excitable. Neuronal shape and size vary depending upon the function. But they have common functions to change incoming information into electrical potential. This process is known as *impulse generation*.

Neuronal anatomy

- Soma (cell body) (perikaryon) responsive of metabolic maintenance of the cell.
- Processes to receive and conduct information. They are of two types
    - *Dendrites*
    - *Axons*
- Most of the neurons posses multiple dendrites and a single axon. These anatomical structures are arranged in specific zones and play a different role in neural signaling.

Neuronal soma

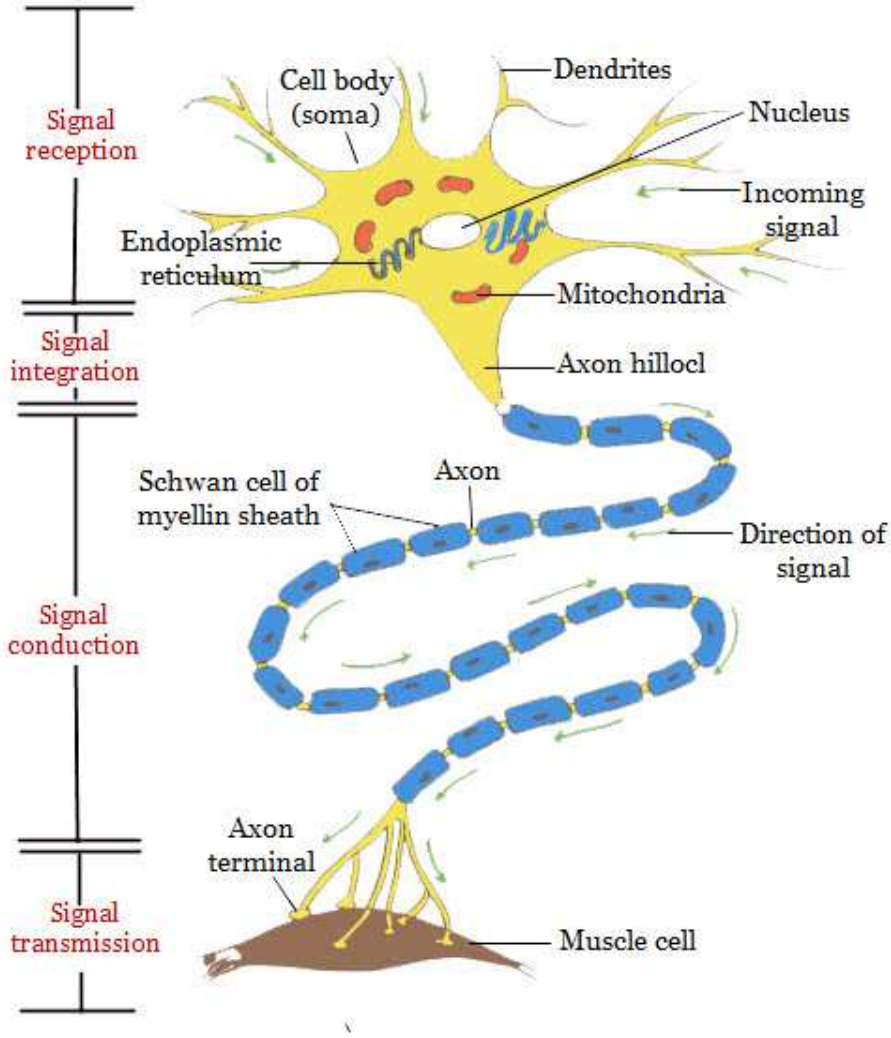
- Body of the neuron consists a nucleus surrounded by cytoplasm.
- It contains neurofibrils, nissil bodies (basophilic granules) or tigroid bodies, golgi apparatus and mitochondria. The nucleus is large, circular with large spherical nucleolus. Nucleolus has a ribonucleic acid (RNA) and no centrosome. Absence of this attributes to the failure of normal regeneration, it looses the cell division.

Neurofibrills

- Fine filamentous meshwork in cytoplasm connecting dendrites and axons.

Nissil granules

- Clumped basophilic granules in the cytoplasm concerned with the protein synthesis & have the ribonucleic proteins. These granules are not present in the areas of origin of axon. Their size, number and shape vary.



Dendrites

- They are fine branching extensions of the neuron originate from cell body and serve as the receptive surface.
- These show heavy branching and are responsible to sense and gather incoming signals. As they receive, are converted into an electrical signal. This in turn is propagated as change in membrane potential.
- The dendrite with extensive arborizatrion receives many inputs.

Axons

- This is otherwise known as nerve fibers. This is the long extension from soma to conduct information away from cell body. Nerves are general term referring bundles of axons running through the tissues. Axon has neurofibril and mitochondria. Axon arises from an area in the caudal soma known as **axon hillock** which is devoid of nissil granules. This area is said to be **trigger zone** where action potential are initiated. Axon terminates into numerous branches called TELODENDRIC or TERMINAL BUTTONS which allows the signals to pass on simultaneously to many structures. Telodendria contain vesicles to store synaptic transmitters.
- Both axon and dendrite are nourished by the cytoplasm of the cell. They transfer nutrients to the terminal axons. Cytoplasmic flow depicts ambeoid movement of the phagocytic cell. The flow in axon is popularly known as axoplasmic flow.

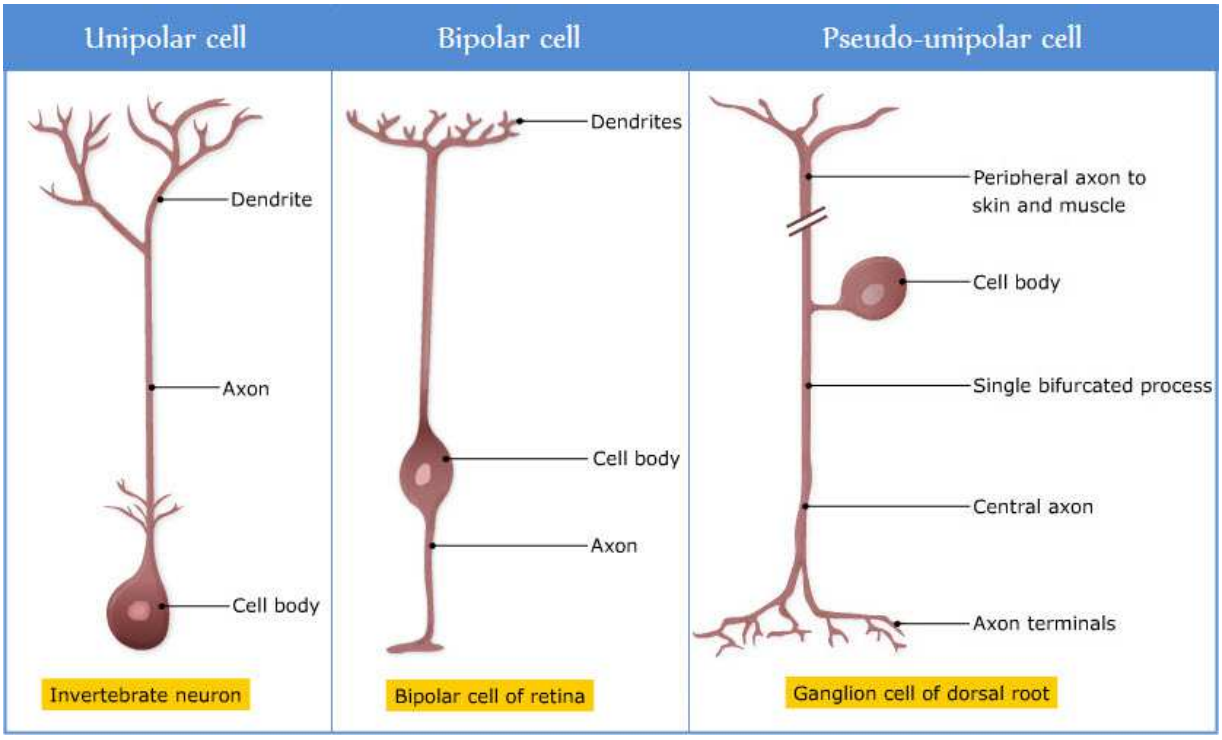
CLASSIFICATION OF NEURONS

Classification of Neurons

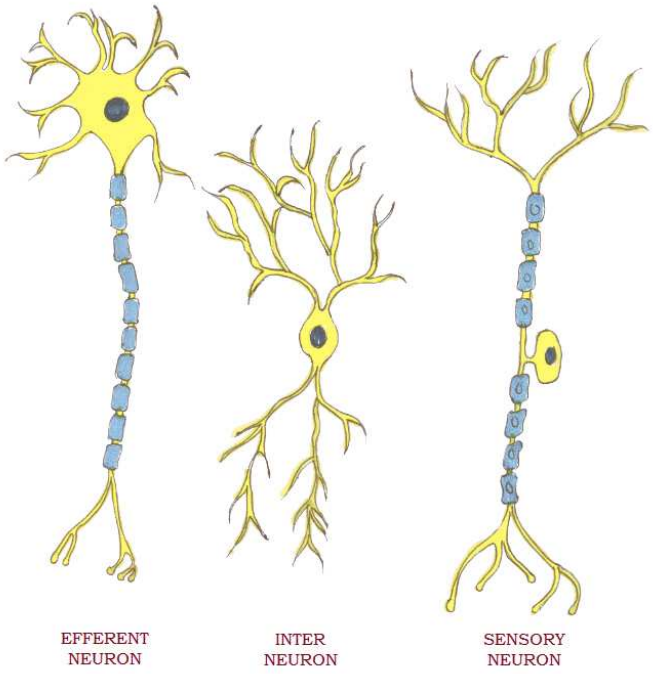
- Anatomical classification
- Functional classification

Anatomical classification

According to the number of processes that arise during embryological development. All developing neuroblast pass the stages where processes are formed.



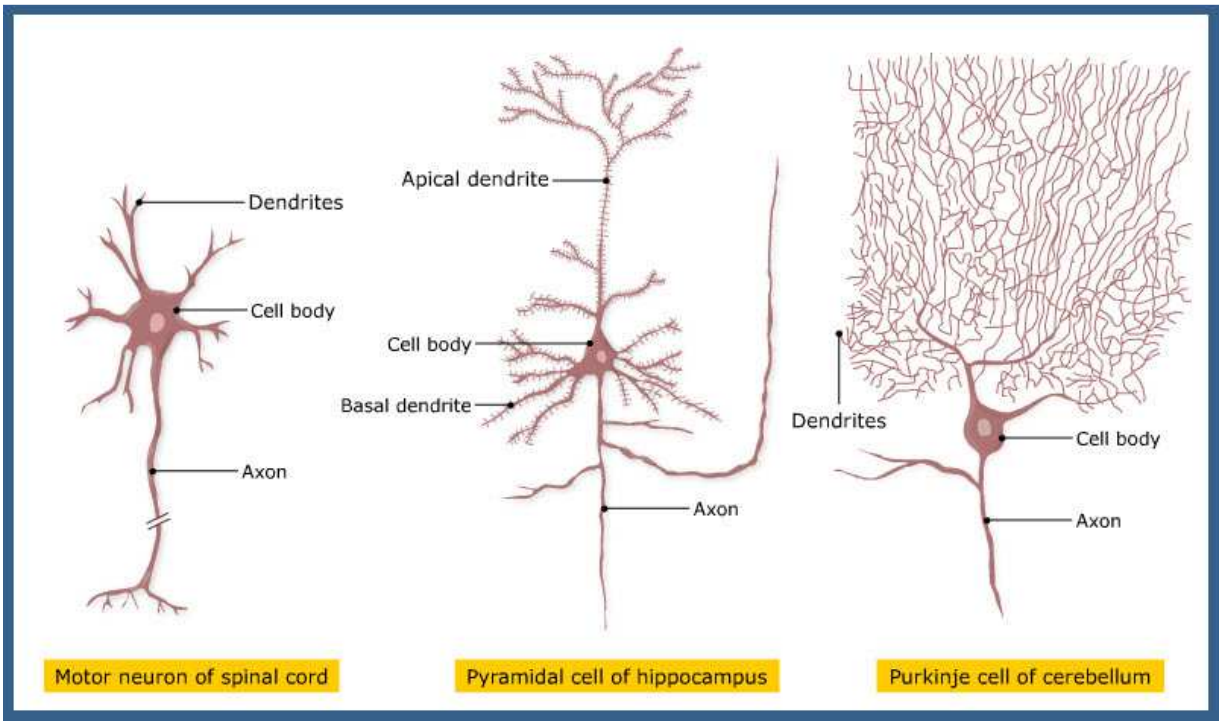
- *Apolar neuron:* There are no processes
- *Unipolar neuron:* They have a simple processes developing from soma. This process may split into two, one conveys towards the cell body and other conveys away from cell body. Hence they are referred to as pseudounipoloar neuron.
- *Pseudounipolar neuron:* A typical bipolar neuron. Develop as bipolar neuron but as it develops they fuse to form one process. Sensory in nature. Pseudo unipolar cells are "T" shaped cells whose peripheral branch is physiologically a dendrite but serve as axon to conduct nerve impulse centrally. Located in the ganglia outside the CNS.
- *Bipolar neuron:* These are the neurons are spindle shaped with axon at one pole and dendrite at the other pole. These are the neurons with the two processes emerging from opposite sides of the soma. Commonly seen in vertebrate retina to transmit light impulses to retinal ganglion cells, as well as olfactory neuroepitthelium. One of the two main processes extending from the cell body, is highly branched & conveys information to the cell body, similar to the function of dendrite.



Multipolar neuron: These have many cellular processes extending from the cell body. One of these is an axon, and other constitutes highly branched dendrites. Most commonly seen in vertebrates, in cerebellar cortex (purkinje cells), motor coterx (pyramidal cells) and ventral horn of the spinal cord.
There are two types of mutipolar neuron. Classified depending upon the length of the axon.

- *Motor nerve cell:* cells with long axon. E.g. neurons in the ventral grey column of spinal cord. Axons of these cells arise from the CNS to become peripheral motor nerve fiber.
- *Type II Golgi cell:* cell with short axons. The axon divides into many branches in the area of cell body. Because of this arrangement sensory neurons are placed in contact with motor neurons. Hence they are known as internerurons or internuncial neurons. They lie within

the CNS. They remain between afferent and efferent neurons. For more complicated actions more interneurons are involved.



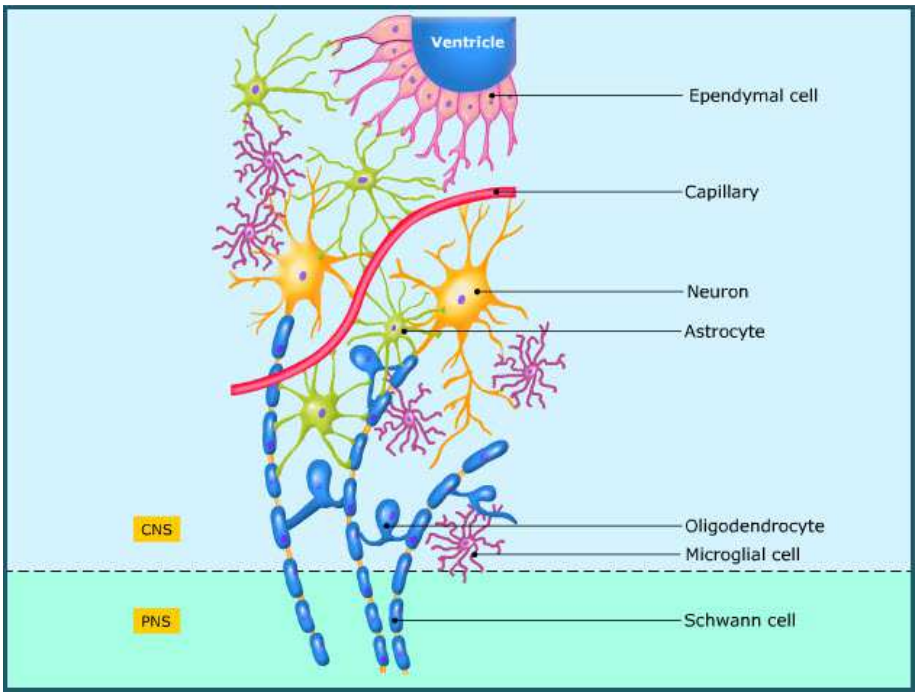
Functional classification

- *Sensory or afferent (towards/ carrying):* Information from sensors to CNS intimating CNS of external environment (sensory stimuli) and provides status reports on internal activities (visceral) that are regulated the nervous system.
- *Internuncial neurons:* remain in CNS act as a relay between two neurons.
- *Motor or efferent (from/ exit):* instructions from CNS to the effector organs to provide appropriate action.
- This outflow is by two means
    - *Somatic nervous system* - to skeletal muscles. Fibers of motor neurons that supply skeletal muscles.
    - *Autonomic nervous system* - using sympathetic and parasympathetic divisions control the effector organ such as smooth muscle, cardiac muscles and glands.

SUPPORTIVE CELLS

Supportive cells of *CNS are known as Glial cells or Neuroglia*. Neuroglia is a special type of interstitial tissue which provides nourishment to the neurons. Glial cells serve as a connective tissue of the CNS and provide support the neurons, both physiologically and metabolically. They

constitute 90% of cellular population, and they do not branch as neurons. They don't participate in the initiation and conduction of nerve impulses. These cells homeostatically maintain composition of specialized environment surrounding the neuron.



They are classified according to shape, size and number of processes into three main types.

- Astrocytes (star shaped, ectodermal origin)
  - *Protoplasmic astrocyte- grey matter of brain and spinal cord. They have branching of protoplasmic processes.*
  - *Fibrous astrocyte- seen in white matter. They are long and have unbranched fibers.*
- Oligodendrites(ectodermal origin)
- Microglia(smallest cells, mesodermal origin)
- Ependymal cells.

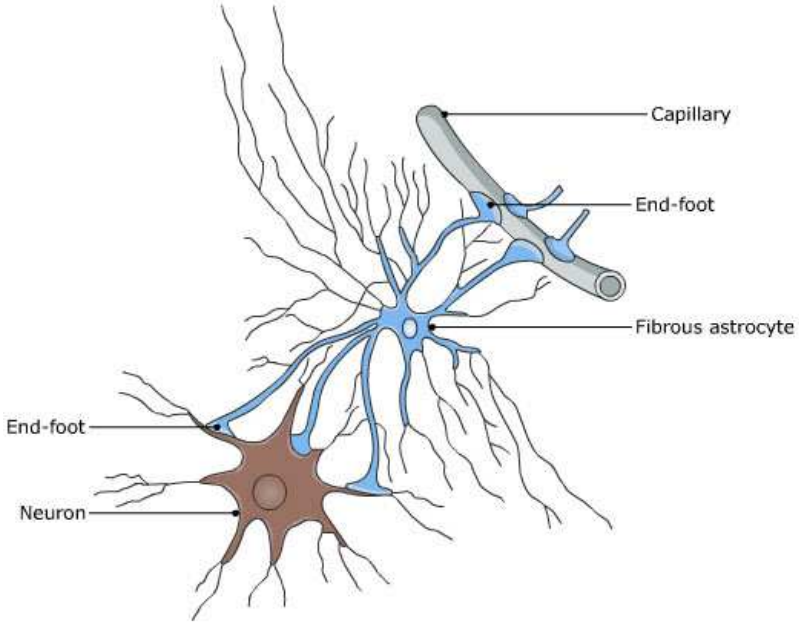
General functions of neuroglia

- They provide support, insulation and phagocytosis. They support physically, metabolically and functionally. They provide biochemical and biophysical environment for proper functioning of neurons.

Astrocytes

- Most abundant in number and they are star shaped. They posses number of critical functions.
  - During fetal development, they guide neurons for development.
  - These cells are responsible for establishment for blood brain barrier.

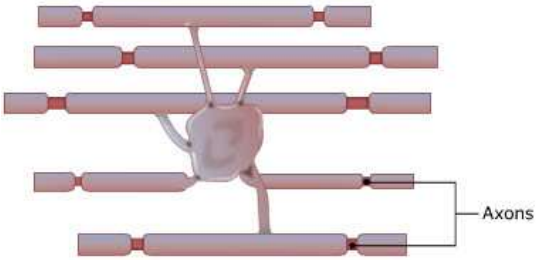
- They act as a glue to hold the neurons together and confine their spatial relationship.
- They regulate the neurotransmitter activity by degrading and reabsorbing the neurotransmitter.



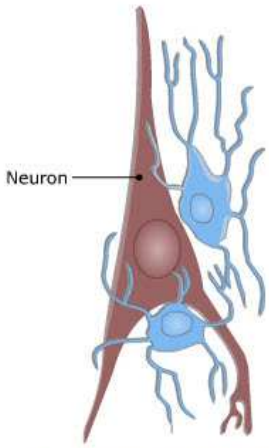
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- They absorb excess potassium ion by activating sodium–potassium pump and maintains ionic environment suitable for neural excitability.
- They enhance synapse formation and potentiate synaptic transmission.
- Astrocytes communicate via gap junctions between themselves and neurons.
- Gap junctions are small tunnel like connecting units helping in signal communication.
- Astrocytes have receptors for neurotransmitter and they release stored calcium ion prior to the release of neurotransmitter by the presynaptic neuron.
- The release of calcium strengths the synaptic activity by accelerating the release of neurotransmitter.
- They secrete neurotrophic factors that protect neurons against excitotoxicity and oxidative insult.

Oligodendrocytes

- They form the insulation by providing myelin sheath around the axons in the CNS.



Oligodendrocyte in white matter



Perineural oligodendrocytes

Microglia

- They are the immune defense cells of the CNS. They derived from the parent tissues of monocytes and they migrate to CNS during development. They are active when there is an infection in the CNS otherwise the resting microglia releases nerve growth factor that help neuron and other growth cells to survive.
- They have primary responsibility of phagocytosis in the CNS. These cells have long branches radiating outward during phagocytosis. They are getting activated. They retract the branches with their high motility and move towards the infective area to remove the invaders. Activated microglia releases degrading chemicals to neutralize the tissue debris.

Ependymal cells

- They line the internal cavities of vertebral CNS. They are of important as they contribute to the formation of CSF. Among the glial cells, they are ciliated and by the action of cilia the flow of CSF throughout the ventricles is regulated. An important characteristic of ependymal cells is that act as the stem cell which has the potential of forming glial cells and new neurons.

Supporting cells of PNS

- *Schwann's cell:* They are helpful in the formation of myelin sheath and provide insulation.

NERVE FIBRES

- Nerve fibres are *axons of neurons* which can be either insulated or naked. Insulation is provided by sheaths which may be either myelin sheath of lipoid layer or thin tubular covering of neurilemma or schwann sheath.
- The nerve fibres with myelin sheath are known as *myelinated* and those without myelin sheath are known as *unmyelinated fibres*. Myelinated fibres are seen in both CNS and PNS whereas fibres with schwann cell, sheathing core characteristically seen in fibres lie outside CNS.

Structure of Myelinated nerve fiber

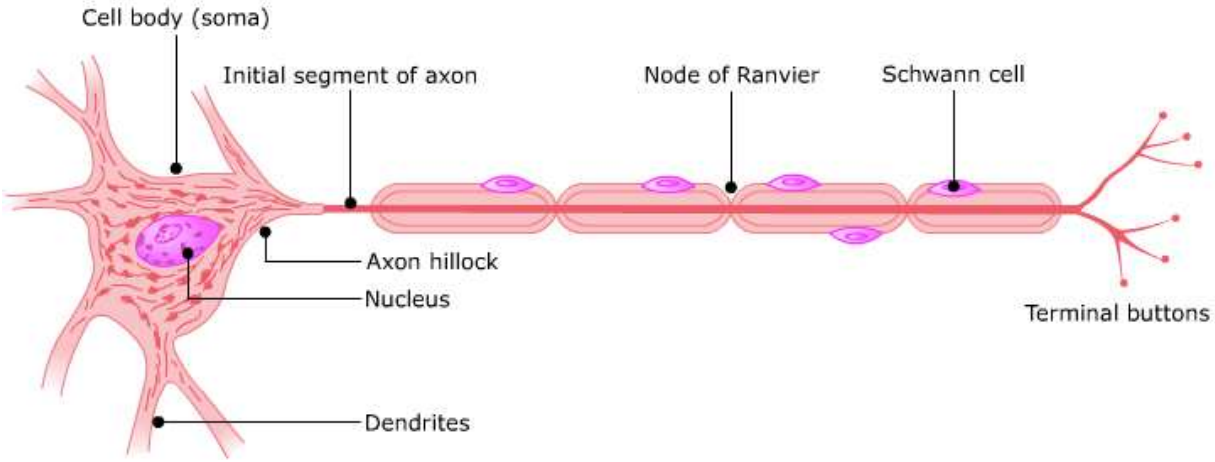
- Axon is composed of delicate neurofibrils embedded in an axoplasm along with mitochondria and endo plasmic reticulum. Axon has a cell membrane known as Axolemma separating the axon from the surrounding structures.
- Axolemma is surrounded by thick myelin sheath made-up of nucleated membranous Schwann cell sheath. This structure poses the axon to be highly refractive and provides white colour to the axon.
- Myelin sheath consists of lipid-Sphingomyelin. The myelin sheath is interrupted at regular intervals by constriction which appears to be a node and is known as Nodes of Ranvier. At each node, there is a constriction of neurilemmal sheath towards the axon. These sheath cells are absent in the myelinated fibers of CNS.

Myelinogenesis

- It is the process of myelin formation. It begins with crowding and excessive growth of Schwann cells completely encasing the whole length of the axon.
- The cell membrane of the Schwann cells surrounding the axons rotate in a manner to wrap around the axons and forms closely packed helically arranged layers of double membrane. Each membrane is composed of bilipid layer sand witched between the protein layer. Thickness of myelin sheath is determined by the number of membrane layers wrapped round the axon.
- Myelination of the sensory tract precedes the motor tract. Myelination is influenced by Vitamin B 12, Folic acid and Thyroxin hormone.

Stages of Myelination

- First stage begins with enclosure of axon with myelin forming cells. Mesoaxon is formed by the union of enveloping processes from outer surface.
- Elongation of mesoaxon in a spiral manner which coils around the axon occurs. Loss of cytoplasm of turns of mesoaxon tightens the spiral turns around the axon.



Conduction of impulse in myelinated fiber

- Nerve impulse travel at rapid rate in myelinated fibers. Electrical membrane has effective contact with the interstitial fluid only at the nodes of Ranvier.
- In myelinated fibers the local circuit flow is from one node to the other, known as Saltatory conduction.

- Due to this the conduction velocity in these fibers is greater as the conduction is restricted to patches of the membrane which are then depolarized. The conduction speed is 20 times greater than the unmyelinated fibers.

Unmyelinated nerve fiber

- Lacks myelin sheaths and are found in grey matter of spinal cord, some parts of brain and in ANS.
- Conduction velocity is about 1 meter and is dependant upon the diameter of the fiber.

CLASSIFICATION OF NERVE FIBRE

- Classified as *Type A, Type B* and *Type C fibers* based on the diameter, myelination and propagation speed. According to the diameter of the fiber, a physiological property of nerve fiber varies. As diameter increases, the conduction velocity increases which requires lower threshold of excitation with greater magnitude of response. Time of conduction is shorter with shorter refractory period.
- Based on the characteristics, nerve fibers are classified into 3 groups viz A, B and C fibres. Large fibers conduct impulses rapidly whereas small fibers are slow in conduction. The speed of conduction can be achieved by 2 nodes
    - By employing myelinated fibers
    - By increasing the diameter of the fibers.
- Group A consists of largest fibres (up to 30 microunit in dia) with conduction rates of 50-100m/sec. The fibres of this group are myelinated. This group is further typed into *alpha, beta, gamma and delta*.
    - Alpha is subtype is concerned with proprioception, to carry impulses from muscle spindle and golgi tendon organs.
    - Beta fibres are concerned with touch, pressure and other somatomotor impulses carrying from muscle spindle.
    - Gamma fibres are concerned with motor functions to muscle intrafusal fibers.
    - Delta fibres are similar to beta in function.Group B fibres are which are rich in preganglionic autonomic fibers and group C functions in post ganglionic sympathetic fibers.

PROPERTIES OF NERVE FIBRE

- Ability of nerve fibre to respond to stimulus.

Stimulus

- It is an external agent that brings about visual response. It can also be from within and can elicit response which can not be visualized.
- Types of stimulus

- *Electrical:* this is the most common stimulus to bring about neuronal response.
- *Chemical:* osmotic also come under this. Hypo, hyperosmotic, acid or alkali.
- *Mechanical:* touch, thermal, light, sound, pressure changes are included in this.

All or none law

- In a nerve fibre, action potential are equal in amplitude and velocity. Because amplitude and velocity are independent of intensity of the stimulus. This is the basis of all or none law. Nerve fibres obey all or none law.
- When adequate strength of stimulus is applied to a nerve, the action potential propagates in the same magnitude all along the nerve. If the stimulus is low in threshold, there is no action potential and if the stimulus is above the threshold the action potential is uniform, magnitude of action potential does not increase with the stronger stimuli. This is known as all or none principle. As the action potential travels along it triggers the subsequent axon to fibre.

Refractory period

- However, for a period following the passage of an impulse, there is unresponsiveness, though a second stimulus is strong axons are incapable of generating new impulses. This interval known as **absolute refractory period**. This period varies wit the type of nerve and with species.
- In mammalian nerve it is about 1.04 m s. absolute refractory period is due to complete closure of sodium gates which resists the membrane to produce an action potential most of the axons exhibit 1 ms as their absolute refractory period.

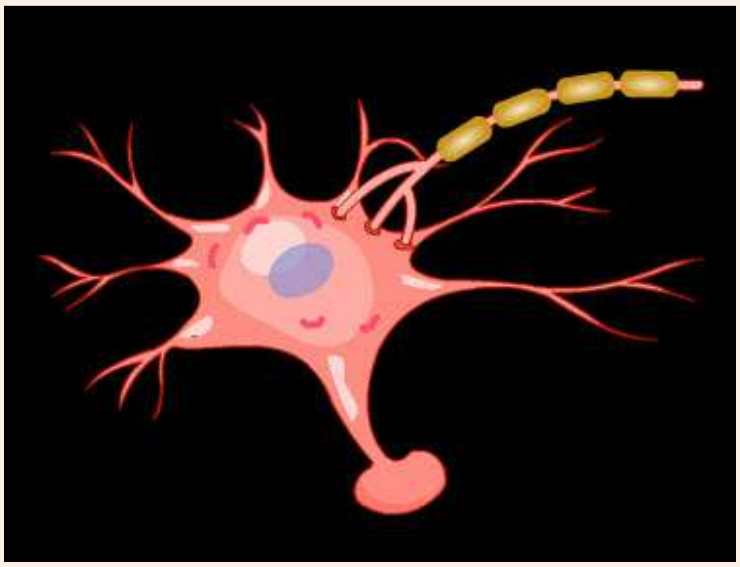
Relative refractory period

- It follows the absolute refractory period. In this, the response to second stimuli of threshold strength is absent but if the second stimuli is of higher strength than threshold response is appreciated.
- The excitability gradually rises to normal. Because of this the nerve fibre can be stimulated for hours without excitability fatigue.
- Relative refractory period is due to the opening of K^+ gates with Na^+ gates resting their usual state. Due to free flow of K^+, stronger than usual stimulus is necessary to initiate action potential.
- Relative refractory period in axons experimented was 2-4 m s. The refractory period governs the rate at which a membrane can fire.

Accomodation

- Strength of the stimulus applied bears relationship with length of the time of its application to produce a response.
- Short duration, strong stimuli and long duration weak stimuli fail to produce a response.
- Rate of rise of stimuli when applied rapidly produces better response. But slow rise in stimuli for longer duration fail to provoke a response due to adaptation by nerves to weaker stimuli which is known as accommodation. This phenomenon is due to partial inactivation of sodium carrier by nerve cell.

MODULE-2: IMPULSE GENERATION AND PROPAGATION



LEARNING OBJECTIVES

This module deals with,

- generation of an action potential and impulse and
- the propagation of an impulse in myelinated and non-myelinated fibres and conduction velocities in them.

PRINCIPLES OF MEMBRANE POTENTIAL

Basic Physics

- In the tissue the fluid outside (interstitial fluid) and inside (intracellular fluid) are electrolytic solutions, containing 150 to 160 mEq / litre of positive ions (cations) and the same concentration of negative ions (anions).
- Generally a very small excess of negative ions accumulate immediately inside the cell membrane along it's inner surface, and an equal number of positive ions accumulate immediately outside the membrane, the effect of which is the establishment of membrane potential between the inside and outside of the cell membrane.
- Such a membrane whose inside is more negative than the outside is said to be a polarized membrane. The basic means by which membrane potentials develop are
    - Passive diffusion of ions i.e. diffusion along concentration gradient,
    - Active transport of ions against concentration gradient and creation of imbalance between negative and positive charges.

RESTING POTENTIAL

The resting potentials of nerve and muscle Fibers

- In the resting state of membrane i.e. the state under which the membrane of the nerve or muscle fibers is not transmitting signals, the fluid inside the nerve or muscle cells has greater concentration of K^+ and the fluid outside the Na^+.
- The potential across the membrane is a negative potential and is known as resting potential. The resting potential of the nerve as well as the muscle fiber is -75mV i.e. the potential inside the fiber is 75mV more negative than the potential outside.

Sodium - Potassium Pump mechanism

- The mechanism is present in all cells of the body. The negative resting membrane potential is established by this mechanism. It transports Na^+ from the cells to the exterior and K^+ from the exterior to the cell. Hence it is responsible for maintaining the normal Na^+-K^+ concentration difference across the cell membrane.
- The pump mechanism uses a carrier protein, which is a complex of two separate globular proteins, one of which is large with a molecular weight of about 100,000 and the other is small with a molecular weight of about 45,000. The function of the latter is not known. The large one has the following 3 specific features that confirm the pump function:
    - As the carrier protein complex exists across the cell membrane the large protein protrudes on either side of the membrane.
    - It has 2 receptor sites on the outside protrusion for K^+ and 3 on the inside protrusion for Na^+.
    - The site adjacent or close to the Na^+ receptors has ATPase activity.
- In the operation of the pump mechanism $3Na^+$ bind (each at a receptor) from interior of the cell and 2 K^+ (each at a receptor) from exterior of the cell. The binding of Na^+ at the receptors activates the ATPase. A molecule of ATP is cleaved and reduced to ADP. Energy is liberated from a high energy phosphate bond, cause conformational change in the carrier protein molecule and this change extrudes the Na^+ to the exterior and K^+ to the interior of the cell.
- Since all the while the concentration of Na^+ at the exterior and K^+ in the interior of the cell is high, both these ions are transported by the pump mechanism against very large concentration gradients.

Electrogenicity of the Na^+ - K^+ pump mechanism

- At each revolution of the pump 3 Na^+ are transported to exterior for every 2 K^+ to the interior of the cell. This amounts to a net of one positive change removed from the interior to the exterior.
- This obviously creates positivity outside the cell, leaving (owing to the deficit of positive ions inside the cell) negativity on the inside and thus an electrical potential is established across the cell membrane. Hence, the pump mechanism is said to be electrogenic.

ACTION POTENTIAL OF NERVE AND MUSCLE FIBERS

- The action potentials are the rapid changes in the resting membrane potential. Nerve signals are transmitted by action potentials.
- Each action potential begins with a sudden change from the normal resting negative potential to a positive potential, and ends with an equally rapid change back again to normal negative potential.
- The action potential initiated moves along the fiber until it comes to the end of the fiber and thus spreads the impulse or signal along the fiber.

THE STAGES OF ACTION POTENTIAL

Depolarization stage

- The resting membrane suddenly becomes very permeable to sodium ions (stimulus applied causes this permeability) allowing tremendous number of Na^+ to flow to the

interior of the fiber. *The normal resting state of -75mV* is lost with the potential rising or changing rapidly in the positive direction. This is called depolarization.
- In large fibers the potential over shoots beyond the zero level and becomes slightly positive (reversal potential). In some smaller fibers as well as in many CNS neurons the potential approaches the zero but does not overshoot or reverse.

Repolarization stage

- Within a fraction of a second (few 10,000 ths of a second) after the membrane becomes highly permeable for Na^+ the channels for the Na^+ close almost as rapidly as they had opened. Then rapid diffusion of K^+ to the exterior re-establishes the normal negative resting potential. This is called *repolarization of the membrane.*

THE VOLTAGE-GATED SODIUM CHANNEL

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Voltage-gated channels of the Cell Membrane

- There are two separate and specific voltage-gated channels, one for Na^+ and the other for the K^+.
- The principle operator of the stages of action potential i.e. causing the depolarization and repolarization of the membrane is the so called voltage-gated Sodium channel. However, the voltage-gated potassium channel also plays an important role in certain nerve fibers.

- The channel or the passage has two gates one at the exterior end (opening to the outside of the cell) and the other at the interior end (opening to the inside of the cell). These gates may be referred to as external or internal gates of the channel. The external gate is known as 'Activation gate' and the internal as 'Inactivation gate'. The potency of the gates at different stages of an action potential is explained as follows:
    - *Resting Stage:* The activation gate is closed and the inactivation gate is opened. The Na^+ cannot enter the channel.
    - *Activated stage (Depolarization stage):* Both the gates are opened and the Na^+ can freely pass through the channel from the exterior to the interior of the cell.
    - *Inactivation Stage (Repolarization stage):* The inactivation gate is closed and the Na^+ cannot enter the channel from the interior of the cell.
        - When the activation gate is opened it is referred to as *"activation of the channel"* and when the inactivation gate is closed it is referred to as *"inactivation of the channel"*.

Activation

- Is effected when the resting membrane potential (-75mV) drops or becomes less negative and reaches a level between -60 to -50mV. Then sudden conformational change in the activation gate flipping it to the open position (Activated state). Then Na^+ literally pours into the cell through the channel. At that moment the depolarization process of the membrane begins. The movement of Na^+ through the membrane is referred to as Sodium permeability of the membrane and it is increased as much as 500 to 5000 fold during activation.

Inactivation

- The above said drop in negativity otherwise increase in voltage that opens the activation gate also closes the inactivation gate. However, the closure of the inactivation gate occurs a fraction of a second (few 100,000ths of a second) after activation gate opens ie. The conformational change that flips the inactivation gate to closed state is comparatively a slow process. Therefore, during the time lapse between the activation and inactivation of the channel which is also a fraction of a second (few 100,000ths of a second) there is passage of Na^+ to the interior of the cell and at the end the inactivation gate suddenly closes and the Na^+ no longer pour to the inside of the cell or membrane. At that moment the repolarization process begins.
- Important characteristics of the inactivation
    - The inactivation gate will not reopen until the disturbed membrane potential returns either to or nearly to the original resting membrane potential level. Therefore, without the repolarization or nearing completion of repolarization of the fiber it is not possible for the Na^+ channel to open again.

THE VOLTAGE-GATED POTASSIUM CHANNEL

- This channel has only one gate at the interior of the membrane and closes or opens to the interior of the cell. The opening of the gate is referred to as activation. The gate operates in two stages:
- *Resting stage:* During the stage the gate is closed and the K^+ is prevented from passing to the exterior of the cell through the channel.

During the stage of action potential manifestation

- When the membrane potential, rises from -80mV toward zero (becomes less negative). This voltage change causes conformational opening of the gate and allows K^+ diffusion outward. But, this process operates in a slow manner hence, it is timed, that the opening of the K^+ channel is initiated at the same time that the Na^+channels are becoming inactivated (closing). The later decreases the Na^+ entry into the cells.
- Thus the decreased Na^+ entry to the cell and concomitant increase in K^+ exit from the cell increases the intracellular negativity and greatly speeds the repolarization process. Therefore, in a fraction of a second (few 10,000ths of a second) there is attainment of the resting membrane potential.

GENERATION OF THE ACTION POTENTIALS

- When a fiber is stimulated at a point the action potential is initiated at the point. A local current or EMF (Electromotive force) is produced at the point.
- A "*local circuit*" of current is established between the depolarized point and the adjacent resting point.
- Positive electric charges flow through the circuit into the fiber through the depolarized membrane and continue to flow along the core of the fiber. This kind of flow increases the voltage (to above threshold value) along the core through a distance of 1-3 mm. This strong impulse promptly activates the Na^+ channels present on the course and the explosive action potential spreads.

Spike potential

- The initial very large action potential is called the spike potential. In large myelinated nerve fibers the spike potential lasts for about 0.4 seconds. It is also called as *nerve impulse.*

Negative after potential

- At the termination of the action potential the membrane potential sometimes fail to return to its resting level for an additional few milliseconds. Often instances will be there after a series of rapidly repeated action potentials. The situation is due to the build up of K^+ immediately outside the membrane and the side becomes more positive than normal and this increase the time for the potential to return to the normal resting level. This less negative membrane potential through the additional few milliseconds is called the negative after potential.

Positive after potential

- Once the resting value of the membrane potential is reached, it further becomes little more negative than the normal resting value. This excess negativity is from a fraction of an mV to a few volts than the normal resting level and this potential is known as **"Positive after potential".** The after potential state can last from 50 milliseconds to as long as many second. It is principally the recharging process (the electrogenic pumping of excess Na^+ outward).

PROPERTIES

All or nothing principle

- The principle operates in the membrane excitation of the fibers i.e. once an action potential has been elicited at any point on the membrane of a normal fiber, the depolarization process travels over the entire membrane. This is called the all or nothing principle of membrane excitation.
- It applies to all normal excitable tissues. However, when the fiber is in an abnormal state the action potential may reach a point in the membrane at which it does not generate sufficient voltage to stimulate the adjacent resting point of the membrane and therefore the spread of depolarization may stop.

- But under normal conditions, such stoppage will never occur and the impulse generated stops only when it reaches the fiber end.
- Therefore, for normal propagation of impulse, the ratio of action potential to the threshold for excitation of the fiber called “Safety factor” at all times be greater than one.

Conduction (Signal transmission) in nerve fibers

- A typical small nerve trunk is constituted by both the large nerve fibers and small nerve fibers.
- The large fibers are myelinated and the small fibers are unmyelinated.
- The ratio of distribution of these myelinated and unmyelinated fibers in such a trunk is 1:2 respectively.

STIMULUS

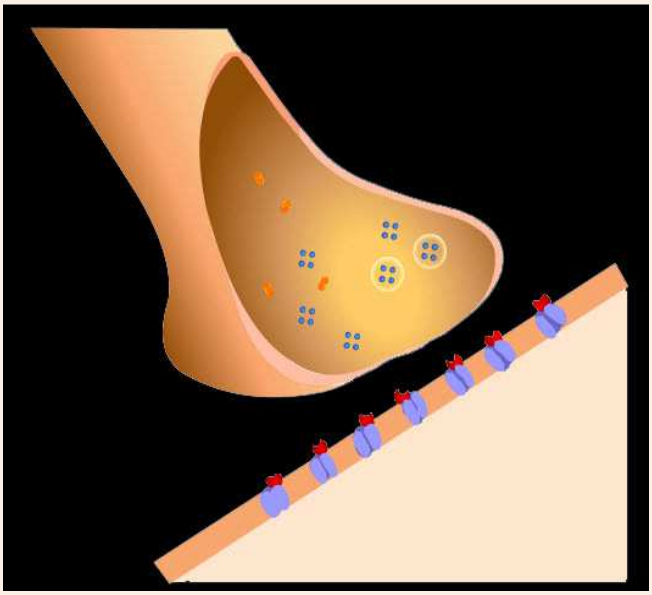
Classification of Qualitative stimulus

- *Chemical stimuli:* Strong acid or alkali or any chemical irritant inducing excitability.
- *Thermal stimuli:* Heat applied to the skin causing the excitability.
- *Mechanical stimuli:* Pinching, pin prick causing the excitability.
- *Osmotic stimuli:* Tissue when subjected to hypo or hypertonic solutions will be stimulated.
- *Electrical stimuli:* Exposed to shock.

The electrical stimulus is more or less similar to the natural stimulus present in the body. Moreover, the strength of the electrical stimulus can be regulated as required. So it is often preferred for conducting experiment. Based on the strength (Quantitative) of the electrical stimulus it is classified as follows

- Sub-minimal or sub-threshold strength
- Minimal or threshold strength
- Sub-maximal or above threshold strength
- Maximal strength
- Above maximal strength (destructible to the tissues)

MODULE-3: SYNAPSE AND ITS PROPERTIES



LEARNING OBJECTIVES

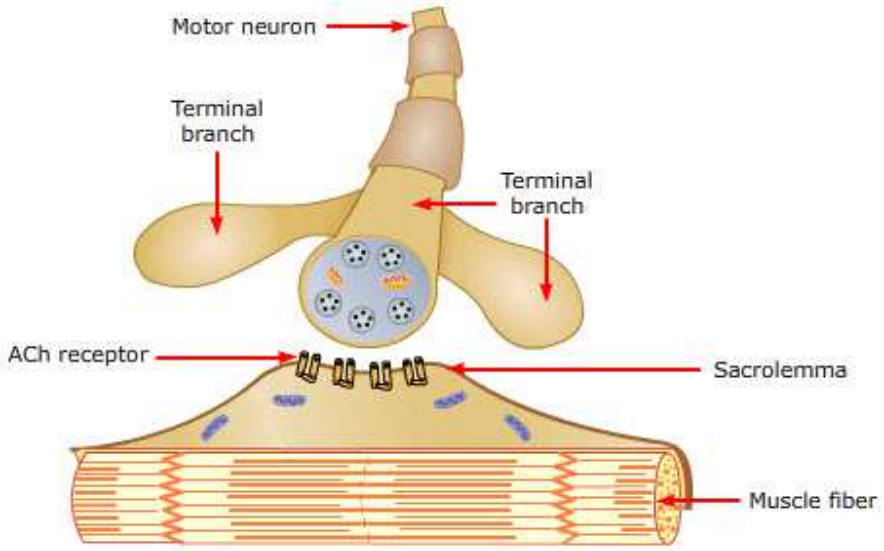
This module eloborates,

- the definition of a synapse and its types,
- the structure of a synapse, neurotransmitter release and synaptic transmission of impulse,
- the role of neurotransmitters on post synaptic membrane with special reference to chemically activated ion channels and second messenger activators and the fate of neurotransmitters,
- the difference between chemical and electrical synapses and
- the classification of neurotransmitters, its synthesis, effect and metabolism of a few common neurotransmitters.

SYNAPSE-ANATOMY

Functional anatomy of the synapse

- A typical motor neuron (efferent neuron) shown under electron microscope numerous (average about 6000) small knobs spread over the surface of the dentrites (80-90%) and soma (10-20%). These knobs are the ends of nerve fibrils that have their origin in many other neurons (usually not more than a few derived from any single previous neuron). The knobs are referred to as presynaptic terminals or terminal knobs, buttons, end feet and synaptic knobs.



- Neurons in other parts of the cord and brain differ markedly from the motor neuron in the following features :
  - Size of the soma
  - Size, length and number of dentrites (ranging in length as long as many centimeters)
  - Size and length of axon
  - The number of presynaptic terminals (range from few to more than one hundred thousand).
- These differences are responsible for the neurons of the different parts of the nervous system to react diversely to incoming signals in their functions.

TYPES OF SYNAPSES

Types of synapses

- Chemical synapse
- Electrical synapse

Chemical synapse

- All the synapses present in the central nervous system are chemical synapses.
- In these, the first neuron secretes a chemical substance called as a neurotransmitter at the synapse.
- The transmitter travels and acts on receptor (protein in nature) present in the membrane of the next neuron.
- According to the functional nature of the receptor there may be excitation or inhibition or modification of the sensitivity of the neuron.
- There are about 30 such different transmitter substances reported.
- The chemical synapses always transmit the signals in one direction (unidirectional transmission) that is from the presynaptic neuron to the post synaptic neuron.

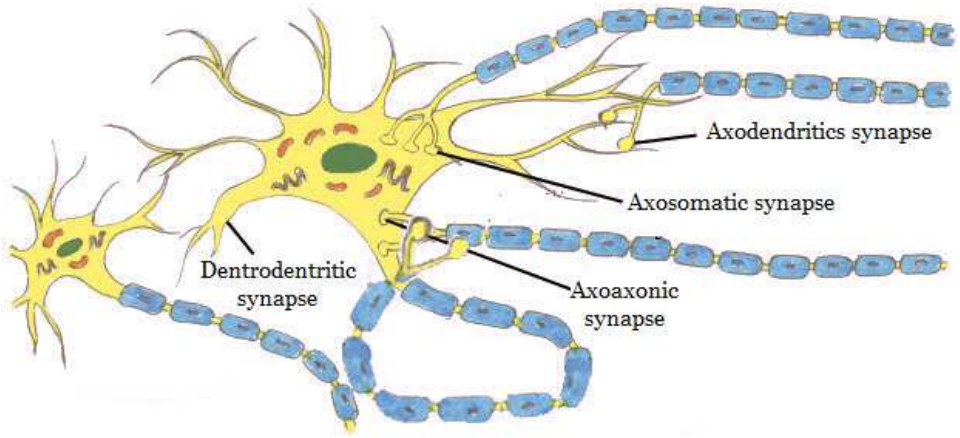
- The unidirectional conduction allows signals to be directed toward specific goals, discrete and highly focused areas of the nervous system to perform its varied functions of sensation, motor control , memory and many others.

Electrical synapse

- Have channels directly connecting one neuron to the other and electricity is being conducted from one nerve cell to the other.
- The channels are usually small tubular structures made up of protein and referred to as gap junctions.
- The gap junctions allow free movement of ions from the interior of one cell to the other.
- Only few such synapses are found in the central nervous system.
- In smooth muscles the action potentials are transmitted from one fiber and in the cardiac muscle from one cell, to the next by way of the gap junctions.
- The conduction through electrical synapse is by-way or bi-directional i.e. the signals are transmitted in either direction.

Synaptic junctions of the nervous system

- *Axosomatic :* It is the synapse between the axon of one neuron and the soma of the next. Found in spinal cord and autonomic ganglia.
- *Axoaxonic :* It is a synapse between axons present in the (mammalian) spinal cord between axons of interneurons.
- *Axodentritics :* It is a synapse between axon of a neuron and dendrite of another neuron. Present in the dorsal horn of the spinal cord.
- *Dentrodentritic :* Synapse between the dentrites of different somas. Present in the cerebellum.



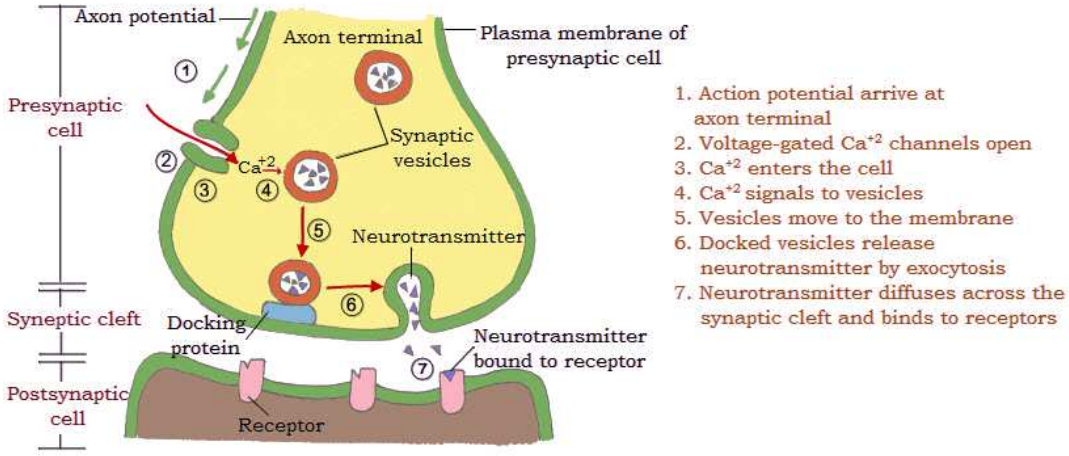
SYNAPTIC TRANSMISSION OF IMPULSE

Structural properties common to all synaptic functions

Presence of synaptic cleft

- There is no continuity of the cytoplasm across the synapse. There is a synaptic cleft (200-300 micron width) between the pre and post synaptic membranes.
- Thickened and modified portions both on the synaptic knob membrane and post synaptic membrane.
- Rich presence of mitochondria in the cytoplasm of the synaptic knob.
- Presence of synaptic vessicle (300-600 micron dia.) storing the chemical transmitter in the synaptic knob. The vessicles are absent in the post synaptic cell.

Mechanism of synaptic transmission



Transmitter release

- The membrane of the synaptic knob contains large number of voltage gated calcium channels (the other area of the nerve fiber has very few channels). The action potential travelling along the fiber depolarizes the terminal membrane. Large number of Ca^{++} along with Na^{+} from the extracellular fluid enter the synaptic knob. It is postulated that when the Ca^{++} enter the knob bind themselves with the protein receptors present on the inside of the knob membrane (synaptic membrane). The receptor sites are known as release sites. This Ca^{++} receptor combination causes the nearby vesicles to adhere and fuse with the synaptic membrane, and finally the vesicles to open to the exterior (into the cleft) by a process called exocytosis. A single action potential often opens several hundred vesicles and release their transmitter into the cleft.

Fate of the released transmitter

- It is accompanied by three ways
  - Reincorporation of the transmitter by or into the vesicles. The opened vesicle invaginates towards the inside of the knob and pinch off to reconstitute the vesicle. The appropriate transport proteins still present in the vesicle membrane shift into and concentrate the transmitter in the vesicle.
  - Resynthesis of the transmitter in the cytoplasm of the knob and their immediate absorption into the vesicles. This is a continuous process. By this mechanism, the

exhaustion of the transmitter is minimised. As the vesicles by these methods of replenishing are used again this method is referred to as recycling of the vesicles.
    - But, when both the vesicle and the mitochondria are aged they disintegrate. Both, new vesicles and the mitochondria are continually transported from the soma down the axon to the presynaptic knob. They move along the axon at a velocity from 1cm to 40 cm / day.
- Action of the transmitter on the post-synaptic membrane
    - The transmitter acts on the receptor proteins present on the post synaptic membrane. The receptor have two functional components
        - A binding component, which protrudes from the membrane to the exterior into the cleft and binds with the neurotransmitter released into the cleft.
        - An ionophore component which through the membrane protrudes to the interior of the post-synaptic membrane or neuron.

IONOPHORE COMPONENT

There are two types of ionophores

- Ion channels (chemically activated)
- Second messenger system (enzymatically activated internal metabolic system)

Chemically activated ion channels (ligand-activated channels) are of three types

- *Sodium channels:* Allow mainly Na^+ (Some K^+ as well)
- *Potassium channels:* Allow mainly K^+
- *Chloride channels:* Allow Cl^- and few other anions to pass through. The transmitter that selectively open the Na^+ channels excites the post synaptic neuron and called as excitatory transmitters. Those, open K^+ and Cl^- separately or both of these channels together inhibit the neuron, and referred to as inhibitory transmitters.

Enzymatically activated internal metabolic system

This type of receptor causes, in the post synaptic cell,

- Activation of cellular genes and such genes cause manufacture of additional receptors for the post synaptic membrane
- Activation of protein kinases and effects disintegration of receptors and their reduction in number in the post synaptic membrane. These effects on the receptor number of the post synaptic membrane can alter the reactivity of the synapse for minutes, days, months or even years. The transmitters that act on the enzymatically activated system of the cell is called as modulators. The modulators are important in operating memory processes.

The duration of action and fate of the transmitters

- When either an excitatory or inhibitory transmitter is released into the synaptic cleft and excite or inhibit the post synaptic receptors by opening the specific chemically activated ion channels and these channels remained open for only 1 to 2 milliseconds. The reason for these channels be opened through a very short duration is that the transmitter agent is rapidly removed from the cleft in the following ways :
    - By diffusion of the transmitter out of the cleft, into the surrounding fluid.

  - By enzymatic destruction, within the cleft itself.
  - By transmitter re-uptake by the pre-synaptic knobs.
- The extent to which each of these methods of removal is operating is different for each type of transmitter.

SYNAPSE-NEUROTRANSMITTERS

- A transmitter cause excitation or inhibition is determined by
  - The transmitter released at the synaptic terminal
  - The nature of the receptor present in the post synaptic membrane
- A single neuron may either be excited or inhibited. When excited, a transmitter is released at the synaptic knob which acts on the post synaptic excitatory receptors of the knob (eg. Acetyl choline). When inhibited, a transmitter is released at certain other knobs which acts on the post-synaptic inhibitory receptors of the knob (eg. glycine). Similarly, in the CNS norepinephrine released in the synaptic knob when acted on excitatory post synaptic receptors of the knob causes excitation and when the receptors are inhibitory causes inhibition.
- Usually, only a single type of transmitter substance is released at the nerve terminal by each neuron and the concept is known as *"Dale's principle"***.** However, there are exceptions in which the same nerve terminal might secrete another transmitter which may be excitatory or inhibitory, and yet another that acts as a modulator.
- For example, the *excitatory* transmitter might be *glutamate* which causes immediate excitation as it secretes one of the *neuropeptides* might also be secreted as a *modulator* and increases the number of glutamate receptors and this increases the sensitivity of the synapse for days or weeks.

The transmitter substances

- More than 30 different biochemical substances have been identified. These substances are grouped as follows :
  - Class 1: Acetylcholine
  - Class 2: Several amines
  - Class 3: Several amino acids
  - Class 4: Neuroactive peptides.
- The following are the important transmitters :
- *Acetylcholine :* It is secreted by neurons of the following areas:
- Parts of the brain such as
  - large pyramidal cells of the motor cortex
  - neurons in the basal ganglia
  - motor neurons that innervate the skeletal muscles.
- Preganglionic neurons of the autonomic nervous system.
- Post ganglionic neurons of the parasympathetic nervous system
- Some of the post ganglionic neurons of the sympathetic nervous system.
- *Glycine:* Glycine mainly secreted at synapses in the *spinal cord* and probably always acts as an *inhibitor* transmitter.
- *Gamma-aminobutyric acid (GABA):* In addition to its secretion in the neurons of the *spinal cord*, it is secreted in the neurons of the cerebellum, basal ganglia and many areas of the cortex. It always causes inhibition.
- *Glutamate:* Glutamate is secreted by the presynaptic terminals in most of the *sensory pathways* and also in many areas of the *cortex*. It is an *excitatory transmitter***.**

- *Substance P:* It is released by *pain fiber terminals* present in the *dorsal horn* of the *spinal cord*. It is also found in the *basal ganglia* and *hypothalamus*. Generally it is a *excitatory transmitter*.
- *Enkephalins:* Enkephalins are secreted by the nerve terminals, in the *spinal cord, brain, stem, thalamus and hypothalamus*. Act as *excitatory transmitters* to the *system* that *inhibit the transmission of pain.*
- *Serotonin :* serotonin is secreted by the nuclei that present in the *median rapne of the brain stem* and the fibers of the nuclei project to many brain areas such as *hypothalamus* and *dorsal horn* of the *spinal cord*. Serotonin acts as an *inhibitor* of the pain pathways.
- *The esterase:* Hydrolyses both, the noradrenaline and the adrenaline into the principal end product *3-Methoxy, 4-hydroxy mandalic acid* which is secreted along the urine. The esterase, is found in the adrenergic nerve ending.
- *Post-ganglionic:* The transferase is found predominantly in the adrenal medulla as the norepinephrine is the precursor of epinephrine in the medulla and the transferase-methylates the former to the later. On generalised sympathetic stimulation adrenal medulla releases 80% epinephrine and 20% norepinephrine into the circulation.
- In the nerve endings the enzymatic inactivation of the norepinephrine accounts only 15% of the total transmitter released on stimulation. The rest 85% is reincorporated into the storage vesicles of the nerve endings. The receptors bind the transmitter and prevents it's diffusion away from the nerve terminals, the later help the reincorporation of the transmitter.
- Examples of neurotransmitters are
    - Dopamine
    - Norepinephrine

Dopamine

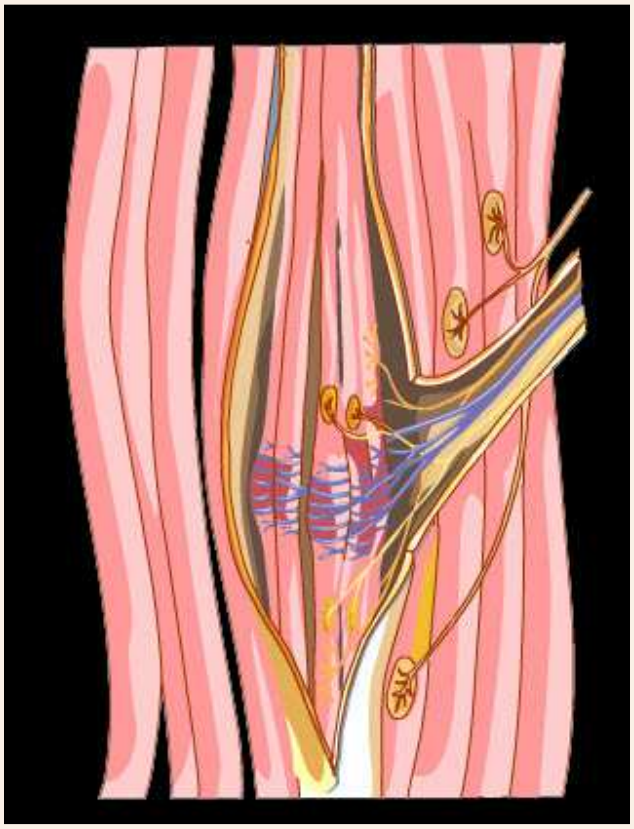
- It is secreted by the *neurons of* the *substantia nigra* and the *basal ganglia*. The effect is usually *inhibition*.
- There are about 3000 molecules of acetylcholine are present in each vesicle that store the transmitter. Enough number of such vesicles are present in the presynaptic terminal of a neuron to transmit few thousand impulses.
- Acetylcholine is synthesized from acetyl-CoA and choline with the help of the enzyme *choline acetyl transferase* which is present in abundance in the cytoplasm of the cholenergic type of presynaptic terminal. The released acetylcholine present in the cleft is rapidly split again to acetate and choline by the enzyme cholinesterase which is present in the proteoglycon reticulum filling the cleft. The choline is transported back into the knob and is reutilized for resynthesis of new ach. molecules.

Norepinephrine

- It is secreted by many neurons of the *brain stem, hypothalamus and pons*. The later sends nerve fibers to extensive areas of the brain and function in controlling the overall activity and mood of the mind. Mostly it causes excitation in the areas and however some inhibition too. Most of the post ganglionic nerve endings of the sympathetic system secrets norepinephrine and excite some organs and inhibits the others. The adrenaline acts on the other post ganglionic nerve endings.
- Catechol-O-methyl transferase and catechole-O-methyl esterase are the two enzymes that metabolise the noradrenaline and adrenaline respectively, in the cord and also helps to control the mood and cause sleep.

- Other substances such as *peptides*, other *amino acids, histamine, prostaglandins, cyclic AMP* and many others also act as neurotransmitters.

MODULE-4: RECEPTOR AND ITS TYPES



LEARNING OBJECTIVES

This module deals with,

- sensory function of skin and
- response, sensory transduction, generator potential, receptor adaptation and modality of sensation with reference to the somato-sensory nervous system.

DEFINITION OF A RECEPTOR

- Receptors are specialized as neuronal structures to detect diverse stimuli from the internal and external environment.

MECHANISM OF ACTION OF A RECEPTOR

- Incoming stimuli transduce the membrane potential into the receptor or generator potential.
- The protein structures within the sensory receptors absorb energy from stimuli to undergo conformational change which in turn activate signal transduction pathway.
- This allows change with appropriate voltage gated channel to induce the impulse generation through the proper neurotransmitters.
- Sensory reception include signal reception, signal transduction, signal amplification, signal transmission to higher centres and perception at the integrating centre.

Properties of Sensory Reception

- Primary afferent neuron is the cell whose axon carries the information to higher centres.
- Sensory receptor cells are present in the sense organs.

Generator potential vs Receptor potential

- If the potential developed by sensory receptor which also serve as primary afferent neuron, is termed as Generator Potential.
- If sensory receptor cell is different from primary afferent neuron the potential developed in the sensory receptor cell is termed as Receptor Potential.
- Mechanism of signal transduction varies in each of these situations.

Receptor characteristics

- Receptors have the ability to convert other form of energy in to electrical energy i.e. action potential, causes transduction if adequate stimuli is applied.
- They have differential sensitivities to different stimuli. The permeability of the receptor is modified and ends up in graded receptor potential.
- They show different adaptability to a stimulus and speed of adaption varies with receptors. Each receptor has specific acuity i.e. discriminative ability.

CLASSIFICATION OF SENSORY RECEPTORS

Sensory receptors are classified based on

- Function of the receptors
- Location of receptors
- Source of the stimulus
- Type of stimulus or modality
- Location of receptors

Function of the receptors

- *Phasic receptors:* Produce action potential during part of stimulation hence they initiate response. It does not provide clue for duration of stimulus.
- *Tonic receptors:* Generate action potential upon the application of stimulus and propagation of action potential goes on till the stimulus is removed. This gives information regarding the duration of stimulus. In this type, the possibility of weakening of the impulse irrespective of the maintenance of constant strength of stimulus due to prolonged exposure leading on to a phenomenon known as RECEPTOR ADAPTATION.

Location of receptors

Exteroceptors: Detect stimuli from outside the body in animals' micro or macro environment (change in temperature, air pressure). They can be divided into:

- *Superficial cutaneous sensations:* Touch, light superficial pressure, temperature, cold, warmth, and pain.

- *Deep sensations:* Receptors in the muscle tendon and joints together known as kinesthetic sensation mediated through the muscle spindle receptor and Golgi tendon organ to sense position and movement.
    - Special Sensory receptors
    - Skin receptors.
- *Interoceptors:* Detect stimuli within the body (Blood pressure, O2 uptake). They are,
    - *Proprioceptors:* Convey information from the muscles, tendons, and joints.
    - *Viscероceptors:* Carrying information from the viscera.
    - *Specialized receptors:* Baroceptors, osmoceptors, chemoceptors, and volume receptors.
    - *Baroceptors:* Present in carotid sinus and aortic arch to detect BP change.
    - *Osmoreceptors:* Present in hypothalamus to detect osmotic change.
    - *Chemoreceptors:* Sensitive to chemical change. Present in tongue to detect taste and in nose to detect smell. Hypothalamus central chemoreceptors detect pH change, blood glucose, and hormonal levels.

Source of the stimulus

- *Telereceptors:* Detect the stimuli at a distance (Photo and auditory receptors)

Stimulus or modality

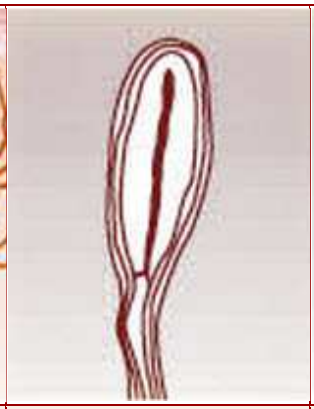
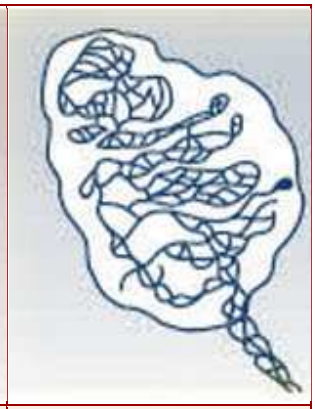
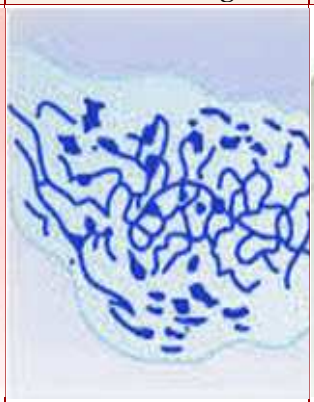
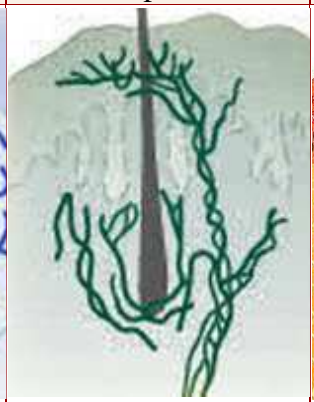
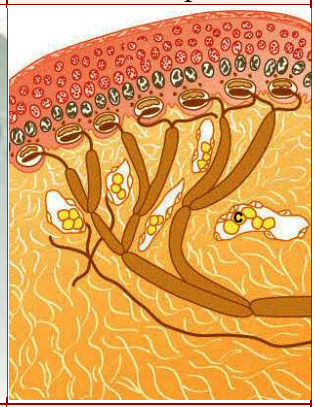
- *Photoreceptors:* Detect change in intensity of light mediating sense of vision.
- *Thermoreceptors:* To detect change in temperature.
- *Electroceptors and Magnetoceptors:* (Mostly in insects and birds). They detect electrical and magnetic field respectively.
- *Mechanoreceptors:* Involved in sensing touch, pain, hearing, balance (equilibrium).
- *Chemoreceptors:* Detect chemical signals, sense of gustation, olfaction, blood O2, tension, and change in pH (internal environment).

Location of receptors

- *Superficial cutaneous sensations:* Touch, light superficial pressure, temperature, cold, warmth, and pain.
- *Deep sensations:* Receptors in the muscle tendon and joints together known as kinesthetic sensation mediated through the muscle spindle receptor and Golgi tendon organ to sense position and movement.

CUTANEOUS RECEPTORS

- These receptors mediate sensation of touch, light, pressure, heat, and cold.
- These consist of lamellated connective tissue capsule that surrounds soft core where the axon ends.
- Pain sensations are mediated through the fine free nerve ending.

			
Expanded tip receptor	Free nerve endings	Krause corpsucle	Meissners corpuscle
			
Pacinian corpuscle	Ruffinis end organ	Tactile hair	Markel's disc

- Sensing light touch
  - Mediated by Meissner's corpuscles, Merkel's disc and basket arrangement of nerve fibers at the base of the hair follicle.
- Meissner's corpuscles
  - Located in the skin papillae beneath the epidermis. It has irregularly coiled nerve endings with capsules of connective tissue. The density of distribution is uneven.
- Merkel's discs
  - Associated with enlarged epidermal cell with small receptive field. It has group of 3 or more cup shaped discs. The discs have reticulated appearance. Nerve fibre branch out and supplies to each disc. They are found in the snout of pig and other mammals. They are used for fine tactile discrimination. They are slowly adaptive tonic receptors.
- Basket arrangement of nerve fibers or Root Hair Plexus
  - Most sensitive. They are located at the base of the hair follicle as wrappings and monitor movement across the body surface. It has short vertical nerve filaments that ends in bulged portions and gets stimulated by slight movement of hair. They adapt rapidly and are phasic receptors. Displacement of hair causes movement of hair follicle. This causes stretching of the sensory nerve ending and stimulates mechanoceptor protein stretch on the dendritic membrane.
- Touch receptors
  - Mediates through 3 qualities.
    - Tactile localization.
    - Tactile discrimination-Recognition of two different stimuli simultaneously

        - Stereognosis-Recognition of stimuli by touch without visual assistance.
- Pacinian corpuscles
    - Located in the deep layers of the skin, subcutaneous tissue, periosteum, muscle, tendon, joints, and blood vessels respond to stimuli of unequal firm, deep pressure. They have lamellated appearance similar to the cross section of an onion. They help out in recognizing the location of the stimulus with high degree of accuracy (Tactile localization). It mediates pressure sensation and they adapt quickly.

THERMOCEPTORS

- The hypothalamus of the animals has central thermoceptors to monitor their internal temperature.
- Two different receptors to mediate warmth and cold.
    - The *warmth receptors* are situated near deep blood vessels.
        - They are Ruffini end organ which can also serve to detect tactile sensation (Poly modal receptor).
        - Ruffini's corpuscles are located in the connective tissue of the skin in association with collagen fibers. Hence detect skin stretch and are involved in proprioception.
        - Warmth receptors detect change in temperature of 20° C to 47° C.
        - The frequency of action potential increases with increase in the temperature. Cold sensation is mediated by the end organ of Krause.
    - The *Cold receptors* detect and are stimulated at the temperature between 10° C to 35° C.
        - These are extremely sensitive and can detect change in even 0.5° C.
        - There are also another thermoceptor which detect painful hot stimuli. The temperature ranges above 45° C.
        - The frequency of action potential generated is in accordance with the amount of pain sensation.

Mechanism of Action

- Free nerve ending of the thermoreceptor neurons have specific thermoreceptor proteins. They are known as Thermo TRPs. The thermo TRPs are many, variable to different temperature range, and some are to detect distinct temperature. They get stimulated and act through ion channels and gating mechanism by modifying the protein configuration.

NOCIOCEPTORS

Nocioceptors

- These detect pain sensation by naked nerve endings. This sensation does not have any special structure to mediate. These nociceptors are available in the superficial layers of dermis, which are represented by non-medullated fibers. They act as protective response to an injurious stimulus.

Vibratory sense

- It is mediated by tactile and pressure receptors namely Merkel's discs and Pacinian corpuscles.

Itching and tickling sense

- It is mediated by stimulation of both tactile and pain receptors.

PROPRIOCEPTORS

- Receptors are located in the skeletal muscle, tendons, and joints known as the kinesthetic receptors. These inform the CNS about the movement and position of the limbs to maintain posture and equilibrium.
- The joint sense includes sense of movement and sense of positions.
- The sense of movement is the ability to feel the movement with closed eyes. Sense of position is to feel the position of the body in relation to space. These two are mediated by receptors in and around the joint.
- Receptors involved are the Spray nerve endings, Pacinian corpuscles, touch receptors, and free nerve endings located in the synovial membrane, ligaments, and tissues near the joint.
- The impulses generated from these reach cerebral cortex through dorsal tracts, medial lemniscus, and thalamus to end in conscious perception of the position of the body in relation to space.
- The impulses from muscle spindle organ, Golgi tendon organ, and vestibular apparatus of the internal ear do not reach cerebral cortex.
- They are concerned with reflex adjustment of the muscle tone with the involvement of spinal cord to maintain posture and equilibrium. These are known as non-sensory proprioceptive impulses as they never bring out conscious sensations.

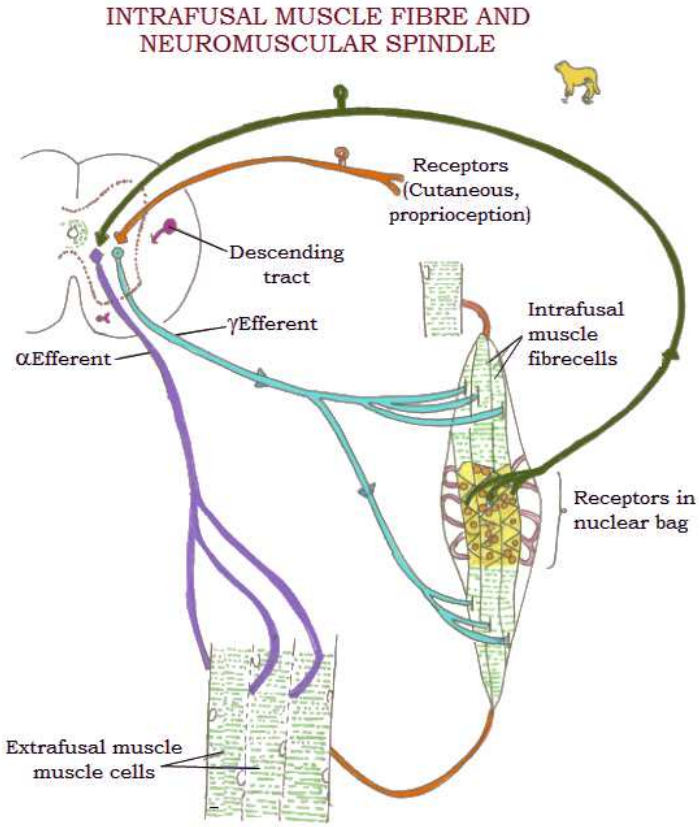
Proprioception and Respiration

- Some of these proprioceptors sends signals to the respiratory neurons about changes in the force exerted by the respiratory muscles and the movement of the ribcage.
- The functioning of diaphragm with respiratory control is also mediated by one of the tendon organs in the external costals which inhibit inspiration upon stimulation by small change in the force of contraction of these muscles.
- The stretching of diaphragm results in the inhibition of its contraction and inspiration acting through the tendon organs.
- The intercostal reflexes are significant due to the presence of numerous muscle spindle organs in the external intercostals.
- The muscle spindles help to coordinate breathing during changes in the posture and stabilize ribcage during reduction in lung compliance.

MUSCLE SPINDLE ORGAN

- These are located within the belly of the skeletal muscles.
- They are in the shape of a spindle and are made up of specialized striated muscle fibres which are known as Intrafusal fibers. These intrafusal fibers are placed within the connective tissue capsule attached to the perimysium of the skeletal muscle.

- The longitudinal axis of them are oriented parallel to that of skeletal muscle fibre which contains them.
- Structure of muscle spindles
    - There are two types of intrafusal fibers. One of them is known as *Nuclear bag* where their centre portion is interrupted and dilated which houses many nuclei. Nuclear bag region tapers towards the pole. Another intrafusal fibre is known as the *Nuclear chain*. They have uniform diameter throughout, but near the middle small, but appreciable increased diameter due to the increased number of cell nuclei.
- Nerve Supply
    - It is innervated by the Gamma motor neurons or Intrafusal fibers. The cell bodies of the Gamma motor neurons are located in the ventral horn of the grey matter of the corresponding spinal segment. The skeletal muscles are innervated by α-motor neurons. These are referred to Extrafusal fibers.



- The Gamma motor neurons are of two types,
    - Gamma plate neurons.
    - Gamma trail motor neurons.
- The Gamma plate neurons innervate nuclear bag intrafusal fibers and are tonic neurons generating sustained action potential.
- Gamma trail motor neurons supply nuclear chain intrafusal fibers. These are phasic neurons and produce action potential of short duration with high frequency.

MECHANISM OF ACTION OF GAMMA MOTOR NEURONS

- Excitation of these neurons ends up in contraction of intrafusal muscles.
- This in turn stretches the nuclear bag fibers and nuclear chain fibers.
- Sensory receptor of the muscle spindle in limited to these areas and can detect the tension upon these receptors.
- Stretch initiates depolarization in the sensory receptor membrane.
- Intrafusal fibers have no control on the tension developed in the extrafusal fibers.
- The nuclear bag fibers are tonic and sarcolemma of these are not excitable.
- No appreciable action potential is generated, hence no muscle twitch occur.
- On contrary nuclear chain fibers respond to neural excitation and are phasic in nature. It is excitable when stimulated and generates action potential in response to gamma trail motor neurons.
- This end up in the twitch type of contraction.

Nature of Sensory fibers

- Afferent fibers from nuclear bag spiral around the intrafusal muscle and form *Annulospiral or Nuclear bag endings* of the muscle spindle. These fibers are large and myelinated, responsible for initiation of myotatic reflex.
- Sensory fibers associated with nuclear chain intrafusal fibers terminate in plaque like ending namely the *Floewer Spray ending*. These fibers are small and are myelinated and are responsible for flexor reflex. These system functions to maintain posture of an animal and muscle tone of the skeletal muscles.
- The effect exerted by Gamma plate motor neurons is referred to as Gamma loop mechanism.

GOLGI TENDON ORGAN

- It is located within the tendons of the skeletal muscles.
- It is present at the junction of muscle fiber and its collagenous tendon. As they are closely associated with the tendon, the stretch of the tendon causes depolarization of the receptor organ.
- Action potential is generated within the sensory fibers which are large myelinated fibers. Golgi tendon organ act as stretch receptor. These receptors are excited by muscle contraction that imposes tendon stretch.
- The stimulation of Golgi tendon initiates excitation of α-motor neurons supplying to antagonistic muscle of the contracted skeletal muscle. These receptors initiate reflex activity to reduce the tension in the tendons of the stretched muscles.

VISCERAL RECEPTORS

Mucosal receptors or Epithelial receptors

- Located immediately below the mucosal epithelium. These are well distributed in the stomach and small intestine. They are rapidly adapting mechanoceptors, but are slowly adapting to chemical stimuli.

Tension receptors

- Located within the stomach, small intestine of cats, sheep's rumen and urinary bladder. These are slowly adapting mechanoceptors and they monitor the distention of the above mentioned structures.

Serosal receptors

- Located under the serosa or in the mesentry. They are slowly adapting mechanoceptors. They monitor visceral filling and distortion in the shape.

RECEPTOR SPECIFICITY

- Adequate stimulus is preferred, if large magnitude of stimuli that can provoke response from many receptors.
- Some receptors are designated by nature to respond to different stimuli. Example: Nociceptors to detect pain which may be due to temperature, pressure, or chemicals. These are termed as Polymodal receptors.
- Receptive field : It is the region in the sensory surface of afferent neuron which detects the stimuli to generate response upon stimulation.

MODULE-5: CENTRAL NERVOUS SYSTEM - CEREBRAL CORTEX



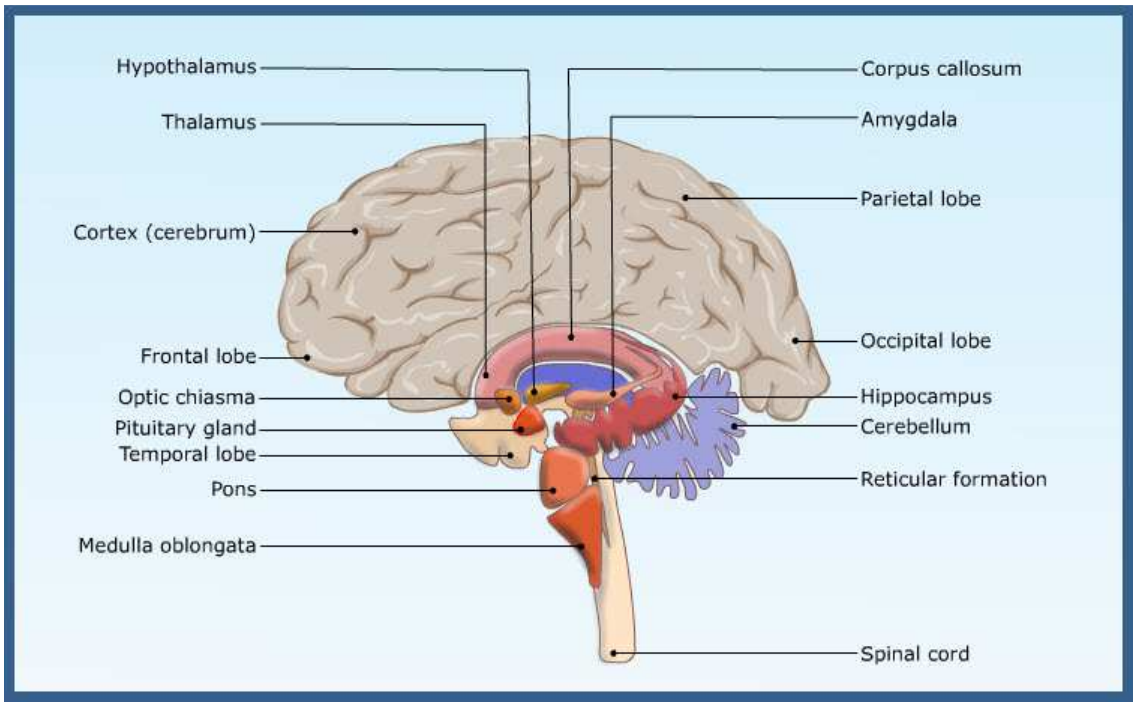
LEARNING OBJECTIVES

This module helps to understand conscious sensation voluntary movement of the animal by exploring,

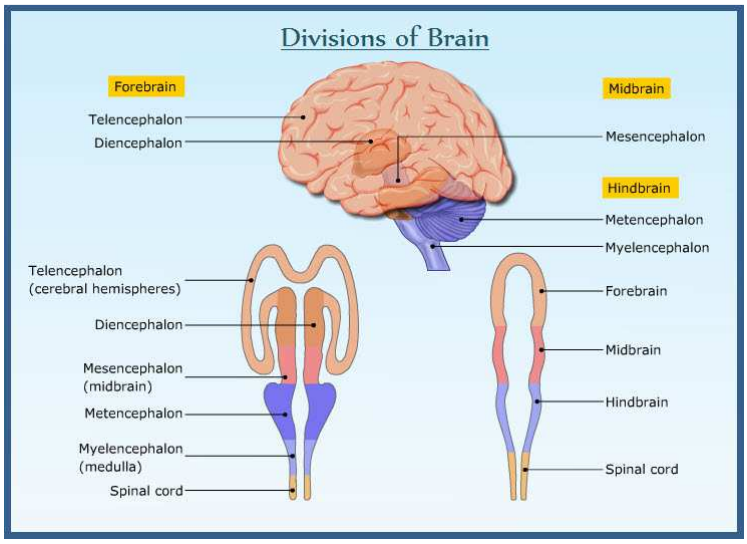
- the organization of the brain and basic anatomy of the cerebral cortex,
- different areas of cerebral cortex and
- the functions of the basal ganglia and its components namely the corpus striatum, globus pallidus and the amygdala.

CENTRAL NERVOUS SYSTEM

- The CNS comprise the brain and the spinal cord.
- The brain consists of structures situated intra cranially and includes cerebrum, mid brain, pons, medulla, and cerebellum.

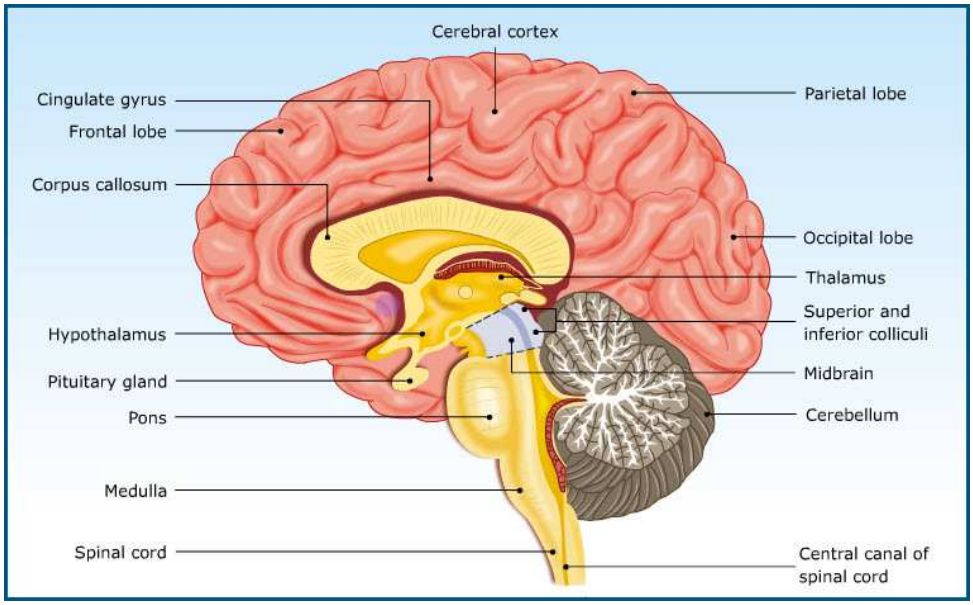


- Brain is divided into distinct regions,
  - The fore brain [prosencephalon] which is subdivide into telencephalon [two cerebral hemisphere] and diencephalon or inter brain [consists epithalamus, thalamus, subthalamus and hypothalamus, posterior pituitary and pineal glands]
  - The mid brain or mesencephalon
  - Hind brain or rhombencephalon comprising pons, medulla, and cerebellum.



CEREBRAL CORTEX

- Provides necessary neural connections and orders to skeletal muscles required for proper response based on emotion.
- As Ganglionic blanket covers the brain, has unexcelled structure function millions of nerve fibres are originated from here.
- The cerebral hemisphere is the predominant portion of the brain .
- Each cerebral hemisphere is composed of a covering of grey matter [cortex or pallium], a cerebral mass of white matter and the basal ganglia [mass of grey matter.
- The cortex is the highest integration centre in the somatic nervous system to regulate complex skeletal muscular movement and its functions are under voluntary control.
- It acts as a centre for consciousness, and storages of experience. The bilaterally situated cerebral cortex of telencephalon is divided into a primary olfactory portion called the allocortex and a non olfactory portion called the neocortex, in mammals the allocortex is often called as the limbic system



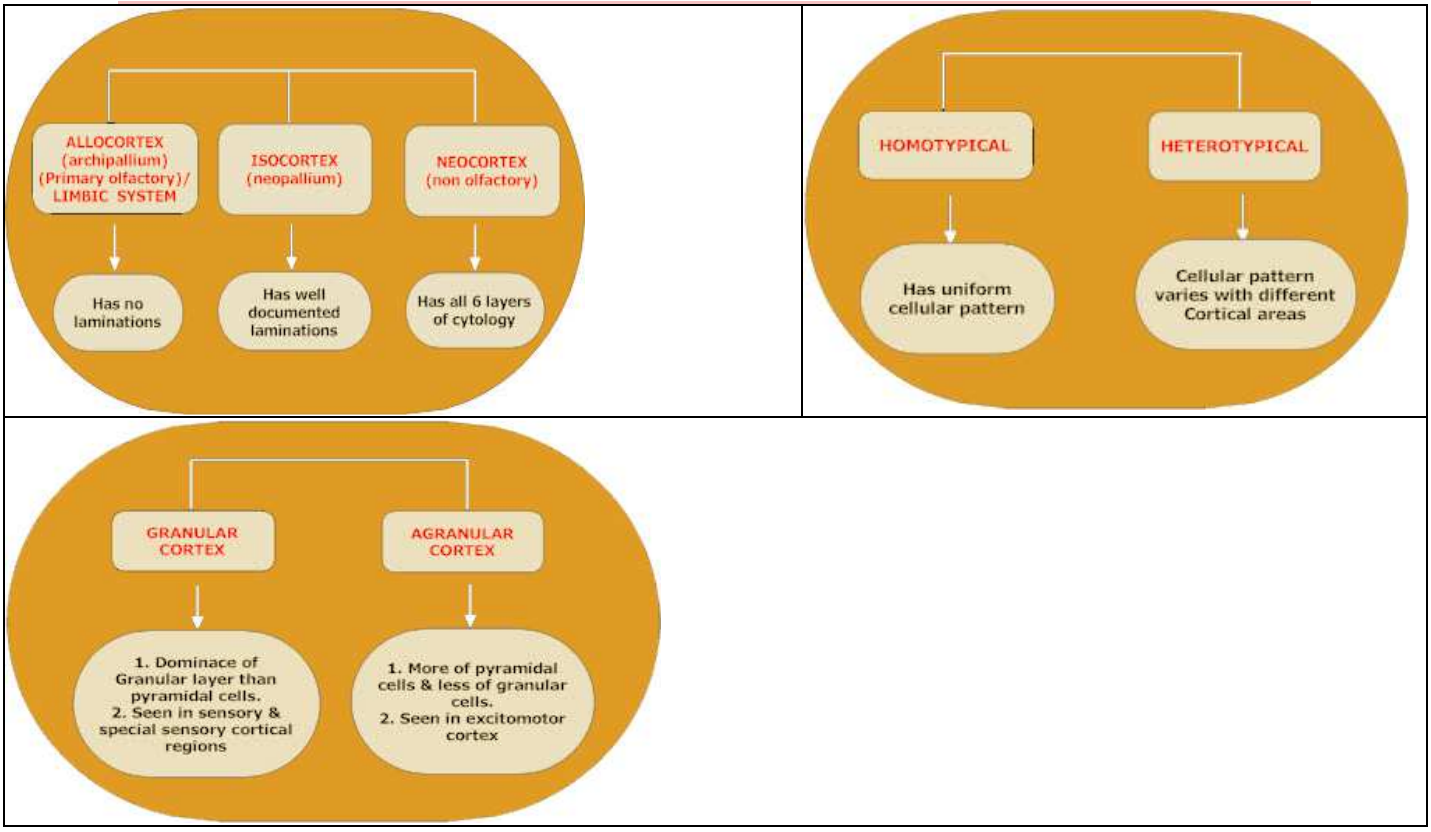
Isocortex

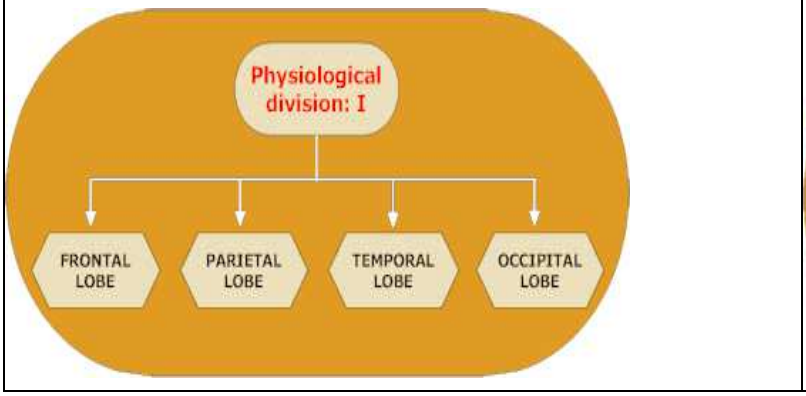
- It is the outer layer, that integrates and interprets sensory information to initiate voluntary movements.
- There are outer folds namely GYRI and grooves namely SULCI to increase the surface area of isocortex.
- Increase in surface area increases number of neurons, there inter connections enhance functional complexity.
- In other vertebrates like reptiles, birds and amphibians the neocortex is absent or poorly developed.
- Allocortex regulates emotions. Cortex has definite cellular architecture and divided into six layers.

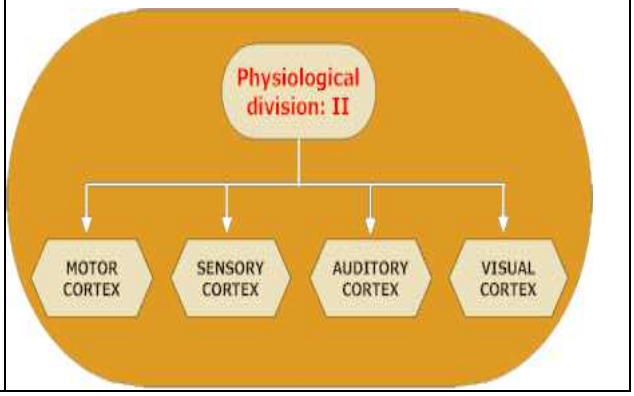
CYTO ARCHITECTURE OF CEREBRAL CORTEX

- Six functionally distinct layers have specific pattern of neuronal cell bodies.
- Variation exits between shape and density of neurons located in each layer.
    - *Molecular layer:* Most superficial layer has many dendritic, axons and glial cells from a felted mat like appearance. Few cell bodies and few connections among cells.
    - *External granular layer:* Has many different shaped [angular, polygonal, triangular, round cells] whose axon end in deep layers. Cells are many, fibers are less and integrates signals within in the cortex.
    - *External pyramidal layer:* Has pyramidal cells whose size increase gradually.
    - *Internal pyramidal layer or ganglion layer:* Has graded sizes of pyramidalcells-giant pyramidal cells viz. BETZ cells are seen .This type of cellular architecture is evident is precentral cortex. Communication with other parts like thalamus, brain stem and spinal cord.
    - *Internal granular layer:* Has number of small stellate cells.it has rich nerve fibres, horizontal crisscrossing fibres pose a white band like appearance.
    - *Fusiform cells layer:* Has small spindle shaped cells that are matted perpendicular to the surface.

- **DIVISION OF CEREBRAL CORTEX**

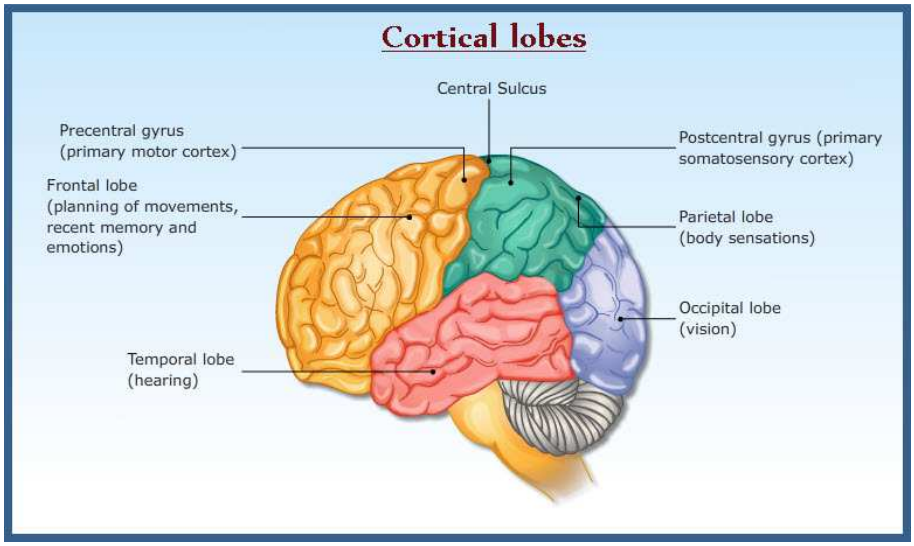






REGIONS OF CEREBRAL HEMISPHERE

- *Frontal lobe:* Voluntary motor activity, vocal ability and higher mental functions.
- *Parietal lobe:* Associated with movement, orientation, recognition, and perception of stimuli.
- *Occipital lobe:* Involved in visual processing
- *Temproal lobe:* Involved with perception and recognition of auditory stimuli.



Topographical organisation of cerebral cortex

- Visual cortex
- Auditory cortex
- Somatosensory cortex
- Motor cortex

Frontal lobe

Motor areas

- Primary motor cortex which is the exact motor cortex
    - Primary motor cortex is formed by precentral gyrus and has small narrow zone which posses BETZ cells. Primary motor cortex is a centre for voluntary movement, and initiates voluntary movement this is origin of corticospinal tract and project fibres to pons, corpus striatum red nucleus and subthalamus. Most of the skeletal movement are under the control of motor cortex.Stimulation of excitomotor cortex causes quick and phasic movement.
- Supplementary motor area
    - Present in the medial surface of the hemisphere,has greater threshold stimulation of this part causes show movement which causes some change in posture for brief period of time.There are other areas in primary motor cortex which includes secondary motor cortex which compensates loss function of primary motor cortex. Adjacent to this area is that, the one which control mainly postural and stretch reflex which on stimulation causes inhibition of reflex induced movement of cortically induced movement.It exerts such inhibitory influence.It has connection with caudate nucleus and reticular formation.
- Premotor cortex
    - Fibres of pyramidal and extra pyramidal tracts original from here, this area is concerned with
        - Excitatory effects
        - Complex coordinated movement of contra lateral side of the body.
            - Rotation of head and trunk to the opposite side simultaneous movement of flexion and extension of contra lateral limbs.
            - Adjacent to the premotor cortex is the area concerned with movement of eyes, pupillary dilation and secretion of lachrymal glands.

Prefrontal areas

- This area is concerned with socializing in animals. Stimulation of this area causes low frequency of autonomical, respiratory, circulatory renal and gastrointestinal responses.
- This area is principally connected to thalamus. Hypothalamus sub cortical and other cortical region.

Parietal lobe

- Parietal lobe have areas concerned with sensory function

Somesthetic area or primary sensory cortex

- All somesthetic sensation is perceived here except pain and crude temperature with are perceived at the thalamus.
- This is the area which helps in discriminating the qualities of external objects like size, shape, texture, weight and position of limbs in relation to space.
- Judging of the particular sensation and nature is done by this areas
- Its function includes

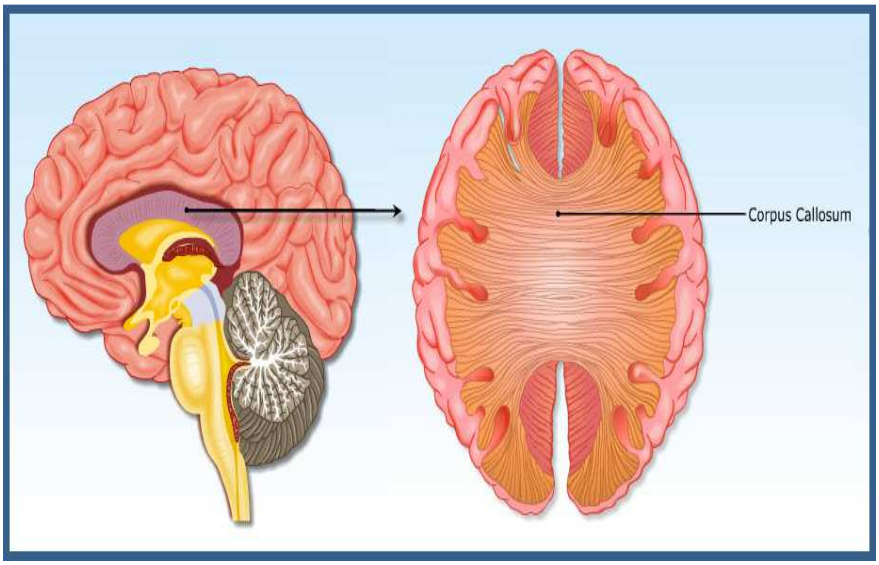
- Discrimination and localisation of touch
- Recognition and discrimination of different stimuli at different intensities
- Spatial recognition –to relate the movement and position of limbs
- Recognise the stimulation and dissimilarities [size, shape, weight]
- Taste sensation –different taste in relation with different taste receptors. This area is adjacent to the motor cortex governing the masticatory muscles.

PHYSIOLOGICAL FUNCTIONS OF CEREBRAL CORTEX

- Centre for voluntary movements-exerted by primary motor cortex Viz. Excitomotor cortex and is also responsible for highly skilled movements.
- Centre for quick and phasic movements
- Centre for postural maintenance-exerted by supplementary motor area
- Centre for stretch and postural maintenance
- Centre for major motor control-mediated by pyramidal and extra pyramidal tracts
- Centre for complex co-ordinated movements
- Frontal eye field of cerebral cortex is concerned with eye movement, pupillary dilatation and tear secretion
- Centre for reflex vocalization
- Centre for learning, memory processing and memory storage-compares past experiances and interprets the present experience
- Centre for mood, emotions and behaviour-like socializing
- Centre to receive specific sensations-crude sensations have sub cortical destination(except pain and crude temperature perceptions at thalamus level itself) finer sensations of touch temperature joint and vibratory senses are relayed to cortex
- Centre for discriminative ability
- Centre for spatial recognition
- Centre for tactile discrimination and localization
- Centre to compare and recognise relative intensity of different stimuli
- Centre to recognise similarity and differences stereognosis
- Centre for gustation-lower end of somestheti c cortex, lie next to motor cortex governing muscles of mastication
- Centre for visual senses and visuo-psychic sensations
- Centre for auditory senses and audito-psychic sensations
- Centre for olfactory sensations

CORPUS CALLOSUM

- It is a thick band consist of million of neuronal axons transversing and helping communication between two hemispheres.
- They are myelinated fibres.

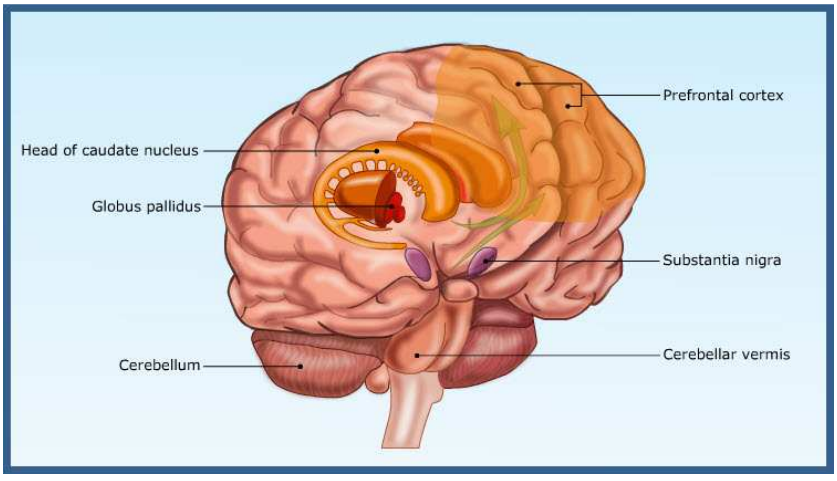


CORPUS STRIATUM

- It consists of CAUDATE NUCLEUS and PUTAMEN and initiate gross intentional movements of the body.
- It regulates the position of the body.It derives the name due to its straited appearance.

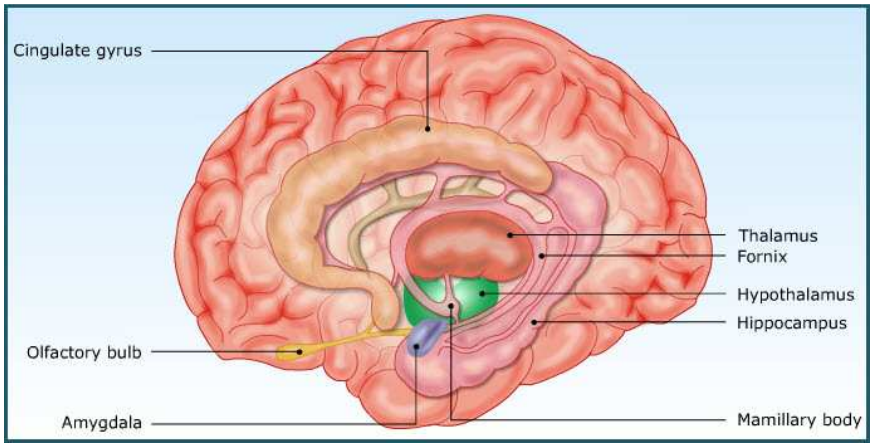
GLOBUS PALLIDUS

- Most of the outflow of signals from the basal ganglia is channelled through globus pallidus en route to cortex or lower brain centres.
- Its activity is associated with tonic contractions of voluntary movements to effect discrete activities. The Globus pallidus and the putamen collectively form a structure, the Lenticular nucleus which resembles lens.



LIMBIC SYSTEM

- Consists of cortical and subcortical areas which are structurally and functionally related to forebrain. It forms a ring of forebrain structures that surround the brainstem and are interconnected by intricate neural pathways.
- The components of the limbic system are hippocampus, amygdala and cingulate gyrus.
- It is concerned with emotions and visceral functions viz. VISCERAL BRAIN. It helps out in basic survival and sociosexual behavioral pattern. The subcortical relay nuclei comprises of septal nuclei, mammillary bodies, medial and lateral hypothalamic nuclei associated with limbic system.
- Functionally, it includes lobes of cerebral cortex, basal nucleus, thalamus, and hypothalamus.
- Limbic system has certain areas designated as 'reward and punishment' centres which functions in the programmed way based on the previous experiences.
- Limbic system is not well defined in birds.

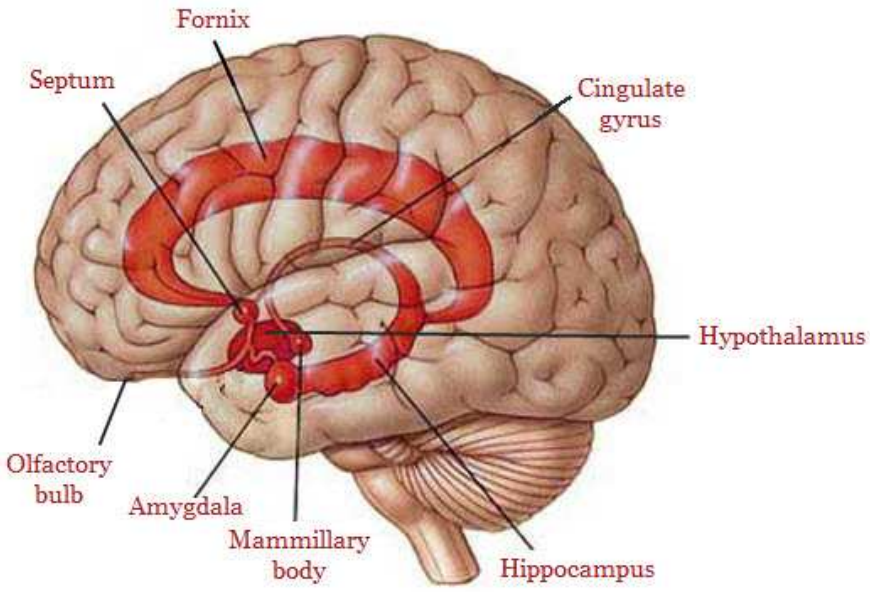


Functions of limbic system

- Limbic system is the neural substrate for emotional experience and expression through somatic and visceral changes.
- Limbic system is involved in 3 levels of behavior.
    - Drive
    - Emotions.
    - Goal directed behavior. (Motivation: It is the ability to direct behaviour towards specific goals and is called as goal direct behaviour.)
- Expression of drive is concerned with rhinal system comprising of amygdala, prepyriform cortex and septal nuclei. Emotional behavior is mediated via hippocampus. Goal directed behavior is mediated by cingulate gyrus, prefrontal cortex, and the hypothalamic structures.
- Visceral automatic control: Regulation of BP, respiratory and vasomotor mechanism.
- Olfactory processing.
- Neuro-endocrine control.
- Formation of memory process (Short-term-Long term memory by hippocampus)

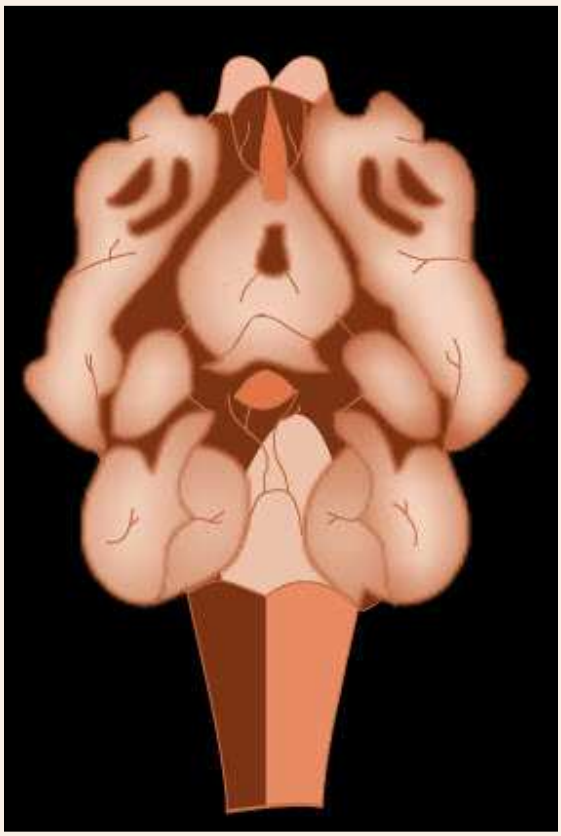
AMYGDALA

- It morphologically forms the part of the basal gangalia and has bi-directional connections with hypothalamus. Amygdala receives input signals from all the portions of the limbic cortex, the orbital surface of the frontal lobe, cingulate gyrus, hippocampal gyrus, neocortex of the temporal, parietal and occipital lobes especially from the auditory olfactory and visual associated areas. Hence it is referred to as window of the limbic system. This forms the behavioural awareness area that operates at a subconscious level. A portion of the amygdala has olfactory function.



- Amygdala in turn transmits signals back to the cortical area, hippocampus, septum, thalamus and hypothalamus.
- Septum functions in sexual and maternal behavioural pattern. Stimulation of amygdaloid nuclei causes a pattern of rage, escape, of being punished and reactions of reward and pleasure.
- Functions: It forms the part of the visceral control centre of the limbic system.
    - Located in the interior of the basal temporal bone.
    - Concerned to process inputs that generate fear response and to avoid danger.
    - Deals with emotional memories and prepare the animals to set the goal for fight-flight system.
    - The circuits of amygdala releases GABA, an inhibitory neurotransmitter, so as to initiate fight responses after receiving stronger stimuli.
    - Amygdala due to the presence of GABA filters unthreatening stimuli.

MODULE-6: INTERBRAIN, THALAMUS AND HYPOTHALAMUS



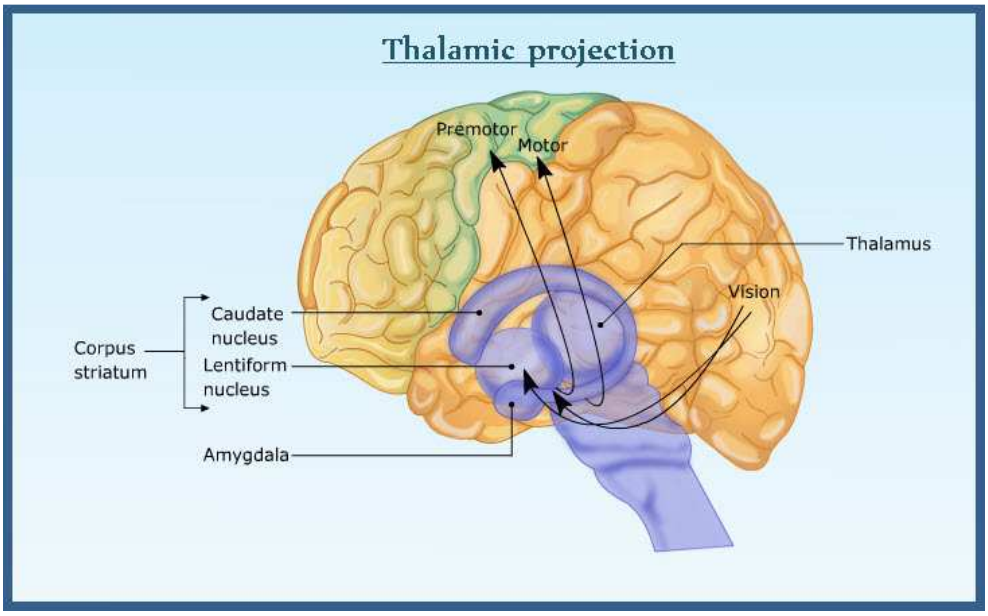
LEARNING OBJECTIVES

This module assists to equip the knowledge on,

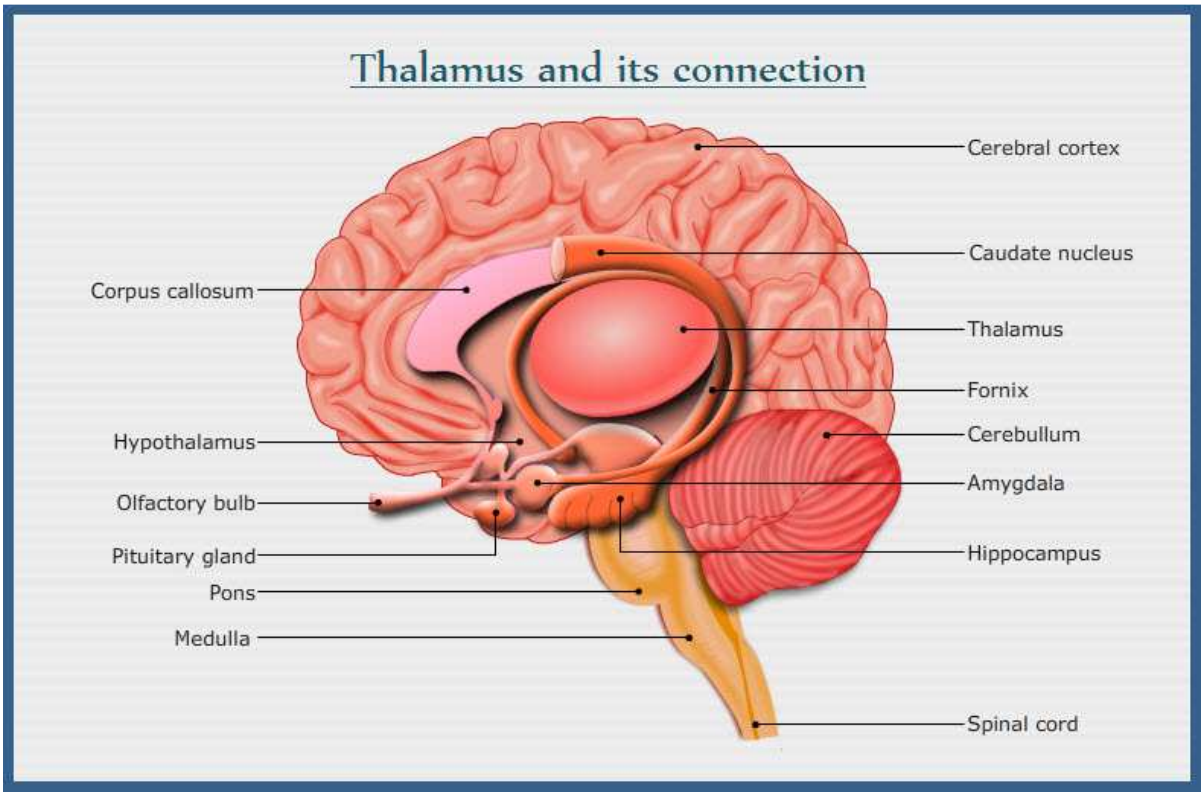
- the function and components of the interbrain,
- the projection systems and functions of the thalamus and the limbic system,
- the importance of hypothalamus and its control over the vegetative functions of the body and
- the function of the pineal gland in regulating photoperiod.

THALAMUS

- It is a sensory relay station and is important in motor control and play a major role in control of cerebral excitability.
- Located deep within the brain near the basal nuclei forms the part of diencephalon.
- Represented by large grey mass related medially to the third ventricle and at the base of lateral ventricle.
- It mediates connections between cerebral cortex, basal ganglia, hypothalamus, cerebellum, brainstem, reticular formation, and spinal cord. Physiologically thalamus can be considered to have three systems.
- Specific projection systems-transmits information to and from cortex.
- Nonspecific or diffused projection system-has widespread action on cortex.
- Acts as pacemaker that regulates the cortical activity.
- Nuclei with subcortical connections.



- *Specific projection system:* Receives specific afferents from the periphery, which are processed in the thalamus and relayed to specific areas of the cortex for conscious perception.
- *Lateral geniculate body:* Receives the visual signals through optic tract and projects the outputs to the visual or the striated area of the cortex.
- *Medial geniculate body:* Receives the auditory signals through the cochlear nerve and projects the efferents to the temporal lobe of the cortex.
- *Ventrobasal complex:* Receives sensory impulses through medial lemniscus, spinothalamic and trigeminaol nerve and projects its output signals to the somesthetic and gustatory areas of the cortex.
- *Posterior nuclear group:* Receives the pain sensations through medial lemniscus and spinothalamic tract and projects them to the cortex.
- *Ventrolateral nuclei:* Acts as a motor relay station that receives the input signals from the basal ganglia and cerebellum and projects the efferents to the motor cortex.
- *Anterior nuclei:* Receives input signals from the mammillary body of the hypothalamus and projects the efferents to limbic cortex and cingulate gyrus of the cortex.
- Non - specific projection system: It forms a widespread action on the cortex and thereby regulates cortical activity and consciousness.



Functions

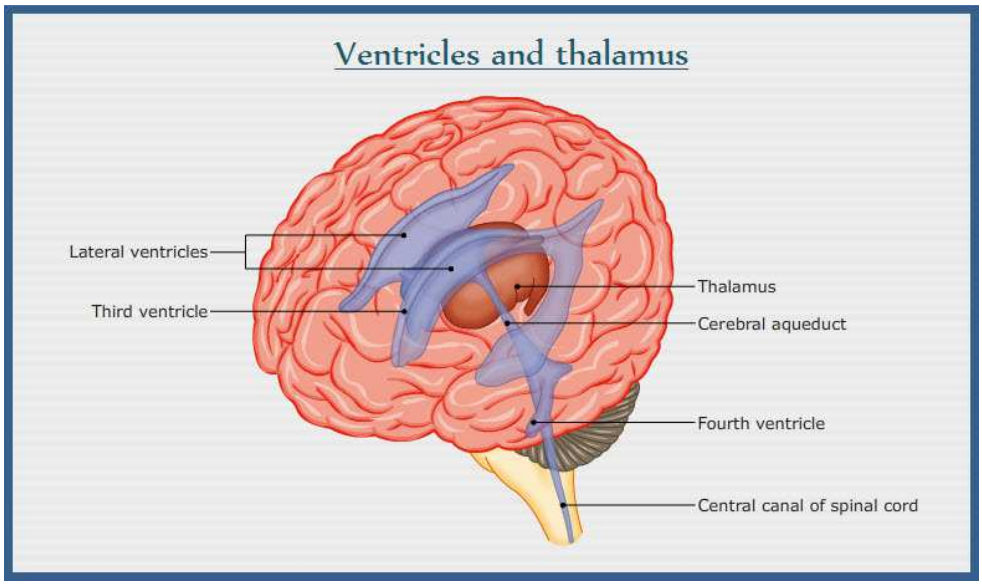
- It acts as a synaptic integrating centre for preprocessing of all sensory information to cortex except olfaction.
- It screens the sensory impulses to appropriate areas of the somatosensory cortex.
- It is capable of recognizing crude awareness of various sensations, but unable to identify and distinguish strength, location, and intensity.
- Somesthetic sensory impulses reach the thalamus through medial, spinal lemniscus to posteroventral nucleus of the thalamus.
- Taste fibers are relayed in arcuate nucleus.
- Arousal, alert, or wakefulness is maintained by part of the thalamus due to the fibers from forebrain and midbrain reticular formation and hypothalamus.
- Act as higher centre for crude sensation and of pain and temperature.

FUNCTIONS OF THALAMUS

Nonspecific nuclei

- Plays a dominant role in the control of cerebral excitability by antagonizing ascending reticular formation.
- Nonspecific thalamic nuclei are responsible for the production of α-rhythm in an awake, resting animal.
- These nuclei integrate the sensory inflow and elicit motor activity.
- These nuclei maintain relationship with hypothalamus and hence modify autonomic and endocrine functions.

- These nuclei functions in the crude conscious perception of sensory information.
- These nuclei represent the brainstem association centre and control many involuntary adjustments according to the environmental influences.



Specific nuclei

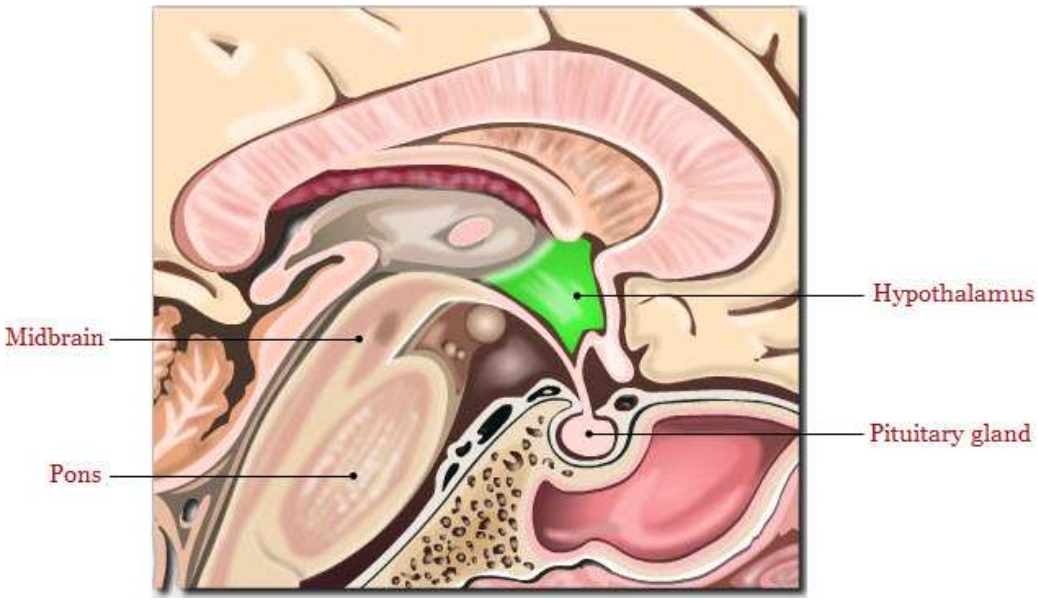
- Functions to relay sensory or motor information to and from the cerebral cortex, hypothalamus, and basal ganglia, and the reticular formation.
- Many sensory information reach here after converging through ascending systems like spinothalamic tract, lateral or medial lemniscus. Because of the convergence of mixed sensation, which lack specific receptor organ or neural pathway could be perceived.
- Specific sensation like from visual system project to lateral geniculate body and auditory system project to medial geniculate body of thalamus.
- Pain signals end in posterior group of nuclei.
- Tactile and gustatory pathways project to ventroposterior nuclei.
- Ventrolateral nucleus act as relay nucleus for sensory information from basal ganglia and cerebellum. They control voluntary motor activity.
- A largest nucleus of thalamus is dorsomedial nucleus functions to relay two way system of communicating information from prefrontal cerebral cortex and hypothalamus. They receive information from putamen, caudate nucleus, amygdaloid, and para-olfactory area of cortex.

HYPOTHALAMUS

- Responsible for involuntary internal response of body systems and prepares for appropriate action in a particular emotional state. It occupies the ventral portion of the brain stem.

- It serves as the major control of visceral motor activity namely CONSUMMATORY MOTOR BEHAVIOUR.

- The hypothalamus contribute to rage behaviour, affects emotions like fear response, arousal, exploration and orientation.
- The hypothalamus and its related structures control many internal conditions of the body such as body temperature, osmolality of the body fluids, the drive to eat and drink etc. known as "the vegetative functions of the body".
- It receives connection from amygdala, hippocampus, dorsomedial nucleus of thalamus septal area, ascending recticular formation, globus pallidus and cerebral cortex.



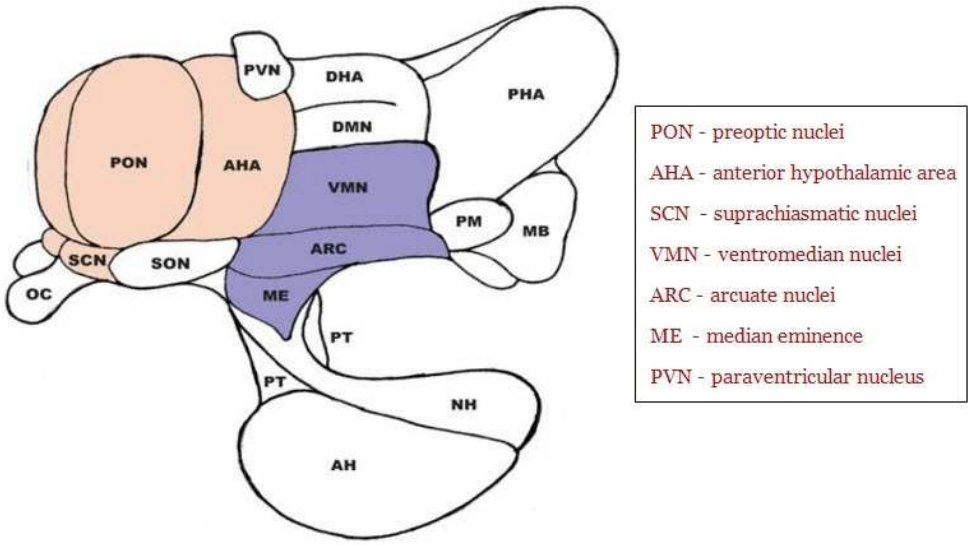
- Hypothalamus sends output signals downward through the brain stem reticular formation of mesencephalon, pons and the medulla as autonomic outflow and upward to the areas of the anterior thalamus and limbic cortex.
- It regulates emotional and visceral motor activity by passing efferent fibres to dorso medial nucleus of thalamus, anterior group of nuclei, superior colliculus and tegmentum (mid brain).
- The hypothalamus includes optic chiasma, tuber cinereum, mamilary bodies, median eminence, infundibulum and neurohypohysis.

It forms that portion of the rhinencephalon (limbic system) which are closely related to the behaviour.

VEGETATIVE FUNCTIONS OF HYPOTHALAMUS

- Cardio vascular regulation
  - Stimulation of the posterior and lateral hypothalamus increases the arterial pressure and the heart rate, but the stimulation of the preoptic, area depresses the blood pressure and heart rate.
- Regulation of body temperature
  - The preoptic and anterior hypo-thalamus is concerned with regulation of body temperature, panting and sweating. Temperature is monitored peripherally and centrally by thermosensitive neurons . Hypothalamic receptors sense both cold and warm and monitor core body temperature. These receptors are present in the

preoptic area of the anterior hypothalamus . Posterior hypothalamus integrates peripheral and central thermal sense and modify the heat loss or heat generating mechanisms. Hypothalamus is more sensitive to central thermoceptors.

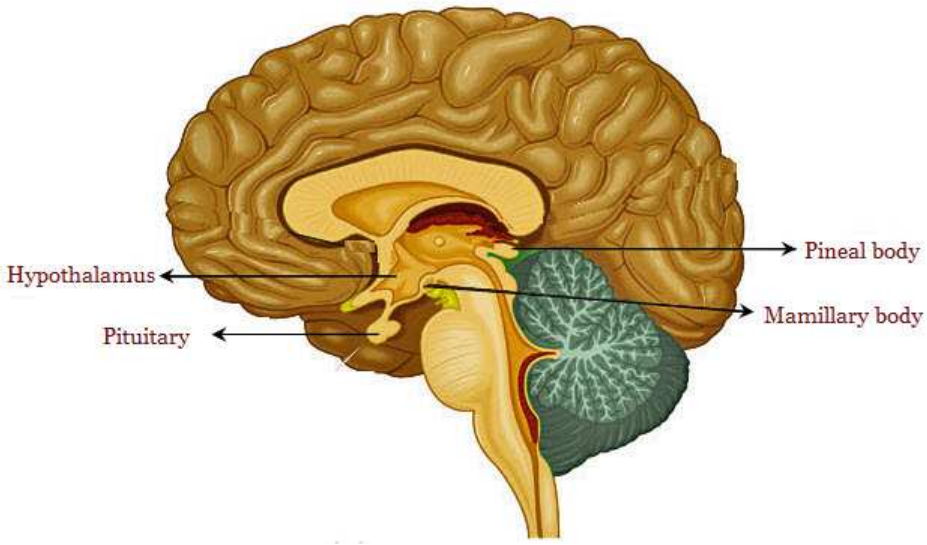


- Regulation of body water
  - Hyperosmolality of the body fluid stimulates the thirst centre in the lateral hypothalamus. The hyperosmolality (hemoconcentration) in turn stimulates the ADH secretion from the supraoptic nuclei, that aids in water reabsorption from the collecting tubules and ducts of the kidney.
- Regulation of thirst and water balance
  - Hypothalamus senses the external condition and control thirst. This region of the brain detects circulating hormones due to hyperosmolarity posed by Na+ as they lie in the zone of incomplete blood brain barrier system. Subfornical organ (SFO) , Organum vasculosum of lamina terminalis (OVLT) in the median preoptic nucleus are collectively known as circumventricular organs that detect osmotic conditions. Detection is mediated by both osmotic and hormone receptors. Osmotic receptors monitor the osmolarity of CSF that cushions the hypothalamus. Hormone receptor to bind Angiotensin II in this area exerts renal water sparing action. Perception of osmolarity is mediated by above said mechanism. They send signals to the thirst center of dorsomedial hypothalamic nuclei and drives thirst behaviour.
- Regulation of uterine contraction and milk ejection
  - Neuroendocrine reflex stimulation of paraventricular nuclei, (entry of fetus into the birth canal stimulates sensory nerves, which passes via the spinal cord to hypothalamus and releases oxytocin) causes increased uterine contraction. Tactile stimulation of udder sends sensory impulses to hypothalamus causing release of oxytocin, which aids milk ejection by stimulating the contraction of the myoepithelial cells surrounding the alveoli of the mammary gland.
- Regulation of feed intake
  - The ventromedial region of the hypothalamus, is referred as feeding centre. Stimulation of this centre causes hyperphagia, whereas the lateral region of hypothalamus known as satiety centre, on stimulation reduces feeding behaviour. Stimulation of optic chiasma area of hypothalamus causes increased gastric secretions.

- Sleep and wakefulness
  - Stimulation of the anterior hypothalamus induces sleep, while the stimulation of mammillary body (caudal hypothalamus) activates the ascending reticular activating system (RAS) to produce wakefulness.
- Reproductive functions
  - Median eminence of the hypothalamus releases GnRH in a pulsatile manner, which regulates the FSH and LH release from the pituitary gland. Estrogen and progesterone acts on the hypothalamus and causes negative feed back effect on GnRH release; the estrogen also produces a positive feed back effect on GnRH release. Thus the functions of ovary and testes are regulated by the hypothalamus.In seasonal breeders and birds hypothalamus plays a key role to effect photoperiodism and thereby the breeding performance is regulated.
  - Melatonin released from the pineal gland regulates the duration of light, and activates/inhibits GnRH release from hypothalamus in short and long-day breeders. Until puberty, hypothalamic GnRH release is highly sensitive to oestrogen and is kept in check by the negative feed back effect of oestrogen. At the time of puberty, the sensitivity of hypothalamus to oestrogen is reduced resulting in pulsatile GnRH release and puberty occurs.
- Regulation of anterior pituitary functions
  - Hypothalamus controls the anterior pituitary gland secretions through neuro-secretory peptides and amines known as the releasing and inhibitory hormones. E.g. TRH, GHRH, GHIH, PIH, CRH, GnRH, PRH. The TRH and CRH are released from paraventricular nucleus, GnRH from preoptic area, GHIH from anterior hypothalamic area, GHRH and PIH from arcuate nucleus. These regulatory hormones pass from the hypothalamus to adenohypophysis through the hypothalamic-hypophyseal portal system.
- Regulation of emotional status
  - Hypothalamus controls both fear and aggressiveness. The posterior hypothalamus regulates sympathetic response of the emotional status via catecholamines.

PINEAL GLAND

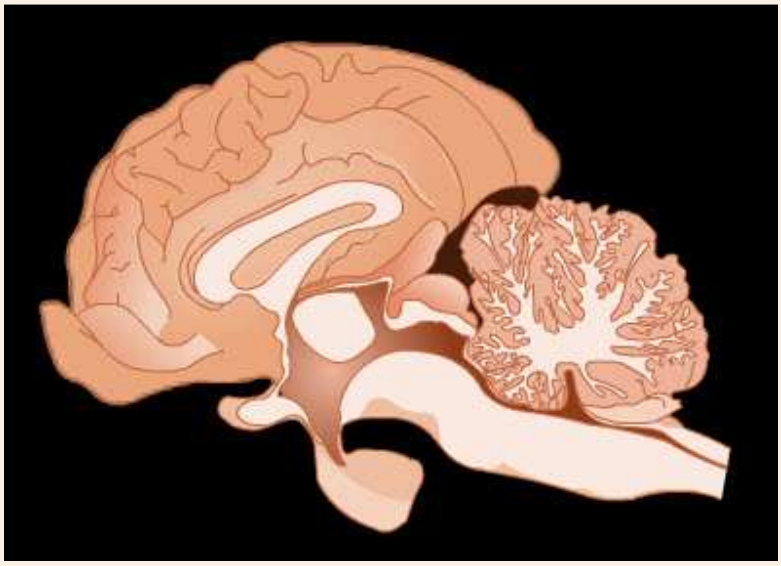
- It is the main translator of photoperiodic effect in animals.
- The pineal gland produces a hormone, melatonin in response to darkness. Light passes from the retina to supra chiasmatic nucleus of the hypothalamus, superior cervical ganglia and to pineal gland.
- The melatonin inhibits gonadal activity.
- Cat and horse are positively affected with increasing light period and goat and sheep are positively affected by decreasing photoperiod.
- The pineal gland acts to relay light-dark information to the hypothalamus and regulates seasonal breeding in animals.



INTERBRAIN (DIENCEPHALON)

- Structures including epithalamus, subthalamus (ventral area), thalamus (dorsal area), hypothalamus, pituitary gland and pineal gland constitute the interbrain.
- The epithalamus functions as olfactory centre. Subthalamus regulates motor activity through reticular nucleus.

MODULE-7: MID BRAIN AND CEREBELLUM



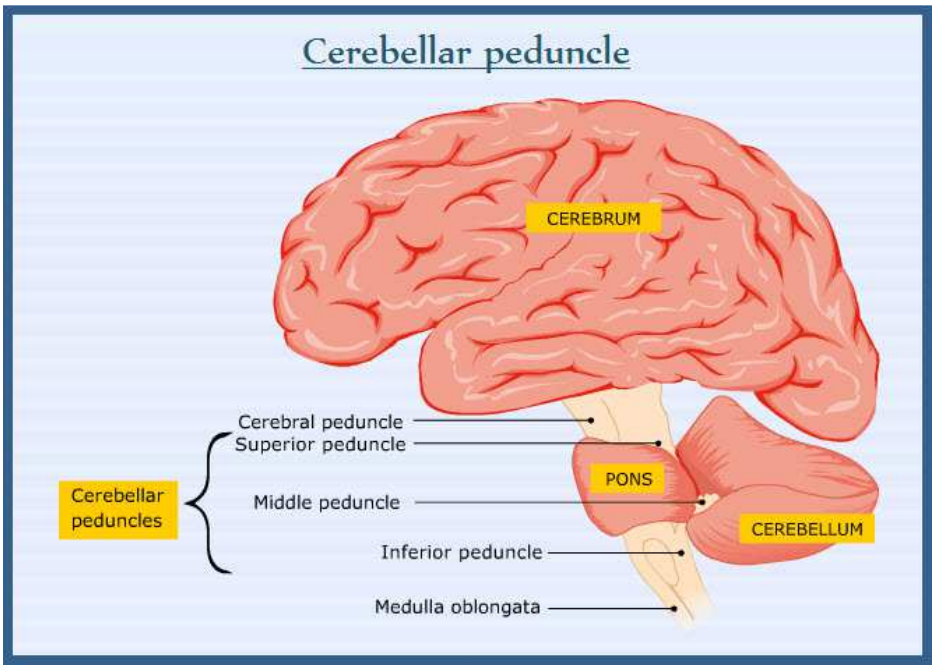
LEARNING OBJECTIVES

This module enables the learner to understand,

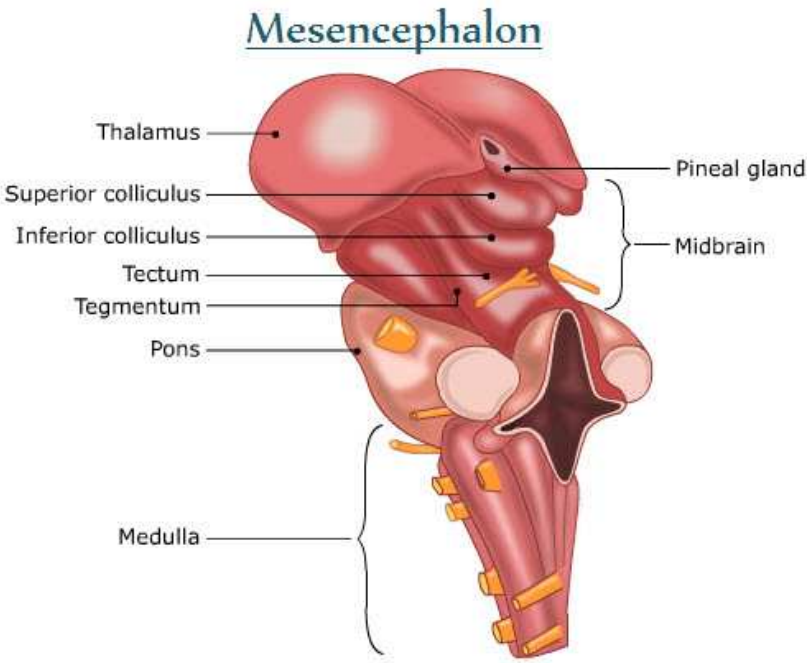
- the parts of the midbrain and the significance of the corpora quadrigemina and cerebral peduncle and
- the basic anatomical organization of the cerebellum and its functions.

CEREBELLUM

- It is encased in cerebellar hemisphere and is located the back of the brain.
- Responsible for motor coordination by integrating sensory inputs from receptors located from muscles eyes and ears with motor orders of the fore brain.
- It maintains equilibrium and posture.
- Present in the anterior part of the brainstem, dorsal to the pons and medulla, and caudal to the cerebral cortex so called as little brain. It contains many neurons.
- It is present in all mammals. Jawless vertebrates lack cerebellum. It is concerned with proper positioning of the body in the space. It maintains subconscious coordination of the movement and coordinates fast phasic motor activity.
- Cerebellum is extensive in birds where they need to balance while flying.
- The anatomy of the cerebellum is as follow,
    - Made up of outer grey matter layer known as the CEREBELLAR CORTEX.
    - White matter of cerebellum is represented by stalks known as CEREBELLAR PEDUNCLE.



    - Three pairs of the cerebellar peduncle functions as sensory and motor fibers to and from cerebellum and through which it is attached to the brainstem.
        - Brachium conjunctivum-to thalamus and mesencephalon. It is the rostral cerebellar peduncle.
        - Brachium pontis-to pons. It is the middle cerebellar peduncle.
        - Restiform body-to medulla and spinal cord. It is the caudal cerebellar peduncle.



- Group of cerebellar nuclei known as subcortical nuclei is embedded in the cerebellar white matter, whose axons leave the cerebellum. They are dentate nucleus, globus emboliformis and fastigial nucleus.

CEREBELLAR HISTOLOGY

Cerebellar cortex shows 5 types of neuronal cells. They are similar to cerebral cortex in this region.

- Stellate
- Basket
- Golgi
- Granule
- Purkinje cells

They are arranged in 3 layers

- *Outermost layer:* *Molecular layer*. Made up of 2 types of cells. Outer stellate cells where axons confines to this layer. Basket cells whose dendrite remain in this layer and axons reaches Purkinje layer and make synaptic contact with cell bodies of Purkinje cells. Axons of the stellate cells synapse with primary and secondary dendrites of Purkinje. It has axons of granule cell (parallel fibers), neuronal dendrites of deep layers, few inhibitory neurons, stellate, and basket cells.
- *Middle layer:* *Purkinje cell layer*. It is composed of simple layer of ovoid/rounded cell bodies of Purkinje neurons. Uniqueness of those cells are extensive branching of their dendrites (multi polar neurons) which are oriented at right angles to parallel fibers. These cells are the output neurons, uses GABA as their neurotransmitter hence they are inhibitory in nature. Axons of Purkinje cells travel outside the cerebellar cortex to reach the cerebellar nuclei. Purkinje cell comprises of primary, secondary, tertiary, and quaternary dendritic branches. The primary and secondary are smooth but tertiary and

quaternary have numerous synaptic spines and are rough surfaced. They synapse with the neurons of the granular layer beneath this.

- *Inner layer: Granule cell layer*. It has cells which have darkly stained nucleus and little cytoplasm giving this layer granular appearance. It has many small granule cells with few Golgi neurons. The Golgi neurons are located at the junction between granular and Purkinje layers. Dendrites of granular cell limit within this layer and receive synaptic influence from sensory fibers to the cerebellar cortex (mossy fibers) and axons of Golgi neurons. Axons of granular cell project to the molecular layer through the tertiary and quaternary Purkinje branches and synapse their spines. Among the cells of the cerebellar cortex, only the granule cells have excitatory neuron.

Cerebellum consists of 3 functionally distinct parts in birds and mammals

Vestibulocerebellum/Archicerebellum

- The classification is based on the embryological basis. The archicerebellum is an initial evolutionary structure. It deals with control of tonic or postural reflexes of the spinal cord. Important for maintenance of balance and coordination with eye movement. It mediates coordination of vestibular reflexes.
- It is present in the flocculonodular lobe.
- It receives sensory information from vestibular system. It sends back orders to vestibular nuclei via cerebellar nuclei (fastigial nuclei), to coordinate the axial and proximal muscles controlling balance. These orders are mediated by vestibulospinal tracts which in turn help to coordinate head and eye movements.

Cerebrocerebellum/Neocerebellum

- It involves in planning and initiating voluntary activity.
- It provides input to motor cortex concerned with graceful, intricate, appropriately timed voluntary movements. It occupies most of the cerebellum representing in the lateral cerebellar hemisphere.
- It receives inputs predominantly from supplementary and premotor cortex via the corticopontine cerebellarly system. They have no clue of the sensory information from the peripheral receptors. Their output goes back to motor cortex from a communication loop for planning and preparation of movement ahead of actual execution and enhances appropriate timed transitions of a movement sequence.

Spinocerebellum/paleocerebellum

- This portion represents the medial portion of the cerebellum.
- It is concerned with maintenance of muscle tone, coordinates skilled voluntary movements. It controls the tonic or postural motor activity of the spinal cord. They are concerned with accurate timing of muscular contractions. It maintains coordination between muscles of different parts to complete an action, by receiving information from peripheral somatosensory system that constantly inform higher centres about the body movement.
- The cerebellum receives information from higher centres about the orders sent and at the same time they receive information from muscle spindle, vestibular, visual, and other sensory receptors about the movement the body is performing.
- It predicts the position of the body in next minute and to make adjustments needed. This mediates the intentions of the motor cortex to the execution of muscular movements

happening. It is important to mediate the corrections on the intended movement. The general pathway is that it receives inputs from muscle and cutaneous receptors, spinal reflex circuits through the spinal cord. The information from corticospinal, descending brain stem pathway, primary motor cortex, and somatosensory cortex reaches this area. The information from high centres encode the movement to be executed by the skeletal muscle and sensory inputs inform what exactly executed by the muscles.

- Output from these areas are mediated through deep cerebellar nuclei, reticular nuclei of brainstem that controls antigravity muscles, red nuclei of the brainstem which controls musculature of distal limbs, to adjust the timing and coordination of the movement and muscle tone. Inhibition of this area, activity of medullary lateral vestibular nuclei is enhanced by way of fastigial nucleus. This nucleus is one of the cerebellar nuclei that lie over the roof of the fourth ventricle. Enhanced activity of the medullary nucleus excites α-motor neurons of antigravity muscles of the limb and results in rigidity of the antigravity muscles.

The degree of tonic inhibition exerted on the fastigal neurons and medullary neurons is depended on orders from cerebral cortex (cortico-ponto cerebellar pathway), basal gangilia and from cutaneus and propioceptive systems of the body. Spinocerebellum exerts flexible controls over the skeletal muscle tone on the posture.

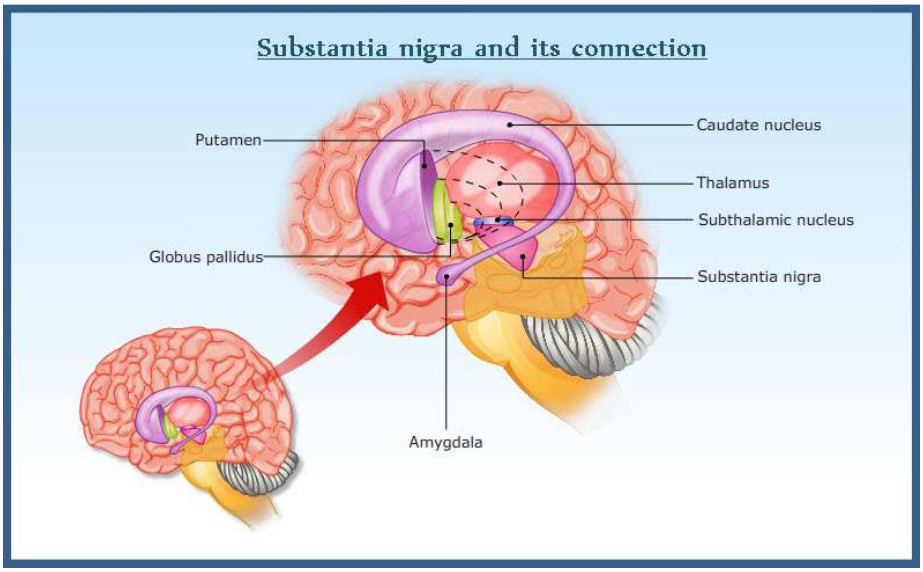
INPUT TO CEREBELLUM

- Represented by group of fibers namely Mossy fibers and Climbing fiber axons
- They use excitatory neurotransmitter to cause excitatory post synaptic potential in the subsequent cells of the cerebellar cortex. The climbing fiber axons exhibit one to one relationship with Purkinje cells.
- These fibers carry information from higher motor centres and peripheral sensory receptors.
- Both climbing and Mossy sensory fibers are excitatory. When peripheral receptor or CNS is stimulated, it causes activation of these two fibers. The CNS organization facilitates early excitation of Mossy fibers 5-10 millisecond prior to the excitation of climbing fibers. Hence rapid initiation of the cerebellar cortex activity is appreciated.
- Excitation of the granule cells and Golgi neurons of cerebellar cortex by Mossy fibers.
- Depolarization of dendrites of Purkinje cells, outer stellate, basket cells, and Golgi neurons through parallel fibers from excited granule cells.
- Mossy fibers influence the duration of discharge of impulse in the granule.
- Inhibition of Golgi neurons are potentiated by stellate and basket cells by way of parallel fibers.
- Degree of inhibition is inversely related to the excitation of climbing fibers that reaches the cerebellar cortex and efferents of the Purkinje cell reflect the cortical activity.
    - Cerebellar functions are mediated through disfacilitation of the structures of CNS.
    - α-RIGIDITY: It is the rigidity that results from hyper excitation of the α-motor neurons of the spinal cord.
    - γ-RIGIDITY: It is the result of the hyper excitation of the γ-motor neurons of the spinal cord.
- In birds, cerebellum is concerned with regulation of spinal cord and brain stem mechanism to help in flying and maintenance of equilibrium. Cerebellar pathology is associated with persistent rigidity of the antigravity muscles.

MIDBRAIN

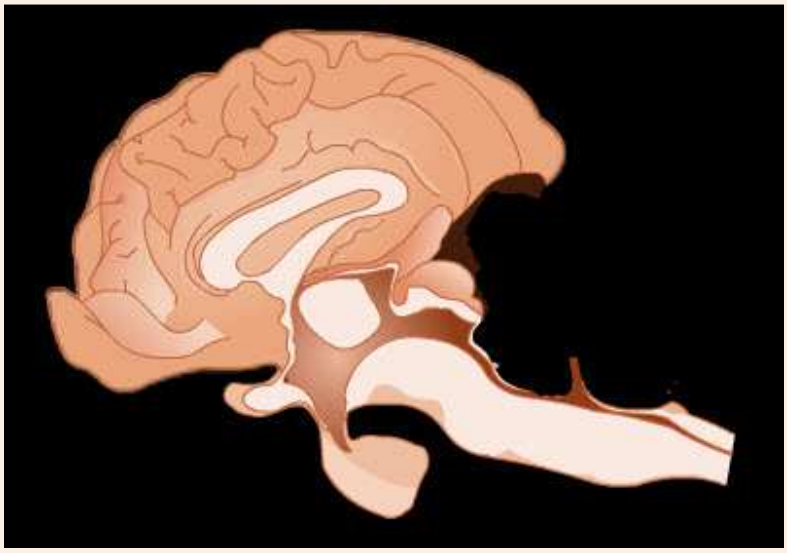
- It is a relay centre. Extends from pons to thalamus concerned mainly with auditory and visual relay system. It has dorsal portion namely the corpora quadrigemina which consists of two pairs of colliculi and ventral portion namely the cerebral peduncle. Roof of the midbrain is known as Tectum which has pair of brain centres vis namely optic lobes in non mammalian species. It has a pair of inferior colliculi.

- *Corpora quadrigemina:*Represented as paired colliculi, anterior pair concerned with visual relay is known as Superior colliculi and posterior pair concerned with auditory relay is known as Inferior colliculi.
- *Superior colliculi:* It is similar structure as optic lobes in mammals functions as reflex optical response (orientation towards visual stimuli, focusing etc.) Visual reflex centres houses many nuclei. Optic tract via lateral geniculate body of thalamus leaves collaterals to the suprachiasmatic nucleus of the hypothalamus, pretectal nucleus and superior colliculus. The pretectal region immediately anterior to superior colliculus receives inputs from retina, superior colliculus, lateral geniculate body and frontal eye field. Through these they modify pupillary light reflexes, accommodation, and ocular fixation reflexes. Through hypothalamic connections they modify the activities of limbic system and visual control. Through the connections with hypothalamus they also influence light-related endocrine secretions, circadian rhythm, sleeping cycle, and eating and drinking. Through connections with accessory optic system with flocculonodular lobe and inferior olivary nucleus, eyeball movements in relation to position and movements of head in space is integrated.Superior colliculi do receive collaterals through optic tract which connects visual and non-visual information via circuit. These neural circuits integrate visual information with non-visual information for coordinating head, eye, and body movements (postural reflexes).
- *Inferior colliculi:* It is concerned with hearing. It is an important relay station for the integration of ascending and descending auditory signals. These provide functional link between brainstem and telencephalic organization of auditory information. It receives the afferents from superior olivary nucleus which forms the major ascending auditory relay from cochlear nuclei. It involves lateral lemniscus to exhibit reflex vocalization and correlation of information of equilibrium.
- *Cerebral peduncle:* consists of tegmentum, substantia nigra, and basis peduncle.
- *Tegmentum:* It is a posterior part of midbrain that posses fine control of muscles. Act as a relay in integrating functions of sensory and motor functions. It forms the ascending reticular formation of the pons. It contains key structure of somatomotor neuron RED NUCLEUS along with the nuclei of trochlear and oculomotor nerves.
- *Red nucleus:* Located within the rostral portion of the midbrain and extend up to the posterior part of thalamus and hypothalamus. It is well connected with other structures of CNS via corticorubral (from cortex), pallidorubral (basal ganglia), and cerebellorubral (cerebellum-dendate nucleus) through the respective tracts and through axons to subthalamus. Motor fibers from red nucleus are projected to spinal cord via rubro-spinal tract, to reticular formation via rubro-thalamic tract, to olivary nucleus through rubro-olivary tract. They do send efferents via axons to III, IV, and VI cranial nerves.Red nucleus maintains muscle tone and equilibrium. It act as a centre for all righting reflexes except visual righting reflex. It integrates various impulses and transmits them to spinal cord to control somatic activity.
- *Substantia nigra:* Functions as integration centre for skilled muscular activity by receiving sensory information from the body surface, ear, eye, and nose. It inhibits extra-pyramidal motor movements.



- *Basis peduncle:* Forms the passage for corticospinal and corticopontile tracts.

MODULE-8: BRAIN STEM, RETICULAR FORMATION AND SLEEP



LEARNING OBJECTIVES

This module enables the learner to understand,

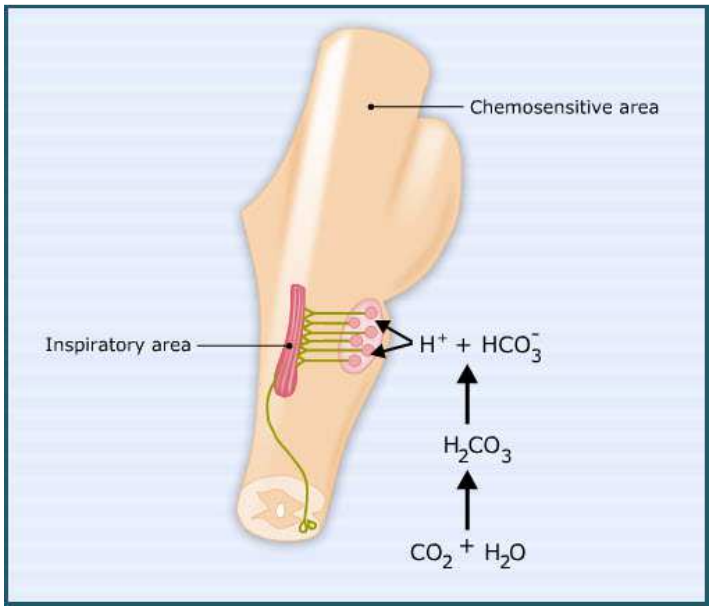
- the components that make up the brain stem,
- the medullary centers that control the body's functions, nerves that arise in the region and the nuclei present here,
- the function and reflex activities of the brain stem,
- the formation of the BSRF and its functions and
- sleep, its types, theories and brain waves on the various stages of sleep.

BRAIN STEM

- It is the group of CNS structures continuous with the spinal cord. It forms a vital link between the spinal cord and higher centres of the brain. It comprises of medulla oblongata and pons that controls many of the life sustaining process and baseline activities of skeletal muscles. These comprise of caudal most portion of the brainstem. It maintains relationship with fourth ventricle.
- Difference exists in the distribution of the white and grey matter in the brainstem. Grey matter of brainstem is represented by nuclear group separated from each other by fiber system.
- Largest representation of the brainstem is the reticular structures namely the brainstem reticular formation. This extends from anterior end of the spinal cord to thalamus and hypothalamus. This system controls the consciousness and perception, motor and sensory activity of an animal.

NUCLEI OF BRAIN STEM

- These are associated with cranial nerves and are organized into columns, representing different functional types. They are three motor columns (somatic, visceral, and branchial motor columns) and two sensory motor columns (somatic and visceral sensory systems).
- Somatic motor column is placed in the dorsomedial position of the brainstem. It consists of α-motor neurons. These neurons supply the muscles of head, extrinsic muscles of eye, and muscles of tongue.
- Visceral motor column contains neurons similar to the intermediolateral cell column of the thoracic and lumbar spinal cord. They supply fibers to the intrinsic muscles of the eye, vagal fibers to the cervical, thoracic, and abdominal regions.



- Branchial motor column is located ventrolateral to the somatic sensory column and are similar to the α-motor neurons of the somatic motor column. They supply motor fibers to the skeletal muscles of mastication, facial muscles, and muscles of larynx, pharynx, and trapezius.

- Somatic sensory column consists of neurons of sensory fibers of trigeminal nerve supplying to head and face. As they function to innervate muscle spindle of the cheek muscles, they are responsible for control of muscle tone of masticatory muscles and reflex control of chewing. These neurons are responsible for perception of delicate tactile sensation, pain, and temperature in the areas of head and face are processed via the nucleus gracilis and nucleus cuneatus.
- Visceral sensory column is represented by a single nuclear structure at the medulla and is known as nucleus of fasciculus solitarius. Sensory information from mouth, pharynx, larynx, thoracic, and abdominal viscera are received by this via visceral afferents of vagus and glossopharyngeal. This nucleus relays taste afferents within the facial, glossopharyngeal, and vagus nerves.

Other nuclei of brainstem

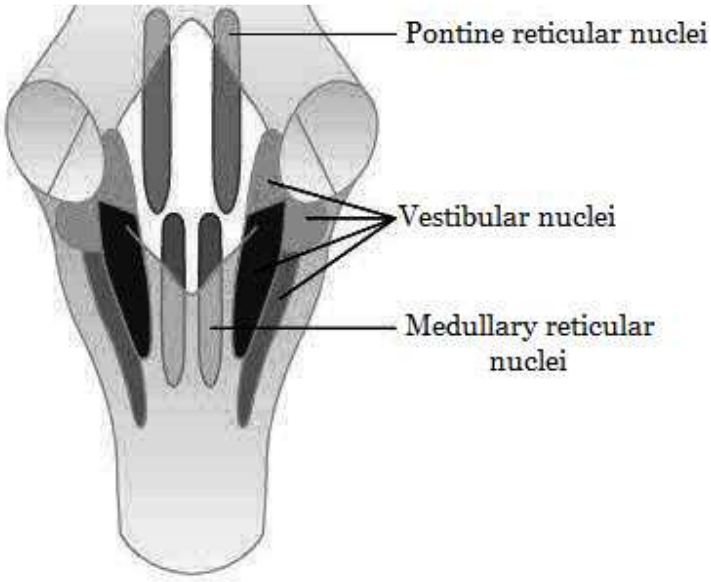
- These nuclei serve as relay of sensory or motor functions. and cerebral cortex are also received by these nuclei as a part of messaging system of CNS to the cerebellum for the control of somatic motor activity.

- Nucleus gracilis and nucleus cuneatus lie within the medulla and relay sensory information of tactile and kinesthetic sense of joint to higher centres of the brainstem. Fibers of this nucleus at the medullary level form medial lemniscus by arching and decussating to the opposite side and terminate within thalamus.
- Accessory cuneate nucleus forms relay nuclei for tactile, joint angle, and muscle stretch of areas of neck via the spinocerebellar tract. These nuclei lie lateral to nucleus gracilis and cuneatus.
- Caudal olivary nuclear complex is similar to the accessory cuneate. They receive neural information from spinal cord and project them to the cerebellum. Fibers from basal ganglia
- Pontine grey matter lies within pons and serve to relay information for pontocerebellar fiber system to control somatic motor activity. Nerve fibers of this decussate to form Brachium Pontis, third peduncle in the cerebellum.
- Red nucleus is a prominent somatic motor nucleus of midbrain. Midbrain comprises of structures concerning auditory and visual relay system. This nucleus receives sensory fibers from cerebellum, motor cortex, globus pallidus, thalamus, vestibular nuclei, and brainstem reticular nucleus and give rise to RUBROSPINAL TRACT to control somatic motor activity by ending in the internuncial neurons to excite somatic and visceral motor neurons of the spinal cord.
- *Nuclear mass*
- Pre cerebellar reticular nuclei projects to cerebellum.
- Non-cerebellar reticular nuclei: Raphe nuclei: Inhibits the reticular activation system (RAS) and induces sleep.
- Central group of nuclei and Lateral group of nuclei -These two nuclei contain respiratory and cardiac centres.

MEDULLA OBLONGATA

- Located in the top of the spinal cord.
- Has centres that control reflectes activities such as respiration, heart rate, vaso dilation and blood pressure.
- Houses neural part ways that connect cerebral cortex and spinal cord

- The cranial nerves glosso-pharyngeal (IX), vagus (x), spinal accessory (xi) and the hypoglossal (xii) originate from the medulla.
- The nuclei gracilis, and cuneatus transmit the informations of the spinal cord to the somato sensory regions of the cerebral cortex through the thalamus.
- The nucleus gracilis and cuneatus give rise to fibres, which pass to the contralateral side and form the medial lemniscus; this medial lemniscus ascends to thalamus.
- The inferior olivary nucleus input signals to cerebellum. Within medulla there are centres for cardiac, respiratory (inspiratory and expiratory), vasomotor, swallowing, vomiting, cough and sneezing activities.
- Most of the tracts between the spinal cord and the higher levels of the brain pass through medulla.



- Ascending tracts: A portion of dorsal funiculus of the spinal cord terminates in accessory cuneate nucleus from which fibres project to cerebellum as external arcuate fibres. Other ascending fibre tracts of the medulla are spinothalamic, dorsal and ventral spinocerebellar, spino-olivary, spino-tectal tracts.
- Descending tracts: includes Vestibulo spinal, rubro spinal, tecto spinal tracts.

PONS

- Structures present above the medulla function as bridging and communicating information between medulla cerebellum and forebrain structures.
- Pons contain pontine nuclei.
- It also includes superior olivary nucleus of auditory system and nuclei of visceral centres to regulates respiratory and cardio vascular system.
- It is concerned with control of alertness and initiates sleep.
- It serves as integration centre and relay motor centres.
- It also houses cranial nerves 5,6,7 and 8.
- It has two paired respiratory centres known as pneumotaxic centre and apneustic centre that control respiratory rate and rhythm.

FUNCTIONS OF BSRF

- It is a neuronal network extending the length of the medulla, pons, midbrain and then projects into thalamus and hypothalamus.
- Caudally it is continuous with the internuncial neurons of the spinal cord.
- The efferent fibres of the reticular formation are organised into ascending and descending reticular formation.
- The ascending reticular formation projects into other areas of brain, including brain stem, cerebrum and cerebellum; the descending reticular formation projects into the spinal cord.
- The BSRF regulates the sensory, motor and the endocrine functions. It is constituted by very small to very large sized sensory and motor neurons.
- The small neurons form multiple connections within the reticular formation.
- The large size neurons are motor in function which bifurcate with one division extending upward to the thalamus or the basal regions of the diencephalon or the cerebrum.
    - Nuclear mass
    - Pre cerebellar reticular nuclei ------> projects to cerebellum.
    - Non-cerebellar reticular nuclei:
        - Raphe nuclei: Inhibits the reticular activation system (RAS) and induces sleep.
        - Central group of nuclei and
        - Lateral group of nuclei -These two nuclei contain respiratory and cardiac centres.
    - The brainstem reticular formation receives sensory input signals from all sensory systems–somesthetic, vestibular, visual, auditory, gustatory and olfactory sensory systems, through spino- reticular tract and also from the vestibular nuclei, cerebellum, and basal ganglia and motor area of the cerebral cortex.
    - It integrates the sensory information within the CNS and controls the activities of both the motor neurons of skeletal muscles and the autonomic neurons (cardiovascular, respiratory and the G. I. tracts). The BSRF plays a key role in somatic and visceral motor control.
    - The BSRF may either causes consciousness (wakefulness) by providing continuous stimulatory signals to the cortex or it may induce sleep by its inhibitory signals through raphe nucleus to the ascending reticular activation system (ARAS).

Ascending reticular formation

- It alters the consciousness through the sensory inputs from the auditory, visual, olfactory, tactile, pain and proprioceptive sensory systems. It regulates the neuronal activity within the brain and contributes to wakefulness. It controls the activity of the cerebral cortex, hippocampus, basal ganglia and cerebellum and it is referred to as behavioral arousal.
- Ascending Reticular Activating System (ARAS) of the BSRF is involved in alertness/ wakefulness of the animal. They receive sensory information from all sensory receptors and also have inputs from the brainstem, cerebellum and cerebral cortex. These sensory inputs activate the RAS neurons and this in turn activates wide areas of the cerebral cortex. Reduction in the activity of the RAS can lead to sedation, sleep and coma.

Descending reticular formation

- It influences the motor activities such as flexor and extensor reflexes and decerebrated rigidity. It alters the activity of the alpha and gamma motor systems through reticulospinal tract. It also contains centres that inhibit spinal motor neurons.
- The reticular formation contains the centres that promote wakefulness (midbrain RAS). Through pontine raphe nuclei it promotes sleep.

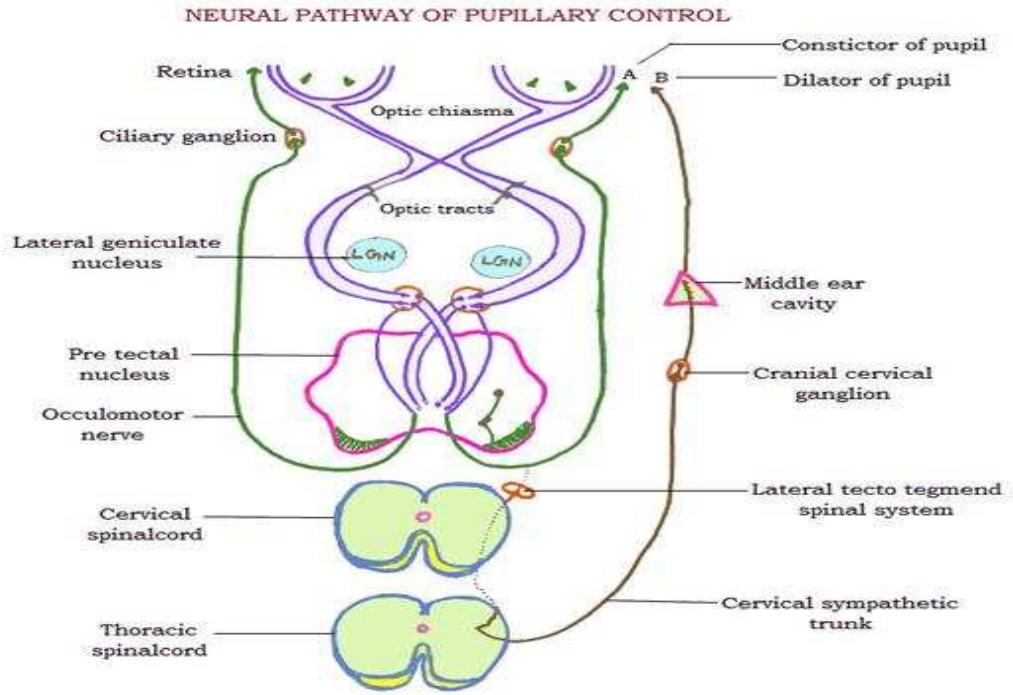
Other functions of BSRF are - Respiratory control, cardiovascular, micturition, emesis, rumination, deglutition, mastication control etc.

REFLEX ACTIVITIES OF BRAIN STEM

- Brain stem exhibits reflex activities.

Light reflexes - Pupillary light reflex

- When light strikes the retina of either eye both the pupils constrict to reduce the amount of light passing through the pupil.
- Receptor organ is retina
- Sensory fibres pass via optic nerve that terminate in the pre tectal nucleus.
- This nucleus lie between the mid brain and thalamus.
- The fibre synapse with internuncial neurons of the pre tectal nucleus from here the fibre project to occulomotor nuclei to cause excitatory post synaptic potential.
- The induced stimulus on occulomotor nerve causes generation of action potential in the circular smooth muscles of the iris-effector organ.
- Pupillary construction is the effect produces by this reflex.



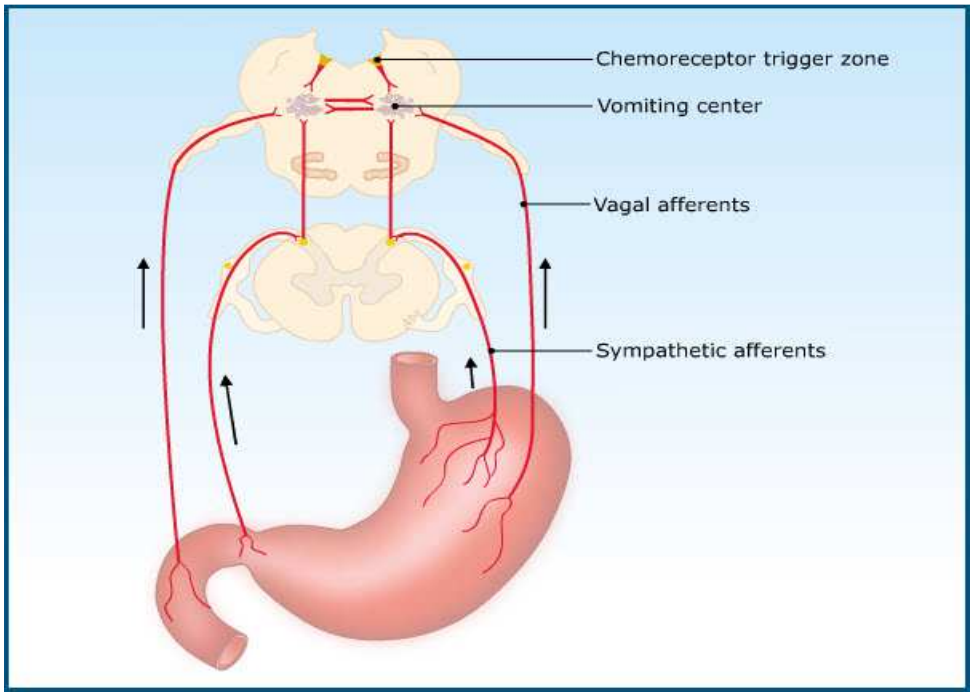
Corneal reflexes

- Corneal blink reflex
  - It is a protective reflex preserving cornea from injurious foreign material.
  - Receptor organ is the cornea.
  - Sensory fibres passes via opthalmic division of trigeminal nerve.
  - It involves internuncial neurons located within the BSRF.
  - Motor neurons are the alpha motor neurons of the facial nucleus that supply muscles of orbicularis oculi, that cause closure of the eyelid.
  - In most occasion blink reflex is acccompanied with lacrimation reflex.
  - If not associated with lacrimation it is known as palpebral blink reflex.

- Lacrimation reflex
  - It is a protective reflex preserving cornea from drying and preserves vision.
  - Afferent fibres or through opthalmic division of trigeminal nerve.
  - Internuncial neurons are involved.
  - Cell bodies are located within the reticular formation of the brain stem.
  - The efferent neurons comprise a portion of rostral salivatiry nucleus.
  - These neurons are parasympathetic pre ganglionic fibres to the lacrimnal gland.
  - The efferent fibres pass via facial nerve from BSRF joint to the petrosal nerve.
  - At this point the synapse with pterygopalantine ganglian.
  - From here fibres are projected to lacrimnal gland
  - This reflex modify the rate of lacrimal flow and aids the removal of foreign material from the surface of the cornea.

EMETIC REFLEX

Emesis is the regurgitation of materials from stomach and upper intestine through oesophagus to oral cavity.



- This reflex is initiated by irritating stimuli from vicera.(pharynx, stomach, duodenum, heart, uterus) or from external auditory canal.
- Protective mechanism to remove the irritating stimuli.
- Motion sickness results due to irritation of the vestibular receptor organs. This particular event involves the participation of flocculonodular lobe of cerebellum.
- This reflex is preceded by salivation & swallowing.
- The nerves of afferent limbs are glossopharyngeal, vagus or vestibular and visceral afferent of the spinal cord (for abdominal/pelvic visceral).
- Emetic center is located in the caudal medulla.
- Cells present in chemo receptor triggerzone of medulla can also initiate emesis on stimulation by specific chemical (apomorphine, ergot alkolids morphine, glycosides or metobolites of systemic acidosis, alkolosis, uremia)
- The central neural mechanism of efferent limb includes nerves supplying to
    - Smooth muscles of G.I tract.
    - Skeletal muscle of larynx, pharynx, thorax, abdominal wall.
- Effernt nerve of emeticreflex causes closure of glottis with inhibition of inspiration; These together produce negative pressure in the intrathoracic and oesophagus.
- Relaxation of cardia, fundus, body of the stomach and oesophagogastric cardiac sphincter prior to act of vomiting.
- Followed by forceful contraction of the abdominal muscle.
- This result in movement of content of stomach to oesophagus.
- Closure of nasopharynx prevents the entry of contents to respiratory organs.
- Powerful antiperistalsis with negative I/oesophageal pressure expels the content to oral cavity.
- Creation of negative pressure by forceful inspiration against closed glottis and strong abdominal muscle contraction is important than antiperistalsis.
- Relaxation of pyloris allow contents from duodenum to oesophagus and then to oral cavity.
- Efferent fibers also reaches salivatory nucleus to initiate salivation before emesis to prevent oral mucosa from acid contents.

- Fibers project to respiratory and cardiac centre to modify appropriately the B.P and the rate of respiration.

SUCKLING REFLEX

- More prominant in newborn animals to fecilitate withdrawal of milk from teat canal and mammary gland while feeding.
- Suckling is a physiological process involving coordinated synchronus movement of tongue and buccal musculature.
- It is termed as a reflex since it involves set pattern of neuronal inputs to complete this action.
- Afferent fibers pass through trigeminal and facial nerve.
- Involves internuncial neurons located within the brain stem reticular formation.
- Motor fibres with orders to the specific muscles reach via hypoglassol, facial and branch of trigeminal that is involved in mastication.
- This reflex is initiated by creating a vaccum which is assisted by withdrawal of tounge from the heart palate.
- The creation of partial vaccum within the oral cavity help to withdraw milk from the mammary gland.
- This reflex is always associated with reflex closure of reticular groove of the young animal to direct the flow of milk into the omasum and abomasum.

SALIVATORY REFLEX

- Protective reflex preventing oral mucosa from drying.
- Elicited when oral and lingual mucosa is stimulated by foreign substances.
- Afferent fibres pass via trigeminal, facial, glossopharyngeal and vagus.
- These fibres end in brain stem reticular formation by synapsing with intenuncial neurons.
- Efferent fibres reach salivary glands depend upon primary stimuli via different routes.
- They arise from neurons of rostral and caudal salivatory nuclei.
- Fibres via facial nerve reaches mandibular and lingual salivary gland.
- Parotid secretion is increased by way of glossopharyngeal motor fibres.
- Feeding and presence of food in the oral cavity influences through this reflex mechanism ends in production of mucinous saliva from sublingual, sub maxillary, pharyngeal and labial glands to facilitate swallowing.
- Noxius when consumed end up in copious flow of serous saliva so as to wash away them without being injested or they are diluted by the saliva from parotid, inferior molar, buccal and phalangial gland as a protective mechanism.
- Salivatory reflexes is initiated by many difference stimuli ans is potentiated by olfactory, gustatory, auditory and visual mechanism.

SWALLOWING REFLEX

- Swallowing centre is located in the brain stem (medulla oblongata)
- Stimulation of the receptors in the soft palate, pharynx (posterior wall) and epiglottis (dorsal surface) by food material initiate this reflex.
- It is a complex multi synaptic reflex involving pharyngeal and oesophageal stage.
- Sensory fibres pass through trigeminal, glossopharyngeal and vagus nerve.
- This reflex involves internuncial neurons.

- Efferent fibres emerge through glossopharyngeal, vagus and hypoglossal nerves which supply to the muscles of myohyoid and hypoglossal.
- The muscles press the tounge against hard palate.
- This causes the tounge to draw backward with the elevation of soft palate.
- Now the tounge forces the bolus into opened oesophagus.
- Opening of the oesophagus is by the pulling action of hyoid bone and layrnx.
- As it happens the epiglottis closes so as to shut the layrnx.
- Swallowing centre also activate through its efferent limb the neighbouring neurons that control respiration.
- This interupts respiration during swallowing as a preventive measure to avoid aspiration of food particles into respiratory passage.

MASTICATION REFLEX

- Reflex activity under the control of brain stem.
- Mastication is basically voluntary but usually takes place involuntarily.
- Mastication reflex or chewing reflex is by the rhythmic movement of the mandible.
- Lowering of the mandible due to the extension of the tongue is known as linguo mandibular reflex.
- Stimuli for masticatory reflex is the presence of food in the oral cavity
- Receptors in the tongue and oral mucosa initiate mastication by the sense of food.
- Sensory impulses are carried via trigeminal, facial and glossopharyngeal nerves to brain stem.
- Motor orders reach masticatory muscles via trigeminal nerve
- This inturn causes rhythmic movement of mandible in relation to the maxilla.
- This results in shearing and crushing a food.
- The main masticatory muscles supplied by trigeminal nerve include Temporalis with assistants from Massetor muscle for shearing and Massetor with assistants from Pterygoid muscles for grinding.

COUGH REFLEX

- It is a protective reflex to remove thr irritant from the respiratory tract.
- Receptor for this reflex is from respiratory tract.
- Irritation of any part of the respiratory tract serve as sensory impulse.
- Vagus is the sensory nerve to endup in BSRF via Fasciculus solitarius nucleus.
- Efferent nerve via vagus project to respiratory centre and laryngeal muscle to effect cough.

SNEEZE REFLEX

- It is a protective reflex to remove the irritant stimuli.
- Receptors are present in the upper respiratory pathway.
- Sensory information passes via trigeminal to the BSRF.
- The efferent via vagus, trigeminal and facial effect sneezing.

MICTURITION REFLEX

- Micturition is the complex phenomenon involves initiation from stretch receptors in the bladder wall.

- Axons have their cell bodies located in the sacral spinal ganglia and dorsal grey column of the spinal sacral segments.
- Stretching of the bladder stimulate the sensory fibres which in turn causes the generation of action potential that passes via spinal nerves to the spinal cord.
- They may employ sacral inter neurons prior to terminate in the higher centre or travel straight to end in the higher centre.
- If sacral inter neurons are involved, spinal cord completes the reflex arc by synapsing with pre ganglionic parasympathetic neurons.
- These parasympathetic system causes contraction of smooth muscles of urinary bladder.
- There is involvement of sympathetic system and is evident in the early stages of filling of the bladder by activating beta adrenergic preganglionic sympathetic neurons by the synapsing of interneurons. These fibres inhibit smooth muscle contraction allowing still more stretching to hold the urine.
- Spino thalamic tract from spinal segments at the level of bladder can also serve as a afferent fibers to end in fasciculus gracilis of medulla where they synapse in the nucleus gracilis. These fibers pass via thalamus and reach somesthetic cortex to cause conscious perception of bladder sensation.
- Stimulation of cell bodies in pelvic plexus especially in bladder wall via axons of post ganglionic para sympathetic system causes contraction of detrusor muscle and relaxes the striated sphincter of the bladder.
- This in turn causes evacuation of urine from bladder and allows the flow via urethra.
- Sympathetic activation via alpha receptor system potentiate contraction and smooth muscle tone of the bladder allows urine output.
- Involvement of brain stem via tecto spinal and reticulo spinal tract results in voluntary micturition.

ROLE OF RETICULAR FORMATION

- Ascending projections to the BSRF to cortex involves in maintenance of cortical excitability and consciousness.
- Wakefulness depend on an adequate impulse from BSRF through the *Ascending Reticular Activating System(ARAS) and Diffuse Thalamo Cortical Projection System(DTPS)*
- Neuro humoral theory suggests the involvement of noradrenaline and serotonin in sleep-wakefulness rhythm.
- Noradrenaline from nerve endings of medullary cells have positive influence on ARAS and maintain wakefulness. It also responsive for phasic events of sleep like rapid eye movement(REM).
- Serotonin produced by the midbrain raphe nucleus is a depressant acting on the BSRF and diminishes the ascending reticular drive, ultimately end up in sleep.
- Acetyl choline balances the action of NE and serotonin in sleep – wakefulness rhythm.
- Melatonin- pineal hormone causes sleep and act as a trigger switching the reticular system to act through other neurotransmitter .

SLEEP

- It is a state of reversible unconsciousness and relative immobility of an animal.
- It is due to temporary inhibition of the reticular activation system of the ascending reticular formation, which in turn reduces the normal excitability of the cortex.
- Sleep centres are located in the pons and medulla known as raphe nuclei. The nerve fibres of this area secrete serotonin. These nuclei send fibres to reticular formation, upward to

thalamus, neocortex, hypothalamus and many areas of the limbic cortex. Fibres extend downward to spinal cord where they inhibit the incoming pain signals. The raphe nuclei provide inhibitory signals to the ascending reticular activation system (ARAS) and thereby induce sleep.

- Diffused Thalamic Projection System, (DTPS), the sleep-inducing centre is present in the thalamic reticular area, projects its ascending fibres to cortex and descending fibres to reticular formation. DTPS regulates cortical activity and is antagonistic to ARAS. Based upon the period of rest and activity, the animals may be classified as
    - *Monophasic animals:* Have prolonged period of rest during night and are active during daytime (Man, birds).
    - *Polyphasic animals:* Show several alternate periods of rest and activity (wild animals)
    - *Intermediate type:* Has alternate periods of rest and activity during daytime followed by brief period of rest at night (domestic animals).

Changes during sleep

- Decrease in sensitivity and responsiveness of the individual to environmental stimuli.
- Reduction in metabolic processes.
- Fall in body temperature and BMR.
- Decrease in heart rate, B.P and respiratory rate.
- Decrease in alimentary secretions and motility.
- Pupillary constriction
- Decrease in muscle tone.

Endocrine secretions

- Decrease in corticoids
- Episodic increase in GH, melatonin and testosterone.

Theories of sleep

- Sleep is due to depression of the cortical centre by acid metabolites accumulated during active periods.
- Increased physical efforts during active hours demand increased skeletal blood flow and relative cerebral ischemia which inhibits brain function.
- Sleep producing substance in the brain is stimulated follwing ischemia.
- Cortical excitation and cortical inhibition rhythmically alternate with each other, conditioned to an environmental cue (day, light/ darkness) internal signals. This rhythm brings about the sleep wakefulness cycle.
- Hence sleep is defined as rhythmic phenomenon which involves the entire cortical excitability.
- Sleep centre- anterior hypothalamus - trop hotrophic area
- Waking centre- posterior hypothalamus - ergotrophic area

PHASES OF SLEEP

- Two distinct phases alternates with each other.
    - Slow wave sleep or orthodox or NREM sleep: 70% of sleep period
    - REM sleep or paradoxical sleep- interprets the slow wave sleep given by RES

- During slow wave sleep, EEG records characteristically slow, synchronized theta waves and delta waves. Muscle tone and responsiveness to stimuli are decreased.

ELECTRO ENCEPHALO GRAM

- It is the graphic record of electrical activity of the brain recorded from the surface.
- Technique is known as Electro Encephalography
- It is the recording of the potential difference as rhythm (frequency) and magnitude.
- Rhythm of the EEG are designated as alpha, beta, theta and delta rhythm

Alpha rhythm

Frequency	Amplitude	Characteristics
8-13 cycles/sec	50-100 mv	Seen in subject when awake relaxed state and closed eyes.

- Replaced by faster wave <---- alpha blocking <----> when respond to stimuli and wave disappear (Not responding to any stimuli)
- Alpha rhythm represents state of relaxed wakefulness of reflex the optimal state of brain excitability namely synchronized rhythm.

Beta rhythm

Frequency	Amplitude	Characteristics
More than 20-30 cycles/sec	25-50mv	Seen in alert cortex wakefulness and responding to a stimulus knowns as desynchronized rhythm.

Theta rhythm (slow rhythm)

Frequency	Amplitude	Characteristics
37 cycles/sec	50-100 mv	dominant in wakefuness is young subjects. seen in early stages of slow wave sleep from hippocampus.

Delta rhythm

Frequency	Amplitude	Characteristics
Less than 3 cycles/sec	High amplitude < 100 mv	stages of slow wave sleep

- Lamda wave - have high amplitude (from fronto-temporal regions)
- Mu waves - from motor cortical areas

- Spike and wave pattern: indicates abnormal electrical activty of neurons seen in epileptic cases.

Mechanism of EEG

- Wave pattern is due to the Oscillation of alternating excitatory and inhibitory post synaptic potentials produced by impulses of non-specific thalamic nuclei.
- Frequency of the rhythm is by the activity of non-specific thalamic nuclei.
- At low intrinsic frequencies (8-13 cycles/sec) the EPSP's are summated & give rise to low frequency synchronised EEG rhythm (alpha).
- When thalamic neurons are driven by the reticular activating system, the frequency becomes too high for summation and becomes desynchronized to become fast EEG rhythm – beta rhythm results.
- Normal ECG
    - 8-13 cycles/sec - alpha wave
    - 20-30 cycles/sec - beta wave
    - 3-7cycles/sec - theta wave
    - Less than 3 cycles/sec - delta wave

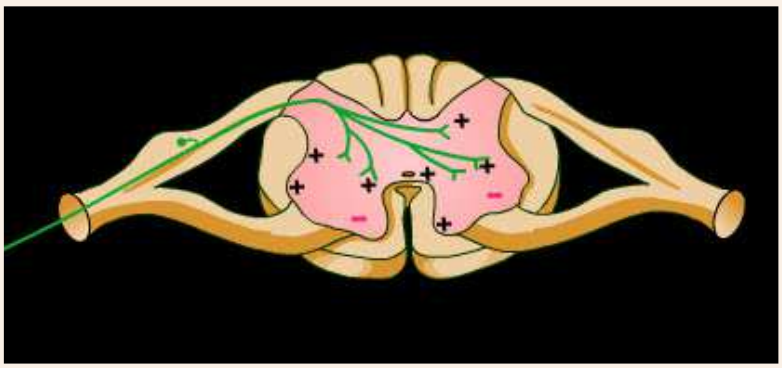
EEG and sleep stages

- EEG studies help in understanding the different stages of sleep Each stage of sleep manifest different wave pattern in EEG. 4 stages of slow wave sleep can be distinguished with different wave pattern, frequency and amplitude.

Drowsiness

- In the relaxed awake state alpha waves predominant 1st stage of slow wave sleep is characterised by fluctuations between alpha wave and low amplitude wave (3-7 cycles/sec).
- As sleep deepens into stage 2 wave pattern changes.
- In stage 3 and 4 EEG shows gamma waves (less than 3 cycles/sec).
- In REM sleep low voltage fast waves are seen.

MODULE-9: SPINAL CORD



LEARNING OBJECTIVES

This module explores,

- the structure and organization of the spinal cord,its nuclei ,tracts and their functions and
- the understanding of involvement of higher nervous

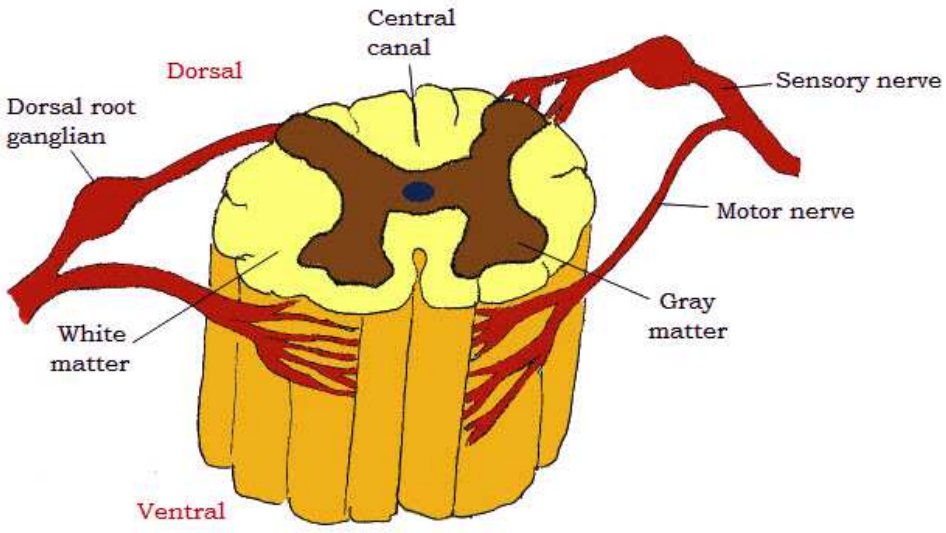
	system to complete the prescribed action.

SPINAL CORD - AN INTRODUCTION

- It is a caudal extension of the medulla oblongata present throughout the length of the vertebral canal. Each spinal segment provides a pair of spinal nerves that are formed by the fusion of dorsal root (sensory) fibres and the ventral root (motor) fibres.

STRUCTURE OF SPINAL CORD

- Cylindrical
- Extends from medulla oblongata at level of atlas for varying distances down the spinal cord



- Divided into many parts : Sensory fibers enter the spinal cord by dorsal roots of spinal nerves along the dorsolateral sulcus and motor fibers leave the spinal cord by way of the ventral roots of spinal nerves along the ventrolateral sulcus
- This arrangement divides the spinal cord into 4 compartments
- Cellular components are distributed primarily within the center of the cord where they form an H shaped column of cells called Gray matter.
    - The grey matter has two types of multipolar neurons called the root cells, and the tract cells.
    - The motor neurons are the principle component of the root cells, whereas the interneurons or the fusiform neurons form the tract cells.
- Nerve fibers are found predominantly outside of the gray matter where they form the white matter of the cord.

- The white matter is divided into components : paired dorsal funiculi (funiculus = a bundle of nerve fibers) , two lateral funiculi and paired ventral funiculi
- The paired dorsal fasciculi are separated by a dorsal median septum and the ventral fasciculi are separated by a ventral median fissure
- Size and shape of the spinal cord vary at different lengths
- Two distinct spinal cord enlargements are seen in domestic animals : a cervical enlargement which serve the pectoral limbs and a lumbar enlargement which serve the pelvic limbs
- Spinal cord and the vertebral column grow at the same rate during fetal life but the vertebral column grows faster in the post natal life
- This causes the spinal cord to be pulled forwards in the vertebral column.
- Spinal cord extends for different lengths along the vertebral column in different species
- Stretching out of spinal nerves arising from the caudal portion of the spinal cord forms the cauda equina
- Spinal cord is organized into segments each of which is associated with a single spinal nerve
- There are as many spinal cord segments as there are spinal nerves, But the spinal nerves and spinal cord segments do not actually correspond in adults due to differential growth. The relationship however is necessary in determining the locations of pathological processes in the spinal cord.

GRAY MATTER

Two types of cells

- Motor neurons
- Internuncial neurons

Motor neurons

- It project their axons into the PNS
- They are of 3 types
    - *Somatic or Alpha motor neurons* – supplying skeletal muscles
    - *Autonomic neurons* – supplying visceral organs
    - *Intrafusal or Gamma motor neurons* – supplying special receptors called muscle spindles
        - Somatic or Alpha motor neurons show Nissel substance in their cytoplasm
        - Internuncial neuron processes remain within the confines of the CNS
        - They vary considerably in their shape, size , organization
        - Two classes according to the course of their projection of their fibers within the nervous system
        - Commissural Internuncial neurons whose fibers cross the midline
        - Non Commissural Internuncial neurons or association neurons whose fibers do not cross the mid line
        - Association neurons are called so because by their connections they correlate activities within various portions of the nervous system
        - Gray matter of the spinal cord is divided into many nuclei on anatomical basis
        - Many portions of the CNS are said to composed of cell laminae which are groups of similar cells arranged together in a particular order

INTERNUNCIAL NEURONS

- Classified into 2 types based on their synaptic capacity :
    - *Excitatory Internuncial Neurons* – exert a depolarizing influence on the postsynaptic with which they have contact
    - *Inhibitory Internuncial Neurons* – exert hyper polarizing influence on the neurons with which they have contact
- With the production of presynaptic inhibition, Internuncial Neuron produces inhibition by depolarizing the presynaptic terminals of other neurons.
- The presence of two types of Internuncial neurons ensure that activity and signaling inside the spinal cord remain flexible
- Reciprocal innervation
    - A phenomenon exhibited by afferent nerves to the spinal cord on somatic efferent nerve activities
    - When motor neurons supplying a skeletal muscle are depolarized by excitatory interneurons or by dorsal root afferents, the motor neurons supplying muscles that are antagonistic to this muscle are hyperpolarized by inhibitory interneurons.
    - All voluntary motor activity is made meaningful because of this integration by inhibitory and excitatory interneurons
- Interneurons are classified based on their pattern of convergence into 5 basic groups
- They can have one or two or three muscle groups projecting muscle spindle fibers to them
- One synergistic muscle group :
- Some of these are excited by exteroceptive afferent sources especially from the foot pads or hair in between the toes
- Two different muscle groups : Usually antagonists acting on the same joint
- Or they may be synergists acting at neighboring joints
- Or antagonists acting at neighboring joints
- Few are influenced by sensory information from the skin (cutaneous afferents)
- Three muscle groups are also seen
- Interneurons form the system that is within the spinal cord
- Interneurons are capable of prolonged rapid discharge of action potentials following afferent stimulation
- Interneurons play an important role as convergence systems for sensory (afferent) fibers of sensory modalities
- The convergence systems are not necessarily anatomical but rather are ones that are active at that time
- Spinal reflexes usually involve interneurons except some autonomic reflex arcs through the spinal cord and the myotatic reflex
- The corticospinal, rubrospinal, vestibulospinal, tectospinal, interstitiospinal and reticulospinal motor pathways for somatic and autonomous controls exert their influence through these interneurons
- Many interneurons are organized to perform specific functions
- Alpha motor neurons give rise to large fibers which innervate skeletal muscle
- These fibers also give rise to collateral branches which reenter spinal gray matter to synapse upon inhibitory interneurons
- These Internuncial neurons are called Renshaw cells
- They control the duration, intensity and distribution for motor neuron discharge
- Thus they function in localization of reflex motor activity
- Disinhibition : The above neurons in certain cases increase the firing of motor neurons inhibitory to the alpha motor neuron

- Discharge of Renshaw cells is inhibited by stimulation of the cutaneous and muscle afferents
- A group of neurons within the nucleus intermedialis and nucleus proprius give rise to the ascending fiber system
- These neurons give rise to ventral and rostral spino cerebellar tracts , the spino cervical tracts, the spino olivary tract and spino reticular tracts

NUCLEI OF SPINAL CORD - SENSORY NUCLEI

- Spinal cord has 9 laminae - sensory nuclei

Nucleus Dorsomarginalis

- Corresponds to lamina I
- Extends the length of the spinal cord
- Covers the dorsal and lateral surfaces of the dorsal horn of the spinal cord
- Cells are spindle shaped
- Neurons receive afferent fibers from the dorsal root fibers and project their axons into dorsal and lateral funiculus from which they may ascend or descend to form inter segmental pathways for intraspinal integration of neural activity
- These fibers form the dorsolateral fasciculus of the propriospinal fiber system.
- It receives the pain and thermal sensation .
- It act as a low threshold mechanoceptors.

Nucleus Substantia Gelatinosa

- Corresponds to laminae II and III of the spinal gray matter
- Cells have small , spheroid or spindle shaped nucleus and little cytoplasm
- They also have a rich dendritic tree
- Axons of these cells turn either dorsally into dorsolateral fasciculus or ventrally. Some turn ventromedially and thereby reach the Substantia Gelatinosa of the other side
- Substantia Gelatinosa is a closed neuronal system because all of its axons turn back and terminate within the nucleus of origin
- Long range connections between different regions a re established over the lateral part and the dorsal part of the lateral propriospinal tract
- These connections may bridge 5 to 7 segments of the spinal cord
- Ipsilateral (same side) short range connections of 2 to 3mm distances are also seen
- Axons originating in the Substantia Gelatinosa are always traced back to the Substantia Gelatinosa of the ipsilateral or contralateral side
- The Substantia Gelatinosa has its outlet in the large cells that occupy the underlying lamina IV and project dendrites into the nucleus Substantia Gelatinosa.
- It forms a chief association centre for pain, thermal, tactile sensory perception mediatory reflexes.

Nucleus Proprius

- Forms the nuclear column that extends the length of the spinal cord and comprises lamina IV and lamina V

- This nucleus serves as a origin of both crossed and uncrossed ascending tracts
- The lateral portion of nucleus proprius is referred to as the reticular formation of the spinal cord.
- It acts as a sensory pathway for light cutaneous mechanical stimulations (tactile, pain and thermal perception) and also from the Golgi tendon organ.

Nucleus dorsalis

- Extension is limited to the thoracic and upper lumbar levels of the spinal cord
- Occupies a medial portion of lamina VII
- In cat the nucleus begins at the most cranial thoracic segments of the spinal cord
- Cells of dorsal nucleus gives rise to ipsilateral dorsal spinocerebellar tract which terminates in the vermis of the cerebellum
- This system has proprioceptive functions related to Golgi tendon organ and muscle spindle activity
- The system also has two sub divisions activated exclusively by afferents from cutaneous receptors :
- One informs the CNS about pressure on foot pads
- The other has neurons activated by light touch , pressure and pinching of cutaneous areas
- An additional dorsal spinocerebellar tract subdivision which is neither exteroceptive nor proprioceptive is seen. These are activated by the flexor reflex afferents
- Proprioceptive cells of the nucleus dorsalis receive afferent information from the class Ia fibers from the muscle spindles and from class Ib of the Golgi tendon organs of skeletal muscles
- A given neuron receives inhibitory influences mediated by group I afferents from both synergistic and antagonistic muscles
- Synapses upon the cells of the dorsal nucleus by group Ia afferent fibers from muscle spindles are large compared to the synapses of group Ia fibers with alpha motor neurons
- Though these synapses use the same neurotransmitter according to Dale's principle , they have widely varied effects
- Synaptic activity upon the dorsal nucleus requires less summation to discharge to discharge action potential
- One bouton gives rise to a small series of action potential because of larger amount of transmitter agent released per action potential in the pre synaptic fiber or less diffusion of transmitter agent away from the reactive site
- Both alpha motor neurons and neurons of dorsal nucleus exhibit long positive after-potentials
- Discharge rate of alpha motor neurons as about 200 impulses per second and that of the dorsal nucleus neurons about 700 impulses per second
- Action potential is brief – about 0.5 to 0.75 msec
- Repetitive firing of neurons of the nucleus dorsalis is important in CNS function.
- In primates it extends from the upper lumbar to lower cervical spinal cord, while in other animals it extends from upper lumbar to lower thoracic levels.
- It receives the input signals from the muscle spindle through the dorsal root ganglion of the spinal cord, caudal to T1 spinal segment.

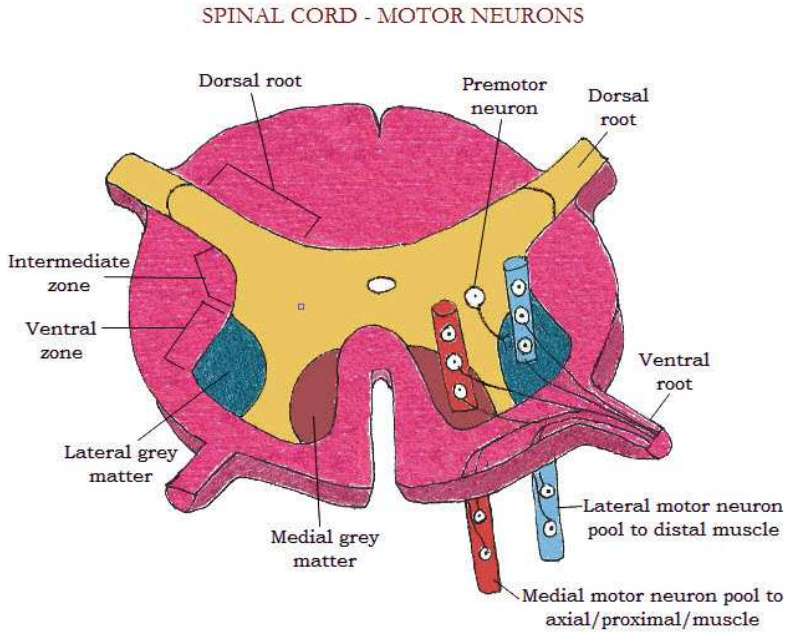
Nucleus Intermedialis and Nucleus intermediolateralis

- It receives the sensory signals from Golgi tendon organs up to cervical level in primates and up to thoracic level in other animals.
- It also senses tactile, thermal and pain sensations along with nucleus proprius.

NUCLEI OF SPINAL CORD - MOTOR NUCLEI

These nuclei are placed in the medial, lateral and intermediolateral columns of the spinal grey matter and controls somatic and visceral motor activities. They include

- Medial motor nuclei
  - Extend throughout the length in the medial motor column of the spinal grey matter and controls the motor functions of skeletal muscles.
- Lateral motor nuclei
  - This group of nuclei is placed in lateral motor column of the spinal grey matter and they extend the entire length of the spinal cord. They control the motor functions of the muscles in forearm, hind limb, shoulder, hip and lateral portions of the trunk.
- Intermediolateral nuclei
  - They are located in the inter-mediolateral column of the spinal grey matter from first thoracic to third lumbar spinal segments that form the pre ganglionic sympathetic fibres which control the motor activities in the visceral organs.



- Large motor neurons (alpha motor neurons) supply extrafusal muscles and small motor neurons (gamma motor neurons) supply either extrafusal or intrafusal muscle spindles
- Motor nuclei are placed somatotropically – laterally located cells supply distal extremity muscles and medially placed ones supply muscles of upper limb and trunk
- The cells of the motor neurons are excited monosynaptically by group Ia afferent fibers from muscle spindle receptors

- Multisynaptic activation or inhibition occurs by means of internuncial neurons of gray matter.

WHITE MATTER

- It forms the tract systems of the spinal cord containing myelinated nerves, and they transmit sensory impulses through ascending tracts and the motor impulses through the descending tracts between the spinal cord and the higher centres of the brain.
- The white matter is divided into paired dorsal funiculi, ventral funiculi and two lateral funiculi
    - Ascending (Sensory) tracts of spinal cord
    - Dorsal funiculus
        - Is composed of two ascending fiber tracts
        - These terminate in the medulla oblongata
        - The fasciculus gracilis terminates on the nucleus gracilis of the medulla
        - The fasciculus cuneatus terminates on the nucleus cuneatus and nucleus cuneatus lateralis of the medulla
        - But many of the fibers of the Dorsal funiculus terminate in the gray matter of the spinal cord before the rest enter the medulla
        - These fibers support intersegmental reflex activity and a portion them terminate on the cells of the nucleus proprius , nucleus dorsalis and nucleus intermedialis, and these further give rise to another system that ascends like the spinocerebellar, spinobulbar and spinothalamic tracts
        - The fasciculus gracilis and fasciculus cuneatus are also arranged somatotropically
        - Ascending fibers from the coccygeal, sacral, lumbar and thoracic levels comprise the fasciculus gracilis
        - Fibers from the cervical and upper cervical (T1) comprise the fasciculus cuneatus
        - As these fibers represent sensory information , the ipsilateral surface of the body is represented topographically within the dorsal funiculus
        - The fibers of the Dorsal funiculus convey impulses from joint receptors, muscle spindle and golgi tendon organ, cutaneous receptors for tactile and pressure sensations and so relay sensations related to joint movement and position.
        - Fibers from joint afferents and skin afferents terminate in the spinal cord for reflex mechanisms or within the nucleus gracilis and nucleus cuneatus of medulla for relay to thalamus and then cortex.

LATERAL FUNICULUS OF SPINAL CORD

- Ascending fiber tracts of lateral and ventral funiculi.
- These are not as clearly seen as that of the dorsal funiculus.
- It contains the following spinal tracts
    - Dorsal spino cerebellar tract
    - Ventral spino cerebellar tract
    - Rostral spino cerebellar tract
    - Spino cervical tract
    - Spinothalamic tract (Ventral and Lateral)
    - Spino-olivary tract

- Spino reticular tract
- Spino vestibular tract
- Spino tectal tract

DORSAL SPINOCEREBELLAR TRACT

- Occupies a portion of the lateral funiculus between the lumbar vertebrae and medulla
- This fiber tract originates from the ipsilateral dorsal nucleus and ends within the ipsilateral cerebellar cortex
- The fibers terminate in two regions of the cerebellar cortex.
- Conduction velocity of these fibers range from 30 to 110 m /sec.
- It carries afferent impulses from muscle spindle through dorsal root ganglion of the spinal cord and project to cerebellar cortex.

VENTRAL SPINOCEREBELLAR TRACT

- Arises from the cells occupying the lateral part of nucleus proprius and nucleus intermedialis
- This tract is not excited by afferents from the forelimbs
- The axons are mostly contralateral
- The tract reaches the cerebellum exclusively by brachium conjunctivum
- Most terminate contralaterally which make them ipsilateral to their origin
- Conduction velocity of these fibers range from 70 to 120 m /sec
- The tract contains afferent fibers of both exteroceptive and proprioceptive systems (Golgi tendon mostly) .

ROSTRAL SPINOCEREBELLAR TRACT

- Forearm equivalent of Ventral spinocerebellar tract
- Activated from ipsilateral forelimb nerves and uncrossed
- The tract reaches the cerebellum through the restiform body and the bracium conjunctivum
- The tract is activated monosynaptically by Golgi tendon organ afferents and polysynaptically by flexor reflex afferents

SPINOCERVICAL TRACT

- The tract arises from the nucleus intermedialis
- The tract terminates in the lateral cervical nucleus from where it is relayed to the ventero – postero-lateral nucleus of the thalamus
- Mostly involved with tactile afferents.

SPINOTHALAMIC TRACT

- The lateral and ventral Spinothalamic tracts which are present in primates, are absent in domestic animals.
- Instead these exist as a single tract concerned with tactile , thermal and pain sensations.
- The fibers project through the lateral lemniscus of the brain stem to the ventro – postero-lateral nucleus of the thalamus.

- The fibers also project through the reticular formation and central tegmental fasciculus to the intralaminar nuclei and the nucleus medialis dorsalis of the thalamus.
- In primates, the fiber system is very important for the relay of pain sensations but in domestic animals transection of this tract does not affect pin perception because pain perception continues due to the diffuseness of the propriospinal fiber which crosses and re crosses.
- Hence unilateral hemisection of the spinal cord in primate alleviates pain from contralateral lower extremity. This is not seen in domestic animals.
- Spinothalamic tracts arise from cells located in the dorsal horn in the region of the nucleus proprius of the spinal cord.

SPINO OLIVARY TRACT

- Arise from cells of the nucleus proprius of the spinal cord.
- The tract terminates in the caudal olivary complex of the medulla (medial and dorsal accessory olivary nuclei).
- This forms another pathway to the cerebellum for exteroceptive and proprioceptive afferents.
- Action potentials reach the cerebellar cortex ipsilateral to their origin within the peripheral nerves.
- This tract is important in informing the cerebellum of the position of the limbs and the external environment.

SPINO RETICULAR TRACT

- Arise bilaterally from all levels of the spinal cord.
- Terminate within many portions of the nuclei of the Brain stem reticular formation. These nuclei all receive afferents from the spinal cord.
- *Nucleus lateralis reticularis:* Projects fibers ipsilateral to cerebellar cortex
- *Nucleus reticularis paramedianus:* Projects fibers both ipsilateral and contralateral to cerebellar cortex. This forms another spino reticular cerebellar pathway
- *Roller nucleus and Nucleus intrafascicularis:* Project fibers ipsilateral to cerebellar cortex and forms a third spino reticular cerebellar pathway
- *Nucleus reticularis tegmenti pontis:* Forms a fourth spino reticular cerebellar pathway

Functions

- Important in many facets of normal activity.
- Carry sensory information from the skin and deep tissues and play an important role in the regulation of somatic and visceral motor activity.
- In maintenance of consciousness and perception .
- Accommodation of motor activities occur if this is transected bilaterally in domestic animals.
- Relay ascending activity to higher centers of the brain.
- Significant in pain perception.
- Modify respiration by terminating some of the fibers on nucleus of the pneumotaxic center.
- Control micturition.
- The tract terminates on neurons of the BSRFand serves as a feedback system for control of spinal cord activities.

SPINO VESTIBULAR TRACT

- Fibers ascend ipsilaterally
- Serve as part of a feedback system for the control of activity within the lateral vestibular nucleus and modifies activities within the spinal cord
- Plays an important role in regulating muscle tone

SPINO TECTAL TRACT

- Arise from cells located within the nuclei proprius and nuclei intermedialis of the gray matter.
- They decussate to the contralateral side and project to the cerebellum through rostral colliculi.
- *Function:* Convey tactile and thermal information and the sensation of pain.
- Also modifies activity within the descending tectospinal tract which is important in the control of motor activities of neck muscles.
- Also functions as a part of a multisynaptic pain pathway.
- Descending fiber tracts of the lateral and ventral funiculi.

PYRAMIDAL AND EXTRA PYRAMIDAL SYSTEM

Descending (Motor) tracts of spinal cord

It consist of pyramidal and extra pyramidal system

- Pyramidal system
  - This system controls the voluntary and fine motor movements in association with pyramidal motor cortex.
  - This tract is well developed in primates and carnivores, but absent in birds and reptiles.
  - It consist of cortico spinal tract

- Extra pyramidal system
  - It is a complex network consisting of three important motor nuclei, red nucleus, vestibular nucleus and reticular formation and the inter-connecting tracts passing the motor signals from the motor cortex to the motor neurons of the spinal cord. This system generates *gross involuntary movements* by providing muscle tone of the extensor muscle and thereby regulates posture and locomotion.
  - Cortical activity descending in the extra-pyramidal pathways to the *lateral vestibular nuclei* of the brain stem (strong excitatory for spinal alpha -motor neurons) is primarily inhibitory.
  - This system is constituted by four major descending tracts - the reticulospinal, vestibulospinal, and rubrospinal and tectospinal tracts that leave the brain stem to influence the spinal motor neurons.

CORTICO SPINAL TRACT

- It originates from the pyramidal (Betz) cells of the motor cortex, (60% from primary motor cortex and 40% from somasthetic motor cortex).

- Enter the internal capsule, pass through the cerebral peduncles, form the pyramids at he base of the medulla and then enter the spinal cord.
- In the medullary region, it divides into two unequal portions.It partially decussates before entering the spinal cord so the crossed ventral, uncrossed lateral, uncrossed ventral tracts are also formed.
- 80% of this tract decussate to the opposite side (contralateral) and descends as lateral corticospinal tract, in the lateral funiculus of the spinal cord.
- It extends throughout the length of the spinal cord in primates and carnivores and regulates fine motor control both in the fore and hind limbs by its direct synapse with the spinal alpha motor neurons. While in other animals it synapses with the internuncial neurons, which in turn control, the motor activities of the alpha and the gamma neurons.
- The remaining 20% of the tract descend from the medulla as uncrossed (ipsilateral), forming the ventral corticospinal tract in the ventral funiculus of the spinal cord and descend up to caudal cervical and cranial thoracic spinal segments.
- At this point, it decussates to the opposite side of the spinal cord and synapse with the alpha motor neurons, ventral corticospinal tract is totally absent in dogs.
- Four in number in domestic animals :
    - Crossed lateral
    - Uncrossed lateral
    - Crossed ventral
    - Uncrossed ventral

- All four tracts degenerate in the cervical and thoracic portions of the spinal cord
- The differential termination make them important in various function that can be grouped as :
    - Involving components from the sensory cortex : function primarily to alter activities of ascending sensory systems such as spinocerebellar, spinothalamic, spinotectal and spinoreticular tracts
    - Involving components from the motor cortex: function primarily to alter somatic and sensory motor activities
- There is always one Internuncial neuron between the corticospinal tract fibers and spinal alpha motor neurons
- Thus a high frequency discharge is necessary to overcome synaptic resistance and produce a motor activity
- In ungulates (horse, cow, pig, sheep and goat) the Corticospinal tract terminates upon the Internuncial neurons of the spinal cord. From here it is carried to the thoracic and lumbo sacral levels of through propriospinal systems.
- Primary function is the control of spinal cord reflexes for the elicitation of motor mechanisms in animals
- Produces excitation of alpha motor neurons supplying flexor muscles and inhibition of alpha motor neurons supplying extensor muscles of the limbs
- Also exerts control over the ANS.
- The pyramidal tract is involved in the voluntary motor control of the skeletal muscles.
- The pyramidal system is important in maintaining muscle tone.

RETICULO SPINAL TRACT

- Originate from Pontine and medullary reticular formations.
- Pontine reticular fibres are ipsilateral.

- Medullary reticular fibres descend bilaterally.
- They exert an influence upon the motor neurons (alpha and gamma) of the spinal cord without the involvement of interneurons.
- But the majority use interneurons for their influence upon somatic motor activity
- Visceral functions are mediated by this tract include the control of respiration , micturition, defecaton,cardiovascular alterations and gastrointestinal activity.
- This tract is tonically active and influences lower gamma motor neurons, the gamma loop mechanism mediated by this tract causes reflex contraction in the anti gravity muscle to maintain postural muscle tone

VESTIBULO SPINAL TRACT

- Consists of two descending fiber tracts :
  - Vestibulospinal tract and
  - Medial longitudinal fasciculus.
- Oringinates from the lateral vestibular nucleus and descends ipsilaterally through th length of the spinal cord.
- Medial longitudinal fasciculus fibres terminate monosynaptically upon alpha motor neurons.
- The tract ipsilaterally exerts an excitatory influence upon extensor muscle tone and an inhibitory influence upon flexor muscle tone and contralaterally exerts an inhibitory influence upon extensor muscle tone and an inhibitory influence on flexor muscle tone.
- Also alters the tonus of the muscles of neck and forelimb.
- Exert considerable influence upon the ANS by the production of motion sickness involving the constant acceleration and deceleration of the head.

RUBRO SPINAL TRACT

- This is the more important motor tract in domestic animals.
- Arises from the red nucleus of the midbrain
- Decussates near its origin and enters the spinal cord contralateral to its origin
- Descends in the lateral funiculus throughout the entire length of the spinal cord
- There is always involvement of Internuncial neuron between the rubrospinal tract fibers and spinal alpha motor neurons
- They are rapidly conducting.
- This controls semi skilled movements.
- It functions to stimulate the flexor motor activity and inhibit the extensor and motor neuron activity.

TECTO SPINAL TRACT

- Arises from the deep layers of rostral colliculus.
- It deccussates within the dorsal tegmentum.
- Descends caudally in the brain stem.
- In cat it descends contralaterally.
- Terminates in the rostral region of cervical spinal cord.
- Plays a major role in mediation of auditory and visual reflexes and modifies neck reflexes.
- There is always involvement of Internuncial neuron between the tectospinal tract fibers and spinal alpha motor neurons.

MAJOR MOTOR CONTROL

- Many fibres connect adjoining and distant segments of the spinal cord with each other and do not project outside the spinal cord , called the Propriospinal fiber systems.
- They make up a major mass of the white matter of the spinal cord.
- They originate from the spinal gray matter except the motor nuclei.
- They are important in the interrelation of neural activities within the spinal cord.
- Reflex ats and complex reflexes involve several such relays.
- During walking or running, they are responsible for coordination of motor activity required to maintain equilibrium and allow the proper sequence of limb movement for propelling the animal forward.
- In spinal animals after a period of weeks , if assisted in maintaining balance , exhibit stepping movements similar to those observed in normal locomotion.
- The central motor control system includes the *motor cortex, basal ganglia* and the *cerebellum*. The motor cortex transmits its signals through caudate nucleus, putamen, globus pallidus, subthalamic nucleus, substantia nigra and reticular nuclei of the brain stem into the spinal cord.
- The motor cortex controls *pyramidal* and *extra pyramidal systems*. The motor cortex-pyramidal system is associated with voluntary performance of fine skilled movements. Gross involuntary movements are generated by the motor cortex-extra pyramidal system.
- The primary motor cortex controls the contraction of individual muscles and evokes discrete movements. The secondary cortex regulates complex movements of head, neck, trunk and limbs. The premotor cortex controls fine movements of hands, fingers, mouth and tongue, co-ordinated movements of eyes and head.
- Both the basal ganglia and the cerebellum are concerned with co-ordinated movements with pyramidal and extra-pyramidal systems. Co-ordination of *slow or ramp movements* is associated with basal ganglia. The cerebellum co-ordinates the *fast or ballistic movements*.

MOTOR CONTROL OF THE BRAIN

Demonstration of the motor control of the brain

- Motor control system of the brain is studied by removal of a particular portion and observing the alterations in the somatic motor activity.

Decorticate preparation

- Demonstrated in cat or dog.
- In such preparations cerebral cortex with underlined white matter will be removed.
- Such animal will exhibit normal standing posture.
- When the animals are suspended in space they extend their limbs.
- This extension disappears when they contact the supporting surface.
- In such preparations tactile placing and visual placing reflexes are absent.
- Vestibular placing, tonic reflexes and body righting reflexes are present.
- Righting reflex
    - To perform righting reflex and progressive movement certain brain structures are necessary.
    - They are sub thalamic nucleus, Red nucleus, substantia nigra with inter connecting fiber tracts, basal ganglia, reticular formation and cerebellum.

- These enters responsible for righting reflexes are located within the rostral mid brain. This can be demonstrated in amphibians/animals when it is put on its back it immediately rights itself.
- When it is placed in water it swims. Frog response to various stimuli by jumping and crawling.

Decerebrate preparation

- This preparation emphasis the importance of the motor activity of structures or parts of the brain anterior to the pons and lower mid brain.
- This preparation is done by transection of the brain stem at the level of mid brain.
- Such animal exhibit pronounced antigravity rigidity that is depended on hyper excitability of the segmental myotatic reflex.
- This is known as "Decerebrate rigidity".
- This is primarily due to hyper excitability of gamma motor neurons. Hence, otherwise known as "GAMMA RIGIDITY".
- Transecting the dorsal root supply to the particular limb will be abolished by this rigidity.
- The higher centre responsible for the development of rigidity is located within pontine reticular formation.
- During this preparation pontine reticular formation is relieved from inhibitory influences of cerebral cortex.
- Other portion of brain stem reticular formation is also renowned of inhibitory influences from brain stem.
- These causes increase in reticulo spinal excitation on gamma motor neurons supplying intra fusal fibres.
- To maintain the decerebrate rigidily cerebellum and verticular system play a major role.
- In the decerebrate preparation, tonic labyrinth and tonic neck reflexes are elicited.
- Vestibulo spinal tracts are major system contributed to decerebrate rigidly.
- Righting reflexes like vertibular lighting reflexes are absent in these preparation. They stand when placed in a standing position but they cannot be moved.
- Decrebrated animal will have increase in muscle tone of both flexors and extensors of the fore limbs.
- The limbs become pillars while standing and tail become erect and horizontal.
- Head is extended and body shows hyper extension (hobby/cortoon/caricature animal)
- Increase in muscle tone is due to intact vestibulo spinal tract but, removal of inhibitory reticulo spinal tract.
- Normal standing posture is exaggerated.
- Stretching postural reflexes like stretch reflex, supporting reactions, tonic reflexes labyrinth reflexes are present.

GAMMA LOOP MECHANISM

- To initiate all somatic motor actions the influence of motor system upon gamma motor neuron is significant.
- The motor pathways depolarized dominantly gamma motor neurons which in turn initiate the contraction of intrafusal fibres of muscle spindle organ. These results in activation of group Ia afferent fibres.
- These fibres monosynaptically connect with alpha motor neurons and causes extra fiscal muscle contraction. This is known as "*Gamma loop mechanism*"
- Cutting across the dorsal root blocks this mechanism and causes immobility and uncoordinated somatic motor activity.

- Excitability of gamma motor neurons in proportion to alpha motor neurons is determined by particular spinal afferent under the influences of BSRF.
- BSRF is under the influences of basal ganglia and cerebral cortex.
- This indicates cerebellum and basal ganglia alter the motor activity directly through the involvement of red nucleus primary motor cortex and vestibular nucleus and indirectly through BSRF.

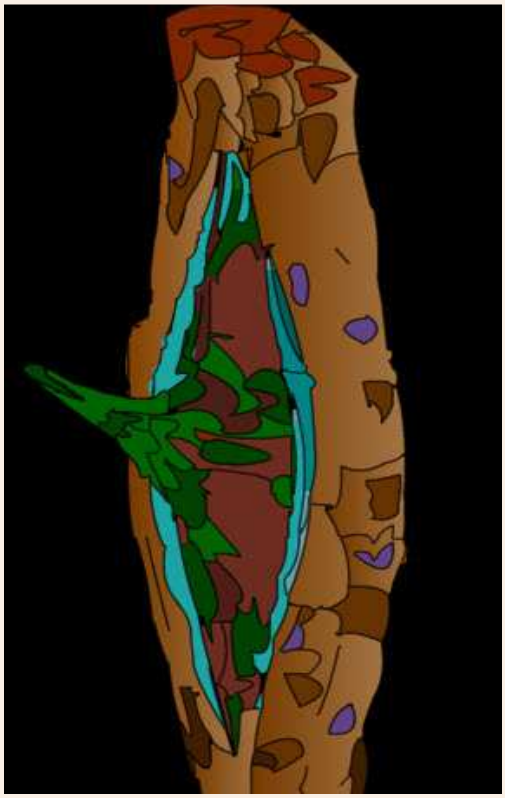
SPINAL ANIMAL (FROG)

- Spinal animal is prepared either transecting at the spinomedullary junction or by ligating both carotid arteries and the mid-vertebral artery to arrest the brain activities.
- Frog's head behind the tympanic membrane is transected to separate spinal cord from the brain. Transection is to be done below the foramen magnum.
- In these preparation when frog is palced on its back there is no spontaneous movement due to complete paralysis of the movement.
- When a portion of a skin is pricked with the needle no withdrawal reflex is noticed.
- This is known as “Spinal shock depression of the spinal function and is reversible.
- Spinal shock is removed in few minutes in a frog and few hours in a dog.
- Regaining from the spinal shock certain reflex activity are seen.
- When one of the limb is pulled down it is quickly drawn up.
- Withdrawal of the limb is noticed on pinching. (flexor reflex)
- If one of the limb is stimulated with the greater force all the limbs are flexed.
- However, respiratory movement voluntary movement righting reflexes are absent (they can’t swim when they place in the water).
- Except basic stretch reflex and supporting reactions all the other reflexes are absent.

Thalamic animal

- Cerebral cortex, white matter some portion of the basal ganglia are removed.
- Thalamus and hypothalamus with the caudal portion of the basal ganglia remain intact.
- These animals show righting reflexes and temperature regulation.
- Reflexes associated with emotional states fight and anger are present .
- Tonic labyrinth and tonic neck reflexes and other righting reflexes are present.
- Propriokinesthetic, righting reflexes placing reactions and hopping reflexes are absent.

MODULE-10: REFLEXES OF SPINAL CORD



LEARNING OBJECTIVES

- The outcome of this module help to,
  - understand basic operation of reflexes,
  - components involved,
  - participation of various levels of system,
  - spinal reflexes with an indepth analysis of its importance to animal,
  - properties of reflex arcs and study the classification of reflex arcs and
  - conditioned reflexes and their mechanism.

POSTURAL REFLEXES

- This is also known as postural reactions.
- It is defined as the position or arrangement of the body and its limbs.
- It is controlled by the activity of the skeletal muscles of the neck and limbs.
- Control of posture involves mechanisms that alter the excitable activity of α-motor neurons of the spinal cord.
- Structures concerned for maintenance of posture are located at all levels of brainstem, medulla and pons.
- It is divided into
  - Supporting reflexes
  - Attitudinal reflexes

Supporting reflexes

- These reflexes involve activity that initiates co-ordinated skeletal muscle activity to cause the limbs to be fixed into supporting columns against the pull of gravity.
- The supporting reflexes are further classified as local supporting reflexes and segmental supporting reflexes.
- Local supporting reflexes
  - These reflexes of the limbs produce a fixed standing posture, which prevents collapse under the force of gravity.

  - They involve only a few segments of the spinal cord for reflex actions.
  - Local supporting reflexes include the myotatic reflex, and extensor thrust reflex. These reflexes are initiated by the stimulation of muscle spindle or cutaneous receptors of the limbs involved or by stimulation of the opposite limbs with noxious stimulus.
- Extensor thrust reflex
  - When pressure is applied to the footpads of a dog, that limb is extended into a supporting column by the extensor muscles of the elbow, which are initiated by myotatic reflex.

Segmental supporting reflex

- These reflexes use many segments of the spinal cord. Segmental supporting reflex arcs are not solely limited to the spinal cord, but also involve the brain stem.
- The reflex is represented by crossed extensor reflex. These reflexes involve sensory input to a local area and produce reflex activity over a wide region of the spinal cord.

☐ The same reflex occurs in the opposite limb. When a hind limb is extended, the opposite forelimb also is extended.

ATTITUDINAL REFLEXES

Attitudinal reflexes

- It involves modification of posture as a result of varying positions of the head.
- Displacement of one part of the body is followed by postural changes in other parts so that a new posture is assumed.
- This is classified into two types,
  - Cortical reflexes
  - Extra cortical reflexes.

Cortical reflexes

- It is mediated through cerebral cortex and can be divided into
  - Placing reflex
  - Hopping reflex
  - Vestibular placing reflex

CORTICAL REFLEXES - VESTIBULAR PLACING REFLEXES

Vestibular placing reflex

- Blind folded animal if dropped head down position towards the ground; the animal assumes specific supporting postures.
- The forelimbs are greatly extended.
- Toes are spread apart.
- Tries to get support against an expected contact with the ground.
- Here, direction of acceleration within the field of gravity is not a deciding factor to initiate this reflex.

CORTICAL REFLEXES - HOPPING REFLEX

Hopping reflex

- If the animal is made to stand on one limb and if simultaneously is moved horizontally the supporting limb will hop to maintain support for the body.
- This is supported by spinal reflex, myotactic reflex and tactile placing reflex.
- This reflex is temporarily abolished by transecting the corticospinal motor pathways. But due to involvement of myotactic reflex of the limbs, there will be weak exhibition of this reflex.

CORTICAL REFLEXES - PLACING REFLEX

Tactile placing reflex

- Blind folded animal if held in space in a dropping position near a supporting surface to facilitate either of the parts of the body like feet, ventral surface of the body, neck or chin in contact with the surface, the animal quickly assumes a posture to place either of the limbs with the supporting surface.
- This reflex requires involvement of both spinal and cerebral cortex by way of corticospinal motor system.
- Decorticate animals or animals wherein descending motor pathways are abolished are not in a position to exhibit this reflex.

Visual placing reflex

- Similar to tactile placing reflex in a normal animal which is mediated through visual integrity system to reach the supporting surface prior to involvement of the cutaneous contact.
- Descending corticospinal motor system do participate in this surface.

EXTRACORTICAL REFLEXES

- It is mediated through spinal cord, medulla and pons. It is of two types - the tonic neck reflexes and tonic labyrinthine reflexes.

Tonic neck reflex

- Neck receptors include muscle stretch receptors and receptor organs that sense the position of joints between cortical vertebrae related to head position.
- Extension of the head to dorsal position causes flexion of the hind limbs, but extension of the forelimbs occurs when the head is moved downwards.

- Turning of head to right side causes the extension of both the right limbs and the flexion of both the left limbs to support the body against gravity; this helps to prevent falling.
- If the turning is on the left side, it results in the extension of the left limbs and the flexion of the right limbs.

EXTRACORTICAL REFLEXES - TONIC LABYRINTHINE REFLEXES

Phasic labyrinth reflex

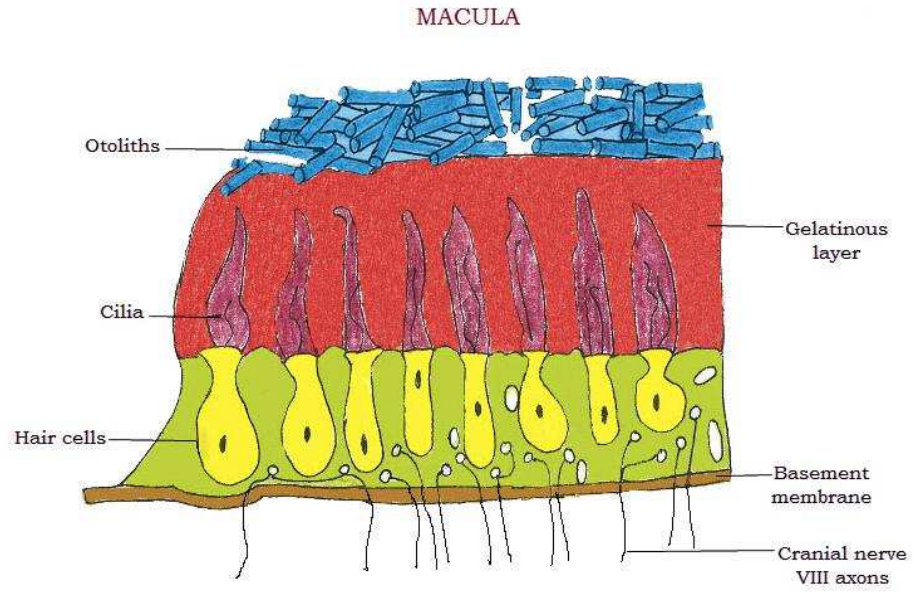
- These reflexes are initiated by receptor organs of the semicircular canal.
- Rotational acceleration of the head stimulates vestibular semicircular canal.
- Maintenance of upright posture on rotation if kept on rotating platform and altering the position of head and body to the opposite of the rotation.
- On stoppage of the rotation, the animal's head will be turned and animal lean towards the opposite direction.
- Semicircular canals help to maintain the animal's balance on sudden change in the head's direction.

Righting reflex(RR)

- This is the involvement of tonic labyrinthic reflex to maintain the position of the head upon the neck.
- If the animal is maintained in a supine position. Further due to the involvement of tonic neck reflex to later the body's position to an upright position.
- This reflex response to body position is known as righting reflex.
- If a cat is dropped in a space from a supine position, as in the process of dropping, the position of the head is immediately altered to an upright position due to tonic labyrinth reflex. Due to this, other parts of the body of the body position so as to support which is due to tonic neck reflex and assist the animal to land in upright position.

VESTIBULAR APPARATUS

- The vestibular apparatus located in internal ear that detects sensation related to equillibrium (linear accelaration of head).
- Popularly known as OTOLITHIC organ.
- It consist of bony labyrinth (semicircular canal with utricule and saccule) and membranous labryinth. This is the functional part of the vestibular apparatus and is composed of cochlear duct, three semicircular canals, the utricle and saccule.
- Membranous labryinth with an enlargement at one end called as is filled with endolymph and surrounded by perilymph.
- The ampulla is the enlargement of the semicircular canals at it base and is filled with endolymph.
- The bag-like structures (vestibular sacs) the utricle and saccule have sensory area known as *maculae*. This detects the orientation of the head according to the direction of gravitational pull.



- Utricule and saccule are located within the petrous portion of the temporal bone.
- The maculae is covered by a gelatinous layered and has otoconia or statoconia which are small crystals of calcium carbonate.
- The gelatinous layer has numerous hair cells projecting with their cilia with synapse with the sensory axons of vestibular nerve.
- Change in the orientation of head alter the bending of the celia due to the weight of otoconia accordingly signals are transmitted to cerebellum.
- Maculare detect linear acceleration of the head and act as the receptor for righting reflex.

- The ampullae of the semicircular canals do not have otoliths, but has the cristae which contains hair cells inside the cup shaped gelatinous mass called as cupula. This houses a sensory organ called crista ampullaris that will be excited by the flow of endolymph.
- Hair cells do project their cilia into cupula and the base of these hair cells are connected to axons of vestibular nerve.
- Bending of cupula with the flow of endolymph alters the position of the head.
- The cristae detect angular/rotary acceleration of the head, while the ampullae of the vestibular sacs detect linear acceleration. Most of the vestibular nerve fibres from the macula and semicircular canals end in vestibular nuclei present at the junction of medulla and pons; some fibres pass directly to cerebellum.
- The fibres that end in the vestibular nuclei send second order neurons to the cerebellum, cerebellar cortex, vestibulospinal tract and other areas of the BSRF.

Process of equilibrium maintenance

- When the head is tiled to one side, the otoliths and gelatinous mass of the ampullae of the vestibular sacs induce a bend in the hair cells. This excites or stimulates the hair cells.
- Change in head movements causes movement of otoliths and inertia cost subsequently induces generation of action potential and through vestibular nerve cerebellum receives the information relating the position of head in space.
- The macula of the utricle is placed horizontally in the ear and therefore can detect movement in the horizontal plane.

- The macula of the saccule is placed in vertical position and therefore detects vertical movement of the head.
- But within these maculae, the hair cells are oriented in opposite direction and thereby detect movements of forward, backward, and side to side directions.
- Since the hair cells placed in one plane are stimulated to tilt of the head in one particular directfion, the brain can perceive the direction of the movement, which sends motor signals to maintain the body in posture to maintain the position of the body.
- The semicircular canals are located at right angles to each other and therefore the ampulla in these canals detects angular movement of the head. Each semicircular canal detects the acceleration in one single plane pertaining to it.
- If the head is turned in one particular plane of one particular canal, then there is fluid motion inside the canals causing push on the ampulla and stimulating the hair cells.
- Since each canal is placed in one plane, the degree of movement is sensed by the stimulation of the hair cells in the canals.
- The whole body position is maintained by inputs from visual, auditory and proprioceptive systems.

NYSTAGMUS

- It is the movement of the eye in relation to the movement in the head of the animal.
- It consists of rhythmic movements of the eyes in a dorsoventral, lateromedial, or oblique directions.
- It is characterized by the slow movement of the eye in one direction and then quick return to its original starting position.
- These movements are continued till the rotatory acceleration is maintained.
- The receptor organs of the nystagmus are the cristae of the semicircular canals.
- The afferent neurons for the nystagmus are present within the VIII cranial nerve.
- These fibers synapse with internuncial neurons within the vestibular nuclear complex of the medulla and pons.
- The motor activity is relayed to the extrinsic muscles ot the eye via the brainstem reticular formation.

REFLEXES

- A reflex is a specific, stereotyped, involuntary response to peripheral stimulation.
- The spinal cord is a reflex neural system and its activity is modified by influences transmitted from the brain via descending fiber system
- The regulating system of the brain are informed of the reflex activity initiated within the spinal cord and is able to initiate descending influences to modify this reflex activity so that it is of maximum economy to the animal
- Reflexes involve several components of CNS and PNS connected in series
- Skeletal motor system and visceral motor system (ANS) are closely related to spinal cord reflexes.

- The components of a reflex arc are,
    - Receptor organ
    - Afferent neuron
    - Efferent neuron
    - Effector organ
- Sometimes one or more internuncial neurons

- Only the reflex arcs utilized by the muscle spindle stretch receptor afferents do not contain internuncial neurons Eg : The myotatic reflex arc in which fibers terminate directly on the efferent neurons . Such an reflex arc is known as an monosynaptic reflex arc because only one synapse is involved
- All other reflexes are referred to as polysynaptic reflex arcs indicating that more than one synapse is involved
- Reflex arcs that do not involve a synapse either in the PNS or CNS involve in reflexes referred to as axonal reflexes
- Many autonomic reflexes take place within peripheral ganglia without spinal cord component and are known as ganglionic reflexes which are mostly monosynaptic
- All autonomous reflex arcs that pass through the spinal cord are polysynaptic or atlest bisynaptic in nature.
- Alpha motor neurons are the final common pathway to skeletal muscle.
- Alpha motor neurons possess large cell bodies which are filled with well-developed, granular endoplasmic reticulum called Nissel substance.
- Before emerging out of the spinal cord , the axon gives rise to form one to six branches which terminate within the confines of the spinal cord called the recurrent collaterals of Alpha motor neuron axons.
- They terminate upon neurons that exert an inhibitory effect on Alpha motor neurons called the Renshaw cells.
- According to Dale's principle, a given neuron can produce only one type of transmitter agent at all its terminal synapses. The Alpha motor neurons produce acetylcholine at all its neuro effector junction.
- Prolonged discharges of action potentials upto 50 msec at high frequencies of 500 to 1000 per sec are possible at these synapses.

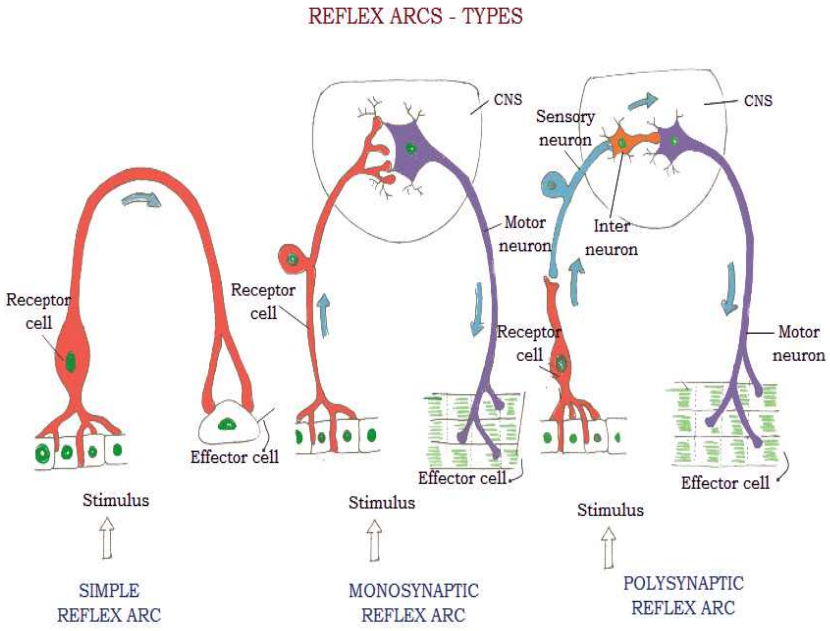
CLASSIFICATION OF REFLEX ARCS

Components of reflex arc

- The neural circuit for reflex is referred to as reflex arc. Five components commonly make up a reflex arc. They are the receptor organ, sensory (afferent) neurons, interneurons in spinal cord or brain, motor neurons (efferent) and an effector organ (muscle or glands).

Classification of reflex arcs

- Neurons may combine to give at least five types of reflex arcs.

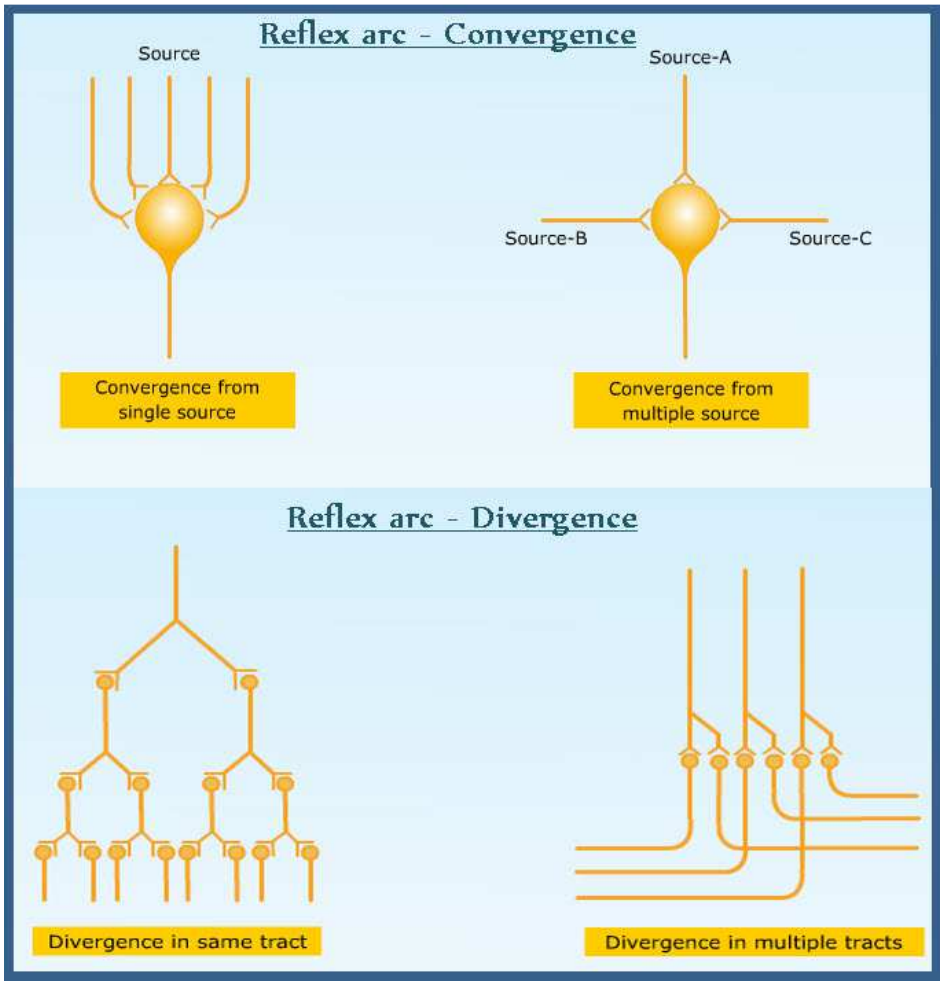


Monosynaptic reflex arc

- There is a single synapse between sensory and motor neurons of the spinal cord (without interneuron) e.g Myotatic reflex

PROPERTIES OF POLYSYNAPTIC ARCS

- *Series synaptic reflex arcs:* Several simple reflex arcs are united in series to give a chain of reflex arcs, e.g. flexor and extensor responses related to muscle tone.
- *Convergent effector reflex arcs:* Two or more afferent neurons are in synaptic relation with only one efferent neuron. Hence, stimulation of many receptors converges to excite a single effector.
- *Divergent effector reflex arc:* One afferent neuron is synaptic connection with two or more efferent neuron. Stimulation of a single receptor diverge through many neurons to excite more than one effector
- *Internuncial reflex arcs:* Two or more afferent neurons are connected with two or more effectors. Stimulation of either of the two receptors causes reflex response in one or both effectors. The connections are established through interneuonrs. It provides reciprocal excitation and inhibition for smooth co-ordinated activity.



PROPERTIES OF REFLEX ARC

One way conduction (Bell and Magendie law)

- The dorsal root fibres of the spinal cord are afferent (sensory), while the ventral root fibres are efferent (motor).
- The sensory impulses are centripetal (towards the CNS) and motor impulses are centrifugal (away from the CNS).

Slow speed of conduction

- Slow build up of EPSP to threshold stimulus causes the synaptic delay, thus slows the speed of conduction.
- *Fatigue:* Due to exhaustion of the neurotransmitter substance at the synapse.
- *Refractory period:* Show both absolute and relative refractory periods.

- *Reinforcement:* (Successive induction): By discharging signals from many neurons. It is of two types,
    - *Positive successive induction:* Irritation at a point of the skin by a tick leads to scratch reflex
    - *Negative successive induction:* Reciprocal stimulation and inhibition; e.g., biting reflex, reflex flexion and extension of the limbs.
- *Rebound:* Increased reflex response following the withdrawal of inhibitory stimulus.
- *After discharge:* Continuous impulse discharge from the centre, even after the cessation of the original stimulation.
- *Summation:* Subminimal stimuli at rapid succession develop effective EPSP, the temporal summation.
- *Recruitment:* Prolonged and altered intensity of stimulus progressively activates more number of motor neurons.
- *Irradiation:* Gradual increase in the intensity of stimulus excites the additional neurons, and the effectors.

MYOTATIC REFLEX

- Sudden stretch of a muscle causes its reflex contraction.
- Commonly known as "KNEE-JERK" reflex.
- It forms the basis of the maintanence of *"Muscletone"*
- Receptor organ concerned with this reflex is *Muscle –spindle organ.*
- These receptors are encapsulated in the belly of skeletal muscle.
- CNS receives sensation from these receptors and coordinates *Posture and Locomotoin* .

MUSCLE-SPINDLE ORGAN

Muscle-spindle organ -Complex receptor organs

- Made up of specialized striated muscle called *intra fusal muscle fibres*, present within a connective tissue capsule and are attached to the skeletal musclel.

Extra-fusal-fibres

- These are the muscle fibres which produces physical shortening of the muscle.
- Alpha motor neurons innervates them.

Speciality of intra fusal muscle fibres

- Have contractile proteins at their polar ends only and not in the centre.i.e. polar ends can contract without the contraction of centre portion.
- Centre portion is known as equatorial region from where sensory nerve arises and carries information to CNS from muscle spindle via peripheral nerves.
- The contractile ends of intra fusal fibres are innervated by gamma motor neurons (ventral horn of spinal cord).

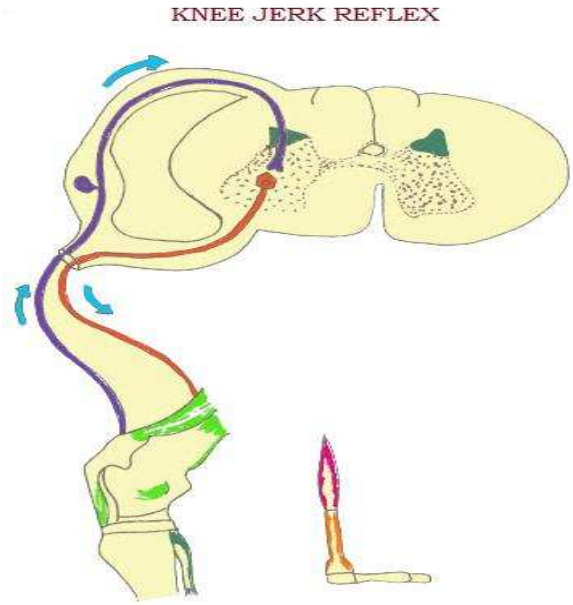
IMPORTANCE OF EQUTORIAL REGION

- Action potential are generated from this region by stretching

- Muscle spindle receptors lie parallel to the extra fusal fibres.
- Functionally these receptors are connected to the origin and insertion of the tendon i.e. lengthening of the whole muscle will stretch the equatorial region.
- Contraction of polar ends also generate action potential, which passes in the sensory nerve, in proportion to lengthening of the equatorial region.
- Sensory fibres are capable of detecting the changes in length during the dynamic phase of muscle lengthening and during resting stage.
- When CNS receives the sensory inception, it causes excitatory mono-synaptic connection with alpha-motor neurons (that supplies extra fusal fibres of the same muscle)
- This leads to contraction of the whole muscle which results in shortening of the muscle spindle (especially the equatorial region)
- This shuts-off the action potential from spindle receptor and muscle contraction is stopped.

KNEE-JERK REFLEX

- Strech of quadriceps muscle results when patellar tendon is struck with a blunt object. (the muscle spindle is stretched).
- Action potential from muscle spindle go to lumbar spinal cord and causes excitation of alpha motor neurons of quadriceps which causes contraction of the muscle and extension of knee joint.(muscle stretch reflex).
- Action potential passing on the gamma motor neurons causes shortening of polar region of intra fusal fibres and not the equatorial region.Hence the equatorial region stretches and initiates contraction.
- CNS initiates contraction of extra fusal fibres reflexely by gamma motor neurons viz gamma-loop. Gamma-loop has gamma motor neurons, intra fusal fibres (including middle receptor), sensory nerve of the spindle which forms excitatory mono synaptic synapse and alpha-motor neurons back to the same muscle and extra fusal fibres



Co-activation

- Each time when CNS transmits signals to alpha-motor neurons, the Gamma-motor neurons are also stimulated.This causes the contraction of both extra fusal and intra fusal fibres around the same time.
- This mechanism of simultaneous contraction check the length of the muscle spindle receptor and to maintain proper body movements without jerks.

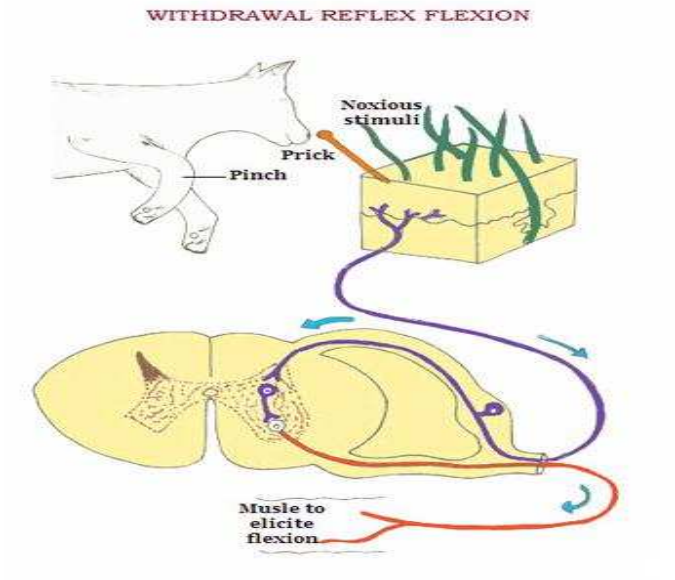
GOLGI TENDON REFLEX

GOLGI tendon REFLEX/ inverse myotatic reflex / clasp-knife reflex

- Sensory receptor organ : golgi tendon organ
- Present in tendons of somatic skeletal muscle
- Present in pheripheral terminal and large of mylinated, class I_b nerve fibres.
- These receptors are excited during muscle contraction as contraction causes stretch of the tendon.
- These recptors are sensitive to change in the muscle tension.
- Class Ib fibres terminate in the internuncial neurons of spinal cord after forming a mono synaptic reflex arc.
- When muscle contracts, muscle tension increases which also causes increased tension of muscle tendons which in turn ends in the stimulation of golgi tendons.
- Sensory fibres from this receptor transmit signals to spinal cord to inhibit muscle contraction.
- It is a protective reflex to prevent over stretching and tear of muscle fibres and coordinate skeletal muscle activity.

WITHDRAWAL REFLEX

- Protective reflex, helps to avoid noxious stimuli by flexion, hence known as flexion reflex. Ex. Withdrawal of limb if noxious stimuli is applied to the limb. This reflex is poly synaptic
- Withdrawal is achieved by contraction of flexors (the muscle should be excited) and inhibition of extensors (occurs simultaneously).



CROSSED EXTENSOR REFLEX

- To avoid noxious stimuli there will be withdrawal of limb and is associated with alteration of posture involving extensors of unaffected limbs to support the body.
- Neurons involved are same as that of withdrawal reflex and also include neurons that control the extension of opposite limbs.

EXTENSOR THRUST REFLEX

- When pressure is applied to the footpad it causes simultaneous contractions of flexors of the digits and the extensors of elbow which extend the limb into a supporting column which are initiated by myotatic reflex.

SCRATCH REFLEX / MULTISYNAPTIC REFLEX

- Initiated by combination of tactile and pain receptor organ.
- Afferent enters the spinal cord and end in internuncial neurons.
- Preventive reflex helps to remove the irritating stimuli from skin (dorsal lateral surface of the thorax and neck) with one rear limb and out affecting the posture of the animal.

CONDITIONED REFLEXES

Conditioned reflexes (Conditioned responses)

- These reflexes are acquired or reflexes that are gained naturally or artificially by the individual during the course of their lifetime.

- Salivary and the gastric secretions produced in response to food is a natural and inborn reflex. The sight or the smell of the food forms the natural stimulus, which causes the natural conditioned response in the form of salivary or gastric secretions.
- If the animal is fed in association with the bell sound or music (neutral stimuli) and this practice is frequently repeated, it causes conditioned salivary secretion to the bell sound or music even in the absence of food. This type of response is referred to as artificial conditioned reflexes. Pavlov, a Russian Scientist proved this reflex mechanism in the dogs.
- Conditioned responses develop in relation to an already existing mechanism, which can be an inborn reflex or another conditioned mechanism. These reflexes have been developed in relation to cardiovascular, alimentary, pupillary and secretary reflexes.
- Visual, olfactory, auditory, gustatory, tactile and proprioceptive stimuli have been used to evoke conditioned responses. The afferent side of the conditioned mechanism of conditioned stimuli becomes attached to an already functional efferent side of the reflex.
- Cerebral cortex plays an important role in the formation and maintenance of conditioned responses.

REQUIREMENTS OF CONDITIONED REFLEXES

- The natural stimulus and the conditioned stimulus (neutral stimulus) must be in coincidence with each other. E.g. Food supplies (natural stimuli) along with a bell sound (neutral stimuli). The animal exhibits simultaneous conditioned response only for the bell sound even in the absence of food. But if the application of the neutral stimulus (sound) precedes the natural stimulus (food) the animal develops delayed conditioned response.

PROPERTIES OF CONDITIONED RESPONSES

Reinforcement

- If two neutral stimuli (e.g. sound and light) are applied simultaneously, the conditioned response will be greater than by a single stimulus.

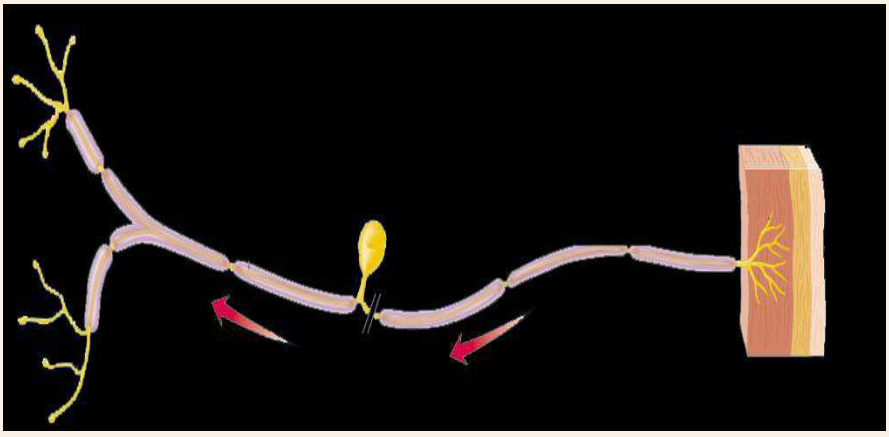
Abolition

- The neutral or the conditioned stimuli must be applied at a constant strength and duration to produce a conditioned response. Otherwise, this may gradually weakens and finally disappears.
- Frequent rapid succession of the conditioned stimulus (neutral) without the natural stimulus (e.g. food) the developed conditioned response will gradually weaken and finally disappears completely.

Differentiation

- Animal that has the ability to differentiate two independent neutral stimuli (sound and the flashes of light), develops two types of conditioned responses which are very specific. Hence, the other neutral stimuli cannot produce the same conditioned response.

MODULE-11: PERIPHERAL NERVOUS SYSTEM



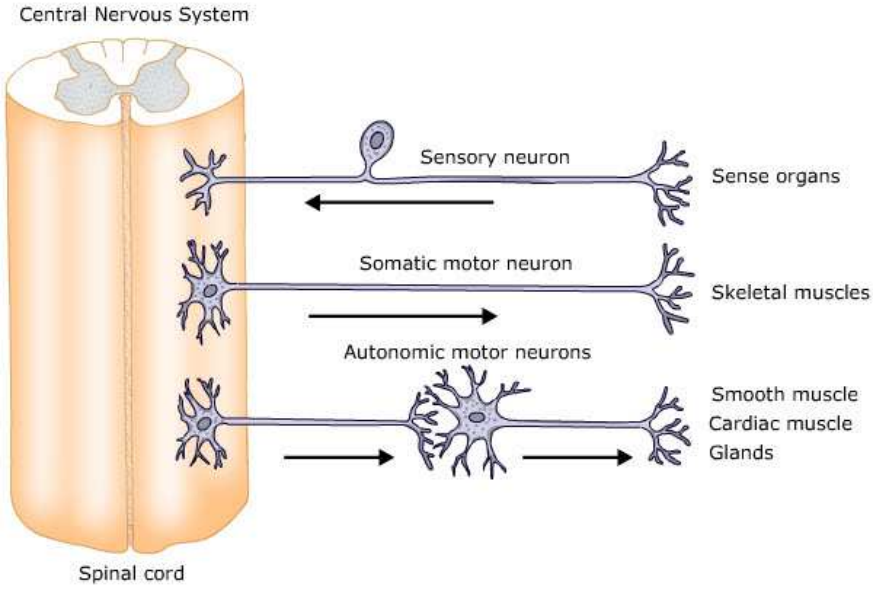
LEARNING OBJECTIVES

Learning this module fulfills,

- the understanding of the nervous system and the learner will appreciate how an animal reacts instantaneously to an external stimulus with the linking of peripheral nervous system to central nervous system.

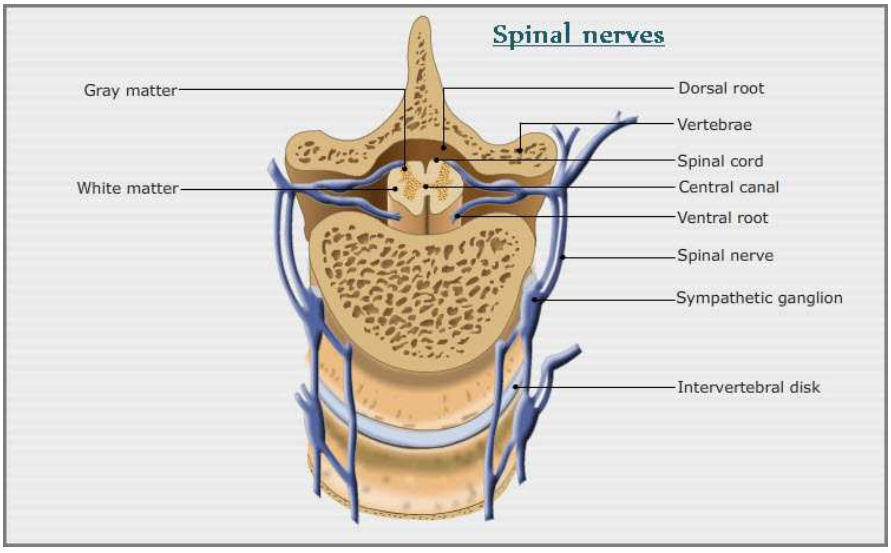
INTRODUCTION TO PERIPHERAL NERVOUS SYSTEM

- This includes all nervous structures, the peripheral ganglia, spinal nerves, cranial nerves and the autonomic nerves, located out of the brain and the spinal cord.
- It functions to provide communication between the receptor organ and the CNS (sensory), from the CNS and the effector organs (motor).
- Peripheral nerves are myelinated.
- It is divided into motor (efferent) and sensory (afferent) subsystems.
- The motor peripheral nerves that supply to skeletal muscles are referred to as somatic motor nerves and those that supply the cardiac and smooth muscles and exocrine glands are referred to as autonomic nerves.
- The afferent (sensory) system is of two types: Somatic and visceral. Somatic sensory nerves carry impulses from the photoreceptors (eye), auditory receptors (ear) and stretch receptors (skeletal muscles), whereas the visceral sensory nerves carry visceral sensations from the chest and abdomen.



SOMATIC COMPONENTS

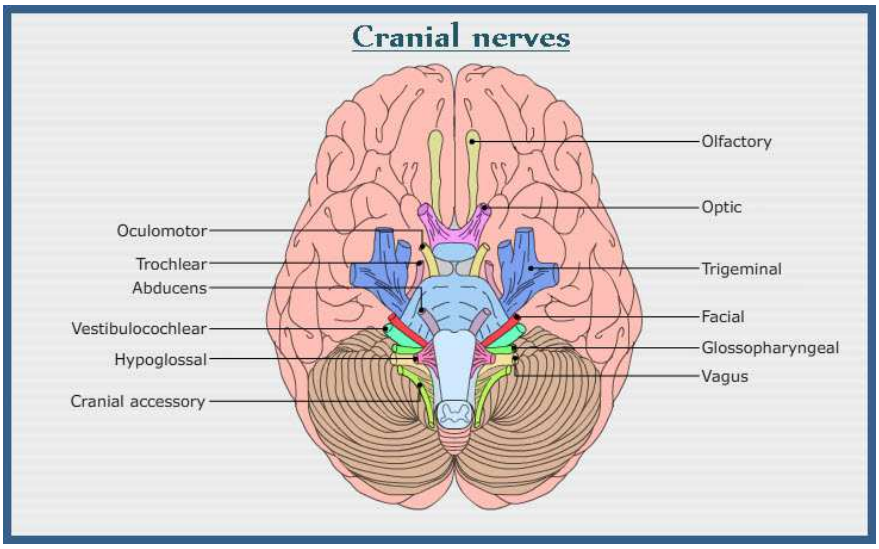
- The spinal nerves emerge as a pair of nerves from each spinal segment through the inter-vertebral foramina, which lies between the two adjoining vertebrae.
- Thus, the number of pairs of thoracic, lumbar and sacral nerves is similar to the respective vertebrae.
- Although the cranial portion has seven cervical vertebrae, eight pairs of cranial nerves emerges, while the caudal vertebrae have fewer pairs of coccygeal nerves.
- The terminal portion of the spinal cord, the meninges and the nerves are collectively referred to as cauda equinae.



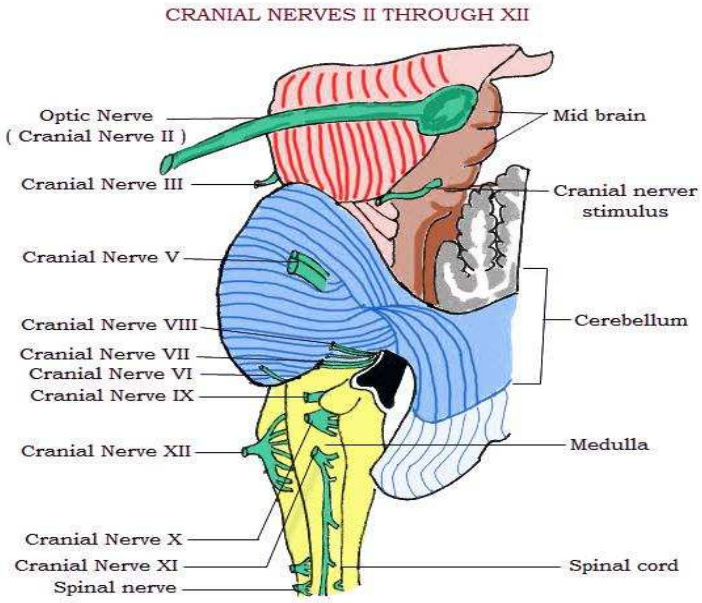
- The dorsal spinal root is the afferent neuron and it propagate sensory impulses from the periphery towards spinal cord and the cell bodies of these neurons are located in the dorsal root ganglion.

- The somatic motor nerves or the efferent nerves emerge from the cell bodies of ventral roots, located in the ventral horn of the spinal cord. Both the dorsal and ventral roots unite to form the spinal nerves that emerge through the inter-vertebral foramen.
- Thus, the spinal nerves are mixed nerves, containing both sensory and motor fibres. However, outside the vertebral canal it divides into dorsal and ventral branches.
- The nerves from the last three cervical and the first one or two thoracic nerves form a net work, the left and the right brachial plexus, supply to the respective fore limbs.
- The ventral branches of the last 3 to 5 lumbar and first 2 to 3 sacral nerves unite to form the right and left lumbo-sacral plexus, supplying to the respective hind limbs.

CRANIAL NERVES



- Twelve pairs of cranial nerves emerge through the foramina of the skull; hence they do not have dorsal and ventral roots.
- Some of the cranial nerves are purely sensory and some are strictly motor in function, while some nerves are mixed in functions.



AUTONOMIC NERVOUS SYSTEM

- Described as visceral efferent system by Langley in 1898.
- All efferent axons leaving the CNS other than those of skeletal muscle belong to ANS.
- ANS supplies efferent information to smooth muscle, glands & cardiac muscle.
- ANS modifies motor activities that have been initiated with in the involved effector organs.
- Neural activity of ANS may either be excitatory or inhibitory.
- It has two divisions
    - Sympathetic (thorocolumular)
    - Parasympathetic (craniosacral divisions)
- ANS as the name implies is not under the control of structures of other nervous systems.
- Unique difference from somatic nervous system is the presence of ganglia along their nerve trunks or in the wall of the effector organ. Ganglia are clusters of neurons of peripheral autonomic neurons which form swelling on the nerve trunk.
- It provides mechanism of synapsing for efferent fibre prior to innervating effector organs.

CLASSIFICATION OF GANGLIA

Vertebral ganglia

- Located along vertebral column and are associated with sympathetic trunk.

Collateral ganglia

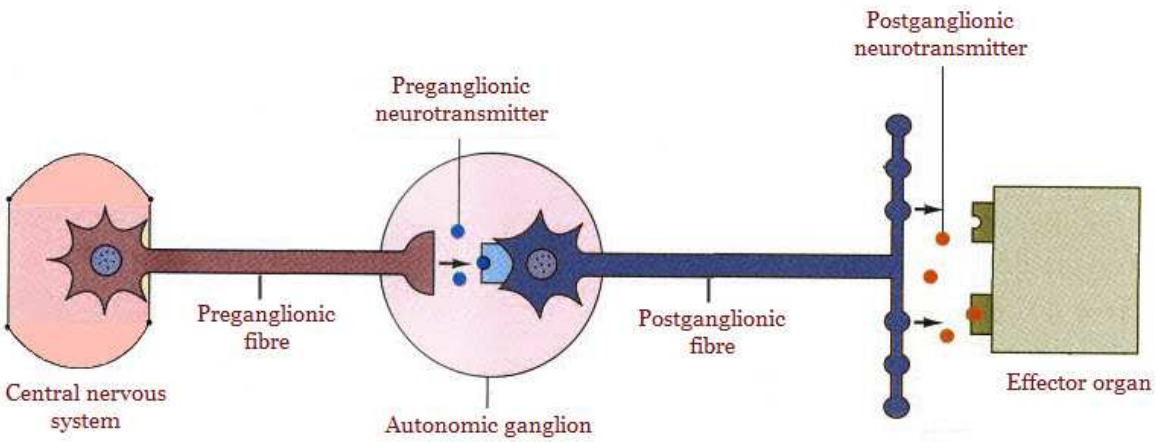
- Located in between CNS and viscera, associated with visceral blood vessels. eg. celiac ,pelvic and cranial & caudal mesentric

Terminal ganglia

- Located in the wall of the visceral organ eg. Cardiac, pulmonary & enteric ganglia.

NERVE FIBRES

- Ganglia possess two types of nerve fibres
  - Pre ganglionic nerve fibre
    - Nerve fiber from CNS to cells of ganglia for synapse.
    - They are small, myelinated with conduction velocity of 3-15 m/sec, representing class B fibre.
  - Post ganglionic nerve fibre
    - Nerve fibre from ganglia to effector organ.
    - These fibres are non-myelinated, small , class C fibers with conduction velocity of 0.7-2.3 m/sec.



SEGMENTS OF ANS

- The ANS is divided into small segments after its origin from CNS.
- Three segments are evident:
  - Cranial segment
    - From brain stem with nerve fibers of oculomotor, facial, Glossopharyngeal & Vagus
  - Thoraco lumbar segment
    - From spinal cord with ventral root of spinal nerves from T_1 or L_{5or6} segments.
  - Sacral segment
    - From spinal cord and the ventral root of S_1-S_3 Sacral nerves.
    - Preganglonic axons of the parasympathetic division originate in the cell bodies of the brain stem (cranial out flow) and in the terminal segment of (sacral out flow) of the spinal cord .

NEURO TRANSMITTER OF ANS

Acetyl choline (Ach)

- This is the neuro transmitter produced by *preganglionic* fibers of both sympathetic & parasympathetic systems.
- Post ganglionic fiber of parasympathetic system produce Ach at their neuro effector site in visceral organ

Adrenaline/Nor-adrenaline

- Post ganglionic fibers of sympathetic system produce noradrenaline or adrenaline at their neuro effector site on the effector organs.
- Exception
  - Isolated preparation of sympathetic neuro effector junctions of cat's sweat gland secrete Ach.
  - HYPOGASTRIC sympathetic supply to the uterus of bitch receives Ach as post ganglionic sympathetic neuro transmitter.
  - Stellate ganglionic sympathetic fibers at their post ganglionic end secretes Ach in corony supply of cat and dog.
- Other
  - Sympathetic vasodilators secreting Ach is seen in the buccofacial region of dog, hind limb muscles of cat, coronary arteries of dog and cat, and penis in dog.

PERIPHERAL DISTRIBUTION OF NERVE FIBERS

	Sympathetic	Parasympathetic
Preganglionic	Short fibers and ganglia are close to spinal cord or brainstem with in the vertebral or collateral ganglia	CNS to collateral or terminal ganglia
Post ganglionic	Long fibers	Short fibers

SYMPATHETIC NERVOUS SYSTEM

- Cell bodies of preganlionic sympathetic fibers are located on the intermedia lateral column of thoracic & upper lumbar spinal cord
- Fibers from these nucleus emerge out through respective spinal nerves from ventral root of the spinal cord.
- Short fibres of pre ganglionic system synapse with autonomic ganglia.

SYMPATHETIC TRUNK

- Composed of chain ganglia viz vertebral ganglia.
- This extends from upper cervical to coccygeal vertebrae.
- Sympathetic trunk is represented as a paired structure having a ganglion for each thoracic, lumbar, sacral & upper coccygeal nerve.

- Segmental ganglia in the cervical region is fused into three large ganglia (cranial cervical, middle cervical & caudal vertical ganglia)
- Stellate ganglion is the fusion of caudal cervical ganglia with upper two thoracic ganglia.
- Fibres from trunk composed of pre ganglionic & post ganglionic system.
- Sympathetic pre ganglionic fibers in thoracic cavity, neck , head, blood vessel and sweat glands synapse with vertebral ganglia.
- Those supplying abdominal and pelvic organs do not synapse within the vertebral ganglia
- But collateral ganglia are formed by the synapse formed by these pre ganglionic fibers with post ganglionic neurons.

- **FUNCTIONS OF SYMPATHETIC AND PRASYMPATHETIC DIVISIONS**

Effector organ	Sympathetic	Receptor	Parasympathetic
Heart S A node	Increased in heart rate	□□□□□□□□□□□□□□□$_1$	Decrease in heart rate
Atria	Increase contractility and conduction velocity	□$_1$	□ contractility & conduction velocity
AV node and Conduction system	conduction velocity	□$_1$	□ conduction velocity
Ventricles	contractility & conduction velocity	□$_2$	---
Arterioles Coronary, skeletal, Pulmonary, renal, Abdominal & skin	Vasoconstriction Dilatation	□ □$_2$	Dilatation
Systemic veins	Constriction or Dilatation	□ / □$_2$	---
Respiratory tract Bronchial muscles	Relaxation	□$_2$	Contraction
Bronchial glands	Inhibition?		Stimulation
Smooth muscles G. I. Tract	Decreased motility Sphincter constriction Inhibits secretion?	□, □$_2$ □	Increased motility Sphincter relaxation Stimulates secretion
Gall Bladder	Relaxation		Contraction
Urinary bladder Muscle Sphincter	Relaxation Contraction	□ □	Constriction Relaxation

Ureters	Increase motility	☐	Increase?
Uterus	Variable Contraction	☐, ☐$_2$	Variable
Male Sex organs	Ejaculation	☐	Erection
Sweat glands	Localised secretion		Generalised secretion
Secretions Salivary	Vasoconstriction Thick, viscous secretion	☐	Copious and watery
Pancreatic	Decreased secretion ☐ insulin & glucagon secretion	☐ ☐	Increased secretion insulin & glucagon Secretion
Gastric	viscous secretion		secretion
Juxtaglomerular cells	Increased renin secretion	☐	--
Pineal gland	Increased melatonin secretion	☐	--
Milk letdown	Inhibition		--
Ciliary muscle	Relaxation (far vision)	☐	Constriction (near vision)
Pupillary muscle	Dilatation (dim light)		Constriction (bright light)
BMR	Increased		---

FUNCTIONS OF ANS

- Preserves cellular environment of animal to maintain homeostasis
- Sympathetic and para sympathetic system makes fine adjustments to rectify the effect of minor internal and external stress in a resting
- It involves specific reflex activities of an organ or a part of the organ.
- In stress, sympathetic systemic activity is modified. They involve accelerated response in whole of the visceral motor system
- Referred by Cannon as FIGHT OR FLIGHT RESPONSE
- This ends up in the immobilization of energy and reroute the circulation to vital organs for nutrients.
- Psychological modification evident in fight/flight response are due to sympathetic activity are
    - Increased Heart rate
    - Increased B.P
    - Vasocostrictor of blood vessels of skin and viscera
    - Vasodilatation and increase in blood supply to skeletal muscles
    - Spleenic contraction to increased oxygen carrying capacity of blood
    - Increased bronchodilation and vasodilatation of lung facilitate increased oxygen uptake and used up Carbondioxide release

  - Pupillary dilation to widen the visual receptivity
  - Piloerection to conserve body heat (calorigenic and glycogenic effect)
  - Muscle glycogenolysis and hepatic glycogenolysis (to some extent) to mobilze glucose to skeletal muscle for energy need.
  - Increased activity of many receptor organs to sense changes in homeostasis like mechanoceptors of skin and mesentery, muscle spindle response, gustatory and olfactory activity are evident.
  - Potentiated by adrenal medullary involvement as an autonomical counterpart - sympathetic adrenal system
- Parasympathetic system plays an important role in co-coordinating specific functions of digestive tract.

REFLEX ACTIVITIES OF AUTONOMIC NERVOUS SYSTEM

Autonomic reflexes

- These reflexes arc, similar to somatic reflex arc have
  - Sensory part
  - Motor part.
- Visceral sensory fibers enter the dorsal horn of spinal cord, branching into spinal sensory column.
- They employ interneurons.
- Interneurons transmit impulses to the preganglionic neurons in the spinal cord or to the brain stem.
- <liEfferent part represented by preganglionic axon to carry the visceral motor stimuli to ganglionic cell.</li
- Post ganglionic axons innervate visceral effectors.
- Examples of Autonomic reflexes are
  - Peristaltic reflex and
  - Micturition reflex.

Peristaltic reflex

- Peristalsis is the motor activity of the intestinal musculature to cause onward propulsion of the contents.
  - Sensory information by way of stretching of the wall.
  - This intum stimulates intra mural sensory neurons.
  - Sensory neuron either directly or through interneuron synapse with Post ganglionic parasympathetic neuron.
  - This reflex is unique, that it can be seen in isolated preparatious also and is known as a type of Extra spinal reflex.
  - In normal animals, this reflex has an additional influence from brain stem.
  - Control is mediated through preganglionic vagal, pelvic nerves.

Micturition Reflex

- Act of voiding urine is known as micturition.
- It is a reflex act, under the control of higher centres.
- But the uniqueness is that this reflex persist even after the transection of the spinal cord.

- This reflex involves stretch receptors of the bladder wall, primary axons of cell bodies in the sacral spinal segment and dorsal grey column of sacral segments.
- The urinary bladder is the storage component and allows urine to be accumulated.
- Contractions of the smooth muscles in the walls of renal pelvis and ureter by way of action potentials move down towards urinary bladder. The contractions are initiated by stretch and distension as a result of continued production of urine.
- This peristaltic activity is autonomous and myogenic.
- Junction of uretar and urinary bladder is not provided with any anatomical sphincter.
- There is a functional sphincter developed by the bladder muscle viz Detrusor muscle.

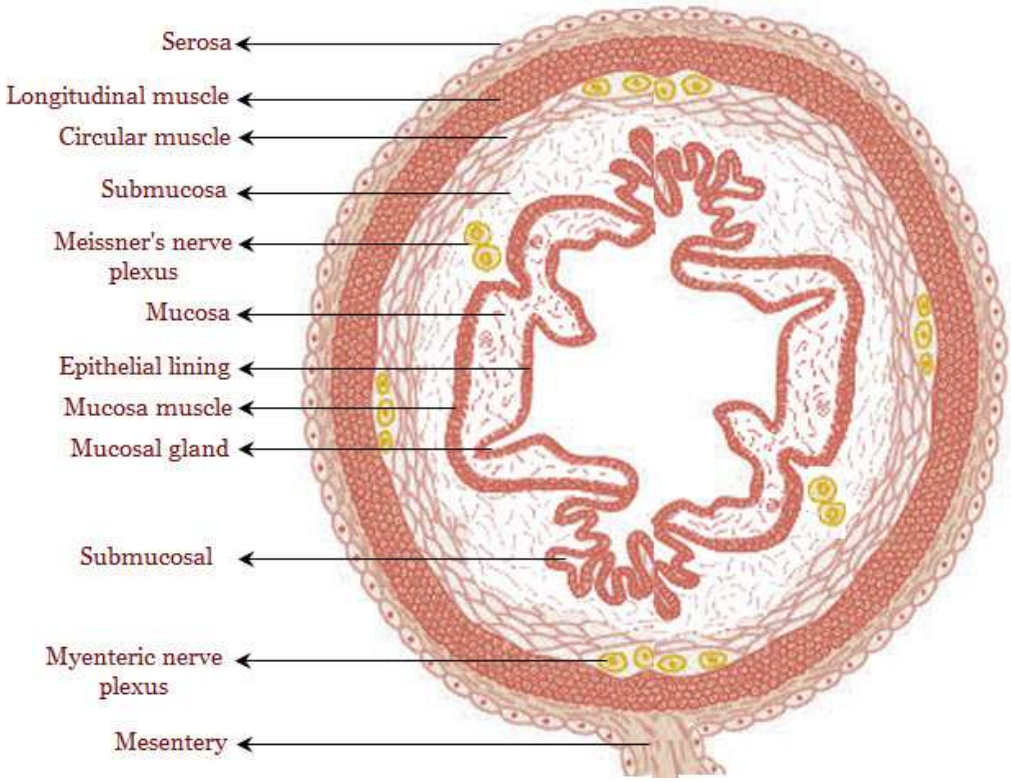
- The functional sphincter act as a valve to prevent reflux of urine into ureter.
- The smooth muscle fibers of Detrusor muscle interlaces and converge at the urethral ori face.
- It allows the bladder as a leak proof compartment while it is filling, that is as the bladder expands , the neck of the bladder at the level of urethra constricts due to the convergence of smooth muscle fibers and vice versa as it relaxes.
- Detrusor muscle is controlled primarily by parasympathetic and sympathetic innervation.
- Activity of Detrusor muscle in expulsion of urine is a reflex mechanism.
- This micturition reflex centre is located within the sacral spinal card and this centre is intum controlled by the brain stem and cerebral cortex.
- Filling of urinary bladder activate stretch receptors.
- Sensory neuron via pelvic nerve which has its cell bodies in the dorsal root ganglia of sacral spinal nerves.
- They employ inter neurons.
- Fibers take two routes from here, one is for local reflex and other passes to fasciculus gracilius to the nucleus gracilius of medulla.
- Some fibers through another set of interneurons synapse with spinothalamic tract.
- Thus, information about the stretch reaches medulla / brainstem and also to cerebral cortex for conscious perception and voluntary emptying of the bladder.
- Motor pathway is mediated via pelvic nerves, whose cell bodies are in intermediate mucleus of the sacral spinal segment.
- When motor activity is initiated to contract the bladder Detrusor muscle, the neck of the bladder relaxes simultaneously with contracting of bladder wall allowing the urine to flow down.
- Pudental nerve serve as motor nerve to external urethral sphincter which is a skeletal muscle.
- As micturition reflex is in progress, the external urethral sphincter relaxes allowing voiding of urine.

ENTERIC NERVOUS SYSTEM

- Innervations are from sympathetic and parasympathetic divisions of ANS, which control on intramural plexus (myenteric/auerbach plexus and meissner's plexus)
- Intramural plexus made up of 10^8 neurons which is almost equal to number of neurons in spinal cord.
- Sympathetic division made of thoracolumbar outflow has got preganglionic cholinergic nerve fibres.
- They end upon coeliac ganglia , superior and inferior mesenteric ganglia or hypogastric plexus.
- From these ganglia, post ganglionic adrenergic nerve fibres arise these nerves end up on intramural plexus and bring about inhibitory action (secretion & motility reduced). Some

branches of post sympathetic nerves end upon sphincters and when sympathetic is stimulated it will inhibit the myenteric plexus in sphincters. As a result sphincters relax-defaecation occurs.

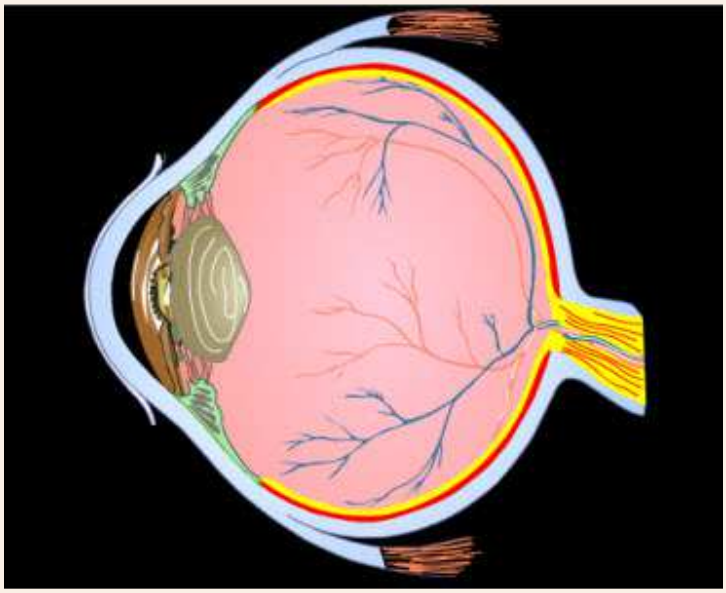
- Inhibitory action includes reduced glandular secretion, reduced blood supply (constricted blood vessel), reduced motility, (when there is sympathetic dominance).
- Parasympathetic outflow is from cranio sacral division (3,7,9,10 cranial nerves and sacral) has got preganglionic cholinergic nerves which are long enough to reach intramural plexus (ganglia).



- The post ganglonic parasympathetic nervous system arise from these ganglia and thus are present within the structure or organ.
- Post ganglionic PSNS use Acetyl choline as neurotransmitter .
- Post ganglonic SNS use Nor epinenephrine as neurotransmitter.
- Sympathetic stimulation decreases GI function and Parasympathetic stimulation increases GI function.
- Organo phosphorous compounds are parasympathomimetics.
- In case of Organophosphorous poisoning in animals, there will be excess salivation, shooting of diarrhoea, body temperature is reduced.
- Sympathetic post ganglonic nerves act on neural circuits in the intra mural plexus that provide signals to smooth muscles of GI tract.
- Intramural (intrinsic) plexus can control much of the coordinated activity in the absence of extrinsic innervation of GI tract.
- Parasympathetic nervous division (Vagus and Sacral spinalcord outflow) with their cholinergic preganglionic fibres act up on intramural plexus (i.e. Intramural plexus act as ganglia for preganglionic PSNS).

- From intramural plexus, postganglionic nerves arise which are seen within the structure of innervation. On stimulation motor and sensory activities are improved.
- There are local and central reflex pathways that affect the functioning of GI tract organs.
- Meissners plexus contains neurons with are intimately attached with circular muscles and glands and blood vessels of GI tract (Synaptic cleft is very much shortened). The neurons of their synaptic cleft is widened. So it takes more time for axon potential to reach post synaptic structure. The circular muscles of GI tract contract ahead of longitudinal muscles as a result gut becomes longer and thinner. When longitudinal muscles contract, gut becomes shorter and wider. Preganglionic parasympathetic nerves sometimes get innervations from post ganglionic sym nerves and adrenoreceptors are present on both pre or post ganglionic sympathetic nerves and α_1 adrenoceptors are present on both pre and post ganglionic PSNS. In such cases Ach (acetyl choline) will not be discharged thereby Parasympathetic activity getting reduced. NE (norepinephrine) inhibits Ach release through its interference via α_1 adrenoceptors on axon terminals of PSNS.
- Atropine is the cheapest and effective antidote for OP (Organo phosphorous) poisoning. (Atropin- from Atropa belladonna)
- In OP poisoning -excess PSNS dominance.
- Atropine inhibits Ach to occupy post synaptic membrane receptors. In these two ways (ie NEP and Atropine) the vagal refex can be controlled.
- The Vagovagal reflex (generally is stimulating in nature (i.e. Increase motility, secretory and vasodilatory) carries both afferent and efferent through vagus, whereas only ENS is independent of CNS generally inhibitory involving only ENS and is intestine intestinal reflex is generally inhibitory involving only ENS and is independent of CNS.
- Central reflex pathway eg. Vagovagal reflex.
- Local reflex pathway eg. intestino intestinal reflex.
- Most of the reflexes are mediated by Ach and NE There are other neuromodulatory substances/neuro transmittors which include enkephalins, 5-hydrooxytryptophan (5HT), cholecystokinin (CCK), glycine, motilin, angiotensin, secretin, galanin, somatostatin, VIP (vasoactive intestinal polypeptide), substance-P, GABA, bombesin, gastrin, histamine, GRP (Gastrin releasing polypeptide) PG (Prostaglandin), GIP (Gastrin inhibitory polypeptide), enteroglucagon etc. Out of these, 7 are produced from enteroendocrine cells whereas others are produced from gut cells as well as from nerve endings, so called brain gut peptides.
- The chief neurotransmitters of ENS are Ach , NEP, enkaphalin (or opioid compounds) and VIP (Vasoactive intestinal peptide).
- Enkephalins decreases gastro intestinal motility by inhibiting release of Ach.
- VIP acts directly on smooth muscles to cause muscle relaxation. It is localised with cholinergic nerves in vagus.
- VIP is in local neurons (ENS) and is released when vagal fibres excite these inhibitory neurons to cause relaxation.
- Enkephalins, VIP, nitric oxide (NO-only gaseous neuro transmitter) serotonin are all neurotransmitters produced by specialised neurons which are localized along with Ach or NEP secreting neurons in ANS.
- Depending on the nerve, whenever Ach or NEP are released, these neurons release their neurotransmitters.

MODULE-13: PHYSIOLOGY OF VISION



LEARNING OBJECTIVES

At the end of this module, the learner will be able to understand,

- the structure of the eye,
- the visual cycle and photoreceptor functions,
- the theory of vision and types of vision and
- the Accommodation of eye and problems in vision.

PHYSIOLOGICAL ANATOMY OF EYE

SCLERA	Covering of the eye – Protective in function, Tough in nature.
CORNEA (stratified squamous)	Transparent Structure, anterior modification of sclera.
CHOROID	Vascular and Pigmented area of sclera in its posterior part.
RETINA	Innerside of choroid, has PHOTORECEPTORS.
DARK PIGMENT	Animal with day light vision, to absorb light which has to pass receptor without stimulating, present between CHOROID and PHOTORECEPTORS.
TAPETUM	Causes nightshine in nocturnal animal by reflecting the light viz REFLECTING PIGMENT. (allows use of optimum light by retina)
LENS	Causes accomadation, made up of elastic capsule filled with jelly like substance, suspended by suspensory ligament.
SUSPENSORY LIGAMENT	Muscular structure connecting lens and ciliary body.
CILIARY BODY	Convexity of the lens is altered to focus the images of varying distance and is achieved by ciliary muscles and its contraction increases convexity and focusses near objects.

ANTERIOR CHAMBER	Between CORNEA and LENS
POSTERIOR CHAMBER	Between IRIS and Suspensory Ligament
AQUEOUS HUMOR	clear watery fluid, from ciliary processes of ciliary body in the posterior chamber, flows through PUPIL to anterior chamber. Absorbed into veins, passes through CORNEA and IRIS. Accumulation of aqueous humor due to obstruction in absorption causes increased intraoccular pressure viz GLAUCOMA.
IRIS	Diaphragm separating Anterior and Posterior chambers. Pigmented structure has dilator and constrictor muscles (smooth)
PUPIL	Hole in the Iris, whose diameter can be modified by smooth muscles and allows the varying intensity of light to strike retina
VITREOUS HUMOR	HYDROGEL with mucopolysaccharide, hyaluronic acid and collagen fibers filling the chamber behind IRIS, nourishes retina.
NEURAL RETINA	Present behind vitreous humour.
OPTIC DISK	Area where axons of retinal ganglia passes to brain. Has optic nerve (cranial nerve), arteries and veins. Nutrition of retina is taken care by CHOROID VESSELS.
LACRIMAL GLANDS	Production of tears from this, near lateral canthus. Parasympathetic stimulation ends in tears and are drained into nose by NASO-LACRIMAL DUCT.
NICTITATING MEMBRANE	3 EYE LID, Present in medial carithus, protects eye and also produce tears.
HARDERIAN GLANDS	Lacrimal gland, absent in carnivores, found in birds and mammals.

RETINA

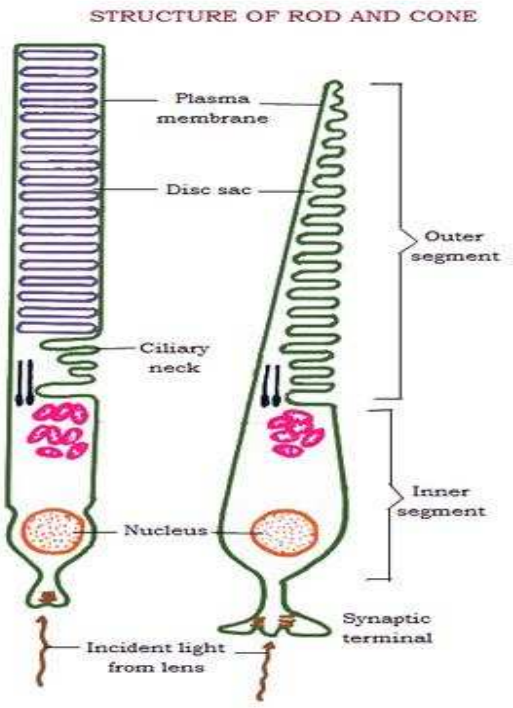
RETINA (part of CNS)	Has 5 cell types, • Photoreceptor cells has two types RODS, CONES. • BIPOLAR CELLS with INTERNEURONS – Connects rods and cones with ganglion cells by synapsing. • GANGLION CELLS : Ganglion cell layer-single layer of multipolar ganglion cells with scattered neuroglia cells with some blood vessels. o Transparent and Unmyelinated carry impulses to brain via optic nerve.

	• AMACRINE CELLS and HORIZONTAL CELLS : are interneurons lie between the bipolar and the ganglion cells. • The light has to pass all the other layers like the nerve fibre layer, ganglion cell layer, etc., to reach the layer of rods and cones.
FOVEA	• Area present in the back of the retina – Light rays are not distorted in this area, devoid of ganglion and bipolar cells. • Light rays directly fall on retinal photoreceptors. Provides sharp visual image. • FOVEA IS ABSENT IN DOMESTIC ANIMALS.
CONES	• They are present in dense in the central portion of the retina - (in primates) cones are best in day light – PHOTOPIC VISION. • They are color sensitive. • Similarly to fovea more no. of cones are present in MACULA or CENTRALIS in other animals.
RODS	• Extremly sensitive to light, present in the periphery of retina. • Responsible for NIGHT/SCOTOPIC vision, Sensitive to blue – green light.
BLIND SPOT	• No photoreceptors at the origin of optic disk which is viz Blind spot.(Free from Rods and cones).

STRUCTURE OF RODS AND CONES

Rods and cones have 3 parts

- *Outer segment* – Photoreceptor part, has stacked membrane discs with VISUAL PHOTOPIGMENTS.
- *Inner segment* – Has nucleus, mitochondria and cytoplasm.
- *Synaptic Termial* – Synapse with bipolar cells.



BIOCHEMISTRY OF RODS AND CONES

- Photosensitive pigment – *rhodopsin / visual purple :* Present in rods
- Made up of proteins viz *opsins*
- In Rods it is called *scotopsin* which consist of an aldehyde of vitamin A called *retinal.*
- As light strikes, Rhodopsin is transformend from CIS to TRANS and by hydrolysis, gets splits into scotopsin and retinal.
- Retinal is enzymatically changed into retinol (alcholic form) and is stored as retinyl ester in the pigment cell
- Stability of rhodopsin and its functions depends on the presence of CIS form of vitamin A. Formation of Rhodopsin is continuous, in the presence of Isomerase, retinal joins with scotopsin to form rhodopsin.
- During night/darkness- rhodopsin accumulates in rods, and have maximum sensitivity to light, this is known as *dark adaptation*. When exposed to light concentration of Rhodopsin decreases in rods and so rods become insensitive to light and the vision is caused by cone stimulation known as *light adaptation*. A vitaminosis causes night blindness viz nictalopia.
- *Cones:* Colour sensitive photoreceptor responsive for day vision photopigment in cones are IODOPSIN consists of opsin and RETINAL IODOPSIN varies from SCOTOSIN, in their protein part.
- Cone opsins are of 3 types
    - Blue cone (430 nm)
    - Green cone (535 nm)
    - Red cone (575 nm)
- Difference cones respond to different wavelength. The primary colours are specific with cone. Various colour sensation are made possible by the combination of different cones. Colour blindness endup is due to lack of different cones.
- Domestic animals has more rods and birds have more cones.

- *ELECTRO RETINOGRAM – ERG :* Recording of electrical changes in the retina by flashing the light. Has 3 waves a,b and c;
- a wave : originates from rods and cones.
- b wave: from glial cells, Amacrine and bipolar cells.
- c wave: from Pigment epithelium.

PHOTO TRANSDUCTION

- The light falls on the retina by the adjusting the refractive media of the eye.
- As the light falls on the retina, the rods and cones absorb energy and convert it into electrical signal.
- The rods and cones are located at the back of the retina synapse with bipolar cells which in turn synapse with the ganglion cells.
- Many rods cells synapse with single bipolar cell and many bipolar cells synapse with single ganglion cell.
- The rods provide hazy images, but the visual field is much larger.
- Each cone synapse with single bipolar cell and single bipolar cell synapse with single ganglion cell. Thus cone provide very fine detailed image of the object, but the receptive field of a cone photoreceptor is very small.
- The receptive fields of ganglion cells contain both 'on' and 'off' regions. In both regions, the 11-cis retinal is converted to all-trans retinal when the light activates the photoreceptors.
- The all-trans retinal in turn activates transducin, a G-protein which decreases the cGMP found inside the photoreceptor cell. As the cGMP is decreased it closes the Na+ channel, thereby hyperpolarizing the cell.
- The hyperpolarized cell releases neurotransmitter, the glutamate. The activity of the glutamate determines the 'on' and 'off' region of the receptive field.
- The glutamate depolarizes the bipolar cells, which synapse with the 'on' region photoreceptor and increases neurotransmitter release from the bipolar cell which in turn depolarizes the ganglionic cell.
- The glutamate hyperpolarizes the bipolar cells, which synapse with 'off' region photoreceptor, and decreases the neurotransmitter release from bipolar cell, which causes hyperpolarization in the ganglion cell.
- The complex receptive fields of ganglion cells improve the ability to detect the difference between light and dark. When both the receptive fields of the ganglion cell are exposed to light, both 'on' and 'off' regions are activated and result in weak stimulation.
- The horizontal cells enhance the contrast through lateral inhibition of the bipolar cells. The lateral inhibition increases contrast, sharpen borders and edges as it is excited strongly when the light is focussed on the 'on' region photoreceptors. Another step in increasing the integration of the image is by the synapsing of the bipolar cells with amacrine cells via gap junctions.
- It modifies the inputs from many bipolar cells and alters the neurotransmitter release in to the ganglion cell.

MECHANISM OF PHOTORECEPTION

- As the light strikes the photoreceptors, transformation of photo pigment occurs with the resultant HYPERPOLARIZATION of receptor cells.
- During darkness, the Na^+ channel are opened / remain open on rod and cone cell membrane.

- Na^+ leak into rods and lower the membrane potential. As light strikes the rods retinal is released from rhodopsin and causes closure of many Na^+ channels.
- This causes HYPERPOLARIZATION of the receptor cell membrane and a decrese in the transmitter released at the synapse with the bipolar cells.
- Breakage of rhodopsin to RETINAL + OPSIN by light is temporary and Rhodopsin is resynthesized afterwork.

Activity of photoreceptors

In dark

- The cell membrane of the photoreceptors contains chemical messenger-gated Na^+ channels. They respond to the second messenger cGMP.
- The Na^+ channels are open when the cGMP is bound to them. In absence of light, the concentration of cGMP is high; therefore it is bound with the Na^+ channels, keeping them open.
- The inward leak of the Na^+ ions depolarizes the photoreceptors. This in turn keeps the Ca^{2+} channels opened, which triggers the release of the neurotransmitter from synaptic terminal.

In Light

- There are other biochemical cascades that take place in the light. The concentration of the cGMP is reduced.
- The reduction in cGMP is by the cascades as follows: The retinene absorbs light and changes the conformation and activates the photopigment.
- The rods and cones contain G-protein called transducin, activates the enzyme phosphodiesterase. which degrades the cGMP causes the closure of Na^+ channels. As the Na^+channels closes, the photoreceptor cell gets hyperpolarized which in turn reduced the release of neurotransmitter.

VISUAL PATHWAY

- Hyperpolarization of rods and cones alter the activity of bipolar cells.
- Impulse frequency of ganglion cells too get altered.
- Horizontal cells influence Bipolar cells, Amacrine cells influence ganglion cells to detect contrast and contour.
- Axons of ganglion cells project as OPTIC NERVE to the lateral geniculate nucleus of THALAMUS (visual relay area) and the cells of this area project to VISUAL CORTEX (The Occipital Lobe)
- Axons of ganglion cell travel with optic nerve to optic Chiasma where crossing of right and left optic nerve occurs. So they travel to the respective side of lateral geniculate body. Images from left field is perceived at right visual cortex and right field is perceived at left visual cortex.
- Few fibers of optic nerve ends in the Superior Colliculi of mid brain and pretectal nucleus of the brainstem for Reflex activity.

TYPES OF VISION

Monocular or periscopic

- Laterally placed eyes (frog and reptiles) viewing the object independently due to wide visual angle between optic axis and mid line of the eye.

Binocular vision

- Anterio laterally placed eyes (primates, carnivores, birds), both the eyes view the same object simultaneously, overlap of field of vision due to parallel optic axis and mid line of the eye.

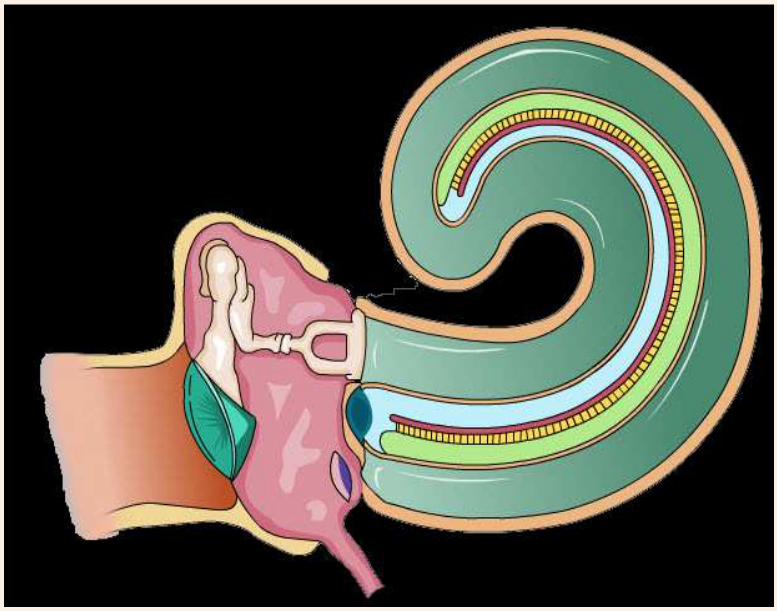
Stereo vision

- Anteriorly placed eyes (cats) Dissimilarity in viewing the object by left and right eyes, three dimensional view due to very small angle between optic axis and mid line of the eye, fusion of the image by the visual centre.

ACCOMMODATION OF THE EYE

- Ability of the eye lens to change its focal length by changing its radius of curvature.
- In near vision contraction of ciliary muscels causes decreased tension to the suspensory ligament and causes bulging of the anterior curvature of the eye.
- To cause distant vision ciliary muscles are relaxed and causes increased tension of the suspensory ligament which flattens the anterior curvature of the eye.
- *Presbopia :* It is the gradual loss in the power of accommodation of eye lens due to aging.
- *Refraction errors :* Inability of the eye lens to adjust its focal length.
    - *Myopia (short sight) :* Due to abnormal increase in the anterior – posterior diameter, image fall just before the retina
    - *Hypermetropia (long sight) :* Due to reduction in the anterior-posterior diameter, image fall just behind the retina
- *Astigmatism :* No common point of focus due to difference in the radius of curvature of cornea, overlapping of images.

MODULE-14: PHYSIOLOGY OF HEARING, OLFACTION AND TASTE



LEARNING OBJECTIVES

This module enable the learner to appreciate,

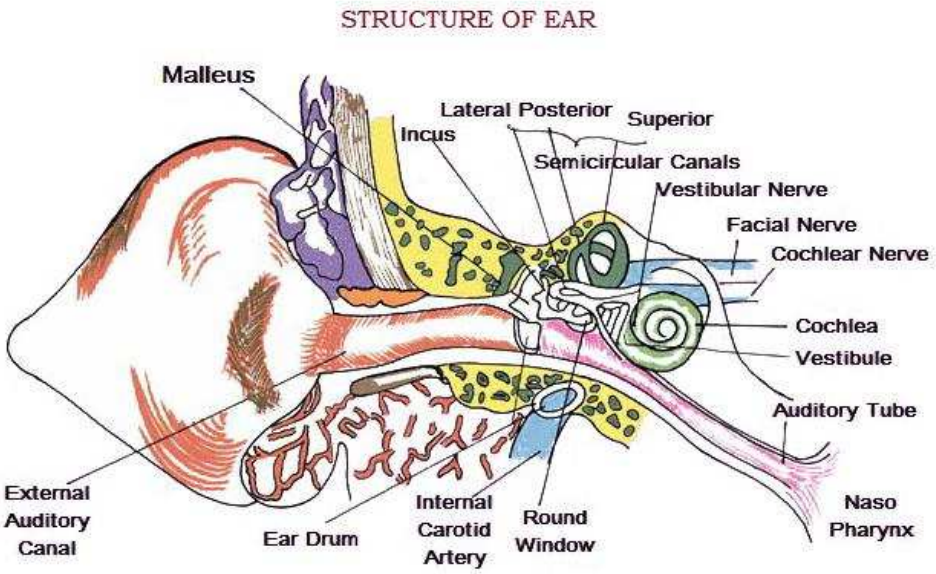
- the structure of the ear,
- the mechanism of hearing and the auditory pathway,
- the major component of olfactory system and basic mechanism of the sense of smell and
- the basic structure of taste buds and the gustatory pathway.

HEARING

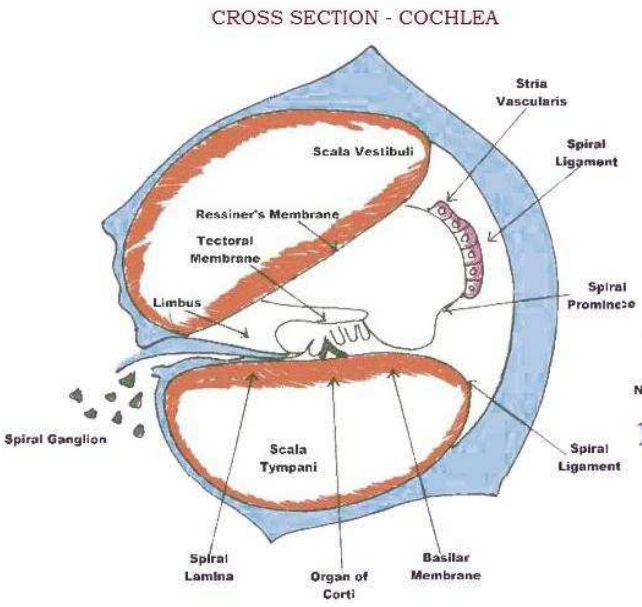
Auditory apparatus

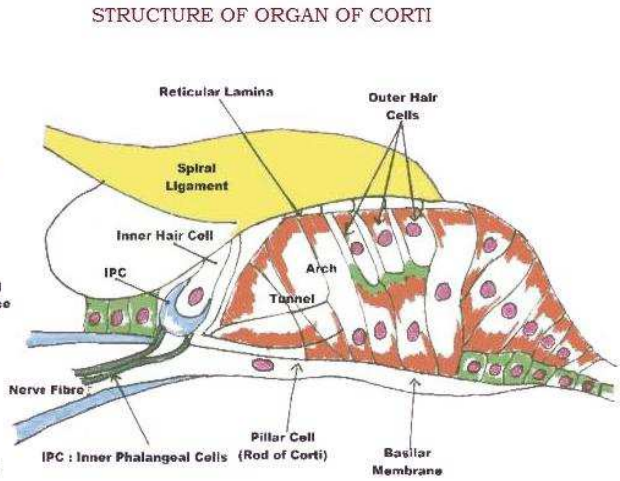
- The sense of hearing is the process of detecting and interpretation of sound waves.
- It is done by the organ called as the ears.
- The ear consists of three parts: The outer ear, middle ear, and inner ear.
- The outer ear is of the external pinna and the auditory canal.
- The pinna is of different shape and sizes in various species of animals and helps in focussing the sound waves to the auditory canal.
- The auditory canal conducts the sound waves into the middle ear. It is lined by waxy secretion called as cerumen. This filters the air entering the ear to avoid infections.
- There is a separation between the outer ear and the middle ear called as the tympanic membrane or commonly called as the ear drum.
- The ear drum transfers the sound waves to the vibrations which are conducted towards the inner ear by the middle ear.
- The middle ear consists of the ear ossicles. The ear ossicles are bony structures namely malleus (hammer), incus (anvil), and stapes (stirrup). They conduct the vibrations toward the inner ear.

- These ear ossicles are connected in a way to amplify the sound waves to evoke or stimulate the hair cells of the inner ear.



- The middle ear and the inner ear is separated by the oval window
- The inner ear consists of fluid filled membranous sacs and canals.
- The whole inner ear is covered by a fluid perilymph.
- The inner ear has both vestibular apparatus, which is the organ of equilibrium and hearing apparatus.
- The hearing apparatus is the coiled tubular structure called as the cochlea.
- The cochlea is replaced by cochlear duct in birds, which is the hearing apparatus and it is small in size.
- The cochlea is filled with a fluid known as endolymph, which has high K+ concentration than the Na+ concentration.
- The cochlea is formed by the coiling of three fluid-filled tubes known as *scala vestibuli*, *scala media* (cochlear duct), and *scala tympani*.





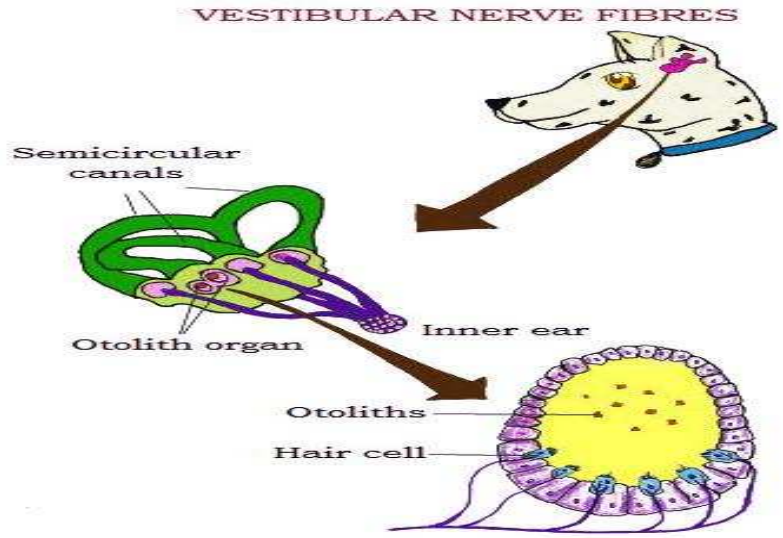
- The scala media is separated from the scala vestibubi by the *Ressiner's membrane* or *vestibular membrane*, while the scala media is separated from the scala tympani by the *basilar membrane*.
- Along the floor of the scala media, on the basilar membrane lies the *hair cell* receptor system, the *Organ of Corti*.
- The basilar membrane of the cochlea consists of the real organ of hearing called as the Organ of Corti. It contains hair cells.
- The hair cells mainly are of two types: The inner hair cells and outer hair cells.
- The inner hair cells detect sounds and outer cells amplify the sound waves.
- The cochlea contains two fluid systems *perilymph* (similar in composition to extracellular fluid) is the fluid in the scala tympani and scala vestibuli and *endolymph* (similar to intracellular fluid) is the fluid in the scala media.
- The oval window is located at the proximal end of the scala vestibuli, and the round window is located at the proximal end of the scala tympani.
- The hair cells are modified epithelial cells and they synapse with the sensory neurons.
- The hair cells contain projections called as cilia. Among these, one cilium is much longer than the other and is present in one side and is named as kinocilium. The shorter cilia are called as stereocilia. These stereocilia are arranged in increasing order of size gradually till they reach the kinocilium.
- The stereocilia and kinocilium are interconnected with tip-links and acts as one unit.
- The hair cells present within a cupula, a gelatinous mass covering over the hair cells with the synapsing the sensory neurons are known as neuromast.
- These hair cells are present both for hearing as well as maintaining equilibrium.
- The hair cells, tectorial membrane and basilar membranes with *cochlear nerve* terminals form the *organ of Corti*.
- *Stereocilia* that project into the endolymph filled scala media. The tectorial membrane, the specialised flap of the basilar membrane, over hangs the cilia of the hair cells. The terminals of the cochlear nerve fibres synapse with the basal ends of each hair cell. The auditory impulses are transmitted through *vestibulo - cochlear nerve, (cranial nerve VIII)* to higher brain centres.

PROCESS OF HEARING

- The pinna concentrates the sound waves to the auditory canal which conducts the sound waves to the ear drum.
- The sound waves strike the ear drum causing it to vibrate. These vibrations are conducted by the ear ossicles to the inner ear through the oval window.
- The vibrations cause the perilymph to vibrate causing waves of that fluid. These waves push on the basilar membrane of the cochlea causing it to vibrate.
- The vibrations are transferred to the organ of Corti. Thereby the inner hair cells bend in response to the vibrations.
- The bending of the stereocilia causes opening of the ion channels and depolarize.
- The inner hair cells release glutamate as neurotransmitter, which excites the synapsing sensory neurons.
- This action potential is propagated via the auditory nerve.
- The neurons from each part of the basilar membrane synapse with neurons in particular area of the brain's auditory cortex. Since the tension of the basilar membrane differs from the centre and the periphery of the membrane, which in turn differs in frequency of the

vibrations, the specific areas of the auditory cortex respond to particular frequencies. This phenomenon is known as place coding.

- The round window of the ear bulges outward in response to the increasing pressure in the inner ear thus avoiding backflow of the fluid and increasing the sound clarity.
- The outer hair cells change in shape in response to the sound waves.
- The outer hair cells depolarize just like the inner hair cells, but sends signal to voltage-sensitive motor protein unlike neurotransmitter release of the inner hair cells. This protein changes the shape of the outer hair cells.
- Due to the change in shape, the outer hair cells pull the basilar membrane and ending in increasing tension which in turn results in increased basilar membrane movement.
- Since basilar movement is related to the sound loudness, increased basilar membrane movement causes increase in the stimulus for the inner hair cells.
- The ears can also detect the location of the sound by perception of the brain about the time lag and sound intensity. In mammals, outer ear also helps to localize the sounds. But this is not efficient, so the animals move their head to localize the direction of the sound.



AUDITORY PATHWAY

View animation...

- The vestibulo cochlear nerve endup in the cochlear nucleus of the medulla.
- The impulses from here are transmitted to superior olivary nuclei, inferior colliculus and median geniculate body of thalamus.
- From thalamus it endsup in temporal lobe of cerebral cortex (auditory cortex)
- Temporal lobe sends efferent to limbic system for behavioural modification and to muscles of larynx for reflex vocalization.

OLFACTION

- It is defined as the detection of chemicals that are carried in the air. It is the sense of smell.

- The vertebrates have greater capacity in distinguishing the variety of odorants (chemicals detected by the olfactory system).

Olfactory system

- It comprises of the olfactory receptors, supporting cells, and basal cells.
- The supporting layer, Bowman' glands secrete mucus which covers the nasal passage.
- The basal cells are precursors for new olfactory receptors, which are replaced once in every two months.
- The olfactory receptors found in dogs are about 50 million in number.

Process

The olfactory system of the vertebrates is located in the roof of the nasal cavity.

- Olfaction starts when the odorant enters the nasal cavity and comes into contact with the mucus layer that lines the olfactory epithelium of the nose.
- These mucus contain odorant binding proteins which allow lipophilic odorants to dissolve in the aqueous mucus layer.
- The olfactory receptor cells are bipolar neurons where one end of the neuron ends in the olfactory epithelium and the other ends as a synaptic terminal with the neurons of the olfactory bulb of the brain.
- The olfactory receptor cells are covered by cilia which project into the nasal passage. These are non-motile, and contain odorant receptor proteins.
- The odorant receptor proteins are G-proteins. These proteins are of multigene family and contain many numbers of genes coding for odorant receptors. In mouse about 1000 potential odorant receptor genes have been identified.
- Each receptor expresses single receptor protein. Thereby the ability to distinguish even the slightest change in the odorants.
- As the odorant molecules bind with the receptor, the receptor undergoes conformational change that sends signals to the G-protein associated with it. The activated protein act through the adenylate cylase pathway and cause generator potential. There are also other mechanism of causing action potential like the IP3 pathway.
- If the depolarization is large enough, then the action potential is triggered which travel to the bipolar neuron. In this, the action potential moves toward the cell body and not away from it as in other motor neurons.
- The action potential is transmitted from the bipolar neuron to the synaptic terminal with the neurons of the olfactory bulb and they carry the information to the olfactory bulb.
- The olfactory bulb is a complex neural structure. It has a lining of ball-like neural junction known as glomeruli. Each glomerulus receives signals only from a single type of receptor cell. Thereby, these glomeruli act as smell files to sort out the type of smell of the odorant.
- The fibers leaving the olfactory bulb pass through two routes.
    - *Subcortical route:* It goes to the regions of the limbic system mainly to the medial sides of the temporal lobe which is concerned with olfaction and known as primary olfactory cortex. It includes hypothalamic involvement and coordinates smell with behavioral reactions.
    - *Thalamic-cortical route:* This route is important for conscious perception and fine discrimination of smell.
        - The odorants are cleared from the nasal passage very quickly and the olfactory system adapts quickly, so the sensitivity to the smell of a particular odorant diminishes after a short exposure to the smell. The

odorants are cleared by enzymes named as odorant-clearing enzymes. They resemble the detoxification enzymes in liver.

Pheromone

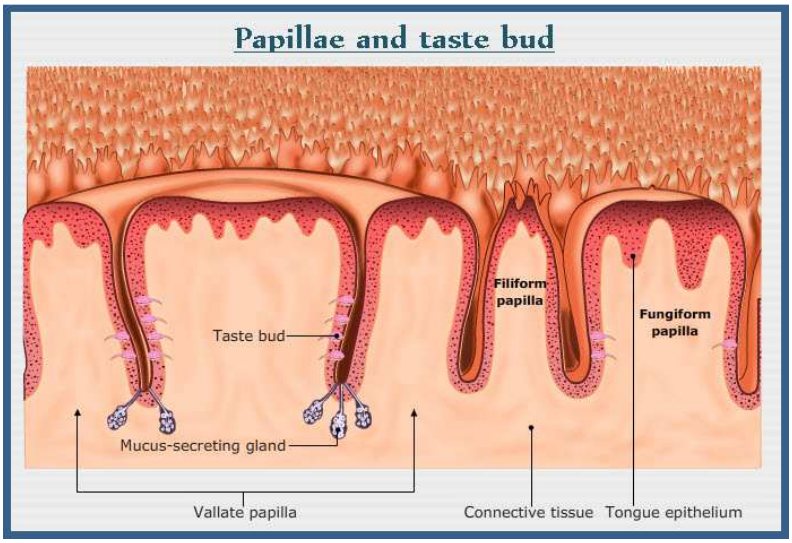
- It is a chemical that causes a natural behavioral response from the another member of the same species (especially reproductive behaviour).
- These pheromones are detected by separate organ known as Vomeronasal organ. It is also known as Jacobson's organ. It is found on each side of the base of the nasal cavity near the nasal septum. A narrow tube is found between the vomeronasal organ and the nasal cavity. The vomeronasal organ has receptors that act via the phospholipase C-based signal transduction.

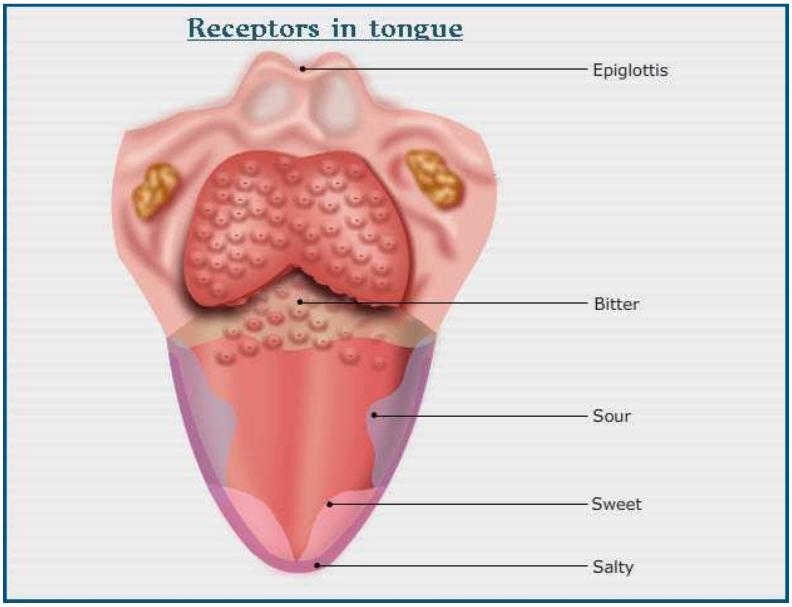
Variation in olfaction

- Animals differ in their ability to detect odours. Animals like dogs, are very sensitive in detection of odour and are called as macrosmatic; those that can detect odour but with less sensitivity like birds are termed microsmatic; those that lack olfactory apparatus like dolphins and whales are termed anosmic.
- Ungulates and rhesus monkeys detect the female in heat by the odour of the vaginal secretions. The olfactory system involved in maternal recognition, regulation of reproduction, mother-young interaction, establishment of dominance.

GUSTATION

- It is the process of detecting dissolved chemicals emitted from the ingested food. It is the sense of taste.
- This is carried out by receptor known as taste buds. There are almost 15,000 taste buds in the porcine oral cavity and throat, while 1 550 in the lizards, but only 24 in chickens.
- The taste buds are not able to discriminate between different molecules as olfactory system.
- Taste buds are onion shaped structure that contains numerous taste receptor cells. It has a pore that opens onto the surface where the tastants (chemicals from the food) enter and contact the receptors.
- On its apical surface, the taste bud has number of microvilli, which also contain receptors and ions channels to mediate taste signal.





TRANSDUCTION OF TASTE

- There are primarily five tastes like salty, sour, sweet, bitter, and umami. The salt taste is mainly due to the Na^+ ions. So the salty food causes opening of the Na^+ channels. The

substance namely Amiloride inhibits the receptor channels and diminishes the salt sensation of the receptors.

- The sour taste is due to the H^+ ions in the food. The sour perception is done via the Na^+ channels, which are permeable to H^+ ions, but compete with them and so, it is done only in species which have less Na^+ ions in their saliva. In other animals, it is by acid-sensing ion channels. It causes opening of the Na^+ channels in response to the change in pH.
- The sweet perception is by the binding of the sweet substance to the G-protein which activates G-protein gustducin, which signals the adenylate cyclase pathway. High sweet containing substances also use IP3-mediated signal transduction.
- The umami is caused by L-glutamate and other amino acids present in the food. There are two different types of receptors. One resembles that of the glutamate receptor in the brain. When glutamate binds to the G-protein, it changes it conformation and releases phosphodiesterase enzyme that degrades cAMP to AMP and that causes increase in neurotransmitter release. The other receptor resembles that of the sweet perception.
- The bitter perception is complex and it is estimated that at least 25 genes code for bitter-taste receptors in human beings.
- Each taste receptor cell expresses more than one taste receptor protein. The taste receptors are not neurons instead they are epithelial cells which secrete neurotransmitter onto a primary afferent neuron. Single taste neurons synapse with more than one taste receptor cells. Olfaction and gustation work together closely and that is the reason why the taste perception is dependent on smell perception of an item.

GUSTATORY PATHWAY

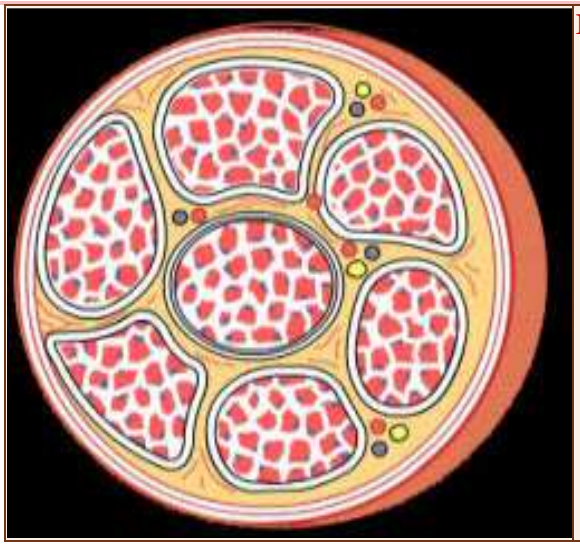
- Taste impulses from the anterior two third of the tongue pass through chorda tympani branch of the facial nerve, thence to nuclei tractus solitarius of the brain stem, while the posterior one-third sends the taste sensations through the glossopharyngeal nerve to nuclei tractus solitarius. Vagus receives the sensory inputs from the base of the tongue and other parts of the pharyngeal region.

- Thalamic area adjacent to ventrobasal complex receives the second order neurons from the nuclei tractus solitarius. The third order neurons originate from the thalamus to the lower tip of the post central gyrus in the parietal cortex, the gustatory cortex. Some of the third order neurons also project to the nearby opercular - insular area, located deep in the sylvian fissure.

Taste reflex

- Tractus solitarius transmits impulses to the superior, and the inferior salivatory nuclei, thus reflexly regulates the secretions of the submandibular, sublingual and the parotid glands.

MODULE-15: STRUCTURE OF MUSCLE, TYPES AND EXCITABILITY



LEARNING OBJECTIVES

- This module helps to,
  - explore the structure and types of muscles,
  - appreciate the ionic basis of muscle membrane potentials and
  - know about inner capabilities of muscle fibres and their compositions.

ELECTRON MICROSCOPIC STRUCTURE OF MUSCLE FIBRES

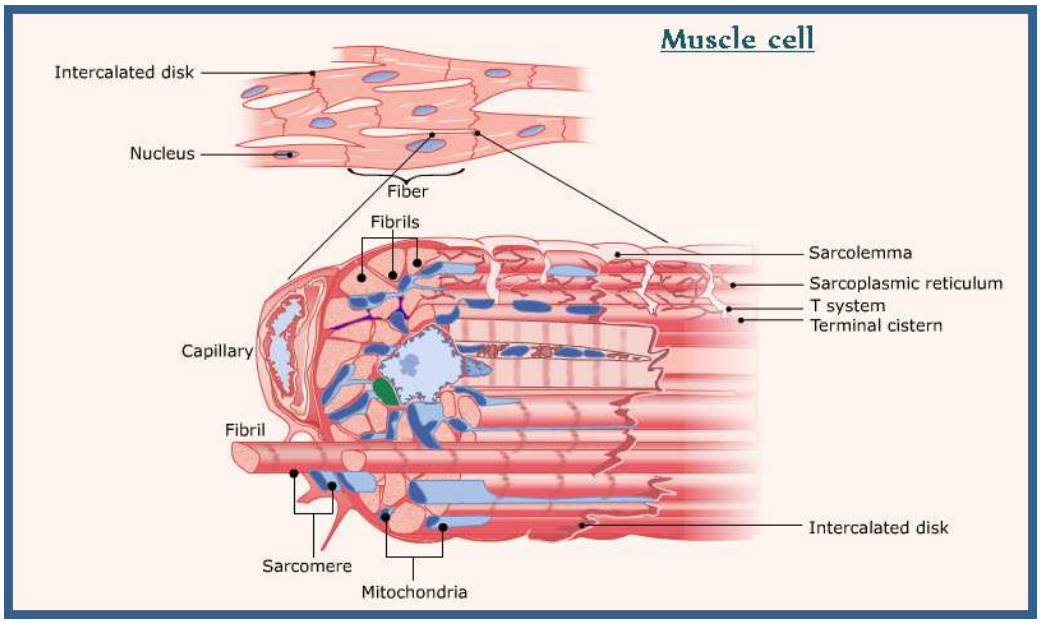
- *Myofibril*
  - Muscle fibres are made up of several hundred to several thousand of parallely arranged myofibrils having the special function of *contraction*. Each myofibril posses the contractile proteins *myosin (thick)* and *actin (thin) myofilaments* results in regular repetition of dark and light bands in the skeletal and cardiac muscles. Each muscle fibre contains about 1500 molecules of myosin and 3000 molecules of actin.
- *Myosin* is a large protein molecule composed of 6 polypeptide chains 2 of which are 100% α - helically exist. Well distinct Head with binding site for Actin and is equipped with ATPase system.They have Hinges to alter the position of the head. The twisted end show the tail.
- Properties of myosin molecular are
  - It has enzymatic ability to split ATP and release energy. Enzyme is myosin ATPase. Actin and Mg^{2+} are required for activating these enzymes.
  - Myosin binds with actin and the actin myosin complex is specifically dissociated by ATP.
  - Myosin spontaneously aggregates to form dimers which can aggregate again to form native thick filaments.
    - The active sites for the myosin ATPase are activity and the sites on myosin that bind to actin are both located in the heads of the myosin molecule and each of the myosin molecule has two such sites.
- *Actin:* much smaller molecule than myosin and contains only one polypeptide chain per molecule. Actins are found in aggregations which form a double stranded helical filament. It has been calculated that one thick filament contains around 500 myosin molecules and each thin filament contains 340-380 actin monomers.

- *Tropomyosin:* molecules are rod shaped they lie in the 2 grooves of the double stranded actin filament and aggregate end to end to produce two strands of tropomyosin running enter length of this filament.
- *Troponin:* More ellipsoidal or globular molecule, binds to a particular site on the tropomyosin molecule. Troponin has an important role as switch for muscle contraction. It has got 3 sub unit polypeptide chains called,
    - *Troponin – T* (This chain has site for attachment with tropomyosin)
    - *Troponin – I* (Inhibits the actin activated myosin ATP ase)
    - *Troponin – C* (binding sites (4 numbers) for Ca^{2+}).
        - Troponin – I attach not only with troponin C and T and but also with actin in the absence of Ca^{2+}. Troponin C attaches itself to Troponin T and I but not with actin.
        - The electron microscopic view of light bands indicates actin filaments represented as *"I" bands* because they are isotropic to polarised light.
- The dark bands of myosin is represented by *"A" Band*, which are anisotropic to polarised light. "H" zone represents partial overlapping the myosin filaments with actin filament. Small projections of myosin filaments called *cross-bridges* protrude out of the myosin filaments along the entire length except the very centre of the myosin filament.
- The centre of I bands contain a dark line called the "*z line*" or "z disc" where the ends of the actin filaments are attached. The z-discs are filamentous proteins, located in a vertical position in the myofibril. That portion of the myofibril between two successive z-discs is called as *"sarcomere"*. The sarcomere is the physiological contractile unit of muscle.
- *`H' Band* of the myofibril is a dark line seen at the centre of "A" band is rarely seen in normally functioning muscle. It is well marked only during relaxation beyond physiological limit.

TYPES OF MUSCLES

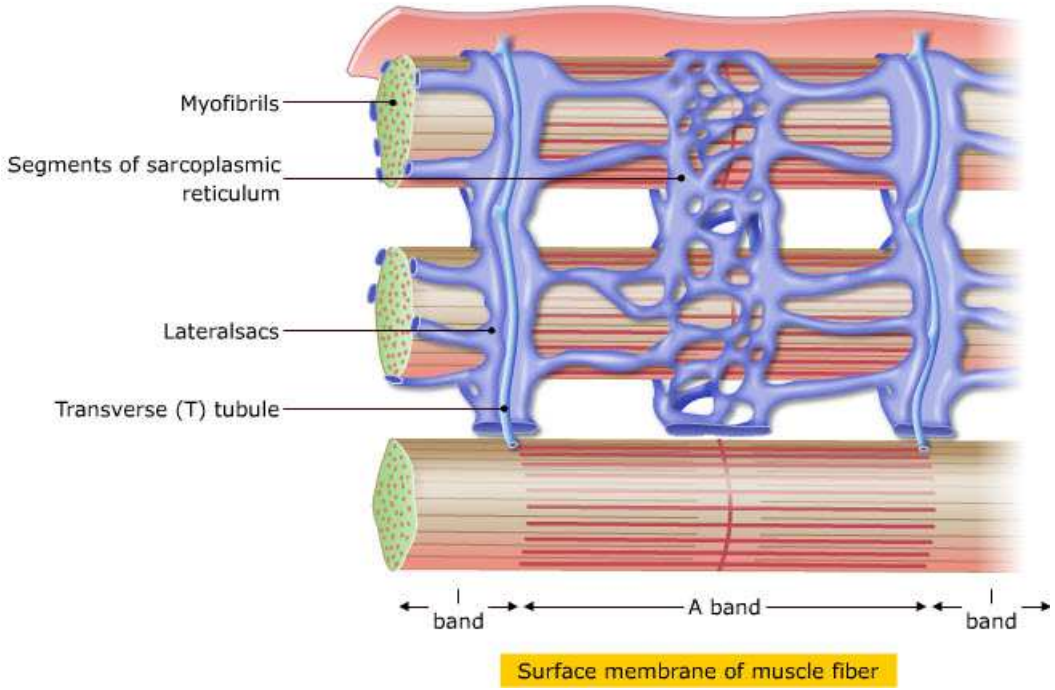
- In mammal three kinds of muscles tissue are found
    - Skeletal muscles
        - Cross striated and attached to skeleton
        - 40% of the total body weight
        - Voluntary neural control.
    - Cardiac muscle
        - Cross striated
        - Involuntary in function
        - Under autonomic nervous system control.
        - Myofibrils are similar to skeletal muscle except at the point of "Z" line cardiac muscle forms Intercalated Disc which forms the muscular network of "Syncytium".
        - T tubular system is present at the level of Z line and not at the level of A and I bands.
        - Triads are simple due to ill developed sarco plasmic reticulum.
        - cardiac muscles have rich mitochondrial supply and many Ca-Na exchangers.
    - Smooth muscle
        - No cross striations found, poorly organized Actin and Myosin filaments.Ratio of Actin and Myosin is 15:1
        - Absence of Troponin complex.
        - Involuntary under ANS control
        - 10% of the total body weight

- Found in the walls of blood vessels, lymph vessels, urinary tract, digestive, respiratory, reproductive organs and gizzard of chickens and turkey.
- Exist as Multi unit and single unit muscle fibres.
- Muscles cells, like neurons can be excited chemically, electrically and mechanically to produce an action potential that is transmitted along their cell membrane Sarcolemma. They contain contractile proteins which are activated by action potential.



STRUCTURE OF MUSCLE

- A muscle has an origin (less movable), belly and the insertion (highly movable) parts.Muscles are surrounded by a sheet of connective tissue called *epimysium* (site for fat deposition).
- From epimysium smaller sheets of connective tissue called *perimysium* arise and divide muscle into several small bundles. This perimysium contains blood vessels, nerves and muscles spindles. Perimysium envelopes a bundle of 50 to 200 fibres. At the end of each fibre, the sarcolemmal sheath fuses to form tendon fibre. Bundle of tendon fibres in turn forms the muscle tendon, which aids the attachment of the muscle into the bones. Each muscle fiber is enclosed by *endomycium* which is a fine sheet of connective tissue arising from perimysium.
- A muscles fiber is termed as a muscle cell- which is elongated tubular structure with tapering ends. Muscle cell is covered by the plasma membrane called *sarcolemma* (Gk: Sarkos – flesh). Each muscle cell is around 20-30 mm in length with a diameter of 10-100 µm. Length and diameter varies with exercise, plane of nutrition and maturity. In a muscle, the muscle fibers are arranged as number of muscle bundles known as "*fasciculus*".



Muscle cells can be grouped into 3 categories as per its metabolism

- *FG* cells (fast twitch glycolytic cells) – Type IIIb (white) – absent in canines
- *FOG* cells (fast twitch oxidative glycolytic cells) – Type IIa
- *SO* cells (slow twitch oxidative cells) – Type I (red)

Sub cellular components

- Muscle cell contains organelles like sarcolemma, nucleus, Golgi apparatus, lysosomes, mitochondria or sarcosomes, endoplasmic or sarcoplasmic reticulum, perioxisomes and ribosomes and myofibrils.
- *Sarcolemma* is about 70 – 100 nm in thickness composed of 60% proteins, 20% phospholipids, 15% cholesterol and 5% collagen fibrils, can propagate an action potential.
- Skeletal muscle cells are *multinucleated* and an average of 100 - 200 nuclei per cell. Because skeletal muscle cell in a mature muscle originates form fusion of 100 – 200 different embryonic muscle cells called myoblasts, with each myoblast contributing one nucleus. They lie peripherally just under sarcolemma in mature vertebrates skeletal muscle. Nuclei of skeletal muscle lack mitotic ability where as smooth and cardiac muscle cell contains only one nucleus, located in the interior of the cell.
- Cardiac muscles cells and FOG and SO types of skeletal muscle cells contains numerous mitochondria and they are concerned with aerobic metabolism for energy and are capable of prolonged and sustained contractions. FG skeletal muscle cells contain a low no of sarcosomes.
- Muscle cells contain small spherical glycogen granules which are numerous in cells of well fed rested animals than in muscle cells of starved or exhausted animals (1% of the muscle mass).

SARCOPLASM AND MYOFIBRIL

Sarcoplasm

- It is the fluid present inside each fibre in which the organelles like *sarcosomes* (mitochondria), sarcoplasmic reticulum and myofibril are suspended.
- Large number of sarcosomes is placed in between the myofibrils, indicates the large quantity of energy (ATP) requirement for the contraction of fibrils.
- Cytoplasm of the muscle cell is called sarcoplasm which contains myoglobin. It is an O_2 storage protein and has a higher affinity for O_2 than haemoglobin. Myoglobin binds with O_2 from oxygenated hemoglobin in the blood stream. This O_2 is used by mitochondria for aerobic metabolism. Oxygenated myoglobin is red in colour and myoglobin concentration varies from 1-4 mg/g in mammals.

Myofibrils

- *Myofibrils* which are elongated protein threads are the contractile organelles.
- They lie parallel with the long axis of the muscle cell. Myofibrils occasionally branch and this branching is a growth mechanism.
- Not covered by any membrane and are insoluble at the ionic strength of the sarcoplasm. 12 to 14 proteins have been identified in the myofibrils: Actin (15% by wt), Myosin(45%), Tropomyosin (4%), Troponin (4%), C- Protien (2.5%), Actinin (3%), M-Protein (3%), Creatine kinase(1%), Desmin (1%), Filamim (1%), Titin (10%), Nebulin (3%), Gelsolin (1%) and Para myosin (6.5%).
- Myofibrils are cross striated, i.e they consist of alternating light and dark bands – transversely striated.
- The dark band is anisotropic or birefringent called 'A' band. The light band is isotropic called 'I' band. The I band is bisected by a dark disc called 'Z' disc or 'Z' line. (The distance from 'Z' disc to next is called sarcomere and is 2.5 to 2.8 µm in resting mammalian muscle.
- During contraction the sarcomere length reduced to 2 or even to a maximum of 1.8m). A contractile unit of the myofibril is the sarcomere. In the middle of the dark A band there seen H zone or Hensons band.
- Myofibrils consist of two kinds of filaments – thick and thin in an interdigitating fashion. Thick filaments got double the thickness over thin filaments (14-16nm) and have a length of 1.5 µm. Thin filament got 1 µm length anchored to the Z disc.
- Thick filaments got cross bridge or protrusions projecting outside. Thick filament – myosin and thin filament – actin, tropomyosin and troponin. Z disc composed of actinin, actin and tropomyosin. 'M' line is present which is a transverse structure connecting thick filaments at their centre is made up of M protein and creatine kinase.

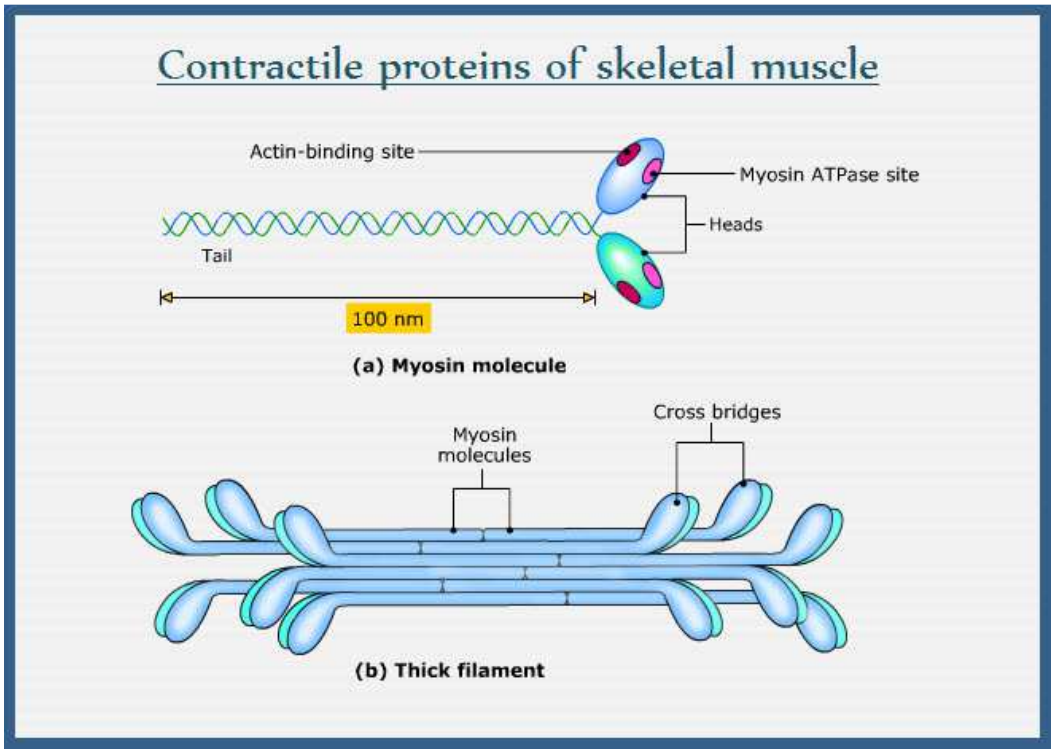
MYOSIN AND ACTIN FILAMENT

Myosin Filament (View animation)

- It is composed of multiple myosin molecules, each molecule having a molecular weight of 480,000. It has two polymerized portions, *Heavy meromyosin* and *Light meromyosin* . Heavy meromyosin is formed by HMM – S_1 and HMM – S_2 molecules.
- The myosin molecule is made up of six polypeptide chains: *2 heavy chains* and *4 light chains*. The two heavy chains wrap spirally around each other forming a double helix. At one end of these chains get folded into a globular structure called *myosin head*. Thus each

myosin molecule has two free heads lying side by side at one end of the myosin molecule. Myosin head is made up of HMM – S_1 molecule, has ATPase activity and readily binds with the "G" actin. The elongated coiled portion of the myosin molecule is called *tail* formed by HMM – S_2 molecule has no ATPase activity and binding property with "G" actin. Out of the 4 light chains, outer two are folded and form part of the myosin heads whereas the inner two form the tail and body.

- The heads of the myosin molecules hang outward to the sides of the body. Along with the head, "*arm*" extends outward from the body of the myosin filament. The flexible points of the arm are *hinges* which allow the head to be extended outward or brought closer to the body of myosin filament. The protruding arms and heads together are called *"cross-bridges"*. There are no cross-bridges at the centre of the myosin filament.
- The myosin head possess ATPase enzyme, which helps to cleave the ATP to release energy for the contraction process. The binding ability of the myosin head with the binding site *G actin* is regulates by ATPase. Only in the presence of actin, Mg^{++} ions enhance the ATPase activity of the myosin head.



Actin Filament

- It is composed of three proteins; actin, tropomyosin and troponin
- Actin
  - It is a double helix structure formed by the basic unit "*F" actin* molecule. Each strand of F-actin helix has hundreds of "*G" actin* molecules which have ADP molecule. G-actin provides the active binding site to myosin head. *Tropomyosin* and *Troponin* are the polymer of actin.
  - Tropomyosin
    - During resting stage, it covers "G" actin and prevents the binding of myosin head with the binding site.

- Troponin
  - It is a complex made up of 3 proteins namely Troponin "T", *Troponin "I", and Troponin "C".*
    - *Troponin "T"* has high affinity to tropomyosin and binds the troponin complex with tropomyosin.
    - *Troponin "I"* readily binds with actin and tropomyosin. In resting muscles, they inhibit ATPase activity in myosin head.
    - *Troponin "C"* has a very high affinity for calcium ions. Each molecule can bind up to 4 Ca^{++} ions. The combination of Ca^{++} ions with troponin is the triggering factor to initiate muscular contraction.
  - The presence of *troponin and tropomyosin (tt-complex)* in actin filament inhibits the binding reaction of actin with myosin. In a relaxed muscle fibre, the tt - complex physically covers the "G" actin for binding with myosin.
- When a muscle is stimulated, a high concentration of Ca^{++} ions is released from the cisternae into the sarcoplasm. The Ca^{++} ions readily combine with the troponin `C' of the TT – complex causes conformational change in the troponin complex and moves the tropomyosin molecule deeper into the groove between two actin strands. This exposes the "G" actin filament to myosin to initiate the contractions of the myofibrils. Only the skeletal muscle and cardiac muscle require troponin and tropomyosin, for execution of contraction coupling.
- In cardiac muscle during excitation-contraction coupling, Ca^{++} ion is released not only from the cisterna of 'L' tubules, but also from ECF through "T" tubules and the strength of cardiac fibre contraction depends to a great extent on the ECF Ca^{++} concentration.
- In smooth muscle instead of troponin another regulatory protein "*calmodulin*" reacts with four Ca^{++} ions. The calmodulin-Ca^{++} complex activates myosin kinase and Ca^{++}dependent phosphorylation of light chain myosin molecule to activate myosin cross-bridge reactions.

Sarcoplasmic reticulum - Tubular system

- Muscle cells also contain a special structure called sarcotubular system which has two sets of tubules. Longitudinal tubules (L-Tubules) and transverse tubules (T-Tubules).
- `L' Tubule system
  - Present parallel to myofibrils. At both the ends of the tubule, bulbous structures known as *terminal cisternae* function *as a* storage place for the Ca^{++} ions and play a key role in muscle contraction.
- `T' Tubule System
  - The T system consist of set of tubules formed by the invaginations of the plasmalemma. This tubule run perpendicular to the long axis of the muscle cell and lie between the cisternae of two successive "L" tubules.
  - The "T"' tubules are located at the junction of "A" and "I" bands which pass through the fibres, open into the inter-fibre space, permits the flow of extracellular fluid.
  - For every sarcomere 1 or 2 T-tubules are observed (preferably at Z disc region). Lumen of T-tubule is actually the extra cellular space. T - Tubule brings action potential from sarcolemma to the interior of the muscle cells. As sarcoplasmic reticular membranes approach the T-tubule, they coalesce to from 2 large sacs called the lateral cisternae, one on each side.

Triad

- It is a paired cisternae and a "T" tubule in between the cisternae. One T- tubule and 2 lateral cisternae associated with it forms a 'Triad'. They have remarkable ability to

accumulate Ca^{2+} against a concentration gradient. Calcium pump an energy dependant - operating in the L-tubules which can transport 2 Ca^{2+} with an expense of one ATP from the sarcoplasm to the lumen of sarcoplamic reticulum.

- The ATPase enzyme found associated with ca – pump, keeps free of Ca^{2+} concentration in the sarcoplasm of a resting muscle at a very low level (10^{-8}M) and the most of Ca^{2+}found in the lateral cisternae bound to a protein called *calsequestrin*. In the cytoplasm, the Ca^{2+} is bound with *calmodulin***,** which permits the transports cytoplasmic Ca^{2+} to its site of action.
- When an action potential pass along with T-tubule, Ca^{2+} from the lateral cisternae become disgorged go to the sarcoplasm thereby rising the free Ca^{2+} concentration from (10^{-8}M) to (10^{-5} to 10^{-6}M) and this intracellular Ca^{2+} triggers muscle contraction.

Proximate composition of mammalian skeletal muscle

- Water: 55 – 78% by wt: Water content varies inversely with lipid content.
- Protein: 15-23%
- Lipid: 1 to 20%
- Carbohydrate: 1-2%: mostly glycogen or lactic acid
- Ash: 1% mostly K^+, less Na^+, Cl^- and Mg 2^+
- Nucleic acid: < 1% - 25 – 30 mg DNA/100g and 100mg RNA/100g.
- Other soluble organic composition: 1% - 8-5 mM ATP, 20mM phosohcreatine, 350 mg carnosine /100g.

EXCITATION CONTRACTION COUPLING

- The arrival of an Action Potential at Sarcoplasmic Reticulum causes release of Ca^{2+} down the concentration gradient to sarcoplasmic reticulum.Released Ca^{2+} causes conformational changes to effect contraction. As action potential passes,$Ca^{2}$are pumped again to sarcoplasmic reticulum and relaxation happens. This cycle is known as Excitation Contraction Coupling.

EXCITABILITY

- It is the ability of any living cell or tissue to exhibit an electrochemical change (action potential) to a stimulus. However, nerve and muscle cells are highly excitable cells than other cells.

Events of action potential

Action potential of nerve and muscle fibres

- The action potential is caused by a sequence of changes or events occurring in the membrane permeability to Na^+ and K^+ ions. When an excitable cell is stimulated, action potential aids the transmission of impulses through nerve and muscle fibres. Action potential is the rapid changes in the membrane potential from its normal negativity to positive potential inside the cell membrane which last for few milliseconds, and then returns back to its original resting negative potential level.
    - *Polarised membrane* is the resting cell membrane with a normal negative resting membrane potential

- *Depolarization stage*: It is the *first event* of action potential is characterised by rapid increase in the permeability to Na^+ ions (5000 folds) to interior of the cell generating more positive electrical potential inside of the cell.

- This is followed by a gradual inactivation of Na^+ channels (closure) that occur within another few milliseconds and the membrane becomes impermeable to Na^+ ions. It is associated with gradual opening of *voltage gated K^+ channels* to allow K^+ ions outflow to the exterior of the cell membrane. The potential inside cell is re-established to its normal resting level (- 75mV). This stage is called as the *repolarisation stage*.

- Higher concentration of K^+ ions in the exterior of the cell towards the end of the action potential continues for a short period creates more negativity inside referred to as *hyper-polarised state*. At this state, re-excitation of the cell will not occur.
- The *final event* is characterised by *electrogenic pump mechanism*, which aids in the transport of three Na^+ ions to the exterior for every two K^+ ions to interior of the cell and create the normal resting potential (- 75 mV) on the inside of the cell membrane.
- The whole action potential lasts about 1 - 2 milliseconds in most nerves, but longer in many muscle cells.

- Channel systems of cell membrane
  - Na^+ – K^+ leakage channels
  - Voltage gated Na channels and K channels
  - *Na^+ – K^+ leakage channels*
  - These channels are 100 times more permeable to K^+ ions than Na^+ ions and are exceedingly important in determining the level of normal resting membrane potential.
- Voltage - gated channels of the cell membrane
  - The *voltage-gated Na^+ channel* causes both depolarisation and repolarisation of the nerve/muscle membrane during the action potential. The *voltage-gated K^+ channel* also plays an important role in establishing the repolarisation of the membrane. These two voltage-gated channels are present in the cell membrane.
- Voltage - gated Na^+ channel
  - The voltage gated Na^+ channels have two gates, the *external gate or activation gate* which opens to outside of the cell. The other gate is at the interior end and opens to inside of the cell referred as *internal gate or inactivation gate*.
  - The activation gate is closed during resting stage (- 75 mV), while the inactivation gate is opened. This prevents free passage of Na^+ ions from outside to interior of the cell.During depolarisation stage the resting membrane potential drops from – 75 mV or to - 70 to - 50 mV causes opening of the activation gate due to conformational changes results in increased Na^+ ion permeability as much as 500 to 5000 folds into the cell through the channel system.
  - A gradual increase to positivity following the opening of the activation gate also closes the inactivation gate comparatively at a slow speed. The time lapse between the activation and inactivation of the channels causes the passage of Na^+ ions to the interior of the cell for few milliseconds.
  - During repolarisation stage, the inactivation gate is completely closed and prevents Na^+ ion entry from outside to the interior of the cell membrane. The inactivation gate will not reopen until the disturbed membrane potential returns nearly to the original resting membrane potential level of -70 to- 80 mV.
- Voltage - gated potassium channel

  - This channel has only one gate at the interior of the membrane. It may either close or open to the interior of the cell. During resting stage this gate is inactivated and prevents the passage of K^+ ion to the exterior of the cell through this channel system. When the electrical potential drops towards zero, Na^+ channels get inactivated causes the activation of K^+ channels to allow increased K^+ ions diffusion to outside of the cells.
  - Electro-chemical changes during action potential Spike potential (over shoot)
  - It is the steep change in the negativity of the cell from - 75 mV to more positivity of + 40 mV in the membrane potential due to rapid increase in the permeability of the Na^+ions to the interior of the cell.
  - In large fibres, the electrical potential during depolarisation stage overshoots slightly positive beyond the zero level whereas in the small fibres and many CNS neurons the potential does not over shoot beyond zero.
- Positive after potential
  - During repolarisation, the membrane potential drops little more negative than the normal resting value of – 75mV due to excess of K^+ ion diffusion to out side the cell. This is caused by the prolonged opening of K^+ channels for several milliseconds even after repolarisation.

MEMBRANE POTENTIAL

- The interstitial fluid and the intracellular fluid contain 147 mEq/litre of positive ions (cations) outside the cell, and about 155 mEq /L, of negative ions (anions) inside the cell. Higher concentration of Na^+ ions (142 mEq/L) and lower concentration of K^+ ions (5 mEq/L) characterise the interstitial fluid.
- The intracellular fluid has more of K ions (140 mEq/L) and less of Na^+ ions (14mEq/L).
- This concentration difference of ions across a selectively permeable cell membrane creates an electrical potential difference.
- This electrical potential difference across the cell membrane is called as *"membrane potential"*.

Basic mechanisms of membrane potential

Three major factors cause the membrane potential.

- Differential permeability of the membrane to diffusion of ions.
  - During resting state the cell membrane K^+ ions is 50 to100 times more odiffusable to out side than to Na^+ ions through *non-gated leak channels due to* their concentration gradient.
  - This contributes accumulation of positive charges immediately outside the membrane.
  - The resting membrane is almost completely impermeable /very slightly permeable to Na^+ ions.
- The Na^+, K^+ electrogenic pump
  - It generates positive membrane potential out side the cell by actively pumping 3 Na^+ ions out of the cell for every 2 K^+ ions pumped into the cell against their concentration gradient.
- Trapping of negatively charged anions inside the cell
  - Many intracellular anions are large molecules proteins, organic SO_4 and PO_4 are trapped within the cell and are attracted to the inner surface of the cell membrane generate negative electrical potential inside the cell.

- These three factors are the primary cause of membrane potential. The magnitude of this potential produced by diffusion of ions can be predicted by the *Nernst equation*.
  - $E_x = (-60mV/z) \times \log[X_{inside}/X_{outside}]$ E_x = equilibration potential for ion X
  - z = valence of the ion – 1 for Na & K
  - [X] = Concentration of ion X
- The diffusion potential caused by K and Na ion diffusion would give a membrane potential of about. An additional giving a net resting membrane potential of – 90 mV.

The resting membrane/ diffusion potentials of skeletal muscle fibres

- During resting stage, the muscle fibres have an electrical potential of - 90 mV inside the fiber (range from - 75 to - 90 mV) of which – 86 mV is determined by K diffusion and the remaining – 4mV is contributed by the electrogenic Na^+- K^+ pump.

SERIES ELASTIC COMPONENT OF A MUSCLE

- The components such as sarcolemmal sheath at the end of the muscle fibres, the tendon and the hinged arms of the cross bridges of the myosin filaments are known as *series elastic components* of the muscle. stretch slightly as tension increases. Consequently, the contractile unit must shorten an extra 3-5% to make up for the stretch of these elements. During isometric contraction, these series elastic components develop greater tension that opposes the contraction of the myofibrils. Hence the contraction becomes zero, but the tension is very high.
- In isotonic contractions, the contractile elements shorten and stretch the series elastic components. This causes the tensionto rise just to exceed the force of contraction due to the effect of the weight. Thereafter the tension in the muscle remains constant.

Types of muscles

Slow Muscle fibres		Fast Muscle fibres	
1	Smaller fibres	1	Much larger fibres
2	Innervated by small nerve fibres	2	Innervated by comparatively large nerve fibres
3	Have extensive blood supply, hence referred as *"Red muscle"*, shows prolonged performance of work (Long contraction time).	3	Have less blood supply, hence called as "*White muscle"*, shows quick and repetitive contractions (Short contraction time).
4	Increased number of sarcosomes	4	Fewer sarcosomes
5	Muscle metabolism by aerobic or oxidation of glucose, fatty acids and amino acids, causes the release of 38	5	Glycolysis or anaerobic type of metabolism liberating 2 molecules of lactic acid and 2 molecules of ATP.
6	Large amount of myoglobin in sarcoplasm.	6	Lack of myoglobin.
7	Less sarcoplamic reticulum	7	Extensive sarcoplasmic reticulum
8	Less Ca^{++} ion release	8	Rapid Ca^{++} ion release
9	Has large motor units (More muscle fibres/ neuron), no fine degree ofcontrol.	9	Has few motor units (less muscle fibres/ neuron), higher degree of fine control.

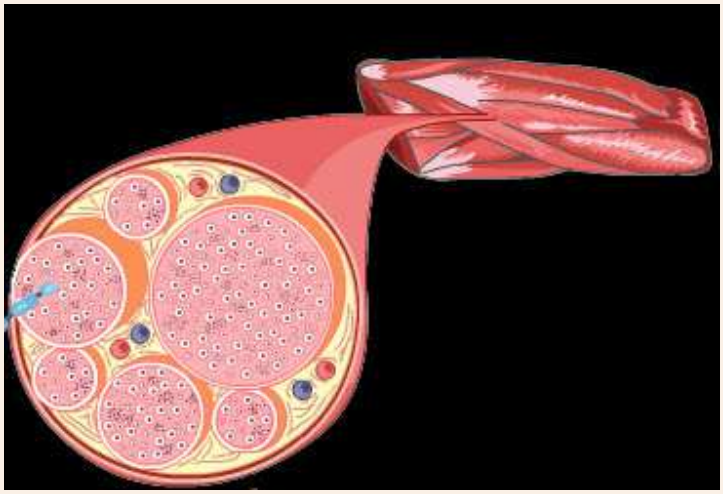
10	Adapted for prolonged, continued muscle activity – like support of the body against gravity	10	Adapted for rapid and powerful muscle contractions – like jumping, running

- Based on the duration of contraction, the *muscle fibres are classified* as slow (tonic) and fast (twitch) fibres. Every muscle in the body is composed of slow and fast muscle fibres.
- Based on energy source and rate of contraction, skeletal muscle fibres can be classified into slow-twitch (type-I) /oxidative and fast-twitch (type-II) /glycolytic fibres. Type-I fibres are always oxidative and slow muscle fibres. Type-II is either oxidative or glycolytic and rate of contraction varies from slow to fast and superfast.

The motor units of the muscle

- Each motor neuron leaves the ventral horn of the spinal cord and innervates many muscle fibres. All the muscle fibres in a muscle that are innervated by a single motor neuron (all the muscle fibres supplied by a single motor neuron) are called as *motor unit*.
- The ratio of motor nerve to the number of muscle fibres innervated in a given skeletal muscle is called *innervation ratio***.** The number of muscle fibres per motor unit differs as per the function for which the muscle is used in the body.
- Small muscles which act rapidly and which require finer control of movement have an innervation ratio of 1:2 to 1:4. Large muscles that do not require finer control have an innervation ratio of 1:800 or more.

MODULE-16: MUSCLE CONTRACTION



LEARNING OBJECTIVES

- This module helps to,
  - learn about properties of muscle contraction,
  - appreciate thermal changes occurs in muscle during work and
  - understand rigor mortis.

TYPES OF WHOLE MUSCLE CONTRACTION

- Muscle contraction is principally of two types
  - Isometric contraction
    - Muscle contraction without shortening in length that occurs during the manifestation of the process of muscle contraction.

        - In isometric contraction, the muscle contracts between two fixed points. Hence during the process of contraction it does not shorten in its length and develops the force of contraction.
        - Therefore in isometric (same length) contraction, there is no work performance, less force of contraction and low Fenn effect due to liberation less energy.
    - Isotonic contraction
        - On stimulation, the muscle contracts between one fixed and one moveable points (eg. Lifting the weight) shows shortening in length, but the tension on the muscle remains constant.
- Contractions of muscle in the body
    - In the body, most of the muscular contractions are a mixture of both isometric and isotonic contractions. During standing posture the quadriceps shows isometric contraction, to tighten the knee joint to keep the legs stiff. Contraction of the biceps is isotonic when the hand lifts the weight.
    - Alternate isometric and isotonic contractions of the quadriceps and gastrocnemius (calf muscle) help to effect running or walking. The quadriceps contracts isometrically when the foot hit the ground, while the calf muscle contracts isotonically when the foot is lifted off the ground.

MOTOR END PLATE

- The axon terminals of the motor neurons innervate each muscle cell. The innervation ratio is the number of muscle fibers or cells innervated by a single motor neuron (motor unit).
- In muscles where delicate control is required the innervation ratio is small (3 to 6) as in extrinsic eye muscles. In the muscles of limb where the fine control is not necessary innervations ratio may be as high as 1000.
- The muscle cell plasmalemma at the motor end plate has got large numbers of infoldings - 0.5 to 1 µm deep 50 – 100 nm width.
- The axon terminal (contains 1,00,000 synaptic vesicles) sends finger like processes into these infoldings but they remain separated by a space of 40 – 60 nm.
- The nerve impulse propagated along the motorneuron and its axon collaterals reaches the neuromuscular junction.

Motor end plate or neuromuscular junction

NEUROMUSCULAR JUNCTION

- The motor neuron branches at its end and each branch comes into a close opposition with the skeletal muscle at a specialised area called the *neuromuscular junction* or *motor end plate*. This synapse has a *presynaptic membrane,* a narrow space between the nerve and muscle, called *synaptic cleft* and a *post synaptic membrane sarcolemma*.
- The presynaptic knob is the terminal portion of the axon of the motor neuron; the extends from the CNS to the muscle cell. The axoplasm contains a large number of vesicles called *synaptic vesicles*; contain the excitatory neurotransmitter substance *acetylcholine*. These vesicles are in clustered located very close to the presynaptic membrane. Acetylcholine is synthesised in the cytoplasm of the nerve terminal and stored in the synaptic vesicles.
- The presynaptic terminal also contains many mitochondria, required for the synthesis of neurotransmitter substance.

- Synaptic cleft is a narrow space of 20 to 30nm wide separates the presynaptic membrane and postsynaptic sarcolemma membranes. The synaptic cleft is filled with extracellular fluid and spongy reticular filaments called basal lamina. The synaptic cleft contains an enzyme, *acetylcholinesterase*, to hydrolyse acetylcholine.

- The postsynaptic cell membrane has a series of invaginations called *junctional folds* that increase the surface area of the postsynaptic membrane. Acetylcholine receptors are located on the postsynaptic membrane at the entrance of these junctional folds.

Transmission of impulse across the neuromuscular junction When the action potential reaches the presynaptic terminal of the motor neuron, the depolarisation opens *voltage-gated* Ca^{++} *channels* ; and extracellular Ca^{++} diffuses into the presynaptic nerve cytoplasm. The increased intracellular Ca^{++} level causes the synaptic vesicles to fuse with the presynaptic membrane; the fused synaptic vesicle opens and releases the acetylcholine into the synaptic cleft by exocytosis.

ACETYLCHOLINE RECEPTORS

- The released acetylcholine diffuses across the synaptic cleft, reaches the postsynaptic membrane (sarcolemma), and binds with acetylcholine-specific receptors. This causes opening of *ligand-gated ion channels* in the sarcolemma. And diffusion of sodium ions into the muscle cell generating local potential change called *end plate potential*.

- Depolarization in the end plate potential opens voltage-gated Na^{+} channels at the postsynaptic membrane and leads to the generation of an action potential on the muscle cell membrane. Presence of the neurotransmitter in the presynaptic nerve terminal permits the transmission of action potential unidirectionally from the motor nerve to the skeletal muscle fibre.

- The binding activity of acetylcholine with its receptor remains for a very brief period. The acetylcholine is rapidly removed from the receptors by
    - Enzymatic hydrolysis of *cholinesterase*, located on the postsynaptic membrane hydrolyses acetylcholine into acetyl and choline molecules.
    - Diffusion of acetylcholine out of the synaptic cleft and is no longer available to act on the muscle.
- *Myasthenia gravis* is a neuromuscular disorder in which autoantibodies are produced against acetylcholine receptors. Hence, neuromuscular transmission is impaired and leads to paralysis.

MUSCLE CONTRACTION

- It is accomplished by a sliding together or telescoping of the interdigtating thick and thin filaments and as a result narrowing and eventual disappearance of the H zone and shortening of I band happen.

Summary of events

- Before contraction ie in the absence of Ca^{2+} in the sarcoplasm, troponin T strongly binds with tropomysion, troponin I and troponin C. At the same time troponin I firmly attached

to actin. Here actually the tropomyosin blocks or hides out the binding site of myosin with actin. This is achieved mainly by the association of troponin - I with the actin.

- An action potential travels along a motor nerve to muscle fibre
- At the neuro muscular junction the nerve secretes a neurotransmitter – acetylcholine (Ach).
- Acetylcholine acts on the muscle sarcolemma and opens Ach-gated ion channels
- Flow of Na^{+} ions to the interior of the muscle fibre membrane at the point of the nerve terminal initiates an action potential in the muscle fibre
- The action potential travels along the muscle fibre membrane
- The action potential also travels deeply into the muscle cell through the sarcoplasmic reticulum releases Ca^{++} from the cistern into the myofibrils
- When action potential reaches triad then Ca^{2+} released, Ca^{2+} binds with troponin – C leading to some confirmational change (Structural change in the tt – complex and exposure of "G" actin to myosin head) on the C molenucle leading to troponin – I to loose its affinity towards actin.
- When troponin I looses linkage with actin, troponin – T can no longer hold the tropomyosin strand out of the groove. Only when tropomyosin is placed outside the groove formed by the double stranded actin helix, then only myosin biding site on actin is marked.
- When actin groove occupied by tropomysin, binding site exposed to myosin.

- Myosin heads or cross bridges attach to actin's binding site and swivel it simultaneously and then dissociate in a sarcomere at a time only 5-20% of the total cross bridge attached to the thin filament at a given instant.
- Now the thin and thick filament slide part each other at a uniform rate without any jerk. Time required for one cycle of cross bridge is 0.1 milliseconds.
- Swiveling or rotating of the cross bridge in an inclined way cause actin filament to move towards the center of the sarcomere.

- Now the thin and thick filament slide past each other at a uniform rate with out any jerk. Time required for one cycle of cross bridge is 0.1 ms.
- Swiveling or rotating of the cross bridge cause actin filament to move towards the center of the sacromere.
- Once swiveling is over ATP is hydrolyzed and energy liberated and this energy used dissociation of cross bridge from the actin. So ATP acts as a dissociator and soon reorientation of the utilization of energy. In other words for swiveling action, no energy is needed but for only dissociation and reorientation of myosin head energy is utilized.
- This series of event continue till Ca^{2+} is rebound to sarcoplasmic reticulum.
- New ATP get bound to the spent cross bridge and hydrolysis of ATP happens only after swiveling.
- After death all myosin cross bridge stop attach with actin at an angled position and results in rigor mortis.

SLIDING MECHANISM OF MUSCLE CONTRACTION

"Walk- along theory" of contraction (Sliding mechanism of contraction)

- In the relaxed state, actin filaments lie adjacent and parallel to the myosin filaments. In contracted state, the actin filaments extending from the z-discs are also pulled inward by head portion of the free ends of myosin filaments.
- Thus, muscle contraction occurs by sliding action actin over myosin filament.

- In the relaxed muscle, the troponin -tropomyosin complex covers active binding sites "G" actin filaments and prevents its attachment with myosin heads to cause contraction.
- In the presence of large amounts of Ca^{++}, troponin-C readily binds with Ca^{++}, the troponin complex undergoes a conformational change and moves the tropomyosin molecules deeper into the groove between two stands of actin strands.
- This effect "uncovers" the active site "G" actin and the myosin head binds with the active sites and initiates muscle contraction.

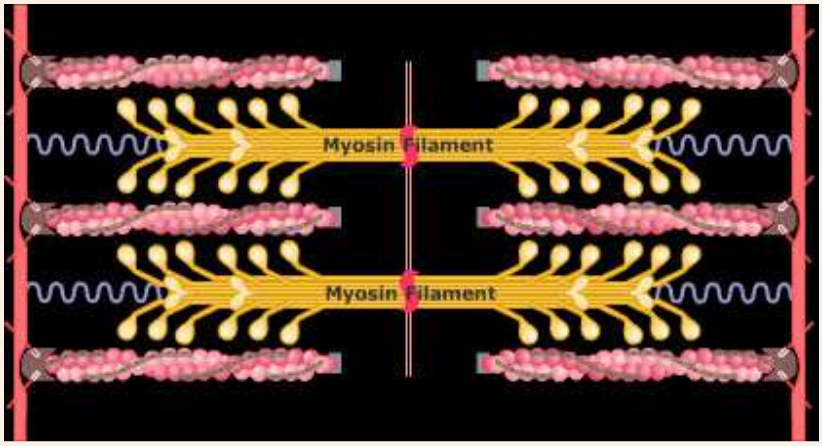
Interaction between actin and myosin filaments

- As soon as the actin filament becomes activated by the Ca^{++}, the heads of the cross bridge of the myosin filaments are immediately attracted to the active sites, "G"actin. This causes sliding movement of actin filament over the myosin filament, which is known as "*Walk along*" theory (*"ratchet" theory*) of contraction.
- ATP molecule binds with heads of the cross-bridges of myosin molecule. ATPase activity in the myosin head causes a partial hydrolysis of ATP to ADP and the inorganic phosphate (Pi). The head extends perpendicular to the actin filament and shows high affinity to actin filament
- The troponin - tropomyosin complex binds with Ca^{++}. Active sites the "G"actin are uncovered.
- The bond between the head of the cross-bridges and "G" actin cause a conformational change in the head causes complete hydrolysis of ATP and release of ADP and Pi. The head shows tiling towards the arm of the cross-bridge. This known as the *"power-stroke"* which pulls the actin filaments in wards. The actin filaments slide over the myosin filament.
- Once the head of the cross-bridge is tilted, ADP and Pi are released. Now a new molecule of ATP binds with the head of cross-bridge detaches the myosin head from the "G" actin.
- The new ATP is cleaved to begin the next cycle leading to power-stroke, i.e. the energy '*cocks*' the head back to its perpendicular position ready to begin another power-stroke cycle again.
- Similar repeated back and fourth movements of the cross bridge makes the heads walk along the actin filament step by step thus pull the actin filament towards the centre of the sarcomere, until the z-membrane reaches the ends of the myosin filaments.

The role of Ca^{++} in muscle relaxation

- The Ca^{++} does not remain in the myofibrils for more than few milli seconds. As the action potential is over, the `L' tubules almost immediately reabsorb the Ca^{++} from the sarcoplasm, and stored it in the cisternae.
- When the Ca^{++} concentration drops in the sarcoplasm, the strong affinity of Ca^{++} for troponin `C' is lost. This causes the movement of tropomyosin molecule to its original position to cover the active binding site, the "G"actin results in relaxation of the muscle.

MODULE-17: PROPERTIES OF MUSCLE CONTRACTION



LEARNING OBJECTIVES

- To learn about properties of muscle conraction.
- To appreciate thermal changes occurs in muscle during work.
- To understand rigor mortis.

CHARACTERISTICS OF MUSCLE CONTRACTION

Stimulus

- It is an external agent, when applied to an excitable tissue provoke a visible *response* .
- Types of Stimuli (Based on strength)
  - *Sub threshold :* sub minimal/ low intensity, unable to produce a response
  - *Threshold :* minimum intensity, strong enough to initiate an action potential
  - *Sub maximal :* higher strength than threshold
  - *Maximal :* highest threshold strength
  - *Supra maximal :* destructive to tissues
- Classification of stimulus
  - *Mechanical :* Pain, Pressure, and Touch
  - *Chemical :* Acid, Alkali; *Thermal*: Warmth, Cold
  - *Osmotic :* Hypertonic solution, Hypotonic solution

PROPERTIES OF MUSCLE CONTRACTIONS

- When a muscle fibre is stimulated by minimal or maximal effective stimuli, the whole fibre contracts to the maximum or it will not contract at all. This law is applicable to single motor unit (motor nerve and all the muscle fibres it supplies).
- *Whole cardiac muscle obeys this law because of syncytium.*

Recruitment

- Because different motor neurons in a nerve has different threshold levels, an increase in intensity of the stimulus applied to the nerve evoke action potential in additional motor neurons and contraction in additional muscles fibers in the muscle served by that nerves..This increase in tension development of a muscle caused by gradual increase in intensity of the electrical stimulation is due to recruitment of additional motor units to the contractile state.

- Recruitment occurs because a nerve contains many different motor neurons having different threshold levels for initiating an action potential. Generally motor neurons having small diameters have low threshold values and are the first motor units recruited by a given stimulus. These small motor neurons are associated with somatic fibres. Largest motor neurons innervate FG fibers have highest threshold levels.

Simple muscle contraction *(Single muscle twitch)*

- Application of single threshold level of stimulus (electrical stimulation) causes a single muscle contraction followed by relaxation. The single muscle twitch has three phases.
    - Latent period
        - It is the period between the application of stimulus and beginning of contraction (about 10m.second).
    - Period of contraction
        - This period indicates the actual shortening (about 40m.second).
    - Period of relaxation
        - It is the period between the point of maximum contraction and the period of complete relaxation (about 50m.second).

Refractory period

- It is a brief period during which a muscle undergoing contraction for a first stimulus is unable to respond to a second stimulus. The earliest chemical change during muscular contraction is the breakdown of ATP. So long as the breakdown ATP is not resynthesised in adequate amounts, the muscle cannot be excited.
- Refractory period is that period during which this resynthesis of broken ATP takes place. Following stimulation of a motor neuron or a muscle cell a brief period exists during which the motor neurons or sarcolemma is unresponsive to the II stimulus is called resting potential.
- It is of two types
    - Absolute refractory period
        - It is the time interval between the twin impulses (less than 3millisecond) during which the muscle can not be stimulated even by stronger stimuli. This is due to the rising phase of the action potential when Na^+ conductance to the interior of the cell is high.
    - Relative refractory period
        - It is the period of reduced excitability, which requires increased intensity of second stimulation to generate another action potential. It is characterised by after depolarization stage of action potential during which the K^+ permeability is very high and the Na^+ channels are recovering their excitability.

Treppe *(Staircase phenomenon)*

- When a stimulus of constant strength and duration is repeated once or twice per second (below tetanising frequency)causes increased contractions during the first few stimulations, which finally reach a constant response. This is due to development of physical and chemical changes i.e. reduced viscosity of the sarcoplasm, increased heat production, increased Ca^{++} availability for binding with troponin during first few contractions, referred as *beneficial effects*.

SUMMATION OF MUSCLE CONTRACTION

- Summation is the added effect of individual muscle twitches to get strong and powerful muscle contraction.
- Summation occurs in two different ways
    - By increasing the number of motor units to contract simultaneously.
    - By increasing the rapidity of contraction of individual motor unit of a contracting muscle.
- Multimotor unit summation (Spatial summation)
    - The strength or the force of contraction of a muscle increases progressively with respect to increasing number of contracting motor units by increasing the strength of stimulation.
    - The strength of resulting contractions in a skeletal muscle is proportional to the size or the number of motor units contracted at the time, which in turn is directly proportional to the strength of the stimulus; i.e. increasing the strength of stimulus increases stimulation of motor units.
- Wave summation (Temporal summation)
    - The frequency of stimulation is increased to a motor unit or units in such a way that the successive stimuli stimulate additional motor units during its contraction phase.
    - When a motor unit is excited by a rapid succession of weak stimuli, it may evoke stronger contractions. Applying stimuli in rapid succession to a muscle increases the excitatory potential of the postsynaptic nerve ending. This increased excitatory potential is often referred to as *excitatory postsynaptic potential (EPSP)*.
    - The second stimulus arrives before the completion of the first contraction, which initiates the second contraction. This gives greater strength or force, as well as duration to the already progressing contraction.

FACTORS INFLUENCING THE STRENGTH OF CONTRACTION

- Size and the number of motor unit stimulated.
- Rapidity with which individual motor unit of a muscle is stimulated.

Starling's law

- The first factor is governed by the strength of stimulus. The second is mostly by the rate of effective stimuli. The last is by the initial length of the muscle fibre.
- In cardiac muscle, mechanical *summation is impossible* because of prolonged action potential (250msec.) almost completely overlaps with long twitch (300 m sec.).
- In smooth muscle, the action potential is relatively short (5 -10 m. sec.), hence no overlapping of twitches, thus *summation is possible*.
- Starling's law of muscle contraction
    - The energy of contraction of the muscle is directly proportional to the initial length of the fibre. This law can be proved by the load and after load experiment on the gastrocnemius muscle of the frog.

Tetanisation

- It is the fusion of successive twitches when the frequency of stimuli is given at a rapid rate.
- Experimental tetanus
  - The experimental tetanus of a muscle may be defined as fusion of individual contractions caused by repeated experimental stimulation of a viable muscle, by which two kinds of tetanus can be produced by using gastrocnemius muscle of the frog.
- Complete tetanus
  - If the frequency of the successive stimulation to a muscle is so adjusted that the subsequent effective stimuli fall during the preceding contraction phase, a complete tetanus or complete fusion of contractions can be obtained ***(tonic contraction)***.
- Incomplete tetanus
  - When the subsequent effective stimuli are applied to fall during the relaxation phase of the preceding contraction, then incomplete tetanus or incomplete fusion of contractions can be obtained **(*clonic* contraction)**.
  - Cardiac muscle fibres do not functionally tetanise, since, mechanical summation is not possible due to prolonged action potential (250msec.), which almost completely overlaps with long twitch (300msec.) duration.
- Critical frequency
  - It is the lowest frequency required to produce a complete tetanus.

Factors responsible for tetaniszation

- Viscose property
  - Viscose inertia of the sarcoplasm, sarcolemma, epimysium, perimysium and facia provide a resistance to change in the length immediately after contraction.
- Fusion of activation process
  - Successive pulsatile stimuli at rapid rate causes fusion of activation or the manifestation of muscle contractions by providing free Ca^{++} continuously to the sarcoplasm.

MUSCLE FATIGUE

- Fatigue is the decrease in the working capacity of a muscle or tiredness of the muscle when it is continuously stimulated.
- It is characterised by diminished force of contraction, increased latent period and contraction period and prolonged relaxation period. The relaxation is incomplete i.e., the muscle is in a state of contracture, (the muscle contracts even without the presence of action potential). At this stage the muscle cannot be excited or do any more work.
- In the locomotor system of the skeletal muscle, there are three sites, which are easily fatigued.
  - Synapses of CNS.
  - Neuromuscular junction.
  - The muscle.

Causes of fatigue

- Continuous stimulation of the nerve fibre or continuous stimulation of the muscle that too under ischemia (impaired blood supply) causes,

- *Lack of energy sources:* Depletion of O_2 from myoglobin; depletion of muscle glycogen; depletion of ATP
- *Depletion of neurotransmitters:* Prolonged muscle activity diminishes neurotransmitter acetylcholine concentration in the neuromuscular junction, thereby reduces the transmission of the impulses to muscle which causes diminished muscle contraction.
- *Accumulation of metabolic end products such as lactic acid:* In oxygen lack, the muscle glycogen broken down by anaerobic glycolysis to give pyruvic acid which is then reduced to lactic acid by the action of lactic dehydrogenase. But in an intact muscle the blood supply minimises the above effect by continuously supplying the O_2 and nutrients to a functioning muscle.

RIGOR MORTIS

Muscle exhibits

  - In-excitable
  - Shortens in length
  - Increases in thickness
  - Become viscous and losses translucency
  - Acidic pH 5.8
  - Stiff
  - Glycogen disappears
  - Gives off H_2CO_3
- Rigor mortis starts in the 2nd hour and is completed in 3 hrs after death. Rigor mortis disappears 24 – 36 hrs after death due to autolysis.
- Deficiency of ATP causing establishment of permanent link between actin and myosin. When ATP falls to 85% and creatine PO_4 to 30% of its initial value, symptoms of rigor mortis begin to appear.
  - Ca^{++} pumps run out of ATP
  - Ca^{++} cannot be removed
  - Continuous contraction
  - Eventually tissues break down
- This is major physiological different between cardiac and skeletal muscles. Skeletal muscles ordinarily contracts only when stimulated by nerve impulse.
- In contrast cardiac muscles tissue can contract with out our nerve stimulation. Its source of stimulation is conducting tissue of specialized muscle with in the heart.
- Nerve stimulation causes the conducting tissue to increase or decrease its rate or discharge. Cardiac muscle tissue has got an extra long refractory period.

THERMAL CHANGES DURING MUSCLE CONTRACTION

- Out of total energy liberated by the hydrolysis of ATP in a contracting muscle, only 40-45% can be converted into work energy and the rest is evolved as heat; i.e. the efficiency of muscle contraction is about 45%. It is recorded by using *thermopile*. Thermopile has plates of antimony and bismuth connected to galvanometer. In a single twitch 0.001° to 0.005° C temperature produced.
- Total heat Produced = Wt. of muscle (gm) × Sp. Heat × Extent of temperature.

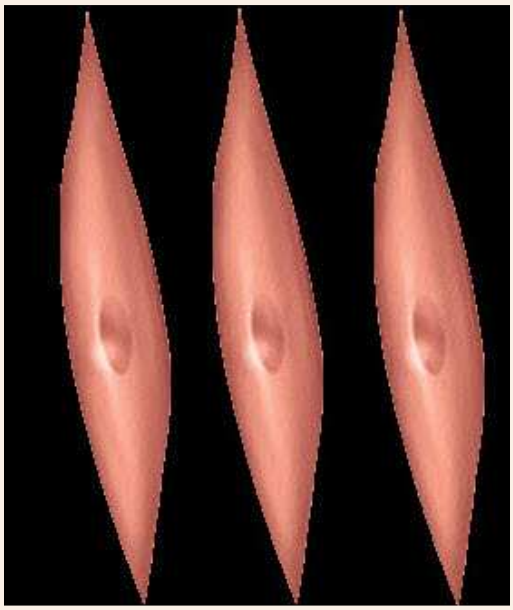
Fenn effect

- When a work is performed by the muscle and energy is required. Heat generation by a muscle is directly proportional to its work performance due to increase hydrolysis of ATP. This Of the total energy liberated from ATP, the muscle utilises 40 to 50% energy in performing work and the rest is liberated as heat to maintain body temperature.
- The liberated heat is classified as
    - *Resting heat:* Used for biochemical reactions during resting state
    - *Initial heat*: It is produced both at initiation and during the course of muscle contraction. It is dependant on aerobic change in the muscle due to breakdown of ATP and creatine PO_4 and formation of lactic acid. It consists of
        - *Activation heat:* produced due to Ca transport associated with initiation of contraction
        - *Shortening heat:* Liberated by sliding reaction, muscle lifts load and does external work.
        - *Maintenance heat:* Heat liberated in a muscle that is stimulated but does not lead to physical work (isometric contraction); it is proportional to duration of contraction and tension developed during contraction.
        - *Recovery heat:* Heat liberated after contraction of muscle fibre. It is due to pumping of Ca^{++} back into the tubules and resynthesis of ATP for next cycle.

Effect of temperature in the muscle

- An increase of temperature shortens the latent and contraction periods whereas relaxation is not much affected.
- A decrease of temperature has the opposite effect causing increased latent and contraction period and prolonged relaxation period.
- When temperature exceeds a few degrees above 40 ° C, irreversible changes take place in the sarcoplasm.
- There is a minimum temperature (near 0°c) is required for muscle contraction, below which the muscle fibres fail to respond to stimulation; but the loss of excitation is reversible.

MODULE-18: MUSCLE METABOLISM AND SMOOTH MUSCLE PHYSIOLOGY



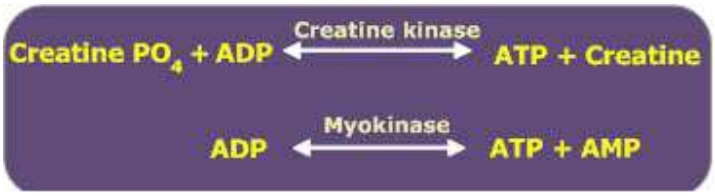
LEARNING OBJECTIVES

At the end of this module, the learner will learn the following,

- energy for muscle metabolism and work done by them,
- smooth muscle and
- muscle pathologies.

MUSCLE METABOLISM

- Sarcoplasm contains creatine kinase and also myosin kinase. ADP degraded to AMP and ATP. AMP produced is degraded to IMP – inositol monophosphate by adenylic deaminase, which again converted to uric acid and excreted.

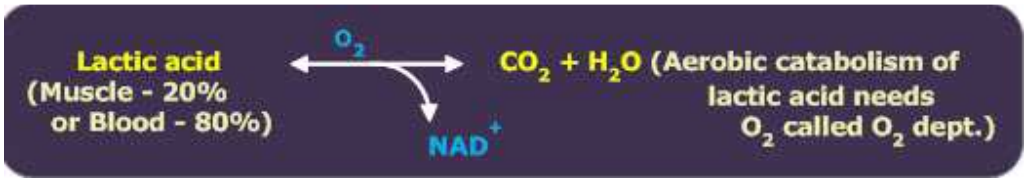


- Hydrolysis of ATP produces 7.3 kcal/mol



METABOLISM OF GLUCOSE IN MUSCLE

- ATP formed during metabolism used for phosphorylating creatine, so creatine PO_4 formed. This creatine PO_4 used during contraction for phosphoylating ADP to form ATP and later ATP is hydrolyzed.
- ATP is also produced from FFA. Eg Palmitic acid generates 140 ATP
- FFA ----------> CO_2 + H2O + ATP
- Hence creatine kinase is used for restoring creatine PO_4 as well as to restore ATP. Approximately 0.3 µmol ATP is hydrolysed per gram of muscle during single twitch. Liver and skeletal muscle store glycogen of the animal body in equal proportion (9 to 16g of glycogen from 1 kg of skeletal muscle).
- FG muscles rely mostly on anaerobic glycolysis as the product of ATP is rapid and happens with out O_2. SOG and FOG cells relay on anaerobic metabolism for their energy production.
- Longer time required to obtain ATP through aerobic metabolism of glucose. Such muscle can sustain activity for longer periods of time (heart: only brief rest periods).
- During initial stages of contraction pH of sarcoplasm increases due to libration of creatine from creatine PO_4. But later pH decreases due to accumulation of lactic acid especially in anaerobic conditions.
- In anaerobic breakdown of glucose, lactic acid production is more which can alter the pH by altering the buffering capacity. So this type of metabolism is self limiting one. Lactic acid brings about fatigue.
- Liver can prepare glycogen from lactic acid more quickly than it is done by the muscles. Muscle can form glycogen more readily from glucose that it is done by the liver. Pyruvic acid partly converted to lactic acid in anaerobic condition.



TYPE OF MUSCLE FIBRES

- FG cells have a
  - Shorter contraction time (5 – 50ms).
  - Less amt of myoglobin as FG cells go for anaerobic meth myoglobin with in the stage for O_2 is less needed.
  - Have more phosphorylase needed for degraded glycogen).
  - Larger cell diameters (fast twitching).
  - Fewer mitochondria (more numbers needed for aerobic metabolism).
  - More glycogen needed for anaerobic glycolysis.
  - Much less lipids (lipids are oxidized aerobically by β-oxidation).
  - Less blood flow.
  - Slower rate of protein turnover do not depend AA metabolism for energy.
  - Narrower Z disc 66 nm compared to 144 nm by SO cells – rapid contraction.
  - Fairly less red in colour (less myoglobin).
  - A more extensive and highly organized sarcotubular system (more Ca^{2+} storage for stronger contraction).
  - Less resistance to fatigue, and are innervated by large motor neurons,
  - Higher contractile speed (myosin dependant ATPase activity of FG fibers has 2 – 3 times more than SO fibers. FOG cells have intermediary characteristics between FG and SO cells. 2 disc width is 88 nm. Speed of contraction higher than SO but lower than FG, have shorter contraction periods.

WORK DONE BY THE MUSCLE

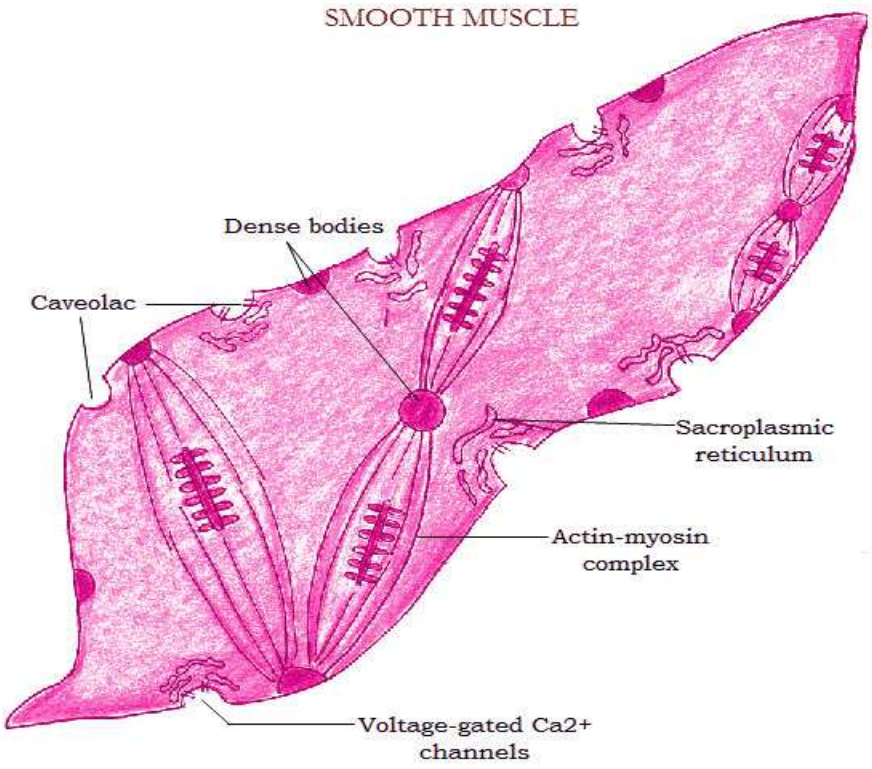
Work done (W) = (Hxl)/L

  - L – Length bet fulcrum and the tip of the lever
  - L – Length bet fulcrum and the weight load
  - H – Observed HT on contraction
- The chemical energy liberated during contraction is party converted into mechanical energy with manifibers ency indicated paction of total energy liberated is converted into work.
- Untrained animals = 20%
- Trained – 25 – 30%
- Isolated muscle – 40%

SMOOTH MUSCLE TISSUE

- Contains action and myosin filaments with are not orderly arranged. So no striations.
- **Two kinds**

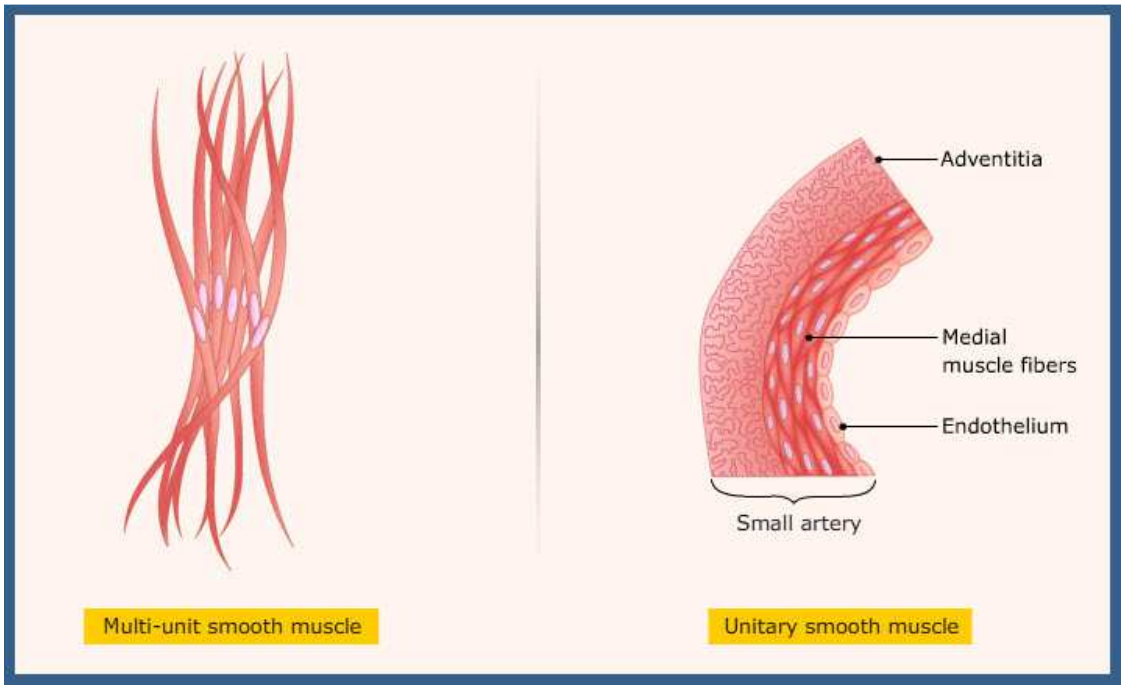
- *Visceral smooth muscles:* About 99% of the smooth muscles are of the unitary type. The fibers are tightly bound together top form continuous network called '*syncytium'* and wrap around sheets that form part of the walls of hollow visceral such as uterus and stomach, intestines, urinary bladder etc. When a neuron stimulation a fiber impulse travels over the other fibers via '*gap junctions'* so that contraction occurs in wave over many adjacent fiber direct electrical conduction
- *Multi unit smooth muscle tissue (Direct electrical conduction):* consist of individual fibers each with its own motor nerve endings, similar to that of skeletal muscles. About 1% of the smooth muscle is present as multiunit smooth muscle. Seen in the walls of blood vessels (sympathetic stimulation), erector pili muscles that attach to hair follicles, intrinsic muscles of eye such as iris, nictitating membrane.



- Both kinds of smooth muscles can contract and relax more slowly than skeletal muscles tissue (peculiar arrangement of thin and thick filaments). Smooth muscles can maintain forceful contraction for longer time (labour pain of uterus) than skeletal muscles. Smooth muscles can be stretched with out developing tension eg., Urinary Bladder, stomach wall etc.
- The actin filaments are attached to protein – dense bodies which in turn are attached to cell membrane. There are only 1 myosin filaments to every 12 – 15 actin but each myosin filament for more numbers of cross bridges – 3 kg/cm^2 pressure. Latent period 150 – 200 ms and total time 1 to 3 sec for contraction and relaxation with in about 30 times as long as the single twitch of skeletal muscle. Swiveling action of cross bridge is much slower ie., less ATPase activity. This economy of energy utilization by smooth muscle is important to

overall function of the body, because organs such as intestines, UB, GB, must maintain moderate tone throughout life.

- Contration duration is longer in smooth muscles.
- Muscle proteins loosely organized so no myofibrils but force of contraction is almost equal to skeletal muscles.
- Extra cellular materials like collagen, elastin are produced by smooth muscle.
- Smooth muscles are seen as sheets (GIT tract) or bundles (Erector pili) and have no tendons except gizzard (Central tendon)
- Actin molecules are held by dense bodies (similar to Z-lines)
- Contain α- actin similar to Z line
- No tropomyosin, instead Calmodulin
- In smooth muscle, myosin and actin happens only when light chain is phosphorylated.
- More of voltage gated Ca^{2+} channels - slow channels. These channels allow entry even in the absence of action potential.
- ECF (Ca^{2+}) has got strong influence on force contraction.
- Similar to cardiac muscle.
- Prolonged contraction and relaxation period (200-300 msec.).
- No T-tubules
- Stretching – contraction because of single unit smooth muscle depolarised and spread over large area (Gut).
- Multi-unit smooth muscle found in ciliary body, iris, ductus deference and walls of large arteries. Here each muscle fiber in innervated separately & contracts only when it receives synaptic stimulation.



Action Potential in smooth muscle

- They are two types
  - Spike potentials and

- AP with plateaus ie., prolonged repolarization causing sustained contraction as in uterus.

ACTION POTENTIAL IN SMOOTH MUSCLE

Slow wave rhythm

- Some smooth muscles are self excitatory ie; elicite AP with out an extrinsic stimuli; due to waxing and waning of pumping of Na^+ outward – if rapid sudden decrease in electronegativity, if slow gradual decrease in electronegativity and once reaches – 35 mv causing action potential. ie., most of the peaks of slow waves, one or more AP occurs leading to rhythmical contraction of smooth muscle (Pacemaker waves). This type of activity is prominent in gut, ureter etc.
- Excitation of visceral smooth muscle by stretch also possible, here slow wave potential and decrease of membrane potential caused by stretch are the causes. Eg., gut loaded by ingesta --------> stretch --------> peristalsis; Uterus at full term -------> delivery.
- In case of multi unit smooth muscle, depolarization is cased by neurotransmitters contraction with out generation of its own action potential. Because the fibers are two small to generate an action potential where as in visceral smooth muscle 30 to 40 smooth muscle fibers getting depolarized simultaneously ie., self propagating AP ensures. Yet even without an AP in the multiunit smooth muscle fibers the local depolarization (caused by NT) spreads itself electronically over the entire fiber and it is all that is needed to cause the muscle contraction.
- Mechanism of Ca^{2+} in smooth muscle contraction: An effective troponin system is lacking Ca^{2+} itself excites contraction in smooth muscle fibers by activating ATPase activity of myosin head.
- Local tissue factors and hormones causing smooth muscle contraction with out AP generation: In fact probably half or most of all smooth muscle contraction is initiated not AP but by stimulatory factors like non – nervous and non-AP stimulatory factors like local factors & hormones. Eg., Contraction of met-areterioles, pre-capillary sphincters where virtually nervous connection is not there (smaller the size, little or no nervous supply), local changes like lack of O_2,excising of CO_2, excess H^+ cause vasodilatation. Hormones such as nor-epinephrine, epinephrine, Ach, angiotensin, vasopressin, oxtocycin, serotonin, histamine can cause vaso-constriction/dilatation depending upon the receptes in the smooth muscle (Excitatory receptor or Inhibitory receptor).
- Some of the hormones viz., nor-epinephrine, vasopressin and angiotensin have powerful excitatory effect causing spasm, of blood vessels for hours together.

Action potential in smooth muscles

- The smooth muscles have relatively lower resting membrane potential and short action potential (5 - 10 m sec.) than the skeletal and cardiac muscles. The resting potential of the smooth muscle cells range from – 50 mV to – 60 mV. The action potential is regulated by Ca^{++} pump.
- Action potential in visceral smooth muscle occurs in two forms: (1) spike potential (2) action potential with plateau.
- *Spike potential:* Typical spike potential as in skeletal muscle occurs in smooth muscles which is elicited by electrical stimulation, hormones, transmitter substance from nerves, by stretch or spontaneously.
- Action potential with *plateau* occurs in visceral smooth muscle cells similar to cardiac cells to prolong the period of contraction in smooth muscles of ureters, uterus and vascular

smooth muscles. In smooth muscle cells, the voltage-gated Ca^{++} channels are comparatively more in number than the Na^{+} channels. Ca^{++} generate the action potential in the smooth muscles. Since the Ca^{++} channels are slow to open and close, the on set of smooth muscle contractions are slower.

CARDIAC MUSCLE

- Involuntary, single centrally located nucleus, numerous mitochondria. The T-tubular of mammalian cardiac muscles are larger (esp. in Ventricles) than those of skeletal muscles and are located at the Z lines rather than at the A – I band junction as in skeletal muscles fibers. While the groups of skeletal muscle fibers an arranged in a parallel fashion, those of cardiac muscles branch freely with other fibers to form 2 separate net works.
    - Muscular walls and septum of atria
    - Muscular walls of septum of ventricles
- When a single fiber of either network is stimulated all the fibers in the network become stimulated as well. Thus each network contracts as a functional unit. Each fiber in a network is separated from the next fiber by an irregular transverses thickening of sarocolima called intercalated disc. Thus cardiac muscle can act as a *Syncytium* (which provides more rapid spread of impulses between fibers).
- Under normal conditions cardiac muscles tissue contracts & relaxes rapidly continuously and rhythmically with out stopping.
    - AP spreads cell to cell through "*Gap Junctions*", and in this way to entire heart (in skeletal muscles each fiber must be separately stimulated by a motor neuron).
    - T – tubules poorly developed in atrial musculature. In verticals, they are much larger in diameter as compared to skeletal muscle.
    - Location: They lie over Z-lines sarcoplam reti is not well developed;
    - No lateral cisternae – So no Triads (ie., have *diads*).
    - AP = 100 – 250 ms duration (Skeletal muscles = 2-3 ms duration.
    - Catechols – increase the Ca^{2+} release which causes force and frequency increase
    - Purkinje fibres specialized cardiac fibers for impulse conduction, part of the pace maker
    - Have extensive Ca^{++} - Na^{+} exchange system.

Rhythmicity

- Repetitive discharge of impulses normally occurs in the pace maker cells of SA node, smooth muscle and many neurones of CNS in which the resting membrane potential is only - 55 to - 60 mV which is not enough to close the Na^{+} and Na^{+}- Ca^{++} channels. This cause inward flow of Na^{+} and Ca^{++} creates still less negativity inside these cells and generates spontaneous and cyclic action potential. More over their cell membranes are naturally leaky to Na^{+} to generate the action potential at a faster rate.

MEMBRANE POTENTIAL IN CARDIAC MUSCLE

Resting membrane potential in cardiac muscle

- The resting membrane potential in cardiac muscle is – 90 mV, but has long action potential (250msec).
    - Cardiac muscle = - 85 to - 95 mV
    - Conduction system = - 90 to -100 mV

- Ventricle muscle = -100 to -105 mV
- S.A. node = - 50 to - 55 mV

Action potential in cardiac muscle

- Cardiac muscle has an inherent property of generating its own action potentials rhythmically, independent of nerve stimulation. This occurs in the peacemaker cells of the S.A. node, which depolarises faster than any other parts of the heart.
- The onset of action potential in cardiac muscle is slower but prolonged than skeletal muscle which lasts for 150msec. in atria and 300 m.sec. in ventricle. In cardiac muscle, repolarisation does not occur immediately after depolarisation The positivity generated as the spike potential by the depolarisation remains as a plateau near the peak. This plateau lasts for a few-hundred m.sec prolongs the contraction of the cardiac muscle.
- *The reason for the prolonged action potential* in cardiac muscle cells is due to the presence of two separate channel systems
  - *Voltage activated Na channel (fast channel).*
  - *Voltage activated Na - Ca channel (slow channel).*
- Activation of the fast Na channels causes the spike potential of the action potential, whereas the slow channel prolongs the passage of Ca^{++} and Na^{+} into the interior of the cell, thus establishes the plateau in the action potential.
- The inflow of Ca^{++} in to the cardiac muscle cells decreases K^{+} permeability by about 5 fold. This delays the K^{+} permeability to outside which in turn delays the re-polarisation process of the action potential in cardiac muscle. This prolonged action potential (250 – 300 m.sec) provides longer contraction period in the cardiac muscle cells than the skeletal muscles. Hence cardiac muscle is not functionally tetanized.

MUSCLE PATHOLOGIES

- Skeletal muscle pathologies are classified into 3 categories
  - Muscle atrophies that have a neural involvement that derive form disease of ventral root cells, motor nerve roots or peripheral nerves
  - Disorders of neuro muscular transmission including myasthenia gravis, poisoning and various acetyl cholinesterase compounds botulism and tick paralysis
  - Disorder of muscle cell itself which are genetically determined myopatheis, muscle damage by external agents including physical damage, toxins and drugs such as steroids with chloroquine, inflammatory reactions due to viral or bacterial or parasite infestations and autoimmune disease (polymyositis), muscle disorders associated with endocrine or metabolic disease, myopathies associated with malignant disease such as rhabadomyeloma, tumors, amyloidosis and paroxysmal myoglobinuria.

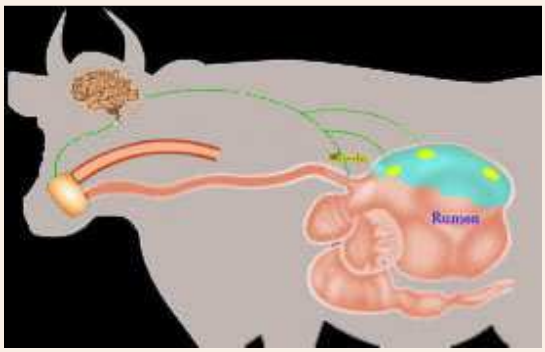
- **CHARACTERISTICS OF SKELETAL, SMOOTH AND CARDIC MUSCLES**

Property	Skeletal	Smooth	Cardiac
Cell shape, size and interactions and cell organelles	Cylindrical 10 - 100 µm diameters 2-3 cm long, no electrical coupling (cell to cell conduction), no branching, prominent T- tubules,	Elongated fusiform 1 – 12 µm dai.,	Multi branched 2 –20 µm dia., 20-200 µm long, intercalated disc provide electrical coupling. Abundant mitochndria. No lateral

	multiple nuclei, varying degree mitochondria present, triad is prominent.	15 – 500 µm long, functional syncytium, no branching. Moderately developed sarcoplasmic reticulum system, less prominent, T- tubules, few mitochondria	cisternae, fairly developed sarcoplasmic reticular system.
Rhythmicity	Nil	Present	Present & characteristic
Conductivity	Very fast	Slower	Slower
All or none law	True for single fiber	True for single fiber	True for the whole heart.
Contractility	Simple muscle curve	Slow- all periods of the curve longer	Contraction longer than relaxation
Refractory period	Short (with in latent period)	Longer	Longest (whole contraction period is absolute refractory)
Tetanus	Possible	Not so	Impossible (long refractory period ensures recovery)
Fatigue	Possible	Possible	Impossible (long refractory period ensures recovery)
Tonicity	Depend on nerve	Independent of nerve	Independent of nerve
Action Potential	RMP = -70 mV, AP is 1 msec long, AP depends on Na^+ and K^+ gradient. No spontaneous activity.	RMP = – 55mV (-50 to -60), larger resting Na^+permeability. AP depends on Ca^{2+} gradient. AP 10 ms long, AP is Ca^{2+} permeability dependent (SLOW CHANNEL). Stretch and pacemaker activation.	RMP = -90 mV. AP dependent gradients of Ca^{2+} K^+ and Na^+. Spontaneous pacemaker activity, no stretch induced activity, AP 250 msec long.
Actomyosin ATPase	Well developed	Lowest actomysion ATPase.	Moderate amount of actomysin ATPase.
Ca^{2+}	Ca^{2+} removes inhibition due to troponin. $Ca^{2+}$from sacrcoplasmic reticulum only, Ca^{2+}increases the activity of ATPase during contraction.	No troponin system. sparse sarcoplasmic reticular Ca^{2+}. Ca^{2+} form ECF (by diffusion). Ca^{2+} so latent period is more prolonged. Ca^{2+}pump works slowly causing relaxation, so more contraction time.	Ca^{2+} from ECF and sarcoplasmic reticulum. Troponin system present. Extensive Ca^+, Na^+exchange system. Ca^+ ↑es strength of contraction and duration of systole.

Na^+	Excitation	Probably same	Initiates and maintains heart beat.
K^+	Reduce excitability and hastens fatigue	Probably same	Inhibits contraction and produces relaxation
Protein	Maximum	Less	Less
Glycogen	Less	More	More
Carnosine	Maximum	Less	Less
Fats	Mostly neutral fats	Mostly neutral fats	More phospholipids and cholesterol than in other
Inorganic	Na^+/K^+ = 1/5 Ca^{2+}	About same	Na^+/K^+ = ½ Ca^{2+} ie., more Na^+
Energy metabolism	Glycolysis and oxidative metabolism.	Primarily glycosysis	Oxidative metabolism with high O_2 need. Also glycolysis & fatty acid metabolism.
Lactic acid	Oxidized less easily than glucose and often incomplete	Same as skeletal	Completely and more readily than glucose
Glycogen	Decreased metabolism of glycogen during starvation	Same as skeletal	Increased glycogen metabolism during starvation.
Blood supply and O_2consumption	Moderate	Less	High
Control	Voluntary	Not so, involuntary	Not so, involuntary
Innervations	Single innervation with Peripheral NS A-ch as neurotransmitter.	Dually innervated by Autonomic NS. Excitation and inhibition modulates ongoing activity	Dually innervated by ANS. Parasympathetic slows down activity and sympathetic increases.
Electrical events	Short AP (1 ms) does not over lap with twitch event (30 ms). Summation possible.	Relatively short Ap (5-10 ms), does not over lap with prolonged twitches. Summation possible	Very long AP (250 ms), almost completely overlaps with long twitches (300 ms). No tetanus and no summation possible.

MODULE-19: DIGESTIVE SYSTEM OF MONOGASTRIC ANIMALS AND RUMINANTS



LEARNING OBJECTIVES

At the end of this module, the learner will be able to know,

- the difference between ruminant and monogastric gastro intestinal tract,
- factors of digestion and
- functions of saliva and it's importance in the ruminant animal will be known.

PHYSIOLOGY OF DIGESTION AND ABSORPTION

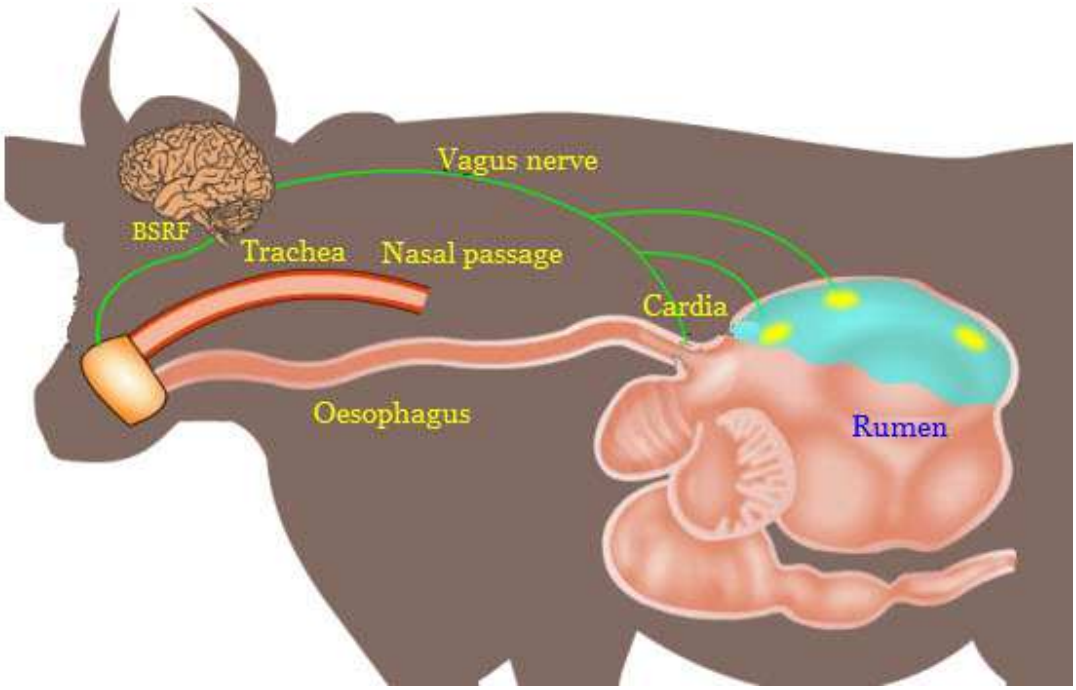
- Digestion is the process of breakdown of complex food into simpler form by the activities of the alimentary tract and glandular secretions for absorption of nutrients and the rejection of their residues.

Food

- Food is a complex mixture of substances like carbohydrates proteins, fats, vitamins, inorganic salts and water to meet the nutritive requirements of an animal.
- *Herbivorous animals* (*Cattle, sheep, horse, goat* etc.,) derive their nutritive requirements from plant sources.
- *Carnivorous animals* (dog, cat etc.) obtain their food from animal sources.
- *Omnivorous animals* (*man, pig, etc.*) get their foods from both animals and plants sources.

Alimentary tract

- It extends from the lips, mouth, pharynx, oesophagus, stomach, small intestine (Duodenum, jejunum and ileum), large intestine (Caecum, colon and rectum) and anus.
- In carnivores, alimentary tract is small and simple.
- The simple stomach herbivores (horse, rabbit), are also referred as *hind-gut digesters* because they have relatively simple stomach portion, but the large intestine is much more complex and voluminous than the carnivores.
- In ruminants (cattle, sheep and goat), the stomach is extensively large and complex, whereas the large intestine is relatively small, hence they are known as *fore-gut digesters.*



Functions of GI tract

- To assist in the acquisition of nutrients.
- To prepare nutrients for digestion.
- To digest the nutrients.
- To felicitate absorption of products of digestion.
- Absorption of water.
- As excretory organs to help in elimination of waste products.
- As an endocrine gland to influence digestion and other metabolic functions.
- Motility of the GI tract.

Prehension of food

- Prehension is the seizing and conveying of food into the mouth. In bipeds (Primates) the hands are the prehensile organs. Dogs and Cats hold their pray with the forelimbs which is passed into the mouth by the head and jaw movements. In horse, upper lip, tongue and the incisor teeth are the main prehensile organs to collect the food. The clefted upper lip in sheep favours close grazing, on contrast the unclefted upper lip in goat. In cattle, large strong, rough protrudeable tongue and incisor teeth of the lower jaw are the prehensile organs. The pointed lower lip functions as a prehensile organ in swine.

FACTORS OF DIGESTION

Mechanical factors

- Mastication (Grinding the food)
- Deglutition (Swallowing)
- Regurgitation (Return of the cud from stomach to mouth)

- Rumination (Chewing of cud)
- Remastication (Regrinding)
- Redeglutition (Reswallowing)
- Intestinal motility
- Eructation (Elimination of gases)
- Defecation (Elimination of fecal matters)

Chemical factors

- Digestive enzymes and hormones
- Non-enzymatic chemicals: HCl, HCO_3

Microbiological factors

- Bacteria
- Protozoa
- Fungi

MASTICATION

Definition

- Extensive chewing of the feed causes mechanical reduction in the size of the food. The finely divided food particles provide greater surface area for enzymatic action and proper mixing with saliva for easy swallowing.
- This act is achieved by diduction and occlusion of both the upper and lower molar teeth.
- Temporo mandibular joint is involved. Hence, lateral forward and backward movement is possible.
- Most of the animals masticate on one side at a time due to the anatomical arrangement of the dental table (wider upper jaw and narrower lower jaw).
- Molar teeth are chisel shaped with the arrangement of inner most lower teeth and outer most teeth. Both are sharp edged.

Importance of mastication

- In herbivores mastication is of greater significance due to the coarse bulky nature of food.
- In ruminants, only during remastication the food is thoroughly ground.
- In herbivores, mastication of food material is by lateral movement of the lower jaw and to-and-fro movement.
- In herbivores, the upper jaw is wider than the lower jaw and mastication of food occur on only one side at a time.
- The molar teeth are chisel shaped with sharp edged lower teeth pointed towards inner side and that of upper teeth towards outer side. Incisor teeth are used to cut and lacerate the food
- In ruminants, the upper incisors are absent, but modified as *dental pad* and in the lower jaw they are loosely and obliquely placed in the sockets.
- Grazing
    - *Traditional grazers* (sheep, cattle)
        - Grass and roughage eaters.
        - Mostly graze on monocotyledons

        - Use their small mouth and long tongue to grasp grass and hay.
        - Tongue is curved around the grass
    - Browser ruminants (deer, giraffe and moose)
        - Graze on dicotyledons
        - Have large mouth opening and use their incisors to gnaw on fruits and flowers, trees.
    - *Intermediary* – opportunistic feeders (goat)
        - Adapt to both the pattern of feeding.
        - They prefer low fiber dicotyledons.
        - Preferred food can be sought by climbing
- Bovine
    - Keratinized papillae of the bovine helps in the passage of food in the oral cavity.
    - As the tongue pushes the cell grass between the lower incisors and upper dental pad, simultaneous mandibular severing reduces the size of the particles.
    - As rumination is the predominant activity, they spend less time in initial mastication.
- Sheep and goat
    - Mandibular grinding
- Horse
    - Incisors thrust the feed into the oral cavity with the propulsive movement of mandible, maceration of grass at its base is achieved.
    - Masticate thoroughly prior to swallowing.
- Omnivores occupy a position between herbivores and carnivores in the aspect of timing.
- Pig
    - Use incisors, tongue and propulsive movements of head for mastication.
    - Rhodents have the unique property of mastication due to their ability to shift their teeth in a proplineal pattern.
- In carnivores, mastication is imperfectly performed by vertical movement of the lower jaw.
- Dog and cat
    - Mastication is done by teeth with food in oral cavity by propulsive head and mandibular movements.
    - In cats, papillae of the tongue (dorsal lingual spicules) help in pushing the feed into the oral cavity.
- In carnivores and omnivores, the upper and lower jaws are equal width and the teeth are relatively simple.

Drinking

- *Dogs and cats*
    - Fluid is drawn into the oral cavity by the rapid extension and retraction of the tongue by making the free end of the tongue as laddle like structure to convey the liquid into the mouth.
- *Horse, Bovine, Sheep and Goat*
    - Drinking of fluid is by suction by creating negative pressure by keeping closed mouth beneath the fluid and tongue.

Suckling

- It is effected by creating negative pressure in the mouth largely by the back ward movement of base of the tongue.
- Milk is forced from the teat into the mouth due to pressure gradient.

FUNCTION AND MECHANISM OF MASTICATION

Function

- Physical breakdown increases the surface area of food, which improves the microbial digestion.
- Assist in appreciating the flavor of the food.
- Improves salivary and gastric activities.
- Initial digestion of carbohydrates is facilitated.
- Contributes to dilution and buffering of ruminoreticular fluid.

Mechanism

- Reflex activity under the control of CNS.
- Mastication is basically a voluntary act but usually takes place involuntarily.
- Mastication reflex or chewing reflex is the rhythmic movements of mandible.
- Lowering of mandible due to extension of tongue (mainly in bovine) is known as LINGUO MANDIBULAR REFLEX.
- Stimuli for masticatory reflex is the presence of food in the oral cavity.
- These are receptors in tongue and oral mucosa which initiates mastication by the sense of food.
- Sensory impulses are carried via trigeminal, facial, and glossopharyngeal nerves to brain stem.
- Efferent via trigeminal supplied to mastigatory muscles.
- This in turn causes rhythmic movement of the mandible in relation to the maxilla.
- This results in shearing and crushing of food.
- The main mastigatory muscles include temporalis with the assistance from massetor muscle for shearing and massetor with the assistance from pterygoid muscles for grinding.

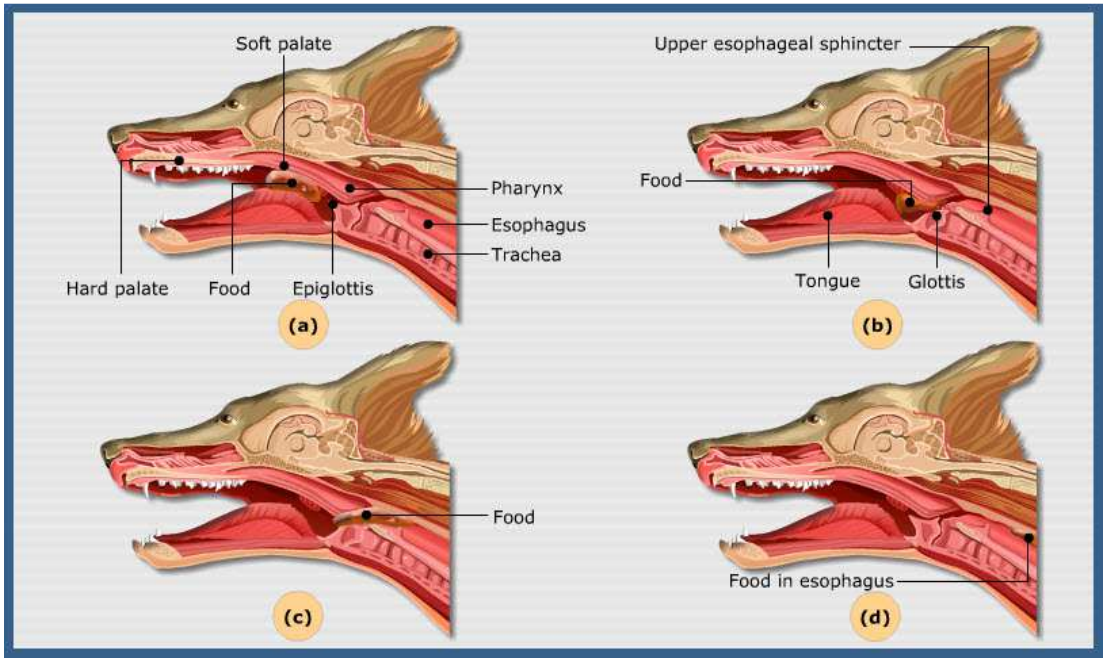
DEGLUTITION (Swallowing)

- Deglutition is the act of passage of food from the mouth to the stomach through pharynx and oesophagus. It starts, as a voluntary act then becomes an involuntary reflex during its execution. The afferent nerves originate from receptors located in the posterior part of the mouth and pharynx. Deglutition center is situated in the medulla.
    - It is the motor activity involving the integrated movement of muscles of tongue, pharynx and oesophagus. (including the peristaltic movement)
    - It is under direct neural control of brain.
    - Complex action of tongue, to place the bolus centrally between the tongue and the hard palate. Pressure of the bolus in the pharynx stimulates pharyngeal pressure receptors to initiate swallowing reflex.

Act of swallowing

- It takes place in three phases
    - From mouth to pharynx (voluntary act)
    - From pharynx into the oesophagus (reflex mechanism)
    - From the oesophagus into the stomach (reflex mechanism)
- Contraction of mylohyoid and hypoglossal muscles.

- Pressing of the tongue against the hard palate and pulling the root of the tongue backwards.
- The tongue acts like a plunger driving bolus towards pharynx.
- Elevated soft palate is to cut off communication with nasal passage.
- Forward pulling of hyoid bone and the larynx, cause opening of the entrance of oesophagus.
- Closure of the larynx by epiglottis make the bolus to enter directly into the oesophagus.



STAGES OF SWALLOWING

- Pharyngeal stage - Phryngeo oesophageal sphincter
- Oesophageal stage - Significance of lower esophageal sphincter.

Pharyngeal stage

- Relaxation of pharyngeo oesophageal sphincter.
- Peristaltic contractions of the pharynx to propel the food bolus from the base of the tongue into laryngopharynx.
- Propulsion of the bolus via relaxed pharyngeo oesophageal sphincter to the esophagus.
- Pharyngeo oesophageal sphincter
    - It is also known as upper oesophageal sphincter formed by the cricopharyngeal muscle and oesophageal circular muscle.
    - As the bolus proceeds to upper oesophagus, simultaneous contractions of this sphincter to a closed state to initiate next stage of swallowing reflex.
    - The bolus travels from cranial esophagus to caudal oesophagus via gastro esophageal junction.
    - Gastro esophageal junction is guarded by a lower oesophageal sphincter.

- Reflex, contraction of the oesophageal muscles to propel the bolus down.
- This reflex is completed by vagus.
- Accumulation of several boluses in the oesophagus causes local myogenic stimulus initiates more powerful peristalsis to push the food to gastro oesophageal junction.

Oesophageal stage

- Lower oesophageal sphincter
  - It differentiates esophagus and stomach.
  - Physiologically is in a high pressure zone.
  - The incoming bolus exert force to open this sphincter.
  - This is an active reflex mediated by vagus to increase oesophageal pressure.
  - Synchronized act of relaxation of lower oesophageal sphincter with oesophageal wave propel the bolus into the stomach.

SWALLOWING REFLEX

- Swallowing centre is located in the brain stem (medulla oblongata).
- Stimulation of the receptors in the soft palate, pharynx (posterior wall) and epiglottis (dorsal surface) by food material.
- Sensory fibers emerge through vagus, glossopharyngeal and hypoglossal to mylohyoid and hypoglossal muscles.
- Contraction of mylohyoid and hypoglossal muscles press the tongue against hard palate. Backward pulling of the tongue elevates soft palate.
- The tongue forces the bolus in the opened oesophagus.
- Pulling action of hyoid bone and larynx opens the oesophagus.
- Larynx is closed by epiglottis.
- This reflex involves internucial neurons from reticular formation.
- Swallowing centre also activates the neighboring respiratory regulatory neurons.
- This interrupts respiration during swallowing to avoid aspiration of food particles into respiratory passages.

Mechanism of regurgitation

- Common in ruminants.
- Initiated with closed glottis and raised palate.
- Drop in intrathoracic and intraesophageal pressure due to inspiratory effort of the tongue.
- Opening of cardia, clearing of cardia.
- Extra reticular contraction (regurgitation contraction) pushed ruminoreticular cud to the mouth via esophagus.

OESOPHAGUS AND STOMACH

- Oesophagus is a muscular tube like structure extends from the pharynx to the stomach. In dog, cattle and sheep, the muscular layer is striated throughout the length of the oesophagus, whereas in pigs and horse, it begins as striated but becomes smooth muscles at caudal oesophagus.
- The *pharyngeo – oesophageal* junction is normally closed by *oesophageal sphincter*.
- Vagus is the main motor nerve regulates the motility of the oesophagus.

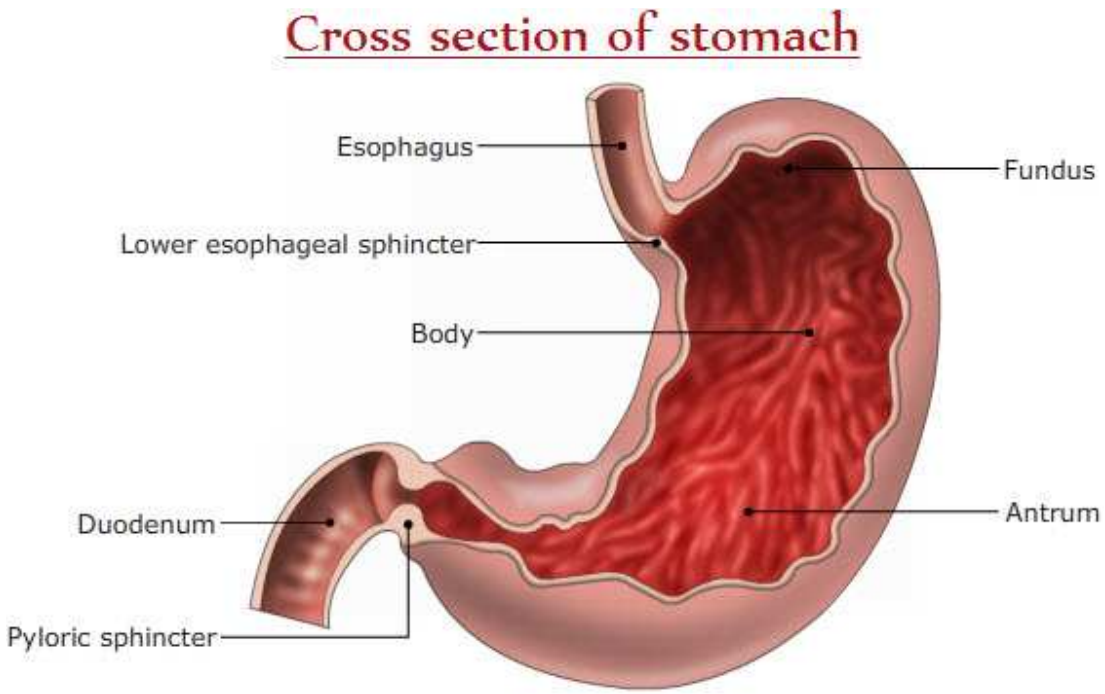
- During swallowing, the peristaltic wave travels from pharyngeo - oesophageal sphincter towards *cardiac sphincter* which is located at *gastro-oesophageal junction*.
- Reverse peristalsis/antiperistalsis is involved in bringing the gastric contents into the oesophagus during belching and regurgitation.
- Peristaltic waves, bucco - phryngeal pressure and gravity are responsible for the movement of food bolus through oesophagus, of which bucco - pharyngeal pressure is important for the passage of liquids.

Cardia

- The point of opening of oesophagus into the stomach is called cardia.
- It is provided with a sphincter muscle known as cardiac sphincter.
- It prevents back flow of food from stomach to oesophagus.
- Cardia is ordinarily closed except during swallowing and regurgitation.
- The cardiac sphincter is well developed and powerful in horse.
- The activity of cardiac sphincter is under the control of CNS.

Stomach

- It functions as a reservoir of food.
- Actively involved in grinding the food to reduce their size
- Initiates enzymatic digestion of food materials
- Controls the rate of passage of food to the small intestine for final digestion and absorption.
- Produces the intrinsic factor for the absorption of vitamin B12 from the intestine, which involves in hematopoiesis.



- Based on structure and function of stomach, domestic animals fall into two general classes.
  - Non-ruminants/ simple stomach animals – Horse, cat, dog, and pig.
  - Ruminants – Cattle, sheep , goat, camel and buffalo
- The stomach of nonruminants is simple consisting of only one compartment, whereas the stomach of ruminants is complex, consisting of four compartments (rumen, reticulum, omasum and abomasum) of which only abomasum secretes the gastric juice
- The stomach is a hollow, sac like organ made up of four layers - serous, muscular, submucosa and mucosa from outside to inside.
- The stomach mucosa of simple stomach animals, is divisible into *oesophageal (glandless)* region and *glandular* region. The glandular area includes *cardiac, fundic (parietal)* and *pyloric* regions.
  - In horse, oesophageal region is extensive up to 1/3 to 1/5 of the surface area of stomach. In the glandular region, the cardiac gland zone is very narrow, while the fundic gland zone is very wide.
  - In pig, the oesophageal region of the stomach is limited to a small area around the cardia. The cardiac gland zone is very extensive, whereas the fundic and pyloric gland zones are similar to those of horse.
  - In dogs, the oesophageal region is absent, the cardiac glands are found as a narrow zone scattered along the lesser curvature of the stomach around the cardia. The fundic glandular zone is extensive occupying about 2/3 of gastric mucous membrane.

RUMINANT STOMACH

- Ruminants are animals capable of regurgitating their food from their stomach and remasticate them.
- Capacity of the stomach varies with age and size of the animal.
- All herbivorous animals have spacious compartments in their G.I tract. It favours retention of bulky fibrous plant for soaking, mixing and microbial fermentation.
- In ruminants and kangaroo, stomach (Rumen) provides an additional space for microbial fermentation.
- Bacteria, protozoa and fungi in the rumen are responsible for extensive fermentative digestion in the rumen. It is supported by the mechanical activity of the three compartments (rumen, reticulum and omasum).
- Only the abomasum, the true stomach secretes gastric enzymes and HCl. Ruminants are animals which can regurgitate and remasticate.
- Abomasum is the largest compartment in new born ruminants.
- As age advances, the rumen and reticulum grow at a faster rate than abomasum.
- In adults, rumen and reticulum occupies 69%, omasum 8% and abomasum 23% of the stomach portion. View image...
- Omasum is not well developed in sheep and goat, it is absent in Camel and Llama (Tylapoda)
- Oesophagus opens into the rumen through cardia.
- Rumen has two sacs, dorsal and ventral sacs that are freely communicating with each other.
- Rumen is connected to reticulum by rumino – reticular folds.
- Reticulum is communicated with omasum through reticulo –omasal orifice.
- Oesophageal/reticular groove extends from the cardia to reticulo – omasal orifice.
- It is a gutter like invagination. This groove is more functional in young ruminants. View image...

- During suckling the receptors in the pharynx and mouth get stimulated causes reflex closure of reticular groove to conduct liquid and milk directly from the oesophagus into reticulo – omasal orifice, bypassing rumen and reticulum.
- In calves, reticular groove/oesophageal groove acts as a bypass route for the passage of milk directly from the oesophagus into the omasum and abomasum.
- Closure of this groove is mediated by behavioural, psychological resposed and also by chemicals.
- Administration of chemicals like NaCl, $NaHCO_3$, $CuSO_4$ and sugar solutions reflexly close this groove, but $CuSO_4$ is less effective in calf and older ruminants, but more effective in sheep.
- Sodium salts like $NaHCO_3$ (60ml of 10% solution) stimulates closure in calves.

OTHER STRUCTURES

Liver

- Liver and gall bladder represent accessory structures of the digestive system.
- Metabolic organ, participate in digestion, metabolism and also in circulatory/immune system.
- Special and unique organ as it receives double blood supply, seat of metabolism and is of diagnostic purpose.

Pancreas

- Representing one of the accessory structures of the digestive system secretes clear, colorless liquid representing water, enzymes and salts like sodium bicorbonate.

Small Iniestine

- Has 3 portions Duodenum, Jejunum and Ileum. Concerned with digestion and absorption.

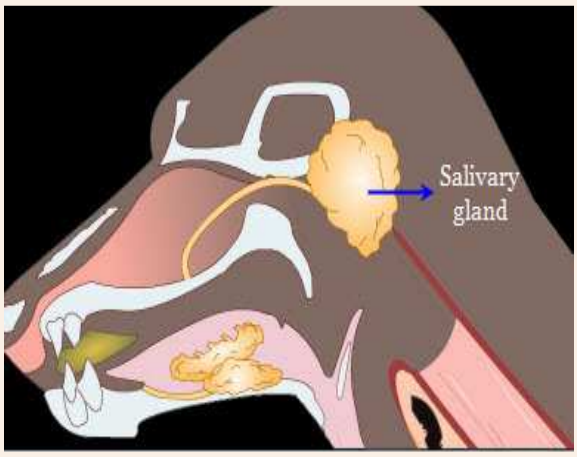
Large intestine

- Has 3 major portions viz. cecum, colon and rectum. Main function includes formation of certain vitamins, completion of absorption, formation of dung and its expulsion.

Gastro-intestinal motility

- The walls of GI tract are muscular and capable of generating slow waves of electrical depolarization which are originated by the presence of food/ingestion as a stimuli from various areas of gut lumen.
- These motility aims to propel ingesta from one area to the other, to promote digestion and to aid absorption.
- Movement popularly known as motility are of propulsive, reventire or mixing.

MODULE-20: SALIVARY SECRETION, FUNCTION AND REGULATION



LEARNING OBJECTIVES

At the end of this module, the learner will understand,

- the basic unit of salivary secretion,
- mechanism of salivary secretion and
- it's control and importance in animals.

SALIVARY GLAND AND ITS SECRETION

- Salivon is the functional salivary unit.
- It begins most proximally as several Acini which converge into a intercalated duct.
- Numerous intercalated ducts unite to form a striated duct. A group of these ducts directly drain their content into glandular tubule which pour their content into fewer excretory ducts.
- Finally, the excretory ducts converge to form a single excretory duct, which leads to oral cavity.

Salivary gland and secretion

- Saliva is the mixed secretion of three pair of main salivary glands, namely parotid, sub maxillary or sub mandibular and sublingual and also many small glands found in the mucous membrane of the mouth.
- Glands in general are divided into serous, mucous and mixed types.
- Serous glands give rise to thin, watery secretion containing protein without mucin, mucous glands produce a secretion containing the glycoprotein in mucin ; mixed glands produce both serous and mucous secretions.
- Cells that gives rise to a serous secretion also secrete enzymes. Zymogens granules are the precursors of enzyme stored in the serous cells.
- Mucous gland is collected in a system of ducts which discharge their content through larger ducts into the mouth.
- The absorptive and secretary activities of the cells lining the ducts affect the composition of the saliva.
- In most of the mammals, parotid glands are serous. In some animals the secretion of this glands is devoid of enzymes .
- The sub maxillary glands is mixed in ungulates, dogs and cats but it is serous in rodents.
- The sublingual gland of the horse, ox, pig, dog and cat is a mixed gland and that of rodents is mucous.

Salivary glands of ruminants

- Parotid, submaxillary and sublingual are the major salivary glands of ruminants.
    - In addition, sheep and cattle have two inferior molar glands, small and numerous buccal and labial glands in cheek and lips, palatine glands in hard and soft palate and pharyngeal glands in the pharynx and roof of the tongue.
    - *Inferior molar is a* serous gland
    - *Buccal, Pharyngeal* and *Palatine* are mucous glands
    - *Labial is a m*ixed gland
- Based on the composition of saliva salivary glands are classified as
    - *Alkaligenic glands* (Parotid, Inferior molar, Buccal and Palatine) secrete more of HCO_3 and low content of mucin.
    - *Mucogenic glands* (Submaxillary, Sublingual and Pharyngeal) secrete more of mucin with low concentration of HCO_3

Pattern of secretion

- The parotid glands secrete spontaneously and continuously. Its secretion is rapid during feeding and rumination on the side of bolus chewing. Its flow is about 2ml/min. at rest and 30 to 50ml/min. during rumination. Mechanical stimulation of mouth, cardia, reticulo-omasal orifice, lips of oesophageal groove and walls of reticulum reflexly stimulate parotid secretion.
- Secretion of inferior molar occur only during feeding, whereas buccal and palatine glands shows slow resting rate of secretion. Flow of saliva in cow is 60 to 160litres/day; in sheep 6.0 to 16litres/day.
- In dogs, submaxillary and sublingual glands show free flow of saliva during chewing of normal meat (no secretion by parotid), whereas dry meat powder excites abundant secretion from the parotid.
- In horse, parotid secretion occurs only during feed intake, whereas in ruminants, parotid secretion is continuous.
- Sheep show slight increase in parotid secretion as a result of feeding.
- Numerous small glands like inferior molar, buccal, labial, palntine and pharyngeal do contribute to the quantity of ruminat saliva. Volume of saliva produced by adult cattle is 90-190 litre per day.
- Salivary secretion is continuous.
- Flow of saliva varies with activity and increase with feeding and rumination to aid deglutition.
- Type of ration influences rate of salivation.
- Saliva is predominantly of two types i.e. serous and mucus.
- Serous saliva is rich in carbonate ions and is secreted continuously.
- Mucus saliva helps to reduce foaming by increasing surface tension and is secreted when animal is feeding and is rich in mucopolysaccharides.

Methods of studying salivary secretion

- Mixed saliva may be obtained directly from the mouth.
- By means of cannulation of the appropriate salivary ducts.

SALIVA - COMPOSITION AND SECRETION

Spontaneous secretion of saliva

- Some salivary glands possess the innate ability to secrete in the absence of any from of stimulation and this type of secretion is known as spontaneous secretion. This mechanism exists for maintaining the moist mucous membrane of the mouth and pharynx. In ruminant, the parotid salivary glands secrete spontaneously.
- Stimulation of salivary buccal receptors by the entry of the food in the mouth reflexly stimulate copious salivary secretion. Normal foodstuffs cause salivary secretion rich in mucin and enzymes to facilitate easy swallowing, while dry or noxious materials cause watery secretion with little mucin.

Composition and formation of saliva

- Saliva like all digestive secretions, is formed from the blood.
- Mixed saliva obtained from the mouth is a colorless, slightly opalescent liquid containing small amounts of electrolytes, protein and in some animals, a carbohydrate splitting enzymes- salivary alpha amylase, desquamated cells from buccal mucosa and lymphocytes derived from the lymphoid tissues in the mouth and pharynx are also present.
- Dog parotid saliva is highly fluid and thin like water,whereas dog sub maxillary glands saliva is highly viscous. The protein of saliva are complex and the term mucin is the galactosamine containing muco proteins. In some animals the resting serous cells of salivary glands contains numerous zymogen granules which are the precursors of the salivary enzymes, alpha amylase, and which disappear after prolonged secretion.
- During para sympathetic stimulation, the rate of secretion of both the sub maxillary and parotid glands is almost linear to the rate of stimulation, some of the inorganic constituents of the saliva change their concentration as the rate of secretion increases, the sodium and chloride concentration in the saliva are greater at high rates of secretion than at lower rates whereas the concentration of bicarbonates and potassium are higher at higher secretory rates.

Composition of Ruminant saliva

Inorganic salts	Nacl, KCl, $CaCO_3$, $NaHCO_3$, PO4
Organic components	Mucin, Ptyalin, Urea,Uric acid, Creatinine and Amino acids.
Suspended organic matters	Leukocytes and epithelial cells
Gases	CO_2, O_2, H_2, N_2 and water vapours

Reaction of saliva

- Slightly acidic in man, slightly alkaline in most of the domestic animals (except ruminants) and distinctly alkaline in ruminants.

Amount of saliva

- Amount of salivary secretion is inversely related to moisture content in the food. Quantity of salivary secretion:
    - Horse- 50 ml / minute during mastication
    - cow- 100 to 200 L/ day
    - single parotid gland of sheep- 930 to 1840 ml/24hrs.

NERVE SUPPLY AND FUNCTION

Innervation

- In addition to neural or hormonal stimuli the secretory cells of the salivary glands secrete saliva spontaneously. Neural stimulation produce more secretion in most species. The salivary glands receive both efferent innervations of sympathetic and para sympathetic nervous system which mainly act synergistically on the salivary glands.

Sympathetic supply

- The sympathetic fibers leaves the spinal cord as pre ganglionic fibers in the ventral roots of the first two or three thoracic nerves and pass through the cervical sympathetic chain to the superior cervical ganglion. The post ganglionic fibers (adrenegic nerves) from the superior cervical ganglion are distributed to blood vessels of the salivary glands and the secretory cells.

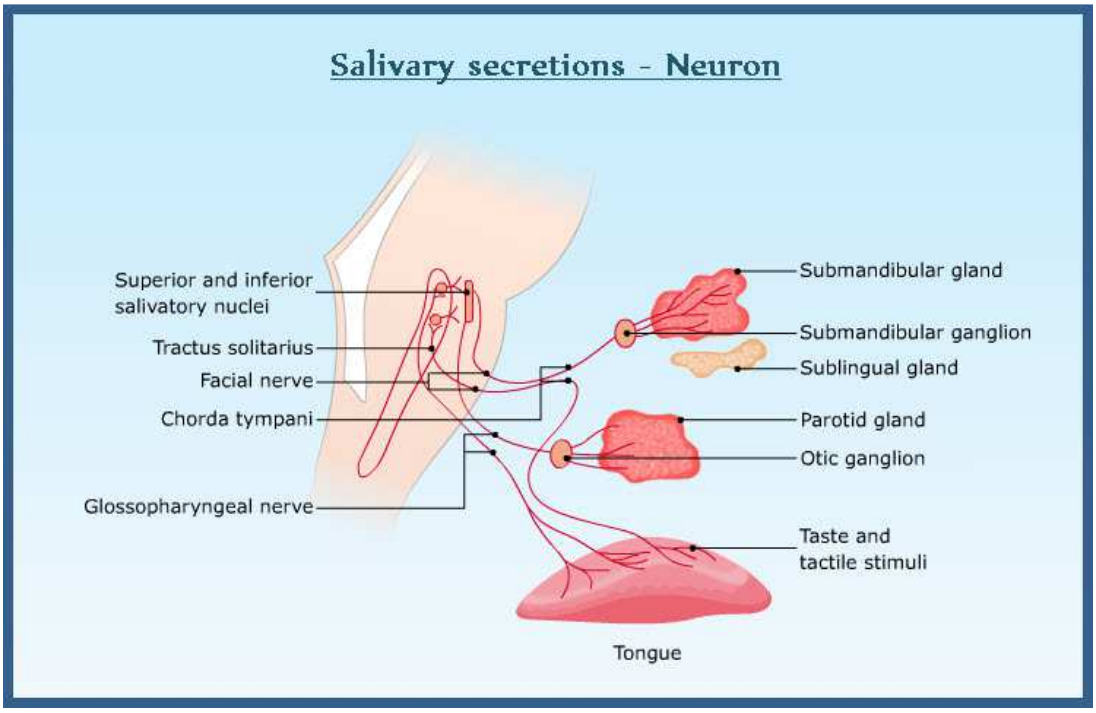
Para-sympathetic supply

- The preganglionic fibers of the para sympathetic fibers leave the brains stem through glosso-pharyngeal and trigeminal nerves, synapse with the otic ganglion which privides post ganglionic fibers to the parotid gland. The preganglionic parasympathetic fibers to the sub maxillary and sub lingual glands leave the brain stem in the facial nerve and travel along the chorda tympani to join the lingual nerve before their synaptic junctions with sub maxillary ganglion. This ganglion provides cholinergic post ganglionic prasysmpathetic fibers to sub maxillary and sub lingual glands.
- Two salivary secretory centers are located in the central nervous system-Brain stem (Medulla), one is concerned with the sub maxillary gland, and the other with the parotid. However stimulation of any one of these two areas causes secretion by both glands.
- The medullary salivary nuclei acts as center of the gustatory reflex, It is also regulated by inhibitory or excitatory signals from higher centers of the brain , particularly from certain regions of the diencephalon and cerebral cortex. Afferent impulses from the mouth, pharynx, and olfactory areas are carried by the trigeminal and glossopharyngeal nerves to the salivary centers.

Regulation of salivary secretion

- Nerve supply
    - *Trigeminal* and *glossopharyngeal* are the sensory/afferent to the salivary centers, located in *medulla oblongata*. The motor/efferent fibers are formed by both sympathetic and parasympathetic nerves of the autonomic nervous system.
    - Stimulation of sympathetic nerves causes vasoconstriction in the salivary glands results in thick salivary secretion rich in mucin and protein.

- *Facial* and *glossopharyngeal* nerves are the parasympathetic efferent fibres to salivary glands. They stimulate cholinergic receptors in the salivary acini, causes vasodilation and copious salivary secretion rich in water and HCO_3, but low content of protein.
- *Bradykinin* a vasodilator polypeptide, formed by parasympathetic stimulation of the salivary glands.
- Anticipation of food intake produces increased parasympathetic response results in increased salivary secretion.



Effects of nerve stimulation

- Stimulation of the parasympathetic efferent fibres results in a copious secretion of saliva, with high mucoprotein content from the sub maxillary glands, and serous or watery secretion from the parotid glands. The amount and composition of the secretion vary with intensity of stimulation. Stimulation of the parasympathetic efferent fibres shows marked vasodilatation which greatly increases the blood flow to the glands.
- Vasodilatation is also brought about by specific vasodilator substance like bradykinin.
- The response of stimulation of sympathetic efferent fibers is variable and differences are found in different glands among different species.In the cat, there is a considerable flow the sub maxillary gland and none from the parotid. Sympathetic stimulation immediately followed by para sympathetic stimulation produce saliva containing much higher proportion of mucus than para sympathetic stimulation alone.

Effect of denervation

- Sectioning of the nerve supply of the salivary glands leads to complete cessation of the salivary secretion except the glands which exhibit the property of spontaneous secretion. Atrophy of the glands develops after denervation accompanied by histological changes by

hypersensitivity to chemical or hormonal agents like epinephrine, nor epinephrine, acetylcholine and pilocarpine. Claude Bernard in 1864 observed increased secretion (paralytic secretion) from the sub maxillary gland after sectioning the chorda tympani by removing the inhibitory action of these gland cells.

REGULATION OF SALIVARY SECRETION

- When food enters the mouth, a copious secretion of saliva takes place by reflex stimulation of the salivary glands through the buccal receptors and secretory centers.
- Normal foodstuffs causes salivary secretion rich in mucin and enzymes to facilitate easy swallowing, while dry or noxious materials causes watery secretion with little mucin.
- The types and volumes of secretion varies with diverse range of stimuli. The chewing of meat cause a flow of saliva from its sub maxillary and sublingual glands but not from parotid gland. While dry bread promoted abundant parotid watery saliva which contains little mucin.

- The saliva produced from one gland differs from that of the other. These may be mixed together in differing proportions. A single gland may be composed of different type of cells and their relative contributions to the saliva produced by that gland may vary. The secretion from a gland composed of only one type of secretory cell also varies in composition at differing rates of secretion .There is a general mechanism by which change in the composition of saliva can occur.
- The secretion of saliva in response to food in the mouth is an unconditioned reflex,and the stimulus producing it is termed as unconditioned stimulus .It is not necessary for food to enter the mouth to provoke a flow of saliva because the sight,smell or even the thought of food may stimulate salivation .The extent to which this psychic secretion occurs varies in different species and it is particularly marked in dog. The nature of the secretion produced varies,the sight of meat evokes a thick,stringy, mucus secretion from the sub maxillary and sublingual glands, while the sight of dry bread results in an abundant secretion of watery parotid saliva.
- Psychic reflexes were termed as conditioned reflexes by Pavlov [1910] because they were dependent on certain conditions for their development known as acquired reflexes.The conditioned reflex is that when a stimulus which in itself does not provoke secretion.
- The optic and olfactory nerves function as afferent/sensory pathways for this reflex activity. Psychic salivary secretion is exhibited by goats and pigs, but it is feeble or absent in horse and sheep. Food is a *natural stimulus* for salivary secretion and salivation during feeding is an *inherited reflex*. When a dog is conditioned by a *neutral stimuli* (bell sound at the time of feeding) for certain period of time, only the bell sound without food is sufficient for increased salivary secretion by way of *acquired reflex*.

FUNCTIONS OF THE SALIVA

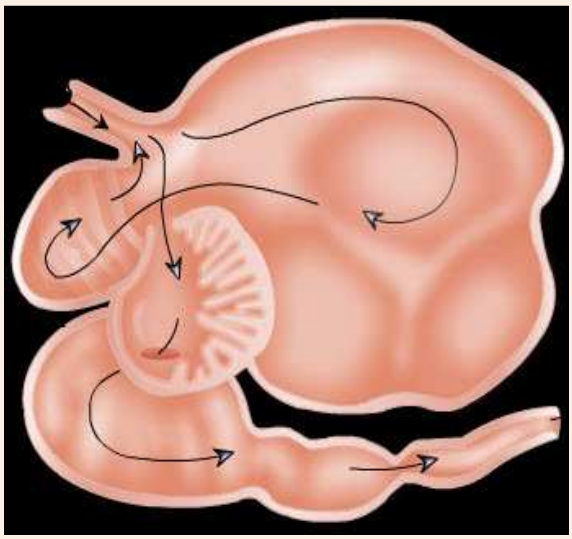
- Saliva has lubricating function that facilitates the mastication and swallowing of ingested food.
- The mucin content provides an adhesive to food to form bolus for swallowing.
- Water soluble components of food get dissolved in saliva which gain access to the taste buds to distinguish different tastes which in turn produces large quantity of saliva.
- Spontaneous secretion of saliva even during sleep has an important protective function by keeping oral mucosa moist and lubricated even in the absence of food.

- Increased secretion during eating also enhanced protection to the mucosa and serous layer by diluting harmful substances.
- Pronounced salivary secretion prior to vomiting protects the mucous membrane during subsequent passage of acid vomitus
- Saliva of pig, rabbit, dog and man contain an amylolytic enzymes *ptyalin or salivary amylase* that hydrolyses alpha 1_4, glucosidic linkages of the statch and leads to the formation of series of dextrin which are finally broken down to maltose (87%) and glucose (13%).
- Salivary amylase is absent in horse, cattle, sheep and goat.
- The optimum hydrogen ion concentration for salivary amylase is around pH 6.2; it is active over a wide range of pH and continues its digestive action in the stomach. Most of the carbohydrates are partly brokendown by salivary amylase in the stomach. In man, amylolytic activity continue up to 30mintues after the entry food into the stomach.
- Many other enzymes like lipase, maltase, peroxidase, mucolytic enzymes are also found in saliva, but they are derived from the bacteria normally residing in the mouth or from the break down of desquamated cells from the buccal mucosa .
- Young animals have another enzyme, *lingual lipase*, for hydrolysing the fat
- Saliva possesses bacteriostatic properties due to the presence of lysozyme to dissolve bacteria
- Saliva is aslo involved in thermoregulation as evoprative heat loss, and is significant in dogs and cats.

Functions of ruminant saliva

- Excellent buffer due to rich contents of bicarbonate and phosphate ions with pH of 8.10
- *In ruminants* , saliva provides a proper media for the bacterial growth and activity in the rumen.
- HCO_3 and PO_4 content of saliva neutralize the Volatile fatty acids produced during microbial fermentation.
- *Urea* is a non protein source supplies nitrogen for the bacterial growth and microbial protein synthesis
- *Phosphates* are utilized for nucleoprotein and phospholipid synthesis.
- Mucin content of saliva has anti-foaming characteristic and reduces foaming tendency of the diet.

MODULE-21: RUMINANT DIGESTION



LEARNING OBJECTIVES

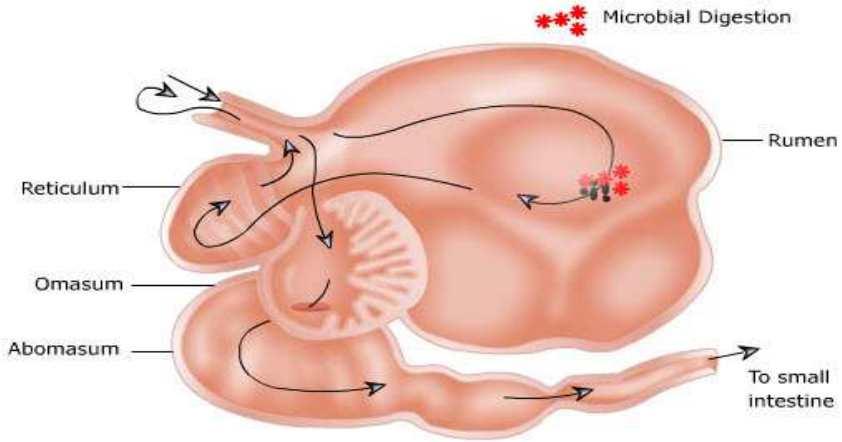
At the end of this module learner are confident to explain the following,

- features of digestion in ruminants,
- mechanical factor of digestion in ruminants,
- nervous control of rumination and
- eructation-its mechanism and significance.

DIGESTION IN RUMINANT STOMACH

Features of digestion in ruminants

- Gastro intestinal tract has colonies of microbes. As the animal grows the type of microbes gets modified according to the type of diet. Host animal and its populations of gut microbes interact. There is a complex interaction between the host animal and microbial population. Host animal does not produce enzymes degrading cellulose and other plant polymers, hence depends on metabolic activities of gut microbes to utilize the fibrous food. Microbial cellulose digestion is a slow process hence they are retained for longer period in the gut.



- Value of microbial products are more to the host (VFA/ Vitamin B Complex)
- They are well absorbed in rumen and lower stomach

- Ammonia and non protein nitrogen ,metabolized by microbes to synthesize good quality microbial protein which is later digested by the host.
- Large fibrous particles are selectively retained in the reticulo-omasal orifice for mechanical breakdown.
- Large quantity of gases produced by metabolism is readily expelled by eructation.
- Large output of saliva provides high buffering action against VFAs and also media for microbial growth.
- Toxic substances are detoxified during fermentation before getting into the small intestine.

Mechanical factor of digestion in ruminants

Mastication	Grinding of the food particles
Deglutition	Swallowing of patially graind food materials (Cud) from the mouth into oesophagus
Rumination	Chewing of the cud. This occurs in four phases
	Regurgitation: Reentry of cud, the liquid portion from the reticulum in to the mouth
	Remastication: Regrinding of the cud
	Reinsalivation: Additional salivary secretion to the bolus
	Redeglutition: Reswallowing of the bolus from the mouth into oesophagus
Eructation	Elimination of CO_2 and CH_4 from GI tract

RUMINATION

- This consists of four processes
  - Regurgitation
  - Re-mastication
  - Reinsalivation
  - Redeglutition

- Rumination is a reflex initiated by mechanical stimulation of mucosal receptors of reticulum, ruminireticular fold, rumen and cardia.
- Particles of the feed influences the time on rumination.
- Feeding of hay diet requires eight hours of rumination. Feeding well chapped smaller pieces of grass reduces time of rumination.
- Rumination starts with third contraction of reticulum viz regurgitation contraction which aids in clearing the cardia for opening.
- Regurgitated bolus is primarily of liquid ingesta.
- From reticulum it is pushed reflexly to opened cardio oesophageal sphincter.
- Lower intra oesophageal pressure at this point causes the pressure gradient to flood the ingesta from the stomach to cardia.
- Forceful inspiration with closed glottis causes decrease in intrapleural and intra oesophageal pressure.

- The presuure gradient is purely achieved by forceful inspiratory effort mediated by the contraction of costal muscles and not by ruminal/ reticular contraction.
- Reverse peristalsis moves the regurgitated bolus from esophagus to mouth. In the mouth, the liquid ingesta from the bolus is squeezed and swallowed. This is followed by reinsalivation and remastication.
- Vagus provides the afferent fibres and the rumination center is located in the brain stem.
- Efferent fibres supply to salivary gland, esophagus, reticulum, muscles for inspiration, mastication and deglutition.

MECHANISM AND NERVOUS CONTROL OF RUMINATION

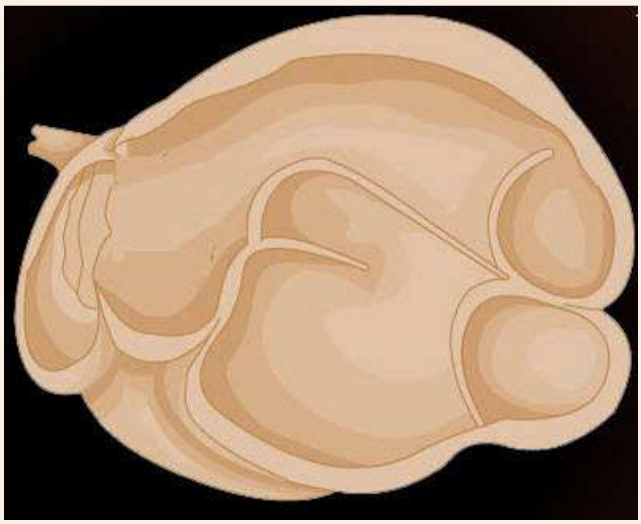
- Rumination
  - It is the process of bringing back the *cud* from the rumen and reticulum to the mouth for further *chewing*.
- Regurgitation
  - The cud that returns to the mouth is mainly the liquid from the rumino reticulam.
  - The regurgitated cud consists of small particulate matter highly mixed with liquid and in advanced stage of fermentation.
  - Freshly eaten forages whose particle size is too great to be suspended in the rumen fluid for extensive maceration and are not regurgitated.
  - The glottis is closed by the elevated soft palate and inspiratory effort with tongue causes a drop in intra-thoracic and intra-oesophageal pressure.
  - The cardia and caudal oesophageal sphincter get open, simultaneously extra-reticular contractions force the cud from the retuculum into the oesophagus due to negative intra-thoracic pressure.
  - An antiperistaltic wave of oesophagus carries the cud to mouth.
- Remastication
  - In the mouth, the liquid portion of the cud is squeezed and swallowed. The remaining solid mass is chewed with slow, regular chewing movements for about 40-seconds.
- Reinsalivation
  - During the process of remastication saliva is added from parotid gland followed by reswallowing
- Redeglutition
  - It is the act of reswallowing of the cud
  - Time spent in rumination varies in different animals and with different rations.
  - Average daily duration for rumination in cattle is 10 hours on hay diet.
  - The proportion of grain and roughage in ration influences rumination time.
  - With low roughage diets or finely ground roughage,total rumination time may be reduced to 3 hours/ day.
  - Peak rumination occurs during afternoon and in the middle of the night.
- Nervous control of rumination
  - Rumination is a reflex act, but it can also be influenced by voluntary control.
  - The receptors for rumination are in the reticular wall, cranial pillars of rumen and rumino-reticular fold.
  - Vagus provides afferent nerve fibres to the rumination centre in the medulla.
  - The efferent nerves are the motor nerves controlling muscles of larynx, oesophagus, and reticulum.
  -

ERUCTATION

- It is the expulsion of fermentation gases like CO_2 and methane that has accumulated in the rumen.

- Receptors for eructation reflex are the *tension receptors* located in the reticulum, cardia, and cranial rumen sac, which are stimulated by accumulation of gas.
- Vagus provides both afferent and efferent fibres. Eructation centre is present in medulla.
- Volume of gas produced in rumen of a dairy cow is ½ to 1 litre/minute. The gas layer is moved cranially by the secondary contraction of the dorsal sac.
- The cardia and lower oesophageal sphincter open, and the gas enter into the relaxed oesophagus. By an antiperistaltic wave in the oesophagus and elevation of soft palate the gas is expelled through the mouth.
- Free gas found at the top of the ingesta in the dorsal rumen sac move cranially and ventrally to the cardia. Cardia remains closed by contact with watery ruminal content. Cardia gets free of ingesta during the contractions of dorsal sac, cranial pillar and caudal pillar.
- Accumulation of gas in the dorsal sac is the primary stimulus of eructation. Increase in the pressure in the dorsal sac of the rumen increases the eructation frequency and amount of gas expelled.
- Eructation occurs once in two ruminoreticular contractions. Inhibition of eructation isobserved during blokage of ruminoreticular contraction. Block of eructation leads to bloating of the rumen, called as TYMPANY which can also happen when gas production is higher than the elimination.
- Rumen distension decreases the lung volume and alters breathing pattern. Mechanical activity of the diaphragm is inhibited. Oils and non absorbable surfactants are used to treat bloat.
- Biphasic increase in pressure in the esophagus is observed during eructation. In the first phase, gas enters the esophagus. In the second phase,sharp increase in pressure accompanied by esophageal contraction.
- High grain feeding and feeding of lush alfalfa results in trapping of gases in the rumen called " feedlot bloat" or "legume bloat". This is due to formation of foam in the ruminoreticulum which traps many small gas bubbles. This can be prevented by reducing feeding of grains or lush pastures.

MODULE-22: RUMEN CONTRACTIONS AND ACT OF VOMITING



LEARNING OBJECTIVES

- At the end of this module the knowledge on
  - the basic mechanism of ruminoreticular motility,
  - control of rumination will be gained and
  - vomiting, its mechanism and the significance in non-ruminant.

MOTILITY IN RUMEN AND RETICULUM

- There are two patterns of motility are seen in reticulo-rumen
  - Primary or *mixing contractions* called primary cycle
  - Secondary or *eructation contractions* known as secondary cycle
- Primary contractions
  - The primary cycle consists of
    - a biphasic contraction of reticulum
    - a caudal moving single contraction of dorsal sac
    - a contraction of ventral sac of rumen.
  - In the primary cycle, rumino-reticular contractions begin from the reticulum as biphasic contractions of which the *first contraction known as mixing contraction.*
  - It then relaxes.
  - The second contraction with strong force is called as evacuation contraction.
  - During feeding, the contractions occur at 75 seconds intervals.
  - Functions of reticular contractions
    - Direct liquid flow caudally to the cranial ruminal sac
    - Direct low density ingesta into the dorsal sac
    - Assist in regurgitation
    - Regulate the flow of ingesta from reticulo-rumen to omasum
- Secondary contraction
  - During second contraction of the reticulum, the rumen starts contracting from the dorsal sac of the rumen as caudally moving contraction towards caudal sac.This is followed by caudally moving ventral sac contractions.

  - Then the dorsal and ventral sacs show cranial moving contraction called the primary cycle. It is involved in mixing of ingesta and separation of large and small particles.
  - Secondary contractions is cranially moving begin from the caudo-dorsal blind sac moving over the dorsal sac.
  - The function of the secondary contraction is to expel the gases out of the rumen.
  - The reticulo-rumen contractions occur at a frequency of 1 to 3/ min. It is increased during eating.
  - During rumination, triphasic contractions occur in the reticulum in which the stronger extra contraction precedes the normal biphasic contraction.
  - The extra contraction in the reticulum as primary wave is referred as regurgitation contraction.
  - During rumination, the primary wave of the ventral sac is absent, but shows strong and prolonged ventral sac movements as secondary wave.
- Nervous control of stomach motility in ruminants
  - Tension receptors and epithelial/mucosal receptors are sensory receptors of fore-stomach stimulated by both mechanical (coarseness of rumen material) and chemical (decrease in pH, increase in VFA) stimuli.
  - Sensory afferents through vagus regulates the major contractions of rumen and reticulum.
  - Total vagotomy abolishes rumen and reticular contractions, rumination, eructation and reticular groove reflex.
  - Vagus is also the motor nerve to abomasum.
  - Sympathetic nerves are inhibitory to ruminant stomach, but their actions are not very important.
  - The intrinsic nerves of reticulo-rumen also cause weak and non cyclic contractions.

RUMINAL CONTRACTIONS

Contractions of rumen

- Cyclic, spontaneous contractions occur in the first three compartments. Frequency of cycle varies with the type of activity. Three types of activity prevails normally during
  - resting
  - eating
  - ruminating
- Resting activity is the contractile activity of the stomach when the animal is neither eating nor ruminating. During rest, cyclic motility is less rhythmic with low frequencies.
- Eating: During feeding, strength of contractions and frequencies of contractions are greatly increased.

Cycle of contraction during feeding follows a pattern

- Cycle of contraction begin at reticulum.
- Two contractions occur during feeding or at rest.
- First contraction is mixing contraction and is weak.
- Second contraction is evacuation contraction which is strong.
- Between these two contractions, a complete relaxation of reticulum occurs.

- Third contraction follows the second strong contraction during rumination. This third contraction is the strongest of the three and is referred to as *extra or regurgitation contrction of the reticulum.*
- Regurgitation of the bolus occurs at the peak of this contraction
- Primary functions of biphasic/triphasic contraction of the reticulum are,
    - To direct the flow of liquid into the cranial sac of rumen.
    - To direct low density digesta into the dorsal sac.
    - To regulate liquid ingesta flow from the ruminoreticulam to the omasum.
    - To assist in regurgitation by flooding the cardia by the ingesta.

Ruminal contractions

- During second contraction of the reticulum, cranial sac of rumen also contracts.
- Cranial pillars start contracting and form a barrier between the cranial and caudal rumen.
- At the end of the contraction of cranial sac the ingesta in the cranial sac are pushed into relaxed reticulum. At this point, vigorous contraction of entire rumen pillars, sac and other parts of rumen occur to mix the ingesta well with rumen fluid.
- This rumen contraction is said to be primary and secondary contractions occuring in the frequency of 6-8 contractions / 5 minute. Cardia is always submerged in the liquid ingesta, gets cleared prior to eructation by contraction of pillars of the rumen and walls of the dorsal sac.
- Bubble of gas transferred from dorsal sac to the cardia cranially. This process occurs during reticular relaxation corresponding to the time of contraction of pillars of the rumen.

NERVOUS CONTROL OF RUMINANL CONTRACTIONS

- Cyclical contractions of the first three compartments which begins early in life continue through out life without any interruption.
- Cyclical contractions are disturbed during pathological conditions affecting firstly rumen (impaction, intestinal intussusceptions) resulting cessation of rhythmic spontaneous contractions.
- Smooth muscle fibres of rumen receives sympathetic and parasympathetic (vagus) innervations which provide rhythmic propulsive motility.
- Rumination is an indicator of good health because of all the nervous pathways and neuromuscular functions of the stomach must be functioning well forexecution of the complex rumination reflex.
- The ruminal stomach is supplied with an extensive network of afferent nerve fibres that carry information from in the stomach to the CNS. Tension receptors are more in reticulum and cranial sac.
- Epithelial receptors are more in reticulum and cranial sac and pillars of rumen.
- Mucosal receptors are rich in abomasum and duodenum and are sensitive to both mechanical and chemical stimuli.

MECHANICAL FACTORS INFLUENCING MOTILITY

- When animal feeds, heavier boli they pass from oesophagus to cranial sac of rumen and are immediately passed into reticulum and omasum through reticulo-omasal opening. Heavier feed are retained less time for microbial fermentation.
- Coarse particles of hay are retained for more hours than finely chopped hay. Fermentative activity and contractions of the pillars of rumen and reticulum result in physical breakdown of particles. Lighter particles (hay/straw) leave the cardia of the oesophagus to the dorsal sac of the rumen. They remain there for several days get soaked in the ruminal fluid for microbial action (rumen bacteria, protozoa and fungi).
- Mixing and powerful crushing activity of contractions of the pillars help in through microbial digestion. This results in further breakdown because of microbial action and physical breakdown of particles during contraction of pillars and walls of the rumen and reticulum. This results in reduction of hay particles. Due to this the density increases and they settle down in the ventral sac. During contraction, the ingesta is pushed from ventral sac to cranial sac through cranial pillar of the rumen. From cranial sac via reticulo-omasal opening they are pushed into omasum and subsequently into abomasum.

Omasal transfer of substances

- Ingesta from reticulum to abomasum is transferred via omasum. Omasum removes water and electrolyte from ingesta and reduces the size of the ingesta. Omasal canal act as pump and forces the ingesta into abomasum due to omasal contraction.
- Contractile activity of the reticulo- omasal and omaso abomasal opening, walls of reticulum, omasal body and abomasum create the pressure gradient which aids in the flow of ingesta. Increase in reticular pressure dilates reticulo-omasal opening to push the ingesta into abomasum. Abomasum does not show rhythmic contraction similar to other three.

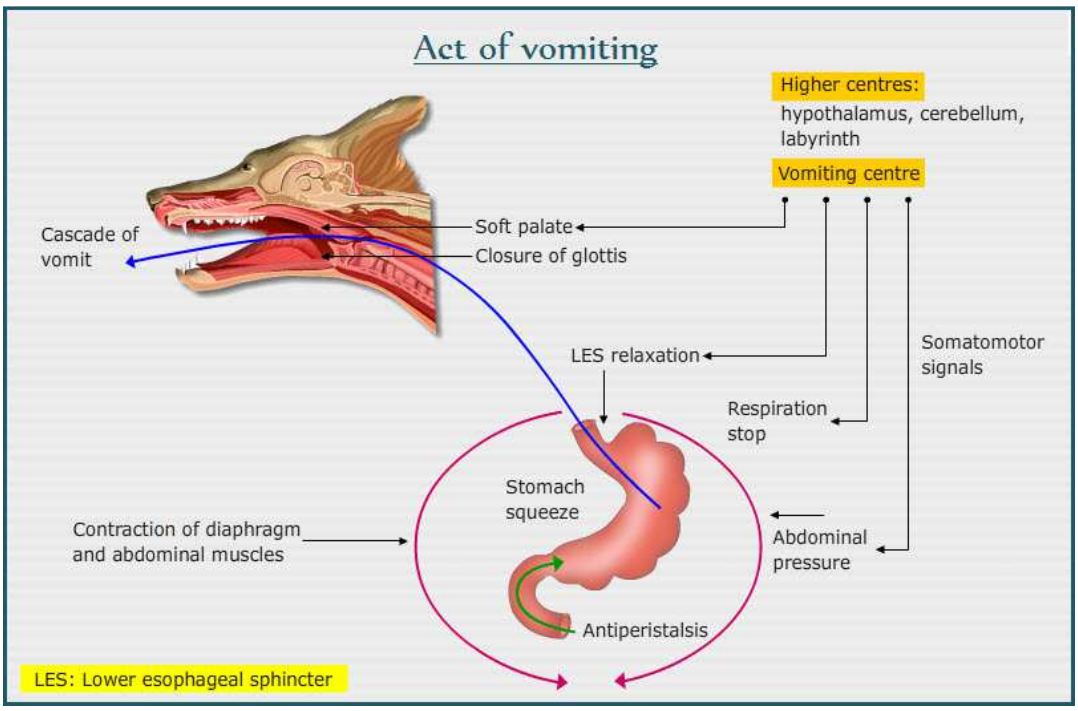
FOREIGN OBJECTS IN RUMEN

- Indigestible objects which are occasionally swallowed without proper mastication pass to the rumen. If the surface of the piece is sharp, it causes inflammatory reaction in the mucosa of reticulum and rumen.
- This leads to infectious spread on pericardium, peritoneum or pleuaral membrane (Traumatic Reticulo Pericarditis).
- High density material pass via cranial sac, sink to the floor of the cranial sac.
- Heavy indigestible objects always remain in the floor of reticulum as reticulo- omasal orifice lie dorsal to the floor of reticulum.

VOMITING (EMESIS)

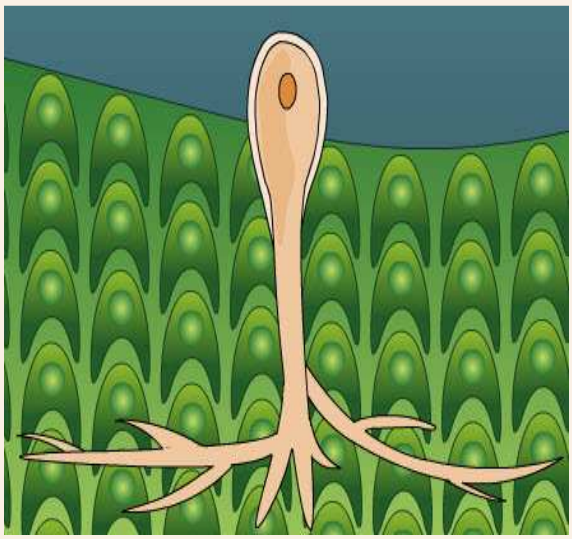
- Vomiting is the spasmodic and forceful ejection of the stomach contents to mouth through dilated oesophagus.
- Vomiting is very rare in horse due to the presence of well developed and powerful cardiac sphincter and distant position of the stomach from the abdominal walls.
- In carnivores and omnivore vomiting is common.
- Herbivores and rodent never vomit.

- The vomiting centre is located in the lateral reticular formation of the medulla which is stimulated by afferent impulses from pharynx, stomach, duodenum, heart and uterus.
- In other non-vomiters, the vomiting center may be absent or rudimentary.
- The cranial nerves 5, 7, 9, 10 and 12 to the upper gastro-intestinal tract and spinal nerves to the diaphragm and abdominal muscles are the efferent nerves involved in the reflex act of vomiting.
- Act of vomiting
    - It is composed of four phases
        - Oesophageal dilatation
        - Gastric emptying
        - Gastric reflux
        - Oesophageal collapse



- After a deep involuntary inspiration, the glottis and the nasopharynx are closed by elevation of soft palate.
- Relaxation of oesophagus, gastro-oesophageal junction and the body of the stomach followed by strong pyloric contraction.
- The intra-thoracic pressure is increased by the contraction of diaphragm and abdominal muscles.
- Additional pressure exerted by the contraction of abdominal muscles and the diaphragm, force the stomach contents into mouth through the relaxed oesophagus.
- Massive antiperistalsis originates from the small intestine which reflux into the stomach.

MODULE-23: RUMEN MICROBES



LEARNING OBJECTIVES

- This module explores,
  - the microbial habitat in rumen,
  - their types,
  - their nature and number and
  - their characteristics and benefits.

RUMEN FERMENTATION

- The unique feature of digestive physiology in the ruminants is the fermentative digestion known as microbial digestion which occurs in the rumen and reticulum. Microbial digestion of food also occurs in the three chambered ruminats, camel, lama and in the stomach of marsupials and in hippopotamus.
- Anaerobic *ciliate protozoa* and non-spore forming *anaerobic bacteria* and *anaerobic fungi* are the major microbes in rumen.Some facultative bacteria are also present. The microbes have a volume of 3.6 % of strained rumen liquor of which 50% is contributed by cilitate protozoa and the remaining 50% by the bacteria.The number of rumen fungi is negligible but their activity is of great importance.
- The bacteria and protozoa grow on the substrates of the food in the fore stomach of the ruminants.

Establishment of bacteria in young ruminants

- Development of bacterial flora in young ruminants occurs at a very early age.
- The nature and rate of development is affected by type of diet fed and degree of isolation of young from adult animals. Under normal conditions, bacteria of adult type establish at about 6th week of age in young animals.
- The ciliates may not become established in young animals unless they are maintained in close contact with animals harbouring them or are inoculated.
- Rumen microbes digest all major carbohydrates of ruminant diet such as cellulose, hemicellulose, pectin and xylose etc. These group of carbohydrates are not digested by mammalian digestive enzymes. The fermentation of cellulose is a slower process and is incomplete in rumen, but the digestion of cellulose and hemi-cellulose is almost completed in the rumen.
- Microbes derive their energy for their growth by fermenting cellulose, hemicellulose, pectin, soluble sugars, starch, the carbohydrates of major plant constituents .

- Rumen microbes are responsible for the digestibility of about 70 to 85% dry matter of the diet in ruminants, results in the production of *volatile fatty acids* (Acetic, Butyric and propionic acids) gases like CO_2, methane (CH_4) , ammonia (NH_3)with small amounts of N_2, H_2, and O_2 and microbial cells.
- From carbohydrates, certain organic isoacids, NH_3 and minerals, the microorganisms can synthesise good quality microbial proteins and B complex vitamins required for their growth and metabolic activities.
- They also hydrolyses lipids, unsaturated fatty acids. Proteins and NPN substances of dietary and salivary sources get degraded by the microbes and the released amino acids, $NH_3$are utilized for microbial growth and microbial protein synthesis. Thus the ruminant animal can be maintained on diets free of essential amino acids.
- In animals maintained in green pasture, the number of bacteria is higher than those fed dry rations. When ciliate protozoa are absent, viable bacterial count increases. Rate and *method of feeding also affects bacterial count*.
- The pH of the rumen liquid ranges from 5.8 to 7.0, which get decreased after feeding.

Rumen as microbial habitat

  - Maintenance of constant temperature of 40°C
  - Maintenance of constant pH of 6 – 7 by the HCO_3 & HPO_4 buffers of saliva
  - Aqueous environment by continuous salivary secretion
  - Continuous supply of substrates for microbial activity
  - Mixing of the ruminal contents by rumino-reticular contractions
  - Continuous removal of the fermentative end products.
- Abomasal secretions are dependent on volume of material flowing into it and pyloric distension. A rise in abomasal pH, short chain fatty acid level stimulate HCl and gastrin secretion. Parasympathetic nerves and gastrin also involved in abomasal secretion.
- Microbes derive their energy for their growth by fermenting cellulose, hemicellulose, pectin, soluble sugars, starch, the carbohydrates of major plant constituents .They result in the production of acetic, butyric propionic and lactic acids and gases like CO_2, methane and H_2.
- Abomasum receives a continuous flow of these materials and functions as a true stomach by secreting gastric juice. More of gastric secretion is from the fundic glands but the pyloric secretion is low. Gastric juice in the fundus region contain pepsin and HCl, has a pH close to 1.0. Pyloric secretions are slightly alkaline with little peptic activity.

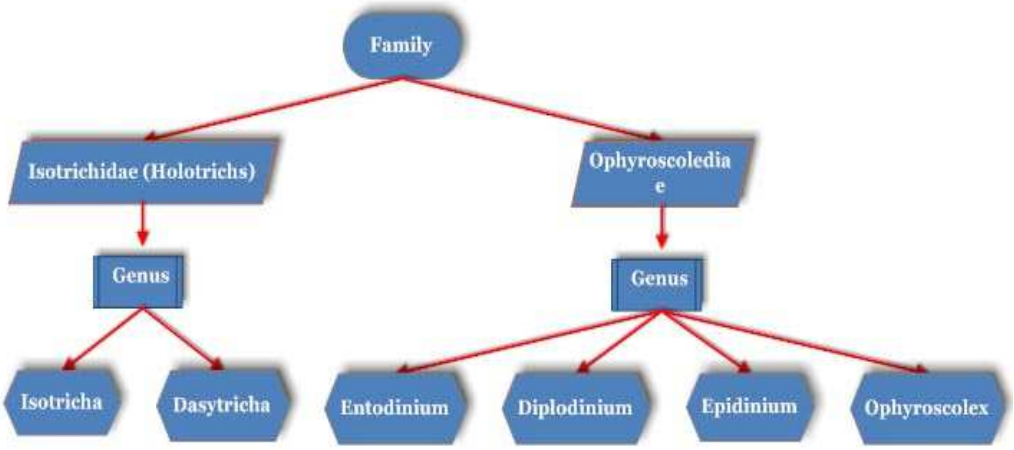
RUMEN BACTERIA

1. Cellulolytic Species	• Fibrobacter succinogenes • Ruminococcus albus & R. flavifaciens • Butyrivibrio fibrisolvens
2 Hemicellulolytic Species	• Butyrivibrio fibrisolvens • Ruminococcus sp. & • Bacteroides ruminicola
3. Pectinolytic Species	• Butyrivibrio fibrisolvens • Bacteroides ruminicola • Succinivibrio dextrinosolvens

4. Amylolytic Species	• Bacteroides amylophilus • Streptococcus bovis • Succinomonas amylolytica
5. Ureolytic Species	• Succinivibrio dextrisolvens • Selenomonas sp. • Butyrivibrio sp • Bacteroides ruminicola
6. Methane producingSpecies	• Methanobrevibacter ruminantium • Methanobacterium formicicum • Methanomicrobium mobile
7. Sugar-Utilising Species	• Treponema bryanti • Lactobacillus sp.
8. Acid-Utilising Species	• Megasphera elsdenii • Selenomonas ruminantium
9. Proteolytic Species	• Bacteroides amylophilus • B. ruminicola • Butyrivibrio fibrisolvens • Streptococcus bovis
10. Ammonia Producing Species	• Bacteroides ruminicola • Megasphera elsdenii
11. Lipid-Utilising Species	• Anaerovibrio lipolytica • Butyrivibrio fibrisolvens • Micrococcus sp.
Bacterial number is about 10^{10} to 10^{11} per ml rumen fluid	

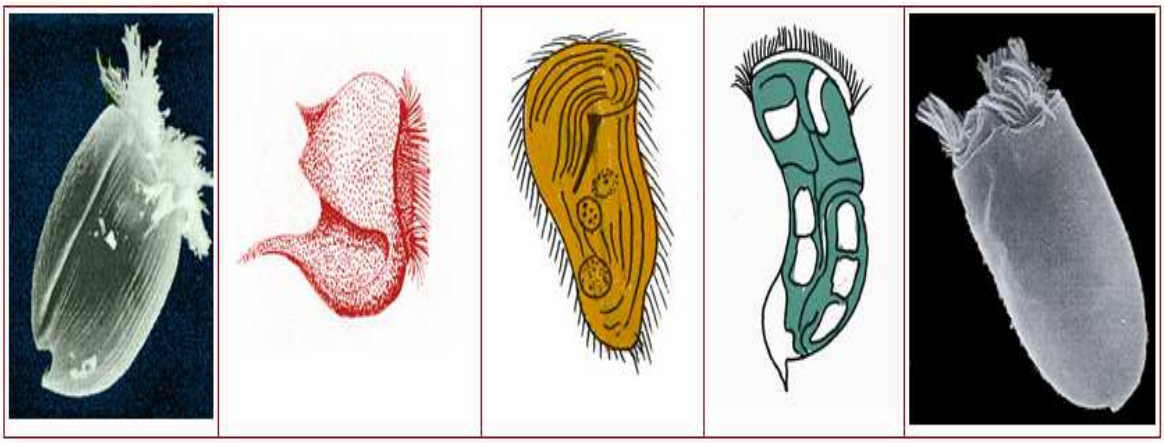
RUMEN PROTOZOA

- Many species of protozoa are active in rumen.
- All protozoa are strictly anaerobes and found only in ruminants.
- Ciliates form the bulk of protozoa and flagellates are present to a very limited extent.
- The ciliates belong to two families.

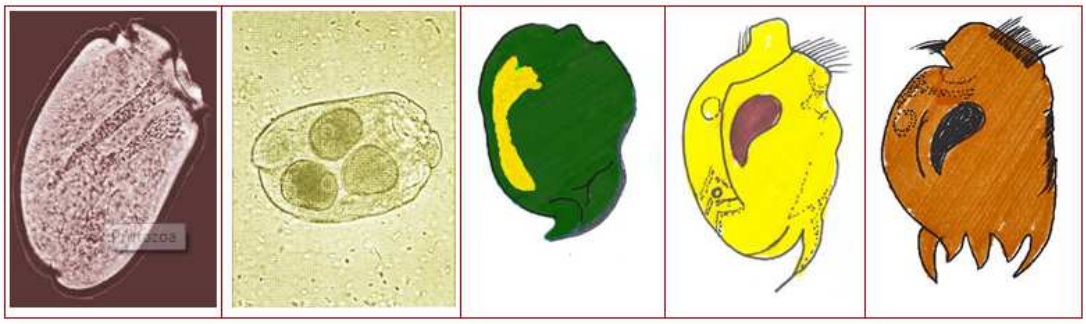


- Many species of micro-organisms are present in the rumen and variations occur in number of certain species with time after feeding, dietary regime and individual differences.
- Protozoal number is affected by pH (<5.5), type and composition of diet, season and frequency of feeding. Highly digestible diets increase their number.
- Protozoa store large quantities of reserve starch, which is used when exogenous energy supply is exhausted. Some facultative bacteria are also present.
- The microbes have a volume of 3.6% of strained rumen fluid and this volume contains 50% ciliate protozoa and 50% bacteria.
- Metabolic activity of bacteria is far greater than protozoa though total volume of small bacteria might be same as ciliate protozoa, which is due to greater surface area provided by the bacteria.
- Protozoal number range between 10^5 and 10^6 per ml fluid. Most of the rumen bacteria and protozoa are strict anaerobes.

Isotrichidae family



Ophyroscoledia family



Role of protozoa

- Increase in digestibility and weight gain is observed in faunated animal than in defaunated animals. But the absence of protozoa does not affect the animal performance.
- The protozoa help to stabilise the rumen fermentation, by ingesting feed particles and storing reserve polysaccharides and they may control the availability of substrates by sustaining uniform fermentation between feedings.
- Mixed protozoal-bacterial protein is better in quality than bacterial protein alone in contributing essential nutrients to ruminant animal.

RUMEN FUNGI

- Anaerobic fungal zoospores are found in rumen. Several are flagellate organisms. *Neocallimastix frontalis, Sphaeromonas communis* and *Piromonas communis* are identified in rumen. Presence of plant cellwall carbohydrates facilitate increase inthe fungal population (Spores) and thus further help in lignocellulosic digestion.
- The fungi are important in the digestion of plant cell wall, their number ranges from 10^5 to 10^7 per gram of rumen contents.
- Feeding of high roughage diet increases their number as shown in animation below.

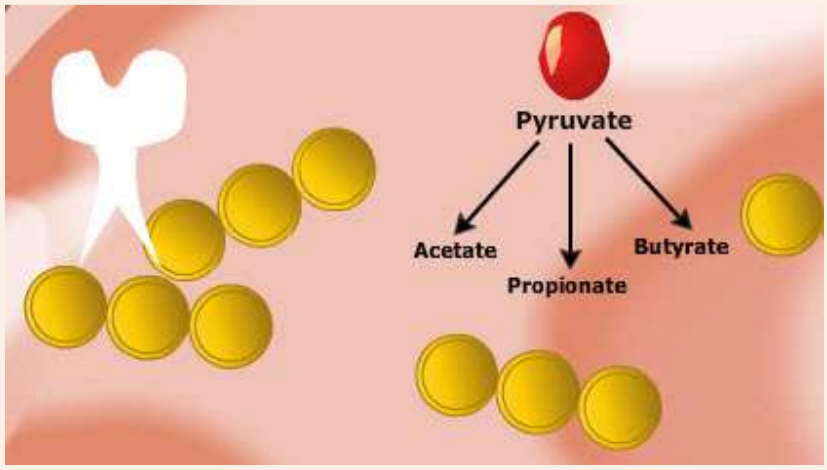
- Fungi play a significant role in adhesion with plant cell wall.
- It is difficult to culture fungi *in-vitro* as they are strict anaerobes.

FUNCTIONS OF RUMEN MICRO-ORGANISMS

- Ferment the dietary carbohydrates including the cell wall constituents of plants, which can not be enzymatically digested by the mammals.
- Produces short chain fatty acids like acetic, butyric, propionic and lactic acids and gases such as CO_2, Methane, and H_2.
- The short chain fatty acids are absorbed from the rumen and are the major energy source of the ruminants.
- By microbial enzymes the dietary proteins are broken down in to ammonia and branched-chain VFAs.
- The short chain fatty acids and NH_3 are utilised for growth of microbes and microbial protein synthesis.
- The microbial proteins are utilised by the host animal as a source of quality protein.

- Triglycerides are hydrolysed to glycerol and fatty acids. The glycerol is converted to propionic acid. The unsaturated fatty acids are hydrogenated .
- Microbes synthesize vitamin K and B complex vitamins.

MODULE-24: DIGESTION OF CARBOHYDRATES IN RUMINANTS



LEARNING OBJECTIVES

This module explains,

- fermentation of carbohydrates in the rumen,
- the pathways leading to the formation of major VFAs,
- role of VFAs with energy production cycles and
- methods to improve ruminal fiber digestibility.

FERMENTATION OF CARBOHYDRATES

- Speed of carbohydrate fermentation varies with their availability and solubility.
- Soluble sugars are rapidly fermented but starch is less rapidly fermented.
- Cellulose and hemi-cellulose are slowly fermented.
- Starch is a glucose polymer with alpha 1, 4 glucose linkage.
- The fructosans are polymer of fructose units with beta linkage.
- In roughages, most of the carbohydrates are structural and found in plant cell wall (cellulose, hemicellulose and pectin).
- Cellulose is beta 1,4 glucose linkage polysaccharide and hemicellulose is polysaccharide composed of beta linked xylose units and few hexoses.
- Pectin, found in cell wall and intracellular is beta linked galacturonan (polysaccharide based on galactose with uronic acid).

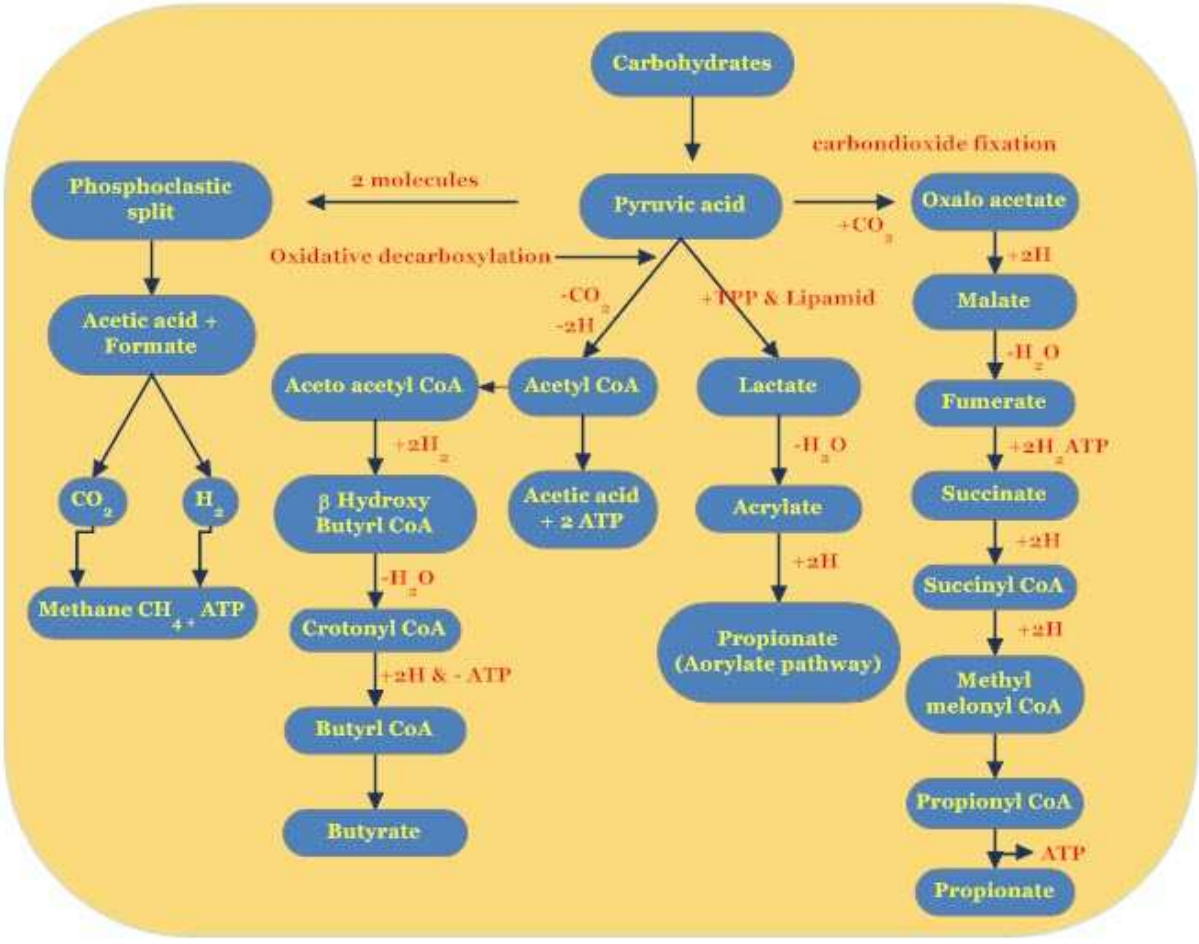
- The alpha linkage can be readily hydrolysed by amylases, whereas the beta linkage can be digested only by microbial enzymes, but not by mammalian enzymes.
- Lignin, a phenolic compound, a constituent of plant cellwall is resistant even to microbial enzymatic digestion. Only a small portion of dietary lignin is digestible due to the action of rumen fungi.
- The lignin reduces the digestibility of cell wall carbohydrates.
- Lignin and cellulose content increase with the age of the plant and temperature.
- Fructosans are more in young grasses.
- Xylose is the most abundant sugar in grasses.

DEGRADATION OF DIFFERENT CARBOHYDRATES

Four steps are involved in the degradation of carbohydrates.

- Adherence
  - This is the most important factor in fibre digestion.
  - Coccoids show preferance to get attached to plant cell wall.
  - Adherance helps the bacteria to be retained for a longer time and fecilitates sustained action.
  - This attachment helps the cell bound enzymes of adhered bacteria to come into intimate contact with substrate and ensure that resulting degradation products are preferrentially available.
- Factors affecting the adherence
  - pH: At a pH below 5.0 adherence decreases and is fecilitated at the pH of 5.5 to 7.8
  - Temperature: Adherence is maximum at 40 ^{o}C.
  - Lignin and lignified portions of the plant cell wall are inhibitory. Soluble cellulose derivatives like carboxy methyl cellulose and methyl cellulose are inhibitory.
  - Certain compounds like phenyl propionic acid increases attachment with plant cell wall.
- Disaggregation
  - The fibrous feeds are soaked in the rumen fluid and are broken to small pieces.
  - Starch granules are easily attacked when ground. Lignin inhibits degradation of structural carbohydrates.
- Extra-cellular degradation
  - The rumen liquor is the best source of bacterial and protozoal enzymes, which are secreted by the microorganisms and released into the rumen liquor.
  - The food substances that are soaked in the rumen fluid are acted by these microbial enzymes and degraded into short chain oligosaccharides and sugars.
- Intra-cellular degradation
  - The oligosaccharide and disaccharides are degraded to simple sugars prior to engulfement by bacteria.
  - The simple sugars enter into the bacterial cell for further metabolism by the microbes.The sugars are metabolised in the microbes by phosphoroclastic cleavage by intracellular enzymes and leads to the formation of pyruvate, phosphoenol pyruvate, short chain fatty acids,CO_2 and methane (CH_4). During this process, 2 NAD are reduced to NADH.
  - Starch is degraded by bacterial enzymes to maltose and some glucose. Maltose is further fermented to glucose. Glycolysis is the major pathway of fermentation of glucose and other monosaccharides. Conversion of hexose to two molecules of pyruvate yields two ATPs, the main energy source for growth and maintenance of bacteria.

- Cellulose is converted to glucose by cellulases and then to pyruvate by a complex process. Cellulose exists in amorphous and crystalline forms; crystalline form is the most difficult to degrade in the rumen. Hemicellulose is also degraded by cellulases similar to cellulose. Pentose sugars are produced from hemicellulose.Pectins are degraded to galacturonic acid, methyl esters of galacturonic acid and finally converted to short chain fatty acids.
- Pyruvate is the intermediate product of carbohydrate metabolism in ruminants which is converted to short chain fatty acids, CO_2 and CH_4. During this process, ATPs are formed which are utilised for microbial growth and multiplication.
- The microbes do not require CO_2 and CH_4 for their growth, whereas the short chain fatty acids are the major energy source for the host animal (ruminants). Concentration of short chain fatty acids in the rumen is within a range of 60 to 120mmol/L of rumen liquor.



FORMATION OF VOLATILE FATTY ACIDS

Acetic acid formation

- Oxydative decarboxylation of pyruvic acid

- Pyruvic acid is converted into Acetyl-CoA by the removal of CO_2 and H_2, in the presence of thiamine pyrophosphate (TPP) and lipomide. The Acetyl-CoA yields acetic acid.

- Phophoroclastic split
  - Two molecules of pyruvic acid yield one molecule of acetic acid and formic acid. The formic acid is converted to CO_2 and H_2. The methanogenic bacteria utilises a portion of this H_2 for CH_4 production, whereas the other portion of H_2 will be utilised for the production of succinate, propionate, butyrate, lactate as well as hydrogenation of unsaturated fatty acids.

$2CH_3\text{-}CO\text{-}COOH \; CH_3\text{-}COOH + HCOOH \;\; CO_2 + H_2 + ATP + CH_4$

Propionic acid formation

- By CO_2 fixation
  - CO_2 combines with pyruvic acid to form oxalo acetic acid (CH_2COOH CO COOH) which is then reduced by hydrogenation (+2H) to malic acid;
  - On removal of one water molecule malic acid is converted to fumaric acid (CHCOOH CHCOOH).
  - Addition of H_2 and one molecuel of ATP to fumaric acid results in the formation of succinic acid
  - Decarboxylation (- CO_2) of succinic acid yields propionic acid ($CH_3\,CH_2\,COOH$).
- By acrylate pathway
  - Pyruvic acid on hydrogenation (+2H) forms lactic acid (CH_3CHOH COOH)
  - On removal of water lactic acid is converted to acrylic acid (CH_2CH COOH) on hydrogenation (+2H) it yields propionate.
- Butyric acid
  - Two molecules of acetyl-CoA condense to form acetoacetyl-CoA and $2H_2$. On reduction (+2H) acetyl-CoA is converted to beta hydroxy butyrl CoA, which by the removal of one molecule of H_2O is converted to crotonyl CoA. Reduction of crotonyl CoA leads to formation of Butyrl CoA along with one molecule of ATP . The butyrl CoA yields butyrate ($CH_3CH_2CH_2COOH$).
  - During the formation of acetate and butyrate NAD is generated.The production of acetate leads to generation of ATP and formation of excess NADH. During the formation of propionate NAD is regenerated with a release of free hydrogen and this H is subsequently used to reduce CO_2 to CH_4 and H_2O.
  - Thus there is a direct relationship between acetate and CH_4 production.when more acetate is produced from pyruvate, more CH_4 is also produced.
  - Likewise, there is a reciprocal relation between propionate production and CH_4 formation; as more pyruvate is converted to propionate production, CH_4 formation is reduced.

END PRODUCTS OF CARBOHYDRATE FERMENTATION

Short Chain Fatty Acids (acetic, propionic and butyric acids).

- Isoacids like valeric, isovaleric, isobutyric and 2-methylbutyric acids.
- Gases produced are CO_2, CH_4 and H_2 .
- The CH_4 accounts for nearly 30 to 40% of the total rumen gas production and CO_2 accounts for 50 to 60%.

- The short chain fatty acids are available asrespective sodium salts. In cattle fed a mixed diet, the proportions of VFAs are acetate 60-65%, propionate 15-20% and butyrate 10-15%. The ratio of A: P: B ranges from 70:20:10 for high forage diets to 60:30:10 for high grain diets. Fluctuation in the VFA concentration occurs on daily ration.
- The normal total VFA content of the rumen ranges from 60 to 120mEq/L. The type of diet, quantity and quality of feed, health of the animal alter the concentrations of individual acids. Total VFA production is higher in animals fed high starch diet than high fibre diet. Propionic acid decrease in animals fed on hay.
- The rumen ecosystem adapts to the type of diet. Lactic acid production is increased in the animals fed with large quantities of easily digestible carbohydrates like starch, sugar or rapidly changing the diet from high roughage to high starch diet.
- Un-dissociated acids do get absorbed from rumen and the rate of metabolism follows similar pattern of butyrate to propionate to acetate.
- Rate of absorption of volatile fatty acids increases when the pH of the rumen ingesta decreases.
- This helps in regulating rumen pH by removal of acidic materials and constant pH is being maintained.
- Loss of VFA results in accumulation of carbon di oxide (as bicarbonate and as CO_2) and concentration of HCO_3 ion increases as that of plasma.
- Lactate production is relatively low. High starch diet if fed too high cause accumulation of high concentration of lactate.
- This follows lactacidemia (high lactate in blood) due to absorption from the rumen.
- Metabolism of propionate in the ruminal epithelium increases as the concentration of lactate.
- Increased lactic acid formation in the rumen reduces rumen pH, leading to lactic acidosis.There is an increase in the number of Streptococcus bovis. This causes suppression of the growth of other types bacteria as they are sensitive to pH, causes rumen dysfunction.Dehydration results due to osmotic imbalance.

Significance of short chain fatty acids in ruminants

- The propionate is glucogenic acids and provides glucose through gluconeogenesis by entering into the Kreb's cycle at the level of succinate.
- Acetate and butyrate contribute the energy needs of the ruminant animal by entering Kreb's cycle as acetyl-CoA. They are ketogenic acids cause the formation of the ketone bodies, acetone, acetoacetic acid and beta hydroxy butyric acid. The ketone bodies serve as energy source in tissues like CNS and heart.
- The acetate is the precursor for milk fat synthesis.

Methods to improve ruminal fiber digestibility:

- Physical and mechanical treatments
  - Soaking with water
  - Grinding and pelleting
  - Increase the fiber intake
  - Increase surface area for microbial degradation
  - Irradiation by Heat/steam treatments

These treatment break the chemical bonds

- Chemical treatments

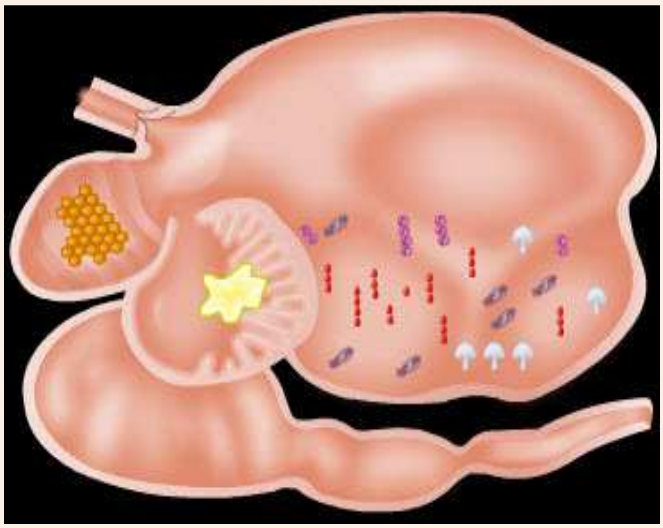
- Alkali treatment
- Treating with Ammonia and urea
- Chlorine compounds – Improves cell wall digestibility
- Ozone - Decreases lignin content and increases cell wall contents
- Sulphur dioxide – Solubilizes hemicellulose
- Alkaline hydrogen peroxide – Solubilizes approximately 50% of lignin

- Treatment with fungi

SIGNIFICANCE OF METHANE PRODUCTION

- The carbohydrate fermenting bacteria produce formate, hydrogen and CO_2.
- The methanogens trasform the H_2 and CO_2 into CH_4.
- The formate is converted into H_2 and CO_2.
- Methanogens preferrentially use formate as substrate for methanogenesis.
- The methanogens act as an electron sink and favoures acetate production.
- This is known as interspecies H_2 transfer that helps to maintain low partial pressure of H_2.
- This favours acetate production from pyruvate and discourages the formation of ethyl alcohol.
- Upon the accumulation of H2,the formation of H_2 from NADH is inhibited and NADH formed during glycolysis is reoxidisde to form alcohol.

MODULE-25: PROTEIN DIGESTION IN RUMINANTS



LEARNING OBJECTIVES

This module eloborates,

- the fate of protein in rumen,
- advantages of non-protein nutrition,
- participation of rumen bacteria in protein degradation and
- end products of nitrogen metabolism and their values.

PROTEIN DIGESTION IN RUMINANTS

- Rumen microbial population have proteolytic activity. The diet of ruminants contains proteins and non-protein nitrogenous (NPN) substances (ammonia, nitrates, urea). Of the

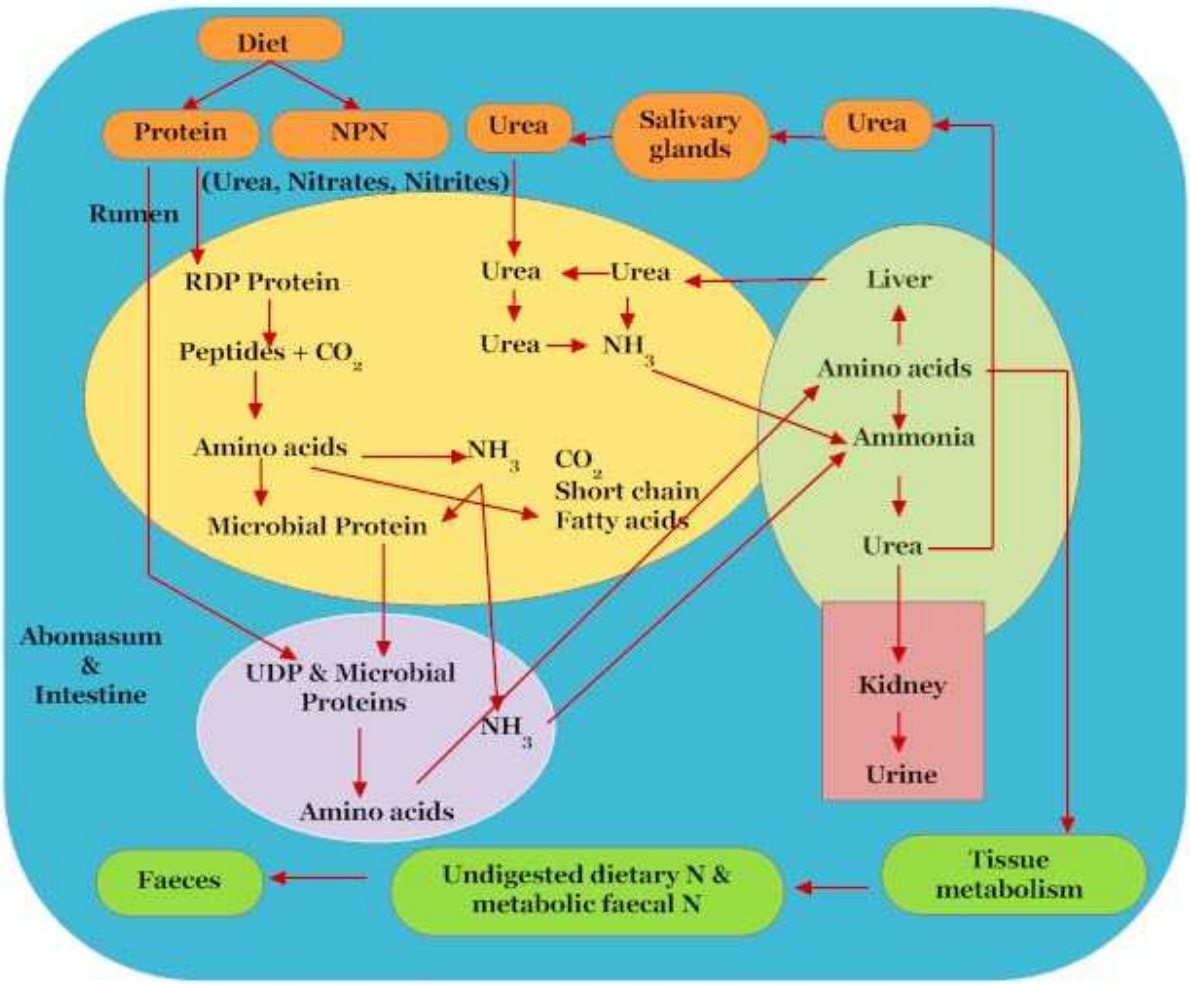
total proteins entering the rumen, 20 to 100% will be degraded to ammonia which are referred as rumen degradable protein (RDP).

- The fraction whch is not degraded by the microbes in the rumen escape and by-pass the microbial digestion, called as rumen undegradable protein (UDP); this fraction reaches the small intestine for enzymatic digestion by the animal.
- When the dietary proteins enter the rumen, the RDP fraction is hydrolysed by extracellular microbial proteases and produce short-chain peptides as end products.
- The peptides are absorbed into the microbial cell bodies. Within the microbial cells, the peptides can either be used for the formation of microbial protein or enter in to VFA pathways for energy production**.**

- The amino acids on deamination yield ammonia and a carbon skeleton; the carbon structures of many amino acids enter directly into various steps of the VFA pathways, leading to the formation of three major VFAs (Acetic, Butyric and Propionic acids).
- Three branched-chain amino acids valine, lucine and isoleucine are involved in the biosynthesis of branched-chain VFAs, the iso acids (isobutyrate, isovalarate and 2–methyl butyrate) which are growth promoters of cellulolytic rumen microbes.

NITROGEN METABOLISM IN RUMEN

- Some proteins are converted to amino acids extra cellularly. The amino acids are absorbed by the microbial cells and are utilised by the bacteria for incorporating into the microbial protein or microbial cell wall or in the nucleic acids.
- The amino acid requirement of the ruminant is met by microbial proteins synthesized by bacteria utilizing dietary protein, NPN substances, urea from saliva,urea through rumen epithelium and un degraded proteins that escapes degradation by rumen microbes (UDP).
- Protein breakdown in the rumen is proportional to solubility, the degree of secondary and tertiary structures, cleavage of disulphide bonds, cross linking between amino acids and also the concentration of ammonia in rumen fluid.
- The merit of conversion of dietary protein to microbial protein,depends upon the composition of food protein.
- Proteolytic activity is well pronounced in Bacteroides sp, Selenomonas sp, Butyrivibrio sp, megasphaera elsdeni and Succinivibrio dextrisolvens.
- Proteolysis results in oligopeptides which undergo degradation to smaller peptides and amino acids. Peptides are more effectively incorporated into bacterial protein whereas amino acids undergo rapid deamination providing NH_3 for bacterial growth. The amino acids on deamination give rise to NH_3, CO_2 and VFA.



AMMONIA NITROGEN IN THE RUMEN FLUID

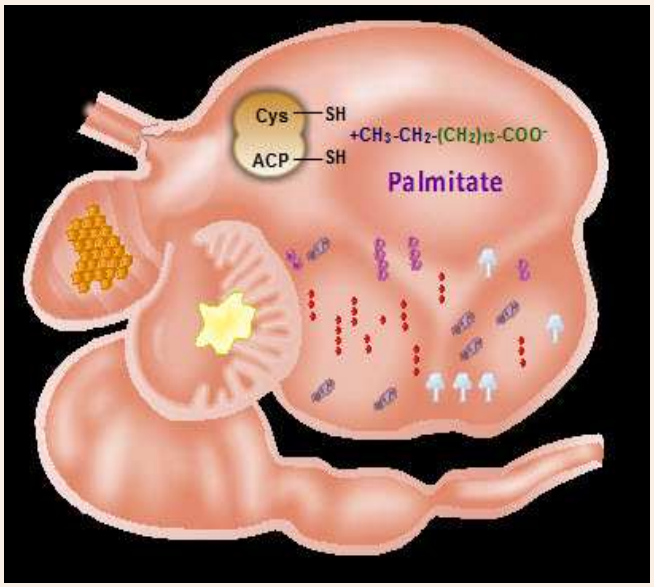
- Ammonia is an important source of N for the growth of rumen microorganisms. About 92% of rumen bacteria utilize ammonia as a nitrogen source.
- Ammonia nitrogen in the rumen fluid varies with diet(2mg to 100 mg/ dl) (normal range is 5-25mg/dl)
- High concentration of starch diet tends to reduce rumen ammonia.
- Urea, frequently used to replace true protein in ruminant ration as a non-protein source is rapidly cleaved in rumen by ureolytic bacteria to ammonia.
- Ammonia toxicity is most frequent sequelae of urea feeding where rumen pH may rise to 7.3. But urea toxicity is tolerable if pH is maintained within 7.0.
- The route by which NH3 utilized depends upon the concentration of NH3 in the rumen fluid. When the NH3 concentration is low, the energy dependent GS/GOGAT systems is utilized whereas when the concentration is high, GDH system is utilized. This is the pathway by which NH3 is converted to microbial protein. When NH3 present in the form of ammonium ion and it requires active transport.

- Glutamine occupies a central role in the nitrogen metabolism of rumen bacteria. As there is no functional TCA cycle (Ketoglutarate is not formed) it is synthesised by reverse TCA cycle.

AMINO ACID BIOSYNTHESIS

- The NH_3 is taken up by the bacteria and fixed to glutamate. Based on the source of carbon atom amino acids are divided into groups like glutamate family (glutamine, proline, arginine).
- Serine family (Serine, glycine, cysteine).
- Aspartate family (Aspartate, lysine, methionine, threonine, isoleucine).
- Pyruvate family (Alanine, isoleucine, leucine, valine).
- Aromatic family (Phenylalanine, tyrosine, tryptophan and histidine).

MODULE-26: DIGESTION OF FAT IN RUMEN



LEARNING OBJECTIVES

At the end of this module knowledge on

- the fate of lipids in the rumen,biosynthesis of saturated fatty acids in the rumen and the synthesis of vitamins in rumen and
- absorptive capablities of fore stomach will be gained.

FATE OF RUMEN LIPIDS

- Dietary lipids occur as structural lipids in forages and storage lipids in oilseeds.
- In forages, less than 50% of total lipids are free fatty acids and the majorityof them are phospholipids.
- In oilseeds, 65-80% of lipids are free fatty acids.
- Rumen microbes hydrolyse the dietary lipids to glycerol and free fatty acids.
- Glycerol is converted to propionic acid by the microbes.
- The metabolism of lipids by microorganisms takes place in 4 steps:
    - Hydrolysis of esterified fatty acids.
    - Biohydrogenation of unsaturated fatty acids.

  - Lipid biosynthesis in the rumen.
  - Metabolism of phytal to phytanic acid.
- Hydrolysis of esterified fat
  - The rumen bacteria produce two hydrolytic enzymes, cell bound esterases and lipases. The triglycerides undergo rapid hydrolysis and give rise to high concentration of unesterified fatty acids especially when concentrates are supplemented in the diet.
- Biohydrogenation of unsaturated fatty acids
  - The bacteria are capable of synthesising long-chain fatty acids and phopholipids from short chain fatty acids.
    - The unsaturated fatty acids like linolenic acid present in grass is rapidly hydrogenated in the rumen yielding cis-trans dienoic acid, cis- trans monoenoic acids and stearic acid. This ability is present in bacteria as well as in protozoa. In a similar way linoleic acid is converted to conjugate dienoic acid, monoenoic acid and finally to stearic acid. The ability for biohydrogenation is greater with organisms like Butyrivibrio fibrisolvens and Ruminococcus albus. The hydrogen for biohydrogenation comes from the metabolic end products of carbohydrate fermentation.
- Lipid biosynthesis of fatty acids
  - Acetate propionate and butyrate serve as the prime molecules for fatty acid synthesis and the malonate is used as extender of carbon chain in the fatty acid.
- Metabolism of phytal to phytanic acid
  - The chlorophyll of the leaf contain the isoprenoid alcohol known as phytal. The phytal is hydrogenated to dihydrophytal which then get oxidized to phytanic acid and incorporated in rumen organisms.

Synthesis of vitamins

- Vitamin K and many B complex vitamins (thiamine, riboflavin, nicotinic acid, pantothenic acid, cyanocobalamine, pyridoxine and biotin) are synthesised in rumen are absorbed from the rumen.
- When a diet is deficient in cobalt, synthesis of vitamin B12 will be inadequate.

Detoxification of toxic substances

- Toxic substances in plants (e.g.oxalates and gossypol) are detoxified in the rumen

ABSORPTION FROM THE FORE-STOMACH

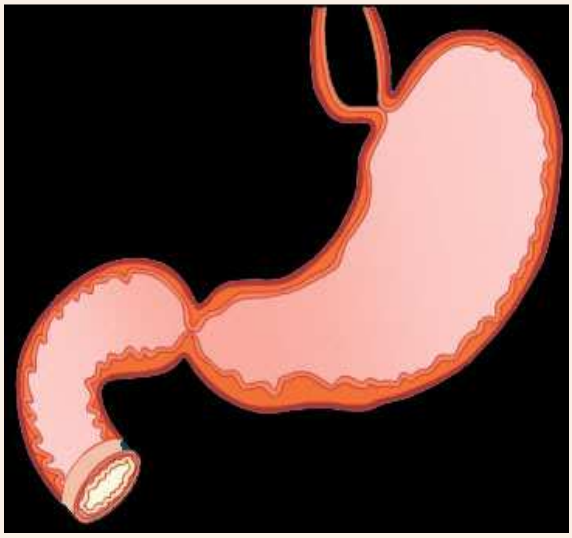
- Absorption in the first three compartments is more rapid than in the abomasum.
- The ruminal papillae increase the absorptive surface area.
- In ruminants, the short chain fatty acids are absorbed from rumen, reticulum, and omasum
- The rate of absorption of acetic, propionic and butyric acids increases as the rumen pH gets decreased.
- The rate of absorption of butyric acid is > propionic > acetic acid.
- Large amounts of lactic acid is produced during excessive feeding of cereals or foods rich in sugars. Lactic acid has a lower rate of absorption (1/10 of VFAs).
- Part of the acetate is absorbed as such, whereas about 30% of the acetate are converted to lactate in the ruminal epithelium.

- All of the butyrate is converted to beta OH butyrate in the ruminal epithelial cells before absorption.
- Propionate is absorbed unchanged.
- Short chain fatty acids and water are absorbed from omasum.
- Ammonia is also absorbed from rumen and its absorption is more rapid at pH 6.5.

Sodium ions are absorbed against the concentration gradient and are actively absorbed.

- Potassium ions are actively secreted in rumen. (Potassium ion is high in rumen compared to plasma and sodium concentration is less in rumen than plasma.)
- Rumen wall is impermeable to phosphate, calcium and magnesium ion. Water is absorbed and hypotonic to plasma.

MODULE-27: GASTRO INTESTINAL MOTILITY



LEARNING OBJECTIVES

- At the end of the module the learner will know the following,
  - the importance of gastric motilityand how it assists in gastric mixing and emptying and
  - the types of intestinal motility and their physiological significance.

GASTRIC MOTILITY

Mechanism of gastric motility

- Function of the stomach is to regulate the flow of food to the small intestine at a controlled rate .
- The proximal region (fundus) near the oesophageal end serves storage function and the distal region serves grinding and sieving function.
- Muscular activity of the fundus shows weak continuous contraction, but capable of relaxation without changing intraluminal pressure as the food enters the stomach . Thus, the fundus serves as a food storage area and not involved much in the mixing of food.
- The body of the stomach serves as a mixing site of the gastric juice with the food.
- The distal stomach, the antrum shows intense slow wave of muscular contractions.

- It acts as a gastric pump and regulates the propulsion of food through the pyloric sphincter into the duodenum.
- The antral contractions also show reverse peristalsis of the pyloric contents, thus delay the passage of solid particles from the stomach.
- The peristaltic waves movement of the stomach begins near the middle of the body of the stomach and runs towards pylorus.
- The sphincter gets closed only when a peristaltic wave reaches it.
- Bile and duodenal contents flow into the stomach through the opened sphincter.
- As gastric digestion proceeds, the peristaltic wave becomes stronger and small quantities of gastric contents are carried into the pylorus.
- As the peristaltic wave comes near the pylorus, the pylorus constricts and blocks the exit of gastric contents excepts those particles smaller than 2mm in size.
- Larger particles are retropelled back into the corpus for further mixing and regrinding.
- Thus, the distal stomach serves to propel the food for mixing and grinding and acts as gastric pump, regulating the rate of gastric emptying (View animation).
- Termination of pyloric peristaltic wave follows as duodenal contraction to carry the ingesta along the duodenum.
- Reverse peristaltic waves are also noticed in the stomach.
- The rate of passage of solids is regulated by the rate at which the solids are broken down to small particles, which in turn is controlled by the motility of distal stomach.
- Greater the motility of antrum, faster the material is broken down.
- Liquids leave the stomach more quickly than solids and it is less dependent on antral motility.
- Emptying of solid material depends on fat content. Low fat meals leaves the stomach in 3 to 4 hours.

Regulation of gastric motility

- The stomach muscle possess a high degree of automaticity.
- Gastric motility is regulated by the vagi and sympathetic nerves.
- Stimulation of vagus increases peristaltic activity of the distal stomach, but suppresses muscular activity of the proximal stomach.
- Anticipation of eating increases vagal activity.
- Entry of food into the stomach provokes peristaltic contractions, through reflex arcs involving, receptors in stomach mucosa, vagus nerves and vagal nucleus in medulla.
- Sympathetic nerves inhibit peristalsis.
- Gastrin increases gastric motility, whereas secretin and CCK suppress gastric motility.

Enterogastric reflex

- The enterogastric reflex involves both extrinsic and intrinsic nervous system and endocrine system.
- This reflex regulates gastric emptying by gastric motility. These reflexes have their afferent receptors in duodenum which are activate by low pH, high osmolality, and presence of fat.
- Vagus is the extrinsic nerves stimulation of which brings about increased gastric motility, whereas inhibition of vagus nerve causes a decrease in gastric motility.
- Intrinsic nerves with their receptors and motor nerves also alter gastric motility through interconnections between stomach and duodenum.

- Osmotic pressure and acidity of duodenal contents are detected by duodenal chemoreceptors which alter the vagal activity. Increase in osmotic pressure and decrease in acidity depresses gastric motility.

Endocrine control of gastric motility

- The endocrine system controlling the gastric motility include secretin, produced by low duodenal pH; CCK released in response to fat in duodenal contents and GIP release in response to carbohydrates. These hormones enter into the blood, carried to stomach and inhibit gastric motility.

Hunger

- Hunger or urge to eat is a discomfort or pain localised in the epigastric region caused by a need for food.
- The feeling of hunger is marked by rhythmic contraction related to motor activity of the stomach which is determined by low blood sugar level.
- Hunger contractions appear even before the stomach has completely emptied itself.
- Hunger sensations mainly act through the hypothalamic hunger center, which determines food intake and the satiety center, which regulates satisfaction of eating.

ABOMASAL MOTILITY

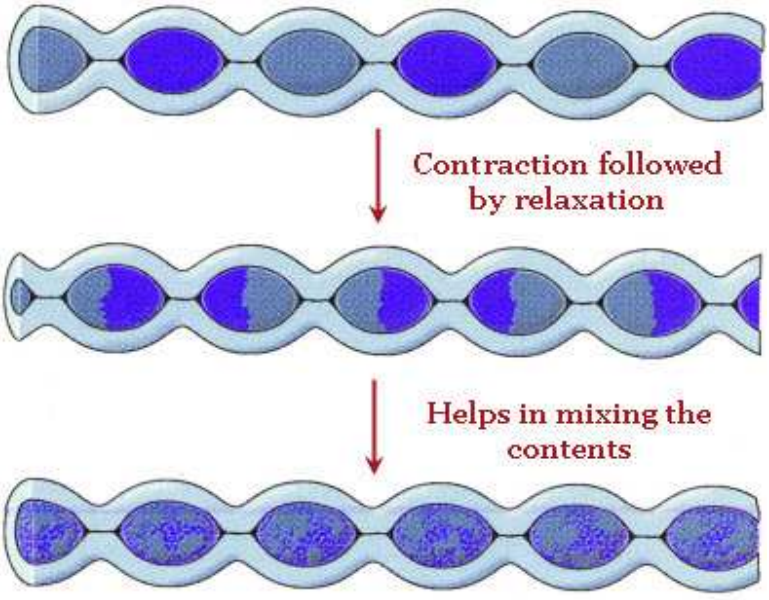
- Does not show cyclical rhythmic activity as that of the three compartments. Emptying of duodenum increases abomasal contractions.
- Enteric hormones (secretin, CCK, pancreozymin) produced by duodenal mucosa decreases abomasal activity and secretion.
- Distension of abomasum has negative effect on other three compartments i.e. distension of abomasum decreases the contractions of other compartments and delays the passage of food from omasum to abomasum.
- Duodenal receptors which are sensitive to composition of abomasal contents play a major role in abomasal emptying. Eg. osmolality of gastric chyme is the primary factor influencing abomasal emptying.
- Abomasum is always filled up each time as omasum contracts. Contraction of abomasum spreads through its walls forces the ingesta to move towards duodenum.
- Abomasum has two compartments pyloric antrum, body of the abomasum. Pyloric antrum of abomasum is more active prior to feeding. Other than vagi, neuromuscular activity in the walls of the abomasum is of importance in maintaining abomasal motility.
- Abomasal motility is ceased when ruminant fed on high concentrate and low roughage diet, due to high VFA production.
- High concentration of butyrate, propionate limits abomasal motility other than the total volume of VFA produced.
- Contraction of abomasum expels abomasal gas back to reticulum.
- Grain feeding increases abomasal methane production. Methane poorly absorbed across the epithelium and is trapped in the abomasum and causes abomasal dysfunction like displacement.

INTESTINAL MOTILITY

- The functions of the intestinal movements are
  - To mix the ingesta with digestive secretions
  - To bring the digested products in contact with intestinal mucous membrane for absorption
  - To move the food masses from place to place in the intestine
  - To expel the residue from the rectum through anus
  - To assist in flow of blood and lymph through vessels of intestinal wall

Motility in small intestine

- Motility of the small intestine occurs in two phases:
  - one during digestive period following food intake and
  - the second phase during interdigestive period, when less food is present in gut.
- During digestive phase, two patterns of motility occurs: propulsive/peristalsis and nonpropulsive/segmentation.



Segmentation - Small intestine

Types of motility during digestive phase

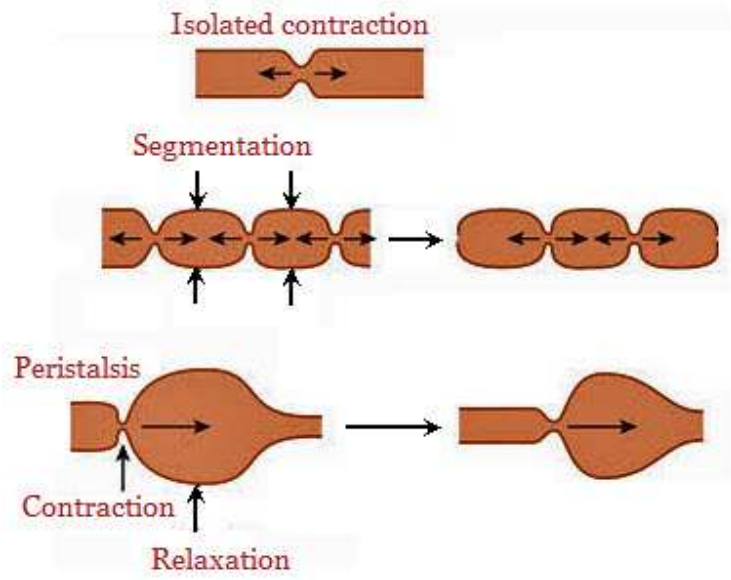
Rhythmic segmentation

- It is produced by contraction of circular muscles.
- During segmentation, a mass of food lying in a length of intestine (3-4cm long) is divided into smaller ovoid pieces by constrictions caused by circular muscles.
- Within few seconds, the constricted portions relax and new areas get constricted.
- Segmentation may be taking place in many different areas of small intestine at the same time.
- The amplitude of segmentation varies and is strong after feeding.

- The effect of segmentation is to mix the food material with digestive secretions and to expose the mixture to the absorptive mucosa.
- Slight onwards movement of ingesta also occurs during segmentation. This segmentation is myogenic in origin and is increased by vagal stimulation and inhibited by epinephrine.
- Segmentation occurs in dogs 17-18 times a minute in the upper jejunum and 12-14 times a minute in the ileum.

Peristalsis

- This is the main mechanism for the onward movements of semisolid intestinal contents.
- It is achieved by creation of a rings of contraction, which pushes the bowel contents to succeeding relaxed areas.
- A stimulus at any point in the intestine can cause contraction above and distension below.
- The wave of contraction and relaxation moves along the intestine as a peristaltic wave, which carries the ingesta towards the lower end of the tract. This movement is neurogenic and is carried out by intrinsic nerves.
- Digestive phase peristalsis in contrast to interdigestive phase peristalsis, pass over only short segments.
- A combination of segmentation and slow wave movement by brief peristalsis ensures complete absorptive process.
- Movements during interdigestive phase: small intestine shows powerful peristaltic contractions at times travels the entire length. These waves are called as migrating myoelectric complexes (MMC) which begins in the duodenum as slow waves leading to spike and muscle contraction.
- It serves to push undigested materials out of the small intestine.



Spiral motility

- Spiral motility exposes the inner contents of the food to the intestinal mucosa for absorbtion.

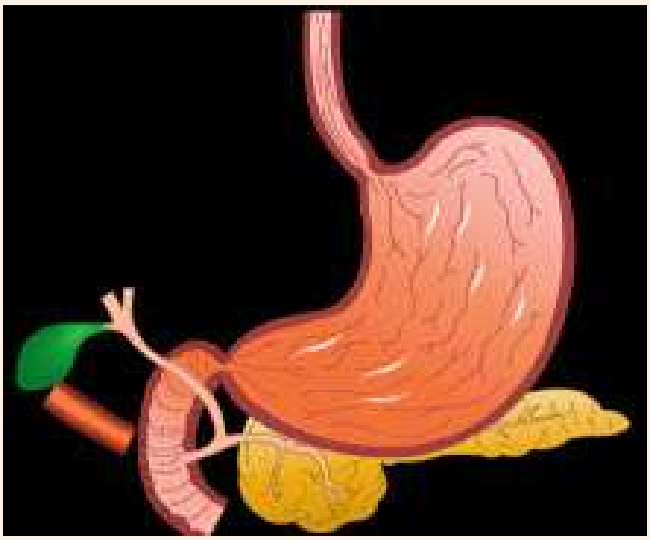
Antiperistalsis

- Peristalsis towards oral direction is designated as antiperistalsis or reverse peristalsis.
- It occurs throughout the digestive tract, but it is not powerful as peristalsis. It helps to
  - Delay the movement of food down the intestinal tract.
  - To ensure adequate mixing.
  - To regurgitate duodenal contents into stomach.
- Ileocaecal sphincter
  - The ileocaecal sphincter is at the junction of small and large intestine to prevent back flow of colon contents into the ileum.
  - It is a circular muscle that remains constricted most of the time.
  - There is a flap of mucosa that acts as one-way valve, which further blocks the back flow of colon contents to ileum.
  - The sphincter relaxes when a wave of peristalsis reaches it, and allows movement of material from ileum to colon.

Motility in large intestine

- These movements are more sluggish than that of small intestine, favours prolonged bacterial attack on cellulose and other substances and also function as a reservoir of faecal matter.
- Sometimes the contractions become powerful leading to effective propulsion.

MODULE-28: GASTRIC SECRETIONS AND REGULATION



LEARNING OBJECTIVES

- At the end of the module the following will be explained,
  - the components of gastric juices and it's celluar origins,
  - the mechanism of HCl secretion,
  - the control of gatric secretion and it's three phases and
  - the functions of

	the components of the gastric juices.

GASTRIC GLANDS

- The glandular mucosa of the stomach has many pores known as *gastric pits*.

- The surface area of the stomach lining the pits is covered with mucous cells which secrete mucus to protect the surface epithelium.
- Secretions of gastric glands
  - Cardiac glands secrete only mucus
  - Parietal/fundic glands secrete HCl and pepsinogen
  - Pyloric glands secrete mucus and gastrin.
- Cell types of fundic glands
  - Fundic gland contain three main types of cells.
    - *Body chief cells or peptic cells* are locate at the base of the gland secrete *proteolytic enzymes (Pepsin and Rennin)*
    - *Parietal or oxyntic cells* are present at the upper third of the gland is the site of *HCl* secretion.
    - *Neck chief cells,* placed near the surface epithelium near the upper part of the gland secret *mucus* and a mucoprotein *"Intrinsic factor"* required for *Vit. B_{12}* absorption from the intestine is necessary for erythropoiesis.

Cells of pyloric glands

- The pyloric glands are structurally similar to the parietal glands
- Has body chief cells and neck chief cells only.
- Parietal cells are absent.
- "G" cells secrets Gastrin (Gut hormone).

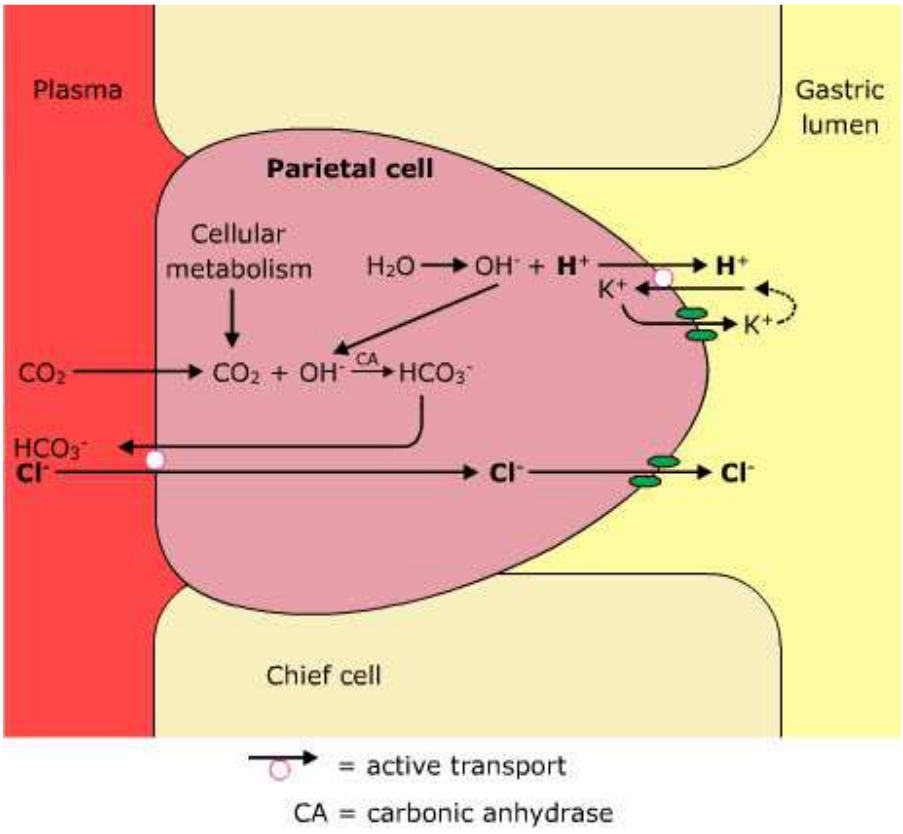
GASTRIC JUICE - COMPOSITION

- It is a colourless fluid, containing HCl, enzymes and mucus. Concentration of HCI in the stomach content varies with the nature of food, stage of digestion, amount of saliva to be neutralized etc

Hydrochloric acid

- It is actively secreted by the parietal/oxyntic cells of the fundic glands involves expenditure of energy, which is derived from the oxidative metabolism.
- Diffusion of CO_2 from the plasma into parietal cells
- Higher concentration of carbonic anhydrase in the parietal cells

- Chemical reaction of CO_2 with cellular water in the presence of "carbonic anhydrase"
- Formation of carbonic acid (H_2CO_3) in the parietal cells
- Dissociation of H_2CO_3 in to HCO_3^- + H^+
- Diffusion of HCO_3^- ions into plasma with a exchange of Cl^- ion from the plasma
- During active digestion, the concentration of HCO_3 increases very high and increases the pH of blood, called as alkaline tide.
- Active exchange of H^+ and K^+ ions between parietal cell and luminal surface of the stomach by of H^+ - K^+ ATPase, enzyme present in the luminal surface of the stomach
- H^+ ion is actively secreted into the luminal surface of the parietal cell with a exchange of K^+ ions into the parietal cell
- Reaction of K^+ ions with Cl^- ion and formation of KCl in the parietal cell
- Release of KCl into the luminal surface
- Reaction of KCl with H^+ ion and the formation of K^+ ion and HCl



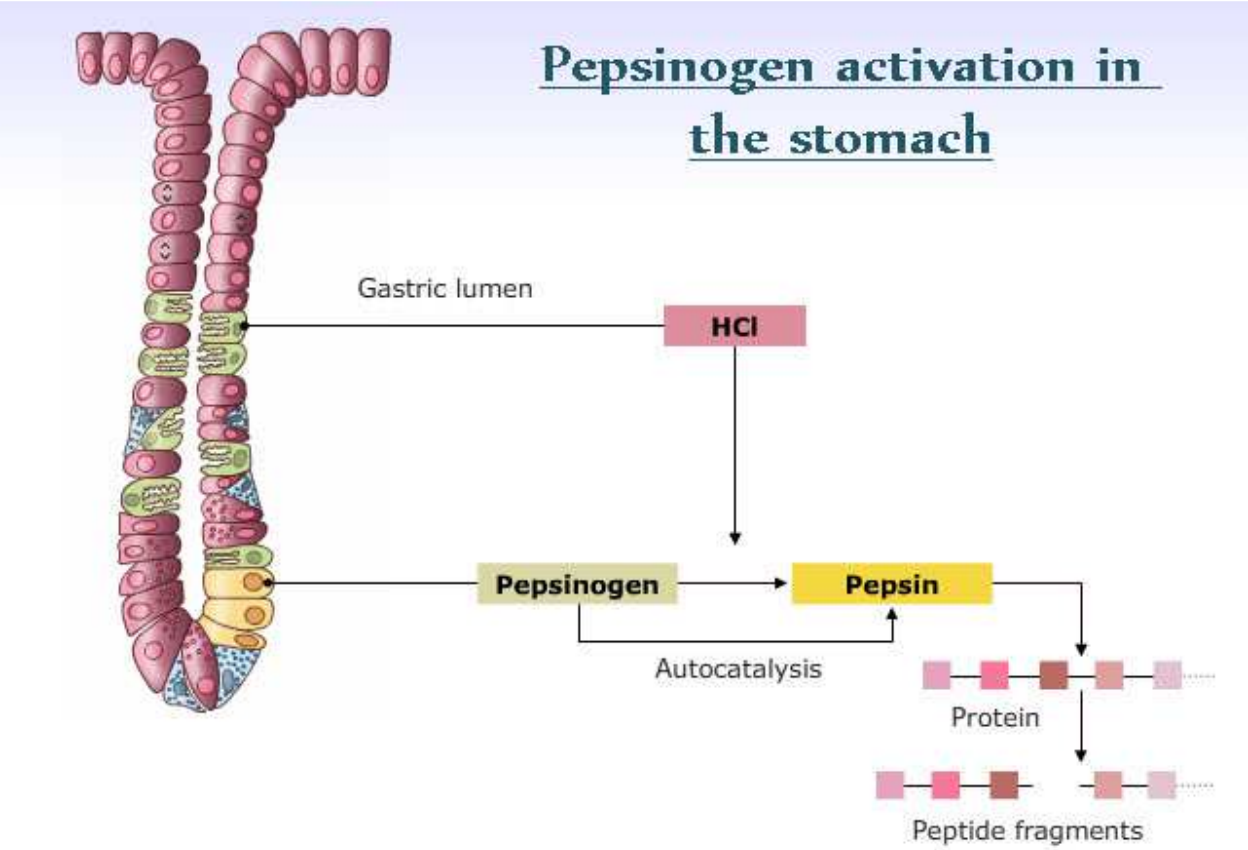
Functions of HCl

- Activates pepsinogen, the inactive form of enzyme into active pepsin, hydrolysing enzyme of protein
- Slight hydrolysis of sucrose
- Also function as antiseptic in the stomach.

GASTRIC ENZYMES

Pepsin

- Proteolytic enzyme, involved in hydrolysis of proteins into polypeptides
- Synthesized as "Pepsinogen" (inactive form of pepsin)
- Pepsinogen is activated by HCl
- Optimum pH of 1.5 – 3 is required for pepsin activity



Rennin (Chymosin, Rennet)

- It is a milk coagulating enzyme, present in young animals(Calf, Lamb, piglet etc.,)
- Secreted as prorennin (inactive form of rennin)
- Prorennin is activated by HCl
- Rennin reacts with casein (Milk protein) in the presence of calcium ion forms calcium paracaseinate to delay the passage of milk through stomach for prolonged action of pepsin on casein

Gastric lipase

- Acts on emulsified fat
- Hydrolysis fat in to fatty acids and glycerol
- Require the optimum pH of 5.5 – 7.5
- Its concentration in carnivores is low
- Absent in birds and ruminants.

CONTROL OF GASTRIC SECRETION

- In man and horse, the secretion of gastric juice is continuous but it is intermittent in dogs and cats.
- The rate of secretion increases during feed intake.
- The secretions of gastric glands are regulated by nervous and chemical mechanisms.
- The gastric secretion takes place in three phases

- Cephalic phase
- Gastric phase
- Intestinal phase

Cephalic phase

- Stimulation of sensory endings in mouth and pharynx or psychic/conditioned reflex due to the impulses originating from the smell, sight of food sensation provoke the cephalic phase of secretion. It contributes about 45% of gastric secretion. The extent of psychic secretion is well developed in dogs and also present in pigs.

Gastric phase

- Entry of food into the stomach causes copious secretion of gastric juice. This contributes about 45% of the gastric secretory response.
- Basically two principle stimuli are responsible for gastric phase
    - Mechanical stimuli
    - Humoral /hormonal stimuli
- Mechanical stimulation
    - Stimulation of intrinsic nerve system
    - Vago-vagal reflex from fundic area
        - Contact of food bolus with the receptors of stomach mucous membrane and the distension of the stomach causes the releases of acetylcholine. Both “G” cells and parietal cells of the gastric glands are stimulated by acetylcholine results in increased secretion of gastrin and HCl respectively.

Hormonal stimulation

- Major portion of gastrin, a gut hormone is produced by the “G” cells of the pyloric glands. Small amount of gastrin is also released from the fundus, duodenum and small intestine. Vagal stimulation during anticipation of eating and distension of stomach stimulates gastrin release and HCl secretion.

- Histamine present in the gastric mucosa is also a powerful stimulator of gastric acid secretion**.**Histamine is secreted by enterochromaffin-like cells (ECL) in parietal mucosa. Acetylcholine produced by parasympathetic nerve endings also stimulate HCl secretion by the parietal cells. Vagus conditioned the parietal cells and potentiates the action of gastrin on HCl secretion. Gastrin stimulates the gastric juice with high HCl content and low pepsin activity. The pepsin release is not hormonally stimulated, but its secretion is enhanced by vagal stimulation.

Intestinal phase

- Accumulation of food in the intestine excite gastric secretion by humoral mechanism due to entry intestinal gastrin and cholecystokinin (CCK) from duodenum into gastric gland through blood stream. This phase contributes about 10% of the gastric secretion.
- Secretion and functions of mucus
    - The gastric mucus is derived from cardiac and pyloric glands, neck chief cells of fundic glands and surface epithelium of stomach. Mucus secretion is independent of secretion of water, HCl and enzymes. Vagus controls the mucus secretion. Mucous functions as a lubricant and protects the to gastric mucous membrane of stomach wall against the highly HCl content.
- Inhibition of gastric secretion
    - Gastric secretion may be inhibited by higher nerve centers via smell or sight of unappetizing food. The autonomic nervous system is involved in marked inhibition of gastric secretion during pain, anger and other emotional states.
    - Accumulation of acid (< pH 2.5) in the stomach inhibits further HCl production by inhibiting gastrin release from pylorus.
    - Entry of fat, sugar, or high acid contents into the duodenum, causes the release of gastric inhibitory polypeptide (GIP), from the duodenum. It is transported to stomach via blood stream and inhibits gastric secretion. Secretin and CCK secretions from the duodenum are also involved in the inhibition of gastric secretion, when acid contents of stomach enter the duodenum.

GASTRIC SECRETION IN RUMINANTS

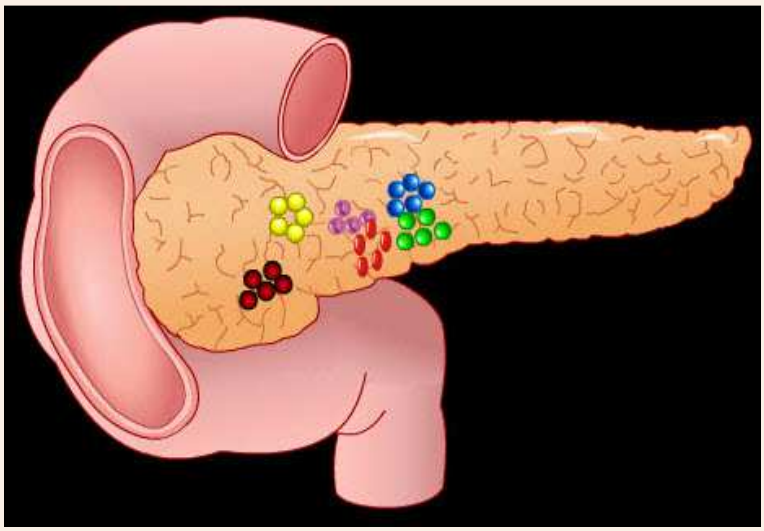
Gastric secretion

- Abomasum is the only part of the ruminal stomach secreting digestive juices.
- Fundic abomasums is rich in HCl, pepsin and in young ruminant rennin- a milk coagulating enzyme is found.
- pH of abomasal juice is approx 1.0 to 1.3.
- Buffered ingesta with volatile fatty acids entering abomasum increases abomasal secretion.
- Regulation of abomasal secretion is compelx. It involves humoral and neural factors.
- Gastrin, G.I tract hormone is secreted by abomasal mucosa in response to cholinergic stimuli.

Absorption

- Ruminal wall is composed of stratified, squamous, highly vascular which facilitates transport of nutrient and other components from rumen to blood and vice versa.
- Water, electrolytes, volatile fatty acids, alkaloids, iodide, sodium cyanide, potassium ion and are readily absorbed from the rumen.
- 60 to 70% of the digestible organic matter is readily absorbed from the rumen and other two compartments.

MODULE-29: SMALL INTESTINE AND PANCREATIC SECRETIONS

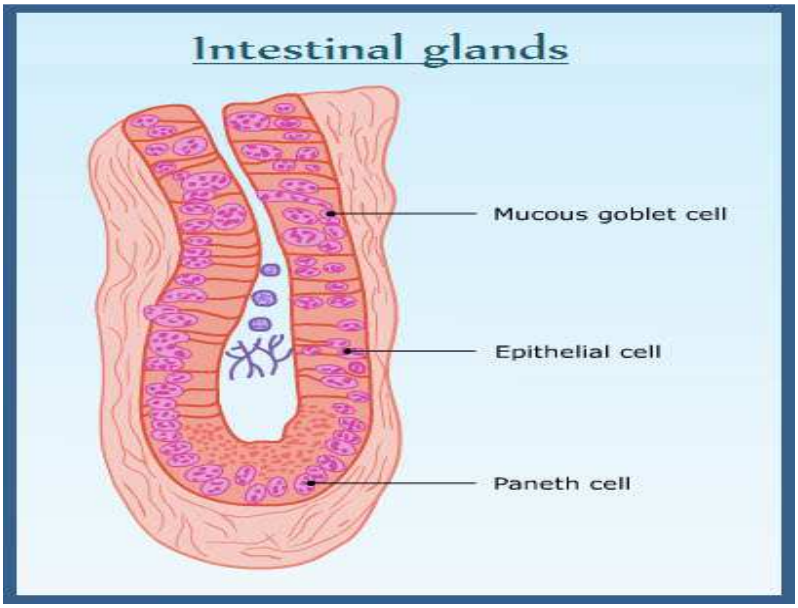


LEARNING OBJECTIVES

- This module explains,
  - the composition of the pancreatic juice and it's nervous and hormonal regulation and
  - composition and functions of intestinal juice.

SMALL INTESTINE

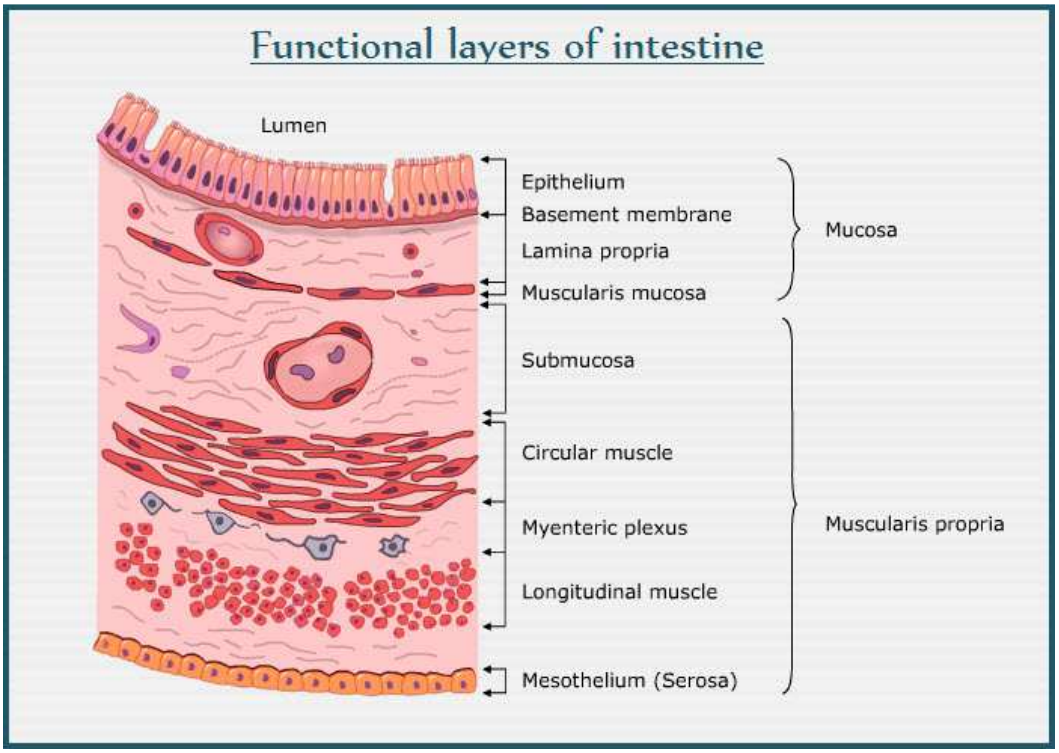
- Small intestine is the predominant site of digestion and absorption. This begins at the end of pyloric sphincter and goes upto large intestine. This is highly coiled structure, has few glands to assist digestion.



- It is composed of 4 layers-serosal (outer), muscular, submucosa and innermost mucosal layer. Mucosa is the functional layer involved in digeston and absorption.
- Has three major portions namely duodenum, jejunum and ileum, with specific functions.

DIGESTIVE ENZYMES IN THE INTESTINAL JUICE

- *Enterokinase:* activates trypsinogen to trypsin.
- *Dipeptidase* : converts proteases to peptones and amino acids.
- *Aminopeptidases* : converts peptones to amino acids.
- *Maltases :* converts maltose to glucose
- *Lactases :* converts lactase to glucose and galactose
- *Sucrase :* coverts sucrose to glucose and fructose
- *Nuclease :* converts nucleic acids to mononucleotides.
- *Nucleotidase and nucleosidases:* convert nucleic acids to purines, pyrimidine bases, phosphoric acid and petose sugars.
- *Lipases :* act on fat liberating fatty acids and glycerol.
- *Amylases :* act on starch resulting in dextrins and maltose.



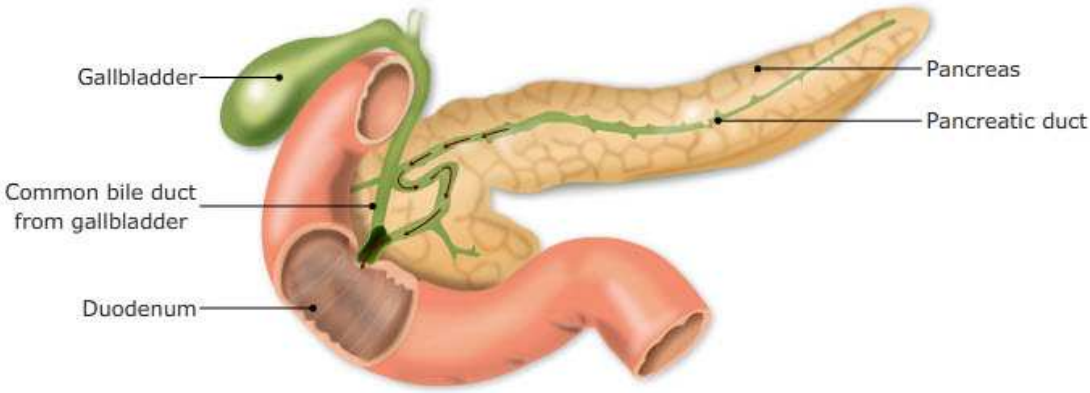
Optimum pH in various compartments of the G.I tract

- Stomach contents: 4.60
- Duodenum : 7.30
- Jejunum : 7.47
- Ileum: 7.55
- Caecum: 7.09
- Colon: 7.09

- Rectum: 6.24

PANCREAS

- Pancreas has both endocrine and exocrine portions. The endocrine portion of the pancreas is made up of *islets of langerhans*, secrets hormones like gulcagon (alpha cells), insulin (beta cells) and somatostatin (delta cells) into the blood. The major portion of pancreas is exocrine in function.



- Secretions
  - The enzymes of the pancreatic juice are synthesized and secreted by the acinar cells.
  - The small duct system contain high level of carbonic anhydrase is responsible for the secretion of water and electrolytes.
  - In most species, the pancreatic duct empties directly into duodenum.
  - Horse, dog and fowl usually have two pancreatic ducts.
  - In sheep, pancreatic duct joins with common bile duct before entering into duodenum.
- Composition of pancreatic juice
  - Pancreatic juice is a clear alkaline fluid made up of the secretions of two separate phases , aqueous and organic phases
  - The aqueous phase causes a higher concentration of HCO_3 and lesser amounts of Cl. The HCO_3 is important for partial neutralisation of the acid chyme from the stomach and maintenance of H^+ ion concentration suitable for digestive activities of pancreatic enzymes.
- The organic phase produces more of pancreatic enzymes. The amount of juice secreted in Horse – 10 to 12, Cattle – 3 to 5, Sheep – 0.5 to 1 L /100kg body weight/day

Pancreatic enzymes

- There are three major groups of enzymes namely *proteases, lipases and amylase* capable of digesting proteins, fats and carbohydrates respecively.
- The following proteolytic enzymes of pancreas are secreted as zymogen granaules or proenzymes.
  - Trypsinogen
  - Chymotrysingon

  - Procarboxy peptidase A and B
  - Proelastase.
- Trypsingoen is converted to active trypsin either by the action of an enzyme enterokinase present in duodenal and jejunal juice or by autocatalysis.
- Chymotrysingoen, procarboxy peptidases and proelstases are activated by trypsin.
- Trypisn, elastase and chymotrypsin are endopeptidases; carbodxy pepetidase is a exopeptidase.
- *Pancreatic amylase* (*amylopsin*) is secreted as an active state. It acts on starch and produces *oligosacharides* and *maltose*. It requires an optimum pH of 6 to 9 and chloride ions for this action.
- *Pancreatic lipase* (*steapsin)* is secreted as an active form.It splits fats into free fatty acids and glycerol. Calcium ions, polypeptides, peptidees and bile salts enhance lipase activity.
- *Cholesterol esterase and phospholipase* act on choleterol esters and phospholipids producing non-esterified fatty acids, cholesterol and lysophospholipid.
- *Ribonuclease* and *deoxy ribonuclease* present in pancreatic juice reduce ribose nucleic acid and deoxy ribonucleic acids to mononucleotides.

REGULATION OF PANCREATIC SECRETION

Nervous regulation

- In three phases
  - Cephalic phase.
  - Gastric phase.
  - Intestinal phase.

- *Cephalic phase:* The cephalic phase of gastric secretions , simultaneously transmit the impulses along the vagi to pancreas; results in pancreatic secretion rich in enzymes, but little water and electrolytes.
- *Gastric phase:* Distension of stomach due to entry of food into the stomach casuse reflex stimulation of pancreatic secretion through vagovagal reflex.
- *Intestinal phase:* Distension of intestine following the entery of food into the small intestine, stimulates the production of pancreatic enzymes and their secretion. This phase is controlled by both intrinsic nerves of the intensine and vagus. Acetylcholine released by the vagal action and local nerve reflexes sensitise the pancreas for the action of secretin and CCK.

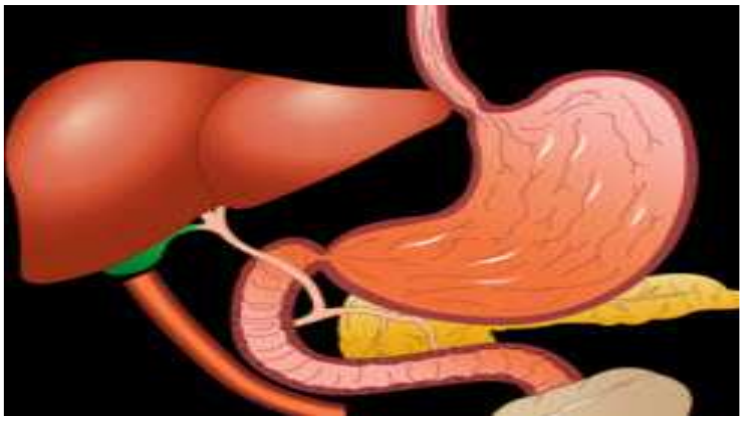
HORMONAL REGULATION

- Secretin and CCK are the important hormones regulate pancreatic secretion.
- Secretin is the gastro intestinal hormone secreted by the duodenal mucosa of small intestine. Presence of acid ingesta, peptides, soaps and amino acids, HCl in the stimulate the release of secretin. When the duodenal pH falls below 4.0 secretin causes increased secretion of thin watery pancreatic juice with high concentration of HCO_3 with

less or no enzyme referred as hydrolytic secretion. This type of HCO_3 release is to neutralise the acidity of duodenal contents. It inhibits gastric secretion.

- Cholecystokinin (CCK) stimulates the pancreatic secretion rich in enzymes referred as ecbolic secretion identical to the action of vagal stimulation. CCK also causes contraction of gall bladder and also delays gastric emptying.
- Vasoactive intestinal polypeptide (VIP) another intestinal hormone, stimulates pancreatic HCO_3 secretion, whereas the pancreatic polypeptide, inhibits pancreatic HCO_3 and enzyme secretions.

MODULE-30: SECRETION OF LIVER AND THEIR REGULATION



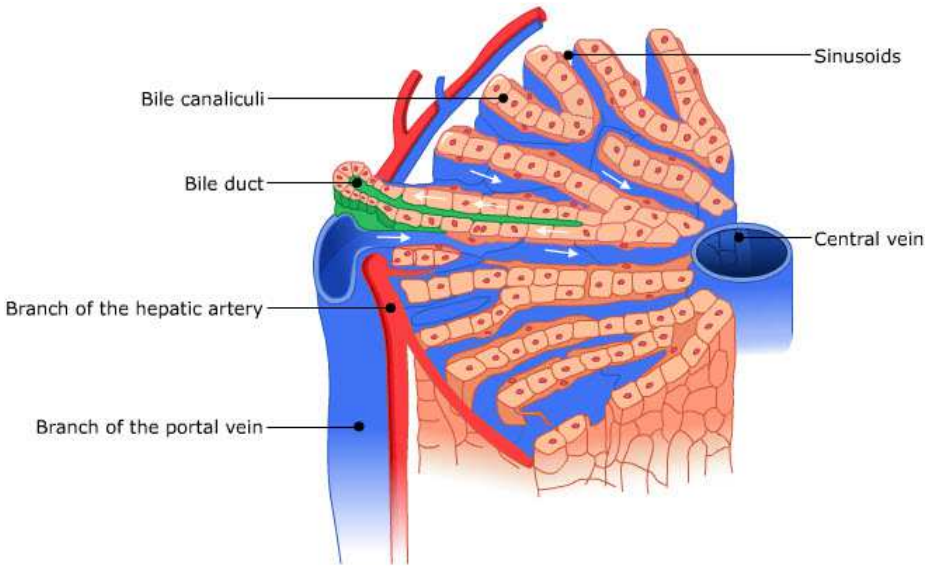
LEARNING OBJECTIVES

This module explains the following,

- the functions of the liver,
- the composition of bile with a brief introduction on jaundice and regulation of bile secretion and it's functions and
- entero-hepatic cycling of bile.

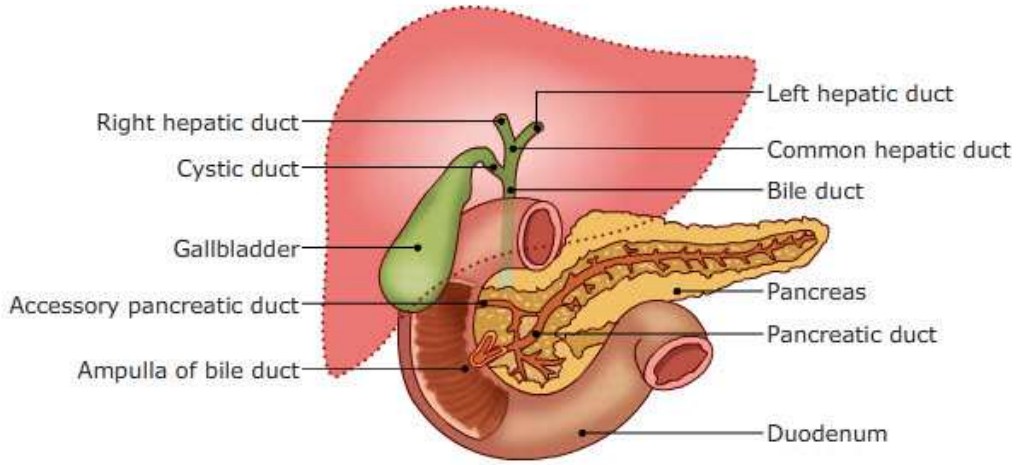
LIVER AND BILE

- Liver is the largest gland in the body and has number of important functions.
  - Secretion of bile
  - Formation and storage of glycogen
  - Regulation of glucose level in the systemic circulation.
  - Deamination of NH_3 and formation of urea.
  - Destruction of uric acid
  - Synthesis of fatty acids from carbohydrates and proteins.



- Phosphorylation of fats, introversion of fatty acids, partial oxidation of fatty acids and formation of ketone bodies.
- Detoxification of poisonous substances brought to it by blood.
- Destruction of old erythrocytes
- Storage and distribution of anti – pernicious anemia factors
- Formation of fibrinogen and other plasma factors.
- Formation of prothrombin
- Destruction of estrogen.
- Metabolism of hormones.

- Secretion of bile
  - Secretion of bile as a source of bile acids which is necessary for fat digestion and absorption in the jejunum
  - Bile pigments provides the excretory route for certain endogenous metabolites and drugs.
  - Additional buffer to neutralize H^+ ions in the proximal duodenum.
- Gall Bladder
  - It is a storage organ of the bile for continuous secretion of bile. The walls of the gall bladder secrete mucin and absorb H_2O from the bile inorder to concentrate the bile.
  - The horses, rat, deer, elk, moose, giraffe, camel, elephant and pigeon do not possess a gall bladder.
  - Removal of the gall bladder, does not result in any great physiological disturbance.
- Bile
  - Bile is both a secretive and an excretive substances. It plays an important role in the solubilization and absorption of fat.



PHYSICAL PROPERTIES OF BILE

- Bile from hepatic duct is a viscid, greenish or greenish yellow with golden tinge liquid with a bitter taste.
- The compositions varies upon whether the bile is obtained from gall bladder or from the ducts of the liver.
- The constituents of the bile are
  - Bile pigments
  - Bile acids or salts of bile acids
  - Cholesterol
  - Lecithine
  - Mucin and alike substances
  - Fat, soap, urea
  - Inorganic substances like Na, K, Ca, Cl and HCO_3

Composition of bile

	Hepatic bile	Gall bladder bile
Water	97.5 gm%	92 gm %
Bile salts	1.1 gm%	6 gm %
Bilirubin	0.04 gm%	0.3 gm %
Cholesterol	0.1 gm%	0.3 - 0.9 gm %
Fatty acids	0.12 gm%	0.3 – 1.2 gm %
Lecithin	0.04 gm%	0.3 gm %
Na+	145 mEq /1	130 mEq/1
K+	5 mEq/1	12 mEq/1
Ca+	5 mEq /1	23 mEq/1
CI	100 mEq/1	25 mEq/1
HCO_3	28 mEq/1	10 mEq/1

pH	7.1-7.3	6.9 – 7.7

Amount of bile secreted in different species

Horse	250-300 ml/hour
Ox	98-111 ml/hour
Sheep	7-154 ml/hour
Pig	70-140 ml/hour
Dog	7-14 ml /hour
Man (Adult)	300-1200 ml/hour

- The volume of bile secretion depends on many factors
  - Blood flow to the liver,
  - Digestive state of the animal
  - Composition of food.

Methods of bile collection

- Bile can be obtained by gall bladder fistula from gall bladder.
- Direct connection of the common bile duct also helps in getting the bile from animals.

FUNCTIONAL ANATOMY OF BILIARY SECRETION

- The basic functional unit of the liver is the liver lobule, which is constructed around a central vein.The lobule is composed of many hepatic cellular plates that radiate away from the central vein. Between the cells in the hepatic plates is the bile canaliculi that collect the bile secreted by the liver cells and empty them into terminal bile ducts. These bile ducts join with the cystic duct to form the common bile duct in the species which have gall bladder.
- The common bile duct terminate as a slight swelling in the duodenal mucosa as bile duct papilla which is controlled by a sphincter muscle known as *Sphincter of oddi*.This sphincter is well defined in carnivores but not so prominent in herbivores.
- In human, dogs, cats and horses the bile and pancreatic ducts open closely together into the duodenum. In sheep and goats the pancreatic duct empties directly into the common bile duct so that the mixture of bile and pancreatic juice enters the duodenum. In pigs and cows bile and pancreatic ducts lie some distance apart.
- Bile is secreted continuously in all species.Bile is stored in the gall bladder in animals which feed once or twice daily. Gall bladder is the storage organ of bile and has a ability to concentrate the bile up to 20-30 times during inter digestive periods by the absorption of NaCl or $NaHCO_3$ and H_2O . So gall bladder bile differs in composition as that of hepatic bile. The degree of concentrating ability varies with species of animals. In ruminants and pigs only slight absorption of water takes place. More over these animals are continuous feeders hence need continuous flow of dilute bile in to the intestine. In horse, gall bladder is absent, hence large and continuous flow of hepatic bile to the duodenum.

Emptying of the gall bladder

- Basic conditions are necessary for the gall bladder to empty.

  - Relaxation of sphincter of oddi allows the bile to flow from the common bile duct into the duodenum
  - Contraction of the gall bladder provides the force required to move the bile along the common duct. Meal rich in fat, shows both of these effects.
- Fat as well as partially digested proteins in the food entering into the small intestine causes the release of cholecystokinin from the intestinal mucosa of upper small intestine. It is absorbed through circulation passes to gall bladder, causes contraction of the gall bladder muscle. This provides the pressure that forces bile towards the duodenum.
- Vagal stimulation associated with the cephalic phase of gastric secretion causes an additional weak contraction of the gall bladder.
- Neurogenic or a myogenic reflex from the gall bladder to the sphincter of oddi inhibits sphincter of oddi. Chloecytokinin may also have a direct inhibitory action on the sphincter.
- Presence of food in the duodenum causes increased peristalsis in the duodenal wall. When peristaltic wave travels towards the sphincter of oddi momentary relaxation referred as "Receptive Relaxation" of sphincter of oddi occurs before the peristaltic contraction of sphincter of oddi adjacent to intestinal wall.

Formation of bile

- Bile secretion involves two components.
- First is the *bile salt dependant* secretary flow.
- Bile is a major route of cholesterol excretion. Bile salts are synthesized in the liver from cholesterol, have cyclo pentano per hydro phenanthrene ring structure.
- The primary bile acids in most species are cholic acid and chenodeoxy cholic acid conjugated with either taurine or glycine which lower the pH of bile salt .
- The secondary bile acids (deoxy cholic acid and litho colic acid) are derived from bacterial de hydroxylation in the colon.
- the pH of intestinal content is close to pH 6, leads to failure of conjugation and accumulation of large amounts of un-ionized, hydrogenated bile acids.
  - Limited H_2O solubility of these acids leads to absorbtion by a passive diffusion. Thus conjugation of bile salt provides concentration of ionized bile salts which is required for normal fat absorption.
  - The bile salts are actively transported from the hepatocyte into the bile canaliculi. Bile is iso osmotic because of passive flow of cations and H_2O. Bile salts are re circulated from the intestine(Entero hepatic circulation) which inturn stimulates hepatic bile flow.
- Second component of the bile secretion is the *bile salt independent* flow and involve the ductular epithelium. Active transport of Na^+ accompanied by Cl, HCO_3, and H_2O provides buffer to the duodenum. This phase is under the control of secretin, cholecystokinin, and gastrin to increase HCO_3 concentration and bile flow.

BILE ACIDS AND SALTS OF BILE ACIDS

- Bile salts are Na or K salts of glycocholic and tauro cholic acids formed in the liver and found in the bile. In carnivores, man, sheep, goat and birds, tauro cholic acid predominate. In pig and rabbit , it is the glyco cholic acid predominates. Glycocholic acid is a compound of glycine (simplest amino acid), cholic acid (derived from cholesterol).

Taurocholic acid is a compound of taurine (sulphur containing nitrogenous substance derived from an essential amino acid cystine) and cholic acid. Chief bile acid is cholic acid. Others are deoxycholic, lithocholic and chenodeoxy cholic acid .

Functions of bile salts

- They have powerful surface tension reducing ability (Detergent effect) and thereby act as a strong emulsifying agent and assist in fat emulsification. This allows the agitation in the intestinal tract to break the fat globules into minute sizes.
- Activates pancreatic lipase.
- Activates cholesterol esterases
- They help in the absorption of fatty acids, monoglycerides, cholesterol and other lipids from the intestinal tract, by forming minute complexes of lipids which are called MICELLES . These are highly soluble because of the electrical charges of the bile salts. The lipids are "ferried" in this form to mucosa where they are absorbed. Without the bile salts about 40% of the lipids are lost into the stools.
- Helps in absorption of fat soluble vitamins

Enterohepatic circulation

- Approximately 94% of the bile salts are reabsorbed by an active transport process through the intestinal mucosa in the dorsal ileum.
- On reaching the liver by portal circulation almost entire bile salts are absorbed by hepatocyte from venous sinusoids and recirculated into the bile so that these salts can circulate about 18 times before being voided in the feces.
- Loss of bile salts are replaced by the liver. This recirculation of the bile salts is called entero hepatic circulation.
- The amount of bile salt in the enterohepatic circulation alters the bile secretion.

BILE PIGMENTS

- The bile pigments are the waste products of the haemoglobin (Hb) breakdown give yellow/green colour to bile. *Bilirubin,*the brownish yellow pigment is readily oxidized into *biliverdin* which gives green colour to the bile of the herbivores.
- Bilirubin is produced from Hb with destruction of RBC in the reticuloendothelial system (RES) of the body (eg. spleen, liver and bone marrow).
- It is normally present in the blood in low concentration. Bilirubin is highly soluble in all the cell membranes. But high concertration of bilirubin is very toxic, needs excretion through bile,one of the very important functions of the liver.
- When the RBC have lived out their life span of 120 days they have become too fragile to exist longer in the circulatory system, their cell membrane rupture and release Hb.
- Released Hb undergo phagocytosis by tissue macrophages (RES). Haemoglobin is split into haem and globin of which globin is readily transported to protein pool of the body for reutilization.
- Haem ring gives (a) free iron that is transported in the blood by transferrin and (b) 4 pyrol nuclei , the substrate for bile pigments. The first pigment formed is biliverdin and is rapidly reduced to free bilirubin , which is gradually released into the plasma.
- Free billirubin immediately and strongly combine with the plasm albumin and transported through the blood and the interstitial fluid. Bilirubin bound with plasma protein called as "Free bilirubin" is insoluble that can't pass through kidney.

- Within hours, this free billirubin pass through the hepatic cell membrane, where it is released from the plasma albumin, but instantaneously combines with one of the 2 protein (Y and Z proteins) inside the liver cell which traps the bilirubin.
- Soon after bilirubin is released from these proteins it conjugated with glucuronic acid, phosphate and sulphate to form bilirubin glucuronide and 10% as bilirubin phosphate and rest as billirubin sulphate respectiely.
- These conjucated forms are excreted by active transport into the bile canaliculi.

Formation and fate of urobilinogen

- In the intestine, ½ the conjugated bilirubin is converted into highly soluble urobilinogen and stercobilinogen by bacterial action.
- Some of the uro bilinogen gets absorbed from intestinal mucosa into the blood.
- Most of them re-excreted into the gut by the liver
- After exposure to air the urobilinogen is oxidized to urobilin and excreted through urine, giving yellow colur to urine.
- Stercobilinogen is converted into stercobilin. and excreted through the feces.

JAUNDICE

- lt is much a symptom than a disease.
- It gives a yellowish tint to the body tissues, including the skin and deep tissues.
- Usual cause of jaundice is the presence of large quantities of bilirubin in the ECF, either free or conjugated form.
- Normal plasma concentration of bilirubin (both free and conjugated) is about 0.5 mg/100 ml of plasma.
- In abnormal conditions, it can rise up to 40mg/100 ml. Skin may appear yellow when the concentration of bilirubin rises to about 1.5 mg/100 ml.
- Physiological classification of jaundice:
    - Haemolytic jaundice
    - Obstructive jaundice

Hemolytic jaundice

- It is due to excessive destruction of RBC with rapid release of bilirubin in to the blood. In this condition excretory function of the liver is not impaired
- Plasma concentration of both free and conjugated form of bilirubin are elevated much above the normal level .
- The rate of formation of urobilinogen in the intestine is also increased.

Obstructive jaundice

- Obstruction in the bile ducts or damage to the liver cells leads to improper excretion of the usual amounts of bilirubin in to the G.I tract.
- In this condition the rate of bilirubin formation is normal.
- The free bilirubin gets into the liver cell for conjugation. Rupture of thebile canaliculi directly empties conjugated bilirubin into the lymph, hence most of the plasma bilirubin are of conjugated type rather than free bilirubin.

DIAGNOSTICS DIFFERENCES

Differntial diagnosis of haemolytic and obstructive jaundice

- Vanden Berg test is the specific test to differentiate free and conjugated bilirubin in the plasma. In the case of *direct Vandenberg reaction* Vandenberg reagent shows immediate reaction with conjugated bilirubin.
- Free bilirubin in the presence of alcohol precipitates the protein and releases the bilirubin, which reacts with Vandenberg reagent. Hence it is called as *indirect Vandenberg test.*
- Indirect Vandenberg reaction is the confirmatory test of haemolytic jaundice due to high content of free bilirubin in the plasma, whereas obstructive jaundice can be confirmed by direct Vandeberg reaction due to increased content of conjugated bilirubin.
- In the case of total obstruction of bile flow , lack of bilirubin in the intestine prevents the conversion of bilirubin in to urobilinogen by bacterial action. Urobilinogen is not available for reabsorbtion through portal blood and the excretion of urobilin through urine, henc test for urobilinogen in urine is completely negative .
- The stools become clay coloured due to lack of stercobilin and other bile pigments.
- Kidney can excrete only highly soluble conjugated bilirubin but not the albumin bound "free bilirubin".
- In severe case of obstructive jaundice large quantities of conjugated bilirubin froms foam and becomes intense yellow colour while shaking the urine.

CHOLESTEROL

They belong to the group of steroid present in animal fat. About 10% of the bile is the cholesterol, secreted by hepatic cells. There are two principle sources of cholesterol in the bile.

- Synthesised by the hepatic cells
- By absorption from the intestine through portal circulation to the liver

Gall stones and their formation

- Two kinds of gall stones are formed
  - Cholesterol and bile pigments
  - Cholesterol,bile pigments and $CaCO_3$ with other fatty substances.
- The cholesterol is insoluble in the water. Bile salts, fatty acids and lecithine present in the bile provide hydrophobic property to cholesterol and keep it as solution in bile . When the bile becomes concentrated, these hydrophobic substances also get concentrated.
- Under abnormal conditions, cholesterol may be precipitated which results in the formation of gall stones in the bladder.
- The following are the conditions that cause cholesterol precipitation:
  - Excessive absorption of water from the bile
  - Excessive absorption of hydrophobic substances
  - Excess secretion of cholesterol in bile
  - Inflammation of the gall bladder epithelium.

- Inflammation of gall bladder epithelium alters the absorption of water, bile salts or other hydrophobic substances through gall bladder mucosa.
- Excessive absorption of water, bile salts or other hydrophobic substances, leads to precipitation of cholesterol .
- Many small cholesterol crystals act as a nidi or nucleus which favours further deposition of cholesterol crystals leads to occurrence of multiple gall stones.

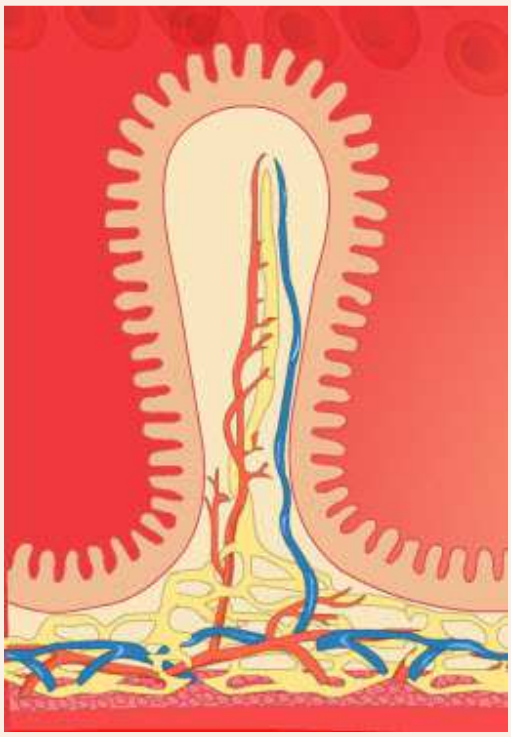
FUNCTIONS OF GALL BLADDER

- Storage of bile and concentrating the bile.
- Delays the absorption of bile salts from the intestine by the addition of mucin to the bile.
- Prevents excess absorption of bile.
- Assisting stabilization of fat emulsion in the intestine.

Function of bile as whole

- Bile acids brings about the absorption of fat and fat soluble vitamins like vit.A,D, E and K.
- High content of HCO_3 in the bile maintains alkalinity for optimal enzymatic reaction in the intestine .
- Mucin and pseudomucin in the bile act as a stabilizers of fat emulsion in the intestine.
- Bile has a marked antiseptic property and regulates the bacterial growth in the bowel.
- Bile has a mild laxative effect.
- In the absence of bile in the intestine, fat absorption is reduced and other food particles like proteins are coated with fat or fatty acids, escape enzymatic digestion leads to putrification of protein which gives offensive odour to feces.

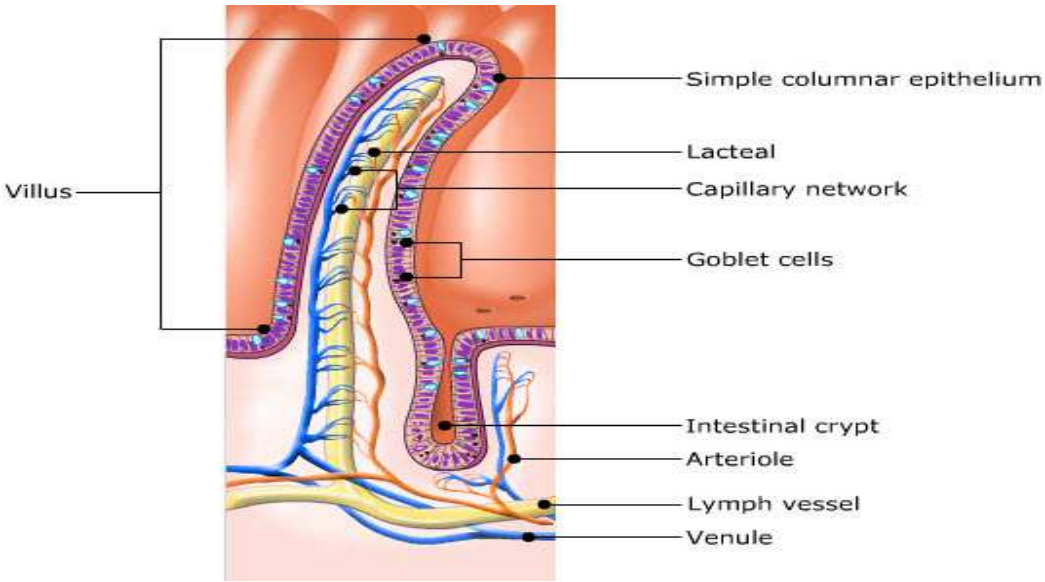
MODULE-31: PROCESS OF DIGESTION



LEARNING OBJECTIVES

- This module deals with,
  - duodenal and intestinal secretions and their composition and functions,
  - the regulation of intestinal secretions,
  - the digestion of carbohyhdrate proteins and lipids at molecular/celluar level and
  - the luminal and membranous phases of digestion and the functions of enzymes.

STRUCTURE OF VILLI



- Surface area of the small intestine is enormously increased by large folds of mucosa known as *plica circularis* to facilitate contact between mucosa and lumen contents.

- The mucosal surface area is covered with finger-like epithelial projections called as *villi* to increase the surface area by about 10 to 14 fold.
- The villi are covered with brush like surface membrane known as *brush border*, composed of *microvilli* causes further increase the surface area. The gland like structure at the base of the villi is known as *crypts of Lieberkuhn***.**
- The villi and crypts are covered with a layer of epithelium.
- The epithelial cells covering the villi and the crypts are called as enterocytes.
- The enterocytes facing towards the lumen is called apex which are covered with apical membrane. This apical membrane is provided with the microvilli (brush border).
- A jelly-like layer of glycoproteins known as *glycocalyx* covers the microvilli . Digestive enzymes and other proteins are attached to microvilli and project into glycocalyx.
- That part of enterocyte facing opposite to lumen is called as the basolateral membrane.
- Nutrients absorbed into enterocytes exit the cell through basolateral membrane before entering into blood.

CHEMICAL DIGESTION OF NUTRIENTS

- Enzymatic digestion of nutrients takes place in two phases
  - Luminal phase
  - Membranous phase
- Chemical digestion of each major nutrient is by hydrolysis *glycosidic* bonds in carbohydrates, *peptide* bonds in proteins, *ester* bonds in fats and *phophodiester* bonds in nucleic acids by the insertion of water molecule.
- Hydrolysis of nutrients in the digestive tract is catalysed by the action of enzymes secreted by salivary, gastric and pancreas glands. These glands pour their enzymes in the lumen of the GI tact for mixing with ingesta hence referred to as *luminal phase* of digestion. This phase of digestion results in incomplete hydrolysis of nutrients leads to the production of short-chain polymers of original macromolecule.
- In the *membranous phase* of digestion, the enzymes are synthesised within the enterocytes and transported to the apical membrane. Direct contact of these enzymes in the epithelium with the substrates derived from luminal phase of digestion exhibit final break down of the products of the short-chain polymers of luminal phase of digestion into monomers inside the epithelium of small intestine, hence referred to as *membranous phase* of digestion.
- It is followed by the absorption of end products of nutrients across the intestinal epithelium. .

Carbohydrate digestion

- Dietary carbohydrates are mainly monosaccharides (Glucose, galactose and fructose) disaccharides (lactose and sucrose) and fibrous carbohydrates (cellulose, hemicellulose).
- Oligosaccarides (maltose, isomaltose and maltotriose are rarely present in the diet, but they are formed in the gut during the course of carbohydrate digestion.
- Starches is present as amylose and amylopectin.
- Amylose contains glucose monomers linked by a -1, 4 glycosidic linkage, whereas amylopectin also contain glucose linked by *a* -1, 4 glycosidic linkage, but the chains are branched, having a -1, 6 glycosidic linkage at branched points.

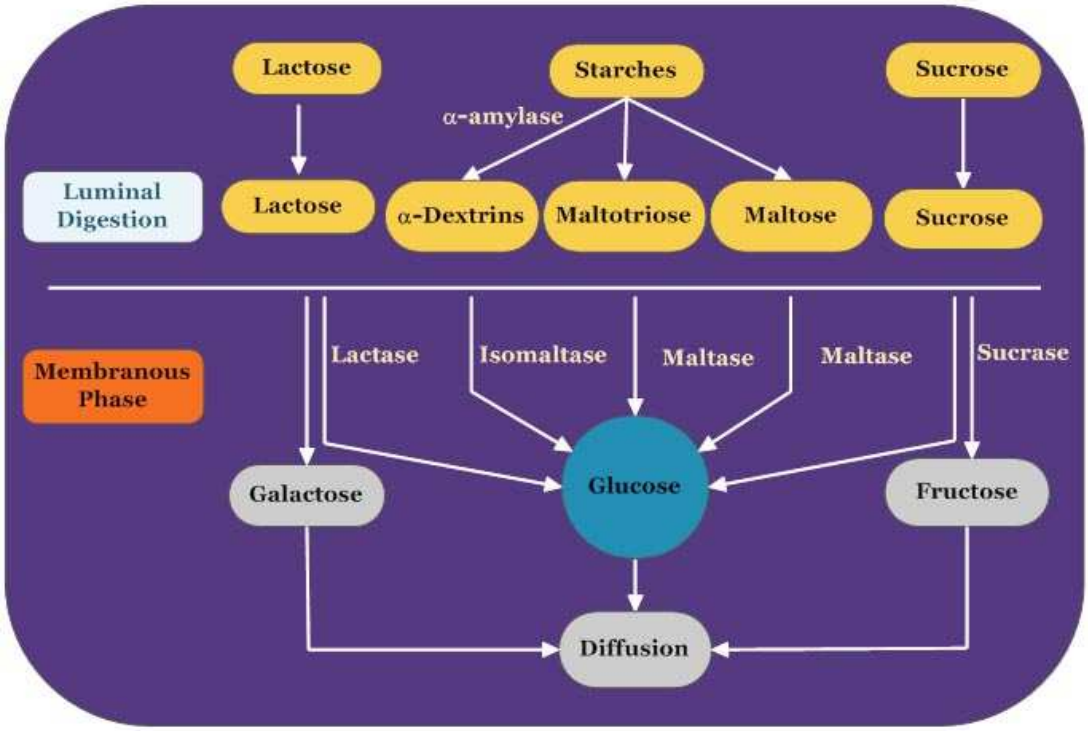
LUMINAL AND MEMBRANOUS PHASE OF DIGESTION

Luminal phase

- In luminal phase of digestion, *alpha amylase* is secreted by the salivary glands (eg. pig, dog and rabbit), and pancreatic amylase hydrolyses starches to yield *oligosaccharides (a -dextrins, maltotriose, maltose* without free glucose),whereas fibrous carbohydrates (eg. Cellulose and hemicellulose) can not be hydrolysed by *alpha amylase of mammals*.
- Hydrolysis of *amylopectin* produces branched-chain oligosacchrides, *limit dextrins* and alpha 1, 6-linked disaccharide, *isomaltose*.

Membranous phase

- The oligosaccharides, di- and tri-saccharides are hydrolysed by saccharidases enzymes like *maltase, isomaltase, sucrase and lactase* in the membranous phase.
- In the glycocalyx of enterocytes, these enzymes hydrolyse the oligosaccharides (eg. maltose, sucrose and lactose) to monosaccharides *(glucose, galactose, and fructose)*.
- Most of the monosaccharides are absorbed into the portal blood and carried to liver. some are also absorbed through lymph stream.



PROTEIN DIGESTION

- Dietary, endogenous, microbial and shed mucosal cells proteins are capable of being hydrolysed by *endopeptidases*, the proteolytic enzymes present in gastric , pancreatic and intestinal juices break the proteins at internal peptide points.
- In the luminal phase of digestion, gastric proteases (eg. pepsin and rennin) and pancreatic proteases (eg. trypsin, chymotrypsin, carboxy peptidase a & b) yield *oligopeptides* and some amino acids.
- These oligopeptides are further hydrolysed by *oligopeptidases* in the glycocalyx to produce *amino acids, di- and tri- peptides*.

Luminal phase of digestion

- The proteolytic enzymes are secreted from gastric glands and pancreas as inactive *zymogens*. In the stomach, *pepsinogen and prorennin (chymosinogen)* are activated by HCl.
- Pepsin also autocatalytically activates pepsinogen.
- Trypsinogen from pancreas is activated as trypsin by the action of *enterokinase,* an enzyme produced from duodenal mucosa.
- Trypsin autocatalytically activates trypsinogen.
- The active trypsin inturn activates other proteolytic pancreatic enzymes (Chymotrypsinogen and procarboxypeptidase a & B) .
- The HCl of stomach also has hydrolytic properties on protein

Membranous phase of digestion

- In the membranous phase of digestion, the e*xopeptidases* present on the enterocyte apical membrane act at the ends of peptide chain to release free amino acids at the surface of mucous membrane.
- Some long-chain peptides are incompletely hydrolysed leading to the production of di- and tri-peptides.
- These di- and tri-peptides are easily absorbed by the epithelial cells; where they are subsequently hydrolysed by the intracellular peptidases, forming free amino acids.
- Thus, the free amino acids are produced at two sites: on the surface of the enterocytes and the second within the cell.

Enzyme	Source	Precursor	Activator	Action
Pepsin	Gastric glands	Pepsinogen	HCl & Pepsin	Endopeptidase
Chymosin (Rennin)	Gastric glands	Pro-rennin	HCl	Endopeptidase
Trypsin	Pancreas	Trypsinogen	Enteokinase & Trypsin	Endopeptidase
Chymotrypsin	Pancreas	Trypsinogen	Trypsin	Endopeptidase
Elastase	Pancreas	Pro-elastase	Trypsin	Endopeptidase
Carboxypeptidase A	Pancreas	Procarboxypeptidase A	Trypsin	Endopeptidase
Carboxypeptidase B	Pancreas	Procarboxypeptidase B	Trypsin	Endopeptidase

LIPID DIGESTION

- Lipids make up a large portion of diet in carnivores, where as they form only a minor portion of diets in herbivores.
- Primary dietary lipid is *triglyceride*; other lipids include *cholesterol and cholesterol esters* from animal sources, waxes from plant sources and *phospholipids* from both plant and animal sources.
- Lipid digestion occurs in four phases; *emulsification, hydrolysis, micelle formation and absorption***.**

Emulsification is a process of reducing lipid droplets to a smaller size for their suspension in water.

- In the gut, lipid globules are broken down to droplets by the mixing and agitating actions of distal stomach.
- Emulsification is completed in the small intestine by the detergent action of bile acids and phospholipids.
- Bile salts reduce the surface tension of the lipid droplets and further, reduce in size of the fat droplets.
- The bile coated or emulsified droplets are subjected to hydrolytic enzyme action.
- *Triglycerides* are the major dietary lipid, undergo hydrolysis by the action of gastric, pancreatic *lipase and co-lipase*, which are secreted as active form.
- The co-lipase "make a pathway" through the bile product coating the emulsified lipid droplet, giving access to the lipase to reach the underlying triglyceride.
- Lipase cleaves the fatty acids from the end of triglyceride molecule resulting in the formation of two free nonesterified fatty acids and a monoglyceride.
- *Cholesterol esterase and phospholipase* are the other lipid digesting enzymes of pancreas.
- The products of these enzymes are *nonesterified fatty acids, cholesterol and lysophopholipid***.** The fatty acids, monoglycerides etc., combine with bile acids and phospholipids to form very small lipid droplets, *micelles* . The micelles are water soluble allow the lipids to diffuse through glycocalyx and into close contact with absorptive surface of the enterocytes.

LARGE INTESTINE

Large intestine of herbivores

- Large intestine is of great importance in herbivores. In simple stomached animals, the enormous caecum and colon are involved in the microbial fermentation hence they are called as ***hind gut digesters***. The absorption of the fermented end products of cellulose and digestible products synthesised by bacteria occur in large intestine of simple stomached herbivores.

Structure

- Caecum and colon of simple-stomached herbivores are very capacious and sacculated, whereas in ruminants, they are simple and non-sacculated. Glands and villi are present in large intestine of herbivores. The glands secretes mainly mucous but has no enzymatic digestive function. In carnivores c olon is short and non-sacculated, caecum is poorly

developed, intestinal glands are present throughout large intestine but villi are absent. The glands of large intestine secretes only mucous not the enzymes.

Bacterial action

- No bacterial fermentation occurs in small intestine of carnivores. However, in large intestine, putrifactive (proteolytic) and cellulolytic type of bacteria act on the ingesta which have escaped digestion in small intestine. The products of bacterial action in large intestine are indole, skatole, paracresol, VFA, NH_3 and gases CO_2, CH_4, H_2S and B complex vitamins. These products may be absorbed into the portal blood and carried to liver.

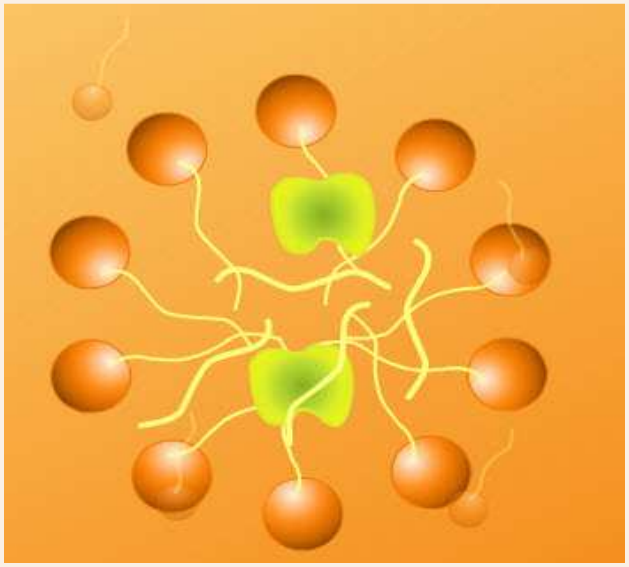
Digestive process

- In monogastric animals, hind-gut maintains conditions favourable for microbial fermentation, which includes substrate availability, control of pH and osmolality, anaerobiosis, retention of substrate and removal of end products of fermentation. There is extensive urea recycling in caecum and colon supplies N_2 for microbial growth.
- Bacterial action in large intestine is mainly on structural and non-structural carbohydrates and proteins. The digestion of cellulose in large intestine yields VFA, CO_2 and CH_4. Ammonia is produced from proteins and urea is absorbed from large intestine of horse.
- In ruminants, enzyme digestion occurs after microbial fermentation and the host animal also utilises the microbial cell bodies. In simple stomached herbivores, microbial fermentation follows after enzymatic digestion and only fermentation products and not the microbial cell bodies are available for absorption by the host. The VFAs absorbed from the large intestine contributes to the energy needs of the host animal.
- Volatile fatty acids acetic, propionic and butyric acids are found in the large intestine of ruminants and horse. Appreciable quantities are also produced in pigs, rabbit and fowl. Very small amounts are found in large intestine of cat and dog. Bacterial synthesis of B complex vitamins occurs in caecum and colon.
- The gases found in large intestine are CO_2, CH_4 and very small amounts of H_2S, O_2 and considerable quantities of N_2 which are expelled in part through anus and in part absorbed into the blood and eliminated through lungs.

Functions of large intestine

- The digestive processes are practically complete in the small intestine. Materials escape absorption in small intestine are gradually propelled into the large intestine through the ***ileocaecal valve***. Large intestine is involved in the functions of reabsorption of water that has been poured out by digestive glands. It also acts as reservoir for waste material that constitutes the *faeces*, which are expelled at intervals by the act of defecation.

MODULE-32: ABSORPTION



LEARNING OBJECTIVES

This module explain the following,

- absorption - basic histology of the villi of intestine and venous and lymph,
- various transport machanism,
- the mechanism of digestion of carbohydrates, proteins and lipids and
- the importance of bile in lipid absorption.

ABSORPTION

- It is the process whereby the products of digestion and the digested foodstuff form the lumen of the gut is transferred to the blood or the lymph across the epithelial cell membrane.

Site of absorption

- No absorption of food or end products of digestion in the mouth and oesophagus
- In the monogastric animal, absorption in the stomach is very limited.
- However certain drugs can be absorbed from the pharynx and to a limited extent from the oesophagus.
- The small intestine is the chief site of absorption in the carnivores and omnivores.
- Villi is the the chief site of absorption in small intestine
- The large intestine is the chief organ of absorption in all herbivores (eg. horse) and to a limited extent in the carnivores and man where it is restricted to the initial colon.
- In ruminants, the digestion and absorption of contents is of special importance in the anterior part of the digestive tract.
- The large intestine has specific absorptive sites with respect to water and electrolytes.
- In herbivores, especially in equines, the large intestine absorbs volatile fatty acids and ammonia.

ROUTES OF ABSORPTION

Small intestine has extremely well developed lymphatic and blood system which function as a route of absorption of digestive products.

- Lymph
  - In the core of the villus lymph capillary originates as *lacteal* near the tip of the villus and enters into a plexus of lymph vessels lying just on the inner side of the muscular coat. Branches of these plexus then enter into the submucosa and form a loose plexus of large lymphatics, finally pass into mesentry.
  - The lymph capillaries drain their content into large lymph vessels, which intern empty into the mesentric vessels. These mesentric vessels are then connected with mesentric lymph nodes. The contents of the mesentric vessels empty into the *cisterna chyli* which is continued forward as thoracic duct, finally empties into the venous system anterior to heart.
  - Glycerides, long chain fatty acids, cholesterol and the immuno globulins during the first 24 hours of life are absorbed by the lymphatic system. The rate of lymph flow increases after a meal.
- Blood
  - Each villus contains several small arteries, which enter the base of the villus and form a dense capillary network immediately under its epithelium. Near the tip of the villus, one or two veins arise from a capillary network and run downward.
  - The venules and veins, drain into the portal vein. The portal vein enters into the liver where its blood is mixed with that of hepatic artery. The hepatic vein conveys the blood from the liver to the posterior vena cava.
  - Amino acids, monosaccharides, free glycerol, water, inorganic salts and short chain fatty acids are absorbed through blood route. After a meal, rapid flow of blood causes increased absorption rate, but this increase is less than that of lymph.

MECHANISM OF ABSORPTION

- The possible mechanisms of absorption are broadly classified into three groups:
  - Non-carrier mediated transport (passive diffusion)
    - It depends on the electro-chemical gradient, occurs through channel pathways of ions.
    - Non-carrier mediated process aids in the absorption of short chain fatty acids, inorganic salts and lipid soluble compounds.
  - Carrier mediated transport – includes
    - Facilitated transport from higher to lower concentration
    - Exchange diffusion Na+/H+ exchanger: transports H+ out and Na+ into the cell
    - Active transport against concentration gradient with expenditure of energy
      - The carrier-mediated process may help water-soluble materials to pass the lipid layer of the cell membrane, whereas glucose and amino acids are absorbed by active transport**.**
  - Pinocytosis transport of intact luminal materials in the form of vacuoles into the mucosal cells. This is important for absorption of intact proteins and intact triglycerides.

CARBOHYDRATE ABSORPTION

Carbohydrate absorption

- Monosaccharides (eg. glucose, fructose) are absorbed by active transport mediated by carrier and sodium pump.

Absorption of glucose

- Glucose gets attached to specific transport proteins that lie on the luminal side of the enterocytes. These transport proteins have two binding sites one for glucose and one for sodium.
- Once glucose and Na^+ occupy the binding sites, the transport protein moves across the cell membrane and unload the glucose and Na^+ into the cell. Hence, this process is referred to as sodium co-transport.
- The transport of glucose will not occur unless Na+ is present.
- Within the cell, glucose moves down by concentration gradient through the basoleteral membrane by facilitated diffusion, to extra-cellular space and then into the blood and finally to liver where the monosaccharides are stored as glycogen.
- Galactose is absorbed more rapidly than glucose but fructose absorption is slower than glucose absorption.
- Fructose absorption is by facilitated diffusion and not energy dependent. Hence, fructose can not be absorbed against concentration gradient.
- Mannose, xylose and arabinose are poorly absorbed by diffusion.
- Maltose, sucrose and lactose as such are absorbed very slightly.
- Disaccharides do not generally enter into the blood stream because of the presence of disaccharidases in the brush border of mucosa, which converts them to monosaccharides.

PROTEIN ABSORPTION

- The free amino acids are readily absorbed chiefly by active transport requiring Na -co transport system.
- Three types of carriers are involved in the transport of acidic, basic and neutral amino acids.
- L-isomer forms of plant and animal protein are more readily absorbed than D-isomers, acidic basic and meutral amino acids
- Some di and tri peptides are also absorbed. Intracellular peptidase hydrolyses these peptides to amino acids.
- Intracellular amino acids diffuse across the basolateral membrane to reach liver via portal blood, whereas intact proteins are absorbed via lymph pathway.
- Immediately after birth immunoglobulins from colostrum are absorbed by a process of pinocytosis particularly in lambs, piglets, kids, calves and pups.
- The immunoglobulin absorption decreases with time after birth and ceases after 24-36 hours.

LIPID ABSORPTION

- Lipid absorption begins in the dental duoderium and ends in proximal part of the jejunum.
- As the micelles come in close contact with surface of enterocytes, the lipid components diffuse through the glycocalyx to the apical membrane by a special fatty acid binding proteins which aids the transport the fatty acids across the cell membrane.

- Other components in the micelle such as monoglycerides, cholesterol and vitamin A diffuse into the apical membrane.
- Bile salts get detached from the micelles during fat absorption and remain in a free state. In the ileum, Na-co transport system function as specific bile acid transport which nearly complete the absorption of bile salts.
- After absorption, the bile salts are transported to liver by the portal blood.
- The liver extracts bile acids and maintains the normal concentration of bile acids in systemic blood. This process of recirculation of between liver and intestinal lumen is referred as *enterohepatic circulation of bile***.**
- Glycerol is absorbed by passive diffusion into the mesentric venous blood.
- Short chain fatty acids up to C^{10} are water-soluble get into mesentric portal blood.
- Monoglycerides and long chain fatty acids enter the microvilli and pass on to the lacteal by simple diffusion.
- Only free form of cholestrol can be absorbed, whereas cholestrol esters must be hydrolyzed by pancreatic brush border hydrolases..
- Cholesterol is absorbed less efficiently than triglycerides. Presence of hydro cholesterol or plant sterols inhibits cholesterol absorption.
- In the epithelial cells, cholesterol is re-esterified before their transfer to lacteals.
- Before absorption phospholipids are hydrolysed to free fatty acids and lysophospholipids by phospholilpase of pancreas and intestinal epithelium.
- Within the epithelial cells, long chain fatty acids are converted into fatty *acyl-CoA* involving co-enzyme A and *ATP*.
- The fatty acyl co-enzyme reacts with monoglycerides to form di and tri glycerides.
- The newly formed triglycerides differ from that of dietary fat. GlycerolPO_4 derived from glucose metabolism provides glycerol residue for the triglyceride synthesis.
- In addition, phospholipids and cholesterol esters are produced in the epithelial cells. Small amounts of proteins are added to the lipid droplet before their transfer from epithelial cells to lymph.
- *Chylomicrons* are the products containing high amount of triglycerides, low level of phospholipids, cholesterol esters and proteins. They leave the cell by *reverse pinocytosis* and enter into the lacteals.
- Lipid absorption begins in the distal duodenum and completed in the proximal jejunum.
- The absorbed fat is in the form of an emulsion and imparts a milky appearance to the lymph,called as *"chyle"***.** This leave the cell by reverse pinocytosis and enter in the lacteal. Though lymphatic channel and thoracic duct it is added into the blood for its distribution to tissues.
- Short chain fatty acids are absorbed by blood from colon and caecum in sheep and horse and ceecum in pigs.

MECHANISM AND ABSORPTION OF MINERAL AND WATER

- Sodium is transported as sodium co-transport along with glucose and amino acids.
- Sodium transport is also coupled transport with chloride ion
- Sodium ion also shows diffusion across the electrochemical gradient.
- Potassium is transported by passive diffusion across the concentration gradient
- Bicarbonate ions are transported actively or rarely with sodium ion.
- Magnesium ion is transported actively, but poorly absorbed.
- Calcium ion is also transported actively by the calcium-binding protein produced under the influence of vitamin D_3

- Phosphorous is actively absored related to calcium absorption.
- Copper ion is absorbed in small quantities.
- Cobalt and manganese ions are readily absorbed.
- Iron absorption is related to its level in mucosal cells and is mediated by apoprotein carriers.
- Water is transported passively

ABSORPTION OF VITAMINS

Absorption of vitamins

- Fat soluble vitamins (A, D, E, K) pass through the intestinal mucosa by passive diffusion.
- Water soluble vitamins (except B12) are also absorbed by passive diffusion but may also be transported actively.
- Vitamin B12 requires an intrinsic factor, secreted by the gastric glands of the stomach, for its absorption by and active transport process.

MODULE-33: AVIAN DIGESTION



LEARNING OBJECTIVES

This module explores,

- the difference between the alimentary tract of chicken and mammals and
- the digestion and absorption of food form the avian G.I tract.

AVIAN DIGESTION -ALIMENTARY CANAL

- In birds, the alimentary canal is a long tube like organ starting from the beak at the head and ending with the vent or cloaca in the abdominal region.

Special features of the avian digestive system includes

- Presence of beak
- Absence of teeth
- Proventriculus

- Gizzard
- Paired caeca
- Vent

MOUTH STRUCTURE

- In the fowl ,the lips and cheeks are replaced by the beak - an area of dense and horny skin lying over the mandible.
- Absence of teeth, but so called egg tooth found on the end of the beak of newly hatched chickens to aid their escape from the egg at the time of hatching which disappears after a day or two.
- The hard palate, forming the roof of the mouth. It has five transverse rows of backwardly pointing, hard, conical papillae.
- Numerous ducts of the salivary glands pierce the hard palate to release their secretions into the mouth cavity.

SALIVARY GLANDS

- A thick layer of stratified squamous epithelium covers the free surface.
- The salivary glands run the whole length of the hard palate, the groups of glands merging to form one mass of glandular tissue under the epithelium.
- Lymphoid tissue is found in most glands.
- The salivary glands are:
    - Maxillary - in the roof of the mouth
    - Palatine - on either side of the nasal opening in the roof of the mouth
    - Apheno-pteryoid glands - in the roof of the pharynx on each side of the common opening for the eustachian tubes
    - Anterior sub-mandible glands
    - Posterior sub-mandibular glands
    - Lingual glands - in the tongue
    - Crico-arytenoid glands - around the glottis
    - Small gland in the angle of the mouth.

PHARYNX AND TONGUE

- The pharynx is continuous with or follows the mouth.
- The combined cavity of the mouth and the pharynx is often referred to as the oropharynx.
- The common opening for the two eustachian tubes is located in the middle of its dorsal wall (roof).
- The tongue is long and pointed and conforms to the shape of the beak in which it operates.
- The epithelium of the tongue is thick and horny, especially towards the tip.
- A transverse row of simple, large and horny papillae with their tips directed towards the rear of the mouth cavity are located on the posterior end.
- The hyoid bone provides the framework to support the tongue. The entoglossal bone extends longitudinally in the median plane.
- Small patches of lymphatic tissue are located throughout the corium. Mucous glands are located in the tongue with short ducts directed towards the rear.
- Some but not others believe that there are taste buds located on the tongue.
- In any case the sense of taste appears to be very weak if at all present.

- The mouth has two major functions:
    - To pick up the food particles and direct the food into the oesophagus.

OESOPHAGUS AND CROP

- Oesophagus is wide which connects the mouth region to the crop.
- The wall of the oesophagus is composed of four layers of tissue, the innermost being mucous membrane.
- The mucous membrane is an important barrier to the entry of microbes.
- It secretes mucous to lubricate and to aid the passage of the food along the alimentary canal.
- The crop is a large dilation of the oesophagus located just prior to the oesophagus entering the thoracic cavity. In some birds there are two crops and carnivorous birds donot have crop.
- The crop provides the capacity to hold food for some time before the commencement of digestion. This capacity enables the bird to take its food as “meals” at time intervals and permits continuous digestion.

    - The crop structure is similar to that of the oesophagus except there are no glands present in fowls.
    - Ducks and geese have glands in the crop mucous membranes.
    - In pigeons, the surface cells of the crop slough off during brooding to form pigeon’s milk - used to feed the baby pigeons in the nest.

- Inside the thoracic cavity, the oesophagus becomes the proventriculus very glandular part of the digestive tract often called as glandular stomach.

PROVENTRICULUS AND GIZZARD

Proventriculus

- The glandular stomach or proventriculus is relatively small and tubular.
- The wall is very thick composed of five layers:
    - Outer serous membrane.
    - Muscle layer composed of three separate layers:
        - Two thin longitudinal layers.
        - One thick circular layer.
    - Layer of areolar tissue containing blood and lymph vessels.
    - Thick layer composed mainly of glandular tissue.
    - Inner mucous membrane.

Gizzard

- The muscular stomach or gizzard is located immediately succeeding the proventriculus.
- It is placed partly between the lobes and partly behind the left lobe of the liver.
- Under the outer layer very powerful masses of red muscle are located.
- The inner surface is lined with a creamy colored, thick, horny tissue raised in ridges.

- The gizzard almost always contains quantities of hard objects such as gravel or other grit that aids in the disintegration of food - the primary function of the gizzard.
- The gizzard consists of number of layers of tissues in some of which straight tubular glands are located.
- The innermost layer is a strong, flexible skin able to withstand the potentially damaging effects of the muscular action of grinding the food often in the presence of stones or other insoluble material.
- The glands of the gizzard produce a liquid, keratinized material that passes to the surface of the horny lining where it hardens to replace tissue worn away by the grinding action of the organ.

DIGESTION IN BIRDS

Enzymatic digestion

- After ingestion, the food is mixed with saliva and mucous to moisten the food. Amylase, is produced by the salivary and oesophageal glands.
- The secretions of the proventriculus or glandular stomach are hydrochloric acid, pepsin that acts on protein and the hormone gastrin that stimulates the production and release of gastric juice in the proventriculus and pancreatic juice from the pancreas.
- *Gizzard* is a very powerful muscular organ actively involved in breaking the food particles into smaller sizes. The enzymes released into the food with the saliva and by the proventriculus are also thoroughly mixed with the food. This breaking and mixing function of the gizzard is enhanced by the presence of insoluble grit such as stones.
- Pancreatic juice and bile from the liver enters via ducts located at the distal end of the duodenum at about the junction of the duodenum and the jejunum. The digestive process starts prior to the entry point to the small intestine due to back flow of pancreatic juice and bile towards the gizzard.
- In addition to enzymes, the pancreas produces insulin and sodium bicarbonate. Insulin is involved in the maintenance of blood sugar levels while the sodium bicarbonate, being strongly alkaline, will increase the pH of the intestinal contents.
- The small intestine also produces enzymes that play their part in the digestive process of reducing the complex food compounds into simple building blocks that can be absorbed across the intestinal wall for the transport to the organ for further processing or storage.
- Food materials that escape enzyme action are subjected to bacterial breakdown in the caeca.

SMALL INTESTINE

- Small intestine begins at the exit from the gizzard and ends at the junction of the small intestine, caeca and colon.
- It is relatively long and has a constant diameter. Of the three parts of the small intestine duodenum, jejunum and ileum, only the duodenum can be easily distinguished in the fowl.
- There is no clear demarcation between the jejunum and ileum.
- Much of the digestion of the food and all of the absorption of the nutrients takes place in the small intestine
- Structures of small intestine

- Outer serous membrane.
- A layer of longitudinal muscle
- A layer of thick circular muscle
- Structures located between the two muscle layers are:
  - Blood vessels.
  - Lymph vessels.
  - A network of nerve fibers.
- An ill-defined sub-mucosa
- Mucous membrane consisting of:
  - Thick muscularis mucosae of longitudinal and circular muscle.
  - Corium - many glands, lymphoid tissue, muscle fibers and a variety of free cells.
  - Inner epithelial surface.

Duodenum

- The duodenum starts at the gizzard and forms an elongated loop about 20 centimeters long. The pancreas lies between the arms of the loop and being attached to each arm of the duodenum actually holds the two arms together.
- Lymphoid tissue in the duodenum is very plentiful and is usually located in the corium.
- Bile ducts from the gall bladder attached to the liver and two to three pancreatic ducts enter the small intestine by a common papilla at the caudal end (closest to the rear) of the duodenum.
- Pancreas is located closely associated with the duodenum being attached to each side of the duodenal loop and lying between the two arms.

Villi

- It is a long flattened, fingerlike projections extending into the lumen of the intestine like flexible fingers.
- A single layer of columnar epithelium together with goblet cells covers the lining. The goblet cells secrete mucous.
- A lacteal (lymph vessels), capillaries, bundles of plain muscle fibres, nerves and other tissues and cells occupy the core of the villus. The villi have the function of providing a vastly increased surface area for the more efficient absorption of the nutrients.
- The villi are very actively involved in the absorption process.
- The efficiency of the absorption is influenced by the surface area available for the nutrients to move through.

Jejunum and the ileum

- The jejunum and the ileum, together about 120 cm long, commence at the caudal end of the duodenum where the bile and the pancreatic duct papilla are located and terminates at the ileo-caecal-colic junction. It is a junction of small intestine, the two caeca and the colon. This portion of the small intestine is similar in structure to the duodenum except that:
  - The villi are shorter
  - There is less lymphoid tissue
- Meckel's diverticulum is a constant feature about half way along the small intestine appearing as a small projection on the outer surface of the small intestine. This projection represent this point of attachement of yolk sac during the development of the embryo.

Functions of small intestine

- Produces number of enzymes involved in the digestive process
- Site of final digestion of the food
- Site of absorption of the end products of digestion

LARGE INTESTINE

- The large intestine is very short and ends at the cloaca.
- This section includes paired caeca, colon and rectum and the rectum being the terminal part.
- The bursa of fabricius is located immediately above the cloaca of young birds, but disappears when the birds have reached approximately one year old.

Caeca

- Paired caeca or blind pouches are about 16-18 centimeters long in the adult.

Structure of the caeca

- Outer serous membrane
- Outer longitudinal muscle
- Circular muscle
- Inner longitudinal muscle forming the muscularis mucosae of the mucous membrane

Cecal function

- Microbial digestion of cellulose.
- Reflux of urine into the ceca exposes the cecal microflora to urea and uric acid, which are then degraded; the nitrogen is not recycled for use by the host.
- Microbial synthesis of B compex vitamins

Cloaca

- It is a tubular cavity opening to the exterior of the body provides a common passage to the digestive and urogenital tracts.
- It divides into three chambers as follows:
    - *The copradaeum* - a continuation of the colon-rectum
    - *The urodaeum* - middle part into which the ureters and genital ducts open
    - *The proctodaeum* - opens to the exterior of the vent. Birds less than one year old have a dorsal opening leading into the blind, rounded sac - the bursa of fabricius.

ABSORPTION AND EXCRETION

Absorption

- Absorption of glucose occurs via both passive and carrier mediated transport system.

- Passive transport appears to predominate in the duodenum and jejunum. Chicken has a sodium dependent carrier system for the active transport of sugars.
- Carrier medicated transport of D -glucose is facilitated by a Na^+-glucose co-transpoter protein.
- Amino acids are also transported by Na^+ dependent and energy dependent carrier mediated process in birds.
- Four system of amino acids tansport have been identified in chicks:
    - Neutral amino acids system
    - Basic amino acids system
    - Acid amino acids system
    - A system for proline, beta alanine and related amino acids.
- Neutral amino acids are transported more rapidly than basic or acidic amino acids.

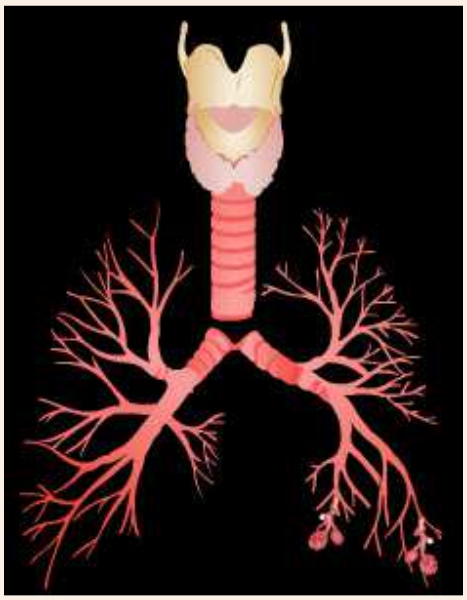
Faeces

- This material consisting of waste and undigested food mixed with the urine in the cloaca and eliminated from the body as faeces.
- Typically it is a rounded, brown to grey mass topped with a cap of white uric acid from the kidneys.

ABSORPTION OF SALT AND WATER

- Sodium, K^+ and Cl^- ions are almost completely absorbed. Na^+ ions are involved in the transport of water, monosaccharides, amino acids, pyramidine and bile salts.
- Na^+ is transported as
    - Na-co transport
    - Cl^- coupled transport and
    - Diffusion by electro -chemical gradient.
- Chloride ions are transferred through Cl^- coupled Na^+ transport
- Association with Na^+ co-transport of glucose
- Diffusion by electro-chemical gradient.
- K^+ is absorbed by passive diffusion due to concentration gradient.
- HCO_3 absorption is by active transport.
- Water is transported passively.
- Intestinal absorption of Ca^{++} is by active transport. Vitamin D induces mucosal Ca^{++} binding protein and increases Ca^{++} absorption.
- Mg^{++} is absorbed by active transport but its absorption is poor. Absorption of phosphorus is active process and related to active transport of Ca^{++}.
- Absorption of Fe^{++} is related to its level in mucosal cell. Absorption is limited by binding capacity of apoprotein (carrier protein).
- Cu is absorbed in small amounts.
- Co and Mn are readily absorbed.
- Fat-soluble vitamins A, D, E and K pass through mucosa passively as also water-soluble vitamins. Vitamin B_{12} requires an intrinsic factor secreted by stomach for its absorption by active transport.

MODULE-34: RESPIRATORY APPARATUS AND PHYSICAL PRINCIPLES OF GAS EXCHANGE



LEARNING OBJECTIVES

This module deals with,

- what is respiration?
- the structures associated,
- basics of gaseous transport and
- terms related to capacity of lung during various phases.

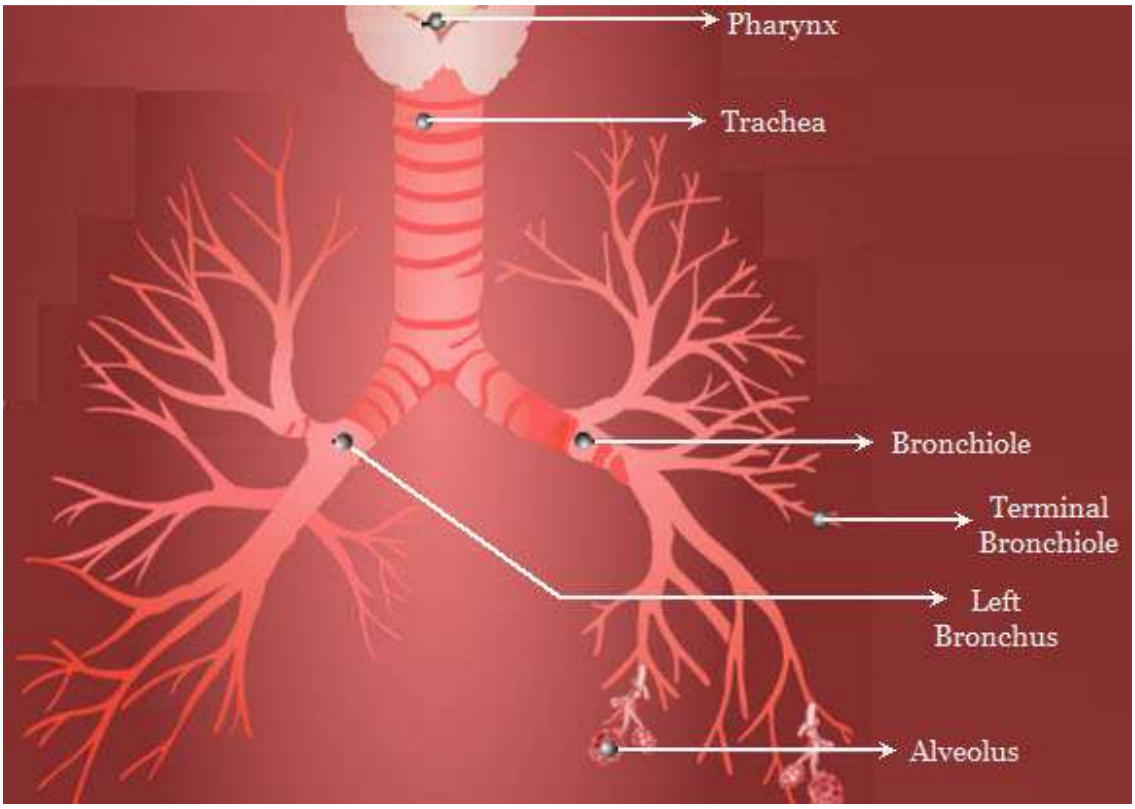
RESPIRATORY TRACT

- It is the conducting portion or airway consists of nose, pharynx, larynx, trachea, bronchi, bronchioles, and terminal bronchioles.
- Pharynx is a common passageway for air and food. The opening from the pharynx leading to the continuation of the airway is the larynx, the organ of phonation (sound production) in mammals.
- It is produced by the controlled passage which causes vibration of vocal cords of larynx.
- The organ of phonation in birds called the syrinx which is located at the junction where trachea divides into bronchi.
- The glottis is the slit-like opening between the vocal cords. Extending craniad from the larynx is the epiglottis.
- It is a leaf shaped plate of cartilage covered with mucous membrane.
- It is located at the root of the tongue, which is passively bent over the larynx during the act of swallowing, thereby preventing the entrance of food into the trachea.

TRACHEA

- The lungs are paired organs located within the thorax.
- In general, the left and right lungs have two and four lobes, respectively. Trachea is the primary passageway of air to the lungs. It is located between larynx and bronchi.
- The tracheal wall contains cartilaginous rings to prevent collapse of the tracheal airway.
- Each tracheal ring is incomplete (not joined dorsally), which permits variations in diameter by the tracheal smooth muscle.

- The right and left bronchi and their subdivisions (bronchioles) continue all the way to the alveoli. The walls of the airways into the alveoli, particularly the bronchioles, create resistance to airflow.



- The larger the diameter of the airway,less the resistance to airflow. The diameter of the airways can be altered by the degree of contraction of the smooth muscle in these airways.
- Stimulation of the sympathetic nervous system causes relaxation of these walls, which allows air to more readily enter the lungs.
- The respiratory portion is the site of gas exchange between the air and blood and consists of the respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli. This extensive branching of the respiratory channels is called the bronchial tree.
- The alveoli are thin-walled sacs where gaseous exchange occurs.

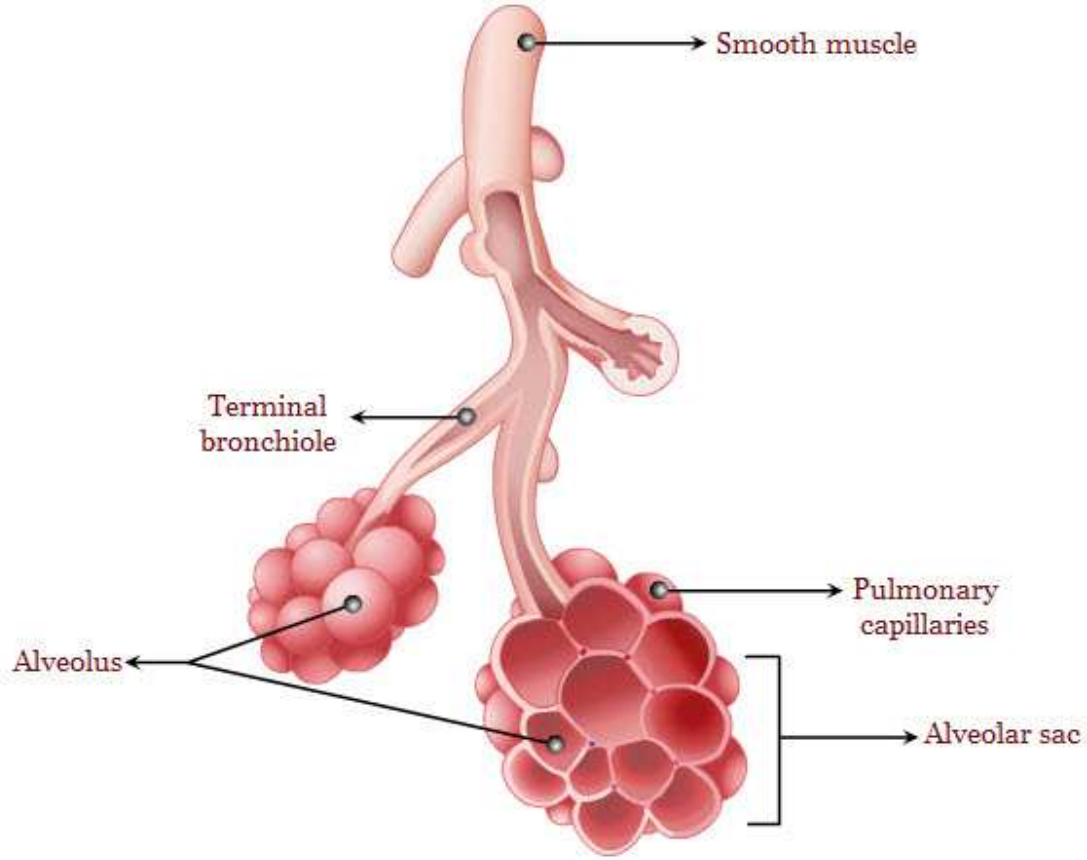
PLEURAL MEMBRANES

- The lungs are surrounded by a serous membrane called as pleural membrane. The superficial layer lining the thoracic cavity is the parietal pleura and the layer closely adhering to the lungs is the visceral pleura.
- The narrow parietal space between these two layers contains a small amount of pleural fluid that allows the two layers to slide over one another during breathing.
- The lungs have an almost friction-free movement within the thorax because of the pleura. Inflammation of the pleural membrane is called pleurisy.
- After the first breath, the lungs become less dense. If the lung floats in water, it indicates that the animal took at least one breath, and therefore was born alive. This fact allows one to determine whether a newborn animal is stillborn or not.

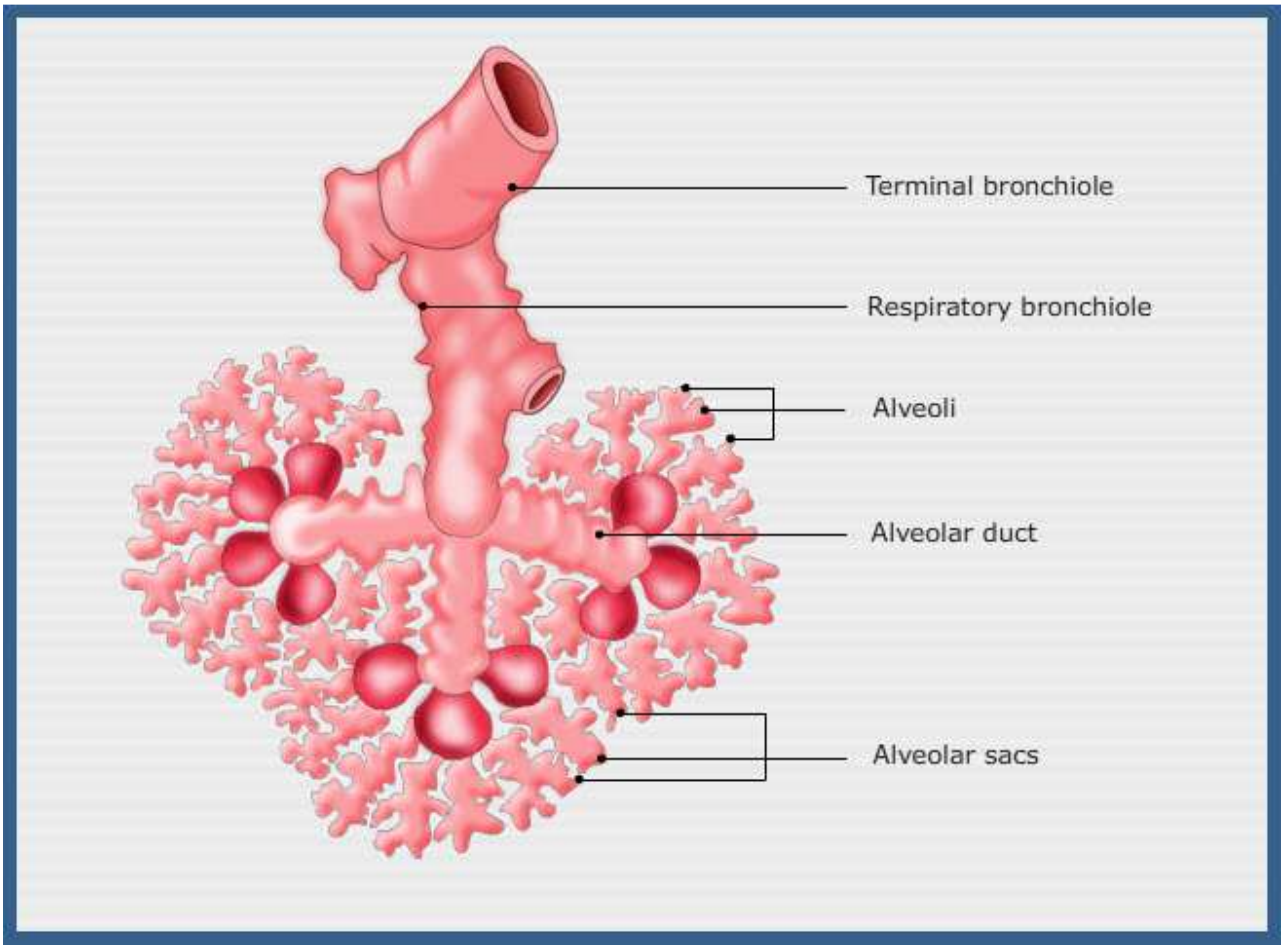
- The pressure inside the intrapleural space is negative due to continuous removal of fluid that is secreted . This negative pressure is vital for the expansion of the lungs.
- If an injury to the chest wall punctures the pleural membrane, it can allow air to enter the intrapleural space, resulting in a condition called as "pneumothorax".

ALVEOLUS

- The pulmonary alveoli are the principal sites of gas diffusion between the air and blood.
- The separation of air and blood, and thus the diffusion distance, is minimal at the alveolar level.



- The alveolar epithelium and the capillary endothelium are intimately associated. Different cells are seen, such as,
    - Type I alveolar cells, predominant,which are simple squamous epithelium and the main site of gas exchange.
    - Type II alveolar (or septal) cells, cuboidal epithelial cells containing microvilli that secrete alveolar fluid containing surfactant.
    - Alveolar macrophages remove debris from the lungs.
    - Fibroblasts produce elastic fibres.



RESPIRATORY MEMBRANE

- The respiratory membrane is where O_2 and CO_2 diffuse across the alveolar and capillary walls. It is a very thin membrane about 0.5 μm thick and consists of four layers. Exchange between alveoli and capillaries.

- Gas moving from alveolus to blood must pass through:
  - Thin layer of pulmonary surfactant
  - Alveolar epithelium : A layer of type I and type II alveolar cells, and alveolar macrophages.
  - The epithelial basement membrane.
  - The capillary basement membrane.
  - Capillary endothelium
  - Plasma
  - Red blood cell membrane
  - Red blood cell to reach hemoglobin

- The pulmonary artery carries deoxygenated blood, while oxygenated blood returns to the left atrium via the pulmonary veins. The darker purple color of venous blood becomes bright red arterial blood during the saturation of hemoglobin with oxygen that has freshly diffused from the alveoli.
- During exercise, cardiac output can increase by as much as eightfold. During this time, blood flow to the lungs must increase. Pulmonary blood vessels dilate. In the horse, pulmonary arterial pressure can be so high as to cause erythrocytes to leak from pulmonary capillaries, a condition called as exercise-induced pulmonary hemorrhage, in athletic horses.
- Pulmonary ventilation
    - Respiration, the process of gaseous exchange, occurs in three steps:
        - Pulmonary ventilation, or breathing, is the mechanical movement of air into (inspiration) and out (expiration) of the lungs.
        - External respiration is the exchange of gases between the lungs and the pulmonary capillaries, which occurs across the respiratory membrane. The blood gains O_2 and loses CO_2.
        - Internal respiration is the exchange of gases between systemic capillaries and tissue. The blood gains CO_2 and loses O_2.

PULMONARY VOLUMES AND CAPACITIES

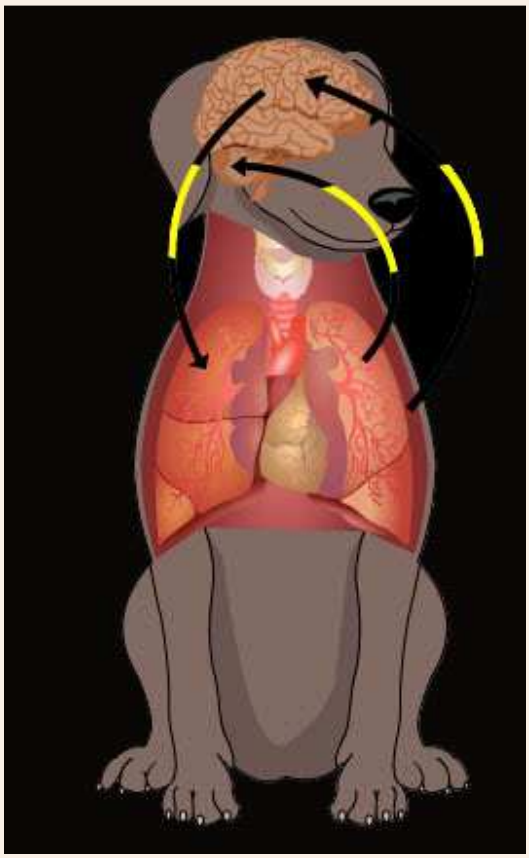
- Pulmonary volumes and capacities are recorded by Spirometer.The recordings are known as Spirogram. It shows the following:
    - *Tidal Volume (VT)* - the volume of air entering or leaving the nose or mouth per breath. It accounts 70% of the total lung capacity. If an animal inhales more forcefully, it can increase the volume of air entering the lungs above normal tidal volume.
    - *Inspiratory Reserve Volume (IRV)* - the volume of gas inhaled into the lungs during a maximal forced inspiration starting at the end of a normal tidal inspiration.
    - *Residual Volume (RV)* - the volume of gas left in the lungs after a maximal forced expiration. Determined by the force generated by the muscles of expiration and the inward elastic recoil of the lungs as they oppose the outward elastic recoil of the chest wall. Dynamic compression of the airways during the forced expiratory effort is also an important determinant of the residual volume - as airway collapse occurs gas is trapped in the alveoli.
    - *Total Lung Capacity (TLC)* - the volume of air in the lungs after a maximal inspiratory effort. Determined by the strength of contraction of the inspiratory muscles in opposition to the inward elastic recoil of the lungs and chest wall.
    - *Inspiratory Capacity (IC)* - the volume of air inhaled into the lungs during a maximal inspiratory effort that begins at the end of a normal tidal expiration.
    - *Functional Residual Capacity (FRC)* - the volume of gas remaining in the lungs at the end of a normal tidal expiration. Balance point between the inward elastic recoil of the lungs and the outward elastic recoil of the chest wall.
    - *Vital Capacity (VC)* - the volume of air expelled from the lungs during a maximal forced expiration starting after a maximal forced inspiration. About 4.5 liters.
    - *Expiratory Reserve Volume (ERV)* - the volume of gas expelled from the lungs during a maximal forced expiration that starts at the end of normal tidal expiration.

GAS LAWS

- Avogadro's hypothesis
  - Equal volumes of different gases at equal temperature contain the same number of molecules.
  - Equal numbers of molecules in the same volumes at the same temperature will exert the same pressure. (One mole of any gas will contain $6.02x10^{23}$ molecules and will occupy a volume of 22.4 L at a temperature of 0°C and a pressure of 760 mmHg.)
- Dalton's law
  - In a gas mixture, the pressure exerted by each individual gas in a space is independent of the pressures of other gases in the mixture.
  - e.g. $PAl_v = PH_2O + PO_2 + PCO_2 + PN_2$ Partial Pressure of O_2= $\%O_2$ content X Total partial pressure
- Boyle's Law: $P_1V_1 = P_2V_2$ (at constant temperature)ie at constant temperature pressure of the gas varies inversly with its volume.
- Charles' law or Gay-Lussac's Law
  - $V_1/V_2=T_1/T_2$
  - $P_1/P_2=T_1/T_2$
  - ie at constant pressure the volume of given mass of the gas increases by 1/273 of its volume at zero degree rise in temperature.
- Henry's Law
  - The weight of a gas absorbed by a liquid with which it does not combine chemically is directly proportional to the pressure of the gas to which the liquid is exposed (and its solubility in the liquid).
- Graham's Law
  - The rate of diffusion of a gas (through the gas phase) is inversely proportional to the square root of its molecular weight.
- Diffusion
  - All gas moves across the blood-gas interface by passive diffusion; that is, net movement is from an area of high partial pressure to low.
- Diffusion of O_2 and CO_2 obey Fick's Law
  - Rate of diffusion or flow of a gas is directly propotional to A X D X $(P_1-P_2)/T$
  - where: A = area , T = thickness, D = diffusivity, P_1-P_2 = partial pressure gradient
  - The volume of gas per unit of time moving across a tissue sheet is directly proportional to the surface area of the sheet, the diffusibility, and the difference in gas concentration between the two sides, but is inversely proportional to the tissue thickness.
  - The surface area is in several square meters depends upon size of the lung. The thickness is generally ½ micron. This large surface area and small thickness are excellent for diffusion.
- Diffusivity or Diffusion constant
  - D = Solubility / Squreroot of Mol.Wt.
  - CO_2 diffuses about 20% slower because of its molecular weight but 24 times faster due to its greater solubility. Therefore the diffusivity of CO_2 is about 20 times that of O_2.
- Dalton's law
  - Dalton's law states that each gas within a mixture exerts its own pressure independent of the other gases present.,

  - Atmospheric air is a combination of nitrogen (N_2),oxygen (O_2), carbon dioxide (CO_2), and water vapor. At sea level, atmospheric pressure is 760 mmHg.
  - Atmospheric air is 78.6% nitrogen, 20.9% oxygen, 0.04% carbon dioxide, 0.06% other gases, and varying water vapor, depending on the humidity. PH_2O presence would cause a dilution of the other gases, and thus their partial pressures would be lowered to maintain the total pressure at 760 mm Hg. The ventilation process does not evacuate the alveoli completely with each breath, but rather it is a gradual replenishment and evacuation
- The approximate composition of alveolar air can be
  - PO_2 = 104 mm Hg
  - PCO_2 = 40 mm Hg
  - PN_2 = 569 mm Hg
  - PH_2O = 47 mm Hg
  - Total = 760 mmHg
- All the components are diluted by water vapor, which is equal to 47 mm Hg in the alveolar air. PH_2O = 47 mm Hg represents 100% humidification of alveolar air at body temperature.
- The PO_2 is lower and the PCO_2 is higher than their respective atmospheric pressures because oxygen is continually diffusing from alveolar air to the tissues (where it is used) and CO_2 is continually diffusing from the tissues (where it is produced) to the alveolar air (where it is expelled).
- The PN_2 of alveolar air is lower than its value in atmospheric air primarily because of its dilution by water vapor.
- Altitude - air pressure at 10,000 ft = 563 mm Hg
- Scuba diving - air pressure at 100 ft = 3000 mm Hg
- Henry's law
  - Henry's law states that the quantity of a gas that will dissolve in a liquid is proportional to its partial pressure and its solubility coefficient.
  - Gases will dissolve in body fluids more readily if they have a greater partial pressure and solubility coefficient.
  - The solubility coefficient of CO_2 is 24 times higher than that of O_2. Therefore, CO_2 dissolves in blood more readily than O_2.
  - In contrast, the solubility of nitrogen is very low, so even though atmospheric air has 79% N2, it has no effect on body functions.

MODULE-35: MECHANISM OF RESPIRATION AND EXCHANGE OF GASES



LEARNING OBJECTIVES

- This module explains,
  - inspiration and expiration mechanics,
  - basic physics prevailing possibility of lung expansion,
  - respiratory passage and its resistance,
  - pulmonary blood flow,
  - vasculr resistance,
  - pulmonary frequencies,
  - physology of ventilation,
  - importance of dead space and
  - physiology of gas exchange.

MECHANISM OF RESPIRATION

- Boyle's law states that at constant temperature and pressure, there is an inverse relationship between volume and pressure of a gas:
  - $P_1V_1 = P_2V_2$
  - where P is the pressure of the gas (mm of Hg), and V is the volume of gas (ml).
- Therefore, the pressure inside a closed container will decrease as the volume of the container increases, and vice versa.
- When this law is applied to the lungs, the pressure inside the lungs decreases as the volume of the lungs increases.
- Air moves into the lungs as a result of a decrease in air pressure within the lungs.
- In most animals, inspiration is an active process.
- The main muscles responsible for quiet inhalation are the diaphragm and external intercoastal muscles. The diaphragm is a dome-shaped muscle innervated by the phrenic nerves.

- As the diaphragm contracts, it increases the horizontal dimensions of the thoracic cavity, thus increasing the volume of the thorax.
- Diaphragmatic contraction enlarges the thorax in a caudal direction, and intercoastals muscle contraction enlarges the thorax in a craniad and outward direction. This causes the volume of the lungs to expand and thereby decreases the pressure inside the lungs. This results in inspiration.
- Contraction of the external intercoastals muscles causes the ribs to move cranially and ventrally, thus increasing the diameter of the thorax. This accounts for about 25% of the entry of air into the lungs.
- In a horse at rest, the intrapleural pressure is negative, about 754 mmHg. As inspiration begins, this pressure drops to approximately 744 mmHg.
- Atmospheric pressure is 760 mmHg at sea level. As the volume of the thoracic cavity increases during inspiration, the parietal pleura is pulled outward, and the visceral pleura is pulled with it. As a result, the pressure inside the lungs, (the alveolar pressure) decreases. Air then flows from an area of high pressure (the atmosphere) to an area of lower pressure (the alveoli).
- As an animal increases the force of inspiration, additional muscles are engaged. These include the sternocleidomastoid muscles that move the sternum rostrally, the scalene muscles that pull the first two ribs forward, and the pectoralis minor muscles that pull several other ribs forward.

EXPIRATION

- Normal expiration is a passive process involving no active muscle contraction. Like inspiration, it is due to pressure gradients, but in an opposite direction. Because of the elastic recoil of the lungs and chest wall, there are two inwardly directed forces resulting 1) the elastic fibers that were stretched during inhalation, and 2) the inwardly directed force due to the surface tension arising from the alveolar fluid.
- As the neural signals to the diaphragm cease, it relaxes and this dome-shaped muscle moves rostrally, thus decreasing the volume of the thoracic cavity.
- The external intercostals also relax, and allow the ribs to move dorsally and caudally, thus further decreasing the volume of the thoracic cavity. This decreases lung volume and causes alveolar pressure to increase approximately 2 mmHg above atmospheric pressure. As a result, air flows out of the lungs to an area of lower pressure.
- During forceful exhalation, the abdominal and internal intercostals muscles contract. This causes the ribs to move caudally and dorsally compressing the abdominal viscera and decreasing the thoracic volume. This increases pressure inside the thoracic cavity and forces air outward.

TYPES OF BREATHING

- There are two types of breathing: *abdominal* and *coastal.*
    - *Abdominal breathing* is characterized by visible movements of the abdomen, Normally in animals the abdominal type of breathing predominates.
    - The *coastal breathing* is characterized by pronounced rib movements. During painful conditions of the abdomen, such as peritonitis, in which movement of the viscera would aggravate the pain, coastal breathing can predominate. Similarly, during painful conditions of the thorax, such as pleuritis, abdominal breathing will be more apparent.
- Eupnea is the normal quiet breathing, with no deviation in frequency or depth.

- Dyspnea is difficult breathing, in which visible effort is required to breathe. The animal is conscious of this state.
- Hyperpnea refers to breathing characterized by increased depth, frequency, or both after severe exercise. The animal is not actually conscious of this state.
- Polypnea is rapid, shallow breathing, similar to panting. Polypnea is similar to hyperpnea in frequency, but not in depth.
- Apnea refers to a cessation of breathing.

SURFACE TENSION OF ALVEOLAR FLUID AIDS IN EXPIRATION

- Alveolar fluid coats the inside surface of the alveoli. The surface tension produces an inwardly directed force causing the alveoli to assume the smallest possible diameter.
- In order to expand the lungs, this surface tension must be exceeded. This surface tension accounts for approximately two thirds of the lung's elastic recoil.
- Surfactant, produced by type II alveolar cells, is a complex of lipids and proteins that reduces the surface tension in much the same way the soap allows lipids to dissolve in aqueous solutions.
- Surfactant decreases surface tension which increases pulmonary compliance (reducing the effort needed to expand the lungs) reduces tendency for alveoli to collapse.
- Since surfactant is one of the last compounds produced during embryonic development, premature animals often have respiratory distress as a result of the underdeveloped respiratory system.
- In the case of sheep, surfactant is released into the alveolar spaces near the beginning of the fourth month of gestation. Its release correlates with a rise in plasma cortisol levels.

COMPLIANCE OF LUNGS

- The distensibility of the lungs is referred to as lung compliance. High lung compliance means that the lungs will expand easily.
- Compliance is seen decreased
    - pulmonary edema resulting from accumulation of fluid in the lungs
    - insufficient of surfactant.

RESPIRATORY FREQUENCY

- Respiratory frequency refers to the number of respiratory cycles each minute.
- It is an excellent indicator of health status, respiratory frequency can be affected by other factors, such as:
    - body size,
    - age,
    - exercise,
    - excitement,
    - environmental temperature,
    - pregnancy, and
    - breed.
- Pregnancy and digestive tract filling increase frequency because they limit the movement of the diaphragm during inspiration.

- When expansion of the lungs is restricted, adequate ventilation is maintained by increased frequency. Thus when cattle lie down, the large rumen pushes against the diaphragm and restricts its movement, and the respiratory frequency is seen to increase.
- It indicates the number of breaths per minute during normal quiet respiratory activity.
- Rate of respiration varies with age, sex, body size, exercise, external environmental temperature pregnancy, fullness of digestive tract, rumination, etc. Pregnancy and fullness of the GI tract increases the frequency because they limit the movement of diaphragm during inspiration.
- Frequency also increases as the environmental temperature rises, to help in the thermoregulation.

Respiratory rate of different animals

Horse	10 – 14 / min (12)
Pig	32 – 58 (40)
Cattle	26 – 35 / min. (29)
Dog	20 – 34 (24)
Fowls	15 – 30 / min
Man	12 – 20
Cat	20 – 40 / min
Rat	97 / min
Sheep & Goat	20 – 34 / min (25)

ARTIFICIAL RESPIRATION

- Artificial respiration is resorted to in cases of
  - cessation of respiration while under general anaesthesia
  - drowning where the animal has been rescued from water – mainly applicable for small animals;
  - poisoning by narcotics or paralytic substances
  - asphyxia from fumes, smoke, gases etc.

Method of administration

In large animals

- The animal is released from all restraints, head and neck are extended to allow a straight passage of air into the lungs and the tongue is pulled out. The elastic ribs are compressed by alternately leaning the whole weight of the body of a person on the hands kept pressed on the ribs and then releasing the pressure about once every 4 or 5 seconds, for the purpose of stimulating normal movements of breathing. As an alternative, a heavy

person may sit himself with some vigour astride the ribs for about 4 or 5 seconds, rise for a similar period and then reseat himself.
- Inhalation of strong of ammonia upon a piece of cotton and held about a foot from the upper nostrils assists in inducing a gasp, which is sign of returning of respiration. The animal is turned on its sides every 2 or 3 minutes to prevent stasis of blood. As long as the heard continues to beat, attempts at resuscitation should be pursued. Resuscitators or mechanical ventilators are useful to give artificial respiration. For pigs and sheep, the same procedures can be applied.

Dogs and cat

- Mouth-to-mouth or mouth-to-nose ventilation, which uses exhaled air to provide 16% O 2 can be used for artificial ventilation. The preferred technique is use of endotracheal intubation and providing ventilation with an anesthetic machine or mechanical ventilator.
- In another technique, the animal is laid on its side with the head at a lower level than the rest of the body. The palm of a hand is placed flat over the upper side of the abdomen and the other hand is placed on the rib cage, weight is placed on the hands for a second or two and then the pressure is released. The motions of artificial respiration should be faster than normal respiration, but a slight pause should always be observed before each rhythmic movement.

Resuscitators

- A frequently used resuscitator consists of mechanism for applying intermittent positive pressure and a facemask or a connector to connect the equipment with endotracheal tube. These apparatus forces air through the mask into the lungs during positive pressure cycle an allow air to flow passively out of lungs during the remaining periods.
- For controlled positive pressure ventilation, either orotracheal intubation or tracheotomy is indicated. The technique is used in comatose, anaesthetized or sedated animals

RESPIRATORY PASSAGEWAY RESISTANCE

- *Upper respiratory passageways* - relatively large, very little resistance to airflow (unless obstruction such as from food lodging or cancer)
- *Lower respiratory passageways* - from medium-sized bronchioles on down, can alter diameter based on autonomic stimulation
- *Parasympathetic* - causes bronchioconstriction
- Sympathetic - inhibits bronchioconstriction
- *Epinephrine* - used to treat life-threatening bronchioconstriction such as during asthma and anaphylactic shock.

PULMONARY BLOOD FLOW

- Dual circulation of the lung
    - Bronchial circulation - arterial supply of tracheobronchial tree (as far as the terminal bronchioles), and other thoracic structures. About 2% of cardiac output.
    - Pulmonary circulation - site of gas exchange between alveolar air and capillary blood. Mixed venous blood in pulmonary artery.

- Main pulmonary artery is much thinner-walled than the aorta. Much less smooth muscle in the walls of the vessels of the pulmonary arterial tree. No highly muscular arteriole.
- As a result low resistance to pulmonary blood flow. Pulmonary vascular resistance (PVR) is about 1/10 of systemic vascular resistance.
- Pulmonary vessels are more distensible than systemic vessels. Blood at a time seen about 30% arteries, 40% capillaries; 30% veins. This can change with lung inflation or during hypoxia.
- Pulmonary capillaries in the adult:280 billion capillaries supply 300 million alveoli with 50-100 m^2 surface area. Diffusion distance normally less than 0.5 micron.

PULMONARY VASCULAR RESISTANCE

- Although pulmonary vascular resistance is very low at rest, it can decrease further, as in exercise, during which blood flow increases. Neural : Sympathetic nerves to larger vessels - NE released.
- Effects somewhat controversial - most likely increases pulmonary vascular resistance and decreases distensibility of larger vessels. Parasympathetic - Ach decreases PVR if tone elevated.
- Humoral : Increase PVR: Norepinephrine, alpha adrenergic agonists, serotonin, prostaglandins F_2 and E_2, angiotensin, endothelin, alveolar hypoxia (hypoxic pulmonary vasoconstriction), alveolar hypercapnia, low pH of mixed venous blood.
- Histamine constricts pulmonary veins. Decrease PVR: Acetylcholine, beta adrenergic agonists, bradykinin, prostaglandin E_1 and I_2 (prostacyclin), nitric oxide.

ALVEOLAR GAS EXCHANGE

- Usually, gas pressure is considered in terms of total pressure, regardless of whether it is a single gas or a mixture of gases. However each gas in the mixture separately contributes to the total pressure.
- It is defined as the pressure exerted by a particular gas in a mixture of gases. The sum of the partial pressures of the gases within a mixture equals the total pressure.
- Because oxygen is consumed and carbondioxide is produced by cells, the venous blood will have a high PCO_2 and a lower PO_2 and vice-versa for arterial blood.
- A more active location would consume more O_2 and produce more CO_2 than less active locations. Because of these differences, the jugular vein blood may not be representative of whole body venous blood (i.e., blood from the right atrium).
- Gases diffuse from an area of their higher concentration to an area of their lower concentration.
- Two gas laws are applicable to gas exchange .
- Dalton's law explains how gases move by diffusion based on pressure differences while Henry's law describes the diffusion of gas based on its solubility.

EXTERNAL AND INTERNAL RESPIRATION

- External respiration, also called pulmonary gas exchange, is the diffusion of O_2 and CO_2 from the alveoli to pulmonary blood. Blood circulating through the body picks up $CO_2$and delivers O_2. As this blood travels through the pulmonary capillaries, CO_2 diffuses into the alveoli while O_2 diffuses from the alveoli to pulmonary blood. The exchange of these gases occurs independently and passively.

- Pulmonary gas exchange is facilitated by a very thin respiratory membrane.
- Ventilation is generally regarded as the process by which gas in closed places is renewed or exchanged. As it applies to the lungs, it is a process of exchanging the gas in the airways and alveoli with gas from the environment. The main function of breathing is to provide gas for ventilation.
- In addition, there is a close association between the amount of gas reaching the alveoli (ventilation), and the blood flow through the pulmonary capillaries (perfusion).
- Internal respiration, or systemic gas exchange, occurs at the tissue level, where there is an exchange of O_2 and CO_2 between systemic capillaries and tissue. O_2 diffuses from the capillaries into the cells; CO_2 diffuses from the cells into the systemic capillaries.

VENTILATION-PERFUSION RELATIONSHIPS (V/Q)

- For optimal gas transfer to occur in the lung, ventilation and perfusion must be matched. In man for the whole lung, alveolar ventilation and pulmonary blood flow are both about 5 L/min. Therefore, for the whole lung V/Q is about 0.8 - 1.2. Local airway responses and hypoxic pulmonary vasoconstriction help match ventilation and perfusion.
- Low Oxygen in alveolus→ vasoconstriction
- High Oxygen in alveolus→ vasodilation
- High CO_2 in alveolus→ dilate bronchioles
- Low CO_2 in alveolus → constrict bronchioles

DEAD SPACE VENTILATION

- The tidal volume is used to ventilate not only the alveoli, but also the airways leading to the alveoli. Because there is little or no diffusion of oxygen and carbon dioxide through the membranes of most of the airways, they compose part of dead space.
- The other part of dead space ventilation is made up of alveoli with diminished capillary perfusion.
- Ventilating these alveoli is ineffective in producing changes in the blood gases. Ventilation of nonperfused alveoli and the airways, because neither accomplishes exchange of the respiratory gases. is referred to as physiologic dead space.
- Physiologic dead space is defined as the volume of gas that is inspired but takes no part in gas exchange in the airways and alveoli. Therefore, the tidal volume has a dead space component and an alveolar component.
- More ventilation of lower (with respect to gravity) regions of the lung than upper regions of the lung. Intrapleural (intrathoracic) pressure is less negative at the bottom of the lung than at the top because of the weight of the lung and the configuration of the chest wall.
- Therefore the transpulmonary pressure gradient is greater at the top of the lung than at the bottom.
- Therefore the alveoli at the top of the lung are at a higher, less compliant point on the pressure-volume curve than those at the bottom.
- Therefore the alveoli at the bottom of the lung increase their volume more with each inspiration and decrease their volume more with each expiration during eupnea (from FRC).
- Dead space ventilation is a necessary part of the process of ventilating the alveoli and is not totally wasted. It assists in humidifying inhaled air and in cooling the body under certain conditions, such as when panting is necessary.
- During panting, the respiratory frequency increases and the tidal volume decrease so that alveolar ventilation remains approximately constant.

PHYSIOLOGY OF GAS EXCHANGE

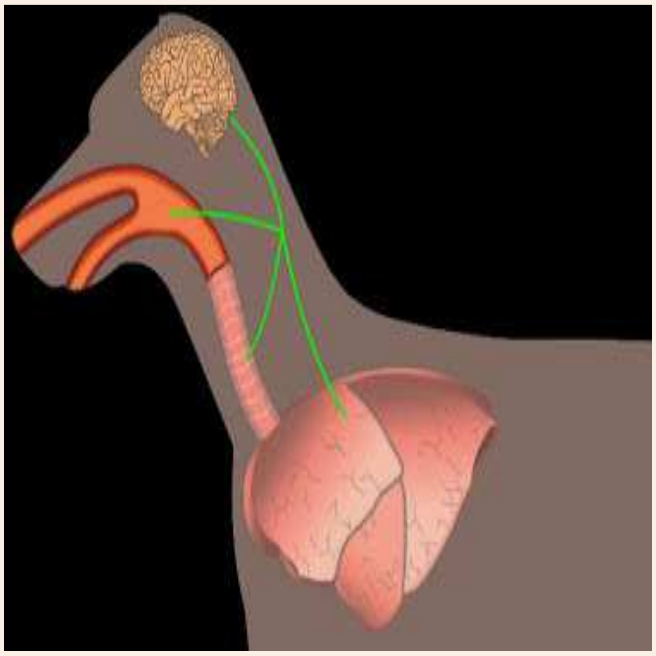
- Ventilation brings O_2 to the alveoli and removes CO_2. Because O_2 is being consumed in the tissues, a pressure difference exists for its diffusion from alveoli to venous blood (which then becomes arterial) and from arterial blood to the tissues. Because CO_2 is being produced in the tissues, a pressure difference exists for its diffusion from tissue to arterial blood (which then becomes venous) and from venous blood to the alveoli. **(Animation831-After title correction)**

Gases	Venous Blood	Alveolar Air	Arterial Blood	Tissues
O_2	40	104	100	30 or less
CO_2	45	40	40	50 or more
N_2	569	569	569	569
Water vapor	47	47	47	47
Total	701	760	756	696

Values are in mm of Hg

- The total pressure in venous blood is somewhat less than atmospheric pressure because the volume of CO_2 produced is lower than the volume of O_2 consumed.
- As because not all of the blood going to the lungs is arterialized (nonperfused alveoli) the difference in PO_2 exists.
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MODULE-36: CONTROL OF RESPIRATION



LEARNING OBJECTIVES

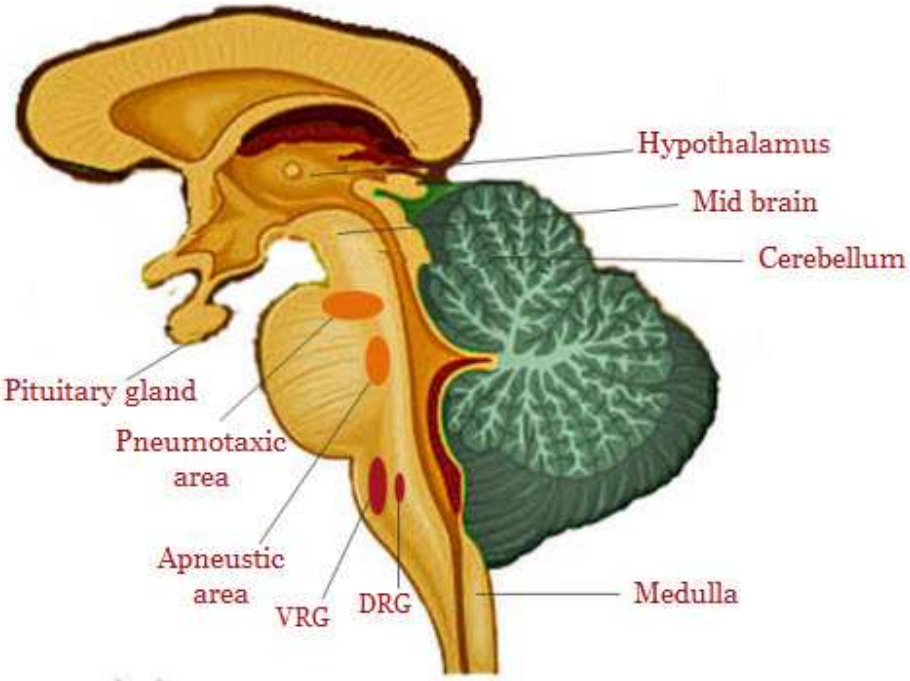
- This module helps to gain knowledge on
  - regulation of respiration,
  - means of regulation,
  - structures associated,
  - mechanical factors assisting the regulation and
  - hormonal participation.

NEURAL CONTROL

- Pulmonary ventilation is regulated closely to maintain the concentrations of H^+, CO_2, and O_2 at relatively constant levels while meeting the needs of the body under varying conditions.
- If either the H^+ or the CO_2 concentration increases or if the O_2 concentration decreases, their levels will be returned to normal by increasing ventilation.
- Conversely, if either the H^+ or CO_2 concentration decreases or if the O_2 concentration increases, pulmonary ventilation will be decreased. This regulatory mechanism is controlled by changes in tidal volume, frequency of respiratory cycles, or both.

RESPIRATORY CENTRES

- The central mediator of these changes is the respiratory center in the brain stem which has four specific regions .
- The medullary rhythmicity area is located in the medulla oblongata, and it controls the basic respiration rhythm.
- It consists of two areas, the inspiratory and expiratory areas, also called the dorsal respiratory group and ventral respiratory group, respectively.
    - *Dorsal respiratory group:* group of neurons predominately associated with inspiratory activity (lung inflation and termination of inspiration).
    - *Ventral respiratory group:* group of neurons containing inspiratory and expiratory neurons (assist in inspiration begun by those in the dorsal respiratory group and also assists expiration).
- The respiratory center is gray matter in the pons and the upper medulla, is very sensitive to the PCO2 in the arteries and to the pH level of the blood.



- The CO_2 can be brought back to the lungs in three different ways; dissolved in plasma, as carboxyhaemoglobin, or as carbonic acid. That particular form of acid is almost broken down immediately by carbonic anhydrase into bicarbonate and hydrogen ions. This process is then reversed in the lungs so that water and carbon dioxide are exhaled.
- The medulla oblongata reacts to both CO_2 and pH levels which triggers the breathing process so that more oxygen can enter the body to replace the oxygen that has been utilized.
- The medulla oblongata sends neural impulses down through the spinal chord and into the diaphragm.
- There are two other ways the Medulla Oblongata can be stimulated.
    - The first type is when there is an oxygen debt (lack of oxygen reaching the muscles), and this produces lactic acid which lowers the pH level. The medulla oblongata is then stimulated.
    - The second type occurs when the major arteries in the body called the aortic and carotid bodies, sense a lack of oxygen within the blood and they send messages to the medulla oblongata.
- The inspiratory area sends signals to the diaphragm via the phrenic nerve and to the external intercostal muscles via the intercostal nerves. These signals cause muscle contraction resulting in inspirations. When these signals cease, inspiration is concluded, which allows the diaphragm and external intercostal muscles to passively relax, during which time the elastic recoil of the lungs and thoracic walls causes the volume of the thoracic cavity to decrease.
- Transection between the spinal cord and medulla oblongata stops breathing.
- Although not active during quiet breathing, forceful expiration requires signals from the expiratory area that cause contraction of the internal intercostals and abdominal muscles.
- Contraction of these muscles further decreases the volume of the thoracic cavity, thus increasing exhalation.

PNEUMOTAXIC AND APNEUSTIC AREA

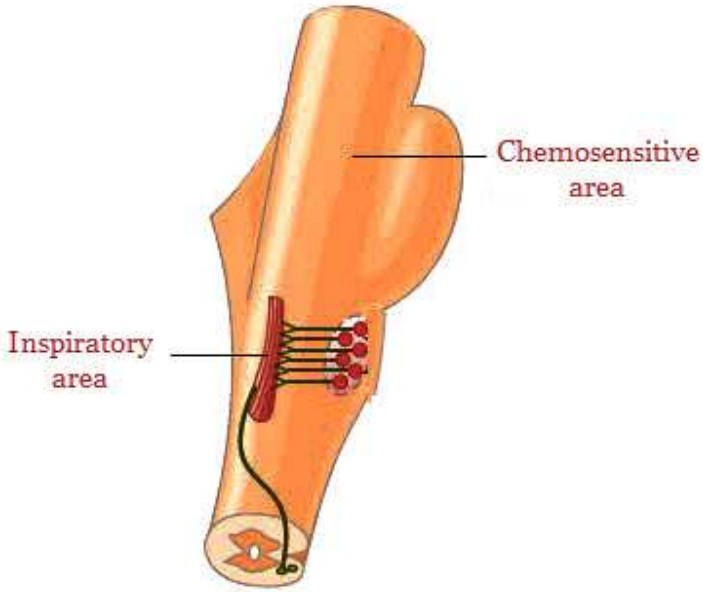
- Pneumotaxic centre believed to modulate respiratory center sensitivity to inputs that activate termination of inspiration and facilitate expiration .
- Apneustic centre believed to be associated with deep inspirations.
- The pneumotaxic area, also called the pontine respiratory group, sends inhibitory signals to the inspiratory area. These signals primarily function to prevent overfilling of the lungs.
- Conversely, the apneustic area located in the lower pons sends stimulatory signals to the inspiratory area that prolongs inspiration.
- The pneumotaxic area can override the apneustic area. Impulses going to the respiratory center (afferent impulses) from several receptor sources have been identified. Importantly the Hering-Breuer reflexes.

HERING - BREUER REFLEX

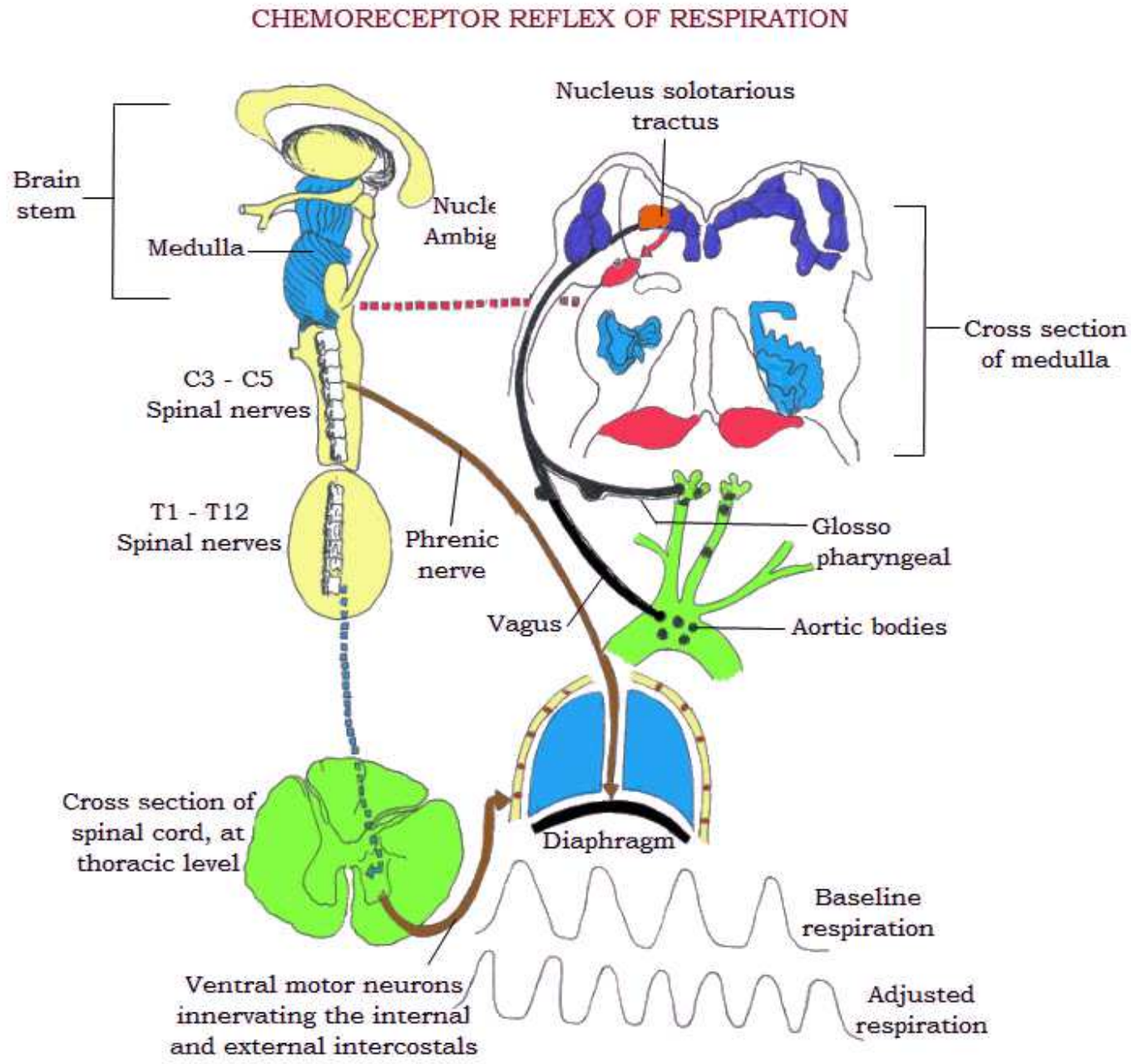
- The receptors for these reflexes are located in the lungs, particularly in the bronchi and bronchioles.
- The nerve impulses generated by these receptors are transmitted by fibers in the vagus nerves to the respiratory center.
- The effect of inflation-receptor stimulation is to inhibit further inspiration (stimulation of neurons in the dorsal respiratory group) and to stimulate expiratory neurons in the ventral

respiratory group. However the tidal volume can be increased by a person with the modulation of pneumotaxic centre.

- There are other peripherally located receptors that modify the basic rhythm. Stimulation of receptors in the skin are excitatory to the respiratory center, and deeper than usual inspiration can be noted.
- Their excitation to the inspiratory area is through the apneustic area because inspiratory gasps are noted. Advantage is taken of these receptors when breathing stimulation is desired in newborn animals.



- Rubbing the skin with a rough cloth often initiates the breathing cycles. Several respiratory reflexes originate from receptors in the upper air passages.
- Stimulation of the mucous membranes in these regions causes reflex inhibition of breathing. Eg: inhibition of breathing that occurs during swallowing; also, in diving birds and mammals, there is a reflex inhibition of breathing when they submerge.



- Stimulation of the laryngeal mucous membrane in the unanaesthetized animal causes not only inhibition of breathing, but also usually powerful expiratory efforts (coughing and sneezing).
- Ordinary respirations proceed involuntarily. They can be altered voluntarily hastened, slowed, or stopped altogether, for a while. Phonation and abdominal pressure in the expulsive acts of defecation, urination, and parturition are examples of complete voluntary control of the respiratory movements.

- Afferent impulses from pressure receptors in the carotid and aortic sinuses have as their principal function a role in the regulation of circulation, but impulses from these receptors also go to the respiratory center.
- The impulses are inhibitory in nature-the higher the blood pressure, the greater the inhibition to respiration. Because of the influence of inspiration on return of blood to the heart, it can be seen that the reduction in inspirations would slow down the return flow of blood to the heart and thus help to lower blood pressure.

HUMORAL CONTROL

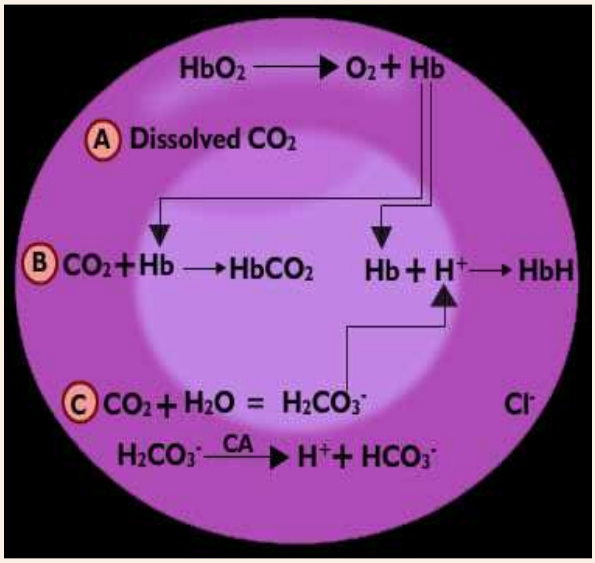
- Humoral control refers to those factors in the body fluids that influence ventilation like CO_2, O_2 and H^+ .
- Their concentrations in the blood affect alveolar ventilation in several ways:
    - Carbon dioxide increase causes alveolar ventilation to increase; its decrease causes alveolar ventilation to decrease.
    - H^+ increase causes alveolar ventilation to increase; its decrease causes alveolar ventilation to decrease.
    - Oxygen decrease causes alveolar ventila tion to increase; its increase causes alveolar ventilation to decrease.

- The respiratory system functions to bring in O_2 and eliminate CO_2 from the body. This function is assisted by specialized receptors called chemoreceptors that monitor the levels of CO_2, O_2 and H^+, and then send such information to the respiratory center.
- These chemoreceptors are located in several locations. There are central chemoreceptors found in the medulla oblongata that respond to changes in cerebrospinal fluid H^+ and PCO_2. Because of the much greater diffusibility of carbon dioxide, as compared with H^+, it is distributed more quickly from the blood to the interstitial fluid of the medulla and to the cerebrospinal fluid than hydrogen ions.
- The H^+ concentration of the interstitial fluid of the brain stem is the deciding stimulus for respiratory drive. The influence of CO_2 is exerted by its conversion to H^+ through the hydration reaction.
- Peripheral chemoreceptors include the aortic bodies and carotid bodies whose removal eliminates a respiratory response to hypoxia.
- The aortic bodies are a cluster of chemoreceptors in the aortic arch; the carotid bodies are oval nodules in the wall of the left and right common carotid arteries, where they bifurcate into the internal and external carotid arteries.
- Axons from the chemoreceptors in the aortic bodies are part of the vagus nerve (cranial nerve X), whereas those of the carotid bodies project in the glossopharyngeal nerves (cranial nerve IX).
- The levels of CO_2 and H^+ are highly correlated.
- Throughout the body, CO_2 is quickly converted to carbonic acid catalyzed by the enzyme carbonic anhydrase. Carbonic acid dissociates into HCo_3^- and H^+. Therefore, increase in PCO_2 lead to increases in H+ while decreases in CO_2 lead to decreases in H^+. As a result, has a large affect on respiration, whereas P O_2 affects respiration only if its levels change substantially.
- Increases in arterial blood CO_2, called hypercapnia, cause an increase in H^+. This has a particularly large effect on central chemoreceptors since there is little protein within the cerebrospinal fluid to buffer the H^+. Activation of the central chemoreceptors causes increased respiration rate, possibly causing hyperventilation. Conversely, low arterial blood CO_2, called hypocapnia, inhibits respiration. Large drops in arterial PO_2 increase ventilation by stimulating peripheral chemoreceptors.

INFLUENCES OF PO_2 IN VENTILATION RATE

- The influence of oxygen is transmitted from the carotid and aortic bodies to the respiratory center. The carotid and aortic body receptors also respond to carbon dioxide and hydrogen ion concentration, but the effectiveness of the carotid and aortic body response to carbon dioxide and hydrogen ions is far less than the response from the brain stem.
- Thus, the carotid and aortic bodies are considered to be the most influential for the regulation of oxygen. These bodies are distinct structures with an abundant blood supply located just outside the aortic arch, at the division of the carotid arteries. They respond to changes in the PO_2 (arterial) of blood.
- Blood with reduced amounts of hemoglobin, and consequently less oxygen, has the same PO_2 as blood with normal hemoglobin and oxygen, and thus no ventilation response would be elicited because there is no change in PO_2.
- In anemia ventilation might be increased, not because of less oxygen, but because of greater H+ concentration caused by reduced buffering associated with the hemoglobin decrease. In the case of carbon monoxide poisoning and lack of oxygen carried by hemoglobin, ventilation is not increased, not only because the PO2 is normal, but also because there is adequate hemoglobin present for buffering hydrogen ions.
- Arterial blood PO2 must be in the range of 30 to 60 mm Hg for the respiratory center to receive stimulation to ventilation from the carotid and aortic bodies. At this stage hemoglobin is still about 90% saturated with oxygen at a PO2 of 60 mm Hg.
- Also, the slowing effect of an increased arterial PO2 would not normally be observed in animals breathing atmospheric air because the arterial PO2 seldom increases above 100 mm Hg.
- The slowing is noted in anesthetized animals breathing an oxygen-enriched atmosphere, in which the arterial PO2 could increase to 350 to 400 mm Hg.
- The regulation of ventilation by oxygen is not ordinarily thought to be important. There is usually no problem in maintaining arterial blood PO2 in the range of 80 to 100 mm Hg, and it is not advantageous to have it higher than 100 mm Hg because hemoglobin is almost saturated at that partial pressure.
- Ventilation could even be reduced to about 50% of normal and hemoglobin still would be considerably saturated. The most important chemical factor in the regulation of ventilation is the concentration of carbon dioxide; relatively small changes can have an effect.
- The regulation of ventilation by oxygen becomes more important in such conditions as pneumonia and pulmonary edema, in which gases are not diffused as readily through the respiratory membrane.
- Decreased diffusion is more noticeable for oxygen than for carbon dioxide because of the smaller diffusion coefficient for oxygen. Hyperventilation caused by oxygen lack can therefore reduce the carbon dioxide concentration (because CO2 readily diffuses) and thus reduced formation of hydrogen ions , so that they become ineffective in stimulating increased ventilation.
- The oxygen deficiency mechanism (originating from the carotid and aortic bodies) continues to function and provides the drive to increase ventilation.

MODULE-37: TRANSPORT OF GASES



LEARNING OBJECTIVES

- This module explains to understand
  - various means of oxygen and corbondioxide transport,
  - their significance and
  - factors modifying their transport.

OXYGEN TRANSPORT

Oxygen is transported in blood in two forms :

- as physically dissolved O_2
- O_2 in combination with haemoglobin

PHYSICALLY DISSOLVED O_2

- The O_2 diffuses across the alveolar membrane and the lung capillary endothelium into the blood plasma where it is physically dissolved according to its solubility co-efficient and partial pressure.
- 0.003ml of O_2 is dissolved in 100ml plasma at a PO_2 of 1mmHg.
- At a PO_2 of 100mmHg, 0.31ml O_2 gets dissolved in each 100ml blood.

TRANSPORT OF O_2WITH HAEMOGLOBIN

- After entry into the blood plasma, most of the O_2 enters across the red cell membrane into the cell. The major portion of O_2 is carried by blood is not in physical solution but is associated with haemoglobin molecule inside the red cells.
- If plasma is exposed to an atmosphere of 100mm.Hg partial pressure of O_2 only 0.3ml of O_2 is taken up, But when blood is similarly allowed to equilibrate at an O_2 tension of 100mmHg, it can carry 19 – 20 ml of O_2/ 100ml of blood.
- Hb present within erythocytes does the extra uptake.
- Each Hb molecule contains 4 heme groups. Each heme molecule contains one Fe^{++} atom in a reduced state.

- Each molecule of Hb can combine with 4 molecules of O_2 depending on the relative concentration of Hb and O_2 in blood. When reduced Hb combines with O_2 it has been oxygenated then it is called Oxy – Hb.
- Oxygen binding with Hb is a four step process; the O_2 affinity of a particular heme is influenced by the oxygenation of other hemes in that haemoglobin molecule. These heme-heme interactions are responsible for the sigmoid shape of the oxyhemoglobin curve.
- Greater the concentration of Hb in blood greater the amount of O_2 carrying capacity.
- At 150 mm Hg PO_2, the blood is nearly saturated.

OXY HEMOGLOBIN DISSOCIATION CURVE

- The Loading and unloading of O_2 ability of the Hb in the form of a graph referred as oxy – hemoglobin dissociation curve.
- This graph is obtained when the Hb is allowed to equilibrate with various PO_2 and the values of percentages of Hb saturation (X axis) against the PO_2 (Y axis), are plotted. For *oxy Hb* the curve is *"S" or sigmoid shaped* and for *myoglobin* the dissociation curve is *rectangular hyperbola* in shape.
- Under normal conditions, at a PO_2 of 100mm. Hg, blood leaving the lungs is 95 – 98% saturated with O_2. Further increase in PO_2 do not increase O_2 carrying capacity of blood but increase the amount of O_2 in physical solution according to Henry's law.
- One gram of Hb can transport 1.34 ml of O_2. The average hemoglobin concentration is 15 g/ 100 ml of blood. In atrial blood at a PO_2 of 100 mm.Hg, Hb is 97.5% saturated with O_2 and transport 15 x 1.34 x 97.5/100 = 19.6 ml / 100 ml or 19.6 volumes percent.

- 100 ml of blood carries 19.9 ml of O_2 of which 19.6 ml in combination with hemoglobin and remaining 0.3 ml in physical solution As the arterial blood reaches the tissue, O_2 is unloaded from the blood to tissues, the O_2 saturation falls to about 72% in venous blood.
- At 72% saturation of O_2, the blood will have 14.5 ml O_2. Hence each 100 ml blood unloads approximately 5 volume percent of O_2 to tissues. PO_2 of venous blood is 40 mm Hg.
- Shift of dissociation curve to right results is greater release of O_2 from oxy Hb due to decreased affinity of Hb to O_2. A shift to left increases the affinity of Hb to O_2 and decreases O_2 released from Hb. Variation occur in O_2 dissociation of foetus and adults.
- The extent of oxygen dissociation from the oxy hemoglobin dissociation curve is influences by.
    - PO_2
    - PCO_2
    - H^+ ion concentration
    - Temperature
    - Concentration of DPG (2 – 3 diphosphoglycerate) in erythrocytes.
- Increase in H^+ ion concentration and CO_2 level shifts the curve to down the right as also increase in temperature and 2 – 3 DPG. The 2-3 DPG level increases in hypoxia. Shifting the curve to down and right causes increase in release of O_2 from the Hb.
- The shift of oxy-Hb dissociation curve to down and right by increased CO_2 tension is termed as *Bohr's effect*. In tissues as O_2 is unloaded, CO_2 is simultaneously taken up by blood and this enhances unloading of O_2.

CARBON DIOXIDE TRANSPORT

- The CO_2 produced during metabolism is a waste product and has been eliminated. The flow of CO_2 is effected under a continuous pressure gradient.

Tissues	Venous blood	Alveolar air	Expired air	Atmospheric air
50 mm Hg	46 mm Hg	40 mm Hg	32 mm Hg	0.3 mm Hg

- The transport of CO_2 is effected in the following ways.
  - As physically dissolved CO_2
  - Transport in chemical combination

PHYSICALLY DISSOLVED CO_2

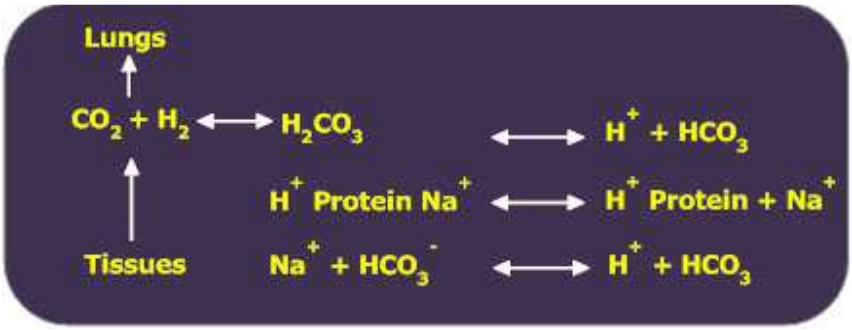
- Compared with O_2, CO_2 is 24 times more soluble in blood plasma. Only about 5 – 7 % of total CO_2 carried by blood is in a simple physical solution.
- The factors that determine the transport are the partial pressure of CO_2 (Henry's law) and temperature.
- Both plasma and cells can transport CO_2 in a physically dissolved state. (The solubility factor per litre for CO_2 at 38°C is 0.03 for plasma and 0.025 for cells).

TRANSPORT IN CHEMICAL COMBINATION

- By far the greater proportion of CO_2 is transported in blood in chemical combination as HCO_3.
- Majority of the CO_2 diffusing into the blood combines with water *forms carbonic acid*, which ten rapidly dissociates into HCO_3ions and H^+. This reversible reaction is kept moving to the right because the H^+ is buffered by plasma proteins / Hb.
  - Formation of HCO_3 in plasma
  - Formation of HCO_3 by RBCs *(Isohydric Transport of CO_2)*

FORMATION OF HCO_3 IN PLASMA

- The Chief buffers of plasma that provides for buffering action by combining with H^+ is protein system (Proteinate / Protein)
- The reaction is plasma is as follows :

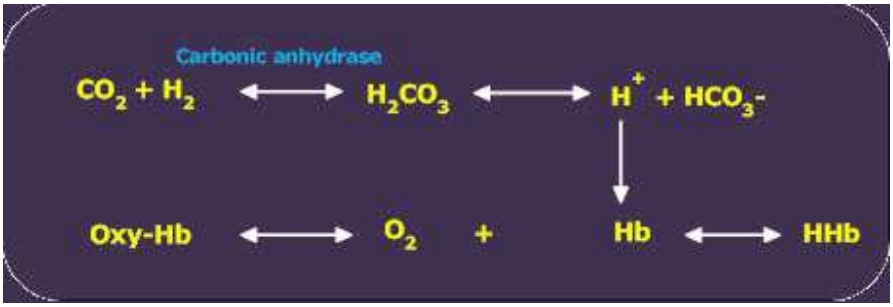


- The PCO_2 in tissues is higher than in arterial blood causes diffusion of CO_2 into blood and forces the preceding reversible reaction to the right causing an increase in HCO_3. This increase in HCO_3 represents transport due to plasma proteins.

- When venous blood reaches the lungs, CO_2 diffuses into alveolar air and is exhaled out. This causes the preceding reaction to shift to left.
- The pH of arterial plasma decreases about 0.025 units in becoming venous plasma.
- The transport of CO_2 as HCO_3^- by plasma proteins represents about 4% of total CO_2 transport.

FORMATION OF HCO_3 BY RBCs

- The CO_2 from tissues enter plasma then into the RBCs. H_2CO_3 formation by the reaction of CO_2 with water is too slow to less of importance in plasma but inside the RBCs, the enzyme carbonic anhydrase catalyses the reaction between CO_2 and H_2O accelerating the rate of about 5,000 fold.
- H_2CO_3 rapidly and spontaneously dissociates into H^+ and HCO_3. Since an increase in H^+ concentration is severely detrimental to an organism, a base buffer must be available to remove the H^+ ions. The Hb provides the buffering, driving the reaction to the right.
- In the tissue Oxy-Hb delivers O_2 and becomes deoxygenated Hb. The deoxygenated Hb is a weaker acid than Oxy-Hb and function as a better buffer readily combines with $H^+$and facilitates the formation of HCO_3 form CO_2.



- Hb is a major buffer in blood that removes free H^+ from blood and an equal quantity of HCO_3 is left dissolved in fluid. The HCO_3 diffuse out of RBC into the plasma due to concentration gradient by exchanging choloride ions from the plasma across the RBC membrane referred as *chloride shift or Hamburger shift*.
- The conversion of CO_2 via H_2CO_3 to HCO_3 ion in erythrocytes accounts for 70% for CO_2transport.

- In the lungs, high PO_2 favours diffusion of O_2 into the red cells and oxygenates Hb. On oxygenation HHb becomes Oxy Hb and releases H^+ ions. This H^+ ion combine with HCO_3 and drives the reaction to the left.

TRANSPORT AS CARBAMINO HEMOGLOBIN

- Haemoglobin contains terminal amino group (NH) reversibly react with CO_2 to from an unstable *carbamino compound*.
- About 20% CO_2 is transported as carbamino haemoglobin (Though plasma proteins can also form carbamino compounds, this reaction in plasma is negligible because free NH group is relatively less in plasma proteins).
-

CO_2 EQUILIBRATION CURVE

- The total quantity of CO_2 combined with blood in all forms of transport of CO_2 depends on PCO_2, which can be expressed through the CO_2 equilibration curve.
- The PCO_2 ranges between 40mm. Hg in the atrial blood and 45mm. Hg in venous blood. At the capillary bed the concentration of CO_2 rises to 52 volumes percent and the level falls to 48 volumes percent as the blood passes though lungs, hence 4 volumes percent of this is actually exchanged in the process of transporting CO_2 from tissues to lungs.
- The combining of CO_2 with Hb is influenced by degree of oxygen saturation of Hb. Greater the O_2 saturation, lesser is the CO_2, carrying power, more the displacement of CO_2referred as *Haldane effect* .
- On oxygenation Hb becomes a stronger acid. (Reduced Hb being a weaker acid and displaces CO_2 from carbamino Hb. Moreover, increased acidity of oxy-Hb causes increased release of H^+ ion which combines with HCO_3^- to form H_2CO_3. In the lungs H_2CO_3 on dissociation releases CO_2 and H_20.

HYPOXIA

- It is a state of inadequate O_2 supply to tissue. Absence of O2 is referred as anoxia. The Cerebral effects seen will be mostly as excitement, hallucination, restlessness and unconsciousness. When blood O_2 below 13%, breathing is stimulated. The anoxia can be classified into the following types.

Anoxic hypoxia (Ambient hypoxia)

- This is due to reduced alveolar ventilation and reduced O_2 tension in blood caused due to obstruction of air passage, paralysis of respiratory muscle, pulmonary disease or congenital diseases of heart. Ambient anoxia is caused by low pO_2 in environmental air in high altitudes or closed space. Symptoms seen are dyspnoea, alkalemia, high cardiac output, increased pulse pressure, dilatation of peripheral vessels.

Anemic anoxia

- Decrease in O_2 carrying capacity of Hb due to low concentration of Hb seen in haemorrhages, anaemia, CO poisoning. In this case the partial pressure of O_2 is normal but insufficient O_2 delivery to tissue results in increased cardiac output and rapid circulation time.

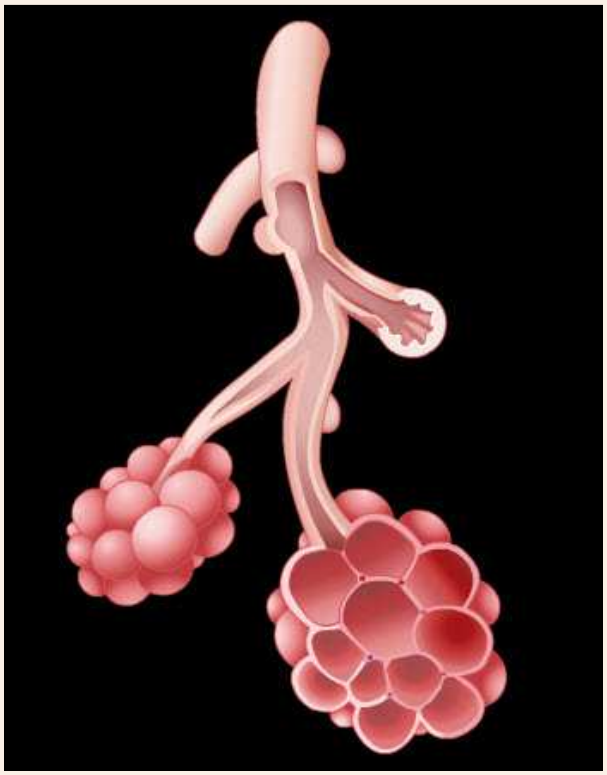
Stagnant hypoxia

- The O_2 content of arterial blood is normal, but the tissues receive low O_2 supply because general or local circulation failure.

Histotoxic hypoxia

- The tissue oxidation is interfered due cyanide poisoning. Paralysis of cytochrome oxidase system is responsible for this condition. The amount of O_2 and pO_2 are normal in arterial blood and above normal in venous blood.

MODULE-38: ROLE OF RESPIRATORY SYSTEM IN ACID – BASE BALANCE



LEARNING OBJECTIVES

- This module deals with,
  - acid-base balance of body,
  - their importance,
  - respiratory support to restore the imbalance and
  - other non respiratory functions of lungs.

ROLE OF RESPIRATORY SYSTEM

- Blood PCO_2 can be varied extensively because the partial pressure of CO_2 in alveoli determines the amount of CO_2 dissolved in blood.
- The respiratory mechanism depends upon the sensitivity of respiratory control systems to change in blood PCO_2 and pH.
- A small increase in PCO_2 or decrease in pH stimulates pulmonary ventilation, so that CO_2 expiration increases.
- If an acid is added to body fluids, chemical buffering results in formation of additional H_2CO_3 and leading to a depletion of HCO_3

$H^+ + HCO_3^- \text{-----} > H_2CO_3 \text{-----} > CO_2 + H_2O$

- When the production of CO_2 is increased or pulmonary hypoventilation occurs, it leads to the accumulation of CO_2 in blood Therefore, the pH falls slightly.
- However, the increase in CO_2 and decrease in pH stimulate respiration, causing a rapid expiration of extra CO_2. Then blood PCO_2 decreases and pH returns to normal.

- When an alkali is added to blood or pulmonary hyperventilation occurs, this causes a decrease in H_2CO_3 {CO_2} level in blood. The respiratory system is inhibited and additional CO_2 is retained in ECF. These increase PCO_2 to balance the increased HCO_3 resulted from added alkali.

METABOLIC ACIDOSIS

- In *metabolic acidosis*, (ketosis, diabetes mellitus, renal acidosis and diarrhoea) blood HCO_3 falls either as a result of a reaction with acid or due to direct loss from ECF and pH falls. This results in fall of the blood buffer bases.
- Usually no change in plasma PCO_2 because of buffer action can be detected. However, a fall in pH results in increased alveolar ventilation and a fall in PCO_2.
- Decreased PCO_2 will bring the ratio of conjugate base to weak acid back to normal . However, the total bases will be less than normal and this requires renal correction – the excretion of H^+ and restoration of plasma HCO_3^-.
- The academia stimulates secretion of H^+ ion by the renal tubule. This ensures reabsorption of all HCO_3 from tubular fluid and the excess H^+ will begin to acidify the urine. For each H^+ ion secreted one HCO_3 will be reabsorbed in to plasma.

METABOLIC ALKALOSIS

- Metabolic alkalosis, (vomiting, K^+ deficiency) involves the gain of base (OH^- or HCO_3^-) or loss of strong acid by ECF. In these conditions, there is an increase in HCO_3 in ECF, resulting in increased base content.
- The response of the body is opposite to the one observed in metabolic acidosis.
- Alkalemia results in rise in pH which will depress pulmonary ventilation and PCO_2 will rise. This respiratory compensation thus will bring pH back to normal.
- Renal correction consists of decreased secretion of pH ions and so increased excretion of HCO_2.

RESPIRATORY ACIDOSIS

- If excretion of CO_2 by the lungs falls below the rate of CO_2 production in the body, respiratory acidosis develops.
- There will be increase in blood PCO_2 *(hypercapnia)* and the primary defect will be in the inability of lungs to expire CO_2 at a normal rate. This may be due to
    - Depression of respiratory centres in CNS.
    - Abnormality of chest wall or respiratory muscles, which prevents enlargement of thorax.
    - Obstruction to gas movement in lungs
- A rise in PCO_2 cause increase in H_2CO_3 and buffer reaction prevents the fall of pH caused by rise in H_2CO_3.
- Renal compensation then follows. Low pH stimulates secretion of H^+ into urine with a rise in plasma HCO_3^-.

RESPIRATORY ALKALOSIS

- When alveolar hyperventilation occur the expiration of CO_2 may exceed the rate of its production within the body and respiratory alkalosis develops. There will be low plasma PCO_2 *(Hypocapnia)* and alkalemia.
- Hyperventilation is caused by abnormal stimulus to respiratory centres either directly as in NH_3 toxicity or through hypoxemia acting through peripheral chemoreceptors. Buffer reaction follows :

$HHb + HCO_3$ ---->$Hb^- + H_2CO_3$ ---->$CO_2 + H_2O$

- The HCO_3 falls. Renal compensation begins ; alkalemia depresses H^+ ion secretion by renal tubules and excretion of filtered HCO_3 rises. This result in further fall of plasma HCO_3 and the ratio of HCO_3 to H_2CO_3 moves back to normal.

NON - RESPIRATORY FUNCTIONS OF LUNG

- Animals are exposed to pollutants like dusts, ammonia, pungent gases carbon monoxide, products of plant and animal origin, parasites, bacteria, viruses, spores, endotoxins, diesel fumes, ozonized oxides of nitrogen.
- During transport the animals and animals reared under intensive housing are exposed to stressor promoting agents like pneumonia/pleuritis.
- Non-infectious stress for prolonged time leads to chronic airway diseases.
- These treats are handled by variety of defense mechanism of the lung.

Non-specific immunity/Innate immunity

- Protects against harmful inhaled substances. This includes mucociliary system; cough reflex, phagocytic activity of the alveoli.

Toll-like receptors

- Proteins on the surface of the cell membrane that can recognize the molecules which are common to many bacteria/fungi. On stimulation of these receptors initiates the activities of proinflamatory cytokinins.

Specific defense against specific agents (Bacteria/virus)

- It takes several days for the activation of the process of immune memory to protect future attack of the same organism.

MUCOCILIARY SYSTEM

Functions

- Aerosol is the collection of particles/lipid droplets which are small enough to remain suspended in the air for a period of time.
- Aerosols suspended in air are inhaled
- Aerosols are classified as inhalable/PM 10 (10μ or less) and respirable/PM 2.5 (2.5μ or less)
- These inhaled particles are trapped by the moist surface of the tracheobronchial tree depends on the size of the particles.
- Large particles (> 5μ) are trapped by inertial impaction
    - Inertial impaction occurs as bends in the large air ways
    - Large particles (> 5μ) travel at high velocity are unable to cross these bends.
    - Inertial impactions are provided by the lymphoid tissues in the tonsils, bronchi associated with lymphoid tissues.
- Particles < 4μ settle down by means of sedimentation on the wall of the airways and remain in the lung for ever.
- Particles that diffused into the alveoli epithelium are exhaled out.
- Type of breathing alters the settlement of the particles in the respiratory tract
- Slow, deep breathing transport the particles deep into the lung
- Rapid, shallow breathing enhances inertial deposition in the air ways
- Bronchoconstriction enhances the deposition in the central airways
- Bronchodilatation favors peripheral deposition

Deposition of particles

- Toxic gases get deposited based on their solubility and concentration
- Highly soluble sulphur dioxide at lower concentration is removed by the nose, but at higher concentration it will go deep into the lung
- Less soluble gases also go deep into alveoli
- Inhalation of toxic gases initiates protective reflexes like cough, sneeze, high mucous secretion and bronchospasm.

Anatomical specialties of mucociliary blanket of the respiratory tract

- It consists of sol and gel mucous layers overlying the epithelial cells
- Particles deposited in the lung are transported by the mucociliary escalator to the pharynx for its swallowing
- Sol layer is a low viscous layer over which the cilia beats and baths surface of the epithelial cells
- Gel layer is more viscous that traps the inhaled particles
- Forward beating of the cilia propels the particles to tracheobronchial system or nasal cavity
- Rate of ciliary beating depends on the surface area and also gravity
- Slow ciliary beating in the bronchiole than bronchi and trachea
-

Respiratory mucous secretion

- Clara cells are the non ciliated portion of bronchioles in the smaller airways
- Goblet cells are located in the large air ways secrete mucous
- Bronchial glands also secrete serous and mucous via sub mucosal glands
- Mucous secretions are under the influence of autonomic nervous system
- On the surface of the epithelial cells, the microvilli assist the ions fluid exchange
- The composition of the mucous varies with the nature of stimuli
- During bacterial overload increased viscosity of mucous secretion, lesser the clearance rate
- Transepithelial movements of water and ions changes the composition of the mucous layer

Cough reflex

- It is a protective reflex initiated by irritation of the sub epithelial layer or stimulation of stretch receptors in the large bronchi by the foreign bodies
- Inflammation of the air passage and injury to respiratory epithelium also causes hyper responsive cough reflex
- It clears the mucous secretions from the trachea and large bronchi

EFFECT OF ALVEOLAR MACROPHAGES

- It is the first line of defense
- In the lungs, macrophages are in the alveolar lining fluid
- Surfactant proteins, complement proteins, opsonins and lysozymes in the respiratory secretions assist in the phagocytosis of particles like bacteria.
- Some of the destroyed particles are removed by microciliary system. Others cross the alveolar wall and enter into the lymphoid tissues of the airways.
- Lung's immune response of the functioning macrophages viz antigen presenting cells
- Hypoxia depressed the macrophage activity
- Endogenous glucocorticoids/synthetic corticoids suppresses macrophages activity
- Excessive administration of synthetic corticoids causes increased susceptibility to bacterial infection in the lung
- Animals transported to a longer distance show stress induced macrophage dysfunction
- Viral infection decreased macrophage functions and contributes the secondary bacterial infection.

Effect of other WBCs

- Neutrophils are also involved in the phagocytosis of inhaled particles
- The invaded bacteria is destroyed by the release of toxic oxygen radicals and proteolytic enzymes
- The free radicals also damage the lung tissues also; Antioxidants and protease inhibitors protect the lung tissue from this effect.

Role of cytokinines and chemokinines

- Cytokinines and chemokinines are proteins released by the macrophages, lymphocytes, epithelial cells during inflammatory process in response to lung injury by infectious agents, allergy or inhalation of toxic gases.

- These proteins attract the inflammatory cells (Neutrophils) to the site of injury and provide communication between neutrophils during inflammatory process.
- These proteins are also involved in tissue remodeling and healing process
- They act as a potent chemo attractant of the inflammatory cells (Neutophils) and produce agents for prolonged inflammatory response

FLUID EXCHANGE

- Continuous exchange of water and solutes between the alveolar epithelium and interstitial cells through interstitial septum
- Hydrostatic and oncotic pressures also favors fluid exchange
- Diffusible particles get exchanged
- Alveolar epithelium is impermeable to certain substances, hence prevents the leak of fluid into the alveoli
- The fluid between pleura has less protein, lubricates the pleura
- Fluid from the interstitium as well as pleural fluids are cleared by lymphatic system and by pumping of the lung during respiration

METABOLIC FUNCTIONS OF THE LUNG

- Endothelial surface is good enough to clear the blood of substances produced in other part of body.
- More over pulmonary capillary blood receives total cardiac output which favours cleansing.
- Endothelial surface is studded with clveolae which houses many enzymes involved in metabolism of vasoactive substances.
- Monoamino oxidases neutralizes serotonin and norepinephrine.
- Angiotensin converting enzymes metabolizes bradykinin and convert angiotensin I to II.Neutralizing enzymes of PGE2, F2 and leukotrienes are present.
- Toxic exogenous substances are also neutralized in lungs.

OTHER NON-RESPIRATORY FUNCTIONS OF THE LUNG

- Sneezing
- Phonation
- Cooling of blood going to the brain
- Olfaction
- Panting in dogs
- Purring in cats

MODULE-39: RESPIRATION IN BIRDS



LEARNING OBJECTIVES

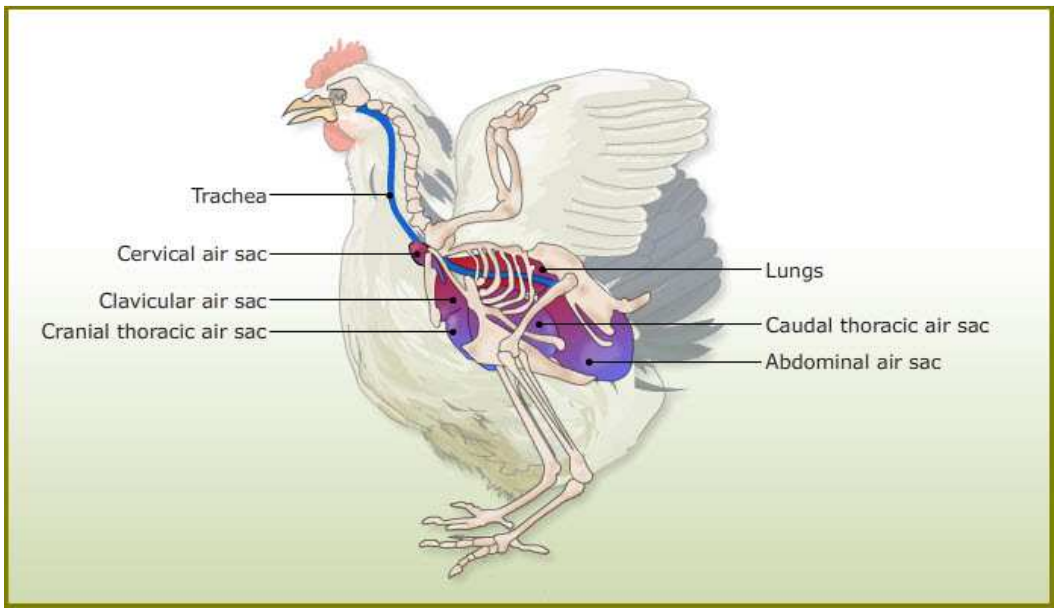
- This session explores,
  - anatomy of avian respiratory tract,
  - its merits and adaptations,
  - mechanism and regulation of respiration in birds.

RESPIRATION IN BIRDS

- The lungs are small and attached to ribs and are rigid; they do not expand or contract during the respiratory cycle. Large discrete air sacs are present. Both inspiration and expiration are active.
- Respiratory system beginning from external nares leads to nasal cavities, which open into pharynx. The air also passes through the mouth in to the respiratory system.
- Trachea begins with glottis and divides into two bronchi. At the division is the syrinx, which is the vocal organ in birds. The length and volume of the trachea is greater in birds than in mammals.
- Each lung contains one main intrapulmonary *primary bronchus*, which divides into these sets of *secondary bronchi*.
  - The Secondary bronchi divide into *tertiary bronchi or parabronchi***.**
  - The three sets of secondary bronchi are
    - **Mediodorsal**
    - **Medioventral**
    - **Lateroventral.**
- The parabronchi connect the mediodorsal and medioventral secondary bronchi and also connect some secondary bronchi to the airsacs.
- Those parabronchi (structures that contain gas exchange tissues in their walls) that passes between mediodorsal and medioventral secondary bronchi are nearly parallel and are called as *paleopulmonic parabronchi*.
- Those that pass from mediodorsal and lateroventral secondary bronchi and intrapulmonic primary bronchus to the caudal airsacs are called *neopulmonic parabronchi*. Exchange of gases between lungs and blood occur in parabronchi.

AIRSACS

- These large thin-walled airsacs arise from some secondary bronchi. They lie outside the lungs in the body cavity. They function as airways. Since they are avascular, no gasesous exchange occurs.
- There are 9 airsacs ; (5 anterior and 4 posterior) unpaired *inter clavicular* sac, paired *cervical* and anterior *thoracic airsacs*) are anterior air sacs. Paired *posterior thoracic* and large *abdominal sacs* are the posterior air sacs.
- Diverticula from some of the airsacs are connected to many of the bones. Hence, many of the bones in birds are pneumatic and the humerus is an important *pneumatic bone*.
- The humerus a bone is connected to interclavicular air sac. Lungs can be ventilated via humerus and birds can suck air through broken and opened humerus.
- Gas volume of the airsacs is approximately 10 times more than lungs.



MECHANICS OF RESPIRATION

- There is no diaphragm in birds. So, there is no division of abdominal and thoracic cavities.
- Pressure occurring during respiratory cycle is referred to as thoraco-abdominal pressure. Respiration is caused by changes in body volume.
- During inspiration, the transverse and dorso-ventral diameter of thorax increase because of the ventral cranial movement of sternum and lateral movement of ribs.
- During inspiration, some gas moves into the medio dorsal secondary bronchi, pass through the paleopulmonic parabronchi and enter into the cranial airsacs.
- The pressure increase in expiration in all air sacs and air is forced out of the air sacs, which moves through lungs to outside. Thus, gas flows through the avian lungs during both inspiration and expiration.
- During expiration, gas from the caudal airsacs passes again through the neopulmonic parabronchi and lateroventral secondary bronchi in the opposite direction and then moves

into the mediodorsal secondary bronchi and passes through the paleopulmonic parabronchi in the same direction as during inspiration.
- Gases from the cranial airsacs move into the medioventral secondary bronchi and out of the lungs without passing through the gas exchange area.
- The unidirectional movement of gas through paleopulmonic parabronchi, which is the gas exchange portion of avian lungs, increases efficiency of ventilation.
- Gas exchange can occur in the paleopulmonic parabronchi during both inspiration and expiration.
- The remainder the gas moves through the neopulmonic parabronchi, lateroventeral secondary bronchi into the caudal airsacs.
- During expiration these diameters decrease. In air sacs pressure drops and volume increase during inspiration and air move through the lungs into the airsacs.

RESPIRATORY RATES

- Respiratory rates of resting birds vary from 5/min in ostrich to 100 in small birds.
- Respiratory rates during flight are 3 – 18 times greater than at rest.
- Chicken, male 17 and female 27, Pigeon 28, House sparrow 59, and Goose 14/min.

TRANSPORT OF BLOOD GASES

- The difference in gas transport and diffusion between mammals and birds are not great. O_2 dissociation curve is same as in mammals except that it is to the right of mammalian curve. i.e. avian blood is less saturated with O_2 than mammalian blood at same temperature and partial pressure and release of O_2 to tissues is greater than mammalian blood.
- PO_2 for the arterial blood ranges 90 - 96 mm Hg. O_2 saturation is 88 – 90% in arterial blood and 40% in venous blood which are lower than mammalian blood. PCO_2 fowl blood is 28 – 34 mmHg. Arterial pH is generally higher in birds (7.44 to 7.58).

REGULATION OF RESPIRATION

- Respiratory centres as are as in mammals, Chemoreceptors are also important for regulation of respiration in birds Most of these receptors are present in lungs and they detect CO_2 levels in lungs and not of blood.
- They are stimulated by low CO_2 level and once stimulated they reduce respiration. The central chemoreceptors of brain respond to PCO_2 or H^+ ion concentration.
- Carotid and aortic chemoreceptors are also present in birds. Stretch receptors are present in lungs and their activity is modified by CO_2 level.
- Mechanoceptors are chemoceptors in muscle or joints activated by movements or build up of metabolites increases ventilation. Thermoceptors of skin, muscle and hypothalamus on stimulation lead to rapid shallow, panting type of respiration.
- During heat stress in birds, respiratory frequency markedly increases, while tidal volume decreases and panting occur (dead space ventilation increase for evaporative water loss).
- There is no marked change in arterial blood gases and pH. However, in chicken, there is severe hypocaprina and alkalosis during panting.
- Birds hyperventilate during running or flying, and arterial PCO_2 decreases.

- In high altitude due to low PO_2 of air, hyperventilation occurs.
- The birds become highly alkalotic but they tolerate this condition much better than mammals.

EXERCISE-1: INSTRUMENTS AND APPLIANCES (used in experimental physiology)

- Sherrington - Starling Recording Drum
  - Drum cylinder 6"x 6" with locking clamp.
  - Drum base carrying
    - Main spindle with contact arms at the base and drum adjustable screw at the top
    - Gears
    - Clutch operating clamp
    - Contact block with contact knob and two terminals
    - Leveling screws
    - Switch
- Myograph stand with adjustable knob at the base
- Cross blocks
- Muscle grip
- Isotonic muscle lever with after load screw and small weight
- Straw holder and light straw lever
- Frog board with cork surface
- Dissection board with clips
- Low voltage terminals.
- Induction coil: Primary coil, secondary coil, K,N,D terminals and Neef's hammer.
- Simple key
- Short circuiting key
- Simple electrode
- Vagal electrode
- Copper wire
- Tuning fork - Electrically operated
- Adjustable vibrating reed with mercury well, needle, adjustable knob and terminals
- Perfusion funnel with three way stopper
- Syme's cannula

EXERCISE-2: COMMON ELECTRICAL CONNECTIONS

Galvanic current

- Connect the simple key and the simple electrode with the battery, apply the electrode to tongue and record the effect of make current (M) Passage of current (C) and break current (B)

Faradic/induced current

- Connect the simple key and primary coil of the induction coil in series with the low voltage terminals.
- Connect the terminals of the secondary coil to the pair of electrodes. Move the secondary coil away from the primary coil and turn its axis to 90 degree angle to the primary coil.
- Apply the simple electrodes on the skin and feel the effect of make current, passage of current and break current.

- Repeat the above procedure by decreasing the angle between the primary and secondary coils by 150 and also the distance by 1cm until the stimulus is felt on the skin. Record the results.
- Slightly increase the current and observe the intensity of make (M) and break (B) current. Adjust for single maximal break induced current.

Introducing short circuiting key

- In the above circuit introduce the short circuiting key between the secondary coil and the simple electrode.
- Observe the effect of short circuiting key in on or off position during make current (M) Passage of current (C) and break current (B)

Replacement of simple key with the following instruments

- Drum base terminals
- Adjustable vibrating reed
- Record your observations

EXERCISE-3: MUSCLE - NERVE PREPARATION PITHING

- It is the manual destruction of the brain and spinal cord at spinomedullary junction to make the frog unconscious to perform various Physiological experiments.
- Hold the frog with the left hand and bend the head down by gentle press with the index finger.
- Insert the sharp pithing needle into the depression in the occipito-vertebral junction.
- Push the needle forward into the skull cavity and destroy the brain by rotary movement of the needle, then turn towards backward into the vertebral column to destroy the spinal cord .
- Pithed frog shows complete paralysis of the limbs and absence of voluntary movements like jumping or crawling.
- Prick the skin of the limb with a needle and observe no withdrawal/ flexor reflex response.

Procedure

- Keep the pithed frog on its back on the dissection board.
- Cut the skin and abdominal muscles upto sternum and expose the abdominal viscera.
- Remove the liver, stomach, intestine and kidneys without touching the under lying sciatic nerve trunk emerging from the vertebral column passing to the hind limbs.
- Hold the vertebral column with a piece of cotton cloth and strip the skin off the legs with the help of another piece of cloth .
- Cut the vertebral column above the origin of the sciatic nerve.
- Discard the anterior portion of frog's body. Separate the urostyle from the surrounding muscle and cut away its attachment with pelvic girdle. With the help of a bone cutter , longitudinally split the vertebral column into two equal halves .
- Cut the pelvic girdle carefully into two equal halves with a bone cutter . Preserve one of these legs in Frog Ringer's solution.
- By holding the piece of vertebral column with forceps, isolate the sciatic nerve upto pelvic girdle.

- Observe the location of the triceps femoris, iliofibula, semimembranosus and small pyriformis muscles on the dorsal aspect of the thigh.
- Make a small incision between the iliofibula and semimembranosus and carefully separate the muscles along the intermuscular groove and expose the sciatic nerve.

PITHING - APPARATUS

- Frog tray
- Pithing needle
- Dissection board.
- *Dissection set:* 6" and 3"Scissors, 6" and 3"Forceps and Bone cutter

COMPOSITION OF FROG RINGERS SOLUTION

Chemical substance	Quantity
Sodium chloride	6.50 gm
Potassium chloride	0.14 gm
Calcium chloride	0.12 gm
Sodium bi carbonate	0.25 gm
Sodium di hydrogen Ortho Phosphate	0.01 gm
Dextrose	1.0 to 2.0 gm
Distilled water to make	1 liter

PROCEDURE

- Keep the pithed frog on its back on the dissection board.
- Cut the skin and abdominal muscles upto sternum and expose the abdominal viscera.
- Remove the liver, stomach, intestine and kidneys without touching the under lying sciatic nerve trunk emerging from the vertebral column passing to the hind limbs.
- Hold the vertebral column with a piece of cotton cloth and strip the skin off the legs with the help of another piece of cloth.
- Cut the vertebral column above the origin of the sciatic nerve. Discard the anterior portion of frog's body.
- Separate the urostyle from the surrounding muscle and cut away its attachment with pelvic girdle.
- With the help of a bone cutter, longitudinally split the vertebral column into two equal halves .
- Cut the pelvic girdle carefully into two equal halves with a bone cutter . Preserve one of these legs in Frog Ringer's solution.
- By holding the piece of vertebral column with forceps, isolate the sciatic nerve upto pelvic girdle.
- Observe the location of the triceps femoris, iliofibula, semimembranosus and small pyriformis muscles on the dorsal aspect of the thigh.
- Make a small incision between the iliofibula and semimembranosus and carefully separate the muscles along the intermuscular groove and expose the sciatic nerve.
- Hold the stump of the vertebral column with a forceps gently raise the nerve and snipping off the lateral branches with the scissors without touching the main nerve.

- Nearing termination it divides into two branches and innervates the muscle at different levels. Both the branches should be preserved.
- Carefully separate the main branch of the sciatic nerve upto its insertion into the Gastrocnemius muscle.
- The nerve should never be touched with hands/ metallic objects.
- Separate Gastrocnemius muscle from the underlying tibia and ligate the tendoachilis at its centre point.
- Cut the tendoachilis well below the ligation. Then cut through the tibia just below the knee joint.
- Remove the thigh muscles and expose the femur bone and cut through femur about 1-1.5 cm of femur above the knee joint.
- Now the Gastrocnemius muscle and Sciatic nerve preparation is ready which consists of knee joint with portions of tibia and femur, then remove the thigh muscles and expose the femur bone and cut through femur about 1-1.5 cm of femur above the knee joint.
- Muscle nerve preparation consists of knee joint with portions of tibia and femur, the Gastrocnemius muscle and the Sciatic nerve with a piece of vertebral column.
- Appropriate electrical stimulus to the sciatic nerve, the Gastrocnemius muscle shows contraction.

EXERCISE-4: EFFECT OF SINGLE STIMULATION ON MUSCLE CONTRACTION MATERIALS

- Dissection board
- Dissection set
- Pithing needle
- Sherrington -Starling recording drum
- Myograph stand
- Cross blocks
- Muscle grip
- Isotonic lever
- Frog board
- Low voltage terminals
- Induction coil
- Simple key
- Short circuiting key
- Simple electrode
- Copper wires
- Tuning fork
- Frog Ringers solutions

SIMPLE MUSCLE CURVE - PROCEDURE

- Muscle-nerve preparation as previous exercise and arrange the electrical apparatus for faradic current to induce electrical stimulation.
- Adjust the distance between the primary and secondary coils to get single maximal break induced stimuli.
- Fix the simple electrode on the moist fiber paper on the frog board. Gently place the nerve on the electrodes without tension .
- Keep the cross block in the myograph stand to hold the muscle grip exactly horizontal to the main axis of the myograph stand and fix the femur .

- Tie the thread from the tendon to the isotonic lever with a distance of 1.5 - 2" and 90° angle.
- Keep the afterload screw in released position and fix the dead weight after the knot of the thread towards the afterload screw.
- Keep the writing point of the lever at appropriate distance from the fulcrum and ensure the free movement of the pointer with optimum pressure over the recording drum about 3" height for every contraction of the muscle.
- Once this adjustment is made the base of the myographic stand should not be disturbed.
- The pointer should be applied to the drum or removed away from the drum only by means of the knob at the base of the myograph stand.
- Switch on the recording drum. When the contact arm touches the contact knob the muscle is stimulated to contract and the pointer of the isotonic lever moves up on the recording drum to draw a curve.
- Switch off the Kymograph and rotate the drum by hand till the contact arm touches the contact knob.
- At this position , circuit was completed and nerve was stimulated.
- Mark a line on the base line by raising the lever about 1 centimeter with finger and label it as A. This point represents the point of stimulus.
- Similarly make arcs
    - at the point where curve leaves the base line, label it as B.
    - at the highest point of curve (C) and
    - at the point where curve retouches the base line (D)
- Move the lever point away from the drum. Set the drum rotating with the same speed as before and record vibrations of a tuning fork of known frequency (100 vibrations per second) below the base line.
- The distance between two peaks will be 1/100 th of a sec. Measure the time interval (seconds) for Latent (AB), Contraction (BC) and Relaxation periods (CD).
- Latent period is the time interval between application of stimulus and onset of contraction.It is due to
    - Time required for conducting nerve impulse along nerve fibre to the motor end plate.
    - Time taken for the release of neuro transmitter i.e. Acetyl Choline from nerve terminal and its diffusion in the neuro muscular junction
    - Time taken for chemical changes to occur in the muscle (true latent period) and
    - The time taken for overcoming the inertia of the lever.
- Latent period – 0.01 second, Contraction period – 0.04 second, Relaxation period – 0.05 second.
- Relaxation is a passive process while contraction is an active process. Hence the relaxation period is longer than contraction period.

EXERCISE-5: EFFECT OF TEMPERATURE ON CONTRACTION OF GASTROCNEMIUS MUSCLE

Effect of heat and cold - Materials

The materials used in this experiment are as follows

- Dissection board

- Dissection set
- Pithing needle
- Sherrington -Starling recording drum
- Myograph stand
- Cross blocks
- Muscle grip
- Isotonic lever
- Frog board
- Low voltage terminals
- Induction coil
- Simple key
- Short circuiting key
- Simple electrode
- Copper wires
- Frog Ringer solution

EFFECT OF HEAT AND COLD - PROCEDURE

- Dissect out a frog and isolate the gastrocnemius muscle and Sciatic nerve complex.
- Fix the muscle-nerve preparation in the frog boar.
- Arrange the apparatus for recording simple muscle curve.
- Record the base line on smoked surface of recording drum.
- Determine the threshold stimulus and adjust the height of contraction to approximately three inches above the base line by adjusting the strength of the stimulus.
- Apply normal frog ringer soultion (30□□C) to the muscle.
- Set the drum for fast speed and record the muscular contraction.
- Drain off the normal Ringer's solution from the muscle and replace it with already warmed Ringer's solution (35□□C to 40□□C).
- Wait for few minutes to equilibrate the temperature and then record the simple muscle curve on the same base line with same point of stimulus.
- Drain off the warm Ringer's solution from the muscle and replace it with cold Ringer's solution (10□ C to 20□□C).
- Wait for few minutes to allow the muscle-nerve complex to equilibrate in this temperature and record a simple muscle curve on the same base line with same point of stimulus.
- Drain off the cold Ringer's solution from the muscle and apply very hot Ringer's solution (>45□ C).
- Stimulate the muscle-nerve complex.
- Label the temperatures on corresponding muscle curves.
- Mark the point of stimulus and height for each curve.
- With the help of tuning fork note the time interval (seconds) for latent, contraction and relaxation periods and height of contraction for each curve and tabulate the results.

EXERCISE-6: EFFECT OF TWO SUCCESSIVE STIMULI - MATERIALS

- The materials used are as follows:
  - Dissection board
  - Dissection set
  - Pithing needle
  - Sherrington -Starling recording drum
  - Myograph stand

- Cross blocks
- Muscle grip
- Isotonic lever
- Frog board
- Low voltage terminals.
- Induction coil.
- Simple key
- Short circuiting key
- Simple electrode
- Copper wires
- Frog Ringer solution

EFFECT OF TWO SUCCESSIVE STIMULI - PROCEDURE

- Arrange the apparatus and the muscle nerve preparation as for recording a simple muscle curve.
- Adjust the point of attachment of the muscle to the lever so as to give a curve of about 3" height.
- Separate the two contract arms sufficiently apart and record two separate simple muscle curves in one revolution of the drum on the lower third of the drum.
- Mark the point of stimulation, point of initiation of contraction, point of maximum contraction and point of complete relaxation. Compare the second curve with the first.
- Using the first curve and lever point as reference bring the second contact arm near the first so that the second stimulus falls within the period of relaxation of the first muscle contraction.
- Lower the drum and record the contractions on the middle third of the drum.
- Mark the point of stimulation, point of initiation of contraction, point of maximum contraction and point of complete relaxation and compare this curve with the first.
- Similarly, adjust the second stimulus to fall with in the period of contraction of the first contraction. Record the contractions on the upper third of the drum.
- Mark the point of stimulation, point of initiation of contraction, point of maximum contraction and point of complete relaxation and compare it with the first.

EXERCISE-7: EFFECT OF FUSION OF TWITCHES ON MUSCLE CONTRACTION - TETANUS

- Tetanus is the fusion of twitches which may either be complete or incomplete.

Materials

- Dissection board
- Dissection set
- Pithing needle
- Sherrington - Starling recording drum
- Myograph stand
- Cross blocks
- Muscle grip
- Isotonic lever
- Frog board
- Low voltage terminals

- Adjustable vibrating reed
- Short circuiting key
- Simple electrode
- Copper wires
- Frog Ringer solution

TETANUS-PROCEDURE

- Arrange the apparatus as for recording the single muscle twitch.
- Dissect out a frog and isolate the gastrocnemius muscle and Sciatic nerve complex.
- Attach the tendon of the muscle to the lever at such a distance to record contraction of about 1 inch height.
- With the help of clutch and gear set the speed of the drum at slow. Substitute the spring key with the adjustable vibrating reed.
- Apply the point of the lever to the drum surface and draw the base line about 1" from the lower edge of the drum. Determine threshold stimulus.
- Apply few stimuli of the vibrating reed by suitably changing the stimuli rate of 5,7,10,15 and 20 per second and take separate recording of contraction for each stimulation rate by manipulating the short circuiting key.
- Replace the adjustable vibrating reed with Neef's hammer in the primary circuit.
- Record the contraction of the muscle at a rate of about 100 stimuli per second.

EXERCISE-8: EFFECT OF FATIGUE ON MUSCLE CONTRACTION

- Sustained repetitive stimulation leads to decrease in the mechanical efficiency of a muscle contraction.

Materials

- The materials used here are as follows:
    - Dissection board
    - Dissection set
    - Pithing needle
    - Sherrington Starling Recording Drum
    - Myograph stand
    - Cross blocks
    - Muscle grip
    - Isotonic lever
    - Frog board
    - Low voltage terminals
    - Induction coil
    - Simple key
    - Short circuiting key
    - Simple electrode
    - Copper wires
    - Frog Ringer solution

EFFECT OF FATIGUE ON MUSCLE CONTRACTION-PROCEDURE

- Arrange the apparatus and the muscle nerve preparation as for recording the muscle twitch.
- Set the drum in fast speed. Apply the point of the lever to the drum surface and take the base line.
- Open the short circuiting key and record the first three contraction.
- By applying the lever on and off the drum surface by manipulating the side rod of the myograph stand record 5th, 10th, 15th, 20th, 25th contractions until the contractions of the muscle are very feeble.
- Mark the point of stimulation, maximum point of contraction and point of relaxation and compare the changes in the recording of $5^{th,}$ 10th, 15th, 20th, 25th contractions.
- Keep the muscle moistened with normal saline and allow it to rest for ten minutes. Now record another curve at another spot. Compare this contraction with the last recording.
- Initially the height of contraction increases due to stair-case effect or Treppe. Afterwards the height of contraction gradually decreases and becomes negligible.
- Latent, Contraction and Relaxation periods are prolonged.
- The muscle fails to relax completely and the lever does not reach the base line. This is known as contracture.
- Contracture is the state of prolonged reversible shortening of muscle. By providing rest and nutrients contracture disappears and the muscle gets relaxed.
- Fatigue of myoneural junction due to depletion of neurotransmitter.
- Fatigue in muscle is due to deficiency of source of energy and accumulation of waste products such as lactic acid.

EXERCISE-9: EFFECT OF LOAD AND AFTER LOAD MATERIALS

- The materials used in this experiment are as below:
  - Dissection board
  - Dissection set
  - Sherrington Starling Recording Drum
  - Myograph stand
  - Cross blocks
  - Muscle grip
  - Isotonic lever
  - Frog board
  - Low voltage terminals
  - Induction coil
  - Simple key
  - Short circuiting key
  - Simple electrode
  - Copper wire
  - Weights
  - Pithing needle
  - Frog Ringer solution

EFFECT OF LOAD AND AFTER LOAD-PROCEDURE

- Arrange the apparatus and the muscle nerve preparation as for recording the muscle twitch.
- Determine the threshold stimulus. Simply keep the drum in neutral gear so that it can be rotated by hand.
- Fix the muscle as after loaded position by making the after load screw to firmly support the isotonic lever.
- In this position the length of the muscle is kept constant independent of weight acting on it.
- Record a contraction of the muscle by applying a single threshold stimulus.
- As the drum is stationary, the record is almost a straight line.
- Add 10 gm weight on the isotonic lever and record another contraction adjacent to the previous one by slightly rotating the drum forwards.
- Go on adding weights and record contractions till such time the muscle fails to contract due to the weights.
- At this stage rotate the drum forwards and make the muscle free loaded by releasing the after load screw. Now apply the threshold stimulus.
- It can be seen that in the free loaded position the muscle lifts the weight, which it failed to lift in the after loaded position.

Calculation of work done

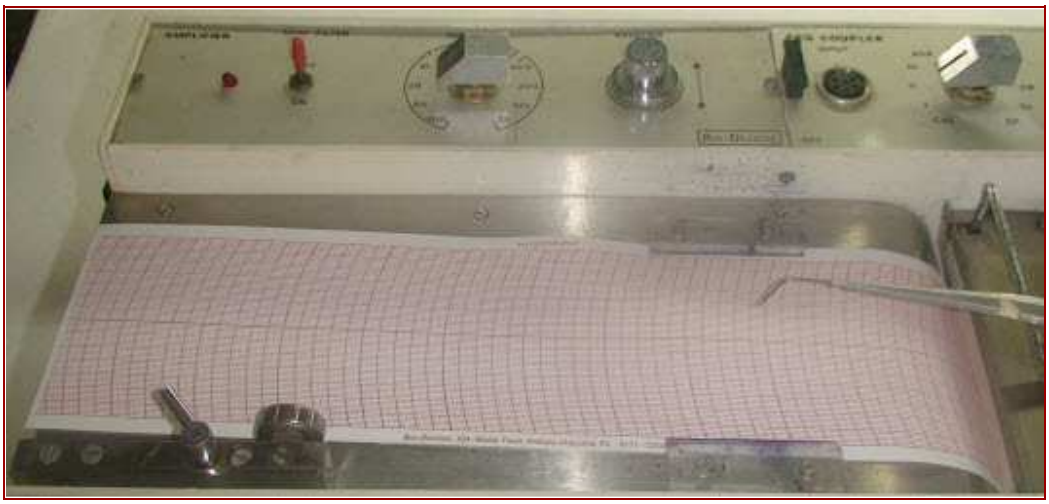
$$W(\text{work done}) = w \times h \quad \begin{pmatrix} w = \text{weight added} \\ h = \text{height through which the weight is lifted} \end{pmatrix}$$

$$h = \frac{H \times I}{L} \quad \begin{pmatrix} L = \text{Length of the lever from the fulcrum} \\ IL = \text{Distance of the load from the fulcrum} \\ H = \text{The height through which the pointof the lever is lifted} \end{pmatrix}$$

	Weight (gms)	Height of contraction (cm)	Work done = Weight x Height
After load	0 gm		
	10 gm		
	20 gm		
	40 gm		
	60 gm		
Free load	60 gm		

EXERCISE-10: PHYSIOGRAPH

- It is an electrical device used in the experimental physiology.
- Many of the experiments done with electromechanical devices as above can be done using the physiograph.
- Many parameters can be recorded on a single apparatus.
- It permits group experiments, i.e. many physiographs can be interconnected and one experimental set up can be used for all.
- Accuracy and sensitivity of the instrument is quite high.



COMPONENTS OF PHYSIOGRAPH

- Physiograph comprises of the following

Main Console

It has three basic components.

- Chart drive
  - It is a multispeed chart drive and provides paper speeds ranging from 0.25 to 100 mm / second. The speed can be adjusted using a selector knob and push button.
- Pen recording system
  - Two recording pens - one for the muscle event recording and the second one for the time / event marker. The contact tension of the pens can be adjusted with the help of cradle springs. A pen lift knob lifts the pens from the paper.
- Main amplifier
  - It is common to all couplers. A 50Hz toggle switch cuts off unwanted interference. A base control knob is available to adjust the position of the pen. Another control is used to select sensitivity of the amplifier ranging from 50 to 500 mV in 4 steps and from 1 to 100 mV in 7 steps.
  - Three screwdriver adjustments are provided on the sides of the main console. Once adjusted they do not require to be adjusted. The side of the console also has IN and OUT jackets. The OUT jacket of one physiograph can be connected to IN jacket of

another unit and both units will record simultaneously from the same experimental set up. Another IN jacket is used to connect electronic stimulator to record Event / Time events.

- Coupler housing
  - Various interchangeable couplers can be just plugged into the coupler housing.

Couplers

- Bio-potential coupler
  - It is useful to record any AC phenomenon like ECG, EEG, EMG, and ERG recording, nerve conduction velocities, movements of the eye etc. A coupler knob selector selects the phenomenon to be recorded. It has a CAL position to check calibration, for which a push button is provided.
- EKG Coupler
  - Used for recording of clinical ECG (EKG). There is a knob for selecting various leads and another knob for calibration.
- Strain gauge couplers and isotonic fine movement
  - These couplers record activity from various strain gauge transducers - pressure volume, muscle force transducers etc. They are useful to record muscle and heart activity, for all experiments on frog sciatic-nerve gastrocnemious preparation like simple twitch, successive stimiuli, tetanus, fatigue, isometric and isotonic contractions, effect of load, muscle stretch reflexes, experiments on frog heart in-vivo, B.P. of cannulated animals.
- Pulse - Respiration Coupler
  - Useful to record arterial pulse with a photoelectric pulse transducer and respiration with respiration transducer. A toggle swtich selects pulse or respiration mode.
- Temperature Coupler
  - Useful to record rectal or surface temperature. The transducer has to be calibrated within a temperature range using water bath.

Transducers

- Used to transform the mechanical activity into electrical activity.

Electronic Stimulator

- Can be used in two modes.
  - As time/event marker.
  - To provide electrical stimuli up to 30 V as asingle pulse, as 2 successive stimuli with time intervals ranging from 0.5 to 100 / sec.

Electrodes

- ECG electrodes, EEG electrodes and EMG electrodes.

Components of Physiograph				
Main Console	**Couplers**	**Transducers**	**Electronic Stimulator**	**Electrodes**

Chart drive	Bio-potential coupler			
Pen recording system	EKG Coupler			
Main amplifier	Strain gauge couplers and isotonic fine movement			
Coupler housing	Pulse- Respiration Coupler			
	Temperature Coupler			

USING THE PHYSIOGRAPH

- Put the recording paper in the paper rack and bring it to the top of the console.
- Lift the pens using pen lift knob. Slide the paper under the pulley guides and under the ball bearing after lifiting it using thumbscrew.
- Fill ink in the inkwells and check the flow of ink by pressing the inkwell lid and closing the opening; ink should flow freely.
- Keep the pens lifted from the paper when not doing any recording.
- Switch off the mains on the console and insert the desired coupler. (Coupler should not be inserted or taken out of the coupler housing while the switch is in ON position).
- Connect the transducer with the coupler and with the experimental tissue or subject.
- Put the main switch of the console in ON position, adjust the sensitivity to the desired level (Varies with different experiments).
- Adjust the position of the pen to the centre. If required, connect the stimulator to physiograph, select a type of stimulus (single, double etc), adjust the voltage of the stimulus.
- Carry out the muscle nerve preparation. Fix the femur with muscle grip and gently lay the nerve on the electrode without any tension.
- The free end of the thread from the tendon is attached to the lever of the isotonic fine movements transducer to convert the mechanical activity into electrical activity.
- Connect the transducer to the strain gauge coupler in the base unit. Stimulate the muscle with stimulator to provide single stimulus, successive stimuli and repetitive stimuil and record the specific contractions in the moving graph sheet.
- For EKG recording, connect the electrodes of leads to the 5-pin junction box and then to the coupler.
- Fix the electrodes of ECG to their respective place on the body after applying conductive jelly.
- Include the bio-potential / EKG coupler to the base unit. Adjust sensitivity to 1 mV.
- Keep the lead selector of the coupler to CAL position, run the paper and press the push button on the coupler 3 to 4 times while adjusting CAL control so that 1 mV produces 1 cm deflection of the pen.
- Record the electrical activity of the heart.
- For recording EEG and EMG from the skin surface or from muscle, small cup electrodes are used. A thick paste is used to keep the electrode in position.
- Needle electrodes are used to record muscle action potential.

EXERCISE-11: EXPERIMENTS WITH ENZYMES - PTYALIN

- Ptyalin is the salivary amylase present in man and pig acts on starch and certain polysaccharides at alpha 1 - 4 glucosidic linkage, the principal product is the maltose.
- Chloride ions are essential for its activity.
- The optimum pH of 6.7 is required for its activity.

Source of Ptyalin

- Rinse the mouth thoroughly with 20 ml warm water at least a minute.
- Collect the content in a conical flask and mix well and filter.

TEST FOR THE PRESENCE OF THE ENZYMES

- Take 5 ml of freshly prepared 1% starch solution and 5 ml of dilute saliva in a clean test tube and mix well (avoid frothing of saliva).
- Place a glass tube in the test tube and keep it in a water bath at 37□C.
- Take few drops of the content from the test tube at half a minute intervals and mix it with few drops of dilute iodine solution.
- A blue colour is developed at first. But with more drops of the mixture is added the colour turns violet, reddish violet and finally reddish brown.
- The development of the various colours is due to the enzymatic conversion of starch to amylodextrin and erythrodextrin.
- When a drop of mixture fails to give a colour with iodine indicates presence of achrodextrin.
- Mix few ml of this solution with a few drops of Fehling's solution causes a marked reduction of Fehlings's solution indicating complete hydrolysis of starch to a reducing sugar, maltose.

DETERMINATION OF THE PTYALIN ACTIVITY

- Take 5 ml of the soluble starch in a test tube.
- Add 2 ml of a buffer solution of pH 6.7 and 2 ml of 0.5% NaCl solution. Keep the tube in a water bath at 37°C.
- Arrange serially numbered test tubes each containing 3 ml of distilled water and two drops of dilute iodine solution.
- Add 2 ml of saliva to the warmed buffed starch solution, mix well and note the time.
- Transfer a drop or two of the mixture to a tube containing dilute iodine at 15 seconds interval and shake well.
- The time at which additions of the solution fails to produce any colour indicates the complete conversion of last trace of erythrodextrin into achrodextrin and maltose which is known as the "Achromic point".
- The time taken to reach this point (chromic period) is the measure of ptyalin activity.

EFFECT OF pH ON PTYALIN ACTIVITY

- Repeat the above procedure with the following modification. Instead of 2 ml of buffer solution of optimum pH 6.7 add 2 ml of 0.4% HCl to a tube (A) and 2 ml of Sodium carbonate to another tube (B).
- The rest of the reagents and procedure are the same.
- The chromic period from reaction tube (A) gives the effect of acid pH (around 2) and that from tube (B), the effect of alkaline pH (around 9) respectively on the enzyme activity.

EXPERIMENTS WITH ENZYMES-PEPSIN

- Pepsin is the proteolytic enzyme secreted as pepsinogen by the chief / peptic cell of the gastric glands of the stomach which is then converted to pepsin by the activation of HCl in the stomach.
- The optimum pH for pepsin activity is 2.4 . Pepsin hydrolyses peptide bonds, releases smaller molecules peptones and peptides.

Action of Pepsin

- Place equal quantities of freshly prepared, washed fibrin in four test tubes labelled A, B, C and D. Then add the following solutions given below.
- Note the change in the tubes after incubation at 37°C for 30 minutes.

Tube	Water (ml)	0.4% Hcl (ml)	Pepsin (ml)	Observation
A	--	3	3	Fibrin swells up, becomes transparent and finally dissolves
B	3	--	3	No alteration
C	3	3	--	Becomes transparent but does not dissolve
D	--	3	3 (Boiled)	As above

Note

- Tube A shows the proteolytic activity of the active enzyme.
- In tube B the enzyme is not active due to lacks the optimal pH.
- In tube C lacks the enzyme itself, Tube D has denatured enzyme.
- Both C and D showing no enzyme activity.

DETECTION OF FIBRIN BY CARMINE FIBRIN METHOD

- To 5 ml of the ferment solution add 5 ml of 0.4% HCl, then divide it into 2 parts, A and B.
- Boil the content of tube 'B' for few minutes and cool. To tube 'A' and 'B' add few pieces of carmine fibrin.
- Place the tubes in a water bath at 37°C for 10 minutes. Shake well and observe the colour of fibrin.
- In 'A' it will be red due to the proteolytic activity. In "B" it will be almost colourless, due to enzyme denaturation.

EXPERIMENTS WITH ENZYMES - TRYPSIN

Reagents

- Casein or fibrin threads
- Trypsin solution
- Sodium carbonate, 1% solution (1.0g sodium carbonate in 100ml of water)
- Ninhydrin solution (1%)

Procedure

- Mix the fibrin threads with 2 ml trypsin solution and 2ml of 1% sodium carbonate solution.
- Adjust the pH between 7.0 and 9.0 with sodium carbonate or NaOH solution.
- Incubate the mixture at 37°C for 30 minutes. Observe the dissolution of fibrin threads.
- The end product, peptides can be tested by the development of pink colour by the addition of 1% Ninhydrin solution.

EXERCISE-12: RECORDING OF RUMINAL MOTILITY AND RETICULAR SOUNDS

- The orderly and synchronized movements of rumen and reticulum help in mixing of ingesta, mix the saliva with the food, in regurgitation, in eructation of gases and to propel the food down the gastrointestinal tract.
- Bacterial fermentation and physical maceration by contraction of the stomach walls are the main functions of the forestomach.
- Ruminal motility is therefore used as an index of digestive function in the ruminant. The rumen is routinely examined clinically as an indicator of the state of the other stomachs.
- The rumino-reticular contractions cause waves or ripples in the left paralumbar fossa, which can be identified by auscultation and palpation.
- The rumen and reticulum show primary cycle (mixing) and secondary cycle (eructation) contractions.
    - Primary cycle consists of a biphasic reticular contraction followed by contraction of dorsal and then ventral rumen sacs. The ventral sac contraction may reoccur after an interval of one minute.
    - The secondary cycle is confined to rumen and consists of dorsal sac contraction.
- The ruminal motility can be assessed by Visual inspection method, Palpatory method.
- The reticular sounds can be assessed by Auscultatory method only.

MATERIALS

- The *materials* used here are as follows
    - Ruminant animal
    - Trevis
    - Phonendoscope
    - Stop watch
- The *methods* used here for recording ruminal motility and reticular sounds are given below

- Visual observation
- Palpation
- Auscultation of Rumino - Reticular Sounds

VISUAL OBSERVATION

- Secure the ruminant animal in a trevis.
- Observe alternate rising and sinking of the left paralumbar fossa in conjuction with abdominal surface ripples/undulations is seen in the hollow left paralumbar fossa and left abdominal region.
- The ripples are caused by rumen motility during both primary and the secondary cycles.
- In the left paralumbar fossa there are two horizontal ripples move from the lower abdominal region up to the paralumbar fossa, where they sink, again move ventrally and disappears at the lower left abdominal region.
- The waves provide information on the intenisty of the rumino-reticular contractions.
- Record the number of ripples moving upper minute.

PALPATION

- Secure the ruminant animal in a trevis.
- Keep your fist over the left flank in the paralumbar fossa, press the fist into the fossa.
- After an initital pressure, withdraw the fist and repress it at the same place.
- Feel the rumen movements on the fist and start counting for five minutes and express your observation as number of contractions per minute.
- Normally 2 ruminal movements / 3minutes are observed.
- In case of impaction, trauma and anorexia, ruminal motility is suspended.

AUSCULTATION OF RUMINO-RETICULAR SOUNDS

- Auscultation of the ruminal and reticular sounds indicate the nature of the intraluminal contents and the frequency and amplitude of the ruminal and reticular movements.

Procedure

- Secure the ruminant animal in a trevis.
- Place the chest piece of the stethoscope and observe fluid gurgling/crackling ruminal sounds in the left paralumbar fossa followed by the pronounced gassy sounds.
- Keep the chest piece of the stethoscope at the ventral aspect of the left side, just 6 to 8 fingers above the sternum near 6th, 7th and 8th intercostal spaces and observe a faint fluid sound of the reticulum
- The rumen sounds of the normal animal consuming roughage are booming-crackling sounds.
- Fluid splashing sounds indicates grain overload, atonic rumen associated with prolonged anorexia.
- In case of bloat, impaction, vagal indigestion and acute traumatic reticuloperitonitis, no sound may be heard.



Auscultating rumino reticular sounds

EXERCISE-13: COLLECTION OF RUMEN LIQUOR RUMEN FLUID EXTRACTION PUMP

It consists of:

- A specially designed suction strainer.
- Three meter nylon tube with perforation at the tip is covered by a stainless steel spiral tube.
- A suction pump
- An air tight sampling bottle with two ways 'T' joint.

PREPARATION

Preparation of the instrument for collection of rumen fluid

- The suction strainer is unscrewed and the perforations at the end of the nylon tube should be clean. The full length of the nylon perforated end is projected out.
- The suction pump for its working condition is checked.
- The two ways 'T' joint is fixed tightly to the sampling bottle to prevent the air leakage.
- The nylon tube from the suction strainer is fixed into the vertical end of the two ways 'T' joint. Horizontal end of joint should be connected to the suction pump.
- The entire set is checked by dipping the suction strainer into a bucket of water and operating the suction pump.
- Liquid paraffin is applied over the stainless steel spiral sound of the instrument which is ready for the collection.

Important points to be observed before and after the collection of rumen fluid

- About 200ml of rumen fluid sample is to be collected for various tests.
- The rumen fluid can be kept in room temperature for five hours and in refrigerator for 24 hours
- After collection, rumen fluid container should be immediately closed air tightly
- The first collection of rumen fluid is discarded, so that error in pH can be minimized

COLLECTION PROCEDURE

Cattle

- The animal is restrained with nose grips.
- The mouth is opened by pulling out the tongue to one side.
- The head of the animal is held high and the strainer steel spiral sound is incubated with suction strainer until it reaches the rumen .
- The rumen fluid is collected by operating the suction pump.
- Rumen fuid is collected upto the brim of the container to minimize the exposure to oxygen and change in rumen pH.
- Can be collected using hypodermic needle.(6 inches long with 18 gauge)



Rumen Liquor collection

Sheep and goat

- The animal is restrained between the knees.
- The head is raised above.
- The mouth gag is inserted across the mouth and the rumen fluid extraction pump is incubated by introducing suctions strainer through the hole in the mouth gag which pass over the tongue and pass the epiglottis into the cranial part of the esophagus .
- The tube passing through the esophagus on the left side of the neck of the animal is visualized .
- When the suction strainer is dipped into the rumen contents, the suction pump is operated and the required quantity of rumen fluid is collected.

RUMEN FISTULA

- In an experimental animal frequent collection of rumen liquor warrants for surgical procedure by which artificial opening is made.
- Rumen fistula (Picture 1) is the PVC made tube like designed to fix onto the rumen of the animal and operable from outside (Picture 2) .



Picture 1



Picture 2

EXERCISE-14: STUDY OF BACTERAL COUNT MATERIALS

The materials used to study bacterial count in strained rumen liquor are given below

- Rumen liquor
- Muslin cloth
- Compound microscope
- Neubauer counting chamber
- Cover slip
- Crystal violet solution
    - Saturated alcoholic solution of crystal violet - 10ml
    - Add Distilled water to make up the volume to 100ml
    - Filter the stain solution before use.

BACTERIAL COUNT PROCEDURE

- Prepare Strained Rumen Liquor (SRL) by filtering the rumen liquor through four layers of muslin cloth.
- Draw this SRL up to mark 0.5 in the R.B.C pipette.

- Carefully draw crystal violet stain solution up to 101 mark without any air bubbles.
- Mix the SRL sample and stain solution by rotating the RBC pipette horizontally between the two palms.
- Discard 3 or 4 drops of the unmixed stain solution from the stem of the pipette.
- Focus Neubaur counting chamber covered with cover slip under high power.
- Charge the diluted SRL between the counting chamber and the cover slip.
- Keep it undisturbed for few minutes. Count the number of bacteria in five medium squares like R.B.C. counting.

BACTERIAL COUNT CALCULATION

- Total number of bacteria in 5 medium squares = Y
- The number of bacteria in 25 medium squares = Y x 5
- Depth factor = 10
- Dilution factor = 200
- 1c.m.m of undiluted SRL sample contains = (Y x 5 x 10 x 200 bacteria)

  = Y x 10,000 bacteria
- 1ml of diluted SRL sample contains = Y x 10,000 x 1,000 bacteria

  = Y x 10^7

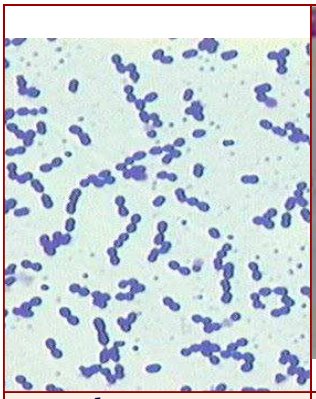
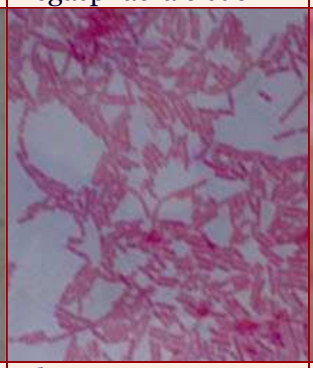
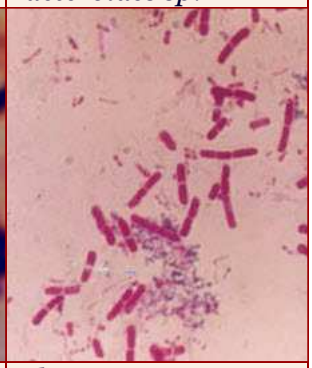
Result

- No. of bacteria per ml of SRL = x 10^7

Normal bacterial count in SRL ruminants (N x 10^{10} / ml of SRL)

- Cattle : 5.4 – 31.4
- Buffalo : 6.9 – 32.7
- Sheep : 18.0 – 88.0

- **DIFFERENT RUMEN BACTERIA**

			
Enterobacter sp.	Megasphaera elsdenii	Ruminococcus albus	*Bacteroides sp.*
			
Ruminococcus flavifaciens	Selenomonas ruminantium	Sterptococcous bovis	*Selenomonas ruminantium*

EXERCISE-15: TOTAL PROTOZOAL COUNT IN RUMEN LIQUOR

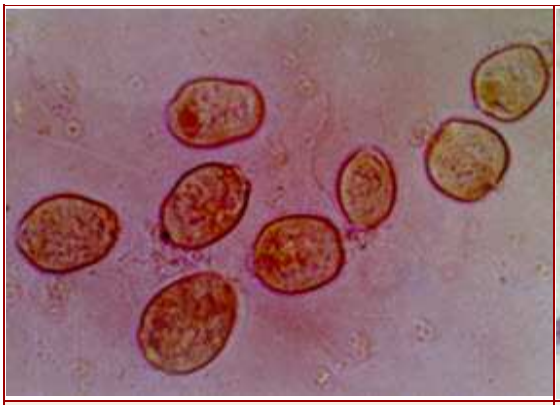
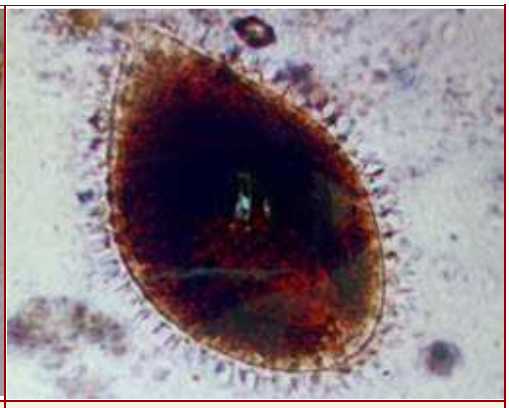
MATERIALS

The materials used in this experiment are as follows

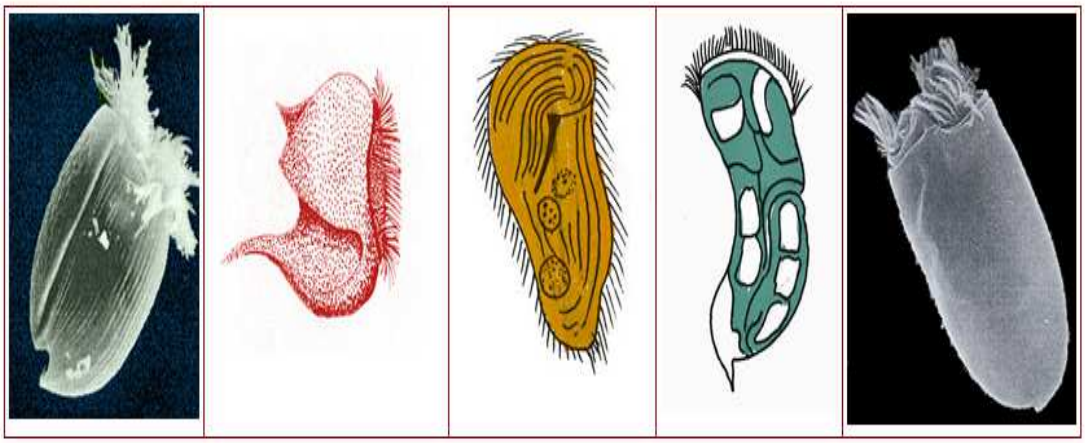
- Rumen liquor
- Muslin cloth
- Compound microscope
- Neubauer counting chamber
- Glass slides
- Cover slip
- Pipettes
- 10% Formal saline V/V(10 ml Formalin with 90 ml of 5% NaCl)
- 2% Eosin solution

TOTAL PROTOZOAL COUNT PROCEDURE

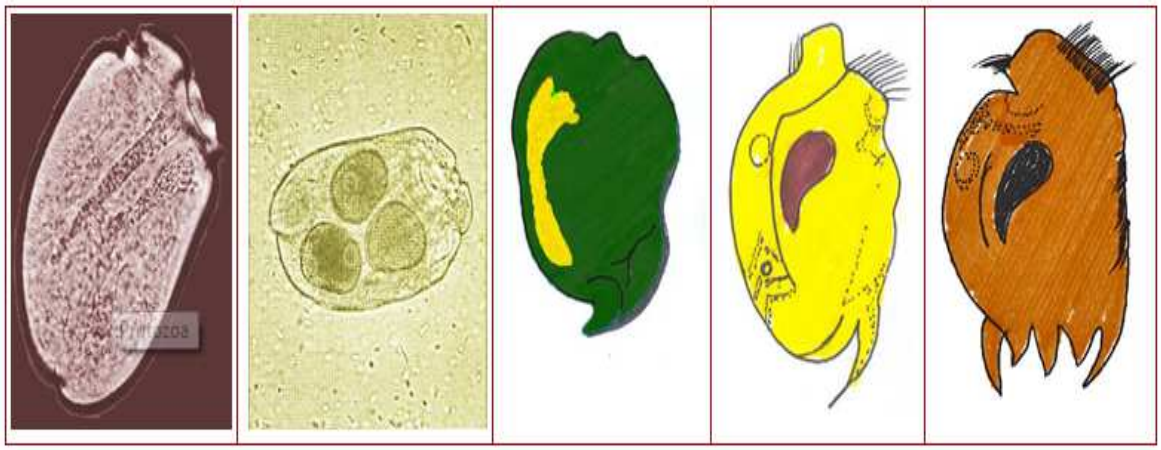
- Prepare Strained Rumen Liquor (SRL) by filtering the rumen liquor through four layers of muslin cloth.
- Dilute 5 ml of SRL with 25 ml of 10% Formal saline.
- Take 5 ml of this diluted SRL and add 5 drops of 2% Eosin solution.
- Focus the chamber under high power and charge the diluted and strained SRL into the Neubaur counting chamber without air bubble.
- Keep it undisturbed for few minutes. Count the number of protozoa in four W.B.C. squares.
- As the protozoal populations tend to vary based on the geographical location which the animal is reared, type of diet being fed, age and other factors, the pictures provided in this lesson facilitate the learners for gross identification of protozoa.

	
Entodinium sp.	Isotricha sp.

Isotrichidae family



Ophyroscolecidae family



TOTAL PROTOZOAL COUNT CALCULATION

- Volume of 1 WBC large squares = 0.1 cu mm
- 0.1 cu mm of SRL contains = X protozoa
- 1 ml (1000 ml) contains = (X x 1000)/ 0.1 = X x 10,000 protozoa
- Dilution rate = 5 (5 ml of SRL is diluted to 25 ml)
- So 1 ml of diluted SRL = 1/5 = 0.2 ml of undiluted SRL
- 0.2 ml of undiluted SRL contains= X x 10,000 Protozoa
- 1 ml of undiluted SRL contains = X x 10,000 x (1/0.2)

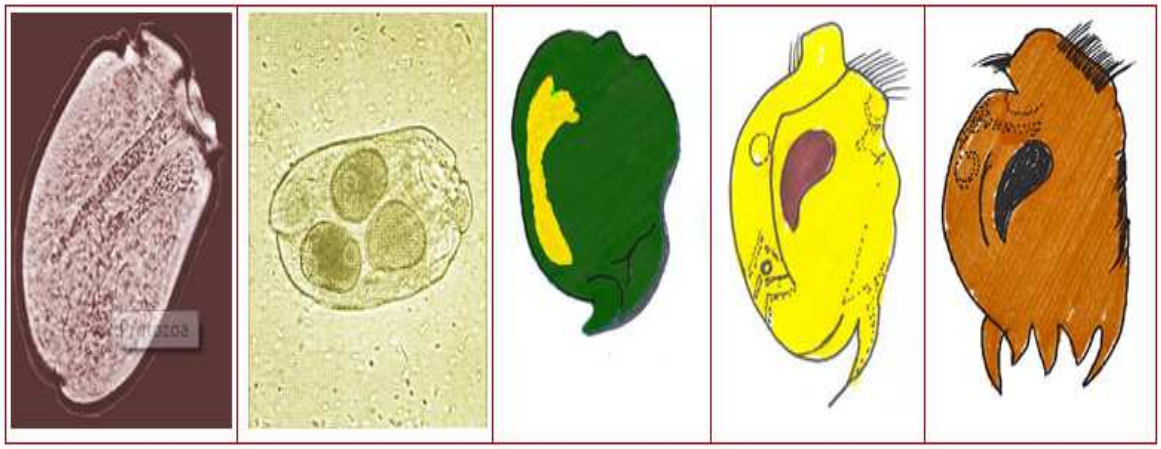
  = X x 10,000 x 5 protozoa

  = X x 5 x 10^4

Result

- Number of protozoa per ml of SRL = x 10^4

OPHYROSCOLECIDAE FAMILY



EXERCISE-16: ESTIMATION OF VOLATILE FATTY ACIDS IN RUMEN LIQUOR BARNETT AND REID METHOD

- Volatile Fatty Acids are collected in distilled water with the help of distillation apparatus and titrated against standard alkali.

Materials

- Distillation apparatus
- Heating apparatus
- Scarisbrick buffer (10% Potassium oxalate and 5% oxalic acid)
- 0.01N NaOH
- Phenolphthalein
- Distilled water
- 250 ml conical flask

PROCEDURE - VFA ESTIMATION

- It consists of two steps. There are
  - Preparation of Strain Rumen Liquor (SRL)
  - Estimation of Volatile Fatty Acids

Preparation of Strain Rumen Liquor (SRL)

- Collect about 200 ml of rumen liquor from the fistulated animal.

- Strain the rumen liquor (SRL) using four layered cheese cloth and centrifuge at 1000 RPM for 5 minutes.
- Arrest the bacterial activity of the rumen liquor by adding saturated solution of mercuric chloride @ the ratio of 1:20.
- Store the solution in the refrigerator at 4°C until further use.

Estimation of Volatile Fatty Acids

- Take 2 ml of SRL and 1 ml of liquid paraffin containing 2% octyl alcohol in the micro Kjeldhal apparatus.
- Add 2 ml of Scarisbrick Buffer (1 ml each of 10% Potassium oxalate solution and 5% Oxalic acid).
- Take 25 ml of distilled water in a conical flask and keep the tip of the glass collection tube deeply into the distilled water.
- Collect about 150 ml of the distillate in the conical flask.
- Titrate the distillate against 0.01 NaOH using Phenolphthalein as indicator.
- Titrate the blank as above using 2 ml distilled water instead of SRL. Subtract the value of the blank from the sample.

CALCULATION

• 1cc of 0.01 N NaOH	= 1cc of 0.01 N acetic acid
• Titer	= 0.01 M. Moles of acetic acid
• Titer value	= X cc
• Volume of rumen liquor	= 2 cc
• For 1000 cc of liquor	= (X x 0.01 x 1000) / 2 = X x 5 = millimoles of VFA/litre
• Result	= ..... millimoles/Litre

EXERCISE-17: ESTIMATION OF RUMINAL AMMONIA CONWAY CELL METHOD

Materials Required

- Conway microdiffusion cell.
- Strained rumen liquor.
- Saturated sodium carbonate solution.
- 2% boric acid.
- Mixed indicator (Tashiro's reagent, 66mg methyl red, 33mg methylene blue or bromocresol green in 100 ml alcohol) 1 ml/100 ml boric acid.
- Incubator.
- 0.01 N sulphuric acid



Conway microdiffusion cell

Procedure

- Microdiffusion technique of conway (1957) is the method used to estimate ammonia nitrogen concentration in strained rumen liquor.
- Rumen liquor collected from the animal is processed through four layered cheese cloth to prepare strained rumen liquor.
- The conway microdiffusion cell has two chambers Viz. outer chamber and inner chamber.
- Two ml of two percent boric acid with mixed indicator is placed in the inner chamber.
- One ml of saturated sodium carbonate solution is placed in the outer chamber in one side and one ml of strained rumen liquor is placed on the opposite side of the outer chamber.
- The lid of the conway microdiffusion cell is fixed and the contents in the outer chamber are mixed rorating the assembly gently.
- The conway dish is incubated at 38° C ± 1° C for an hour.
- Due to mixing of SRL and sodium carbonate, ammonia in the SRL is released and is trapped by the boric acid present in the central chamber during incubation.
- After an hour, the content in the inner chamber are titrated against 0.01N H_2SO_4 using burette.

Calculation

Ammonia nitrogen = 1ml of 0.01N H_2SO_4 = 14 mg NH_3-N / 100 ml SRL

- Other methods used for NH3-N estimation: Microkjeldhal distillation and colorimetric method.

EXERCISE-18: RECORDING OF INTESTINAL MOTILITY MATERIALS

- Organ bath
- Oxygen cylinder
- Kymograph
- Tyrode solution
- Anesthetic
  - Ketamine and
  - Midazolam
- Adrenaline (1 in 1,00,000)
- Acetylcholine (1in 1,00,000)

TYRODE SOLUTION

TYRODE SOLUTION	
NaCl	0.8 g
KCl	0.02 g
CaCl2	0.02 g
Mg Cl2	0.01 g
NaHCO3	0.1 g
NaH2PO4	0.005 g
Glucose	0.1 g and
Distilled water	100 ml

ORGAN BATH

- It consists of an outer container filled with tap water.
- Temperature is maintained at 38 ° C with a help of thermostat and a glass-warming coil.
- The coil lies in the outer bath with one end connected to a central glass vessel and other end to a bottle containing Tyrode solution.
- The central glass vessel has an outlet as its lower end, which is fitted to the bottom of the container and is used to drain off the Tyrode solution.
- A hollow glass tube with a hook at its lower end is clamped to a pillar by a boss head clamp and the glass tube can be lowered into the central vessel containing Tyrode solution.
- The pillar also carries a writing lever.
- A stirrer is provided at one end of the outer vessel to maintain uniform temperature.

RECORDING OF INTESTINAL MOTILITY - PROCEDURE

- Prepare the organ bath to maintain the temperature at 38°C, fill the container with Tyrode solution.
- Anaesthetise the rabbit by injecting Ketamine @ 25mg/kg body weight i/m in combination with Midazolam @ 5mg/kg body weight intra muscularly.
- Shave the abdominal region and clean the area with warm water.
- After cleaning, the skin can be sterilized by swabbing with 70% ethanol.
- Open the abdominal cavity by making linear incision along the linea alba.
- Remove gently a few 3-4cm long portions of jejunum (without mesentery) and oxygenate them frequently.
- Using a needle put a thread onto the wall of one of the isolated intestine place from inside to outside and make a small loop, leaving the thread in place.
- Pass a threaded needle to the other end of the intestinal piece, tie a knot and leave about 15cm thread.
- The long free end of this thread is connected to the lever.
- Do not close the lumen of the gut.
- Pass the loop over the O_2 tube and tie the intestine with it and put this tube into the inner vessel and fix it in position.
- Connect the oxygen tube to O_2 cylinder and allow 2-3 bubbles of O_2 to pass through Tyrode solution per minute.
- Carefully adjust the lever to mark the kymograph. Record the intestinal movements on the kymograph.
- Add 1 ml of adrenaline to the Tyrode solution and record its effect.
- Completely drain the Tyrode solution, replace fresh Tyrode solution, record the normal intestinal motility.
- Add 0.5 ml of acetylcholine to Tyrode solution and record its effect.

EXERCISE-19: SPIROMETRY
INTRODUCTION

- Spirometry is the most common pulmonary function tests evaluating the health of the lung and being practiced as a basic tool to evaluate the respiratory status of the subject.
- It is a quick and simple test. It provides information on the mechanical properties of the tracheo-bronchial tree, interstitum of the lung, pleura, chest wall and respiratory muscles.
- It generates pneumotachograph to assess long dysfunctions.

Basis of Spirometry

- It is the measurement of volume of (amount) and /or flow rate (speed) of air that can be inhaled or exhaled in a given time. It monitors breathing and helps to assess tidal volume and various long capacities.
- Spirometry apparatus consists of volume measuring devices and flow measuring devices.

Medical Spirometer

- It consists of
    - Drum inverted over a water cylinder.
    - Weight counter balancing the drum.

- Oxygen air cylinder
- Facility with a writing point which is fixed to the counter weight.
- Writing point inscribes on a kymograph.
- Corrugated rubber tubing connecting the mouth of the subject with gas chamber and recording drum.

SPIROMETRY - PROCEDURE

- Water chamber is filled with water.
- Air drum is filled with oxygen or atmospheric air.
- Subject is placed on a table in a comfortable position without exaggerating with the mouth piece fixed on the muzzle.
- Subject is facilitated to adapt for oral breathing through mouth piece.
- Mouth piece is connected to air chamber.
- Kymograph is adjusted to move in a slow speed.
- As the animal breathes in the air chamber moves up which is counter balanced by the weight.
- Inspiratory and expiratory movements are recorded.
- Total volume is calculated.
- As the subject inhale and exhale to its maximum capacity inspiratory reserve volume, expiratory reserve volume, inspiratory capacity and functional residual capacities are calculated using the tracings.

APPLICATION OF SPIROMETRY

- To identify the presence of abnormality.
- To determine the nature of the abnormality.
- To grade the extent of functional impairment.
- To monitor the progress of dysfunction.
- To evaluate the response to therapy.

CHAPTER-1: GENERAL ORGANIZATION OF ENDOCRINE SYSTEM

Learning objectives

The outcome of this chapter helps to

- Understand the location of different endocrine glands
- Know the functions of endocrine factors
- Know the interrelationship between endocrine and nervous system
- Understand the involvement of endocrine system in animal reproduction and production.

GENERAL ORGANIZATION OF ENDOCRINE SYSTEM
INTRODUCTION

- Organs and sytems in the body need to communicate with each other so that homeostasis is maintained.

- Cells use either chemical and/ or electrical signals to communicate with each other.

Cell-to-cell communication

- Cells can communicate by direct transfer of electrical or chemical signals through gap junctions (proteins side-by-side with a hole down the middle) that connect cells to each other.
- Chemical signals are three types
    - **Autocrines** are chemicals that are produced by a cell and act on the same cell
    - **Paracrines** are chemicals that are produced by one cell and act on nearby cells by diffusing locally through the ECF
    - **Endocrine** substances are chemicals that travel long distances through blood

ENDOCRINE SYSTEM

- A wide variety of physiological processes are carried out by the endocrine system through chemical messengers called *"hormones"*.
- The endocrine system consists of many ductless glands which secrete hormones directly into the blood stream. These hormones affect processes like metabolism, sex drive etc.
- N ervous system is wired control system of the body whereas endocrine system is the wireless control system
- A hormone is a specific messenger molecule synthesized and secreted by an endocrine gland or a group of specialized cells. These glands are ductless, which means that their secretions (hormones) are released directly into the blood stream and travel to elsewhere in the body to target organs, upon which they act. This is in contrast to other types of glands in the body such as salivary and sweat glands, which have ducts for carrying the secretions to the points/locations of action.

ENDOCRINOLOGY

- A science dealing with ductless (endocrine) glands. Greek word "endo" (within) + "krinein" (separate)
- Modern (broad) definition of endocrinology - the study of the adjustments of homeostatic and other activities accomplished by chemical messengers.

ENDOCRINE GLANDS

Endocrine gland	Location / description of the endocrine gland	Major hormones secreted	Chemical class
Hypothalamus	Part of the brain that acts as the main control center for the autonomic nervous system, re-establishes homeostasis, and coordinates the endocrine system	GHRH, TRH, CRH, GnRH, Somatostatin, PIF	Peptide

	Often called the “master gland”because it secretes hormones that control other endocrine glands		
Anterior pituitary	Located in the 'sella turcica' at the base of the brain	STH or GH, Prolactin, TSH, ACTH, FSH, LH	Polypeptide/ protein
Posterior pituitary	Neurological origin. Does not synthesize any hormones but releases two hormones which are synthesized in the hypothalamus	Oxytocin, Vasopressin/ADH	Peptide
Thyroid gland	“Butterfly - shaped” gland located on either side of 1st and 2nd tracheal rings in the neck	Thyroxine Tri-iodothyronin	Amino acid derived
		Calcitonin	Polypeptide
Parathyroid	Located on the back of the thyroid gland	PTH	Polypeptide
Pancreas	Islets of Langerhans	Insulin Glucagon, Somatostatin, Pancreatic polypeptide	Protein
Adrenal gland	Located like a cap at the cranial border of the kidneys	Glucocorticoids, Mineralocorticoids Sex steroids	Steroids
		Epinephrine Norepinephrine	Amine
Ovaries	Primary reproductive organs of a female	Estrogen, Progesterone	Steroids
		Inhibin, Relaxin, Activin, Follistatin, Oxytocin, GFs, etc.	Protein
Testes	Male reproductive organ that produce sperm. Located in the scrotum outside the abdominal cavity in most mammals” and within the abdominal cavity in birds	Testosterone	Steroids
		Inhibin	Protein
Thymus	Located between the lobes of the lungs in the upper chest cavity	Thymosin	Peptide
Pineal gland	Small pine cone - shaped structure located deep in the center of the brain	Melatonin	Amine
Heart	Located in the thoracic cavity	Atrial natriuretic peptide	Polypeptide

Kidney	Located in the abdominal cavity at sub lumbar level	Erythropoietin	Polypeptide
GI tract	Located in the abdominal cavity	Gastrin Secretin Cholecyctokinin	Polypeptide
Placenta	Temporary endocrine gland, located within the uterus during gestation	Progesterone	Steroids
		Estrogen	
		hCG	Glycoprotein
		PMSG	
		Protein B	
		Relaxin	Polypeptide
		Placental lactogen	

GENERAL FUNCTIONS OF ENDOCRINE SYSTEM

- Maintain homeostasis
- Control chemical and water balance in body
- Control growth and metabolism
- Control embryonic development and preparation for nurturing a newborn
- Influence sexual behavior, stimulate growth and maturation of the genitalia
- Feedback to the nervous system
- Regulating red cell production
- Inducing adaptive changes to stress

EFFECT OF HORMONES ON CELL STRUCTURE AND FUNCTION

- Hormonal stimulus usually produces one or more of the following changes on cell
    - Altering plasma membrane permeability or membrane potential (or both) by opening or closing ion channels.
    - Stimulates the synthesis of proteins or regulatory molecules, such as enzymes with in the cell.
    - Enzyme activation or deactivation (changes metabolism)
    - Induction of secretory activity
    - Stimulation of mitosis or meiosis
    - Activation of genes
    - Smooth muscle contraction

INTER-RELATIONSHIP BETWEEN ENDOCRINE AND NERVOUS SYSTEM

- The nervous system controls homeostasis through nerve impulses that trigger the release of neurotransmitters whose effect results in either excitation or inhibition of other neurons, muscle fibres, or glands. The endocrine system releases hormones into blood stream that travels to the target organ where it alters the physiological activity.

- In nervous system, a sensory receptor senses the change and sends action potentials by its afferent sensory neuron to the integration center in the brain or spinal cord. The efferent path is the motor neuron that takes action potentials to the effectors (muscle or glands) for the response of contraction or secretion.
- In the endocrine system, the gland/ the cells may act as the receptors and the integration centres which are often influenced by neural and other inputs. The response is either to change the rate of an enzymatic reaction, to control the transport of certain molecules across cell membranes, or to control gene expression and thus make new proteins.

Nervous system	Endocrine system
• Nervous system regulates the activities of muscles and glands via electrical impulses transported through neurons	• Endocrine System regulates the body's metabolic activity via hormones that are transported in the blood
• Reflexes of the nervous system are more specific because nerves regulate specific target cells	• Hormones can control multiple organs and systems
• Nervous system uses both electrical (action potentials) and chemical (neurotransmitters) signals	• Endocrine system uses only chemical signals (hormones)
• Neural control is fast; its effects are short-lived	• Hormonal control is comparatively slower; generally its effects are prolonged
• Stimulus intensity in the nervous system is represented by the frequency and amplitude of action potentials.	• Stimulus intensity in the endocrine system is determined by the amount and duration of hormone(s) released.

HISTORICAL BACKGROUND

- How did scientists first find out what endocrine glands do?
  - Scientists first removed the gland and monitored the organism for changes (like in hormone deficiency).
  - Then scientists replaced the gland or gave an extract of the gland to see if the abnormalities disappeared (like in replacement therapy).

  - Then scientists implanted the gland or gave extract to a normal individual to see the symptoms (like in hormone excess).
  - Once a gland was identified as a source of hormones, extracts from the gland were purified and a test was designed to calibrate the activity of the hormone in another organism (known as a ***bioassay***).
- The writings of Hippocratus and Aristotle as early as BC 460 to 322 contain some information that their might be internal control over body functions
- First recorded endocrine experiment was published by Berthhold (1849). He discovered castrated cockerels failed to develop their combs and wattles and also failed to exhibit male behaviour. Replacement of one or both testes back into abdominal cavity restored the lost function.
- In 1889, von Mering and Minkowski demonstrated that surgical removal of pancreas from dog produced a disease later to be known as diabetes mellitus. This led to the recognition of insulin in the control of diabetes mellitus in man and dogs.
- In 1902, Canadian physiologists ***Bayliss and Starling*** discovered that a substance liberated by small intestinal mucosa stimulate the pancreatic juice flow. The active substance was named as *secretin*. They coined the word “hormone” (from the Greek "*I arouse to activity*" or "*I excite*") to this newly discovered substance.
- ***Banting and Best*** (1921) isolated the potent pancreatic extract containing insulin
- In 1949, Hench and co workers isolated a hormone from adrenal cortex- cortisone or compound E that relieved some of the symptoms of rheumatoid arthritis.
- In 1953, Simpson and Tait found that unidentified steroid present in adrenal gland was aldosterone. In the same year, ***Sanger*** established the amino acid sequence for insulin for which he bagged the ***Nobel Prize***.
- In 1953, Du Vigneaud and his co-workers determined the structure of oxytocin and vasopressin.
- In 1955, Harris provided the data suggesting that the release of pituitary hormones was controlled by humoral factors of hypothalamic origin.
- In 1962, Earl Sutherland described the presence of cyclic AMP in biological materials.
- In 1978, insulin was synthesized by a strain of genetically engineered *E.coli*.
- For the development of RIA for hormone analysis, Rosalyn Yalow received the Nobel prize in 1978.

CHAPTER-2: HORMONE - DEFINITION AND CLASSIFICATION

Learning objectives

- To understand the classification of hormones based on chemical structure, physiological function and solubility.

DEFINITION OF HORMONE

- The term hormone means "to excite" or "arouse".
- Hormones are chemicals secreted at variable rates in response to external or internal stimuli by an endocrine gland or a group of cells in very low concentrations (nanomolar to picomolar – 10^{-9} to 10^{-12} M) directly into the blood for transport to some distant target organs/cells where they alter already existing reactions but cannot initiate new reactions.

CLASSIFICATION BASED ON CHEMICAL NATURE

- Hormones are chemically classified into the following categories

    - Peptides and proteins - consist of chains of amino acids. Examples of small peptide hormones are TRH and vasopressin. Peptides composed of numerous amino acids are referred to as proteins. Examples of protein hormones include insulin and growth hormone. More complex protein hormones bear carbohydrate side chains and are called glycoprotein hormones (e.g) LH, FSH and TSH.
    - Steroids - derived from cholesterol and contains basic cyclopentano perhydro phenantherene nucleus. (e.g) Estrogen, progesterone, cortisol. Being non-polar, they are able to pass through the phospho-lipid membrane of a cell and enter the cytoplasm .
    - Amines - are derivatives of the amino acids tyrosine and tryptophan . (e.g) epinephrine, norepinephrine.
    - Iodionated aminoacids – (e.g) Thyroxine
    - Fatty acids derivatives - Eicosanoids - they are a large group of molecules derived from Arachadonic acid. The principal groups of hormones of this class are prostaglandins, prostacyclins, leukotrienes and thromboxanes. These hormones are rapidly inactivated by being metabolized, and are typically active for only a few seconds.

CHEMICAL CLASSIFICATION OF HORMONES

- Chemical structure determines
    - solubility and transport characteristics
    - degradation rate (plasma half-life)
    - storage
    - route of exogenous administration

Hormones	Characteristics
Peptides	• Structure : chains of amino acids; contain amino terminal (N-terminal) and carboxy terminal (C-terminal) ends; linear - ring or two chains - carbohydrate residues (= glycoproteins). • Solubility: hydrophilic • Synthesis: rough ER, packaged in Golgi complex • Storage: secretory granules • Secretion: exocytosis • Transport: mostly as a free hormone • Receptor site: surface of target cell • Action: channel changes or second messenger system
Amines	**Catecholamines (dopamine, epinephrine, and norepinephrine)** • Structure: tyrosine derivative • Solubility: hydrophilic • Synthesis: cytosol • Storage: secretory (chromaffin) granules

	• Secretion: exocytosis • Transport: as a free hormone and bound to plasma proteins • Receptor site: surface of target cell • Action: second messenger system **Thyroid hormones** • Structure: iodinated tyrosine derivative • Solubility: lipophilic • Synthesis: extracellular colloid • Storage: extracellular colloid • Secretion: endocytosis of colloid • Transport: mostly bound to plasma proteins • Receptor site: inside of target cell • Action: direct effects on genes ® production of new proteins
Steroids	**Mineralocorticoids, glucocorticoids, sex hormones, and vitamin D** • Structure: cholesterol derivatives, lipids, steroid ring (3 six-membered and 1 five-membered ring) • Solubility: lipophilic • Synthesis: stepwise modification of cholesterol molecule in various intracellular compartments (in the endocrine gland and also in the blood and target tissues) • Storage: hormones not stored, only precursor (cholesterol) stored. • Secretion: diffusion • Transport: mostly bound to plasma proteins • Receptor site: inside of target cell • Action: direct effects on genes for production of new proteins
Fatty acid derivatives	• Structure: Fatty acid • Solubility: Lipoophilic

PHYSIOLOGICAL CLASSIFICATION OF HORMONES

- Hormones Regulating Energy Metabolism. Some examples are
  - Insulin, Glucagon, Glucocorticoids, Epinephrine, ACTH, Thyroid hormone
- Hormones Regulating Mineral Metabolism. Some examples are
  - Aldosterone, Renin-Angiotensin II & Natriuretic hormone, PTH, CT , Vitamin D3
- Hormones Regulating Growth. Some examples are
  - GH, Insulin, Estrogen, Androgen, Somatostatin, Thyroxine and Cortisol
- Hormones Regulating Reproduction. Some examples are
  - GnRH, FSH, LH /ICSH, Estrogen, Inhibin, Progestrone, Testosterone, Oxytocin, Prostaglandins and Relaxin, GnIH, Thyroxin, Cortisol, Prolactin and Gonadocrnin
- Hormones Regulating Milk Secretion. Some examples are

  - Estrogen, Prolactin, TSH, ACTH, GH, Glucocorticoid, Placental Lactogen, Thyroxin, Insulin and Oxytocin
- Hormones Regulating Blood Pressure. Some examples are
  - Renin-Angiotensin, Epinephrine, Norepinephrine, Vasopressin, ACTH , Vasotocin, ADH and ANP
- Hormones Regulating Water Balance. Some examples aree
  - Vasopressin (ADH), Renin-Angiotensin, Glucocorticoids & Insulin
- Local Hormones . Some examples are
  - PGs, Histamine, Serotonin
- Gut Hormones. Some examples are
  - Gastrin, Secretin, CCK, Motilin, Bombesin Somatostatin, Gastric inhibitory peptide (GIP), Vasoactive intestinal Polypeptide & Substance P

CHAPTER-3: HORMONE - SYNTHESIS AND RELEASE

Learning objectives

- At the end of this chapter the learner will be able to understand the following
  - Synthesis of protein hormones
  - Synthesis of steroid hormones
  - Pattern of hormone secretion
  - Release of hormones into circulation
  - Transport of hormones.

PEPTIDE HORMONES

- Most hormones are peptide in nature and are composed of aminoacids. Like other proteins, they are synthesized on ribosomes where their specific aminoacids sequence is determined (translated) by a specific messenger RNA sequence. The nucleotide sequences of the mRNA are transcribed from specific nucleotide sequences in the DNA. The proteins are then released and transported into the rough endoplasmic reticulum. The initial protein formed is larger than the active hormone and is called preprohormone.
- In the interior of RER, pre portion of the hormone is cleaved to form prohormone. This in turn transported in the transport vesicles to the golgi bodies where they may be altered (e.g) sulphated or combined with carbohydrate moieties and the hormone is in its final form within the Golgi apparatus.
- Vesicles containing the hormone and possibly other products (proteolytic enzymes) are then pinched off the terminal cisternae of golgi bodies. These secretory vesicles are distributed within the cytoplasm. Protein hormones are stored within the gland until needed for release. Although some of the hormones are secreted continuously, most are excreted by a process of exocytosis of granules in response to specific signals. Exocytosis requires ATP and Ca^{2+}

STEROID HORMONES

- Steroid hormones are synthesized within the smooth endoplasmic reticulum. Steroid secreting cells contain large amounts of smooth endoplasmic reticulum within them. However, a complex multiple enzyme system is required for the synthesis of steroids. These enzymes are present within the mitochondria as well as the cytoplasm.

- Majority of the steroids is formed from cholesterol, which is synthesized by the liver. Cholesterol is either utilized immediately for steroid synthesis or stored in granules in ester form within the cells.
- The first step in the synthesis of all steroid hormones from cholesterol involves cleavage of side chain of cholesterol to form pregnenolone, which occurs within the mitochondria. Subsequently modification of the steroid molecules may occur within the mitochondrion (or) may involve the endoplasmic reticulum of the cell. The type of steroid hormone that is eventually synthesized depends upon the specific enzymes within the particular cell type.
- There is no provision for storage of steroid hormones within the cells. On appropriate stimulation, enzymes in the steroid secreting cells can within minutes cause the necessary chemical conversions to the final hormone. They are secreted immediately after formation by simple diffusion across the cell membrane because of their lipophilic nature

THYROID HORMONES

- These iodinated hormones are synthesized on a proteinaceous substrate (thyroglobulin) located extra cellularly, by iodination of tyrosine.
- They are then taken up by endocytosis and transported through the follicular cells of the thyroid gland where they are enzymatically released from the carrier protein prior to secretion.

PATTERN OF HORMONE SECRETION

- Fluctuations of frequency and amplitude of hormone secretion is described as hormone peaks, pulses, spikes or bursts.
- The rhythm of hormone secretion may be one the following
    - Ultradian – Many short pulses occurring every few minutes to few hours each day (Adrenaline and noradrenaline)
    - Circadian – peaks occur approximately once in 24 h (cortisol peaks in early morning)
    - Infradian – pulses take more than a day but less than a year to occur (preovulatory surge of LH in dogs once in 6 months)

HORMONE TRANSPORT IN THE BLOOD

- Most water soluble hormone molecules circulate in the blood in the free from (not attached to a protein) most lipid soluble hormones bind to transport proteins. Binding of hormones to proteins are loose and reversible.
- Transport proteins are albumin and globulins synthesized by the liver and have following functions
    - provide a circulating pool for the hormone
    - improve the transportability of the lipid-soluble hormones by making them temporarily water-soluble
    - limit the availability of the hormone on the target site
    - prolong the half-life of hormone and they protect circulating hormones from being broken down by enzymes in the blood plasma and liver - free hormones may be

broken down or cleared from the blood in minutes, whereas bound hormones may circulate for hours or weeks.
    - provide a ready reserve of hormone, since already present in the bloodstream
- Bound hormones are hormones attached to a transport protein and unbounded (free) hormone is one that is not attached to a transport protein. Only free hormones (unbound/released from the binding protein) can leave a blood capillary to influence the target.
- Therefore only free hormones are physiologically active. There is a dynamic equilibrium between the quantities of free and bound forms of any hormone in circulation. As the free hormone leaves circulation to influence the targets, additional bound hormone is released from the binding proteins.
- Example of transport proteins
    - Thyroid hormone binds to three transport proteins
        - albumin
        - albumin-like protein called thyrectin
        - an alpha globulin named thyroxine binding globulin (TBG)
    - Transcortin is the corticosteriod-binding globulin (CBG), which has high affinity for cortisol and corticosterone,
    - Sex hormone-binding globulin (SHBG) aids transport of estradiol, progesterone and testosterone.

CHAPTER-4: MECHANISM OF ACTION OF HORMONES

Learning objectives

At the end of this chapter the learner will be able to understand the following

- Location of receptors
- Hormone - receptor interaction
- Target cell
- Mechanism of hormone action.

TARGET ORGANS AND RECEPTOR TYPES

- *Target Cell* : The cells that contain specific receptors for a hormone
- Tropic hormone – A hormone that stimulates functions of another organ / tissue. *Tropic*: a nourishing relationship.
- Although all cells are exposed to all hormones, only few cells or organs can respond to all specific hormone, termed as *target cells or target organ*. This is because of *highly specific receptor sites* in the target organ.
- The receptors are large protein molecules. Hormone exerts its physiological actions only after binding to specific receptors.
- Despite the diversity of hormones, all hormone receptors can be categorized into one of two types, based on their location
- Membrane-bound receptors called as *cell surface* (membrane) *receptors*
- Receptors located within the cell called as *intracellular receptors*

Location of Receptor	Classes of Hormones	Principle Mechanism of Action
Cell surface receptors	Proteins and peptides, catecholamines and eicosanoids	Generation of *second messengers* which alter the activity of other molecules - usually enzymes - within the cell
Intracellular receptors – proteins, floating in cytoplasm and/or nucleus	Steroids and thyroid hormones	Alter transcriptional activity of responsive genes and thereby alter the cell's protein (enzyme) content

RECEPTOR PROPERTIES

- The properties of the receptors are
  - Each cell may have receptors for more than one hormone and therefore is a target for several hormones
  - Specificity spill over – hormones with similar structure may bind to one another's receptors usually with lower affinity and ability to activate the receptors.
  - Activation of receptor by a ligand is an all or none phenomena
  - Message carried by the ligand is in its three dimensional structure which is important to bind to the receptors
  - Once a hormone binds to its receptor the message is delivered and the hormone plays no further role
    Hormone receptor combination initiates a cascade of reactions
- All receptors are proteins
  Specificity: The binding site of the receptor is highly specific for each signaling molecule (hormone, neurotransmitter); molecules that bind specifically to receptor are called ligands.
  However, some molecules having structural similarity to ligands can also bind with receptors and produce a physiological response and such molecules are called as agonists. Some molecules having structural similarity with hormone may bind with receptors without producing a response and inhibit the hormone effect are called as antagonists.
  High affinity: Usually the concentration of the hormone in the blood is very low (in pmol/l to µmol/l). Because the receptors have high affinity to hormones, binding occurs rapidly
  Binding capacity: Number of receptors available on the target cells at any given time is the limiting factor that determines number of hormone molecules bound to target organs; i.e. the binding capacity of target cells is decided by the receptor numbers. The concentration of receptors is quite variable
  When a target cell is exposed to high level of hormone for a period of time, the number of active receptors will be reduced (by receptor degradation) known as down regulation. During deficiency of a hormone, there is an increase in the number of active receptors known as up regulation.
  When a receptor becomes bound to a hormone, the receptor undergoes a conformational change (known as receptor activation), which allows it to interact productively with other components of the cells, leading ultimately to an alteration in the physiologic state of the cell.

MECHANISM OF PROTEIN HORMONE ACTION

- The first step in protein or peptide hormone action, the hormone functions as the first messenger and binds with specific cell surface receptor sites in a target organ; the receptors on the target cell selectively "bind" or"trap" or take up a particular hormone on the cell membrane of target organ.
- In the second step the hormone forms a receptor hormone complex which induces receptor activation. The activated receptor initiates a chain of intracellular events
- The membrane receptors can be grouped into the following types based on how the protein hormone signal is transferred from the receptor into the cell
    - receptors that activate G-proteins
    - receptors functioning as ion channels
    - receptors that activate protein kinases
    - receptors that activate guanylyl cyclase

Receptors that activate G-proteins

- The receptors are coupled to G-proteins (consists of α-,β-, and γ- subunits) which are membrane proteins and which can bind with GTP or GDP. G-protein coupled receptors are involved in signal transduction of protein/ peptide hormones, acetylcholine, serotonin, some receptors of epinephrine and norepinephrine.
- Binding of the hormone to the receptor brings about conformational change in receptor. The activated receptor can now bind with G-protein which undergo conformational change and become activated. The activated G-protein releases GDP from α-subunit and binds with GTP. This binding with GTP dissociates α-subunit from β-γ subunits of G-protein
- The released α-subunit functions (1) by activating or inhibiting many intracellular enzymes or (2) by opening ion channels and modifying ionic movement across the membrane
- Generally, the G-protein activates an enzyme that increases or decreases formation of intracellular regulatory molecules called second messenger.
- The second messenger activates enzymes – protein kinases and phosphorylates cellular enzymes by transferring phosphate from ATP to proteins
- The phosphorylated enzymes increase or decrease intracellular enzyme activity
- Some of the important second messengers formed by the G-protein coupled receptors are
    - cyclic adenosine monophosphate (cAMP)
    - diacylglycerol (DAG)
    - inositol triphosphate (IP3)
    - eicosanoids
- Adenyl cyclase is a membrane bound enzyme which converts cytoplasmic ATP to cylic 3, 5-adenosine monophosphate (cAMP). Adenyl cyclase is activated by activated-G-protein.
- The second messenger, cAMP is released into the cytoplasm where it activates cAMP-dependent protein kinase called protein kinase-A. (phosphodiesterase, an intracellular enzyme inactivates cAMP to AMP)
- Intracellular accumulation of cAMP
    - modifies the enzyme activity - phosphorylation ofprotein kinase which is responsible for biological response
    - alters membrane permeability or
    - may stimulate hormone release.

- The protein kinase A phosphorylates key intracellular proteins and alters their activity and produces various biological effects ofthe hormone.
- DAG: The cell membrane enzyme phospholipase-C (PLC) acts on a membrane phospholipid called phosphotidylinositol bisphosphate (PIP3) and cleaves it into AG and IP3. PLC is activated by G-protein.
- The membrane-bound DAG binds to a protein kinase called protein kinase C which phosphorylates intracellular proteins similar to protein kinase A and brings about biological effects
- The IP3 formed along with DAG enters the cytosol, opens Ca 2+ channels in the smooth endoplasmic reticulum (important Ca 2+ storage place) and the Ca 2+ diffuses out to increase the cytoplasmic Ca 2+ concentration.This Ca 2+ binds to calmodulin (calcium binding protein) and the calcium-calmodulin complex activates calcium-dependent protein kinase which then phosphorylates intracellular proteins to alter cellular functions.
- G-protein can also activate a membrane bound enzyme known as phospholipase A2 and the activated enzyme breaks down phosphotidylcholine and releases the long-chain fatty acid arachidonic acid from the membrane phospholipid.
- An intracellular enzyme cyclooxygenase converts the arachidonic acid to prostaglandins and thromboxanes while another intracellular enzyme lipoxygenase converts arachidonic acid to leukotrienes. The prostaglandins, thromboxanes and leukotrienes are the eicosanoids which are produced in most cell s of the body. The eicosanoids function as intracellular or extracellular signalling molecules and bring about autocrine / paracrine effects

Receptors functioning as ion channels

- This type of receptor function is important in neurons and muscles. Many of the ion channels on the cell membrane also act as receptors for the first messenger; binding of hormone will open or close the ion channels.
- Post-synaptic membrane of neurons and their target cells contain ligand-gated channels which bind with neurotransmitters. In some cases the first messenger activate G-protein second messenger which in turn regulate ion channel opening/ closing.

Receptors that activate tyrosine kinase

- Tyrosine-kinase receptor activation is important for the action insulin, growth factors like IGF-I, epidermal growth factor, and platelet-derived growth factor.
- The hormone binds to the membrane receptor and the activated receptor itself acquires tyrosine kinase activity. The tyrosine kinase phosphorylates the amino acid tyrosine present in intracellular proteins which then alters the functions of cytosolic enzymes producing the biological effects of the hormone.
- For hormones like PRL, GH, EPO and cytokines, the binding of the hormone with receptor activates intracellular tyrosine kinases which then phosphorylates tyrosine to bring about response

Receptors that activate guanylyl cyclase

- cGMP formed by the action of guanylyl cyclase on GTPact as second messenger similar to cAMP. cGMP activates cGMP-dependent protein kinase called protein kinase-G which phosphorylates cellular proteins and brings about biological effects
- Atrial natriuretic peptide produces its effect through cGMP
- In all cases, a small signal generated by a hormone binding to its receptor is amplified many thousand times (signal amplification) within the cell and brings about cascade of actions that change the cell's physiologic state.

MECHANISM OF ACTION OF STEROID HORMONES

- Being lipids, steroid hormones enter the cell by simple diffusion across the plasma membrane.
    - The receptors for these hormones exist either in the cytoplasm or nucleus, where the receptors bind with the hormone.
    - Steroids stimulate their target cells by the intracellular action by entering the cytoplasm. Receptors for steroid hormones are present within the cytoplasm of the target cells.
    - When hormone binds to intracellular receptor, a characteristic series of events occurs.
- **Receptor activation** is the conformational change occurring in the receptor induced by the binding hormone. The major consequence of activation is that the receptor becomes competent to bind DNA. The activated hormone-receptor complex moves into the nucleus.
- **Activated receptors** bind to hormone response elements, which are short specific sequences of DNA, which are located in promoters of hormone-responsive genes.
- **Transcription** The gene to which the receptor has been bound is affected. Most commonly, receptor binding stimulates transcription. The hormone-receptor complex thus functions as a transcription factor.
- Gene transcription stimulates the formation of a specific messenger RNA which leaves the nucleus and stimulates the synthesis of specific protein necessary for steroid hormone functions. Changes in the mRNA levels effect a change in the rate of synthesis of various proteins translated.
- The biological response to a protein/peptide hormone receptor interaction is more rapid than for steroid, since, in the former case, the pre existing enzymes are activated, whereas the steroid / thyroid hormone action requires transcription and synthesis of new proteins (enzymes)

CHAPTER-5: REGULATION OF HORMONE SECRETION

Learning objectives

This chapter deals with regulation of secretion of hormone by

- Concentration of metabolite in blood
- Level of hormone in blood (Feed back control)

BY METABOLITE CONCENTRATION

- It is one of the simplest controls of hormone secretion e.g. a rising level of blood glucose signals release of blood sugar lowering hormone, insulin.
- Insulin facilitates glucose movement through the cell membrane, which is either metabolised or stored in the form of glycogen in the liver for later use. When the blood glucose level drops below normal, the blood glucose raising hormone, glucagon is released from the alpha cells of the pancreatic islets.
- It stimulates glycogenolysis of the glycogen stores of the liver, thereby raising the glucose level back to normal.

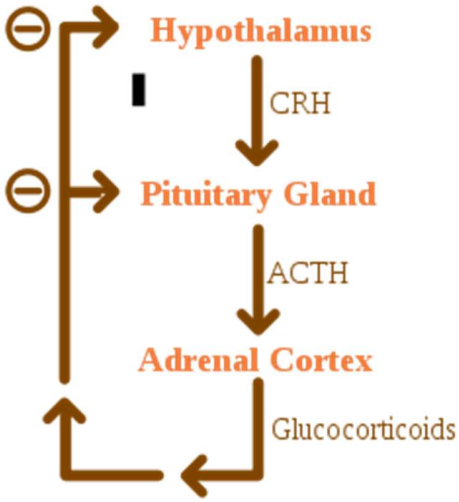
BY FEED BACK MECHANISM (or) SERVO MECHANISM

- It is an automatic means of correcting the hormone output of a gland to a desired level by level-sensing cells in the hypothalamus.
    - The feedback system operates through three pathways
        - Ultra short loop: Between hypothalamic releasing and inhibitory factors; e.g. reciprocal stimulation / inhibition between GHRH and somatostatin
        - Short loop: Between hypothalamic factors and pituitary hormones; e.g. TRH, TSH
        - Long loop: Between hypothalamic factors and peripheral endocrine glands; e.g. GnRH and gonadal hormones
- The feedback control system operates in two mechanisms

Positive Feedback System

- Increase in the level of target organ hormone increases secretion of tropic hormone.
- The hypothalamus controls secretion of tropic hormones from anterior pituitary through the secretion of peptide releasing hormones.Increase in the concentration of target organ hormone stimulates hypothalamus.In the hypothalamus, the releasing hormone output increases to effect increased production of tropic hormones by anterior pituitary and subsequently the secretions of the hormones by the target organ.
- E.g.,preovulatory release of LH, during the late stages of ovarian follicular development causes increased estrogen production which acts on hypothalamus and stimulates LHRH release and increases pulsatile rate of LH secretion from adenohypophysis to effect ovulation.
- NegativeFeedback System It is the most common feedback system.
- An increased target organ hormone concentration above certain physiological limit, causes a shut-down of releasing hormone production in the hypothalamus and also tropic hormone secretion from the anterior pituitary, thus decreases the production of hormone by the target organ.
- E.g., ACTH secretion stimulates cortisol secretion, while increased blood cortisol concentration inhibits ACTH secretion from adenohypophysis and CRH release from the hypothalamus to bring cortisol level to normal.

Negative feedback mechansim



SYNERGISM AND ANTAGONISM

- The activity of one hormone may be influenced by its interaction with one or more hormones involved in the regulation of the same function.
- The actions of different hormones are concerted or additive if they produce the same response and the combined effect of the hormones is the sum of separate actions of the individual hormones. In this case, the two hormones act by different mechanisms.
- In some cases, two hormones can produce the same effect but the effects due to the different hormones are non-additive, i.e. the two hormones act by the same common mechanism.
- Some hormones may act together to produce greater total effect than their individual effects known as synergistic effect. E.g., the toxicity of catecholamines is increased by thyroxine.
- Some hormones exhibit opposite effects of other hormones known as the antagonism; e.g., Insulin and glucagon actions on blood glucose level.
- When one hormone is required for another hormone to exert its effect it is called permissive effect; e.g., small amount of glucocorticoids is needed for catecholamines to produce their calorigenic and lipolytic effects.

CHAPTER-6: HYPOTHALAMUS

Learning objectives

At the end of this chapter the learner will be able to understand the following

- Hormone secretory nucleus in hypothalamus.
- Releasing and inhibitory hormones of hypothalamus and their functions.
-

HYPOTHALAMUS - GENERAL INFORMATION

- The hypothalamus is located adjacent to the third ventricle in the forebrain and is connected by the hypophyseal stalk to the pituitary gland (hypophysis cerebri) immediately inferior to it.
- The hypothalamus receives input from various regions of brain, sensory signals from visceral organs and retina. Hypothalamus controls autonomic nervous system, regulates body temperature, thirst, hunger, sexual behaviour and defense reactions like fear and rage.
- Hypothalamus forms the critical link between the nervous system and the endocrine system. The hypothalamus regulates both autonomic nervous activity and several aspects of endocrine function. The latter role is fulfilled through its links with the pituitary gland, which secretes a wide range of hormones.
- Hypothalamus - monitors the state of the body's internal environment by analyzing negative feedback (eg. various concentrations of chemicals in the blood) and regulates the pituitary gland.
- Secretory neurons in the arcuate, paraventricular, supra optic nuclei and preoptic area of hypothalamus synthesize and secrete seven neurohormones called hypophysiotropins otherwise known as hypothalamus-releasing hormones/factors (RH or RF) and hypothalamus inhibitory hormones or factors (IH or IF). These hypothalamic hormones pass via the hypophyseal portal system to pituitary gland and regulate the release of all the six hormones of the adenohypophysis. Regulation of the anterior pituitary secretions is mediated by the reciprocal actions of RH or IH of the hypothalamus and the concentrations of hormones of the pituitary gland
- The releasing hormones of hypothamus include
    - Growth Hormone-Releasing Hormone (GHRH/ GRH): Increases the release of GH
    - Thyroid Stimulating Hormone–Releasing Hormone (TRH): Increases the release of TSH
    - Corticotropin- Releasing Hormone (CRH): Increases the release of ACTH
    - Gonadotropic - Releasing Hormone (GnRH/LHRH): Increases release of FSH and LH
    - Prolactin-Releasing Hormone (PRH): increases release of prolactin
- The inhibitory hormones of hypothamus include
    - Somatostatin (SRIF/ GHIH): Inhibits the release of GH.
    - Prolactin Release Inhibitory Hormone (PIH-Dopamine): Inhibits the prolactin secretion
- The hormones released by the neurohypophysis – Oxytocin and ADH, are actually formed in the paraventricular and supaoptic nuclei of hypothalamus. They then travel via nerve fibres to the neurohypophysis for storage until release.

GHRH (SOMATOCRININ OR SOMATOTROPIN RELEASING HORMONE)

- GHRH is a hypothalamic peptide hormone containing 44 amino acids.
- It regulates GH release by its intermittent release into the portal pituitary vein.
- Factors that stimulate release of GHRH - estradiol, progesterone, ADH, serotonin, dopamine, a -adrenergics, stress (trauma and fever), deep sleep and exercise.
- Factors that inhibit GHRH release - somatomedins, increased levels of glucose and fatty acids, b adrenergics, melatonin and morphines.
- GH exhibits a negative feed back inhibition to GHRH release.

- As a galactopoitic agent GHRH causes 10 to 25 % increase in milk yield, total milk fat, protein and lactose when injected to cow and ewes.

TRH (THYROID STIMULATING HORMONE - RELEASING HORMONE)

- TRH is a tripeptide and it stimulates release of TSH.
- TRH also stimulates the secretion of ACTH, PRL, vasopressin and GH.
- Circulating level of T_3 or T_4 has negative feedback effect on TRH and TSH.

CRH (CORTICOTROPIN RELEASING HORMONE)

- It is a 41 amino acid peptide that stimulates ACTH synthesis and release by the corticotroph cells of pituitary gland.
- Stress enhances its synthesis and release. CRH secretion is influenced by circadian light-darkness cycle.
- In diurnal animals (cattle, dog, horse) it stimulates ACTH synthesis during morning hours.
- Other functions of CRH include stimulating the sympathetic nervous system, decreasing elevated body temperature, and suppressing reproductive functions and sexual activities, suppressing GH release and altering behaviour.

SOMATOSTATIN (SOMATOTROPION RELEASE-INHIBITING HORMONE - SRIF)

- It is a hypothalamic peptide present in two forms - 14 and 24-28 amino acids, also present in pancreas
- Potent inhibitor of GH, also inhibits TSH, PRL and ACTH, insulin and glucagon.
- Its production is stimulated by GH release from the pituitary, ingestion of carbohydrate, fat, protein, and high level of glucose, amino acids, increased output of glucagon, gastrin, VIP, secretin, CCK, acetylcholine and b adrenergics.
- Somatomedin C (from the liver), dopamine, insulin, estrodiol and a -adrenergics inhibit the release of somatostatin.
- Somatostatin has been isolated from brain, pancreas and intestine.

GnRH (GONDOTROPIC RELEASING HORMONE)/(LHRH)

- It is a **decapeptide** that stimulates the release of both FSH and LH.
- The frequency of GnRH stimulus is the major regulator of the relative proportions of FSH and LH synthesis and release i.e., less frequent GnRH pulse leads to FSH secretion and more frequent GnRH pulse causes LH secretion.
- When estradiol predominates GnRH favours the LH surge to effect ovulation.
- Under the influence of high progestrone level GnRH induces FSH to stimulate folliculogenesis and steroidogenesis.
- Inhibin is a glycoprotein hormone from ovarin follicle and testicular seminiferous tubules inhibit GnRH secretion.
- Prolactin and LH inhibit GnRH via short-loop negative feed back effect.
- GnRH secretion is also influenced by light- dark cycle, pheromones and stress.
- Dopamine, endorphins and increased levels of ovarian steroids provide inhibitory influence on GnRH release.

- In sheep and goat increased production of melatonin during longer dark periods / short day length of autumn initiates GnRH synthesis, thus the puberty in young ones and breeding season in adult. In contrast, lengthening days of spring results in minimal melatonin production and beginning of breeding season in horse.

PIH (PROLACTIN - INHIBITING HORMONE)

- Dopamine is the PIH.
- It also inhibits TSH.
- Apomorphine and ergot alkaloids inhibit prolactin secretion. Bromocriptine – a long acting dopamine receptor agonist reduces prolactin secretion.

CHAPTER-7: PITUITARY GLAND (HYPOPHYSIS CEREBRI)

Learning objectives

This chapter explains the following

- Different lobes of pituitary gland
- Portal system between hypothalamus and pituitary gland
- Growth hormone – its function, regulation and disorder of secretion.

PITUITARY GLAND - GENERAL INFORMATION

- It is divisible into two distinct portions, the anterior pituitary or adenohypophysis and posterior pituitary or neurohypophysis or pars nervosa.
- The pituitary is connected to the brain by means of the pituitary stalk.
- The adenohypophysis consists of three portions, viz., the pars distalis, pars tuberalis and the pars intermedia.
- The pars distalis is the largest of the three parts and secrets the tropic hormones.
- The pars tuberalis does not have any endocrine function.
- The pars intermedia produce melanocyte-stimulating hormone (MSH) and regulate the pigmentation in the body.

ADENOHYPOPHYSIS (Anterior Pituitary)

- Adenohypophysis contains two types of cells - chromophobes and chromophils
- The chromophobes are the inactive precursor or a resting type of cells from which the active (hormone producing) cells known as chromophils arise.
- Chromophils are further divided into acidophils and basophils. The acidophils and basophils include five types that produce six hormones

	Cells	Hormone	Target organ	Biological effect
I	Chromophobes	Inactive cells	-	-

II	Chromophils	Active cells	-	-
	A) Acidophils			
	a) Somatotrophs	GH	All body cells	Growth, Somatomedin synthesis
	b) Lactotrophs/ mammotrophs	Prolactin	Mammary gland	Milk biosynthesis, luteotropic
	B) Basophils			
	a) Thyrotrophs	TSH	Thyroid gland	Thyroxine synthesis and release
	b) Corticotrophs	ACTH & β-LPH	Adrenal cortex	Glucocorticoid synthesis and release
	c) Gonadotrophs	FSH	Testis, ovary	Spermatogenesis, folliculogenesis, estradiol synthesis
		LH (ICSH)	Leydig cells of testis	Androgen production, spermiogenesis
			Graafian follicle of ovary	Ovulation, CL development, progesterone synthesis

VASCULAR SUPPLY TO PITUITARY GLAND

- In general the inferior or posterior hypophyseal arteries supply the neural lobe.
- The superior or anterior hypophyseal arteries from the internal carotid supply the anterior lobe of the pituitary.
- The vessels go to the median eminence of hypothalamus and break into primary plexus in the infundibulum near the lower portion of the hypothalamus. Venous blood from these plexus is collected in two series of parallel veins which pass through the pituitary stalk ending in pars distalis. This constitutes the hypothalamo- hypophyseal portal system.
- Regulating factors from the hypothalamus enter into these portal system and goes to adeno-hypophysis
- The hormones of the adenohypophysis diffuse into veins and reach the target cells.

SOMATOTROPIN (STH)

- Somatotropin or growth hormone (GH) is a complex protein having 191 amino acids and S-S bridges.
- The STH is species specific and most domestic animals respond best to homologous STH and least to heterologous STH.
- STH is structurally related to prolactin and chorionic somatomammotropin of the same species.

Control of STH output

- The control of STH output is achieved by a balance between the growth hormone releasing hormone (GHRH) and somatostatin.
- A sharp drop in blood glucose and fatty acids levels is the primary cause of GHRH release from the hypothalamus and it results in two to ten fold increases in STH secretion.

Stimuli that increases the GH secretion

- Deficiency of energy - hypoglycaemia, exercise, fasting, insulin
- Increase in circulating levels of certain amino acids - protein meal, arginine vasopressin
- Glucagon (mechanism not known)
- Stress (trauma, surgery) pyrogen
- Sleep
- Ghrelin, a 28 amino acid polypeptide hormone secreted from hypothalamus (also from stomach) stimulates GH release

Stimuli that decreases the GH secretion

  - Increase in glucose, cortisol, free fatty acids, GH
  - Obesity results in reduced GH release
  - GIH
- STH appears to be released at a rather similar rate through out the life of the animal.
- Although skeletal growth cease after puberty, STH has a biological role throughout life as an anabolic agent as well as synergistic role by enhancing the actions of ACTH, TSH, LH, and FSH on their target organs.
- Females are more responsive to GH stimuli than males.

PHYSIOLOGICAL EFFECTS OF STH

- Growth hormone has two distinct types of effects
  - Direct effects are the result of growth hormone binding its receptor on target cells. Fat cells (adipocytes), for example, have growth hormone receptors, and growth hormone stimulates them to break down triglyceride and suppresses their ability to take up and accumulate circulating lipids.
  - Indirect effects are mediated primarily by an insulin-like growth factor-1 (IGF-1), a hormone that is secreted from the liver and other tissues in response to growth hormone. A majority of the growth promoting effects of growth hormone is actually due to IGF-1 acting on its target cells.

Normal Effects in the Body

- The STH does not have a specific target organ
- GH exerts its effect on almost all tissues of the body.

- STH causes both increase in size (hypertrophy) and number of cells (hyperplasia), and promotes growth of all tissues of the body which are capable of growing.
- GH enhances the activities of the visceral organs like liver, kidney, intestine, endocrine glands- parathyroid and pancreas.
- It stimulates cardiac output, glomerular filtration and the metabolic activities in liver, skeletal muscle and heart.
- Increases both the soft and osseous tissues of the body and
- Has a profound effect on lactation.

Effects on Growth

- Growth is a very complex process, and requires the coordinated action of several hormones.
- GH does not act directly on bone and cartilage, but indirectly by causing the liver to produce smaller proteins called somatomedins that act on cartilages and bone to promote their growth.
- Two somatomedins, C and A have been identified which resemble structurally to insulin and hence they are also known as insulin like growth factors I and II (IGF I & II) respectively.
- IGF I is transported in blood bound with specific binding proteins called IGF-binding proteins (IGFBP); this binding of IGF prolongs the half-life of IGF I.
- The receptor for IGF is similar to insulin receptors.
- IGF I promotes skeletal and cartilage growth and IGF II is functional during foetal period.
- The metabolic effects of GH on most of the cells are promoted by stimulating the liver and other tissues to secrete *IGF-I* or somatomedin C.
- IGF-I stimulates proliferation of chondrocytes (cartilage cells) resulting in bone growth. GH also seems to have a direct effect on bone growth in stimulating differentiation of chondrocytes. Somatomedin production is inhibited by estrogen and cortisol.
- *IGF*-I also increase muscle growth. It stimulates both the differentiation and proliferation of myoblasts. It also stimulates amino acid uptake and protein synthesis in muscle and other tissues.
- Growth of the long bones continues so long as the epiphyseal lines do not close. In domestic animals, closure of epiphyseal lines soon after puberty signals the cessation of skeletal growth under normal conditions.

METABOLIC EFFECTS OF GROWTH HORMONE

- As a protein anabolic hormone GH enhances almost all facets of amino acid metabolism including uptake, while at the same time reducing the break down of proteins. This includes transport through the cell membrane to the interior of the cells and increased protein synthesis in all cells of the body more specifically the muscle cells. It stimulates extracellular collagen deposition
- Stimulates the formation of RNA by increasing the transcription process in the nucleus.
- By enhancing the ribosomal machinery, produces greater number of protein molecules.
- As a potent protein sparer, it causes decreased catabolism of protein and amino acids for energy. Protein content of the body is increased by GH.
- It stimulates lipolysis and mobilizes large quantities of free fatty acids from the adipose tissue. It enhances the conversion of fatty acids to acetyl Co- A, which is used for energy. Excessive amount of STH at times causes greater amounts of fat mobilization resulting in

excess acetyl Co-A which is converted to acetoacetate, b -OH butyric acid and acetone leading to ketonemia which is called as ketogenic effect of GH. It reduces lipid synthesis and leads to leaner animal. GH decreases body fat content and promotes lean body mass.

- GH causes decreased utilization of glucose for energy (possibly duo to utilization of fat for energy) and enhanced glycogen deposition. It diminishes uptake of glucose by the cells for energy by the muscle and adipose cells, thus increases blood glucose concentration which is known as diabetogenic effect of GH.
- GH shows positive nitrogen balance and phosphorus balance. It decreases urinary excretion of Na^+ and K^+ by increasing the uptake of these ions by the growing tissues.
- In the cow lactation can be induced with STH more easily and with greater regularity than with lactogenic hormone (prolactin). This is due to galactopoietic effect of STH
- In summery, STH favours economical use of proteins and carbohydrates, encouraging the body to retain these building blocks for tissue growth and development.
- Injection of growth hormone increases milk yield by 10-25%, lactose, protein and fat contents of milk. Endogenous plasma concentrations of GH are high in high yielding than low yielding animals.GH partitions nutrients for milk yield and increases fatty acid mobilisation. Exogenous GH increases nitrogen retention and improves carcass quality with improved muscle growth; helps in lean meat production with less adipose tissue in cattle, pigs, sheep.

EFFECTS OF ABNORMAL PRODUCTION

- Panhypopituitarism is decreased secretion of all the anterior pituitary hormones. Underproduction of pituitary hormones in the immature animal results in underdevelopment of the animal referred to as *pituitary dwarfism*. Such an animal shows reduced activity in the other target organs of the pituitary (hypothyroidism, hypogonadism, hypoadrenocortical function).
- The effects of overproduction of STH depend to a great extent on the age of the animal in which this overproduction occurs.
- If the overproduction occurs before the closure of epiphyseal lines in the long bones, it results in lengthening of the long bones and increased deposition of soft tissue, resulting in gigantism. This condition occasionally occurs in domestic animals.
- If excess production of STH occurs after the closure of epiphyseal lines, the long bones of the body are thickened and there is excessive soft tissue and this condition is termed as acromegaly. This is produced in acidophilic tumours.

CHAPTER-8: PITUITARY GLAND

Learning objectives

- This chapter explores the structure, function and regulation of TSH, ACTH, prolactin, gonadotropins and MSH secreted from pituitary gland.

THYROTROPIC HORMONE (TSH)

- The thyrotropic hormone or thyroid-stimulating hormone (TSH) is a glycoprotein with two peptide sub units, the α and β. The a sub unit is non-specific, whereas b -sub unit shows some species specificity.
- TSH has morphological and functional effects on the thyroid.

- Functionally, the effect of TSH is to increase cell size, number of thyroid follicles and also the activity of thyroid gland which includes
- Uptake of iodide (iodine trapping)
- Production and release of thyroxine by increasing iodination and coupling reactions
- Proteolysis of thyroglobulin releasing the thyroid hormones.
- TSH enhances DNA, RNA and ribosome activities, thus favours synthesis of proteins and phospholipids in thyroid follicle cells. TSH stimulates glucose oxidation to generate NADPH needed for the peroxidase reaction**.**
- The functional level of the thyroid is dependent upon stimulation by TSH from the adenohypophysis and the thyroxine level from the thyroid gland depends upon reciprocal inhibition and stimulation between thyroxine and TSH.
- T_3 suppresses not only TSH release but also its synthesis by inhibiting expression of the TSH gene and decreasing the number of TSH receptors.
- TSH secretion is inhibited by dopamine, somatostatin, cortisol and GH.
- Exposure to lower environmental temperature or chilling of the animal causes the release of TRH and TSH, thus the thyroxine formation to elevate basal metabolic rate and heat production by non-shivering thermogenesis.

ADRENOCORTICOTROPIC HORMONE (ACTH)

- Adrenocorticotropin is a polypeptide containing 39 amino acids in a straight chain in sheep, pig, cow and man.
- It stimulates synthesis and immediate release of corticosteriods.
- It helps the uptake of cholesterol by the adrenal cells, cholesterol ester hydrolysis, cholesterol transfer to mitochondria, the rate-limiting desmolase reaction and the critical 11 - hydroxylation steps in cortisol synthesis.
- The primary physiological function of ACTH is to stimulate the secretion of the adrenal cortex, especially of cortisol and corticosterone.
- However the mineralocorticoids show lesser effects to ACTH. During haemorrhage it causes aldosterone release.
- Only in the bird ACTH is needed for aldosterone output. The specific action of ACTH in the adrenal gland is stimulation of cyclic AMP production leading to the production of energy for corticoid biogenesis.
- Stressful stimuli such as haemorrhage, temperature, pain, toxin and emotional states, influence the release of ACTH by stimulating the hypothalamus to release CRH.
- ACTH secretion is influenced by circadian light/darkness cycle. In human and horses ACTH secretion is high during morning hours (6.00 to 8.00 A.M.), in pigs two peaks occur, one in morning and one in late afternoon. In dogs, there is no peak level.
- By long loop servomechanism, the glucocorticoid cortisol regulates CRH release from the hypothalamus, thus decreases ACTH formation in the adenohypophysis.
- Arginine, vasopressin also stimulates ACTH release.
- By short-loop negative feed back action ACTH inhibits CRH release.

PRO-OPIOMELANOCORTIN

- Within the pituitary gland, ACTH is produced in a process that also generates several other hormones. A large precursor protein named *pro*-opiomelanocortin (POMC) is synthesized in the intermediate lobe and in the corticotropes of anterior pituitary and proteolytically chopped into several fragments. Not all of the cleavages occur in all species and some occur only in the intermediate lobe of the pituitary. In the

corticotropes, POMC is cleaved to ACTH and β-LPH and a small amount of β-endorphin. In the intermediate lobe, it is hydrolysed to corticotrophin-like intermediate lobe peptide (CLIP), α-LPH and a large amount of β- endorphins plus α and β MSH

- The major attributes of the hormones other than ACTH that are produced in this process are
    - Lipotropin**:** Originally described as having weak lipolytic effects, its major importance is as the precursor to β-endorphin.
    - β-endorphin, Met-enkephalin**:** Opioid peptides with pain-alleviation and euphoric effects.
    - Melanocyte-stimulating hormone (MSH)**:** Known to control melanin pigmentation in the skin of most vertebrates.

PROLACTIN (LACTOGENIC HORMONE)

- PRL , a lactogenic hormone, is a single-chain peptide consisting of 199 amino acids and three disulfide bridges.
- PRL is structurally similar to GH and has many overlapping biological functions with GH.
- The most important influence of PRL secretion is the combination of pregnancy, estrogen and nursing of newborn. PRL secretion increases steadily during pregnancy.
- PRL secretion rises at night similar to GH.
- Suckling inhibits dopamine (PIF) secretion, thus stimulates PRL release.
- TRH and VIP stimulate PRL synthesis and release,
- Dopamine, catecholamines, LH - RH inhibits PRL release.
- PRL is the principle hormone responsible for lactogenesis (milk production).
- PRL together with estrogen, progestrone, cortisol and GH cause proliferation of duct system of the mammary gland. During pregnancy prolactin, estrogen and progesterone cause development of secretory alveoli in the mammary gland thus causes initiation and maintenance of lactation in domestic animals.
- PRL level increases just prior to parturition and after parturition, milk synthesis and secretion require PRL along with cortisol and insulin. PRL induces enzymes necessary for lactose synthesis. It provokes synthesis of milk constituents including lactalbumin, casein, and lipids. Prolactin increases in the peripheral blood of the cow during milking.
- In pigeon it causes the crop gland to hypertrophy and stimulates the secretion of crop milk (this is used in bioassay of prolactin).
- In ewes, prolactin stimulates corpus luteum referred to as the luteotropic effect; hence it is also referred as luteotropin .
- It is necessary for maintenance of CL in rat, dog.
- An excess of PRL blocks the synthesis and release of LH-RH which inhibits gonadotropin (FSH/LH) secretions from the pituitary and prevents ovulation and spermatogenesis.
- High PRL concentration also inhibits the synthesis of gonadal steroids both in male and female.
- In animals PRL provides productive maternal behaviour toward the newborn.
- Broodiness in birds is controlled by PRL.
- Prolactin may have a growth mediating role similar to GH and reduces lipid synthesis
- Helps in partitioning of nutrients.

GONADOTROPINS (FSH & LH)

- Two hormones from the adenohypophysis, the follicle-stimulating hormone (FSH) and luteinizing hormone (LH/ICSH) in females/males affect the gonadal activity.
- The basal or tonic output of FSH and LH is pulsatile during the reproductive functions in males and females.
- FSH output is under hypothalamic control by GnRH and a negative feedback mechanism involving the gonadal hormones, rising levels of estrogen and inhibin production.
- Follicle stimulating hormone (FSH) is a glycoprotein made up of α- (92 amino acids) and β- (118 amino acids) subunits with carbohydrate moiety.
- Luteinizing hormone (LH) or Interstitial-cell stimulating hormone (ICSH) - LH in female and ICSH of the male is the same hormone.
- LH is a glycoprotein made up of α- and β- subunits having 92 and 115 amino acids with carbohydrate moiety.
- The pituitary content of LH is highest in cattle, sheep and cat but lowest in horse and man.
- Environmental conditions like changing seasons and day light length is mediated from exteroceptors like eye to the hypothalamus to influence GnRH output.
- Basal pulsatile FSH secretion stimulates follicular growth and estrogen secretion in ovaries.
- FSH acts synergistically with LH to effect ovulation.
- In the male FSH acts on the seminiferous tubules and stimulates spermatogenesis.
- The actions of LH and FSH are synergistic.
- Increased production of estrogen by the Graafian follicle stimulates LH surge to effect ovulation by positive feedback effect.
- LH secretion is increased greatly 24 to 48 hours before ovulation to effect ovulation but the increase in FSH is to a lesser extent.
- LH causes maturation of ovum, ovulation and formation of corpus luteum (CL). The CL produces progesterone which inhibits LH secretion and prevents further follicular growth and ovulation, thus preventing estrus during the period of CL.
- In induced ovulators e.g., cat and rabbit, LH secretion is stimulated by neuroendocrine reflex of the hypothalamus.
- In male LH stimulates interstitial cells of testis and causes testosterone production .
- The regulation of LH output from the pituitary gland is dependent upon the hypothalamic GnRH by a negative-feedback control mechanism.
- During pregnancy, placenta also secretes placental luteotropic hormone.
- Estrogen and progesterone in large amounts inhibit FSH and LH release from the adenohypophysis and the feedback effect operates through the hypothalamus and anterior pituitary.
- In addition inhibin secreted from the large graafian follicles in females and Sertoli cells in males inhibits FSH to a major extent and LH to a lesser extent.

MELANOCYTE STIMULATING HORMONE (MSH) AND beta - LIPOTROPIN (LPH)

- Colour change in response to change in illumination and temperature observed in reptiles, amphibians and fishes is caused by the dispersion or concentration of melanin pigments in melanophores which changes the skin colour
- The skin colour change is under the control of MSH or intermedin of pars intermedia of the anterior pituitary.

- In warm- blooded animals, melanophores are not present but melanin pigments are synthesized by melanocytes and MSH accelerates melanin synthesis and darkening of skin.
- MSH and ACTH have overlapping activities.
- α and β - MSH are cleaved from ACTH and β -lipotropic hormone (LPH) respectively.
- The β -LPH may be a prohormone for the endogenous opiates.
- β -LPH may also have a role in stress.

CHAPTER-9: NEUROHYPOPHYSIS (POSTERIOR PITUITARY)

Learning objectives

- This chapter explains the biological function and regulation of oxytocin and vasopressin.

INTRODUCTION TO NEUROHYPOPHYSIS

- The neurohypophysis is composed of pituicytes and fine unmyelinated nerve fibres arising from the supraoptic and paraventricular nuclei of the hypothalamus and terminating in the neural lobe of the pituitary.
- Neurohypophyseal hormones are the secretions of neuro-secretory cells of the hypothalamic nuclei which move along the axons of the hypothalamo-hypophyseal tracts to the neuro-hypophysis, stored until the proper signal arises to cause their release into the circulation.
- Two octapeptide hormones are released from the neurohypophysis
- Antidiuertic hormone (ADH) or vasopressin produced from the supraoptic nucleus of hypothalamus
- Oxytocin produced from the hypothalamic paraventricular nucleus.
- These two hormones after their synthesis are complexed with a polypeptide known as neurophysin and transported as Hering bodies within the axons to the neurohypophysis. Neurophysin I and II aid in transport of oxytocin and ADH respectively.

ANTIDIURETIC HORMONE OR VASOPRESSIN

- The vasopressin is a peptide containing 8 amino acids.
- It shows species specificity in its composition.
- In cattle, man and most mammals the main form of vasopressin is arginine-vasopressin (arginine at position 7 is essential for antidiuretic effect).
- In swine it is lysine-vasopressin.
- In birds it is arginine- vasotocin.
- ADH has a half-life of 18 minutes.
- An increase in osmolality of body fluids due to dehydration of body, hypotension and decreased blood volume, angiotensin II stimulates the osmoreceptrors of the hypothalamus and release ADH.
- ADH acts on the distal tubule and collecting ducts through cAMP mechanism and causes increased reabsorption of water.
- Trauma, pain, anxiety and certain drugs also favour the release vasopressin,
- Diluted body fluid and cold environmental temperature inhibit its secretion.
- A deficiency of this hormone leads to increased urine volume (diuresis) resulting in a condition known as diabetes insipidus in the dog, cat and horse.

Pressor Effect of ADH

- ADH shows less prominent constrictor effect on smooth muscles of the vascular system in normal physiological concentrations; may be involved in regulation of blood pressure. During haemorrhage, vasopressin is released in considerable amounts to cause pressor effect.

OXYTOCIN

- It is a peptide hormone containing 8 amino acids.
- Seven amino acids are common between oxytocin and ADH.
- Oxytocin has a half life of 2 min.
- Oxytocin has specific effects on the smooth muscle of the uterus and the myoepithelial cells of the mammary gland.
- It also functions in the release of prostaglandins from the endometrium.
- Along with estrogen and PGF_2 alpha, it causes powerful contractions in the pregnant uterus and helps in parturition.
- It helps sperm transport in estrogen-primed uterus during follicular phase of estrous cycle.
- In male it is released during mating to cause movement of sperm through the male duct system for ejaculation.
- On the udder it causes contraction of myoepithelial cells which results in milk letdown by a neuroendorcine reflex.
- Suckling stimulates hypothalamus, causes oxytocin release from neurohypophysis.
- In birds it has the ability to lower BP.
- Oxytocin is also secreted by corpus luteum which acts with estrogen and PGF to cause lysis of CL.
- CL also produces oxytocin.
- Gonadotropins, PGF and progesterone stimulate oxytocin from CL. At the end of luteal phase, when progesterone effect on CL is reduced, oxytocin induces luteolysis by initiating release of PGF2 alpha from endometrium.

CHAPTER-10: THYROID GLAND

Learning objectives

This chapter explains the following

- Synthesis and regulation of thyroid hormones
- Morphological effects and metabolic effects of thyroid hormones
- Disorders of thyroid hormone secretion
- Antithyroid compounds.

THYROID GLAND - MORPHOLOGY

- In most mammals, the thyroid lies just caudal to the larynx on the first or second tracheal ring and consists of two lobes connected by a narrow isthmus.

- In birds, the thyroid gland forms two lobes lying on either side of the trachea at the level of the clavicle.
- The thyroid gland is made up of follicles filled with a material called the colloid which contains a protein - iodine complex called thyroglobulin. The thyroglobulin is secreted by the epithelial cells i.e. the follicular cells that line the follicle.

SYNTHESIS OF THYROID HORMONES

- In the body, the iodine is in two forms: the inorganic iodide (ingested / dietary iodine) and organically bound iodine.
- Organic bound form of iodine includes mono-iodotyrosine, di-iodotyrosine, tri-iodothyronine (T_3) and thyroxine (T_4) and reverse triiodothyronine (rT3).
- The first stage in the formation of thyroid hormone is transport of iodides from blood into the thyroid follicular cells. The basal membrane of thyroid cell actively transports iodide by iodide-pump into the interior of the cell, which is called as iodide trapping.
- The iodide pump can concentrate iodide up to 30 times in the gland than the level in blood.
- The trapped iodide is oxidized to iodine (I_2) by thyroid peroxidase enzyme (integral membrane protein present in the apical (colloid-facing) plasma membrane of thyroid follicular cells) and H_2O_2 helps this reaction.
- Thyroid peroxidase catalyzes two sequential reactions
    - Iodination of tyrosines on thyroglobulin
    - Synthesis of thyroxine (or triiodothyronine) from two iodotyrosines
- The I_2 is transported to the lumen of the thyroid follicle, where it iodinates tyrosine present in thyroglobulin.
- The thyroid cells are protein-secreting cells, synthesize and secrete thyroglobulin (TGB) which is a glycoprotein. Each molecule of TGB contains 134 tyrosine amino acids of which 30 amino acids combine with iodine to form iodothyrosines - mono-iodotyrosine (MIT) and di-iodotyrosine (DIT) by the enzyme thyroid peroxidase.
- The coupling of the iodotyrosines occurs to form iodothyronines by two possible routes, the combination of two DIT molecules to form *thyroxine* (T4-3,5,3',5') or the combination of one DIT with one MIT to form triiodothyronine (T3–3,5,3') or to form reverse triiodothyronine (rT3-3,3',5') (has no biological function).These reactions are enzymatically controlled.
- The binding of I_2 with TGB is called organification of TG molecule.

RELEASE AND TRANSPORT OF THYROID HORMONES

- Two of the iodinated amino acids T_3 and T_4 are stored extracellularly in the colloid on the thyroglobulin and secreted into the blood stream when required.
- Thyroglobulin enters the thyroid cells by endocytosis.
- Thyroid hormones are excised from their thyroglobulin by protease digestion in lysosomes of thyroid epithelial cells and diffuse into blood.
- In contrast, the two iodotyrosines (MIT and DIT) are deiodinated within the gland by an enzyme 5' monodeiodinase. This cycle, which can be regarded as intrathyroidal, reclaims the iodine from tyrosine for use in the manufacture of the more active iodinated thyronines-T_3 and T_4.
- Thyroxine is predominant in all animals, i.e., T_4 is 3 to 4 times more than T_3.
- Most of the circulating T_3 are derived from peripheral deiodination of T_4 by the enzyme 5' monodeiodinase present in all the cells of the body.

- The T4/T3 ratio varies from 15:1 to 44:1 in most species of animals.
- In the blood, the thyroid hormone is transported bound with a serum globulin, the thyroid-binding globulin (TBG - present in all animals except cat).
- In many species, thyroid hormones are also transported with an albumin, the *thyroid-binding pre-albumin* (TBPA also termed as transthyrectin – specifically transports T4) and albumin itself is a thyroid hormone carrier;
- Collectively the iodine that is bound to all serum proteins is known as protein-bound iodine or PBI.
- Transthyrectin transports thyroxine and vitamin A and thyroxine helps to convert carotene to vitamin A in liver cells.
- Less than 1% of T4 and T3 are transported as free hormones in the blood.
- T_4 has 3 to 4 times more affinity for the binding proteins than T_3, whereas T_3 in the free form is considerably more active as a thyroid hormone.
- T_4 may be a prohormone or a source of T_3 and T_3 is the active hormone – T_3 is five times more potent than T_4.
- T_4 has slower onset of action than T_3.
- The half life of T3 is about 1 day and for T4 it is about 6 to 7 days, which is very remarkable since other hormones have a half life in minutes only.

REGULATION OF THYROID SECRETION

- TSH from anterior pituitary stimulates the secretion of T_4 and T_3 by the thyroid gland.
- The TSH promotes iodide trapping, increases iodination of tyrosine and coupling of iodotyrosines to form thyroid hormones.
- TSH increases follicular cell size and number thereby thyroid hormones synthesis.
- TSH and T_4 secretions are controlled by the level of hormones in blood by long - loop negative feedback or servomechanism.
- TRH from hypothalamus regulates thyroxine release through TSH release.
- Exposure to cold increases TRH secretion from hypothalamus causes increased secretions of TSH and thyroid hormones and increased metabolic rate.

FUNCTIONS OF THYROID HORMONES

- The effects of thyroid hormones can be roughly divided into two sections
    - Morphological (growth and development)
    - Functional (metabolic).

MORPHOLOGICAL (GROWTH AND DEVELOPMENT) EFFECTS

- Thyroid hormones are required for normal growth and skeletal maturation.
- They potentiate the effects of GH and act in synergy with GH in promoting normal growth of animals. TH is required for maturational changes in vertebrates and helps the growth of the young birds and mammals. Even though GH stimulates growth, for differentiation thyroid hormones are essential.
- TH helps in metamorphosis in amphibians. Thyroxine administration causes differentiation of tadpoles into frogs.
- Thyroxine controls the growth and eruption of the teeth and also horns in sheep and antlers in deer. Hypothyroidism severally retards the eruption of permanent teeth.
- TH is also necessary for the normal development of nervous tissue in fetus and neonates. Thyroid deficiency in young leads to retardation of mental activity.

- TH acts synergistically with sex steroids to function in growth and maintenance of comb in roosters and moulting of feathers in birds.
- In mammals, renewal of epidermis and hair follicles are affected by thyroxine. Thyroidectomized cattle and sheep have thinner hairs, which are coarse and brittle. In growing lamb, thyroxine deficiency may severally impair the quality of the adult fleece.
- In human beings and sometimes in dogs, hypothyroidism causes subcutaneous *oedema* due to accumulation of mucopolysaccharide rich material under the skin. This condition is called as myxedema. Myxedema and *alopecia* have also been observed in calves and pigs born to iodine-deficient mothers.
- TH helps the normal process of spermatogenesis, folliculogenis, and maturation of follicle, ovulation and maintenance of a healthy pregnant state.
- *Reproductive failure* is often a major sign of thyroid deficiency. Iodine deficiency causes birth of excessive numbers of weak or dead young. Abortion, stillbirth, and the live birth of weak young are the major problems of hypothyroidism. Less severe deficiency will result in delayed puberty, irregular estrous, anestrus, and reduced fertility in the females and decreased testicular growth, impaired spermatogenesis and lowered libido in males. In rams, a seasonal reduction in semen quality has been associated with hypothyroidism.
- The accessory reproductive glands, which are most sensitive to the effect of the thyroid hormone, are the mammary gland.
- Thyroxine is a powerful *galactopoietic* agent;
- Thyromimetic agents have been tried as a means of increasing milk production. Iodinated casein fed to cattle increased milk production by 10-30 %.

FUNCTIONAL (METABOLIC) EFFECTS

- Thyroid hormone increases the metabolic rate and stimulates growth in young ones. It increases the rate of O_2 consumption (up to 200%) and heat production which is known is calorigenic effect.
- Thyroid activity increases following low environmental temperature, thus involved in thermo-regulation to *increase internal heat production* by non-shivering thermogenesis
- Thyroid hormone causes increased metabolic activities in almost all tissues of body except the brain, retina, spleen, testes and lungs.
- Thyroid hormone increases the size and number of mitochondria to increase the ATP production. It increases the activities of the respiratory enzymes and enzymes involved with glucose oxidation and gluconeogenis. It stimulates Na^+-K^+ ATPase activities, favours cations transport and thus stimulates mitochondrial O_2 utilization.
- BMR is greatly increased by Thyroid hormone .
- Thyroid hormone increases cardiac output ensuring sufficient O_2 delivery to the tissues. It potentiates the actions of β -adrenergic receptors to catecholamines, thus stimulates the rate and force of contraction of the heart.
- Thyroid hormone increases glucose absorption from the intestine and facilitates insulin-mediated glucose absorption by the cells and produces glycogenolysis.
- Thyroid hormone increases protein synthesis by stimulating mRNA synthesis and the rate of formation of proteins (enzymes) by ribosomes.
- It activates catabolic pathway and inhibits anabolic pathway. It enhances the actions of catecholamines which include non-shivering thermogenisis, lipolytic, glycogenolytic, gluconeogenic and insulin secretion .
- Thyroid hormone potentiates the stimulatory effects of other hormones on metabolism by increasing the uptake of glucose by cells, enhanced glycolysis, gluconeogenesis, and absorption from gastro-intestinal tract lipolysis, ketogenesis and proteolysis.

- Thyroid hormone lowers blood cholesterol level due to increased removal by liver, mobilizes fat for energy, depletes fat stores of the body, increases free fatty acid levels in blood and enhances oxidation of fatty acids. One of the major metabolic results of thyroid deficiency is a marked increase in the serum cholesterol, phospholipids, triglycerides and decreased free fatty acid levels.
- The thyroid hormones influences nervous functions at all levels. Injection of thyroxine causes increased spontaneous electrical activity in the brain.
- Thyroxine enhances the speed and amplitude of reflexes, wakefulness, alertness to various stimuli, awareness of hunger, memory and learning capacity. It regulates creatine phosphate production and storage in the muscle tissue.
- Thyroid deficiency is a major factor for obesity and increased body weight.

HYPERTHYROIDISM

- *Hyperthyroidism:* The entire thyroid gland is hyperplastic, increase in size to two to three times normal; cell number is also increased with increase in secretion rate.
- These changes are brought about by antibodies (thyroid stimulating antibodies -TSI) that bind with thyroid cell membrane receptors similar to TSH and induce continual activation of the cells. The antibodies develop as autoimmunity against TSH receptors. This condition is called as toxic goitre or thyrotoxicosis or Grave's Disease in human beings. Plasma TSH level is usually low.
- Hyperthyroidism occurs in toxic goitre and thyroid tumours of older cats.
- *Symptoms:* Intolerance to heat, increased sweating, weight-loss, diarrhoea, muscle weakness, psychic disorders etc. In hyperthyroidism protrusion of eyeball occurs - called as exophthalmous.
- *Treatment* for hyperthyroidism is surgical removal of the thyroid gland; injection of radioactive iodine (I2 taken by the gland and radioactivity destroys the gland cells).

HYPOTHYROIDISM

- Goitre means enlarged thyroid gland.
- Insufficient I_2 intake in animals and man leads to goitre known as endemic goitre.
- Lack of I_2 reduces thyroid hormone synthesis and the reduced blood TH enhances TSH from adenohypophysis. The TSH stimulates the thyroid gland to secrete excess thyroglobulins into the follicles and the gland grows very large. But the thyroid hormone secretion cannot be increased due to lack of I_2 leading to further increase in TSH secretion. Hence this condition is called as endemic colloid goitre.
- Mild thyroiditis causes slight hypothyroidism which leads to increased TSH secretion and the gland enlarges producing non-toxic or colloid goitre.
- Hypothyroidism is common in dogs.
- Ingestion of goitrogenic substances also causes goitre.

Symptoms

- Fatigue, somnolence, muscular sluggishness, reduced heart rate, decreased blood volume, increased weight, constipation, mental sluggishness, hair falling and edematous appearance of whole body, bagginess of eyes, and swelling of face. Increased quantities of mucoplysaccharides accumulate in the interstitial spaces leading to myxedema.
- When hypothyroidism is severe during foetal life, infancy and in young age especially in humans, there is failure of body growth and CNS development is impaired with brain

hypoplasia and this condition is called as cretinism – physical and mental growth are retarded; skeletal growth is inhibited, soft tissues enlarge greatly and the cretin appears obese and stocky.

Treatment

- Oral ingestion of thyroid hormone.
- Hypothyroidism results in retarded growth while in hyperthyroidism, excessive skeletal growth in young animals.

ANTITHYROID COMPOUNDS

- Drugs or agents that depress the thyroid secretion are called as antithyroid compounds.
- The major cause of goitre in man and farm animals is a simple dietary iodine deficiency.
- Certain foodstuffs contain substances that inhibit thyroid activity. This goitrogenic activity is seen in feeding of cabbage, and other cruciferous plants, e.g., turnip, soybeans, linseed, peas, rapeseed, mustard seed and peanuts.
- A potent goitrogenic compound, goitrin, is responsible for most of the antithyroid activity of the rapeseed, mustard seed. However, the goitrogenecity of many plants is also due to the presence of other goitrogens such as thiocyanate.
- Since thiocyanate and other chemical radicals interfere with the trapping of iodine by the thyroid gland, the goitrogenicity of these radicals can be overcome by feeding excess iodide.
- It is more difficult to overcome the effect of goitrin and related compounds, since they interfere with the organic binding of iodine and not with the iodine trap.
- Other anti thyroid compounds includes sulfonamides, D-amino salicylate and chlorpromazine.
- The compounds exerting the most potent antithyroid activity are the thiocarbamides, thiourea and thiouracil compounds which inhibit either the conversion of iodide to iodine or the organic binding of molecular iodine with tyrosine or the conversion of T_4 to T_3.
- These antithyroid substances are useful in the treatment of hyperthyroidism.

CHAPTER-11: CALCIUM HOMEOSTASIS

Learning objectives

This chapter explains the following

- Role of calcium in body function
- Regulation of blood calcium level by PTH, calcitonin and vitamin D3
- Disorders of the calcium metabolism.

ROLE OF CALCIUM AND PHOSPHORUS

- Calcium is important for
  - Muscle contraction
  - Nerve cell activity
  - Release of hormones by exocytosis
  - Activation of number of enzymes
  - Coagulation of blood
  - Maintaining stability of cell membrane
  - Maintaining structural integrity of bone and teeth
- Phosphate is important for
  - Structure of bone and teeth
  - As part of cell membrane organic PO4
  - As part of number of intracellular components
- About 99% of the body's Ca2+ is in the bone and a very small amount is in the plasma (1%). In most animals and in non-laying birds the blood Ca ranges between 7.9 and 11.7 mg/dl except in horses in which it is between 10.4 and 13.4 mg/dl and in layers 17-39mg/dL.
- The smallest ECF Ca2+ pool is the most vigorously controlled Ca2+ pool in the body and it is composed of interstitial Ca2+, blood Ca2+ and bone Ca2+ in solution.
- The most important regulation of Ca2+ between bone and ECF involves the soluble Ca2+ of bone which is the source of ready exchange of ions with blood and it is located between osteoblasts.
- Plasma Ca2+ exists in three forms – (1) free or ionizable Ca2+ (2) complexed to HCO3, citrate, PO4 (3) bound to plasma proteins. Free or ionizable Ca2+ is predominant (50%) and is in constant exchange with ECF Ca2+. This is the physiologically active Ca2+ which participates in many biochemical functions.
- Bone Ca2+ exists in two forms (1) a readily exchangeable reservoir and (2) larger stable pool which is slowly exchangeable. Two independent homeostatic systems operate on the bone Ca2+ – one is the system that regulates plasma Ca2+ and about 500mmol Ca/day moves into and out of the readily exchangeable pool in bone. The second is concerned with bone remodelling by interplay between bone resorption and deposition and this accounts for bone formation and this pool is about 7.5mmol/day.

BONE CELLS

- In bones there are two types of cells –*osteoblasts* and *osteoclasts* that are responsible for bone formation and bone reabsorption respectively. Osteoblasts are bone forming cells which secrete alkaline phosphatase and collagen for the organic matrix .
- The osteoblasts after the formation of organic matrix are embedded into the matrix and now they are termed as *osteocytes*. Alkaline phosphatase promotes precipitation of calcium phosphate on the organic matrix and thus helps in bone mineralization.
- Osteoclasts are motile, multinucleated, large cells derived from monocytes and they actively reabsorb bone leaving a lacuna in the bone and thus these cells are bone reabsorption cells.

PARATHYROID GLAND
CHEMISTRY

- Parathyroid hormone (PTH) is a single chain polypeptide containing 84 amino acids.

- PTH is synthesized as part of a large molecule containing 115 amino acids called *pre pro-parathyroid hormone* (pre-pro PTH) by the ribosome.
- This large molecule is converted to *pro-PTH* composed of 90 amino acids in the rough endoplasmic reticulum. It moves to the Golgi apparatus and is enzymatically cleaved to produce the biologically active PTH.
- Active PTH may be secreted directly or packaged into secretory granules.

PARATHYROID GLAND
REGULATION OF PTH SECRETION

- Parathyroid hormone has a unique feedback control by the concentration of calcium and to a lesser extent magnesium ion in serum.
- Low blood level of *ionized* Ca^{2+} stimulates rapid and pronounced release of parathyroid hormone
- Calcium has a negative feedback effect on PTH release.
- Magnesium ion has an effect on parathyroid secretory rate similar to that of calcium, but its effects is not equipotent to that of calcium. The concentration of blood phosphates has no direct regulatory influence on the synthesis and secretion of PTH.
- PTH is essential for life.

PARATHYROID GLAND
BIOLOGICAL EFFECTS

- Parathyroid hormone is the principal hormone involved in the *minute-to-minute* and fine regulation of blood calcium level in mammals.
- It exerts its biological action by directly influencing the function of target cells, primarily the bone and kidney and indirectly in the intestine to maintain plasma Ca level

On bone

- *Osteoclastic osteolysis:* PTH activates osteoclasts to cause the breakdown of bone and releases Ca into the blood. Only osteoblasts but not osteoclasts have PTH receptors and hence the osteoblasts bind with PTH. Osteoclasts are stimulated by PTH indirectly through osteoblast-binding with PTH.
- The initial binding of PTH to osteoblasts lining the bone surface exposes the bony matrix followed by attraction of osteoclasts to the resorptive bony surface. The mature osteoclasts are attached to the resorptive bony surface and they cause the dissolution of the bony matrix, thus the bone resorption.
- The plasma membrane of the active osteoclasts is modified into numerous projections known as *ruffled* or *brush border*. The osteoclasts form a sealed chamber within a restricted area at the place of attachment.

- At this restricted area, low pH is created by the accumulation of lactic, citric acids and H^+ released from carbonic acid by carbonic anhydrase.
- Low pH and increased lysosomal enzymatic actions of acid phosphatase and collaginase favour hydrolysis of bony substances, and thus release Ca ions along with organic component, hydroxyproline. These products are transported to plasma by pump mechanism.
- PTH also stimulates conversion of osteoprogenitor cells to osteoclasts, and sustains osteoclastic activity and inhibits osteoblastic activity.

On kidney

- It increases tubular reabsorption of calcium,whereas increases urinary excretion of phosphates, (phosphaturia), K^+,bicarbonate, Na^+ and amino acids and urinary excretion of hydroxyproline. It also causes renal retention of H^+ ions.
- PTH activates 1 α - hydroxylase to convert 25 hydroxy cholecaliferol to 1, 25 dihydroxycholecaliferol (vitamin D_3) in the kidney.

On intestine

- Promotes absorption of Ca and PO_4 from GI tract by activating 1, 25 DHCC.

On Blood

- Elevates the blood Ca concentration and decreases the blood PO_4 concentration.

CHEMISTRY AND REGULATION OF SECRETION OF CALCITONIN

- Calcitonin (thyrocalcitonin) is secreted in the mammalian thyroid gland by *parafollicular* or 'C' cells of thyroid gland which are found in the interstitial tissue between the follicles.
- In reptiles, amphibians and birds, CT is secreted from *ultimobranchial glands*.

Chemistry of Calcitonin

- Calcitonin is a polypeptide hormone composed of 32 amino acid residues arranged in a straight chain with a1-7-disulphide linkage.
- The structure of calcitonin differs considerably between species.

Regulation of Calcitonin Secretion

- Hypercalcemia and to a lesser extent increased levels of blood magnesium will stimulate CT secretion.

- Glucagon, the gastro-intestinal hormones, gastrin, secretin and CCK produced by the presence of food in the GI tract are the stimulants for CT release. Glucagon with gastrin provides the most potent stimulation in the secretion of CT.
- Gastro-intestinal hormones may be important in triggering the early release of calcitonin to prevent the development of hypercalcemia following the ingestion of high calcium meal.
- CT protects the skeleton during Ca stress like during growth, pregnancy and lactation.
- Calcitonin secretion is increased in response to a high calcium meal; often before a significant rise in plasma calcium can be detected.

BIOLOGICAL EFFECTS OF CALCITONIN

- Calcitonin is calcium lowering hormone.
    - With respect to plasma Ca level, CT antagonises the actions of PTH by inhibiting osteoclastic bone resorption. Calcitonin stimulates the development of various degrees of hypocalcemia and hypophosphatemia.
    - Increased blood calcium level is the principle stimulus of CT release. It causes decreased movement of calcium ions from the bone calcium pool (behind the osteoblast-osteocyte barrier) to extra cellular fluid.
    - Osteoclasts have receptors for calcitonin. CT decreases bone resorption through an inhibitory effect on osteoclasts. CT also increases movement of phosphate from the extra cellular fluid into bone, thus helps deposition of Ca^{2+} ions by activating osteoblastic activity .
    - CT increases Ca^{2+} excretion through kidneys.
    - It increases osteoblastic activity, but inhibits the formation of new osteoclasts from osteoprogenitor cells. The CT *prevents the postprandial hypercalcemia*.
    - CT has similar effect as PTH on renal phosphate elimination; i.e., increases phosphate elimination through kidneys by reducing phosphate resorption.

PHYSIOLOGICAL SIGNIFICANCE OF CALCITONIN AND PTH

- Calcitonin and parathyroid hormone provide a *reciprocal inhibitory control mechanism* to maintain the concentration of calcium in extracellular fluids within narrow limits.
- Parathyroid hormone is the major factor concerned with the minute-to-minute regulation of blood calcium under normal conditions.
- *CT is a short-termregulator* of blood calcium and it operates more rapidly compared with PTH.
- CT function more as an emergency hormone to prevent the development of hypercalcemia during the rapid postprandial absorption of calcium and to protect against excessive loss of Ca and phosphates from the maternal skeleton during pregnancy and lactation.

CALCITRIOL

- The third major hormone involved in the regulation of calcium metabolism and skeletal remodelling is *calcitriol* or *cholecalciferol* (vitamin -D_3) or irradiated ergosterol (vitamin -D_2).
- Cholecalciferol is ingested in the diet and also synthesized from 7-dehydrocholesterol by ultraviolet irradiation.

Synthesis

- Epithelial cells of the skin synthesise the precursor of vitamin D, 7-dehydrocholesterol from acetate. Exposure to sun changes 7-dehydrocholesterol to inactive vitamin D.
- After absorption or synthesis, the cholecalciferol is bound to an alpha -2 globulin (vitamin D binding protein – *DBP*) in the blood, transported to the liver for its conversion into 25-hydroxycholecalciferol (calcidiol) by 25- hydroxylase.
- In the kidney 25-hydroxy cholecalciferol is converted into 1, 25 dihydroxy cholecalciferol (DHCC)by a rate limiting enzyme 1-α hydroxylase. 1,25 DHCC exerts strong product feedback inhibition on 1 alpha hydroxylase in the kidney.
- When Ca is normal or in excess, kidneys produce 24,25 DHCC which is inactive.
- Activation of renal hydroxylase is stimulated by PTH, low serum Ca level, low serum PO_4 level, prolactin and estrogen and is inhibited by hypercalcemia, high serum PO_4 level and CT.

BIOLOGICAL EFFECTS OF CALCITRIOL

- Following synthesis in the kidney 1, 25 DHCC is transported in a protein-bound form which is known as *transcalciferin* (a -globulin) to specific target cells in the intestine and bone.
- In the intestine, it stimulates increased synthesisof vitamin-D dependent proteins such as *Calcium Binding Protein* (CaBP) and *calmodulin* by intestinal cells which contributes to increased intestinal absorption of Ca.
- It also promotes intestinal absorption of phosphorus.
- Calcitriol is also necessary for osteoclastic resorption and Ca mobilisation from bone.
- It increases the reabsorption of Ca from kidney.
- An increase in 1, 25 DHCC inhibits the secretion of PTH.
- DHCC induces stem cells to differentiate into osteoclasts to promote osteolysis
- Calcitriol has permissive effect on PTH activity on osteoclasts

PARTURIENT HYPOCALCEMIA IN DAIRY COWS

- *Parturient hypocalcemia* is a metabolic disease of high producing dairy cows characterized by the development of severe hypocalcemia, hypophosphatemia and paresis near the time of parturition.
- Total and ionized calcium levels in plasma decrease progressively beginning several days before parturition. Serum magnesium may increase reciprocally as calcium declines.
- The blood glucose concentration is often increased in response to hypocalcemia due to an interference with the secretion of insulin from b cells. An adequate amount of calcium ion in extracellular fluid is required for insulin secretion in response to glucose.
- PTH level is increased during this condition .
- In certain cows, especially those fed high calcium diets during prepartum show increased secretion of calcitonin which might result in the inability of PTH to mobilise calcium rapidly from bone reserve to maintain blood calcium levels during the critical period near parturition.
- Immediately after parturition the target cells in bone become *temporarily refractive* to the action of elevated levels of PTH.
- Bone turnover, particularly resorption is low and only few osteoclasts are present on smooth inactive trabecular bone surfaces.

- Urinary excretion of hydroxyproline does not increase significantly during late gestation which indicates poor breakdown of bone matrix.
- Calcium homeostasis in pregnant cows fed a high Ca diet appears to be maintained principally by intestinal Ca absorption.
- Feeding balanced or relatively low Ca diets during prepartum (last two weeks of gestation) prevents this disorder by mobilizing the required level of calcium from the bony matrix.

PUERPERAL TETANY OR ECLAMPSIA

- This condition is seen in small hyper excitable breeds of dogs.
- The bitch may develop severe hypocalcemia and often hypophosphatemia near the time of peak lactation (1-3 weeks) postpartum which is the result of failure of calcium homeostasis.
- This failure is due to low Ca^{2+}inflow to the body and high outflow of Ca^{2+} in the milk.
- There will be restlessness, panting, nervousness, trembling, muscular tetany and convulsive seizures in 8- 12hours.
- Administration of intravenous Ca^{2+}combined with temporary removal of pups to reduce lactational drain of Ca^{2+}will correct this problem.
- *Chronic hypocalcemia* from dietary Ca deficiency, vitamin D deficiency or from kidney diseases is more frequent in animals. This hypocalcemia leads to hyperparathyroidism that produces skeletal demineralisation or excess bone reabsorption– e.g. *rubber jaw syndrome* in dogs.
- Vitamin D deficiency leads to poor intestinal absorption of Ca with poor mineralization of bone leading to rickets in young animals and children and *osteomalacia* is adults .

CHAPTER-12: REGULATION OF BLOOD GLUCOSE CONCENTRATION

Learning objectives

This chapter explains the following

- Different cell types in islets of langerhans
- Role of insulin in glucose homeostasis
- Regulation of insulin secretion
- Disorder of insulin secretion.

BLOOD GLUCOSE

- The normal blood glucose concentration varies from 62 to 120 mg/dl in dogs, cats, horses and pigs; in cows, sheep and goats it varies from 42 to 80 mg/dl; adult ruminants have lower level than newborn ruminants.
- The normal blood level is termed *euglycaemia;* when the blood glucose level is increased it is *hyperglycemia* and a decreased level is *hypoglycaemia*.

Control of blood glucose

- There are two major sources of blood glucose

- Dietarycarbohydrates – absorbed from intestine
- Synthesisof glucose in liver – from glucogenic substances – glycogen, glucogenic aminoacids, glycerol and propionate in ruminants,

Role of Liver

- Liver is the major organ involved in regulating blood glucose. After a carbohydrate meal, the increase in blood glucose is prevented by liver by converting glucose to glycogen and fatty acids. Liver can store glycogen up to 10 to 15% of its weight. During decrease in blood glucose level, liver glycogen is broken down to glucose and released into circulation.

Role of Hormones

- The only hormone that produces hypoglycaemia is insulin
- All other hormones – glucagon, growth hormone, cortisol and epinephrine have hyperglycaemic effect.
- When insulin effect is removed by pancreatectomy, IGF-I sustains weak insulin activity (but not sufficient to maintain life).
- Glucagon and epinephrine promotes glycogenolysis while insulin has antagonistic effect of glycogenesis. Molar ratio of insulin: glucagon rather than the absolute concentration is the determining factor of glycogen breakdown or synthesis. Glucagon stimulates gluconeogenesis from amino acids.

Glucose tolerance test

- The capacity of the body's control mechanisms to regulate blood glucose level canbe evaluated by *oral glucose tolerance test*.
- About1 to 2 g of glucose / kg weight is given orally and blood samples are collected 30 min intervals for 120 or 240 min.
- In normal dogs, blood glucose level should reach close to 160 mg/dl in 30 to 60min and the level should reach the baseline normal values by 120 to 180 min.
- In horses, by 120 min, the glucose level should reach 175 mg/dl which should return to normal by 360 min.

PANCREAS

- The pancreas has both endocrine and exocrine functions.
- In 1869, Langerhans described the islet cells of the pancreas. In 1921, Banting and Best isolated pancreatic extract containing insulin from the islet cells. Sanger and his associates in 1955 elucidated the chemical structure of insulin.
- The endocrine portion of the pancreas consists of clusters of cells called *islets of Langerhans*. Four kinds of cells are found in these clusters.

Endocrine Cells and Hormones of Pancreas

- α (A) Cells (20%) -> Glucagon.
- β (B) Cells (70%) -> Insulin.

- δ (D) Cells (<5%) -> Somatostatin
- F Cells (<5%) ->pancreatic polypeptide (PP)

- Insulin and glucagon are often secreted and act reciprocally. Glucagon and insulin are concerned with regulation of blood sugar level.

INSULIN

- Insulin, a protein hormone with 51 amino acids contains two polypeptide chains (A & B) with 21 and 30 amino acids respectively connected by two disulphide bridges with a third disulphide bridge in the A-chain.
- Insulin is synthesised as preproinsulin, processed in the endoplasmic reticulum and released.
- During processing of proinsulin, along with insulin another polypeptide called *C peptide* is produced which also enters the plasma along with insulin.
- Levels of C peptide in plasma indicate beta cell function in insulin deficient patients.
- Insulin has a half-life of 5 minutes.
- Structurally insulin of dogs, cats and pigs is identical. Insulin from cattle, sheep, horse, pig and dog differs only in positions 8th, 9th and 10th in A chain, hence the biological actions of insulin are not highly species specific. Insulin from one species is mildly antigenic when injected into another.

BIOLOGICAL EFFECTS OF INSULIN

- Liver, skeletal muscle and adipose tissue are the principal target organs of insulin where it enhances the entrance of glucose, amino acids, fatty acids, K^+ and Na^+ ions.
- It stimulates glycogenesis, lipogenesis, glycolysis and protein synthesis, whereas it inhibits gluconeogeneis, lipolysis and ketogenesis.
- It lowers blood glucose, fatty acids and aminoacids levels and promotes intra cellular conversion of these compounds to glycogen, triglycerides and protein respectively

On Carbohydrate Metabolism

- *Glucose does not readily penetrate* the cell membranes except in neurons, liver, intestinal epithelium, RBCand WBC, renal tubular epithelium and retina.
- Presence of insulin causes more amount of glucose entry (2 to 5 fold more) through the plasma membrane into the muscle and adipose cells.
- Glucose can be transported down their concentration gradient across the cell membrane by transport proteins called *glucose transport proteins* (GLUTs) which are present in the cell membrane of all cells. Detection of plasma glucose by pancreatic islet cells as well as uptake/ release of glucose from cells involve GLUT. GLUT is present in many isoforms
- When insulin concentration rises in the plasma, number of GLUT-4 molecules increases in the cell membrane and glucose transport into the cell is increased
- In liver, insulin activates the enzyme *glucokinase* to initiate phosphorylation of glucose, thus the *glucose is trapped* inside the liver cells.
- Insulin promotes the activities of *glycogen synthase* to *favour glycogenesis*, while it *inhibits phosphorylase* and prevents the split of glycogen into glucose.
- Liver can store glycogen up to 10 to 15% of its mass.
- The normal resting muscle membrane is almost impermeable to glucose. In the presence of insulin, glucose permeability is increased and muscle glycogen synthase activity is enhanced.

- However during heavy exercise the muscle membrane becomes highly permeable to glucose even in the absence of insulin. Much of the glucose in the muscle is stored in the form of muscle glycogen instead of being used for energy. Approximately 75% of the glucose is converted into glycogen and only 20 to 30% undergo glycolysis.
- In many cells, insulin facilitates glycolysis by activating glycolytic enzymes.
- Insulin inhibits proteolysis in the peripheral tissues, thereby reducing amino acid availability for *gluconeogenesis*.

On Fat Metabolism

- Insulin by increasing rate of utilization of glucose in many body tissues functions as *fat sparer*.
- In the liver when the liver glycogen level goes above 15%, insulin promotes the conversion of glucose to fatty acid and their transport to the adipose cells for triglyceride synthesis and storage. Insulin increases acetyl-CoA formation from pyruvate, the substrate for fattyacids synthesis. Glycolytic break down of glucose by insulin supplies large quantities of a -glycerophosphate which is the source of glycerol. Binding of glycerol with fatty acids forms triglycerides; the triglycerides are transported to adipose tissue and stored as fat.
- Insulin also promotes glucose transport into adipose cells, where they are converted to a-glycerophosphate and to glycerol for triglyceride synthesis.
- Insulin inhibits the action of *hormone-sensitive lipase* and prevents hydrolysis of the triglycerides in adipose tissue and release of fatty acids into the circulating blood.
- Insulin diminishes beta oxidation of fat, thus inhibits ketone body production. In adipose tissue, insulin induces the synthesis of lipoprotein lipase, (promotes movement of fatty acids into adipose tissue) inhibits intracellular lipase and enhances fatty acid esterification. Cholesterol synthesis is also enhanced by insulin.

Ketogenic and Acidotic Effects of INSULIN LACK

- Lack of insulin promotes the activation of *hormone -sensitive lipase* and rapid breakdown of fatty acids from the liver and adipose cells and excessive production of acetyl - CoA.
- Portion of acetyl-CoA utilized for energy in the liver; excess is condensed to form acetoacetic acid, beta hydroxy butyric acid and acetone, which are the *ketone bodies* and their presence in large quantities in the body fluids is called as *ketosis*.

On Protein Metabolism

- Insulin acts along with growth hormone and cause active transport of amino acids into the cells and increases the rate of transcription of DNA in the cell uncles, thus affects the ribosomes to increase the translation of messenger RNA forming new proteins.
- Insulin greatly enhances the rate of protein synthesis and prevents the degradation of proteins for gluconeogenesis and promotes positive nitrogen balance.
- Due to its effects on protein metabolism,insulin is required for growth of the animal; GH and insulin act synergistically to promote growth.

REGULATION OF INSULIN SECRETION

- The most important factor in the control of insulin secretion is the *concentration of blood glucose*. Increased concentration of blood glucose initiates the synthesis and release of insulin by positive feedback mechanism.
- Gastrin, secretin, CCK, gastric inhibitory peptides, ACTH, cortisol, estrodiol, GH, glucagon, progesterone, thyroxine and acetylcholine, amino acids (arginine and lysine are the most potent) and fatty acids in plasma stimulate insulin secretion.
- In sheep the short chain fatty acid butyrate and propionate stimulates the release of insulin, whereas in dog the long chain fatty acids stimulate insulin release.
- Glucagon has direct stimulatory effect on beta cells and insulin secretion, whereas somatostatin and adrenergics inhibit the secretion of insulin.
- Calcium is the final triggering step in the release of insulin from the islet cells; hence hypocalcemic condition in cows and pigs depresses insulin secretion.

DIABETES MELLITUS

- *Diabetes mellitus* is a metabolic disorder of carbohydrates, proteins and fats due to deficiency of insulin and with insulin resistance. It is observed in dogs, humans and cats.
- Diabetes mellitus can be classified into three types – *Type I, Insulin-dependent diabetes mellitus* – (IDDM); *Type II – noninsulin-dependent diabetes mellitus* – (NIDDM) and *impaired glucose tolerance* (IGT).
- Type I diabetes mellitus is characterized by loss of beta cells leading to insulin deficiency.
- Type II diabetes is characterized by insulin resistance which may occur along with reduced insulin secretion. Type 2 diabetes is the most common type.
- Diabetes mellitus is characterized by persistent hyperglycemia
- A fasting blood glucose level of 125mg/dl or more is suggestive of diabetes mellitus.
- Insulin deficiency affects entire metabolism of the organism: fat, protein, carbohydrate, electrolyte and water. It leads to the inability of peripheral tissues to use glucose for energy or for glycogen synthesis, whereas it *stimulates glycogenolysis* and gluconeogenesis which results in hyperglycemia.
- Glucose appears in urine (*glucosuria*) and causes osmodiuresis leading to water and electrolyte loss, polyuria, dehydration and hemoconcentration. Other signs include polyphagia, polydipsia, and hypercholestrolemia.
- Insulin deficiency activates hormone-sensitive lipase activity leading to increased fatty acid mobilization for oxidation and energy, increased acetyl-CoA formation which on conversion to ketone bodies causes ketonemia and ketonuria.
- Protein catabolism is enhanced in insulin deficiency and amino acids are used for gluconeogenesis. There is a negative nitrogen balance with weight loss. BUN is elevated in dogs.
- *Pituitary diabetes* is weakly sensitive to insulin whereas *adrenal diabetes* is moderately insulin sensitive and *pancreatic diabetes* is highly sensitive to insulin.
- When plasma glucose is elevated for a long time, small amounts of Hb are non-enzymatically glycosylated to form HbA1c. Controlling the diabetes with insulin reduces this level; HbA1c concentration is measured clinically as an index of diabetic control for the 4-6 week period before the measurement. Higher level will be observed in poorly controlled diabetes.

CHAPTER-13: GLUCAGON

Learning objectives

This chapter explains the following

- Role of glucagon in glucose homeostasis
- Functions of somatostatin and pancreatic polypeptide.

GLUCAGON

- Glucagon is the *hyperglycemic factor* secreted by the alpha cells of the pancreas and L cells of GI tract. It has 29 amino acids in straight chain and synthesized as preproglucagon.
- There is a considerable homology in the amino acid composition of glucagons of different species. Glucagon is also produced from other sites–from stomach *gut glucagons* (identical to pancreatic glucagons), and from small intestine *glycentin* (immunologically similar to pancreatic glucagons) are produced. This catabolic peptide hormone raise plasma sugar levels. Its secretion from the alpha cells of the pancreatic islets is directly stimulated by:
    - low plasma glucose concentrations: glucagon levels rise in the postabsorptive state some hours after a meal.
    - high levels of circulating amino acids. This may be important in the maintenance of normal plasma glucose levels during absorption of a protein meal, since amino acids also stimulate insulin secretion and this might cause hypoglycaemia in the absence of any opposing action on carbohydrate metabolism.

Regulation of Secretion

- Low blood glucose level stimulates glucagon synthesis and release.
- Glucogenic amino acids–serine, glycine, alanine are powerful activators of glucagon output
- Both insulin and glucagon work *in-tantum* to regulate blood glucose.
- Both sympathetic and parasympathetic systems of ANS, hypoglycemia, gastrin, CCK, gastric inhibitory polypeptide, glucogenic amino acids, protein ingestion and exercise or stress stimulate glucagon release while increased levels of glucose and fatty acids and secretin are inhibitory to glucagon release.
- Somatostatin is more inhibitory to glucagon than to insulin.

BIOLOGICAL EFFECTS

- Glucagon has glycogenolytic, gluconeogenic, lipolytic and ketogenic effects
- Glucagon blocks the synthesis of glycogen by inhibiting the enzyme glycogen synthase. By activating phosphorelase, glucagonstimulates *glycogenolysis* and *gluconeogenesis*. It stimulates adenylcyclase activity in the liver cells and formation of cyclic AMP to activate protein kinase, a regulator protein. Protein kinase activates phosphorylase-b kinase that converts phosphorylase-b to phosphorylase-a and promotes the degradation of glycogen

into glucose-1-PO_4 which on dephosphorylation yields glucose in liver cells. It also promotes lipolysis.
- Gluconeogenesis is enhanced. It decreases the activities of glycolytic enzymes. The ratio between phosphofructokinase and fructose 1-6-diphosphatase determines the relative rate of glycolysis and gluconeogenesis.
- It also promotes lipolysis by stimulating hormone-sensitive lipase activity in the adipose tissue.
- Glucagon secretion begins with ingestion of food and increases with the interval from food ingestion and declining blood glucose level. This helps the animal to mobilize energy stores for maintaining glucose homeostasis i.e., prevent postprandial hypoglycemia.
- In birds, glucagon has a predominant role than insulin on carbohydrate metabolism.
- Glucagon is a ketogenic hormone, activates the hormone-sensitive adipose tissue lipase in adipose tissues to cause lipolysis.
- It inhibits Na^+ resorption from the renal tubules causing natriuresis

SOMATOSTATIN

- It is available in two molecular forms -14 and 28 amino acid peptides.
- It acts as a humoral regulatory agent and found in CNS, pancreas and cells lining the GI tract.
- Released from hypothalamus it inhibits the secretions of GH and TSH,
- It may also acts as a neurotransmitter in retina and spinal cord.
- It inhibits variety of metabolic and digestive functions.
- Pancreatic somatostatin is the secretion from δ cells of the islets of Langerhans
- It inhibits α, β and F cells thereby the secretion of insulin, glucagon and pancreatic polypeptide are inhibited.
- Gastric somatostatin inhibits the secretion of gastrin, secretin, CCK and pancreatic exocrine secretions, HCl and pepsin secretions, GI motility and gall bladder activity, thereby the absorption of nutrients.
- Somatostatin secretion is stimulated by increased level of glucose, amino acids, catecholamines, acetylcholine and glucagon.

PANCREATIC POLYPEPTIDE

- Pancreatic polypeptide is a peptide hormone, containing 36 amino acids and secreted by the stimulation of F (PP) cells in the islets of pancreas.
- It is related to two other 36 amino acid polypeptides – *polypeptide YY (*PYY) of intestine and *neuropeptide Y* (NPY) of brain and ANS.
- It increases gut motility and gastric emptying, stimulates the secretions of the intestinal hormones- gastrin, secretin and CCK.
- It inhibits pancreatic enzyme secretion and gall bladder contraction.
- Protein has stimulatory effect on its secretion, somatostatin inhibits its release.

CHAPTER-14: ADRENAL CORTEX

Learning objectives

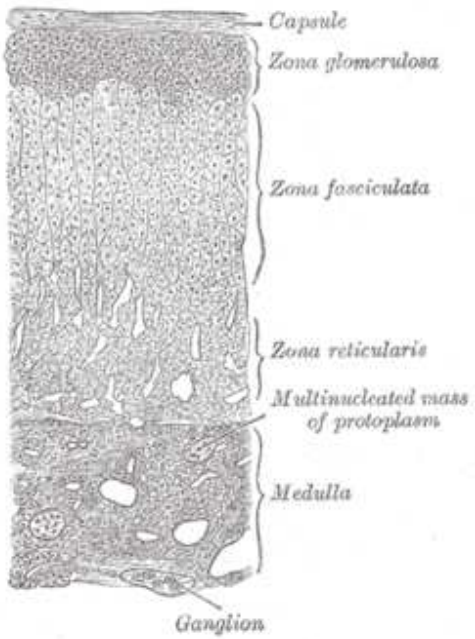
At the end of this chapter the learner will be able to understand the following

- Different layers of adrenal cortex
- Biosynthesis of adrenal steroids
- Biological functions and clinical effects of glucocorticoids
- Regulation of sodium level by aldosterone
- Sex steroids secreted by adrenal cortex.

ADRENAL CORTEX

- The adrenal gland also called *suprarenal glands* is composed of an outer portion- the *cortex* (80 to 90% of gland) which secretes corticosteroids and the inner portion- the *medulla* (10 to 20% of gland) which secretes catecholamines
- In mammals the adrenal cortex is organized into three zones.
- The outer zone, *zona glomerulosa* secrete the hormones *mineralocorticoids*
- The middle zone, *zona fasciculata* secrete the hormones *glucocorticoids*.
- The innermost zone, *zona reticularis* secrete androgens
- The mineralocorticoids play an important role in electrolyte balance and also in the regulation of blood pressure. The major mineralocorticoid is *aldosterone*.
- The glucocorticoids are important in the regulation of metabolism either directly or through an interaction with other hormones. The major glucocorticoid is *cortisol.*
- The adrenal sex steroids are secreted from the inner layers.
- Adrenal cortex is essential for life.

Adrenal cortex layers



SYNTHESIS OF ADRENAL STEROIDS

- Adrenal cortical hormones are steroids having the base structure of cyclopentano perhydrophenanthrine ring.
- All steroid hormones are synthesized from cholesterol. The cholesterol for steroid synthesis is mainly derived from circulating blood and is stored within steroid synthesizing cells in large quantities as lipid droplets or ester form.
- A very small amount of cholesterol is synthesised by cortical cells.
- There are three types of steroids
    - C21 steroids have 2 carbon side-chain at position 17
    - C19 steroids have keto or hydroxyl group at C17 and called as 17-ketosteroids
    - C18 steroids have keto or hydroxyl group at C17, no CH_3 group at C10.
- Adrenal steroids are mostly C21 and C19 steroids.
- C19 steroids have androgenic activity andC21 are either mineralo or glucocorticoids.
- The initial step in steroid formation is the hydrolysis of cholesterol ester by the enzymatic cleavage of carbon side chain from the steroid molecules, leaving a C-21 steroid, the pregnenolone.
- ACTH promotes adrenocortical steroidogenesis through the activation of *cholesterol desmolase* and is important in the conversion of cholesterol to pregnenolone and in C-17 hydroxylation. This step occurs within the mitochondria. This is followed by hydroxylation at C-21 by *C-21 hydroxylase*.
- The difference between the mineralocorticoids (aldosterone) and glucocorticoids (cortisol) is that only the glucocorticoids have a hydroxyl group on C-17, the hydroxylation is done by 17 α -hydroxylase, the enzyme present only in the cells of zona fasciculata but absent in zona glomerulosa.
- Both aldosterone and glucocorticoids have hydroxyl group at C-11. O_2 at C18 position is important for mineralocorticoid activity.
- In zona glomerulosa, 11 hydroxylase and 17 hydroxylase are absent. Deoxycorticosterone is converted to aldosterone through corticosterone by the enzyme aldosterone synthase present only in zona glomerulosa.

- Birds, mice and rats secrete mostly corticosterone, dogs secrete equal amounts of cortisol and corticosterone, cat, sheep, monkeys and human secrete predominantly cortisol.
- 75% of glucocorticoid is carried in plasma in association with specific binding globulins, corticosteroid binding globulin (CBG) called *transcortin*, 20% in *albumin* and 5% in *free form*. 50% aldosterone is carried in *albumin*, 10% with *CBG* and the rest in *free-state*.
- Plasma half life of cortisol is about 60 to 90 min and aldosterone is 20 min.

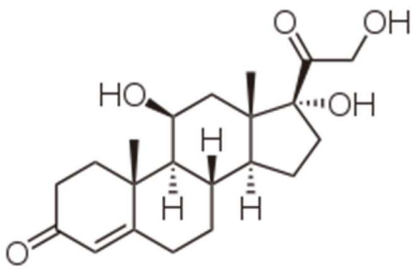
GLUCOCORTICOIDS

Cortisol, corticosterone, cortisone and 11-dehydrocorticosterone are the naturally occurring adrenal cortcal steroids.

Structure of cortisol

Glucocorticoids have primary effect on carbohydrate metabolism.

- Hypoglycemia is a potent stimulus for cortisol release to maintain normal plasma glucose level during fasting.
- Cortisol is very potent and 95% of all glucocorticoid activity is provided by cortisol. Corticosterone provides 4% of all glucocorticoid activity.
- Prednesolone, a synthetic corticosteroid is 4 times more powerful anti-inflammatory than cortisol. Dexamethasone, another synthetic corticoid is 30 times more powerful anti-inflammatory than cortisol.



Cortisol

- Cortisol is a *catabolic* and *diabetogenic* hormone
- It enhances the mobilization of aminoacids from the muscle, amino acids uptake by the hepatic cells for hepatic gluconeogenesis, glycogenolysis and also the conversion of pyruvate to glycogen. The net result is an increase in hepatic glycogen and an *increase in blood glucose*.
- The effect of glucocorticoids on carbohydrate metabolism is *"permissive*" i.e., their presence is required for the gluconeogenic and glycogenolytic actions of glucagon and epinephrine, respectively.
- Cortisol enhances the activities of phosphoenol pyruvate carboxykinase and hence the gluconeogenesis.
- In pregnant animals it is important for the transport of glucose across the placenta.
- Glucocorticoids and insulin have similar effects on liver glycogen metabolism .

- Cortisol stimulates increased *liver glycogen deposition* by inhibiting phosphorylase and stimulating glycogen synthase. However their effects on peripheral utilization of glucose are different.
- Glucocorticoids *inhibit glucose uptake* and *metabolism* particularly in muscles and adipose cells which is referred as ***anti-insulin effect***.
- Glucocorticoids have direct effect *on adipose tissue* where it increases therate of *lipolysis*, thus plasma free fatty acids. It also potentiates the lipolytic actions of GH and epinephrine during fasting. It moderately enhances the utilization of fatty acids for energy. Glucocorticoids increase fat absorption and stimulate gastric acid and pepsin secretions. Cortisol also increases appetite.
- The corticoids redistribute body fat i.e. fat is lost from limbs and accumulated in head and trunk giving a pot belly appearance. In humans it gives a *moon face* appearance.
- *Extrahepatic protein synthesis is inhibited* by glucocorticoids,
- Protein catabolism is enhanced with an accompanying release of amino acids to support hepatic gluconeogenesis. Entry of amino acids into extrahepatic cells is inhibited. The mobilization and incorporation of amino acids into glycogen results in an increased urinary excretion of nitrogen and causes negative nitrogen balance that is accompanied by muscle wasting and reduced protein stores.
- Liver proteins and plasma proteins are increased contrary to protein depletion in other cells.
- Cortisol decreases the synthesis of 1, 25 DHCC and blocks calcium absorption from GI tract, thus increases bone resorption and *decrease bone formation*.
- Hyperactivity of cortisol depresses the development of cartilage and the formation of bone interrupting the growth. Cortisol inhibits secretion of GH from the adenohypophysis. Cortisol inhibits the synthesis of collagen and causes thinning of the skin and walls of capillaries.
- The diurnal increase in cortisol level in early morning (6.00 to 8.00 A.M.) is essential for *normal arousal* and initiation of day time activities. An excess of cortisol interferes with sleep.
- Glucocorticoids stimulates glomerular filtration rate and inhibit vasopressin activity at the level of distal tubule, thus play a role in *water diuresis* i.e., enhancement of water excretion.

Clinical effects of glucocorticoids

- Glucocorticoids have valuable clinical effects particularly concerning *inhibition of inflammatory response* including the prevention of capillary dilatation, extravasation of fluid into tissue spaces, leukocyte migration, and fibrin deposition and connective tissue synthesis.
- Cortisol causes *reduction in circulating lymphocytes,* eosinophils and basophils by causing lysis of these cells and also phagocytosis by WBCs but produces *neutrophilia*.
- Glucocorticoids *suppress immune response* by T lymphocytes and also *reduce the antibody production*.
- Glucocorticoids *inhibit the synthesis of inflammatory mediating compounds* such as prostaglandins, thromboxanes and leukotrienes that arise as a result of arachidonic acid metabolism of injured cells by stabilizing the lysosomal membrane.
- Cortisol inhibits the release of proteolytic enzymes and hyaluronidase from lysosome.
- Glucocorticoids are used to *inhibit allergic reaction* through its inhibitory action on the release of histamine from the granules of mast cells. Hence cortisol is useful in preventing death during allergy or anaphylaxis.

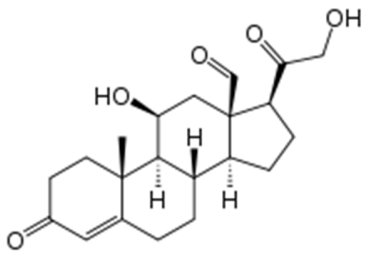
- Prolonged administration of cortisol inhibits proliferation of fibroblasts and synthesis and deposition of collagen fibrils and *prevents normal wound healing* after injury.
- Cortisol facilitates *in-utero* maturation of the GI tract, lung, CNS, retina and skin. It stimulates increased synthesis of surfactant and permits satisfactory breathing immediately after birth. Cortisol facilitates the enzymatic actions of the intestinal mucosa and this permit the new born to digest the di-saccharides present in the milk.
- In both males and females excess of glucocorticoids inhibit LH release, enhances the ability of the sex steroids to suppress gonadotropin secretions through negative feedback mechanism, thus delays the onset of puberty.
- Cortisol from the fetal adrenal cortex acts upon the placenta to reduce progesterone and increase estrogen secretion both of which promotes the synthesis and release of PGF2α and uterine contractions are stimulated to induce labor.

Regulation of Glucocorticoids

- Cortisol level rises in the early morning and declines in the afternoon and evening exhibiting a diurnal rhythm.
- The secretion of the glucocorticoids by zona reticularis is regulated by ACTH.
- A negative feedback system exists whereby glucocorticoids inhibit the release of CRH from the hypothalamus to decrease ACTH secretion by the pituitary. Cortisol has more potent negative feedback effect than corticosterone.
- Another factor that can modify the negative feed back control of glucocorticoids is stress, which stimulates the secretion of glucocorticoids to several folds. ACTH administration depletes the zona fasciculata of cholesterol and vitamin C contents.
- The adrenal cortical hormones are involved in the animal's adaptation to adverse stress known as *general adaptation syndrome* or *fight or flight response*.
- Failure of adrenal cortex to produce cortical hormones lead to a condition called *Addison's disease* in human beings, dogs, cats and horses.
- Hypersecretion of cortisol causes *Cushing's syndrome* where fat is mobilized from lower part of the body and deposited in thoracic, abdominal regions and face (*moon face*). Other clinical signs include polyurea, polyphagia, *hirsutism*, skin pigmentation, poor wound healing and thin skin.

MINERALOCORTICOIDS

- Aldosterone (*electrocortin*), deoxycorticosterone and 17-hydroxy-11 deoxycorticosterone are naturally occurring mineralocorticoid of adrenal gland.



- Aldosterone is very potent and contributes 95% of all mineralocorticoid activity. Deoxy corticosterone is 1/5thas active as aldosterone and secreted in very less amounts. Corticosterone has very slight mineralocorticoid activity.
- 9 alpha fluorocortisol is synthetic compound having potent activity than aldosterone.
- Cortisol and cortisone have sight mineralocorticoid activity.
- *Electrolyte balance* and *blood pressure homeostasis* are the principle physiological effects of mineralocorticoids.
- It promotes *sodium retention* and *potassium and hydrogen ions secretions* at the level of the distal tubules of kidney. Mineralocorticoids stimulates the synthesis of a protein to increase Na^+K^+ ATPase activity in the tubular cells thus increases the permeability of luminal cell surface to sodium influx from the renal filtrate.
- Aldosterone increases the sodium retention in the extra renal tissues like sweat glands, salivary glands and intestine.

Regulation of the Mineralocorticoids

- Aldosterone secretion is primarily regulated by (1) angiotensin-II and (2) K^+ concentration in the extracellular fluid
- Fall in systemic arterial blood pressure or increase in sympathetic nerve activity to kidney increases secretion of the enzyme *renin* from the juxtaglomerular cells of the kidney which is the most important factor in regulating mineralocorticoid secretion
- In the blood renin acts on angiotensinogen, an alpha-2- globulin produced by the liver and converts it to angiotensin-I. It is hydrolyzed by the enzyme angiotensin-converting enzyme (*ACE)* present in lungs and endothelial cells to *angiotensin-II* which directly acts on adrenal cortex to release aldosterone.
- Aldosterone promotes Na retention and K and H secretions in the distal tubules of kidney.
- Hypovolumia increases the output of renin and aldosterone.
- An increase in K^+ stimulates aldosterone secretion. *Macula densa*, a group of specialized cells at the origin of distal tubules of the kidney detects Na ions level and exerts control on aldosterone through the renin-angiotensin system.
- A decrease of Na ion concentration in the body fluids increases renin release.
- β -adrenergics, some PGs, high levels of K ions in the ECF and under some conditions of stress ACTH also stimulates aldosterone release.
- In contrast, natriuretic hormone reduces sodium retention by the kidney by its inhibitory effect on the production of mineralocorticoids and renin. ACTH produces a transient increase in mineralocorticoid secretion.

Mechanism of Action

- Aldosterone acts on the distal renal tubules and collecting ducts and causes
- More numbers of Na channels for Na ion transport via electro-chemical gradient.
- Activation of Na^+-K^+ ATPase

ADRENAL SEX HORMONES

- Zona reticularis produce mostly androgens (androstenedione, dehydroepiandrosterone) and minor amounts of glucocorticoids, progesterone and estrogen.
- About 10% of the circulating androgens in males and 50% in females come from adrenals. The sex steroids formation is controlled by ACTH.

CHAPTER-15: ADRENAL MEDULLA

Learning objectives

At the end of this chapter, the learner will be able to understand the following

- Catecholamine biosynthesis
- Adrenergic receptors
- Role of catecholamines in fight or flight reaction
- Metabolic effects of catecholamines.

STRUCTURE AND REGULATION OF HORMONE RELEASE

Chromaffin cells, neural ganglion cells

- The cells of the adrenal medulla that secrete the catecholamine hormones, are called chromaffin cells and they are innervated by preganglionic cholinergic nerve fibres

Structure and regulation of hormone release

- The central, medullary region of the adrenal gland acts as an endocrine extension of the sympathetic nervous system. It contains cells (chromaffin cells) which lack axons but possess many of the other properties of postganglionic sympathetic neurones.
- These chromaffin cells release the contents of their intracellular vesicles into the bloodstream in response to stimulation by the preganglionic sympathetic nerves which innervate the adrenal medulla.
- As a result, activation of sympathetic nerves leads to an increase in the circulating levels of the catecholamines, i.e. noradrenaline (norepinephrine) and adrenaline (epinephrine).
- The catecholamine hormones are manufactured from the amino acid tyrosine in the chromaffin cells.
- However, in contrast with sympathetic neurones which release noradrenaline (norepinephrine), the adrenal medulla secretes about 80% adrenaline (epinephrine) to

20% noradrenaline (norepinephrine). However, birds, whales and cat mainly secrete noradrenaline

- After release, they are broken down again within minutes in the liver and kidney in reactions involving catechol-O-methyl transferase and monoamine oxidase, so their actions can be switched off rapidly.
- Catecholamine metabolites, such as vanillylmandelic acid, and small quantities of free hormone are excreted in the urine.

BIOSYNTHESIS OF CATECHOLAMINES

- Synthesis of catecholamines begins with either of the amino acids *phenylalanine* or *tyrosine*.
- Biosynthetic pathway begins with the conversion of tyrosine to dihydroxy-phenylalanine (DOPA) by tyrosine hydroxylase (TH) which is the *rate-limiting enzyme*.The products of tyrosine metabolism including DOPA, dopamine, norepinephrine and epinephrine inhibit the activity of TH.
- DOPA is then converted to dopamine in the cytosol by the enzyme DOPA decarboxylase.
- Adrenaline and noradrenaline differ only in the terminal CH_3 group.
- The conversion of dopamine to norepinephrine occurs within the chromaffin granules by a key enzyme dopamine- beta hydroxylase. Norepinephrine moves back into the cytosol where it is converted into epinephrine through the activity of phenylethanolamine-N-methyl- transferase (*PNMT*).
- Adrenal medulla is the primary source of epinephrine (also present in brain) and nor-epinephrine is secreted both by medulla and post ganglionic sympathetic neurons.
- Epinephrine secreting cells also secrete opioid peptides like met-enkephalins.
- About 50% of catecholamines are transported bound with albumin and 50% in free form.
- The catecholamines are degraded in many tissues but mainly in liver and kidney by the enzymes catechol-O-methyl transferase (*COMT*) and mono-amino oxidase (*MAO*).
- The plasma half life of catecholamines is about 2 min.

MECHANISM OF ACTION OF CATECHOLAMINES

- These hormones act in the same way as catecholamine neurotransmitters, i.e. they bind to α- and β-adrenoceptors on the target cell membrane.
- Noradrenaline (norepinephrine) is more effective than adrenaline (epinephrine) at α-receptors, while adrenaline (epinephrine) is the more potent agonist at β- receptors. Activation of these receptors modulates the intracellular concentrations of a variety of second messengers, e.g. levels of cAMP may rise or fall.
- The pattern of second messenger response actually observed varies from cell to cell and depends on the receptor subtype involved. These intracellular signals alter cell function, often through second messenger-dependent kinases which phosphorylate important functional proteins and so alter their biochemical activities.

METABOLIC EFFECTS OF CATECHOLAMINES

- It is mediated mainly by β_2 -receptors.
- Epinephrine is ten times more potent than norepinephrine in the control of intermediary metabolism.
- The effects of epinephrine on glucose metabolism are similar to those of glucagon and opposite to insulin.

- Epinephrine *increases blood glucose* level (hyperglycemia) by promoting hepatic glycogenolysis by activating phosphorylase enzyme.
- Stimulation of α_1 -receptor favours gluconeogenesis. Epinephrine also stimulates glycogenolysis in skeletal muscles resulting inthe production of lactate which is converted into glucose in the liver.
- Through a-receptors, epinephrine inhibits insulin secretion and stimulates glucagon secretion, to increase blood glucose concentration.
- Epinephrine inhibits insulin effect on adipose tissue, thus promotes lipolysis through β_2 -receptors by activation of lipase enzyme and *increases free fatty acids* concentration in blood
- Glucocorticoids potentiate the effect of epinephrine on lipolysis.
- Both catecholamines produce *arterial vasoconstriction* through their a -receptor action. Epinephrine has high affinity for β_2 receptors and causes vasodilatation in heart and skeletal muscles and reduces peripheral resistance.
- Both epinephrine and norepinephrine interact with β_1 -receptors to increase both the *force of contraction* and *heart rate*.
- The action of epinephrine to *increase cardiac output* is a beneficial effect in situations as in *flight or fight*.
- Epinephrine increases systolic and decreases diastolic pressure, whereas norepinephrine increases both systolic and diastolic blood pressures.
- Epinephrine causes splenic contraction and increases RBC numbers in circulation. Epinephrine produces bronchodilatation by its action on b_2 receptor (norepinephrine has little effect) and increases both the*rate and depth of respiration*.
- Epinephrine, through its β-receptor activity, *relaxes the intestinal smooth muscles* and uterine muscles.
- It promotes erection of penis and ejaculation.
- Norepinephrine causes *contraction of uterine smooth muscles*.
- During cold catecholamines enhances the metabolic rate, non-shivering thermogenesis. Epinephrine constricts renal, splanchnic and cutaneous arterioles, but dilates the arterioles of the muscles to shunt the blood flow particularly to coronary and cerebral blood vessel. By vasoconstriction conserve heat during exposure to cold.
- It also improves alveolar gaseous exchange by bronchial dilatation. Epinephrine favours Na^+ reabsorption by stimulating renin-angiotensin-aldosterone mechanism.

CHAPTER-16: PINEAL GLAND, THYMUS GLAND, EICOSANOIDS AND ANP

Learning objectives

At the end of this chapter the learner will be able to understand the following

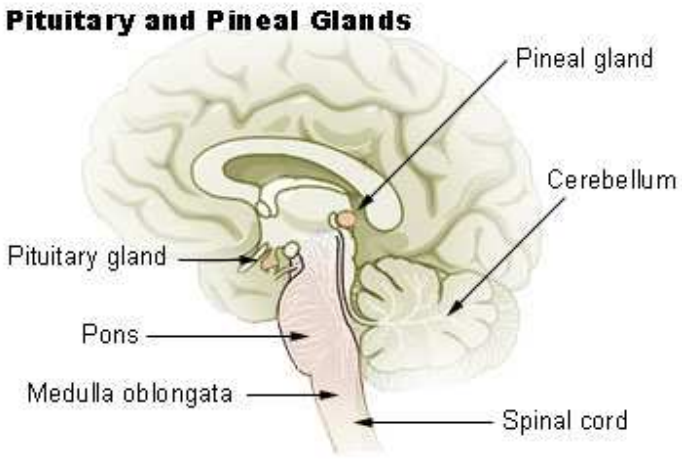
- Melatonin and its biological importance
- Thymic hormones
- Functions of prostaglandin and other eicosanoids
- Role of ANP

PINEAL GLAND

- Pineal galnd is also called as epiphysis cerebri.

Location

- Slightly flattened cone-shaped appendage of the brain.
- Its base is slightly constricted to form hollow peduncle by which it is attached to the roof of the third ventricle.



(Picture source: Wikipedia)

- Narrow prolongation of ventricle called pineal recess .
- It contains the highest content of serotonin and the enzyme hydroxyl-indol-o-methyl-transferase (N-methyl transferase) which converts serotonin to melatonin (N-acetyl 5-methoxy tryptamine).

Biological functions

- Lightening of skin colour in amphibia
  - May be also in mammals
- Inhibits gonadal activity in long day breeders
  - Hamsters, Horse
- Stimulates onset of breeding season in short day breeders
  - Sheep
- Long daily periods of elevated melatonin secretion
  - induce ovarian cycles in ewes
- Continuous administration of melatonin induces cyclicity in acyclic ewes
  - Inhibit ovarian cycles in mares
- Removal of pineal / cutting nerve connections to pineal leads to r emoval of seasonality in seasonal breeders

THYMUS GLAND

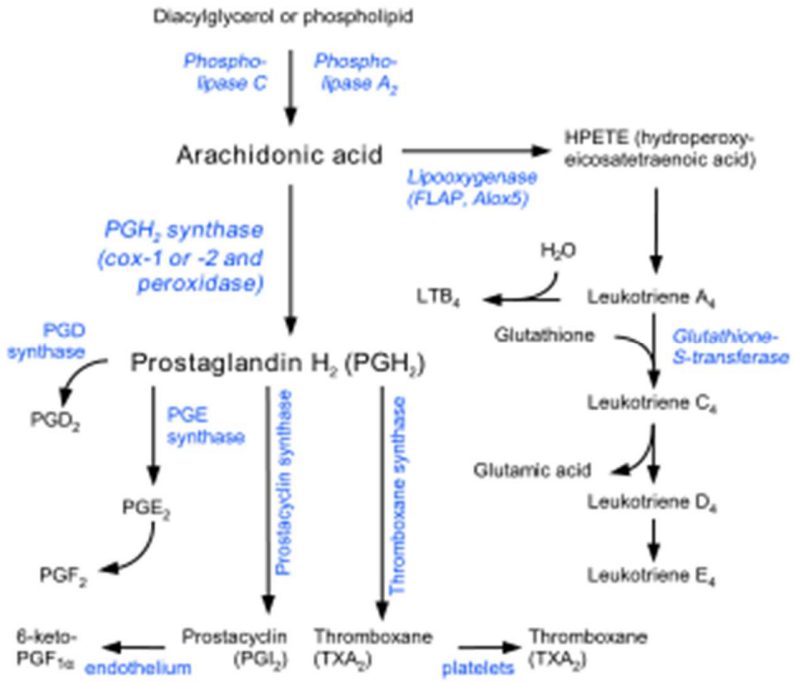
- It is large and active in the newborn, reaches full activity at puberty and thereafter declines in activity slowly.
- The thymus is a lymphoid organ found on both sides of the trachea.
- During its early development, it resembles an endocrine gland and is essential for the normal development and maintenance of immunologic competence.
- Thymic hormones are the homeostatic thymic hormone (HTH), thymic humoral factor (THF), thymosin, and lymphocyte-stimulating hormone (LSH).
- HTH and LSH increase the ratio of lymphocytes to polymorphonuclear leukocytes(PMN). Thymosin and THF stimulate cell-mediated immunity. A thymic steroid substance thymosterin inhibits lymphocytopoiesis and tumor growth

EICOSANOIDS

- Von Euler in 1934 isolated and named a substance isolated from seminal fluid as prostaglandins. Unlike the other hormones, prostaglandins are not localized in any particular tissue. Most prostaglandins act locally at the site of their production on a cell interaction–paracrine effect and also act by autocrine mechanism.
    - *Eicosanoids* are a large group of molecules derived from 20 carbon polyunsaturated fattyacids. The principal groups of hormones of this class are prostaglandins (PG), prostacyclins, leukotrienes (LT), thromboxanes (TX) and lipoxins (LX).
    - Arachidonic acid is the most abundant precursor for these hormones**.** Stores of arachadonic acid are present in membrane lipids and released through the action of various lipases.
    - The specific eicosanoids synthesized by a particular cell are dictated by the group of processing enzymes expressed in that cell.
    - Phospholipase A_2 releases arachidonic acid from membrane phospholipids. The released arachidonic acid is converted to prostaglandin H_2 (PGH_2) by *cyclic oxygenase 1* (Cox 1) and *cyclic oxygenase 2* (Cox 2) enzymes.
    - $PGH_2$is converted to prostacyclins, thromboxanes and prostaglandins by isomerase enzymes present in tissues.
    - ThromboxaneA_2 is synthesized by platelets and promotes vasoconstriction and platelet aggregation.
    - Prostacyclinis produced in the lungs and vascular endothelium, inhibits platelet aggregation and produces vasodilatation.
    - Leukotrienes mediate allergic response and inflammation; released when allergens combine with IgE antibodies on the mast cells and they produce bronchoconstriction, vasoconstriction and increases vascular permeability, attract neutrophils and eosinophils to site of inflammation.
    - Functionsof liopxins are not elucidated.
    - Eicosanoid hormones are rapidly inactivated by being metabolized and are typically active for only a few seconds.

Prostaglandins(PG)

- Prostaglandins have varied functions particularly important in female reproduction, parturition, cardiovascular system.
- They play a prominent role in causing pain, inflammation and fever.
- Glucocorticoids inhibit phospholipase A2 and act as anti inflammatory.
- PGs may be considered to regulate several physiological functions such as contraction of smooth muscles in reproductive organs, and functions in erection, ejaculation, sperm transport, ovulation and also luteolysis.
- They also regulate blood pressure, lipolysis, gastric secretion and blood clotting.
- Thromboxane A2 and $PGF_2\alpha$ are vasoconstrictors.
- Prostaglandins enhance the motility and secretions of the small intestine.
- PGE regulates the renal blood flow from cortex to the juxtaglomerular nephrons during stress. Inhalation of PGE causes bronchodilation while PGF causes bronchospasm.
- PGE 2 exert a characteristic effect in the longitudinal musculature of the oviduct, an increase in tonus in the proximal part, while relaxation in the rest of the organ. PGE2 increases release of GH, glucagon, progesterone and aldosterone. PGE2 inhibits gastric acid secretion, but stimulates contractions of the uterus and dilatation of the blood vessels and has no luteolytic effect. It stimulates the production of plasminogen activator and increases plasmin activity, involved in cell migration, mixing of theca and granulosa cells during CL formation.
- PGE 3 and PGF cause relaxation of the whole oviduct.
- PGE , PGA and PGI are vasodilators and increases renal perfusion leading to increase in Cl, Na and K level in watery urine.
- PGF2 facilitates adrenaline release.
- PGD2 is vasoconstrictor.
- High levels of estrogen promote oxytocin receptors in the endometrium stimulatePGF synthesis and its secretion from the endometrium. PGF_2 alpha potentiates the contractions of the uterus by estrogen during parturition.
- In females PGF2α is involved in the follicular rupture and ovulation, also aids sperm transport in males. From endometrium PGF passes directly through the walls of utero-ovarian vein into artery by counter current mechanism and causes vasoconstriction of the blood vessels supplying the CL, the hypoxia in CL leading to *luteolysis*.



ATRIAL NATRIURETIC PEPTIDE

- The Atrial Natriuretic Peptide (ANP) is synthesised and stored in the atrial muscles as 126 amino acid proANP which is cleaved to become 28-amino acid active hormone.
- Secretion of ANP is stimulated by increase in atrial pressure caused either by pressor agents or by increase in atrial volume.
- On kidney, ANP produces natriuretic and diuretic effects, increases GFR, renal excretion of water, Na, Ca, Cl, Mg and phosphate.
- The ANP blocks the secretion of vasopressin from neurohypophysis and angiotensin II or ACTH stimulated release of aldosterone from adrenal cortex.
- ANP causes decrease in blood pressure. It inhibits renin secretion.

CHAPTER-17: GASTRO INTESTINAL HORMONES AND HORMONE LIKE SUBSTANCES

Learning objectives

- This chapter explains the gastro intestinal (GI) hormones and GI peptides in regulation of GI functions

GI HORMONES

- By size GIT is the largest endocrine organ
- No defined endocrine glands
- Diffuse distribution of entero-endocrine cells through out GIT
    - Endocrine cells decrease progressively towards the colon
    - Enterochromaffin cells contain 5-OH tryptophan (Serotonin)
    - GI hormones show both endocrine and paracrine activities
- No feed back regulation for these hormones
- Presence of food or GIT environment stimulates their secretion
- Removal of the stimulus stops secretion
- There are several polypeptide substances synthesized and secreted by cells of GI tract.
- These include gastrin, secretin, cholecystokinin (CCK) or (pancreozymin), gastric inhibitory peptide(GIP), motilin, vasoactive intestinal polypeptide (VIP), somatostatin, bombesin and substance P.
- Some of these hormones act as paracrine hormones; others enter circulation and act as general hormones.
- Many of these hormones are also found in the nervous system.
- Based on structural and functional similarity, GI tract hormones are grouped into two families
    - Gastrin family – gastrin, CCK
    - Secretin family – secretin, glucagon, glycentin, VIP, GIP

Hormone	Site of production	Stimulus for release	Action
Gastrin	Distal stomach G cells	Protein in stomach, high gastric pH, vagal stimulus	Stimulates HCl, Pepsinogen secretions, gastric motility, growth of stomach epithelium
Secretin	Duodenum S cells	Acid in duodenum	Stimulates $NaHCO_3$ secretion of bile and pancreas, inhibits GI motility and gastric secretions
CCK	Duodenum to Ileum. Also found in brain; CCK 8, 16 & 32.	Protein and fat in small intestine	Stimulates enzyme secretions of pancreas and gall bladder contraction, inhibits gastric emptying
GIP	Duodenum and upper jejunum	Carbohydrates and fat in small intestine	Stimulates pancreatic insulin secretion, inhibits gastric motility and secretion

Motilin	Duodenum and jejunum K cells	Acetylcholine	Regulates gastric motility between meals, tone of lower esophageal sphincter
VIP	Ileum, also found in nerves, brain, ANS	Stimulates intestinal motility, inhibits gastric acid secretion	
Somatostatin	Stomach D cells	Increased level of glucose, amino acids, catecholamines and glucagons	Inhibits pancreatic insulin, glucagons and gastric secretion, gastric motility and gall bladder contraction
Substance P	GI tract	Stimulates intestinal motility	

CHAPTER-18: INTRODUCTION TO REPRODUCTIVE SYSTEM

Learning objectives

To undestand introductory aspects of reproduction, chromosomal sex, gonadal sex and psychic sex.

INTRODUCTION TO REPRODUCTIVE SYSTEM

Sex is the total morphological, physiological and psychological differences between male and female.

Reproduction is essential for continuity of species in living organisms.

Asexual & sexual reproduction both produce new individuals

Asexual reproduction: Progeny genetically identical to parent

Sexual reproduction : Progeny genetically dissimilar to parents

random segregation of genes during meiosis of gametes

recombination of genes at fertilization

Segregation and recombination of genes provides for

Genetic variability

Genetic repair

Disadvantages of sexual reproduction

A synchronous release of male and female gametes

Hazardous internal fertilization & intra-uterine gestation

Sexual reproduction is the most successful form of reproduction

Due to compensating factors High sexual receptivity of the female at the time of ovulation

Male is always ready

In lower forms of animals *reproduction* occurs by mitotic cell division.

In mammals reproduction is carried out by union of gametes produced by dimorphic individuals and this increases genetic variability.

Intersexuality

Freemartin heifer is an intersexual individual.

It is a genetic female which was born as co-twin with a male, leading to modification of female genitalia in the male direction by the masculanising factors of the male. MIH from the male twin enters the circulatory system of the female twin due to anastomosis of placental blood vessels. The male morphogenic agents from the male twin exert their influence before the development of ovary and mullerian system in the female twin and the free martin is stimulated to develop male structures.

The gonads resemble testes, mullerian duct system is inhibited, wolffian ducts remain and differentiate to male duct system.

Freemartins are more common in cattle due to greater degree of placental fusion. It is also reported in sheep, pigs and goats.

Hermaphrodite: The presence of both male and female gonads in one individual animal. Female pseudohermaphrodites have normal internal genitalia but intermediate external genitalia. Clitoris may be enlarged or modified to penis.

Male hermaphrodites are common in animals. True hermaphrodites have both ovarian and testicular tissue with intermediate genitalia and are very rare in higher animals

SEXUAL DIFFERENTIATION

In all living organisms, reproduction is essential to continuity of the species. In mammals, reproduction depends upon the union of anisogametes produced by dimorphic individuals designated as male and female.

Though, the two sexes can be recognized by their morphological features, the complete sexual differentiation includes the total morphological, physiological and psychological differences that distinguish male and female.

With the union of sperm and the ovum there is formation of zygote which undergoes several mitoic divisions. The period of embryo consists of the time from blastocyst development until there is differentiation of the organ systems in the embryo.

The sexual differentiation takes place in consecutive steps, viz. chromosomal, gonadal, somatic and psychic sex differentiation.

1. Chromosomal sex differentiation

Chromosomal sex is determined by a single pair of sex chromosomes. The sex having similar sex chromosomes is the homozygous sex, whereas the one with dissimilar sex chromosomes is heterozygous sex. Among mammals the female is the homogametic sex

(XX) and the male is heterogametic sex (XY). In poultry, hens are heterozygous with ZW and males are homozygous with ZZ chromosomes.

The chromosomal sex of an individual is determined at the time of fertilization itself, though at this stage there is no morphological or physiological differentiation.

Chromosomal sex can be ascertained by examining interphase nuclei of somatic cells. The nuclei of somatic cells containing two X chromosomes contain a characteristic chromatin mass, the 'Barr body'. Thus, the females are chromatin positive (nuclei containing Barr bodies), whereas the males are chromatin negative.

2. Gonadal Sex differentiation

Although the genetic sex of the conceptus is unalterably determined by its sex chromosome complement, each embryo is potentially capable of developing the genitalia of either sex, since the primitive gonad has all the cellular elements to give origin to a testicle or an ovary. The differentiation of sexual characteristics depends upon the quantitative relationship between male and female determining genes and their integration with the internal environment.

Genes associated with female characteristics are believed to be located on the X-chromosome. Thus XX chromosomes are necessary for the differentiation of the normal ovary. Genes for male development are located on autosomes and on the Y chromosome. The presence of a Y chromosome is responsible for the differentiation of the testes.

The mammalian embryo has an inherent tendency to develop as a female. Therefore, castration of male embryo at early stage of embryogenesis results in their development as females.

Gonadogenesis begins with the formation of the genital ridges (in close association with the mesonephros). Primordial germ cells migrate from the yolk sac endoderm to the genital ridges. The gonads at this stage are still sexually bipotent and consist of an inner medulla and an outer cortex. Primordial germ cells invade the medulla to form primary sex cords. At this time, gonadal differentiation takes place, and the gonad loses its bipotentiality.

When genes for masculanity of an XY individual prevail, the medulla persists, cortical development is inhibited, and a testis develops.

When genes for femaleness predominate in the XX embryo, the medulla is inhibited, and the cortex develops. The persisting primodial germ cells in the epithelium invade the cortex as the secondary sex cords, and an ovary develops.

During the indifferent stage the gonad is potentially capable of either testis or ovary formation. The ovaries/testes being the primary sex organs but once there is differentiation of gonads is complete, the individual/organism can be distinguished as MALE or FEMALE.

3. Somatic sex differentiation

Following gonadal differentiation, the accessory genitalia develop under the influence of the gonad. In an indifferent stage, there are 2 duct systems, the Mullerian duct system and the Wolffian duct system.

When the testis has formed, the female (mullerian) duct system regresses to vestigial rudiments and the male (Wolffian) elements develop under the influence of secretions from the newly formed testis.

When an ovary has formed, the absence of the testicular morphogenic agents permit the Mullerian system to develop, and the Wolffian duct and tubules regress to vestigial rudiments.

From these two duct systems eventually there is development of the complete male/female reproductive organs. Thus at the time of birth of individual has morphologically differentiable identification either male or of reproductive organs.

4. Psychic sex differentiation

At the time of birth despite of having either male or female reproductive organs an individual can not behave, feel or act like a male or the female. The organism does not have a sex drive or libido. The psychic sex establishment is the final step in the differentiation of the 2 sexes. The psychic sex the sexual behaviour is basically under the control of gonadotropic and gonadal hormones secreted in proper amount and specific ratio. In addition some external factors like season/ also play an important role.

Prepubertal gonadectomy usually prevents mating behaviour. Ovariectomy of sexually mature females immediately abolishes mating behaviour. Adequate doses of estrogen administered to ovariectomized adult female restores manifestation of estrus and mating behaviour. Similarly in males prepubertal orchiectomy prevents normal mating patterns. Adult castration is followed by a gondal diminution of copulatory responses and sex drive.

HOMOLOGIES OF MALE AND FEMALE - REPRODUCTIVE SYSTEM

Homologies / Homologous organs

During embryogenesis of the genital system the male and female reproductive organs develop from indifferent structures.

Homologous organs are the organs which originate from similar rudimentary structures in the embryonic stage but differ morphologically and/or physiologically during post natal stage.

Homologies of male and female reproductive systems

Indifferent	Male	Female
INTERNAL GENITALIA		
<divGonad </div	Testis Rete testis	Ovary Rete ovari*
Mesonephric tubules	Vas efferens Paradidymis* Vas aberrans*	Epoophoton* Paroophoron*
Mesonephric duct	Epididymis Vas deferens Ejaculatory duct Seminal vesicle Appendix of the epididymis*	Duct part of epoophoron* (Gartner's duct)
Mullerian duct	Appendage of testis* Prostatic utricle*	Appendage of the ovary* (hydatid) Fimbria of oviduct Oviduct Uterus Vagina (all or part)
Urogenital sinus	Prostatic, membranous, and carvemosu urethra Bulbourethral glands (Cowpers glands) Prostate	Urethra Vestibule Vagina (in part) Vestibular glands (Bartholin's glands) Paraurethral glands
EXTERNAL GENITALIA		
Genital tubercle	Glans penis Corpus penis	Glans clitoris Corpus clitoridis
Urethral folds	Raphe of scrotum and penis	Labia minora
Labioscrotal swellings	Scrotum	Labia majora

CHAPTER-19: FUNCTIONAL ANATOMY OF FEMALE REPRODUCTIVE TRACT

Learning objectives

At the end of this chapter, the learner will be able to understand the following

Introductory aspects of female reproduction

Functional anatomy of organs of female reproductive system

FUNCTIONAL ANATOMY OF FEMALE REPRODUCTIVE TRACT

Introduction

Female reproductive system consists of

Pair of ovaries
Duct system
A pair of oviducts / Fallopian tubes
The uterus
The cervix
The internal vagina
External genitalia
EXTERNAL GENETALIA
Outer vagina, vestibular glands, clitoris, labia majora and minora
Hymen may be present at the junction of outer and inner vagina. Usually disintegrates as the animal cycles but may be removed surgically if needed

Primary sex organs – the ovaries (Gametogenic)

Accessory sex organs – the fallopian tubes, uterus, cervix, vagina and vulva (Essential for reproduction but not gametogenic)

Embryonic development in females

Mullerian ducts – Fimbria of oviduct, oviduct, uterus, vagina

Urogenital sinus – Urethra, vestibule, vagina

Genital tubercle – Clitoris

Primary sex organ - Ovary

The ovaries are paired organs that serve as both a gametogenic (produce ovum) and an endocrine function.

The ovaries shape varies according to the species and stage of estrous cycle. The functional ovary of a polytocous animal (sow, bitch or cat) has several follicles or corpora lutea which give the appearance of a cluster of grapes.

The monotocous animal (cow, ewe and mare) has an ovoid-shaped ovary. The mare has a kidney shaped ovary because of the ovulation fossa which is site of all ovulations.

The ovary consists of an inner medulla and an outer cortex region. The medulla of the ovary is the most vascular part while the cortex consists of dense connective tissue interspersed with parenchymal epithelial cells.

The outer layer of cortex is a dense connective tissue capsule, the tunica albuginea. The outmost surface consists of a single layer of germinal epithelium which covers the entire ovary except in mare where it is limited to the ovulation fossa.

TYPES OF FOLLICLE

There are 3 types of follicles present at birth and throughout the life of animals.

Primordial follicle

Growing follicle

Graafian follicle

Primordial follicle or primary follicle

Consists of an oocyte surrounded by a single layer of epithelial flattened granulosa cells with an irregularly shaped nuclei. This stage is reached at birth in the newborn heifer and the ovary may contain as many as 150,000 of these follicles. The number of primordial follicles decrease to a few as 1,000 in a cow by 15 to 20 years of age.

The number of primordial follicles that undergo folliculogenesis to reach mature, graafian stage from puberty is only a small fraction of the number of follicles available in the pool of primordial follicles. Most follicles will either undergo a process of regression called atresia at some stage of folliculogenesis or remain as primordial follicles with no signs of growth.

Growing follicle

Follicles that have left the resting stage as primordial follicles have begun growth but have not developed a theca layer or antrum (cavity).

This follicle has 2 or more layer of granulosa cells surrounding the oocyte; with continued growth, additional layers of granulosa cells appear to surround the oocyte.

A zona pellucida surrounding the oocytes may be seen at this stage.

The number of growing follicle in an ovary at a given time is relatively small in the domestic species and varies with the stage of the estrous cycle.

Graafian follicle

These are follicle in which an antrum (cavity) is clearly visible. The Graafian follicle protrude from the surface of the ovary. As the antrum enlarges, the granulosa cell layer is evened out except at the cumulus oophorus where the oocyte rests in a nest of granulosa cells.

The diameter of the oocyte of the cow is 80-120 μm in this stage and is surrounded by the zona pellucida.

Two layer of theca cells, theca interna and theca externa together with the granulosa cells form the wall of the follicle. Theca externa formed by muscle type cells has actin and myosin contractile filaments for follicular contraction.

The granulosa cells maintain contact with the oocyte during folliculogenesis and in the preantral and antral stages from the cumulus oophorus. The cumulus cells maintain contact with the oocyte even as the follicular fluid fills the antrum and eccentrically displace the oocyte. As the follicle matures, the granulosa cells also undergo morphologic differentiation (increased cytoplasmic organelle) for increased steroidogenic function.

Gap junctions are seen between the granulosa cells, theca interna, cells of cumulus oophorus and the oocyte. These gap junctions serve as a important role in movement of small molecules, icons and nutrients from the basement towards the antrum and as channels for hormonal communication between the peripheral cells of the follicle and oocyte.

Primordial follicles lack an independent vasculature. The capillary bed confined to the thecal layer develops as the thecal cells are formed around the follicle these thecal capillaries increase in size and concentrate in the theca interna in close proximity to the basement membrane. Blood flow through these capillaries also ceases as the follicle matures.

The permeability of the follicular wall increases in the preovulating period in some species. So the composition of the follicular fluid is similar to that of the plasma. However due to the metabolic and secretary activities of the follicular cells, the follicular fluid has a lower concentration of gonadotropins and higher concentration of steroid hormones, than in the peripheral blood.

Embryonic development of ova and follicle

Early in the first weeks of embryonic development, the primitive large germ cells called primordial germ cells are present in the yolk sac. These primordial germ cells migrate by ameboid movement from the yolk sac to the genital ridges. This migration occurs in 35 days of gestation in bovine.

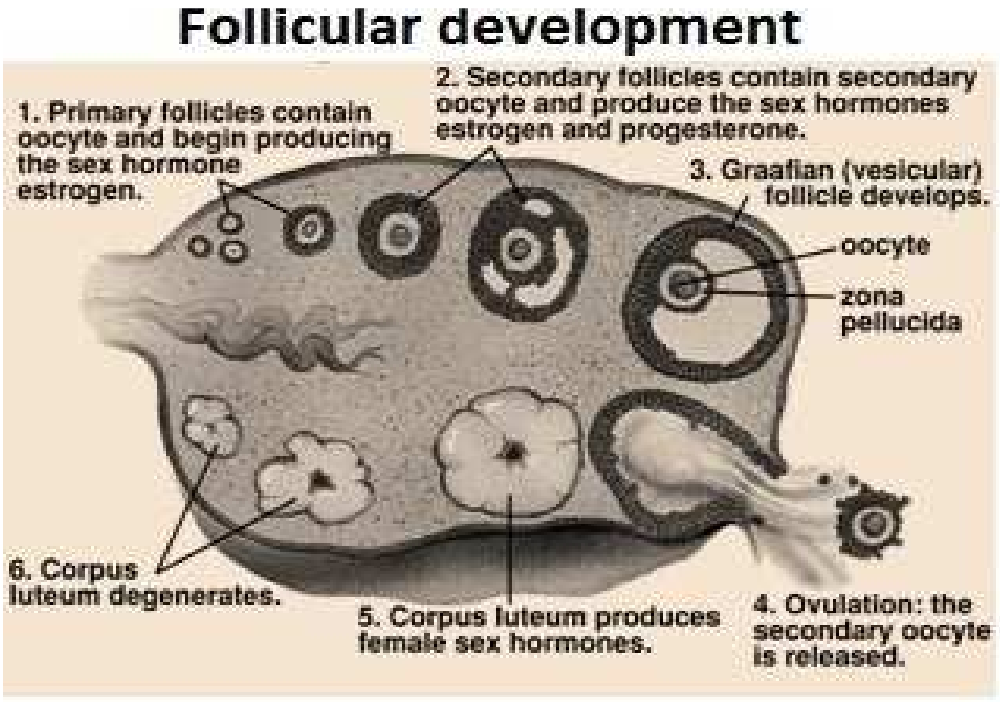
After a few days, the gonadal sex can be distinguished by the formation of superficial tunica albuginea and the peripheral location of oogonia in the primitive ovary. This occurs in the 45 days of gestation in bovines.

These oogonia multiply by mitosis after sexual differentiation and enter the prophase of the first meiotic divisions, when they are called the oocytes. The number of oocytes produced may be determined by the level of FSH in the embryo. Oogenesis is the transformation of oogonia into oocytes. Oogenesis is completed before or shortly after birth in all domestic mammals.

The period of oogonial mitosis signifies the end of the major portion of the period of oogenesis and this occurs at 45-100 days of gestation in cattle.

At the end of this period of oogenesis the oocytes enter the first meiotic prophase. In this stage the ova are surrounded by a single layer of follicular epithelium and these are called primordial follicles.

Thus at birth all female animals are born with their full complement of oocytes in primordial or older follicles and these are never replenished but only decrease in number during the life of the animal. Oocytes that fail to get incorporated into the follicles self destruct by apoptosis.



OVIDUCT

Uterine tubes or fallopian tubes are paired, convoluted tubes which has the unique function of conveying the eggs and sperms in opposite direction.

It connects the ovary to the uterus

Four functional divisions of the oviduct

Fimbriae

Ovarian end of the oviduct with fringed edges

Forms ovarian bursa with mesosalpinx

Ovarin bursa open in all farm animals

Completely covers the ovary in the pig

- Ovarian bursa covers only ovulation fossa in the mare
- Fimbriae move freely
  - Sweep on the surface of the ovary at ovulation

Infundibulum

- Funnels shaped second portion
- Size varies with age and size of the animal
- Has the abdominal opening of the oviduct – ostium abdominale

Ampulla

- Forms more than half the length of the oviduct
- Connects infundibulum to isthmus

Isthmus

- Connected to uterus at the utero-tubal junction

Musculature of the oviduct

- Outer longitudinal and inner circular smooth muscles
  - Thickness increases from ovarian to uterine end
  - Contraction of longitudinal muscles shorten the oviduct
  - Contraction of circular muscles cause annular constrictions
  - Exhibits three types of contractions
    - Localized peristalsis
    - Segmental contractions
    - Worm like writhing contraction
    - These contractions occur simultaneously or in specific patterns
    - Pattern and amplitude of contractions also varies in different parts

- Most vigorous contractions occur at ovulation

Mucosa of the oviduct

- Exhibits primary, secondary and tertiary folds
  - Thrown into high folds in ampulla
  - Height decreases in isthmus
  - Become low ridges at the utero-tubal junction
  - Completely fill the lumen of the oviduct at the ovarian end
- Has one layer of ciliated and non-ciliated columnar cells
  - Proportion of ciliated to non cilaited cells varies along the oviduct
  - Related to the functions of different segments of the oviduct
  - Dependent on ovarian estrogens for their function
    - Ciliated cells are maximum at the fimbria – help pick up of the ovum
    - Non ciliated secretory cells are maximum ampulla onwards
    - Promote interaction of gametes and noursihment of the embryo
    - Direction of oviducal fluids is towardssthe ovary
      - Most of the oviducal secretions are lost into peritoneal cavity for 2-3d after estrus
      - Subsequently fluids move down into uterus carrying the embryo
    - Cilia beat at their maximum at ovulation
    - Cilia usually beat towards the uterus

Functions of the oviduct

- Pick up of the ovum
  - Fimbriae pick up the ovulated egg from the ovarian surface
  - Capable of picking up of ovum from the contralateral side
- Oviducal secretions provide

For Capacitation of sperm

Conducive environment for fertilization

Nutrition to young embryo

Protection to oocyte, sperm and young embryo from attack by maternal immune system

For cleavage of young embryo by secretion of clevage inducing protein

Ciliary movements, muscular contractions and fluid flow help in

Mixing of oviducal contents that improves encounter of the gametes

Denudation of follicular cells from the ovum

Transport of zygote into uterus

Prevention of tubal implantation

Oviduct transports egg and sperm in opposite direction at about the same time but never fails

UTERUS

The uterus has a body and two uterine horns (cornua); species variation is seen.

Cow, Ewe and Mare

Bipartite uterus

Horns completely separated by a septum

Large uterine body particularly in the mare

Sow

Bicornuate uterus

Very long folded horns

Adaptation in litter bearing animals

Short body

Diadelphus uterus

Platypus

- Male has a forked penis

- Duplex uterus
  - Rabbit, G.Pig
- Simplex uterus
  - Primates
- Primitive forms of uterus found some times in higher animals do not interfere with fertility

Uterine histology

Three distinct layers

- Outer serous membrane – extension of the peritoneum
- Middle myometrium – outer longitudinal & Inner circular smooth muscles
  - Vascular layer in between
  - Myometrial contractions synchronized with oviducal contractions
  - Contract towards ovary at estrus and towards cervix subsequently
  - Estradiol increases but progesterone decreases the myometrial contractions
- Inner endometrium
  - Epithelium – simple columnar non ciliated in all domestic animals
    - Tall and actively secretory at estrus due to stimulation by estrogens
    - Low cuboidal during subsequent two days
  - Endometrial Glands
    - Open into uterine lumen
    - Straight at estrus
    - Coiled, complex and increasingly secretory during luteal phase
    - Volume & composition of the secretions varies with the phases of the cycle
    - Important proteins in different species

- Blastokinin, implantation promoting factor, GPC, steroid hormone receptors and uteroferrin

Cotyledons in the endometrium of ruminants

- For attachment of fetal membranes
- Endometrial glands absent at cotyledons

Functions of the uterus

Myometrial contractions

Occur at mating – cause transport of sperm from site of ejaculation to site of fertilization

- Massaging uterus at AI improves fertility

After estrus the uterine contractions directed towards cervix

- Transport the zygote from oviduct into the uterus for implantation

Uterine contractions kept inhibited by high progesterone levels during gestation

- At the end of gestation progesterone levels decrease and estrogens increase which causes uterine contractions that help in parturition
- Pregnancy could be housed only in the uterus since myometrium has the ability grow enormously

Endometrial Secretions

Induce capcitation at estrus

Stimulate sperm metabolism

GPC makes glycerol available from phosphoglycerol

Provide nourishment to pre and early post implantation embryo during luteal phase

- Particularly important in farm animals
- Ex; Cow embryos live freely and proliferate extensively till 30d post estrus

Estrogen and progesterone receptors in the secretions modulate

- Sensitivity of myometrium and proliferation of endometrial epithelium and glands

Blastokinin in rabbit uterine secretions promotes blastocyst formation

Implantation promoting factor present in the mouse uterine secretions

Uteroferrin found in ewe uterine secretions also promotes embryonic growth and implantation

Maintenance /Regression of CL

Sends signals to CL about the presence / absence of the conceptus

Signals carried via the utero-ovarian veno-arterial pathway

Hysterectomy prolongs life span but partial hysterectomy regresses CL

IUDs can also prolong the life span of the CL

CERVIX

Cervix is thick walled sphincter like structures

Separates vagina from the uterus

Connective tissue is the major component

Some smooth muscles also found

Lumen is narrow and exhibits annular rings

Very prominent in cow and ewe – form interlocking rings

Less prominent in the mare

In the sow the rings are cork screw shaped

Corresponding spiral twisting in the tip of penis of the boar

Epithelial lining

Columnar cells and goblet cells

Secrete mucus

Mucosa is intricately folded that increases secretory surface

Mucus is composed of glycoproteins

Mucus has unique biophysical properties

- Ferning, elasticity, viscosity and stickiness
- Physical and chemical properties of the mucus change during different phases of the estrous cycle
  - Ex: estrogens increase the intermolecular space to 2-5 µm
  - Progesterone reduces it

Functions of cervix

- Transport of sperm
  - Ejaculation into vagina near the cervix in most species
  - Sperm oriented towards cervix
  - Hydrolytic enzymes in the semen digest back bone protein in the cervical mucus and remove cross linkages of mucin
  - Channels for sperm passage appear
- Sperm reservoir
  - Large number of sperm lodged into cervical crypts
  - Released slowly later
  - Prevents inundation of uterus by sperm
  - Makes the sperm available at the site of fertilization for up to 24h after mating
- Sperm selection center
  - Several sperm lodged in cervical crypts never get released
  - Particularly those that are less motile
  - Thus cervix selects only vigorously motile sperm for transport to the site of fertilization
- Protection of upper reproductive tract
  - Cervical mucus becomes thick and viscous during the luteal phase and seals the URT
  - This seal persists till next estrus or parturition
- Helps in parturition

Gross biochemical changes increase the size of the cervix enormously to allow the passage of the calf

Cervix during pregnancy

During pregnancy a tightly viscid, non ferning thick and turbid mucus occludes, the cervical canal, acting as a effective barrier against sperm transport and invasion of bacteria in the lumen, thus preventing the uterine infections.

The only other time cervix is open before parturition, at this time the cervical plug liquefies and the cervix dilates to permit the expulsion of the fetus and the fetal membranes.

VAGINA

The vagina consists of two parts - Outer /vestibular portion and Posterior / internal portion

Wall consists of surface epithelium, muscular coat and serosa

No mucosa

Muscular coat less well developed

Thin layer of longitudinal and thick layer of circular smooth muscles

Anterior sphincter in addition to posterior In the cow

Non secretory squamous epithelium

Mucus secreting cells in the vagina of Cow

Synchronous changes in epithelium with the estrous cycle

Stratified squamous at estrous and low cuboidal during luteal phase

Vaginal smear could detect phase of the cycle only in rat and bitch

Plasma cells beneath epithelium secrete IgA and IgG in addition to anti-sperm antibodies

Plasma cells increase during luteal phase and protect against invading organisms

Functions of vagina

Female copulatory organ into which semen is deposited

Acts as semen reservoir for the cervix

Micro-ridges on the surface and muscles in the walls distend to facilitate mating and parturition

Muscular contractions

Important for psychosexual satisfaction

Sperm transport

Absorption of seminal plasma

Seminal fluid constituents absorbed through vagina could influence the activity in other portions of FRT

Excretory duct for the cervix and uterus

Most of the mucus excreted via the vagina is actually produced in the cervix

Incorrectly called vaginal mucus

Fluids exude from the vascular supply although no secretory glands present in the vagina

Along with cervical mucus buffer the pH of vagina

Specific smell of vaginal secretion in cow at estrus

Identified by bulls and dogs

Capacitation of sperm in some species - Rabbit

EXTERNAL GENETALIA

Consists of outer vagina, vestibular glands, clitoris , labia majora and minora

Clitoris

Embryological homologue of penis

Burried in the vestibular mucosa in the cow but prominent in the mare

Large number of nerve endings

Important in psycho-sexual satisfaction

Labia

Homologues of scrotum

Labia minora poorly developed in animals

Labia major

Rich in sebaceous and tubular glands

Swollen, congested and edematous at estrus

Vestibular glands

Sebaceous and bartholin glands

Resemble bulbo-urethral glands in the male

Secrete lubricating mucus to boost copulation

CHAPTER-20: FOLLICULOGENESIS

Learning objectives

This chapter explains the following

Types of follicles in the ovary

Follicular development

Follicular wave.

FOLLICULOGENESIS

Factors that control the initial follicular growth are not known.

When any of the primordial follicles is released from the reserve, it continues to grow until ovulation or until atresia stage.

The development of the follicle is initiated by the growth of oocyte.

The follicle that has an early growing oocyte but still surrounded by a single layer of follicular cells is termed as *primary follicle*. The primary oocyte in *primary follicle* increase in size; at the same time the follicular cells multiply into several layers of cells. This is called *maturing follicle* or *growing follicle (preantral follicle or secondary follicle)* which still has not developed theca or *antrum*.

On further development, an *antrum* (cavity) form by fluid secreted by granulosa cells which collect between the granulosa cells and separating them. When the antrum has formed, the follicle is classified as a *Graafian follicle (antral/tertiary follicle)*.The mature tertiary follicle appears as a fluid-filled bulging on the surface of the ovary.

Two types of cells surround the maturing oocyte – *granulosa cells* on the inside surrounded by *theca cells* on the outside with a basement membrane in between.

The initial phase of follicular growth is termed as *hormone-independent phase* – oocyte increases in size and activity with large amount of RNA synthesis.

Follicular cells grow and divide to become granulosa cells which produce a glycoprotein layer around the oocyte forming the *zona pellucida*. The granulosa cells maintain contact with oocyte by cytoplasmic processes.

This initial growth is followed by h*ormone-dependent phase* with synthesis of FSH and oestradiol receptors on the granulosa cells.

The granulosa cells produce oestradiol by converting androgens synthesised by theca cells. LH stimulates androgen synthesis by theca cells. The androgens diffuse into granulosa cells. FSH stimulates *aromatase*enzyme of granulosa cells to convert androgen to oestrogen. This process is described as *two-cell two-gonadotropin model* of synthesis of follicular oestradiol.

The follicle becomes more responsive to FSH (FSH induces synthesis of its own receptors) and the oestrogens enhance growth and division of granulosa cells which starts secreting fluid into the antrum of the follicle.

Oestradiol inhibits progesterone biosynthesis in the follicle.

When the follicle reaches its maximum size, FSH induces LH receptors in the granulosa cells. When LH surge occurs, LH stimulates the granulosa cells to synthesize progesterone.

The granulosa cells surround the antrum. In addition, a hillock (mound) of granulosa cells is located at one side of the antrum which is called *cumulus oophorus*. The ovum rests upon the cumulus oophorus with other granulosa cells extending around the potential ovum.

The granulosa cells surrounding and in immediate contact with the ovum are termed the *corona radiata*. Both theca interna and granulosa cells are involved in production of oestrogen.

The granulosa cells are the progesterone producing cells in the corpus luteum. They also secrete other compounds that have been identified in follicular fluid which help to regulate the functions of the ovary. When ovulation occurs the follicle ruptures expelling the *liquor folliculi*, some granulosa cells, and the oocyte into the body cavity near the opening to the oviduct.

At the time of expulsion, the oocyte is surrounded by the corona radiata and a sticky mass containing other granulosa (*cumulus*) cells which aid the oviduct in picking up the oocyte and moving it down the oviduct. In some species the corona radiata is present at the time of fertilization. In other species these cells are shed quickly and are not present when fertilization occurs.

FOLLICULAR WAVE

Three to six follicles are initiated to grow together under the influence of increasing level of FSH which is described as *follicular wave*.

The first wave of follicles that develop during mid luteal phase reach antral stage and then regress followed by growth of second wave of follicles.

If CL regression begins at this time one of these follicles become dominant and this follicle reaches ovulatory stage. Rest of the growing follicles regress and become *atretic follicles*.

The dominant follicle produces inhibin and oestradiol which inhibit FSH thereby growth of further antral follicles is blocked.

In sheep, cattle and horses two to three follicular waves occur during luteal phase with the last wave ending in ovulation.

In pigs, rats and human ovulatory follicles develop during follicular phase of the cycle. In these animals, progesterone from CL of diestrus blocks ovulatory surge of gonadotropins. Following luteolysis, serum progesterone falls appreciably low and estradiol level increases favouring ovulatory surge of LH

In *monotocous* animals, at each heat period normally one follicle develops more rapidly than others and releases one ovum. Rest of the growing follicles regress and form atretic follicles. *Polytocous* animals produce two or more offspring per gestation and have several follicles developing and rupturing at the same time.

Following ovulation, the follicular antrum is filled with blood and lymph and forms *corpus hemorrhagicum*. This is gradually converted to *corpus luteum*. After each oestrous cycle the CL decreases in size and finally leaves a scar called *corpus albicans*.

FUNCTIONS OF THE OVARIAN FOLLICLE AND FOLLICULAR FLUID

Follicular fluid is transudate from plasma across the follicular wall

Accumulates in the antral spaces and merges them

Contains crabohydradtes, proteins, amino acids, enzymes and hormones

Gonadotrophins, sex steroids, prostaglandins

Immunoglobulins IgA and IgG are present

Composition modified by the follicular cells

Functions of follicular fluid

Regulation of granulosa cells

Synchronous growth of the follicle and the oocyte

Special components in the follicular fluid

Follicular fluid function as a vehicle for the transport of the oocyte from the rupturing follicle at the time of ovulation

Capacitation of sperm

F ertilization and transport of zygote

Oocyte maturation inhibitor (OMI)	Inhibits premature completion of oocyte meiosis
FSH receptor binding inhibitor	Depresses FSH binding to Granulosa cells
Inhibin (Follistatin)	Depresses secretion of FSH
Lutenization stimulator (Prolactin?)	Stimulates lutenization of granulosa cells after ovulation
Other factors	Promote capacitation & acrosome reaction

CHAPTER-21: OOGENESIS AND OVULATION

Learning objectives

At the end of this chapter, the learner will be able to understand the following

How primordial germ cells develop into oogonia and the oogonia develop into secondary oocyte

How the ovum is released into oviduct from ovary

Formation of corpus luteum and its function.

FORMATION OF HAPLOID OVUM

Formation of haploid ovum in ovary by meiosis is called as *oogenesis*

During prenatal period, mitotic proliferation of primordial germ cells occurs. This ends at the time of birth

Oogonia undergo mitotic division until the final generation of oogonia enters the prophase of meiosis-I. At this point it is called primary oocyte. This is under the influence of meiosis initiating factor

In domestic animals (except bitch and Queen), oogonia develops into primary oocyte shortly before or after birth.

At this stage primary oocyte are surrounded by a layer of follicular epithelium to form primary follicle.

The nucleus of primary oocyte enters the dictyate / resting stage of meiotic prophase I and does not compete until the animal reaches maturity. Resumption of meiosis, beyond dictyate stage depends on preovulatory surge of LH.

After puberty, the rising level of FSH causes follicular cells to develop. The primary oocyte grows by RNA synthesis.

Contact between oocyte and granulose cells prevent maturation of oocyte beyond dictyate stage through oocyte maturation inhibitor (OMI) produced by granulosa cell. OMI concentration declines as follicle matures.

Ovulatory LH surge block the transfer of OMI from cumulus cells to the oocyte allowing meiosis to resume. Half of chromatin and small amount of cytoplasm are extruded as first polar body

First meiotic division completes few hours before ovulation and the cell is now secondary oocyte. In bitch and few mares, completion of first meiotic division and extrusion of first polar body occur after ovulation.

The 2nd day oocyte enters II meiosis. It completes when spermatozoa penetrate Zona pellicida to activate the oocyte. When oocyte is activated 2nd maturation division is completed with the formation of 2ndpolar body.

Now the germ cell is called as ootid

Occasionally first polar body undergoes division in to form two PBs

OVULATION

Ovulation is mechanism by which the ovum is released from mature graffian follicle in response to LH surge.

A few follicles leave the primordial state each day and begin to develop

Developing follicles either will ovulate or may undergo atresia depending on the time of their development, as the follicles continue to develop during all phases of the oestrous cycle including the luteal phase. Several dominant follicles develop during the luteal phase of the cycle at about 10 day intervals

Selection of follicles for ovulation depends on the timing of the regression of the CL. Those follicles developing after the regression of the CL become ovulatory follicles

Accumulation of fluid in the antrum of the follicle exerts pressure on the ovulation wall resulting in bulging and thinning of the surface of the ovary

The granulosa cells loose cell to cell contact as follicle approaches ovulation, probably by dissociate of gap junctions

The oocyte along with cumulus cells separate from the pedicle of the cumulus and freed into the follicular fluid of antrum

Connective tissue of thecal layer dissociated during preovulatory period

Gonadotrophin stimulated preovulatory increase in follicular prostaglandins produced of granulosa cells stimulates ovarian contraction and activate thecal fibroblast to release proteolytic enzymes that digest the follicular wall and basal lamina.

A vascular stigma / papilla appears on the apex of the follicle. As the protrusion progresses, follicular wall layer stretch and thin until it breaks and the follicular contents are released.

FSH also increased before ovulation, which induces LH receptor synthesis in granulosa cells.

LH Surge is caused by the positive feed back stimulation of estrogen on hupothalamus and anterior pituitary. Increasing size of follicle produce more amount of estrogen which stimulates release of LH and FSH.

Preovulatory LH surge occurs about 24 hours before ovulation and it causes maturation of the oocyte

LH surge blocks the production of *meiosis inhibiting factor* and *luteinization inhibiting factor*. Meiosis is resumed and completed before ovulation

In cow and buffalo, ovulation occurs after the end of estrus period and in bitch, mare, sow before the end of estrus period.

In domestic animals ovulation occurs at any point on the ovarian surface except in mare wherein ovulation occurs at ovulation fossa .

TYPES OF OVULATION

Spontaneous ovulators

In many species ovulation occurs spontaneously after normal follicle growth during estrus cycle. Eg: cow, goat, mare, sow, primate

i.e a secretory pattern of estrogen associated with a developing follicle is able to cause a rise in LH release that causes ovulation.

Induced ovulators

Eg: Cat, rabbit, ferret, mink and members of camilidae family.

Induced ovulators are animals that do not ovulate spontaneously even in association with normal follicle development.

Vaginal stimulation during copulation initiates neuroendocrine reflex that results in ovulation. Estrogen priming of the hypothalamus and anterior pituitary is essential prerequisite for copulation to induce GnRH and LH release leading to ovulation.

Ewe	24 hrs	after onset of estrus
Doe	24 – 36 hrs	after onset of estrus
Pig	38 – 42 hrs	after onset of estrus
Mare	1 -2 days	Before the end of estrum
Cat	24 – 36 hrs	Post copulation
Rabbit	10.5 hrs	Post copulation
Cow	12 – 14 hrs	after end of estrus
Dog	Near the beginning of estrus	Fertilization takes place after 4th day

CORPUS LUTEUM

Corpus luteum is temporary endocrine gland, functioning for few days in non pregnant cyclic animal and few months in pregnant animals

The life span of corpus luteum varies with species depends primarily upon the fate of embryo.

Following ovulation, enough hemorrhage into follicular cavity from broken thecal vessels. It act as stopper sealing the residual cavity after discharge of oocyte

The blood filled follicle devoid of oocyte is called as ‘corpus haemorrhgium’

The blood clot serve as physical frame work and a nutritive medium for quick proliferate of granulosa cells. The intact vessel and connective tissue cells from the surrounding theca begins to proliferate

Granulosa layer folding towards the central portion of the residual cavity. Granulosa cells differentiated into luteal cells under the influence of LH and become corpus luteum.

The corpus luteum is one of the most vascular organs of the body. If animals are non – pregnant it is corpus luteum spurium and is destined to regress. If animal is pregnant, it is corpus luteum verum and corpus luteum is continues to function.

The function of corpus luteum is to secrete progesterone. During diestrus phase, maximum level of progesterone is produced which is necessary to

maintain pregnancy

preparation endometrium for implantation

enable the blastocyst to implant

Following implantation of blastocyst, the corpus luteum undergoes extensive enlargement and continues synthesis and secrete hormone.

If fertilization does not occur, the CL regresses allowing other follicles to mature.

Regression of corpus luteum starts (Days from ovulation)

Cow – 14 – 15

Ewe – 12 – 14

Sow – 13

Mare - 17

The regression changes are

Microscopic – Cytoplasmic Vacule

Pyknotic nuclei

Decreased progesterone in luteal cells

Degeneration of capillaries

Gradual replacement of luteal cells by fibroblasts

Macroscopic – The whole organ decreased in size suddenly within short interval

The degenerating a vascular non functional corpus luteum is corpus albicans. For several additional estrus cycle, a visible connective tissue scar remains on the ovary.

The corpora lutea of non – pregnant bitches and Queen continue to function approximately same length of time as for the pregnant animal.

The color of corpus luteum of cow and mare is intense than other species, due to lipochrome pigment- lutein. The ewe and sow are devoid of this pigment. Hence lighter color.

Natural luteolytic substance - $PGF_{2\alpha}$ is secreted by myometrial tissue

Ovarian artery and utero ovarian vein are intertwined and in close apposition. This vascular arrangement favors diffusion of $PGF_{2\alpha}$ into ovarian arterial blood.

Several mechanism are proposed for luteolysis.

Constriction of blood vessels supplying luteal cells causing ischemia and starvation of luteal cells

Interference with progesterone

Competition with LH for receptor site

Destruction of LH receptor site

LH is necessary for luteotropic effect and for progesterone secretion which increases as CL matures.

LH is luteotropic in ruminants and horse

Prolactin is luteotropic in rat

In canines both LH and PRL are luteotropic

In rabbits and pigs oestrogen is luteotropic

Corpus luteum and pregnancy

Progesterone is necessary for maintenance of pregnancy

Progesterone increase resting membrane potential of myometrial cell

Progesterone is serve as immediate precursor to other steroids that also necessary during pregnancy.

CHAPTER-22: ESTROUS CYCLE

Learning objectives

At the end of this chapter, the learner will be able to understand the following

What is estrous cycle

Different phases of estrous cycle

Signs of estrus in different species

Factors influencing estrous cycle

Pseudopregnancy.

ESTROUS CYCLE
INTRODUCTION AND DEFINITION

In all domestic animals that have attained puberty there is a definite physiological functional rhythm of the reproductive system called the estrous cycle. One cycle interval is defined as the time from the onset of one period of sexual receptivity to the next or as the interval between successive estrus.

Estrous cycle can be defined as cyclic period of sexual receptivity exhibited by the females at regular intervals

The name estrus is derived from the greek word oistros meaning "gadfly" and was used to describe the behavior of cows when attached by such files.

Estrous cycle is controlled directly by hormones of the ovary and indirectly by the hormones of the adenohypophysis. The basic pattern of the cycle is same but species differences are found in different stages of the cycle.

Monoestrus animals- animals have only one estrous cycle per year –dogs, wolves, bears

Polyestrus animals - animals have several estrous cycles per year-cattle, pigs, human

Cows and sows - *non-seasonally polyestrus* animals.

Seasonal breeders - animals that have one or more estrous cycles during certain periods of the year

Seasonally polyestrus animals (exhibit many cycles of estrous during certain periods of the year known as breeding season) - ewes, does and mares -

seasonally polyestrus can be s*hort-day breeders* – estrous cycle occur when days are getting shorter (in winter) e.g., sheep, goats, deer; anestrus in spring and summer

Long-day breeders – estrous cycle occur when days are getting longer (spring/early summer) - horses and hamsters; anestrus in winter

Seasonally monoestrus (exhibit a single estrus followed by *anestrus*)- bitches

The queen will undergo a series of ovulatory estrus in each breeding season -spring, summer and early autumn

DIFFERENT PHASES OF ESTROUS CYCLE

The estrous cycle is divided into four marked phases.

Proestrus

Estrus

Metestrus

Diestrus

Based on changes occurring in the ovary, estrous cycle consists of

Follicular phase – also known as proliferative or oestrogenic phase - oocyte matures, ovulation occurs,estradiol from the follicle is dominant - includes proestrus and estrus phases

Luteal phase (secretory phase) – corpus luteum forms, fertilization and embryo development, progesterone from the CL is dominant - includes metestrus and diestrus

PROESTRUS

It is a period of rapid follicular growth under gonadotrophic stimulation and the period in which the CL from the previous cycle in polyestrous specious regress.

The follicle growth (folliculogenesis) occurs through small, medium and large size follicles and is influenced by GnRH, FSH and LH. FSH initiates growth of follicles

LH converts the C-27 steriod cholesterol to C-19 steroid androgen in the theca interna. FSH favours the biosynthesis of C-18 steroid estrogen from androgen by the granulose cells.

In this phase, estrogen concentration increases progressively with simultaneous decline in progesterone level. It is due to the regression of the corpus luteum of the previous cycle by the luteolytic action of the $PGF_{2\alpha}$ in polyestrus species.

In all domestic animals except mare folliculogenesis is depressed during pregnancy because of high level of progesterone

There is marked increased in the growth of the epithelial tissues, the activity of musculature of the reproductive tract, secretion of the mucus and of the vascularity of the endometrium and vaginal mucosa.

In cow and mare, the sticky, dry mucus of the preceding period changes during proestrus to a milky, less viscid discharge (serous) and finally to a clear transparent, stringy mucus late in the proestus period. Proestrus is not clinically evident in most domestic species and lasts only for 2-3 days.

In the bitch, proestrus lasts for 7-9 days and is clearly identifiable by well-defined changes in the external genitaila and behavioral signs due to increasing sexual excitement. The increased endometrial vascularity is characterized by bleeding.

The vulva becomes increasing edematous and swollen. There is a gradual relaxation of the cervix and an increased secretion of viscid, slimy mucus from the goblet cells of the cervix and anterior vagina and from the uterine glands

ESTRUS

It is the period of sexual receptivity during which mating and ovulation occurs in most species and the corpus luteum begins to form

The graafian follicle is large and mature

Ovum undergoes certain maturation changes

There is increasing level of estrogen produced by the synergistic effects of the theca and granulosa cells.

Inhibin and increasing level of estrogen secreted from the graafian follicle inhibit hypothalamic and pituitary gonadotrophin secretion, thus inhibit the FSH release by negative feed back effect

Increased level of estrogen stimulates hypothalamic LHRH release by positive feed back mechanism, thus the ovulatory LH surge before ovulation.

Oviducts – The oviducts are tonic, the epithelium mature and cilia are active, contraction of the oviduct is occurring a close affinity to the Graffian follicle. Increase amount of oviductal fluid is being secreted.

The uterus is erect, turgid edematous. The blood supply to the uterus is increased. The mucosa is rapidly growing and mucus is secreted.

The vagina, vulval and cervical mucosa is pink, congested, due to increased vascularity; the mucus secretion is greatly increased strings of mucus hang from the vulva of the cow form pattern of discharge.

Towards the end of this period, there may be increased leucocytes migrating into the uterine lumen.

In most species the rupture of the mature ovisac or ovulation occurs towards the end of this period of estrum.

Duration of estrus varies from 14 to 18 hours in the cow, 7 to 10 days in mare and bitch

Other external signs – mounting, bellowing, frequent urination, discharge, reduced feed intake and milk yield.

SIGNS OF ESTRUS

Species	Signs

Cow	Bellowing, mounting, Cervical mucus discharge, Frequent urination, Swollen vulva, Reduced milk yield and Reduced feed intake.
Mare	Frequent urination, Exposure of clitoris (*winking of clitoris*).
Doe	Edema and hyperemia of vulva, Frequent and rapid tail wagging.
Sow	Watery vulval discharge, Standing quietly while pressing loin (lordosis).
Bitch	During proestrus, bleeding from vulva and vulva swelling; during estrus-dimishing of vulval swelling, Less turgid vagina.
Queen	Characteristic call and Roll over on its back.

Buffaloes : estrus signs less pronounced than cattle –frequent urination, *teat swelling*, reduction in milk yield, bellowing- many buffaloes come to heat during night and this makes heat detection difficult in these animals

METESTRUS

It is the transitional period between ovulation and full development of the corpus luteum. During this period, the reproductive system switches from estrogen to progesterone dominance.

It is the post ovulatory phase during which the granulosa and thecal cells lining the ruptured follicles begin to grow and reorganize into a structure called the corpus luteum (CL), corpus lutem hemorrhagicum, which is the principal site of progesterone production.

CL grows rapidly from the granulose cells of the ruptured follicles under the influence of LH of the anterior pituitary

Metestrum is largely under the influence of progesterone produced by the corpus luteum. This hormone, by inhibiting the secretion of FSH by the pituitary gland, prevents the development of more Graafian follicles.

In the cow, during the early part of the metestrum, the epithelium over the caruncles of the uterus is very hyperemic and some capillary hemorrhage occurs. This is called post estrual or metestral bleeding. This is due to the estrogen withdrawal.

The mucus secretion decreases and the glands of the endometrium grow rapidly. Towards the end of metestrum. The uterus becomes soft and pliable, due to the relaxation of the uterine muscles.

In the species, cow, sheep and sow and mare, the length of metestrum is about equal to the time it take for the ova to reach the uterus or about 3-4 days.

In dogs and cat the pseudo pregnant period may last about 50-60 days and 30 to 40 days respectively, when, the CL regresses and an anestrum of varying length occurs.

In the queen that is not bred, ovulation, and CL development doesn't occur and the follicle become atretic. Several cycles of this sort may occur in the cat before a period of anestrum sets in. Dogs and Cats do not have diestrual periods.

DIESTRUS

It is the longest period of the estrous cycle in the domestic animals, including, cow, sheep, goat, sow and mare. It is a mid luteal period, it is relatively long period of sexual quiescence (rest) between 2 successive estrous cycles in polyestrus animals.

In this period, the CL is fully developed, actively functional and the reproductive organs are under the dominant influence of progesterone for the preparation of the uterus for implantation of zygote and its growth as fetus.

LH is required for maintaining the CL in domestic animals. In rodents PRL is luteotropic.

The endometrium becomes thicker and glands hypertrophy. The cervix is constricted and the vaginal mucus is scanty and sticky. The mucus membrane of the vagina is pale. The uterine muscle is relaxed.

Late in this period, the corpus luteum begins to show regression. The endometrium and its glands atrophy or regress in size. In species that are not polyestrus, anestrum may occur.

In the non gravid uterus $PGF_{2\alpha}$ is produced and it passes into the uterine vein, transferred to the ovarian artery and reach the CL by counter current exchange at the vein and artery. In CL the PGF interferes with the LH action and produces vasoconstriction and arrests blood supply producing luteolysis.

In the cow, mare, sow, ewe, doe, $PGF_{2\alpha}$ is the natural luteolytic agent.

In the cat, dog and primates $PGF_{2\alpha}$ is not luteolytic and CL is maintained for 35-70 days (pseudo pregnancy)

Following luteolysis, proestrus of the next estrus cycle is initiated.

Corpus albicans represents the fibrosed CL of the previous cycle.

ANESTRUS

It is a stage of sexual quiescence characterized by the lack of estrus behaviour. Anestrus is a normal stage of reproductive function in the prepubertal and in aged animals of all specious. Pregnancy is the most common cause of anestrus in polyestrus specious. After puberty, anestrus is seen

In non pregnant animals in monoestric species such as bitch

In seasonally polyestrus species during non breeding season

During the lactation in females

In all domestic species, anestrus may occur as a pathological condition caused by a variety of factors, including nutritional deficiencies, environmental influences. That cause endocrine imbalance diseases of ovaries and uterus and infectious diseases causing early embryonic death or abortion.

All these factors result in economic losses, due to reproduce failure.

Anestrum is characterized by the quiescent, function less ovaries and reproductive tract.

Uterus is small and flaccid. The vaginal mucosa is pale vaginal mucus is scanty and sticky and cervix and tightly closed.

Some follicular activity of the ovaries may develop but a mature follicle and ovulation seldom occur during the anestrus period

SPECIES, LENGTH OF ESTROUS CYCLE, DURATION OF ESTRUS AND TIME OF OVULATION

Species	Length of estrous cycle (days)	Duration of estrus (hour)	Time of ovulation
Ewe	16-17	24-36	30-36 h from beginning of estrus
Doe	20-21	20-35	30-36 h from beginning of estrus
Sow	19-21	48-72	35-45h from beginning of estrus
Cow	21-22	18-19	10-12h after end of estrus
Mare	19-25 (21)	4-8days	1-2 day before end of estrus
Bitch	In estrus at 7-8 m intervals	7-9 days	1st or 2nd day of estrus
Queen	16 (non bred) (pseudopregnancy 36	5-6 days	Induced – 24-32h after mating

ENDOCRINE CONTROL OF ESTROUS CYCLE

Estrous cycle is regulated by hormones from hypothalamus, anterior pituitary and ovary

The pulsatile release of GnRH from the hypothalamus regulates the release of FSH and LH from the anterior pituitary. FSH and LH are secreted continuously by the adenohypophysis throughout the estrous cycle, but the proportion and levels of each change during different stages of estrous cycle.

FSH promotes follicular growth and maturation, while a synergistic effect of FSH and LH is essential for synthesis of estrogen from the granulose cells

Increased level of estrogen in the blood induces estrous behavior, whereas estrogen and inhibin from the Graafian follicle inhibit FSH release from the anterior pituitary through a negative feed back mechanism

Estrogen favors LH surge from the adenohypophysis Via the positive feed back mechanism by increasing the sensitivity of anterior pituitary to GnRH a large surge of LH and a small surge of FSH synergistically act to initiate ovulation

Immediately after ovulation the concentration of estrogen declines, LH activates the formation and maintenance of corpus luteum in most of the domestic animals

Progesterone increases rapidly during midluteal phase. Progesterone inhibits FSH and LH, thus the follicular growth

In non pregnant animals, $PGF_{2\alpha}$ is released in late luteal phase causes lusis of corpus luteum

Following luteolysis, progesterone decreases. GnRH and gonadotropin level; increases and initiates another cycle

If animal is conceived, progesterone production is continued by corpus luteum verum which inhibits follicular growth and ovulation by negative feed back effect on FSH and LH.

Factors influencing the estrous cycle

FACTORS INFLUENCING THE ESTROUS CYCLE

PHOTOPERIOD

In seasonal breeders, light is the potent stimulatory/inhibitory factor in regulating estrous cycle.

Light acts through hypothalamus and pineal gland.

Melatonin is secreted by the pineal gland during darkness. When the daylight period is decreased (decreased photoperiod-winter season) with increasing darkness, melatonin secretion is increased. This increased melatonin initiates breeding season in seasonal breeders like sheep and goats.

In horses and cats, an increasing photoperiod (spring and summer) initiates breeding activity which may be due to an inhibitory effect of melatonin on GnRH secretion

The farm animals are classified into three groups based on light periods.

Hypophysis is activated by period of short or decreasing day light as in sheep and goat (melatonin / analogue implants are useful to advance the breeding season)

Hypophysis is activated by long or increasing light periods as in horse, donkey, cat (increasing the light period at least 2 months before the expected cycle can cause onset of estrus one month in advance)

Photoperiod sensitivity difficult to perceive, as in cattle and pigs.

FACTORS INFLUENCING THE ESTROUS CYCLE TEMPERATURE AND NUTRITION

Temperature

Hot humid condition depresses reproduction. Protection of ewes from higher summer temperature advances the breeding season. During prolonged summer heat, even the cows exhibit silent heat. Cold does not generally affect reproduction.

Nutrition

Low level of nutrients, i.e., proteins, vitamins, minerals, energy etc., may affect the onset of puberty, breeding season and also the ovarian activity following parturition. Increasing the energy content of the diet from suboptimal to optimal level (especially before the onset of breeding season) increases ovulation rate in pigs, goats and sheep that ovulate more than one ova and this is called flushing.

FACTORS INFLUENCING THE ESTROUS CYCLE PHEROMONES

Pheromones are chemicals that allow communication between animals of the same species.

The pheromones modify the behaviour of males. Vaginal odours of the female attract the males.

Methyl-p-hydroxy benzoate of the vaginal secretion of dogs during prooestrus and estrus attracts males.

Sows assume breeding stance when exposed to urine of boar. Boar odour or boar taint is caused by the pheromone androgen-5α androstenone.

When a male mouse is introduced into a group of female mice, it causes synchronization of estrus in females of that group and this effect is called as **Whitten effect**. This is caused by release of gonadotropins due to introduction of male.

If a pregnant female mouse is placed in a cage with a strange male mouse, pregnancy fails. This effect is known as **Bruce effect** and it is due to blockage of PRL secretion with subsequent CL regression.

Abrupt exposure of females to a male nearing the time of puberty advances the onset of puberty or breeding season. This is known as **Lee Boot Effect.**

When a ram is introduced into a group of ewes which are in seasonal nonbreeding activity, it will advance the onset of breeding season and synchronize the estrus. This effect is called as **ram effect**.

Sight of male as well as physical contact of male with female influences the gonadotropin release thereby affecting ovarian activity.

FACTORS INFLUENCING THE ESTROUS CYCLE

SOCIAL FACTORS

Rearing females in larger groups facilitates puberty earlier than those reared in isolation or in small groups.

Transporting peripubertal animal for a short distance induces oestrous cycle in ewes within a few days due to the effect of mild stress.

Presence of male in a flock intensifies oestrus signs.

PSEUDOPREGNANCY

Pseudopregnancy or false pregnancy can be defined as an exaggerated diestrual response in bitch and rodents.

Related to the extreme sensitivity of the endometrium and mammary glands to progesterone in association with other hormonal factors, including prolactin

Corpora lutea of nonpregnant bitch remain functional for an extended period after ovulation.

The uterus enlarges, abdomen becomes relaxed, and the mammary glands begin to develop in preparation for lactation.

There is a relaxation of the pelvis and external genitalia similar to the symptoms observed during pregnancy.

Pseudopregnancy often lasts as long as or longer than a normal pregnancy.

The bitch may attempt to build a whelping nest in preparation of parturition.

POST PARTUM ESTRUS

The duration of post partum anestrus is affected by various factors – genetic, physiologic, breed, nutrition, suckling, rate of uterine involution, hormones etc.

Time required for post partum uterine involution is 4-6 weeks.

Progesterone withdrawal following parturition can resume estrus and ovulation. Sow exhibits estrus within 48h after parturition without ovulation. In mares there is fertile estrus 1-3 weeks after parturition which is called *foal heat*. Mares can be bred at foal heat.

In cows, ewes and goat, silent ovulation can occur 2-3 weeks after parturition but fertile estrus returns later. Suckling increases PRL secretion which is inversely related to FSH and LH.

In cat, if mating does not occur, the graafian follicle regresses and a new follicular wave occurs. If mating occurs the metestrus is followed by diestrus of almost similar duration as pregnancy, even if the mating is not fertile.

CHAPTER-23: OVARIAN HORMONES

Learning objectives

This chapter explains about steroid and peptide hormones produced by the ovary and their functions.

OVARIAN HORMONES

Introduction

Ovary is the primary female sex organ that produces two main steroid hormones.

Estradiol

Progesterone

These hormones exert their action on the female genital tract and mammary gland.

ESTROGEN – SOURCE

Granulosa cells of ovarian follicle, placenta, corpus luteum. Adrenal cortex also produces estrogen.

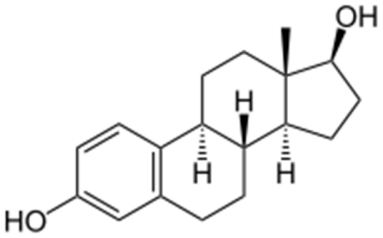
Estradiol 17 β – predominates in non pregnant animals

Estrone – predominates in pregnant animals

Estriol – predominates in pregnant primates

Plant estrogen is also found in legumes – Clover and alfa alfa which cause infertility in animals.

Binding proteins in the circulation carry estrogen.



ESTROGEN – FUNCTIONS

Action on the CNS

Promotes the female sex behaviour and growth of female reproductive tract

To induce behavioural estrus in the female but small amount of progesterone with estrogen is needed to induce estrus.

Regulates the release of FSH and LH from the anterior pituitary by its feed back mechanism.

Action on the uterus

It aids in sperm transport by increasing the myometrial contractions. Estrogen priming promotes the growth of the endometrial glands. Necessary for maintenance of zygote and helps implantation of zygote.

At the time of parturition, it sensitizes the pregnant uterus to oxytocin, thus aid the uterine contraction along with oxytocin and $PGF_2\alpha$. for the expulsion of fetus

Estrogen increase the rate of migration of leukocytes into the uterine lumen, and thus increases the bactericidal activity of the uterus during estrum.

Secondary sexual characters

Estradiol is responsible for the secondary sex characteristic of the female

It causes the growth of the duct system of the mammary gland

It stops growth of long bones by initiating the closure of epiphyseal growth plate

It causes disposition and distribution of body fat.

ESTROGEN - COMMERCIAL USE

In ruminants, estrogen also have a protein anabolic effect to increase body weight gain and growth. The possible mechanism for increased growth may be due to the ability of estrogens to stimulate the pituitary to release more growth hormone.

Diethyl stilbesterol – a synthetic nonsteroidal estrogen was formerly used for growth promotion.

Estrogen have been used to abort cows and sheep because of their luteolytic property.

Regression of CL

Treatment

Post partum metritis

Retained placenta

Mummified fetus

Pyometra

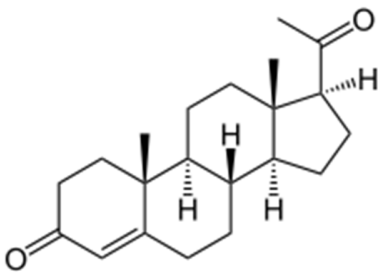
It is epitheliotropic and causes growth of soft skin. So used in ointments in skin ailments, produce cornification of vaginal epithelial in certain type of vaginitis.

PROGESTERONE

Source

Luteal cells of corpus luteum

Also produced from adrenal cortex and placenta.



Functions

Prepares the endometrium for implantation and maintenance of pregnancy by increasing the activity of the secretary glands in the endometrium and by inhibiting the motility of the myometrium. In CL-dependent animals (cow, goat and sow) CL-verum maintains the secretions of progesterone throughout gestation period, while in placental dependent animals (mare and ewes) placenta acts as an additional source of P4 secretion to maintain pregnancy by inhibiting the myometrial activity of oestrogen on uterus

Acts synergistically with estrogen to induce behavioural signs of estrus

By negative feedback mechanism. Progesterone inhibits FSH output and pulsatile LH release from the anterior pituitary. Thus prevents follicular maturation during luteal phase of the estrus cycle in the cycling animals. During pregnancy progesterone exhibits negative feedback inhibition on FSH release and LH surge to prevent ovulation.

Responsible for lobular and alveolar growth in the mammary gland

PEPTIDE FACTORS SECRETED FROM OVARY

Inhibin

Produced by Granulosa cells

It inhibits FSH release from AP without altering LH release

Activin

Present in the follicular fluid and rete testis fluid and have a stimulatory effect on FSH release

Relaxin

It has structural similarity with insulin

Source – Corpus luteum

Dilation of pelvic structure, cervix and vagina prior to parturition

Oocyte maturation inhibitor

Source- Granulosa cells of primordial and developing follicles

Maintains oocyte at dictyate stage of meiosis

Gonadotrophins

Produced by granulosa cells

GnRH like activity

Physiological role not clearly determined, but evidence indicates controlling steroid genesis by thecal cells

CHAPTER-24: PUBERTY

Learning objectives

At the end of this chapter, the learner will be able to understand the following

What is puberty and pubertal age in different species

Mechanism of onset of puberty

Factors influencing puberty.

PUBERTY – INTRODUCTION

Puberty is the age at which the male or female gonad becomes capable of releasing gametes.

In the female, this would be associated with estrus and ovulation. For the female, however, puberty is customarily defined as the age at which she will display the first estrus or heat. At puberty the female becomes sexually responsive to the stimuli and shows visual signs of estrus. Puberty is not a sudden event. It is the result of a gradual process of maturation of the reproductive system. At puberty the organs of reproduction increase in size. This increase in size is a gradual process and it has got a positive relationship with body growth.

In males release of first spermatozoa from the seminiferous tubules precedes by several weeks their reaching the vas deferens for ejaculation. At the onset of puberty, the gonocytes migrate to periphery of tubules and differentiate into spermatogonia.

In domestic animals it takes several weeks before sexual maturity is reached. Sexual maturity is the period when male or female individual is able to reproduce. (The first released gametes are not capable of fertilization). The period between puberty and maturity is known as adolescence.

MECHANISM OF ONSET OF PUBERTY

Puberty is basically the result of a gradual adjustment between increasing gonadotropic activity and the ability of the gonads to assume simultaneous steroidogenesis and gametogenesis.

During prepubertal period, hypothalamus does not secrete sufficient quantity of GnRH and the adenohypophysis is also refractive to GnRH.

As puberty approaches there will be a rise in both the amplitude and frequency of GnRH from the hypothalamus which causes increased responsiveness of adenohypophysis to GnRH. This stimulates gonadotropin release from adenohypophysis.

In the female from birth till puberty the hypothalamus is extremely sensitive to the negative feedback inhibition to even very small amounts of estrogen, which inhibits gonadotropin release from the adenohypophysis. This causes no follicular development.

Only at the age of puberty this inhibitory action of estrogen on hypothalamus is removed, which causes FSH and LH release from adenohypophysis.

Follicular growth begins and by a positive feedback effect of estrogen, the hypothalamus causes increased LHRH release and the LH surge from pituitary to effect ovulation.

Often, the first ovulation at puberty is not associated with behavioural estrus signs (*silent ovulation*) which is due to lack of progesterone (as there is no previous CL) to act in synchrony with estrogen for behavioural estrus

AGE AT PUBERTY (IN MONTHS)

Under normal breeding conditions puberty occurs at about 3 to 4 months of age in rabbits

In buffaloes, the age at puberty is 24 - 30 months in females and 15-18 months in males

Age at puberty of different domestic animals (in months)

	Cattle	Horse	Pig	Sheep	Goat	Dog	Cat
Male	9-10	12-18	5-8	6-8	5-9	5-12	9-12
Female	6-10	12-18	6-8	4-15	4-15	6-7	5-9

FACTORS AFFECTING PUBERTY

Many factors, including interaction with the opposite sex, adequate to high levels of nutrition, favorable climate, and lack of a stressful environment, favor the onset of puberty. Other factors, such as confinement of females, probably because of pheromones, under nutrition, and adverse climate and environment, discourage the onset of puberty.

Breed and Genetic Influences

In general, smaller breeds experience puberty at an earlier age. Bitches of small breeds frequently experience first estrus several months earlier than bitches of large breeds.

Jersey heifers have an average puberty age of 8 months, Holsteins, 11 months and Ayrshires, 13 months. Perhaps the selection for genes controlling breed size was concurrent with the selection of other genetic traits such as age at puberty.

If the animal is in good health and on good diet the age is related to body weight.

Climatic Effects

Puberty in man occurs earlier in the tropics than in the temperate zones. But good comparative data are not available in domestic animals. However, temperate climates, including the interaction of temperature, humidity, diurnal variation and daylight, favor early puberty in all animals.

Seasonal Effects

An unusual situation exists in the seasonal breeding sheep, since age at puberty can be overridden to some extent by the occurrence of the breeding season. If the hypothalamo-pituitary-ovarian axis is sufficiently developed, then puberty can be initiated at an early date. For example, ewe lambs born early in the spring may show first estrus during fall when they are only 180 days of age. But ewe lambs born late in the spring or the early summer may not show first estrus until the breeding season in the fall of the following year when these females have reached 400 to 500 days of age.

Effect of Nutrition

A high plane of nutrition favors an earlier puberty, and a low plane of nutrition delays puberty. This is particularly true for non seasonal breeding animals. Apparently there is an interaction of nutrition, body weight gain, and age, since animals maintained under good nutrition reach puberty at an earlier age. But poor nutrition is not able to prevent the eventual onset of puberty, although severe delays can be caused to the extent that the age of puberty can be doubled.

Effect of Sex

In general females of all species reach puberty at an earlier age than males However, evidence to the contrary is also available. For instance, more bull calves, lambs, goats, dogs, and likely males from other species as well, produce ejaculates containing spermatozoa, and therefore have reached puberty, at an earlier age than the majority of females display the signs of the first overt estrus.

Disease, Hygiene and cleanliness

Poor health and unhygienic conditions are unfavourable for onset of puberty.

Exposure to opposite sex

Favours the early onset of puberty. This technique is used at organized farms.

CHAPTER-25: FERTILIZATION

Learning objectives

At the end of this chapter, the learner will be able to understand the following

How the spermatozoa reach the site of fertilization in the female reproductive tract

Changes takes place in spermatozoa before it penetrates oocyte

How spermatozoa penetrates the oocyte

Union of male and female gamete

Block to polyspermy.

SPERM TRANSPORT

Semen is deposited in vagina in cattle, sheep, primates and it is deposited in the uterus in the horse and in cervix in case of swine. There are 2 phases of transport.

Rapid phase

Prolonged phase

Rapid phase

The rapid phase of transport results in spermatazoa being present in oviducts within after insemination due to

Negative uterine pressure.

Contraction of uterus within and oviduct due to prostaglandins in the seminal plasma in male and oxytocin – due to copulatory stimulation in female.

Transport in oviduct depends on motility of cilia and segmental contraction of the fallopian tubes.

Prolonged phase

Spermatozoa destined to be involved in fertilization are transported slowly and are stored in parts of the tubular genitalia. To animals (eg. ruminants) that deposit semen in the vagina, the cervix serves as a important role as a sperm reservoir, from which spermatozoa are transported to the uterus and oviducts over a period of hours. The spermatozoa disappear from other areas of the genital tract because of phagocytosis by the leucocytes/macrophages, many are trappers in cervix other sperm reservoirs are located in the oviduct near the uterotubal junction.

Spermatozoa pass from the oviductal storage areas into the oviducts, where their numbers are maintained at relatively constant levels.

The passage or transport of spermatozoa from the vagina through the cervix and uterus and into the oviduct is very rapid less than 15 minutes, in all domestic animals.

Times taken for sperm to reach the oviduct

Bitch - 20 min

Mare - 15 min

Ewe - 8 min

Cow - 2-4 min

Spermatozoa retain their fertilizing capacity in the female tract for

Cow, ewe and sow- 24-48 hrs

Bitch - 90 hrs

Mare -5 days

The cervix is considered to be an ideal site for survival of spermatozoa. In the vaginal passage sperm do not survive more than 1-2 hours.

The fertilizing life span of spermatozoa is twice as long as that of ova.

Spermatozoa require a few hours in the female tract to gain the ability to fertilize.

Thus spermatozoa are waiting in large number to ensure the fertilization of ova or an ovum immediately after ovulation.

Billions of spermatozoa are usually ejaculated into the female genital tract of animals at coitus.

In mare and sow most of these promptly enter the uterus. In other species only about 10 or 100 million spermatozoa pass through the cervix into the uterus.

100 - 1000 spermatozoa: ampulla of the oviduct

10 - 100 spermatozoa: vicinity of the unfertilized ovum.

Normally only one spermatozoon is needed for fertilization, but without a total optimum concentration of 10 million sperm/ml in AI, the conception rate declines.

CAPACITATION OF THE SPERMATOZOA

Ejaculated spermatozoa in most species are incapable of penetrating the ovum immediately. Spermatozoa must undergo some physicochemical changes after they are deposited in the female reproductive tract in order to acquire the ability to penetrate the zona pellucida and fertilize the ova. This process is referred to as capacitation. This requires 1.5 h in sheep, 3-6 h in pigs.

Capacitation begins in the uterus and is completed in the oviduct.

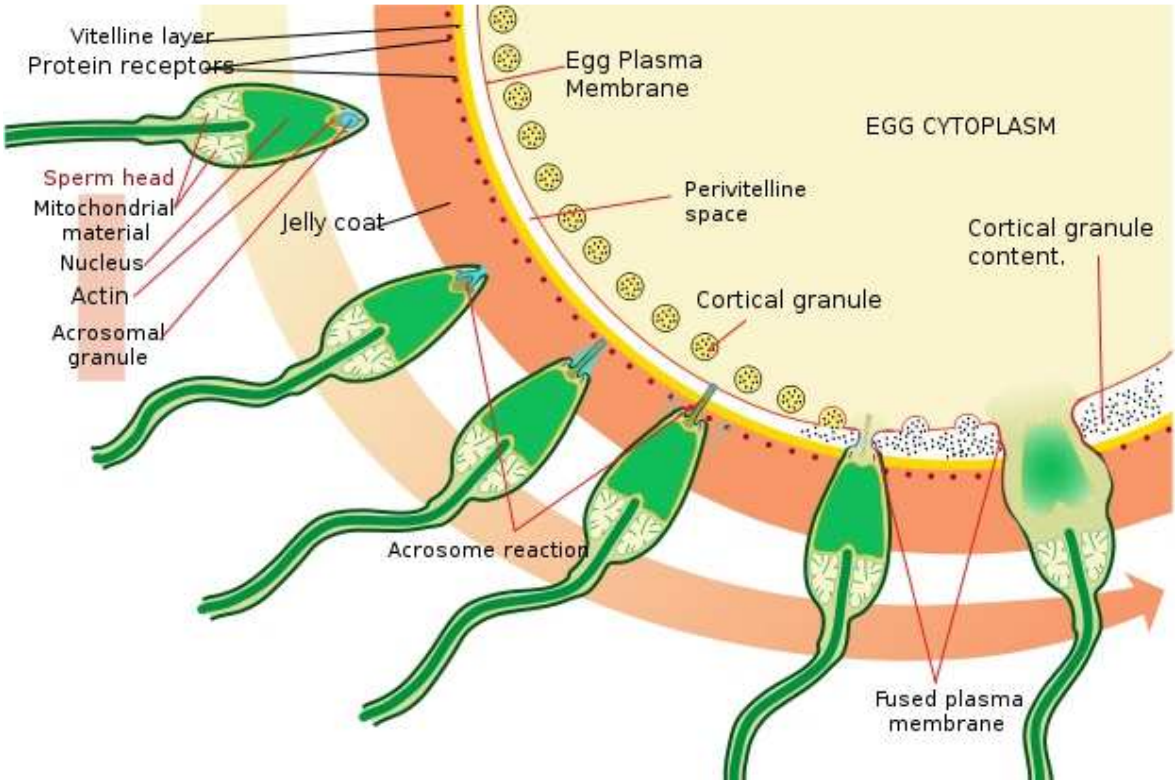
During capacitation

Glycoproteins originating from seminal plasma or epididymal fluid are removed.

Hydrolytic enzymes are activated in the acrosome. The actual process involves a partial breakdown and fusion of the outer acrosome and plasma membranes to form ports through which the acrosomal enzymes hyaluronidase and the trypsin like acrosin escape. These enzymes permit the sperm cell to penetrate the cumulus cell mass around the ovum by depolymerising the hyaluronic acid protein matrix of the cumulus.

i.e. before the sperm cell can penetrate the zona pellucida it loses the acrosome cap and plasma membrane exposing the perforatorium and inner acrosomal membrane which are responsible for penetration through the zona pellucida. These lytic enzymes such as trypsin and hyaluronidase permit the sperm cell to to make oblique tunnels through the proteinaceous zona into the perivitelline space. When the plasma membranes are in contact, the two structures unite and the entire sperm passes into the cytoplasm of the egg.

Capacitation also increases the metabolic activities of the sperm cell by increasing the rate of glycolysis and oxidative mechanism.



OVUM TRANSPORT AND SPERM ATTACHMENT

The fimbriated end of the infundibulum picks up the ovum and the ovum is transported down the uterine tubes into the uterus by the combined actions of the cilia on the mucosal surface of the epithelial cells and contractions of the uterine tubes. The contractions in turn are dependent on.

Ratio of hormones, estrogens and progesterone (high level of estrogens increase the contraction).

The level of prostaglandins.

The degree of stimulation of the uterus by the sympathetic division of ANS.

Immediately following ovulation, the ovum with the vitelline membrane (Cell membrane of the ovum) is surrounded by a thin zona pellucida and by a variable amount of granulosa cells cumulus oophorus. A well defined corona radiata is seen in the bitch, but absent in cow, ewe, sow and mare.

The rate of transport of ova through the ampulla portion of the oviduct to ampullary-isthmus junction is quite rapid and occurs within few minutes (8 to 30 min).

The cilia in the rabbit oviduct beat at a rate 1500 per minute.

The cumulus and corona cells around the zona pellucida of the ovum in most domestic animals only persists for a few hours or less after ovulation and fall away in the oviduct.

In the dog and cat the corona cells may persist for several days. The protoplasmic projections from these cells into the zona pellucida are withdrawn soon after oviduct ion due to contact with fibrinolytic enzymes in the oviductal fluid. The cumulus and corona cells die and separate the ovum aided by the ciliary and muscular activity of the oviduct.

Inside the zona pellucida is the vitelline membrane or vitellus which is similar to the plasma membrane of the somatic cells. The zona pellucida is a homogenous semipermeable structure surrounding the ovum composed of conjugated protein capable of being dissolved by proteolytic enzymes such as trypsin and chymotrypsin. After fertilization the vitellus and a perivitelline space is formed between the zona pellucida and the vitelline membrane into which the polar bodies are extruded.

The fertile life of ovulated ova in domestic animals is relatively short, 12 – 24 hours except in bitch (4 – 8 days)

Cow = 20 - 24 hours

Horse = 6 – 8 hours

Sheep = 16 – 24 hours

Swine = 8 – 10 hours

In cow, ewe and the sow the ovum is released as secondary oocyte, the first polar body has been formed and metaphase of the second maturational division. The second maturational division occurs only after fertilization by spermatozoa.

In the horse, dog and fox oocytes ovulate as primary oocyte ie before the formation of the first polar body and then mature in the uterine tube.

Sperm attachment

The attachment of sperm head to the zona pellucida is regulated by receptor sites on the zona surface. The mature oocyte synthesis a glycoprotein ZP_3, which functions as the sperm receptor to which the sperm with intact acrosome can bind. The presence of glycosyl transferases proteinases and glycosidases on the plasma membrane covering help the sperm head to bind with ZP_3.

FERTILIZATION

Fertilization is the fusion of the male and female gametes to form a one single cell. i.e, to have diploid number of chromosome. Fertilization is the union of viable ovum and sperm. It takes place in the ampullary isthmus junction of the oviduct.

At the time of fertilization, the ovum consists of a nucleus surrounded by a plasma (vitelline) membrane. The ovum is covered with a mucoprotein coat called the zona pellucida. The

special granulosa cells (cumulus oophorus) that surround the zona pellucida are usually lost after ovulation.

Following capacitization and contact with the ovum, when the spermatozoa penetrates the zona pellucida, zona reaction occurs. This protective reaction prevents the penetration of the vitelline membrane by other sperms. This penetration is accomplished by the acrosomal enzymes, hyaluronidase and acrosin and trypsin like enzyme break down the hyaluronic acid – protein matrix of zona pellucida.

The spermatozoa upon penetrating the zona attaches to vitelline membrane and stimulates the second meiosis in the ovum to form the second polar body and the ovum forms the female pronucleus. Once the spermatozoa penetrates the ovum its head enlarges and converts into the male pronucleus.

Both male and female pronuclei enlarge many times before they move towards each other.

The gamete membrane come in contact with each other and fuse incorporating sperm and egg nuclei into a single cell. This process is known as syngamy.

BLOCK OF POLYSPERMY

Immediately following fertilization, the ovum surface changes and prevents further fusion with the additional spermatozoa.

The block to polyspermy (entry of multiple spermatozoa) is at zona pellucida in sheep and swine. With a secondary physiologic block at the vitelline membrane in some species. (eg. rabbit).

Initiation of the block occurs at the point of sperm penetration into the ovum where the central granules are released into the perivitelline space which causes excessive reorganization of the zona pellucida and vitelline surface and this is referred to as the zona or cortical reaction.

This causes the release of enzymes to harden the zona pellucida and thus inactivate sperm receptor (ZP_3).

Proteolysis of female ZP_3 and ZP_2 also alters, physical characteristics of the zona to prevent further penetration of accessory sperm.

CHAPTER-26: PREGNANCY

Learning objectives

At the end of this chapter, the learner will be able to understand the following

How the zygote develops into embryo

How the mother recognizes the presence of embryo in uterus

How embryo gets implanted in the uterus

Gestation length in different species

Hormonal control of gestation.

DEVELOPMENT OF EMBRYO

The mammalian ovum, one of the largest cells of the body after fertilization forms the zygote. By a series of mitotic divisions (Cleavage division) the zygote becomes a two cell, 4 cell, 8 cell and 16 cell embryo and so on. Finally a solid ball of cells called the *morula* is formed.

Most species form a morula at approximately 16 to 32 cells. Morula having a fluid filled cavity (blastocoele) is referred to as *blastocyst*.

The zygote remains in the oviduct for 3-4 days after fertilization.

MATERNAL RECOGNITION OF PREGNANCY

At appropriate time, the conceptus must produce some substances (protein/ steroid) to signal its presence to the mother.

This signal is important for

Inhibition of $PGF_{2\alpha}$ production

Maintenance of CL

Production of Progesterone

Endometrial development and secretary activity

This critical period when the conceptus gives signal is called maternal recognition of pregnancy.

Cow

16 – 19 day of pregnancy conceptus secretes bovine trophoblast protein bTP-1

It makes the endometrium unresponsive to estrogen

Inhibits oxytocin induced PGF2α production

Endometrium produces endometrial PG synthetase inhibitor (EPGS) decreases $PGF_{2\alpha}$ production

Conceptus moves 12 – 14 times/ day between 12 – 14 day of gestation to inhibit $PGF_{2\alpha}$ production

Sheep

Secretary proteins inhibits PGF2α production

Among secretary protein OTP -1 or Type I conceptus interferon

Stabilizes P_4 receptor

Prevents activity of estrogen receptor and formation of oxytocin receptor

Stimulates activity of enzyme – Ketoreductase which converts PGF2α in to PGE2

Pigs

Estrogen of conceptus gives signal. This occur in two phases 11 – 12 day of gestation and 16 – 30 day of gestation

Estrogen not inhibiting PGF2α production. But secretion of PGF2α in to capillaries in endocrine manner is blocked.

Conceptus must be in both the uterine horns and 2 conceptus per horn is essential for pregnancy to continue. If not, PGF2α produced from non gravid horn causes lysis of CL of both ovaries.

Bitch and Queen

Life span of CL of non pregnant bitch greater than pregnant bitch.Hence does not need any mechanism to extend luteal phase.

In queen, the act of inducing the ovulation extend the lifespan of CL to 36 days which is the duration of pseudopregnancy (pregnancy is 65 days). After 36 days of pregnancy some unknown factor lengthens the lifespan of CL

Primates

Secretion of hCG in human and mCG in monkey within 1-2 days after implantation shows luteotropic activity

IMPLANTATION

Implantation is the process whereby the new embryo becomes established at a developmental site on the endometrium to develop into a fetus.

Following fertilization, the new embryo reaches the uterus 4 days post-estrus in cows and ewes. It is slightly sooner in sows and slightly latter in bitches

Estrogen influences the motility of the oviducts and the passage of zygote from the oviduct into the uterus. The rate of transport is controlled by ratio of estrogen and progesterone.

Before the passage of the embryos into the uterus, progesterone from the developing CL has been preparing the uterus to retain and nurses the embryos. Progesterone decreases the muscular activity and tonicity of the uterus and promotes the development of a glandular epithelium, which is responsible for producing uterine milk.

The nutritive requirements of the embryo are supplied by the yolk material and secretions of the oviducts and uterus

The embryo remains free-floating in the uterine fluid after its entry into the uterus until implantation and derives its nutrients from the endometrial secretions.

The rate of transport through oviduct is controlled by the estrogen and progesterone ratio.

After its entry into the uterus the embryo remains free-floating in the uterine fluid until implantation and derives its nutrients from the endometrial secretions.

The embryo continues to grow by mitotic division and the cells become arranged to one side (known as *compaction*) and fluid accumulates in the central cavity known as *blastocoele and the conceptus is called blastocyst*

The blastocyst has two cell types –

an outer layer of cells called *trophoblast* which later forms the placenta and

group of cells surrounded by trophoblast known as *inner cell mass* (ICM) which develops into foetus

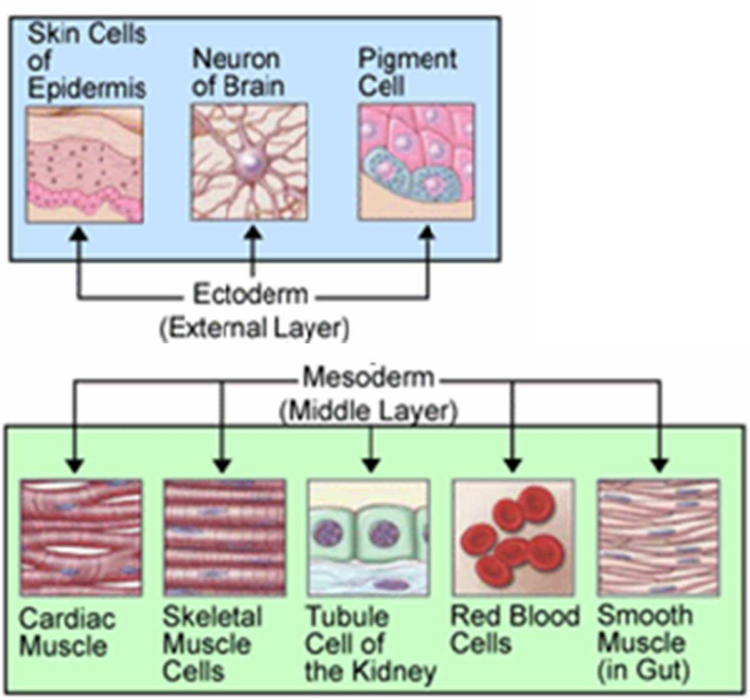
When the blastocyst reaches the uterus, it is still covered by zona pellucida. In 6-11 days after fertilization, the zona pellucida breaks down and release the embryo which is referred to as *"zona hatching". Zona hatching is contributed by blastocyst expansion and contraction and aided by lytic enzymes*

Zona hatching – cattle 9-11 days, horse-8; sheep-7-8 and swine in 6 days after fertilization

The blastocyst elongates because of formation and hyperplasia of extra embryonic membranes

Blastocyst elongation occurs in cows, sheep, and pigs, but not in horses

Concurrent with blastocyst elongation, gastrulation (ICM differentiate into three distinct layers – ectoderm, mesoderm and endoderm) occurs in the embryo



Spacing of embryo in the uterine horn in polytocous animals occurs during this period.

In litter bearing animals, transuterine migration of zygote helps to distribute the embryos which are affected by uterine muscle contraction in the presence of embryos. The spacing of embryos is important for the normal foetal development.

Species	Days of implantation
Cow	30 – 35 days
Doe	20 - 25 days
Ewe	15 – 18 days
Sow	14 - 20 days
Bitch	15 days
Queen	13 days
Primates	6 - 8 days

TYPES OF IMPLANTATION

There are two types of implantation

Invasive

Non-invasive

Invasive

Eg. Primates and rodents.

Blastocyst penetrate the uterine epithelium. This is accomplished by proliferation of uterine epithelium in the viscinity of blastocyst (Decidualization)

Non – invasive

Not penetrating and is superficial

It involves apposition and adhesion of trophoblast and uterine epithelium or interdigitation of trophoblastic villi and uterine epithelium.

Ruminants

Implantation occurs – specialized endometrial protrusion (caruncles)

Horse and pig

Trophoblastic villi interdigitate with uterine epithelium except at opening of uterine glands

But in horse, on day 38 band like structure on the chorion (chorionic girdle) detach themselves, and penetrate endometrium to form endometrial cup.

Dog and cat

Less invasive

PERIODS OF PREGNANCY

Prenatal life of an individual is divided into 3 periods

Period of ovum

Period of embryo

Period of fetus

Period of ovum

It is the period between fertilization to blastocyst formation.

It lasts until the embryo makes loose attachment in the endometrium. Embryo until 4-5 days in most species lies in fallopian tube and nourished by secretions of fallopian tube and small amount of nutrient carried by gamete. After 4-5 days, zygote reaches uterotubal junction and moves into uterus.

In uterus, uterine milk nourishes the embryo.

Period of embryo

It lasts from blastocyst development to organogenesis and development of extra embryonic membranes.

Implantation occurs in this period.

During early part of this period, it derives nutrition from uterine fluids. After that derives from placenta.

Period of fetus

It is the time during which most part of the placenta and fetus occurs and lasts until parturition.

Of the three periods, the period of fetus is longest. Most critical period is period of embryo as most embryonic death occurs in this period.

GESTATION LENGTH

Species	No. of days
Cow	280 days
Buffalo	310 days
Sow	114 days
Mare	340 days
Sheep and goat	150 days
Dog	60 – 63 days
Cat	56 - 65 days

FACTORS INFLUENCING GESTATION LENGTH

Maternal factor

Age of dam – young animals have slightly shorter gestation length than older ones

Fetal factor

Litter size

Monotoccus animals have longer gestation length than polytoccus animals

In monotoccus species, if twin fetuses, gestation length is slightly shorter (5-6 days)

Sex

Male fetus will be carried few days (1-2 days) longer than female fetus

Endocrine factors

Anterior pituitary and adrenal hormones of fetus initiates parturition at end of gestation

Genetic factor

Species –larger size species longer will be the gestation length

Breed

Minor variant due to season, mating time and feeding time

Difference in mutton and wool breed of sheep

Fetal genotype

Prolonged gestation length in dairy cows carrying fetus homozygous for autosomal recessive gene

Mare X Stallion – (320 – 360 days)

Mare X Jack donkey – Mule – 360 – 380 days

Jenny donkey X Stallion – Hinny – Shorter gestation length than horse

Environment

Nutrition – Prolonged deficiency will influence gestation length. Fed animals have slightly shorter than on maintenance ration

Season – If conceived in late summer/ autumn, Shorter gestation length than those conceived in early spring (foals)

HORMONAL CONTROL OF GESTATION

Gestation and parturition – Entirely under hormonal control

Nervous control of uterus is not essential during gestation

- In farm animals
  - 12 – 16 days after estrum/ coitus, presence of embryo/ conceptus causes
    - Persistence of CL
    - Cessation of estrous cycle
  - Due to
    - Continuous release of luteotropin from anterior pituitary
    - Prevent release/ formation of PGF2α
- Progesterone from CL/ placenta essential for
  - Growth of endometrial gland and uterine milk secretion
  - Endometrial growth
  - Attachment of placenta
  - Inhibits uterine contraction
- Certain amount of estrogen is essential to
  - Enhance effect of progesterone
  - In later pregnancy
    - Udder development
    - Relaxation of pelvic structures
    - Cervical relaxation
    - Sensitize the uterus to oxytocin
- LH from anterior pituitary to maintain corpus luteum
- Fetal pituitary and adrenal hormones initiates parturition at the end of gestation

CHAPTER-27: PLACENTA

Learning objectives

At the end of this chapter, the learner will be able to understand the following

What is placenta

How placenta can be classified

What are the functions of placenta

Hormones produced by placenta and its function.

PLACENTA - AN INTRODUCTION

As the embryo increases in size, the extra embryonic or fetal membrane of placenta develops to meet the increasing need for more nutrients to nourish the developing embryo. It also acts as an organ of respiration and excretion.

The trophoblastic cells of embryo participate in the formation of placenta. The outer single layer of cells of blastocyst referred to as the trophoectoderm, contributes to the outer layers of placenta. Inside the trophoectoderm is a group of cells called the inner cell mass which forms the embryo proper also a portion of placenta

The fetal membranes consists of

Primitive yolk sac

The amnion

The allantois

chorion, which when combined with the allantois form the chorio allantois

The Chorion

The chorion is the outermost membrane and is in contact with maternal uterine endometrium.

The chorioallantois or allontochorion is formed by the fusion of outer layer of the vascular allantois and chorion. This structure richly supplied with blood vessels communicating with the endometrium carries metabolic interchanges of gases, nutrients and wastes between the fetal and maternal circulations.

The allantois

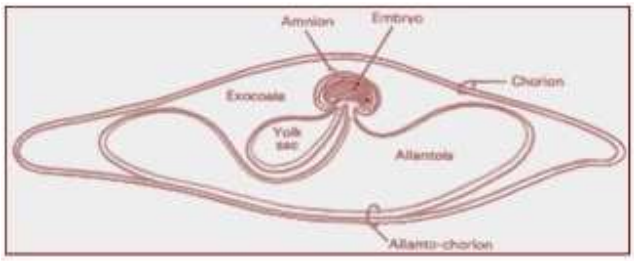
The allantois arises in the second and third week of gestation in bovine fetuses the allantoic cavity stores the waste products of the fetal kidneys. The allantoic sac a fluid filled space formed by 2 layers of allantois is located between the amnion and chorion and is called the first water bag.

The amniotic sac

The amnion forms about 13-16 days after conception in sheep, cow. It is a double walled sac the completely surrounds the fetus except at the umbilical ring. The amniotic cavity plays a role in the care of some waste products. The amniotic sac is also a fluid filled allantoic sac is expelled first then the amniotic sac.

Yolk sac

It is a primitive structure developing early in the embryonic period and dispersing after a short period of time in ruminants and swine but persists for 4-6 weeks in horse.



CLASSIFICATION OF PLACENTA

Based on the number of layers between the dam and the fetus separating the two bloods.

Based on chorionic villi pattern and their shape of attachment to the uterus.

Based on loss of maternal tissues occurring at birth of the fetus

BASED ON THE NUMBER OF LAYERS BETWEEN THE DAM AND THE FETUS

The fetus and the maternal blood are separated by six layers of tissues

Maternal tissues

Endothelium of blood vessels

Connective tissue

Epithelium of uterus

Fetal structures

Chorion (trophoblast)

Connective tissues

Endothelium of the fetus

These form the six layered barrier to separate the maternal blood from fetal blood

Based on the number layers that separate the maternal and fetal blood, placenta can be grouped into the following types

Epitheliochorial

It is the simplest form of placentaion, it consist of all the six layers. Epithelium of the uterus has direct contact with the chorion of the fetal placenta

Eg: Sow, Mare, Cow and donkey

Mother- Endotheilum, Connective tissue, epithelium

Fetus – Epithelium, Connective tissue, Endothelium

Syndesmochorial

It involves the loss of one of six layers - the epithelial lining of the uterus. The fetal structure, the chorion forms a direct contact with the connective tissue and the endothelium of the carnuncles of the dam.

Eg: Ewe and doe

Mother- Endothelium, Connective tissue

Fetus - Epithelium, Connective tissue, Endothelium

Endotheliochorial

Two layers, the connective tissue and the epithelium of the uterus are lost, while the endothelium of blood vessels of the dam directly attaches with the fetal structure.

Eg. Bitch and Queen

Mother- Endothelium

Fetus - Epithelium, Connective tissue, Endothelium

Hemochorial

All the three layers of the maternal tissues are lost, only the three of the fetal structures are left; the chorion (trophoblast), is bathed by the blood of the dam

Eg: Monkey, Women

Mother – blood

Fetus - Epithelium, Connective tissue, Endothelium

Hemoendothelial

Only one tissue layer, the endothelial layer of the fetal capillaries separates the maternal and fetal blood systems

Eg: Guinea pig, rabbit and rat

Mother- Blood

Fetus - Endothelium

BASED ON THE LOSS OF MATERNAL TISSUES OCCURRING AT BIRTH OF THE FETUS

Deciduate

In this type, the deciduate composed of portions of maternal epithelium, submucosa, deciduate cells and the fetal placenta are shed at parturition leaving a portion of the endothelium denuded.

In bitches and queens, there is moderate loss of maternal tissues at birth of the fetus

In monkey and women, there may be extensive loss of maternal tissue during parturition.

Non- deciduate type

There is no loss of maternal tissue at the birth of the fetus (sow,mare,ewe,doe and cow)

In this type, placenta and fetel membranes are expelled at the time of parturition leaving the endometrium intact except in ruminants in which only the surface of the caruncles are devoid of epithelium after the caruncle sloughs about 6-10 days following parturition.

BASED ON CHORIONIC VILLI PATTERN AND THEIR SHAPE OF ATTACHMENT TO THE UTERUS

Diffuse

Chorionic villi cover most of the fetal placenta, projects into the crypts, scattered over the entire endometrium of the uterus except at the openings of the uterine glands.

Eg: Sow, mare

Discoid or Spherical

The placenta gets attached to the uterus in a disk- shaped area only

Eg: Guinea pigs, Rabbit and rat

Zonary

The chorion of the fetal placenta is attached in a grill like band with the uterus of the dam.

Eg: Bitch and Queen

The placenta forms a zonary band in the uterine lumen in the middle of the oval sac.

Cotyledonary

Here the attachment of the portion of the chorion of the fetus (Cotyledons) with the uterine epithelium of the dam in mushroom like area is called as the caruncles.

Eg: Ewe, Doe, Cow and Buffalo

Endometrial caruncle and fetal cotyledon together constitutes Placentomes

There are 75-120 placentomes in the pregnant cow and 80- 90 placentomes in the pregnant ewe.

FUNCTIONS OF PLACENTA

The blood of the fetus and dam never come into direct contact but the two circulations are close together for transfer of substances.

Placenta aids in the transport of nutrients such as carbohydrate, protein, fatty acids, glycerol, water, inorganic salts but lower concentration of vitamin A, D and E from the mother to fetus. Fructose comprises 70-80% of sugar in fetal blood and it is produced by the placenta from glucose. Its function in fetus is not known.

It helps gaseous exchange between maternal and fetal blood

Functions as an excretory organ to transfer the excretory products of the fetus to the maternal blood.

Placenta is a temporary endocrine organ and produces progesterone, placental lactogen, gonadotropin like hormones (eCG and hCG) and at the end of gestation period relaxin

Helps in maintenance of pregnancy

Participates in parturition.

PLACENTAL HORMONES

Placenta is a transient endocrine organ like corpus luteum

It seems to be able to produce hormones either identical to or with similar biological activity to almost any of the known hormones of mammalian system

Placental hormones are

Progesterone

The farm animals can be classified according to the source of progesterone during the latter half of gestation as

placenta dependent – mare, ewe and woman

CL- dependent - cow, doe, sow, bitch and queen

Key hormone in maintenance of pregnancy

Animals in which placental progesterone is low, CL is essential throughout the pregnancy

In the placental- dependent animals, the progesterone production does not require LH

In sheep and primates, luteal production of progesterone is continues throughout the pregnancy, even though placental production of progesterone is dominant.

In mare, CL regression happens at mid gestation. Therefore, placental progesterone is solely responsible for pregnancy by about 150 days of gestation.

Estrogen

Major steroid in addition to progesterone is estrogen

Major estrogen during pregnancy

Primates - Estriol

Domestic animals - Estrone

For pregnancy diagnosis estrogen conjugate in blood/ urine by 40th day of gestation in mare (Cuboni's test)

PMSG or eCG

It was discovered when blood from pregnant mare produced sexual maturity in immature rats.

It is a glycoprotein hormone with α and β subunits, similar to FSH and LH. But with high sialic acid (longer half life)

This placental gonodotropin is secreted by endometrial cups of equine uterus

Secretion from 40th day of gestation to 20th week. Maximum 70 – 85th day

It stimulates follicular development (due to FSH effect)

Some of the follicles ovulates, but most follicles gets lutinized due to LH like action. These additional corpora lutea produce progesterone for maintenance of pregnancy.

Thus eCG has both FSH and LH activity but FSH is dominant

Present in blood. Not found in urine

Used to induce superovulation in domestic animals and follicular development in anestrus animals

Human Chorionic Gonadotropin (hCG)

Glycoprotein with α and β subunits

Has high content of sialic acid

Source – Trophoblastic cells of chorionic villi of placenta of women and primates

Found in both and urine

Detected in urine as soon as 8 days after conception by sensitive RIA.

Has LH and FSH action. But predominantly LH action

Used in the treatment of cystic ovary and induction of ovulation.

Since it appears early in human pregnancy, is detected in the urine under the basis of human pregnancy test.

Placental lactogen

It is a protein hormone with property similar to prolactin and Growth hormone

Demonstrated in Cow, Sheep, Doe and Women

Its level is greatest during last trimester of pregnancy when rapid growth of fetus occurs

Important regulator of maternal nutrients to available to the fetus

GH action

Epiphyseal growth

Protein synthesis/ weight gain

Increase lipolysis

Prolactin

Casein synthesis

Mammary alveolar development

Milk secretion

Its level is higher in dairy cows than in beef cows

Protein B

Half life – 7 days (due to its carbohydrate content)

Source – ovine and bovine placenta

Detected in blood of pregnant cow as soon as 22 days after conception

Not found in milk/ urine

Prevents destruction of CL

First reliable hormonal pregnancy test for the cattle

Relaxin

Polypeptide hormone. α and β subunits connected by 2 disulfide bonds

Structure similarity with insulin

Source – CL, placenta

In mare – from 70 days

Queen – 20 days

Bitch – 20 days

It helps to soften and dilate the cervix

Relaxes the sacroiliac joints in cows and ewes

Widens the birth canal at parturition

CHAPTER-28: PREGNANCY DIAGNOSIS

Learning objectives

This chapter explains about how the pregnancy can be diagnosed by different methods in various species of animals.

PREGNANCY DIAGNOSIS - AN INTRODUCTION

Pregnancy diagnosis is of considerable economic value.

Early diagnosis of pregnancy is required soon after mating for early identification of non pregnant animals so that loss in production time due to infertility may be reduced by treatment.

Pregnancy diagnosis is also required to certify animals for sale, insurance.

To help economic management of animal production.

There are two methods of diagnosis

Clinical method

Laboratory method.

CLINICAL METHODS OF PREGNANCY DIAGNOSIS

It depends on the detection of the conceptus - fetus, fetal membranes and fluids.

Rectal Examination

Accepted method of pregnancy diagnosis in mare, buffalo and cow.

The uterus is palpated through the rectal wall to detect uterine enlargement of pregnancy and the fetus or fetal membranes.

The method can be performed at early stages of pregnancy,

It is accurate and the result is known immediately.

This method is not suited for ewe and sow due to small pelvic cavity. However, fremitus can be felt by palpation through rectal wall, in one or both uterine arteries in sows from 28 days and it is a simple and rapid test.

Radiography

Useful to detect pregnancy in sheep, goat and swine

Based on the identification of fetal skeleton on the X-ray

Can be applied only during the last trimester of gestation, costly, posses radiation hazard and requires restraint of the animal.

Ultrasonography

Ultrasound waves operate at frequencies of 1 to 10 MHz and are inaudible to human ear.

Two types of ultrasound are employed

Doppler phenomenon

Pulse-echo principle

Doppler Phenomenon

In this method, sound waves striking a moving object are reflected to the transmitting source at a slightly altered frequency.

The transducer (probe) is applied to the animal's abdominal wall or inserted into rectum; the transducer emits high frequency waves which are reflected back to the probe with altered frequency by the fetal heart or fetal blood flow;

The reflected waves are converted to audible sound and amplified through speakers or illuminated through oscilloscope.

Pulse-Echo Ultrasound

Pulses of ultrasound generated by piezoelectric crystals in a transducer, when come into contact with tissues of varying impedance, are reflected (echoed) to the transducer; then they are converted to electrical energy and displayed on a cathode ray oscilloscope in two forms

A-mode and

B-mode are currently used.

A-mode (amplitude) is one-dimensional display of echo amplitude versus distance

B-mode produces an accurate two-dimensional image of soft tissues

The ultrasound technique can used to diagnose pregnancy in farm animals.

Species	Earliest day after mating (days)	Accuracy of detecting pregnancy (%)
Horse	9	100
Cattle	20	100
Buffalo	30	-
Sheep and goat	45 – 50	100
Pigs	60	70 – 90

PREGNANCY DIAGNOSIS - LABORATORY METHODS

It depends on detecting changes occurring in the maternal tissues bearing a conceptus or substances produced by the conceptus

Vaginal Biopsy

The vaginal mucosa undergoes changes due endocrine influence during pregnancy with reduction in number of layers. This can be detected by histological techniques.

This method involves sampling, processing and microscopic examinations, which are time-consuming and costly

This method has limited practical applicability.

Immunological Diagnosis

This technique depends on detecting or measuring substances arising in the conceptus, uterus or ovaries and enter the maternal blood, urine or milk.

Pregnancy Associated Substances

Several substances appear in the maternal blood during pregnancy and can be used as indicators of pregnancy.

Early Pregnancy Factor

Pregnancy-Associated Antigens

Hormones

Early Pregnancy Factor

The EPF has been identified in woman, sow, ewe and cow.

It can be detected by the rosette-inhibition test.

Pregnancy can be detected within few days after conception.

It is time-consuming and has limited use in farm animals.

Pregnancy-Associated Antigens

Antigens specific to pregnancy have been reported in sheep, cattle and horses.

They are detected at the later half of pregnancy and have limited value in animals.

E.g. of pregnancy associated antigens–protein-B appear from day 24 of pregnancy in cows

Placental lactogen can be detected at day 16 in ewe, between day 17 and 25 in cows.

HORMONES

Progesterone assay

Most widely used method of pregnancy diagnosis in farm animals.

Progesterone can be detected in blood and milk.

Its level declines in non-pregnant animals. Samples are collected at 22 to 23 days in cows, 17 to 18 days in ewes and 21 days in sows.

P4 level at this time is low in non-pregnant animals and elevated in pregnant animals.

Milk can be used for detecting P4.

These methods are accurate but expensive,

Require laboratory facilities and results can be known after some days.

One step qualitative "cow-side" test kits are available which overcome these shortcomings.

Elevated levels of progesterone do not necessarily signify pregnancy.

In longer than normal oestrous cycle length, progesterone level would be high on the day of sampling (22 days) indicating "false positive" result.

In early embryonic mortality, false positive result may be obtained.

Accuracy of predicting pregnancy with progesterone assay ranges from 70 to 90% and the accuracy of detecting nonpregnancy is 100%, because a cow with low progesterone is not pregnant. Hence, blood/milk progesterone assay is more reliable to diagnose nonpregnancy at a much earlier time.

Oestrone Sulphate

It is the major estrogen produced by the conceptus

Can be measured in the maternal blood, milk or urine in all farm animal species

Estrone sulphate is detectable in the plasma earlier in sow (day 20), and mare (day 40), than in sheep and goat (day 40 to 50) and cow (day 72).

In pigs, estrone sulphate test has more advantages than other methods.

It is more accurate in earlier pregnancy than ultrasonagraphy.

Gonadotrophins

eCG appears in the blood of mares at 40 days of conception, reaches a peak between 50 and 120 d and then gradually declines and pregnancy can be diagnosed using eCG.

CHAPTER-29: PARTURITION

Learning objectives

At the end of this chapter, the learner will be able to understand the following

What is parturition

How parturition is initiated by fetus and completed by the mother

Stages of parturition

Role of hormones in parturition process.

PARTURITION - AN INTRODUCTION

Parturition is a physiological process in which the pregnant uterus delivers fetus and placenta. It is climax of great drama of pregnancy.

Symptoms of approaching parturition

If breeding time is known–prediction of exact time of delivery is possible

Full-term animals segregate themselves from others

Nesting behavior in dog, cat and sow (polytoccus animals)

Relaxation of pelvic ligaments – sinking of croup muscles

Elevation of tail

Vulva edematous and flaccid

Udder – enlarged and edematous

Heifers – from 4th month

Pleuriparous animals – 2-4 weeks before parturition

Swollen teats – Secretions may come out through teat orifice

In mare

Sinking of croup muscles not so profound

Waxing of teat (6-48 hrs before parturition)

In sheep and goat – Udder development is not so predominant

Bitch, so and cat – calm, quite before parturition

Anorectic

Restlessness

MECHANISM OF PARTURITION

Successful parturition depends on Myometrial contraction and Capacity of cervix to dilate

Myometrial contraction

Progesterone blocks myometrial contraction during gestation

As parturition approaches, E_2 rises causes contraction and release of PGF2α

Dilation of cervix

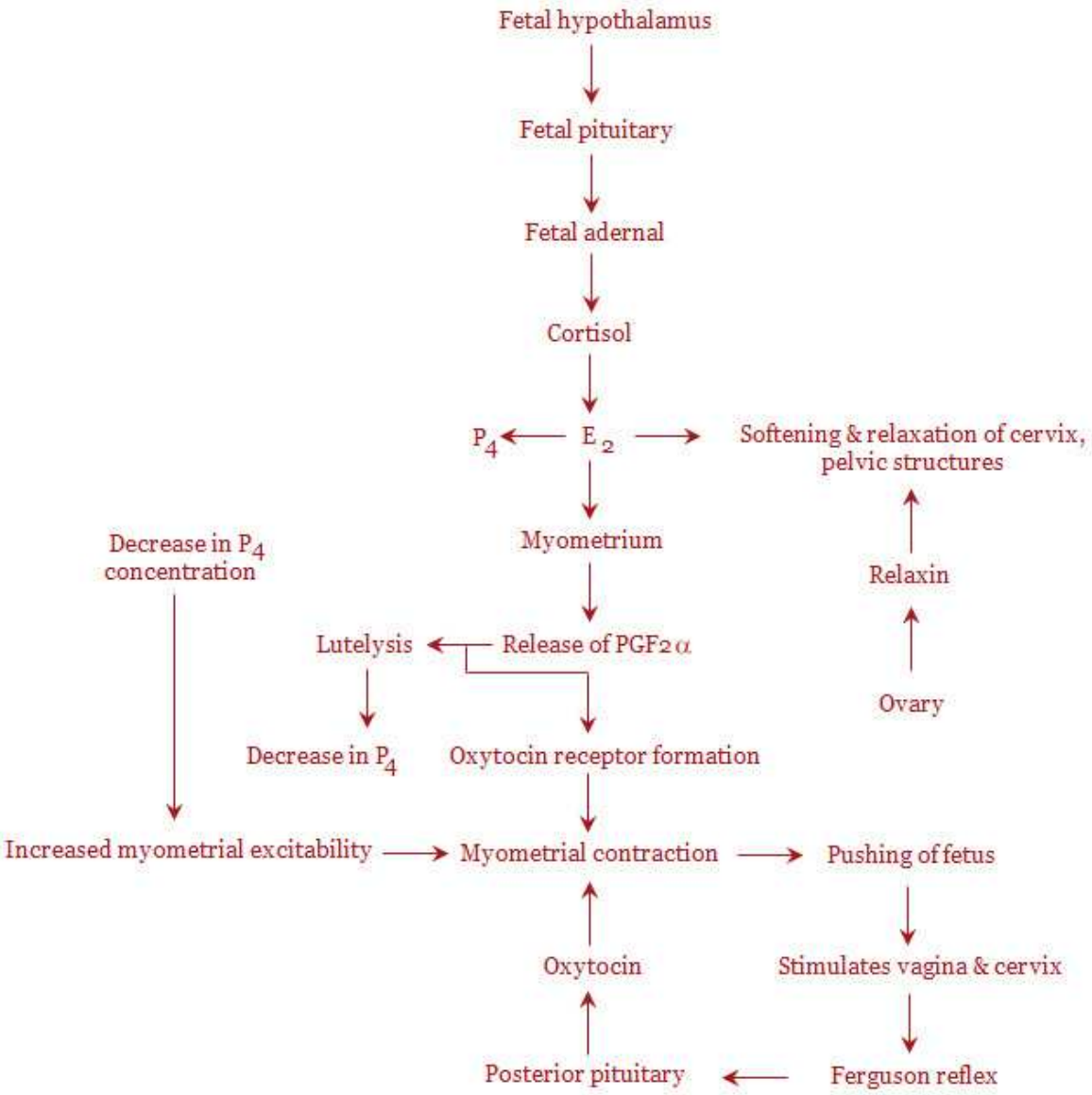
Cervix firm and rigid due to high collagen content

It retains fetus in the uterus

Under hormonal influence changes in physical characteristics of collagen at parturition

Initiation of parturition

Parturition is triggered by fetus and completed by complex interaction of endocrine, neural and mechanical factors.



STAGES OF PARTURITION

Continuous process.

For convenience, it is divided in to 3 stages

First stage

Contraction of uterine muscle and dilation of cervix

Uterine peristalsis starts at apex of the horn

Uterine contraction forces the fetal membrane and their fluid into the dilated cervix.

Rotation of fetus in (mare, bitch) taking place

By end of this stage cervix is completely dilated

This stage is apparently larger in primiparous than pluriparous

Once the portion of fetus enters pelvis, reflex sti

muli results in straining.

Species Duration

Cow and Ewe 2-5 hrs

Horse 1-4 hrs

Sow and Bitch 2 – 12 hrs

Second Stage

Entrance of fetus into the dilated birth canal

Rupture of one or both water bags

Abdominal contraction

Almost in all species, when straining commence it will lie down

As the fetal parts touches cervix – Fergusan's reflex

Combination of uterine and abdominal contraction leads to expulsion of fetus

Species Duration

Cow 3-4 hrs.

Ewe, goat 0.5 – 2 hrs

Mare 5 – 40min

Dog 1 hr

In multiparous animals, the length of second stage is variable depending on the number of fetusus.

Third stage

Expulsion of fetal membranes and involution of uterus

Postpartum events

The time from parturition to occurrence of first estrus postpartum is known as *postpartum interval* or *puerperal period.*

In mare and cow this period is 35 to 60 days.

During postpartum period, the CL of gestation has regressed, there is no follicular growth and ovarian size decreases accompanied with anestrus.

Nursing the young one inhibits ovarian function in most species except mares and rats. Suckling increases opioid peptides which inhibits gonadotropin secretion

The postpartum interval is highly variable and is influenced by species, breed, nutritional status of individuals, environmental conditions, age etc

In mare, estrus is noticed in about 8 to 12 days postpartum which is called *foal heat*.

The sow may have a non-ovulatory estrus in 1 to 2 days postpartum (higher estrogen level at parturition) but they return to ovulatory estrus in about 5 to 7 days after weaning

In cows and ewes at least 4 weeks time is required for uterine involution and repair of endometrium following parturition. Hence, conception can occur only after 30 to 40 days postpartum.

In bitch, the postpartum interval is about 5 to 6 months

ROLE OF HORMONES IN PARTURITION

During the last trimester of pregnancy the fetus shows rapid growth which increases the demand for nutrients. This stress stimulates the release of fetal hypothalamic CRH and adenohypophyseal ACTH which activates the fetal adrenal glands to release cortisol and c-19 steroids (important factor in initiating parturition in cow, ewe, doe and sow) while the equine does not show significant increase in fetal control .

Cortisol and the C-19 steroids diffuse through the placenta and stimulate placental aromatization enzyme system, -17 hydroxylase (17-20 lyase which controls the conversion of progesterone to estrogen and via androgen.

Generally an increase in the estrogon:progesterone ratio stimulates the release of PGs – PGE_2 and $PGF_2\alpha$.

Increased estrogen production is essential

Synthesis of contractile proteins of myometrium - important for development of forceful uterine contractions.

Formation of gap junctions between myometrial cells important for co ordination of contraction in association with the delivery process.

Stimulate development of oxytocin receptor.

Involved in the cervical dilatation.

$PGF_2\alpha$

PGF_2 synthesis and release occurs 24 – 36 hours before term.

Initiates the regression of the CL resulting in progesterone withdrawal allowing the myometrium to become more contractile.

Exerts direct effect on the uterine musculature. This is brought about by its ability to decrease sacroplasmic binding of calcium ions thus freeing them to interact in the contractile process. This initial increase in the contractility of uterine muscles is important for the start of first stage of labour.

Has indirect effect on the uterine contractility by making the uterine musculature more sensitive to oxytocin.

Is involved in relaxation of the cervix by dissolving the ground substance that binds fibroblasts and fibrocytes together.

Also stimulates the release of oxytocin from the posterior pituitary.

Relaxin

Important role in partition by relaxing the cervix.

For softening of the tissues that surround the pelvic canal, which in turn allows the fetus to have maximal space for its passage through the pelvis.

The relaxin and PGs cause the softening of the cervix and the body of the uterus by increasing the proportion of collagen in the tissues, they dilate the cervix and relax the sacroiliac joints. (ewe and cow) and widen the birth canal.

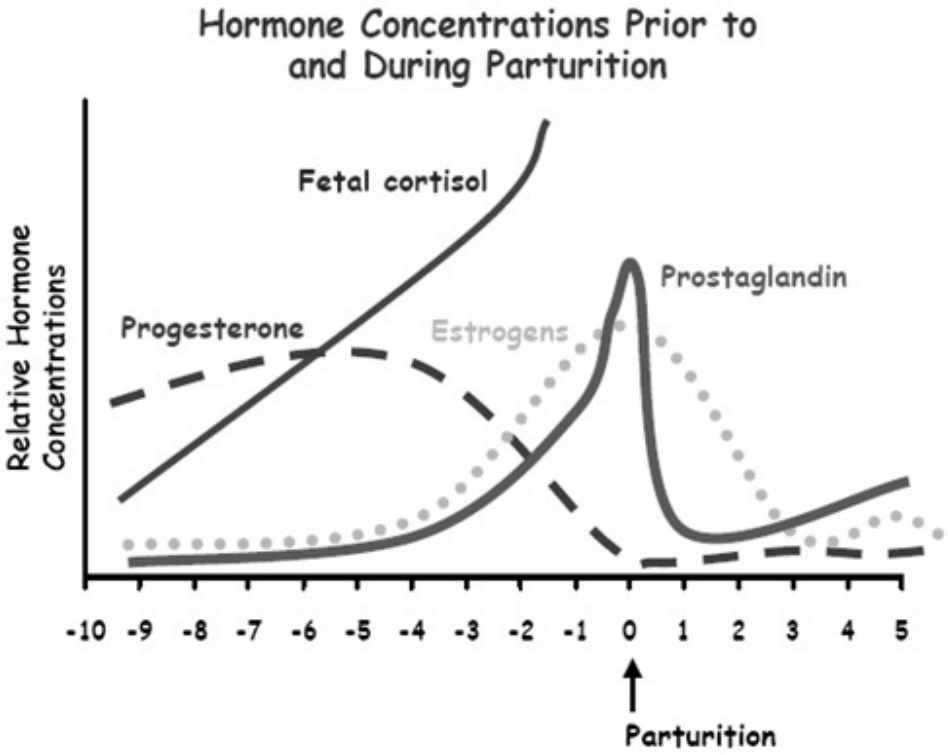
Induction of Parturition

As a managemental tool

Exogenous glucocorticoids can be used in – cattle, goat and sheep.

PGF2α induce parturition in cattle, goat and pig (CL dependent)

Oxytocin can be used only when cervix is dilated.



CHAPTER-30: BUFFALO REPRODUCTION

Learning objectives

This chapter explains the following

limitations in reproduction of buffaloes

approaches to improve reproductive efficiency in buffaloes.

BREEDING ACTIVITY IN BUFFALOES

The breeding and corresponding calving seasons are almost same throughout India - the peak breeding season is from September to February and the calving season from July to November. During this breeding period, the bulls are very active sexually and the quality and quantity of semen is very high particularly during winter (November to February).

The buffalo cow show the maximum ovarian activity and largest percentage of them conceive during this period. This holds the same for river as well as swamp buffaloes. • The endocrine mechanism is responsible for seasonal sex periodicity. During the late autumn

and early winter seasons average day length and daily temperature is associated with the sexual vigour of the animal.

High environmental temperature above 23.6 °C interrupts the normal physiological functions of the buffaloes. High environmental temperature adversely affects spermatogenesis in the buffalo-bulls and ovarian activities in the buffalo cows.

The first post-partum heat varies greatly with season, breed and individual. It can appear within less than 60 days in some cases and over 230 in others. Average post-partum estrus in the Murrah breed is around 100 days. The first post-partum estrus is not always fertile, especially if it comes very near parturition.

At birth buffalo have fewer primordial cells in the ovary than cattle have.

Compared to cows, buffalo suffer from higher atresia of follicles – 20000 versus 100000.

Buffalo have a high proportion of silent estrus and short duration estrus. This is one of the most important problems in buffalo reproductive efficiency. It is even more problematic during the hot and humid months when it is compounded by thermal stress. Short and silent estrus is the main reason why heat is often undetected in buffalo.

A large number of buffalo suffer from post-partum anestrus, a complete absence of estrus cycle and no signs of heat. This is one of the most common causes of buffalo infertility.

REASONS FOR POOR REPRODUCTIVE PERFORMANCE

Climate affects both production and reproduction in all farm animals. However as buffalo are very susceptible to extreme conditions of heat and cold they show a tendency towards better performance during the cool months.

In India 70 to 80% of buffalo conceive between July and February. In Italy the usual calving season is from September to December. In India a lower number of services are needed during the July to February breeding season than in the March to June season. Buffalo are sexually activated by decreased daylight.

Buffalo have poor thermal tolerance on account of an under developed thermo regulatory system and are unable to get rid of excess body temperature. If their housing is not designed to take care of this special species-specific requirement for adequate shade and ventilation, it will affect production and reproduction.

Nutrition plays a major role in the reproductive performance of buffalo, as with other farm animals. Under feeding, over feeding or unbalanced feeding, as well as deficiencies in minerals, vitamins or trace elements will cause reduced fertility in buffalo just as in other farm animals.

A poor body condition score at calving affects fertility, characterized by prolonged post-partum intervals, reduced conception rates, and more services per conception. A very low protein diet can cause cessation of estrus

One of the reasons buffalo suffer from long post-partum anestrus is because their natural behaviour of rolling in dirty water pools, and unhygienic shed conditions, cause buffalo to suffer from a high incidence of endometritis.

The loose broad uterine ligaments and rolling in water cause torsion of uterus cases in buffalo.

Buffalo also suffer from uterine prolapse and retention of the afterbirth. All these lead to uterine infections, delayed involution of the uterus and endometritis in buffalo resulting in the need for repeat breeding.

APPROACHES FOR IMPROVING REPRODUCTIVE EFFICIENCY

Providing the right kind of housing for buffalo to suit their natural behavioural requirements is important for their optimum performance. Free stall as well as tied systems work well for buffalo. However it is important that the housing provides sufficient shelter from both heat and extreme cold. During summer they have to be protected from extreme heat while in winter they have to be protected from extreme cold as well.

Showers or foggers with fans or wallowing tanks should be made available to buffalo during the hottest part of the day. Thermal ameliorative measures such as sprinkling and cooling are known to increase comfort levels and feed intake in buffalo.

Balanced feeding with mineral supplements, plenty of green fodder, and concentrate as per each animal‘s specific need, is necessary to bring buffalo into normal reproductive cycles.

Regular testing of all buffalo and bulls for infectious reproductive diseases like brucellosis and regular culling of infected animals are crucial for good reproductive health in the herd. Attending cases of difficult birth and retained placenta in time and maintaining good hygiene during parturition are also crucial to prevent reproductive disorders such as endometritis.

CHAPTER-31: MALE REPRODUCTIVE SYSTEM

Learning objectives

At the end of this chapter the learner will be able to understand the following

Introductory aspects of male reproduction

Functional anatomy of organs of male reproductive system.

MALE REPRODUCTIVE SYSTEM - AN INTRODUCTION

The male reproductive system includes testes and accessory sex glands. The testes produces male gametes – spermatozoa and accessory sex gland secretes seminal plasma which act as nourishing and transport media for the spermatozoa

Prenatal development

Although the sex of the embryo is determined at the time of fertilization the sexual differentiation doesn't occur until the early fetal period after the primordial germ cells have migrate to the gonadal ridge

The testes develop in the abdomen medial to the embryonic kidney (mesonephors)

Testosterone from the fetus helps in the descent of testes and development of external genitalia penis. It helps in the differentiation of the wolffian ducts to epididymis, seminal vesicle and vas deferens. Mullerian inhibiting substance cause regression of mullerian ducts and descent of testis

MALE REPRODUCTIVE SYSTEM

It Consists of

Gonads – testes

Rete testes

Efferent duct

Epididymis

Vas deference

Accessory glands – Ampulla, Prostate, Seminal vesicles and Bulbourethral organ

Penis

TESTES

Lie outside the abdominal cavity, enclosed with in the scrotum

The position and orientation varies with species. In all domestic animals, it is located between thighs. In cats and boar – Caudal to thighs and caudo ventral to ischial arch

The testis is held in position with in scrotum by

Tunics of scrotal pouch

Spermatic cord, composed of

Spermatic artery

Spermatic vein

Internal cremastor muscle

Lymph vessels

Nerves

Vas deference

Testes covered by a serous membrane – Tunica vaginalis

Beneath tunica vaginalis, a dense thick connective tissue capsule- tunica albugina from which septum radiates into testes to form lobules in the testes. These radial extensions join at mediastinum testis, a connective tissue cord running through testes.

Seminiferous tubules are located with in the lobule

Interstitial / Leydig cells lies between seminiferous tubules

Sertoli cells are located with in the seminiferous tubule

Descent of testes

During fetal life testes are intra abdominal

Testis originates in the sublumbar region immediately caudal to the kidney. The testicles travel a considerable distance from the point of origin to the scrotum, presumably under the influence of testosterone

In bull and ram, testicles enter scrotum half way through fetal life, in boar during last quarter of fetal life and in stallion before or just after birth

Descent is complete at birth / soon after birth

In birds, elephant, rhino, dolphins, whales it is intra abdominal even in adult life

A testis that fails to descend to the scrotum is called a *cryptorchid testis* and animals with such a condition is called a *cryptorchid (testicond)*. In this condition (*cryptorchidism*) spermatogenesis does not occur although the endocrine function of the testis is not affected. These animals show more or less normal sexual desire but are sterile.

In bats, rodents and in some carnivores testes descend into the scrotum during the breeding season and return to abdominal position during non-breeding season

Functions of testes

Leydig cells / interstitial cells – Secrete male hormone

In seminiferous tubules, spermatogenic cells divide and differentiate to form spermatozoa-called spermatogenesis

Sertoli cells

Nourish the developing sex cells

Testosterone is converted into estrogen under the influence of FSH

Participates in spermatogenesis by phagocytosis of residual bodies shed by maturing spermatids

Transport androgen to spermatogenic cells

Also secretes inhibin

Secreting mullerian inhibiting factor

Secreting androgen binding factor which increase the concentration testosterone in seminiferous tubule

Daily sperm production per gram of testis

Bull - 13 – 19 millions

Ram - 24 – 27 millions

Boar - 24 – 31 millions

Stallion - 23.7 – 32.3 millions

Scrotum

It is a double pouch with a median septum, which contains the testes.

Functions

It encloses the testis

It protects the testes from adverse stress conditions.

It helps in thermo regulation of testis for optimum spermatogenesis

Click the following for location of scrotum in different species of domestic animals

Bull

Stallion

Sheep

Goat

Boar

Dog

Cat

Thermoregulation in Scrotum

For effective functioning, the mammalian testes must be maintained at a temperature lower than that of body.

The arrangement of testes and scrotum helps in regulating testicular temperature and testicular temperature is 4 to 7°C cooler than core body temperature.

Thermal receptors in the scrotal skin can elicit responses to lower whole body temperature and on stimulation provoke panting and sweating.

Heat loss

Tortuous spermatic artery runs through the pampiniform plexus which is the venous network leading out of testes and by counter-current mechanism it aids in heat exchange.

Sweat glands are present in the scrotal skin.

Absence of subcutaneous fat promotes heat transfer.

Conservation of heat

Tunica dartos – contains smooth muscle - Dartos muscle on contraction produces wrinkling of scrotal skin reducing the surface area of heat transfer.

Cremaster muscle – skeletal muscle present in the spermatic cord, elevates the testes nearer to the abdominal wall on exposure to cold and relaxes during heat drawing the scrotum away from the body.

Hair on the scrotum.

Ram - thermal infertility can be caused by excessive wool around the scrotum.

Click here to view animation showing thermoregulation in testis

RETE TESTES AND EPIDIDYMIS

Convoluted seminiferous tubules joins with straight tubules called Rete testes

Rete testes come out of testes and joins with efferent duct which converge to form epididymis

Epididymis

It is closely apposed to surface of testes

It is composed of three parts caput (head) , corpus (body) and cauda (tail)

Caput epididymis – Flattened at proximal part of testes

Corpus epididymis – Narrowed portion

Cauda epididymis – It terminate at distal part of testes

Caput, Corpus and Cauda portions of epididymis– Less clarify differentiated in stallion

Motile ability of of spermatozoa is acquired in epididymis during transit

Transport of spermatozoa is about 9 – 13 days

Functions

Transportation of spermatozoa

Storage of spermatozoa

Maturation of spermatozoa

Absorption of fluid in the caput epididymis

VAS DEFERENS AND PENIS

It extends from cauda epididymis to colliculus seminalis

It is thick walled and narrow lumen

Ampulla

Terminal portion of vas deferens, is enlarged and glandular called ampulla

It contributes fluid to semen

Absent in dog and cat. Very small in boar.

Well developed in stallion

Contribute ergothione to ejaculate

Seminal vesicles (Vesicular glands)

Paired gland

Located lateral to terminal part of vas deference

Duct of vesicular gland and ductus deference share a common ejaculating duct that opens into urethra

Absent in dog and cat

The secretary of seminal vesicle contents fructose, PGE, ascorbic acid sorbitol and ergothione.

Prostate gland

Unpaired

More or less completely surround the pelvic urethra

Its secretion give characteristic odor to semen

Its fluid by its alkaline nature, neutralize the acidity of vaginal secretion

It adds its secretion to semen at the time of ejaculation

Cowper's gland (Bulbourethral glands)

Paired gland

It is located on either side of pelvic urethra, caudal gland to other glands

Absent in dogs

In boar, its secretion is white and viscous that forms gel in the ejaculation

The secretion of accessory glands are mixed with fluid suspension of spermatozoa at the time of ejaculation

Penis

Copulatory organ

Terminal portion in gland, this lies free in sheath

UNIQUE FEATURES OF REPRODUCTIVE ORGANS OF MALE ANIMALS

In ram, the penis is characterized by a filiform appendage containing urethra

In pigs, accessory glands are larger in size which contribute remarkable large volume of semen

In penis of stallion, the urethra is protrudes few centimeters from the surface of glans penis

Penis of bull, ram and boar contains sigmoid flexure that is straightened during erection and extension of penis

In dogs, remarkable feature is os penis, locking of penis during coitus

In dogs except prostate and ampulla, all other accessory glands are absent

the presence of spines in the penis of cat

Androgens

Chief site of production -Leydig cells

Includes – Testosterone, Dihydrotestosterone, Androstenedione

Also produced by – Adenal cortex, ovary, placenta

In horse – seminiferous tubules and epididymis also produce testosterone

97-99% of testosterone is bound with carrier protein and remaining is free to enter the cell

In the target tissue, testosterone is converted into dihytrotestosterone in cytoplasm which is biologically active

Functions

Development of fetal testicles and descent of testes before birth

Spermatogenesis and prolong the life span of epididymal spermatozoa

growth and development, and secretary activity of accessory glands

maintenance of secondary sex characters

Muscle mass

Male voice

Hair growth

Thickness of skin

Increase bone growth

Aggressiveness

Sexual behavior and libido

protein anabolic activity by nitrogen retention

Exert negative feed back influence on hypothalamus and anterior pituitary

Application

Synthetic androgen - testosterone propionate is administered to cows that are used as teaser animals for detection of heat

Used by athletes to increase muscle mass and to give a since of well being Bioassay for androgen – Chick comb bioassay

Androgen applied over comb of young male chicken daily for 4 days – comb become larger than non treated birds

CHAPTER-32: SPERMATOGENESIS AND ITS HORMONAL CONTROL

Learning objectives

At the end of this chapter the learner will be able to understand the following

How spermatogonial cells develop into mature spermatozoa

Spermatogenic cycle

Transport of sperm in male reproductive tract

Endocrine control of spermatogenesis.

SPERMATOGENESIS AND ITS HORMONAL CONTROL

Introduction

Spermatogenesis is defined as process by which the spermatogonia undergo a series of cell divisions followed by a metamorphosis resulting in the production of highly differentiated and potentially motile cells called spermatozoa.

The process begins at the wall the seminiferous tubule which is lined with the spermatogonia and ends with the release of mature spermatogoa in the lumen of the seminiferous tubule.

Spermatogenesis is commenced at puberty. It continues throughout the life or until infertility sets, during which atrophy of tubules occur and only few are capable of producing sperm.

In wild species, breeding occur only at certain seasons. In such animals, testes regress completely during non breeding season. In seasonal breeding farm animals it is not clearly defined.

Spermatozoa while passing through the epididymis become actively motile when exposed to the materials secreted by the ampulla, vesicular glands, bulbo-urethral glands and prostate gland

The seminiferous epithelium lining the seminiferous tubule is composed of two basic cell types

The developing germ cells – gives rise to sperm cell

Sertoli cells - provides nourishment to sperm cell

Spermatogenesis takes place in two phases: *spermatocytogenesis* and *spermiogenesis*

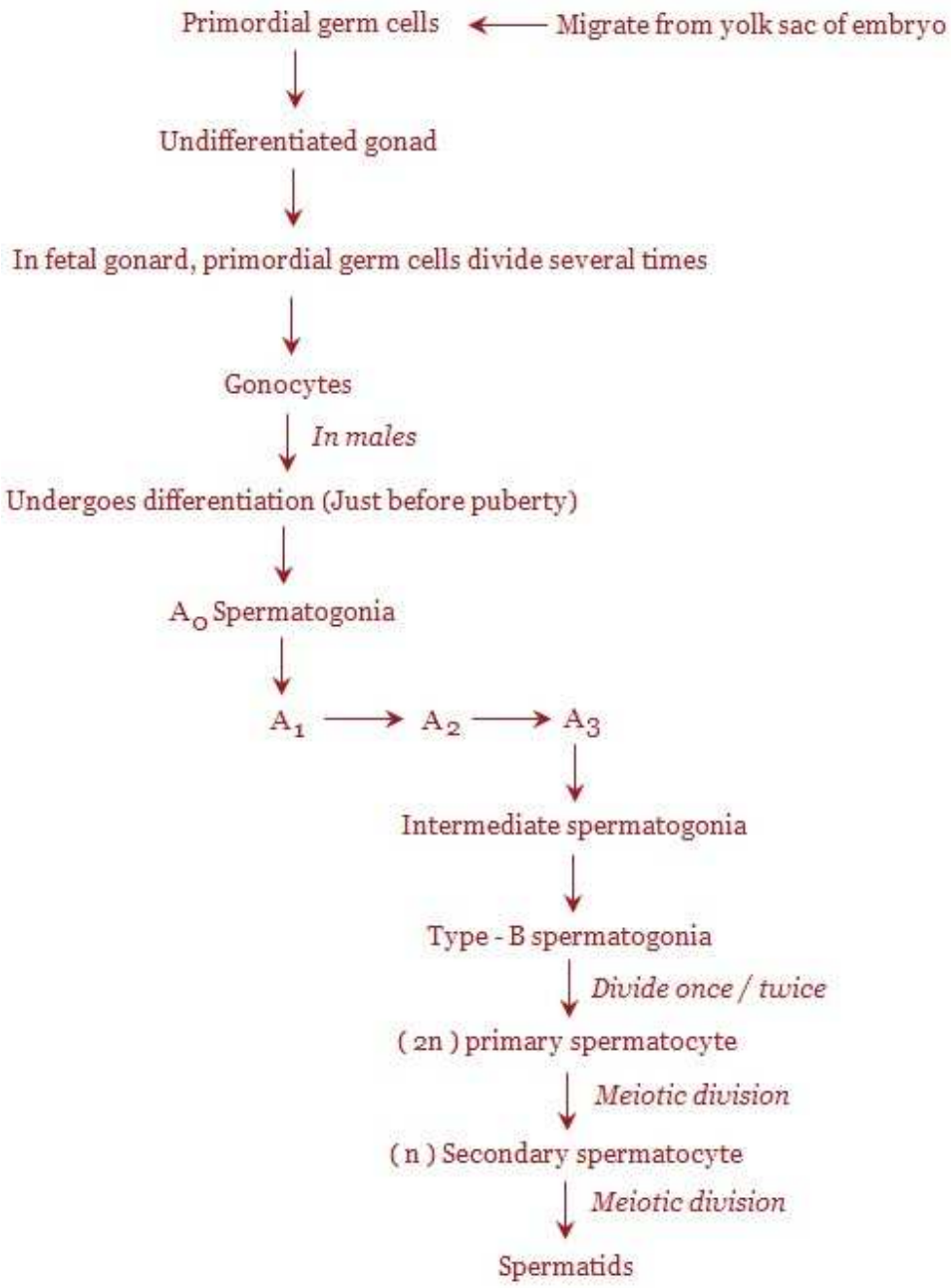
The stem cells called the spermatogonia divide several times by mitosis and then undergoes meiosis thereby reducing the DNA content of the cells to one half that of somatic cells

The haploid cells resulting from this process are called the spermatids. The spermatids undergo a progressive series of structural and developmental changes - the metamorphic changes - to form spermatozoa. The developing germinal cells are closely associated with much larger sertoli cells that surround them during development.

SPERMATOCYTOGENESIS

Spermatogenesis is initiated when the spermatogonia undergo mitosis. One spermatogonium formed by the initial mitotic division does not divide and differentiate further but remains in basal state of differentiation in essence it replaces the parental cells. The other spermatogonium undergoes mitosis. The basic function of the mitosis to ensure the production of large number of germ cells.

During early embryonic development



ach primary spermatocyte that result from series of mitotic division, undergoes a meiotic division forming two *secondary spermatocytes*. With this division the chromosome complement in the nucleus is reduced by half so that nuclei in secondary spermatocytes contain unpaired (n) chromosomes. This is essential to allow the union of haploid spermatozoa and oocytes to form new individuals. This step requires approximately 15 days in the ram. Within a few hours after their formation each secondary spermatocyte will again meiotically divide forming two *spermatids*

16 primary spermatocytes could be produced from one A_3 spermatogonium. By two meiotic divisions, 64 spermatids can be potentially produced. Total number of mature

spermatozoa produced is almost always less than the theoretical number since, certain percentage of potential sperm is lost by degeneration during the course of spermatogenesis.

Following a resting state of several weeks, the dormant (A_1) spermatogonia start another cycle. Even though A_0 spermatogonia (reserve stem cells) can occasionally divide forming new A_0 and A_1 spermatogonia, formation of dormant spermatogonia (A_1) from A_2 spermatogonia is the key to maintaining the continuity of spermatogenesis and thereby not diminishing the supply of potential gametes within the testes

The entire divisional process of spermatocytogenesis from spermatogonia to spermatid takes approximately 45 days in the bull.

SPERMIOGENESIS

Transformation of round, non motile spermatids into potentially spermatozoa by a series of progressive morphologic changes is called as spermiogenesis. These changes include

Development of an organelle, the acrosome that allows penetration of the oocyte

Development of mitochondria to furnish energy during movement in the female tract

Formation of a tail to aid movement within the female reproductive tract

A considerable amount of the cytoplasm is extruded from the cell during spermatogenesis. Remnants of the cytoplasm are sometimes observed attached to the spermatozoa and are referred to as cytoplasm droplets.

The presence of cytoplasmic bridges that have held developing clones of spermatozoa together during development disappear at the end of spermiogenesis.

Spermiogenesis starts in the seminiferous tubules and completes in the epididymis. Spermiogenesis occurs within the cytoplasm of Sertoli cells and then released into the tubular lumen

The various stages of development of spermatid transformation of are divided into 4 phases

Golgi phase

Formation of proacrosomal granules within the golgi apparatus

Coalescence of the granules into a single acrosomal granule

The adherence of the resultant acrosomal granule to the nuclear envelope

The early stages of tail development at the pole opposite that of the adherence of the acrosomal granules

The proximal centriole migrates closer to the nucleus, where it is thought to form a basis for attachment of the tail to the head.

Cap phase

Spreading of the adherent acrosomal granule over the surface of the spermatid nucleus. This process continues nearly two third of the anterior portion of the each spermatid is covered by a thin double layered membranous sac that closely ache adheres to the nuclear envelop.

The developing axonemal (central core of tail 9+2) components of the tail which are formed from the elements of distal centriole elongated well beyond the periphery of the cytoplasm

Acrosomal phase

The acrosomal phase of spermiogenesis is characterized by the major changes in the nuclei, the acrosome and the tail of developing spermatids

The developmental changes are facilitated by rotation of each spermatid so that the acrosome is directed towards the basement or the outer wall of the seminiferous tubule and the tail towards the lumen.

The nuclear changes include condensation of the chromatin into dense granules and reshaping of the spheroid nucleus into an elongated flattened structure.

The acrosomes which is closely adherent to the nucleus, also condenses and elongates to correspond to the shape of the nucleus. These modification in nuclear and acrosomal shape appear to be molded by the surrounding sertoli cells

The cytoplasm is displaced to the caudal aspect of the nucleus where it surrounds the proximal portion of the developing tail within this cytoplasm, microtubules associate to form a temporary cylindrical sheath.

The annulus (ring like structure) first forms near the proximal centriole and then during subsequent development migrates posterior along the tail.

The mitochondria which were previously distributed throughout the cytoplasm of the spermatid begin to concentration close to the axoneme where they form the sheath that characterizes the middle piece of the tail.

Maturation phase

It involves the final transformation of the elongated spermatids into the cells to be released in the lumen of the seminiferous tubules

Sperm head (nucleus and acrosome) becomes its elongated shape characteristic of each species

Chromatin condensation reaches the final stage

A fibrous sheath and underlying nine coarse fibres are formed around the axoneme the fibrous sheath covers the axoneme from the neck to the beginning of end piece.

The annulus migrates distally from its position adjacent to the nucleus along the tail to a point where it will subsequently separate the middle piece from the principle piece of the tail

The mitochondrion become tightly packed into a continuous sheath extending from the neck to the annulus.

The remaining cytoplasm forms a residual body and the sperm is ready for release.

SPERMATOGENIC CYCLE, SPERMATOGENIC WAVE AND SPERMIATION

If a cross section of a seminiferous epithelium is viewed, the developing cells exhibit very definite cell associations among them i.e. each of the sperm-forming developing cell type transform to next stage in synchrony with its nearby cell type.

If a single cross section of seminiferous tubule is viewed continuously, series of cell associations would be seen until the original cell association reappeared. This cycle of developing cells is known as *spermatogenic cycle* or *cycle of seminiferous epithelium*

Each spermatogonium that replaces the parental cell begins to divide at a time interval that is characteristic of each species

The spermatogenic cycle for

Pig – 8 days

Sheep – 10 days

Cattle – 14 days

Human – 16 days

The spermatogenesis is approximately 4 times the duration of the spermatogenic cycle

Spermatogenic wave

The stages of the spermatogenic cycle change not only with time, but also along the length of the tubular loop. A portion of tubule at one stage is usually adjacent to portions of tubule in stages just preceding or following it in time. This sequential change in stage of cycle along the length of the tubule is known as the wave of spermatogenesis

Spermiation

The release of formed germ cells into the lumen of the seminiferous tubules is known as spermiation. The elongated spermatids, which are oriented perpendicularly to the tubular wall are gradually extruded into the lumen of the tubule.

SERTOLI CELLS AND LEYDIG CELL

Sertoli cells

The other major cell type within the seminiferous tubules is the sertoli cell.

These cells have prominent nucleoli are basically situated within the tubule

These cells have long processes that surrounded the spermatocytes and spermatids and provide an intimate interaction with germ cells through out their development

Sertoli cells are important for controlling the development of germ cells with respect to both nutritive and regulatory function

Sertoli cells are the main source of fluid that is secreted into the lumen of the tubule.

The principle permeability barrier between the blood and testis is the complexes at the junctions between the adjacent sertoli cells

Sertoli cells phagocytise the residual bodies remaining from the degenerating sperm cells

Hormonal function of sertoli cells

The secretary activity of the sertoli cells is controlled by FSH

The receptor of FSH is on sertoli cells membrane and the receptor for testosterone are on the cytoplasm and nucleus

The sertoli cells convert testosterone to dihydrotestosterone an androgen of greater biological potency than testosterone

In some species the sertoli cells converts testosterone produced by the leydig cells into estrogen which may have a regularity effect on the production of testosterone by the leydig cells

The sertoli cells synthesis a protein- Androgen-binding protein (ABP) which binds androgens which acts as a means of stabilizing the concentration of androgens in the seminiferious tubules for use in both spermatogenesis and sertoli cell function

Sertoli cells also produce inhibin, a protein that suppresses FSH secretion at the level of pituitary gland

It also secretes mullerian inhibiting factors which inhibits the formation of the falloplan tube, thus helps in the development of testes in the fetus

Leydig cell

The main function of the leydig cell is the production of testosterone which is important for the development and maintenance of spermatogenesis and male characteristics.

Leydig cell production of testosterone is controlled by LH (ICSH). LH has a tropic effect on the Leydig cells – stimulating them to undergo hypertrophy.

Testosterone moves into seminiferous tubules by either simple or facilitated diffusion. High intratesticular concentration of testosterone are required for spermatogenesis and especially for meiosis.

FSH is important for the completion of meiosis of germ cells, through its influence on the sertoli cell activity.

Prolactin is needed for LH binding to receptor on Leydig cells

GH is needed for early divisions of spermatogonic and is necessary for controlling the metabolic functions of the testicles.

Testosterone – Masculine characters, secretary activity of sex glands, sex characters. Leydig cells also produce c-16 unsaturated androgens – Excreted in saliva – pheromones – in urine- undesirable flavour of boar meat, boar taint.

SPERM TRANSPORT IN MALE DUCT SYSTEM

Testicular spermatozoa are transported from the testis through a high convoluted duct known as the Epididymis.

The Epididymis mainly serves three processes

Transport

The passage of spermatozoa is aided by the localized contractions of the duct smooth wall at a frequency of 3 / min

The spermatozoa transport through the epididymis requires about – 7 days in the bull and 16 days in the ram

The transit time may be reduced by 10 -20% by an increased frequency of ejaculation.

Maturation

During the transit in the epididymis, the spermatozoa undergo a maturation process in which they gain the potential ability to fertilize ova.

The functional changes occurring during the epididymal transit of spermatozoa.

Presence of forward motility protein

Maturation of acrosome

Low metabolic activity of sperms

Distal migration and loss of protoplasmic droplet

Attainment of motility

Storage

The head of the epididymis absorbs considerable amounts of fluid that originates in the seminiferous tubules. This absorption results in the high concentration of spermatozoa in the tail of the epididymis. The tail of the epididymis contains 70% of the total number of spermatozoa in the recurrent ducts whereas the vas deference contains only 2%

The spermatozoa contained with in the epididymal duct from head to the tail are termed the extragondal reserves even though only those in the distal section of the tail can be ejaculated.

Disposal of unejaculated spermatozoa one gradually eliminated by excretion into the urine. Those that are not eliminated undergo gradual disintegration.

ENDOCRINE CONTROL OF SPERMATOGENESIS

Leydig and Sertoli cells are the endocrine cells of the testis

GnRH

Episodic release from hypothalamus

Regulates episodic release of LH from anterior pituitary

FSH and LH are stimulated by GnRH

Gonadotropins

LH and FSH are released from the anterior pituitary

LH

Stimulate Leydig cells

Stimulate testosterone production

Required continuously for spermatogenesis

FSH

Stimulates Sertoli cells

Stimulates production of Androgen Binding Protein Modulates androgen function

Also stimulates secretion of oestrogen and inhibin

FSH is required for initiation of spermatogenesis

Androgens

Secreted from the interstitial cells (Leydig cells) of testis

Includes testosterone, dihydrotestosterone and androstenedione Testosterone is more abundant than the other two

Adrenal cortex, ovaries, placenta secrete very less quantity of androgen

Lack of testosterone causes low sex drive (or) lack of libido

Testosterone

Maintains spermatogenesis - Supports meiosis

Causes differentiation of fetal male reproductive tract and testicular descend

Developing and maintaining libido (sex drive) d. Maintains secretory activity of the accessory glands

Develops and maintains secondary male sexual characteristics Heavier bones; Greater muscling; thicker skin; deeper voice, aggressiveness (especially bull)

Anabolic hormone – positive nitrogen balance, Stimulates erythropoesis, increases BMR Stimulates pheromone secretion (cat, boar and many animals)

Dihydroxytestosterone (DHT)

Testosterone is converted to DHT in androgen sensitive cells

DHT is the actual steroid that binds to the androgen receptor to elicit and action

C-16 unsaturated androgens - 5α-androstenone

Produced from Leydig cells

Primarily in the boar

Acts as a pheromone in saliva and urine

Responsible for boar taint

Negative feedback mechanisms

Androgens

Feedback on hypothalamus and pituitary to decrease FSH, LH production

In seasonal breeders (ewe) photoperiod affects testosterone production due to variation in GnRH and LH release

Inhibin

Produced by Sertoli cells in response to FSH stimulation

Protein hormone c. Feedback on pituitary to decrease FSH only

Activin

Antagonistic to inhibin

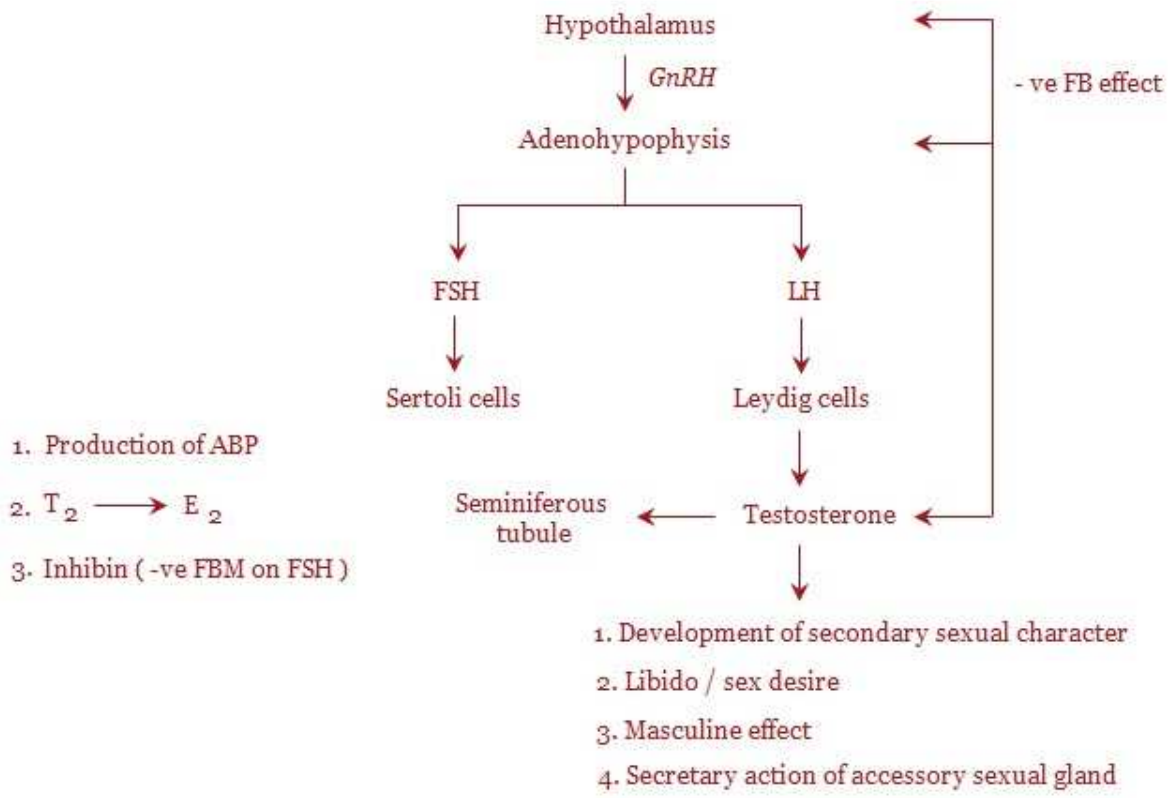
Produced by Sertoli cells

Positive feedback effect on pituitary FSH production

ABP from Sertoli cells helps to maintain a high concentration of testosterone within the seminiferous tubules and also facilitates androgen transport into epididymis where the androgens influence sperm transit and maturation.

Prolactin enhances LH in maintaining testosterone production.

Growth hormone specifically promotes early divisions of spermatogonia and is necessary for controlling metabolic functions of the testicles.



FUNCTIONS OF TESTOSTERONE

The peripheral functions of testosterone include the development and maintenance of

Libido

Secretary activity of accessory organs

General body features

Libido

Testosterone is essential for libido and castration is usually effective in eliminating the male sex drive.

Castration usually refers to the removal of the testis to make the animal more docile.

Vasectomy refers to the removal of a section of each of the vas deference to prevent the passage of spermatozoa from the epididymis

Secretary activity of accessory organs

Testosterone is important for the development and maintenance of secretory epithelium of the accessory sex organs. The route of transfer from the leydig cell is via the general circulation and not via the seminiferous tubule as occurs for the maintenance of the epididymis.

General body features

Testosterone is also important for the development of various general bodies characteristic

Myotropic or anabatic effect that involves an increase in the muscle mass through the retention of nitrogen (thickening of muscles of vocal cords of the larynx which decrease. The pitch of the male voice.

Promotes hair growth patterns

Calcification of antlers in seasonal breeders that shed horns on a yearly basis

Testosterone affects the feather pigmentation and growth in poultry

Increased of size, bone growth, BMR

Behavioral changes

Urinary pattern of male dogs which involve the raising of a hind limb during urination

Aggressiveness that may be more of a characteristics of the male than of female

The marking of a territory by substance known as pheromones which can be produced by the kidney under the influence of testosterone.

CHAPTER-33: MAMMARY GLAND

Learning objectives

This chapter explains about functional anatomy of mammary gland.

MAMMARY GLAND - AN INTRODUCTION

Mammary glands produce milk which serves to nourish the young mammal from birth to weaning. During the first few days after parturition, the secretion from mammary glands is termed *colostrum*. This transmits passive immunity from the mother to her offspring via immunoglobulins. These immunoglobulins are secreted in the colostrum and during the first few hours after birth, they can be absorbed through the gut wall. Colostrum is also

a natural laxative and aids in the passing of the meconium (a bolus formed in the fetal gut by the sloughing off of epithelial cells).

MAMMARY GLAND

Position and number of mammary glands vary with species.

Number of glands (and therefore individual teats) varies with the number of young reared.

The mammary gland and teat of domestic animals is collectively called *udder*.

Location of mammary gland is described as inguinal, abdominal or thoracic.

Goat, ewe, mare have 2 glands and 2 teats - 1 teat opening or *streak canal* / gland (mare-2 opening / teat).

Cow – 4 glands and 4 teats

Sow - 12 to 14 teats (usually) which are small, long and cylindrical /2 inguinal pairs, 3 abdominal pairs, 2 thoracic pairs (thoracic are most productive) - 2 (or 3) teat openings.

Cat-8 teats (usually)/1 inguinal pair, 1 abdominal, 2 thoracic pairs (abdominal are most productive)-3 teat openings on tip of teat. Additional sinus opens on the side of the teat.

Bitch-10 teats (usually)/1 inguinal pair, 2 abdominal pairs, 2 thoracic pairs (abdominal are most productive)-similar to cat. About 12 teat openings situated on tip of teat.

The mammary glands of cattle, sheep, goat, horse and whales are located in the *inguinal region*

Those of primates and elephants in the *thoracic region*

Those of pigs, rodents and carnivores along the ventral surface of *both the thorax and abdomen*

· The mammary glands are greatly modified and enlarged sweat glands.

In the inactive mammary gland, the interstitial spaces between lobes and ducts are filled mainly with fat. There are a few alveoli present compared to those in the lactating gland and the ducts and lobes of the gland are smaller than when the gland is in its active phase.

During early pregnancy, the epithelial cells of the glandular tissue proliferate rapidly to form the buds which eventually enlarge to form the alveoli.

As pregnancy develops, the fat and connective tissue between lobes and ducts are replaced by secretory tissue.

The mammary gland's milk-secreting structures are *alveoli* which empty directly into the larger ducts and also directly into the gland cistern and the teat cistern.

A circular muscle around the streak canal functions as a sphincter which helps to retain milk in the gland cistern.

At the junction of the streak canal and the teat cistern, the *rosette of Furstenberg* is present, which is a double layered epithelial lining that helps to retain milk in the teat.

A group of alveoli surrounded by a connective tissue septum forms a distinct unit called ***lobule***.

A group of lobules surrounded by a large amount of connective tissue form a ***lobe***.

The lobule empty into small ducts called the *intralobular duct* within the lobule. It drains into a central collecting space from which the *interlobular duct* emerges.

Within the lobe, the interlobular ducts unite to form a single *intralobar duct*. The interlobar duct may enter into the gland cistern directly or it may join with one or more other interlobar ducts before entering into the *gland cistern*.

More amount of milk is stored in ducts than in cistern.

Contractile *myoepithelial cells* also called as basket cells which resemble the smooth muscle fibres surround the alveoli and ducts.

On contraction of these cells the milk is letdown from the alveoli and ducts.

Approximately 11% of mammary development occurs before pregnancy, 41% during pregnancy and the reminder takes place during early lactation.

Surface anatomy of the bovine udder

The appearance of the udder varies depending on maturity and functional status.

In dairy cows udder is very large and can weigh up to 60 kg.

The udder is divided into two halves, each half having two quarters, thus four quarters correspond to the four glands - each gland bearing a principle teat.

A median intermammary groove marks the division of the udder into right and left halves.

The division between fore and hind quarters is less distinct

The structures supporting the udder are:

Skin - mobile, flexible and covered in fine hairs. Skin does not provide any mechanical support to udder.

Suspensory apparatus - The right and left halves of the udder are separated by median suspensory ligament containing elastic tissue. Lateral suspensory ligaments composed of non elastic fibrous strands give rise to numerous lamellae that penetrate into the gland and continue with interstitial tissue.

Right and left medial laminae are separated by a layer of loose areolar connective tissue allowing easy surgical division of one side of the udder from another.

Blood supply to udder

Caudal aorta gives off right and left common iliac arteries that branch to external iliac and pudic arteries, the latter gives rise to cranial and caudal mammary arteries to supply the cranial and caudal parts of the udder respectively

The major drainage route from the udder follows the same course as arteries – from mammary veins to external pudic to external iliac veins that join the caudal vena cava.

Another potential drainage route from udder is through *subcutaneous abdominal veins* to cranial venacava. In lactating animals, a large quantity of blood is drained via this route

In dairy cow the ratio of volume of blood flow through mammary gland to volume of milk produced is about 670:1

Nerve supply

The udder is supplied with both sensory and motor nerves. The sensory nerves are important for the milk ejection reflex (does not require efferent nerve).

The motor nerves to udder have both sympathetic and parasympathetic innervations. These nerves do not influence the rate of milk secretion or milk composition

CHAPTER-34: LACTATION

Learning objectives

At the end of this chapter the learner will be able to understand the following

Initiation and maintenance of lactation

Hormonal control of mammary gland development

Milk let-down reflex

Factors affecting lactation.

LACTATION - INTRODUCTION

Lactation refers to the combined process of milk secretion and removal.

M*ammogenesis* is development of mammary gland.

Lactogenesis is initiation of milk secretion.

Galactopoiesis is maintenance of milk secretion.

MAMMOGENESIS

At about 35 days age of embryo in bovines, a mammary line forms from the stratum germinativum parallel to ventral midline on either side of the ventral line.

At birth, the udder has teat and gland cisterns but ducts are developed to a limited extent. Most of the udder structure is formed of adipose and connective tissues.

From birth to puberty, the mammary growth rate is similar to the whole body growth rate (*isometric growth*); increase in size at this time is due to connective tissue and fat.

At the time of puberty, mammary parenchyma begins to grow at a much faster rate than whole body growth (*allometric growth)*. This allometric growth again occurs after conception up to early parturition.

During each oestrous cycle, the mammary gland is stimulated by oestrogen from ovary and GH and PRL from adenohypophysis; the mammary growth by oestrogen involves ductular development.

Most of the mammary growth occurs during pregnancy and this growth occurs in an exponential manner. Alveolar secretory cells increase in number with extensive ductular development; alveoli are formed. Estrogen with GH and adrenal steroids are responsible for proliferation of duct system and progesterone along with PRL is responsible for the development of alveoli. About 10% (cow) to 50% (rats) growth of mammary gland occur during early lactation involving alveolar cell proliferation.

The ratio of progesterone to estrogen as well as the absolute amounts of each is important for complete development of the mammary gland.

Cessation of suckling or milking brings about mammary involution with decrease in number of epithelial cells and secretory activity. Myoepithelial cells remain. Adipose cells increase. If the dry period occurs at later stages of pregnancy the decrease in secretory epithelial cells is reduced. Too short a dry period (<6 weeks) reduce the increase in mammary cells that occur during subsequent lactation.

HORMONAL CONTROL OF MAMMARY GLAND DEVELOPMENT

FSH and LH, the hormones of adenohypophysis control the ovarian hormones, oestrogen and progesterone, and their mammogenic effects in the mammary gland.

Placental lactogen also functions as mammotrophic hormone.

The adenohypophysis influences the mammary gland directly with prolactin and STH and indirectly by its control of the hormones of the thyroid and adrenal cortex.

Hormones involved in mammary gland growth and functions

Endocrine organ	Hormone	Function related to mammary gland
Anterior pituitary	FSH	Oestrogen secretion from ovarian follicles
	Prolactin	Mammary growth; initiation and maintenance of lactation
	GH	Stimulates milk production
Posterior pituitary	Oxytocin	Milk ejection
Thyroid gland	T3, T4	Stimulates O2 consumption, protein synthesis and milk yield
	Calcitonin	Calcium and phosphorus metabolism
Adrenal cortex	Glucocorticoids	Initiation and maintenance of lactation
Adrenal medulla	Epinephrine, norepinephrine	Inhibition of milk ejection
Ovary	Estradiol	Mammary duct growth
	Progesterone	Mammary alveolar –lobule growth, inhibition of lactogenesis
Placenta	Estradiol	Mammary duct growth
	Progesterone	Mammary alveolar –lobule growth, inhibition of lactogenesis
	Placental lactogen	Mammary growth

LACTOGENESIS AND GALACTOPOIESIS

Lactogenesis

It is the process of differentiation in which the mammary alveolar cells acquire the ability to secrete milk.

Lactogenesis includes two stages:

Stage I occurs before parturition in which alveolar cells differentiate and acquire a limited capacity for milk secretion

Stage II begins shortly before parturition and continues for several days postpartum in which the alveolar cells are able to secrete copious amount of milk components.

Many hormones are required for lactogenesis:

Progesterone of pregnancy blocks lactogenesis. Its secretion starts to decline during late pregnancy and mammary gland becomes responsive to the lactogenic hormones – insulin, glucocorticoids and prolactin.

Mammary alveolar differentiation is stimulated by these lactogenic hormones and secrtory apparatus develop in alveolar cells

The lactogenic hormone insulin stimulate gene expression in alveolar cells and thus required for protein synthesis and induces sensitivity to other hormones

Glucocorticoids induce differentiation in alveolar cells for casein synthesis

Prolactin stimulates transcription of genes for milk proteins

Progesterone inhibits prolactin receptor synthesis and prolactin stimulation of milk protein synthesis. It also inhibits the PRL effect on α-lactalbumin (an enzyme component of lactose synthase) synthesis.

Prepartum milking can initiate lactation – neural impulse to hypothalamus initiate PRL and ACTH secretion leading to mammary cell secretion.

Galactopoiesis

It is the maintenance of lactation.

It involves maintaining the alveolar cell numbers and function

After parturition, milk yield increases in cows, which reaches a maximum in 2 to 8 weeks and then gradually declines, which is due to decline in the number of secretory cells.

If milk is not removed frequently from the mammary gland, synthesis of milk will not persist even with adequate hormonal levels.

Maintenance of lactation requires GH, ACTH (glucocorticoids), insulin, thyroid hormones, prolactin and PTH.

The thyroid hormones influence the milk synthesis and duration of secretion;

Insulin is required for glucose uptake in the mammary gland;

ACTH maintains mammary cell numbers and alveolar cell metabolic activity.

Suckling stimulates the prolactin secretion at a high level along with the other galactopoietic hormones from the adenohypophysis. PRL has an important role in milk secretion; it is released during suckling or milking and peak values are reached 30 min after initial stimulus.

INDUCTION OF LACTATION

Lactation can be artificially induced in dairy cattle by the use of variety of hormones when the mammary gland contain sufficient numbers of alveolar cells.

Subcutaneous injection of a total daily dose of 0.1-mg/kg body weight of oestradiol-17 and 0.25mg/kg body weight of progesterone dissolved in 100% ethanol in divided dose at 12 hours interval for 7 days will initiate lactation in about 60 to 70% of heifers and cows. The percentage of success rate increases to 100% if 5mg of reserpine (a tranquilizer) is administered on day 8, 10, 12 and 14 to increase the prolactin level in these animals. Lactation starts between days 14 and 21 after initial injections of estradiol-17 and progesterone.

ACTH or glucocorticoids in large doses inhibit lactation in ruminants and rats

GH produces a dose-dependent increase in milk yield in dairy cows. The increase in milk yield by GH is produced by *nutrient* partitioning from body tissue requirements to milk synthesis. Recombinant bovine GH (bST) has been employed to increase milk yield in lactating animals. Injection of GH three times a week stimulates short term increase 13 to 18% in the milk production and 15% feed efficiency, whereas reduces the feed intake by 29% in high producing cows.

Injection of low dose of synthetic glucocorticoids stimulates milk yield by 14 to 18%.

Iodocaesin and L-thyroxine have been used to induce milk production. Iodinated casein, (0.5 percent crystalline thyroxine) has been fed to dairy cattle at a rate of 1 to 1.5 g per 45kg of body weight daily, resulting in an increase in milk production of 10 to 30 per cent.

Extra feed must be provided to prevent weight loss and milk production declines abruptly following withdrawal of thyroproteins. Thyroproteins increase the need for more nutrients, reduces body weight and interferes with the ability to withstand thermal stress.

Thyroprotein fed to dairy cows at the peak of lactation stimulates milk production by about 10%, whereas during declining phase of lactation it stimulates the milk production by 15 to 20%. Usually a greater increase in milk production occurs in high producing cows and older cows. The maximal increase usually occurs during the first 60 days of thyroprotein feeding. However, beneficial effects disappear within 2 to 4 months

MILK EJECTION (MILK LET-DOWN)

Milk ejection or milk let down can be achieved by the contraction of the myoepithelial cells when stimulated by oxytocin.

Removal of milk is a *neuroendocrine reflex*.

This process involves activation of neural receptors in the skin of the teat.

Mechanical stimulation of teats like milking or suckling the teats by the young initiates the neural reflex which travels from the teats to spinal cord to hypothalamus to posterior pituitary, where oxytocin is released into the blood. Oxytocin reach the myoepithelial cells of the mammary alveoli and contract them expelling millk from the alveoli

The milk ejection reflex can also be conditioned to stimuli associated with milking routines such as feeding, sight of calf and washing the udder

The milk ejection reflex is inhibited by emotional states like loud noise, pain, excess muscular activity and stress which stimulates the sympathetic nervous system and causes the release of epinephrine which blocks the reflex.

FACTORS AFFECTING LACTATION

Among the essential nutrients of growth and lactation, energy is the most limiting factor because it is necessary for metabolism of all the other nutrients. Adequate supply of energy at the proper time and amount is important in lactation; inadequate energy at the beginning of lactation, when demands are greatest will decrease milk production to a greater extent than energy deficiency at the latter part of lactation. Cows at peak lactation are in negative energy balance, which is desirable for efficient milk production because body fat reserves are used to meet energy requirements.

Changes in protein intake have little effect on milk yield and composition especially milk protein.

Milk fat content is influenced by heredity and it is the most variable content of milk. Milk yield and fat are influenced by diet; diets which produce low ruminal acetate/ propionate ratio (high concentrate diets) causes low amount of milk fat. Fat content of the diet does not influence milk fat content. A minimum of 19% fibre in the diet of lactating animals is necessary for optimum rumen fermentation and to maintain milk fat level. Milk fats are highly saturated and to increase polyunsaturated fatty acid content of milk fat, vegetable oils can be fed in a protected form.

Stage of lactation influences the milk yield and composition. Milk yield continues to increase from parturition for about 4 to 8 weeks and after a peak is attained, milk yield gradually decline. Fat, protein and lactose contents in milk increase slightly as lactation progresses.

Pregnancy reduces milk yield. Ideal dry period recommended for cows is 6 to 8 weeks; shorter or longer dry period will reduce milk yield of the subsequent lactation.

Milk yield increases up to about eighth year of age of cows, which then decreases; larger cows produce more milk than smaller ones but yield does not vary in direct proportion to body weight.

Milking twice-daily increases yield by about 40% than once daily milking; milking three times daily yields 5 to 20% more milk than twice daily yield; four times a day milking may

produce another 5 to 10% more milk. The factors that contribute to more milk with increasing frequency of milking are – less intra mammary pressure, increased stimulation by hormones (PRL), less negative feed back inhibition of milk constituents on cellular synthesis.

Environmental temperature alters milk yield and composition by affecting BMR, food intake, digesta passage, and maintenance requirements of nutrients; this effect depends on breed – Bos indicus tolerate heat better than *Bos taurus*. Holsteins are more tolerant for low temperature and jerseys are more tolerant of high temperature

CHAPTER-35: BIOSYNTHESIS OF MILK

Learning objectives

This chapter explains about constituents and functions of milk.

BIOSYNTHESIS OF MILK

Mammary epithelial cells take precursors for milk synthesis from blood through the basal and lateral membrane and discharge the milk through the apical membrane into the lumen of alveolus.

Mammary alveolar cells synthesize fats, proteins and carbohydrates

Fat first accumulate in the basal cytoplasm of the cell, then move to the apex where the droplets protrude into the alveolar lumen.

Bovine milk fat consists of large proportions of short-chain fatty acids (C4-C16). The fatty acids and glycerol are synthesized in the cytoplasm and the triglycerides are formed in the endoplasmic reticulum of the mammary epithelial cells.

The fatty acids are synthesized from three major sources

Acetate and β-hydroxy butyrate–primary source formed in rumen, transported to mammary gland. Acetate contributes to C4 toC14 fatty acids of milk. In non ruminants acetyl CoA from glucose is the major source for milk fatty acids

Fatty acids of the triglycerides present in the circulatory system as low-density lipoproteins and chylomicrons are another source. They contribute fatty acids C16 and above. Half of milk fatty acids are derived directly from blood triglycerides

Cytoplasmic acetyl CO-A from glycolysis and TCA cycle is the third source

Glycerol for triglyceride synthesis is derived from glycerol 3-phosphate of glycolytic pathway or from lipolysis of triglycerides in the mammary gland.

Major milk proteins are caseins (α, β and κ), β-lactoglobulin and α-lactalbumin.

Lactoferrin and lysosomal enzymes are other proteins present in milk.

Milk immunoglobulins are derived from blood as preformed proteins.

Milk proteins are synthesized on the endoplasmic reticulum; the casein molecules pass to the golgi apparatus for phosphorylation, forms micelles within the golgi vesicles.

Several genetic variants of milk proteins occur; the genetic polymorphism of milk proteins are useful for genetic identity of the animal

Lactose , a disaccharide composed of glucose and galactose is found exclusively in milk. It is synthesised from two molecules of the precursor glucose. One glucose is converted to galactose; lactose synthase catalyses the reaction between glucose and galactose to form lactose.

The *lactose synthase* is composed of two subunits–galactosyl transferase and *α-lactalbumin*.

Lactose is synthesized within golgi vesicles, released in conjunction with milk protein. The golgi vesicles fuse with the cell membrane and release protein and lactose by exocytosis.

During pregnancy progesterone blocks the secretion of α-lactalbumin, a milk whey protein, which is also an enzyme needed for synthesis of lactose.

The major minerals in milk are Ca, P, K, Cl, Na and Mg and they are derived from blood.

Vitamins cannot be synthesized by the mammary gland and they are derived from blood.

FUNCTIONS OF MILK

Milk is essential for growth and development of young animal

It provides energy (lactose and fat) and amino acids to the young ones

Provides antibodies, vitamins and minerals

COLOSTRUM

Colostrum, the first milk produced upon delivery of the newborn, is an important source of essential nutrients to off spring.

Contains lymphocytes and monocytes to protect against exposure to infection

Contains a greatly increased amount of protein, especially casein and globulins – the globulins are mainly immunoglobulins and in cow IgG is predominant

Rich in vitamin A, E, carotene and riboflavin.

However, lactose, vitamin D and iron are all low compared to normal milk.

Colostrum has laxative effect and helps to expel meconium from foetal digestive tract.

Neonates have a limited time (24 – 36 h) in which the immunoglobulins (γ-globulins) can be absorbed through the gut. Thus, feeding of colostrums within this period is important to ensure the presence of antibodies in the newborn. In domestic animals this Ig transfer is important for the neonate immunity because of poor placental transfer of antibodies.

Colostrum is formed in the mammary gland before parturition and it is the first milk secreted after parturition.

Milk Composition

Animal	Water	Fat	Protein	Lactose	Ash
Buffalo , water	82.8	7.4	3.8	4.8	0.8
Camel	86.5	4.0	3.6	5.0	0.8
Cow (Bos Taurus)	87.3	3.9	3.2	4.6	0.7
Cow (Bos indicus)	86.5	4.7	3.2	4.7	0.7
Dog	76.4	10.7	7.4	3.3	1.2
Dolphin	58.3	*33.0*	6.8	*1.1*	0.7
Donkey	88.3	1.4	2.0	7.4	0.5
Goat	86.7	4.5	3.2	4.3	0.8
Horse	88.8	1.9	2.5	6.2	0.5
Pig	81.2	6.8	4.8	5.5	1.0
Rabbit	67.2	15.3	*13.9*	2.1	1.8
Seal	34.6	*53.3*	*8.9*	*0.1*	0.5
Sheep	82.0	7.2	4.6	4.8	0.9

Composition varies greatly among species

Protein varies from less than 1 to 14%, fat from traces to more than 50%, lactose form traces to 7%, ash from 0.2 to 2 % energy from 40 to 330 kcal/100g.

Generally, marine mammals have a high fat and hence high-energy value (seals 40-50%, whales-30%, dolphins-40%) and milk fat content is low in equines-mares-1.9%.

Rapidly growing species like rat and rabbit have high protein content.

Among and within breeds of cattle there are major differences in milk composition which are influenced by genetic selection.

CHAPTER-36: AVIAN REPRODUCTION

At the end of this chapter the learner will be able to understand the following

Features of reproductive tract of hen

Follicular growth

Formation of egg

Ovulatory cycle

Broodiness

Oviposition

Features of male reproduction.

OVARY

Reproduction in birds is characterized by ovoviviparity where the embryonic development occurs for a very short period in the mother's body and complete development occurs outside the mother's body. To meet the needs of the developing embryo and fetal growth, the avian egg contains nutrient materials in the yolk, albumin and shell.

Ovary

Consists of outer cortex containing follicles and inner medulla having connective tissue, blood vessels and nerves

In domestic birds, only the left ovary and the left oviduct are functional; it is attached to the dorsal wall anterior to the left kidney posterior to the left lung

The ovary has distinct follicles protruding from the surface. The largest of the follicles is the most mature and is called F1, the next most mature F2 and so on down to F5. This arrangement of group of follicles is known as follicular hierarchy; each follicle differs from the adjacent by about 1 day growth.

The ovary also contains intermediate sized follicles and many other small follicles which are microscopic.

The immature follicles consist of an oocyte surrounded by follicular cells. The oocyte contains nucleus, cell organelles and surrounded by a membrane. As the follicles mature, yolk material is added to the ovum and the yolk is surrounded by perivitalline (vitalline) membrane secreted by follicular cells. The mature follicles consist of an oocyte and several layers of tissue – vitalline membrane and zona radiate (innermost layers), perivitalline layer, granulosa layer, theca interna and theca externa, connective tissue and epithelium (outermost)

The thecal layers are well vascularized and the thecal and granulosa layers are steroidogenic. The stigma is avascular and ovulation occurs through the stigma. After ovulation, the remnants of the follicles remain for a short time but no CL is formed.

OOGENESIS AND FOLLICULAR GROWTH

Oogenesis

After puberty, individual follicles begin to grow; a small white disk is seen on the surface of each follicle called blastodisk that contains the chromosomes of the ovum.

The first meiotic division is completed before ovulation and when spermatozoan penetrates the blastodisk, meiosis is completed and the blastodisk is called blastoderm; the blastoderm is the embryo that can be seen with naked eye when eggs are opened.

Follicular growth

The smallest visible follicle is whitish whereas the intermediate and large follicles are yellowish due to deposition of yolk by the follicular cells. Up to about 6mm diameter of follicle size, the yolk deposition occurs very slowly over a period of 60 days. Then some of the follicles undergo rapid growth phase to attain a size of 30 – 40 mm in 9 to 11 days while many become atretic.

OVARIAN HORMONES

Steroid hormones are secreted by follicular granulosa and thecal cells. Before first ovulation, ovary contains small white follicles and no yolk filled follicles. Thecal cells of these small follicles are the major source of estrogen and androgens. Granulosa cells of the follicular hierarchy (F5 to F1) are the source of ovarian progesterone which enter blood. Some of these progesterone enter nearby cells and converted to estrogen; this conversion diminishes as follicular hierarchy increases from F5 to F1; the F1 follicles releases only progesterone. The increasing level of plasma progesterone stimulates LH release for ovulation.

LH stimulates granulosa and thecal cell steroidogenesis. The role of FSH in birds is not clearly established; PRL inhibits ovarian steroidogenesis

Granulosa cells of the F1 follicles also produces PG involved in uterine contraction and oviposition; inhibin and activins are also released from ovary – inhibin inhibits FSH and the function of activin is not known.

OVIDUCT

It includes the complete tubular genitalia of the female and has 5 parts – infundibulum, magnum, isthmus, uterus (shell gland) and vagina. The vagina is attached to cloaca.

The anterior end of the infundibulum is a thin membranous tissue with no smooth muscles; in the posterior part it is highly muscular and glandular. It is highly motile at the time of ovulation, engulfs a follicle before ovulation and pick-up the released ovum. The infundibulum is the site of fertilization.

The magnum secretes the albumen (egg white) and is the longest portion of the oviduct. Mucosa contains tubular glands. Albumen is made up of many proteins and serves as a reserve for embryonic growth. The magnum secretion is regulated by estrogen, androgen and progesterone

The isthmus, separated from the magnum by a nonglandular band secretes shell membrane that forms an enclosing sac to hold the egg contents and provides support for the deposition of hard shell.

The shell gland (uterus) adds fluid to the developing egg, secretes hard shell, adds a proteinaceous cuticle over the hard shell and in some birds adds shell pigments. It is thick walled muscular organ, has mucosal folds covered with ciliated and non-ciliated epithelial cells. Tubular glands underlie the epithelium.

The uterus is separated from the vagina by uterovaginal sphincter. The vagina contains tubular glands (Sperm storage tubules) that are capable of prolonged storage of spermatozoa. It functions in transport of spermatozoa and transport of eggs. Sperm is stored for up to 32 days in chicken and 70 days in turkeys.

EGG FORMATION

Approximately 25 to 26 h is required from the time an oocyte is released from the ovary till the egg is released from the body; time taken for the passage of egg in oviduct compartments – infundibulum – 15 min, magnum -3h, isthmus - 1½ h, shell gland - 20 h, vagina - less than a minute.

The infundibulum helps in egg transport and is the site of fertilization; has no role in egg formation

The magnum secretes and stores albumen before the egg formation and releases the albumen as the ovum passes. The stimulus for the release of albumen is mechanical distension due to passage of yolk. Two shell membranes are laid down around the albumen as it passes through the isthmus. The two layers are closely attached except at the blunt end of the egg where air cell is formed after egg is laid.

Fluid is added to the albumen during the first 5 h the egg spends in the shell gland and the albumen volume doubles. The added fluid causes stratification of the egg white.

Shell is secreted during the last 15 h the egg remains in the shell gland; it is made up of calcium carbonate (98%) and glycoproteins (2%); contains pores to allow gas exchange. Outside the shell is a proteinaceous layer – cuticle, which blocks bacterial entry.

OVULATORY CYCLE AND ITS REGULATION

Ovulatory Cycle

The cycle of follicular growth and hormonal changes ending with ovulation is called as ovulatory cycle. The ovulatory cycle of the hen is about 24 to 26h; the cycles may occur for many days without interruption is called a clutch. The continuous ovulatory cycle is interrupted by anovulatory days after which ovulation is resumed. In young high yielding hens, the time required to form an egg is 24h with long clutches; as age advances, the time required to form an egg or interovulatory interval becomes 25 to 26 h with short clutches.

Chicken clutches range from 1 to 30 eggs. About 10 – 20% of the ovulated ova are released into the abdominal cavity and no egg result from these ova.

The first egg of a sequence is laid in the morning; subsequent eggs in this clutch are laid later and later until the last egg in the sequence is laid in the late afternoon. Subsequent ovulation occurs 30 min after oviposition of the preceding egg. Egg formation requires more than 24h hence ovulation and oviposition occurs at later times on successive days.

Regulation

LH is the key hormone of the ovulatory cycle; it induces ovulation. Its level increases 4-8h before ovulation. LH surge occur during the dark period; with 15hL: 9hD, ovulatory cycle 25 –26 h in length, LH surge occur progressively later in the dark period with subsequent ovulation of the clutch. When the LH surge due comes to light period, no LH release occurs.

Progesterone and LH are in positive feed back. F1 follicles release mostly progesterone which induce LH release which in turn stimulates more progesterone synthesis and both hormones reach a peak 4-8h before ovulation and following ovulation their levels fall.

Androgen level also rises along with LH and progesterone, which also stimulates LH release. There is a smaller rise in estradiol levels in plasma. Role of FSH in ovulatory cycle is not known.

Prolactin level may rise when LH and progesterone levels are decreasing which may have a turn off mechanism for LH release.

BROODINESS

It is the incubation behaviour in birds.

Observed in broilers, turkeys but have disappeared in layers.

During broodiness ovaries regress and egg production ceases and broodiness becomes undesirable.

Prolactin is the major hormone responsible for broodiness; its level increases before broodiness which causes ovarian regression and interferes with steroidogenesis.

OVIPOSITION

The act of laying eggs is a coordinated activity.

The smooth muscles of the shell gland contract and the muscle separating the shell gland and vagina relax; the abdominal skeletal muscles contract (bearing-down reflex- change in respiratory rate, stance and abdominal muscle contraction). These activities expel the egg.

The posterior pituitary hormones, avian vasotocin (AVT), PGs especially PGE and PGF released from the ruptured ovarian follicle help in oviposition.

PHOTOPERIODISM

Day length regulates breeding season in many wild and domestic birds.

Sexual activity increases with longer days and decreases with shorter ones. Light period of about 12 h stimulates gonads; short, correctly spaced intervals of light also stimulate the gonads with less than 12h total duration of light.

When the brain is exposed to light 12 or more hours after dawn, it causes the hypothalamus to release the GnRH.

The GnRH stimulates the release of LH and FSH which causes growth of follicles and production of estrogen and androgen from the small follicles.

Estrogens stimulate oviductal development and formation of medullary bone; the androgens cause expression of secondary sex characters.

Egg production follows 2 – 4 weeks after photostimulation.

In general, laying begins at the age of 18 to 20 weeks in chicken and 28 to 30 weeks in turkeys.

The influence of light on the sexual activity is perceived by nonretinal photoreceptors located in the hypothalamus. In many avian species, exposure to short photoperiods during the prepubertal period is essential before they can respond to stimulatory photoperiods. This photorefractory period varies with species. Turkey hens should receive a short photoperiod followed by photostimulation of at least 11h per day for normal laying whereas in young chicken, exposure to light more than 8h per day initiates laying without short exposure to light.

TESTES

Males have internally placed testes located anterior to the kidneys; the left testis is generally larger than the right.

Birds have no seminal vesicles, bulbourethral and prostate glands.

A small structure called epididymis is found with each testis which contains efferent tubules and carries sperms from testis to a single epididymal duct. Each epididymis is continued as ductus deferens that pass along the dorsal body wall and terminates with a receptaculum which ends on a small papilla in the cloaca.

The vas deferens and receptaculum are the storage place for the spermatozoa.

The phallus is small and does not function in intromission. The semen is transferred to the female by touching the phallus to the everted vagina. The sperms taken from the testes are functionally mature and are capable of fertilizing the egg.

Hormonal regualtion

The LH acts on Leydig cells to promote their development and testosterone production. FSH acts on sertoli cells.

Testosterone and LH levels have negative feed back regulation

SEMEN

It is a highly concentrated fluid with low volume.

Average volume is 0.5ml in cocks with a concentration of 4 billion ($4x10^9$) sperms per ml.

The sperm head is narrow and long with a small acrosome.

Poultry sperms are immotile before ejaculation and become motile after ejaculation. The vagina of the hen acts as sperm storage tubules which help to supply viable sperms to fertilize the oocytes for many days.

Spermatogenesis involves mitotic division followed by meiosis to produce haploid cells. The spermatogonium produces spermatocytes and spermatids; spermatogenesis occurs at the core body temperature of 40-41C.

Gonadotropins and gonadal steroids are essential for spermatogenesis

Testosterone produced by Leydig cells under the regulation of LH has a role in spermatogenesis.

The immotile sperms after spermiation (release from seminiferous tubules) are carried by seminiferous fluid through the tubules towards the excurrent ducts of testes. Sperm taken from excurrent ducts have fertilizing ability.

Poultry sperm do not undergo capacitation before fertilization.

Polyspermy is very common

CHAPTER-37: THEORIES OF GROWTH

Learning objectives

To make the students to be aware of different theories explained by different scientists.

GROWTH

Growth is a biological synthesis of new biochemical units. It is the aspect of development concerned with increase in living substance or protoplasm which includes one or all of the three processes,

Cell multiplication, **e.g.** cleavage of egg into daughter cells, formation of blood corpuscles, monocytes, hair follicles, and ectoderm cells.

Cell enlargement **e.g.** nervous tissue and skeletal tissue

Incorporation of material taken from the environment **e.g**. proteins and minerals.

THEORIES OF GROWTH

The theories of growth are based on

Development (Needham)

Growth (Hammet, Brody and Maynard)

Needham's theory of Development

Development refers to the directive coordination of the diverse processes into an adult in an organized heterogeneity.

Some types of growth are not developmental e.g. teeth development, teratoma (jumbles of hair), cancerous growth, growth of tissue cultures in vitro due to self-multiplication without directive significance etc.

Needham subdivided differentiation into

Increase in the number of kinds of cells

Increase in the morphological heterogeneity

Cell multiplication

He subdivided growth into

Intussusception/increase in size of cells

Accretion/increase in the amount of non-living matter.

According to him, metabolism includes

Respiration (oxidation)

Fermentation (glycolysis)

Catabolism of protein

Catabolism of fat

Chemical activity as pigment formation, glycogen synthesis etc.

Hammet's theory of growth

Growth is a coordinated expression of incremental and developmental factors and functions.

Brody's theory of growth

Accoding to him, growth is a relatively irreversible change in the measured dimension i.e. increase in size and weight.

Irreversibility changes exclude fluctuating feed supply with consequent fattening and leaning with gestation, lactation etc.

Maynard's theory of growth

He differentiated true growth and false growth (Deposition of fat in the adipose tissue).

True growth is increase in the size and weight of the structural organs and tissues such as muscle, bone except adipose tissue.

Chemically true growth is characterized by increase in protein and mineral deposition in the body.

Increase in weight due to fat deposition and water accumulation is not considered as true growth.

CELL AS A UNIT OF GROWTH

Growth process involves both increase in the size of cells called as ***hypertrophy*** and increase in the number of the body cells called as ***hyperplasia.*** The process by which a single cell grows from within cell is called as ***intussusception***.

During hyperplasia there is an increase in the DNA and most of the cells contain only one nucleus and each nucleus contains the same amount of DNA. The amount of DNA present in a tissue can be used as an index of number of cells present in the tissue.

During hypertrophy cell size increases but DNA content remains constant. In most of the cells increase in the size of cells causes an increase in cell proteins. Hence the protein/DNA ratio is an index of cell size and growth. RNA/protein ratio indicates capacity for protein synthesis.

The protein/DNA ratio varies with tissue, age, species and other factors. Under optimal conditions and nutrition protein/DNA ratio is maximum.

During a short period of starvation tissue protein level decreases while DNA remains constant.

When feeding is resumed, cellular protein level increases rapidly to reach the previous protein/DNA ratio. This rapid rate of growth after a period of starvation is called as ***compensatory growth.*** This type of growth occurs when part of an organ stops growing or when a part of it is experimentally removed.

During the embryonic life all the cells of the body undergo hyperplasia and hypertrophy, whereas in adults, 3 different types of cell show proliferation

Permanent cells or static organs

These cells lose their mitotic activity after differentiation and stop dividing early in their prenatal life and their number remains constant e.g. muscle cells, nerve cells.

Stable cells or expanding organs

These cells undergo mitotic activity and continue to divide and increase in their number during their growth period but they stop dividing in the adult. Their number is fixed and the cells have prolonged longevity, e.g. all exocrine glands, lungs, kidneys, and most of the organs of the body.

Labile cells or renewing organs

These cells continue to divide through out the life of the animal, e.g. blood cells, epithelial cells etc.

Periods of growth

Growth can be divided into two periods

Prenatal growth

Postnatal growth

CHAPTER-38: PRE-NATAL GROWTH

Learning objectives

To learn the different processes from fertilization to birth of a young one and the factors influencing the prenatal growth.

PRENATAL GROWTH

The period of growth between the time of fertilization and the birth of the young one is called **prenatal growth**. In poultry 21 days are required for the fertilized egg to become a fully functional chick, but in farm animals it takes a long time.

A new individual arises from the union of a sperm and ovum, forming one cell with diploid number of chromosomes. The single fertilized cell divides as it passes down the oviduct but there is little increase in total size. In this cell division, the mother cell produces two daughter cells, which are capable of further division to form new cells.

Somatic cells produced from cell division have a diminishing power to grow by hyperplasia. In highly differentiated cells, the power to produce is lost e.g., mature RBCs entering the circulation do not have nucleus and cannot divide and dies after about 120 days but the bone marrow cells produce the RBCs continuously. But, the differentiated nerve or brain cells lose their ability to divide and there is no mechanism to create new cells.

The developing fertilized egg undergoes a process of differentiation in which the mother cell produces different kinds of daughter cells like brain cells and heart cells is called **cytogenesis or histogenesis** . This process is irreversible in which the brain cells cannot form the egg cells. As the development of embryo proceeds many organs are formed and then differentiation ceases. The organization of various dividing cells into different organs each with a particular structure and function is known as **morphogenesis or organogenesis**. During morphogenesis, the differentiated cells synthesize new proteins and enzymes, which are specific for each tissue formed.

Cleavage is the process by which a zygote or fertilized ovum subdivides into smaller cells called blastomeres. When a new embryo has divided into a ball of sixteen or more blastomeres (but has not yet formed layers of blastomeres) it is called a morula. The morula starts to form various organs from a clump of cells called the inner cell mass. The morula then forms itself into a layer of cells called the trophoblast surrounding a fluid filled space called the blastocoele. The trophoblast develops into the placenta and is lost at birth, whereas the inner cell mass develops into the embryo proper. The inner cell mass

together with the trophoblast forms the blastocyst. The embryo developing from the inner cell mass becomes roofed over by amniotic folds that later fuse to form a complete layer- the amnion. The blastocyst then becomes a gastrula, which is successively and three layered. Ectoderm (primary outer), endoderm (inner) and mesoderm (middle) layers develop. These three primary germ layers give rise to the organs of the embryo.

The outermost layer (ectoderm) forms a mid dorsal ridge along the antero-posterior axis of the blastocyst quite early in the development. The innermost layer (endoderm) forms inner lining of gut, its glands and the bladder.

This elongated ridge of neural ectoderm subsequently gives rise to brain, spinal cord and other derivatives of the nervous system. Ectodermal cells lateral to the neural ectoderm develop into the skin and its derivatives such as mammary glands, hair and hoofs.

The mesodermal layer gives rise to muscles, bones, connective tissue and vascular systems.

The outer (somatic) layer of the mesoderm gives rise to a series of somites (body segments) along each side of the spinal cord. In cattle, the differentiation of somites starts on 19th day of pregnancy the number of somites gradually increases to 25 on the 23rd day, 40 on the 26th day and 55 on the 32nd day. Number of somites can be used to determine the pre-natal growth.

Cell division is very rapid during the first few weeks after implantation than at any other time of life. The rat and rabbit conceptus grow to a weight of l g in 2 weeks, pig and cat in 4-5 weeks and in humans 8 weeks. The rate of growth after implantation determines subsequent growth in uterus. The human fetus remains in the uterus for about 280 days to have a birth weight of 3.4kg. The lamb is born with a birth weight of 4 kg, which is attained in about 148 days, and the foal reaches this weight at 160 days of conception.

FACTORS INFLUENCING PRENATAL GROWTH

Heredity

The ultimate growth of fetus depends on its own genotype and that of its mother and littermates determine the ultimate size of the fetus. There are variations in the species, individuals and breeds. The maternal contribution to variability in fetal size is greater than the paternal contribution.

Foetal hormones

Several hormones are secreted by the fetal endocrine glands. The presence of a hormone does not necessarily indicate that it is actually secreted into the blood and plays a physiological role. As these hormones should have receptors on the target system to bring about any physiological actions.

Pituitary gland

In some species like cow, mare, sheep and goats growth hormone is physiologically active wherein the young ones born are relatively mature with their eyes open and have the ability to stand up immediately after birth. These young ones are called as precocial young ones.

In certain species like dogs and laboratory animals, the growth hormone produced by the fetal pituitary is physiologically inactive and the young ones born are immature with their eyes closed. Even though the fetal pituitary produces most of the hormones they are not important for fetal weight gain but mainly involved in fetal metabolism and parturition e.g. Cortisol.

Thyroid gland

The fetus can synthesize and secrete its own thyroid hormone. This hormone is essential for morphogenesis, differentiation and development of nervous system in the foetus e.g. in the man and rats brain fails to develop when thyroxine is deficient.

Gonadal hormones

The fetal gonadal hormones are involved in the development of both the sexes. In the males androgen produced by the fetal testes is important for the descent of the testes into the scrotum.

Size and age of the dam

Foetal growth is directly related to the maternal size. Larger the maternal size, faster is the prenatal growth. Size of the dam plays an important role in spite of a large sire i.e. if the size of the dam is small the fetus size will also be small, e.g. mules and hinnies. The mule with a large horse as a dam is much larger in size than the ninny with the small donkey as a dam. Further the difference in size also depends on the length of the gestation period. A long gestation means more fetal growth than in species with a short gestation period.

Age of the dam influences the size of the fetus. As the age of the dam increases the fetal size also increases. But in the aged and older animals the young born are small because of excessive internal fat deposition, which prevents the full expansion of the pregnant uterus.

Maternal nutrition during pregnancy

The fetus is really privileged so far as nutrition is concerned.The fetus will continue to grow even if the dam is undernourished but will have a reduced growth rate and decreased birth weight. Under certain circumstances the weight of the fetus is proportional to the caloric intake of the mother, e.g. increased feeding of sheep during the later stages of gestation will have a marked effect on the birth weight of the fetus. Under nutrition or deficiency of vitamins and minerals will induce developmental abnormalities in the fetus.

Poor nutrition of mother affects the glycogen content of the fetal muscle and liver. Fetal glycogen serves as a source of energy immediately after birth and normally builds up during later stage of pregnancy. So when the mother is poorly fed, poor synthesis of liver glycogen may be the cause of more neonatal mortalities.

Restriction of diet of mother produces different effects in different organs of fetus. The nervous system skeletal and heart are least affected, whereas kidneys, lung and muscles are moderately affected. The skin, thymus, spleen and liver are highly affected.

Litter size

Increase in the litter size decreases the birth weight of the fetus, e.g. In polytocus species, increased litter size and in the monotocus species, multiple births reduces the rate of prenatal growth because of variations in the functions of the placenta and the duration of pregnancy.

Sex of the fetus

Male fetuses generally grow at a faster rate and have a higher birth weight than the female fetuses. In case of twins the male is always heavier.

Placental size

The placenta determines the prenatal growth, as it is being regarded as a fetal organ and the placenta transports all the nutrients, which reach the fetus. Growth rate of the fetus is directly proportional to the placental size. Smaller the placenta, smaller will be the fetus, the growth of the fetus will be retarded. Anatomically, placenta acts like a blood reservoir in the maternal circulation. Therefore, the blood flow to the uterus will be lower when there are several placentas in large litters than in a single litter. This will reduce the oxygen supply to at least the more distantly placed placentas. This will affect the growth of the fetus.
Ambient temperature

It has a direct effect on the on the growth of the fetus. Either a high or low ambient temperature will reduce the fetal growth, e.g. exposure of pregnant ewes to heat stress will reduce the fetal growth and the degree of reduction is proportional to the length of exposure.

Further variations between species, strains etc. depend upon genetic set up. In some species, breed differences can be detected even during very early stage, though there is no difference in their egg size.

TYPES OF ORGAN GROWTH

Isometric growth

When an organ grows at a same rate as that of the general body growth it is known as isometric growth.

Allometric growth

When an organ grows at a different rate than the general body growth it is known as allometric growth.

All the body organs follow both isometric and allometric growth during different periods of life. In the prenatal life, CNS reaches its maximum size followed by circulatory system, bones, muscle and finally the adipose tissues. During the postnatal life loins grow first followed by pelvis, thorax, head and finally the legs. This type-if differential growth may be dependant on the functional needs of the body but at different stages of development.

CHAPTER-39: POST-NATAL GROWTH

Learning objectives

To study growth of the young one into an adult, from birth to puberty/maturity and factors influencing the post natal growth.

POSTNATAL GROWTH

It is defined as the correlated increase in the body mass at definite time interval.

Growth curve

Growth generally begins slowly, which then undergoes a period of rapid increase followed by slow down or stagnant of growth. Plotting the body weight of an animal on the 'y' axis and the age of the animal on the 'x' axis produces growth curve. It is normally ***sigmoid shaped*** in all animals and man. The course of growth after birth is almost similar in farm animals. But the juvenile or the prepubertal period is very long in human when compared to that of the animals. In humans,puberty is attained when the body weight reaches 60%, whereas in animals puberty occurs when 30% of the adult weight is reached. There are few exceptions to V curve i.e. organs like gonads and mammary glands are showing cyclic growth.

The growth curve has two phases:

Accelerating phase

It is the increasing slope of the curve, in which the growth rate accelerates to the maximum. Here the steep slope of the curve extends from the beginning of growth until l/3rd to ½ of the mature weight is reached. The following two forces acting on the growth rate which determine the shape of the growth curve:

Growth accelerating force

Growth accelerating force is present in the body cells and is due to hyperplasia, hypertrophy and inclusion of exogenous substances.

Growth retarding or decelerating force

When growth cannot continue definitely due to lack of space or food supply, the growth rate retards from this point onwards and the force that act upon is called as the growth retarding force which is found in the environment surrounding the cells.

During the initial linear phase of growth the two opposing forces are in balance.

Decelerating phase/Retarding phase

It is the decreasing slope of the curve. During this period growth rate declines and ultimately ceases. It is the final phase of growth and it occurs when the animal "approaches its mature body weight. There is an in built restraint on further growth, which progressively reduces the proportion of feed intake. There is stabilization of feed intake and reduction in the increase in body weight until intake equals the maintenance requirement. This may be due to the production of hypothalamic somatostatin.

At the junction of the accelerating and retarding phase, where the growth rate stops to increase and which it begins to decrease it is called as the **point of inflection**. Since in all species puberty occurs at this point it is also known as point of pubertal inflection. This point indicates the time of maximum velocity' of growth, age of puberty, beginning of increasing specific mortality and point of reference to determine age equivalents of different animals. This point occurs at 14 years in humans, 9 - 12 in cattle and 6 -7 months in sheep.

Negative growth phase

In the old age the starts degenerating and this phase is referred as negative growth phase.

Growth can be expressed in many ways

Actual weight or growth curve

Percentage increment method

Weight gained per fixed unit of time

For practical application of growth studies, the first two are used and the last method is used to study the theory of growth.

In actual growth curve, the live weight is plotted against age and it has the typical 'sigmoid' shape. It is very similar in shape for all farm and laboratory animals except for man where it is different due to prolonged prepubertal period.

The actual gain per unit time is always slow at first, the increases to a maximum and slows down gradually again, giving 'bell-shaped' curve.

Growth can be expressed in percentage also. The increase in live weight as percentage of previous weight of the individual is always great at first, falls rapidly as the weight of the animal increase and the curve is opposite in shape to actual growth curve.

In early stages of growth in utero, the developing embryo doubles in size over a 24-hour period. However, the doubling rate slows down and the percentage growth rate falls. The percentage growth is particularly important for comparing the relative growth rates of" different tissues or parts of the body within the same organism. The percentage growth rates differ for each component or tissue of the body and it is the reason for conformational changes occurring during growth.

Growth curves of individual tissues and organs are also sigmoid in shape; however, different tissues develop at different times and individual tissue growth curves cannot be superimposed. The order of development of different organs is skeleton, muscle and then fat.

FACTORS INFLUENCING POSTNATAL GROWTH

Various factors influencing the postnatal growth are

Plane of nutrition

The influence of plane of nutrition is very important because of its relation to economics of milk and meat production. Varying the nutritional status of the animals during various periods of growth can alter the carcass composition of meat animals.

The knowledge of this helps the farmers to control the carcass quality of meat animals in order to increase the market demands. The effect of plane of nutrition depends on the stage of growth. If a low plane of nutrition is fed during the accelerating phase of growth it induces retardation of growth.

All vitamins especially Vitamin B complex, A and C are needed for normal growth.

Milk supplied by the dam

Milk has growth promoting ability in young animals in addition to its known value as a complete food. For this reason mammalian young ones grow at a extremely rapid rate than the birds.

Among the birds, pigeon chicks grow at a faster rate than the others because of crop milk production in pigeons.

Sex of the animal

It depends on the genetic makeup and the sex hormones. Generally males grow faster and reach better rates man females of equivalent ages even if both the sexes are provided with same nutrition.

Nervous System

It controls growth by directly regulating the feed intake via hunger and satiety centers and indirectly by the release of hormones needed for growth.

Litter size

Single have better birth weight than the twins, twins are better than the quadruplets. This difference increases during the first month after birth due to limited milk supplied by the dam. But once these animals are fed with concentrate rations the twins overtake singles, the quadruplets overtake twins. The twins when reared as singles they grow better.

Environmental Factors

Climate influences growth by influencing indirectly the availability of fodder and directly the animal itself. Summer in the tropics offer poor quality and quantity of fodder thereby reducing the growth rate. In the temperate zone fodder is scarce in winter therefore decreasing the growth rate.

Further in animals heat tolerance also play a major role eg., African Short horn cattle when maintained in a hot climate with abundant feed, they failed to grow because they cannot consume feed although it was plenty. Similarly Jersey cattle perform better in the tropics due to small body size and low maintenance requirements and better heat tolerance than the H.F cattle.

Effect of Photoperiodism

Photoperiodic control of growth is most evident in seasonally breeding animals in which the growth and development are seasonally interrelated. There is seasonal variation in sheep, in feed intake, and growth, both are highest in summer and lowest in winter even when fed with concentrates alone. In castrated male the change is less market than the intact ' male.

The pineal gland is involved in the Photoperiodic regulation of growth. In sheep pineal gland involved in many events such as secretion of reproductive hormones and prolactin. These, events are mediated by secretion of melatonin during periods of darkness. When photoperiodic effect on growth is positive pinealectomy reduces this effect. Long day

lengths compared to short day lengths increases serum prolactin and growth hormone levels and increases the growth rate in sheep and goats.

Effect of immune system

Diseases retards growth of animals by directly interfering with metabolism and absorption or indirectly by affecting the appetite of the animal. If the animals are immunized it will prevent occurrence of diseases and help in optimum growth of animals.

Growth regulators

These are substances, which have substantial and direct effect on growth. It includes tissue specific growth factors and certain hormones.

CHAPTER-40: GROWTH REGULATION AND THEORIES OF GROWTH REGULATION

Learning objectives

To make the students to understand about hemorrhesis (regulation of growth) and theories related to hemorrhesis.

DIRECT GROWTH REGULATORS

Tissue specific growth factors

These act as cellular growth regulators by affecting the proliferation and development of specific types of cells.

Insulin like growth factor

Epidermal growth factor (EGF)

Nerve Growth Factor (NGF)

Erythropoietin (EP)

Promine and Retine

Chalones

INSULIN LIKE GROWTH FACTOR

In recent years many efforts have been undertaken to elucidate the complex interactions between mediators of the endocrine system and the immune system. The main effector of growth hormone (GH) is insulin-like growth factor-1 (IGF-1), an endocrine mediator of

growth and development under physiological conditions. Besides this important function, IGF-1 also plays a prominent role in the regulation of immunity and in ammation.

Insulin-like growth factor (IGF-I, IGF-II) action is influenced by until today known eight forms of insulin-like growth factor binding proteins (IGFBPs). They have been obtained not only from some human and animal tissues and body fluids but also from conditioned medium of cell cultures. An important biological property of the IGFBPs is their ability to increase the circulating half-life of the IGFs. They are able to act as potentiators of cell proliferation. As IGFBPs bind to cell surfaces, they may act either to deliver the IGFs to those surfaces for activation of specific receptors or to activate cell responses independently of receptor activation. Phosphorylation, glycosylation and proteolysis of IGFBPs influence their affinity to IGFs. The IGFBPs in the role of inhibitors may block the activity of the IGFs and be used for antimitogenic therapy. In the last time measuring of IGFBPs levels can be used for diagnosis of determination of some endocrine diseases or in differential diagnostics.

High-affinity, soluble IGFBPs are specific proteins able to form complexes with IGFs in extracellular and interstitial fluids of live organisms. They influence IGFs transport to receptors on cell surfaces and also IGFs effects on cell proliferation. IGFBPs are proteins with different size, they are produced by many different tissues, especially by the liver and they bind to IGF-I, IGF-II, but not to insulin (with the exception of IGFBP-7).

IGFBPs modulate IGFs effects by endocrine, paracrine and autocrine mechanisms of regulation. The origin of this regulation may depend on IGFBPs recognition such as glycosylated binding protein or posttranslation modification such as phosphorylation. IGFBPs modify the IGFs effects by regulation of IGFs transport, IGFs concentration in specific tissues, IGFs interaction with cell surface receptors, potentiation or inhibition of IGFs function.

Insulin and insulin-like growth factors (IGFs) are well known as key regulators of energy metabolism and growth. There is now considerable evidence that these hormones and the signal transduction networks have important roles in neoplasia. Epidermiological, clinical and laboratory research methods are being used to investigate novel cancer prevention and treatment strategies related to insulin and IGF signalling.

Pharmacological strategies under study include the use of novel receptor-specific antibodies, receptor kinase inhibitors and AMP-activated protein kinase activators such as metformin. There is evidence that insulin and IGF signalling may also be relevant to dietary and lifestyle factors that influence cancer risk and cancer prognosis. Recent results are encouraging and have justified the expansion of many translational research programmes.

Insulin-like growth factor 1 (IGF-1) has been shown to rescue the aging-related or inactivity-induced loss of muscle mass through the activation of satellite cells. However, the signalling pathways and the mechanism by which IGF-1 affects satellite cells have not been not completely identified.

peptide included in the insulin superfamily, as its structure and function are the closest to those of insulin and IGF-I. The last decade investigations revealed the biological properties of IGF-II which distinguish it from related peptides. The primary sequence of the IGF-II structure has peculiar differences from those of insulin but insignificant ones from IGF-I. The tertiary structure of IGF-II is similar to that of

the related peptide molecules, but a peculiar receptor-binding domain in the IGF-II molecule provides for its unique capability of interacting with receptors.

EPIDERMAL GROWTH FACTOR (EGF)

It was first isolated from the salivary gland extracts of mice. It is found in significant amount in the submaxillary gland of male mice. Injection of saliva of male mice into new born mice results in

Precocious opening of eyelids as early as seven days; instead 12-14 days.

Precocious eruption of teeth (incisors) at 6-7 days instead of 10 days.

EGF acts directly on the epidermis rather than through secondary control mechanisms. It stimulates epidermal cell proliferation and keratirdzation

EGF was discovered by Cohen and subsequently his group expanded our understanding of EGF and its receptor EGFR.9 EGFR is a 170 kDa membrane glycoprotein which is activated by multiple ligands including EGF, transforming growth factor (TGF)-a, heparin binding (HB)-EGF, amphiregulin, b-cellulin and epiregulin, all of which are synthesised as transmembrane precursors (proligands) that are inserted in the cell surface and cleaved by proteases to release the mature soluble growth factor. The EGFR signalling pathway has been known to be involved in a variety of physiological responses including proliferation, differentiation, motility, and survival.

EGF (Epidermal growth factor) is found in varying concentrations in milk, saliva, urine, plasma and also in most other body fluids. Cells in various organs, including brain, kidney, salivary gland, and stomach, produce this factor. EGF is a globular protein of 6.4 kDa consisting of 53 amino acids. It contains three intramolecular disulfide bonds essential for biological activity. EGF controls and stimulates the proliferation of epidermal and epithelial cells, including fibroblasts, kidney epithelial cells, human glial cells, ovary granulosa cells, and thyroid cells in vitro. The proliferation of some cell lines have been shown to be inhibited by EGF. EGF is a strong chemoattractant for fibroblasts and epithelial cells. EGF alone and also in combination with other cytokines is an important factor mediating wound.

NERVE GROWTH FACTOR

Nerve Growth Factor (NGF)

It was discovered in two types of mouse tumours which enhanced growth and differentiation of sympathetic ganglia in the chick embryo. It is also found in small amount in tumour tissues of numerous vertebrates, in mouse and in human serum and in sympathetic ganglia of several mammals.

In several animals administration of antiserum to NGF causes complete atrophy of sympathetic ganglia without affecting other nerves or organs. It also stimulates amino acid incorporation and RNA synthesis.

Neurotrophins are critical for the development and maintenance of the peripheral and central nervous system. These highly homologous, homodimeric growth factors control cell survival, differentiation, growth cessation, and apoptosis of sensory neurons.

NGF can be obtained in large quantities from the mouse submandibular salivary gland,Several findings obtained in recent years suggest that NGF, aside from its well-established function as a neurotrophic factor for peripheral sympathetic and sensory neurons, also has trophic influence on the cholinergic neurons of the basal forebrain.

NGF is able to affect survival of central cholinergic neurons after axonal transections in adult rats.

Erythropoietin (EP)

It is a hormonal substance circulating in the blood, which stimulates the formation of RBCs. The production of EP increases in response to anemia, hypoxia, etc., The kidney is important in the production of EP but is also produced by other tissues. EP is present in all species and its action is not species specific.

It stimulates the synthesis of RNA, hemoglobin and several enzymes in erythroid cells. It facilitates the entry of iron into bone marrow before the stimulation of hemoglobin synthesis, and it increases the release of erythroid cells into the circulation.

Promine and Retine

Certain extracts of calf thymus glands inhibit malignant growth in tumour bearing mice while other extracts promote growth. The growth inhibitor is called retine and the growth promoter is called promine. These materials are widely distributed in nature. Retine is present in the thymus gland, muscle, tendon, urine and in large blood vessels.

Especially high concentration of retine is found in mushroom and clams. Promine and Refine are present together in tissues. The mechanism by which they influence growth of cancer cells is not known, They elicit their individual stimulatory and inhibitory effects On the cells and are important in the regulation of cell division.

Chalones

These substances selectively inhibit the mitotic activity of specific tissues. These are extracted from the liver, kidney, epidermis and granulocytes. These mitotic inhibitors are tissue specific and not species specific.

They bring about their effect in the presence of epinephrine. Epinephrine increases mitotic inhibition in the "epidermis in the presence of chalones.

DIRECT GROWTH REGULATORS

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THEORIES OF CONTROL OF GROWTH

In animals cell division is regulated by sensitive control mechanisms resulting in harmonious growth of the body and organs of young animals and the maintenance of body and organ size within narrow limits. Several theories on the specific control of organ growth have

been proposed and the fundamental difference between these theories has been whether cell division is regulated by inhibition, stimulation, or by a combination of both.

Bullough Theory (1964)

He worked on chalones and proposed that only tissue specific inhibitors control the mitotic rate in organs. He stated that it is the nature of cells to prepare for and undergo mitosis whenever possible and it is opportunity, not stimulus that is needed for cell division.

He suggested that high concentrations of chalones suppress the genes responsible for mitosis and permit the genes for tissue function to be active. Decreased chalone concentrations release the mitotic inhibition allowing mitosis to proceed. According to this theory, regulation is by inhibition only.

Weiss and Kavanau Theory (1957)

This theory proposes that a balance of stimulatory and inhibitory factors regulates organ growth. According to this theory, each. cell type produces key compounds characteristic of itself called as 'templates' which stimulate cell division in a particular organ.

Growth rate is proportional to the concentration of these intracellular 'templates' present in the cell under normal conditions. Each cell also produces 'antitemplates' which inhibit the templates by forming inactive complexes. These antitemplates may diffuse out of the cell into the circulation. Their, total concentration is proportional to the amount of organ mass . As the concentration of antitemplates in the circulation increases, intracellular concentration likewise increases, inactivating the templates and causing a decreased organ growth.

When a stationary equilibrium between intracellular and extracellular concentrations of antitemplates is reached , organ growth will cease, but it will maintain its size through a balance of these stimulatory and inhibitory factors.

CHAPTER-41: CONTROL OF GROWTH BY HORMONES

Learning objectives

To learn about the hormones and growth factors involvement in growth regulation.

HORMONAL CONTROL OF GROWTH

Probably all hormones directly or indirectly influence growth by altering biochemical reactions and many of them influence the size of specific tissues and organs. However, only somatotropin, thyroxine, sex steroids and glucocorticoids are generally considered to have a direct effect on whole body growth. These hormones affect body mass and dimensions primarily by altering skeletal and or nitrogen balance.

Regulation or control of growth is known as homeorhesis. It is the co-ordination of the metabolism in the body tissues in support of a dominant physiological process with the

partitioning of nutrients to support developmental process. It is a long term control brought about by growth hormone and prolactin which acts as coordinator of nutrient partitioning.

Somatotropin (Growth hormone)

It is secreted by the anterior pituitary and is essential for normal growth in young animals and for the maintenance of normal metabolism in adult. Removal of anterior pituitary in the young animals results in dwarfism- cessation of growth and decrease in body weight. Long-term injection of growth hormone in young animals produces extremely large individuals-giants, and in mature animals head, hands and feet become enlarged-acromegaly.

Growth hormone has both anabolic and catabolic effects.

Anabolic Effects

It enhances the formation of protein and nucleic acids by stimulating the ribosomal RNA synthesis.

It increases the uptake of amino acids into the cells and exerts a direct effect on skeletal muscle growth, liver, heart, fibroblasts, lymphoid organs etc.

It controls nutrient partitioning during growth and lactation.

Exogenous administration of Growth hormone to lactating cows increases milk yield, lactose, protein, and fat content of milk.

Catabolic Effects

Lipolytic effect is produced through its direct effect on the adipose tissues. It reduces the amount of fat stores, increases free fatty acids in the circulation, which are released from the adipose tissues and are used as a source of energy.

Injection of growth hormone to normal growing animals increases carcass protein and water while it reduces the fat content.

Release of GH is regulated by GHRF (somatocrinin) and GHIF (Somatostatin) from the anterior pituitary.

Somatomedins

Major effects of GH are mediated via somatomedins. These are polypeptides also called as Insulin like Growth Factors (IGF) of which two are important, IGF I and II. They are produced from the liver and also from the kidneys/ other tissues of the body. These are present in high concentrations in the blood and have hypertrophic (insulin like) and hyperplastic properties.

IGF I not only mediates GH effect but also have direct effect on many cells to promote growth. IGF II is mostly functional during the fetal stage.

Thyroid hormone

Thyroid hormones are essential for normal growth and development in animals indicating that they pal a permissive role in growth regulation. It is essential for maintenance of BMR. Growth stimulation by thyroid hormone is brought about via regulation of somatomedin receptors.

Hypothyroidism decreases the BMR and the ability of the animals to respond to GH. Decrease in appetite leads to decreased growth rate. Hyperthyroidism causes weight loss, general catabolic changes including increased nitrogen excretion and increased BMR.

In swines feeding of thyroprotein (iodinated casein) increases the body weight. But after fattening begins in growing pigs, thyroprotein retards the rate of gain and decreases the feeding efficiency.

Thyroprotein stimulates wool growth in sheep and milk production in ewes resulting in heavier lambs.

Feeding of thyroprotein to dairy cattle stimulates milk production by 10-25% and fat content by 0.2-0.3%. It should be fed during the first 3-4 months of lactation and additional feed; is necessary to support increased production.

Goitrogen in moderate doses have limited uses in increasing the feed efficiency during 3- 4 weeks of fattening in swines. Higher doses leads to decreased feed efficiency and weight gain.

Glucocorticoids

Glucocorticoids have a catabolic effect. It decreases the growth of epiphysis by decreasing the stimulatory effects of GH on the bone.

It increases the protein and amino acid degradation and inhibits protein synthesis in the liver.

When given in large doses inhibit growth in adult but not in young animals.

Sex steroids

Androgen

Males tend to grow faster than the females and this is directly related to androgerr secretion, hence androgens are classified as anabolic or growth stimulatory compounds. The chief source of androgens is the interstitial cells of the testes and die adrenal gland. Testosterone is the chief androgenic steroid secreted from the testes.

Testosterone has a slight stimulatory effect on growth and an inhibitory effect on epiphyseal fusion. Castration of males inhibits skeletal growth. Androgens increases the protein synthesis, nitrogen retention and enhances skeletal muscle growth. Small amount of testosterone stimulate GH to produce maximum growth response. After puberty large amounts of androgens secreted enhances epiphyseal closure.

Androgenic compounds improve the carcass quality by decreasing the amount of fat and increases the proportion of edible tissue especially in swine. In beef cattle limited beneficial effects of moderate doses of androgens include increased nitrogen retention and a lower carcass fat content. Females respond more readily than the males to androgen administration.

Estrogen

The major estrogens which are produced by the ovary, placenta and to a smaller extent by the adrenal cortex of both the sexes are estradiol, estrone and estriol. Estrogens are somatic growth inhibitors because sterilization of female results in greater growth than would otherwise occur and sizable doses of estrogens depress growth of male.

Estrogens inhibit skeletal growth by decreasing the growth of long bones. After puberty it promotes rapid epiphyseal closure. At high doses it increases nitrogen retention in ruminants and fat deposition in poultry. Its catabolic effect is due to the stimulation of secretion of corticosteroids which have catabolic properties.

It promotes growth of sex organs. Stimulates the growth of uterine tissue by hypertrophy and hyperplasia. It increases the mitotic rate in the skin.

Other supportive hormones

Insulin

It is secreted by the beta cells of islets of langerhans of the pancreas. It is an anabolic hormone. It increases the uptake of amino acid and glucose into the muscle cells, stimulates the growth by increasing the RNA synthesis and protein. It plays a supportive role rather than a direct effect in influencing growth. It promotes lipogenesis and is positively correlated with the adiposity of the body and thus helps to maintain the body weight. It regulates the binding of somatomedins to adipocytes.

Prolactin

It plays a role in partitioning of nutrients for growth and act as a homeorhetic control. It decreases fatty acid synthesis and increases lipolysis.

EFFECT OF FEED ADDITIVES AND OTHER GROWTH FACTORS

Synthetic hormone like compounds

Synthetic estrogens e.g. Diethyl stilbestrol (DES) and progestogens increases production especially in fattening of beef cattle. It is effective in females, castrated males and in growing bull.-. DES in ruminants is used for greater nitrogen retention, muscle growth and water uptake.

Mechanism of action: It stimulates ACTH secretion from the pituitary and this in turn elevates adrenal androgen secretion which stimulates growth.

Naturally occurring growth factors in feeds

Some plant materials are found to produce estrogenic effects in animals and their amount varies considerably in different plants, e.g., alfalfa and clover contain appreciable quantities of estrogen and produce positive growth response in ruminants. Alfalfa seeds, soyabean, cotton seed and linseed oil meals contain estrogenic compounds which are not usually detectable in grasses.

Normally occurring growth inhibitors are also found in livestock feeds. E.g., trypsin inhibitor in soyabeans and gossypol in cotton seeds. These act by interfering with the availability of other nutrients. Other compounds like alkaloids and specific toxins reduce the growth of farm animals under certain conditions.

Antibiotics

These are used extensively as growth stimulants especially in poultry and swine production. These substances increases growth and feed efficiency but do not alter mature body size of animals. They are effective in eliminating subclinical infections and diseases and thus create a better environment for rapid growth. They act by inhibiting the bacterial growth which compete for nutrients. Some antibiotics act by increasing the absorption of ^{v} vitamins. E.g., bacitracin, erythromycin, penicillin, oxytetracyclin, furazolidone etc.

Anthelmintics

These are used as growth promoters. They control parasitic infections. E.g., DDVP (2,2-dimchlorovinyl dimethyl phosphate) is used to stimulate growth in fattening cattle. DDVP has got both anthelmintic and growth stimulatory effect in cattle.

Tranquilizers

Feeding of. tranquilizers sometimes causes a slight increase in weight gain and feed efficiency in fattening beef and sheep and has no beneficial effect in other animals e.g. Receptive, chlorpromazine.

Paracrine and autocrine growth factors

Growth is affected by growth factors produced locally in a particular tissue. These have autocrine and paracrine effect e.g., IGF I , fibroblast growth factors, transforming growth factors etc.

Fibroblast growth factors are. heparin-binding polypeptides. They stimulate proliferation of myoblasts but inhibit differentiation of myogenic cells, stimulate proliferation and inhibits differentiation of preadipocytes,

Among the transforming growth factors (TGF) , TGF- Bl and TGF-B2 are more abundant.. TGF B causes differentiation and proliferation of cells of epithelial and mesenchymal origin and inhibits differentiation of muscle cells and adipocytes and plays a role in the development and growth of the bone.

CHAPTER-42: MEASUREMENT OF PRE AND POST-NATAL GROWTH

Learning objectives

To study the different methods available to measure the prenatal and postnatal growth.

MEASUREMENT OF PRENATAL GROWTH

There are certain measurements that can be made on the foetus to determine the age or growth rate of the fetus. They are

CR Length - measured from the highest point of skull to the rump

CVR Length- curved crown to rump length.

VR Length- length of the vertebral column

BCVRT Length- total bent length of the fetus i.e. from mouth to the point of the tail.

VRT Length- vertebral column to the tail.

CR length provides a more reliable measure to estimate the age of the fetus.

Foetal length can also be measured by radiography or ultrasound technique.

MEASUREMENT OF POSTNATAL GROWTH

Live weight

Measuring the live weight is the most widely used technique in determining the growth of the animals. It is useful for computation of rations and to make adjustments in the feeding

regimes and to determine the end point of production cycle. In experimental work, it is used to find out the response of a particular treatment.

Recording of live weight is easy and economical. The validity of recorded live weights depends on 3 factors

precision of the weighing machine

human error

fluctuations in weight due to the quantity of food present in the GI tract eg. In growing pigs the proportion of live weight to the content of the GI tract may vary between 0.02 for 90kg pig and 0.05 for 120kg pig. Inorder to reduce the influence of GI tract contents on the live weight fasting of the animals before weighing can be followed or it can be taken at the beginning of the day.

Body measurements

Linear body measurements are taken using various types of measuring rods or sticks or with calipers. These measures reflect primarily the length of the long bones of the animal.

Measurement of circumferences is done using a flexible tape.

These body measurements when taken over a period of time predict the animal live weight and carcass composition. Mostly they are used in cattle and horses and to a smaller extent on sheep and very much limited to pigs and poultry. The accuracy of such measurements also depends on a no, of factors such as,

Correct identification and location of end reference points in linear measurements.

Anatomical distortion produced by the animal changing either position or posture or by changing muscle tone.

Minimum error for caliper measurements whereas greater error when using flexible tape over surface particularly if the are, concave. The body measurements which exhibit the highest correlation with body weight is heart girth.

Visual Appraisal of live animal conformation or Condition Scoring

This method uses the sense of touch and sight in assessing the degree of finish of cattle, sheep and pigs to meet certain market requirements. This method reflects the deposition of soft tissues, in particular fatty tissues in the animal. Many systems of scoring are available but they are of limited value when the difference between the animal is small. This technique reflects a bias towards fatty tissue growth and is of little use in predicting growth in animals where muscle accretion rates are of great greatest importance. It is useful as a management tool to formulate feeding regimes

Dilution technique

It is used to measure water and indirectly by difference estimates fatty tissue because water is inversely proportional to fat: Dilution of water is estimated by introducing a known amount of tracer into the animal.

The tracer is allowed to equilibrate with body fluids and the concentration of the tracer is determined in a sample of blood. Tracers used are either liquid or gaseous forms. E.g. of gaseous tracer, krypton. If gaseous tracer is used, the animal is enclosed in a chamber and the amount of gas given which gets absorbed into the fatty tissue is measured. Liquid tracers used commonly are heavy water (D20) and tritiated water (TOH).

The disadvantages of this technique are

it is difficult to apply

it can be used only as tools for experiment only

many tracers are radioactive and has a long half life as a result the animals carcass at slaughter is unfit for human consumption and has to be wasted.

Neutron Activation Analysis

This technique is used in human medical field and is costly. It requires whole body counting chambers. This method allows the measurement of gamma radiation or photon production, from neutron irradiation in H2, N2, 02, Na, Cl, and Ca and in the naturally occurring radioactive isotope of potassium (^{40}K) and is used in the predictions of muscle content.

Probes

It is mostly used in pigs to measure back fat thickness and has a high precision in predicting the muscle back fat thickness.

The demerits of this method are that local anesthesia has to be induced in the areas where the skin incisions are made for the insertion of the probe.

Balance Studies

It gives a short-term estimate of protein and fat deposition in the animal body. A balance sheet is drawn for a particular element entering and leaving the body and the retention of that element is calculated by difference.

A combined carbon and nitrogen balance give an estimate of storage of both protein and fat in any particular period of time. The animal must be in a positive balance i.e. the excreted amount of carbon and nitrogen must be less than the intake.

Demerits of this technique are that it is expensive and requires sophisticated equipments. It is not possible to predict accurately either the total muscle deposited or its location within the animal. Fat and protein deposition may be either in the carcass or in the non carcass tissues or in both.

X-ray and computed tomography

This method is used to measure growth, tissue mass or volume. It is determined by serially scanning the body and by feeding the details of the pictures obtained into a computer which integrates the series into a whole (computerized tomography-CT).

CT is a valuable invivo method for predicting body fat and energy content but less valuable for predicting the contents of water and protein in the body.

Nuclear magnetic resonance (NMR)

This technique examines proton distribution and binding in the body. NMR-CT relies on the inducion of resonance in protons in the body by placing it in a strong magnetic field.

The signals that are emitted are a reflection of the body's reaction to the high frequency disturbance that the magnetic field induces and are product of the matter of the body itself. This technique is non-invasive, non-ionizing radiation, no hazards, and electromagnetic radiation penetrates bone tissue without significant attenuation.

Ultrasonic technique

In this technique pulses of ultrasound pass through different tissues at different rates, specific for particular tissue. An interface between two tissues partly reflects and partly transmits the ultrasound pulse.

The sound pulses are converted to electrical pulses which are displayed on suitable oscilloscopes as spikes on a baseline. It helps to measure the back fat thickness which is closely correlated with carcass composition and is more useful in pigs than cattle.

CHAPTER-43: ENVIRONMENTAL PHYSIOLOGY

Learning objectives

- To make the students to know about different terms related with environmental physiology.

ANIMAL ECOLOGY

- Ecology is the branch of biology that deals with the relation of living things to their environment, their habits and their mode of life. When concerned with the domestic animals, ecology includes their relationship to human society and economy and in particular their productivity.
- Ecology is divided into two branches
    - Autecology/species ecology/self ecology
        - It is a study of interaction between a single species with its environment. It is primarily experimental.
    - Synecology/population ecology/biocenology/biosociology
        - It is the study of group or community of organisms and their relationship with each other and to their common environment.
- *Environmental physiology* deals with the surrounding conditions that affect the structures and organ functions of humans and animals. Environment includes physical, chemical and biological elements that surround the animals and is therefore included in the study of animal ecology. The components of the environment either promote or impair the performance of the animal.
- *Heredity and environment*
    - Animals are the products of heredity and environment. Heredity pattern is a product of environment as the environment conditions mutation and selection of survival rate. This makes the animal to adapt to different climatic regions, e.g., animals evolved in colder regions are adapted to cold and have abundant wool and subcutaneous fat. Animals adapted to hot weather have sparse wool and subcutaneous fat. The environment of an animal affects the degree of the genetic potential determining the optimum climate and environment required for maximum production of the animals.
- *Bioclimatology* deals with the inter relationships between climate, soil, plants and animals.
- *Weather*
    - It is a short term day to day fluctuations of the meteorological variables.
- *Climate*
    - It is a long term average condition of the meteorological variables, in a given region.
- *Macroclimate*
    - It is the general large scale climatic conditions of the open atmosphere in a large area or a country.
- *Microclimate*
    - It is the climatic condition directly surrounding the animal.

CONTRIBUTION OF CLIMATE TO AGRO-ECOLOGICAL ZONE

- There is a closer relationship between climate and vegetation. Classification of climate helps to recognize relationship between climate and land forms and hence serves as a guide to the introduction of pasture plants and crops that provide forage for grazing an imals, recognizing relationships between climatic stress and functional and reproductive responses of domestic animals. Climatic factors such as rainfall, air temperature, humidity, solar radiation and day length exert a controlling influence over the formation of the main agro -ecological zones.
- Classification of climate is important for adaptation of domestic animals to specific climatic zones. Climatic factors influence the distribution and growth of plants and those which directly influence the animals. Mainly temperature relative humidity and rainfall d etermine the distribution of livestock.

DISTRIBUTION OF DOMESTIC ANIMALS

- Ruminants are the most important domestic animals useful to man. Reindeer is mostly confined to a narrow belt in the extreme Northern areas of Europe, Asia and North America.
- Ilamas and alpacas are also limited in their distribution to the elevated plateau regions of South America.
- Buffaloes are exclusively confined to hot areas particularly those associated with paddy production. They are also found in the drier areas where there are irrigation facilities.
- Camels are distributed to the hot deserts, semi arid areas and in dry cold regions.
- Horses are found in the continents, which have cooler and temperate climate.
- Donkey predominates in hotter and more arid areas.
- Cattle and sheep are the most widely distributed livestock. Higher proportion of sheep is found in pastoral continents of Africa and Australia. Cattle are found in Europe, North America and Asia.
- Pigs are found in the forests of Northern Europe, Asia, and in semiarid Australia.
- Poultry is found in Africa and Asia
- Goats are distributed in the grasslands and semi-desert regions.

CHAPTER-44: ADAPTATION, ACCLIMATION AND ACCLIMATIZATION IN ANIMALS

Learning objectives

- To study about the day to day, short term and long term physiological adjustment of animals to the extreme environmental conditions.

ADAPTATION

- It is the adjustment of an animal to given environment. When the animals are continuously exposed to major environmental changes, they may develop functional and structural changes that results in an increase in their ability to live without stress in a new environment. This is called as adaptation.
- There are different types of adaptation:
-

- *Genetic adaptation*
  - It refers to the heritable animal characteristics that are transformed from one generation to the other, which favour survival of a population in a particular environment. This may involve evolutionary changes over many generations (selection by nature) or acquiring specific genetic properties (selection by man).
- *Physiological adaptation*
  - It is the capacity and process of adjustment of the animal to itself, to other living material and to its external physical environment.
- *Biological adaptation*
  - It refers to the changes with respect to morphological, anatomical, physiological, biochemical and behavioral characteristics of the animal which promote welfare and survival of the animal in a given environment.
- *Phenotypic adaptation*
  - These are the changes that occur during the life time of an individual. These are non heritable and it is the irdividuals response to the environment with accompanying changes in the genotype. Phenotypic adaptation can be physical as in callous formation on the hands and feet or behavioral as in taming of wild animals.
- *Nutritional adaptation*
  - It is associated with both genetic and physiological adaptation. It occurs as a result of climatic and ecological changes. For instance, in the coastal areas of the tropical regions the soil and water are low in calcium and due to heavy monsoon rains the soil is leached out with a loss of calcium. This is reflected in the pastures with low levels of this mineral.
  - The low calcium intake by the animals lead to low serum calcium concentrations, but the tropical animals appear to be tolerant of this and surfer no ill effects.

Acclimatization

- It refers to long-term adaptive physiological adjustments, which results in an increased tolerance when exposed to continuous or repeated climatic stressors (normally produced under field conditions). E.g., if an animal voluntarily migrates from a mountain valley to a high altitude, its lung ventilation rate typically will increase initially to acquire adequate oxygen.
- Within few days or weeks, lung ventilation begins to drop back towards the sea level rates as other physiological mechanism that facilitates gas exchange at high altitude begins to operate. After several days the individual is said to be acclimatized to a new high altitude conditions.
- If an animal physiologist places that same animal in a hypobaric chamber, thus simulating high altitude conditions the animal becomes acclimated to experimental "conditions in a few days.
- Characters of well adapted animals:
  - There must be minimum weight loss when exposed to some stressors like nutritional deficiency, high milk production even when animals are transported.
  - It must have high reproductive rate.
  - It must have high resistance to diseases.
  - High longevity and low mortality.

Acclimation

- It refers to the adaptive changes that take place in response to a single climatic variable.(normally produced in a climatic chamber).

NATURAL ADAPTATION TO ENVIRONMENT

- Desert animals have relatively long legs which enable them to traverse long distances in search of food, camels have large flat foot which enable them to move over the shifting sand. Animals adapted to cold climate have relatively small surface area relative to body mass (European cattle). Animals adapted to warm climate have large surface area relative to body mass (Zebu cattle). Coat thickness and hair characteristics in mammals and plumage in birds are also influenced by climate.

Natural - Morphological, Anatomical, and Functional adaptations of animals

S.No	Environmental Conditions	Adaptive Mechanism	Species
1	Solar Radiation.	Long limbs, and short reflective coat.	Camel
2	High Temperature	Increase in surface area by the skin folds, small body, loose coarse wool, long cars, hair shedding in summer	Tropical cattle, sheep and donkeys.
3	High Humidity	Decreased hair coat, Dark pigmentation.	Buffalo
4	Cold Temperature ,	Long hair .intermixed with fine hair, Minimum exposed ' extremities, increased s/c fat, abundant brown fat, thickest heavy coat.	Temperate cattle, sheep, Yak.
5	Desert	Thick skin, increased water drinking capacity, conservation of metabolic water, hard tissue and thick skin around the mouth.	Camel
6	High Altitude	Increased oxygen carrying capacity in blood through increased concentration of RBC. Ability to transfer oxygen from capillary blood tissues cells at less pO2. High efficiency in extracting nutrients from feed.	llamas, Alpaca.

Adaptation of cattle

- Animals are transported from temperate to tropical zones and from semitropical zones to cold environment.
- Transportation does not change the inherent genetic characteristics of the animal for adaptation to new environment.
- Indian cattle are more heat tolerant than the European cattle
    - Indian cattle have long dewlaps and sheaths (naval flap in females) and long ears have excellent heat dissipation mechanism. These animals have smaller body, resulting in larger surface area per unit weight and their skin has little hair whereas, European cattle have fat, hairy, tight hides, under dewlaps and small ears.
    - In zebu cattle, the hump in high in fat less blood supply and has less sufficient fat than European breeds. Storage of fat makes zebu more heat tolerant.
    - Indian cattle have more extensive system for evaporative cooling and allows them to adapt better to hotter climates.
    - Indian cattle are more resistant to ticks, flies and mosquitoes.
    - Hair colour of Indian cattle is lighter and thus reflects more sunlight than that of European cattle.
    - Cold tolerant animals have lower critical temperature for European cattle it is between 30-60°F, Indian cattle - 50-80°F. Therefore above 60°F European cattle and above 80°F in Indian cattle, thermo regulatory mechanisms become active - Increased in respiration and evaporation.

THEORIES OF MORPHOLOGICAL ADAPTATION

- Some ecological rules explain morphological adaptations in domestic animals. They are summarized by Brody
    - **Bergman's rule** (1847): It relates body size to climate. Small light animals have large surface area per unit weight and hence lose heat more rapidly than large heavy animals, which neve small surface area per unit weight. Larger breeds of cattle (Holstein) are better adapted to cooler areas and smaller (Jersey) breeds are better adapted to warmer areas. The larger Brahman cattle inhabit hot areas and the smaller Shorthorn cattle inhabit cooler climate. The reasons for the variation is that the Brahman cattle have larger periphery - ears, dewlap, naval flap and so large surface area, which increases heat dissipation and these animals, are surface area is smaller in relation to their body weight and so they are cold tolerant The Bergmann rule is based on Newton's law of cooling, which states that the larger the surface of a given body, the greater the rate of heat transfer.
    - **Allen's rule** (1877): In a given species animals that live in cold areas tend to have shorter extremities than those in warm climate.
    - **Wilson's Rule** (1854) relates insulating coyer to climate. Breeds in cold climate have a dense, heavy, thick external coat and those from warm climate have short, glossy and thin hair.
    - **Gloger's Rule** (1833) relates colour to climate. Skin pigments protect against UV radiation. Animals in cooler of hot, dry climate have light coloured skin. Skin pigmentation darkens with increasing temperature and humidity.
    - **Claude Bernard Rule** (1876) relates climatic changes to internal body changes. The vasomotor dilation and contraction regulates rates of blood flow to skin

capillaries arid by using these mechanisms, animals can tolerate very low temperatures at their extremities.

CHAPTER-45: REGULATION OF TEMPERATURE

Learning objectives

- To learn how the heat is exchanged from animal body to environment and vice versa.

TEMPERATURE REGULATION

- Body functions mainly depend on body temperature. Either increase or decrease in temperature will alter the chemical processes occurring in the body. This makes a relatively constant temperature necessarily to be maintained for efficient functioning of all the tissues in the body.
- Based on the stability of the body temperature, animals are classified as
    - Homeotherms (or) Warm blooded animals
        - These animals have the ability to control their body temperature, whereas in a narrow range even when the external temperature vary. These are called as temperature regulations. They have a high rate of heat production of 8-10 times greater than poikilotherms of the same body size and at the same body temperature. E.g. Mammals and birds. The body temperatme of homeotherms is higher than the autrient temperature. This also means that heat production must be higher in a cold environment.
    - Poikilotherms (or) Cold blooded animals
        - These are animals whose body temperature varies with the temperature of the environment. They lack physiological control of body temperature. These are called as Temperature Conformers. Eg. Reptiles, Amphibia, Fishes and Invertebrates.
- Based on the source of body heat, the animals are classified as:
    - *Endotherms:* These animals generate their own heat through metabolic heat production. Eg. Birds, Mammals. Some lower vertebrates and insects. They can maintain their body temperature above the ambient temperature. They have well isolated fur, feathers or fat which make them to conserve heat against high temperature. They have higher resting metabolic rate of about 5 times more than that of ectotherms of same body size and temperature. (Low thermal conductance)
    - *Ectotherms:* These animals depend entirely on the environment for their heat. They generate low metabolic heat and have high thermal conductance. They are poorly insalated and allows minimum loss of metabolic heat to cooler environment. They regulate their body temperature by absorbing heat from the surroundings. Eg. Crab.
    - *Heterotherms:* Are animals are capable of varying their degree of endothermic heat production. They do not regulate body temperature in the narrow limits. Eg. Monotremes (egg laying mammals), few large fishes.

Body Temperature

- Different parts of the body have different temperature complete uniformity of temperature is possible if no heat exchange occurs between the body and its environment. This

difference is due to the variation in the metabolic rate, blood flow and the distance from the surface. The core (or) interior temperature is higher than that of the temperature of the extremities. The factors, which determine the variation of temperature is heat produced, heat gain and heat loss which vary from location to the other. The temperature of brain, liver, heart and other active musculature may be 1-2°C higher than that of carotid blood, which is 0-2°C, lower. Rumen temperature can exceed rectal temperature by as much as 2°C. Rectal temperature is an overall good indicator of core temperature.

- Many factors such as age, sex, feed intake, season, twice of the day, exercise, digestion, pregnancy etc. will affect the normal body temperature. The fetus, which has its own metabolism, will have the body temperature slightly higher than its mother. After birth the body temperature in many cases relatively labile until thermoregulatory much in fully developed.

Normal body temperature of different species

Cattle	Sheep	Goat	Horse	Pig	Dog	Cat	Chicken
38.6°C	39.1°C	39.1°C	37.6°C	39.2°C	38.9°C	38.6°C	41.7°C

- Body temperature is elevated during feeding, exercise, estrus, and at the end of pregnancy. It is decreased during starvation and may fall after shearing and ingestion of large amounts of cold water.

Diurnal variation (24 hr)

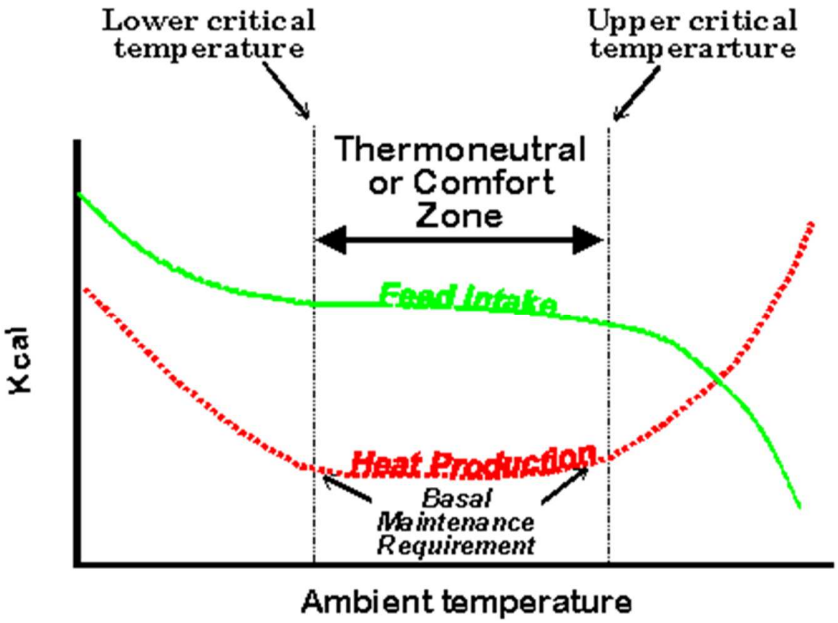
- The domestic animals exhibit a diurnal rhythm of body temperature. This rhythm largely reflects the period of rest and activity of the animal. The animals, which are active during the day, have their minimum temperature in the early morning and maximum temperature in the late afternoon. A reverse of this is found in animals, which are active during the night. The diurnal temperature rhythm can be modified by various routine factors of management.

Thermoneutral zone (Zone of thermal comfort)

- It is the temperature zone at which the performance of the animal is maximized. At this zone, the environmental temperature does not affect the metabolic heat production and circulatory adjustment is sufficient to maintain a thermal constant body temperature. Above and below this thermoneutral zone, circulatory adjustments are no longer enough for the maintenance of heat balance. In high temperature, they must be supplemented by increase of evaporative heat loss and in cold temperature by increasing the metabolism.
- The comfort zone varies with age, body size, protective covering, acclimatization, capacity for evaporative cooling, feed intake and activity of the body. In case of animals with high productivity increase in temperature will reduce the heat production (because of less heat dissipation) and a cold environment will stimulate the heat production.
- Critical temperature: It is the lowest ambient temperature at which a mammal or bird can maintain its body temperature at a normal resting metabolic rate.
- Lower critical temperature: The environmental temperature below which lieat production begins to increase in defense against cold is called as lower critical temperature. The

normal metabolic rate is inadequate and the body has to produce extra heat when the environment temperature falls.

- In man and tropical animals the Lower Critical temperature is 25-30°C.



HEAT EXCHANGE MECHANISM

- To maintain a constant body temperature, heat produced and received in the body must be equal to the heat lost from the body. The pathways of heat gain or loss are:
  - Non evaporative or insensible heat loss via radiation, conduction and convection.
  - Evaporative or sensible heat loss as evaporation of water from the skin and respiratory passages.

Radiation

- It is the energy transfer through electromagnetic waves (heat exchanges between two objects that are not touching). Radiant energy does not heat the air directly but indirectly by heating solid surfaces like soil, water, trees, clouds and animals. Thus radiant energy is changed into thermal energy, which in turn heats the air by conduction and convection.
- Loss or gain of heat- from a body occurs by infrared waves and it depends on temperature. All objects radiates energy. Heat rays leave the body and it is heat loss. Animals receive infrared thermal radiation from their surrounding.
- Heat loss/gained by the radiation is influenced by surface area of the animal, temperature of the animal skin, temperature of the air surrounding the animal and emissivity (ability to absorb and emit heat) of the animals skin.

Conduction

- It is the heat dissipation from warmer to colder objects by direct physical contact and it depends on the temperature radiant between the contacting surfaces, area of the contacting surface and thermal conductivity.
- Heat loss can be minimized by insulation of fur and clothing. Animals can increase heat dissipation in a conduction by resting on the cold floors.

Convection

- Heat is transferred from one molecule to another and is then carried away i.e heat is transferred to air which rises and takes the heat which it cooler air comes into replace warmer air.
- Higher air velocity increases heat loss by convection. Effectiveness of convective heat loss is effected by:
    - Body surface area
    - Velocity of air movement
    - Surface temperature
    - Air temperature
- The air adjacent to the body surface is usually warmer than the surrounding environmental air and replacing the warmer air by the cooler atmosphere air removes heat from the animal by convection. Convection is enhanced by wind. Warm air blowing over the animal provides convective heat gain to the animal.

Evaporation

- Most important mechanism of heat dissipation during severe heat exposure when the temperature and radiant between the body surface and environmental air becomes narrow. Evaporation of water from the skin depends on:
    - Temperature and moisture of the skin
    - Skin covering - hair, wool, etc.,
    - Humidity, velocity, and temperature of the surrounding air
    - Respiration rate and volume
    - Water available for evaporation
    - Surface area of the animal
- Thick hair coat may reduce air movement over the skin and entrap layer of moisture over the skin surface and reduces evaporative loss. When the environmental temperature is above that of the body, water evaporation in the principal mechanism that covers the body.

Heat Gain

- Animals gain heat through internal body actuates and may also gain heat from external sources through radiation, conduction, and convection. Heat gain to the body through metabolism occurs by oxidation of nutrients. Animals require heat / energy for mamtaining circulation, respiration, excretion, muscle tension etc, which is called as Specific Dynamic Action (SDA) heat. This maintenance energy is given off as heat.

- In cold condition, animals have to conserve/produce heat. They absorb heat from the surrounding objects if their temperature is higher than their own. Some heat is also gained by ingestion of food. Heat production varies among animals per unit weight. Zebu cattle produce 20% less heat than European cattle, which makes them to be more heat tolerant and less cold tolerant. Males have a higher BMR than females and castrated males. Lactating cones produce more heat than non-lactating dairy cones of the same size.

Heat Dissipation / Loss

- To maintain homeothermy animals dissipate excess heat produced by internal sources or received from external sources.
- Heat produced in animals is dissipated by conduction, convection, evaporation, and radiation and through respiration tract and factices and urine.

CHAPTER-46: BODY TEMPERATURE REGULATING MECHANISMS

Learning objectives

- This chapter explains how the homeotherms regulate their body temperature to heat and cold conditions.

PHYSIOLOGICAL RESPONSES TO HEAT

Circulatory Adjustments

- Cutaneous vasodilatation causes a rise in skin temperature, this increases heat loss. This cutaneous vasomotor reactions in response to thermal changes is mediated mainly by sympathetic vasoconstrictor-nerves. Peripheral vasodilatation is brought about by inhibition of sympathetic vasoconstrictor nerves.
- The stimulus for this is the temperature of the blood circulated to the brain. Thermosensitive cells are present in the anterior hypothalamus that will respond to warmth and evoke thermoregulatory mechanism for heat loss. Thermoreceptors present in the skin may also bring about this response.

Evaporative heat loss

- Evaporation of water will cause an effective cooling of the body..Three types of evaporation (perspiration) are common
    - Insensible perspiration- diffusion of water,
    - thermal sweating from the sweat glands
    - non-thermal or emotional sweat.
- Insensible perspiration
    - It is continuous, occurs by diffusion of water through the skin and from lungs. Thermal sweating occurs from the sweat glands. Humans may perspire 2 liters of moisture/ hour under dry desert non-shaded conditions. Evaporation of water is an effective way of cooling the body.
    - The amount of heat loss by evaporation of 1.0 g of water is proximately 580 calories. At ordinary temperature and humidity, about 25% of the heat produced

in resting mammals is lost by evaporation of water from the skin and respiratory passages.

- Sensible heat loss
  - Sweating and panting increase heat loss.
- Sweating
  - There are 2 types of sweat glands:
    - *Eccrine glands:* These are found mostly in humans and are scarce in domestic animals (found in the foot pads of dogs). This gland produces an aqueous secretion and is under the control of cholinergic sympathetic fibres.
    - *Apocrine glands:* They produce a protein containing secretion, with loss of the free end of secretory cells and develop from hair follicles. These glands are not supplied by secretory nerves and are sensitive to epinephrine carried in the blood. In the goat and the sheep the apocrine sweat glands are under direct nervous control, In man the eccrine sweat glands alone are responsible for thermal sweating. In many domestic animals, apocrine sweat glands are important for evaporative heat loss.
  - Thermoregulatory sweating is brought about in two ways:
    - By a rise of hypothalamic temperature
    - Reflexly by stimulation of warmth receptors in the skin and other parts of the body outside the CNS.
  - In larger animals sweating is important whereas in panting is more important in smaller species like dogs and sheep. In case of horses sympathetic nerves control sweating in response to heat with epinephrine as a transmitter substance. In dogs sweating is insignificant in heat regulation and polypnea and panting are more important:

Panting and polypnea

- Panting is an effective way of dissipating heat load. In many species heat load will evoke polypnea and in some polypneic panting. Panting consists of increase in respiratory frequency with a decrease in tidal volume. It is more effective in dogs. Breathing occurs at a frequency of 200-400/minute with an opened mouth. Panting is accompanied by increased salivation thereby aids in evaporative cooling. Panting is increased dead space ventilation without change in respiratory alveolar ventilation. Panting is stimulated reflexly and centrally.
- In birds sweat glands are absent and increased evaporation occurs not only by panting but also by gular flutter. Gular flutter is the rapid oscillation of the thin floor of the mouth and the upper part of the throat.

Wallowing

- Pigs and porcine species and water buffaloes have a relatively poor evaporative loss mechanism and depend on wallowing for evaporative heat loss.

Saliva spreading

- Spreading of saliva on the fur when they are heat stressed for evaporative heat loss is seen in rodents and macropod marsupials.

PHYSIOLOGICAL RESPONSES TO COLD

- When there is a fall in the environmental temperature, a drop in animal's body temperature is prevented by a reduction in heat loss by physical regulation. When this physical regulation becomes insignificant to maintain the body temperature heat production will be increased as a line of defense by means of chemical regulation.

Reduction in heat loss

- Behavioral response
  - A reduction of heat loss is brought by adoption of posture such as curled up position of animals at rest. So that the surface area exposed to cold will be reduced.
- Increased fur insulation
  - In case of prolonged cold exposure, the animals will develop increased fur growth and subcutaneous fat as a means of increasing peripheral insulation.
- Piloerection
  - It occurs to reduce the insulation value of hair or fur. In this process the hairs become more erect and is brought about by the erector pili muscle of the hair follicle.
- Circulatory Adjustments
  - When exposed to cold vasoconstriction occurs in the skin and superficial tissues mediated by the vasoconstrictor nerves there by reducing the heat loss in 2 ways:
    - Vasoconstriction decreases the peripheral blood flow causing a drop in skin temperature there by reducing the temperature gradient between the skin and the environment.
    - It increases the functional insulation of the skin because of reduction in convected heat loss of the perfusing blood.
  - An increase in vasoconstrictor tone may be elicited reflexly by stimulation of skin cold receptors or centrally by a lowered hypothalamic temperature. The other important mechanism by which heat is gained is by the arrangement of deep arteries and veins that run close together helping in counter current heat exchange. Hence the cold venous blood is transported centrally adjacent to warm arterial blood coursing peripherally. By continuous heat exchange, the returning venous blood is cooled which minimizes the heat loss to a cold environment. e.g., Pampiniform plexus to maintain the testicular temperature.

Increase in heat production

- When the reduction in heat loss is not sufficient to maintain a constant body temperature, heat production has to be increased called as lower critical temperature and it varies in different animals. Among the farm animals, cattle and sheep have the lowest critical temperature and are therefore able to withstand cold.
- Increase in heat production is brought about in 2 ways:
  - Shivering thermogenesis
  - Non shivering thermogenesis
- Shivering thermogenesis
  - During sudden exposure to cold, shivering is the major contributor to enhanced heat production (increases the BMR 5-10 times the normal). It may increase the oxygen consumption to 400%. It is an involuntary function of the body.

- It consists of muscle tremor with a frequency of about 10/second. The heat is generated through shivering to withstand cold where in 75 % of energy released for muscle contraction is converted into heat. It is initiated by peripheral and central thermoreceptors.

- Non shivering thermogenesis
  - Heat production occurs in the absence of muscular activity. Heat is generated apart from shivering by means of enhanced mobilization of fat from white adipose tissues, increases triglycerides uptake, lipogenesis and lipolysis when the animal is exposed to cold. This non-shivering thermogenesis is due to the release of epinephrine and nor epinephnne during cold exposure.
  - Epinephrine and nor epinephrine has got calorigenic effect which is potentiated by thyroxine. ACTH also plays a role. In birds glucagon enhances/calorigenic effect. In neonates, non-shivering thermogenesis is higher than the adult.
  - In new born animals of several species and in hibernating animals /brown adipose tissue is an important source of heat. When the animal is exposed to cold, it causes a high increase in metabolism and blood flow. During cold adaptation, hypertrophy of brown adipose tissue or fat occurs whereas it decreases in warm adaptation.

Adjustments during prolonged cold exposure

- The physiological adjustments to prolonged cold exposure is divided into 3 categories:
  - *Cold acclimation:* The changes occurring during cold exposure for a few weeks when other environmental factor remains unchanged. It involves a shift from shivering to non-shivering thermogenesis occurring during the first 2 or 3 weeks of cold exposure. Increase in non-shivering thermogenesis enables the animals to survive longer than non- acclimated animals in severe cold. In cold acclimated animals carbohydrate and fat metabolism increases, storage of brown fat increases and the calorigenic effect of epinephrine and nor epinephrine is potentiated by thyroxine.
  - *Cold acclimatization:* Animals living in cold climate maintain a normal body temperature mainly by adjustments involving improved insulation. There by they have a lower critical temperature than the animals of the same species not acclimatized to cold. Heat production is not elevated in cold acclimatized animals. It involves modifications developing during slow seasonal change from summer to winter. In many arctic animals the metabolism in winter is less than that during summer when measured at a given external temperature.
  - *Climatic adaptation:* Genetic adaptations of animals occur as a result of natural selection so that they are best suited for survival in a cold climate. The body temperature of homeotherms does not show any adaptive changes. Arctic mammals maintain their body temperature at a high level by most efficient insulation and do not increase their metabolism until the environmental temperature is very low. The rectal temperature of arctic and tropic mammals will not vary significantly during extreme cold climate. Lower critical temperature for Eskimo dog is -10°C, arctic fox is -30°C, and man and large tropical animals is 25-27°C.
  - *Cold habituation:* Animals that are experiencing a mild cold stress for extended period, when suddenly exposed to severe cold, for several hours, they delay their thermogenic reactions to the acute cold. This is cold habituation and it is enables the animals to withstand acute temperature of short duration without increasing

their metabolic rate and conserves energy. If the cold stress prolongs for larger duration, then heat production increases to maintain body temperature.

TEMPERATURE PERCEPTION AND REGULATION OF BODY TEMPERATURE

- Temperature perception is mediated by peripheral thermo-receptors and CNS thermodetectors. Their joint action seems to be necessary to obtain the maximal temperature regulation against heat or cold.

Peripheral thermoreceptors

- The sensations of warmth and cold originate from thermo-receptors in the skin and certain mucous membranes. Warmth receptors detect rise in temperature and cold receptors detect drop in temperature.
- Certain areas of the skin are of greater importance than others in the peripheral control of body temperature e.g. local warming of the scrotal skin in the ram elicits polypnea much more readily than any other skin area in this animal.

CNS thermodetectors

- The anterior hypothalamus is sensitive to an increase or decrease in body temperature. There are thermosensitive neurons located in the hypothalamus that detect the - changes in the temperature. These neurons increase their rate of discharge with increase in local temperature from 34 to 41°C.
- When the temperature of blood perfusing the hypothalamus is above normal, the anterior hypothalamus activates physiological and behaviour heat loss mechanisms and inactivates heat gain mechanisms. When the temperature of hypothalamic blood is lower than normal, the anterior hypothalamus initiates heat-gain mechanisms. Spinal cord also contains thermal sensitive areas, which is more important in birds.

Interaction between neural and hormonal thermoregulatory mechanisms

- In addition to neural mechanisms, hormonal mechanisms also participate. Epinephrine, norepinephrine and thyroxine are important in cold exposure. Epinephrine, norepinephrine and thyroxine are of major importance in cold stimulated non-shivering thermogenesis. An increased secretion of hormones occurs in mammals during cold stress. The thermoregulatory centre in the anterior hypothalamus participates in the control of these hormonal cold defense mechanisms.
- When warm blood passes through anterior hypothalamus, it inhibits the activation of the sympathico-adreno-medullary system and the thyroid activation. Cold blood passing through the thermo-regulatory centre on the other-hand causes sympathico-adreno medullary activation and increased secretion of thyroxine. The latter response is mediated by the release of thyrotropic hormone (TSH) from the anterior pituitary. Thus, changes in deep body temperature influence both neural and hormonal thermoregulatory mechanisms by altering the activity of central thermo-detectors in the anterior hypothalamus.

CHAPTER-47: REGULATION OF TEMPERATURE IN BIRDS

Learning objectives

- This chapter provides knowledge on specific features of birds in regulating body temperature during heat and cold conditions.

TEMPERATURE REGULATION IN BIRDS

- Birds are homeotherms and maintain constant body temperature.
- Birds are also endotherms and have special mechanism of increasing their body temperature by heat generation within their tissues. In birds, even though similar thermoregulatory mechanisms are available as that of mammals, certain distinguishing differences make the bird more adaptive and precisely control thermoregulation.
- Examples
    - Plumage of birds is helpful for flight as well as insulation;
    - site of deposition of fat differs in birds;
    - availability of salt glands as in ducks prevents dehydration and evaporative heat loss;
    - absence of sweat glands.

Core Temperature

- Core temperature varies during the day and is dependent on environmental temperature. Circadian rhythm of body temperature reveals the extent to which the bird stores heat and the degree to which the thermo-regulatory control mechanisms operate.
- Factors that affect the core temperature are the amount of plumage, size of the bird, physiological activity of the bird and intensity of light; photo period is a major factor that alters the body temperature and ultimately the productivity of the bird.
- Circadian rhythm of the body temperature is reversed by changing the time and degree of illumination and is beneficially used in poultry industry to enhance productivity. Hormones mainly control circadian rhythm of the body temperature of birds.

Heat production vs heat Loss

- Body temperature increases as the air temperature increases. At low temperatures, shivering causes thermogenesis to maintain core temperature. In young birds, core temperature is not constant even within the thermoneutral zone.
- During exposure to hot environment, there will be increased rate of heat storage, which raises the core temperature. Birds gain heat from environment. This can be seen in desert environment where the temperature of air and sand will exceed the bird's body temperature; due to increased heat storage, there will be reduction in the temperature difference between the body and the environment and heat gain from the environment is diminished.
- Storage of heat within the body is directly related to saving of water in the body, which helps to dissipate the moisture evaporation. Desert birds become more hyperthermic when they are dehydrated. Following dehydration, there is an increase in the body

temperature in all birds, which is due to reduced respiratory evaporative cooling. Food deprivation reduces body temperature, which may be due to decreased metabolism.

- Non-shivering thermogenesis is absent in birds since they do not have the browm fat- an important thermogenic tissue.

Heat dissipation

Radiation

- Skin and plumage are the major factors that participate in radiation heat transfer. Colour of plumage and positioning of plumage alters the degree of radiation.

Conduction

- Rate of heat conduction depends on thickness of the layers of tissues. Tissue fat has the lower conductivity but it does not play a major role in heat conduction.in birds because most birds do not have substantial fat layers under the skin.
- Fat is deposited in the abdominal cavity. Plumage of birds provides a very effective barrier to heat loss from skin surface to environment. Feathers help trapping air and act as windproof covering; coating of the feathers with oily secretion of preen glands makes the plumage resistant to wetting.
- Some of the birds adapted to freezing temperature have a "cold vasodilation ", by accelerating the blood flow to the foot and extremities that are exposed to cold. The cold vasodilatation prevent freezing of extremities by increasing the heat loss from these parts. Winter fattening occurs in small birds exposed to cold climate.
- Thermal conductance of small birds is greater than large birds because of change in the thickness of the skin and fat. Surface area in proportion to body weight is the major factor that modifies the amount of heat that is lost.

Convection

- Birds exposed to moving air or bird moving in air loses much of heat. This ' phenomenon is termed as forced convection and it is approximately proportional to the square root of the air velocity.
- Non-evaporative heat loss is a special process of convection and conduction occurring in the upper respiratory tract; it is similar to counter current exchange pattern and it helps in effective cooling of nasal passage.
- Another feature of birds is fluffing of feathers during exposure to cold. This increases insulation provided by feathers and decreases the thermal conductance of the plumage. Fowls with poor plumage have accelerated heat production due to scanty insulation. Thermal polypnoea is another feature that increases evaporative heat loss by increasing respiratory minute volume, and it is characteristic of many hyperthermic panting animals and birds.
- The extensive air sac system helps the birds to get more ventilation during panting, to accelerate the evaporative heat loss. This is one mode of stress management, which aids in evaporative respiratory .cooling without much change in blood gases and acid base balance. In few species, thermal polypnoea would result in mild alkalosis.

- Another feature observed in many birds when exposed to heat is the movement of bucco-pharyngeal area to accelerate ventilation that results in evaporative cooling (gular flutter).

BEHAVIOURAL THERMOREGULATION

- Migration of birds from cooler to warmer areas and vice versa is a thermoregulatory behaviour.
- Hunching - Birds exposed to cold try to reduce the surface area exposed to cold, to reduce heat loss and it is called hunching. They may tuck their head under the wings to cause further reduction in heat loss. They prefer to sit than stand which minimizes heat loss. Sunning behaviour is observed (birds expose themselves to sun to gain heat) during cold.
- Huddling reduces heat loss, most commonly seen in chicks; the chicks huddle with their mother seeking warmth.
- On exposure to hot environment, they avoid the hottest part, reduce their activity and seek shades. They orient their back of the body to sun feet facing the shades, wings kept apart and away from the body to facilitate convective heat loss.
- Domestic fowl splashes water over its comb and wattles, tend to soak its ventral feathers (belly soaking) and try to keep themselves cool in hot environment.
- The behavioural thermoregulatory response in birds is regulated by preoptic region of the brain, anterior hyppthalamus and by peripheral receptors. The peripheral and the CNS neuron: regulate the' body temperature by regulating heat production and heat loss in birds." Cold receptors are present in the tongue, beak, bill and feathered skin areas in the back.

THERMOREGULATION DURING FLIGHT

- During flight, there is a tremendous increase in heat production, which increases the core temperature. Thermal conductance increases during flight as well as accelerated counter current mechanism in the brain blood vessels. The respiratory minute volume increases during flying to meet the increased 02 demand and to enhance heat loss.

CHAPTER-48: CLIMATIC ELEMENTS

Learning objectives

- To explain various climatic elements/environmental variables affecting growth, production and reproduction of animals.

AIR TEMPERATURE

- The air temperature surrounding an animal's body is important to its comfort and functioning of physiological processes. Heat normally passes by conduction from warm skin to the cooler air around it. As the air temperature rises above the comfort zone, the heat loss decreases and when air temperature exceeds skin temperature, heat will flow in reverse direction. This is a serious problem in hot and dry areas. When air temperature is

lower than 5 C, heat flow from the animal's body is accelerated leading to discomfort and lowered performance.

- Animal may also be heated or cooled by temperature of objects in the surroundings. Most important is the heat from the ground. In dry soil, by mid afternoon, ground surface temperature may go >40°C and after sunset, the ground soil cools rapidly. Green vegetation or moist soils heat more slowly; speed, duration and source of wind affect air temperature. If an animal is grazing in a field where the temperature of air is 25° C, it will not experience discomfort, but by mid afternoon, if wind comes from passing over dry land whose temperature is 40° C, the animal's heat load will be markedly increased.
- Altitudes also influence air temperature, which decreases by 0.65°C/ 100m elevations.
- Air temperatures are measured as maximum and minimum and mean daily (average of these two) temperatures. If the difference of maximum and minimum temperature is 8°C but if the minimum is above 24° C, the animal is under heat stress for the entire 24-hour period; if the temperature difference is 22°C or more, it indicates lower night temperature, which can allow the animal to restore thermal balance.

ATMOSPHERIC HUMIDITY

- It indicates water vapour content .Rate of cooling by evaporation from skin and respiratory tract depends upon the humidity of air. Low humidity helps in rapid evaporation (sometimes lead to dehydration). In high humidity, evaporation is slow and reduces heat loss, thereby endangering the animal's heat balance. Heat retention in hot humid climate is more acute than excessive water in hot dry climates.
- High humidity is conducive to diseases, lowers feed quality, and enhances mineral deficiency. In warm humid areas when air temperature is above 21°C and relative humidity above 60%, it becomes difficult for the livestock to loose heat. In dry areas when temperature is above 32°C with high wind velocity and low humidity (< 20%), it is hazardous for animals due to dehydration problem.
- Atmospheric humidity is expressed as relative humidity (RH), which is the ratio between the amount of water vapour present in air and the amount it would hold if saturated at the same temperature. The RH does not indicate the ability of the atmosphere to accept water vapour given out from the skin and the lung of animals. Vapour pressure indicates the concentration of water vapour in the air and it is expressed as the fraction of total barometric pressure contributed by gaseous water vapour; it is a better measure of humidity because it gives a precise estimation of water content of the atmosphere.
- The RH can be measured directly with a hygrothermograph or indirectly as wet bulb or dew point temperature. Vapour pressure can be determined from meteorological tables using air temperature and wet bulb, dew point or RH.
- Dew point is the temperature at which the air could be saturated by the amount of water present in it.
- Daily maximum RH occurs in the early morning hours and minimum in the early afternoon; however the vapour pressure may be constant throughout the day.

RADIANT ENERGY

- Radiation from the sun, sky and sorroundings add to the animal's heat load. Providing shades or shelters that cut off the direct solar energy can reduce it.
- An animal in the open field is exposed to (1) direct solar radiation from the sun, part of which is reflected according to colour and proportion of coat and the remainder is absorbed as heat (2) solar radiation reflected from clouds and other particles in sky, part

of which is reflected by hair coat (3) solar radiation reflected from the ground and surrounding objects; the first two contributes 50 % of heat load and the third contributes the remainder.

- The level of radiant energy is negatively correlated with humidity but positively correlated with maximum temperature. White surface reflects high proportion of visible radiation but little of long infrared; polished aluminum reflects very little of both and absorbs most of the heat. Longer wavelengths are absorbed through skin.
- The energy of radiation absorbed by the animal's body is changed into heat, raising the temperature of the aniinal.
- Measurement of the source of radiation and the amount of radiation absorbed or reflected by the animal are complex process. When the sun is overhead in a clear sky radiation is high; the surface area of the animal presented to radiation varies among species. At noon, in man only a little more than the head is exposed. Due to posture, the greatest surface area exposed in man is between 8 A.M and 10 A.M.; in sheep it is from 10 A.M. to 12 noon. The intensity of radiation decreases with angle of the sun; thus man receives less direct radiation during a day than a sheep or cow.
- Measurement: Approximate estimate of the total radiation exchange of an animal exposed to a particular environment is by using Black Globe Thermometer, which is a sphere fitted with a thermometer, thermistor or thermocouple for recording temperature. The black globe temperature is about 6° C above atmospheric temperature and it is used to estimate the rate of heat gain by the animal in a particular environment.
- For 6 to 12 months in a year in N-S 30° C latitudes, the intensity of direct and reflected radiation results in the animal receiving a significant amount of heat for 5 - 10 hours/day. The radiant heat load is the greatest in the afternoon when the ground radiation and air temperature are high. But the duration of stress period from the radiant energy may extend beyond the day light hours.
- The animal's body absorbs all the radiation coming from infrared radiation irrespective of skin colour or hair coat. On the short-wave (visible) radiation, hair and skin surfaces affect reflectivity depending on the colour.
- The type of instrument used for measuring radiation depends up on the kind of radiation to be measured - ultraviolet, visible, and infrared or total radiation. Two groups of instruments can be used (1) those dealing with the intensity of direct solar radiation and some of the reflected radiation (2) those dealing with all the three components - direct, reflected and thermal radiation.
- If there is a high solar radiation, animal absorbs heat from environment. Animal has to spend more energy to loose the extra heat obtained from solar radiation. Consequently, less energy is available for growth and production.
- Harmful ultraviolet radiations are injurious to underlying soft tissues. Melanin pigment stored in skin of tropically adapted animals protects from their ill effects.

AIR MOVEMENT

- The rate of air movement over the skin of an animal affects heat loss; increased airflow enhances heat loss by evaporation when moisture is present on the skin. But with low moisture, heat loss from the animal is limited. At temperature of < 29°C, faster air movement enhances heat loss, at temperature of > 29°C the reverse is true i.e. there will be heat gain.
- Air movement aids in heat loss from the skin by conduction when air temperature is lower than skin temperature; when the air temperature is higher than skin temperature, there will be heat gain from the surrounding.

- Airflow is described as directional or turbulent (rotational). In hot dry area, air flows around 8 kmph or less during the day does not result in drying and heating the animal body. However, in hot dry climate, the rate of air movement is high, which requires providing protection.
- Airflow rates of 8 to 20 kmph do not produce serious problems. During dust storms of several days, sheep avoid grazing.
- Measurement: Wind vane is used to find wind direction and anemometer is used to measure the speed of wind.

RAINFALL

- Rainfall influences the livestock indirectly by influencing the food supplies. The amount of rainfall, its duration and condition of the soil influence the amount of water going to the subsoil. The soil moisture available for crop production depends on total annual rainfall, its seasonal distribution, and intensity, condition of the soil, vegetation cover and rate of evapo-transpiration. If particular area receives sufficient amount of rainfall, green fodder production will be optimum which ensures better food supply to the animal for growth and production.
- The seasonal rainfall pattern determines the amount of feed that can be produced, grazing practice to be employed, and requirement for storage and type of feed preservation system to be followed. In heavy rainfall season, water content of the forage is increased, which limits nutrient availability to the animal. Rainfall also has direct effects on livestock; it helps in heat dissipation but interfere with feeding and health. Animals cease to graze during rain. In hot environment, rainwater may be trapped by the hair coat, which will reduce thermal stress. Rainfall increases humidity. Sheep can thrive in hot, dry climate but hot, wet climate limits sheep husbandry.
- Measurement: Rain gauge is used to measure rainfall. Day to day levels may vary within a small area due to movement of clouds, but monthly of annual totals are reliable for determining patterns of rainfall over a wide area.

LIGHT

- The period of light during the day is called photoperiod; it is the time between sunrise to sunset. Photoperiod varies with latitude and season and it is related to the path of sun. Length of photoperiod near the equator varies only by few minutes; by ±2 hours at latitude 30° and by ± 19 hours at latitude 60°.
- The eyes in mammals and eyes and skull in poultry perceive photoperiod. The information is relayed to the hypothalamus and pineal via superior cervical ganglion; the pineal encodes this information by producing more melatonin in dark than in light (circadian rhythm).

Effects of light on production

- Providing supplementary light of 4-8 hours with natural day light to cows increases milk yield by 7-16%; in ewes, birth weight of lambs is increased; in pigs, the litter weight is increased.
- In heifers and lambs, growth rate increases by extending the photoperiod; in growing pigs, feed intake or weight are not affected. In poultry, providing 14L, 10D enhances growth rate up to 12 weeks of age; after 12 weeks, very little effect is produced on growth. Low light intensity or blue light, which is perceived as less bright by birds than red or white is preferable to stimulate growth up to 18 weeks.
- In ewes, timing of puberty is determined by photoperiod and nutrition. With adequate nutrition, GnRH and photoperiod regulates puberty. During short day with longer duration of melatonin GnRH pulse frequency increases with subsequent LH release and ovulation. Long day length with short melatonin duration causes refractoriness and ovaries become inactive.
- Sheep is the most affected animal by changes in photoperiod. Seasons of high temperature and long photoperiod cause anestrous and decreased fertility and sperm production in ram. These effects are related to GnRH and LH production.
- In cattle, photoperiod does not affect reproductive activity. Growth and shedding of hair coat in cattle is influenced by length of photoperiod.
- In long day breeders like horse and birds require more light hours for growth, reproduction and production.
- In short breeders like sheep and goat require less light hours for growth, reproduction and production.

CHAPTER-49: ADAPTATION TO EXTREME CLIMATIC CONDITIONS

Learning objectives

- To make the students to learn about various species of animals surviving in the extreme climatic conditions.

ADAPTATION TO EXTREME CLIMATIC CONDITIONS

Dormancy

- It is a general term for reduced body activities, including reduced metabolic rate. The types of dormancy are:
    - Sleep
    - Torpor
    - Hibernation & Winter Sleep
    - Estivation

Sleep

- It is studied intensively in human beings and other mammals and involves extensive adjustments in brain functions. In mammals slow wave sleep is associated with both the

hypothalamic temperature sensitivity and a drop in body temperature as well as changes in respiratory and cardiovascular reflexes.

- Human beings and many other mammals sleep for hours at a time. Many of the big carnivores (lions and tigers) sleep for as long as twenty hours a day especially after a meal.

Torpor

- It is a stage in which animals or birds make their metabolic activities to decline and reduce their body temperature e.g., Humming birds and small birds. They frequently feed to meet their high daytime metabolic activity and go into a stage of torpor during night when they cannot feed. This is an adaptive mechanism to maintain bodily activities to avoid running out of energy stores.
- Before the animal becomes active, its body temperature rises as a result of a burst of metabolic activity especially through shivering or oxidation of brown fat stores or both. Several species of small mammals also undergo torpor but large mammals have too much thermal mass to cool down quickly for short periods of torpor.

Hibernation & winter sleep

- If the period of torpor is prolonged for weeks or even several months in cold climates in a protected environment especially in higher animals it is known as hibernation and winter sleep. Example Bat, Ground squirrel, Dormouse, Woodchuck, Hedgehog, Groundhog (Marmoset) and Opossum are some of the animals that hibernate. During hibernation all the physiological functions continue at a reduced rate - heart rate, respiratory rate and metabolic rate decreases. Body temperature is lowered comparatively to a level compatible with survival for the speciesi.
- The hibernating animals arouse from their dormant stage periodically example for self defense, urination etc., A protective mechanism against profound cooling exist i.e during hibernation arousal occurs if the body temperature declines to the level near freezing or when the environmental temperature falls below -2°C or rises above 30°C.
- The ability of the hibernators to increase their body temperature from reduced level is facilitated by brown fat. Brown fat contributes 80% of the total heat production during the initial phase of awakening. It is found in the inguinal, auxiliary and sub scapular regions and around the deep blood vessels. Brown fat differs from white fat in that they have fat metabolizing enzymes. The cells of brown fat tissue are sympathetically innervated and are rich in mitochondria and cytochrome. They are extensively vascularised. The blood sugar in some hibernators is considerably lower during winter sleep and in all hibernators, the serum Mg^{2+} is elevated during hibernation.

Estivation

- It is summer sleep seen in animals e.g., frog, crocodiles and alligators. When the animals are exposed to hot dry summer, they sleep. The physiological difference between estivation and hibernation is the inability of estivators to withstand low body temperature during dormancy. It is characterized by shallow torpor. Sleeping animals differ from estivating animals in that the estivating animals have prolonged periods with no visible respiratory movements, while animals, which are sleeping, have regular respiratory movements without periods of apnea.

- Hibernators become quiescent during winter and aestivators become quiescent during summer. Hibernators are sometimes also aestivators. Hibernation is a method of avoiding the need for considerable quantities of food that are required to maintain summer levels of metabolism during winter. Aestivation avoids the difficulty of thermoregulation with high levels of metabolism during hot summer e.g., Desert rodents aestivate i.e. become dormant during hottest part of summer with a fall in body temperature, decreases metabolism and as a result a sparing of evaporative loss.

FEVER, HYPERTHERMIA AND HYPOTHERMIA

Fever

- Fever may be defined as any condition of positive heat balance not solely due to food, exercise, or environment. Fever is disturbance of the normal equilibrium between thermo genesis and thermolysis.
- Fever occurs in 3 stages:
    - *I - stage:* heat production is increased but heat loss mechanisms fail to eliminate the excess heat, and heat gain and heat loss is equal.
    - *II- (mid) stage:* once fever reaches to a particular height, body temperature is maintained at that high level; heat gain and heat loss are equal.
    - *III- (termination) stage:* all the heat loss mechanisms are activated and they are able to bring the body temperature to normal.
- Disease organisms, chemical pyrogens like dinitrophenols, thyroxine, surgery etc, may cause fever. Bacteria are found to produce endotoxins. These endotoxins (or exogenous pyrogens) induce a febrile response in man and animals. They are removed from the blood and inactivated by the reticulo-endothelial system.
- The endotoxins induce a release of another agent-endopyrogen from the monocytes and tissue macrophages. These endopyrogens act on the thermo-regulatory centre and facilitate cold defense mechanisms and inhibit the heat loss mechanisms. Endopyrogen may also be released under certain non-infectious conditions. The endopyrogen is Interleukin-I (IL-I). The IL-I causes the release of prostaglandins in the hypothalamus. The PGs cause release of IL-II from the thermtregulatory center, which activates more T-lymphocytes into circulation to enchance, the defense capacity of the body by producing interferon.
- The sensitivity of the hypothalamic temperature sensing neurons to these pyrogenic molecules leads to an elevation in the set point to a higher temperature than the normal. The result is thus body temperature1 rises several degrees and the animal experiences fever.
- During the development of fever, the heat balance is positive due to increased heat production. Later, when the body temperature reaches a certain height, the balance between heat loss and heat production is restored, and the temperature is again precisely regulated at the new high level. Under these conditions, there is still an increased heat production, but this may be only secondary to the elevated temperature, since the velocity of the biochemical reactions follows von't Hoffs Law and increases 2 to 3 times for a temperature rise of 10°C. A rise of 1°C in body temperature should therefore, cause an increase of 10 to 20% in the basal metabolism. Humans can tolerate a temperature limit of about 107 to 110°F. The elevated metabolic rate during the steady state of fever may be more the result than the cause of high temperature. With termination fever, various heat loss mechanisms are activated and heat balance is restored to its normal level.

- Fever is beneficial in many diseases. The rise in body temperature is to confer protection to the animal against bacterial infection. The protection is provided in 2 forms:
    - The antiviral and antitumor agent interferon is more effective at higher temperature.
    - Elevated temperature also diminishes the growth of the microbes.

Hyperthermia

- It is characteristic of heat stroke and develops when the heat production exceeds the evaporative capacity of the environment, or when the evaporative mechanisms become impaired because of loss of body fluid and reduced blood volume.
- The core body temperature increases and there is a decrease in heat loss and increase in heat gain but the set point of the body thermostat is not altered. Here antipyretics are ineffective and a reversal of body temperature can be achieved only by whole body cooling.
- Nonfebrile hyperthermia may also develop in response to drugs that impair the heat dissipation mechanisms and drastically increase the metabolic rate. E.g., genetically determined malignant hyperthermia seen in swines and humans.
- The malignant hyperthermia can be triggered by inhalation anaesthetics and muscle relaxants that cause an abnormal increase of myoplasmic calcium concentration followed by a drastic rise in muscle metabolism and heat production. Once developed malignant hyperthermia is lethal unless intense body cooling is initiated.

Hypothermia

- Hypothermia is a reduction of deep body temperature below normal in non-hibernating animals. It usually develops because of exhaustion of the metabolic cold defense mechanisms.
- Shivering may persist for a long time, causing depletion of skeletal muscle and liver glycogen reserves and also a fall in the glycogen content of cardiac muscle. There is a gradual slowing of heart and haernoconcentration occurs as a result of shift of fluid from the blood to the tissues.
- The lethal low level of body temperature varies among individuals and species. In humans and in dogs, cardiac arrest followed by respiratory depression and death may occur at a rectal temperature of 25°C.

COLD STRESS

- Environmental temperature has a direct effect on the animal's energy expenditure and voluntary food intake and hence affects the production performance of animals. The environmental temperature range in which the heat loss from the animal is minimal is known as the thermo neutral zone; the lower limit of this range is called lower critical temperature (LCT). Below LCT the animal has to increase heat production to maintain heat production to maintain homeothermy leading to reduction in conversion efficiency of feed to animal products .
- The newborn are more susceptible to cold. In heavier animals, cold is not much of a problem as they are fed ad libitum. 2 to 3 days following birth is the most critical period for the survival of new born animals.

Effect of cold on production

- Growth
  - Cold exposed animals increase feed intake to meet the increased energy expenditure. In pigs, a decrease in temperature from LCT (20-25°C) by 1°C is associated with 1.5% increase in feed intake; growth rate remains almost constant between 8 and 20°C. In birds also, within a temperature range of 7-22°C, growth of cattle is less obvious.
- Production
  - Milk production and composition is constant over a temperature range of 0 to 21°C. However, milk production may be reduced due to reduced availability of feed during cold season. In laying hens, at temperatures between 10 and 28°C, the egg weight is increased with improved shell quality; egg production is not affected; however, feed required per egg is higher.

Meat quality

- Prolonged exposure to cold in pigs under restricted feeding conditions results in leaner carcass; with ad libitum feeding, no effect is observed. In broilers, ad libitum feeding produces leaner carcass.

Heat stress

- Heat is the major constraint on the animal productivity in tropics. Growth and production are impaired during heat stress.
- As the ambient temperature increases, heat loss in animals is enhanced through respiration, evaporation of sweat, and by panting. When these mechanisms are not able to maintain homeothermy, the animal reduces heat production, feed consumption and secretion of thermogenic hormones are decreased to lower* the basal metabolism and these effects lead to decline in productivity. If all these mechanisms fail to balance the excessive heat load, the body temperature rises and the animal enters into an acute phase of heat stress. If the exposure to heat is prolonged further, a rapid acclimation takes place and the animal returns to chronic phase of heat stress. Productivity in this phase will be higher than the acute phase but lower than the normal productivity.

Effect of heat on production

- Appetite
  - During heat exposure, feed consumption is depressed. The peripheral warmth receptors transmit impulses to the hypothalamic appetite centre to decrease the feed consumption; this reduces heat load. Hypothalamic releasing factors are suppressed with decrease in pituitary hormones; insulin and thyroxine levels also decrease. Metabolic pathways slow down; protein synthesis reduces negative nitrogen balance.
- Animal production
  - *Growth*

- Growth performance - growth rate and daily weight gain are impaired at elevated temperature; these are caused by decreased feed intake, decreased anabolic activity (due to decreased voluntary feed intake of nutrients) and increase in catabolism mainly occurring in fat depots. In Brahman cattle, there may be increase in body weight with raising environmental temperature.

- Milk production
  - Milk Yield in cows is unaffected within a temperature range of 0 to 21° C. Milk production is impaired in cows in hot climate. A rise in temperature by 1.6°C above normal (21°C), decreases milk yields by 4.5%. The reduction in milk yield is greater in high producing than in low producing cows. Total solids, fat protein, ash and lactose are lowered during hyperthermia in high producing animals maintained at 38 C.
  - The decrease in yield and constituents of milk during exposure to high environmental temperature may de due to decline in protein, carbohydrate, lipid, mineral and vitamin metabolism leading to negative nitrogen balance. Many hormones like insulin, thyroxin and cortisol also decline and may decreases milk yield.
- Reproduction
  - In males, high environmental temperature exerts a negative effect on sexual libido, despite the thermoregulatory mechanism of testes. Semen fertility and spermatogenesis are reduced, and semen volume is also reduced. These effects reduce male fertility.
  - The detrimental effects of temperature on reproductive performance of cow are caused by the external temperature acting through thermoreceptors, photoreceptors, hypothalamus, CNS, endocrine glands and gonads. During heat stress, less gonadotropin production leads reproductive failure. Reduced thyroid activity also affects the reproduction. Deficiency in minerals - P, Cu, Co, I, Zn and vitamin A in heat stressed cows also affects reproduction.

Egg production in birds

- Egg production in birds is not reduced. Exposure to temperature of 38°C decreases both the egg weight and shell quality.
- Fertility and hatchability of the eggs are decreased by high temperature.
- Blastoderm viability is optimum between 10 and 12°C for weeks. At 27°C, hatchability of eggs is decreased.

ADAPTATION TO HIGH ALTITUDE

- High altitude refers to a height of 7500ft or more above mean sea level. Animals living at this altitude suffer from a condition known as mountain sickness. The symptoms seen are distress and generalized fatigue, disturbances in the circulatory, respiratory and reproductive systems and acid base balance disturbances.

Circulatory disturbances

- Increase in blood volume, erythrocyte count (polycythemia)

- Increased heart rate and decrease in haemoglobin -O2 saturation (73%), increased viscosity of the blood- heart has to pump very hard, decreased cardiac output.

Respiratory disturbances

- Increased rate and depth of respiration.
- In extreme altitudes cheyne-stokes breathing is noticed. Acid-base disturbances
- Decreased CO2 level in blood with increase in pH. Reproductive disturbances
- Generally in females fertility is not affected.
- Sheep are less susceptible to high altitude than cattle.
- Horses are more susceptible.
- Mules are best adapted to high altitude
- In rabbits and rats male reproduction is affected i.e., testicles become atrophied and germinal epithelium is replaced by connective tissue and spermatogenesis is inhibited. This process can be reversed if these animals are brought back.

Species specific symptoms at high altitude

- *Dogs:* more affected with mountain sickness, symptoms include sleepiness, vomiting, laboured breathing, muscle weakness and unable to stand. These symptoms are reversed after acclimatization at high altitude.
- *Cattle:* Brisket disease (Right heart failure)

Adaptation at lower altitude

- The physiological changes at lower altitude occur in under water diving, in mines and in tunnels. There is high pressure in lower altitude and is called as *hyperbarism*.
- Physiological effects of hyperbarism
    - Building up of CO2, the levels may go up to 3% and in extreme cases may go up to 10%. This high level acts directly on the brain and causes narcosis leading to unconsciousness.
    - O2 supply to tissues is reduced due to low atmospheric O2 content.
    - High pressure on the body especially on the thorax.
    - More of N2 gets dissolved iri the 'blood, which acts at brain level leading to nitrogen narcosis. Initially there will be convulsions with seizures finally causing unconsciousness.

Decompression sickness or dysbarism

- This condition is seen when animals are taken from lower altitude to a higher altitude. Dysbarism is also known as Bends, Caisson's disease, Divers paralysis. This is mainly caused by N2.
- When the animals ascend to high altitude, N2 escapes from the dissolved state in the blood by forming bubbles- small N2 bubbles cause pain and nerve block where as large bubbles cause air embolism, pulmonary edema and brain damage.
- Bends is a condition of joint pain usually in the legs, arm and neck region.

CHAPTER-50: ANIMAL BEHAVIOR

Learning objectives

- This chapter decribes the information on meaning and physiological concept of behaviour (ethology).

ANIMAL BEHAVIOUR

- Behaviour is the reaction of animals to a stimulus or it is the manner in which the animals reacts with its environment. Each individual or a species react to the same stimulus in a different manner.
- The behaviour of animals as a species determines their ability to survive in the wild state.
- The study of animal behaviour in their natural habitat is known as Ethology.
- The behaviour of livestock to a stimulus is of particular importance to the producer. The livestock producer can know with some accuracy how an animal is going to react in a particular circumstance, what makes it to act in that particular way and how the expected reaction can be changed to obtain best result. This will help to produce more efficient breeding, feeding and management of farm animals.
- Behavioural patterns influence decisions on size and shape of pasture, number of animals to run peer group, placing watering facilities or type of grazing.

Uses of studying behaviour

- It is an indicator of health and welfare.
- Enable to design housing and husbandry systems for the animals.
- To design the size and shape of the pasture, number of animals that can be kept in a group, type of grazing, watering facilities etc.
- For training the animals.

PHYSIOLOGICAL CONCEPT OF BEHAVIOUR

- Behavior is regulated by both the nervous system and the endocrine system.

Nervous system

- The chief function of the brain, spinal cord and peripheral nerves is production of behaviour. Information about the internal and external factors of animal is transmitted through a range of receptors (exteroceptors or interoceptors) to the CNS. From the CNS through specific neurons they reach the glands and muscles leading to expression of behaviour. The stimuli for a particular behavior may be chemical, mechanical or thermal.
    - *Cerebral cortex*

        - Cerebral cortex have localized regions where sensory impulses are received and redirected. They possess four main sensory areas:
            - Somesthetic /body sense area- receive nerve impulses from many parts of the area.
            - Visual area- receive nerve impulses from the retina to the optical nerve to the occipital part of the cerebrum
            - Auditory area- located in the temporal lobe of the cortex and is involved in hearing.
            - Olfactory area- deals with smell and plays an important role in breeding behaviour.
        - Apart from this, nerve impulses are also received into the brain via reticular activating system. Storage of information, learning, thought and conscious control movement.
    - *Hypothalamus*
        - Cerebral cortex inter relates closely with the hypothalamus. Some of the regions of the cortex directly communicate with the hypothalamus. It is involved in anger, excitement, sexual activities, pleasure and hunger. Much of the influence of hypothalamic activity is directed at the production of hormones from the pituitary gland.
    - *Brain stem*
        - It regulates subconscious activities necessary to maintain life-B.P, heart rate, respiration, and co-ordination of movement.

Endocrine system

- It regulates metabolism, reproductive cycles, broodiness etc. The various hormones involved are:
    - *Pituitary gland*
        - The hormones involved are
            - Anterior pituitary- gonadotropic hormones- FSH and LH
            - Posterior pituitary- Oxytocin
    - *Thyroid gland*
        - Thyroxine is involved in metabolism, and is important in the regulation of maternal activities, female cyclic sexual activities and sex drive in both the sexes.
    - *Adrenal gland*
        - Androgen secreted from the adrenal cortex are important in male reproduction. Adrenaline secreted from the medulla is involved in threat behaviour, fight or flight behavior, fighting among the males and in maternal protective behaviour.
    - *Sex organs*
        - Testes- Testosterone is involved in typical male behaviour. Ovary- Estrogen is involved in typical female behaviour.
    - *Pineal gland*
        - Melatorin regulates the cyclic breeding behaviour.

CAUSE OF BEHAVIOUR

- Behaviour is a result of the sum of all the muscle contractions and hormonal secretions. It is due to the interaction of nervous system with the external and internal conditions.
- Most of them genetically programmed while certain behaviours are stereotyped that is they cannot be changed.
- Behavioural responses in animals are determined by heredity or internal factors and by external factors or learning experiences. Some responses are caused by interaction between these two factors.
- Breeds of horses have been developed for running, pacing or for draft purposes but a single breed does not excel in all three. This emphasizes that certain inherited factors must be present before an animal can learn from experience to excel in a given area.
- Heredity
    - Produces behavioural responses in several ways. It can affect the growth and development of many body parts like sensory and/or motor organs. All animals including humans have almost the same components of brain, but only a few species (e.g. parrot) can be taught to speak a human language. Hence, an animal can show only those behavioural responses for which it has appropriate neural mechanisms.
- Learning and experience/ external factors
    - Certain behaviours will be inherently present at birth. These are called as Instinct or reflexes. These are behaviours will be exhibited by the animals in the absence of opportunity to learn e.g., ability of animals to swim without learning; maternal behaviour etc.
- Animals are divided into two groups based on the instinct:
    - *Precocious:* Animals which have skills and abilities at birth itself, they are able to get up and walk immediately after birth e.g., calf, lambs.
    - *Altricial:* Newly born and helpless e.g., rats, rabbits.
- These instinctive behaviours can be modified by experience called as learning. E.g., Many birds possess nest-building capacity, but the quality of nest building improves with succeeding attempts-due to learning.

CHAPTER-51: LEARNING, INTELIGENCE AND MOTIVATION

Learning objectives

- To make the students to know about different terms related with behavioural physiology.

LEARNING

- It is a process, which produces adaptive changes in individual behaviour as a result of experience.
- Various types of learning are

Habituation

- It is the simplest type of learning. It is the permanent weakening of a response as a result of repeated stimulation unaccompanied by reinforcement.

- In this, the animal will learn not to respond to irrelevant stimuli, i.e., upon repeated exposure to a stimulus, the animal gradually decreases its natural response until it may disappear entirely. Thus an animal will orient towards a moderate sound but with each successive exposure, its orientation is less and less until it orients no longer.

Conditioning (Associative learning)

- It is classified into two types
  - Classical conditioning
  - Instrument or operant conditioning

Classical conditioning

- Classical conditioning is that the animal learns to respond to an artificial stimulus (conditioned stimulus) in the same way (conditioned response) as for the normal of natural stimulus (unconditioned stimulus). E.g. the classical Pavlov experiment with ringing of bell when he fed meat to a dog, which eventually produced saliva with the sound of the bell alone without seeing, smelling or tasting the meat, because the animals are conditioned to perform reflex activity. This type of learning is the basis of training the farm animals. E.g. lactating cows learn to "let-down" milk when they enter the milking area.

Instrument or operant conditioning/ Trial and error learning

- In this type the animal learns by associating its own behaviour with the particular outcome in its own environment e.g., if the outcome is rewarding- access to food, the animal repeat the behaviour that resulted in food access.
- It therefore learns a conditioned response to obtain food. When animals are rewarded when they perform certain behaviour correctly (positive reinforcement) or are punished when they do it incorrectly, they associate the behaviour with the results.
- Animals learn to stay away from electric fence when they touch it and receive shock (Avoidance behavior). A behaviour that is learned by reward will be eliminated if the reward is withheld for a long time, which is called as extinction.

Insight learning (Reasoning)

- It is mostly prevalent in higher mammals. It is the ability to respond correctly at the first time when an animal encounters a certain situation; it enables the animals to solve a new problem by reasoning without trial and error.
- Severe malnutrition is very early life can result in life-long behavioural modifications - retarded ability to solve complex problems.

Imprinting

- It is seen most clearly in birds like chickens, ducks, geese and turkeys during their early period of their lives following hatching.
- E.g., after hatching, the chicks will have a tendency to approach and follow their mother based on her vocalization where as in other animals it is based on olfactional clues from the mother.

- The imprinting usually can occur within 36 h following hatching. If the bird is not imprinted with in the few days of hatching, it may never imprint at all.

INTELLIGENCE

- It is the ability of animals to adjust to certain situations successfully (or organization of behaviour, i.e. the ability to learn from experience and solve problems).
- Degree of intelligence varies with species and among the individuals of the same species.
- Primates are most intelligent mammals followed by carnivores, ungulates or grazing animals (cattle, elephants, horse and swine) in the order.
- Memory plays an important role in determining intelligence. There are two types of memory. 1. Short term memory and 2. Long term memory. Memory is related to protein synthesis in the brain.

MOTIVATION

- Motivation is the drive or tendency to respond in a particular manner. Hypothalamus of the CNS controls many types of behaviours; e.g. hunger, thirst, sex drives etc.
- Motivation often arises as a reflex response to cues or signals that correspond. e.g., to a prey an enemy or a social or sexual partner. Other factors that affect motivation are level of arousal, level of autonomic activity, hormone levels and emotion producing functions within the limbic system.
- Motivation usually leads to a certain goal directed behaviours. e.g., if there is a drive for quenching thirst, an animal will go in search of water, if water is found, the goal is reached and water is consumed. Therefore the drive for water is reduced.
- Many of these drives are expressions of physiological regulatory systems e.g., in body water regulation, a rise in osmolality will be detected by the osmosensitive neurons in the hypothalamus which activates the water conservative mechanisms and triggers the drive of thirst.

CHAPTER-52: TYPES OF BEHAVIOUR

Learning objectives

- To learn different types of behavioural responses exhibited by the animals in their life time.

INGESTIVE BEHAVIOUR

- This includes eating and drinking and it is closely related to the growth and health of animals.

Feeding behaviour

- Various species chew their food in different in different ways:
    - Rabbits and rodents chew their food with their sharp incisor teeth.
    - Ungulates use their tongue, lips and teeth during feeding.
    - Cattle mostly use their tongue to encircle the grass and pull it into their mouth.

  - Sheep graze more closely than goatbecause of the flexibility of their upper cleft lip.
  - Carnivores gulp down their food.
- Special feeding patterns which are seen characteristic to certain species:
  - Rabbits are coprophagus
  - Dogs bury or carry bones from one place to the other. Pigs root and turn over the soil.
  - Social influences also affect the feeding in that group feeding will tend to eat more and dominance relationships often develop causing dominant animals to eat more than subordinate animals.

Grazing behaviour

- Grazing animals eat the tender parts of the plants and reject the coarser parts. They first eat the plants which are more desirable ,then they will graze the less desirable ones.
- Most grazing animals do not eat poisonous plants when forage supply is adequate, but when overgrazing is practiced, they may consume poisonous weeds.
- Cattle and sheep graze mostly during daylight. If the day temperature is very high, cattle will graze more at night. Grazing species functions as social unit.
- Goats are browsers and have no bitter taste receptors and enjoy a variety of plants.
- Cattle graze about 3 miles per day. Zebu cattle graze longer and travel farther than European cattle. Sheep travel - about 3 to 8 miles per day.

Rumination behavior

- Calves will start ruminating as soon as they start eating grass or hay.
- Rumination can occur during standing or walking but more prominent when lying down and they do not ruminate while asleep

Dietary selection/ nutrient selection

- Smell and taste usually affects the palatability and consumption of foodstuffs. E.g., If given a choice, pigs usually select a nutritionally adequate diet. In most of the species intake of sodium is well regulated.
- Decrease in sodium in the blood leads to an increase in appetite for sodium. In case of phosphorus deficiency, cattle, which graze on phosphorus poor land, may eat bones of dead animals.
- Certain aversions also develop, e.g., Bait shyness of rats. Rats that survive after eating poisoned bait will refuse to eat the bait again. In birds food intake is mostly determined by vision.
- Young animals are more likely to learn to adapt to a new feed than the adults.

COMMUNICATION BEHAVIOUR

- Animals communicate with each other by means of visual, auditory and chemical communication systems. The ways of communication are:
  - Visual
    - Some of the behavioural responses to visual stimuli are genetic, e.g. hens will crouch in fear when they see a flying image resembling hawk.

        - Some visual responses are learned -animals hurry towards feed trough when a person approaches them during the anticipated time of feeding; sexual attraction of females birds to the colorful plumage of males.
    - Auditory
        - Sound is an important means of communication. Female farm animals respond to the distress calls of their young. Bats use ultrasound echolocation to fly in dark.
        - Auditory communication has advantages - sounds can travel around corners and can be readily detected.
    - Chemicals
        - Olfaction is a kind of chemical communication. The stimuli can travel over long distances and are not affected by visual or auditory conditions and messages conveyed are highly specific.
        - Olfactory bulb plays a special role.

Pheromones

- These are chemicals secreted by one animal and influence the behaviour of another animal. Territorial marking behaviour of many mammals depends on pheromones released through urine or glands, e.g., territorial boundaries marked by dogs.
- One well-known pheromone of mammals is n-butyl mercaptan of the skunk.
- Pheromones generally modulate sexual behaviour e.g., attraction of male dogs to bitches in estrus. They accelerate puberty in young rodents and regulate maternal behaviour (rejection of newborn of another animals by a mother).
- Olfactory mucosa and vomeronasal organs play an important role in detecting pheromones. The species -specific curling of the upper lip known as flehmen helps in detecting the pheromones by the VNO and it is commonly observed in cats and ungulates.

SOCIAL BEHAVIOUR

- It deals with characters like aggressive/agonistic, gregarious and dominant behaviour.

Agonistic (combat or aggressive) behaviour

- This behaviour involves self defense (submission), offence (aggression), escape and passive activity. Males are more likely to fight than females, but females also exhibit fighting behaviour under certain circumstances. Castrated males are submissive, which indicates the involvement of testosterone in this type of behaviour.
- Threat gesture at times help to prevent actual fighting and body damage; e.g. barring of teeth by the dogs, pawing of the ground by angry bulls. The response to aggressive behaviour evokes counter aggression or appeasement signals. Dogs signal appeasement by rolling over and spreading the hind legs to expose abdomen.
- Bulls, boars, rams and stallions that live together from a very young age seldom fight because they have settled their social rank among themselves. In a group of young bulls, many will drive a single animal to death, if-he is not removed from the group - this phenomenon is called as butter syndrome.
- When two sexually mature males of the same species are bought together, it will always result in fight. Sometimes one becomes submissive after a short fight, at other times fighting becomes intense, which may lead to injury to one or both or even death of one.

- The aggressive behaviour involves limbic system areas like hypothalamus and amygdala.

Gregarious behaviour

- This behaviour indicates preference to live in flocks or herds observed in certain species. It is advantageous in detecting enemy and to obtain food especially in the wild state.
- Cattle roam in group of various sizes depending on the size of pasture. Cattle breed promiscuously (engaging in sexual intercourse indiscriminately).
- Sheep have a tendency to run in flocks and usually follow a leader, which can be an older ewe.
- Swine are gregarious in nature in the wild, but in confinement, this behaviour is not well observed.

Social dominance

- Within a group of gregarious animal species, there is a well-organised social rank order. One individual is dominant over all others; while others are ranked as second, third so on down to one, which is subordinate to all. This order is seen in all species of farm animals; since males are kept away from females, the social rank order is important only in females.
- The social rank order is called peck order in chickens. When two strange hens meet for the first time, they fight vigorously, till one hen finally runs away, ending the fight; this fighting goes on whenever they meet again until one becomes subordinate to the other dominant hen. If a rooster is allowed into a flock of hens, the rooster assumes command of the flock, followed by the usual hierarchy among the hens.
- Social rank affects food consumption; when feed is limited, the dominant individual get more than its share of feed and the subordinates starve and perform, inefficiently. Amygdala is involved in determining social ranking.
- Several factors influence social rank order. When a male chicken is castrated, it goes to the bottom of the rank; injection of testosterone of hens enhances its social rank. Age also increases the social rank and young ones have a low rank. In cattle, size, weight and aggressiveness also influence the ranking. The social dominance is heritable.

SEXUAL BEHAVIOUR

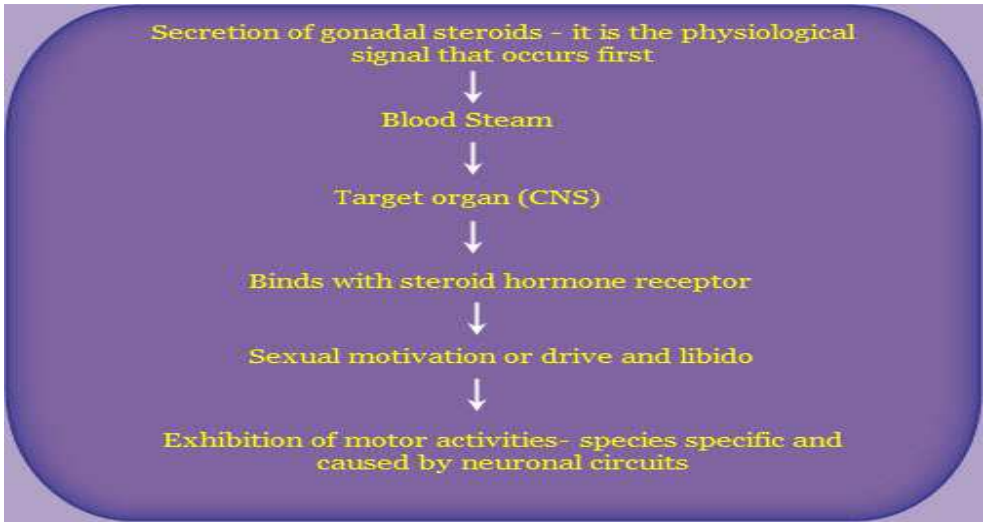
- This behaviour mainly involves courtship and mating. Hormones mainly control it though some amount of psychological (learned) factors is also involved.
- Female farm animals are restless during estrus (heat) and have intense sexual desire. Mares in estrus sequel and urinate profusely. Cows and sows will stand when mounted by other females (this behaviour is not exhibited by mares and ewes).
- Males detect females in heat by sight or smell. Courtship behaviour involves a linked sequence of stereotyped behaviours, e.g. mating behaviour of a bull begins with his approaching the genital area of a female (probably in response to pheromones), nuzzles the vulva and the underside of tail, which elicits urination in the 4ow. The bull then licks the vulva, the underside of the tail followed by flehmen behaviour. In response, the cow signals her willingness to be mounted. Sexual behaviour is influenced by social facilitation; members of the opposite sex stimulate each other hormonally.
- The stallion bites or teases the mare in heat, preparatory to mating. The boar nudges the sow around the: head and nose and "grunts". The rooster will spread one of his wings and perform a dancing act before the hen.

- Male sexual behaviour is influenced by the social environment, sensory ability and sexual stimuli and testosterone. Female are sexually active during estrus and it is regulated by estrogen. During proestrus and estrus of dogs, vaginal secretions contain pheromones called as methyl-p-hydroxy benzoate, which is responsible for attraction of males.
- Boar taint- urine of boar contains 5d androsterone. This causes the sows in estrus to assume a reflex known as rigidity reflex- mating posture in sows.

Pheromones are the cause of two classic phenomena in mice

- Whitten Effect
  - When a male mouse is introduced into a group of female mice, it causes synchronization of estrus because of release of GnRH.
- Bruce Effect
  - When a male mouse is introduced into a cage of an already bred mouse it leads to failure of pregnancy because introduction of male mouse blocks the release of prolactin.
- In domestic animals similar to the Whitten effect, Ram effect is noticed. Introduction of a ram into a flock of anestrus ewes, leads to synchronization of estrus due to the release of GnRH.

Physiological mechanism/neuroendocrine basis



Role of nervous system in sexual behaviour

- Hypothalamus- by releasing GnRH, regulates FSH and LH and, estrogen: as a part of limbic system, involved in regulating sexual behaviour.
- Spinal cord and medulla- to maintain postural reflexes; mating posture during copulation.

- Parasympathetic nervous system- involved in erection of penis and clitoris; parasympathetic nerves supply almost all reproductive organs.

Motor patterns during courtship/mating/copulation

- Sexual arousal
- Sexual display
- Erection
- Penile protrusion
- Mounting
- Intromission
- Ejaculation
- Dismounting
- Refractoriness

Sexual behaviour in males

- Sniffing, licking the female, Flehmen reaction, nuzzling-rubbing the female body surface with nose

- Sexual behaviour in females
    - During estrus- restless, loss of appetite, attempting to mount, bellowing Sniffing- particularly the scrotal region of male Mating posture- allow the males to mount

Species-specific characters

- Cow and goats- bellowing and bleating; mount on other animals and allow other animals to mount; goat turns its head back during mounting; sheep - flattening of tail and also turns itsfoack.
- Sows - grunting sound produced from guttural pouch; lordosis- binding the back like an arch is the mounting posture specific in pigs; cocking(erection) of ears are seen.
- Mares - frequent urination, winking of vulva and clitoris, vaginal mucus exudation

Duration of courtship and copulation

- Sheep- less than a second to few seconds.
- Cattle- few seconds
- Horse- 40 seconds
- Pigs- 2-5 minutes (minimum) to 20-30 minutes (maximum)

Factors affecting sexual behaviour

- Breed end strain affect genetic factors - variation in libido.
- Physiological factors- These are determined during fetal and postnatal life depending on neflnatal androgen.
- Environmental factors- These have more effect on male than the female. Males become more active when new females are introduced in a herd. Season and climate affect the sexual behaviour. Sexual drive is less during the hot climate.

- Previous experiences

Abnormal estrus patterns

- Nymphomania - also called "Buller cow"- cow acts like bull. It is a hereditary- disorder in cattle; cystic follicles are common in this condition, which produces more estrogen leading to persistent estrus; voice and behaviour will be like male.
- Split estrus - common in mares. There will be estrus for short period with interruptions.
- Quite estrus - there will not be estrus signs but ovulation will occur and if inseminated it will conceive.
- Prolonged estrus - also common in mares with prolonged estrus far. a period of 1-6 weeks.
- Weak estrus — estrus remains for a short period and conception rate is less.

Abnormal sexual behaviour in males

- Homosexuality - males exhibit mounting on male; common when young males/ are housed in pen
- Hypersexuality — increased sexual activity; increased frequency of copulation
- Hyposexuality - few animals fail to mount and abnormality in ejaculatory patterns
- Autoerotic behaviour - self sexual arousal; seen in stallions- it rubs its penis against hypogastrium; in bulls- pelvic movements of penis in and out of preputial orifice resulting in ejaculation

MATERNAL BEHAVIOUR

- Maternal behaviour begins shortly before birth and end after the young are weaned. This behaviour varies among different farm animals.
- In the **cows** after the young is born the mother rises, inspects and starts licking to clean the calf s hair coat clean and shiny, removes the placenta and starts to nurse the calf. The mother is possessive with her young during the nursing period. If the calf encounters any trouble, the mother comes to its rescue, and may even injure a person who comes too close.
- The mother and young uses visual, tactile, taste and olfactory communications to identify each other. At the time of weaning, the cow and calf bellows almost continuously for 2 or 3 days after which they become quite. After several days of separation, the bond between them is severed and they show no sign of recognition.
- Before parturition, the **sow** prepares nest and marks an area contoured to her body. It lies in such a way that soon after birth, the piglets can easily approach the mammary gland. Some sows are nervous during farrowing and may eat piglets. Sows are very protective of their young and will defend them when they squeak. The sow emits sharp barking grunts in rapid succession, when an intruder approaches the sow. She cares for her young till weaning, and 2 to 3 days after weaning she loses interest in them. Sows will accept piglets from other litter, if placed with them during the first 1 or 2 days of farrowing.
- Within few days after birth, each piglet in the litter locates a teat, and the same teat is used during the remaining period of nursing. Runt piglets get rear teats where availability of milk is lesser than the front teats.
- The **ewes** after parturition lick the newborn lamb to clean it. The lamb staggers to its feet and try to find teat. The ewe stands quietly during this process. The lamb wags its tail during nursing. Ewe will accept and suckle orphan kids and vice versa.

- In **hens**, broodiness (inclination to sit on eggs) is the maternal behaviour and it is inherited. It is influenced by prolactin. Broody hen will sit on nest with eggs or even without eggs and they spend very little time to obtain food or water during the incubation period.

Abnormal maternal behaviour

- Cannibalism - common in carnivores and birds
- Placentophagy - common in herbivores and carnivores, absent in horses and camels
- Stealing the young ones - is common in sheep, when a ewe is in prepartum it will have a strong maternal motivation which makes it to steal the young ones of other ewes.

THERMOREGULATORY BEHAVIOUR

- These behaviours are controlled by hypothalamus.

Hot weather adjustments

- Animals seek out comfortable environments. E.g., Grazing cattle seek out shade; camels alter their body orientation to minimize sun exposure; animals can reduce their metabolic heat by decreasing food intake.

Cold weather adjustments

- Young animals huddle against each other and against their mother to conserve heat when exposed to cold. Such behaviour leads to social attachment.

CHAPTER-53: BIOLOGICAL RHYTHMS

Learning objectives

- To study about meaning of the biological rhythms and their related terms with examples.

CIRCADIAN RHYTHM

Biological rhythms

- Most of the animals can measure the time of the day accurately. The biological clock is an integral part of animal physiology. Environmental factors by acting on the organisms set at the clock.
- Many physiological processes vary according to the time of the day.
- Daily variations observed in physiological processes are variation in behaviour, body temperature, heart rate and secretions of G.I. tract and endocrine glands.

Circadian rhythm

- It is the biological rhythm occurring every 24 hours. It consists of circadian period in constant conditions i.e., the duration of light and darkness is not altered. Circadian rhythm is self-generated or endogenous in nature.
- Circadian period or the period of rhythm is also called as FRP- Free Running Period. The self-generated or endogenous nature of the rhythm mainly depends on light intensity and temperature.
- Most FRP do not differ from 24-hour period by more than 1 or 2 hours on either side. Such periods are referred to as circadian (about a day); biological rhythms, which have circadian periods, are called circadian rhythms.

Factors influencing circadian rhythm

- Light signal, light cycle, day light or photoperiod.
- Temperature
- Bird song
- Social interaction
- Other poorly defined factors like psychological factors and social factors in humans.

Free running period

- It varies according to species and the magnitude of FRP depends upon the intensity of light- it varies by few minutes in different species.
- Squirrels- 23-24.3 h; sparrow- 24.3-26.2h; mice- 25h; birds- 23.3h
- Aschoff s rule- Increase in light intensity causes lengthening of FRP in nocturnal animals and increase in light intensity shortens FRP in diurnal animals.
- In birds and mice if providing 100-lux candlelight increases the light intensity, the FRP in mice is increased to 25.6h from 25h whereas in bids it is reduced to 22.2h from 23.3h.
- Pineal gland is involved in regulating circadian rhythm. It is called as the natural clock, internal physiological clock or internal clock. Photoperiod acts on the animal through the pineal gland; it contains serotonin, which is the precursor for melatonin synthesis by the enzyme hydroxyl indole methyl transferase.
- Melatonin is released only at limited intervals and depends on photoperiod (day length); light inhibits melatonin production and blood concentration of melatonin is found during dark(night). Melatonin inhibits gonadal function. During long days, reproductive processes are augmented. Long days promote egg production in chicken.

Other changes taking place during circadian rhythm

- ACTH secretion is increased before awakening
- STH is more during early hours of sleep. Cattle will not sleep for long hours; i.e. from midnight to dawn, cattle remains in a quiescent state.

Jet lag sickness

- It is also known as time zone effect or pilot's fatigue. It is the disturbing effect in the circadian rhythm due to flying over long distances covering different zones. E.g., a flight

leaves New York at 7 pm and arrives in Amsterdam at 1 am. That person is ready for sleep but the local time is 7am and he loses one night sleep. This disturbing effect is jet lag sickness.

- *Circadian rhythm:* Biological activities which are taking place over a day.
- *Diurnal rhythm:* activities taking placej at a particular time of the day
- *Ulradian rhythm:* rhythm taking place for a short time in a day
- *Intradian rhythm:* rhythm that will take more than a day to occur but less than a year.

EXERCISE-1: ESTROUS CYCLE IN FARM ANIMALS

Introduction

- Estrus (oestrus) is derived from the Latin word "*oestrus* " meaning *frenzy* or *gadfly* or *sexual desire* (original Greek work "*oistros*" has the same meaning)
- Estrous cycle, or *oestrous cycle*, is the reproductive cycle found in most mammalian placental females during which there are recurring periods when the female is sexually receptive (estrus) and fertile and interrupted by periods in which the female is not fertile and sexually receptive (anestrus).
- Estrous cycle comprises recurring physiologic changes that are induced by reproductive hormones in most mammalian placental females.
- Estrous cycles start after puberty in sexually mature females and are interrupted by anestrous stage or pregnancy.
- Most female placental mammals (dogs, cats, elephants, New World monkeys so on) undergo an estrous cycle; humans and the great apes (chimpanzees, gorillas, and orangutans) experience a true *menstrual cycle*.
- Animals that have estrous cycles reabsorb the endometrium if conception does not occur during that cycle. Animals that have menstrual cycles shed the endometrium through menstruation when pregnancy does not occur and in which the female may be sexually receptive at any time during the cycle. The term *menstruation* refers to the periodic discharge of blood and cells from the lining of a non-pregnant uterus. (Some animals that undergo estrus - domestic cats and dogs, do produce a very short and mild flow of blood)
- There are many variations among animals in their estrous cycles.
- Animals that undergo estrous cycle only one time a year during a particular season are called *seasonally monoestrous* animals (bears, white-tailed deer, foxes, wolves). Some animals exhibit estrous twice a year – e.g. dogs are *diestrous* animals
- *Polyestrous* species exhibit estrus several times a year if not pregnant - cats , cows , and pigs .
- S*easonally polyestrous* animals undergo more than one estrus during a particular season of the year if they do not become pregnant – (horses, sheep); these animals can be divided into short-day and long-day breeders.
- *Short-day* breeders are sexually active during short day-length as in winter–sheep, goat *Long-day* breeders are active sexually during increasing day-light period as in summer – horses, hamsters.
- According to the nature of ovulation as spontaneous or not, mammals can be broadly grouped into

- Spontaneous ovulators: those that do not require an act of coitus to induce follicle rupture and ovulation, rather by a series of hormonal events ovulation occurs; e.g. most mammals including rats, mice, guinea pigs, cows, horses, monkeys, sheep and humans
- Induced ovulators or "reflex ovulators,": those that require the act of copulation to stimulate ovulation. Coitus stimulates GnRH release that leads to release of LH and induces ovulation. Sexual receptivity occurs in cycles that are spontaneous, but ovulation does not occur without copulation; they require the act of copulation to stimulate ovulation, which occurs about 12 hours after mating e.g. cats, ferrets, minks, and camels.
- Induced estrus and ovulation: A few species undergo behavioral estrus and ovulation only in response to certain stimuli; females require physical contact to begin behavior estrus and mating to promote ovulation –e.g. prairie voles, montane voles *(voles and moles are small rodents resembling mouse* also called field mouse in North America*)*, and meadow moles.

ESTROUS CYCLE

- The estrous cycle is conveniently divided into 4 phases:

Proestrus

- One or several follicles of the ovary start to grow - number is specific for the species; lasts from one day to 3 weeks, depending on the species.
- Endometrium starts to develop under the influence of estrogen. Some animals may experience vaginal secretions that could be bloody. The female is not yet sexually receptive.

Estrus

- Estrus refers to the phase when the female is sexually receptive ("in heat," or "on heat").
- Ovarian follicles mature under the influence of estrogen which is regulated by gonadotropic hormones and estrogen secretions exert their biggest influence.
- The animal exhibits a sexually receptive behaviour that is exhibiting physiological changes visibly. A signal trait of estrus is the lordosis reflex, in which the animal spontaneously elevates her hindquarters.
- In some species, the vulvae are reddened.
- Ovulation may occur spontaneously in some species (e.g. cow), while in others it is induced by copulation (e.g. cat). If there is no copulation in an induced ovulator, estrus may continue for many days, followed by 'interestrus,' and the estrus phase starts again until copulation and ovulation occur.

Metestrus

- During this phase, the signs of estrogen stimulation subside and the corpus luteum starts to form.
- The CL begins to secrete small amounts of progesterone. This phase is brief and last 1 to 5 days. In some animals bleeding may be noted due to declining estrogen levels.

Diestrus

- Diestrus is characterised by the activity of the corpus luteum that produces progesterone.
- In the absence of pregnancy the diestrus phase (also termed pseudo-pregnancy in cats, dogs) terminates with the regression of the corpus luteum.
- The lining in the uterus is not shed, but will be reorganised for the next cycle.

BEHAVIOURAL SIGNS OF ESTRUS IN MARE

- The mare may act different to other horses, and may be more nasty to them, but act interested in stallions or geldings
- Vicious temperament
- Frequent urination
- Brownish yellow discharge from vagina
- Tail head is often raised and clitoris is frequently exposed by frequent "*winking*" of vulva
- Vulvar mucous membrane is congested and orange or scarlet in colour
- During estrus, mare turns her back to stallion and stands quietly with tail raised to one side

BEHAVIOURAL SIGNS OF ESTROUS IN CATTLE AND BUFFALO

- Restlessness
- Bellowing frequently
- Standing for mounting by a bull (heterosexual behavior)
- Standing for mounting by another female (homosexual behavior)
- Being mounted by another female but moving away rather than standing (homosexual behavior)
- Attempting to mount another female (homosexual behavior)
- Smelling other females, trailing other females,
- Depressed appetite,
- Nervous and excitable behavior,
- Mud on hindquarters and sides of cattle,
- Roughed up tail hair, vulva swelling and reddening,
- Clear Mucus discharge flowing as a string from vulva, and mucous smeared on rump.
- Spontaneous sinking of the loin
- Reduced milk yield
- Frequent micturition
- When the skin of lumbar or sacral region is held tightly, all animals arch their back but animals in estrus in addition to arching its back, raises its tail to one side – this is called loin reflex or tolerance reflex
- When an animal allows other animals to mount her while she remains standing - called standing heat. Cattle may be willing to mount others but may not stand to be mounted when outside of standing heat - usually indicates the animal is in early or late heat.
- observation at least twice daily, typically early in the morning and late in the evening makes better detection of heat
- Buffaloes are shy or poor breeder; seasonally polyestrous - expression of heat limited to eight months of the year and sexually inactive from March to June.
- Estrus symptoms in buffaloes are much less obvious than those in cattle - less than a third of buffaloes in estrus can be detected by homosexual behaviour.

- Symptoms such as swollen vulva, mucous discharge and increase frequency of urination are not regarded as reliable indicators of estrus.
- Estrus can be detected on the basis of reactions to a vasectomized male/ teaser or an androgenized female buffalo.
- Silent estrus is the common problem in the buffaloes and estrus in the buffaloes commences towards late evening, with peak sexual activity occurring during the hours of darkness.
- Temporary engorgement of teats (for 3 days) is exhibited by majority of buffaloes prior to the onset of heat and this proestrous behaviour is a useful tool of ensuing heat
    - Mucus discharge
    - Vulval edema
    - Sniffing the vulva

BEHAVIOURAL SIGNS OF ESTROUS IN EWE AND DOE

- Restlessness
- Anorexia
- Homosexual behaviour occasionally exhibited in goats not in sheep
- The doe walks restlessly, try to reach out for the buck or stand near the fence
- Frequent and peculiar sound – bleating good sign of heat in doe
- Reduction in milk yield
- Swelling of vulva – vaginal discharge – clear and colourless at the beginning of estrus and becoming whiter and opaque towards the end of estrus
- Wagging of tail continuously in doe – even in the absence of male - good sign of heat- probably helps to spread pheromones
- Signs of heat more clear in does than ewes
- Ewes exhibit a strong ram-seeking behaviour
- Signs of estrus in the ewe are much less pronounced than in the cow or doe and can usually not be detected unless a ram is present
- seek out the ram and stand still for him to mount them. Sometimes they wag their tails vigorously.
- They may nuzzle the ram around the belly or scrotum and even try to mount the ram.
- Ewes not in heat urinate when a male approaches it but ewes in estrus do not urinate when a male approaches it – probably to signal that they are not in heat.

BEHAVIOURAL SIGNS OF ESTROUS IN SOW

- Restlessness
- Mounting on other animals- the most definitive behavioral sign of estrus is standing to be mounted by the boar
- Lordosis - s ows in estrus will often assume this rigid stance, called the "*lordosis reflex*", when pressure is applied on the rump ("back-pressure") by the herdsman.
- Vulvar swelling
- Pink red colour of vulva
- Mucus discharge occasionally
- High-pitched grunting
- Onset of frequent tail-wagging at the onset of estrus
- The group -housed sow actively seeks out the boar.
- Other signs of estrus: depressed appetite, alertness, pacing, and gnawing of the jaws.

- Back pressure test or riding test or lordosis test is the most efficient and practical method for estrus detection in sows

BEHAVIOURAL SIGNS OF ESTROUS IN BITCH

- The most notable sign during proestrus is vaginal bleeding
- Vulva swollen and reddish
- Female paying increased attention to her rear end.
- Female dogs will become attractive to male dogs but the female shows little interest in male or even become aggressive towards males
- Restlessness
- Frequent urination with small quantities of urine. The urine contains both pheromones and hormones signalling to any interested males
- During estrus, vulvar discharge becomes less hemorrhagic, or even colourless than proestrus bleeding
- Vaginal swelling is maximum
- Attracts males and shows signs of courtship towards male
- When a male attempts to mount, she will move the tail to side to allow breeding. When the perineal area is massaged, the estrus bitches respond with tail movement with deflection of tail to one side – *tail turning reflex*
- Estrus averages 9 days in duration, but can be as short as 3 days or as long as 21 days.

EXERCISE-2: VAGINAL CYTOLOGY

Introduction

- The vaginal epithelium is responsive to sex steroids, particularly estrogen, and undergoes predictable changes through the estrous cycle in response to changes in blood concentrations of ovarian hormones.
- Rising levels of estrogen cause the vaginal epithelium to become "*cornified*" - the surface cells become large and flattened, with small or absent nuclei.
- Vaginal cytology is a *type of endocrine assay*. Tracking changes in the morphology of desquamated vaginal epithelial cells provides a convenient means of assaying changes in estrogen levels. The technique is used widely in managing canine breeding programs
- Vaginal cytology has utility in evaluating the reproductive function in several species-dogs, cats and rats.

PREPARING A CANINE VAGINAL SMEAR

Materials required: microscope slides, methanol or ethanol for cytology, blunt glass pipette with rubber bulb or cotton swabs (6”), saline solution, Leishman’s or methylene blue stain

Pipetted method

- Take 1ml of saline in a glass pipette, insert it into the vagina of a bitch
- Squeeze the rubber bulb many times – to mix the vaginal discharge with saline
- Suck the vaginal discharge and take the pipette out of the vagina

- Take a small drop of fluid from the pipette on a slide
- Spread the drop of the discharge over the slide by gently tipping slide vertically
- Remove the excess fluid from the slide by blotting
- Air-dry the smear

Swab method

- Moisten a cotton swab with saline
- Insert a speculum into the vagina
- Insert the moist swab into the vagina - after 1-2 inches of the swab has been inserted, the angle of insertion can be altered to roughly 45 degrees and insertion continued.
- When the swab is fully inserted, rotate the end through 2-3 revolutions, which will allow the cotton tip to pick up an adequate load of cells. The swab can then be gently withdrawn
- Prepare the smear immediately after withdrawal of the swab by *rolling* (not sliding or rubbing) the cotton tip along the length of a glass microscope slide
- Dry in air
- As so on as the smear is prepared, dip it 5 to 10 times in a container of methanol
- Stain with Leishman's stain solution – pour the stain over the slide and allow for 1 min, dilute the stain with equal volume of distilled water and allow 10-15 min time, wash the slide and dry it

VAGINAL EPITHELIAL CELLS

- Majority of cells observed in a normal vaginal smear are not vaginal epithelial cells - leukocytes, erythrocytes and bacteria are usually seen, and small numbers of other contaminating cells and microorganisms can also be observed
- The epithelial cells of vaginal smear can be classified into three fundamental types: parabasal, intermediate or superficial cells.
- Parabasal cells are the smallest epithelial cells - round or nearly round and have a high nuclear to cytoplasmic ratio
- Parabasal cells can be on smears taken during diestrus and anestrus, and also during early proestrus. Parabasal cells are conspicuously absent during estrus.
- Intermediate cells vary in size and shape, but have a diameter two to three times that of parabasal cells; these cells can be further subgrouped into
- small intermediates : nearly round or oval shape with large, prominent nuclei
- large intermediates : polygonal shape with a small nuclear/cytoplasmic ratio
- Intermediate cells are prevalent during all stages of the cycle except estrus.
- Superficial cells are the largest cells on a vaginal smear. They are polygonal in shape and distinctly flat. Their nuclei are either absent or pyknotic (very small and dark). Superficial cells without nuclei are referred to as "fully cornified".
- Superficial cells are seen in large sheets or strings - fully cornified cells.
- They are not seen during anestrus and increase in number during proestrus.
- The presence of large numbers of superficial cells or only superficial cells is the definite characteristic of cytologic estrus; their abrupt decline in number marks onset of diestrus.

Other cells

- In addition to the epithelial cells, a number of other cells are seen on vaginal smears.

- Erythrocytes are usually observed in large numbers during proestrus. In some bitches, they are seen through estrus and even into early diestrus.
- Neutrophils are abundant in smears taken during early diestrus, and are also seen at other stages, but rare during estrus. Moderate numbers of neutrophils are a common feature of normal canine vaginal smears and not indicative of vaginitis.
- "Foam cells" are non-descript epithelial cells containing numerous vacuoles that are seen on smears prepared during anestrus.
- Bacteria are often seen on vaginal smears in large numbers, covering cells and spilling onto the background.

CYTOLOGIC CHANGES DURING CANINE ESTROUS CYCLE

- Cytologic changes reflect the ongoing endocrine events of the cycle-they are always a better predictor of "fertile time" and gestation length than behavioral or physical signs
- Estrogen levels rise prior to and during proestrus and fall in conjunction with the preovulatory surge of LH. Rising levels of estrogen induce "cornification" that is characteristic of smears examined during estrus. Ovulation occurs two days after LH surge

Anestrus

- Intermediate and parabasal cells predominate in smears taken during anestrus. Superficial cells are absent or found in very small numbers. Neutrophils may also be present or absent.

Proestrus

- Concentration of estrogen rise during proestrus, leading to capillary breakage and leakage of RBCs through uterine epithelium, as well as proliferation of the vaginal epithelium.
- from early to late proestrus - a gradual shift from intermediate and parabasal cells to superficial cells.
- RBCs are present in large numbers and neutrophils are commonly observed. Large numbers of bacteria are also often present.

Estrus

- The characteristic of cytologic estrus is the predominance of superficial cells.
- Most, but not all, bitches will undergo full cornification, and the smear will reveal a pattern composed almost exclusively of anucleate superficial cells.

Diestrus

- The onset of diestrus is marked by precipitous decline in the number of superficial cells and reappearance of intermediate and parabasal cells.
- The cellular profile changes within a single day from 100% superficial cells to less than 20% superficial cells.
- Identifying the onset of diestrus is more accurate predictor of the time of ovulation, and hence gestation length, than sexual behavior. Dogs ovulate 5-7 days prior to onset of diestrus (7-9 days after the preovulatory LH surge), and hence, gestation length is usually 57 ± 1 day from the onset of diestrus day 1.

- The period of behavioral estrus is variable, and extends up to several days before and/or after cytologic estrus. Gestation lengths calculated from the onset or cessation of receptivity is correspondingly inaccurate. Onset of diestrus also correlates with loss of fertility, and breeding after diestrus are rarely fertile.

STUDY OF VAGINAL CYTOLOGY IN RATS

- The duration of estrous cycle in rats is 4–6 days. The four stages of the estrous cycle – proestrus, estrus, metestrous and diestrous can be easily identified cytological study of vaginal smears collected from the breeding female rats.
- About 6 – 10 female rats along with 1 – 2 males are brought to the laboratory three to four days before the beginning of recording the estrous cycle. Female rats are separated from the males, but kept close to males such that visual, tactile and olfactory sensations are maintained between the sexes without the possibility of mating. Such sensory stimulations from the males are essential for females to exhibit regular estrous cycle, otherwise the females enter into anestrous stage. The rats are to be maintained without stress and water and feed are provided *ad libitum*.

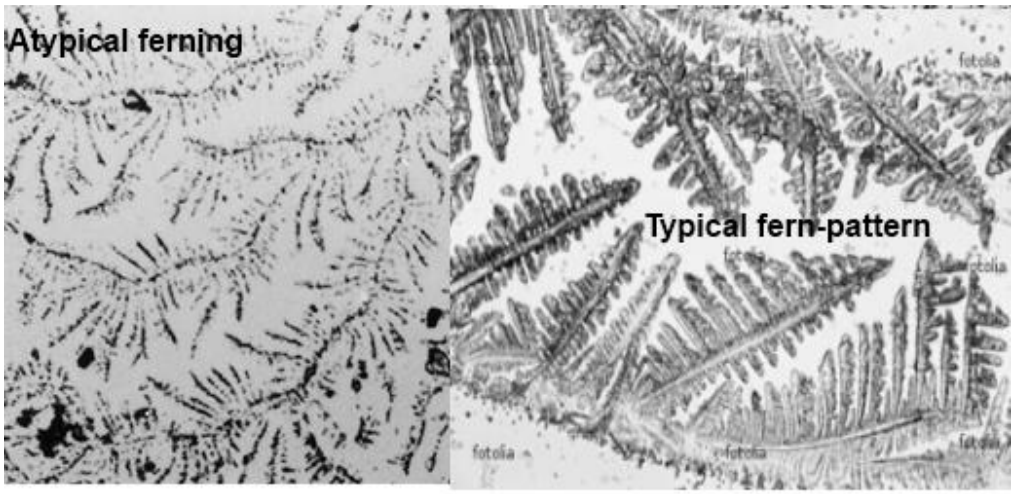
Procedure

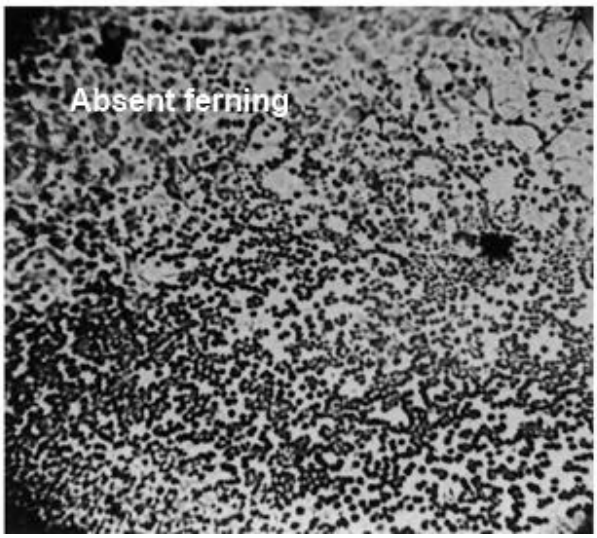
- By careful handling, vaginal fluid is aspirated from each female by flushing the vagina with few drops of normal saline using ink fillers. The aspirated vaginal fluid drops are smeared on the slide and observed under low power of microscope. The stage of the estrous cycle is identified based on the vaginal epithelial cells and leukocyte distribution, as per the criteria given below
- Diestrus: Only leukocytes and complete absence of epithelial cells (parabasal and intermediate cells)
- Proestrus: Few leukocytes and triangular or square shaped – intermediate and parabasal cells
- Estrus: Full of only square, large sized vaginal epithelial cells – predominantly superficial cells
- Metestrus: few leukocytes with appearance of ruptured or disappearance of vaginal epithelial cells
- Daily recording of the different stages of the estrous cycle for each rat has to be continued for a minimum of 8-10 days. Based on cytology, the rats can be classified into regularly or irregularly cycling rats.

EXERCISE-3: FERN-PATTERN OF CERVICAL MUCUS IN COWS

- "Crystallization" or arborization or fern phenomenon of cervical mucus produced by the stratified squamous epithelial cells of the cervix is one of the most sensitive to changes in the level of estrogen. The "ferning" is useful for the detection of ovulation and for assessing corpus luteum function, in the early diagnosis of pregnancy. Sodium and potassium ions along with colloids present in the mucus when become dried acquire the shape of fern-leaf
- Collect cervical mucus through a glass rod passed into the cervix, aspirate the mucus and collect it in a test tube or
- Obtain a piece of cervical mucus hanging from the vulva of a cow / buffalo
- Place the mucus on a glass slide and spread it evenly
- Dry the smear in air
- Examine the slide under a microscope

- Observe the different types of fern-pattern
- The crystallization is absent in mucus drawn during luteal phase and from pregnant animals.





EXERCISE-4: RECTAL PALPATION

Introduction

Examination of the reproductive tract of cattle and buffalo via rectal palpation is possible because of several anatomical features of the cow:

- The reproductive tract of the bovine is suspended from the dorsal body wall (as in other species) by the *broad ligament*.
    - The posterior part of the tract (vagina, cervix, sometimes part of the uterus) is located in the pelvic cavity

    - The anterior part of the tract hangs over the pelvic brim (uterus, oviduct, ovaries and, sometimes part of the cervix) and is suspended in the abdominal cavity.
    - The tract is located immediately below the rectum.
- The rectum of the cow is large enough to allow insertion of the hand and arm of the palpator and the rectal wall is pliable to allow one to grasp and reproductive tract structures through rectal wall.
- The rectal wall of the cow is strong so that tearing is rare. (in other species, esp. in mare the rectal wall is easily damaged during palpation).
- Rectal examination can be carried out in mare with greater care; however, in small animals, genital tracts can be examined by abdominal palpation.

FEATURES OF THE REPRODUCTIVE TRACT EXAMINED BY RECTAL PALPATION

Cervix

- The cervix can be grasped and manipulated to insert an artificial insemination pipette through the rings of the cervix and into the uterus. The cervix is also sometimes used to pull the reproductive tract into the pelvic cavity during palpation so as to make palpation of ovarian and uterine structures easier.

Uterus

- The uterus can be palpated to get a rough estimate of endocrine status of the cow.
    - Cows under the influence of estrogen have a uterus with "*tone*" where the uterus feels turgid and muscular.
    - Cows under the influence of progesterone have a more *flaccid*-feeling uterus.
    - A very small uterus can be an indication that a cow is reproductively quiescent or sterile.
- After calving, the uterus can be palpated to determine involution (return of the uterus to a pre-pregnant condition). The major criteria are size (reduction in size of uterine horns and a return of the horns to a more uniform size (i.e., the horn previously containing the fetus is no longer larger than the other horn), location in the body cavity (more forward into the pelvic cavity) and tone [i.e., large, soft and pliable and more turgid].
- During pregnancy (after ~ d 35-40), the presence of conceptus can be palpated by feeling for the "*fetal slip*". The fetal slip refers to the feel of the placental membranes moving through the fingers as the palpator first grasps the uterus and then lets it gradually slip through his fingers.

Ovaries

- Follicles can be felt as fluid-filled pliable structures; (only a fraction of number of follicles actually present are usually detected by rectal palpation) - ultrasound examination is required to detect all follicles on the ovary.
    - The corpus luteum is a hard structure that protrudes from the smooth curvature of the ovary.
- Cystic ovaries are identified by rectal palpation as very large ovaries (twice or more in normal size). *Cysts* are follicular structures that have grown very large in size (>2.5 cm) which persist for 10 or more days and which lead to inhibition of ovulation and

nymphomania. Follicular cysts can be luteinized or not. Luteinized cysts (about 30% of cysts) have a thicker wall than nonluteinized cysts. Luteinized follicular cysts can be confused with a fluid-filled corpus luteum which is a normal tructure that does not disrupt the estrous cycle.

USES OF RECTAL PALPATION

- Artificial insemination
- Non-surgical recovery of embryos for subsequent embryo transfer
- Pregnancy diagnosis
- Postpartum reproductive exam to determine suitability of the animal for breeding and to estimate whether the cow is cyclic.
- Clinical diagnosis of cystic ovarian disease and uterine infections.

PROCEDURE FOR EXAMINATION OF REPRODUCTIVE TRACT BY RECTAL PALPATION
MATERIALS AND PRECAUTIONS

Materials required: Full-arm plastic gloves, crate, apron, gumboots, lubricant, soap, water and towel

Precautions:

- Palpators nails should be properly trimmed
- Rings, watches should be removed
- Protective clothing, cloves and gumboots should be worn

PROCEDURE FOR EXAMINATION OF REPRODUCTIVE TRACT BY RECTAL PALPATION
ENTERING THE RECTUM

- The cow is restrained by placing it in a crate
- The cow is approached in such a way that minimizes fright to the animal. Do not make sudden movements or act in a loud, frightening way. Dairy cows will usually not kick although some will. One can approach the cow slightly from the side to avoid kicks. The cow can be gently touched on the flank or back to make it aware of the palpator's presence and to evaluate the cow's temperament.
- Before beginning palpating, the arm used for palpation (either arm can be used) should be covered with a full-arm gloves.
- After donning the palpation gloves, it should be lubricated with K-Y jelly.
- By approaching the cow sideways, grab the tail with the ungloved hand and push it aside (for cows prone to kicking, the tail can be raised perpendicular to the cow to prevent kicking). The anus can be identified as the upper of the two openings under the tail (the lower one being the vulva).
- To enter the rectum, form the hand into a cone shape by bringing the fingers and thumb together and use a slight rotary motion to insert the fingers and hand into the rectum. Entering the rectum takes some physical effort because of the strength of the anal sphincter muscle and because of peristaltic contractions in the rectum.

  - To enter the rectum while strong contractions are taking place, the palpator must use enough force to overcome the contractions while being careful to avoid injuring the cow.
  - Do not use excess force during strong peristaltic wave of rectum or when the animal is straining or tenesmus is present
- After the rectum has been entered, the palpator should remove faecal material from the rectum by using the cupped hand as a rake. Removal of faecal matter is not always necessary if the faeces are not abundant or are very soft.
- Insert the hand up to pelvic rim
- It is possible for the cow to suck air into the rectum, especially if the palpator makes many rapid in-an-out motions with the arm. When this occurs, it is nearly impossible to palpate the reproductive tract because the rectum balloons outwards. To reduce the problem, air can be removed by grasping a fold of the rectal wall and slowly moving it backwards to the anus.

PROCEDURE FOR EXAMINATION OF REPRODUCTIVE TRACT BY RECTAL PALPATION

FINDING AND PALPATING STRUCTURES IN THE REPRODUCTIVE TRACT

- For a new palpator, the cervix is usually the best landmark to help find the reproductive tract.
  - The cervix can be identified as a rope-like or chicken-neck-like structure about 3 inches in diameter that is usually in the pelvic cavity along the midline.
- The cervix can be picked up and held in the hand. Do not attempt to grab the cervix between thumb and the first two fingers of the hand because it is hard to hold the cervix in this manner. Rather, grab the cervix from the side while placing the fingers underneath the cervix and the thumb on top. If the reproductive tract is located far forward in the abdominal cavity, one can use the cervix to pull the reproductive tract into the pelvic cavity.
- The uterus can be identified and examined by following the cervix forward.
  - The cervix and uterus is grasped in the hand so that the hand lies over the tract with the palm down and the thumb underneath the tract. The external bifurcation can be identified as the point where the two uterine horns branch from the uterine body. The two horns should be roughly the same size (during late pregnancy and early postpartum period uterine horns will be unequal in size).
  - The uterine horns can very greatly in size depending upon the reproductive status of the animal. At estrus, the horns tend to be in a turgid, muscular state. Acyclic and sterile animals will have a very small reproductive tract. After calving, the uterine horns will be very large, with one uterine horn larger than the other and the reproductive tract displaced into the abdominal cavity.
  - Empty uterus usually contracts reflexly in response to palpation which can interfere with palpation examination, uterus should be palpated before attempting to palpate the ovaries
- The ovaries are either placed slightly underneath the uterus or located to the side of the uterus at a variable distance. Sometimes, the ovaries can be found by grasping the uterus, and then, after releasing the uterus, turning the hands counterclockwise while using fingers to probe for ovary.
- Once located, the ovary can be palpated with the fingers while the ovary is held in the palm.
  - The fingers can be moved along the entire surface of the ovary.
  - Follicles appear as pliable, fluid-filled structures. Care must be taken to avoid rupturing preovulatory follicles.

    - Corpora lutea appear as hard structures that protrude from the edge of the ovary.
- The oviducts are usually not found during palpation unless some pathology is present (i.e., oviductal inclusion or infection).

EXERCISE-5: SEMEN EVALUATION

- The purpose of semen evaluation: to estimate its fertility level

Semen evaluation

- Semen quality influence greatly on the fertility level and its evaluation is to be completed before insemination
- Laboratory tests on semen quality are NOT ACTUAL MEASURES OF FERTILITY

(I) Gross examination

- *Semen volume*: Health factors and inadequate collecting procedures: decrease the volume
- *Semen color*:
    - Bull: creamy white
    - Ram and buck: creamy white and thick
    - Boar: milky
    - Stallion: greyish white and thin
- Semen density: Too thin: from azoospermia or oligospermia
- Granular appearance of the layer along glass tube or semen sample on the slide glass: due to sperm movement

(II) Progressive motility

- See Experiment No.6

(III) Direct cell count (Hemocytometer):

- See Experiment No.7

(IV) Photoelectric colorimeter

- Based on light absorption by sperm in semen sample
- Used by commercial AI centers due to fast determination
- Calibration: By using samples of known concentration estimated by hemocytometer
- Dilution of semen: With 2.9% sodium citrate buffer

(V) Electronic particle counter

- Principle: When only one cell at a time can pass through a capillary, sperm head of diluted semen causes an abrupt increase in resistance between the electrodes at both sides of the capillary. Frequency of resistance change is registered on a counter

- Uses: Most accurate determination but needs an expensive instrument

(VI) **Sperm cell morphology**

- Abnormal sperm %:
  - Should be examined once a month
  - Abnormality of 20 to 25%: not affect fertility

Preparation of slide

- Place a small drop of 2.9% sodium citrate solution on a warm slide
- Mix a semen sample with the citrate solution using a stirring glass rod
- Place another slide on sample slide and both are gently pulled apart and dried

Staining slides

- Stain slides with eosin, along with a background stain as nigrosin

Examination and counting

- Examine several fields on each slide with a magnification of 400x and count a total of 100 sperm per slide

(VII) **Differential staining of live and dead sperm**

- See Experiment No. 8

(VIII) **Speed of sperm**

- Speed at which sperm travel: can be measured by electronic equipment objectively
- But the equipment is not used routinely because it is too expensive and speed of sperm is not highly correlated with the fertility level
- Average speed of bull sperm: 6 mm /min.(up to 21 mm / min)

(IX) **Evaluating frozen semen**

- Freeze-thaw sperm survival rates: Greatly vary in different males and different ejaculates from the same male
- Post-thaw % progressive motile sperm should be measured accurately because CR in AI depends greatly on number of motile sperm inseminated
- Between 5 and 15% of ejaculates from bulls meeting the standards for processing are discarded after freezing, with the highest rate in summer
-

(X) Photographic method of determining progressive motility. Time-exposure dark-field photography:

- An objective method of determining motility using a dark-field microscope equipped with a camera
- Dilution of semen: 10-million/ ml
- Chamber used: Petroff- Hausser counting chamber (20u in depth)
- Warming samples: for 30 sec. on 38°C warming plate
- Taking pictures: 6 parts at the periphery of the ruled area with exposures for 2 sec. and with magnification to the film: 75x
- Motile cell counts: Number of tracks on the film following development:
- Highly repeatable results, but time-consuming: 30 to 35 determinations/ technician/ 8-hour day

(XI) Changes in acrosome

- Acrosomal cap deterioration: From aging of sperm or injury to sperm
- Examination:
    - Incubate thawed semen in a 37°C water bath for 2 - 4 hrs.
    - Slide preparation: Similar to slide for measuring % progressive motile sperm
    - With differential interference contrast (DIC) or phase contrast optics at 800x
- Changes in acrosome by freezing:
    - Good quality dairy bull semen: Sperm with intact acrosomes : 90%; 60 to 65% after freezing
    - Most beef bulls: 50 to 55% intact acrosomes in post-thaw samples
- Uses:
    - Correlations between NR rates and intact acrosomes: higher than those between NR rates and post-thaw motility
    - More widely used than the photographic method of determining motility

(XII) Other tests

- Resistant of sperm to clod shock
- Measuring metabolic activity:
    - Oxygen uptake
    - Fructolysis
    - Methylene blue reduction time
    - Resazurin reduction time
    - pH change

EXERCISE-6: ESTIMATION OF MOTILITY OF SPERMATOZOA

Principle

- Sperm progressive motility : % sperm those are motile in a straight line under their own power
- Most important test: No. of motile sperm inseminated is highly correlated with fertility
- Range in fresh ejaculates: 0 to 80%

- Use multiples of 10% in reporting motility because of its subjective measurement
- Ejaculates with initial motilities of 50 to 80%: little different in fertility level, if desired no. of motile sperm are present at insemination
- Semen with less than 40% initial motility: not suitable to use, if not from exceptionally superior bull

ESTIMATION OF MOTILITY OF SPERMATOZOA
MATERIALS REQUIRED

- Neat and diluted semen
- Glass slides
- Cover glasses
- Microscopes with stage thermostat.

ESTIMATION OF MOTILITY OF SPERMATOZOA
PROCEDURE

Observation of Swirls and Eddies in Neat Semen (mass motility)

Preparation of slide

- Gently invert the semen tube 2 or 3 times
- Dilute small subsamples with isotonic solution
- A uniform small semen droplet on a microscope slide, placed on a microscope stage heated to 38°C
- Cover the semen droplet with glass cover slip to spread to a uniform thickness and to prevent drying

Use of microscope

- Use stage incubator to prevent changes in temperature
- Examine in several fields under low power objective lens
- Examination of semen diluted with egg yolk, milk and glycerol: Speed of sperm is reduced but % motility is not altered by viscosity
- Circular motility: due to cold shock or extension with hypotonic diluters

Under low power, look for the vigour of swirls and wave formation in undiluted semen and grade them as follows:

Number of swirls (Per second)	Approx. motility (%)	Grade

4-5	80 and above	+ + + +
3-4	70 and above	+ + +
2-3	60 and above	+ +
1-2	50 and above	+

Individual Spermatozoa Motility (initial motility)

- Put a drop of diluted semen on the clean glass slide and cover it with the cleaned cover glass.
- Focus it under low power and then high power of microscope at 38 ° C temperature.
- Record the motility pattern as follows:

S. No	Motility pattern	Motility of spermatozoa (%)	Grade
A	Forward and progressive	80% and above 70%	Excellent
B	Forward and progressive	70% and above 60%	Very good
C	Forward and progressive	60% and above 50%	Good
D	Forward and progressive	above 50%	Fair
E	Circular or static movement and less progressive and forward movement	Below 50%	Poor

EXERCISE-7: ESTIMATION OF CONCENTRATION OF SPERMATOZOA

- Accurate determination of concentration i.e. the number of spermatozoa per unit volume (per millilitre) of semen is important because dilution ratio of semen for AI depends on it.

- Concentration of spermatozoa can be assessed by tests based on consistency or appearance of semen or by the use of haemocytometer , colorimeter or electronic particle counter.
- The haemocytometer method is slower but more accurate. The colorimeter method is both fast and accurate, but the equipment may not be available at all places.

Direct Cell Count Method

- In this method haemocytometer is used. It is a direct visual counting of sperms in a standard volume of semen under a microscope.

ESTIMATION OF CONCENTRATION OF SPERMATOZOA
AIM

- To estimate the sperm concentration to know the quality of semen and to fix the extension ratio or dilution rate later on. It is also helpful to know the semen producing ability of a bull.

ESTIMATION OF CONCENTRATION OF SPERMATOZOA
MATERIALS REQUIRED

- Haemocytometer set (thick glass slide with grid lines, cover glass and Thoma Red cell pipette)
- Watch glass
- Microscope
- Eosin containing diluting fluid
- Glass rod for mixing the diluting fluid with the semen

ESTIMATION OF CONCENTRATION OF SPERMATOZOA
PROCEDURE

- Count Sperm number in 1.0 x 1.0 x 0.1 mm chamber
- Dilution rate (recommended):
    - Bull, ram and buck semen: 200x
      Boar and stallion semen: 100x
- Dilution with pipettes for red blood cell counting:
    - Take a drop of semen in a clean watch glass.
    - Add a speck of eosin powder just to stain the semen to a pale pink color.
    - Take the R.B.C. pipette and draw the stained semen up to 0.5 mark
    - Draw 1 per cent formal saline into the pipette exactly up to 101 mark, which results in a standard dilution rate of 1:200
    - Rotate the pipette with quick wrist motion holding the pipette between the thumb and forefinger. This ensures proper distribution of sperms.
- Sample preparation on counting chamber:
    - Focus the R.B.C. counting chamber of the haemocytometer under high power objective of the microscope
    - Place a cover slip over the counting chamber

- Take the R.B.C. pipette, mix once again, discard the initial three to four drops and gently charge the counting chamber by placing a drop of the diluted semen near the cover slip. Avoid air bubbles or floating of the cover slip with diluted semen.
- Wait for sometime for uniform dispersion of sperm cells and then count the total number of sperm cells lying within the five big squares (normally all the four corners and the center square).

ESTIMATION OF CONCENTRATION OF SPERMATOZOA
CALCULATION

Total Spermatozoa Count/mm^3	= Number of cells counted (5squares) x depth of chamber x dilution rate x area of square
Total number of sperm cells counted in 5 big squares	= X
Number of cells in 1mm area	= 5X (5 squares counted/ total no. squares of 25)
Depth of the chamber	= 0.1mm (when cover slip is placed over haemocytometer)
Correction factor for 1 mm depth	= 0.1/1 = 10
Dilution factor	= 200
[0.5 mark of semen is diluted to = 101 mar	
After discarding diluent's part in stem = 100 mark]	
Number of sperm cells /mm^3 of neat semen	= 5X x 10 x 200
	= X x 10, 000 / m l or mm^3
Number of sperm cells in 1 ml of undiluted semen	= X x 10,000x 1000
	= X x 10 million / ml

EXERCISE-8: ESTIMATION OF LIVE AND DEAD SPERMATOZOA

The aim of this estimation is to find out the proportion of live and dead spermatozoa in the given semen sample.

Principle

- Principle
  - Eosin cannot pass through living cell membranes but can pass through nonliving cell membranes
  - A background stain such as nigrosin helps make the unstained sperm heads visible
- % live sperm will always be somewhat higher than % progressively motile sperm
  - Depending upon the degree of membrane disintegration the sperm will be colored from pale pink to deep red.

ESTIMATION OF LIVE AND DEAD SPERMATOZOA
MATERIALS REQUIRED

- Clean microscopic glass slide
- 5 per cent eosin stain
- 10 per cent nigrosin stain
- Microscope
- Glass rod

ESTIMATION OF LIVE AND DEAD SPERMATOZOA
PROCEDURE

Preparation of slide

- Take a clean glass slide and place a small drop of 5 per cent eosin solution and 2 drops of 10 per cent nigrosin. In between these two stains, place a drop of semen immediately after collection. Ensure that the semen, stains and glass slide are at the same temperature.

Staining slides

- Stain slides with eosin, along with a background stain as nigrosin
- Take a glass rod with blunt ends and in a quick wrist motion mix the semen eosin mixture with nigrosin.
- The mixing should be uniform and should be done quickly within a few seconds (30seconds).
- With the help of another slide (using it as a spreader) draw the mixture of stains and semen uniformly as a thin smear and draw several such slides.
- Immediately after drawing, wave the slides in hot air or place them on a warm table to ensure quick drying.

Examination and counting

- Examine the slides under oil immersion at a total magnification of 1000 X (10X eyepiece and 100 X objective).
- Dead spermatozoa will appear red or pink in the head region due to the penetration of the stain eosin and live spermatozoa will appear white in color or partially white.
- Partially stained spermatozoa have to be counted as dead ones.
- The field for examination of spermatozoa should be chosen at random and at least a total of 200 spermatozoa should be counted. There should not be any preference for particular field for the ease of counting. Normally, in artificial insemination programs semen samples of bulls with more than 20 per cent dead spermatozoa are not used.
- Live and dead spermatozoa count is used before a bull is used for artificial insemination program and periodically to ensure that good quality semen is being used for artificial insemination. Occasionally, if the bull is encountered with any stressful condition such as vaccination or outbreak of disease, the estimation of live and dead spermatozoa is recommended.

ESTIMATION OF LIVE AND DEAD SPERMATOZOA
PRECAUTIONS

- The stains and the semen should be at the same temperature at the time of mixing.
- Mixing of semen and stain should be done uniformly and quickly for correct estimation or live spermatozoa.
- The slides drawn should be dried immediately and examined that day itself.
    - Examine several fields at random and count 300 to 500 total cells
    - Partially stained sperm: should be counted as dead sperm

ESTIMATION OF LIVE AND DEAD SPERMATOZOA
CALCULATION AND OBSERVATION

- **Calculation**
- $$\text{The percentage of live sperm} = \frac{\text{Total number of live sperm}}{\text{Total number of sperm counted}} \times 100$$
- **Observation**

Field. No	No. of Live sperm counted	No of Dead sperm counted	Total No of sperm counted
1			
2			
3			
4			
Total			

EXERCISE-9: ESTIMATION OF ABNORMAL SPERMATOZOA

Morphology of spermatozoa

- Morphological examination of the spermatozoa is a detailed test for assessing the semen quality.
- Every ejaculate contains some abnormal spermatozoa and samples with a high proportion of abnormal spermatozoa can be expected to result in poor fertility.
- Periodical morphological examination of sperm from individual bull is a valuable tool for assessing any normal deviation from normal.

- Overall length of bull spermatozoa: 68 - 74 ☐m

Head	Length : 8 - 10 ☐m
	W idth : 4 - 5 ☐m
Neck	0.3 - 1.5 ☐m
Mid-piece	8 - 10 ☐m
Tail	Principle piece : 45- 50 ☐m
	End piece : 2 - 4 ☐m

- 60 per cent of the anterior head is covered by acrosome or galea capitis.
- *Sperm abnormalities are produced by* :
  - Inheritance
  - Diseases
  - Adverse environmental effects
  - Improper handling of semen
- The sperm abnormality increases with:
  - Extremes of temperature
  - Infection of the reproductive tract
  - Congenital testicular hypoplasia
  - Acquired testicular degeneration
  - Epididymal dysfunction
  - Ephemeral fever
  - To a subtle extent with advancing age.
- There are several methods of classifying sperm abnormalities and the most convenient one is the classification of abnormalities according to the region of the spermatozoa.

ESTIMATION OF ABNORMAL SPERMATOZOA
CLASSIFICATION OF SPERM ABNORMALITIES

Based on the location of abnormality

- Head abnormality - micro, macro, pyriform, narrow, double, detached knobbed heads and detached galea capitis.
- Neck abnormality-broken, abaxially placed neckd protoplasmic droplet in neck
- Mid piece abnormality - swollen, narrow, broken, double, abaxially placed mid- piece and protoplasmic droplet in mid-piece.
- Tail abnormality - coiled, double, forked bent, looped, detached tail and presence of protoplasmic droplet in tail

Based on the cause of abnormality

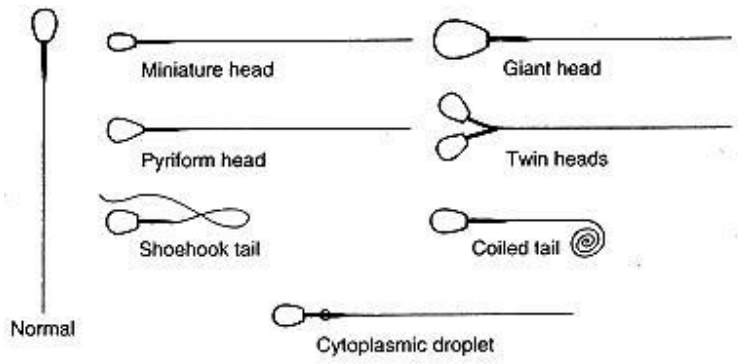
- Primary: due to impaired spermatogenesis or spermiogenesis.
- Secondary: due to damage during travel in epididymis.
- Tertiary: due to damage at the time of collection or preparation of smear.

Based on essential or specific and non-essential or non-specific abnormality:

- Non-specific abnormality indicates sub-normal fertility.
  - Knobbed acrosome defect - hereditary -recessive gene - HF-bulls - sterile
  - Dag defect - hereditary - Danish Jersey bulls
  - Diadem defect - due to disturbed spermiogenesis
  - Cork-screw defect - due to progressive testicular degeneration
  - Pseudo - droplet defect - Holstein Friesian (HF)
  - Decapitated defect - hereditary - Guernsey breed of cattle – bulls - sterile.
- Morphology of spermatozoa is studied by suitable staining techniques viz. Indian ink or Eosin (Y)/ (B) - Nigrosin stain or 3 per cent Rose Bengal stain or 1 per cent Mercurochrome stain.

METHOD- I (Eosin - Nigrosin method)

- Place one or two drops of eosin - nigrosin stain and a small drop of semen separately at the one end of a clean, warm microscopic glass slide. The drops should be allowed to reach the same temperature before mixing.
- The semen and stain should be mixed and drawn out with the edge of another slide which serves as a spreader so that, a thin film is formed. Push rather than pull the edge of the slide.
- Allow the smear to dry before examination and examine the slide under oil immersion objective of the microscope.
- Count and examine at least 200 spermatozoa from different fields of the slide. Increasing the number of spermatozoa assessed increases the accuracy of the test.



METHOD - II (Rose Bengal or Mercurochrome method)

- Dilute the semen sample with Tris buffer or sodium citrate buffer or clear normal saline at the rate of 1:10 (take 0.1 ml of neat semen and mix with 0.9 ml of buffer).
- Place a small drop of diluted semen on a clean dry glass slide.
- Prepare a smear with the help of another glass slide held at an angle of 45 °
- Allow the smear to dry completely in air.
- Stain the smear with 3 % Rose Bengal or 1% Mercurochrome stain for 5 to 10 minutes.
- Wash the slide with tap water and dry it in air.
- Examine the slide under oil immersion objective of the microscope.

- Count a minimum of 200 spermatozoa and tabulate the observations.
  - Click here for method of smear preparation for determination of abnormal spermatozoa
  - Click here for procedure for sperm abnormality estimation
  - Click here for determination of abnormal spermatozoa percentage
- Click here for head abnormalities - Double head ; Giant head ; Pyriform head ; Knobbed acrosome ; Acrosomal lost ; Micro head
- Click here for mid piece abnormalities - Broken neck ; Abaxial placement ; Mid piece defects
- Click here for tail abnormalities - Coiled tail ; Proximal droplet ; Distal droplet ; Double tail ; Dag defect

ESTIMATION OF ABNORMAL SPERMATOZOA CALCULATION AND PRECAUTIONS

Calculation

The percentage of

$$\text{Abnormal sperm (\%)} = \frac{\text{Number of abnormal sperm counted}}{\text{Total number of sperm counted}} \times 100$$

Precautions

  - Avoid artifacts at the time of preparation of the smear on the slide.
  - Avoid cold shock and use of hypotonic solution if diluted semen is used.
- Semen should not be used if the total abnormality exceeds 20 percent. Out of this, head abnormality should not exceed 3 to 4 per cent, whereas mid-piece abnormality should not exceed 4 to 10 per cent and tail abnormality of about 10 per cent. Generally head abnormality is considered to be more serious than other sperm abnormalities.
- Normally, semen containing more than 15 per cent abnormal spermatozoa is not recommended for use in artificial insemination.

EXERCISE-10: BASIC METHODS IN ENDOCRINOLOGY AND REPRODUCTION

Introduction

- The basic methods in endocrinology include both Classical Methods that are still used and the Modern methods, which have helped in the vast development that has taken place in this field in recent times.

THE CLASSICAL METHODS IN ENDOCRINOLOGY AND REPRODUCTION

- **Ablation by Surgery and Replacement Therapy:** This method is based on the principle that upon removal of an organ, the functions performed by that organ will be lost

and upon replacement therapy, the lost functions are restored and this confirms the functions of the organ. This method involves removal of the organ under investigation and observing the consequences. This is followed by replacement therapy by administering the extract of the gland or transplantation of the removed gland.

  - (E.g.) Caponized cockerels followed by retransplantation of the testis to the body cavity; diabetes mellitus could be produced by removal of pancreas; ovariectomy in pregnant animals leads to abortion and administering progesterone prevents abortion.

- **Histochemistry:** The hormones being chemical, certain endocrine cells preferentially take up some particular stains and this helps to identify the hormone secreting cells. (e.g.) Masson trichrome stains the prolactin secreting cells of the anterior pituitary cells to red color, while gonadotropin and thyrotropin secreting cells are stained blue.
- **Immunohistochemistry** is advancement over histochemistry. This involves raising antibody against a particular hormone. These antibodies are tagged with some fluorescent compound and the tagged compounds are applied to tissue section and allowed to attach with the hormone if it is present in the tissue. If fluorescence is seen, it is indicative of the presence of that particular hormone. Even small quantity of hormone, if present in the tissue can be detected and even intracellularly located hormone can be detected.

BIOASSAYS

- This method helped in detecting and quantifying hormones before the development of radioimmunoassay. The basis of bioassays is measuring the animal's physiological response to a hormone. (e.g.) ovarian weight gain response to FSH; ovulatory response to LH;
- An advantage of this method is that it measures only the biologically active form of the hormone and not the inactive forms that may be present in the sample. The disadvantages are it is laborious, expensive not sensitive to measure smaller quantities of hormones etc.

IN-VITRO ASSAYS

- **Radioimmunoassay:** The most significant method that lead to tremendous advancement in the field of endocrinology. This technique was developed by Nobel Prize winners, Berson and Yalow in late 1950's and early 1960's and the first hormone measured by this method was insulin.
- **Radioreceptor Assay:** The RIA principle is used to assay and quantify the hormone receptors
- **ELISA:** This method eliminated the use of radioisotope.

ULTRASONOGRAPHY

- Ultrasonography is a non-invasive method to visualize and examine the internal organs. It is useful in the reproductive studies.

SURGICAL INTERRUPTION

- Hypothalamo-hypophyseal disconnection (HPD) is done to remove the hypothalamic control of pituitary functions. This method helped to detect the role of releasing and inhibiting hormones from the hypothalamus on the pituitary functions.
- HPD animals are used as animal models in many reproductive studies.

NIMAL TRANSLOCATION

- Translocating animals from one geographical location to another helps to study the effect of environment on the physiology of the animal.
- This method helped has helped in the studies on the seasonality of reproduction in animals.
- Now, this kind of studies is not necessary since different environmental factors can be created in environmental chambers.

EXERCISE-11: HORMONE ASSAY BY RIA

- S.A. Berson and Rosalyn Yalow first developed the radioimmunoassay technique in 1959-60 for the hormone insulin.
- Since then this technique is being used to measure various hormones and other bio molecules.

Advantages

- Highly specific
- Highly sensitive, capable of measuring very small quantities without need for elaborate samples preparation or purification.
- Large number of samples can be analysed at a time.
- Simple procedure.

Disadvantages

- It involves radioisotopes, which calls for utmost care in handling and disposal of the radioactive substance.
- Does not measure biologically active forms of hormones

PRINCIPLE OF RIA

- Radioimmunoassays are based on the interaction between an antigen (Ag; in this case, a hormone) and an antibody (Ab) specific for that antigen. The binding of these two is a reversible or equilibrium reaction.

$$Ag + Ab \square AgAb$$

- If we fix the amount of antibody present in the reaction, the distribution of the antigen between the bound and free phases is directly related to the total amount of antigen present. Thus, we can quantify the amount of antigen

The requirements for a radioimmunoassay are:

Purified hormone to act as standards

- Standards have a known amount of antigen and are used as a reference for determining the amount of antigen in tubes where the amount of antigen is unknown (i.e. the samples of blood, urine, etc.)

Labeled hormone ("Tracer")

- In a radioimmunoassay, purified hormone is "*labeled*" with radioactivity to form a *tracer*.
- Most commonly radioactive tracers consist of one or more ^{125}I molecules added to purified hormone. Less commonly, ^{14}C or ^{3}H are added.

Antibody

- This needs to be specific for the antigen (hormone) of interest. Often, it also needs to be species specific.

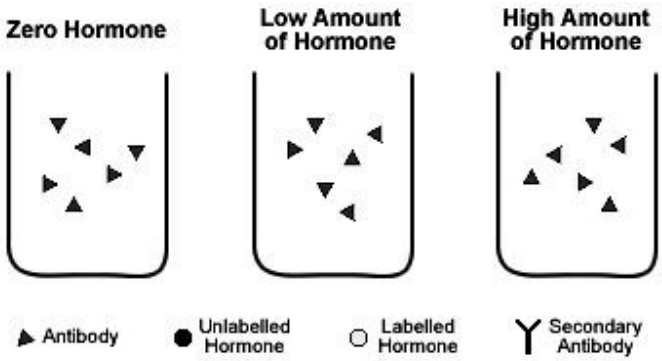
Method for separating bound and free hormone

- Most commonly, bound hormone is precipitated using a second antibody but there are also many other methods that can be used such as gel filtration, adsorption to charcoal or polyethylene glycol precipitation.

Extraction

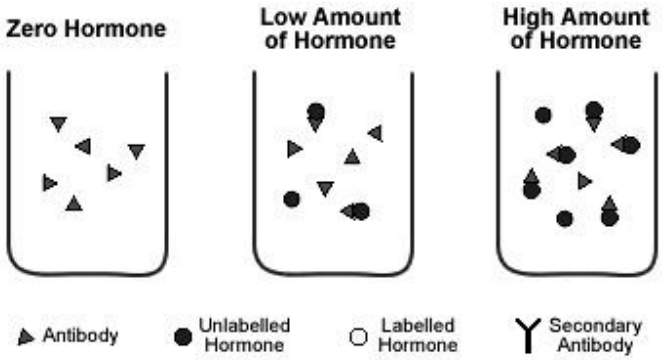
- Sometimes when a hormone is present in blood or urine at very low concentrations, or the hormone must be separated from other substances such as binding globulins, the hormone must be "extracted" from the sample.
- A large variety of methods may be used, but separation with organic solvents is most common, especially for extraction of steroid hormones.

RIA PROCEDURE



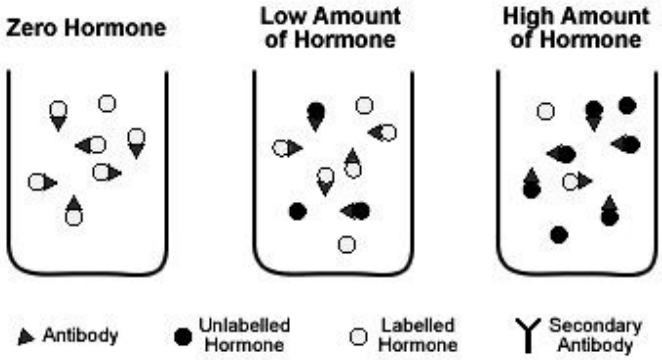
Step 1

Add a known and limiting quantity of antibody specific for the antigen (hormone) of choice.



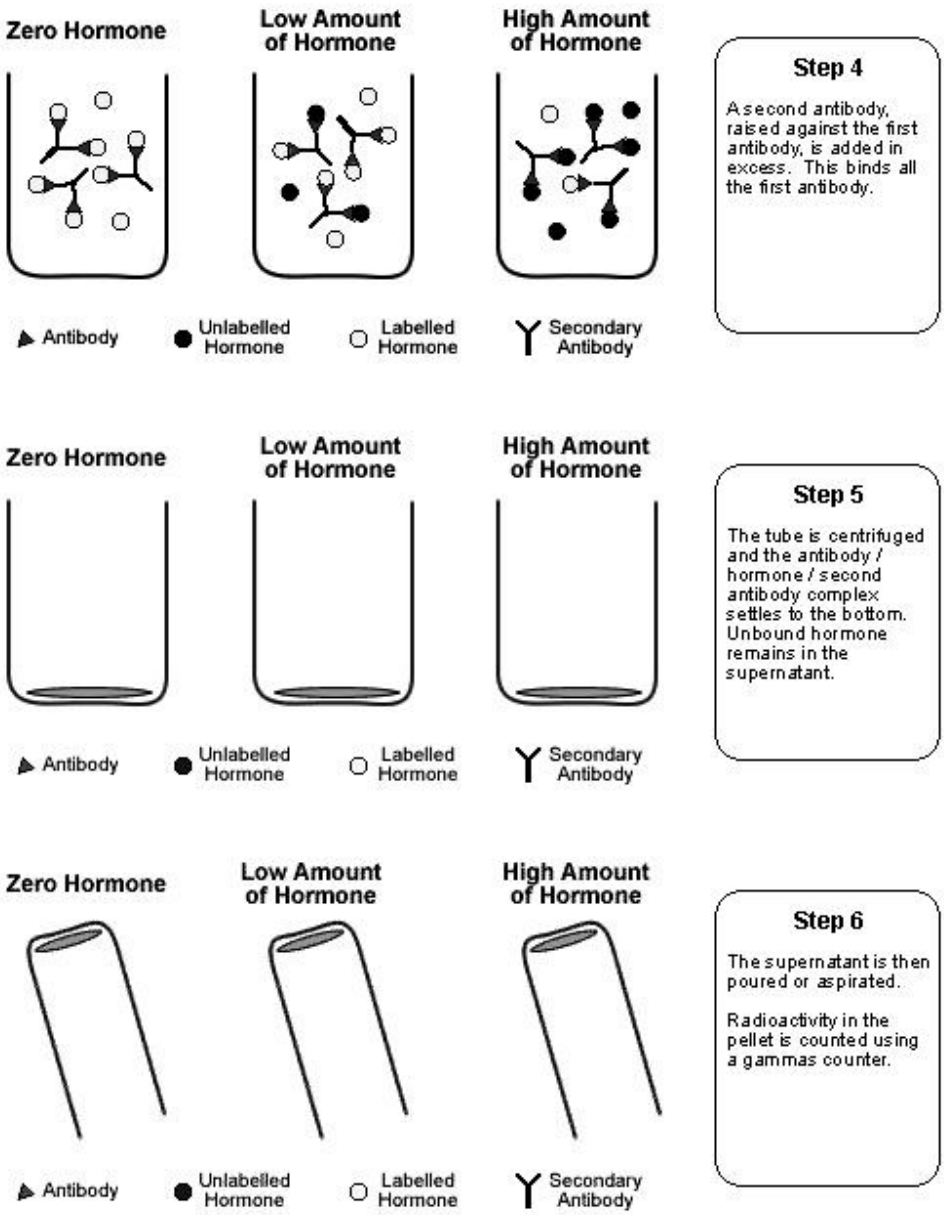
Step 2

Unlabelled hormone in standards or in unknown sample will bind to the antibody



Step 3

A known amount of labelled hormone is added, which will also bind to the antibody, competing for binding sites. The more unlabelled hormone present, the less labelled hormone is bound.



Incubation for antigen – antibody reaction

- Sample or standard plus known amounts of antibody and tracer are added to a tube (step2). As it can see from the figure when there is no unlabelled hormone present, all of the antibodies bind to tracer and so the amount of radioactivity that is bound to the antibodies is high.
- Both standards and the samples are set in duplicates. The test tubes are incubated at 4 ° C for about 24hours depending upon the type of assay.

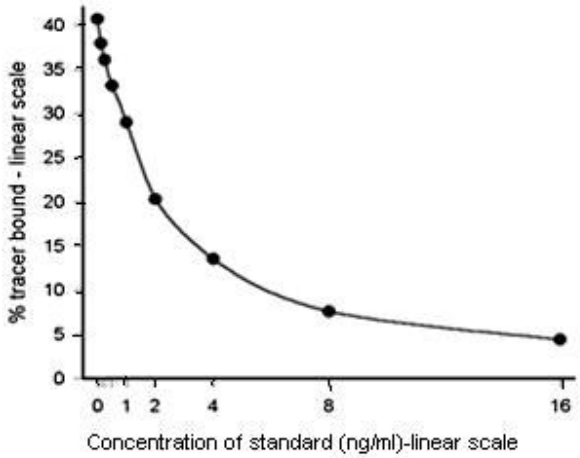
- The concentration of the antibody used should be to get 30 to 50 % binding of the tracer. The amount of tracer (labelled antigen) used is such that it gives a specified number of counts per minute per unit volume – usually 5,000 -- 15,000 counts per minute per 100 m l.
- Once sample or standard is added, these will compete with the tracer to bind with the antibody. The greater the amount of unlabelled hormone (from sample or standard) that is added to the reaction, the lesser the amount of tracer that is able to bind to the antibody. As a result, the amount of radioactivity that is bound to the antibodies will be less (step3).

Separation of bound and free radio labelled antigen

- There is a limiting amount of antibody in the reaction and so there will always be some hormone (both labeled and unlabelled) that is not bound to antibodies. This must be removed from the tube before counting the level of radioactivity in the bound complex
- Most commonly this is achieved by adding an excess of a second antibody, raised against the first (primary) antibody. This binds to the first antibody, creating a large, heavy complex. The assay tubes are then centrifuged and this large antibody complex settles in the bottom of the tube, forming a pellet (step5). The unbound hormone (including tracer) and excess second antibody remains in solution in the supernatant, which can then be poured off (step6). The radioactivity in the pellet is then measured using a gamma counter.

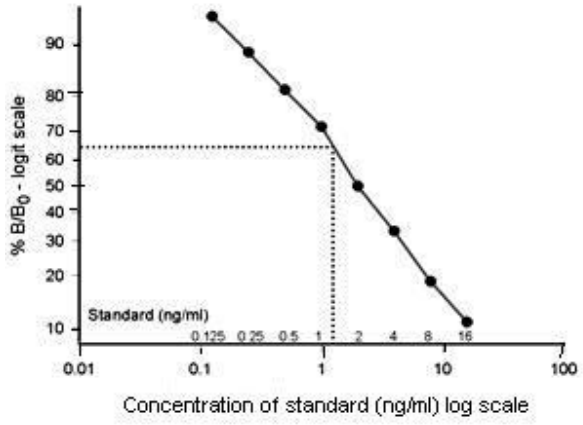
CONSTRUCTING STANDARD CURVES - I

- A standard curve is constructed using increasing amounts of purified hormone of known concentration. The concentration of each standard is plotted against the percentage binding. This can be done in one of two ways.
- In the first method, the number of radioactive counts per minute (cpm) for each standard is divided by the total number of counts of tracer added (determined by the total counts tubes).
- For example, if a particular standard tube produced 3000 cpm and the total counts produced by the tracer is 15000 cpm then the binding would be (3000/15000) × 100 = 20%. An example of a standard curve plotted in this manner is shown in the figure.



CONSTRUCTING STANDARD CURVES - II

- Alternatively, all values can be converted to a percentage of the total possible binding (Bo). Total binding is the binding achieved in tubes where no standard has been added (zero standard). The total binding is then assigned a value of 100% and the other standards are assigned as an appropriate proportion of this.
- For example, if the zero standards (Bo) had 3000 cpm and one of the standards gave 1500 cpm then the binding of that standard as a percentage of binding of the zero standards (B/Bo) would be 50%. An example of a standard curve calculated in this manner is shown in the figure below.
- The range of the standards used to construct the curve depends on the sensitivity of the particular assay and the expected range of values in the test samples. For example, in a bovine- LH assay, the standards used can be 0.0625, 0.125, 0.25, 0.5, 1.0, 2.0, 4.0, 8.0, and 16.0ng/ml. If the expected values are beyond the sensitivity of the assay, appropriate dilution on the test samples is done, before taken for the assay.
- The concentrations of the various reagents are set in such a way, that the required amount is present in 100 m l of the solution. Thus 100 m l each of the standard (pure antigen) / test sample, labelled antigen and antiserum and 200 m l of the buffer are all taken in the assay tubes and incubated.



EXERCISE-12: MEASUREMENT OF HEIGHT OF ANIMALS

- Measurement of height and weight of animals are essential for identification and to issue the certificates and to know the growth and health status of animals.

Horse

- They are generally measured in “hands” and “inches” (1 hand = 14 inches)

Measuring stand

- A *measuring stand* usually consist of an upright pole or stick graded up to 18 or 19 hands and a sliding cross bar carrying a spirit level. A levelled hard surface is chosen for

measuring the height of a horse and a horse will be measured to the highest point of the withers when standing square on level ground. The horses head and neck should be held at right angles to the upright part of the measuring stand.

- It should be measured from the left side and measuring is normally done soon after the horse has been shoed. With the held of the spirit level the cross bar is to be adjusted. The point is then read in hands or inches.

Pocket measuring tape

- It is a weighted tape, graded in hands and inches and this can be used on nervous horses.

Cattle

- The height of the cattle is expressed in inches or centimetres. It is taken immediately behind the hump or at the highest point of croup and one should mention from where the point of height is taken. While measuring the height, the animal is allowed to stand on a level ground with proper standing position.

MEASUREMENT OF WEIGHT OF ANIMALS

- The weight of the animal is essential for the following purposes:
  - To find out whether the animal maintains normal health or not.
  - To study the weight gain during experiments.
  - For computing rations.
  - For calculating the correct dosage of medicine

Direct method

- Usually adult animal are weighed once in a month.
- *Weighing bridge:* The animal is made to stand on the platform of the weighing bridge and it records the weight in pounds (lb) or kilogram (kg)
- *Spring balance:* Sheep and goat may be weighed by using the spring balance.
- *Platform balance:* Small ruminants and pigs can be weighed by using the platform balance.
- *Table balance:* It can be made use of to record the weight of poultry and rabbits

Indirect method (By using formula)

- *Shaffer's formula:* This is the most common formula used for cattle and buffaloes. This formula tends to under-estimate weights of very young and very heavy animals.

$W = LG^2/300$

  - In this formula the *weight of the animal is expressed in pounds.*
  - L - Length of the animal from point of shoulder to point of buttocks in inches

- G - is the chest girth of the animal in inches.

- *Minnesota formula:* In this formula the *weight of animal is expressed in kilograms*.

$W = LG^2/660$

- Where,
- L is the length of the animal from point of shoulder to point of buttocks in inches.
- G is the chest girth of the animal in inches.

- *Aggarwal's modified Shaeffer's formula:* Formula for Indian cattle is as follows

Live weight in seers = (Girth x Length) / Y

- Where Y is equal to 9.0 if girth is less than 65 inches
- Y is equal to 8.5 if girth is between 65-80 inches
- Y is equal to 8.0 if girth is over 80 inches. (One seer is equal to 0.93 kg)

- Mullick's formula for Buffaloes
  - X = 25.156 (Y) – 960.232
  - X = Estimate of body weight in pounds
  - Y = Heart girth in inches

EXERCISE-13: MEASUREMENT OF GROWTH IN FARM ANIMALS

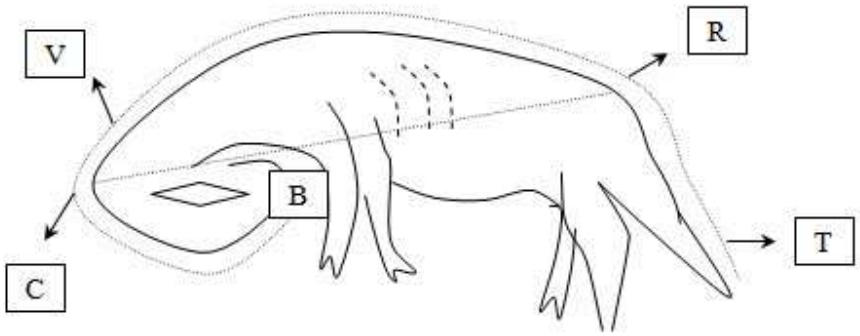
- Growth is an irreversible increase in the metabolically active protoplasm. Two fundamental components of growth are hyperplasia (increase in number of cells) and hypertrophy (increase in cell size). These processes cause change in body weight and body size (daily fluctuations like feed and water intake and disease conditions are not considered as growth). Monitoring of growth is essential to provide proper quality and quantity of feed economically for a viable animal industry.
- Growth in animals can be measured quantitatively by plotting a growth curve which indicates the rate on increase of mass or length over a period of time.

MEASUREMENT OF PRENATAL GROWTH

- It refers to the growth of sex cells, embryo and fetus.

Measuring foetal growth

- The size of the foetus is judged by the total foetal length, crown-rump length, curved crown-rump length or foetal weight.



BCVRT – Total Bent length of fetus
CR – Crown-Rump length (measured from highest point on skull)
CVR – Curved Crown-Rump length
VR – Length of Vertebral column
VRT – Vertebral column + Tail length

Average CR length of bovine conceptus at different ages

Age (days)	CR length
30	1
60	5
90	13
120	30
150	38
180	56
210	71
245	81
280	86

Weight of the bovine fetus

Stage of gestation	Weight of embryo/fetus
0-30	0.5g
31-60	5.9g
61-90	72.6g

91-120	531g
121-150	1.6 kg
151-180	3.8 kg
181-210	9.5 kg
211-240	17.7 kg
241-270	28.6 kg
271-300	39.9 kg

- In live conditions, actual weight of the fetus is difficult to measure. The foetal body confirmation can easily be assessed by ultrasound or X-ray techniques. These foetal measurements can also be employed for estimating the growth rate or the age of the fetus.

MEASUREMENT OF POSTNATAL GROWTH

Different types of growth rRates

- Absolute growth rate
  - Absolute weight gain per unit time; i.e.
    - $= W_2 – W_1 / t_2 – t_1$
  - Where,
    - W_1 = initial weight of the animal
    - W_2= final weight of the animal
    - $t_2 – t_1$ = time interval between two measurements
- Relative growth rate
  - Weight gain in given time in relation to initial weight and expressed in percentage
    - $= W_2 – W_1 / W_1 \times 100$
- Cumulative Growth Rate
  - It is the average of relative growth rates and expressed in percentage
    - = (X1+X2+X3+ X4 + ...Xn) / n
- Instantaneous growth rate
    - = (dw / dt) /W
  - Where as, dw/dt is the instantaneous weight gain and W is the initial weight.
- Practically it is impossible to measure the instantaneous weight gain dw/dt, so the following derived formula can be employed for easy calculation
  - $K = (\ln W_2 - \ln W_1) / t_2 – t_1$
  - ln – natural logarithm
  - W_2 & W_1 – final and initial body weights
  - t_1 & t_2 – final and initial recording time
- Time required to double the body weight = $\ln_2 / K$

Various methods of measurement of growth

Growth curve

- Growth after birth generally begins slowly, then undergoes a more rapid period and then slows down again or even may stop. When body weight of an animal is plotted against the age the curve obtained is known as growth curve.
- Body weight measurements
    - Body weight is taken to measure growth in almost all the farm animals. The sources of error in this method are
        - Variation in the gastrointestinal tract contents and urinary bladder
        - Adaptation of the animals to the experimental conditions
        - Sub clinical infectious diseases like bacterial, viral parasites etc
        - Frequency of weighing
        - Accuracy of weighing machines
- Measurement of weight of the animal
    - By measuring certain specific lengths, the body weight of the animal can be calculated by using formulas

Body weight in kg = L x G x G / 660

    - L – Length in inches from shoulder to pin bone of the base of the tail
    - G – Girth measurement in inches at the heart region. (Circumference of chest behind the hump)
- Estimation of Body Composition
    - This is used to find the nature of growth in animals. It can be employed in both living and slaughtered animals. Whether the growth is mainly due to deposition of water or fat or protein can be estimated.
    - In living animals it can be done by determining the body water content using tritiated or deuteriated water or body's specific gravity using air displacement method or helium dilution method and analysing the tissue biopsies for protein and fat content or by ^{40}K counting respectively. Although K is the most abundant intracellular ion, the accumulating lipids or growing animals are poor in K content. Large amounts of K are seen in muscles compared to other tissues in the body and the whole body K can be used as a quantitative index of the muscle.
    - In slaughtered animals, the carcass can be chopped as a whole or dissected separately into bone, muscle, adipose tissue, skin and other organs for estimating moisture, protein, fat, total ash and total carbohydrates.

EXERCISE-14: MEASUREMENT OF BODY PROTEIN AND FAT IN FARM ANIMALS

- Estimation of protein and fat deposition in the animal body is done by means of balance studies. This method involves drawing of a balance sheet of particular elements that enters and leaves the body and the retention calculated by the difference of the two. A combined carbon and nitrogen balance will give an estimate of the protein and fat stored in the body at any particular period of time. For this, the animal must always be in a positive balance that is the excreted amount of carbon and nitrogen must be less than the intake.
- The amount of body protein and fat stored is calculated as follows.

- Assumptions
- All body proteins contain 160 g nitrogen/kg and
- Amount of carbon present in the protein 520 g carbon/kg
- Amount of carbon present in the fat is 746 g/kg

Calculation

- Amount of protein stored = Nitrogen retained x 6.25
- Amount of carbon stored as protein = Protein stored x 0.520
- As the amount of carbon in the fat is 746 g/kg, the remaining carbon stored as fat is calculated as
  - (Carbon balance - Carbon stored as protein) /0.746
- The disadvantage of this method is that it involves sophisticated equipments and is expensive. This method is suitable only for estimating the deposition of protein and fat under experimental conditions.

EXERCISE-15: MEASUREMENT OF BODY SURFACE AREA

Objective

- To measure the body surface area of bovines by surface integrator

Significance

- The skin, which covers the body surface, performs not only protective function but act as an agency of thermal exchange and thermoregulation, particularly in tropics. Greater body surface area is indicative of better adaptability of animals to hot climatic conditions.

Materials required

- Surface integrator
- Experimental animals
- Measuring tape
- White chalk.

METHODS OF MEASUREMENT

- On the live animal a satisfactory method to measure the body surface area is by the use of the surface integrator. This instrument was initially fabricated by Brody and subsequently modified by McDowell and his associates for bovines.
- It consists of two wheels, the circumference of which is known [27.94 cm], mounted in a fixed distance apart [8.12 cm]. One wheel is recording wheel, geared to a revolution counter.
- The other wheel has a roughened surface, which rubs off chalk from a stick fixed in a holder and leaves a white line on the surface over which it is running. The mechanism is mounted on a handle by which it can be pushed along.
- The area covered by the instrument can be calculated using the following formula

- A_{si} = (Circumference of wheel) X (Distance between two wheels) X (Number of revolutions)
  - where A_{si} = Area measured by surface integrator in cm^2

- The surface area is measured on one side of the animal, with the animal standing squarely on all four legs, as follows
  - Starting from the base of the tail, with the counter set at zero, the integrator is pushed along the back, with the plain wheel running along the centre line of the spinal column.
  - When the shoulder is reached, the integrator is lifted off, and a second run is started from the centre line under the tail with the plain wheel now running along the chalk line made by the marking wheel during the first run.
  - This process is repeated until midline of the belly and chest is reached. The counter is now read.
  - The counter is re-set at zero and run made over the area between the shoulder and head in vertical strips from the lower midline to the centre of the spinal column. At the time of measurement the animal's head is turned in the opposite direction to remove the wrinkles in the skin and the hand is held in opposite side of the dewlap.
  - The area of fore legs and hind legs are measured with integrator by placing the feet wide apart and the integrator run spirally in horizontal strips around the legs down up to the vicinity of knee and hock joint respectively. The counter is now read.
  - Lower part of legs below hock and knee joint are treated as cylindrical. The length is measured from last mark of integrator to the top of the hoof. Four circumferences first just below the last line of the integrator, the second and third just above and below the middle of the shank and fourth at the pastern is to be recorded and averaged.
    - Area = Circumference x Length
  - The tail is also considered as a cylindrical part. The length is measured from the base to the tip of the last coccygeal vertebrae. Five circumferences, first at the base, second and fourth just above and below the middle of the tail and third at the middle and fifth at just above the switch is to be recorded and averaged.
    - Area = Circumference x Length
  - The surface area of head is calculated by assuming the shape of the head as geometrical triangular. The length of the head is measured from the top of the poll to the tip of the mouth and the width determined from base of one ear to the other.
    - Area = ½ Width x Length
  - The ear is considered as possessing a rectangular shape. The area is determined by measuring outside longitudinal length from base to the tip and of the flattened ear at the middle of its length, where it is widest.
    - Area = Width x Length
  - The total surface area (cm^2) of the animal is determined by the equation:
    - Total surface area = $2 [A_{si} + F + R + E] + T + H$
    - where
      - A_{si} is the area measured by surface integrator
      - F is the area of the fore leg
      - R is the area of the rear leg
      - E is the area of the ear
      - T is the area of the tail
      - H is the area of the head.
      -

EXERCISE-16: DETERMINATION OF ANIMAL ENVIRONMENTAL CONDITIONS

- *Climatology:* It is the science that seeks to describe and explain the nature of climate, why it differs from place to place, and how it is related to other elements of the natural environment and to human activities.
- *Meteorology:* It is a branch of physics, which deals with atmosphere.
- *Weather*: It is the condition of the atmosphere at any particular place for a given time.
- *Climate*: It is the condition of the atmosphere over a period of time prevailing in larger area. It is the aggregation of the weather conditions during the period.
- Various environmental weather elements are
    - Temperature
    - Wind velocity
    - Humidity
    - Precipitation

Meteorological instruments

- The important instruments employed in recording the environmental conditions are grouped as
    - Eye reading instruments
    - Self-recording instruments

Classification of the Instruments

	Eye Reading	Self Reading
Temperature of air	• Maximum Thermometer • Minimum Thermometer	• Thermograph Bourdon Tube • Thermograph Bimetallic type
Temperature of soil	Soil Thermometer	Soil Thermograph
Temperature of grass	Grass minimum and maximum thermometers	
Wind direction	Wind vane	-
Wind speed	Robinson's cup anemometer	Dines pressure tube anemometer
Humidity	• Wet & dry bulb thermometers • Hand Whirl Hygrometer • Assman psychrometer	Thermo hygrograph
Rainfall	Symons Rain gauge	Casella's self recording Rain gauge

INSTRUMENTS USED FOR MEASURING TEMPERATURE

- Temperature: It is the degree of hotness or intensity of heat energy.
- Temperature influences all forms of life by altering the vital activities of living organisms like metabolism, growth, reproduction etc.

Temperature scales

- Centigrade or Celsius
- Fahrenheit.
- Kelvin or absolute temperature.

Units	Freezing point	Boiling point
Centigrade	0°C	100°C
Fahrenheit	32°F	212°F
Kelvin	273°K	373°K

Relation between these scales

- °C to °F; °F = (°C X 9/5) + 32
- °F to °C; °C = (°F - 32) X 5/9

Classification of temperature measuring instruments

- Based on the principle of physical effects of co-efficient of expansion of metals. e.g. (a) Liquid in glass thermometers (b) Bi-metallic strip
- Based on the principle of resistance to flow of electricity. e.g. Thermocouple.
- Based on the emission of infra-red rays. e.g. Infra red thermometer.

Liquid in Glass Thermometers

- Maximum Thermometer
  - It is used to measure the maximum temperature of the particular day. When the temperature increases, mercury expands and is forced through the constriction and flows through the stem.
  - When the temperature falls or drops, the constriction prevents the back flow of mercury keeping the mercury level at the highest point. After the reading is taken, the instrument is set by giving a jerk.
- Minimum Thermometer
  - It is used to measure the minimum temperature in a day. The liquid used is alcohol. It facilitates measuring of temperature even below the freezing point.
  - When the temperature drops the alcohol contracts and pulls the metal index towards the bulb edge by the effect of surface tension.

- When the temperature increases, alcohol expands and flows through the stem leaving the index at the lowest point. After the reading is taken the instrument is set by giving a tilt upwards.



Bi-metallic strip Thermograph

- It is an automatic self-recording instrument for measuring the temperature continuously throughout the day and night. It consists of a bi-metallic strip made up of two metals invar and bronze, which have differential co-efficient of expansion of heat.
- Change in temperature causes change in curvature in the strip due to differential co-efficient of expansion. The curvature is magnified by the lever system to the pen-point.
- The pen point marks on the graph sheet, which is fixed on the drum driven by the clock mechanism at the rate of one rotation/day or week.

INSTRUMENTS FOR DETERMINING DIRECTION AND VELOCITY OF WIND

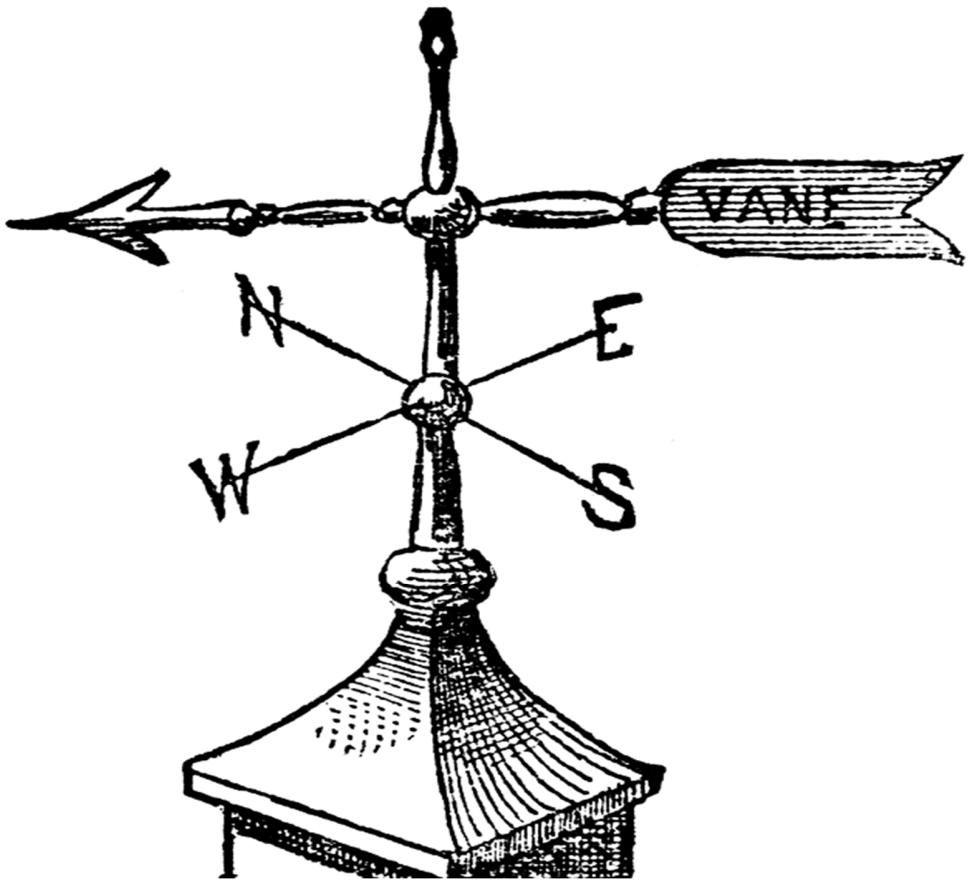
- *Wind:* Wind is the movement of air in the horizontal plane, caused by pressure differences. The direction to which the wind blows is termed as *Leeward Side* and the direction from which it comes is known as*Windward Side*.
- Decreased wind velocity builds up humidity and heat stress on the animals. Continuous exposure of animals to increased wind speed leads to dehydration. Birds seek shelter during high wind speed.

Wind vane

- It is an instrument that indicates the direction of the wind. It consists of a pointer in which one end is arrow headed that point to windward direction and the other end is broad called SAIL (fang) that faces leeward side.
- The pointer is attached to the central axis, which is fixed to the spindle of a ball bearing system enabling the pointer to move in a horizontal plane indicating the direction of the

wind. Four cross arms fixed rigidly a little below the vane are directed towards east, south, west and north.

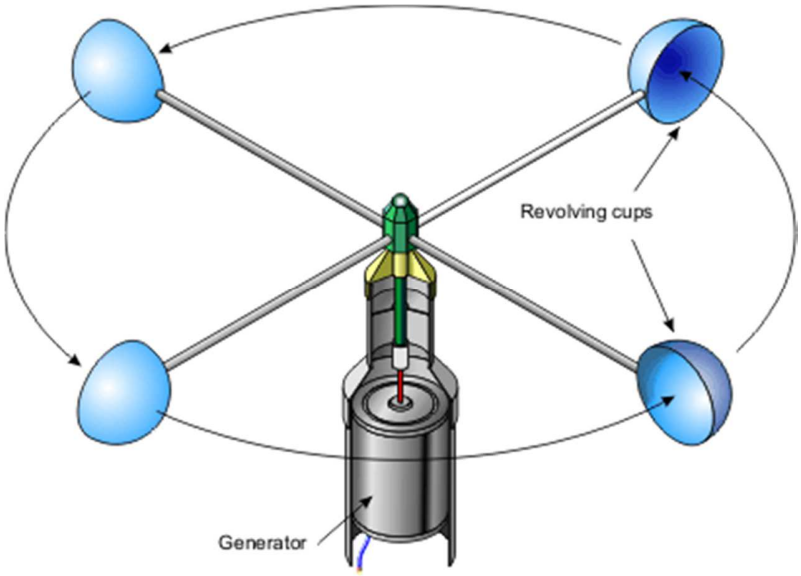
- The vane is fixed at a good height on the top of vertical post in spots not subject to obstructions to the movement of air. If exact direction of the wind is to be known, the direction is indicated by means of degrees taking the 'North' as 0 ° and the other directions marked clockwise.



Anemometer

- It is an instrument used for measuring the speed of wind. The wind speed is expressed in Km/hr or Knots (1 Knot - 1.854 km/hr), or meters per second (m/sec).
- Robinson's cup anemometer
    - Three light aluminium or copper hemispherical cups are fixed to the arms of a steel cross, mounted on a vertical axle. The axle is connected by gear arrangement to a dial, from which the velocity of wind can be read off.
    - Wind is deflected by the convex side of the cup, while the concave side offers resistance to the wind and takes its impact. This sets the cross arms and the central shaft moving with the convex side foremost when there is wind.
    - To determine the wind speed at the time of observation, take two successive readings of Anemometer at an interval of 3 minutes, Subtract the first reading from the second one and multiply the difference by the factor 60/3 and the wind speed will be obtained.

- The wind speed per day is given from the difference between two successive readings at an interval of 24 hours taken at a fixed time.



INSTRUMENTS USED FOR MEASURING HUMIDITY

Humidity

- Humidity is the amount of water vapour present in the air. Various terms are used to measure the amount of humidity of which the expression *relative humidity* is commonly used.

Relative Humidity

- Relative humidity is the ratio of water vapour actually present in the air to the maximum water vapour air could hold at a particular temperature and pressure. It is expressed in percentage.
- The instruments used for the measurement of humidity are
  - Wet and dry bulb thermometer
  - Hygrometer and
  - Hygrograph.

Wet and dry bulb thermometer

- It consists of two identical thermometers with mercury. The wet bulb is covered with a fine muslin cloth connected by a wick to water held in a small container below.
- As water evaporates from the muslin cloth, the bulb is cooled and the thermometer gives a lower reading. The depression of the reading (wet bulb depression) is in proportion to the quantity of water evaporated from the muslin cloth. This in turn depends on the temperature and humidity of the atmosphere.

- When the relative humidity is low, the evaporation of water from the wet bulb is high and the depression in the reading will be more.
- When there is more moisture in the atmosphere, the depression is less. This difference in temperature can be translated into relative humidity with the help of hygrometric tables.

Hand whirl hygrometer

- This instrument consists of a wooden frame on which a wet bulb thermometer and a dry bulb thermometer are suitably attached.
- To one end of the frame a small handle is attached, holding which, the whole frame can be rotated for a few minutes and then the dry bulb and wet bulb readings are noted. The process is repeated 3 or 4 times to ensure correct reading of the wet and dry bulb temperatures. With the help of the table supplied with the hygrometer the relative humidity can be found out.
- This instrument is useful for finding out the relative humidity at various places like outside or inside of a room, at different heights, inside a crop etc.

Hair hygrometer

- It consists of human hair, mounted between two points attached to an indicator over a calibrated dial. As the relative humidity increases, the hair lengthens and if the water vapour in the air is less, the hair shrinks.
- Fluctuation in the hair length is transmitted to the indicator, which shows the relative humidity in terms of percentage. It gives direct reading of the relative humidity.

Hygrograph

- It is an automatic self-recording instrument used for measuring the relative humidity. The expansion or contraction of human hair is magnified by a lever (connected to the hair on one end) is transmitted to a pen arm connected to the other end.
- The pen arm marks on the graph sheet giving a direct reading of the per cent of relative humidity. The graph sheet is fixed on a drum, which is operated by a clock mechanism. The hygrograph may be of weekly or daily type.

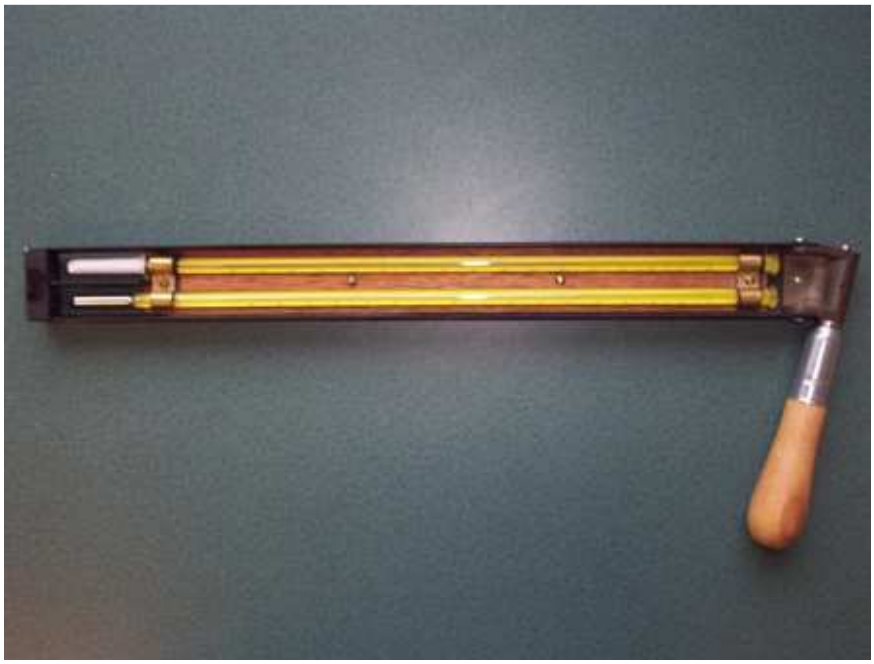
Thermo hygrograph

- It is an automatic self-recording instrument for recording temperature and relative humidity continuously during day and night.

Sling psychrometer

- It consists of two identical thermometers one of which has a thin muslin wick. The cloth is wetted with water and a continuous current of air is passed over it, which evaporates water and the temperature of wet bulb drops according to the rate of evaporation, that is

recorded and humidity is calculated by using separate table. It is also useful to find out the micro-climatic condition.



INSTRUMENTS USED FOR MEASURING RAINFALL

- The water vapour present in the atmosphere condenses on the dust particles and are held against gravity by the ascending currents of warm air. As they go above, there is further cooling and condensation. The droplets coalesce together.
- When the size of the droplets is more than 0.2 mm in diameter the ascending currents are not able to support them and as a result they fall to the ground as raindrops.

The instruments used for the measurement of rainfall are,

- Ordinary rain gauge



Self-recording rain gauge

EXERCISE-17: FEEDING BEHAVIOR IN ANIMALS

- Animals fed in groups are likely to be less fearful and hence more contended, healthier and more productive. The common practice of feeding and milking of dairy animals in groups has sound psychological impact on production of milk.

Aim

- To demonstrate the effect of feeding behaviour in animals (sheep).

Procedure

- Select group of ewes of same age with known weight (kg) and divide them equally into two groups. Withdraw the feed for three hours. Feed concentrate to the ewes in common pen.
- Observe for social order, dominance and feeding behavior of the animals.

MILKING BEHAVIOUR IN ANIMALS

- Understanding the cow's behavioural and sensory characteristics makes the handling easier. Cows entering the milking area calmly and voluntarily will have a natural milk ejection reflex. Fearful cows are more difficult to handle and these stressed cows suffer a decrease in milk yield and quality.
- *Social behaviour:* Cows need to have a social space. Hence enough space should be provided for the cows.
- *Fear:* Cows might get fear when there is unfamiliar sound, or sudden changes in lighting. These fearful cows are more difficult to handle and show exaggerated response to handling. Hence the animals should be familiarised with changes in environment. Cows, if handled bad, will lead to fear for human beings.
- *Milk-let down:* This is a natural process by which the milk is ejected from the udder with the action of oxytocin, released from the posterior pituitary, following suitable stimuli eg. The suckling of teats by calf.
- In fearful cows, the secretion of adrenalin will block the let down of milk. Gentle handling of the cow animal maximises milk production.

REPRODUCTIVE BEHAVIOUR IN ANIMALS

- Observation of animals in oestrus cycle will help to improve conception rate and animal productivity. Detection of oestrus sign sometimes go unnoticed in animals resulting in failure of reproduction.

Aim

- To observe the reproductive behaviour of domestic animals

Procedure

Sexual behaviour

- Male
  - The act of copulation involves sequential order of behaviour in male. It involves the following processes.
    - 1. Sexual awakening, foreplay and courtship.
    - Penile erection and protrusion.
    - Mounting and penile intromission. 4. Orgasm and semen ejaculation. 6. Dismounting and refractoriness.
  - During courtship, different species express different behaviour - Flehmen reaction, is strongly expressed in bulls, butting of female neck is seen in stallion. Except in swine, the males of domestic ungulates smell the female's urine and raise its head with lips curled – this is called *Flehmen reaction*. The sexual behaviour in males can be estimated by grading sex libido. Reaction time is the period used by males from its exposure to female till the male ejaculates semen into the female genetalia. Reaction is inversely proportional to sex libido- short reaction time indicates the excellent libido.
  - In cattle, male sexual behaviour is observed throughout the year. The male buffalo also has similar behaviour as cattle; however, it is more sluggish than cattle, reaction time ranges from 28-122 seconds compared to 1-5 seconds in cattle. In females, estrus animals show increased motor activity, become restless and moves at the slightest disturbance. Receptive animals bellow or bleat and sows utter typical grunt. Cows, sows and goats tend to mount and allow other animals to be mounted (except in ewes and mares). Estrus mares tend to urinate frequently in the presence of stallion. When approached by male, the female assumes a mating posture with the deviation of its tail.

Factors affecting sexual behaviour

- Genetic, environment (season, climate, presence of other animals), previous experience etc

Maternal and neonatal behaviour

- *Suckling:* The young may reach the udder within a few seconds after birth, while the umbilical cord is still attached. They express the suckling response as soon as their nose makes contact with other objects. Suckling excites nervous reflex that augments secretion of hormones from the posterior pituitary. Fright or adrenaline inhibits milk let down.

Parent- Young interaction

- After birth the young are able to move about almost immediately. During this important period the young normally establish a species bond to their species.

- Recognition of young is established in sheep and goats soon after parturition and other young may be adopted at that time. Continuous contact is important and females may reject their young ones if separated from them for few more hours immediately after parturition. During and immediately following birth, the mother chews and licks the placenta (*placentophagy*). The mother may consume a part of it and groom the neonate. Cows to tend to eat the placenta after birth more frequently than do sheep. Neonate calves may take 45 minutes to stand and may start the first suckle about 4 hours after birth.
- Observe the female animals, which are in “heat” (estrus) and note down the behaviour of the animals of various species.
- Similarly, observe the males for mating behaviour and record it.